mcfet

process design review · 859 steps · generated by CarbonEDA v0 on 2026-11-09 19:58 EST

743.1

kgCO2e per wafer start

506.4
EPW energy · 68%

67.6
MPW materials · 9%

169.0
GPW gases · 23%

wafer 300 mm · yield 1 · carbon intensity 0.43 kgCO2e/kWh · facility overhead 0.4 x tool energy
EPW 506.4 = tool 361.7 + facility overhead 144.7 kgCO2e (841 + 337 kWh)

flow technologies/logic/mcfet/mcfet_final.csv
layout technologies/logic/mcfet/mcfet_layout.gds
structure images from the spx run of 2026-11-09 18:17:22

CarbonEDA parameters

config/carbon_config.yaml and materials.yaml as used for every number in this report

EPW energy

(tool + 0.4 x tool facility overhead) x 0.43 kgCO2e/kWh

process area	kWh / step	steps
RIE	1.33	144
Ion_implantation	1.33	7
Diffusion	1.33	38
Metallization	2	69
Wet_etch	0.28	4
Wafer_clean	0.28	170
Resist	0.28	72
CMP	0.28	49
CVD	1.09	229
EUV	3.25	32
ArF-i	0.36	28
ArF	0.33	8
KrF	0.33	3
Hg	0.33	6

GPW process gases

kg x (1 - utilization) x (1 - abatement) x GWP, plus byproducts kg/kg x (1 - abatement) x GWP; defaults: utilization 1.0, abatement 0.9

area	gas	g / step	upstream*	utilization	abatement	byproducts (kg per kg gas)
CVD	NF3	2.54	12.5	0.8	0.95	CF4 0.046, C2F6 0.045, CHF3 0.025, CH2F2 0.00086,...
RIE	C4F8	0.16	12.1	0.82	0.98	CF4 0.045, C2F6 0.031, C4F6 0.018, CHF3 0.012, CH...
RIE	SF6	0.439	125.6	0.71	0.95	CF4 0.04, C2F6 0.044, CHF3 0.0037, CH2F2 2.9e-05

* kgCO2e per kg of gas made, counted in MPW

GWP (kgCO2e per kg, 100-yr) and abatement of every gas emitted

gas	GWP	abatement
CF4	7390	0.89
C2F6	12200	0.95
C4F6	0	0.98
C4F8	10300	0.98
CHF3	14800	0.98

gas	GWP	abatement
CH2F2	675	0.98
CH3F	116	0.98
NF3	17200	0.95
SF6	22800	0.95

MPW materials

blanket film over a 300 mm wafer: thickness x area x density x GWP; the wafer itself 26000 gCO2e (300 mm); yield 1

material	g/cm³	gCO2e/g
TiN	5.21	43.98
W	19.3	75.81
TEOS	0.94	2.86

material	g/cm³	gCO2e/g
TaN	13.7	25.82
Cu	8.96	6.09
Co	8.9	44.5

Step #20: Wafer

substrate · Wafer

26	52	3.5%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

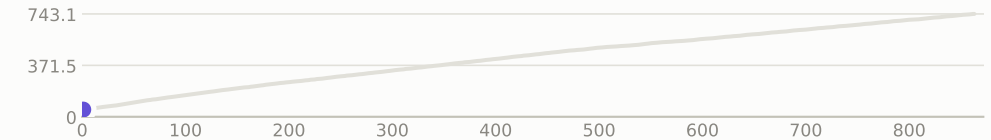
this step

process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	26 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**26 kgCO2e**

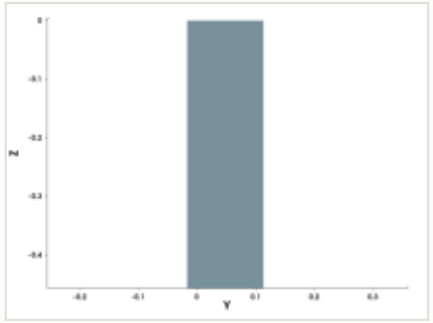
Wafer step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

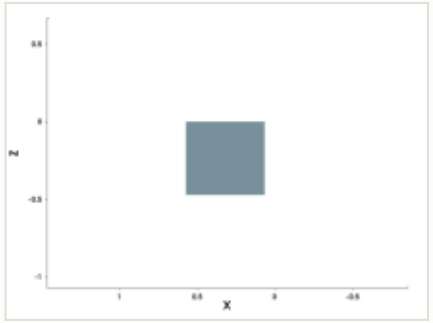
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

steps by process area

materials deposited (g)

gases emitted (g)

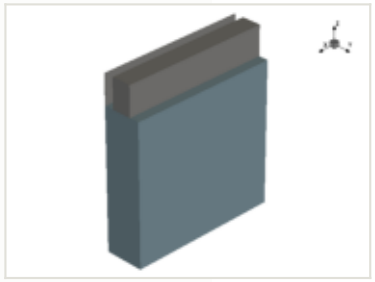
Step #40: KrF

litho_patterning · KrF

0.199	52.2	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

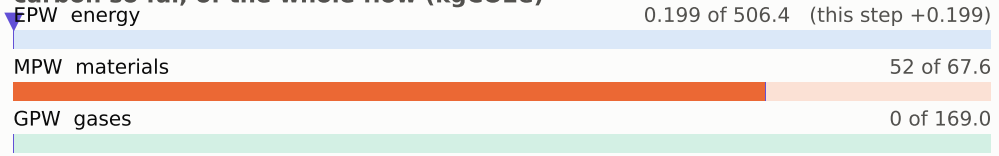
zoomed to fit



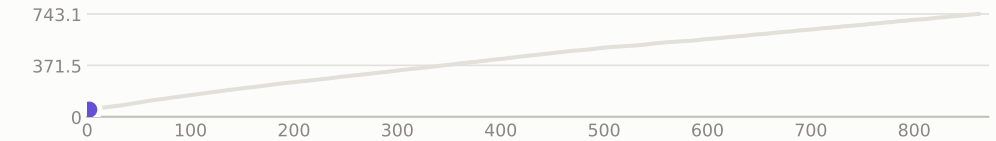
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

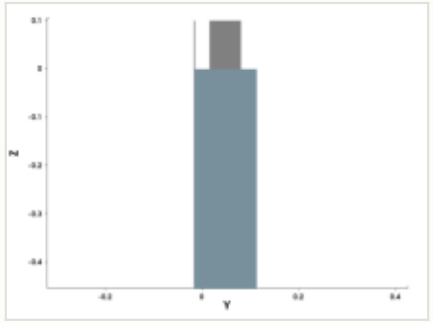
carbon so far, of the whole flow (kgCO2e)



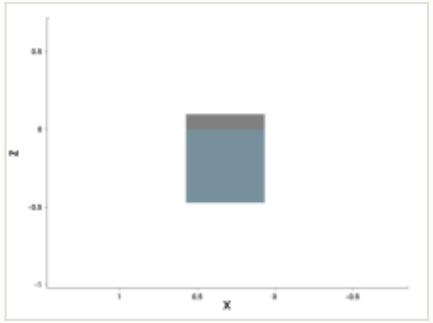
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

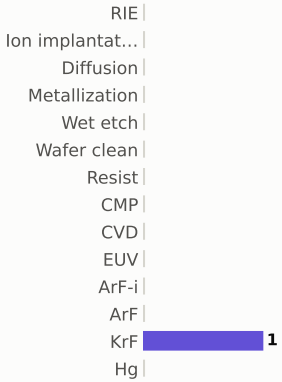


y cut at 0.049 µm

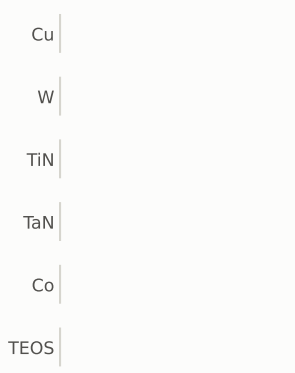


mask: layer 8:0 (GDSII)

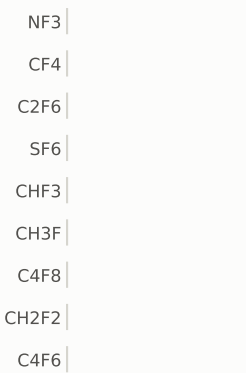
steps by process area



materials deposited (g)



gases emitted (g)



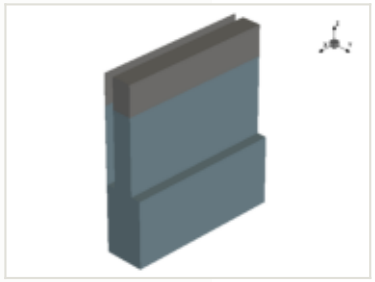
Step #50: RIE

etch_anisotropic · RIE

1.04	53.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

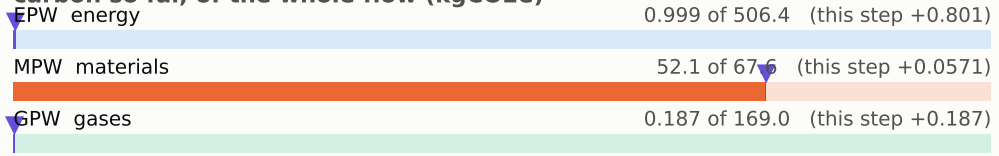


this step

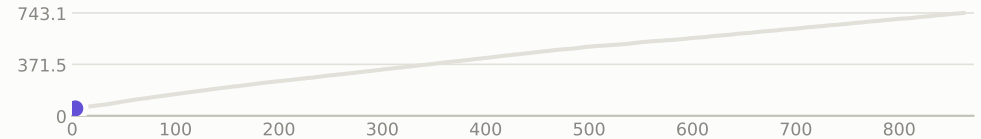
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

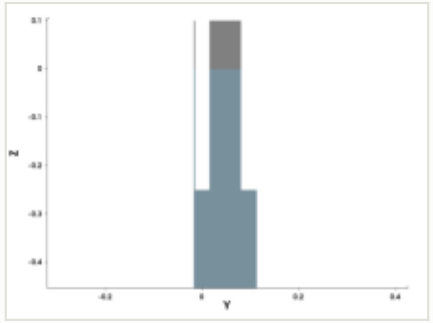
carbon so far, of the whole flow (kgCO2e)



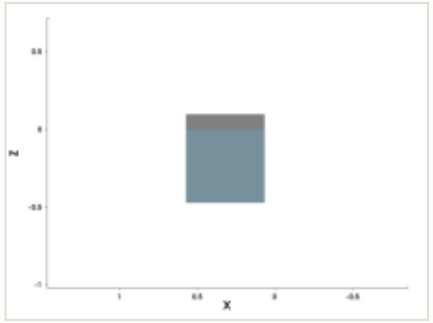
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



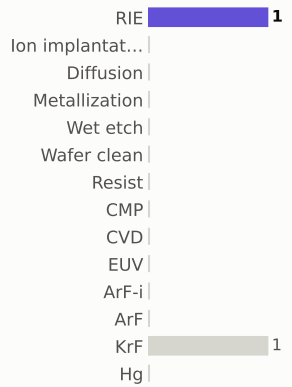
y cut at 0.049 µm



mask for this step

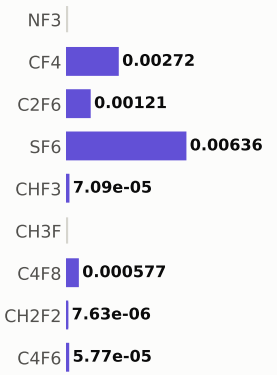
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



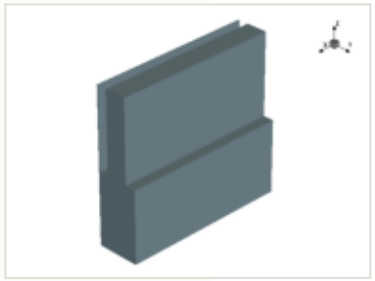
Step #60: Resist

etch_strip · Resist

0.169	53.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

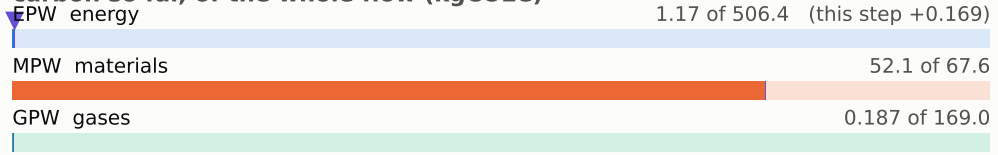


this step

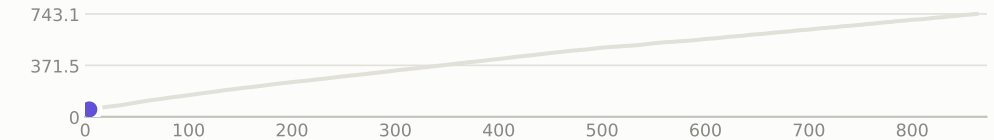
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

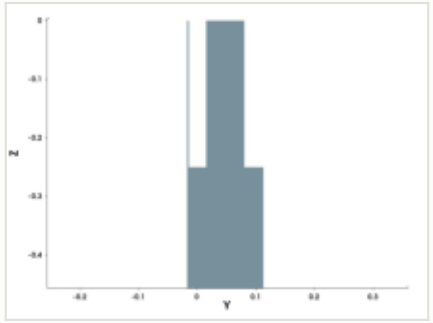
carbon so far, of the whole flow (kgCO2e)



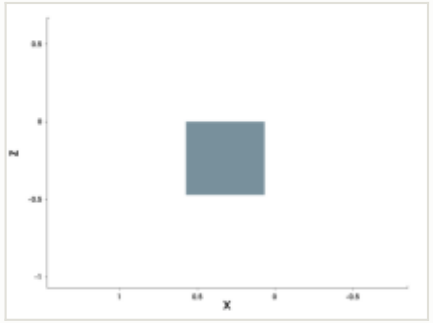
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



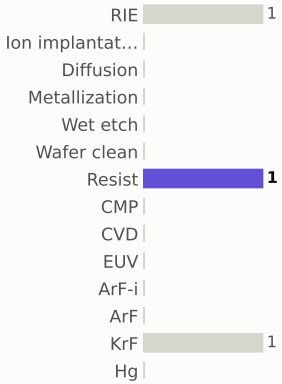
y cut at 0.049 µm



mask for this step

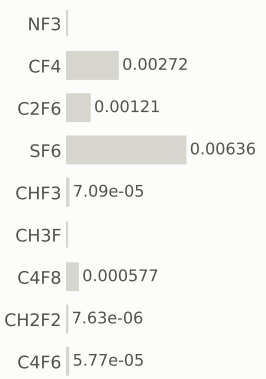
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #70: CVD

depo_isotropic · CVD

1.31	54.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

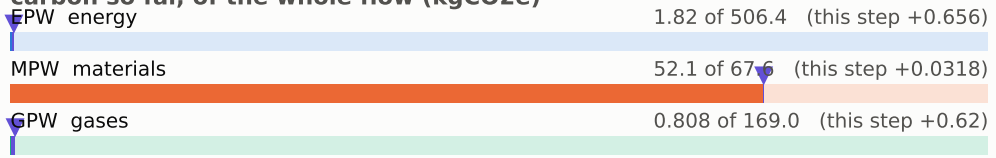
3D structure after this step

zoomed to fit

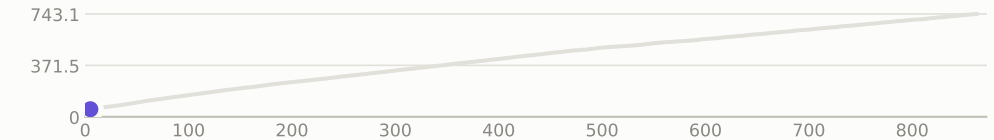
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.005 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

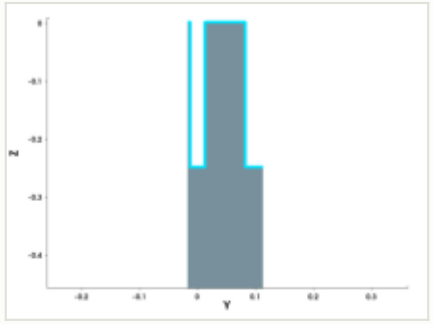
carbon so far, of the whole flow (kgCO2e)



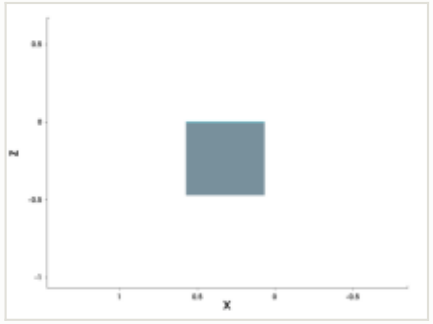
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



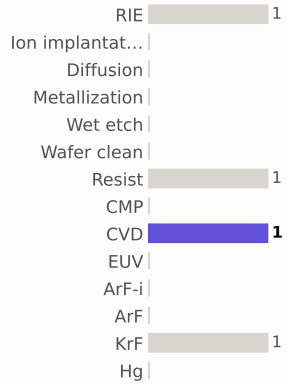
y cut at 0.049 µm



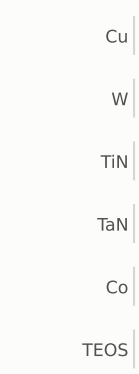
mask for this step

no mask for this step

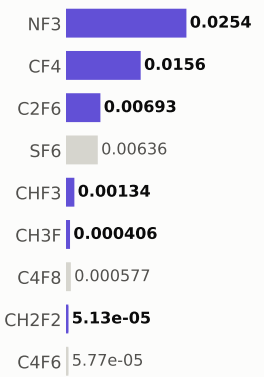
steps by process area



materials deposited (g)



gases emitted (g)



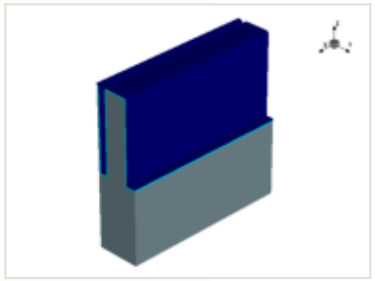
Step #80: Metallization

depo_isotropic · Metallization

1.2	55.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

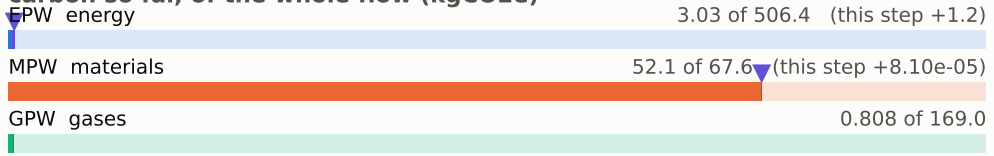


this step

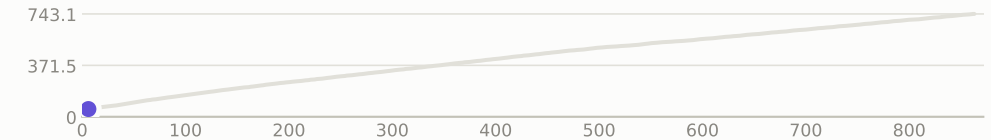
process area	Metallization
material	TiN
thickness	0.005 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	8.10e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

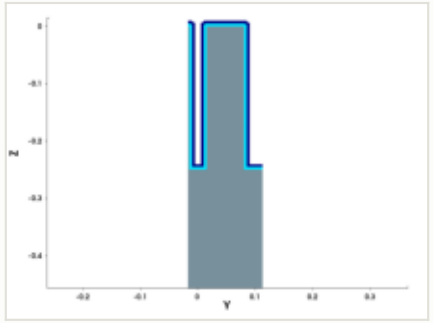
carbon so far, of the whole flow (kgCO2e)



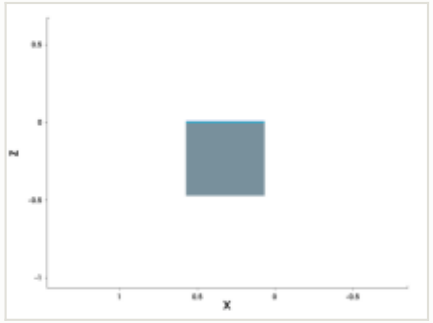
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



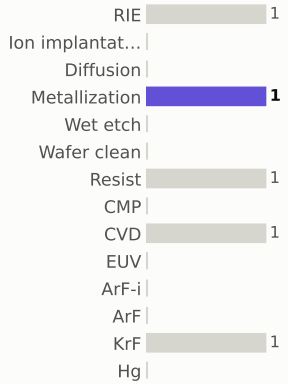
y cut at 0.049 µm



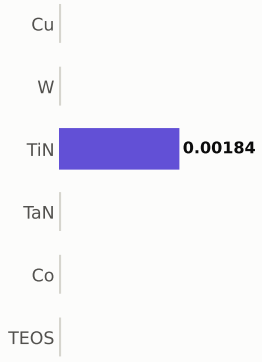
mask for this step

no mask for this step

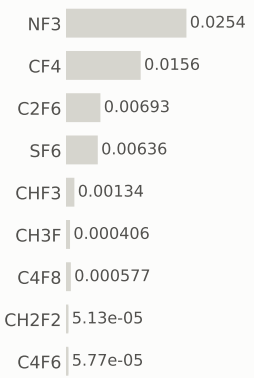
steps by process area



materials deposited (g)



gases emitted (g)



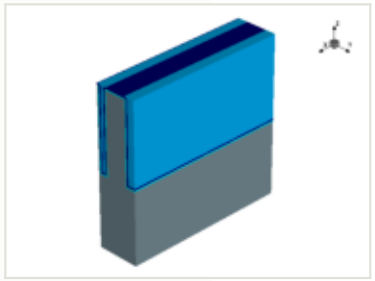
Step #90: Metallization

depo_fill · Metallization

1.2	57.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

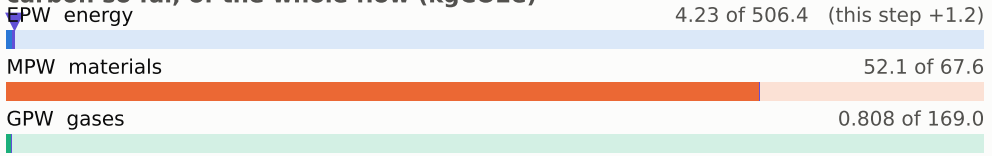


this step

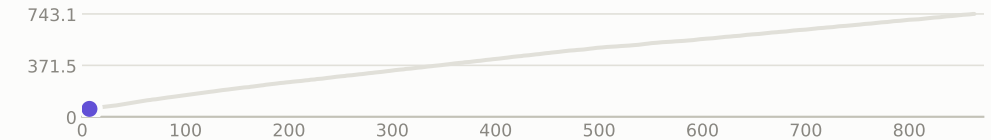
process area	Metallization
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

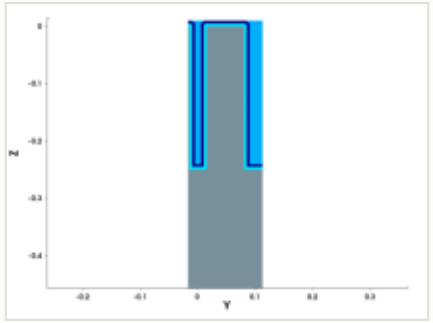
carbon so far, of the whole flow (kgCO2e)



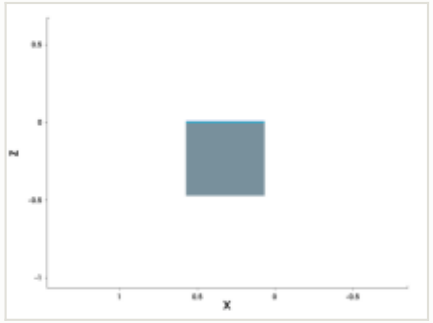
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



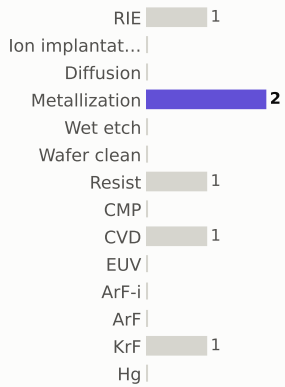
y cut at 0.049 µm



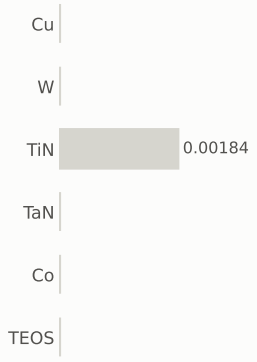
mask for this step

no mask for this step

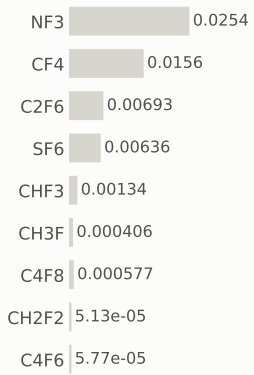
steps by process area



materials deposited (g)



gases emitted (g)



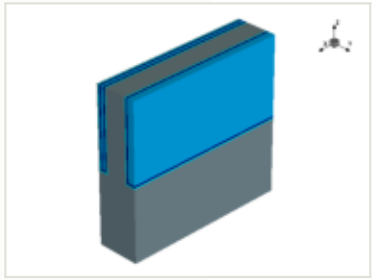
Step #100: CMP

etch_cmp · CMP

0.169	57.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

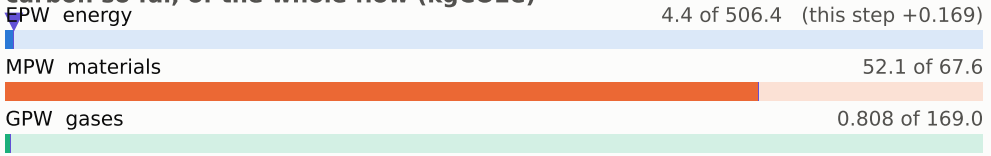


this step

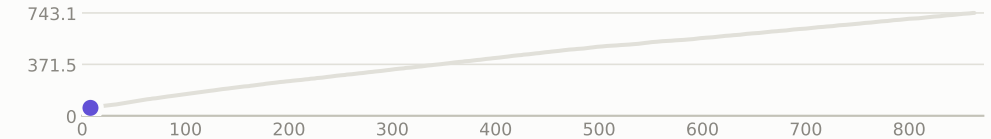
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

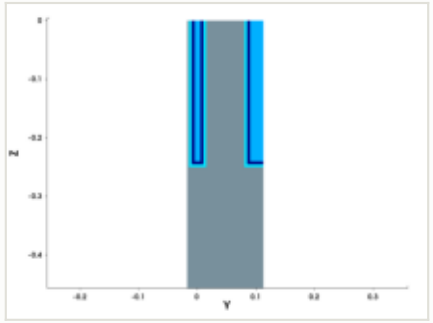
carbon so far, of the whole flow (kgCO2e)



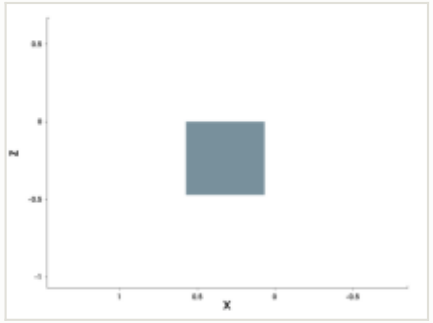
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



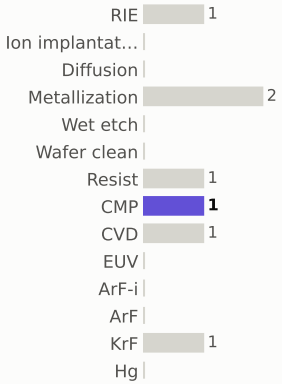
y cut at 0.049 µm



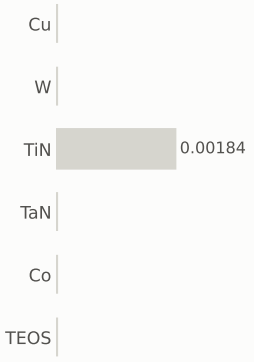
mask for this step

no mask for this step

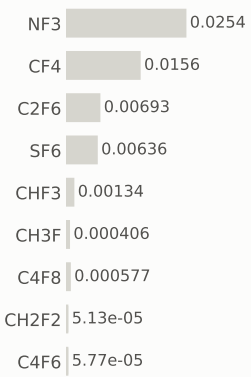
steps by process area



materials deposited (g)



gases emitted (g)



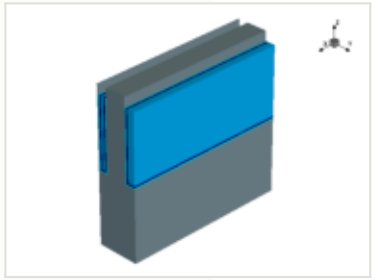
Step #110: RIE

etch_anisotropic · RIE

1.04	58.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

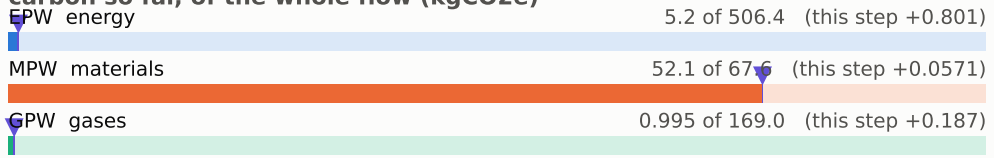


this step

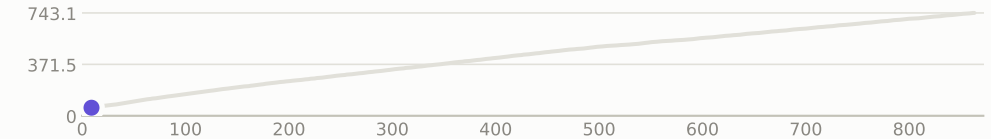
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

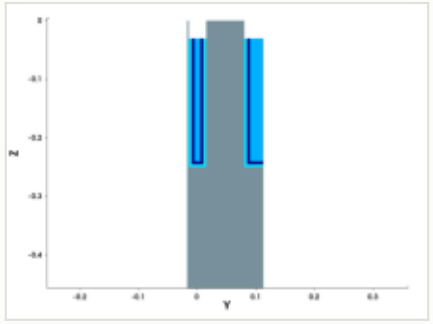
carbon so far, of the whole flow (kgCO2e)



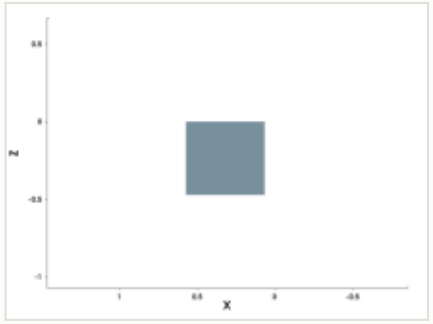
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



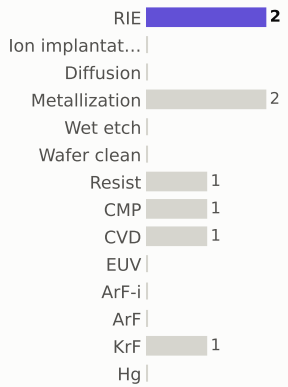
y cut at 0.049 µm



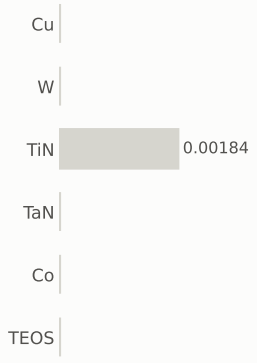
mask for this step

no mask for this step

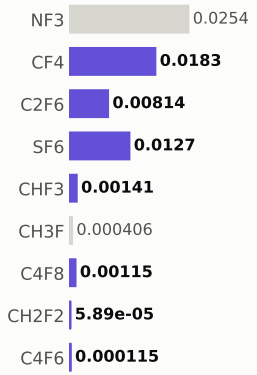
steps by process area



materials deposited (g)



gases emitted (g)



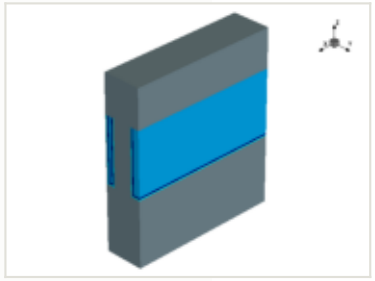
Step #120: CVD

depo_fill · CVD

1.31	59.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

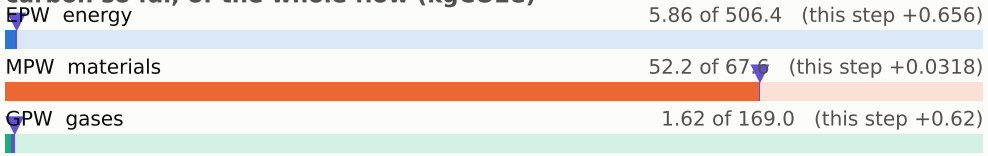


this step

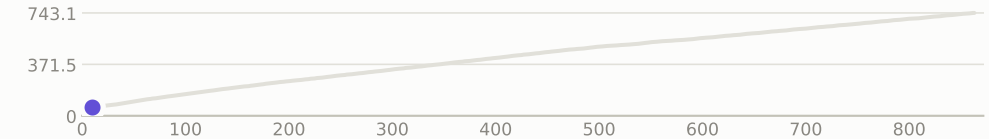
process area	CVD
material	—
thickness	0.1 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

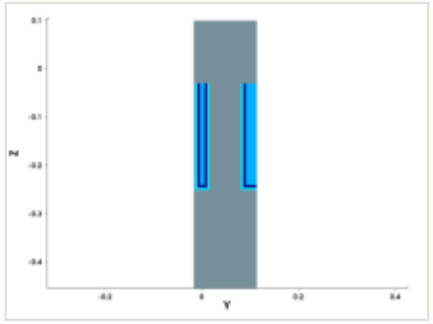
carbon so far, of the whole flow (kgCO2e)



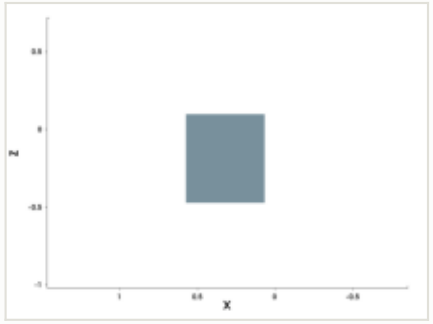
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



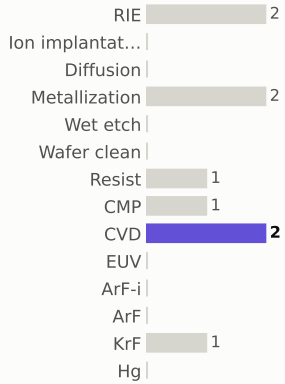
y cut at 0.049 µm



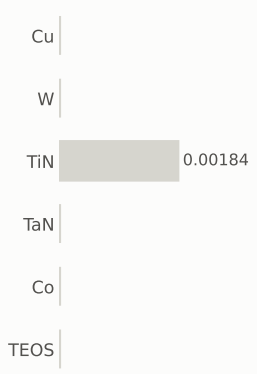
mask for this step

no mask for this step

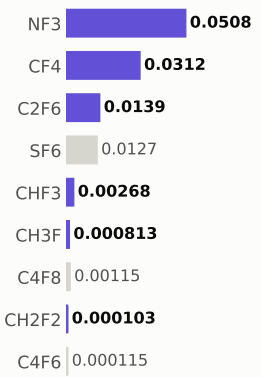
steps by process area



materials deposited (g)



gases emitted (g)



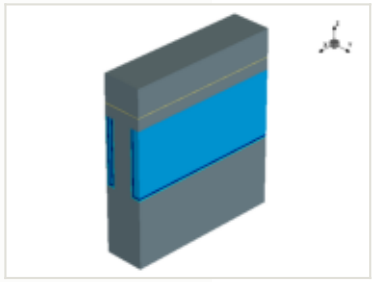
Step #140: Ion_implantation

implant_ellipsoid · Ion_implantation

0.801	60.5	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

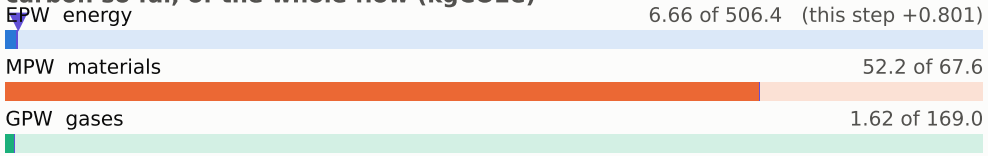
zoomed to fit



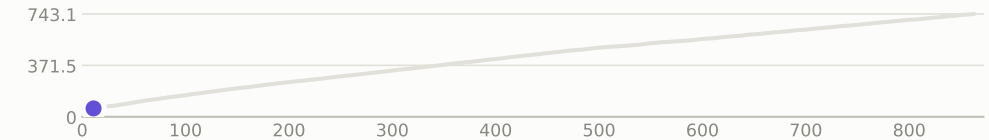
this step

process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

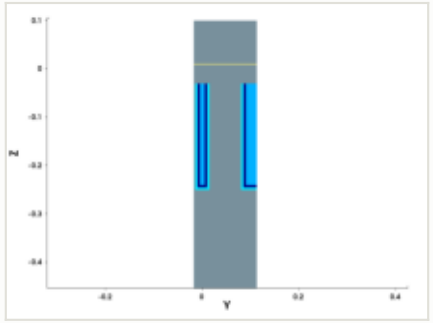
carbon so far, of the whole flow (kgCO2e)



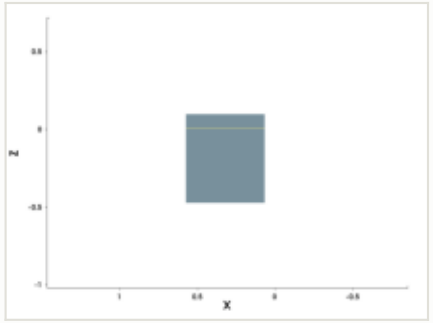
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



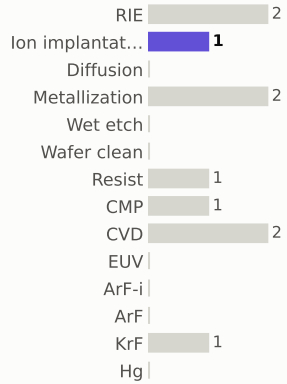
y cut at 0.049 µm



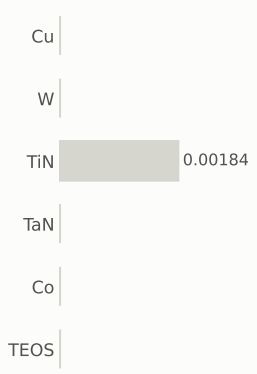
mask for this step

no mask for this step

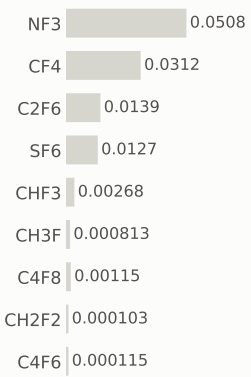
steps by process area



materials deposited (g)



gases emitted (g)



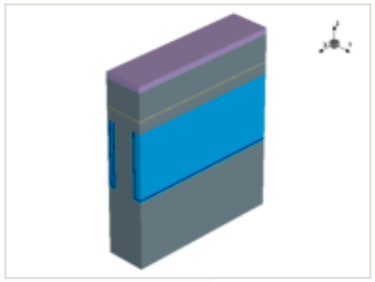
Step #160: CVD

depo_fill · CVD

1.31	61.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

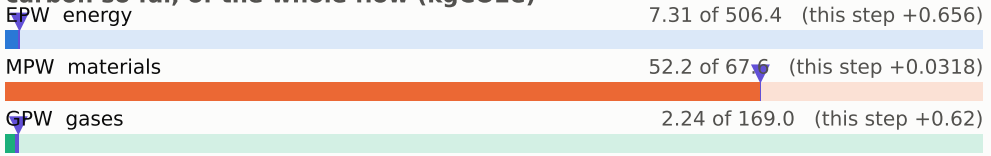


this step

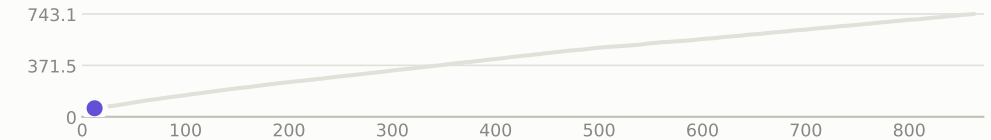
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

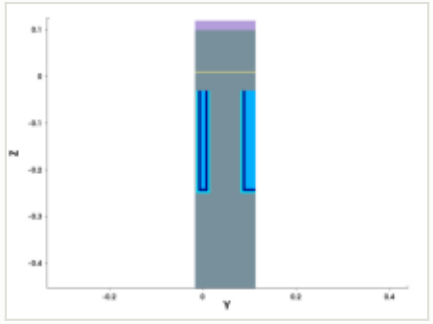
carbon so far, of the whole flow (kgCO2e)



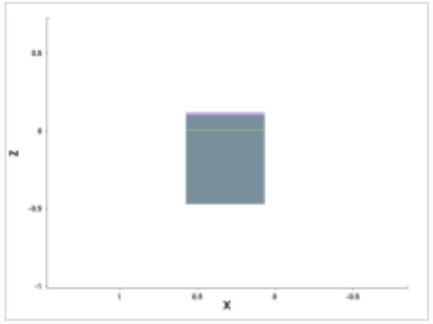
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



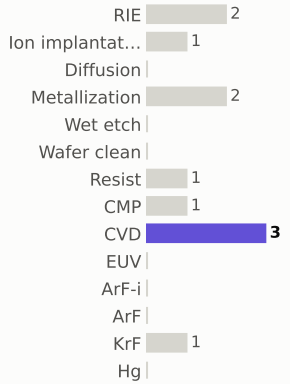
y cut at 0.049 µm



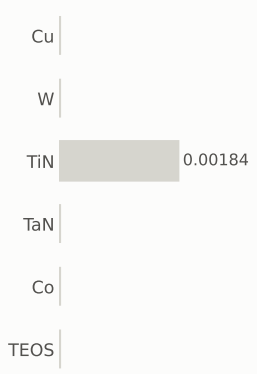
mask for this step

no mask for this step

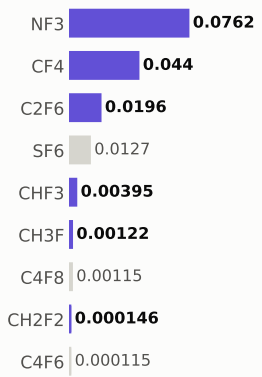
steps by process area



materials deposited (g)



gases emitted (g)



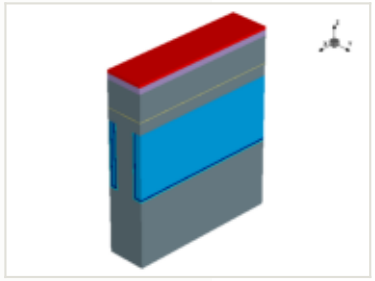
Step #170: CVD

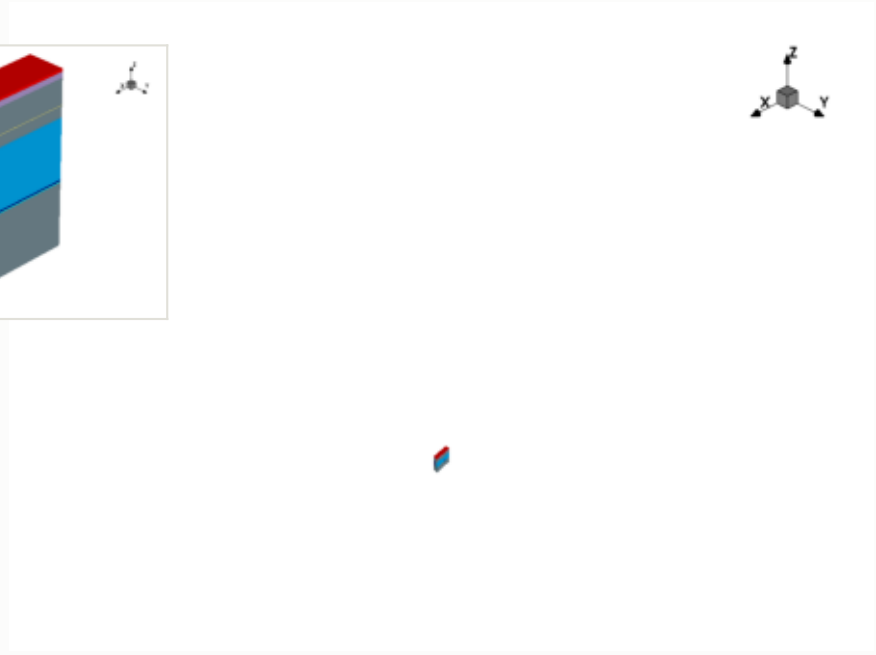
depo_isotropic · CVD

1.31	63.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



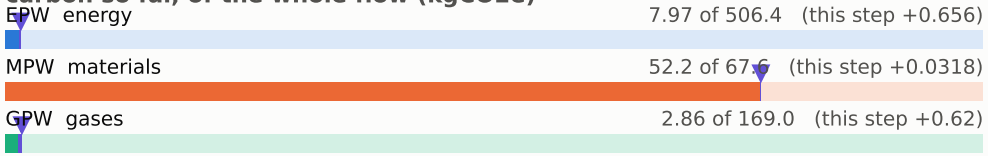


this step

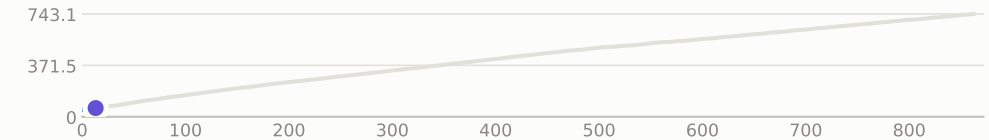
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

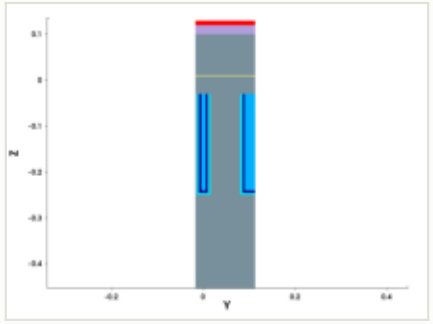
carbon so far, of the whole flow (kgCO2e)



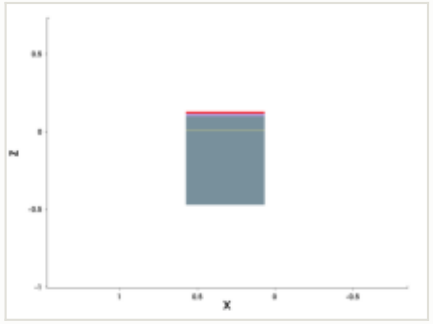
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



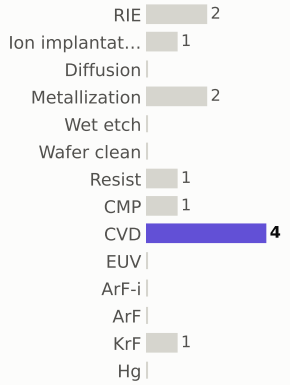
y cut at 0.049 µm



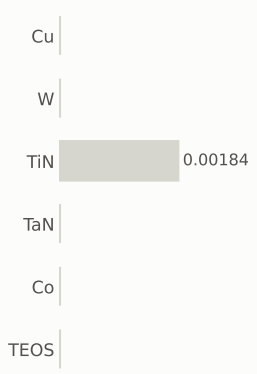
mask for this step

no mask for this step

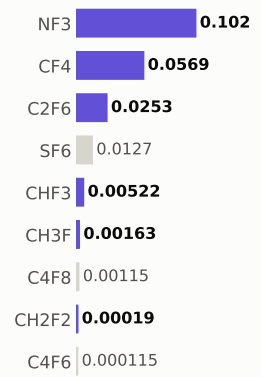
steps by process area



materials deposited (g)



gases emitted (g)



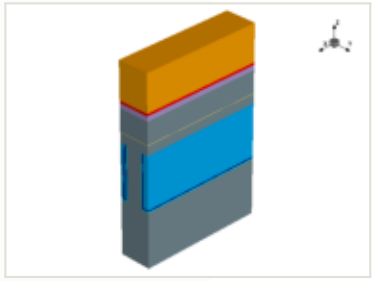
Step #180: CVD

depo_fill · CVD

1.31	64.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

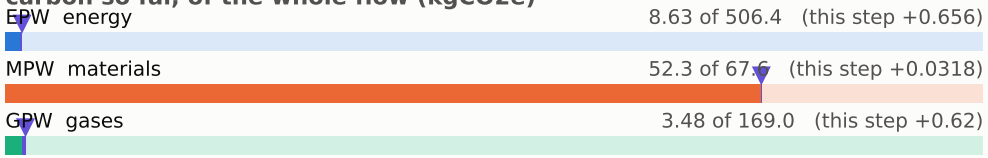
zoomed to fit



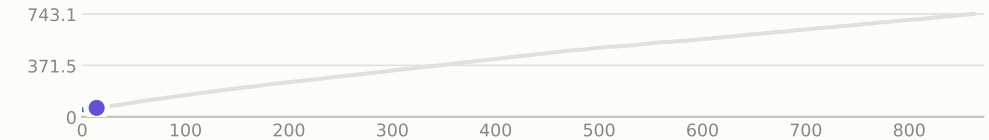
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.15 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

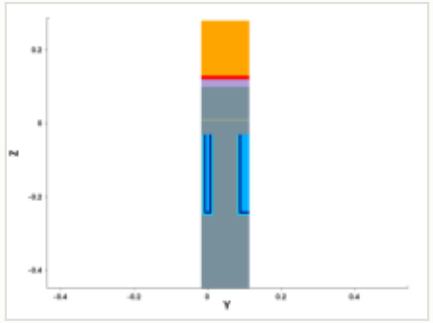
carbon so far, of the whole flow (kgCO2e)



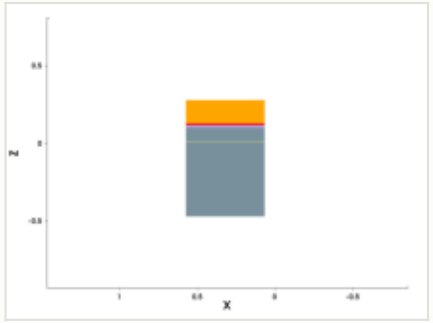
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



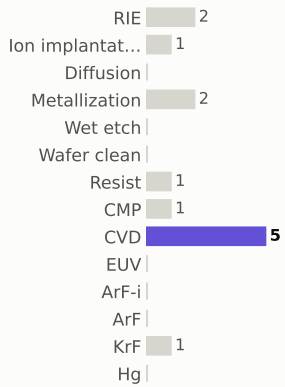
y cut at 0.049 µm



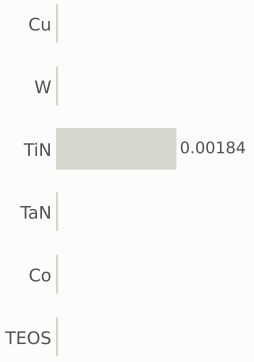
mask for this step

no mask for this step

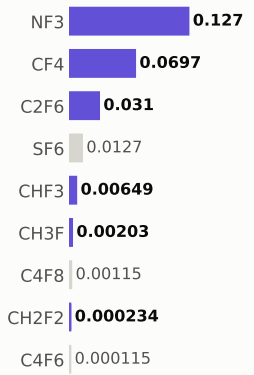
steps by process area



materials deposited (g)



gases emitted (g)



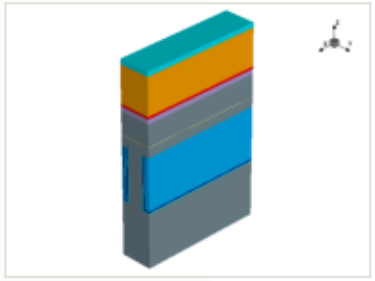
Step #190: CVD

depo_isotropic · CVD

1.31	65.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

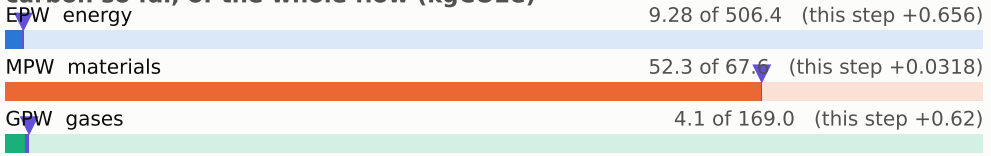
zoomed to fit



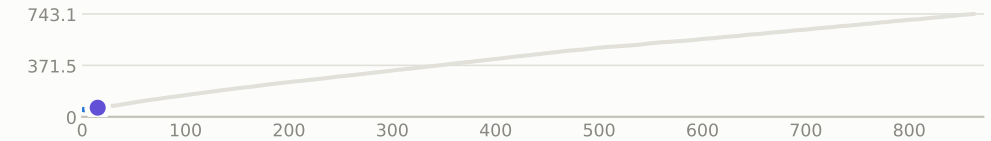
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.02 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

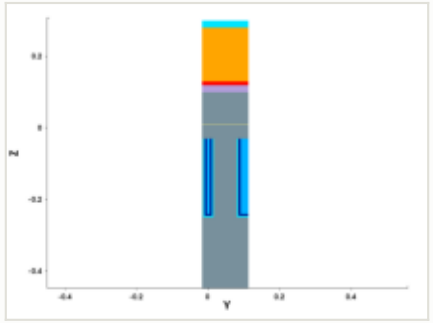
carbon so far, of the whole flow (kgCO2e)



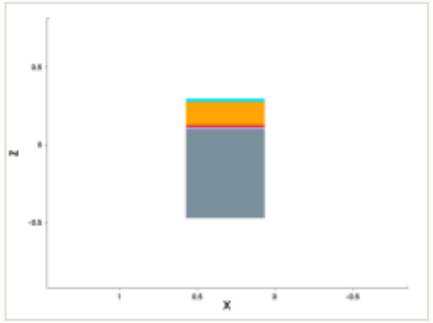
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



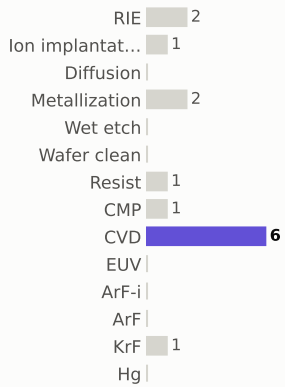
y cut at 0.049 µm



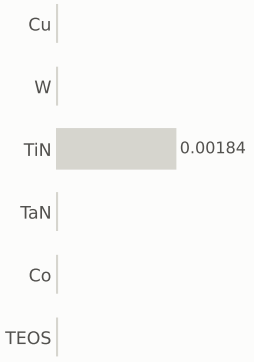
mask for this step

no mask for this step

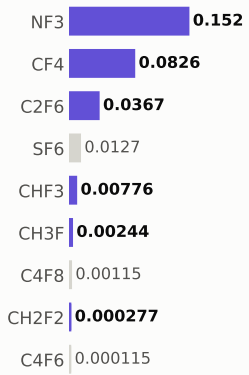
steps by process area



materials deposited (g)



gases emitted (g)



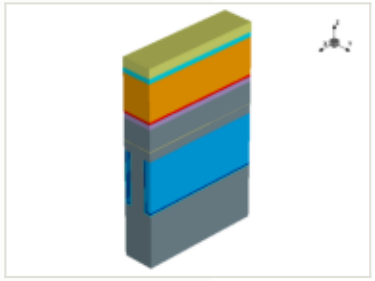
Step #200: CVD

depo_isotropic · CVD

1.31	67	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

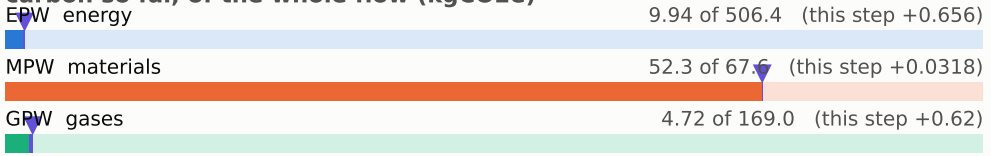


this step

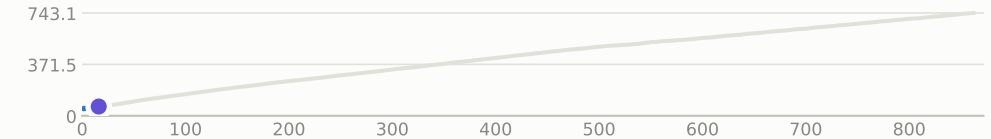
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

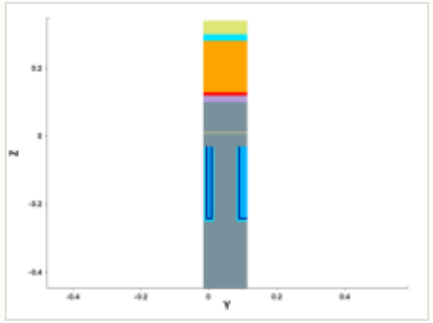
carbon so far, of the whole flow (kgCO2e)



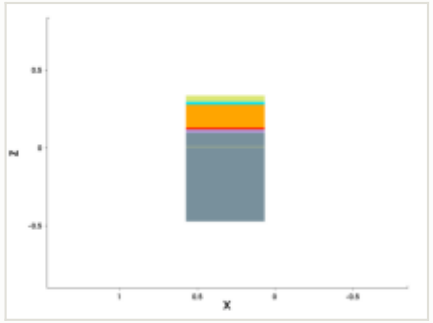
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



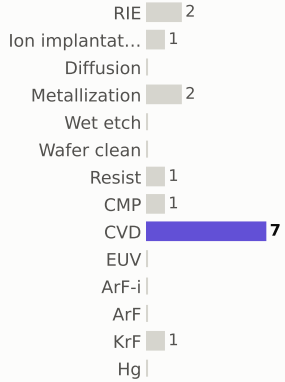
y cut at 0.049 µm



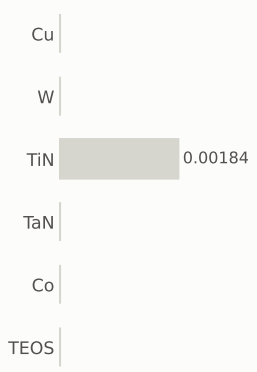
mask for this step

no mask for this step

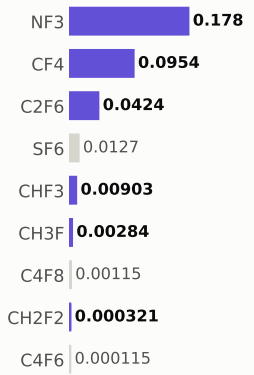
steps by process area



materials deposited (g)



gases emitted (g)



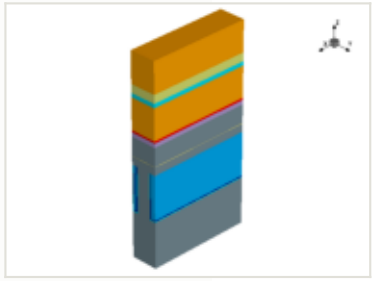
Step #210: CVD

depo_fill · CVD

1.31	68.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

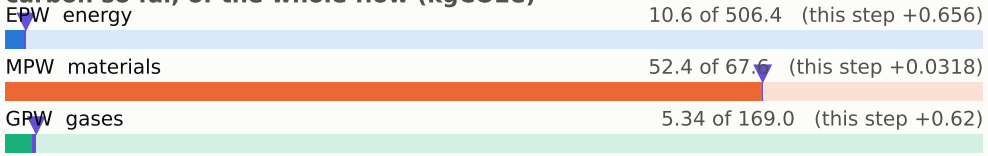


this step

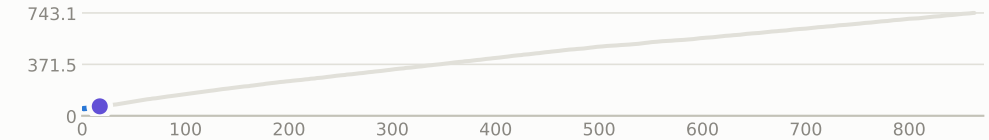
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

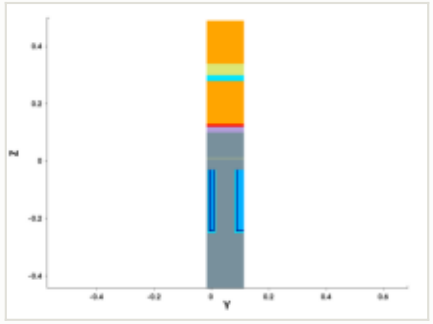
carbon so far, of the whole flow (kgCO2e)



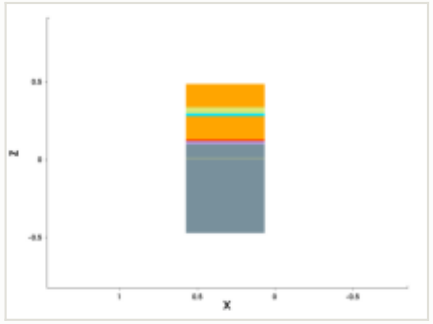
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



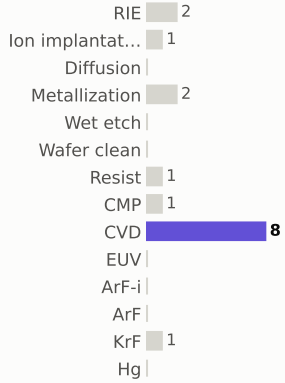
y cut at 0.049 µm



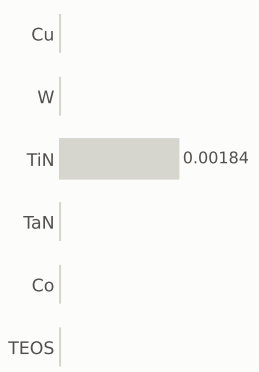
mask for this step

no mask for this step

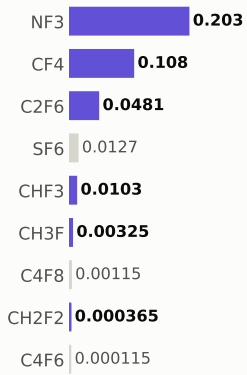
steps by process area



materials deposited (g)



gases emitted (g)



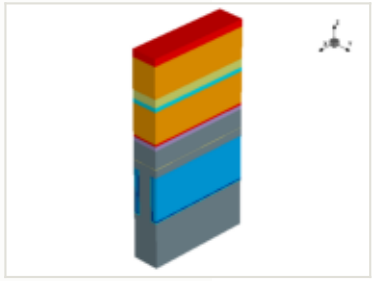
Step #220: CVD

depo_fill · CVD

1.31	69.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

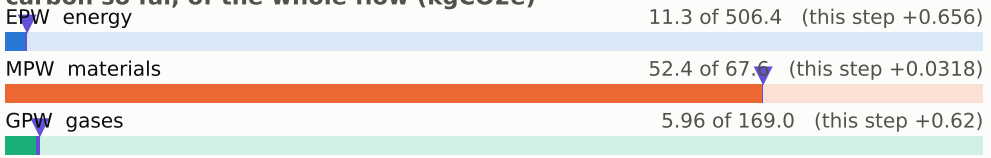


this step

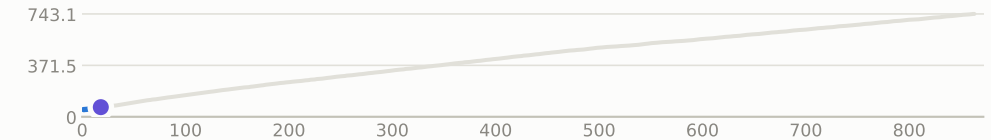
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

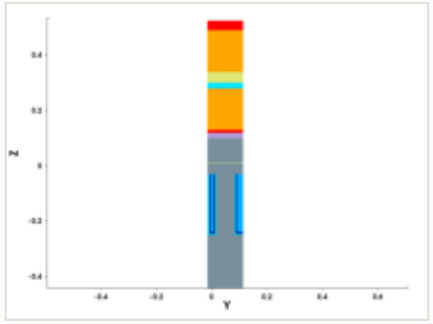
carbon so far, of the whole flow (kgCO2e)



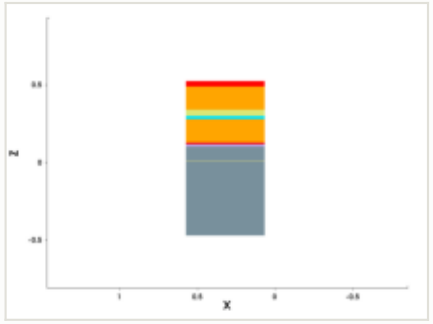
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



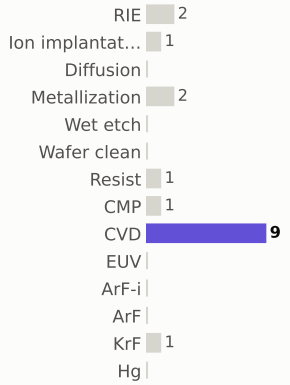
y cut at 0.049 µm



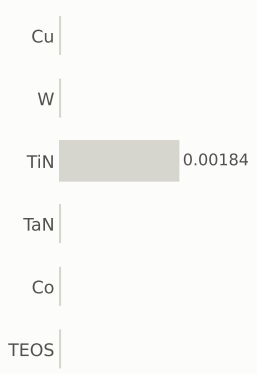
mask for this step

no mask for this step

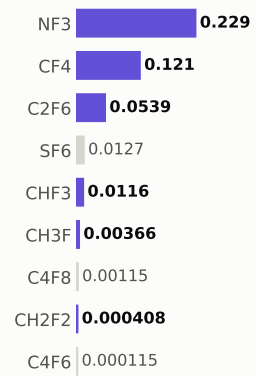
steps by process area



materials deposited (g)



gases emitted (g)



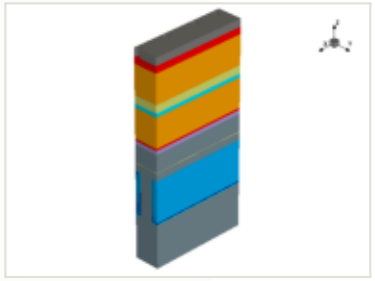
Step #230: KrF

litho_patterning · KrF

0.199	69.8	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

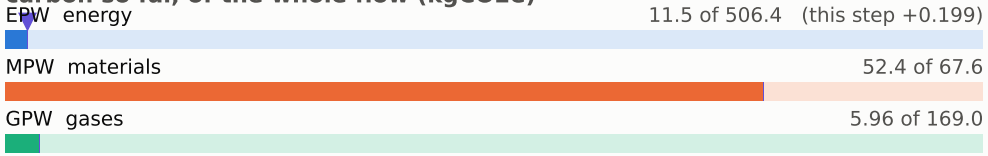


this step

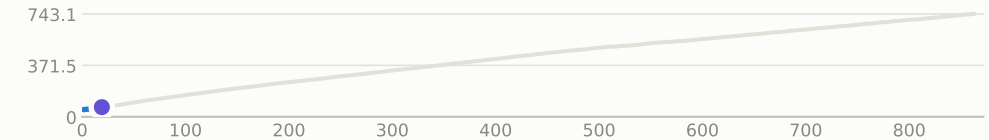
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

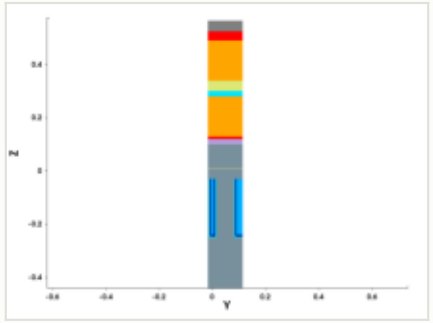
carbon so far, of the whole flow (kgCO2e)



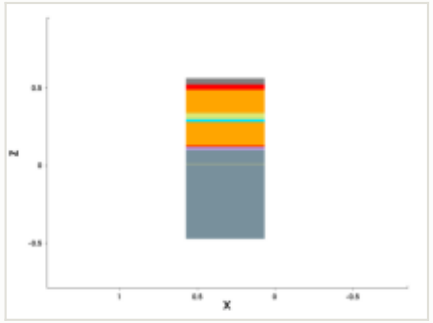
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

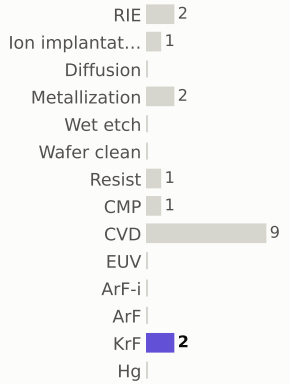


y cut at 0.049 µm

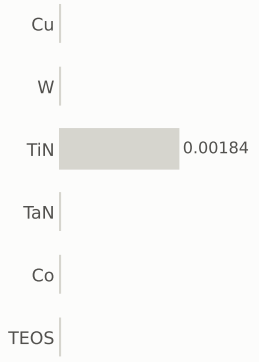


mask: layer 9:1 (GDSII)

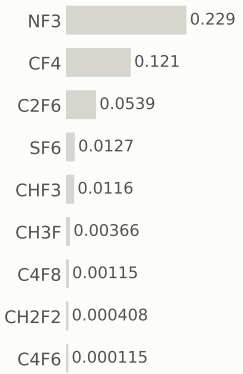
steps by process area



materials deposited (g)



gases emitted (g)



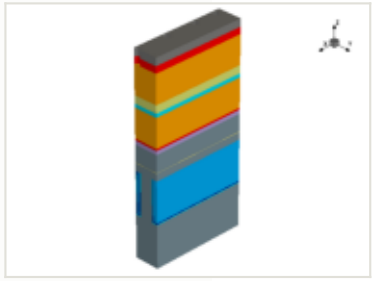
Step #240: RIE

etch_anisotropic · RIE

1.04	70.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

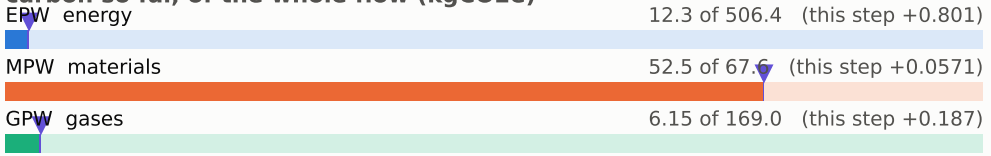


this step

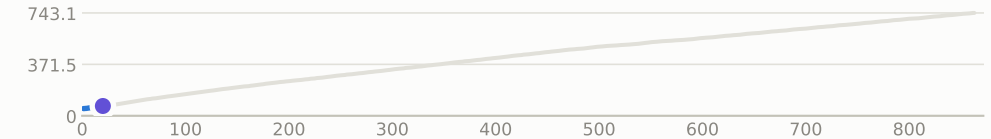
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

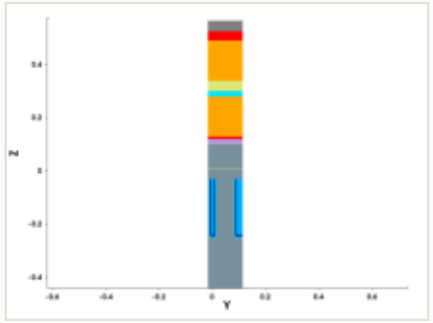
carbon so far, of the whole flow (kgCO2e)



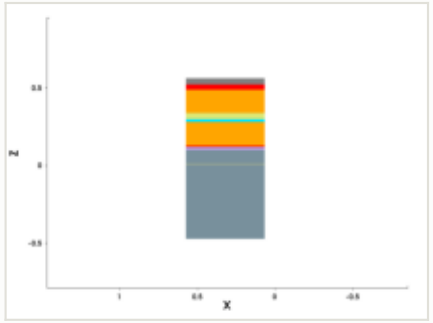
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



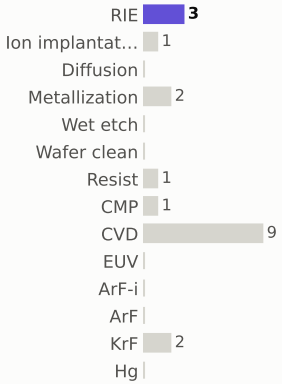
y cut at 0.049 µm



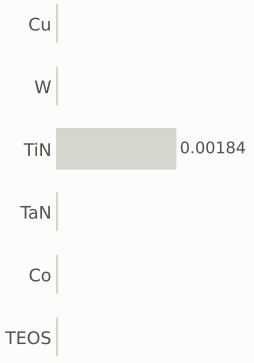
mask for this step

no mask for this step

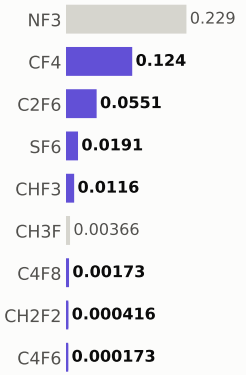
steps by process area



materials deposited (g)



gases emitted (g)



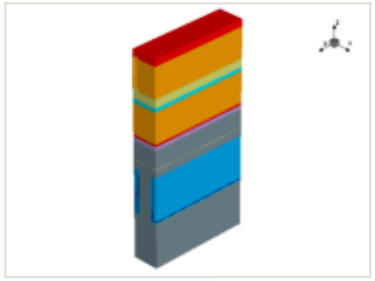
Step #250: RIE

etch_anisotropic · RIE

1.04	71.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

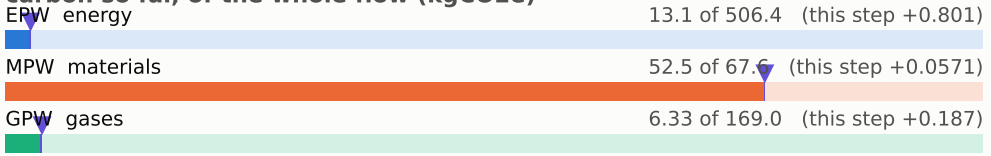


this step

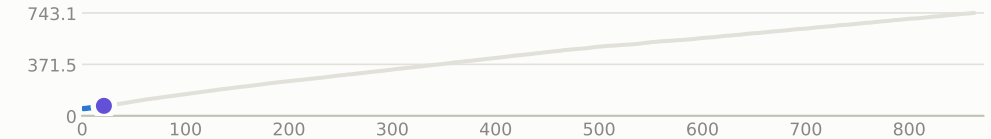
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

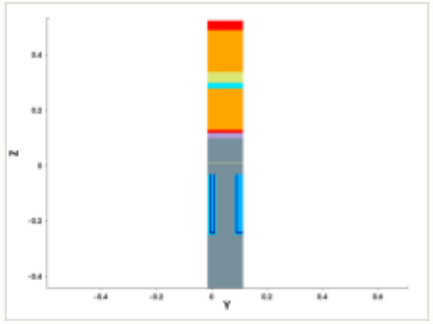
carbon so far, of the whole flow (kgCO2e)



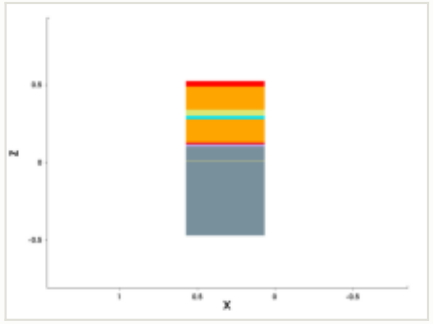
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



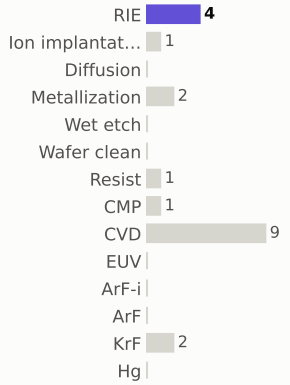
y cut at 0.049 µm



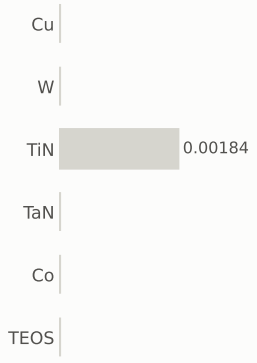
mask for this step

no mask for this step

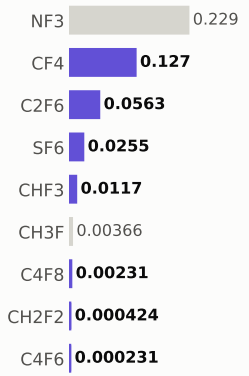
steps by process area



materials deposited (g)



gases emitted (g)



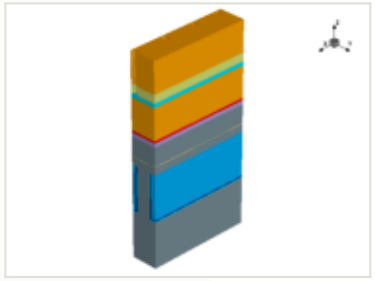
Step #260: RIE

etch_anisotropic · RIE

1.04	72.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

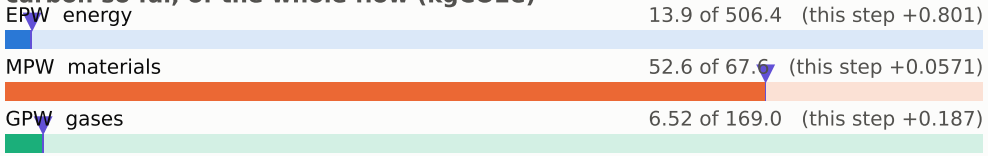


this step

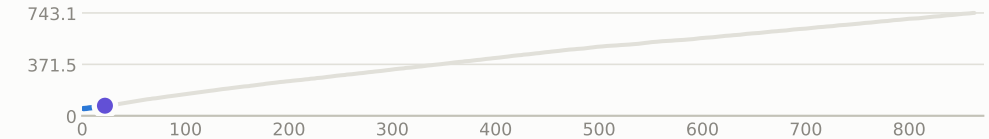
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

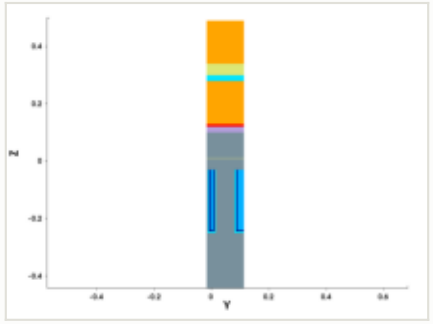
carbon so far, of the whole flow (kgCO2e)



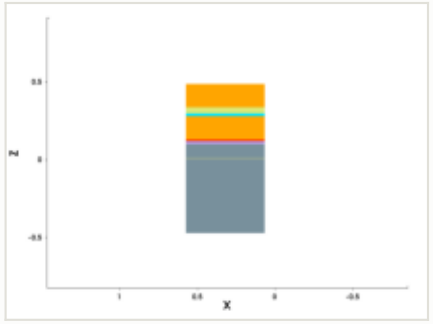
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



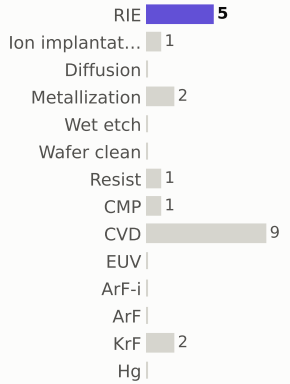
y cut at 0.049 µm



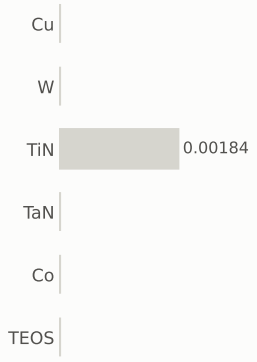
mask for this step

no mask for this step

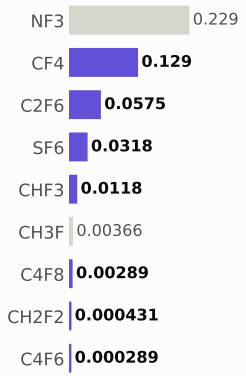
steps by process area



materials deposited (g)



gases emitted (g)



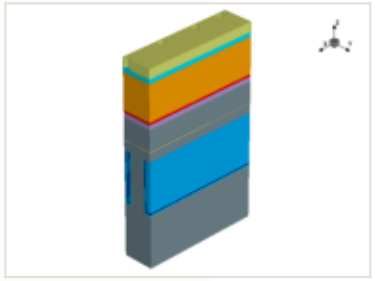
Step #270: Resist

etch_strip · Resist

0.169	73.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

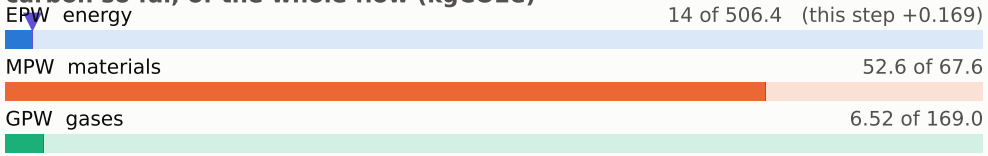


this step

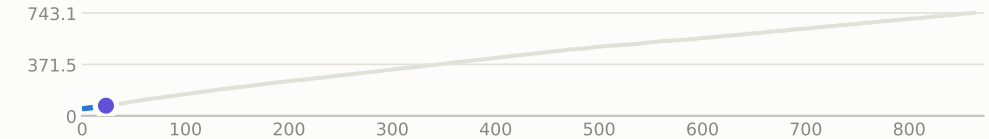
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

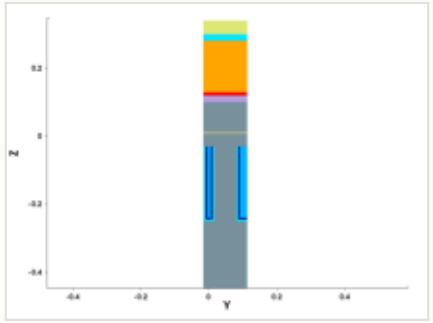
carbon so far, of the whole flow (kgCO2e)



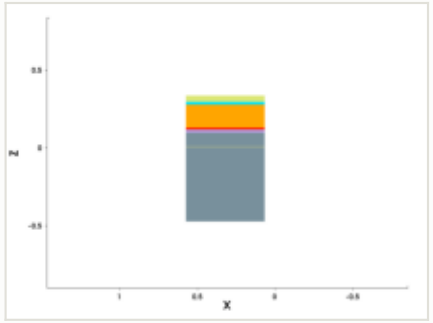
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



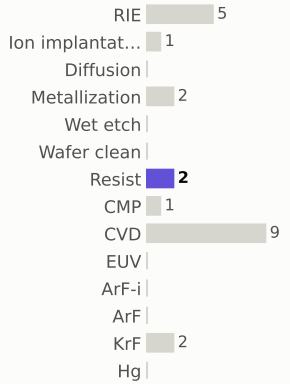
y cut at 0.049 µm



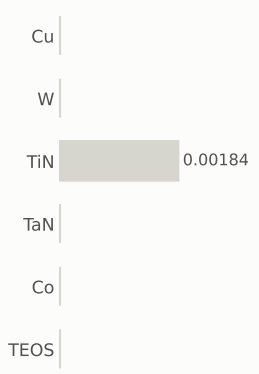
mask for this step

no mask for this step

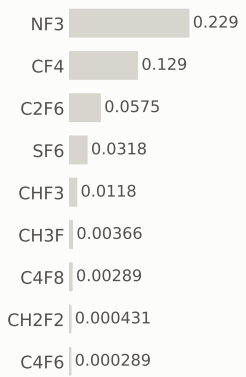
steps by process area



materials deposited (g)



gases emitted (g)



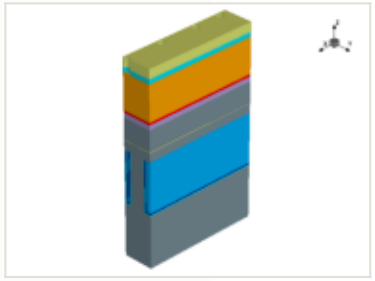
Step #280: RIE

etch_anisotropic · RIE

1.04	74.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

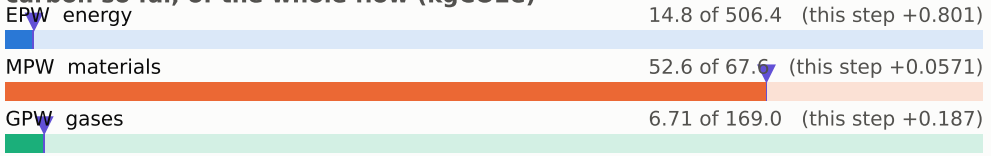


this step

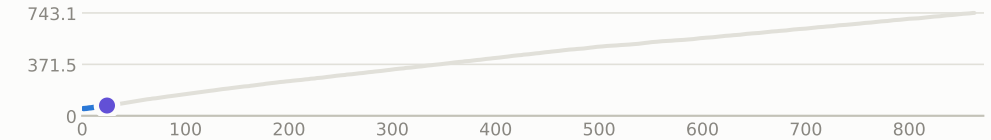
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

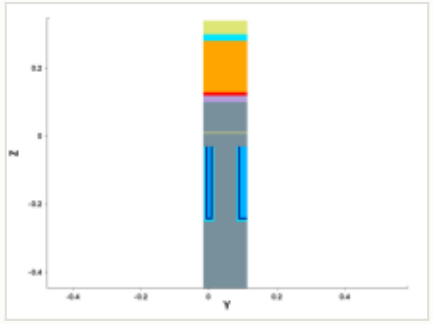
carbon so far, of the whole flow (kgCO2e)



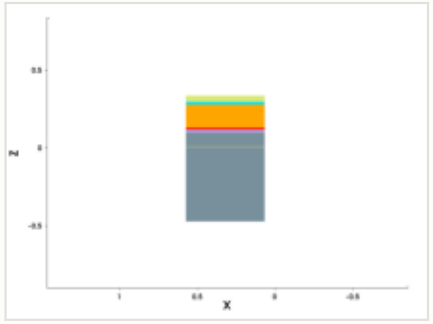
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



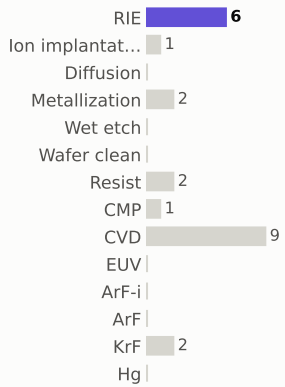
y cut at 0.049 µm



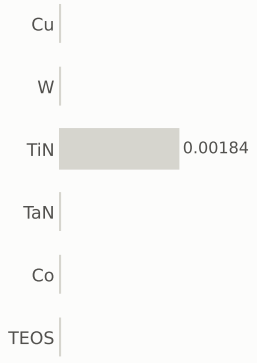
mask for this step

no mask for this step

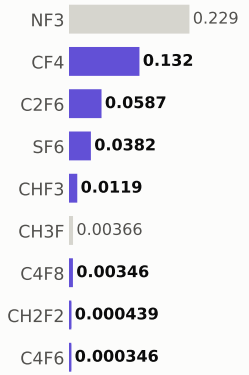
steps by process area



materials deposited (g)



gases emitted (g)



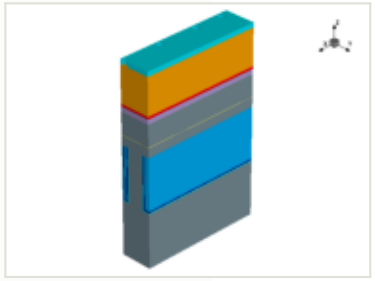
Step #290: RIE

etch_anisotropic · RIE

1.04	75.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

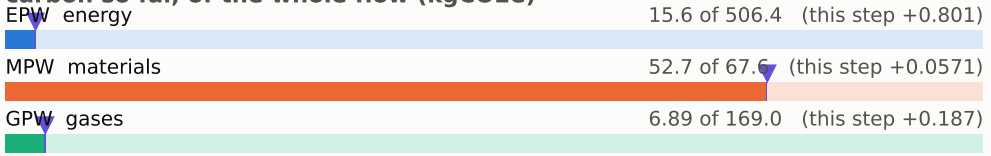


this step

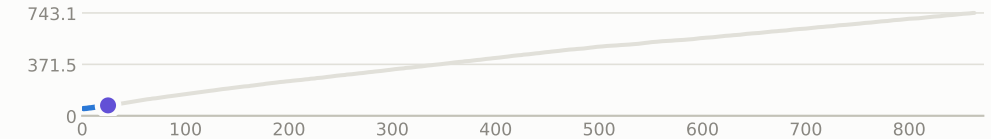
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

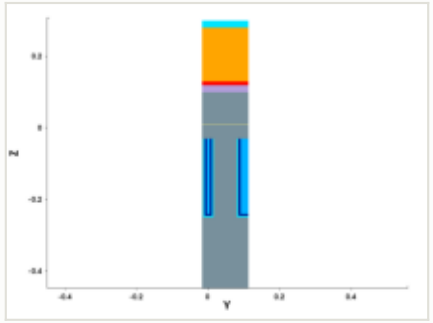
carbon so far, of the whole flow (kgCO2e)



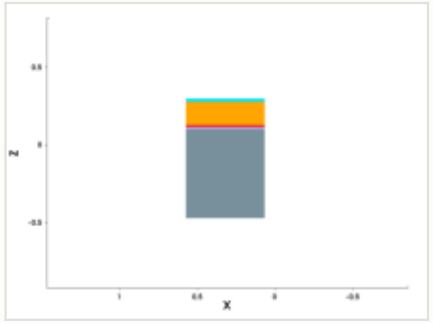
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



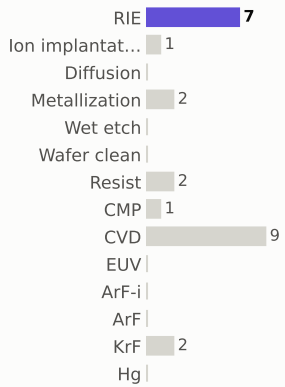
y cut at 0.049 µm



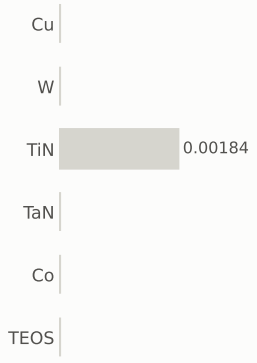
mask for this step

no mask for this step

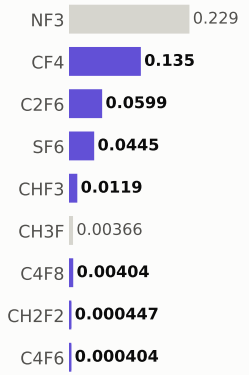
steps by process area



materials deposited (g)



gases emitted (g)



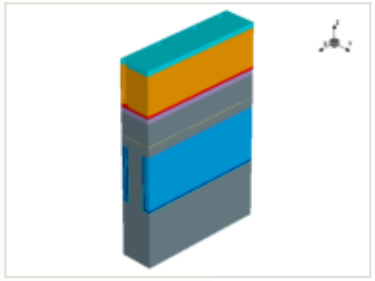
Step #300: RIE

etch_anisotropic · RIE

1.04	76.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

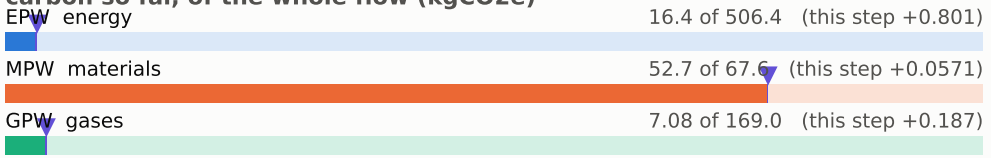


this step

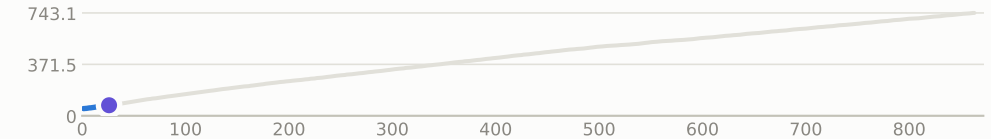
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

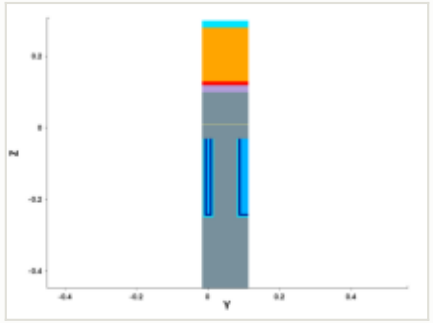
carbon so far, of the whole flow (kgCO2e)



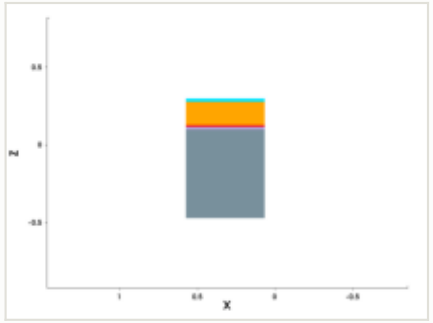
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



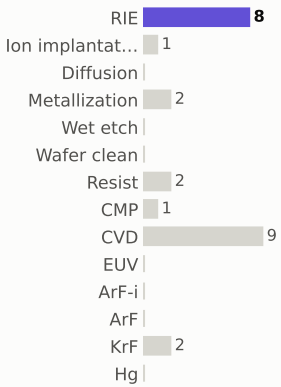
y cut at 0.049 µm



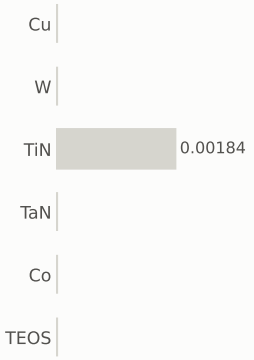
mask for this step

no mask for this step

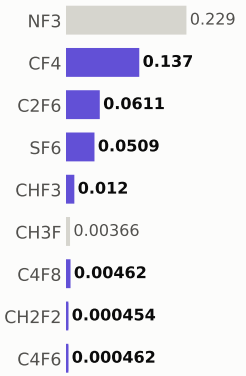
steps by process area



materials deposited (g)



gases emitted (g)



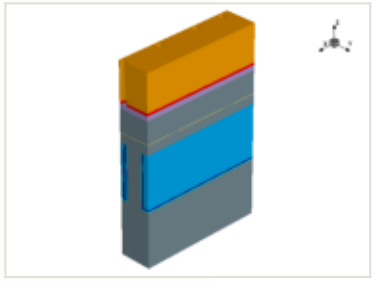
Step #310: RIE

etch_anisotropic · RIE

1.04	77.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

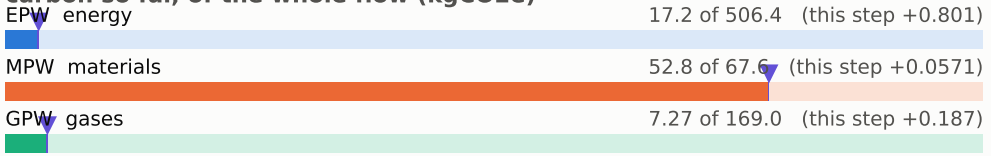


this step

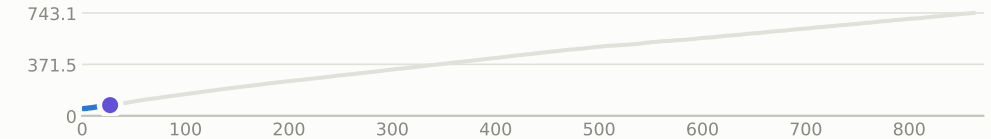
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

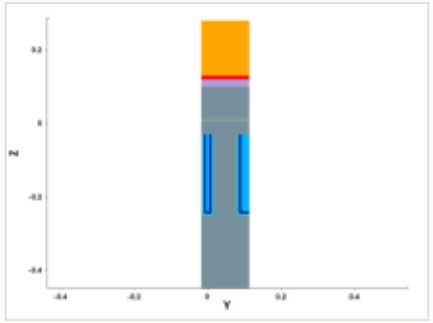
carbon so far, of the whole flow (kgCO2e)



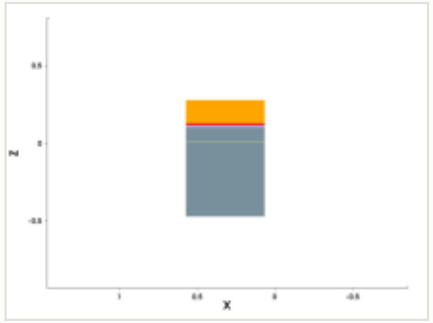
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



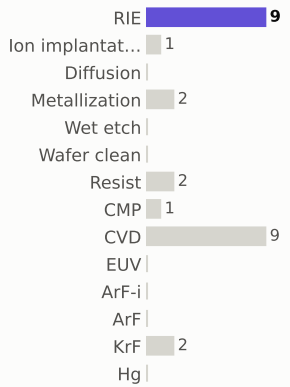
y cut at 0.049 µm



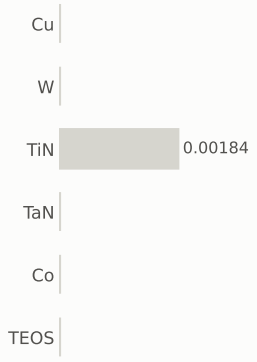
mask for this step

no mask for this step

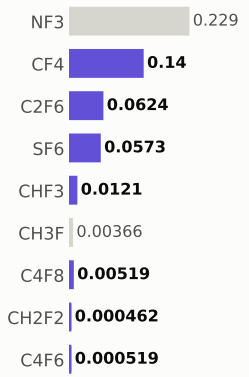
steps by process area



materials deposited (g)



gases emitted (g)



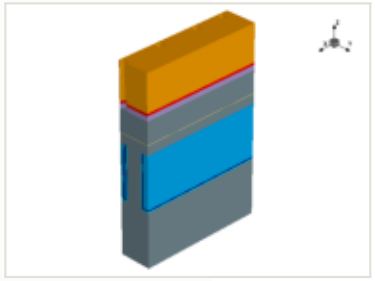
Step #320: RIE

etch_anisotropic · RIE

1.04	78.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

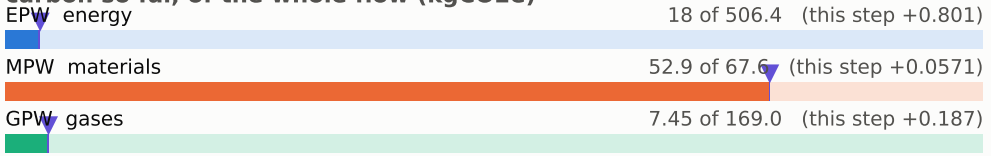


this step

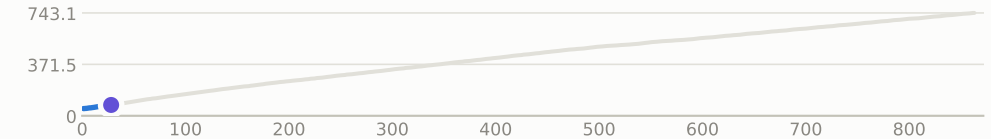
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

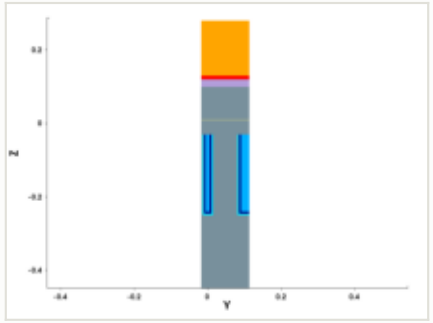
carbon so far, of the whole flow (kgCO2e)



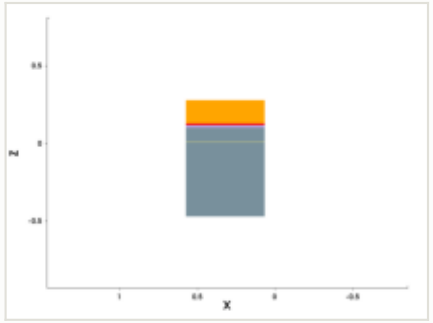
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



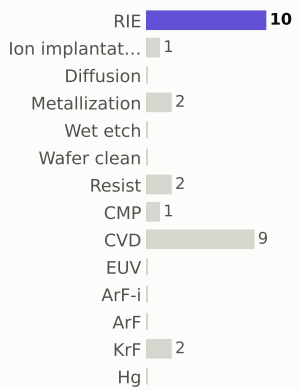
y cut at 0.049 µm



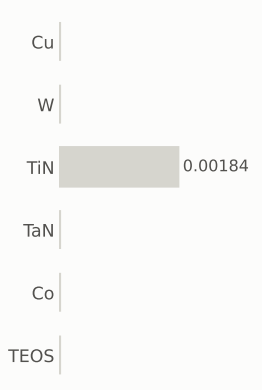
mask for this step

no mask for this step

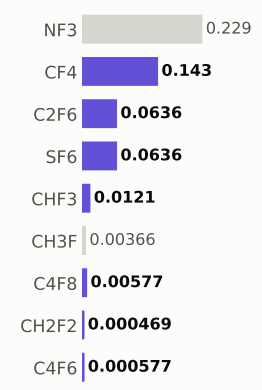
steps by process area



materials deposited (g)



gases emitted (g)



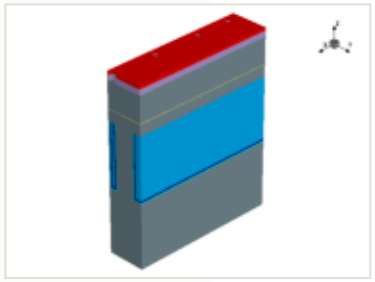
Step #330: Resist

etch_strip · Resist

0.169	78.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

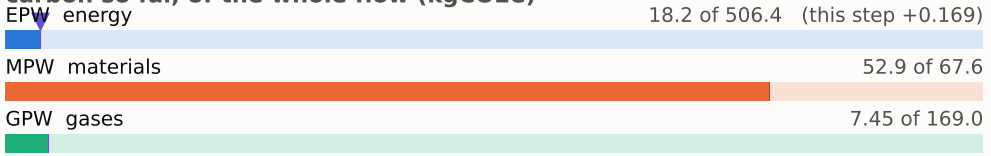
zoomed to fit



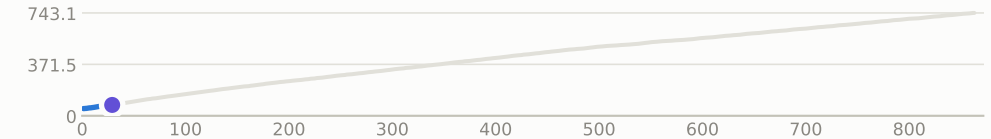
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

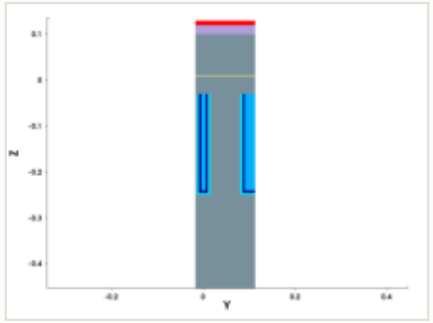
carbon so far, of the whole flow (kgCO2e)



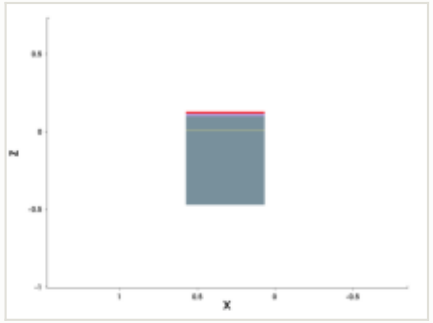
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



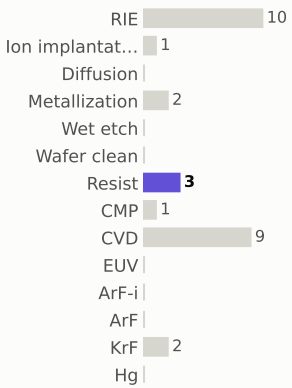
y cut at 0.049 µm



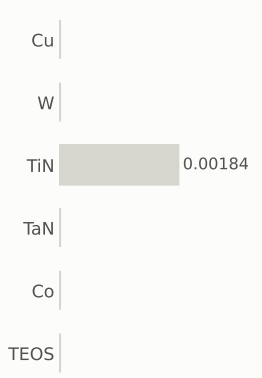
mask for this step

no mask for this step

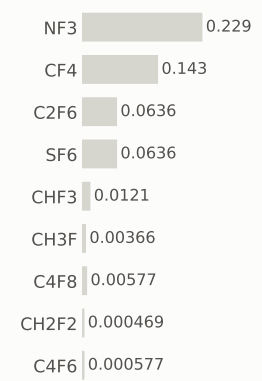
steps by process area



materials deposited (g)



gases emitted (g)



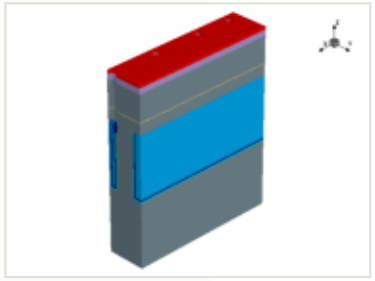
Step #340: RIE

etch_anisotropic · RIE

1.04	79.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

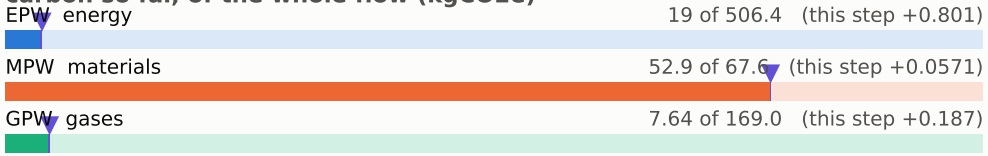


this step

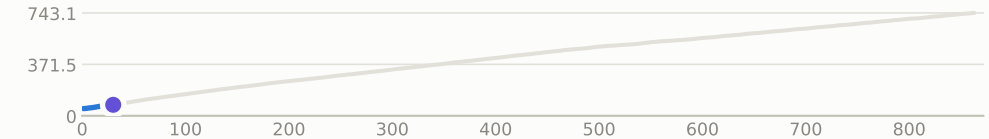
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

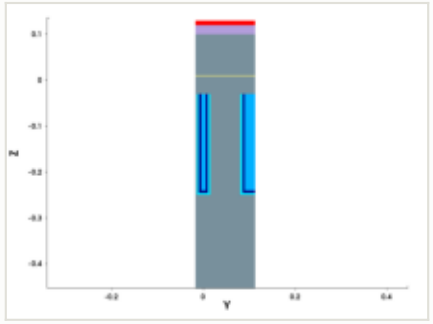
carbon so far, of the whole flow (kgCO2e)



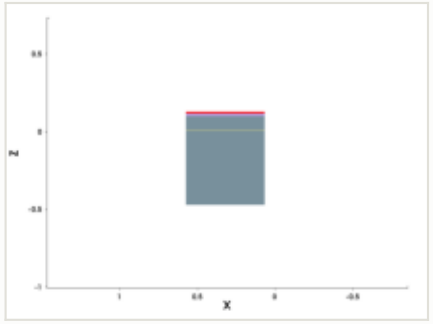
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



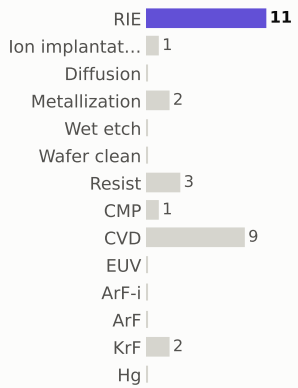
y cut at 0.049 µm



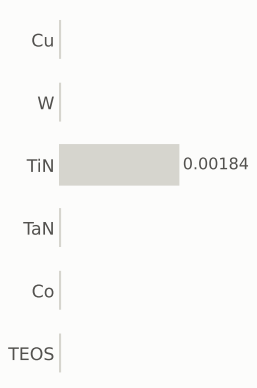
mask for this step

no mask for this step

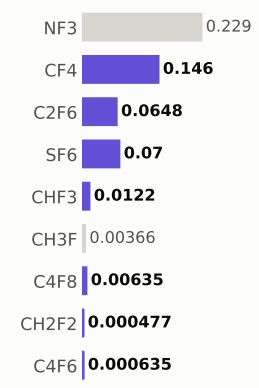
steps by process area



materials deposited (g)



gases emitted (g)



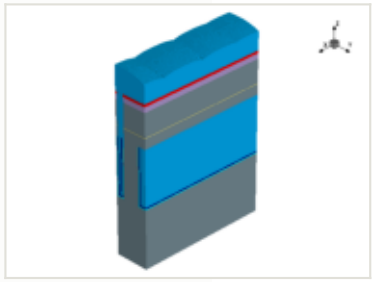
Step #350: Metallization

depo_isotropic · Metallization

1.23	80.8	0.17%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

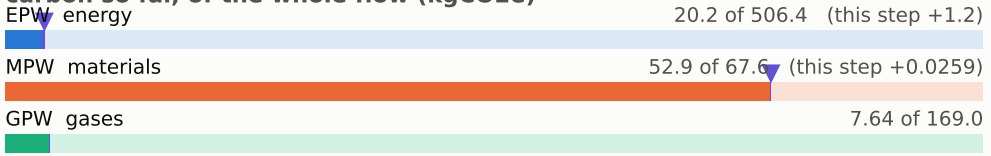


this step

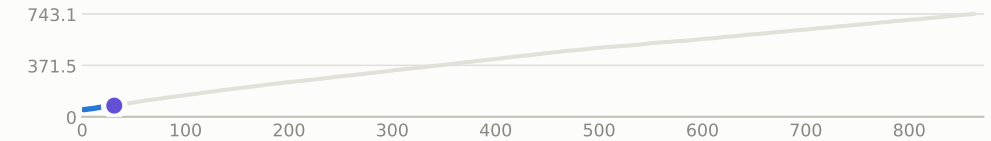
process area	Metallization
material	W
thickness	0.25 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0259 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.23 kgCO2e**

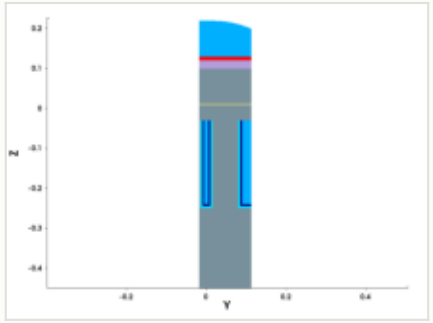
carbon so far, of the whole flow (kgCO2e)



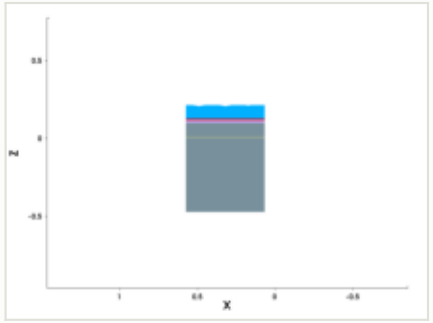
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



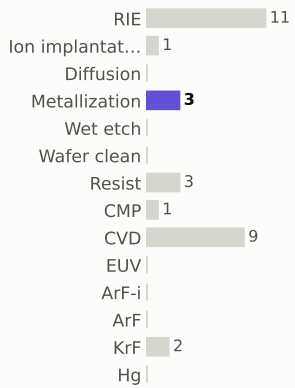
y cut at 0.049 µm



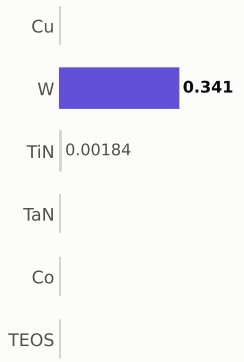
mask for this step

no mask for this step

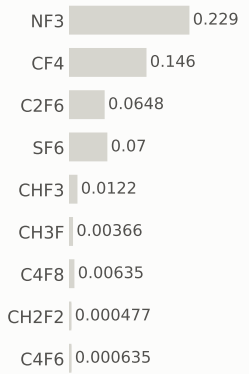
steps by process area



materials deposited (g)



gases emitted (g)



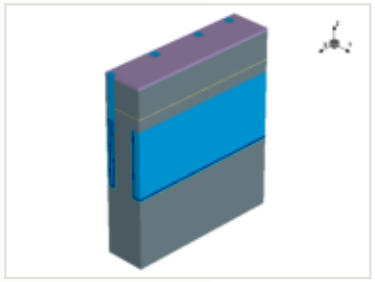
Step #360: CMP

etch_cmp · CMP

0.169	80.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

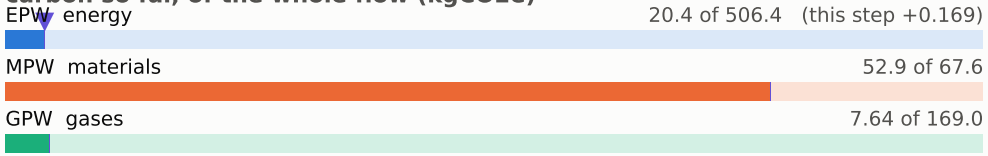
zoomed to fit



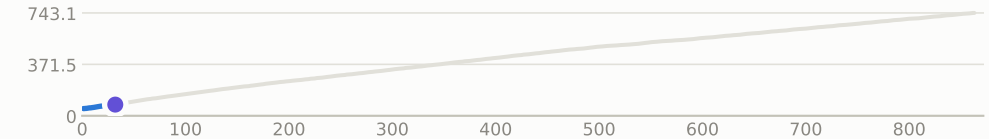
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

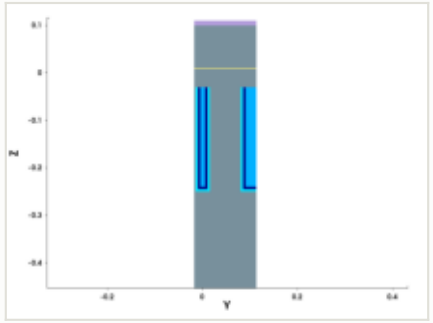
carbon so far, of the whole flow (kgCO2e)



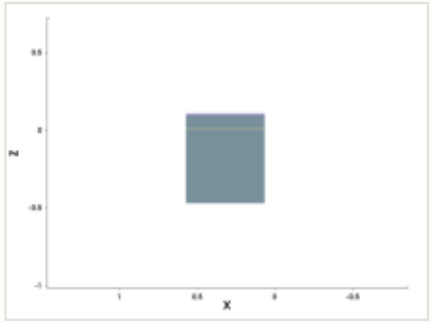
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



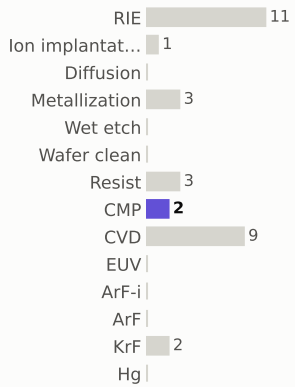
y cut at 0.049 µm



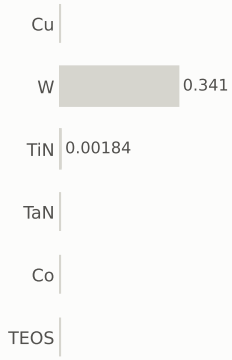
mask for this step

no mask for this step

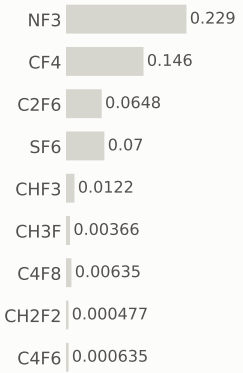
steps by process area



materials deposited (g)



gases emitted (g)



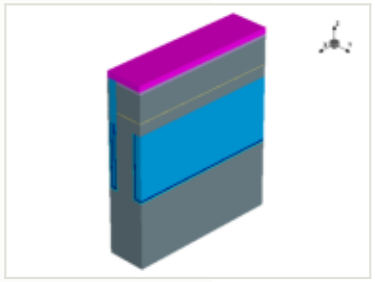
Step #380: CVD

depo_epi · CVD

1.31	82.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

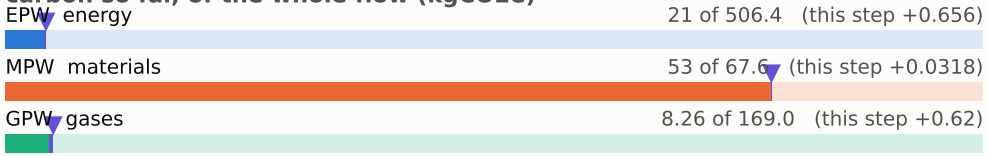


this step

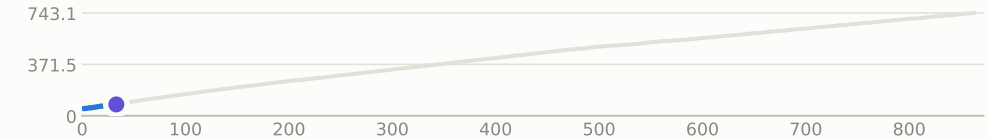
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

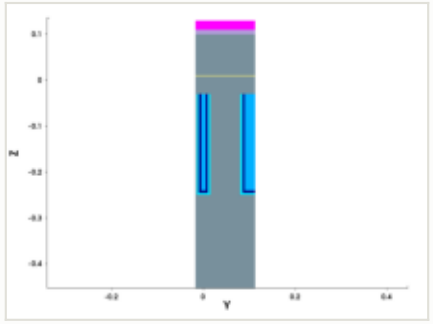
carbon so far, of the whole flow (kgCO2e)



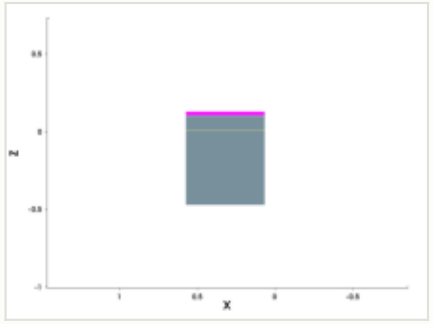
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



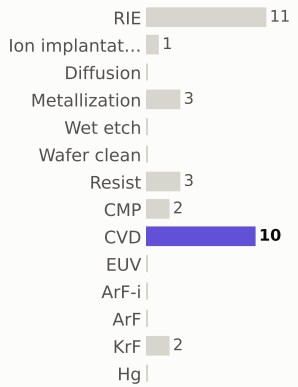
y cut at 0.049 µm



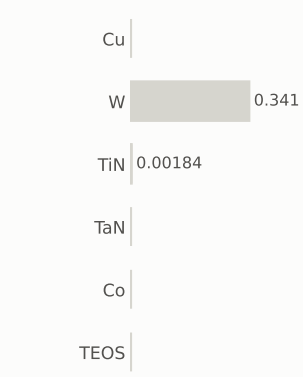
mask for this step

no mask for this step

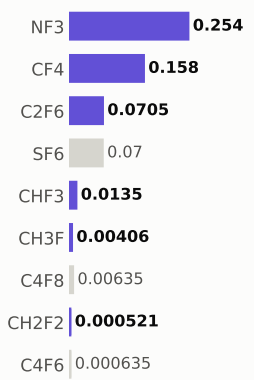
steps by process area



materials deposited (g)



gases emitted (g)



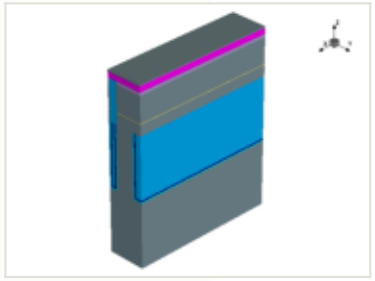
Step #390: CVD

depo_epi · CVD

1.31	83.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

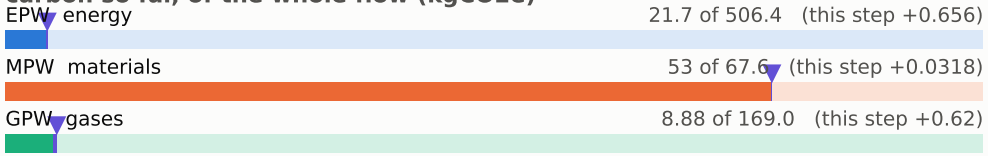


this step

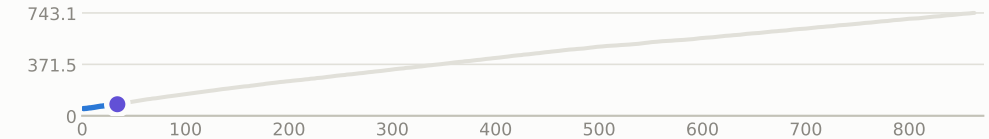
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

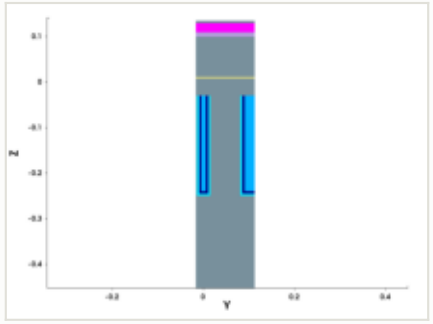
carbon so far, of the whole flow (kgCO2e)



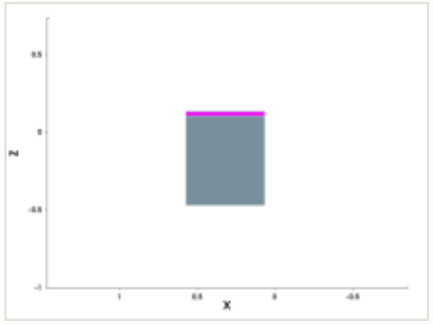
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



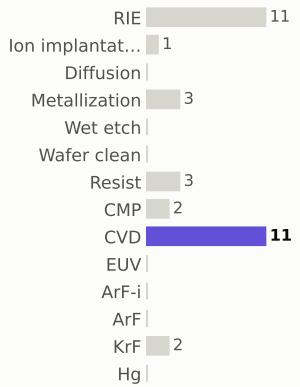
y cut at 0.049 µm



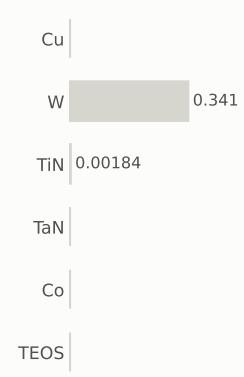
mask for this step

no mask for this step

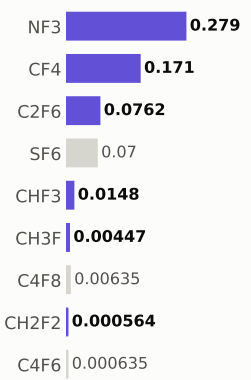
steps by process area



materials deposited (g)



gases emitted (g)



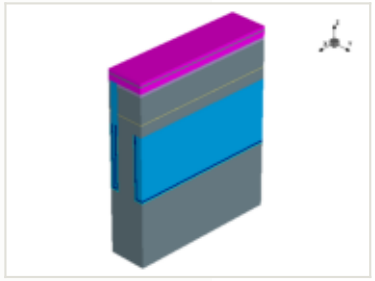
Step #400: CVD

depo_epi · CVD

1.31	84.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

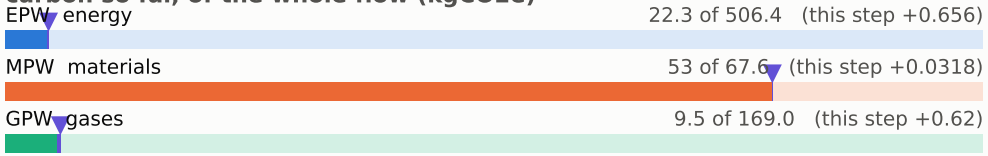
zoomed to fit



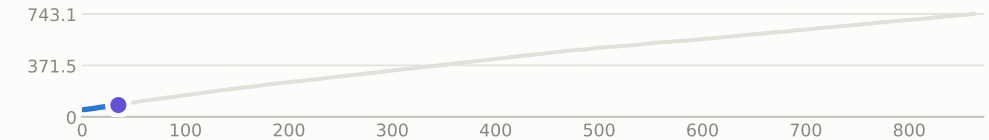
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.01 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

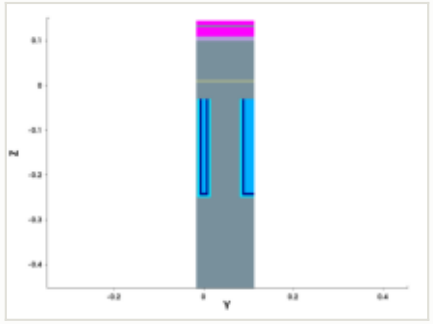
carbon so far, of the whole flow (kgCO2e)



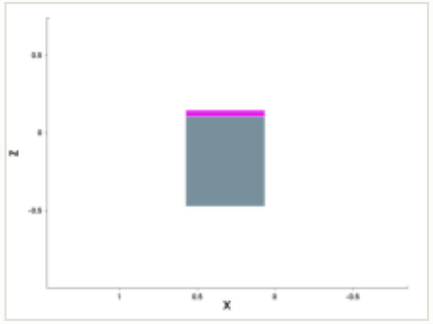
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



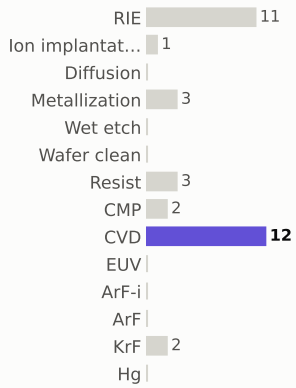
y cut at 0.049 µm



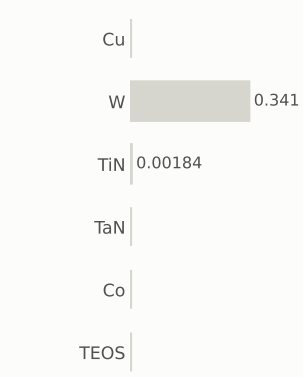
mask for this step

no mask for this step

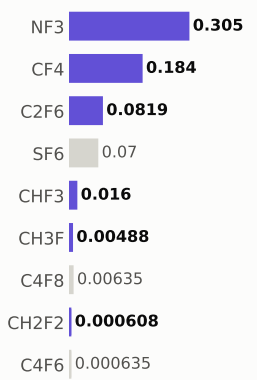
steps by process area



materials deposited (g)



gases emitted (g)



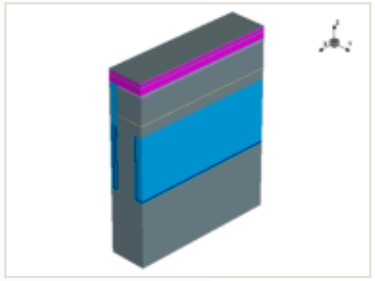
Step #410: CVD

depo_epi · CVD

1.31	86.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

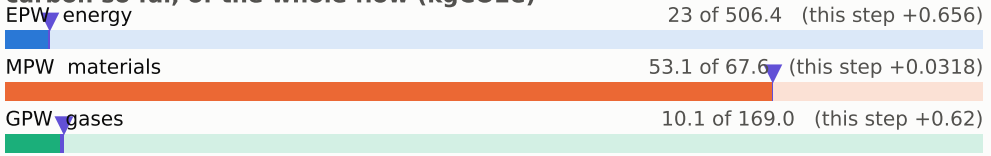


this step

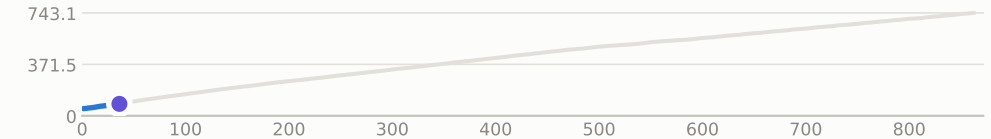
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

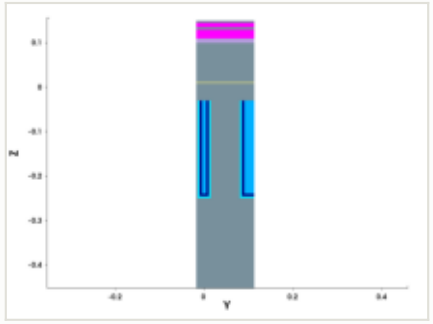
carbon so far, of the whole flow (kgCO2e)



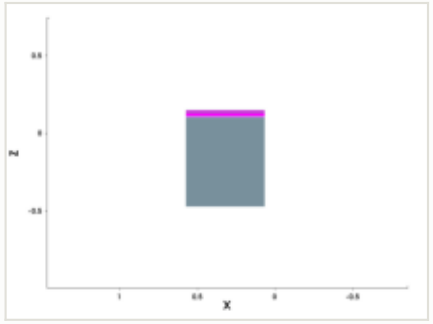
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



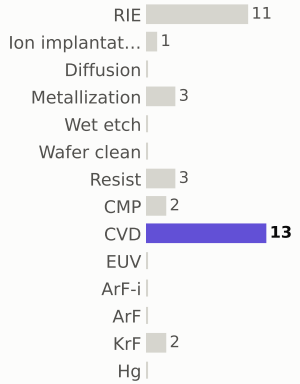
y cut at 0.049 µm



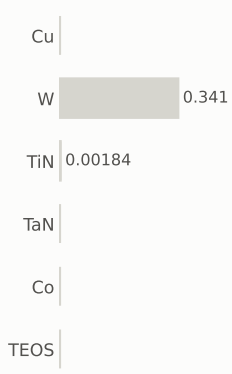
mask for this step

no mask for this step

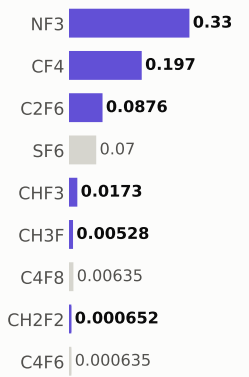
steps by process area



materials deposited (g)



gases emitted (g)



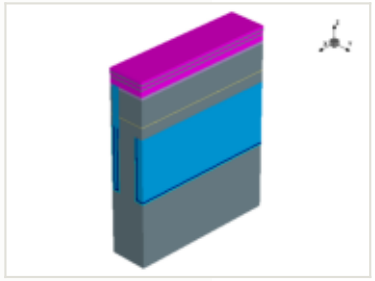
Step #420: CVD

depo_epi · CVD

1.31	87.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

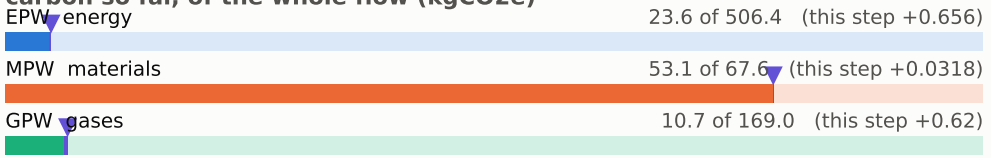


this step

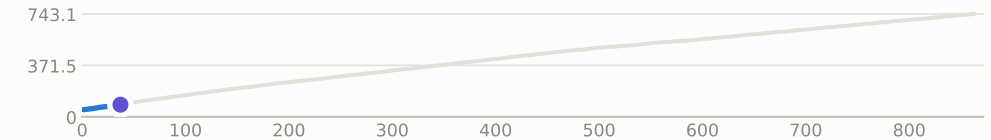
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

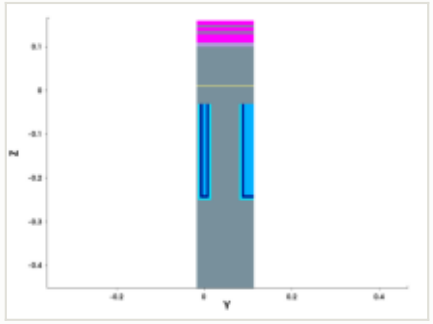
carbon so far, of the whole flow (kgCO2e)



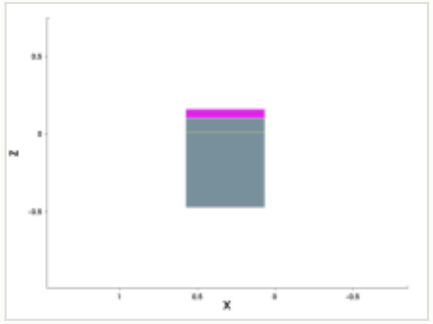
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



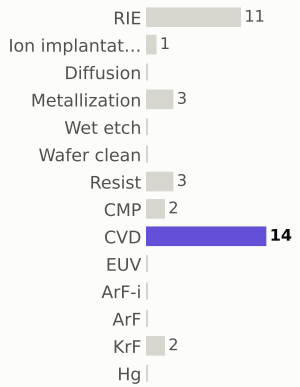
y cut at 0.049 µm



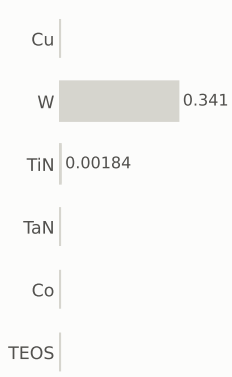
mask for this step

no mask for this step

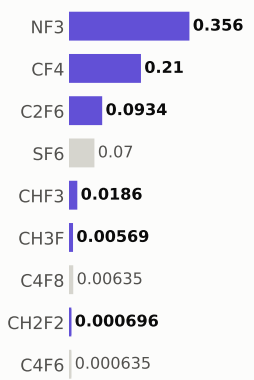
steps by process area



materials deposited (g)



gases emitted (g)



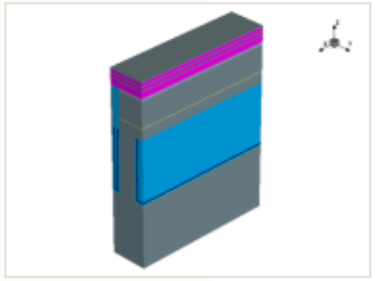
Step #430: CVD

depo_epi · CVD

1.31	88.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

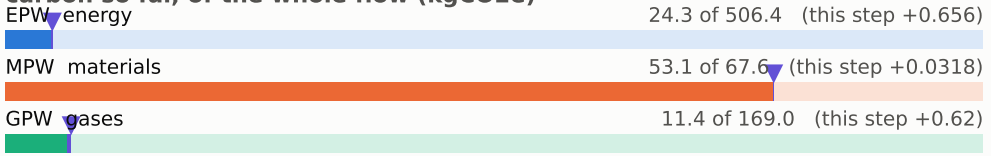


this step

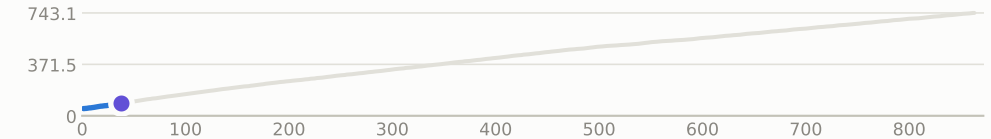
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

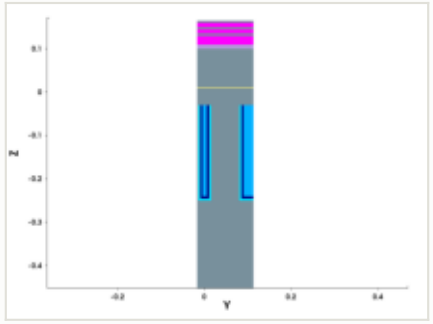
carbon so far, of the whole flow (kgCO2e)



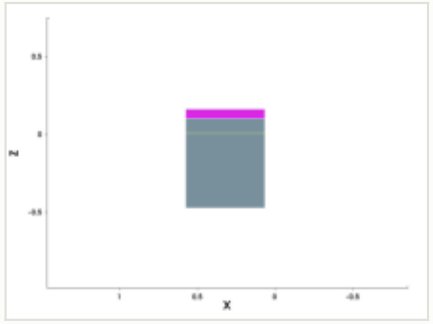
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



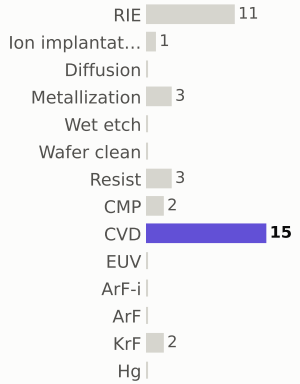
y cut at 0.049 µm



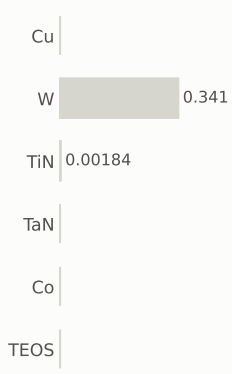
mask for this step

no mask for this step

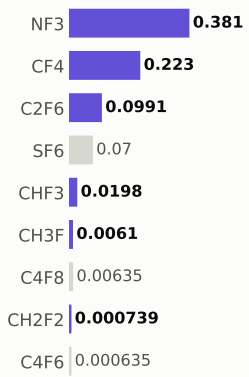
steps by process area



materials deposited (g)



gases emitted (g)



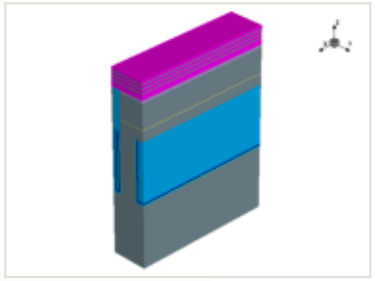
Step #440: CVD

depo_epi · CVD

1.31	90.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

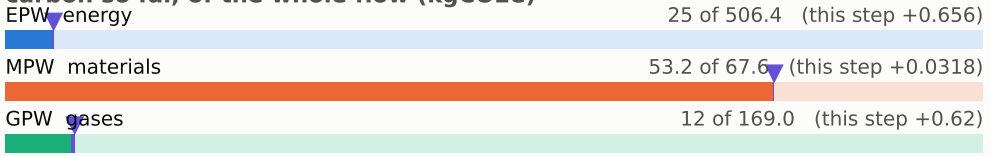


this step

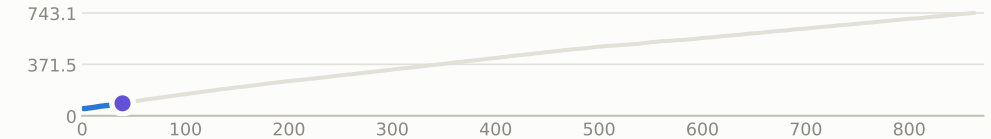
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

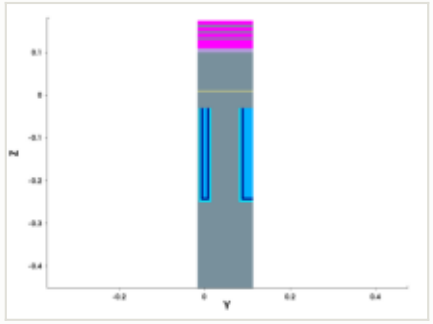
carbon so far, of the whole flow (kgCO2e)



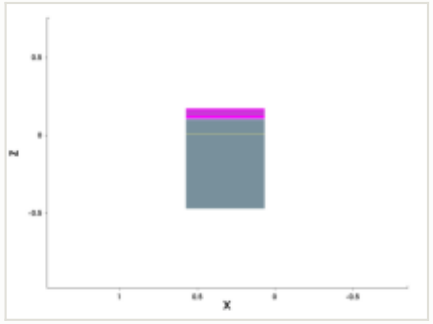
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



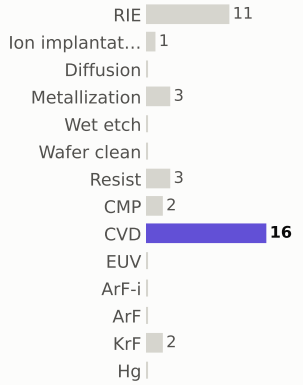
y cut at 0.049 µm



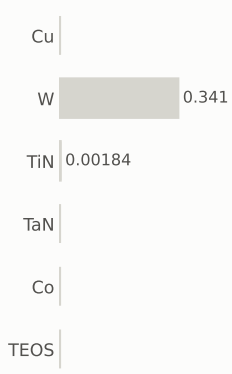
mask for this step

no mask for this step

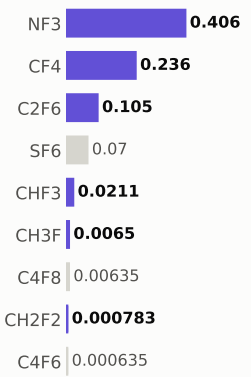
steps by process area



materials deposited (g)



gases emitted (g)



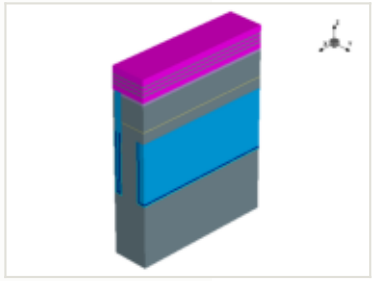
Step #450: CVD

depo_epi · CVD

1.31	91.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

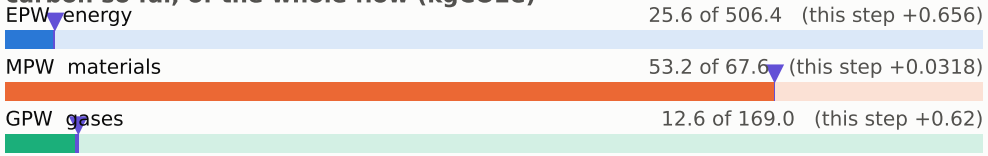


this step

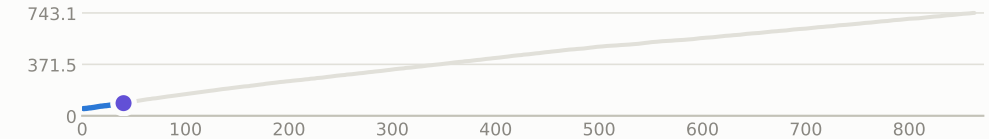
process area	CVD
material	—
thickness	0.015 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

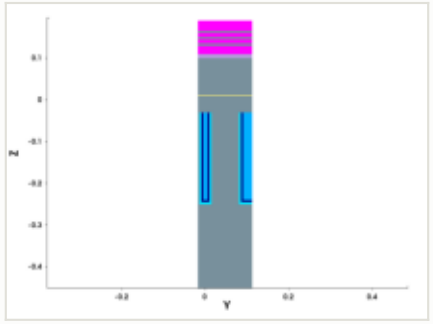
carbon so far, of the whole flow (kgCO2e)



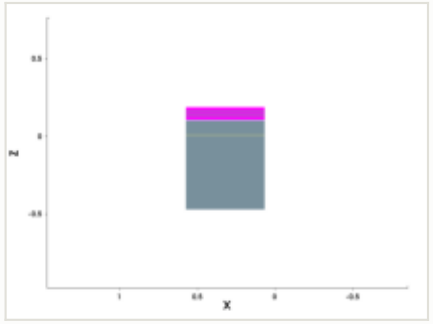
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



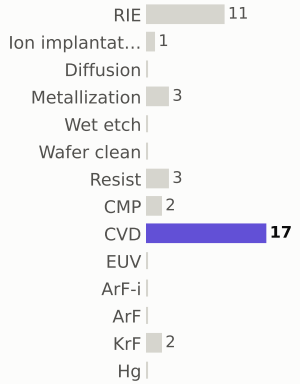
y cut at 0.049 µm



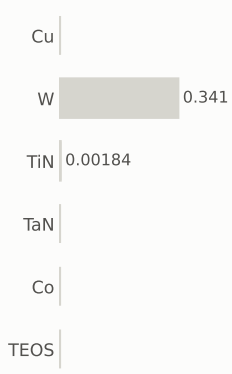
mask for this step

no mask for this step

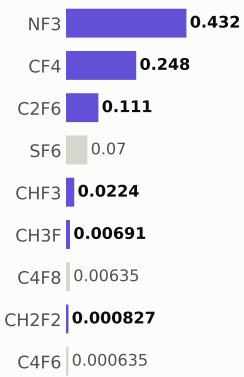
steps by process area



materials deposited (g)



gases emitted (g)



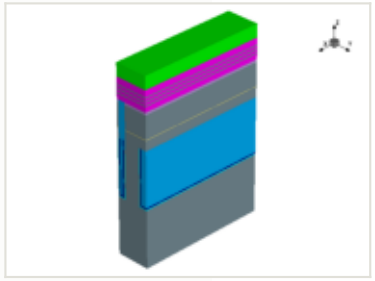
Step #460: CVD

depo_anisotropic · CVD

1.31	92.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

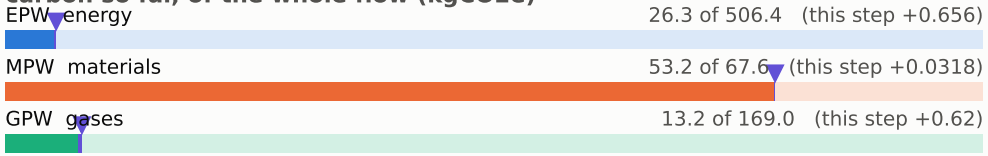


this step

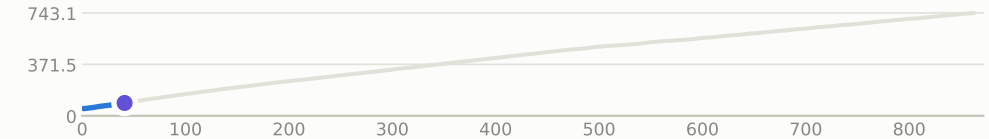
process area	CVD
material	—
thickness	0.05 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

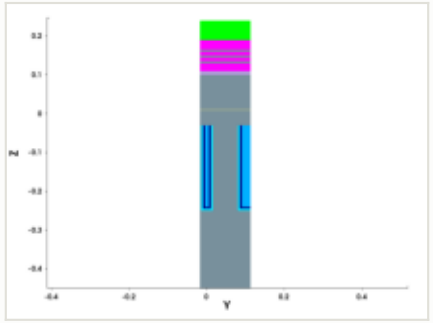
carbon so far, of the whole flow (kgCO2e)



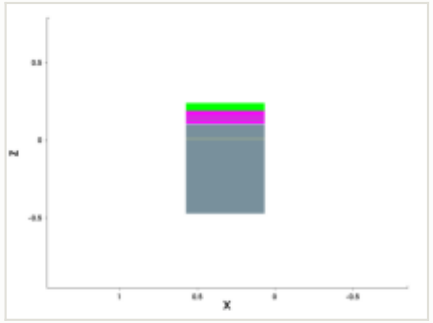
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



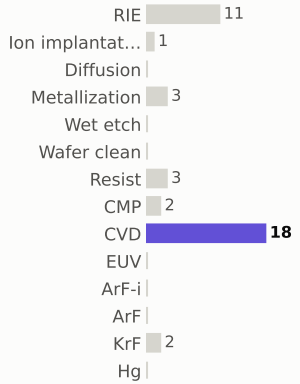
y cut at 0.049 µm



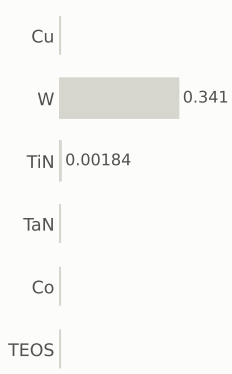
mask for this step

no mask for this step

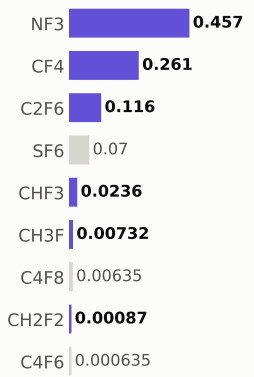
steps by process area



materials deposited (g)



gases emitted (g)



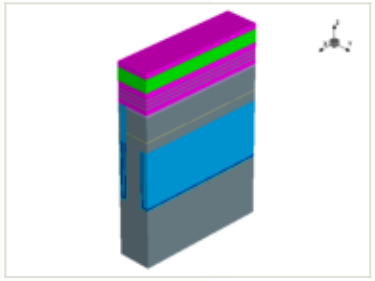
Step #470: CVD

depo_epi · CVD

1.31	94	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

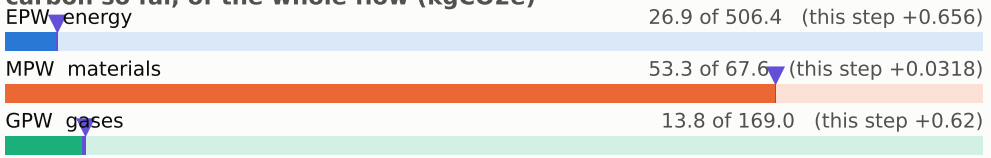


this step

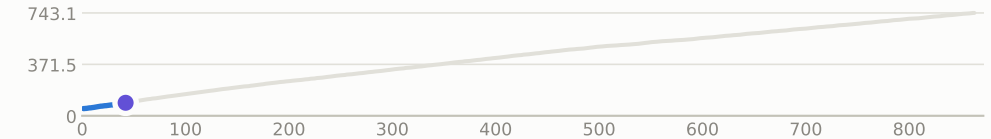
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

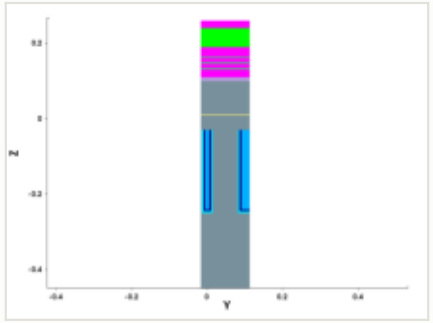
carbon so far, of the whole flow (kgCO2e)



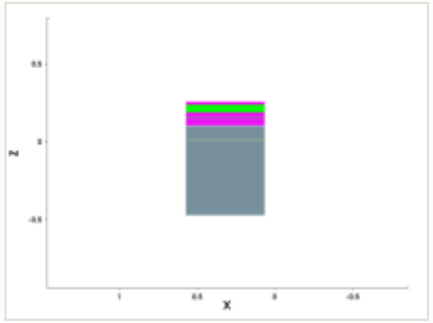
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



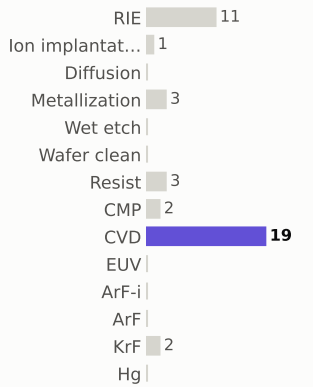
y cut at 0.049 µm



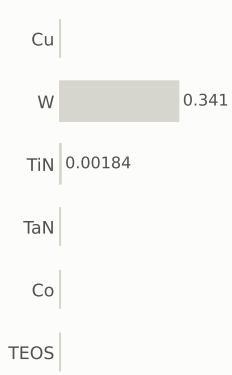
mask for this step

no mask for this step

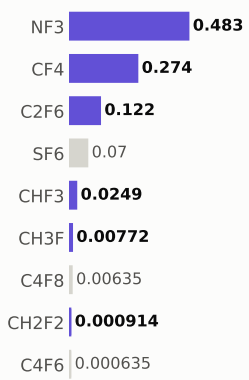
steps by process area



materials deposited (g)



gases emitted (g)



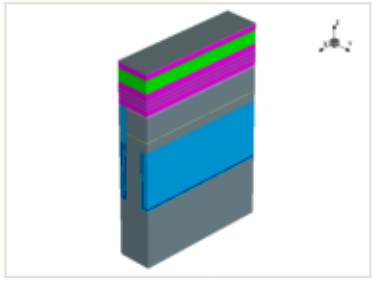
Step #480: CVD

depo_epi · CVD

1.31	95.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

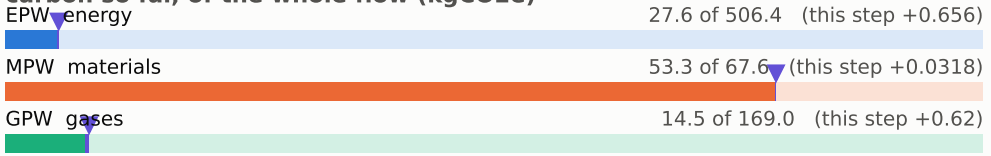


this step

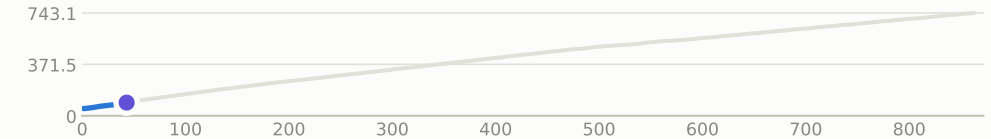
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

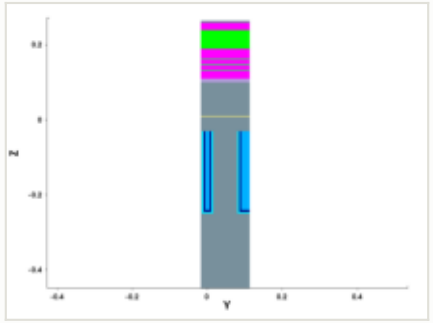
carbon so far, of the whole flow (kgCO2e)



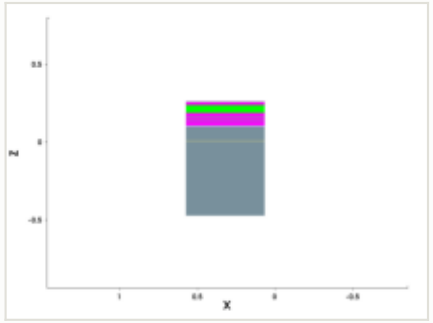
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



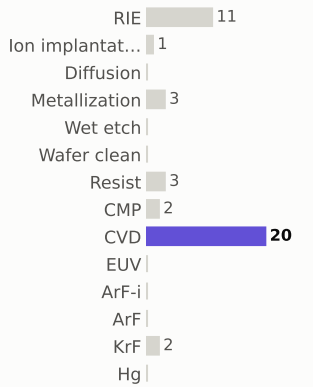
y cut at 0.049 µm



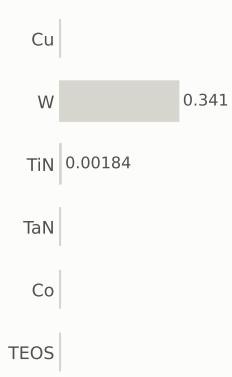
mask for this step

no mask for this step

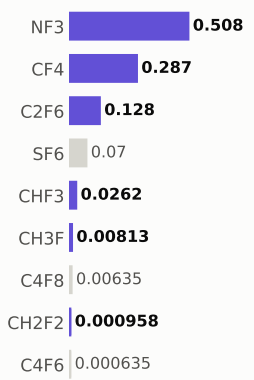
steps by process area



materials deposited (g)



gases emitted (g)



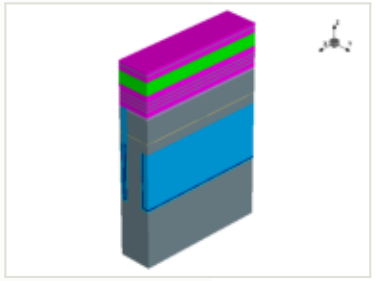
Step #490: CVD

depo_epi · CVD

1.31	96.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

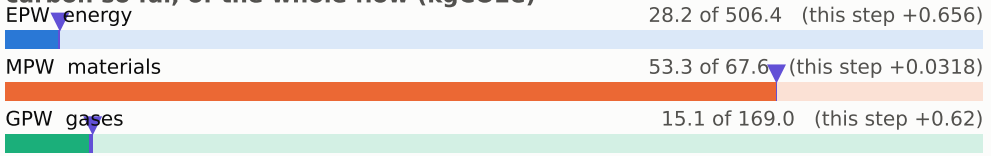


this step

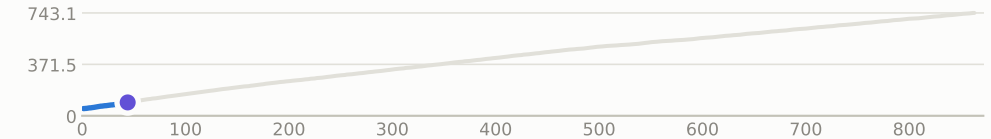
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

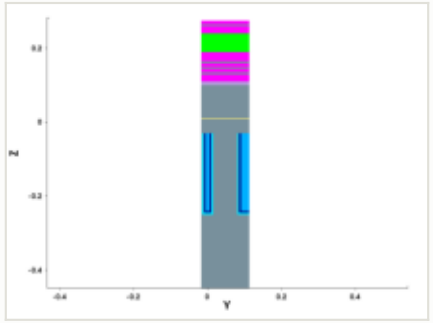
carbon so far, of the whole flow (kgCO2e)



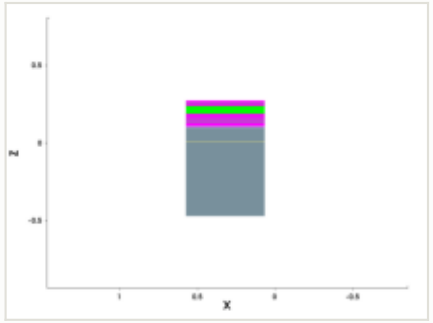
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



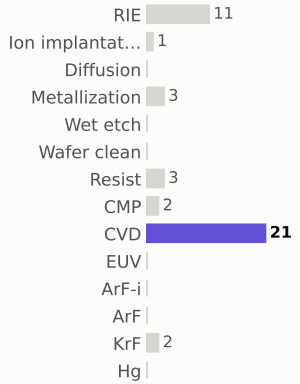
y cut at 0.049 µm



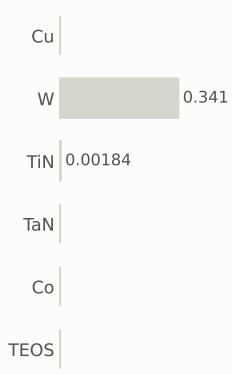
mask for this step

no mask for this step

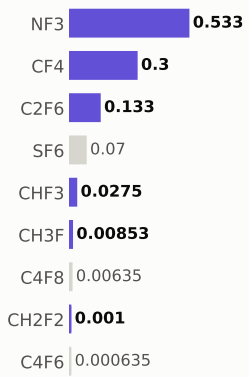
steps by process area



materials deposited (g)



gases emitted (g)



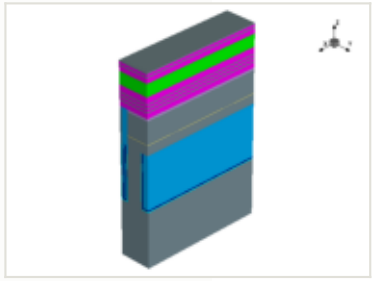
Step #500: CVD

depo_epi · CVD

1.31	98	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

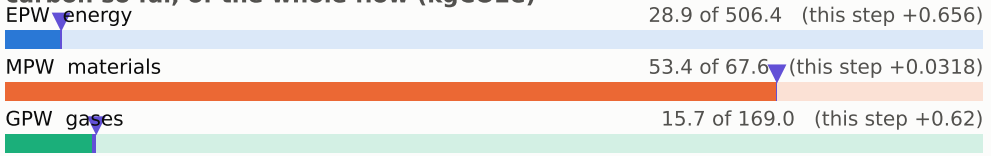


this step

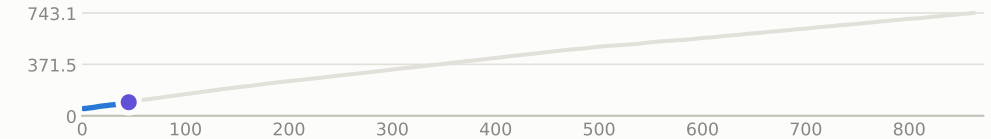
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

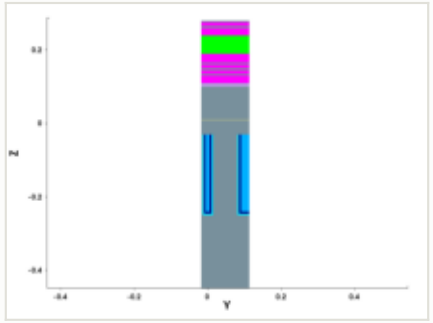
carbon so far, of the whole flow (kgCO2e)



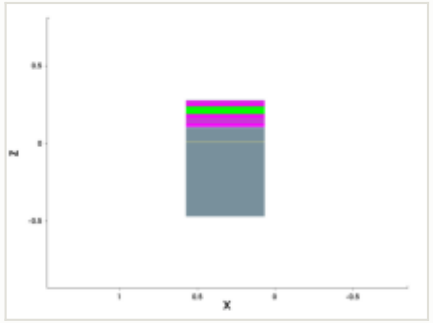
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



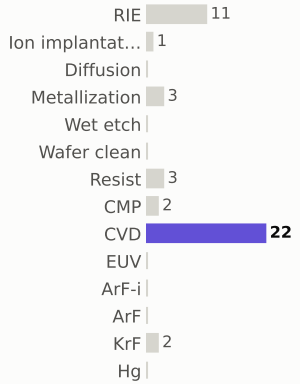
y cut at 0.049 µm



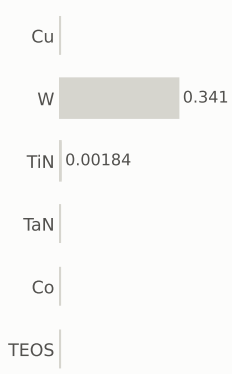
mask for this step

no mask for this step

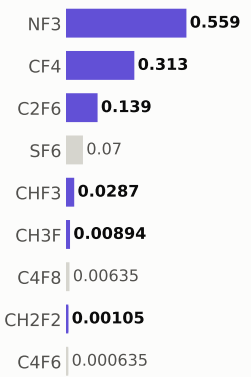
steps by process area



materials deposited (g)



gases emitted (g)



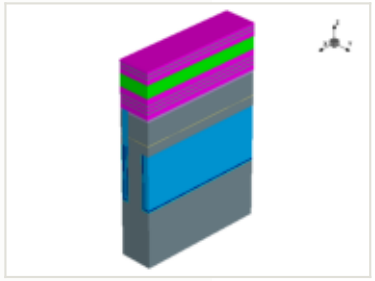
Step #510: CVD

depo_epi · CVD

1.31	99.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

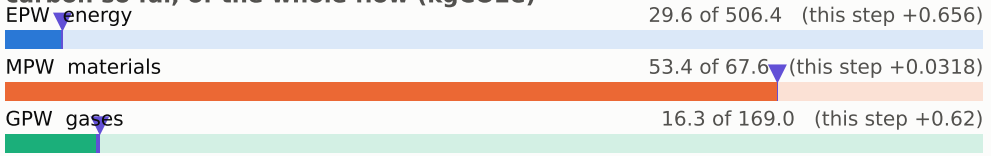


this step

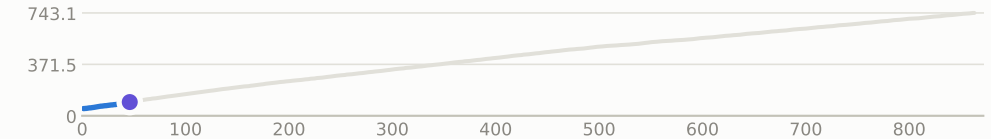
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

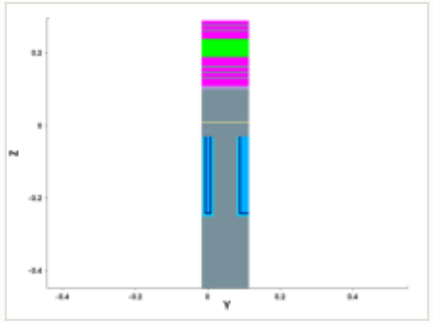
carbon so far, of the whole flow (kgCO2e)



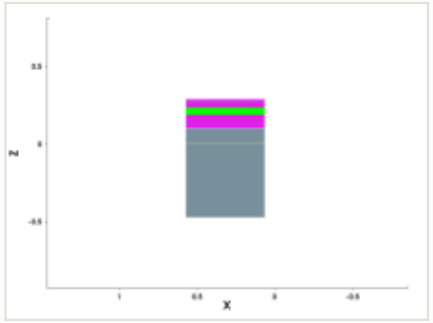
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



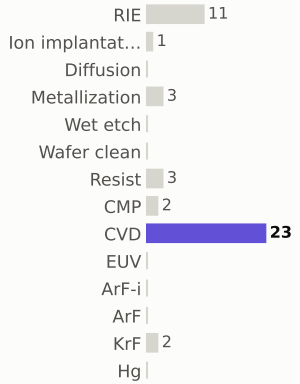
y cut at 0.049 µm



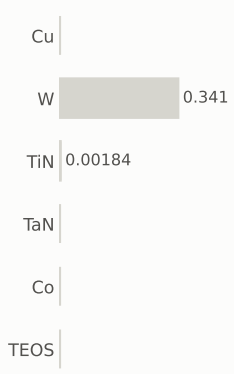
mask for this step

no mask for this step

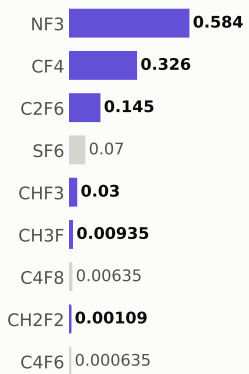
steps by process area



materials deposited (g)



gases emitted (g)



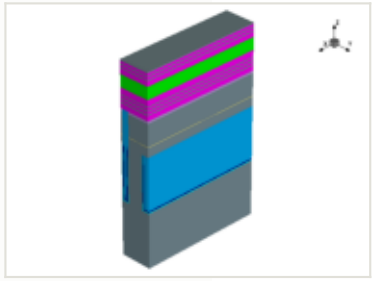
Step #520: CVD

depo_epi · CVD

1.31	100.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

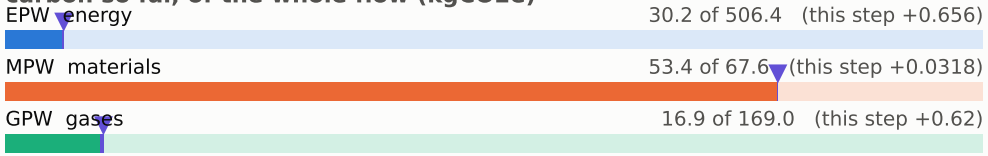
zoomed to fit



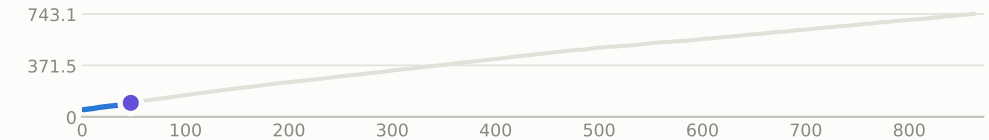
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.005 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

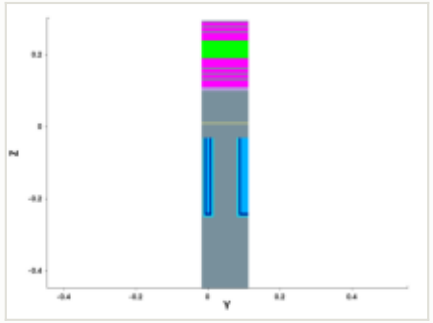
carbon so far, of the whole flow (kgCO2e)



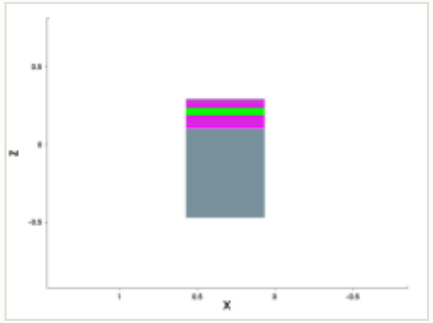
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



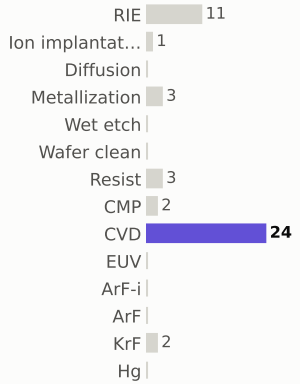
y cut at 0.049 µm



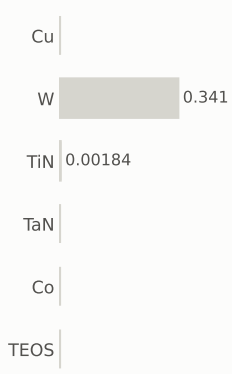
mask for this step

no mask for this step

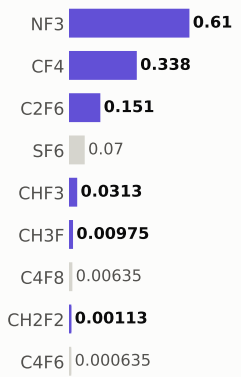
steps by process area



materials deposited (g)



gases emitted (g)



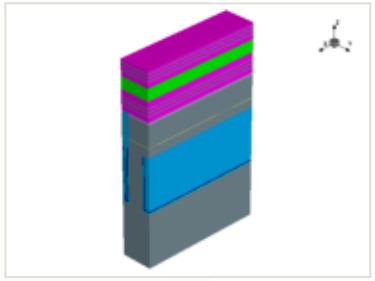
Step #530: CVD

depo_epi · CVD

1.31	101.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

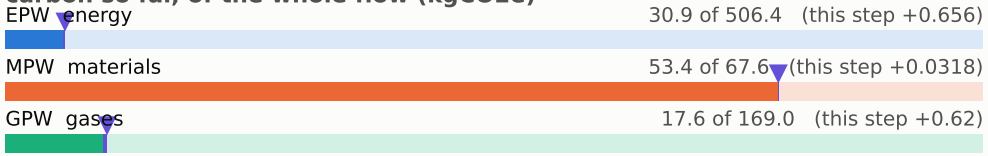


this step

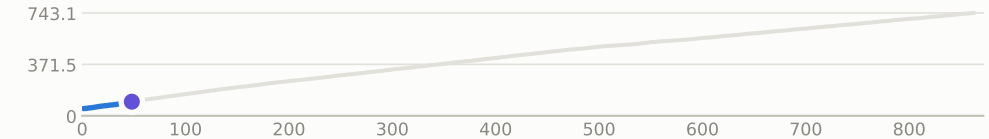
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

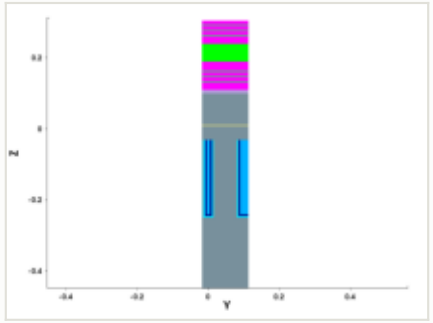
carbon so far, of the whole flow (kgCO2e)



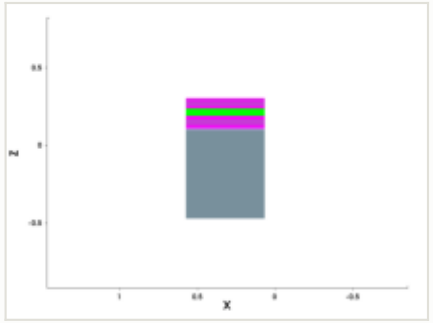
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



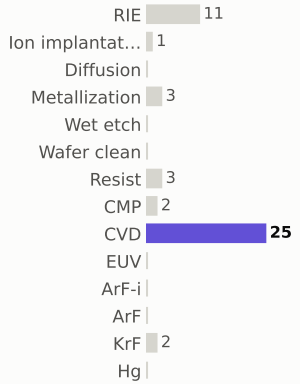
y cut at 0.049 µm



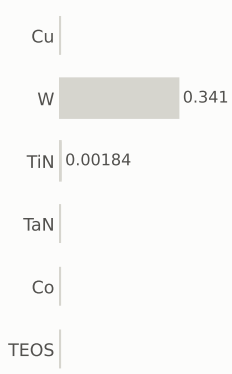
mask for this step

no mask for this step

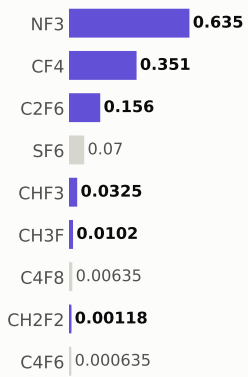
steps by process area



materials deposited (g)



gases emitted (g)



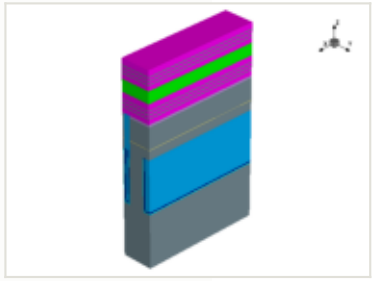
Step #540: CVD

depo_epi · CVD

1.31	103.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

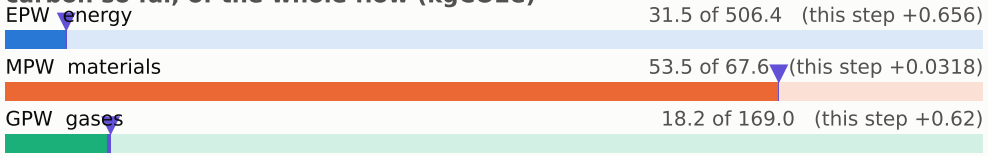


this step

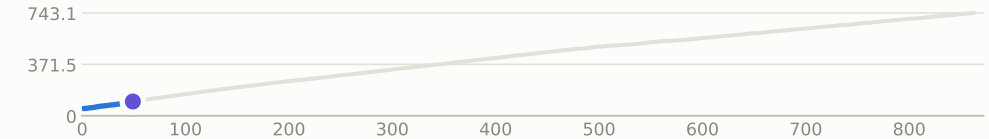
process area	CVD
material	—
thickness	0.015 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

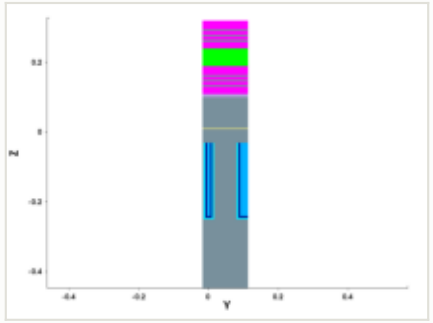
carbon so far, of the whole flow (kgCO2e)



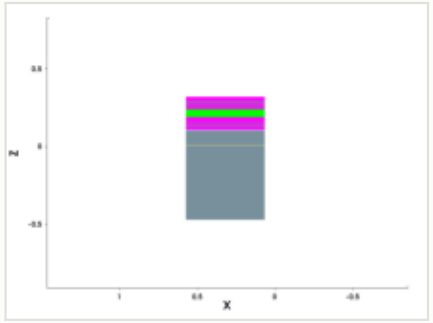
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



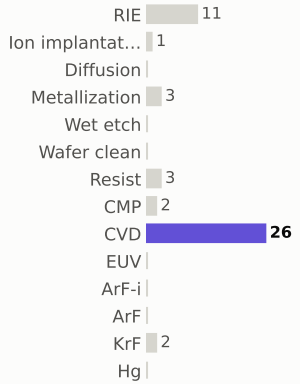
y cut at 0.049 µm



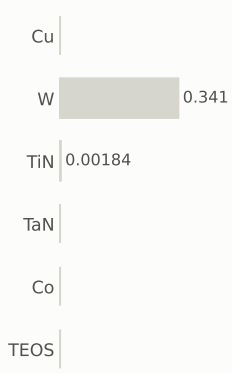
mask for this step

no mask for this step

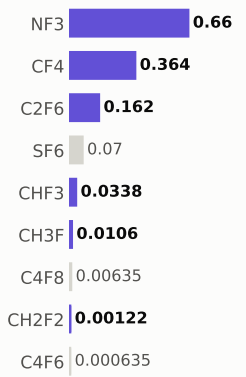
steps by process area



materials deposited (g)



gases emitted (g)



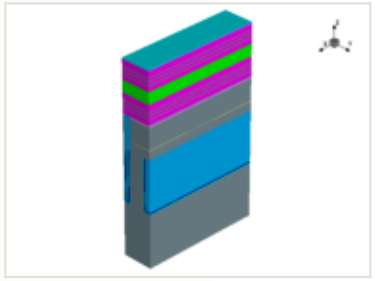
Step #560: CVD

depo_height_dependent · CVD

1.31	104.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

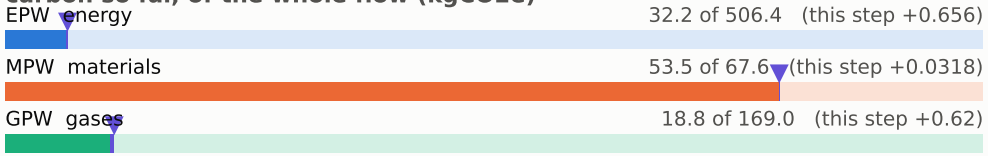


this step

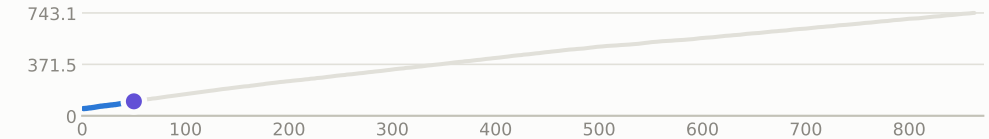
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

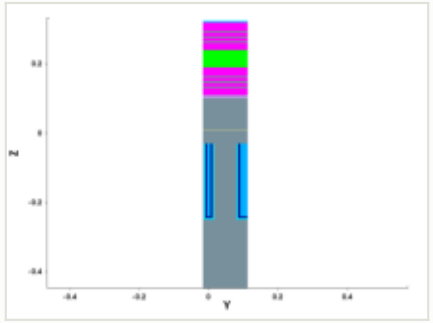
carbon so far, of the whole flow (kgCO2e)



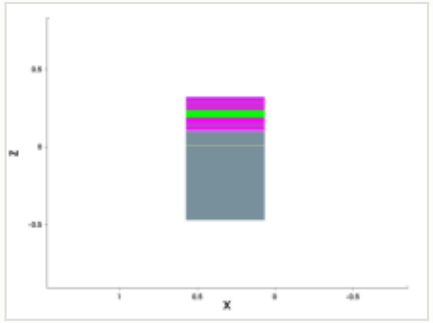
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



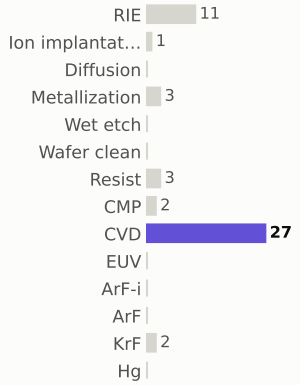
y cut at 0.049 µm



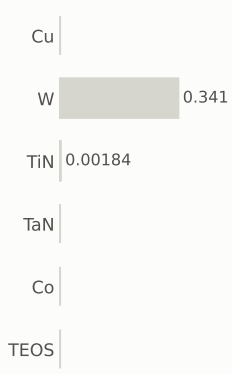
mask for this step

no mask for this step

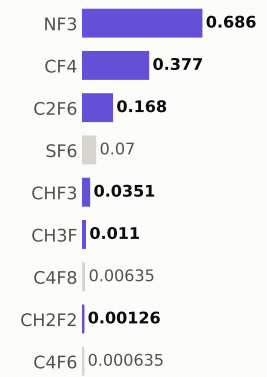
steps by process area



materials deposited (g)



gases emitted (g)



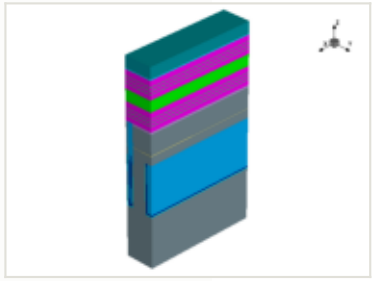
Step #570: CVD

depo_height_dependent · CVD

1.31	105.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

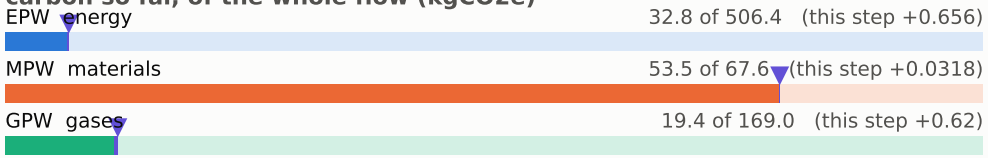
zoomed to fit



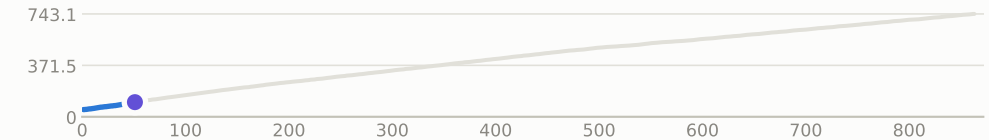
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

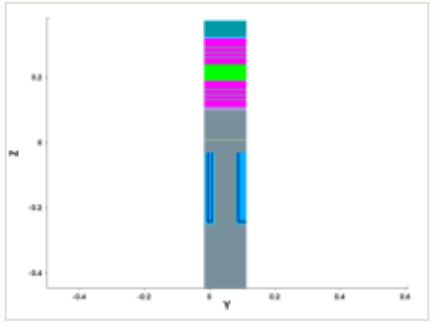
carbon so far, of the whole flow (kgCO2e)



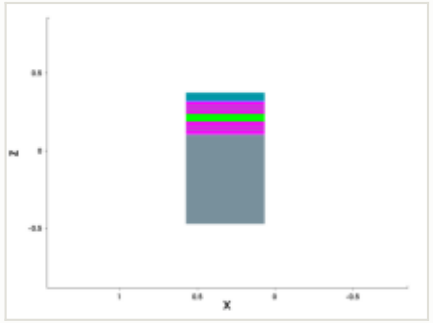
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



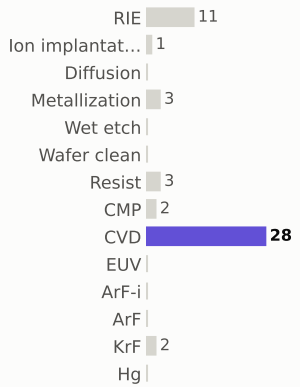
y cut at 0.049 µm



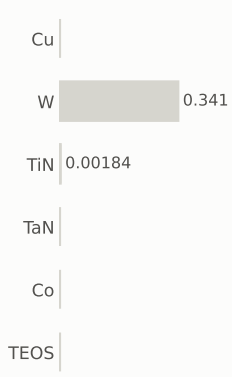
mask for this step

no mask for this step

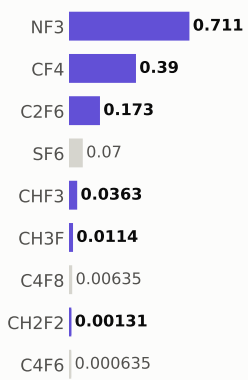
steps by process area



materials deposited (g)



gases emitted (g)



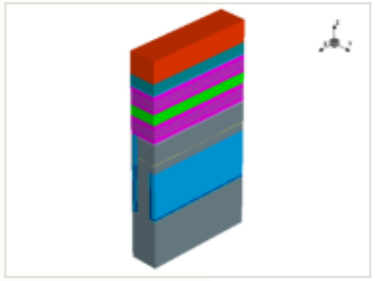
Step #580: CVD

depo_height_dependent · CVD

1.31	107.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

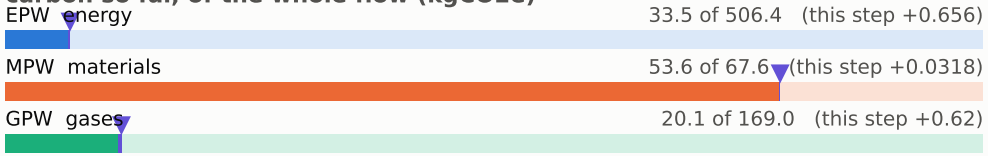


this step

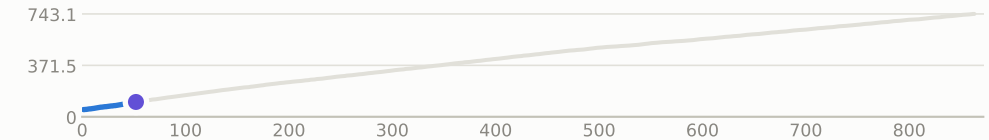
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

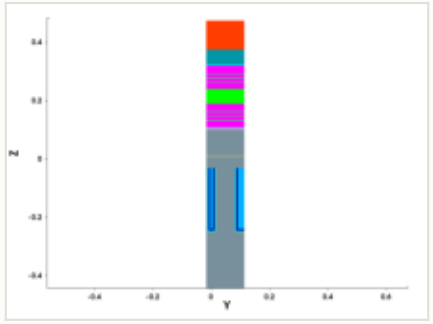
carbon so far, of the whole flow (kgCO2e)



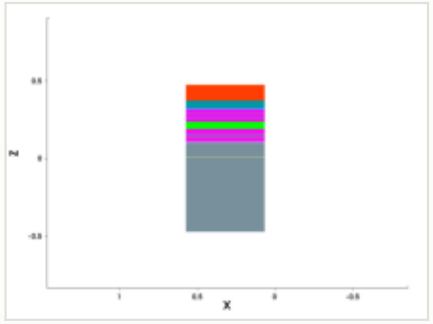
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



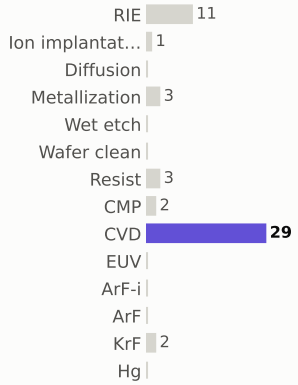
y cut at 0.049 µm



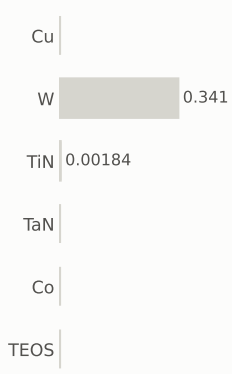
mask for this step

no mask for this step

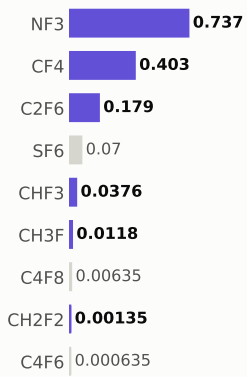
steps by process area



materials deposited (g)



gases emitted (g)



Step #590: CVD

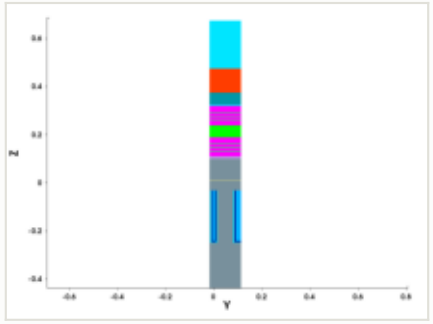
depo_height_dependent · CVD

1.31	108.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

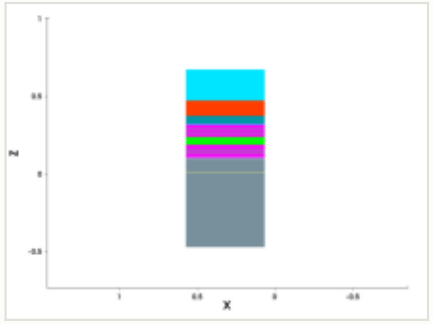
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

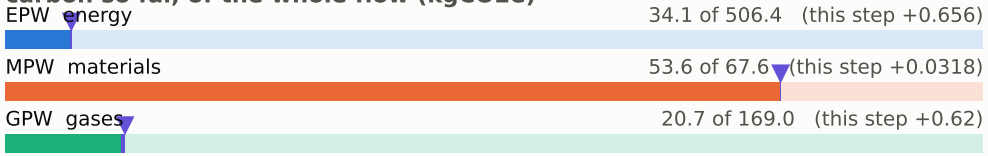
no mask for this step

this step

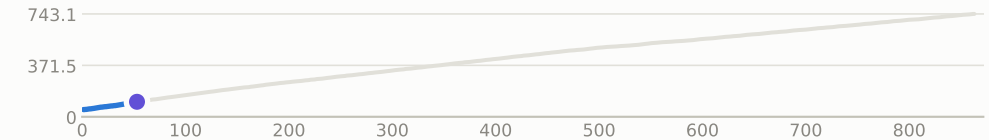
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

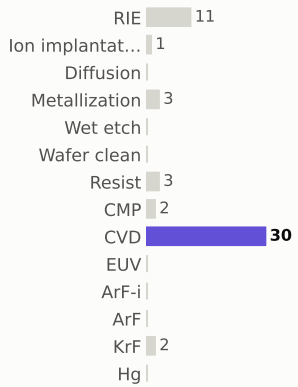
carbon so far, of the whole flow (kgCO2e)



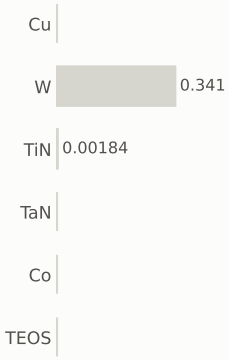
embodied carbon up to each step (kgCO2e per wafer)



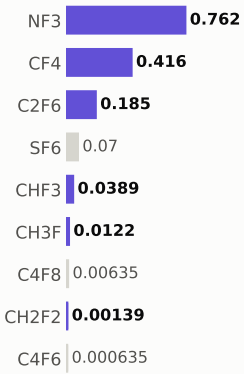
steps by process area



materials deposited (g)



gases emitted (g)



Step #600: CVD

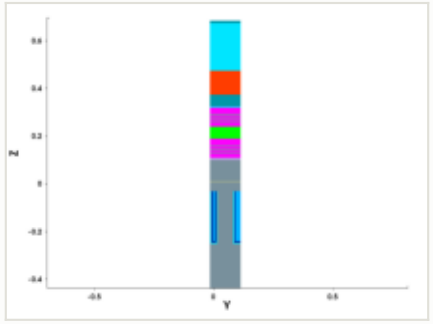
depo_height_dependent · CVD

1.31	109.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

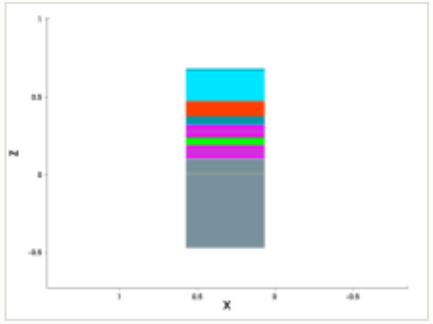
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

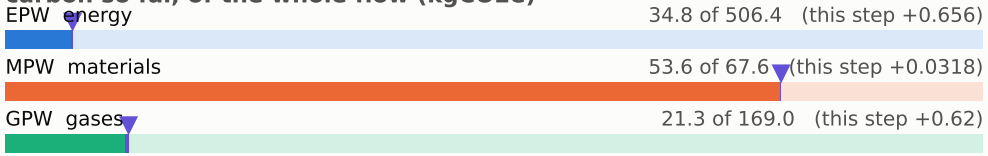
no mask for this step

this step

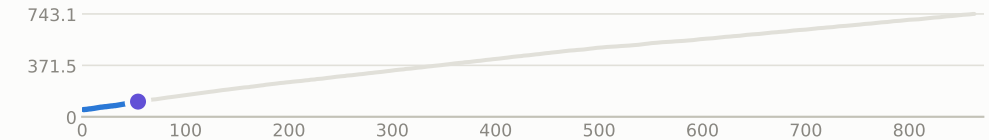
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

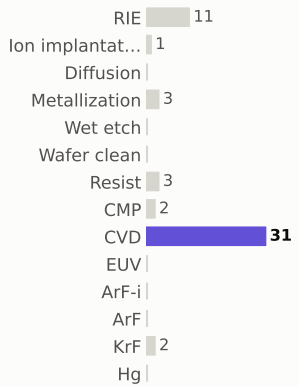
carbon so far, of the whole flow (kgCO2e)



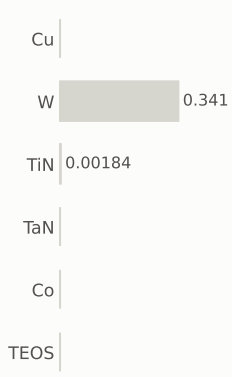
embodied carbon up to each step (kgCO2e per wafer)



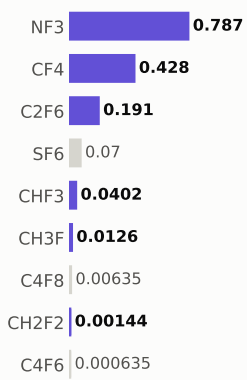
steps by process area



materials deposited (g)



gases emitted (g)



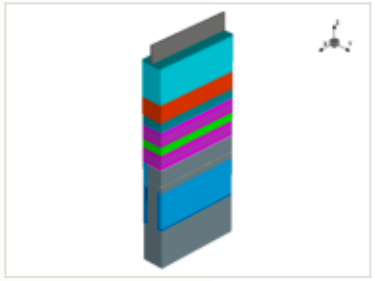
Step #610: EUV

litho_patterning · EUV

1.96	111.7	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

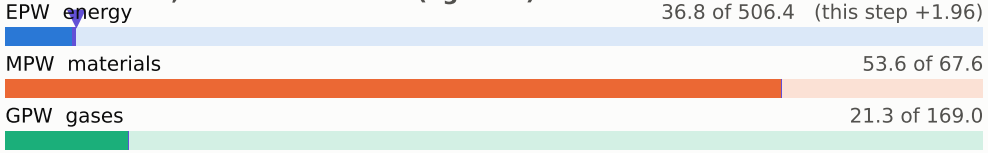


this step

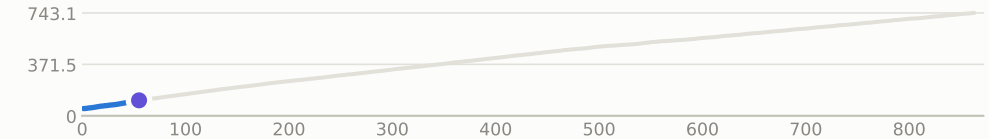
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

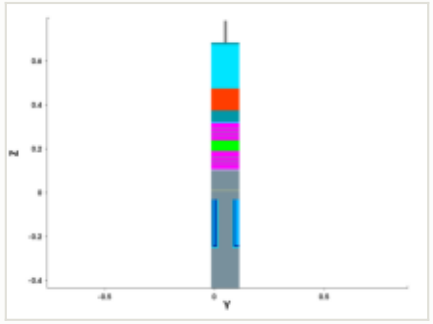
carbon so far, of the whole flow (kgCO2e)



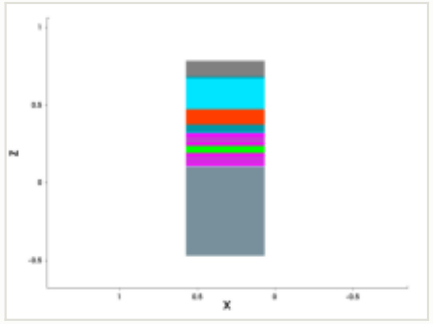
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

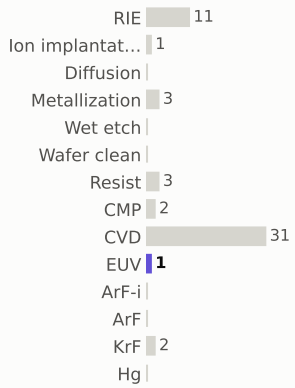


y cut at 0.049 µm

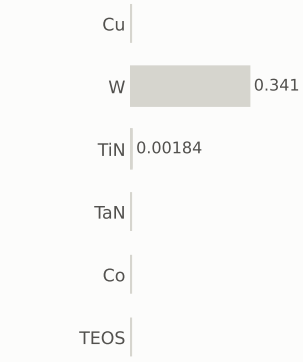


mask: layer 2:0, positive (GDSII)

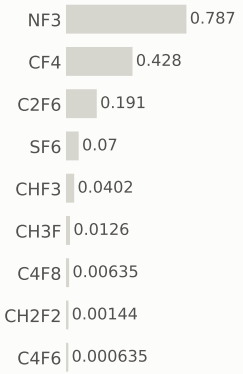
steps by process area



materials deposited (g)



gases emitted (g)



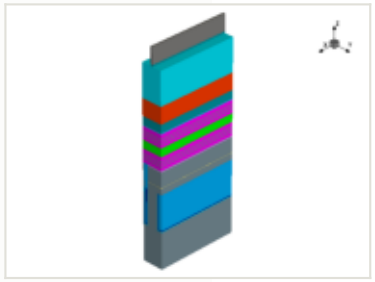
Step #620: RIE

etch_anisotropic · RIE

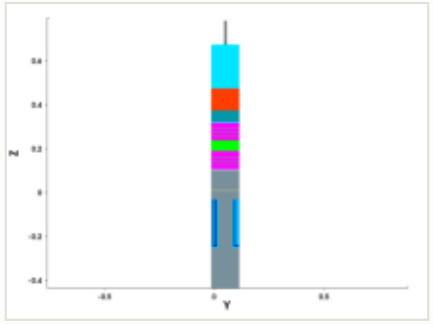
1.04	112.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

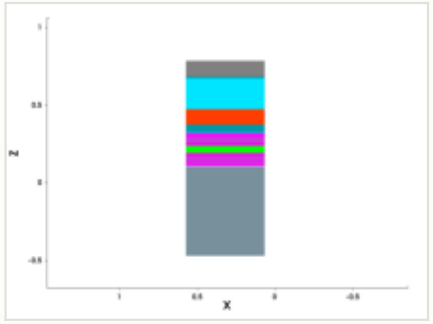
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

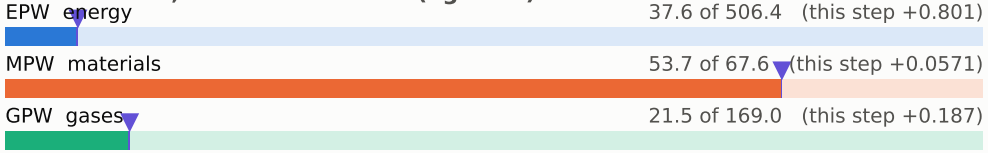
no mask for this step

this step

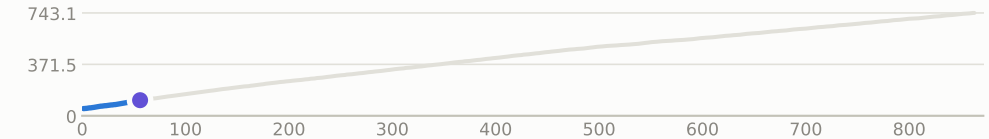
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

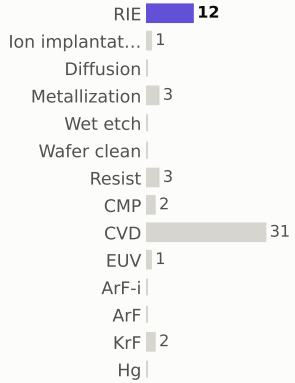
carbon so far, of the whole flow (kgCO2e)



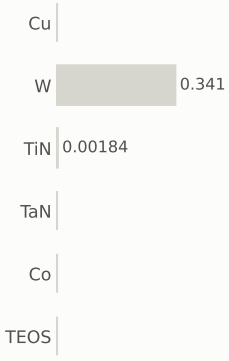
embodied carbon up to each step (kgCO2e per wafer)



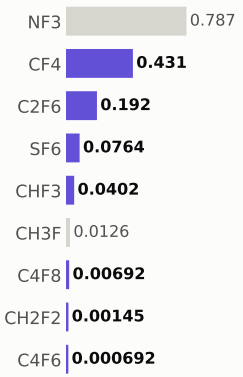
steps by process area



materials deposited (g)



gases emitted (g)



Step #630: RIE

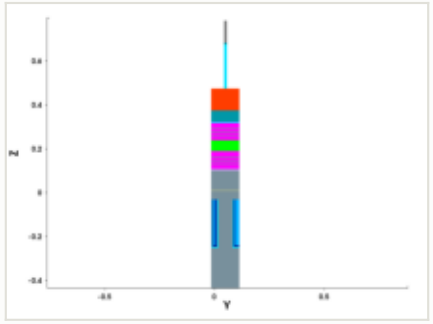
etch_anisotropic · RIE

1.04	113.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

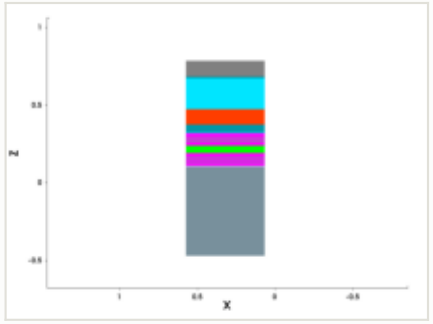
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

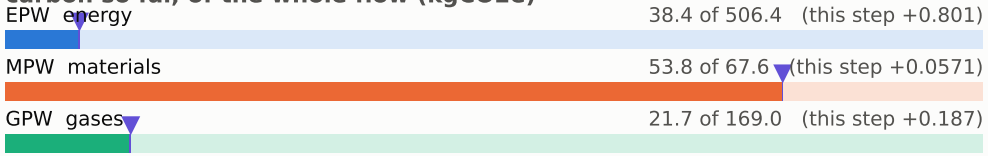
no mask for this step

this step

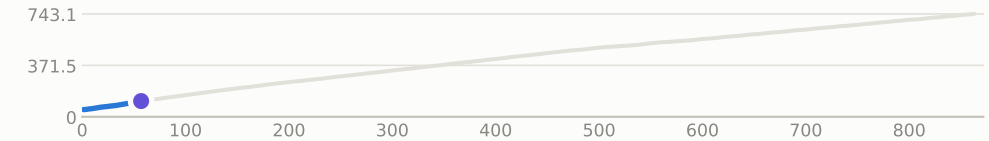
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

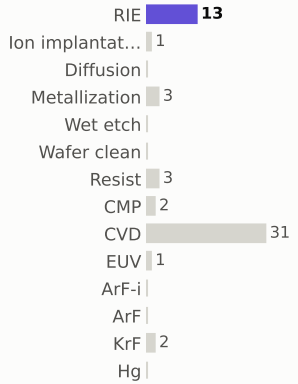
carbon so far, of the whole flow (kgCO2e)



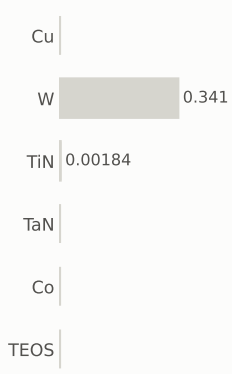
embodied carbon up to each step (kgCO2e per wafer)



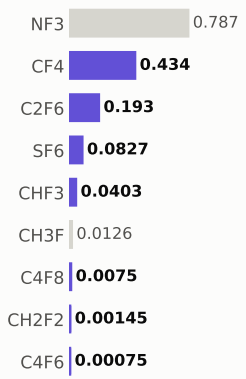
steps by process area



materials deposited (g)



gases emitted (g)



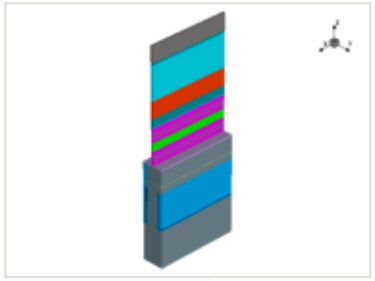
Step #640: RIE

etch_anisotropic · RIE

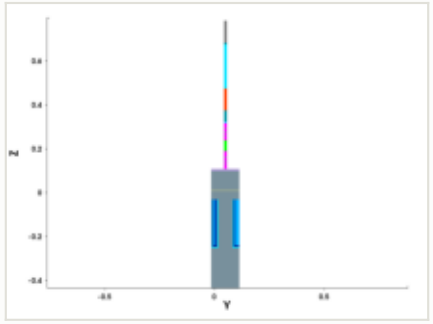
1.04	114.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

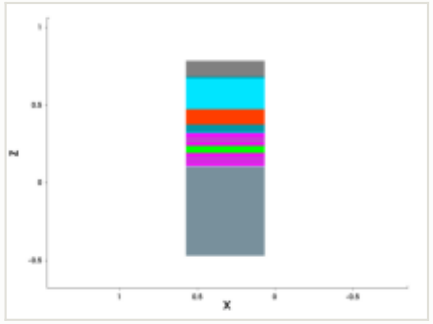
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

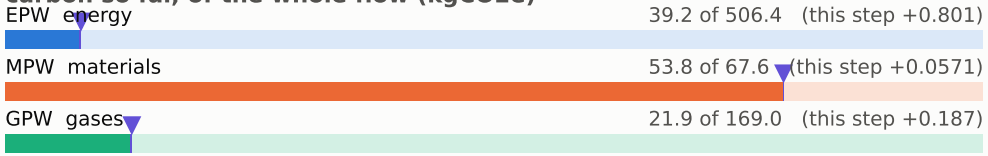
no mask for this step

this step

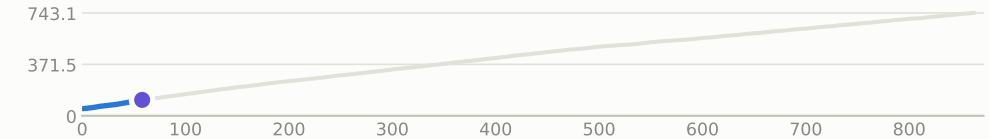
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

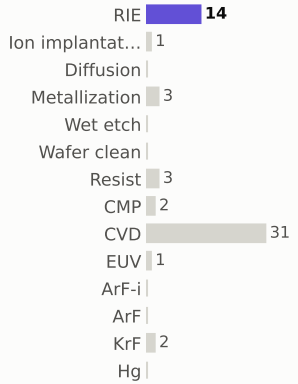
carbon so far, of the whole flow (kgCO2e)



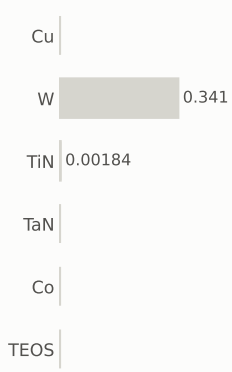
embodied carbon up to each step (kgCO2e per wafer)



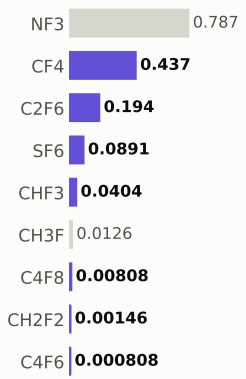
steps by process area



materials deposited (g)



gases emitted (g)



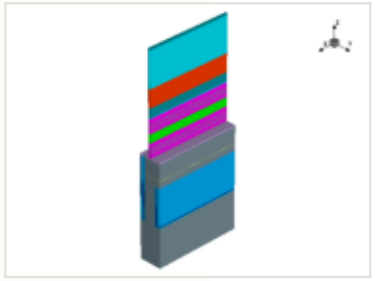
Step #650: Resist

etch_strip · Resist

0.169	115.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

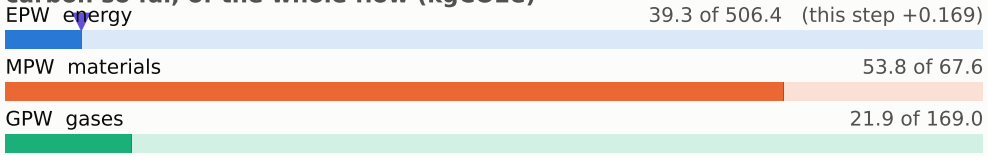


this step

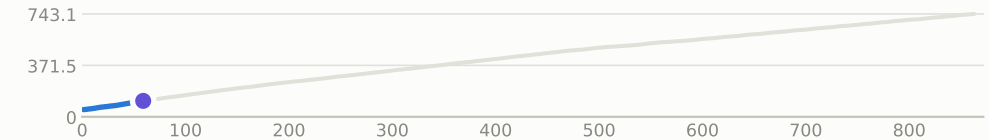
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

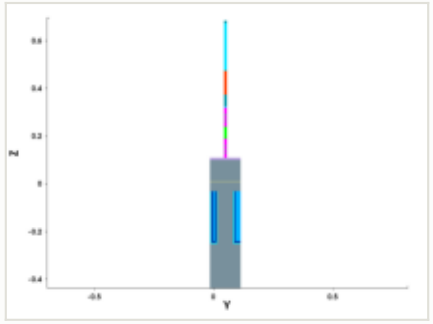
carbon so far, of the whole flow (kgCO2e)



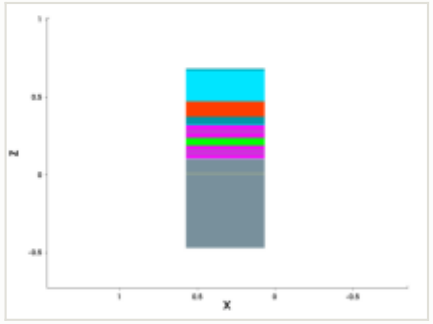
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



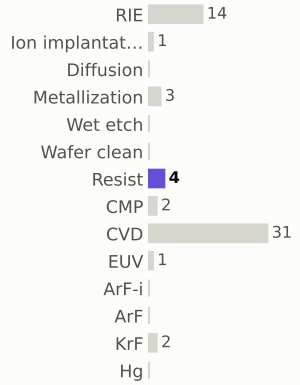
y cut at 0.049 µm



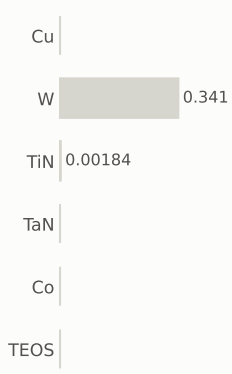
mask for this step

no mask for this step

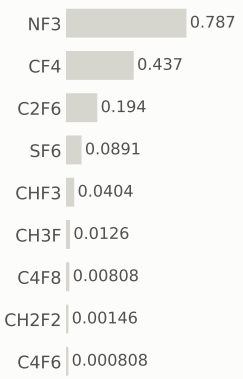
steps by process area



materials deposited (g)



gases emitted (g)



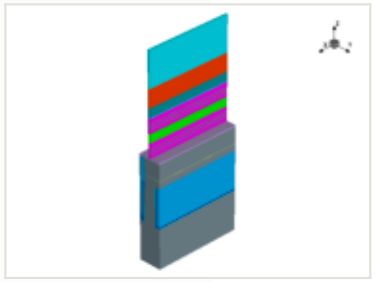
Step #660: RIE

etch_anisotropic · RIE

1.04	116.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

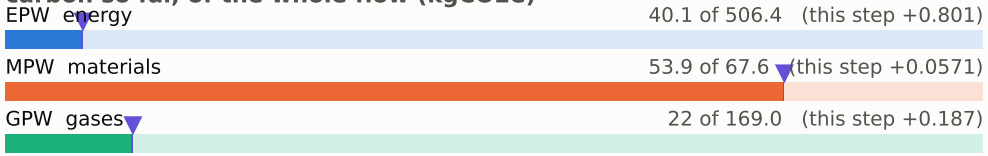


this step

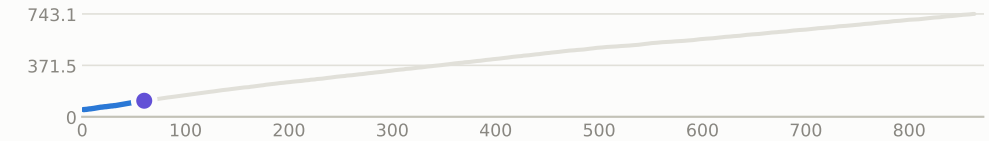
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

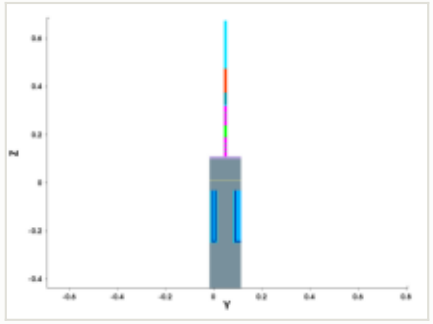
carbon so far, of the whole flow (kgCO2e)



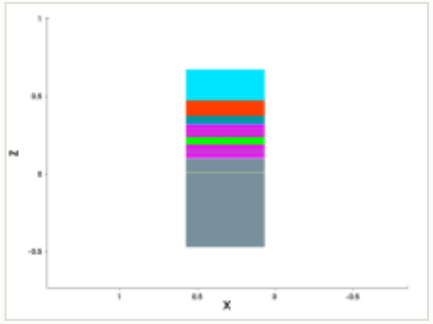
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



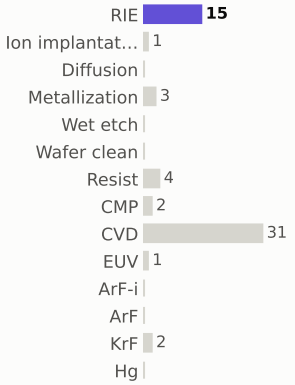
y cut at 0.049 µm



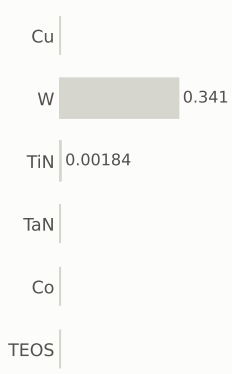
mask for this step

no mask for this step

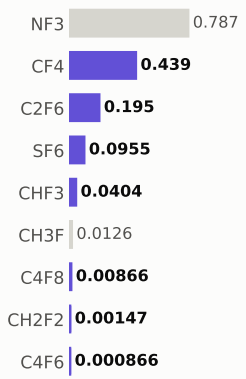
steps by process area



materials deposited (g)



gases emitted (g)



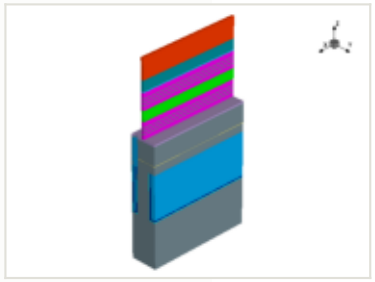
Step #670: RIE

etch_anisotropic · RIE

1.04	117.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

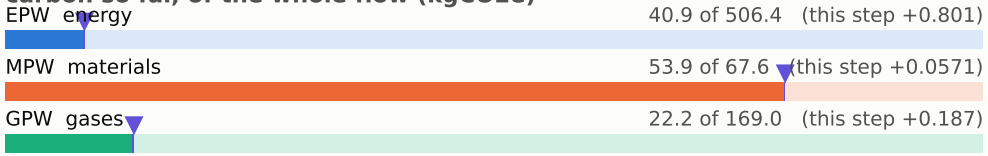


this step

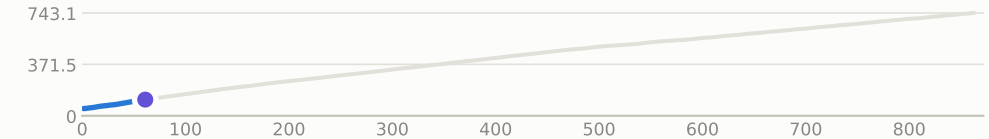
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

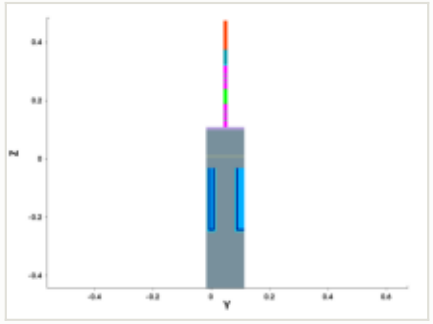
carbon so far, of the whole flow (kgCO2e)



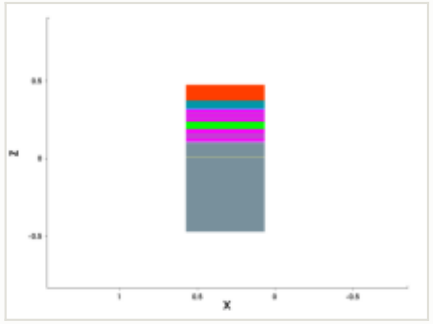
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



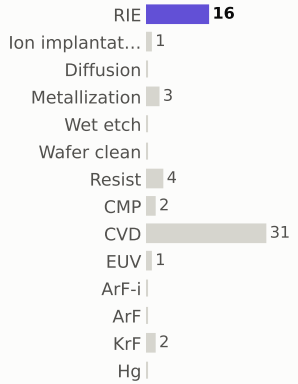
y cut at 0.049 µm



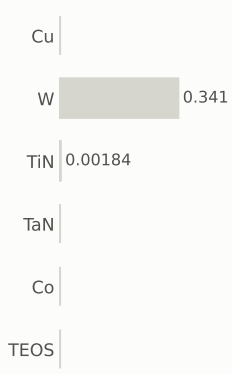
mask for this step

no mask for this step

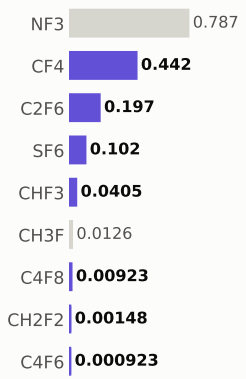
steps by process area



materials deposited (g)



gases emitted (g)



Step #680: CVD

depo_isotropic · CVD

1.31	118.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

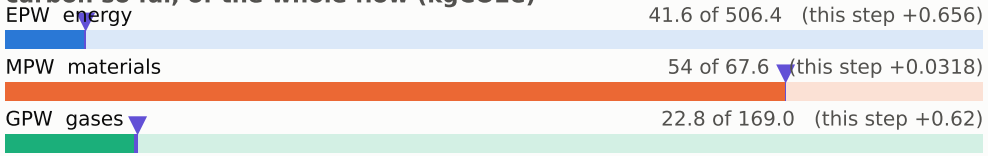
zoomed to fit

this step

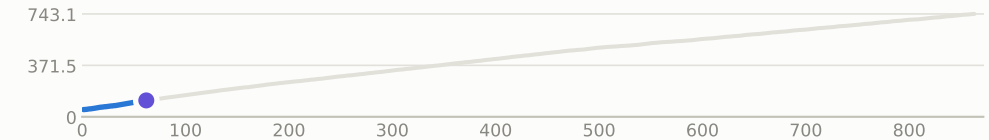
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

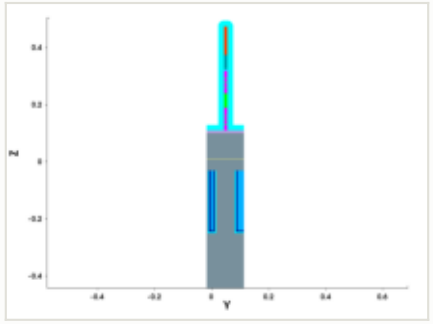
carbon so far, of the whole flow (kgCO2e)



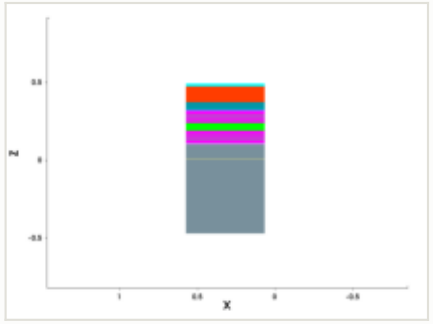
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



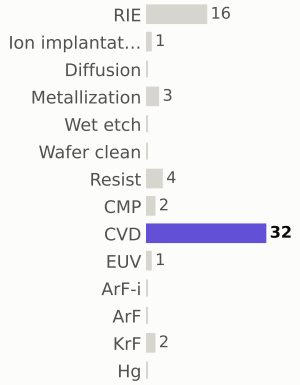
y cut at 0.049 µm



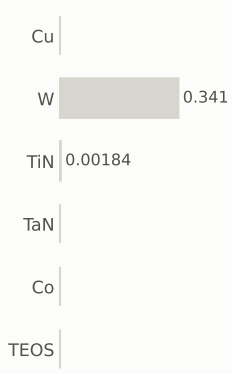
mask for this step

no mask for this step

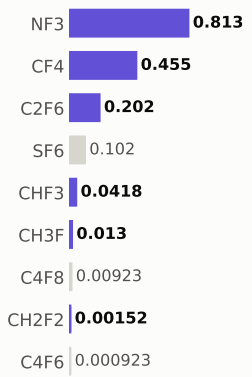
steps by process area



materials deposited (g)



gases emitted (g)



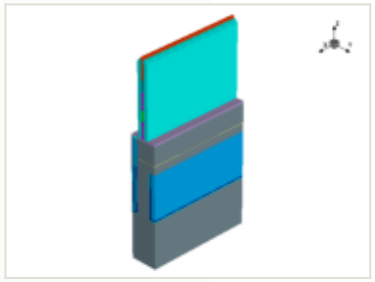
Step #690: RIE

etch_anisotropic · RIE

1.04	119.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

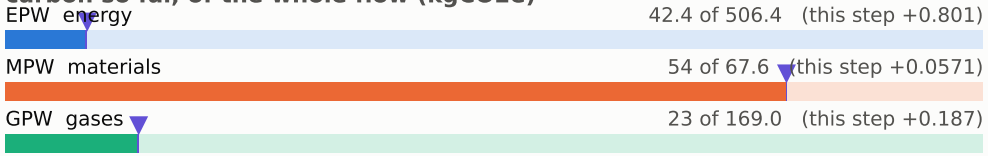
zoomed to fit



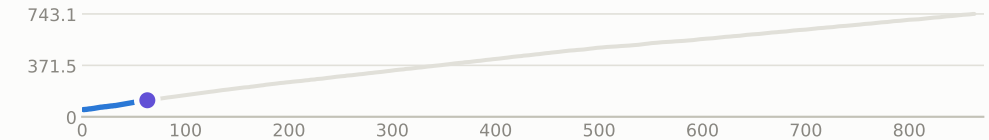
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

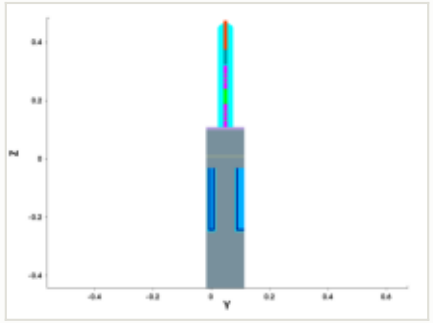
carbon so far, of the whole flow (kgCO2e)



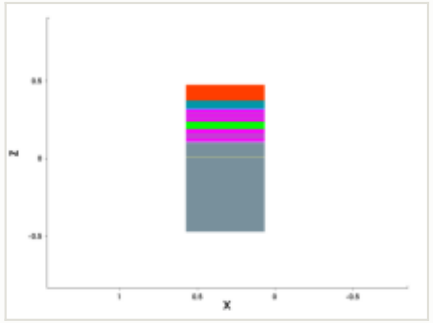
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



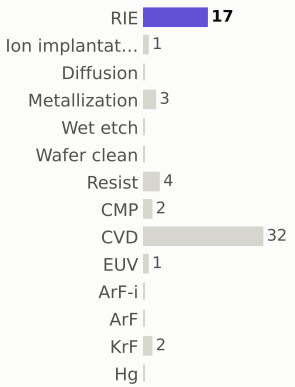
y cut at 0.049 µm



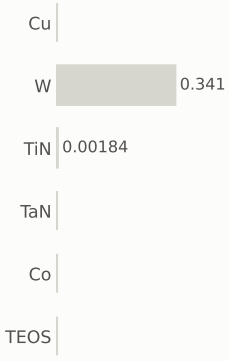
mask for this step

no mask for this step

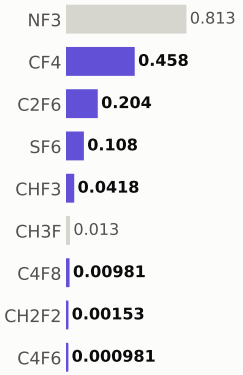
steps by process area



materials deposited (g)



gases emitted (g)



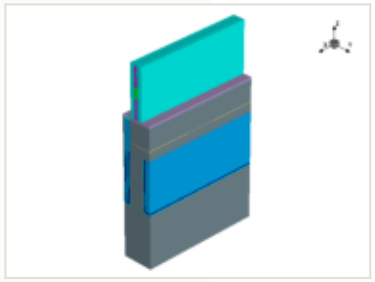
Step #700: CMP

etch_cmp · CMP

0.169	119.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

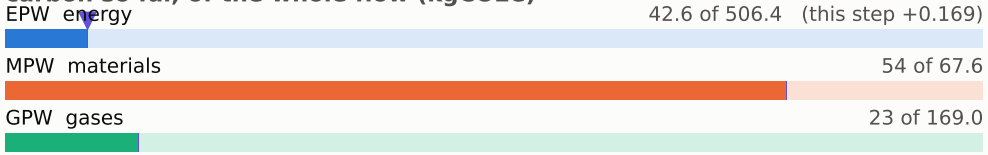


this step

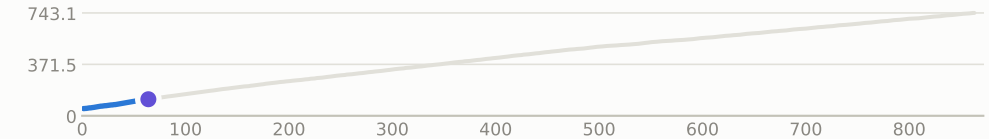
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

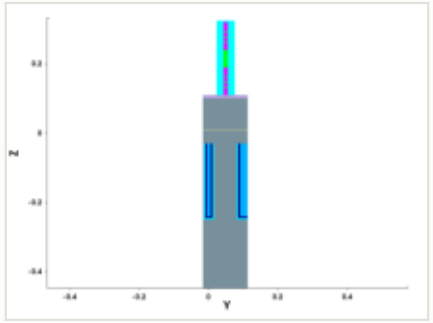
carbon so far, of the whole flow (kgCO2e)



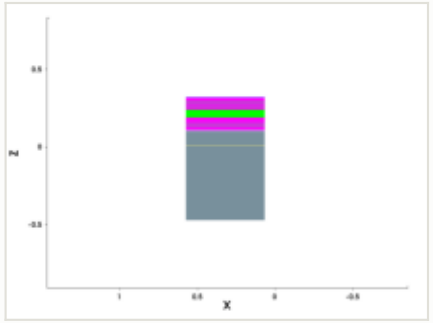
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



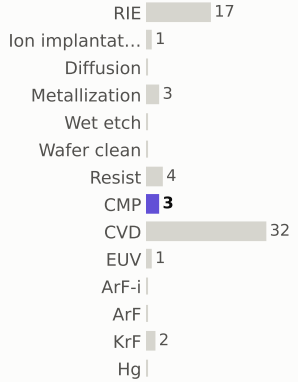
y cut at 0.049 µm



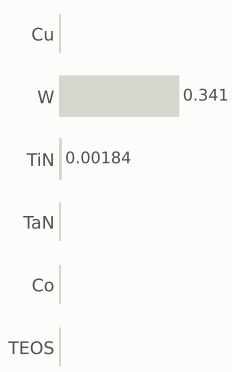
mask for this step

no mask for this step

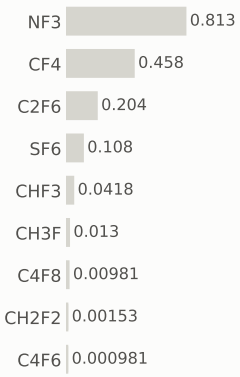
steps by process area



materials deposited (g)



gases emitted (g)



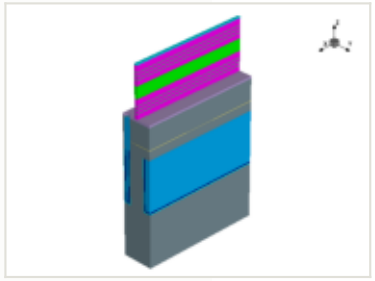
Step #710: RIE

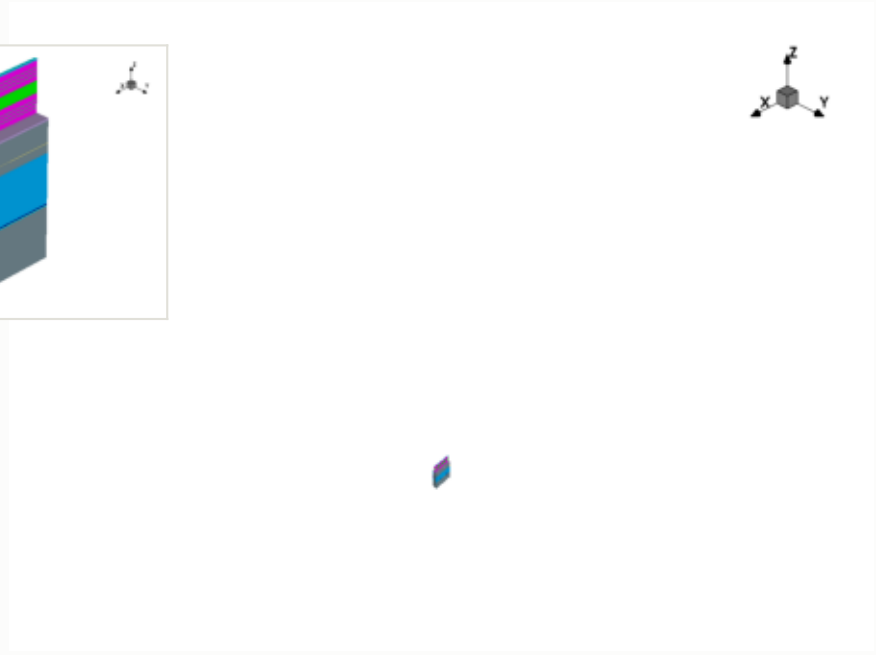
etch_anisotropic · RIE

1.04	120.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



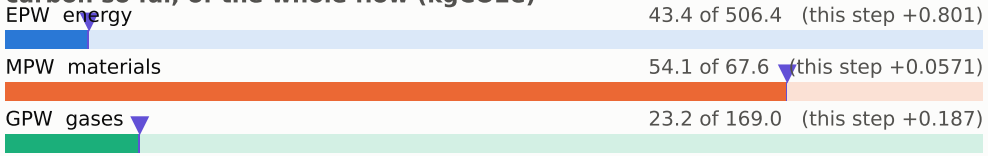


this step

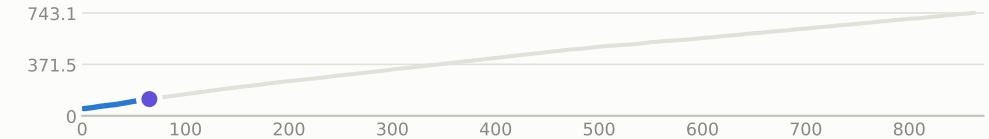
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

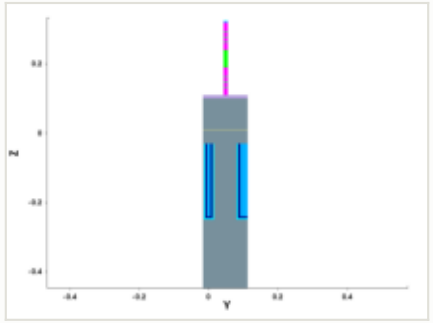
carbon so far, of the whole flow (kgCO2e)



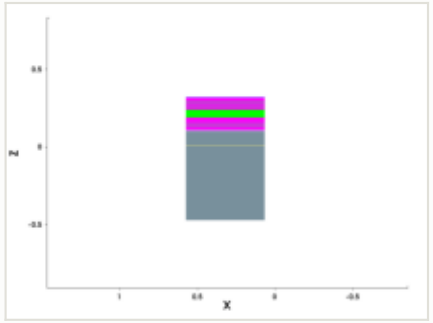
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



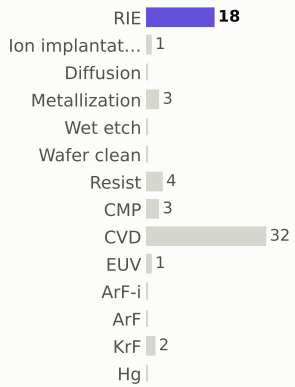
y cut at 0.049 µm



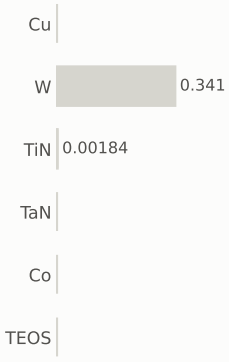
mask for this step

no mask for this step

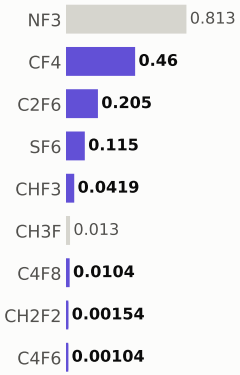
steps by process area



materials deposited (g)



gases emitted (g)



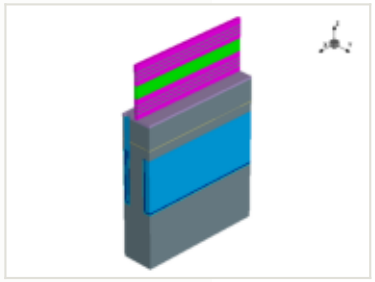
Step #720: RIE

etch_anisotropic · RIE

1.04	121.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

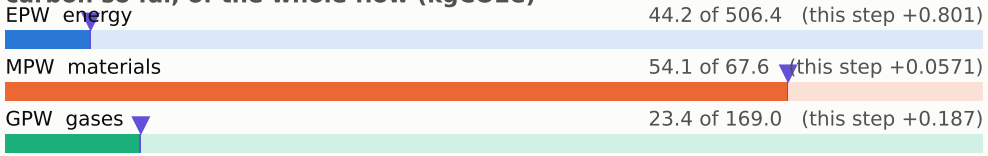


this step

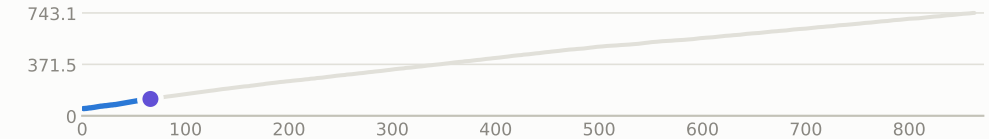
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

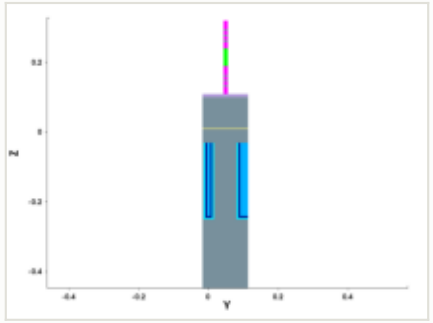
carbon so far, of the whole flow (kgCO2e)



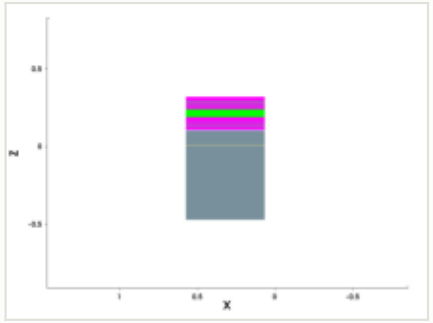
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



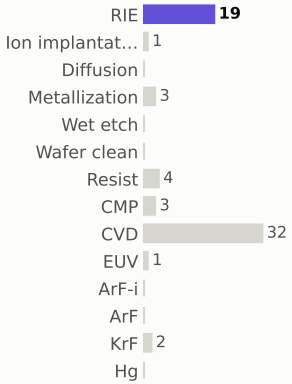
y cut at 0.049 µm



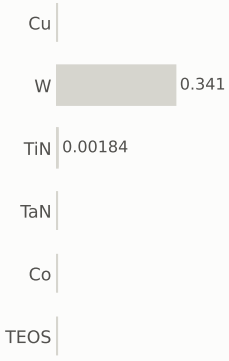
mask for this step

no mask for this step

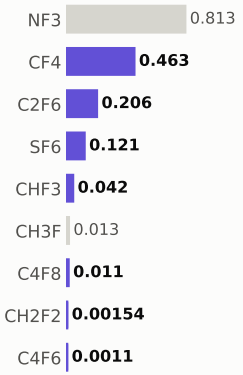
steps by process area



materials deposited (g)



gases emitted (g)



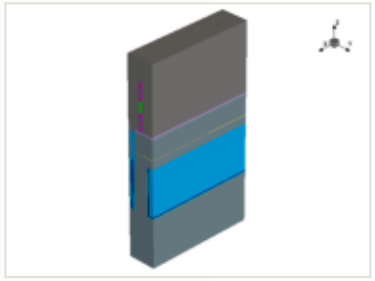
Step #740: EUV

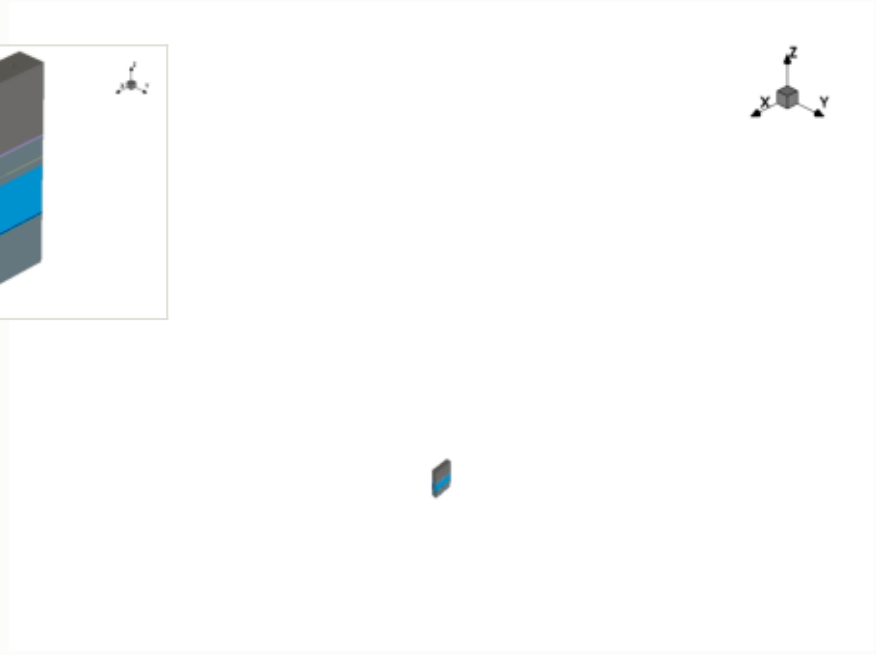
litho_patterning · EUV

1.96	123.6	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



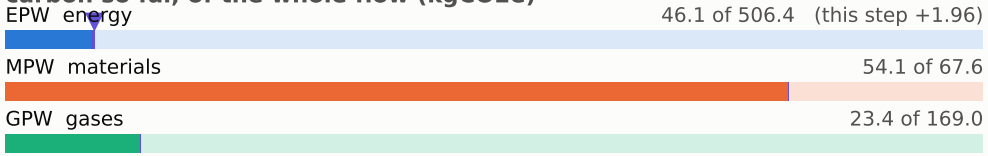


this step

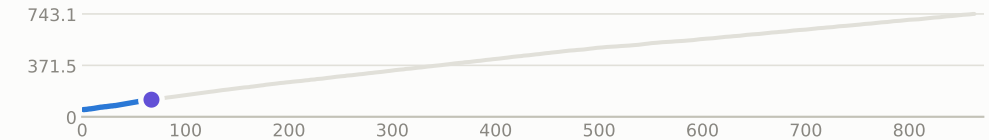
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

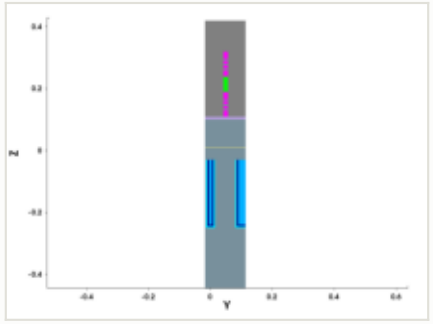
carbon so far, of the whole flow (kgCO2e)



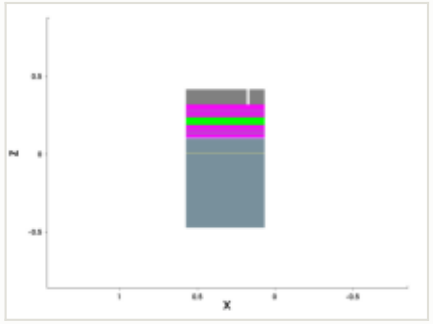
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

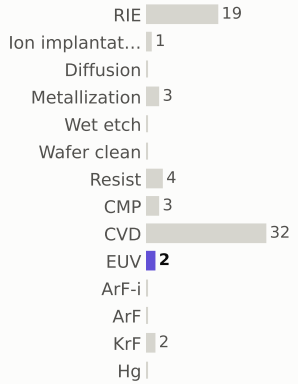


y cut at 0.049 µm

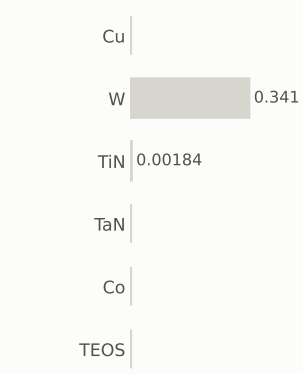


mask: layer 14:0 (GDSII)

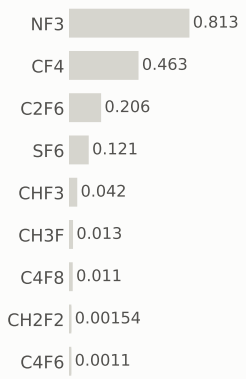
steps by process area



materials deposited (g)



gases emitted (g)



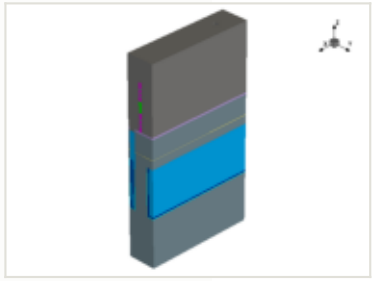
Step #750: RIE

etch_anisotropic · RIE

1.04	124.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

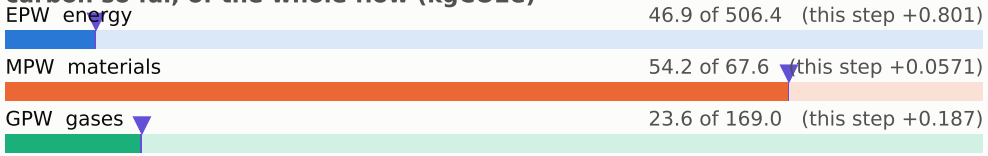


this step

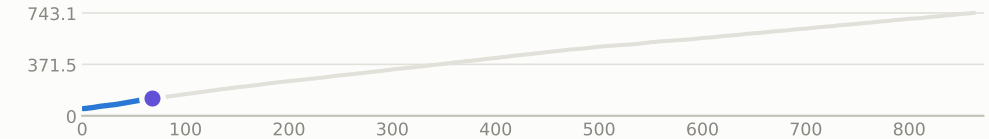
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

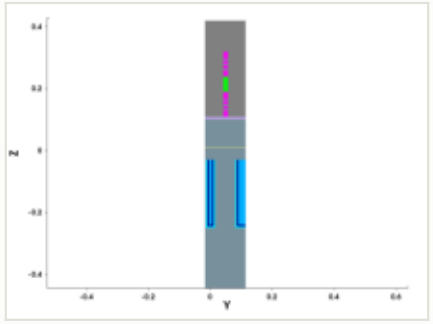
carbon so far, of the whole flow (kgCO2e)



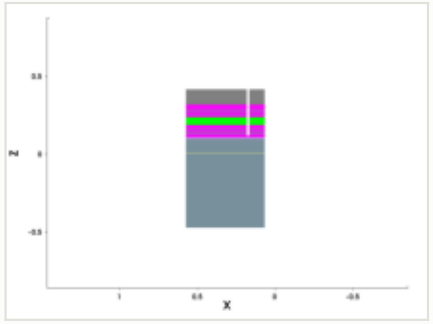
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



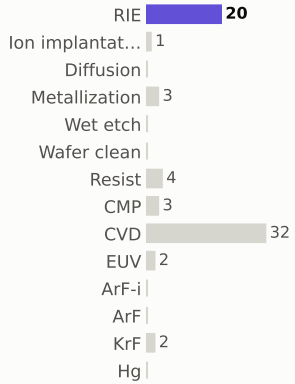
y cut at 0.049 µm



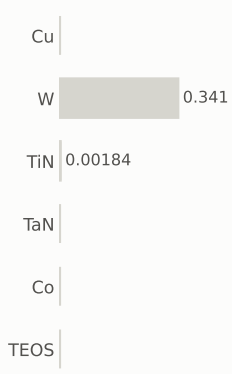
mask for this step

no mask for this step

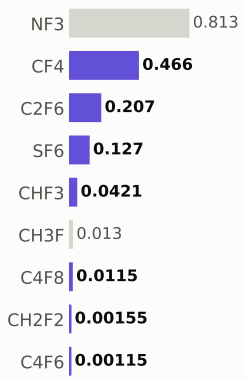
steps by process area



materials deposited (g)



gases emitted (g)



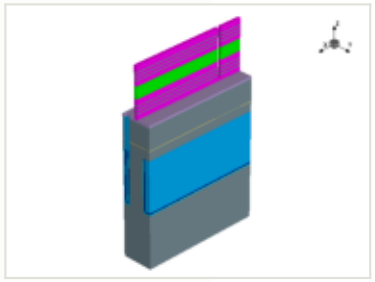
Step #760: Resist

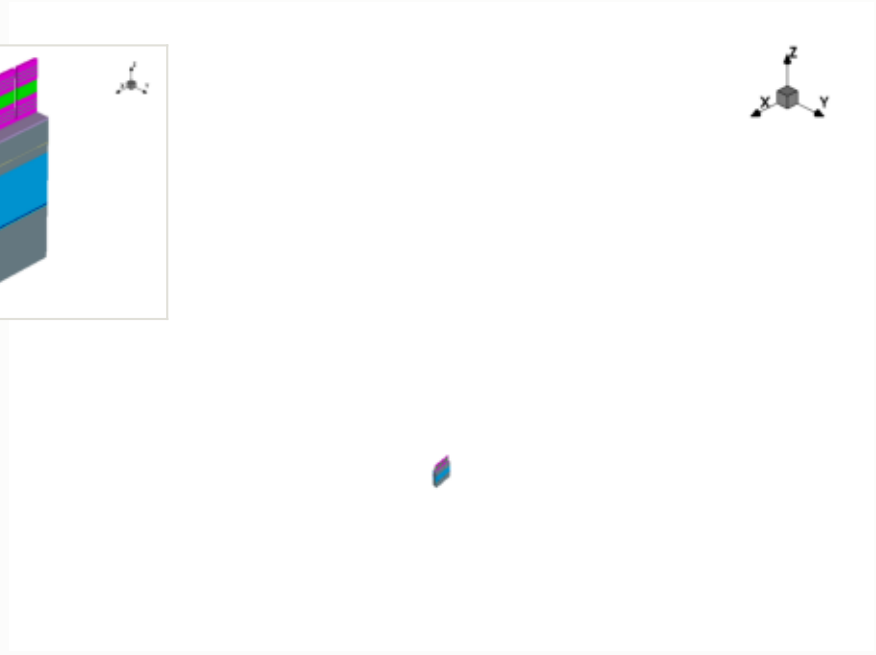
etch_strip · Resist

0.169	124.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



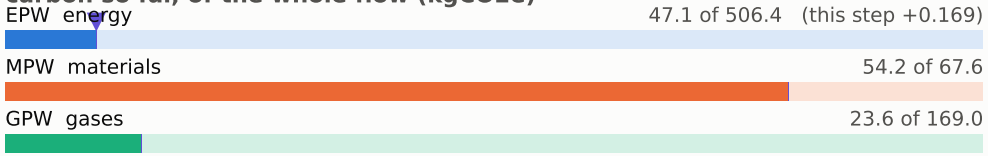


this step

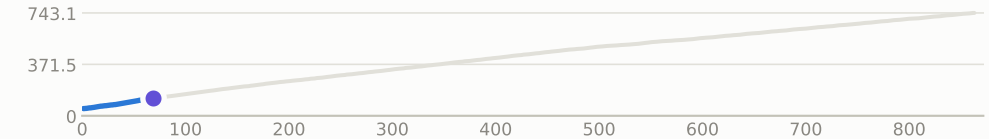
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

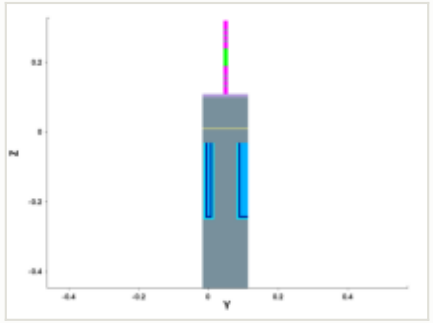
carbon so far, of the whole flow (kgCO2e)



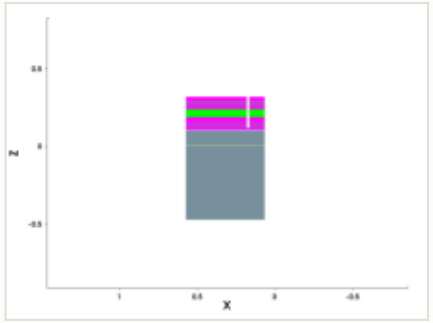
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



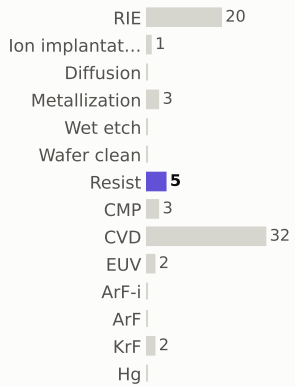
y cut at 0.049 µm



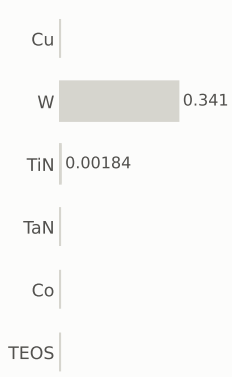
mask for this step

no mask for this step

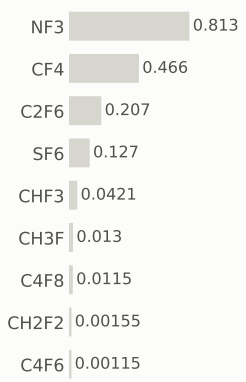
steps by process area



materials deposited (g)



gases emitted (g)



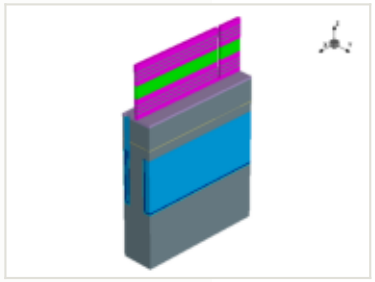
Step #770: CVD

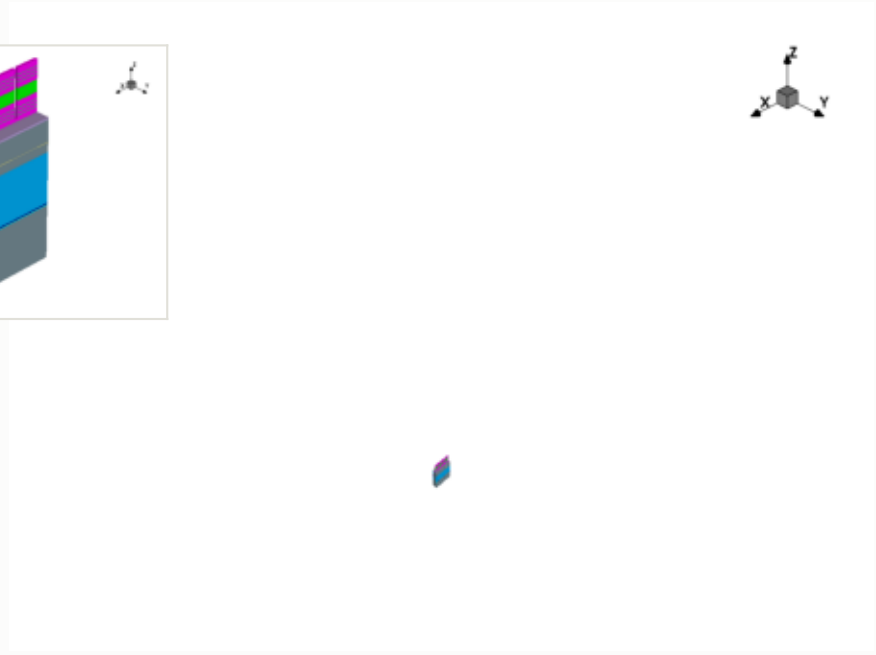
depo_fill · CVD

1.31	126.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



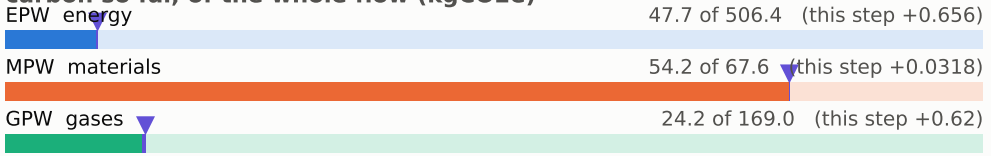


this step

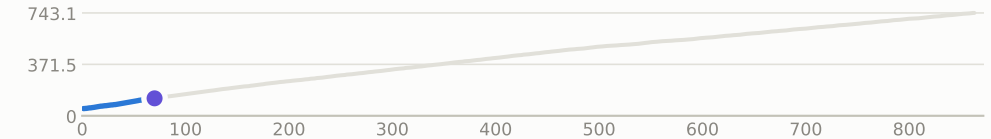
process area	CVD
material	—
thickness	-0.2 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

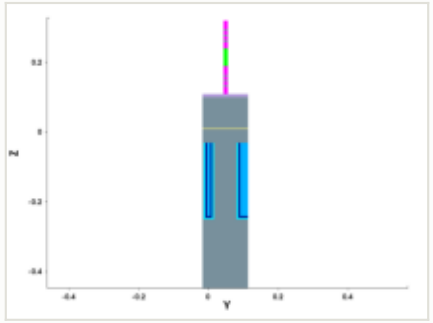
carbon so far, of the whole flow (kgCO2e)



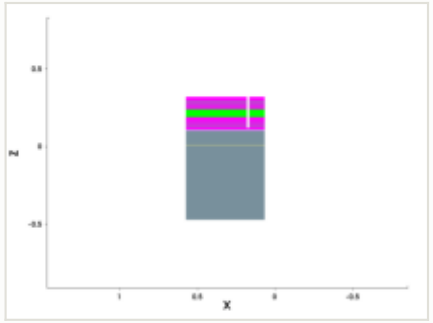
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



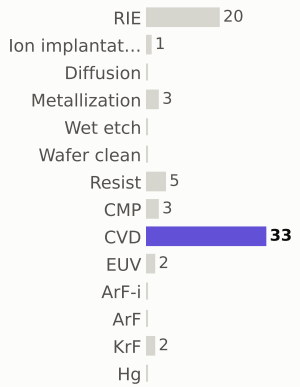
y cut at 0.049 µm



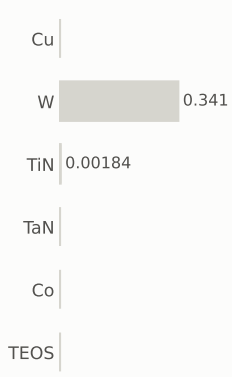
mask for this step

no mask for this step

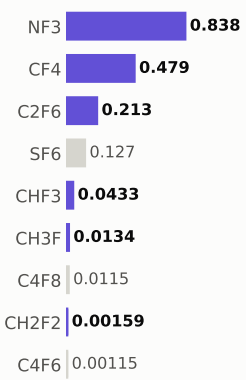
steps by process area



materials deposited (g)



gases emitted (g)



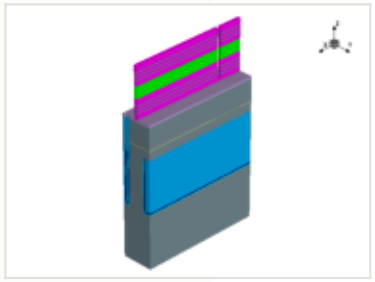
Step #780: CMP

etch_cmp · CMP

0.169	126.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

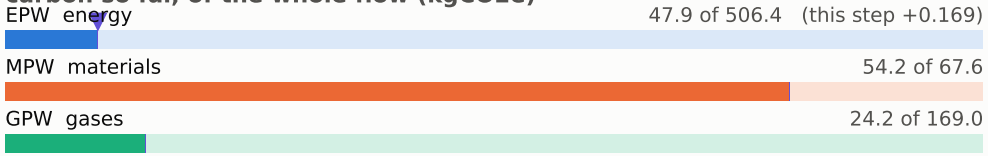


this step

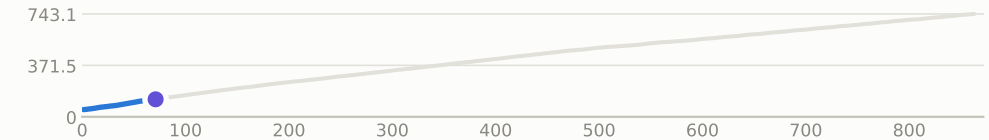
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

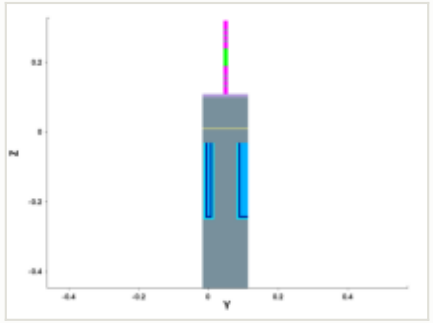
carbon so far, of the whole flow (kgCO2e)



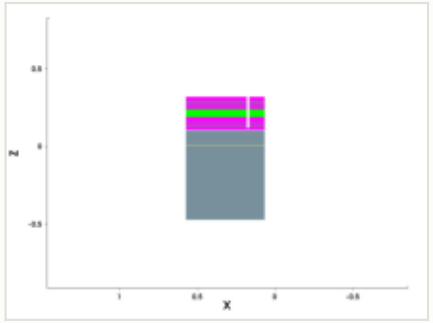
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



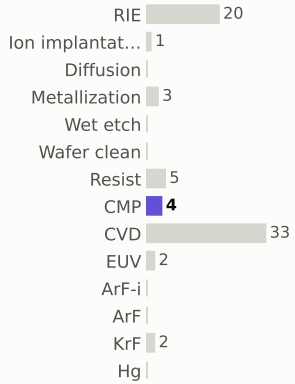
y cut at 0.049 µm



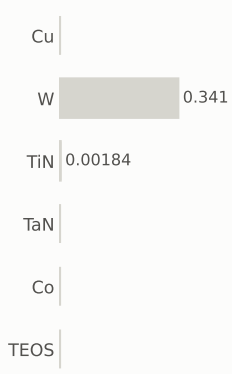
mask for this step

no mask for this step

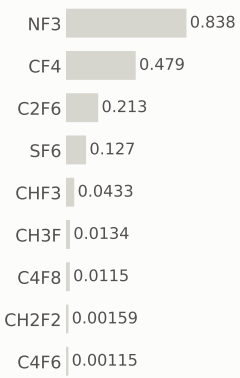
steps by process area



materials deposited (g)



gases emitted (g)



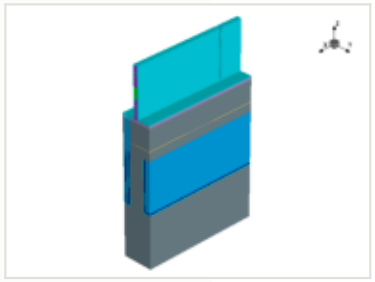
Step #800: CVD

depo_isotropic · CVD

1.31	127.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

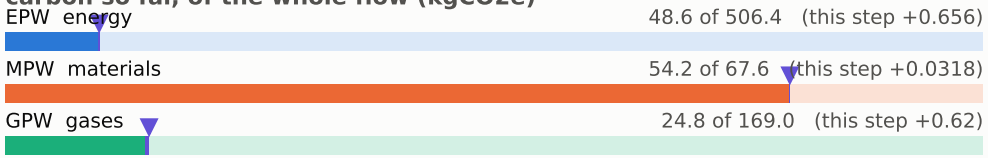


this step

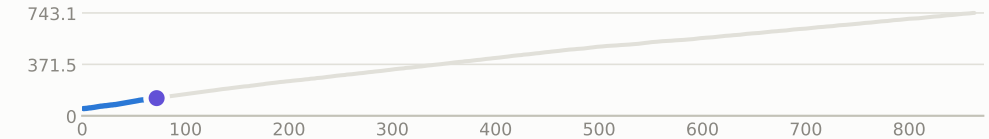
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

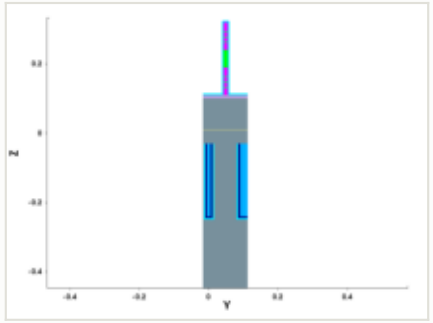
carbon so far, of the whole flow (kgCO2e)



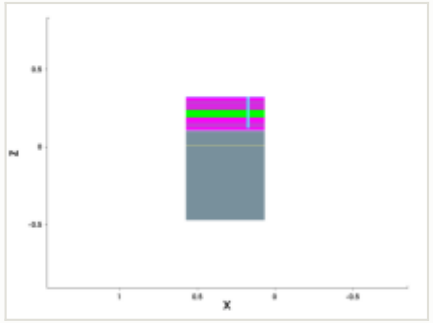
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



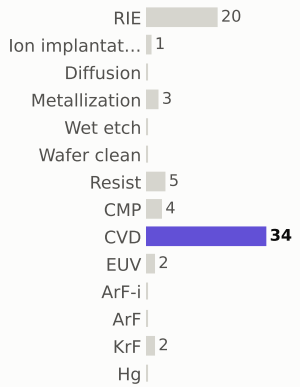
y cut at 0.049 µm



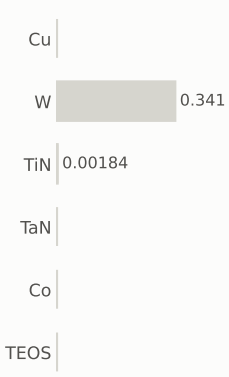
mask for this step

no mask for this step

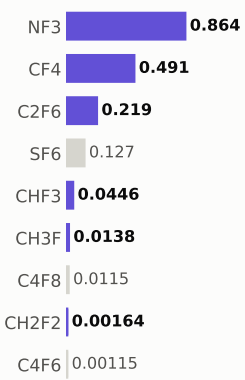
steps by process area



materials deposited (g)



gases emitted (g)



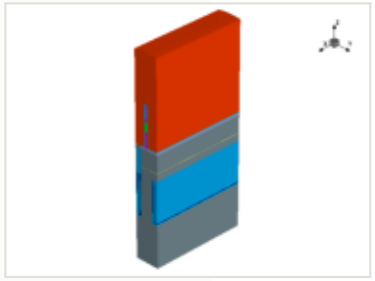
Step #810: CVD

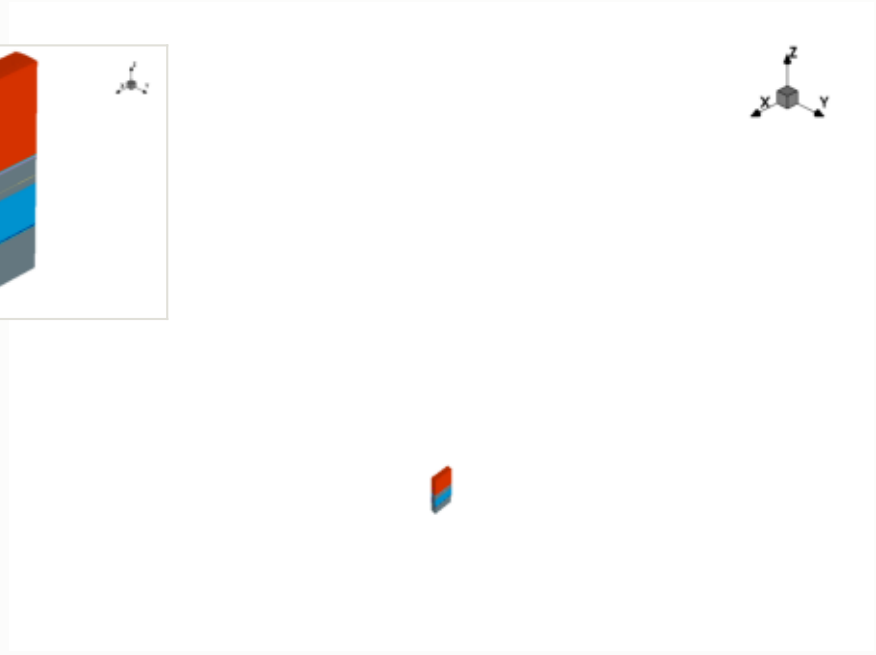
depo_isotropic · CVD

1.31	129.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



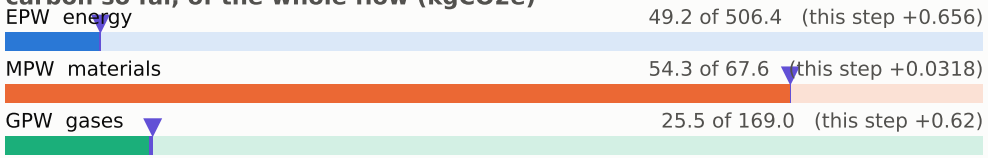


this step

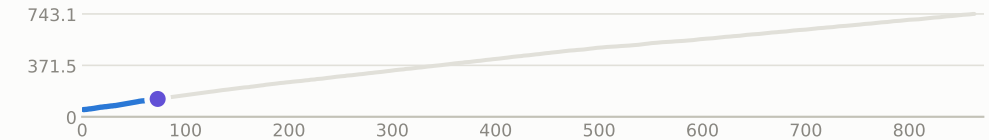
process area	CVD
material	—
thickness	0.25 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

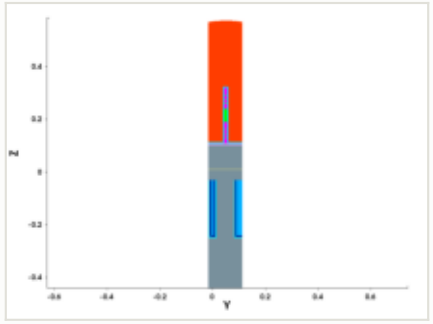
carbon so far, of the whole flow (kgCO2e)



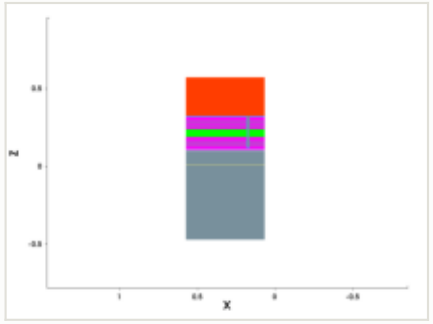
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



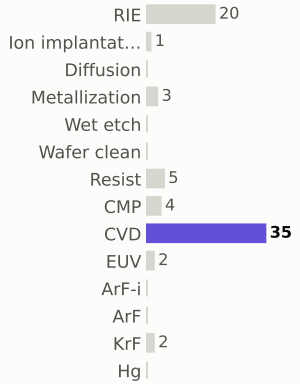
y cut at 0.049 µm



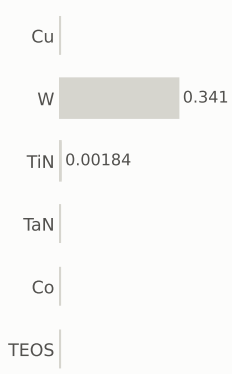
mask for this step

no mask for this step

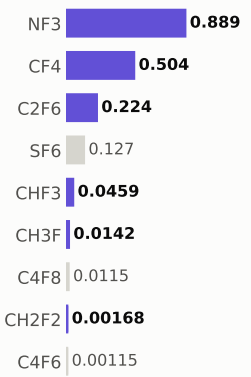
steps by process area



materials deposited (g)



gases emitted (g)



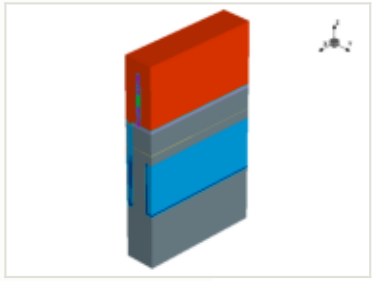
Step #820: CMP

etch_cmp · CMP

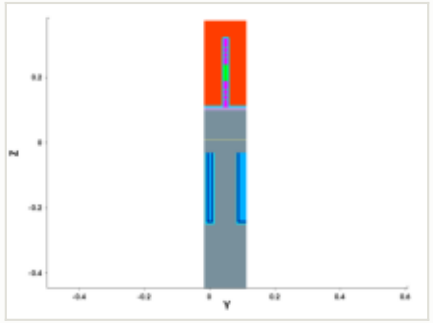
0.169	129.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

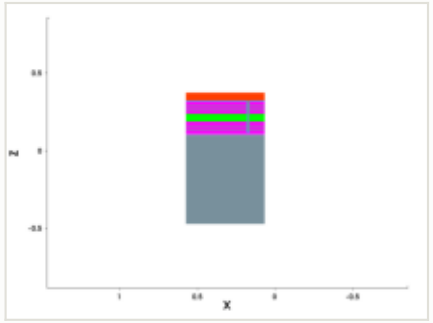
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

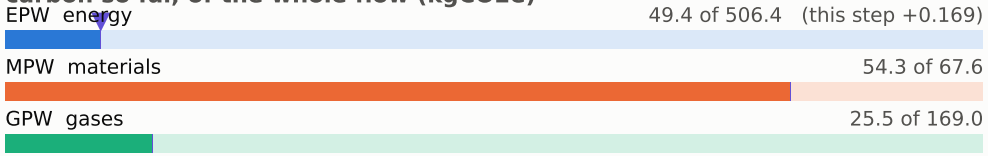
no mask for this step

this step

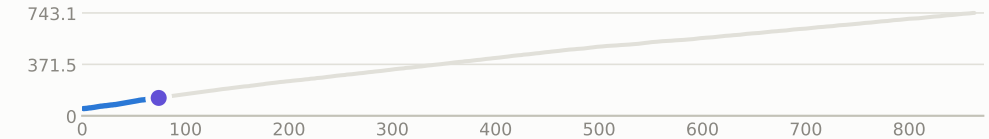
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

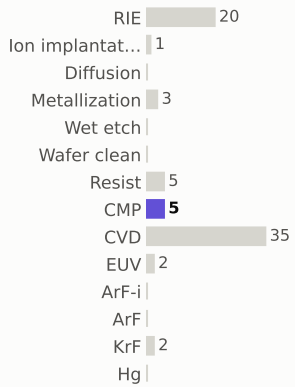
carbon so far, of the whole flow (kgCO2e)



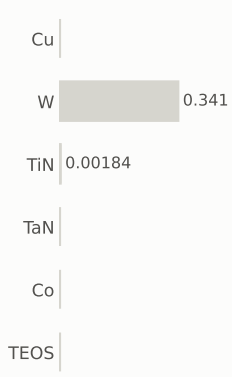
embodied carbon up to each step (kgCO2e per wafer)



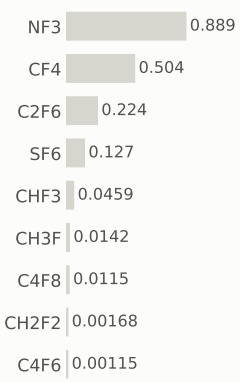
steps by process area



materials deposited (g)



gases emitted (g)



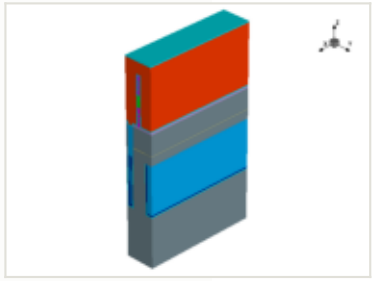
Step #840: CVD

depo_height_dependent · CVD

1.31	130.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

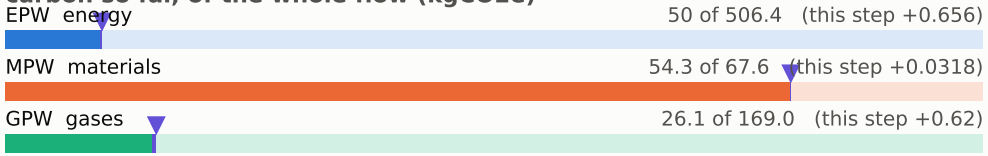


this step

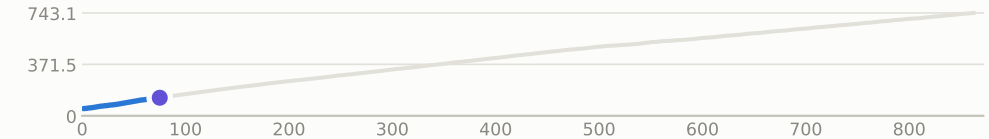
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

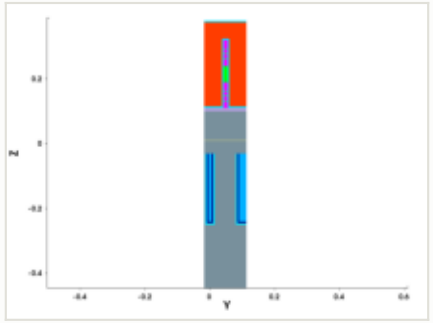
carbon so far, of the whole flow (kgCO2e)



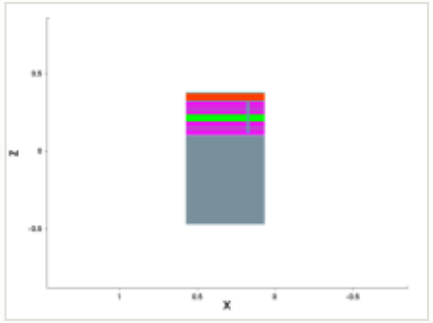
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



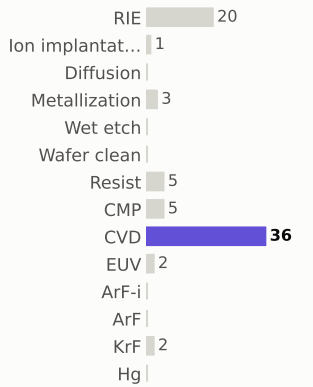
y cut at 0.049 µm



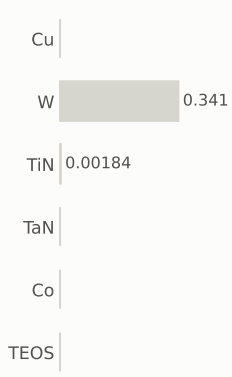
mask for this step

no mask for this step

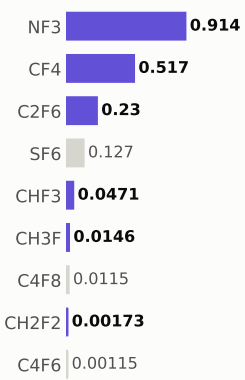
steps by process area



materials deposited (g)



gases emitted (g)



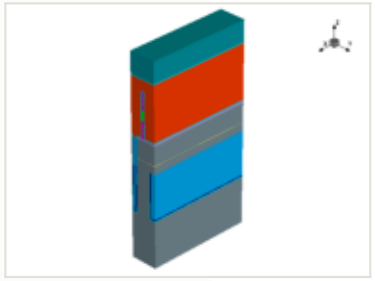
Step #850: CVD

depo_height_dependent · CVD

1.31	131.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

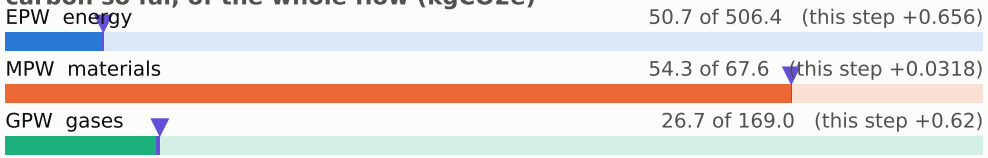
zoomed to fit



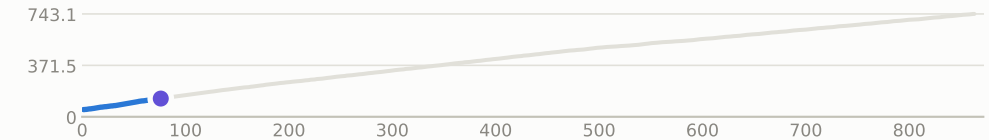
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

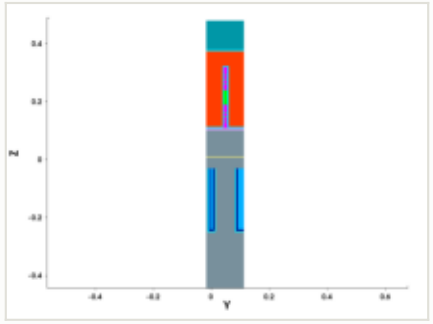
carbon so far, of the whole flow (kgCO2e)



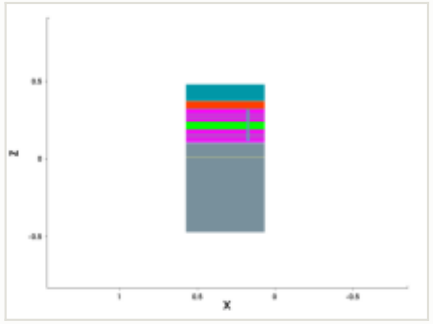
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



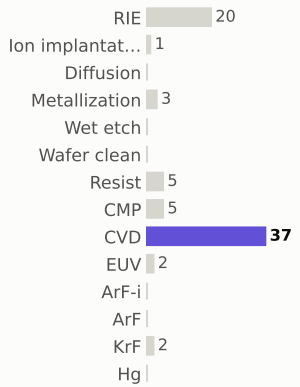
y cut at 0.049 µm



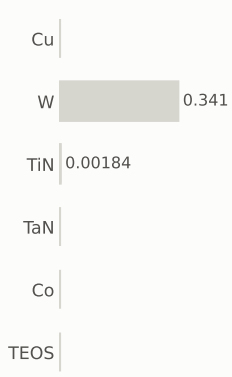
mask for this step

no mask for this step

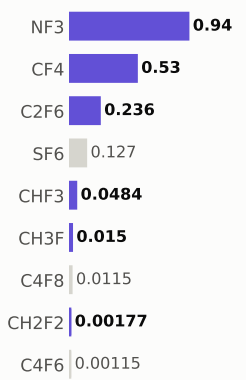
steps by process area



materials deposited (g)



gases emitted (g)



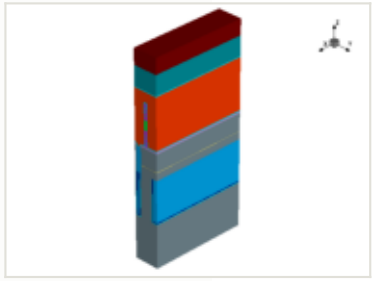
Step #860: CVD

depo_height_dependent · CVD

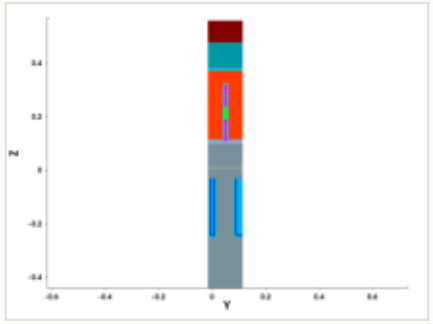
1.31	133.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

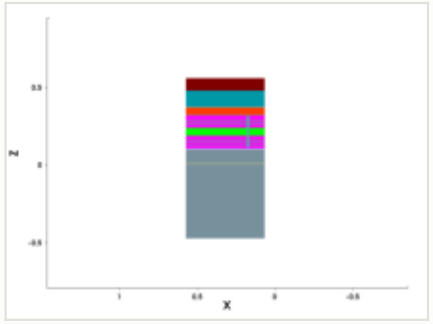
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

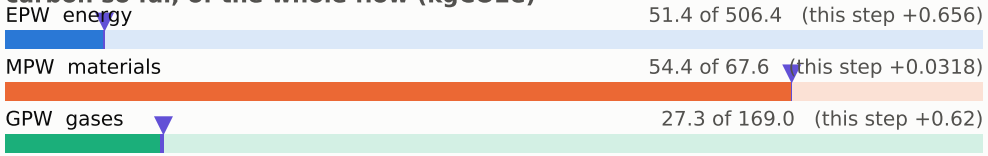
no mask for this step

this step

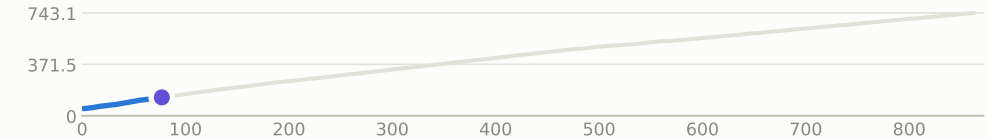
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

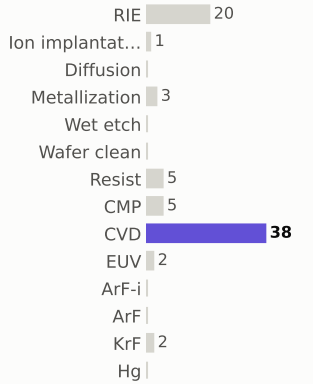
carbon so far, of the whole flow (kgCO2e)



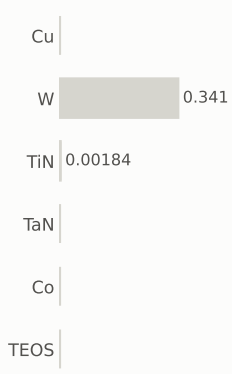
embodied carbon up to each step (kgCO2e per wafer)



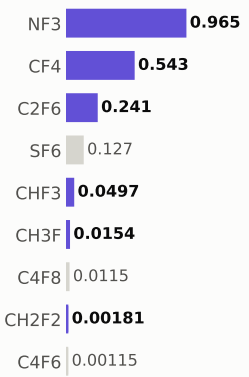
steps by process area



materials deposited (g)



gases emitted (g)



Step #870: CVD

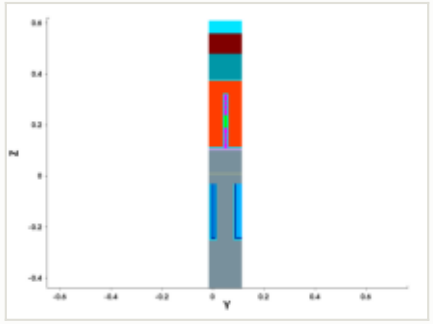
depo_height_dependent · CVD

1.31	134.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

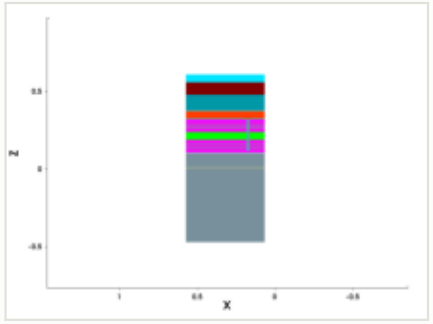
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

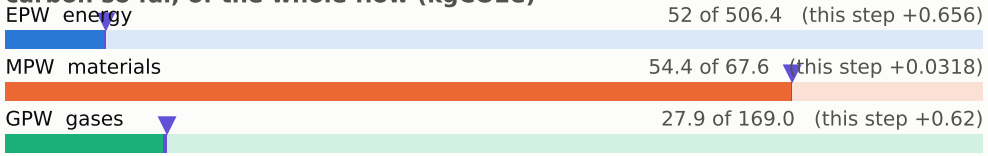
no mask for this step

this step

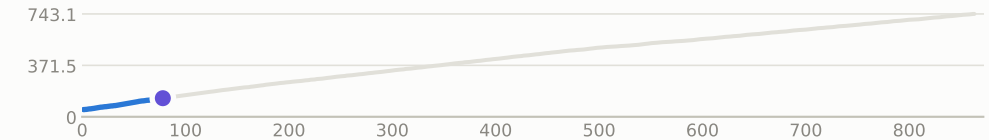
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

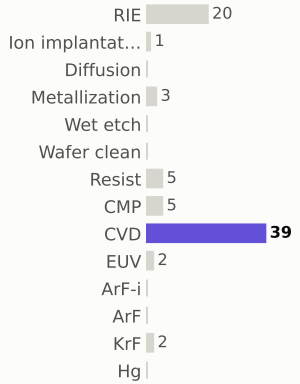
carbon so far, of the whole flow (kgCO2e)



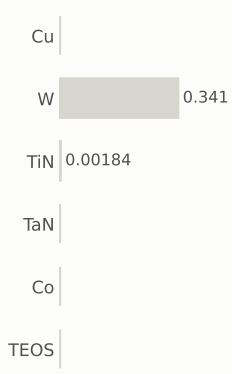
embodied carbon up to each step (kgCO2e per wafer)



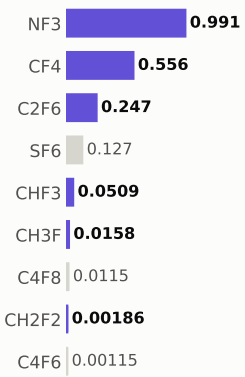
steps by process area



materials deposited (g)



gases emitted (g)



Step #880: CVD

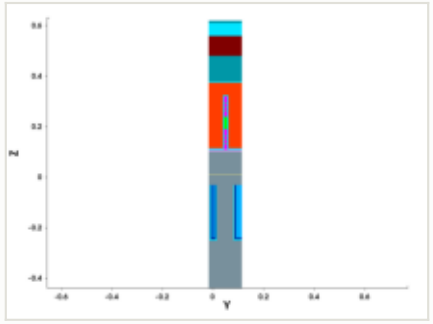
depo_height_dependent · CVD

1.31	135.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

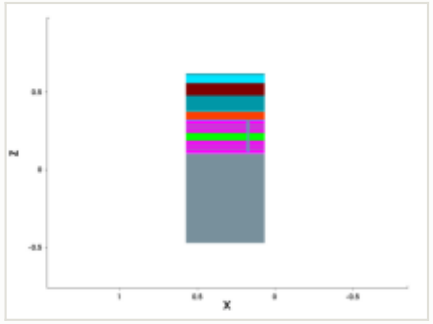
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

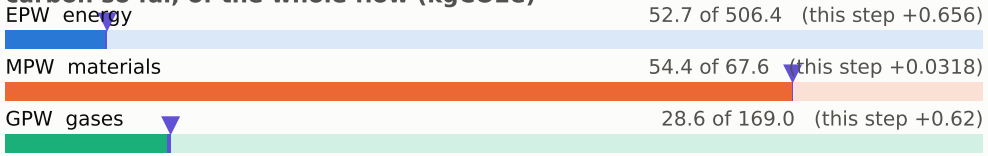
no mask for this step

this step

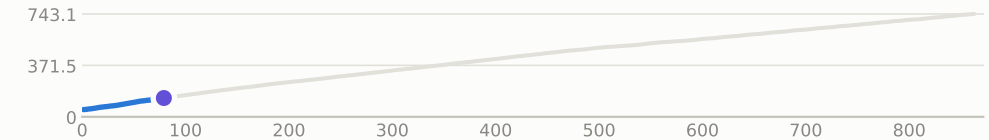
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

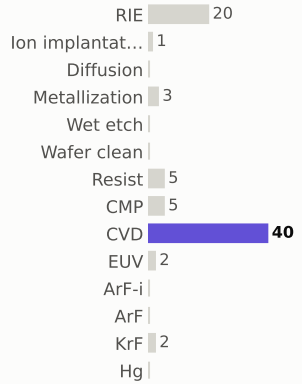
carbon so far, of the whole flow (kgCO2e)



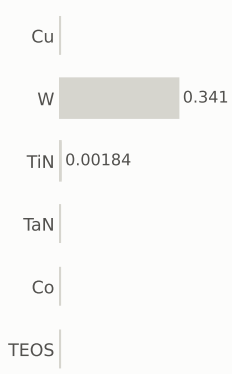
embodied carbon up to each step (kgCO2e per wafer)



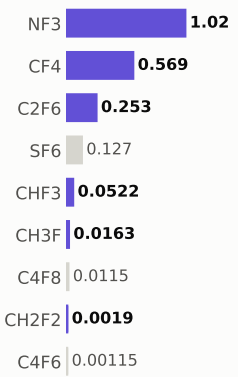
steps by process area



materials deposited (g)



gases emitted (g)



Step #890: EUV

litho_patterning · EUV

1.96	137.6	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

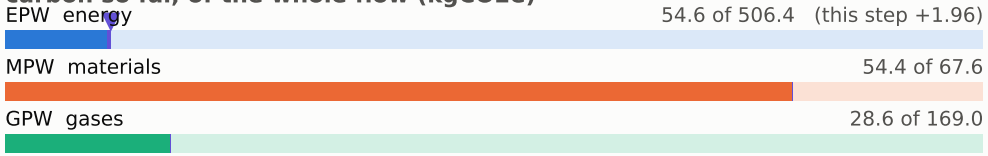
zoomed to fit

this step

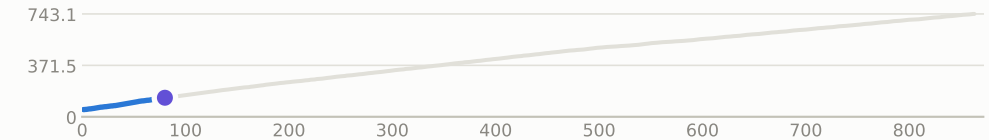
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

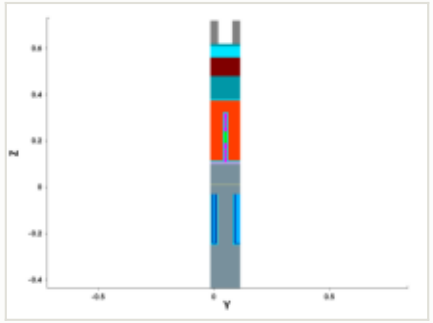
carbon so far, of the whole flow (kgCO2e)



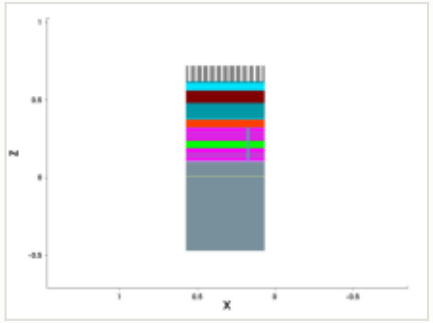
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

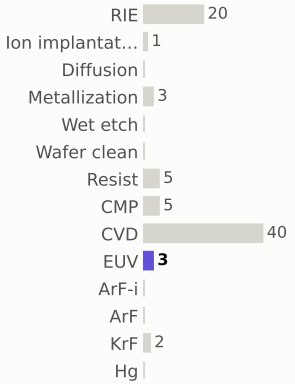


y cut at 0.049 µm

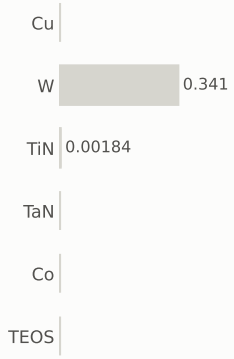


mask: layer 3:0, 3:1 (GDSII)

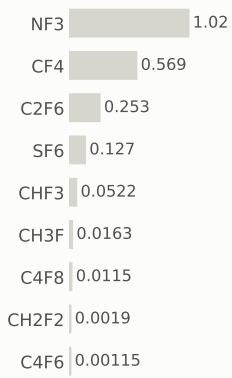
steps by process area



materials deposited (g)



gases emitted (g)



Step #900: RIE

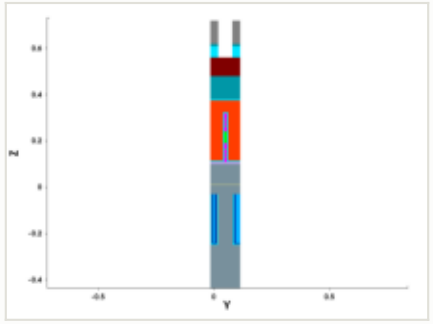
etch_anisotropic · RIE

1.04	138.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

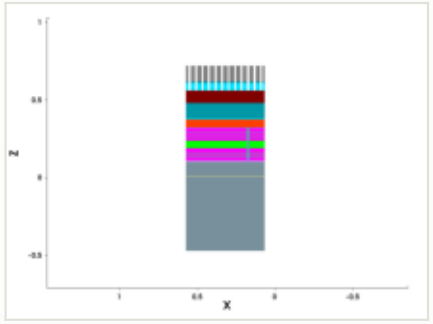
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

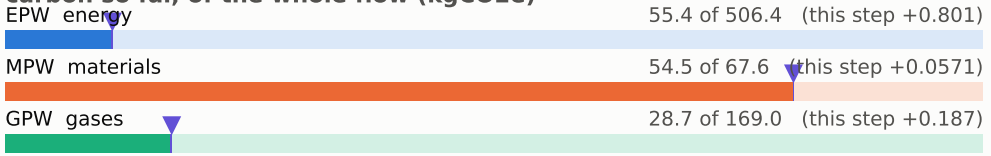
no mask for this step

this step

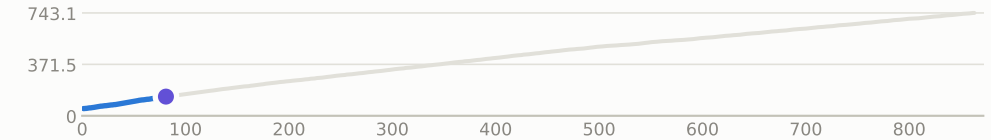
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

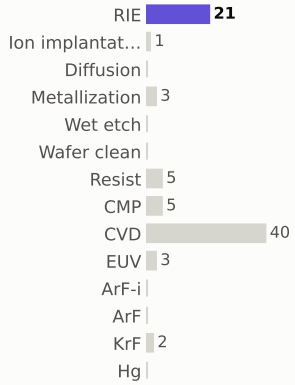
carbon so far, of the whole flow (kgCO2e)



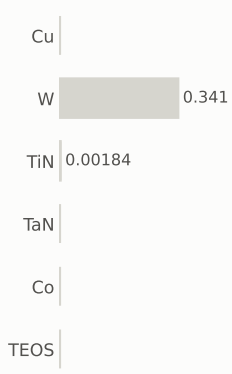
embodied carbon up to each step (kgCO2e per wafer)



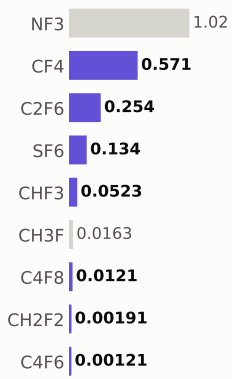
steps by process area



materials deposited (g)



gases emitted (g)



Step #910: Resist

etch_strip · Resist

0.169	138.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

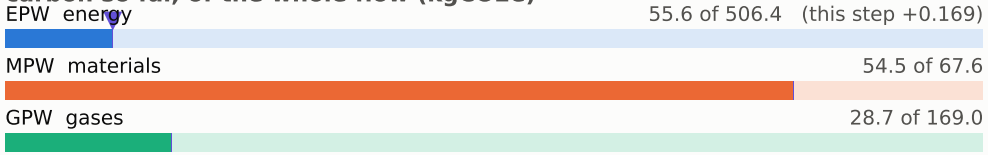
zoomed to fit

this step

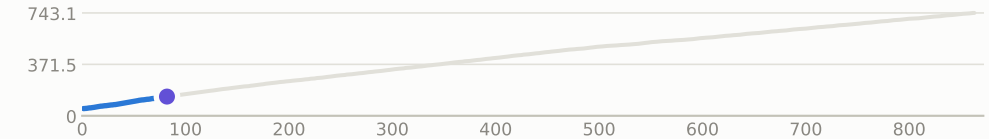
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

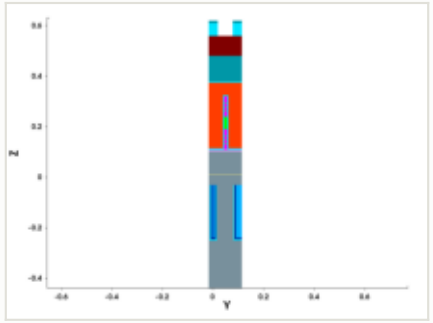
carbon so far, of the whole flow (kgCO2e)



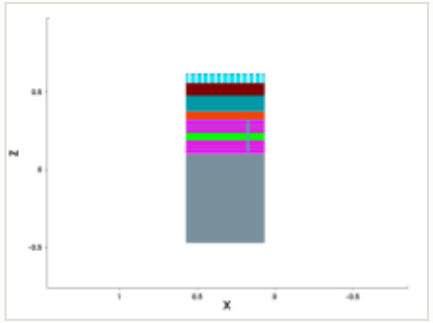
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



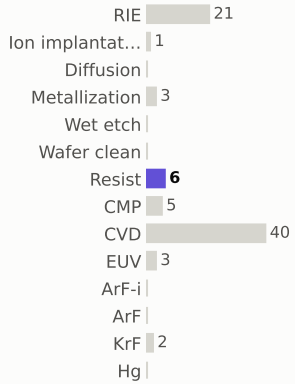
y cut at 0.049 µm



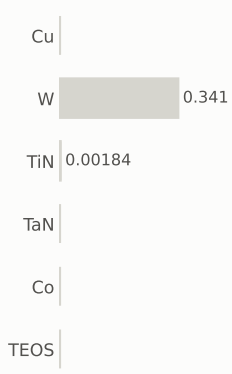
mask for this step

no mask for this step

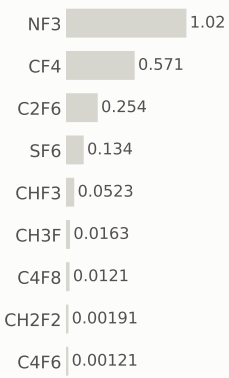
steps by process area



materials deposited (g)



gases emitted (g)



Step #920: RIE

etch_anisotropic · RIE

1.04	139.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

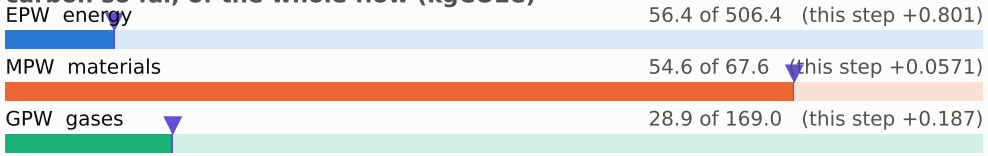
zoomed to fit

this step

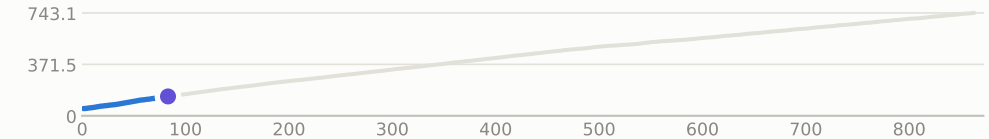
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

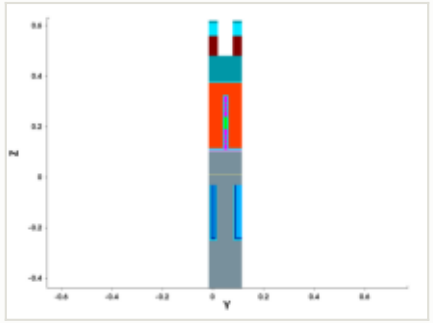
carbon so far, of the whole flow (kgCO2e)



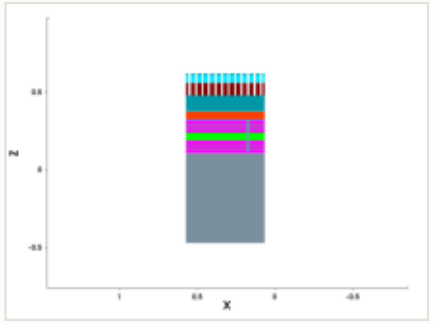
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



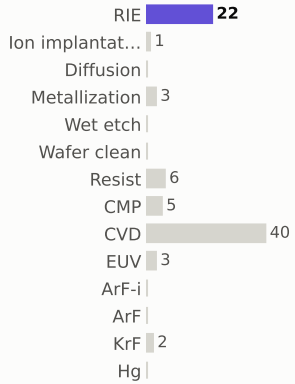
y cut at 0.049 µm



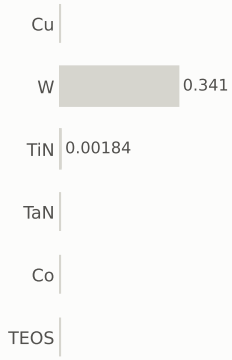
mask for this step

no mask for this step

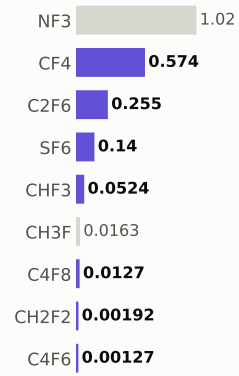
steps by process area



materials deposited (g)



gases emitted (g)



Step #930: CMP

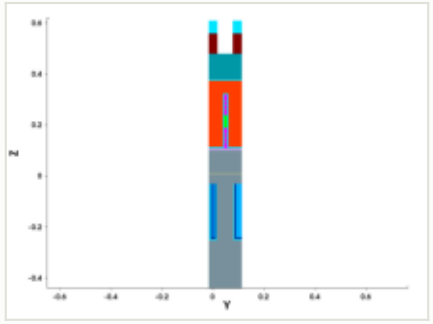
etch_cmp · CMP

0.169	140.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

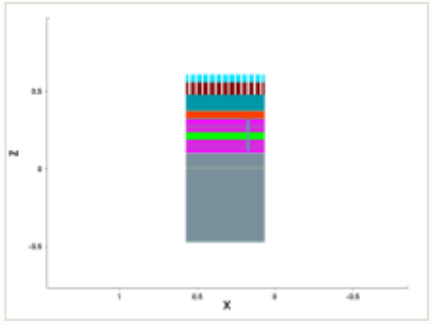
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



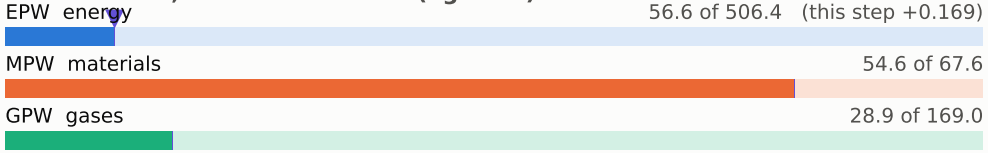
mask for this step

no mask for this step

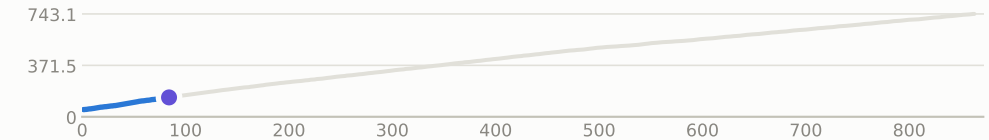
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

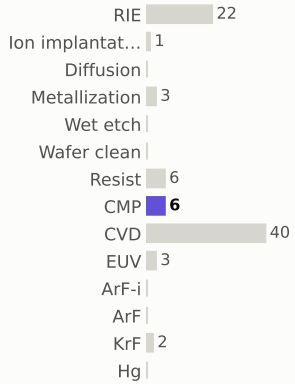
carbon so far, of the whole flow (kgCO2e)



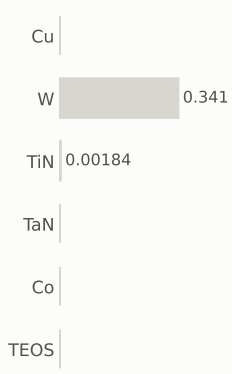
embodied carbon up to each step (kgCO2e per wafer)



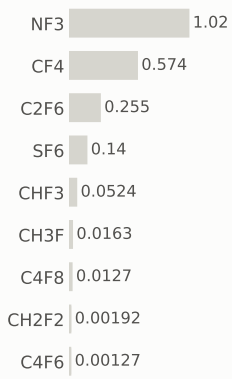
steps by process area



materials deposited (g)



gases emitted (g)



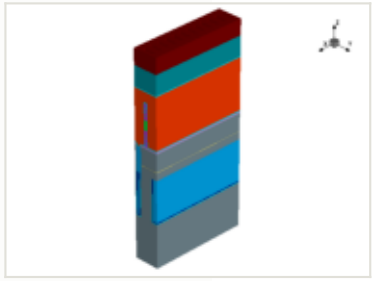
Step #940: RIE

etch_anisotropic · RIE

1.04	141.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

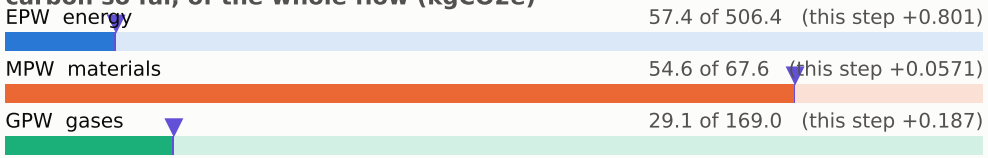


this step

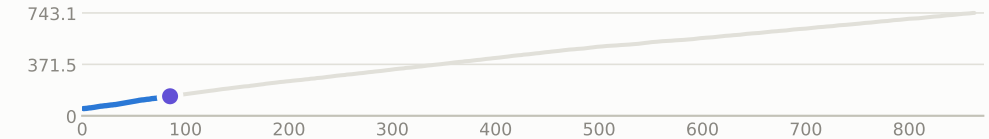
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

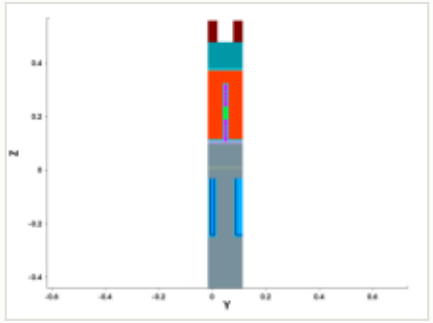
carbon so far, of the whole flow (kgCO2e)



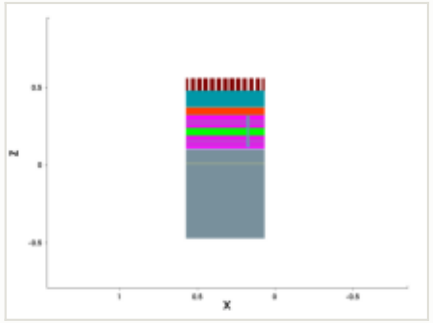
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



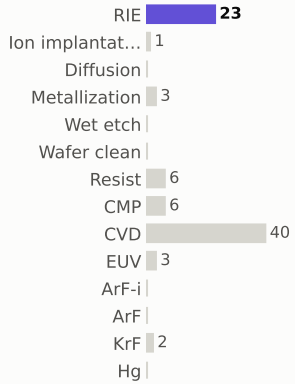
y cut at 0.049 µm



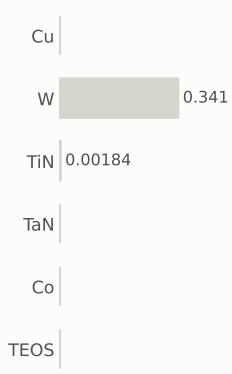
mask for this step

no mask for this step

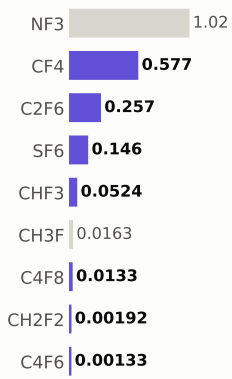
steps by process area



materials deposited (g)



gases emitted (g)



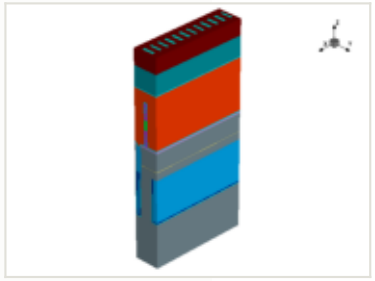
Step #950: CVD

depo_fill · CVD

1.31	142.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

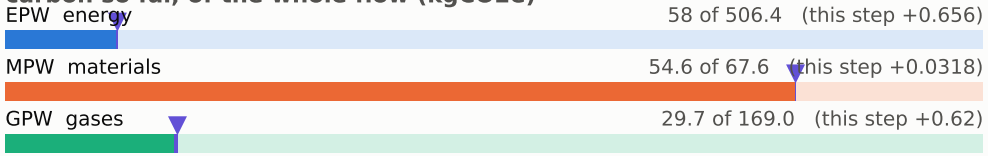


this step

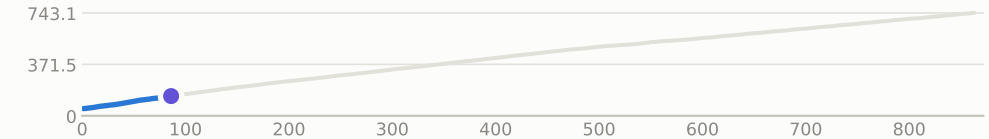
process area	CVD
material	—
thickness	0.08 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

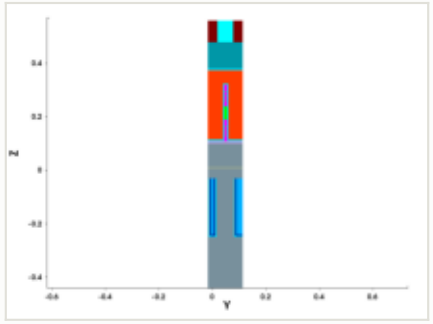
carbon so far, of the whole flow (kgCO2e)



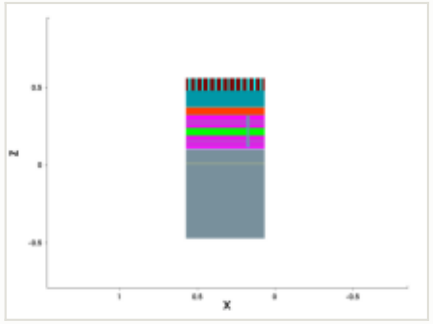
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



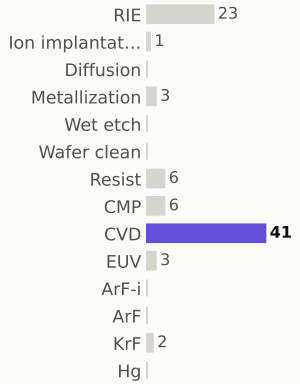
y cut at 0.049 µm



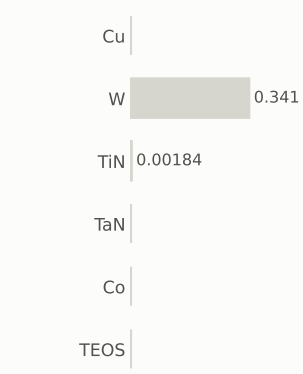
mask for this step

no mask for this step

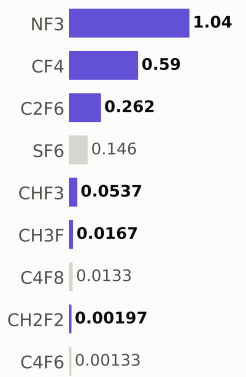
steps by process area



materials deposited (g)



gases emitted (g)



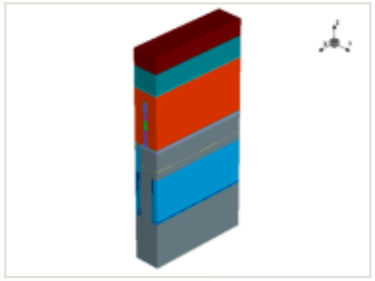
Step #960: RIE

etch_anisotropic · RIE

1.04	143.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

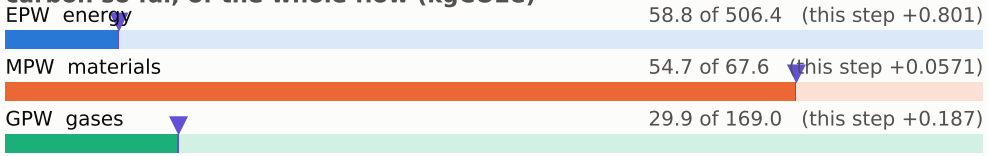


this step

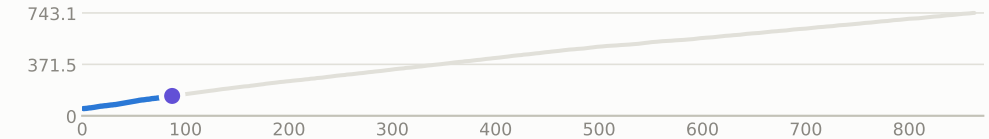
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

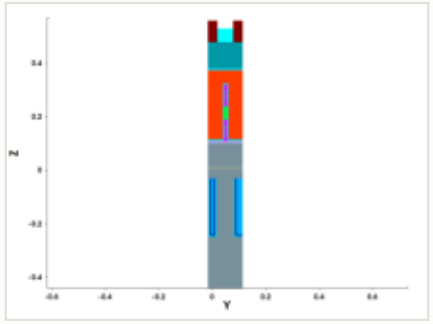
carbon so far, of the whole flow (kgCO2e)



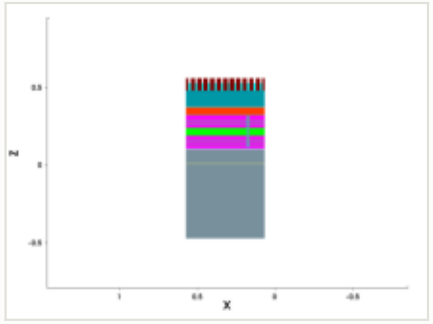
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



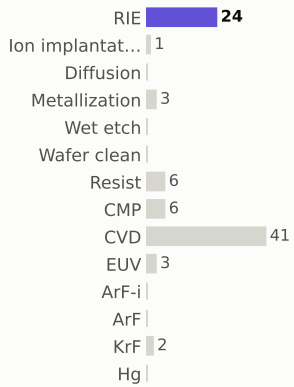
y cut at 0.049 µm



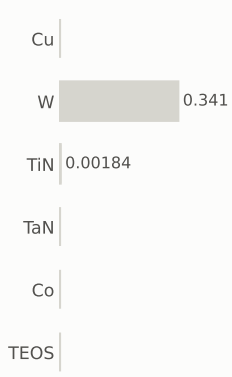
mask for this step

no mask for this step

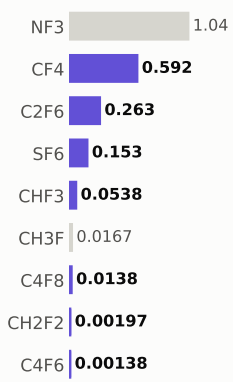
steps by process area



materials deposited (g)



gases emitted (g)



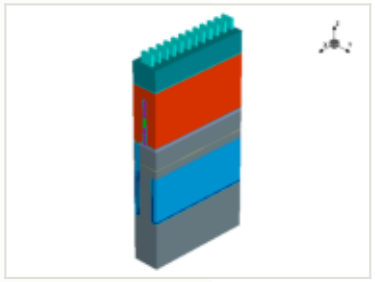
Step #970: RIE

etch_anisotropic · RIE

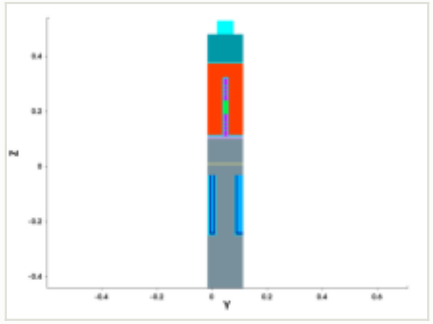
1.04	144.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

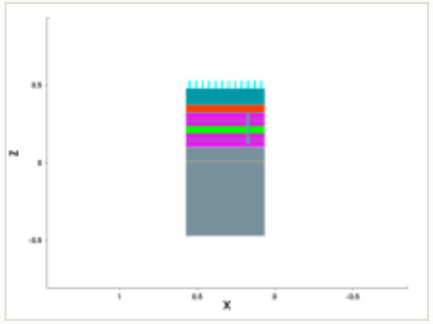
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

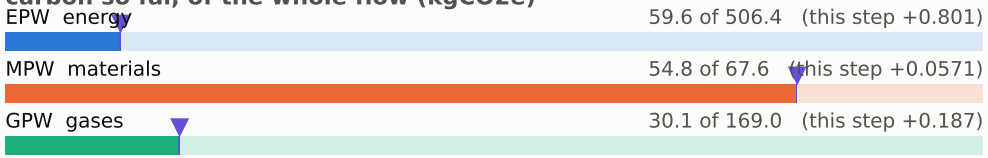
no mask for this step

this step

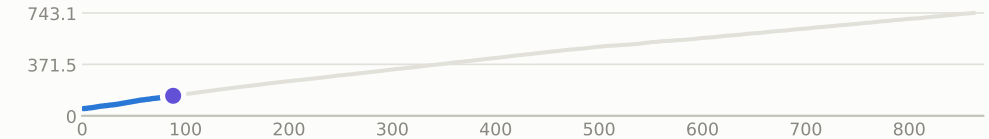
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

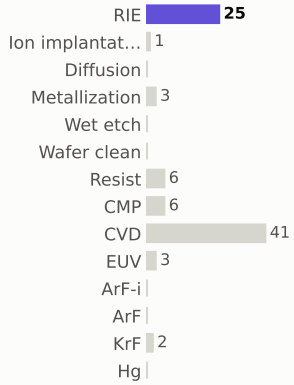
carbon so far, of the whole flow (kgCO2e)



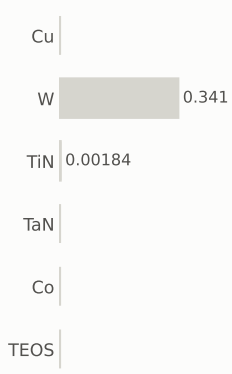
embodied carbon up to each step (kgCO2e per wafer)



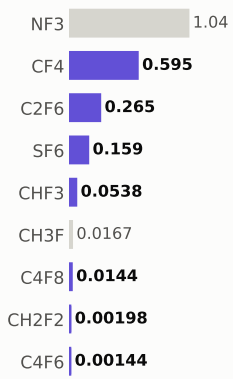
steps by process area



materials deposited (g)



gases emitted (g)



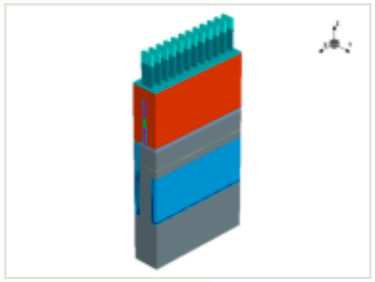
Step #980: RIE

etch_anisotropic · RIE

1.04	145.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

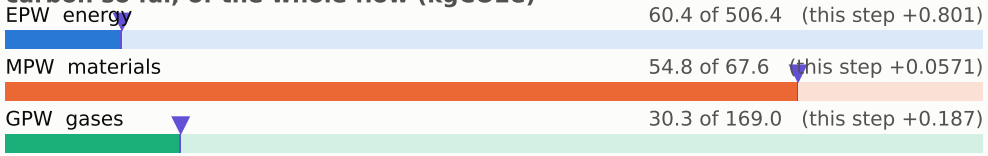


this step

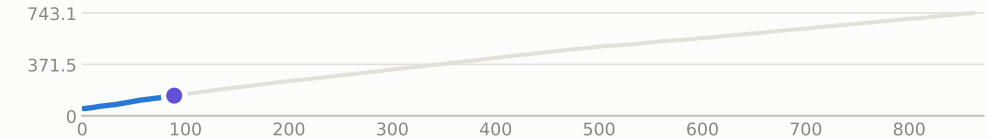
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

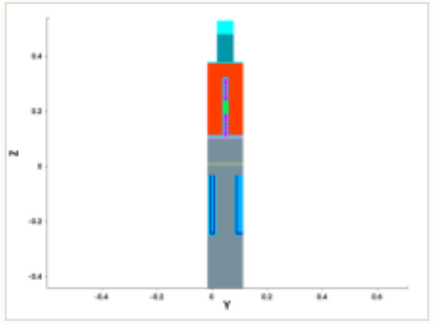
carbon so far, of the whole flow (kgCO2e)



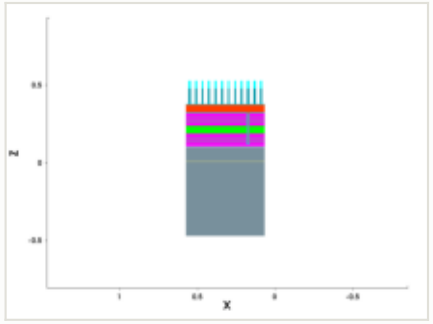
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



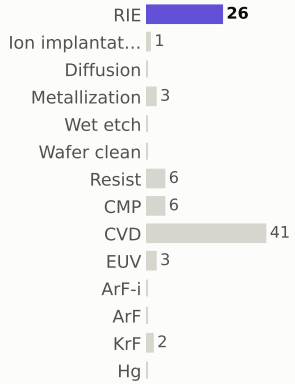
y cut at 0.049 µm



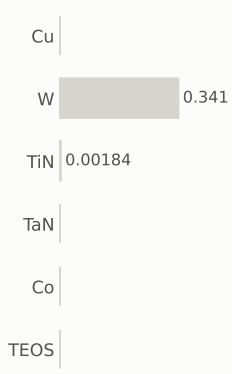
mask for this step

no mask for this step

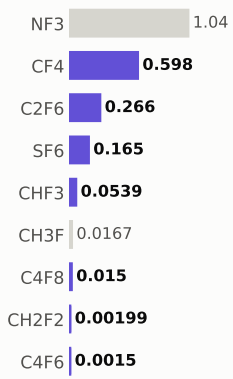
steps by process area



materials deposited (g)



gases emitted (g)



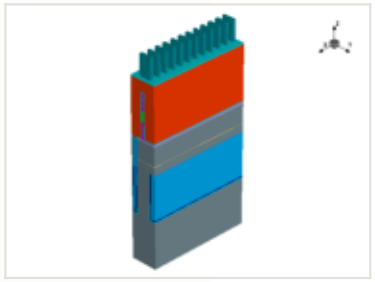
Step #990: RIE

etch_anisotropic · RIE

1.04	146.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

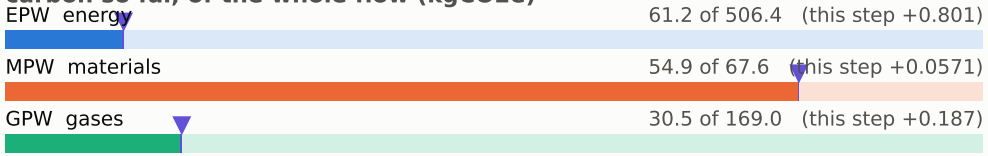


this step

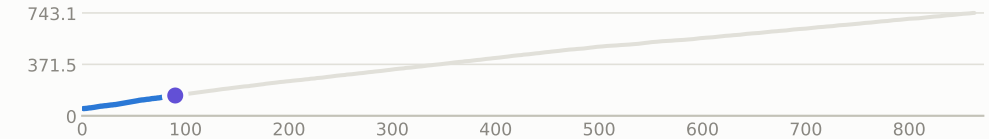
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

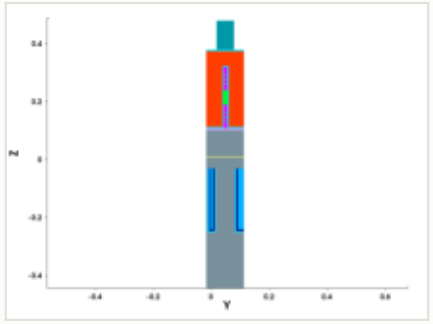
carbon so far, of the whole flow (kgCO2e)



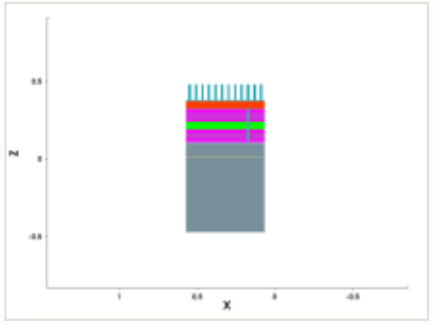
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



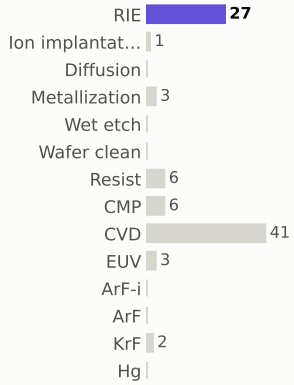
y cut at 0.049 µm



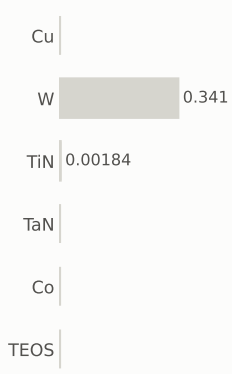
mask for this step

no mask for this step

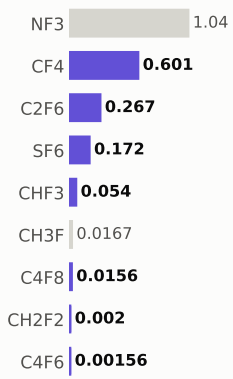
steps by process area



materials deposited (g)



gases emitted (g)



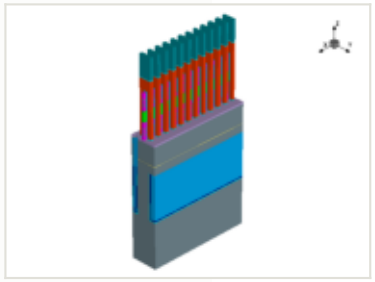
Step #1000: RIE

etch_anisotropic · RIE

1.04	147.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

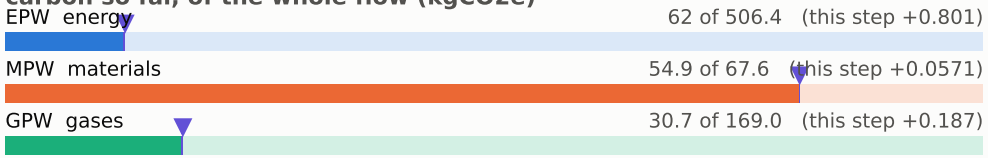


this step

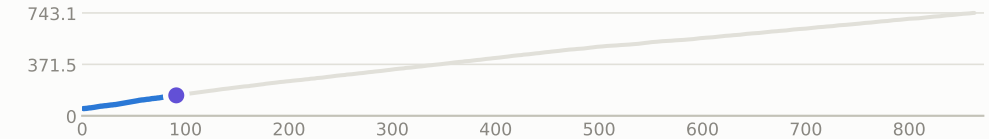
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

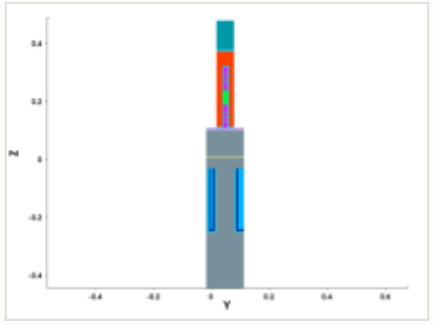
carbon so far, of the whole flow (kgCO2e)



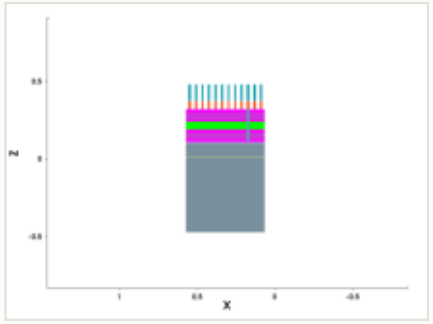
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



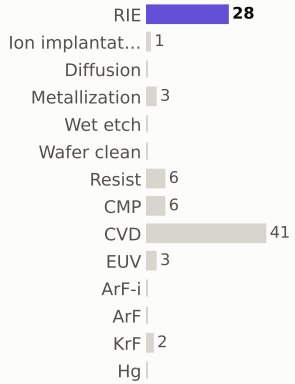
y cut at 0.049 µm



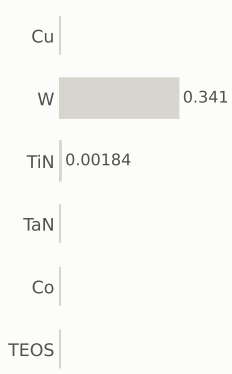
mask for this step

no mask for this step

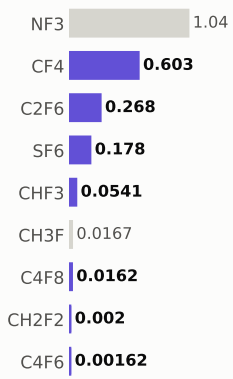
steps by process area



materials deposited (g)



gases emitted (g)



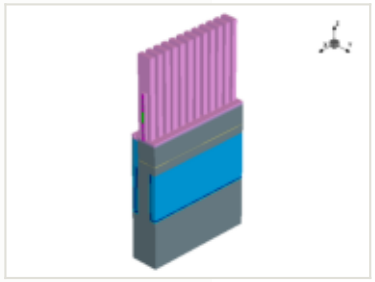
Step #1010: CVD

depo_isotropic · CVD

1.31	148.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

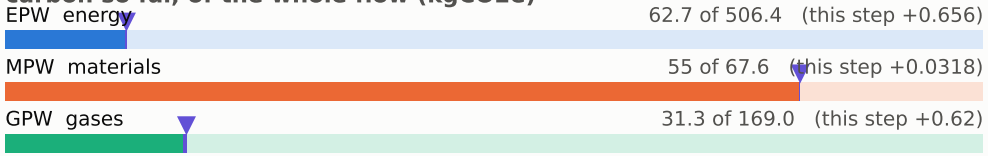


this step

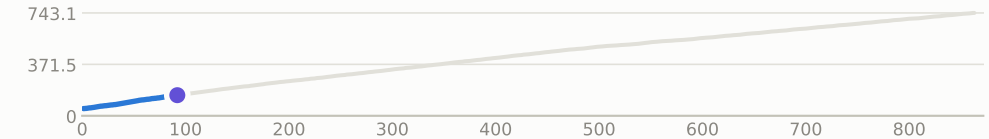
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

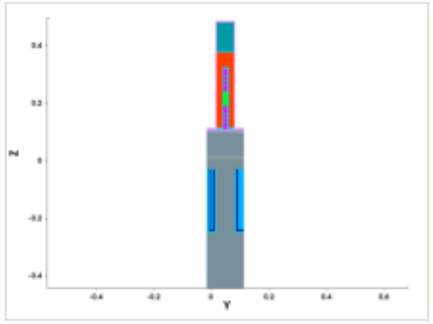
carbon so far, of the whole flow (kgCO2e)



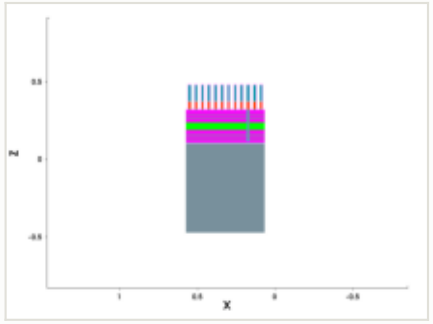
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



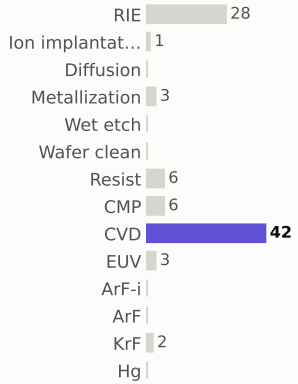
y cut at 0.049 µm



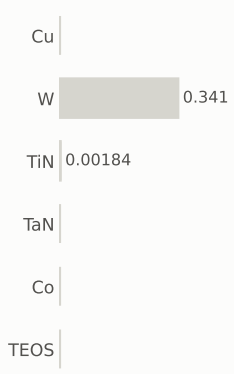
mask for this step

no mask for this step

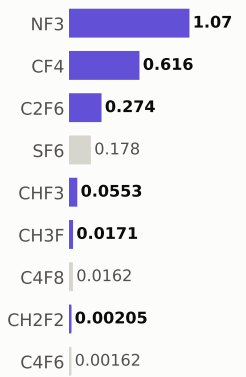
steps by process area



materials deposited (g)



gases emitted (g)



Step #1030: CVD

depo_fill · CVD

1.31	150.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

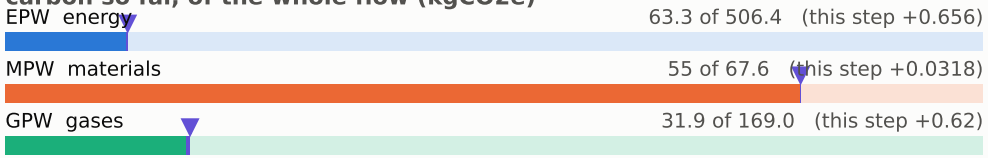
zoomed to fit

this step

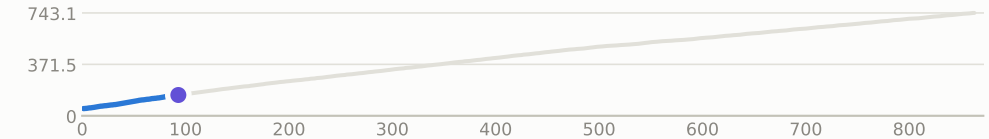
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

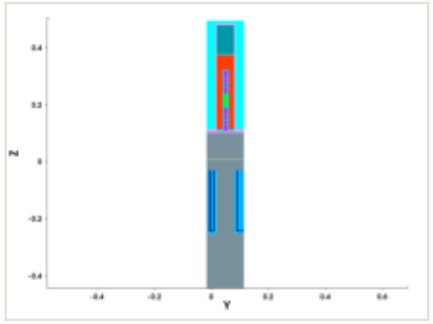
carbon so far, of the whole flow (kgCO2e)



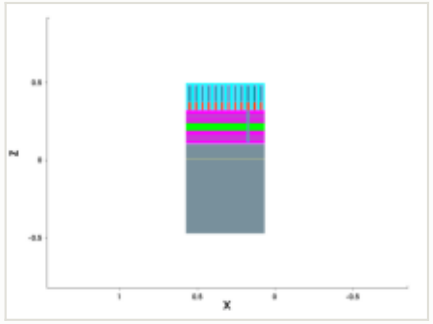
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



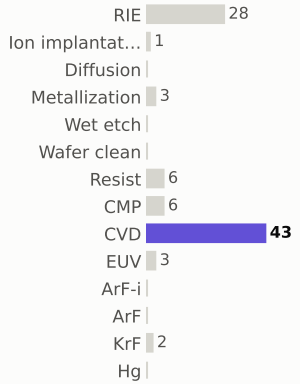
y cut at 0.049 µm



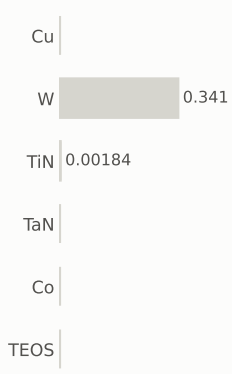
mask for this step

no mask for this step

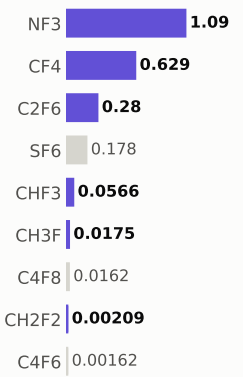
steps by process area



materials deposited (g)



gases emitted (g)



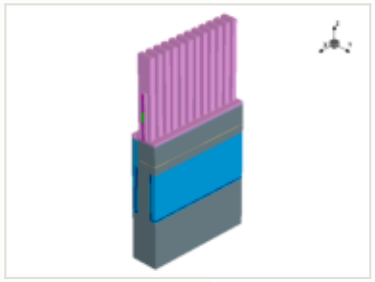
Step #1040: RIE

etch_anisotropic · RIE

1.04	151.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

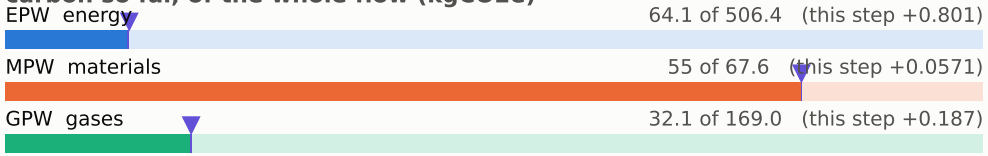


this step

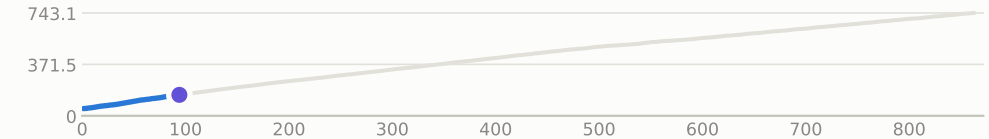
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

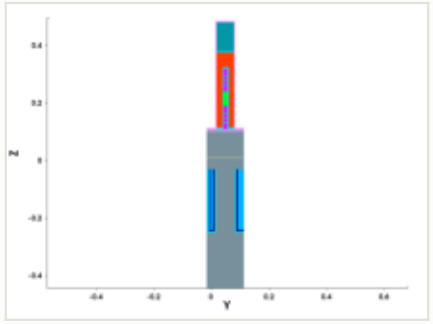
carbon so far, of the whole flow (kgCO2e)



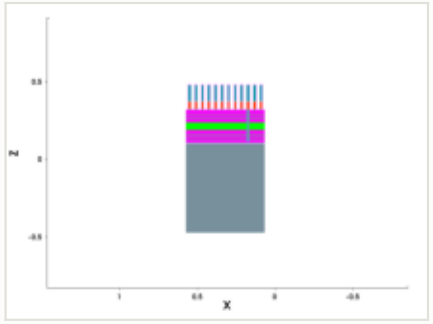
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



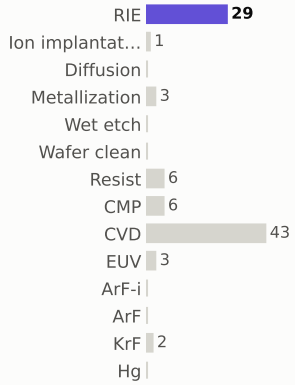
y cut at 0.049 µm



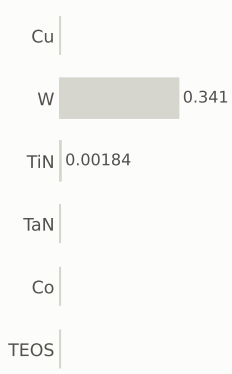
mask for this step

no mask for this step

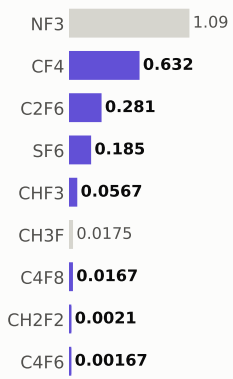
steps by process area



materials deposited (g)



gases emitted (g)



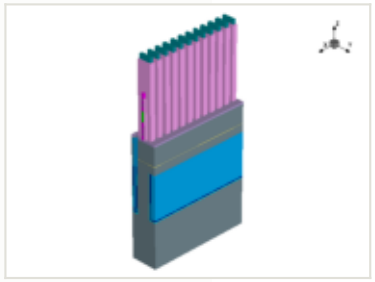
Step #1050: RIE

etch_anisotropic · RIE

1.04	152.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

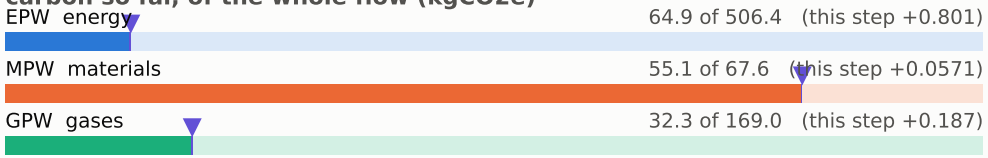


this step

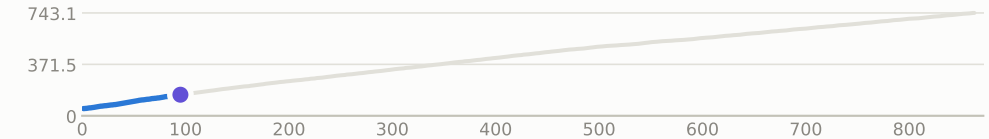
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

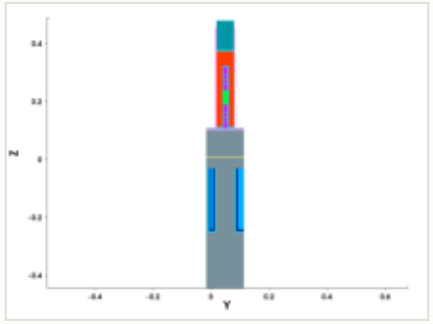
carbon so far, of the whole flow (kgCO2e)



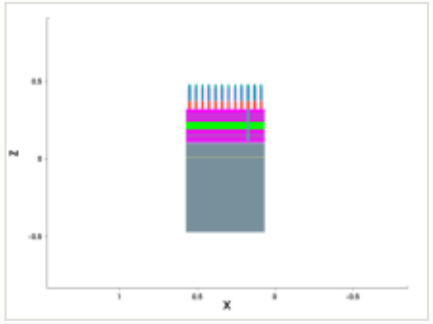
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



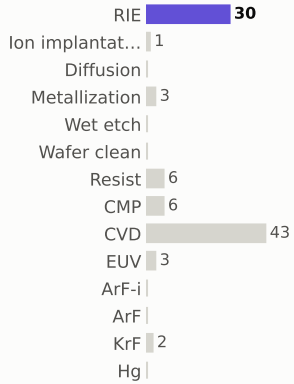
y cut at 0.049 µm



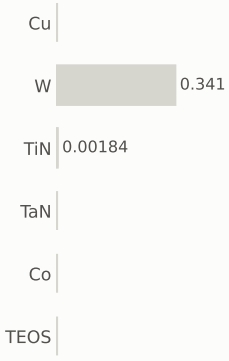
mask for this step

no mask for this step

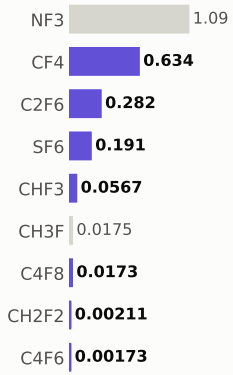
steps by process area



materials deposited (g)



gases emitted (g)



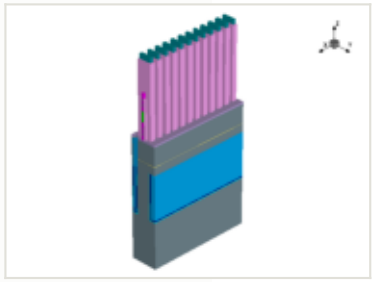
Step #1060: CVD

depo_anisotropic · CVD

1.31	153.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

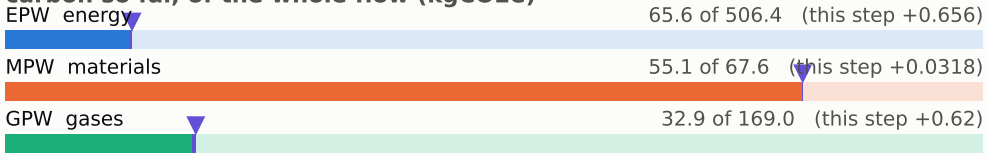


this step

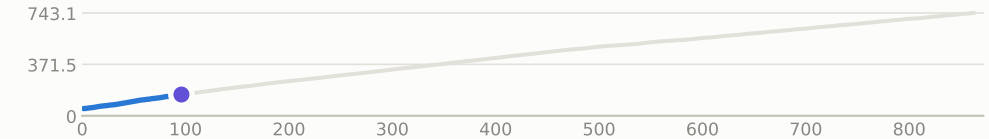
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

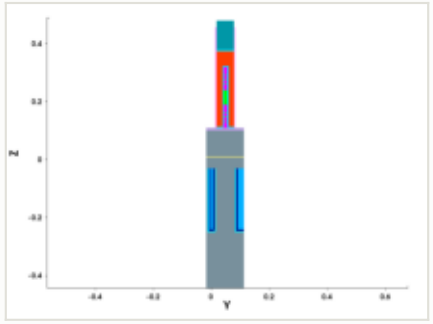
carbon so far, of the whole flow (kgCO2e)



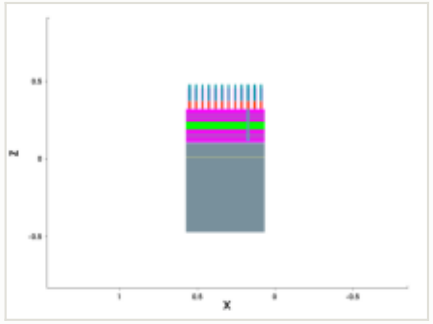
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



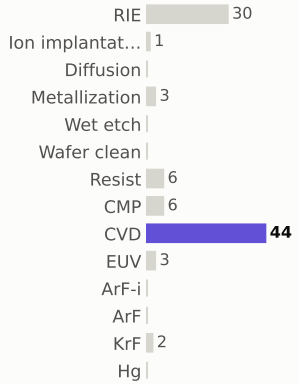
y cut at 0.049 µm



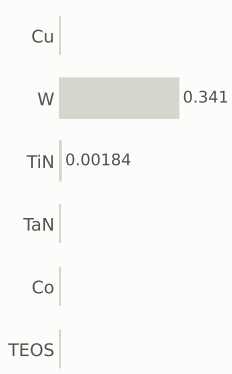
mask for this step

no mask for this step

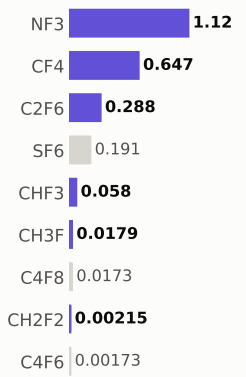
steps by process area



materials deposited (g)



gases emitted (g)



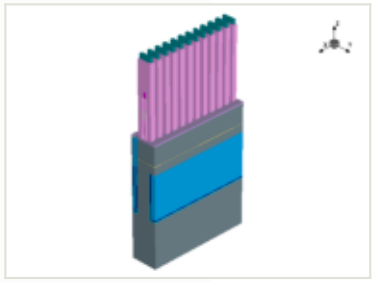
Step #1080: RIE

etch_anisotropic · RIE

1.04	154.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

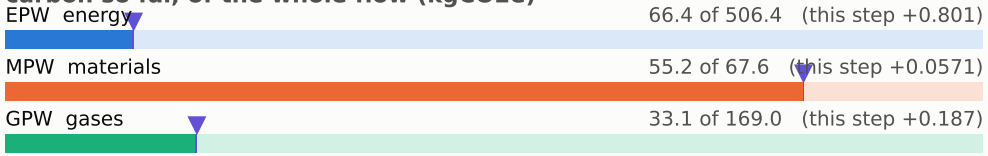


this step

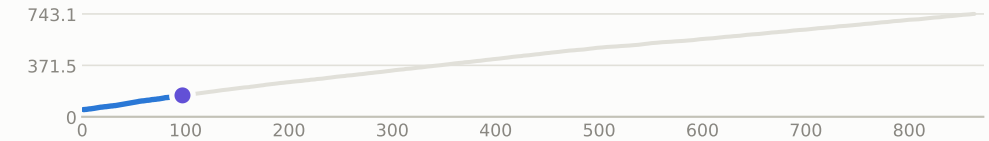
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

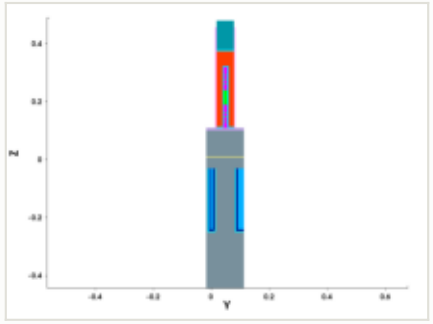
carbon so far, of the whole flow (kgCO2e)



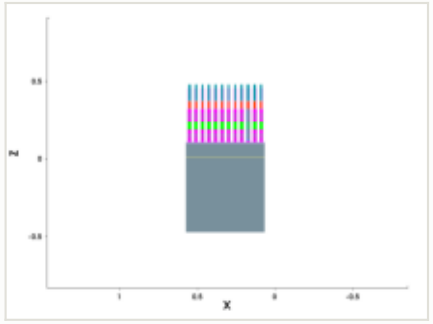
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



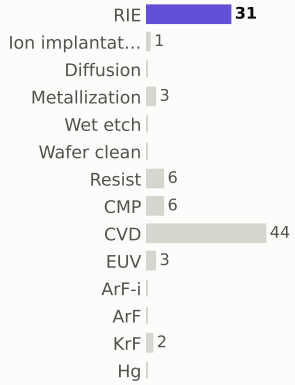
y cut at 0.049 µm



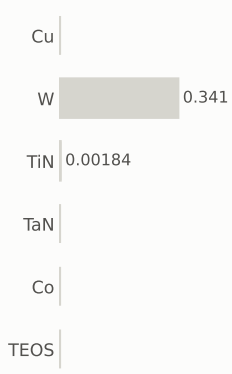
mask for this step

no mask for this step

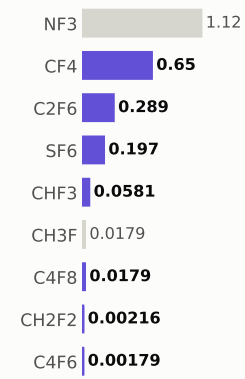
steps by process area



materials deposited (g)



gases emitted (g)



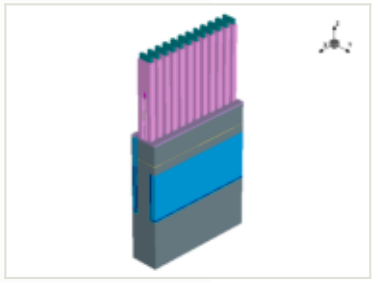
Step #1090: Wet_etch

etch_isotropic · Wet_etch

0.169	154.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

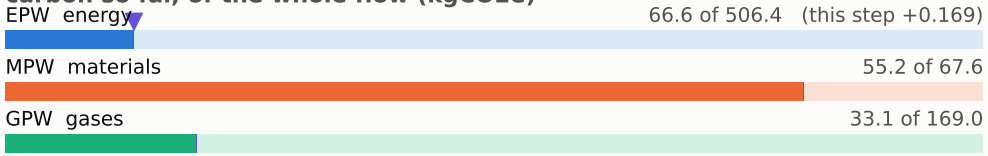


this step

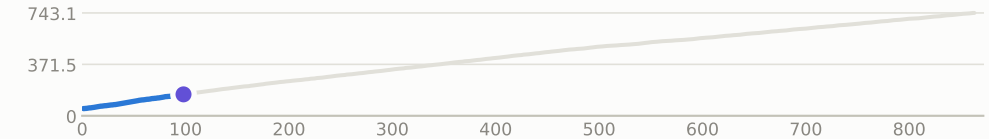
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

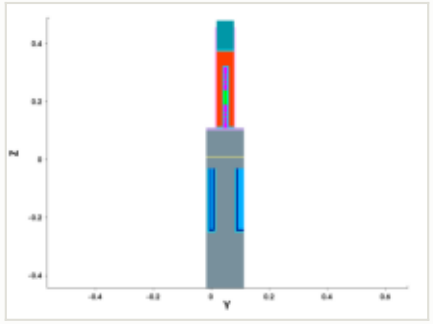
carbon so far, of the whole flow (kgCO2e)



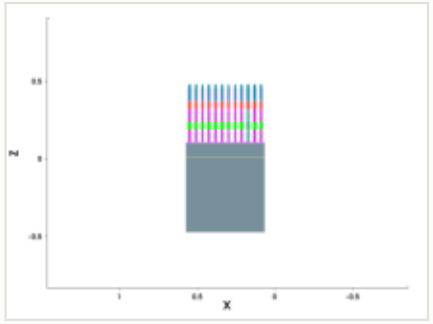
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



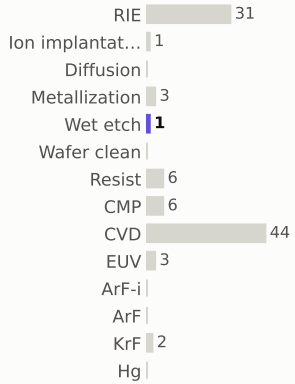
y cut at 0.049 µm



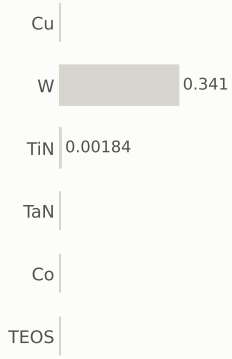
mask for this step

no mask for this step

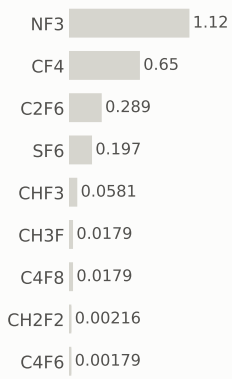
steps by process area



materials deposited (g)



gases emitted (g)



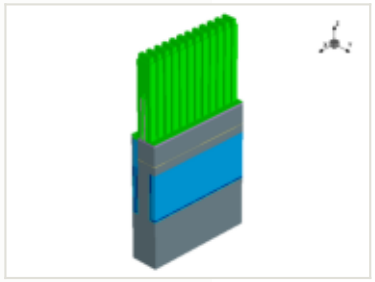
Step #1100: CVD

depo_isotropic · CVD

1.31	156.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

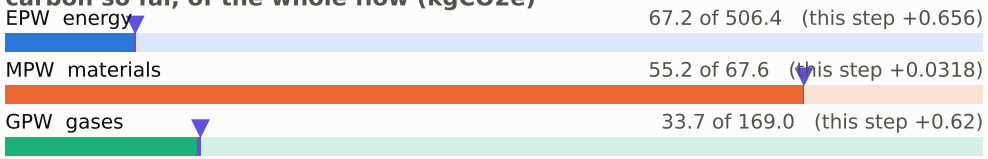


this step

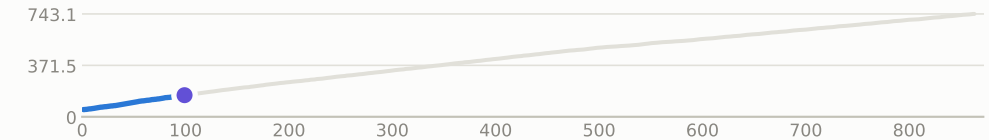
process area	CVD
material	—
thickness	0.003 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

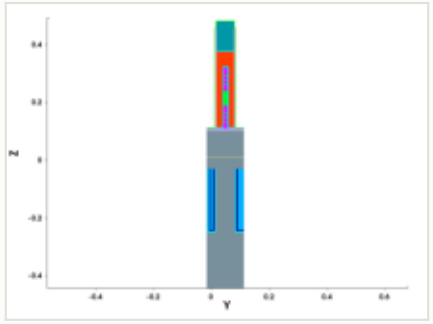
carbon so far, of the whole flow (kgCO2e)



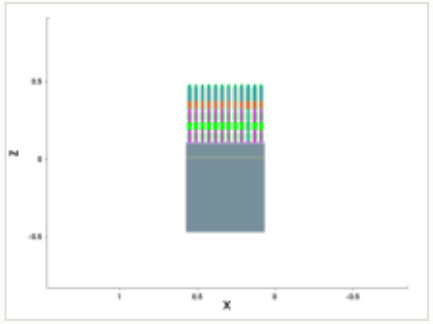
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



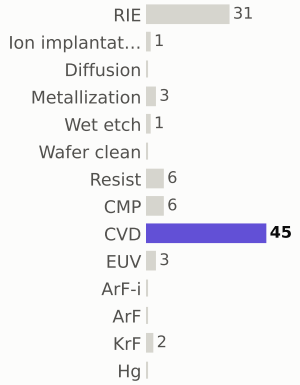
y cut at 0.049 µm



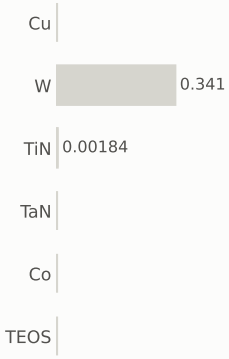
mask for this step

no mask for this step

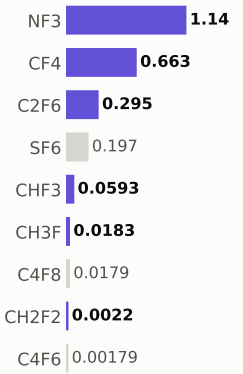
steps by process area



materials deposited (g)



gases emitted (g)



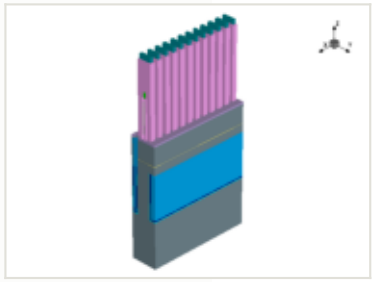
Step #1110: RIE

etch_anisotropic · RIE

1.04	157.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

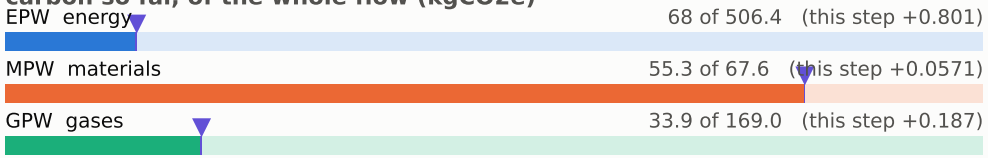


this step

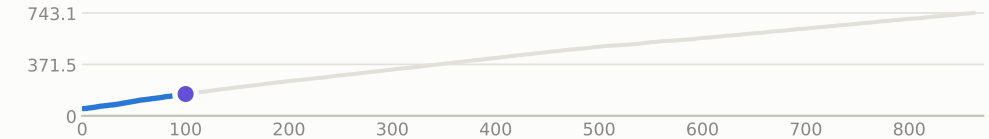
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

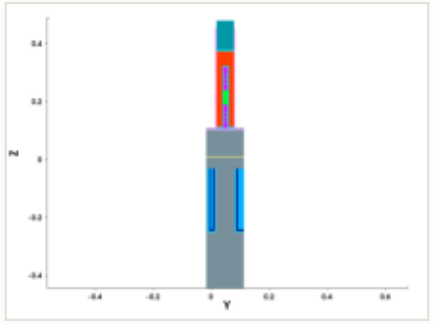
carbon so far, of the whole flow (kgCO2e)



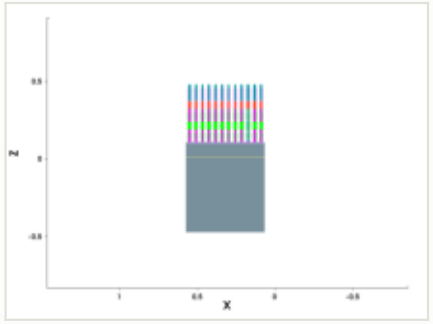
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



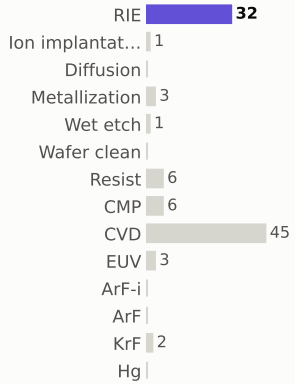
y cut at 0.049 µm



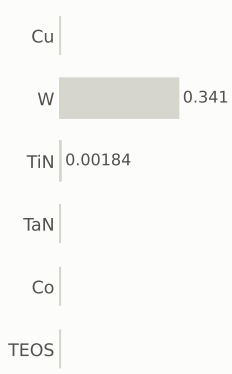
mask for this step

no mask for this step

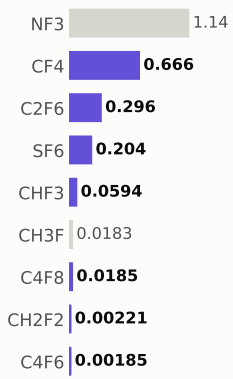
steps by process area



materials deposited (g)



gases emitted (g)



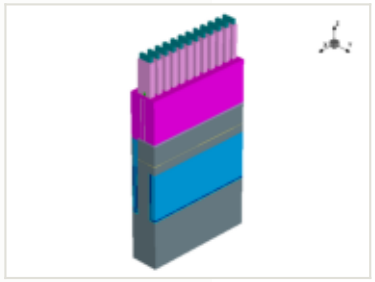
Step #1120: CVD

depo_fill · CVD

1.31	158.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

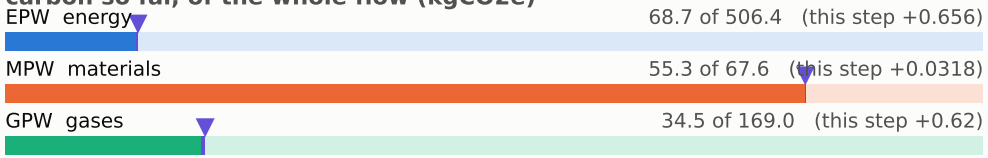


this step

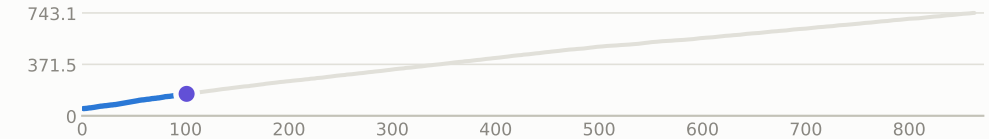
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

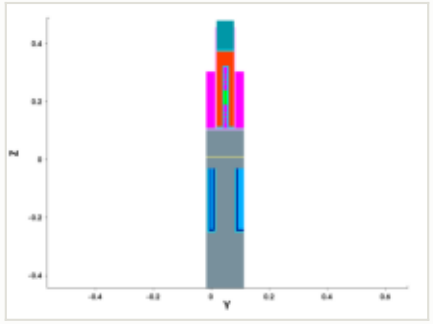
carbon so far, of the whole flow (kgCO2e)



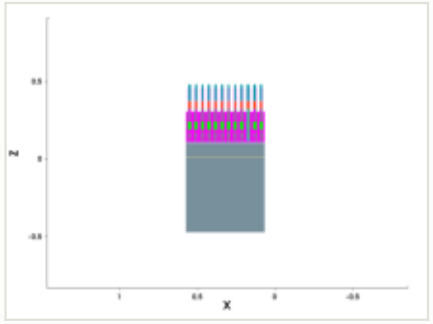
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



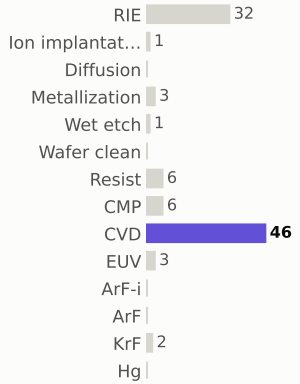
y cut at 0.049 µm



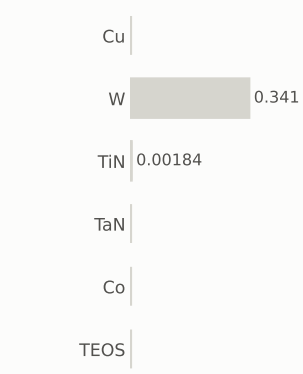
mask for this step

no mask for this step

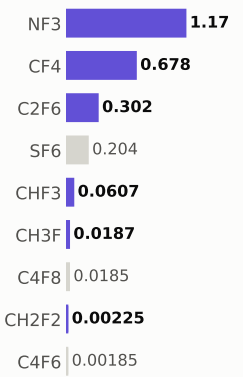
steps by process area



materials deposited (g)



gases emitted (g)



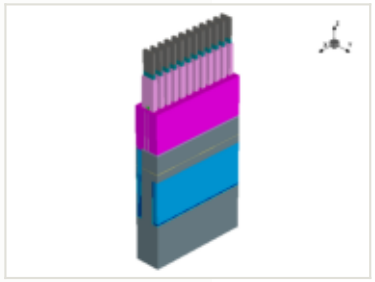
Step #1140: EUV

litho_patterning · EUV

1.96	160.5	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

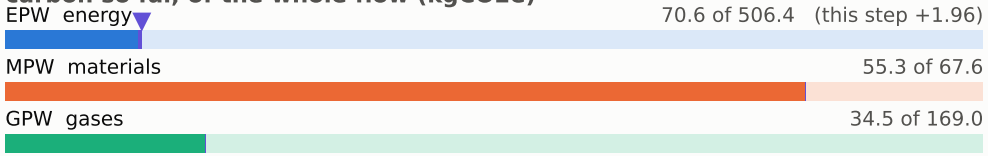


this step

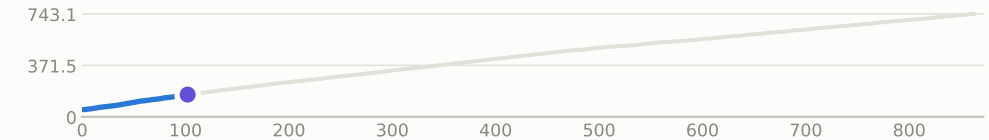
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

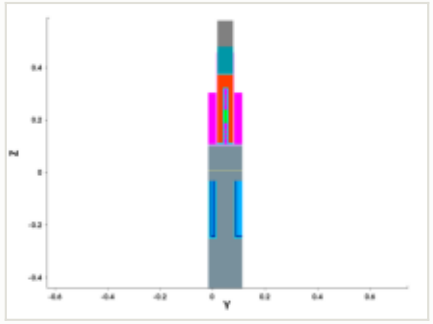
carbon so far, of the whole flow (kgCO2e)



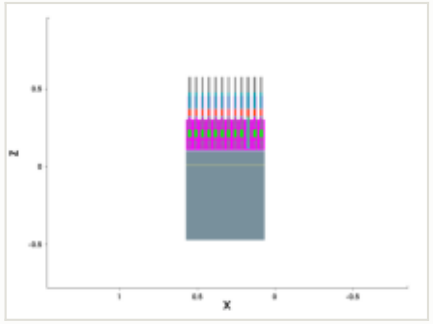
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

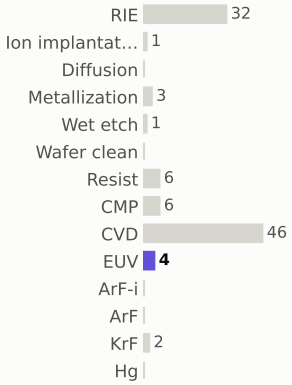


y cut at 0.049 µm

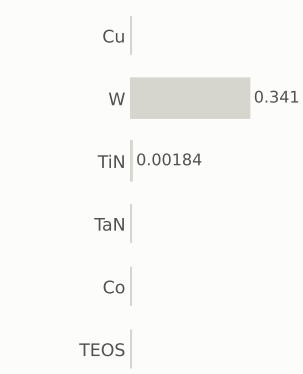


mask: layer 3:0, 3:1, positive (GDSII)

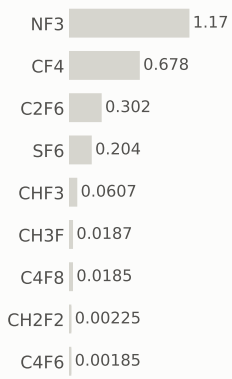
steps by process area



materials deposited (g)



gases emitted (g)



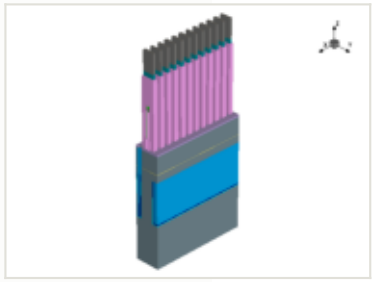
Step #1150: RIE

etch_anisotropic · RIE

1.04	161.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

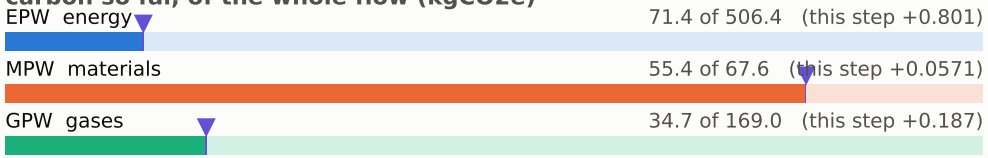


this step

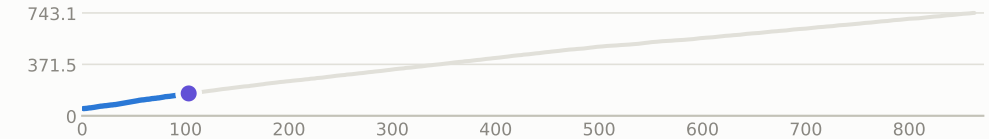
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

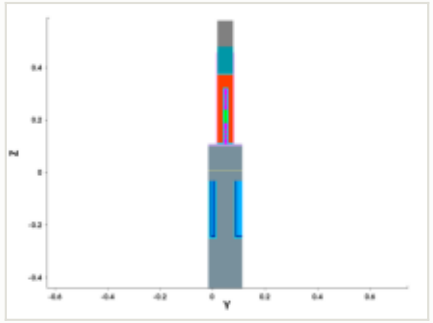
carbon so far, of the whole flow (kgCO2e)



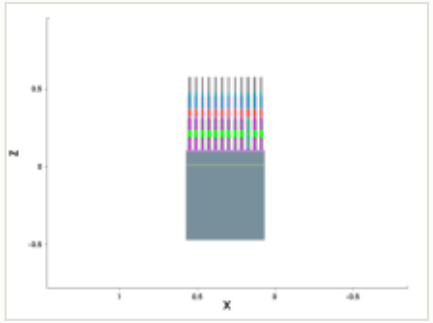
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



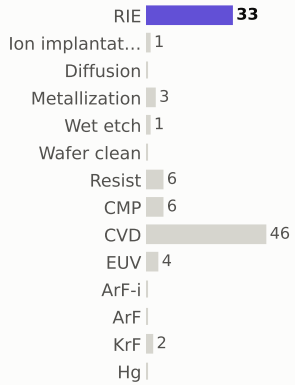
y cut at 0.049 µm



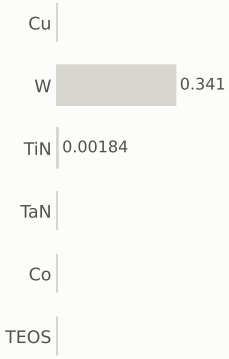
mask for this step

no mask for this step

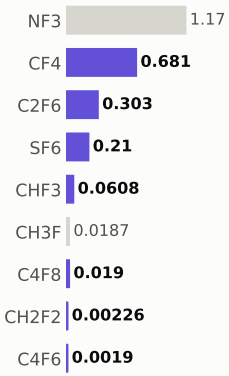
steps by process area



materials deposited (g)



gases emitted (g)



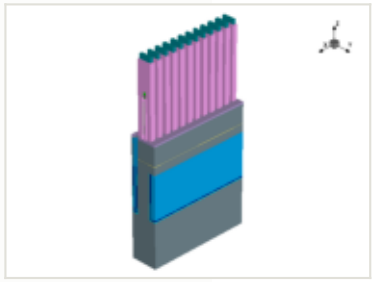
Step #1160: Resist

etch_strip · Resist

0.169	161.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

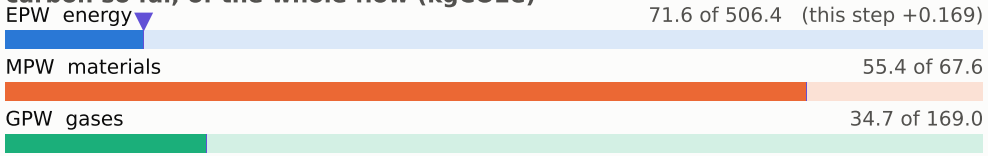


this step

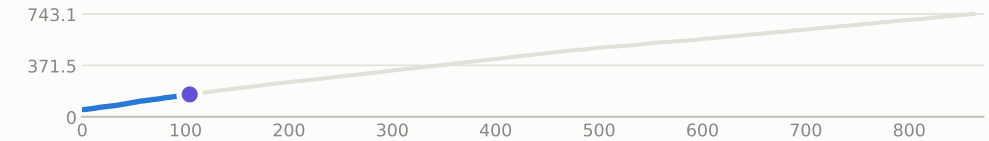
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

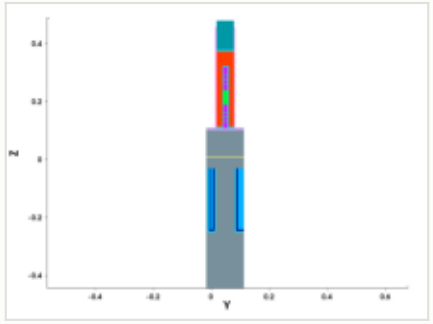
carbon so far, of the whole flow (kgCO2e)



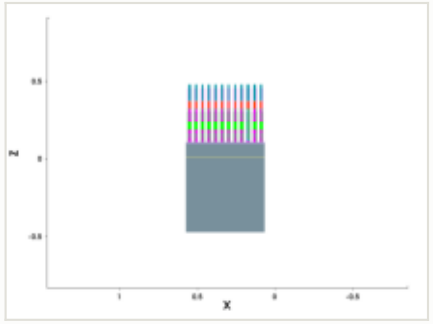
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



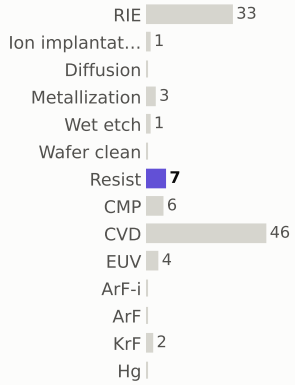
y cut at 0.049 µm



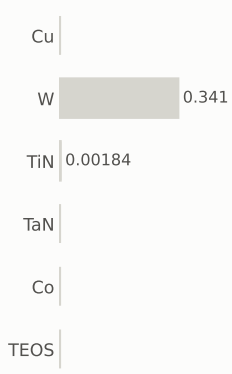
mask for this step

no mask for this step

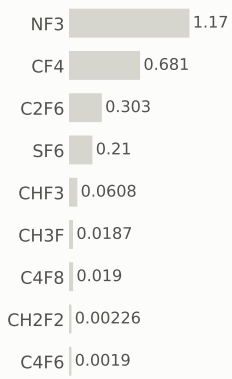
steps by process area



materials deposited (g)



gases emitted (g)



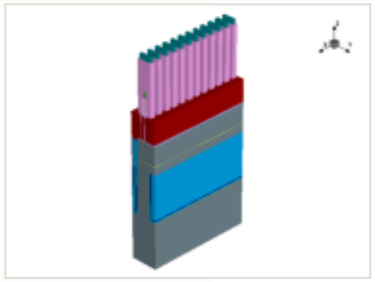
Step #1170: CVD

depo_fill · CVD

1.31	163.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

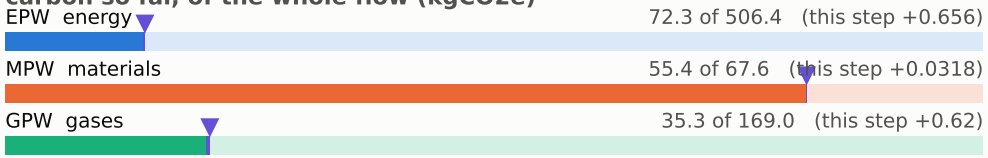


this step

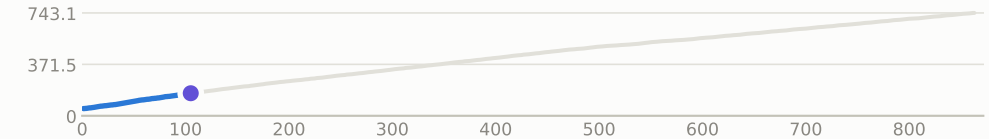
process area	CVD
material	—
thickness	-0.025 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

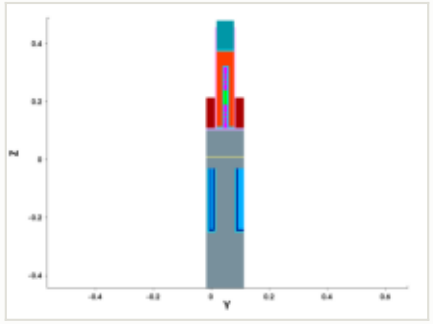
carbon so far, of the whole flow (kgCO2e)



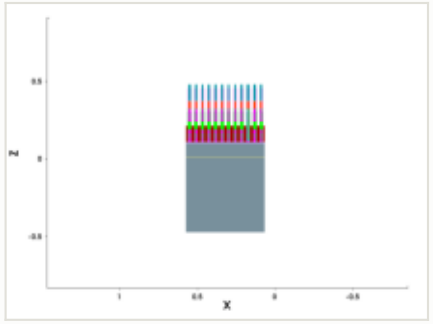
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



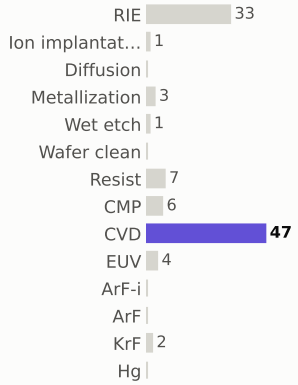
y cut at 0.049 µm



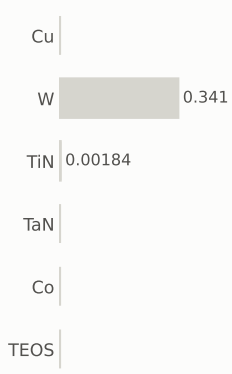
mask for this step

no mask for this step

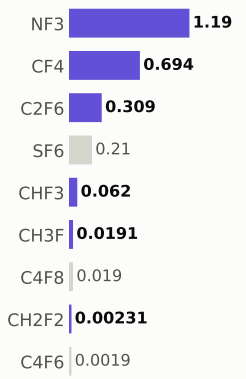
steps by process area



materials deposited (g)



gases emitted (g)



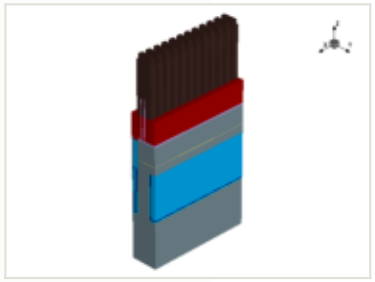
Step #1180: CVD

depo_isotropic · CVD

1.31	164.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

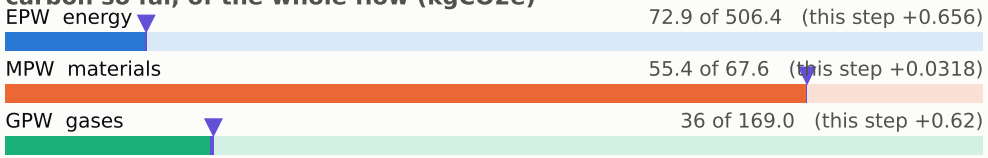
zoomed to fit



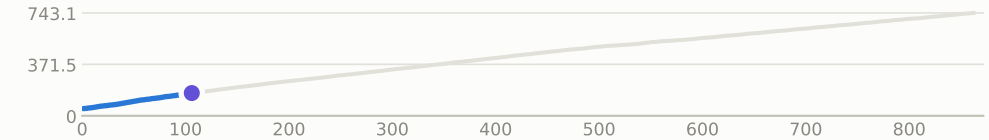
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.005 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

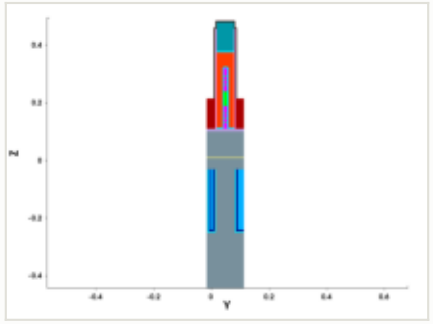
carbon so far, of the whole flow (kgCO2e)



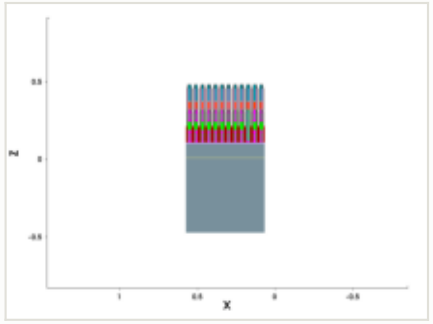
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



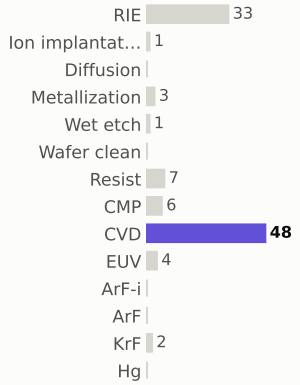
y cut at 0.049 µm



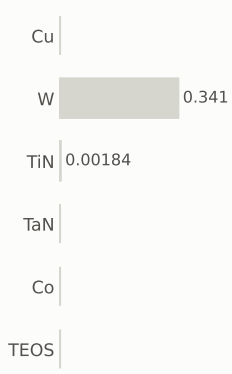
mask for this step

no mask for this step

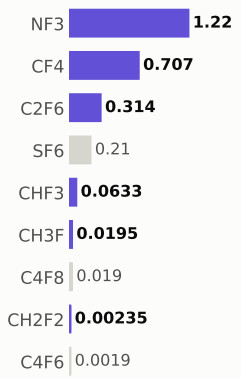
steps by process area



materials deposited (g)



gases emitted (g)



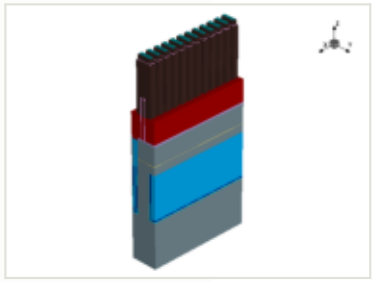
Step #1190: RIE

etch_anisotropic · RIE

1.04	165.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

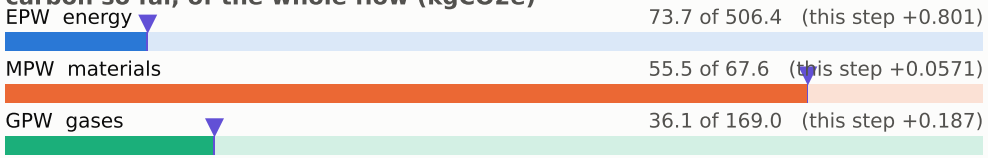


this step

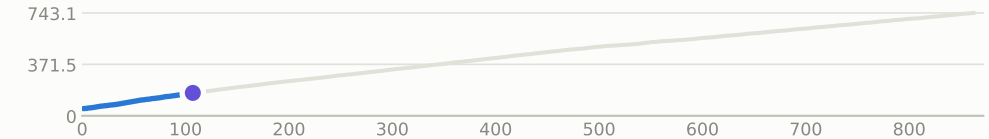
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

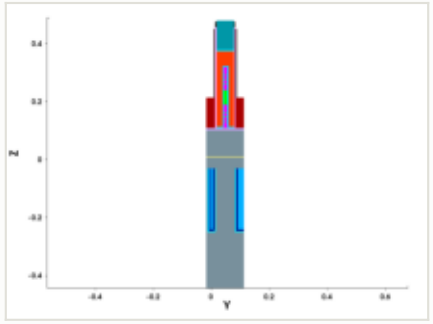
carbon so far, of the whole flow (kgCO2e)



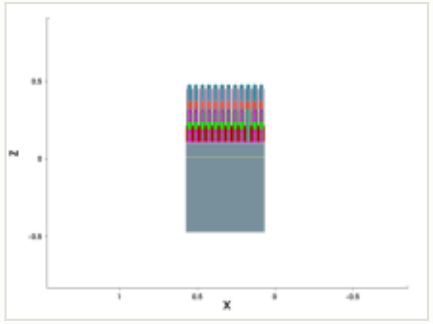
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



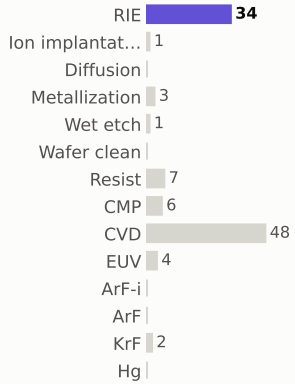
y cut at 0.049 µm



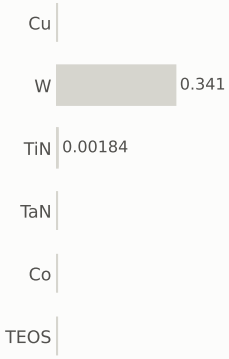
mask for this step

no mask for this step

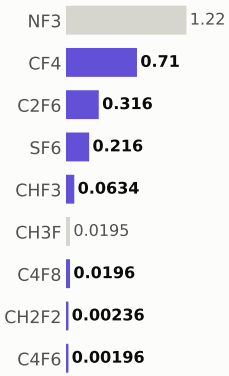
steps by process area



materials deposited (g)



gases emitted (g)



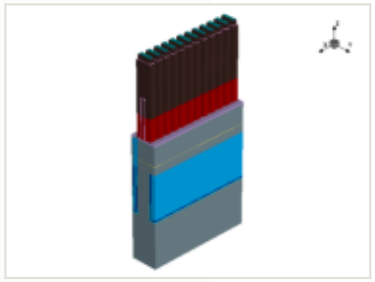
Step #1200: RIE

etch_anisotropic · RIE

1.04	166.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

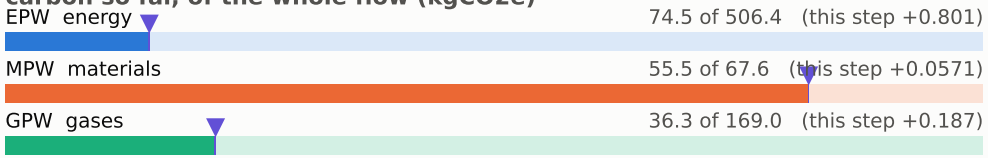


this step

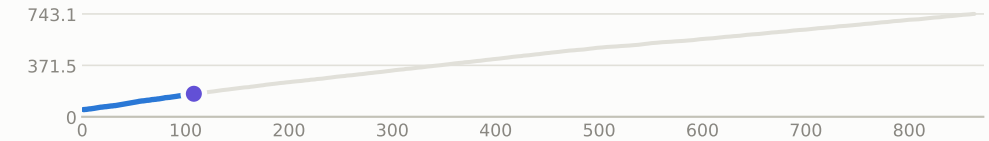
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

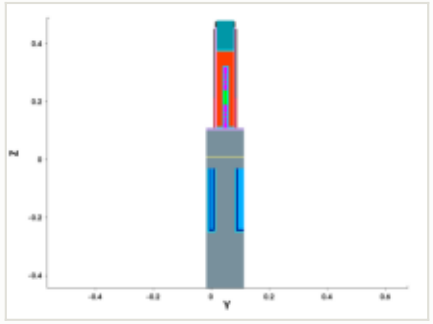
carbon so far, of the whole flow (kgCO2e)



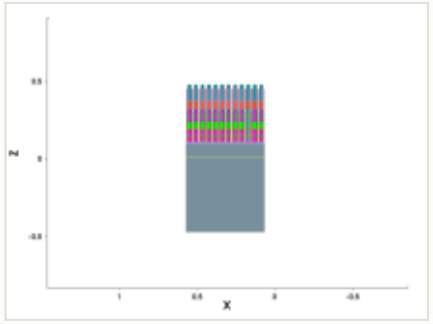
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



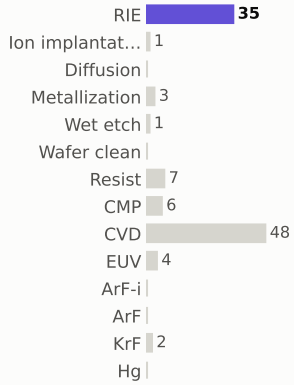
y cut at 0.049 µm



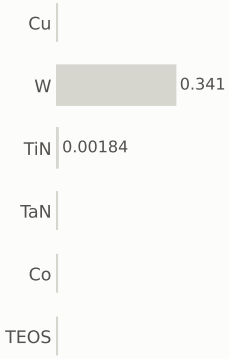
mask for this step

no mask for this step

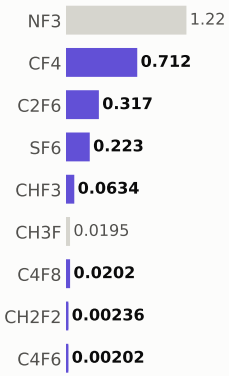
steps by process area



materials deposited (g)



gases emitted (g)



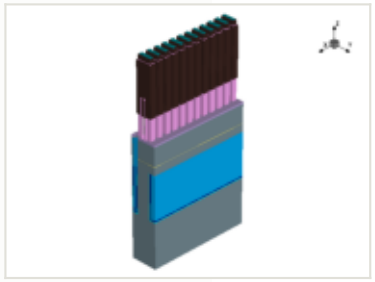
Step #1210: Resist

etch_strip · Resist

0.169	166.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

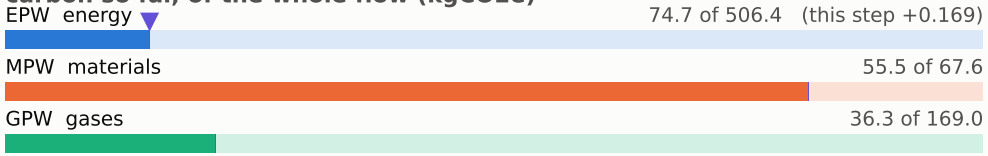
zoomed to fit



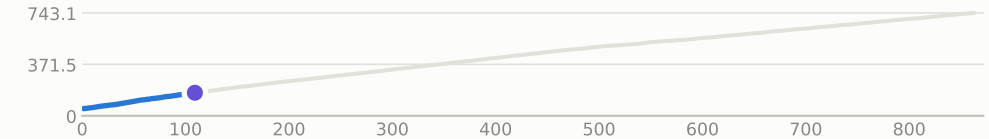
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

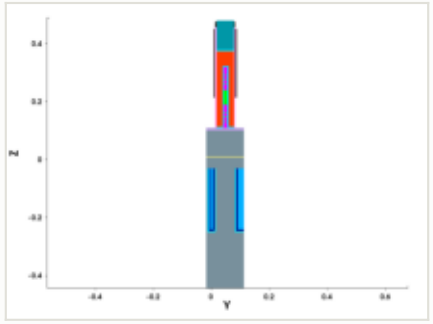
carbon so far, of the whole flow (kgCO2e)



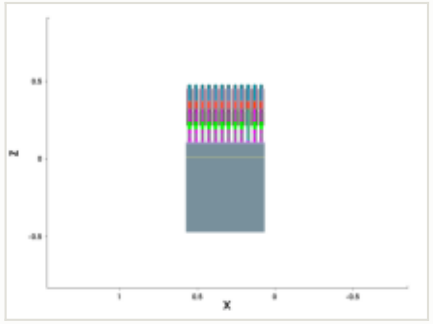
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



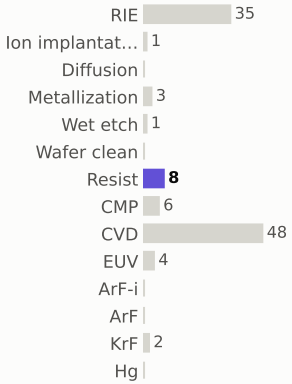
y cut at 0.049 µm



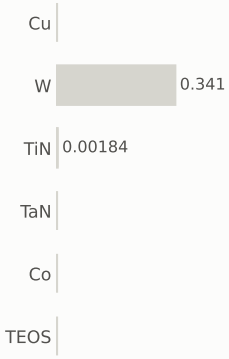
mask for this step

no mask for this step

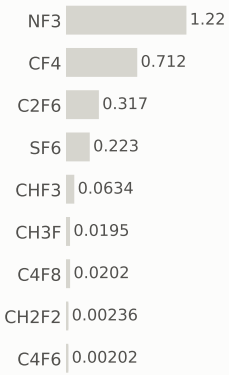
steps by process area



materials deposited (g)



gases emitted (g)



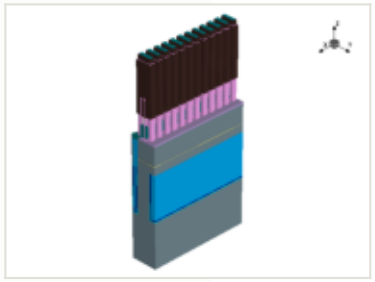
Step #1230: CVD

depo_isotropic · CVD

1.31	167.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

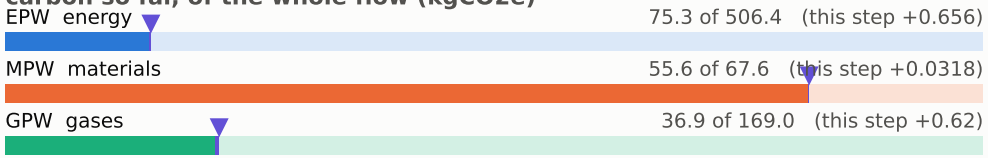


this step

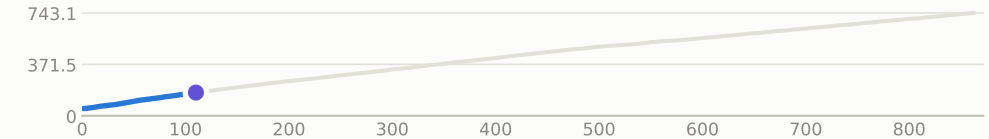
process area	CVD
material	—
thickness	0.014 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

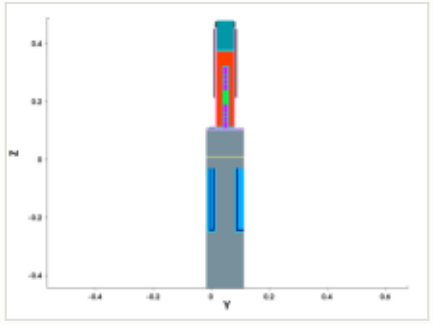
carbon so far, of the whole flow (kgCO2e)



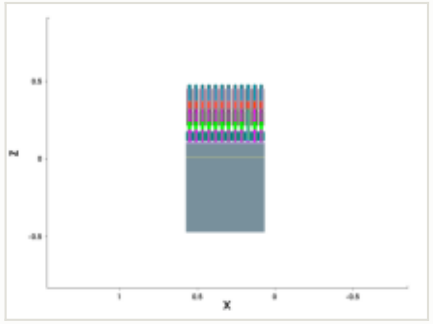
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



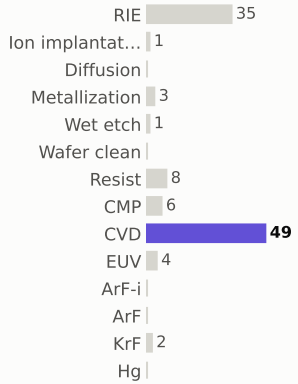
y cut at 0.049 µm



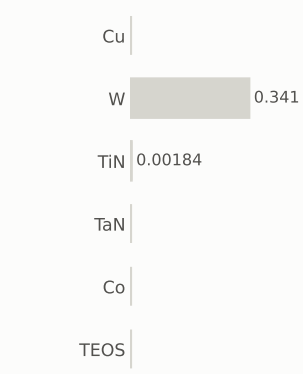
mask for this step

no mask for this step

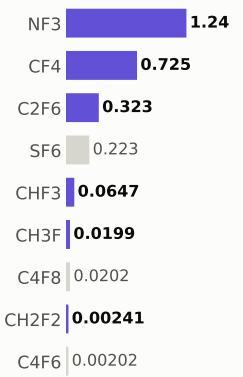
steps by process area



materials deposited (g)



gases emitted (g)



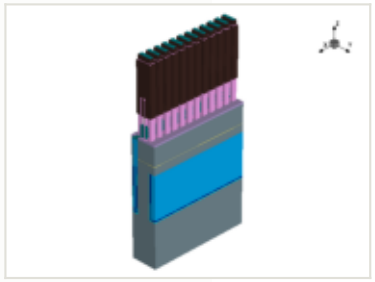
Step #1240: RIE

etch_anisotropic · RIE

1.04	168.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

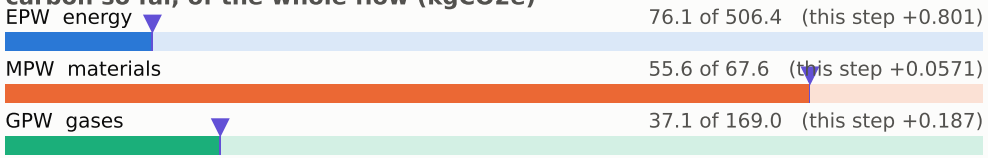


this step

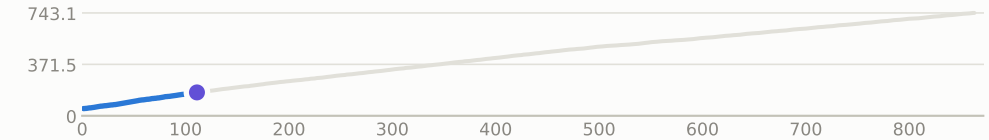
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

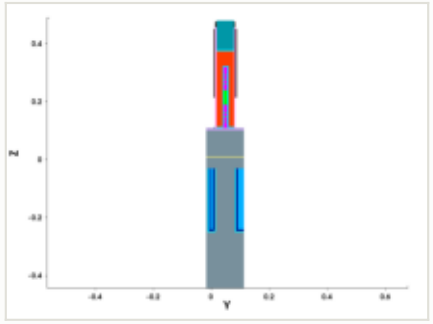
carbon so far, of the whole flow (kgCO2e)



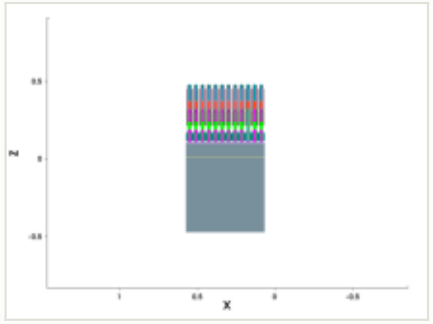
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



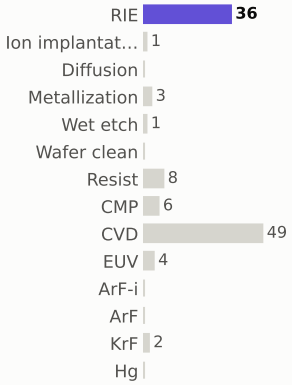
y cut at 0.049 µm



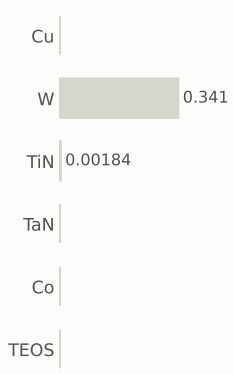
mask for this step

no mask for this step

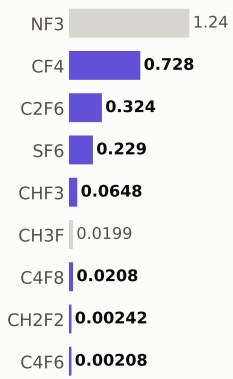
steps by process area



materials deposited (g)



gases emitted (g)



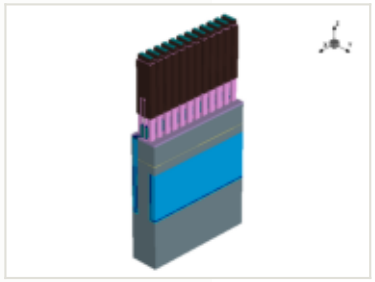
Step #1250: CVD

depo_isotropic · CVD

1.31	170.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

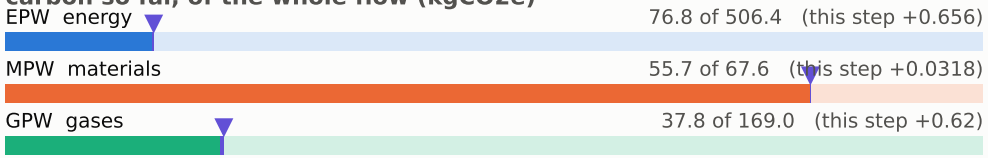


this step

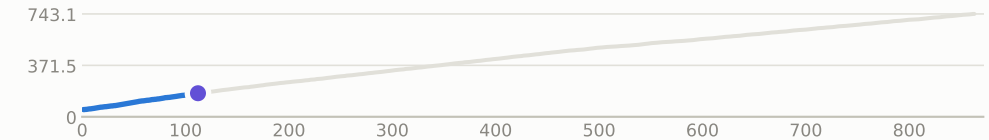
process area	CVD
material	TiN
thickness	0.001 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

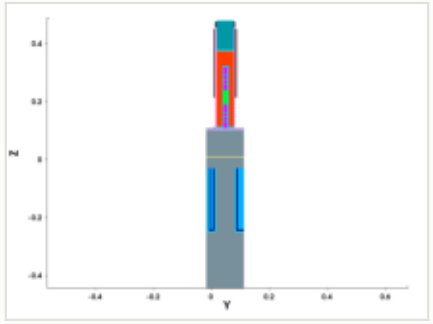
carbon so far, of the whole flow (kgCO2e)



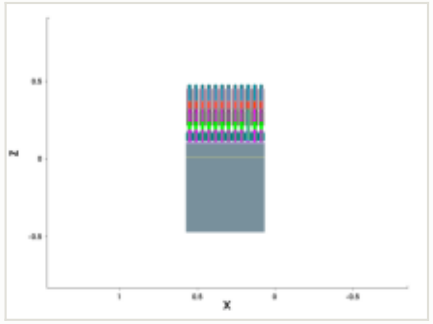
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



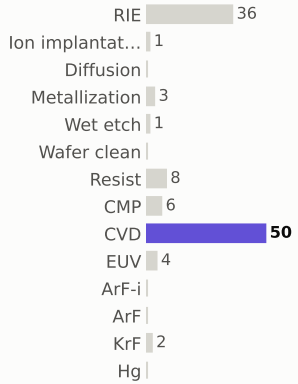
y cut at 0.049 µm



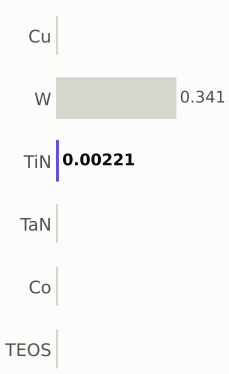
mask for this step

no mask for this step

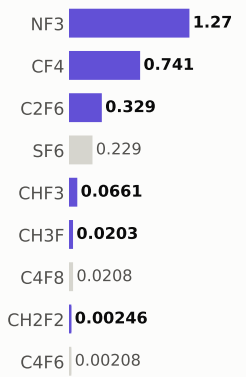
steps by process area



materials deposited (g)



gases emitted (g)



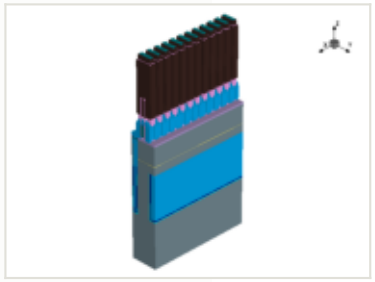
Step #1260: Metallization

depo_height_dependent · Metallization

1.2	171.4	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

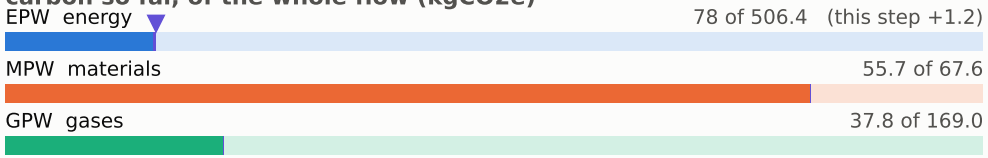


this step

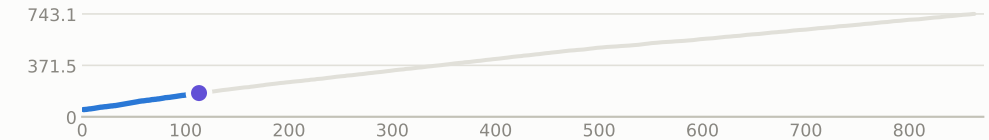
process area	Metallization
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

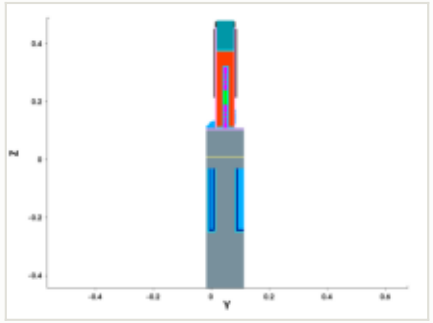
carbon so far, of the whole flow (kgCO2e)



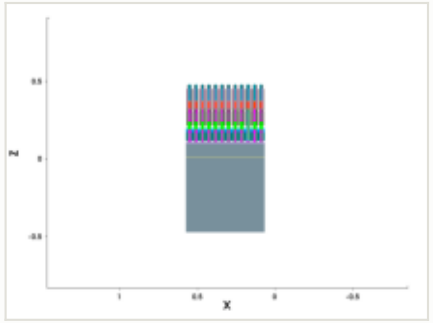
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



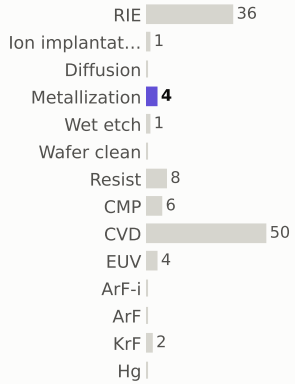
y cut at 0.049 µm



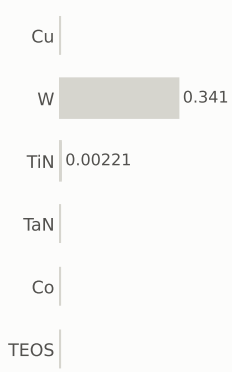
mask for this step

no mask for this step

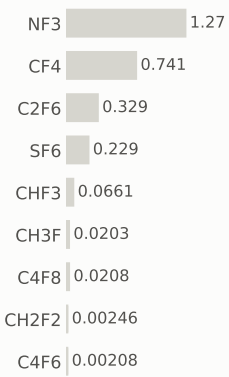
steps by process area



materials deposited (g)



gases emitted (g)



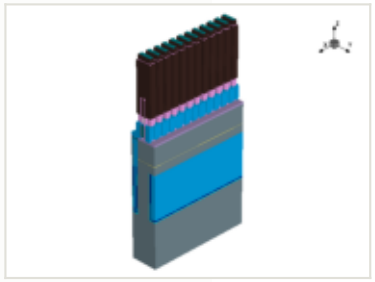
Step #1270: CMP

etch_cmp · CMP

0.169	171.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

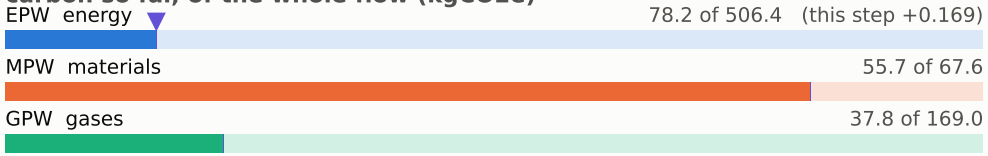
zoomed to fit



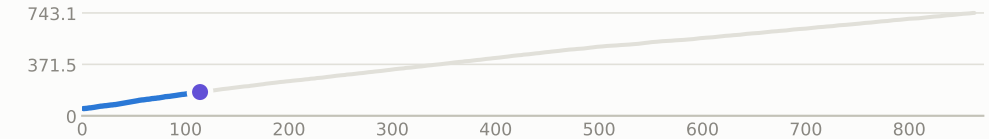
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

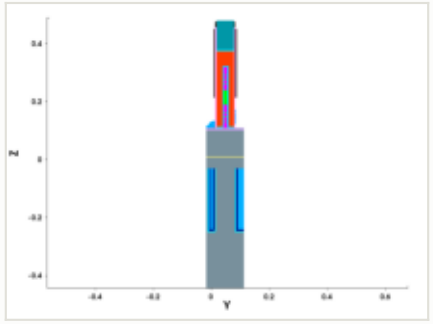
carbon so far, of the whole flow (kgCO2e)



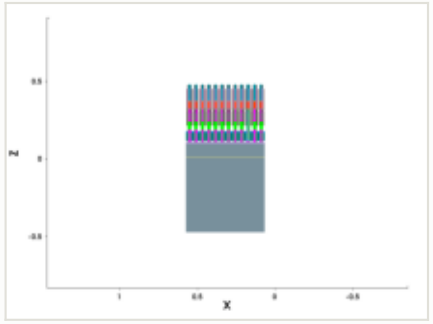
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



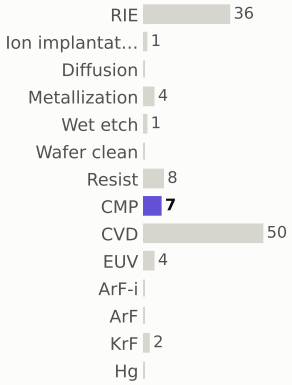
y cut at 0.049 µm



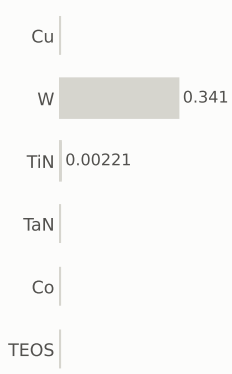
mask for this step

no mask for this step

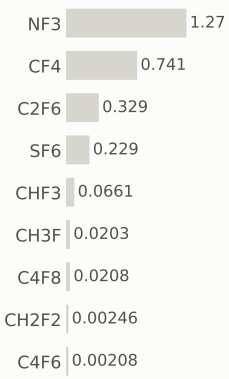
steps by process area



materials deposited (g)



gases emitted (g)



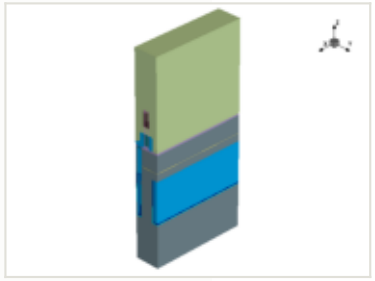
Step #1290: CVD

depo_fill · CVD

1.31	172.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

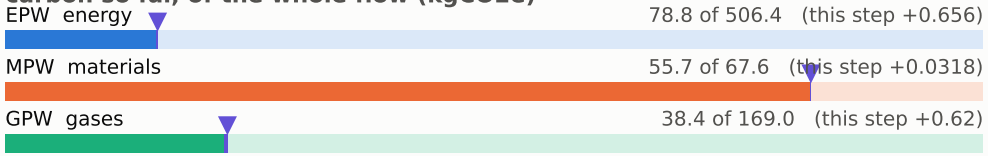


this step

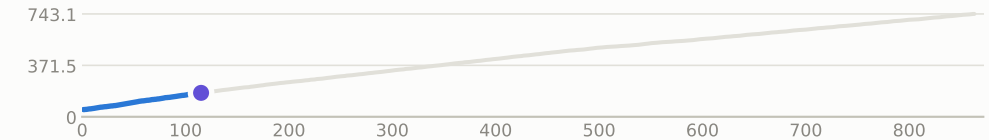
process area	CVD
material	—
thickness	0.1 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

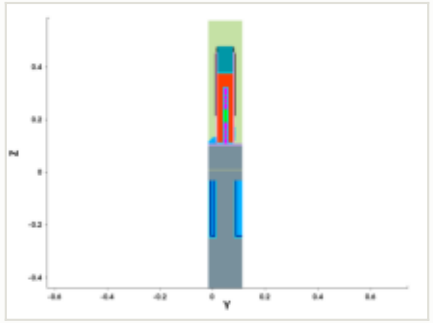
carbon so far, of the whole flow (kgCO2e)



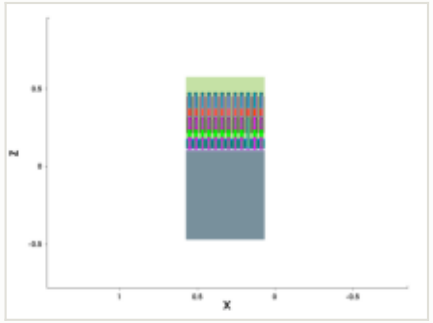
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



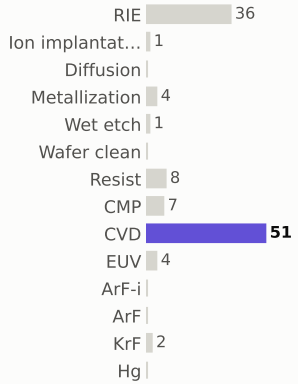
y cut at 0.049 µm



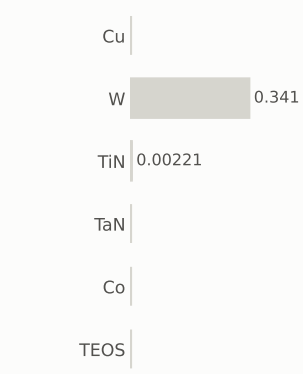
mask for this step

no mask for this step

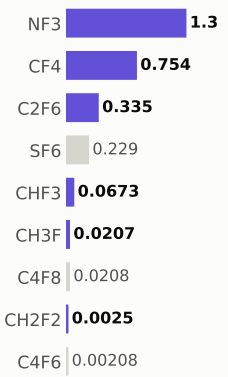
steps by process area



materials deposited (g)



gases emitted (g)



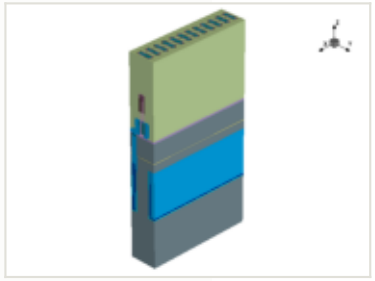
Step #1300: CMP

etch_cmp · CMP

0.169	173.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

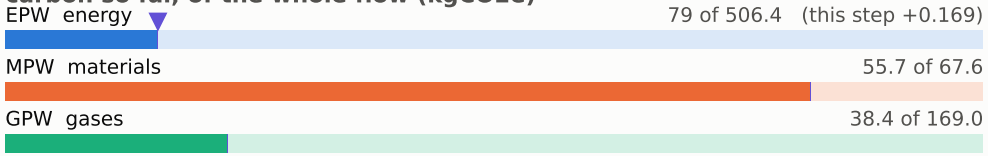


this step

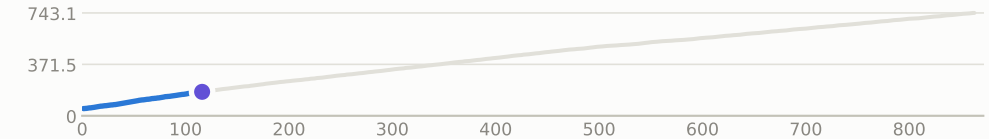
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

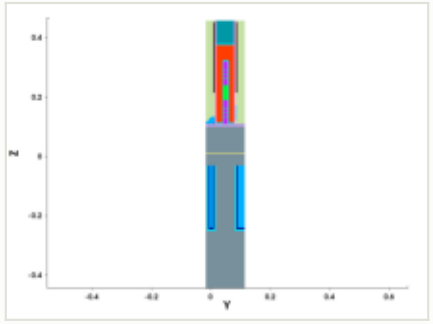
carbon so far, of the whole flow (kgCO2e)



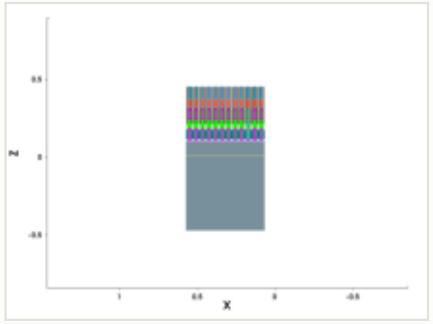
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



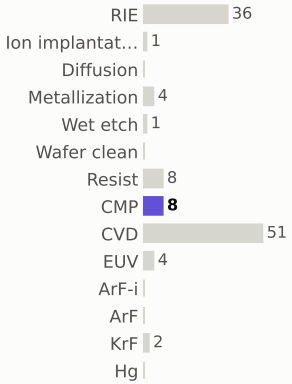
y cut at 0.049 µm



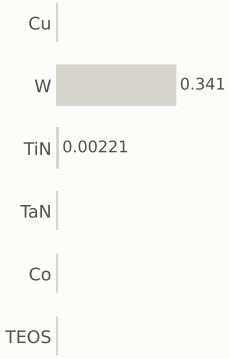
mask for this step

no mask for this step

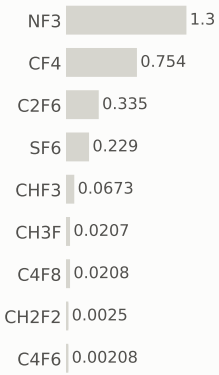
steps by process area



materials deposited (g)



gases emitted (g)



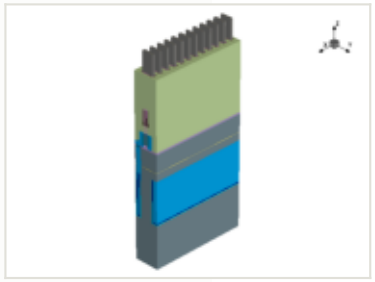
Step #1310: EUV

litho_patterning · EUV

1.96	175.0	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

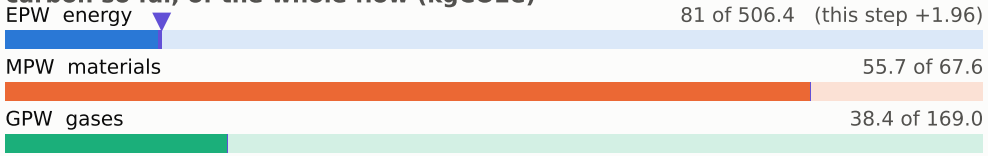


this step

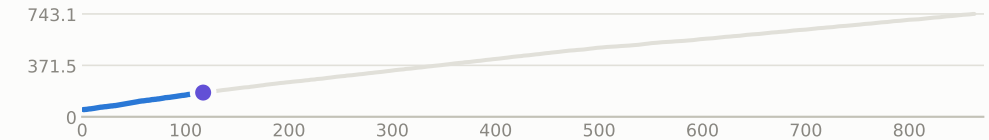
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

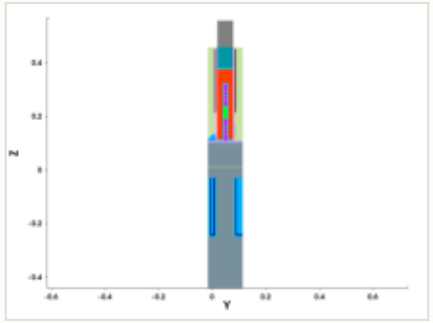
carbon so far, of the whole flow (kgCO2e)



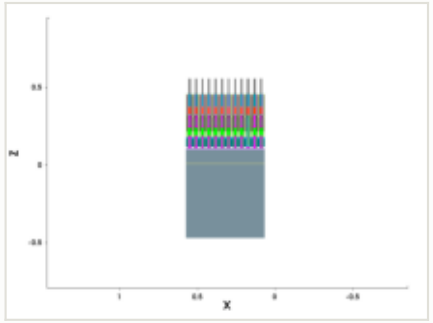
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

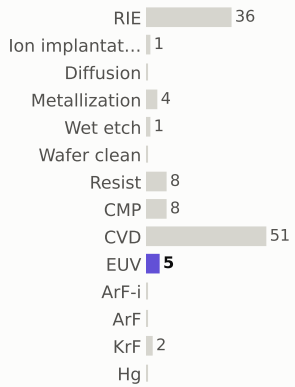


y cut at 0.049 µm

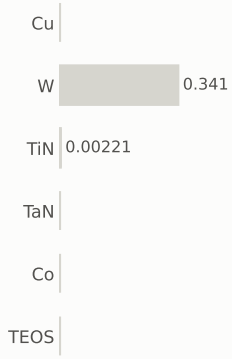


mask: layer 3:0, 3:1, positive (GDSII)

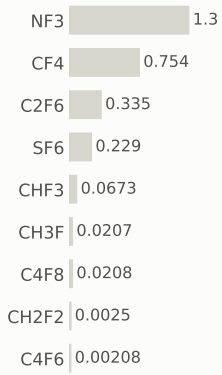
steps by process area



materials deposited (g)



gases emitted (g)



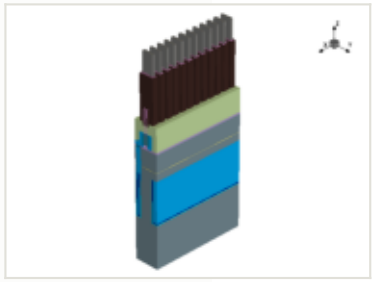
Step #1320: RIE

etch_anisotropic · RIE

1.04	176.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

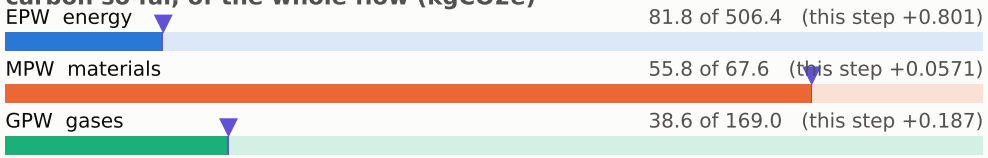


this step

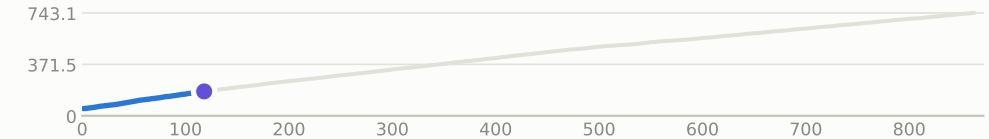
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

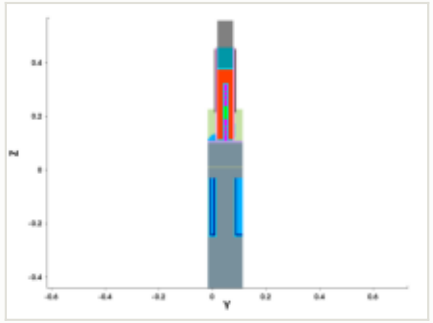
carbon so far, of the whole flow (kgCO2e)



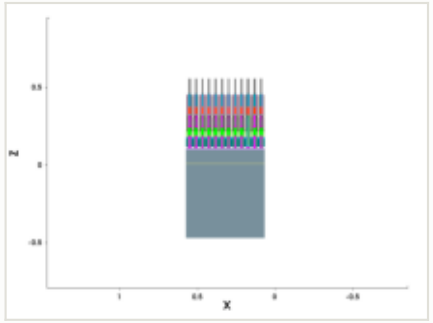
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



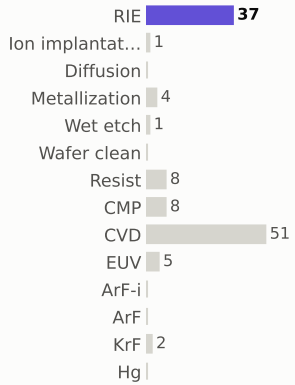
y cut at 0.049 µm



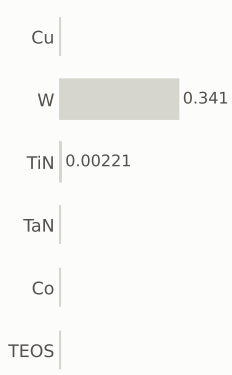
mask for this step

no mask for this step

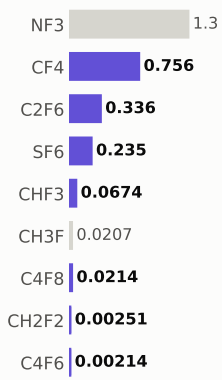
steps by process area



materials deposited (g)



gases emitted (g)



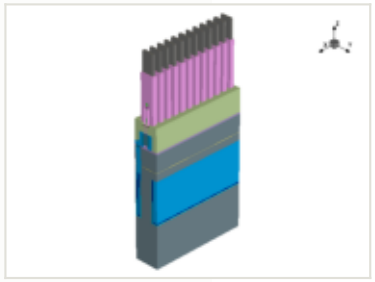
Step #1330: Wet_etch

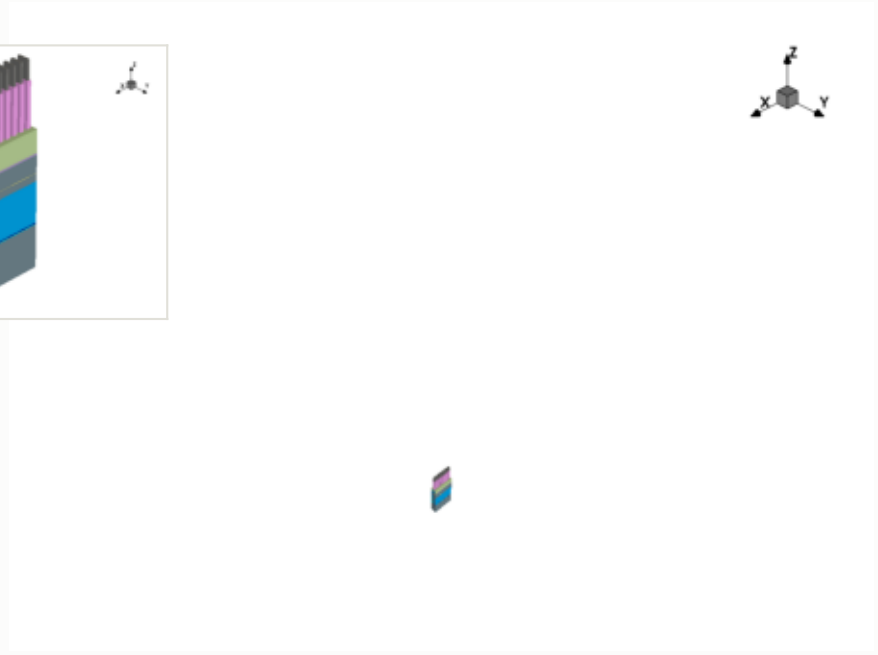
etch_isotropic · Wet_etch

0.169	176.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

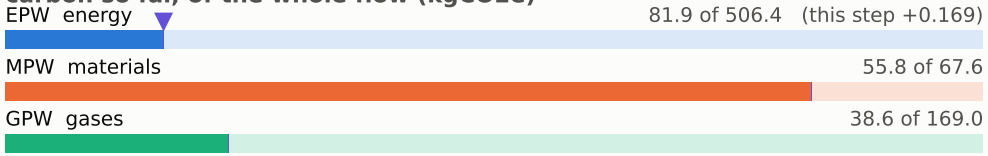




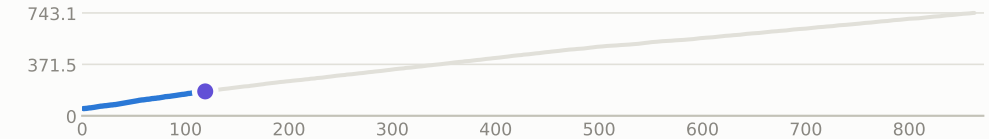
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

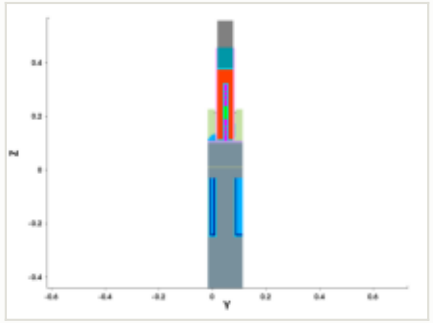
carbon so far, of the whole flow (kgCO2e)



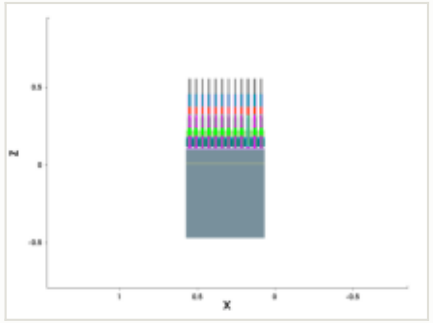
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



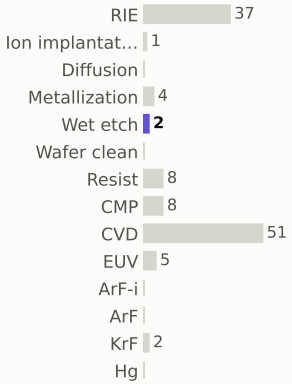
y cut at 0.049 µm



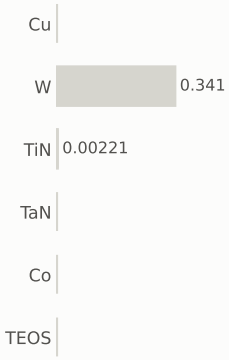
mask for this step

no mask for this step

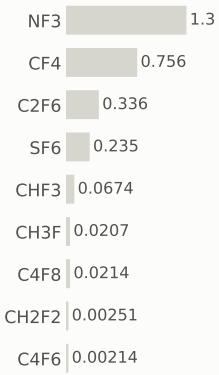
steps by process area



materials deposited (g)



gases emitted (g)



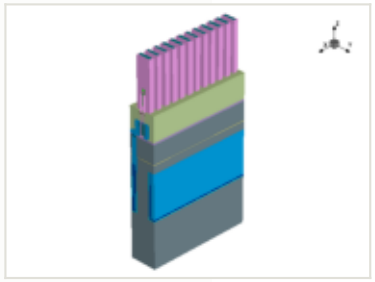
Step #1340: Resist

etch_strip · Resist

0.169	176.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

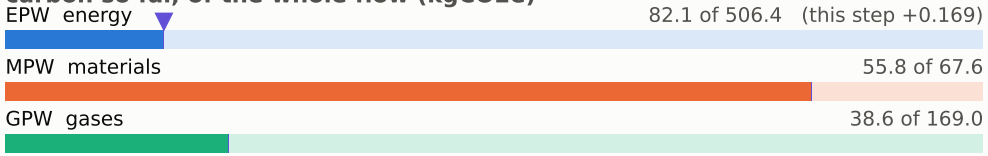


this step

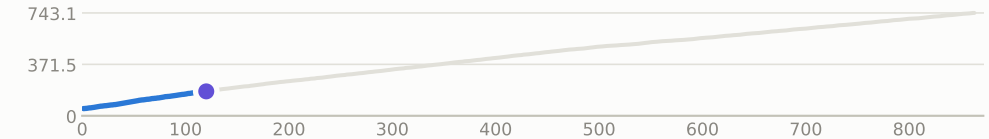
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

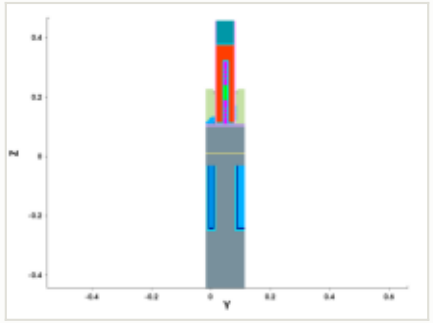
carbon so far, of the whole flow (kgCO2e)



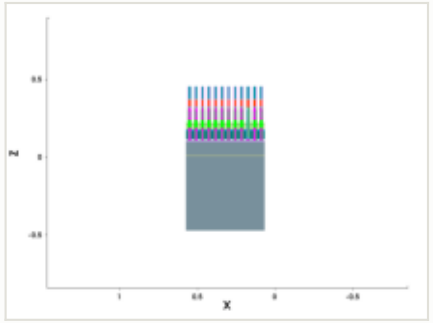
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



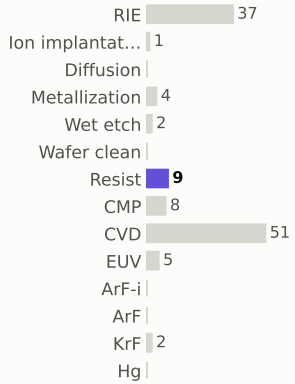
y cut at 0.049 µm



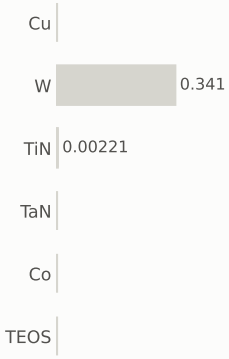
mask for this step

no mask for this step

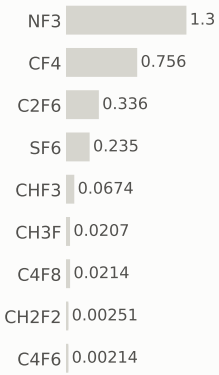
steps by process area



materials deposited (g)



gases emitted (g)



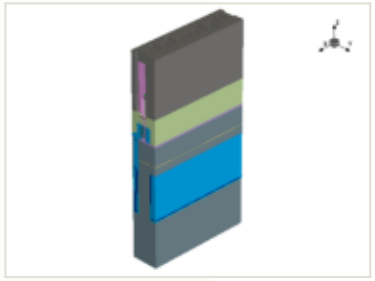
Step #1360: EUV

litho_patterning · EUV

1.96	178.4	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

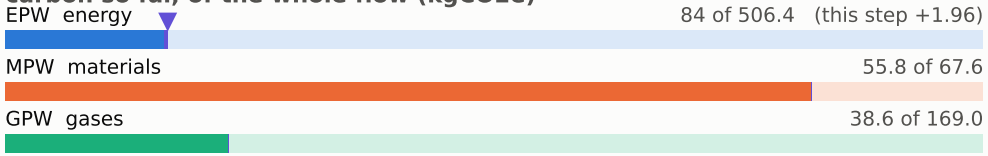


this step

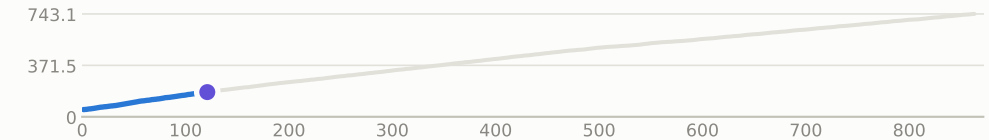
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

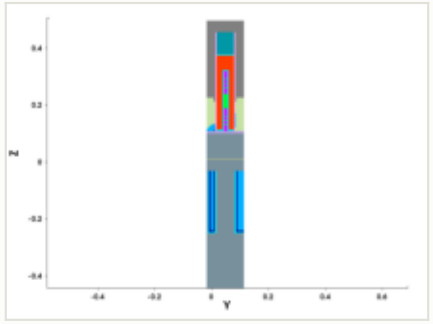
carbon so far, of the whole flow (kgCO2e)



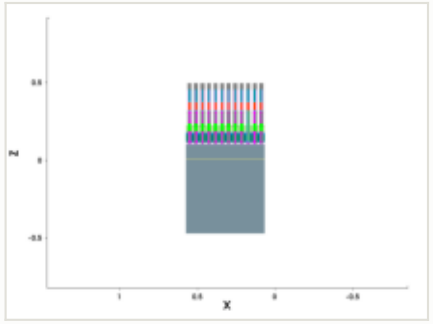
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

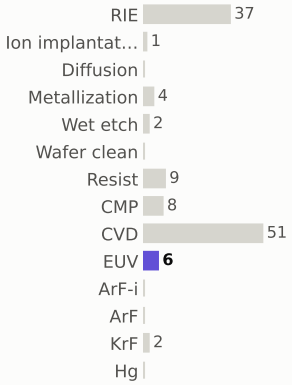


y cut at 0.049 µm

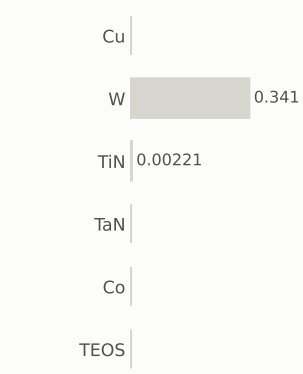


mask: layer 10:1 (GDSII)

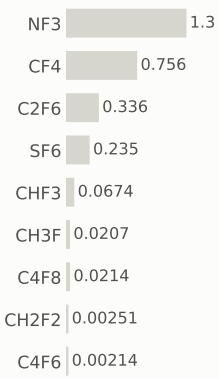
steps by process area



materials deposited (g)



gases emitted (g)



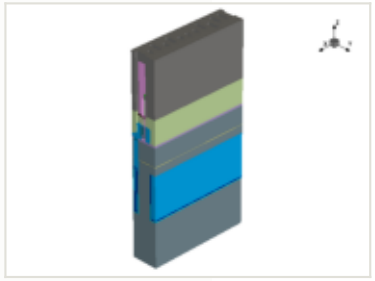
Step #1370: RIE

etch_anisotropic · RIE

1.04	179.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

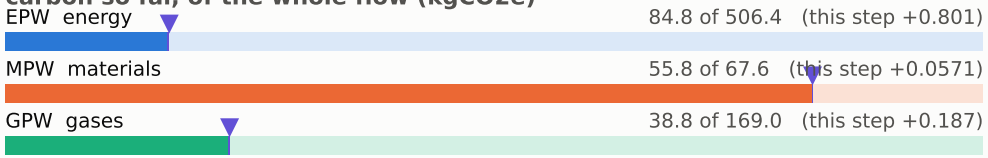


this step

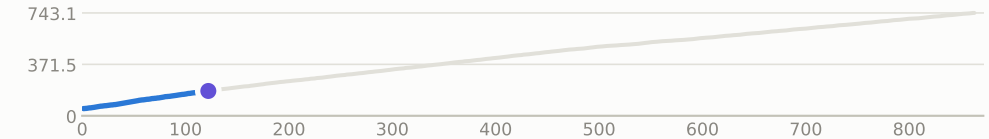
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

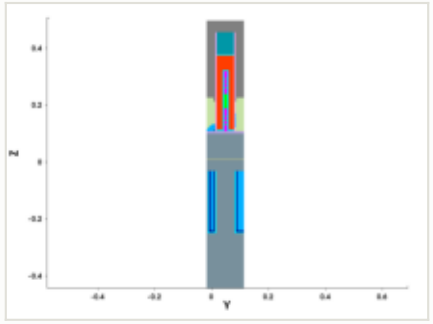
carbon so far, of the whole flow (kgCO2e)



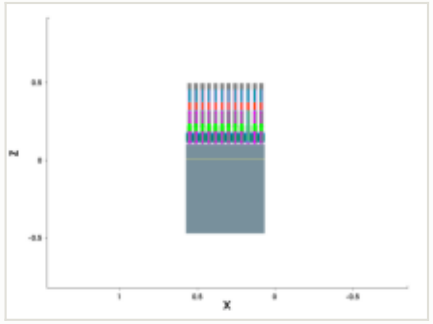
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



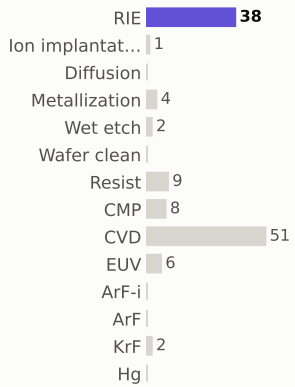
y cut at 0.049 µm



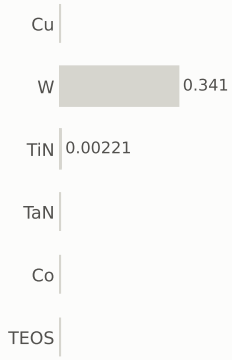
mask for this step

no mask for this step

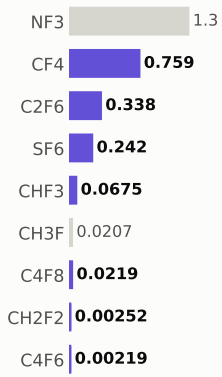
steps by process area



materials deposited (g)



gases emitted (g)



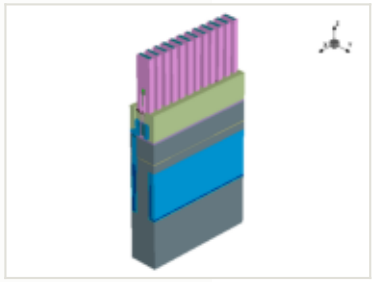
Step #1380: Resist

etch_strip · Resist

0.169	179.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

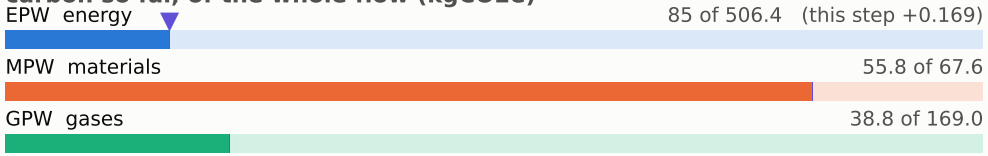


this step

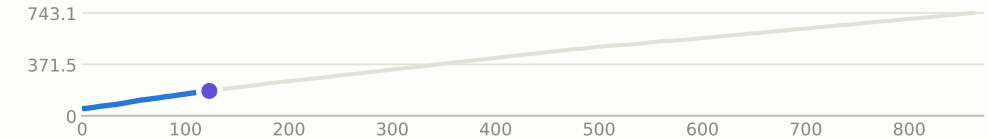
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

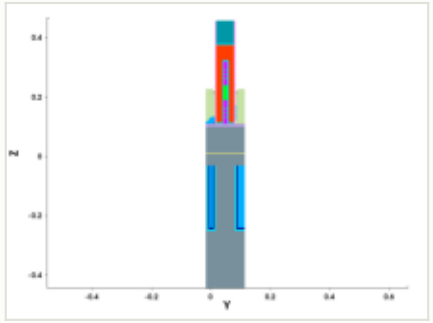
carbon so far, of the whole flow (kgCO2e)



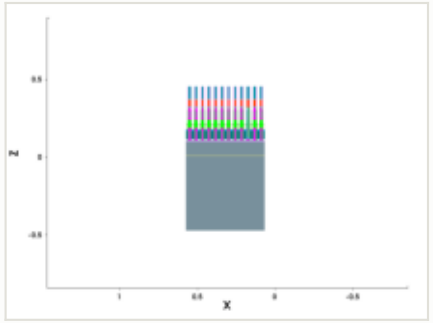
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



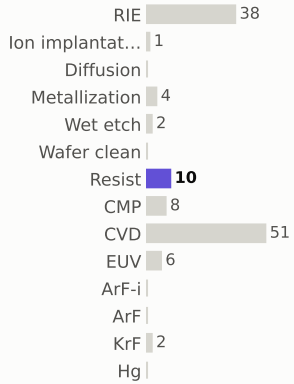
y cut at 0.049 µm



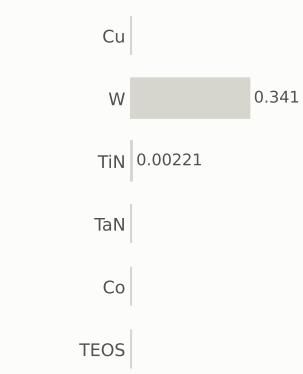
mask for this step

no mask for this step

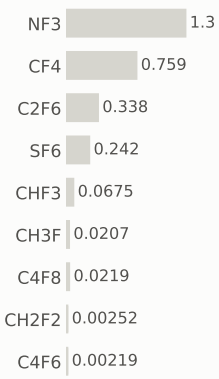
steps by process area



materials deposited (g)



gases emitted (g)



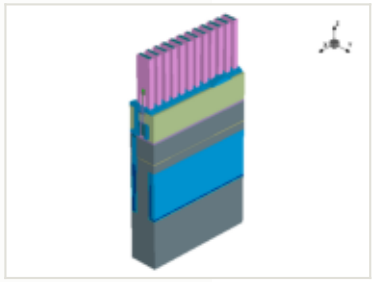
Step #1390: Metallization

depo_isotropic · Metallization

1.21	180.8	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

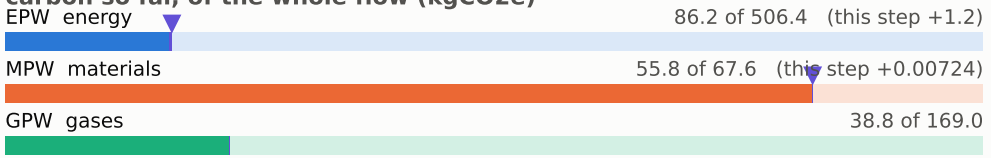


this step

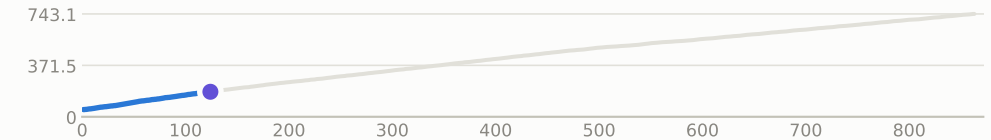
process area	Metallization
material	W
thickness	0.07 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00724 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

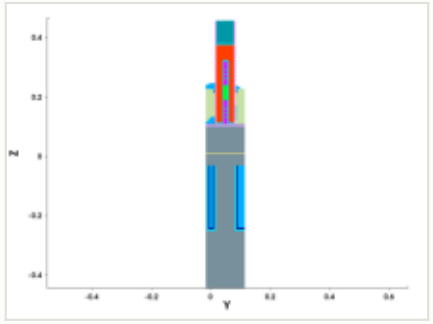
carbon so far, of the whole flow (kgCO2e)



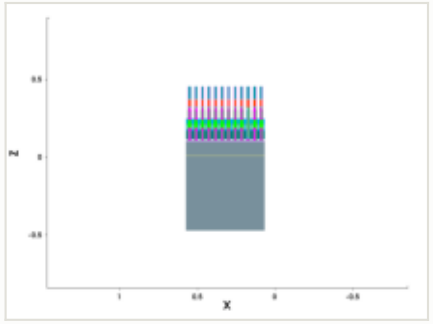
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



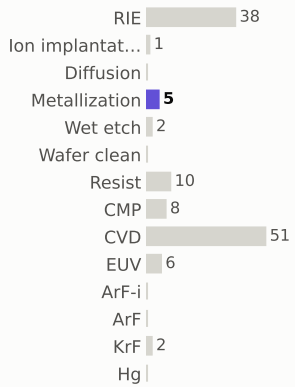
y cut at 0.049 µm



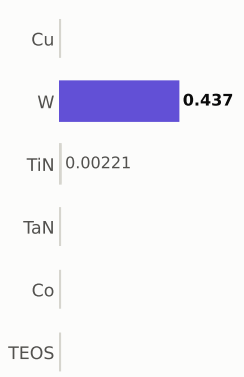
mask for this step

no mask for this step

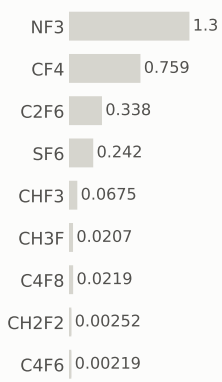
steps by process area



materials deposited (g)



gases emitted (g)



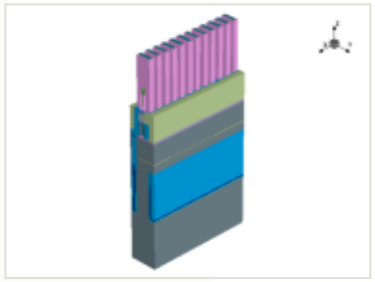
Step #1400: CMP

etch_cmp · CMP

0.169	181.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

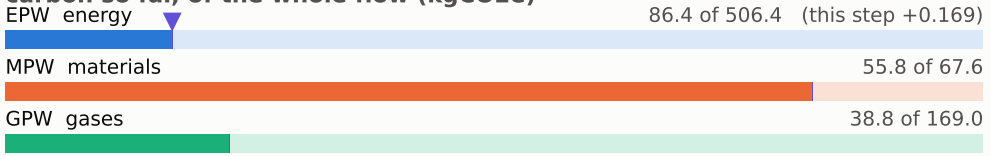
zoomed to fit



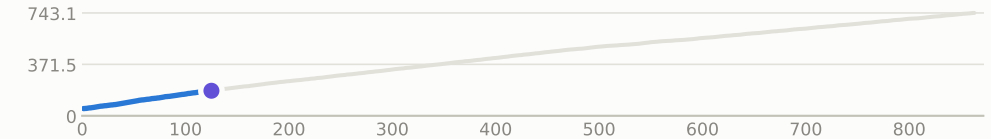
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

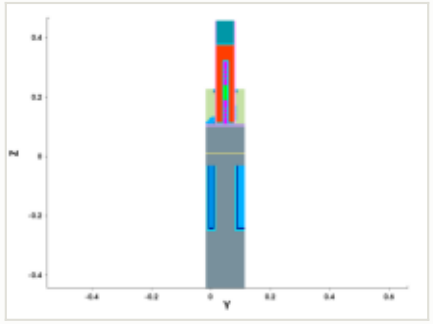
carbon so far, of the whole flow (kgCO2e)



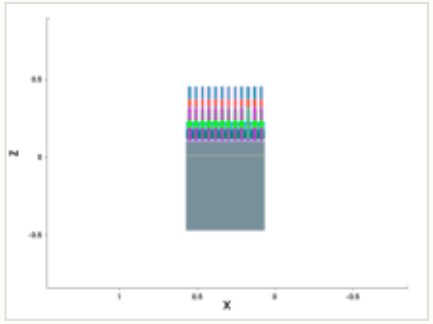
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



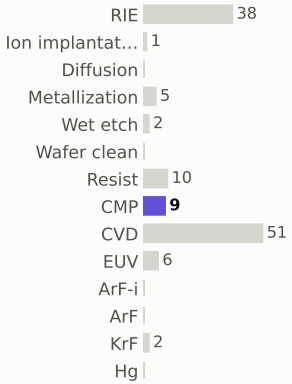
y cut at 0.049 µm



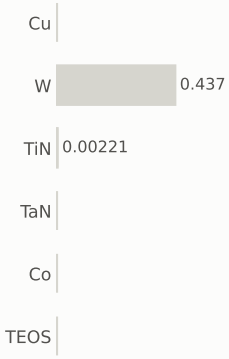
mask for this step

no mask for this step

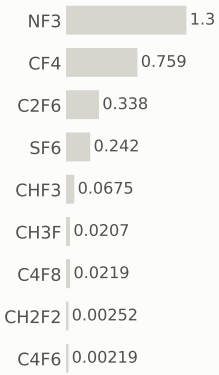
steps by process area



materials deposited (g)



gases emitted (g)



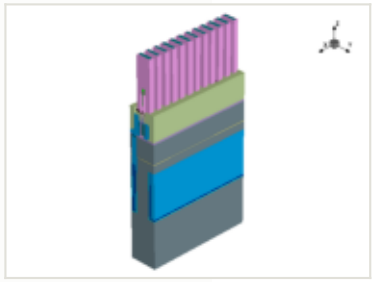
Step #1410: RIE

etch_anisotropic · RIE

1.04	182.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

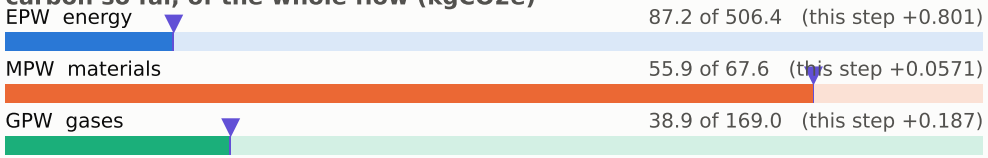
zoomed to fit



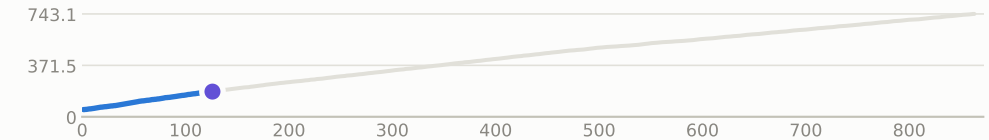
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

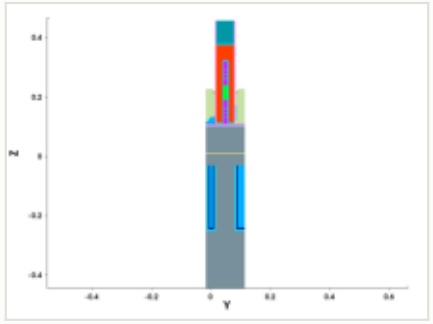
carbon so far, of the whole flow (kgCO2e)



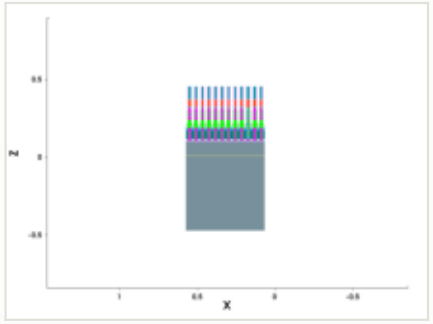
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



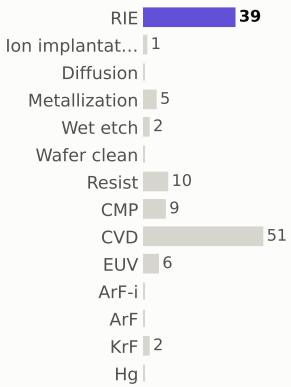
y cut at 0.049 µm



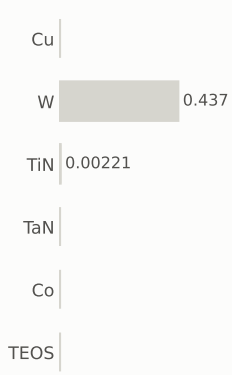
mask for this step

no mask for this step

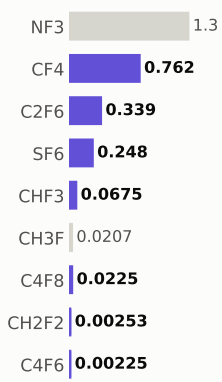
steps by process area



materials deposited (g)



gases emitted (g)



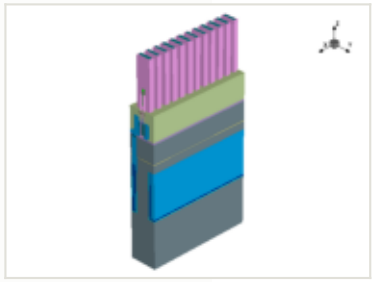
Step #1420: CVD

depo_fill · CVD

1.31	183.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

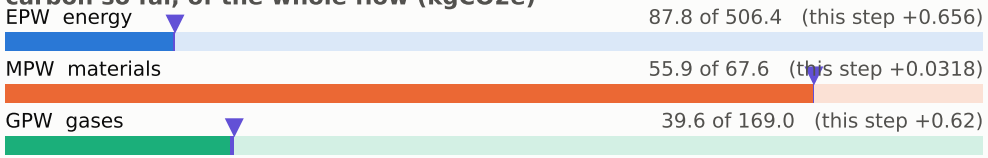
zoomed to fit



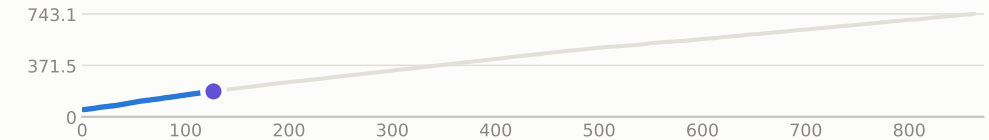
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

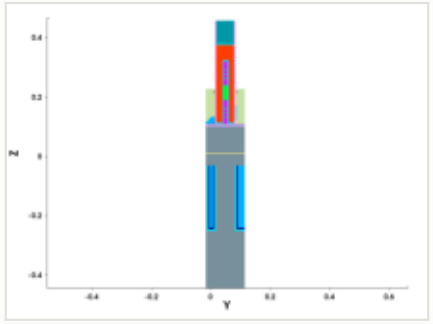
carbon so far, of the whole flow (kgCO2e)



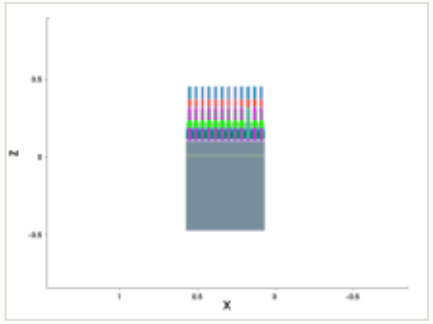
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



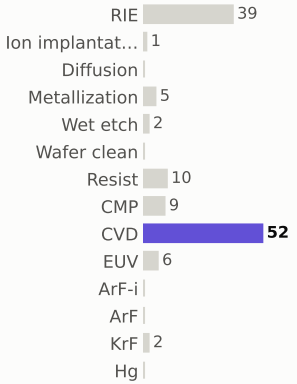
y cut at 0.049 µm



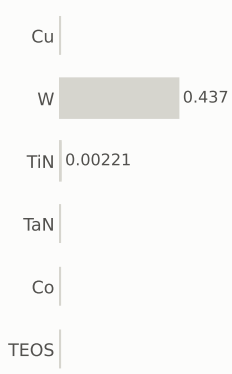
mask for this step

no mask for this step

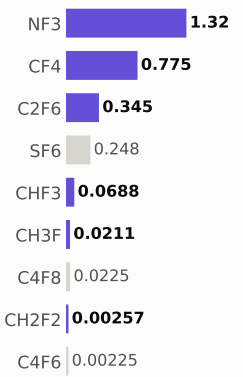
steps by process area



materials deposited (g)



gases emitted (g)



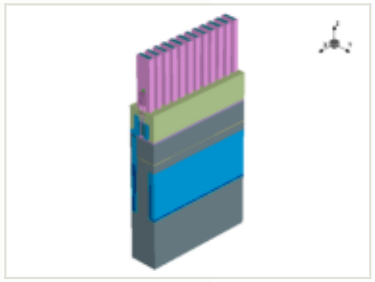
Step #1440: CVD

depo_isotropic · CVD

1.31	184.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

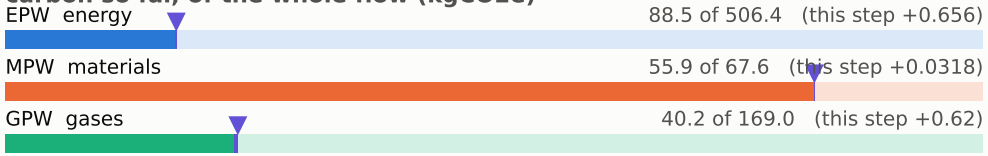
zoomed to fit



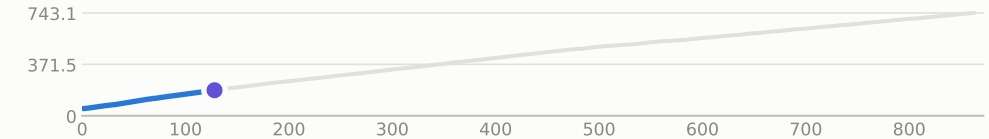
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.013 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

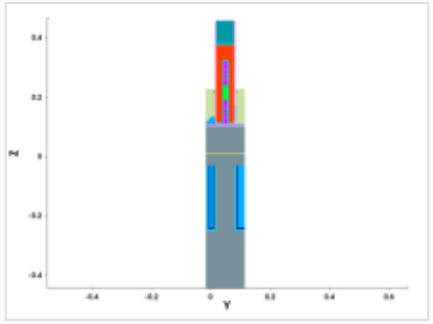
carbon so far, of the whole flow (kgCO2e)



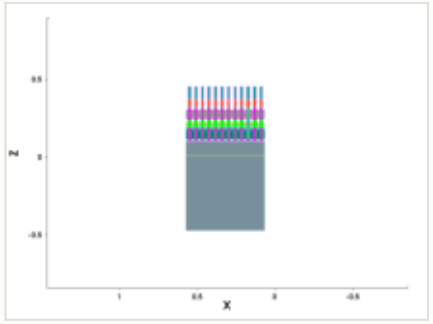
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



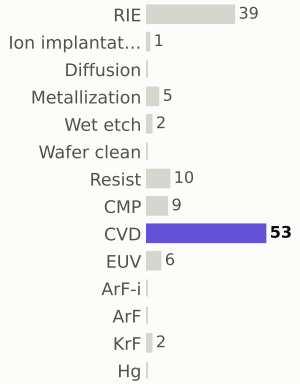
y cut at 0.049 µm



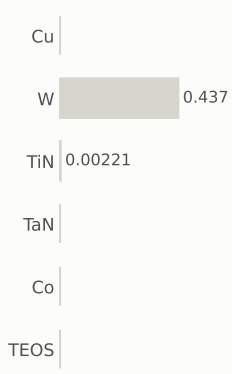
mask for this step

no mask for this step

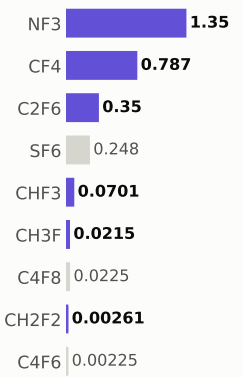
steps by process area



materials deposited (g)



gases emitted (g)



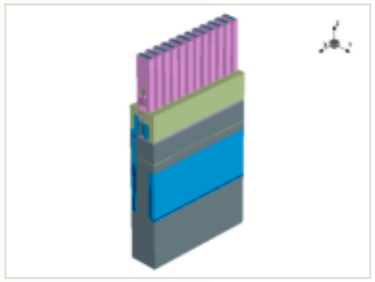
Step #1450: RIE

etch_anisotropic · RIE

1.04	185.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

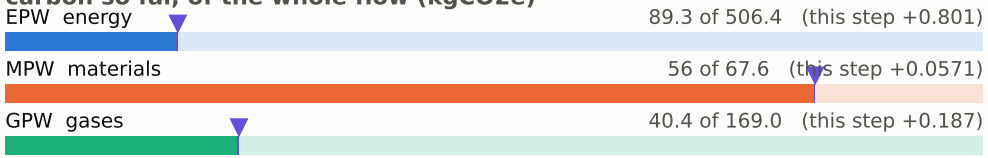


this step

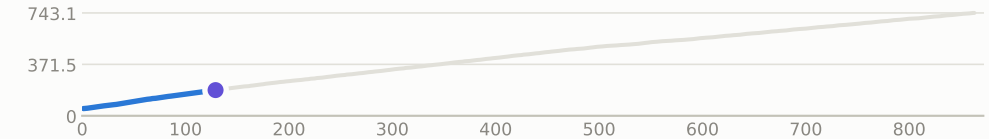
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

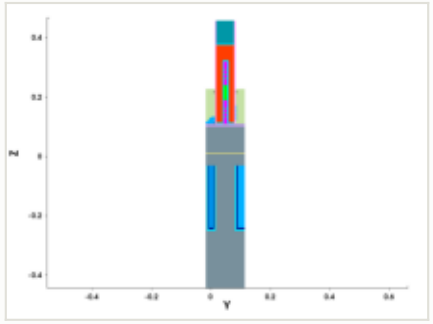
carbon so far, of the whole flow (kgCO2e)



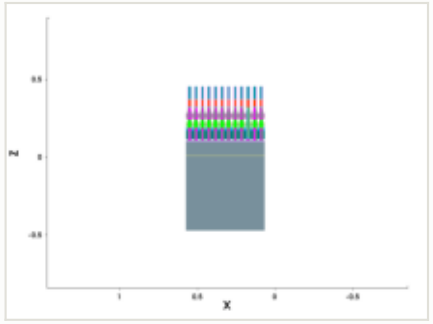
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



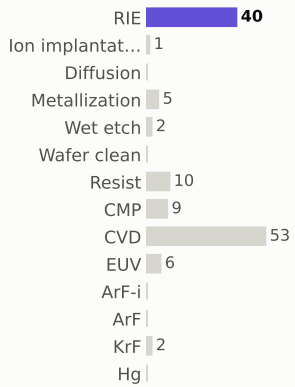
y cut at 0.049 µm



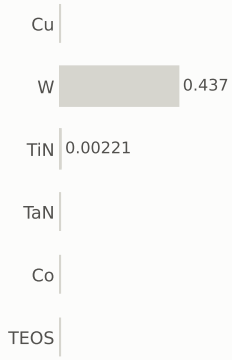
mask for this step

no mask for this step

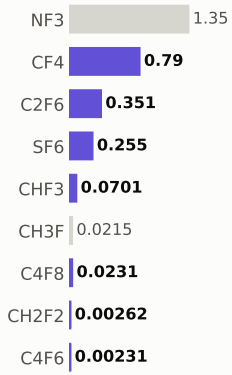
steps by process area



materials deposited (g)



gases emitted (g)



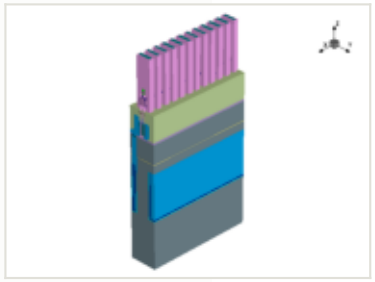
Step #1460: CVD

depo_isotropic · CVD

1.31	187.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

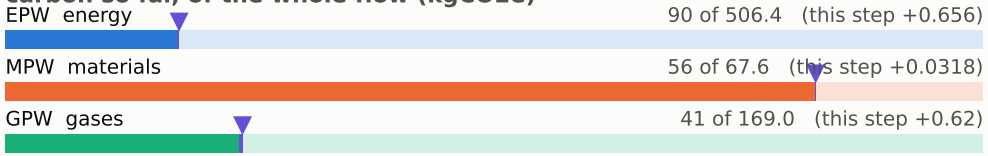
zoomed to fit



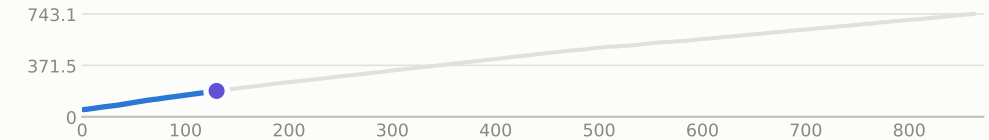
this step

process area	CVD	EPW	0.656 kgCO2e
material	TiN	EPW tool	0.469 kgCO2e
thickness	0.001 µm	EPW overhead	0.187 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

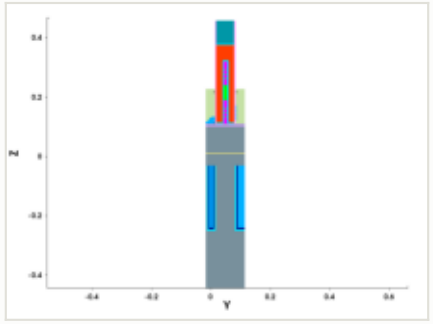
carbon so far, of the whole flow (kgCO2e)



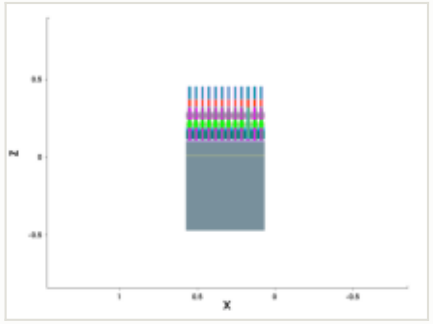
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



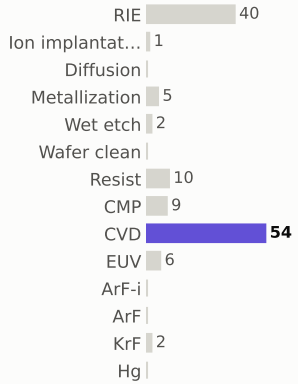
y cut at 0.049 µm



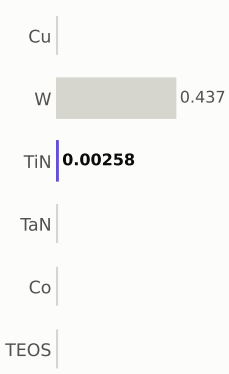
mask for this step

no mask for this step

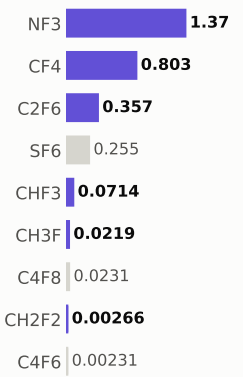
steps by process area



materials deposited (g)



gases emitted (g)



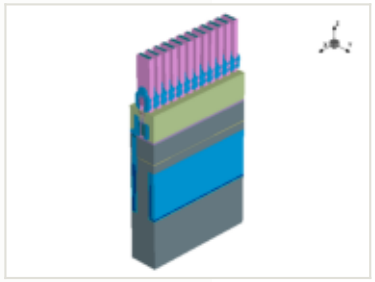
Step #1470: Metallization

depo_height_dependent · Metallization

1.2	188.2	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

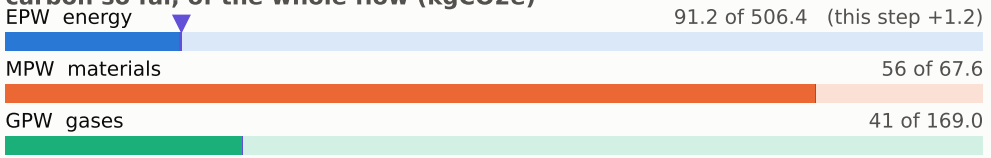


this step

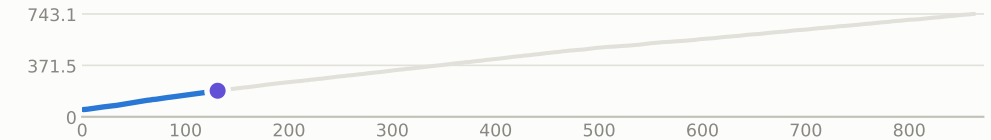
process area	Metallization
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

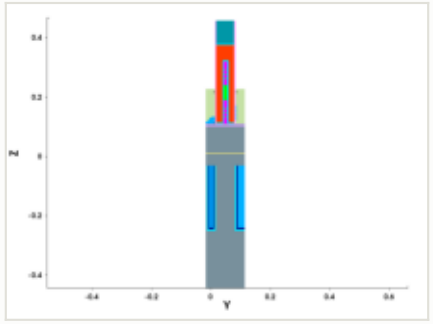
carbon so far, of the whole flow (kgCO2e)



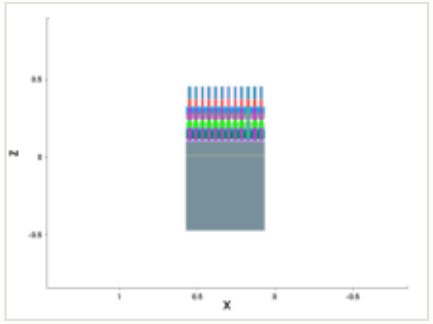
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



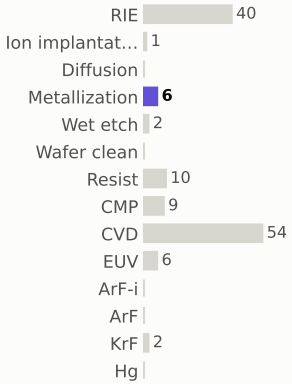
y cut at 0.049 µm



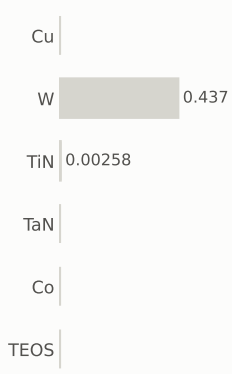
mask for this step

no mask for this step

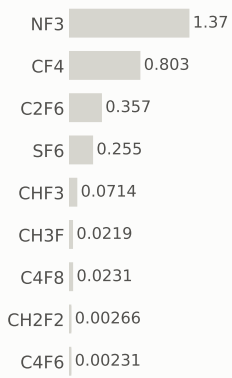
steps by process area



materials deposited (g)



gases emitted (g)



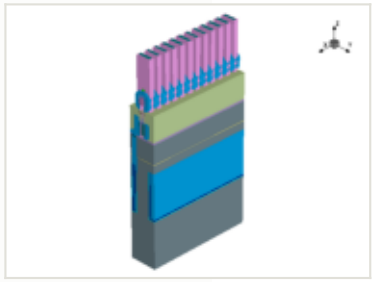
Step #1480: CMP

etch_cmp · CMP

0.169	188.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

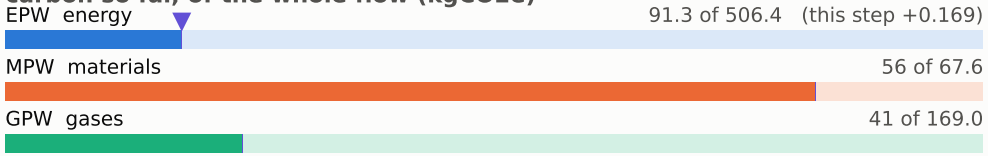


this step

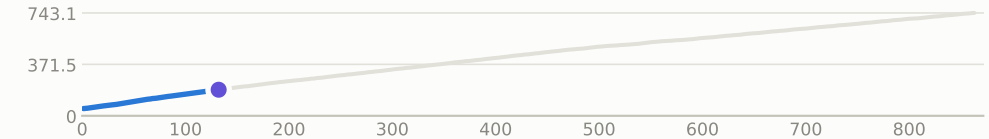
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

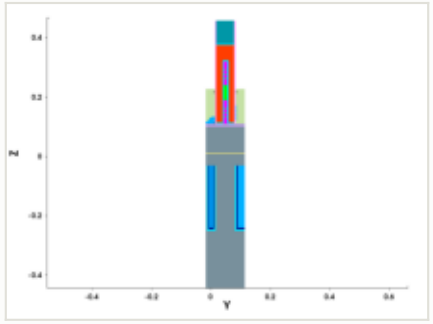
carbon so far, of the whole flow (kgCO2e)



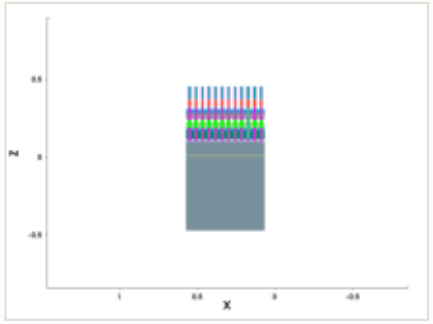
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



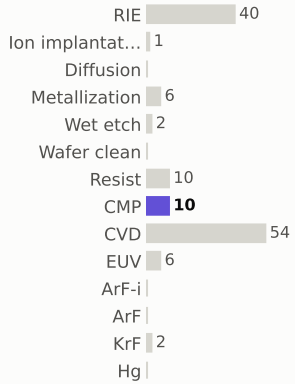
y cut at 0.049 µm



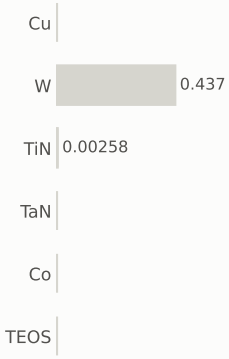
mask for this step

no mask for this step

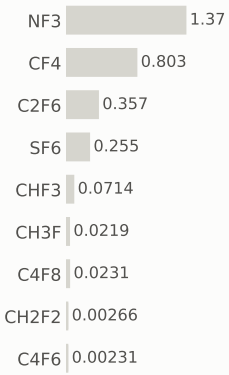
steps by process area



materials deposited (g)



gases emitted (g)



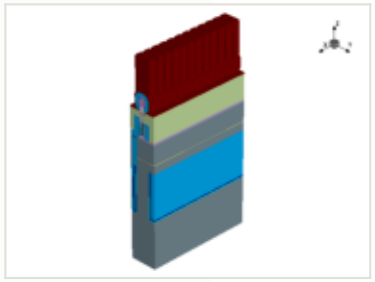
Step #1500: CVD

depo_isotropic · CVD

1.31	189.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

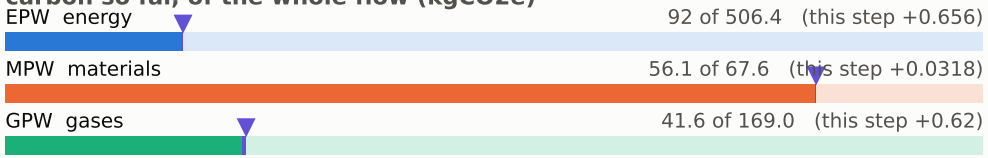


this step

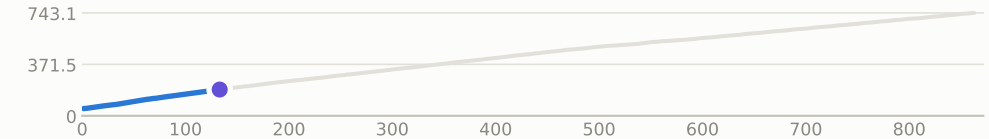
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

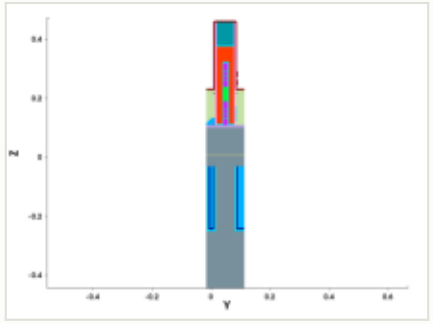
carbon so far, of the whole flow (kgCO2e)



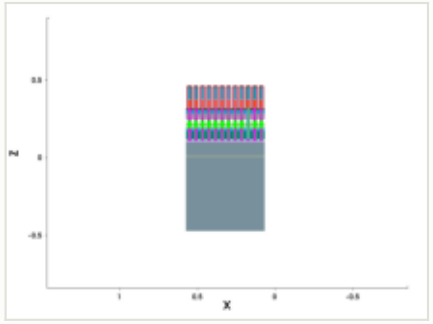
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



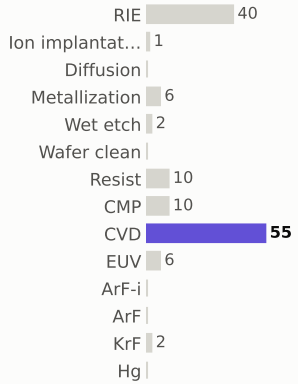
y cut at 0.049 µm



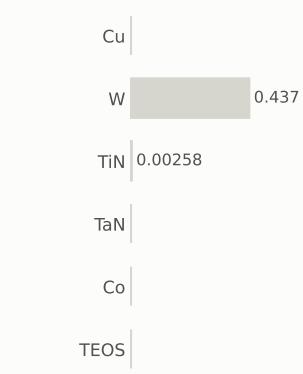
mask for this step

no mask for this step

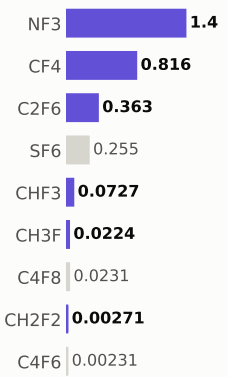
steps by process area



materials deposited (g)



gases emitted (g)



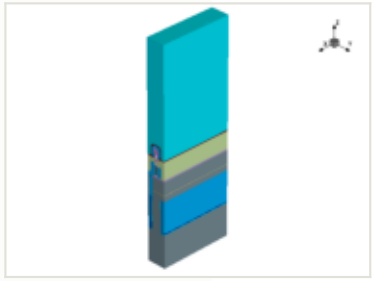
Step #1510: CVD

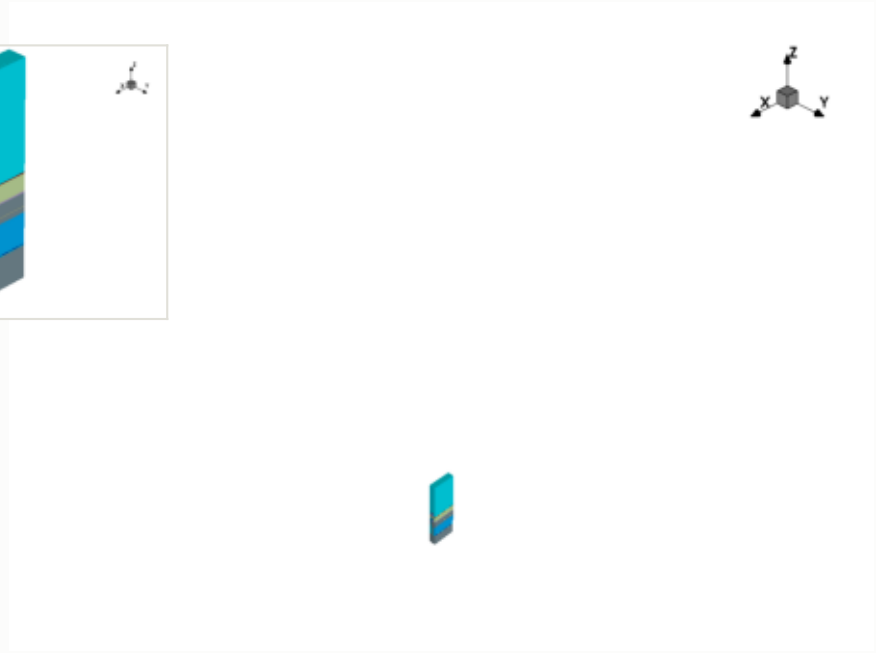
depo_isotropic · CVD

1.31	191.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



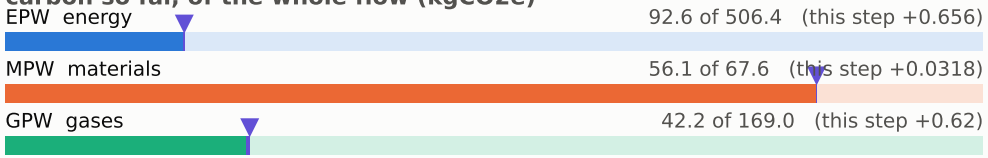


this step

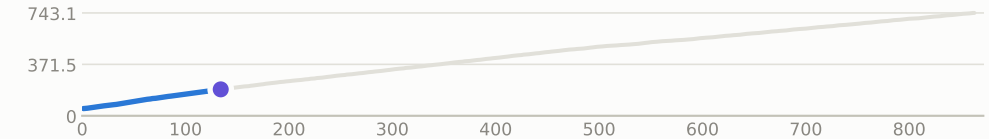
process area	CVD
material	—
thickness	0.5 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

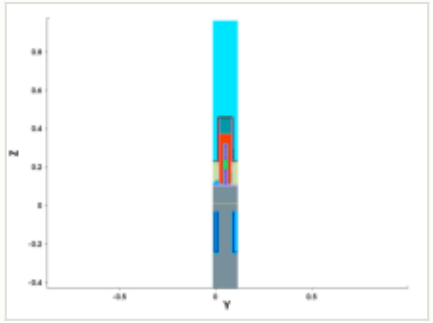
carbon so far, of the whole flow (kgCO2e)



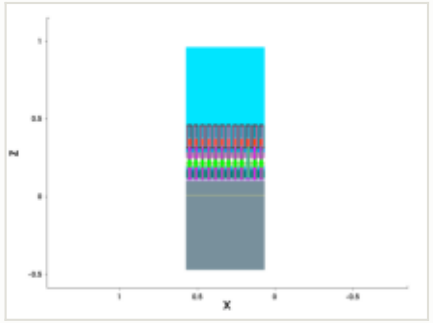
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



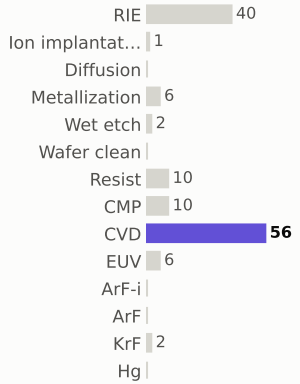
y cut at 0.049 µm



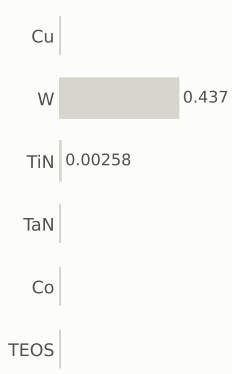
mask for this step

no mask for this step

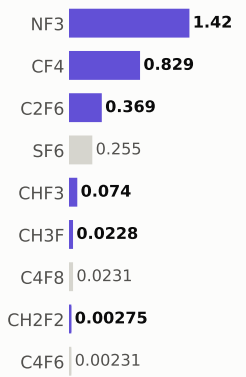
steps by process area



materials deposited (g)



gases emitted (g)



Step #1520: CMP

etch_cmp · CMP

0.169	191.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

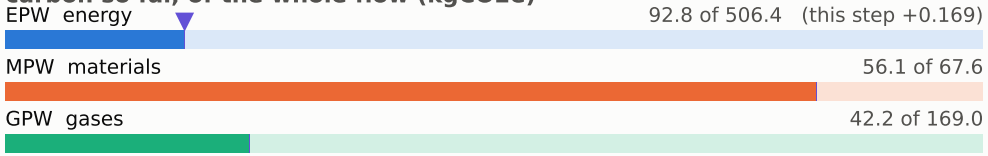
zoomed to fit

this step

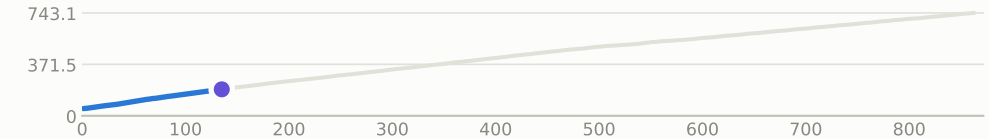
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

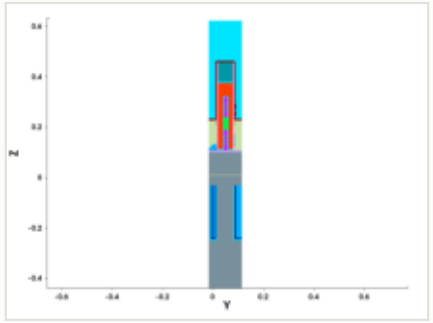
carbon so far, of the whole flow (kgCO2e)



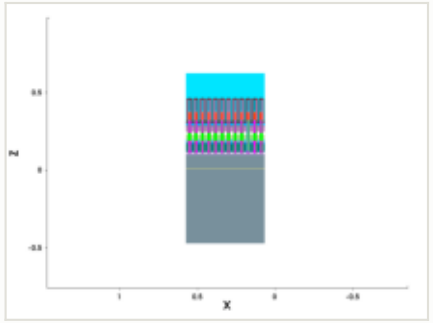
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



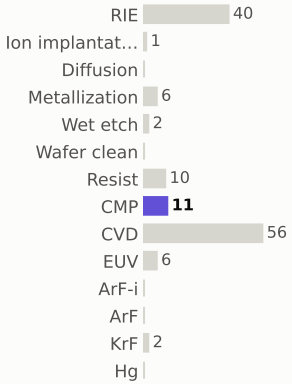
y cut at 0.049 µm



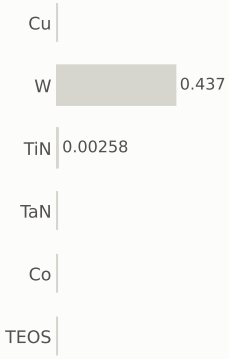
mask for this step

no mask for this step

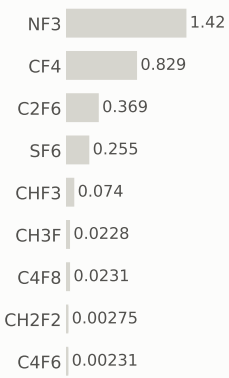
steps by process area



materials deposited (g)



gases emitted (g)



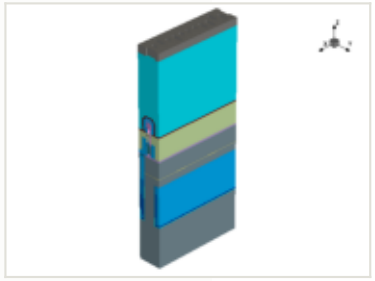
Step #1540: EUV

litho_patterning · EUV

1.96	193.1	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

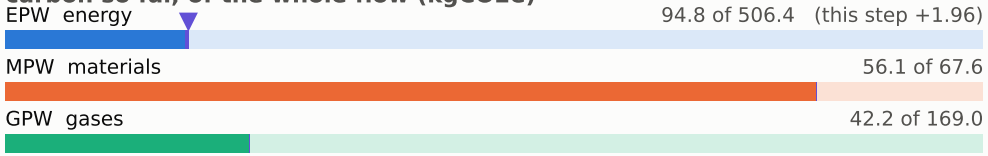


this step

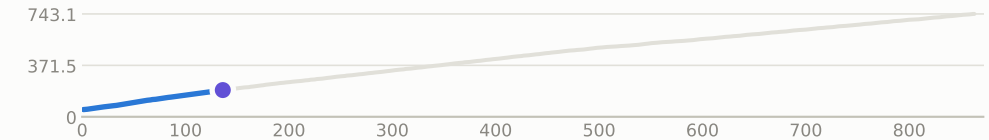
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

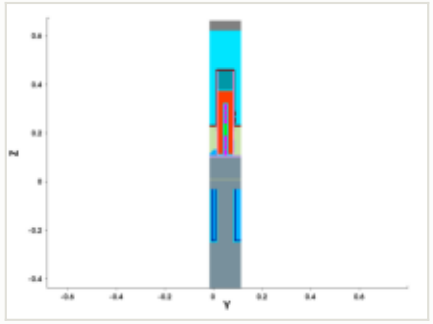
carbon so far, of the whole flow (kgCO2e)



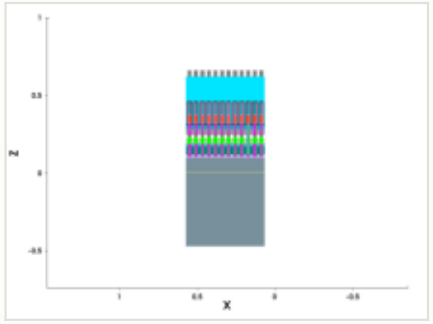
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

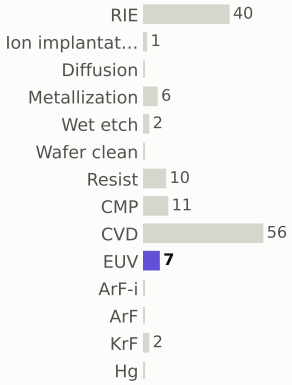


y cut at 0.049 µm

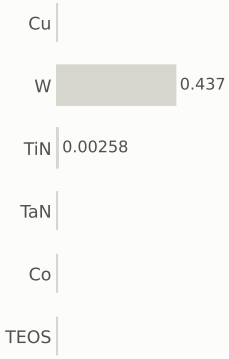


mask: layer 10:2 (GDSII)

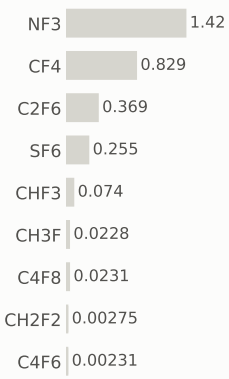
steps by process area



materials deposited (g)



gases emitted (g)



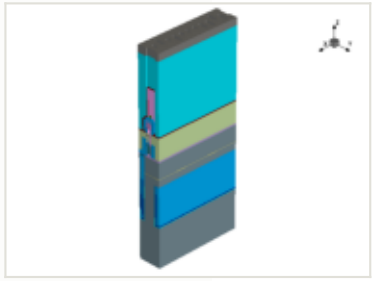
Step #1550: RIE

etch_anisotropic · RIE

1.04	194.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

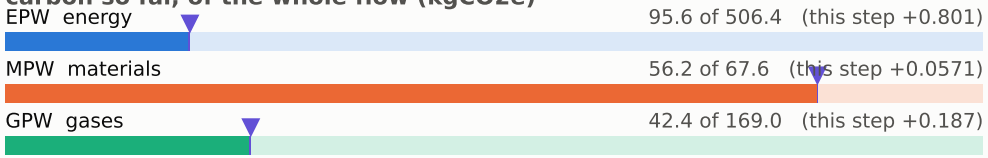


this step

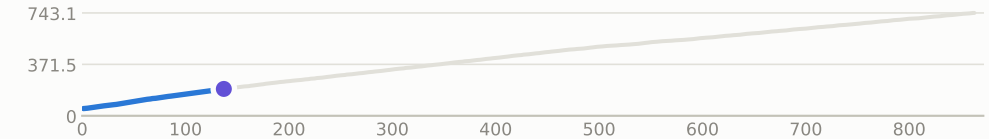
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

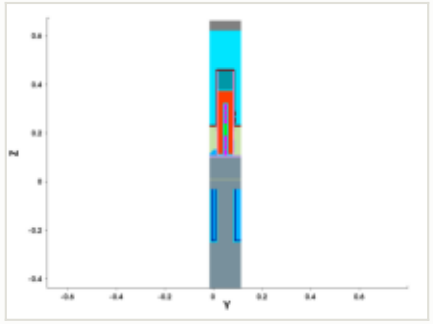
carbon so far, of the whole flow (kgCO2e)



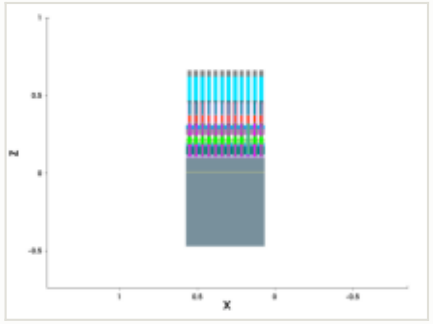
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



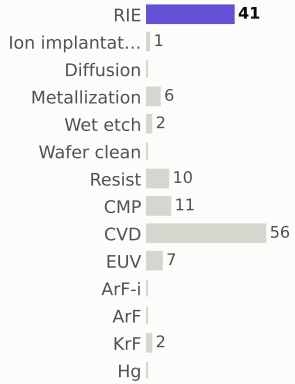
y cut at 0.049 µm



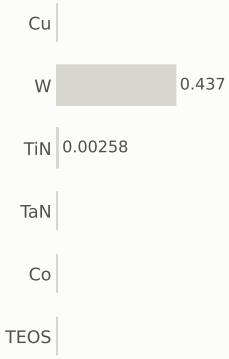
mask for this step

no mask for this step

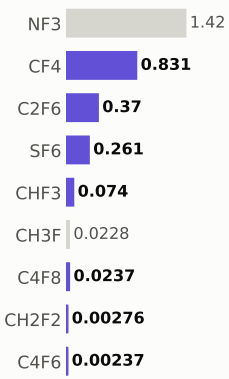
steps by process area



materials deposited (g)



gases emitted (g)



Step #1560: Resist

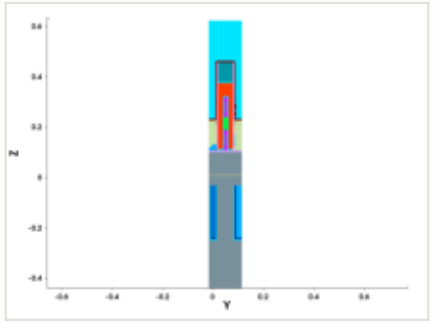
etch_strip · Resist

0.169	194.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

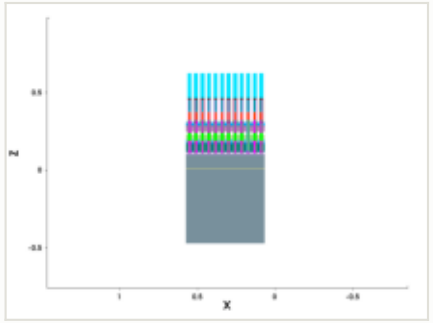
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



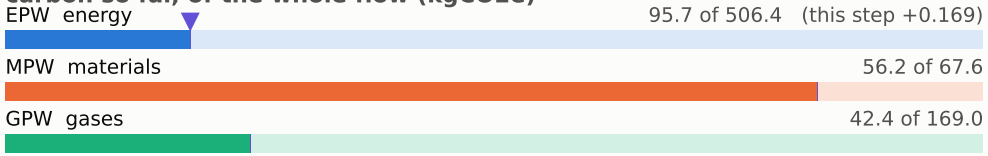
mask for this step

no mask for this step

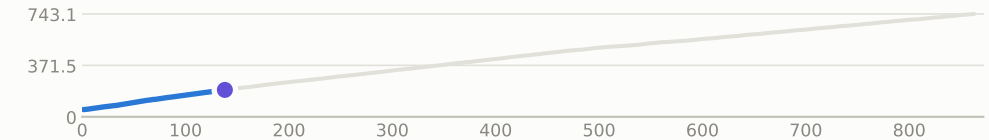
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

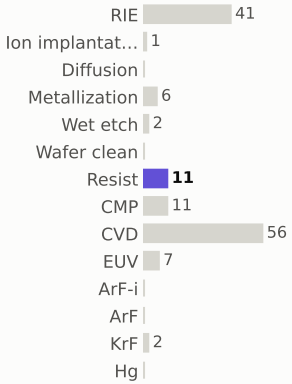
carbon so far, of the whole flow (kgCO2e)



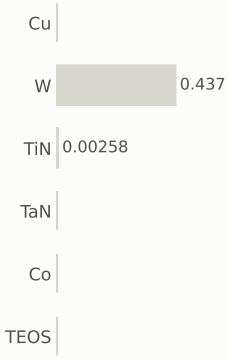
embodied carbon up to each step (kgCO2e per wafer)



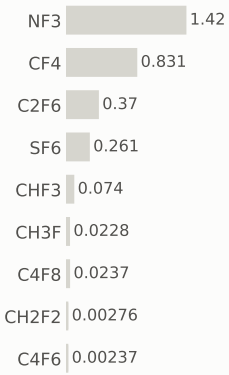
steps by process area



materials deposited (g)



gases emitted (g)



Step #1570: Metallization

depo_isotropic · Metallization

1.21	195.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

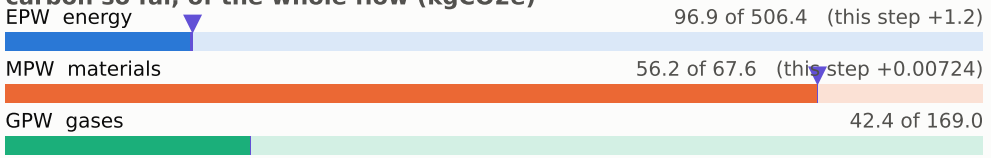
zoomed to fit

this step

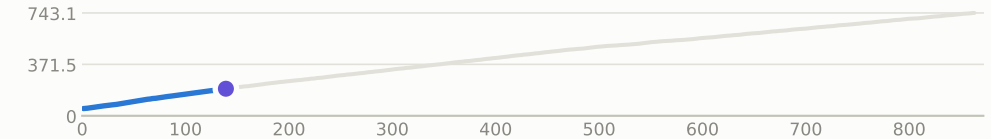
process area	Metallization
material	W
thickness	0.07 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00724 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

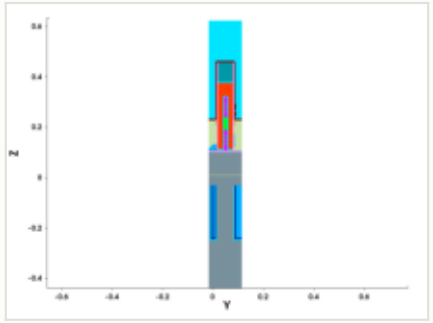
carbon so far, of the whole flow (kgCO2e)



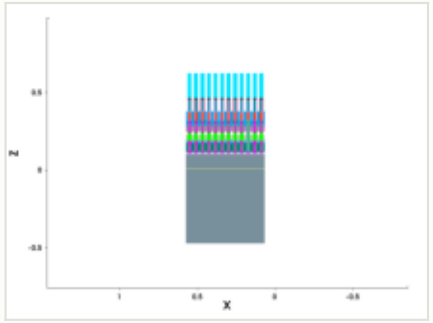
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



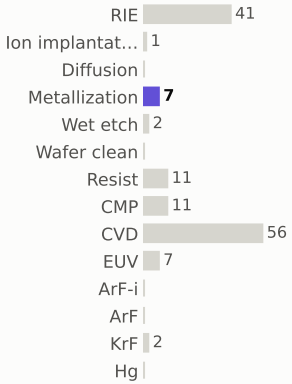
y cut at 0.049 µm



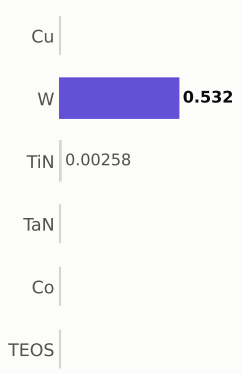
mask for this step

no mask for this step

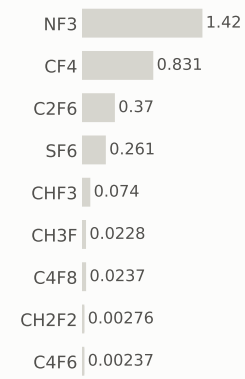
steps by process area



materials deposited (g)



gases emitted (g)



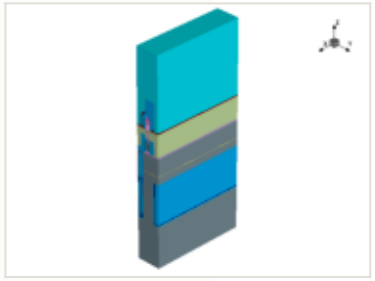
Step #1580: CVD

depo_fill · CVD

1.31	196.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

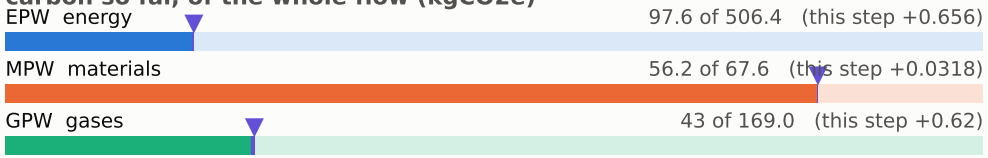


this step

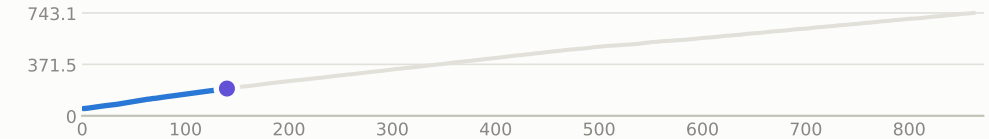
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

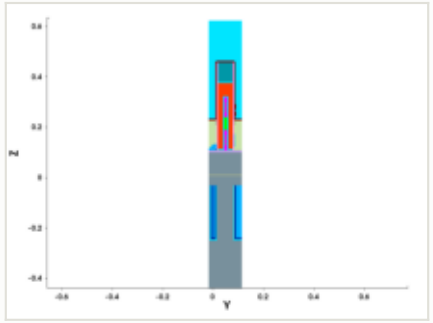
carbon so far, of the whole flow (kgCO2e)



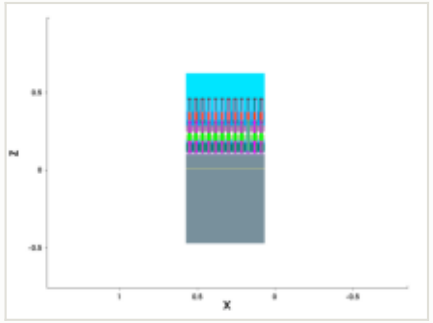
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



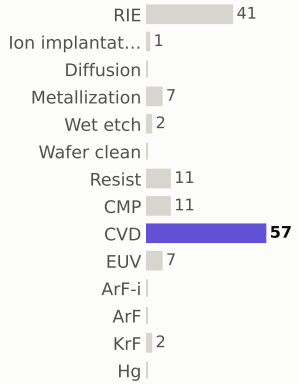
y cut at 0.049 µm



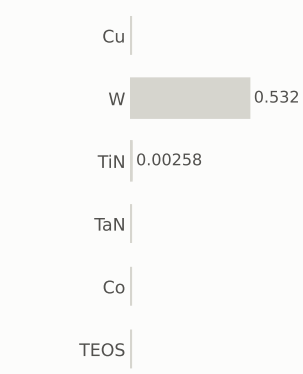
mask for this step

no mask for this step

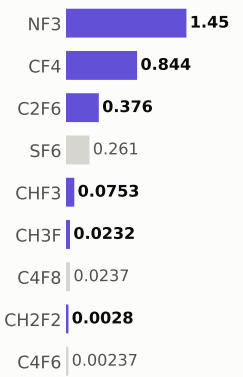
steps by process area



materials deposited (g)



gases emitted (g)



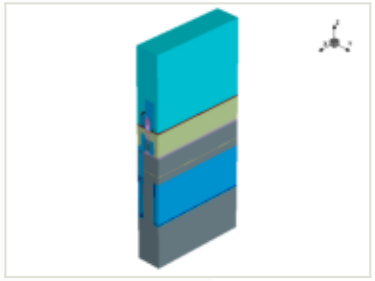
Step #1590: CMP

etch_cmp · CMP

0.169	197.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

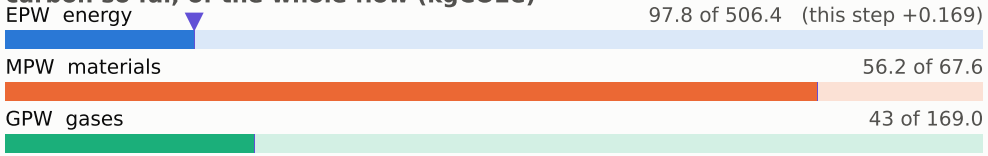


this step

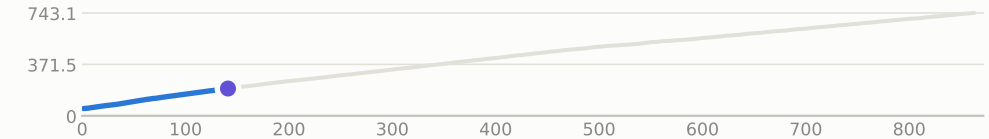
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

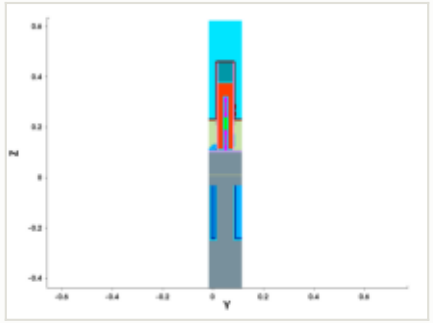
carbon so far, of the whole flow (kgCO2e)



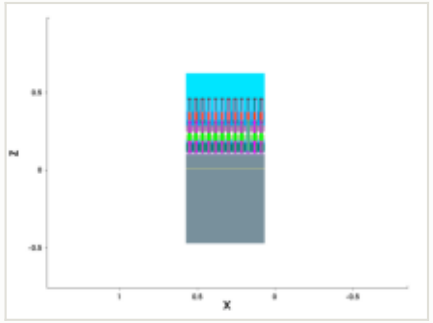
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



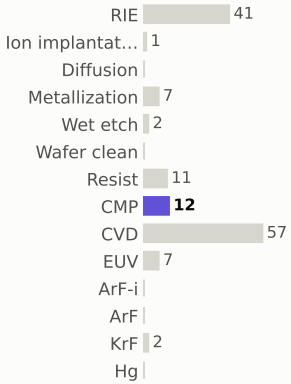
y cut at 0.049 µm



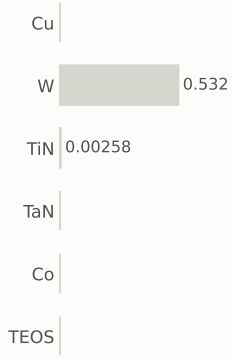
mask for this step

no mask for this step

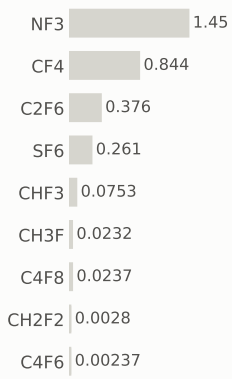
steps by process area



materials deposited (g)



gases emitted (g)



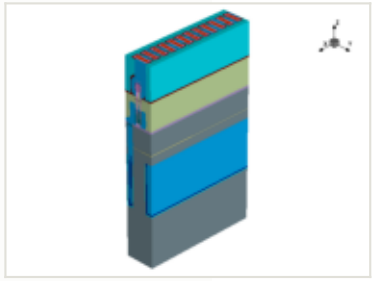
Step #1610: CMP

etch_cmp · CMP

0.169	197.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

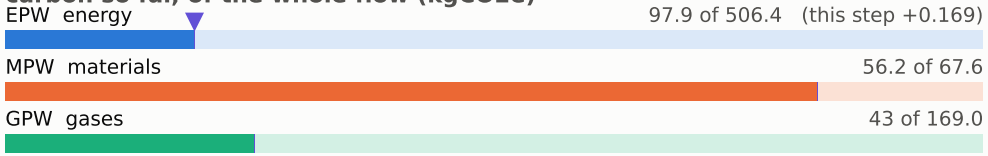


this step

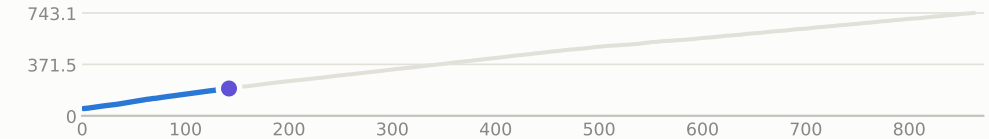
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

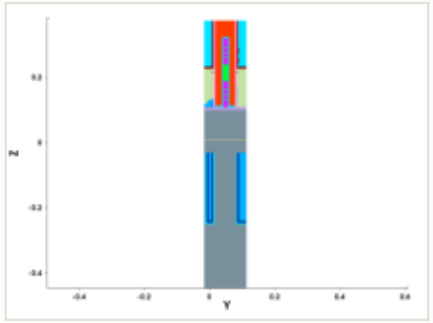
carbon so far, of the whole flow (kgCO2e)



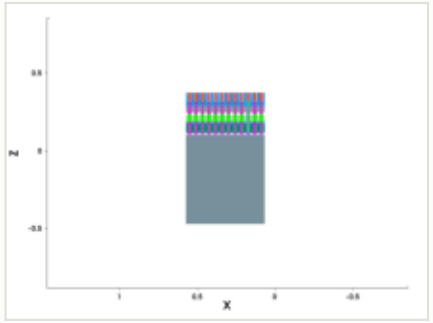
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



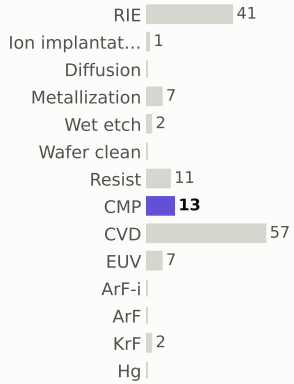
y cut at 0.049 µm



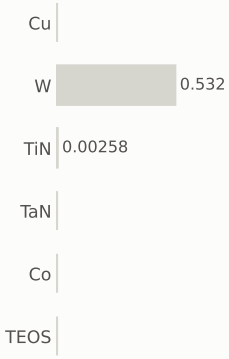
mask for this step

no mask for this step

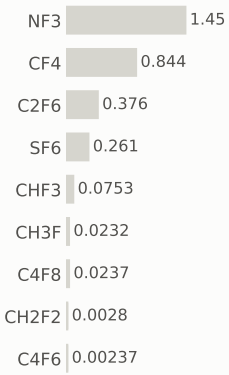
steps by process area



materials deposited (g)



gases emitted (g)



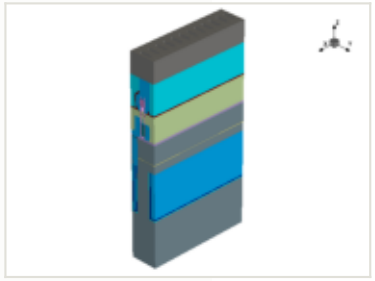
Step #1620: EUV

litho_patterning · EUV

1.96	199.1	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

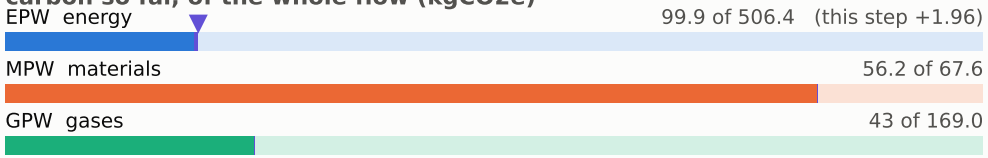


this step

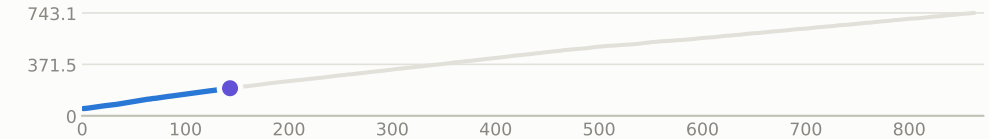
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

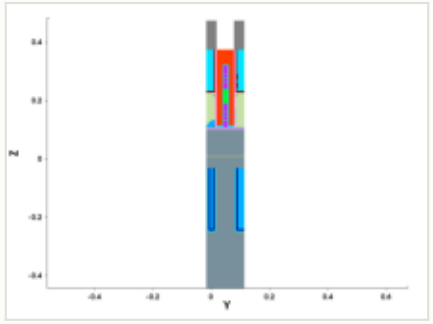
carbon so far, of the whole flow (kgCO2e)



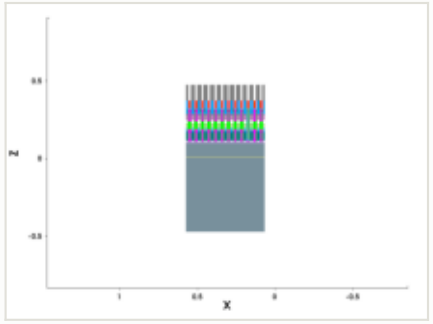
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

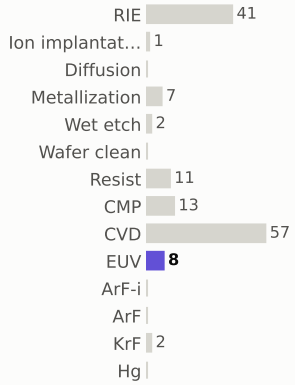


y cut at 0.049 µm

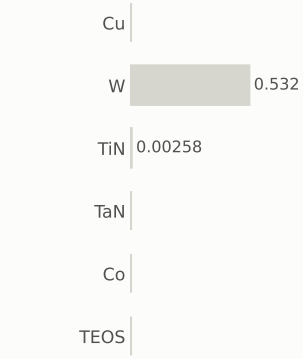


mask: layer 3:0, 3:1 (GDSII)

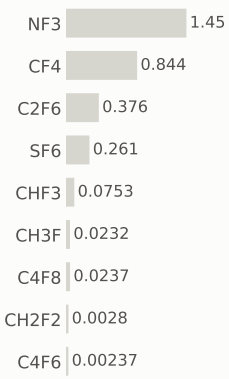
steps by process area



materials deposited (g)



gases emitted (g)



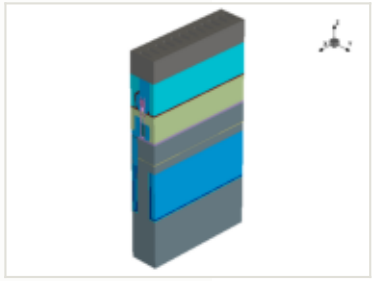
Step #1630: RIE

etch_anisotropic · RIE

1.04	200.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

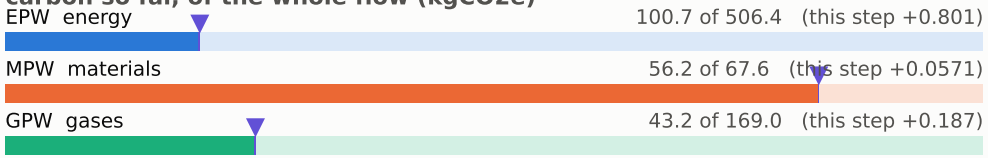


this step

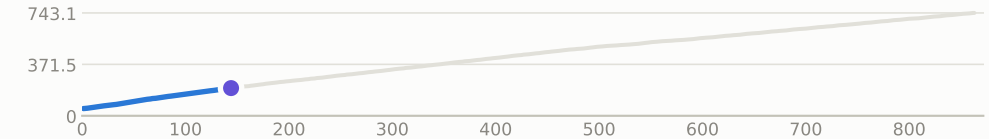
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

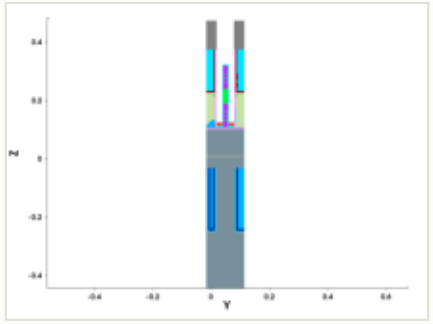
carbon so far, of the whole flow (kgCO2e)



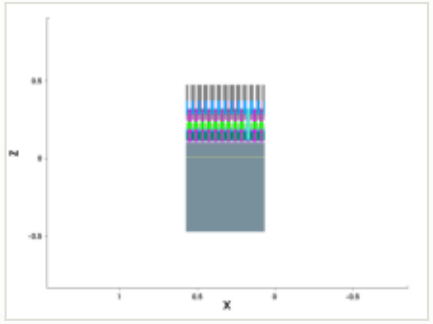
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



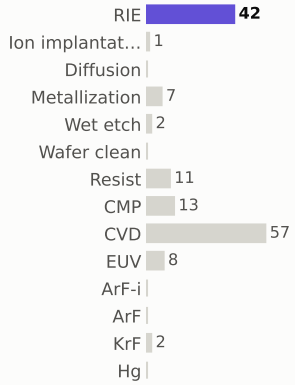
y cut at 0.049 µm



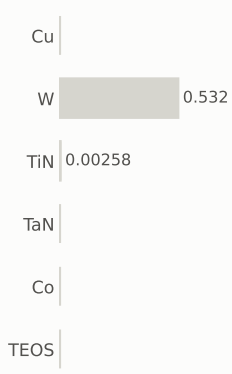
mask for this step

no mask for this step

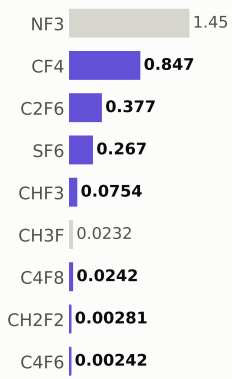
steps by process area



materials deposited (g)



gases emitted (g)



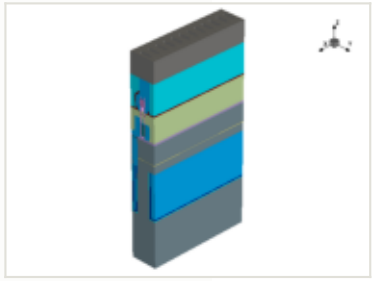
Step #1640: RIE

etch_anisotropic · RIE

1.04	201.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

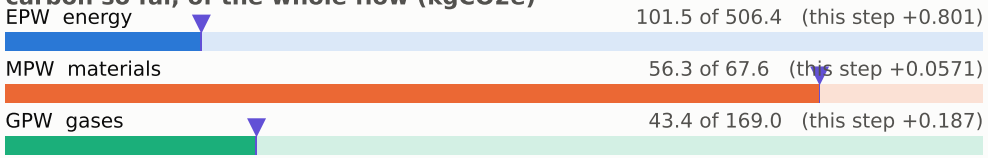


this step

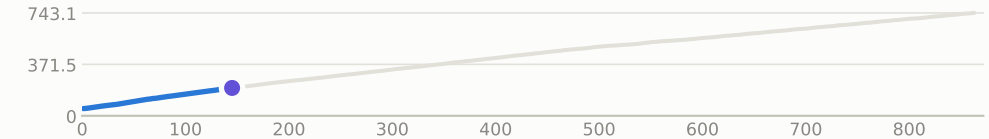
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

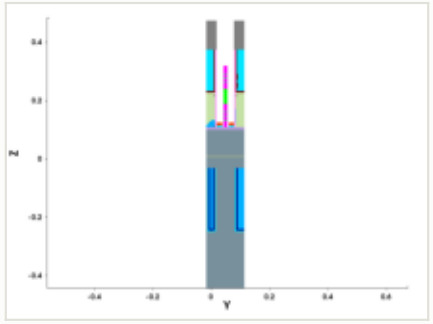
carbon so far, of the whole flow (kgCO2e)



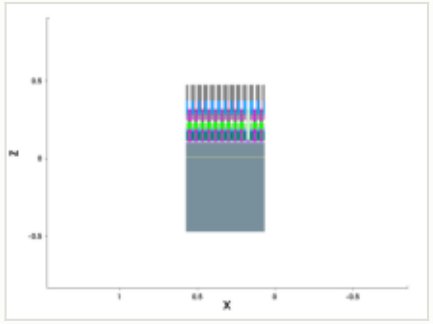
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



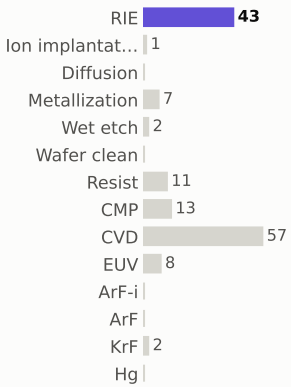
y cut at 0.049 µm



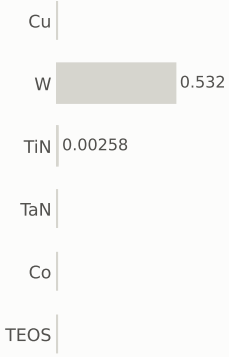
mask for this step

no mask for this step

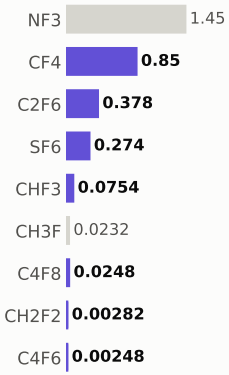
steps by process area



materials deposited (g)



gases emitted (g)



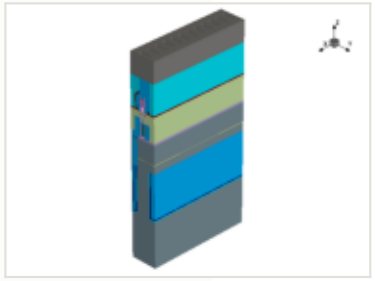
Step #1650: Resist

etch_strip · Resist

0.169	201.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

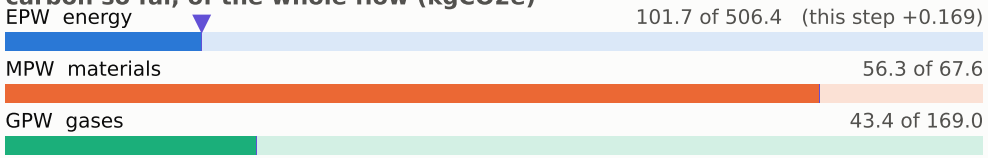


this step

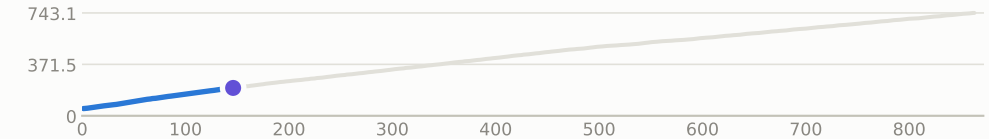
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

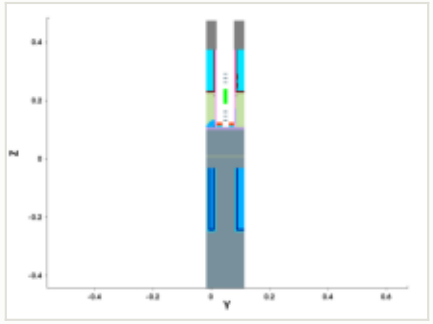
carbon so far, of the whole flow (kgCO2e)



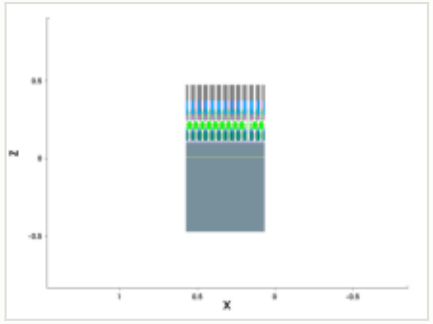
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



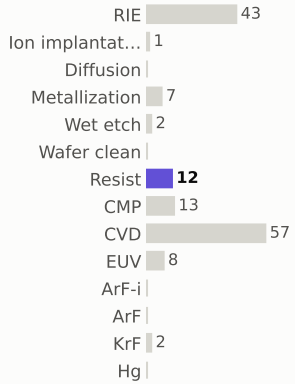
y cut at 0.049 µm



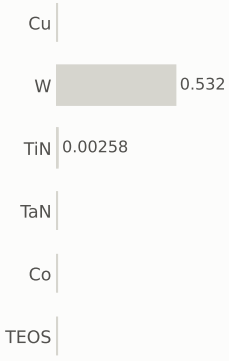
mask for this step

no mask for this step

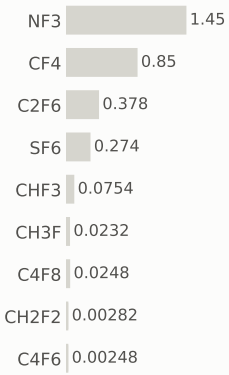
steps by process area



materials deposited (g)



gases emitted (g)



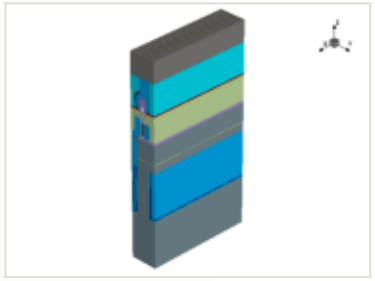
Step #1670: CVD

depo_isotropic · CVD

1.31	202.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

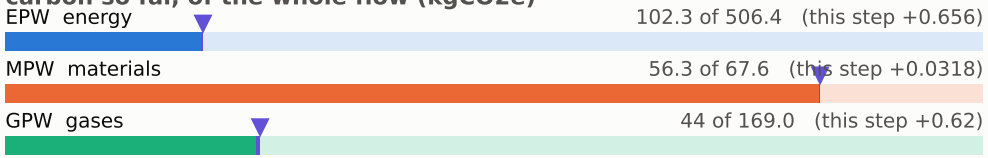
zoomed to fit



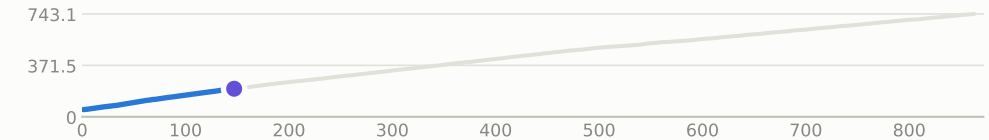
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.0028 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

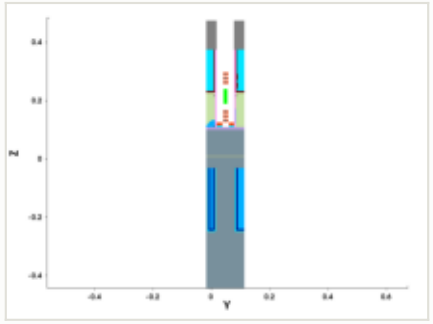
carbon so far, of the whole flow (kgCO2e)



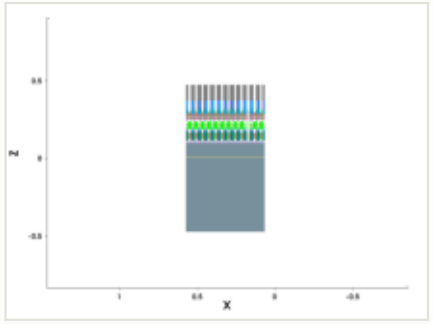
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



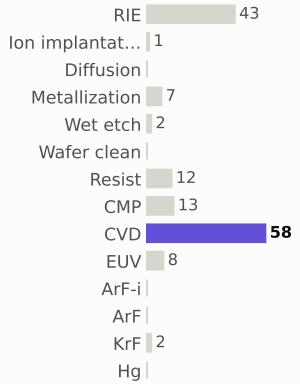
y cut at 0.049 µm



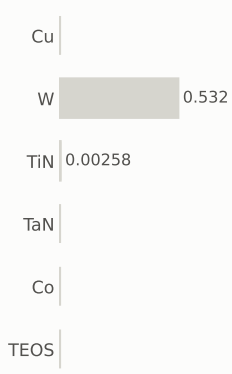
mask for this step

no mask for this step

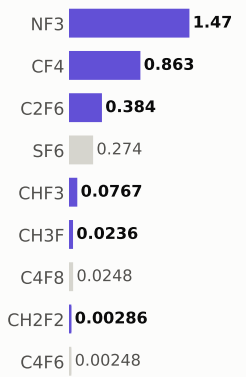
steps by process area



materials deposited (g)



gases emitted (g)



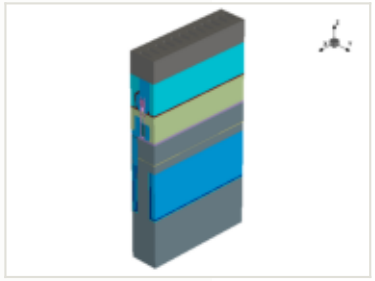
Step #1680: Wet_etch

etch_isotropic · Wet_etch

0.169	202.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

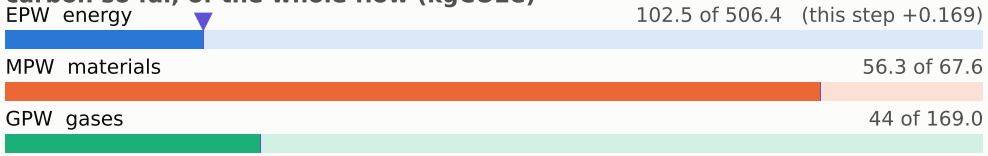


this step

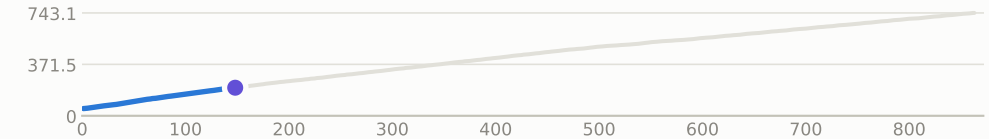
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

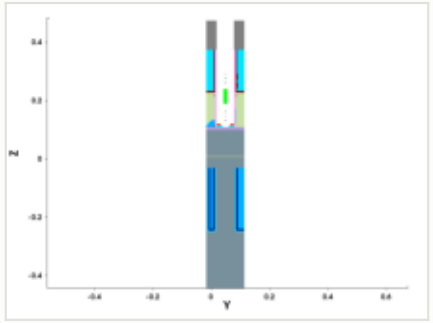
carbon so far, of the whole flow (kgCO2e)



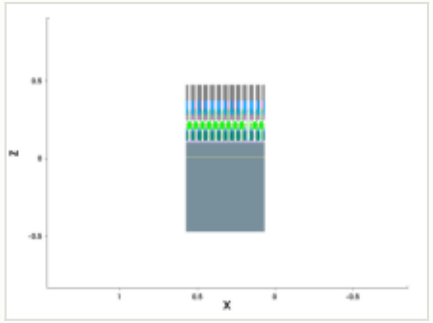
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



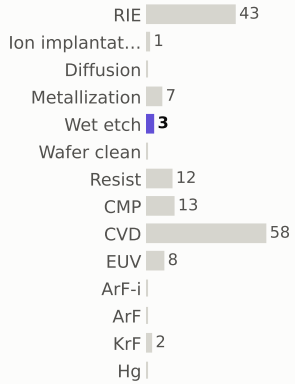
y cut at 0.049 µm



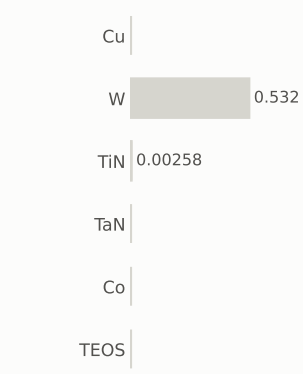
mask for this step

no mask for this step

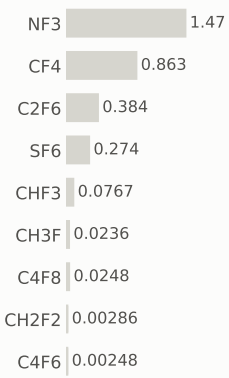
steps by process area



materials deposited (g)



gases emitted (g)



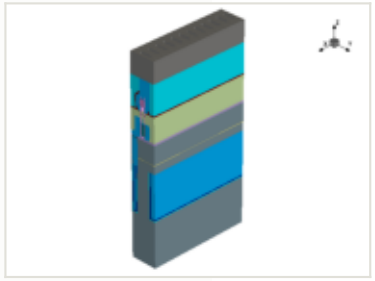
Step #1690: CVD

depo_isotropic · CVD

1.31	204.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

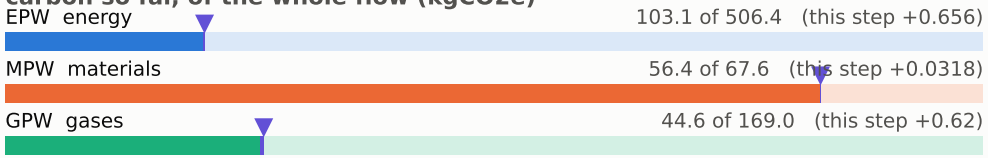
zoomed to fit



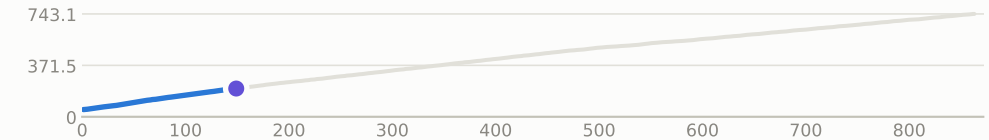
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.0015 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

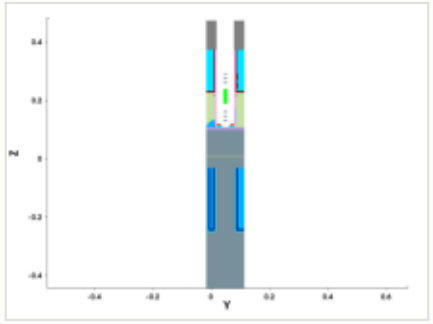
carbon so far, of the whole flow (kgCO2e)



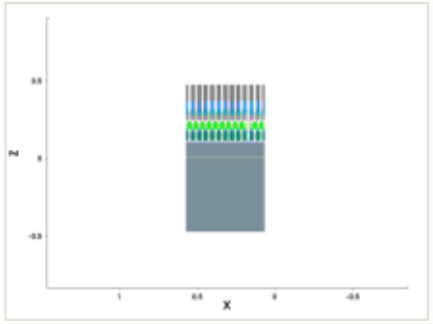
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



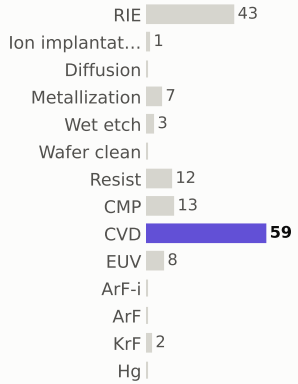
y cut at 0.049 µm



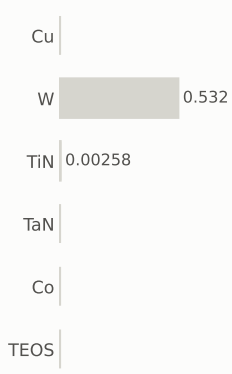
mask for this step

no mask for this step

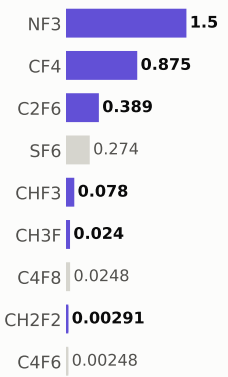
steps by process area



materials deposited (g)



gases emitted (g)



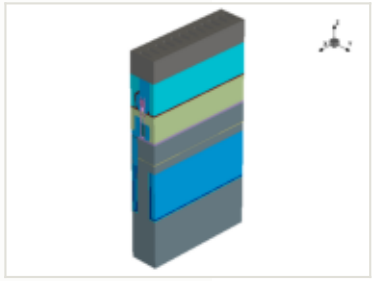
Step #1700: CVD

depo_isotropic · CVD

1.31	205.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

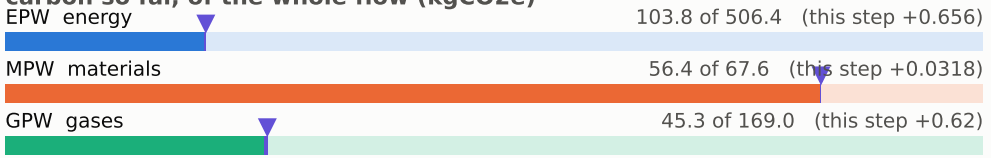


this step

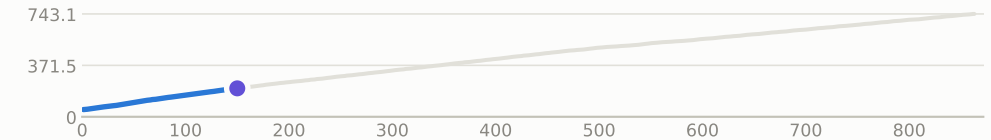
process area	CVD
material	—
thickness	0.00025 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

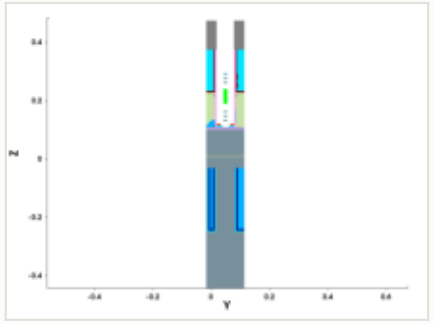
carbon so far, of the whole flow (kgCO2e)



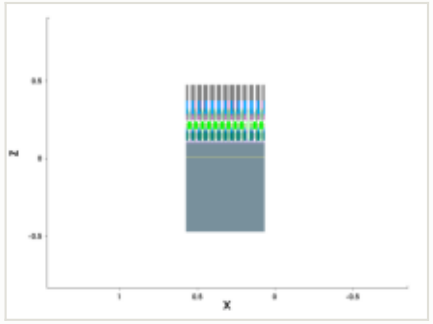
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



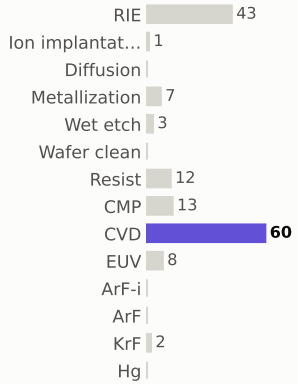
y cut at 0.049 µm



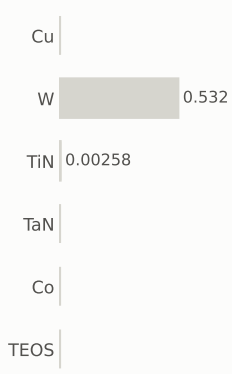
mask for this step

no mask for this step

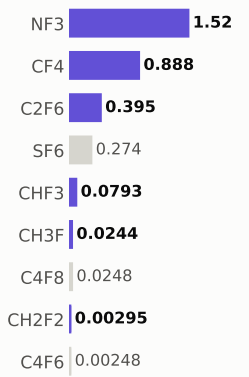
steps by process area



materials deposited (g)



gases emitted (g)



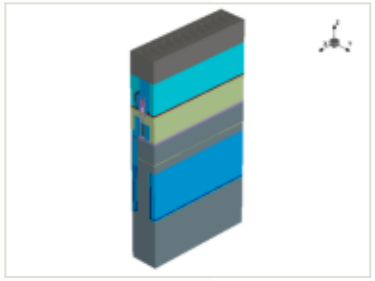
Step #1710: CVD

depo_isotropic · CVD

1.31	206.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

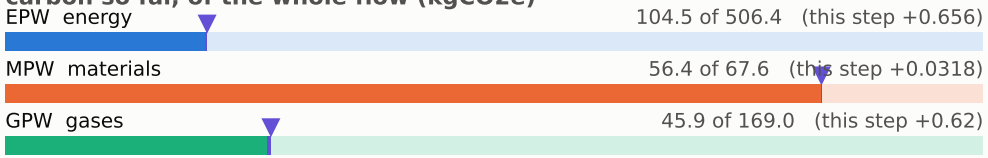


this step

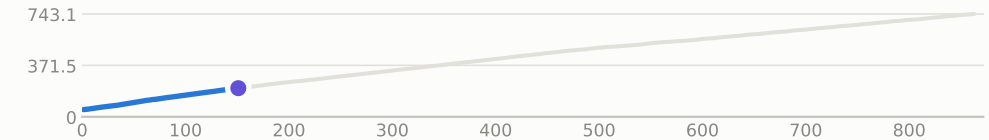
process area	CVD
material	—
thickness	0.00025 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

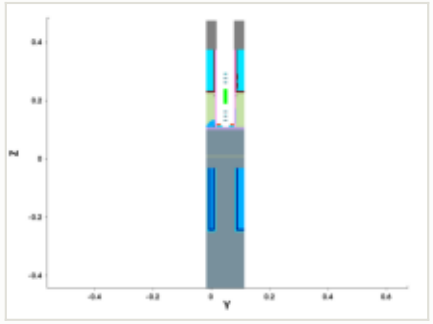
carbon so far, of the whole flow (kgCO2e)



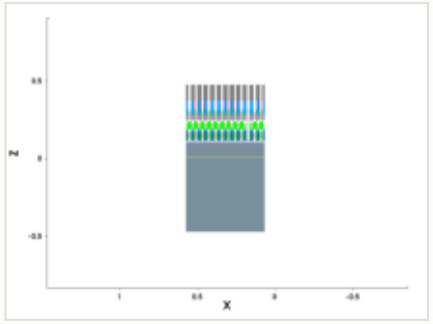
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



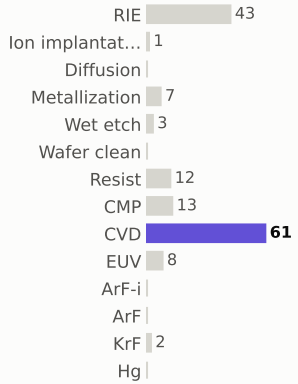
y cut at 0.049 µm



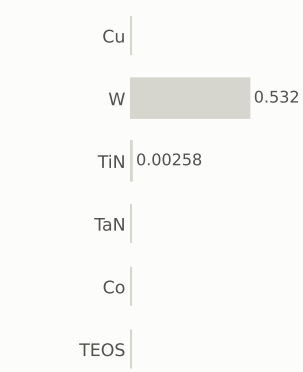
mask for this step

no mask for this step

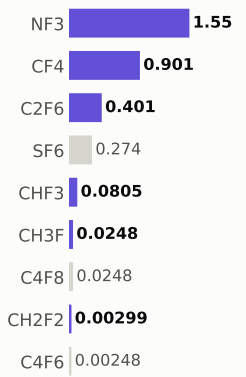
steps by process area



materials deposited (g)



gases emitted (g)



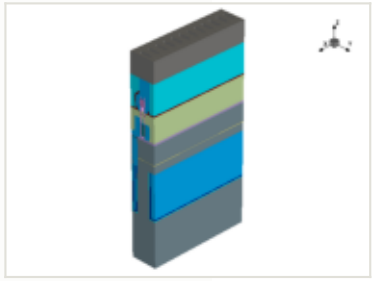
Step #1720: CVD

depo_isotropic · CVD

1.31	208.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

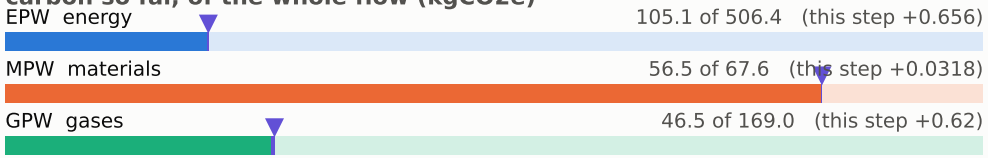


this step

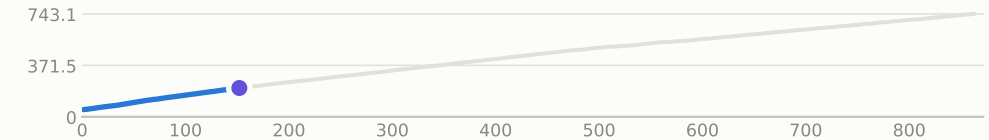
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

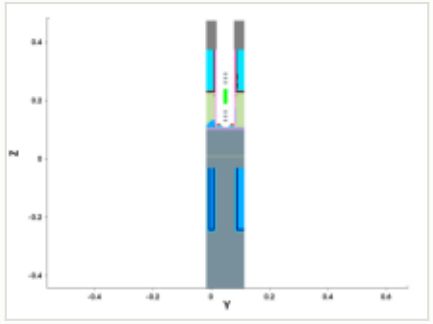
carbon so far, of the whole flow (kgCO2e)



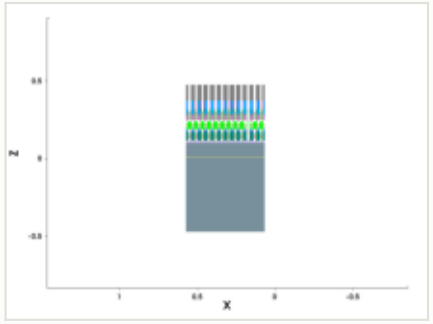
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



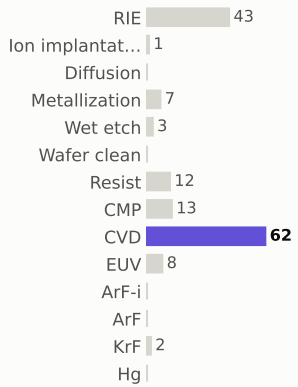
y cut at 0.049 µm



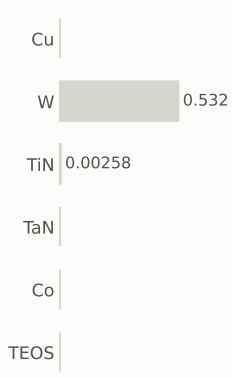
mask for this step

no mask for this step

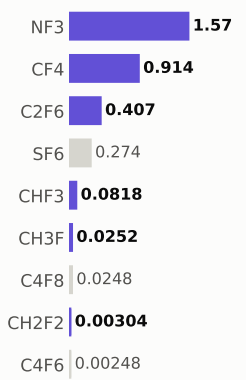
steps by process area



materials deposited (g)



gases emitted (g)



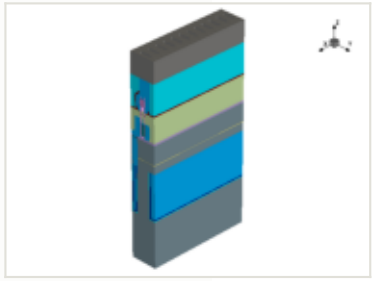
Step #1730: CVD

depo_isotropic · CVD

1.31	209.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

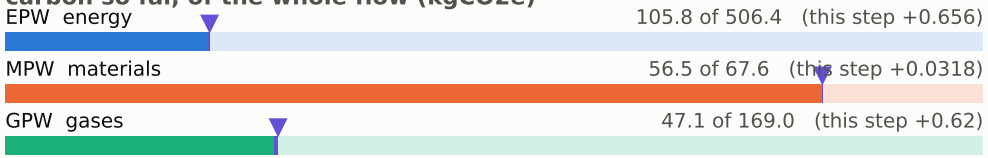


this step

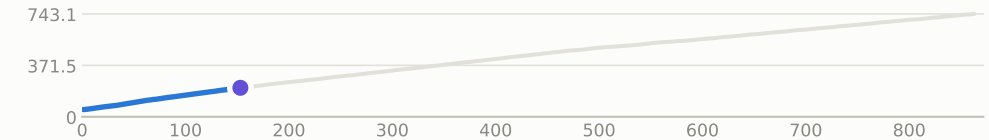
process area	CVD
material	—
thickness	0.0005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

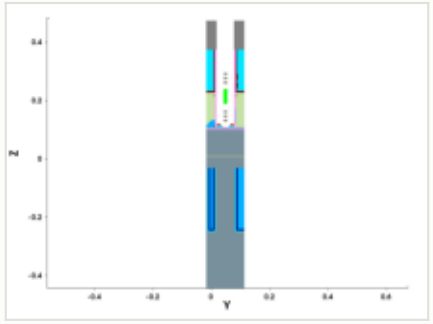
carbon so far, of the whole flow (kgCO2e)



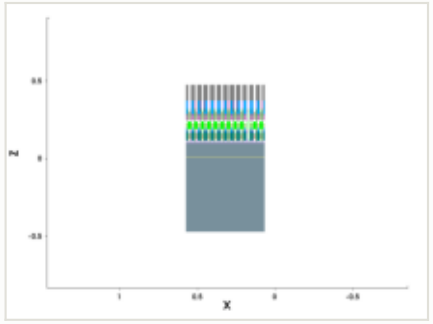
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



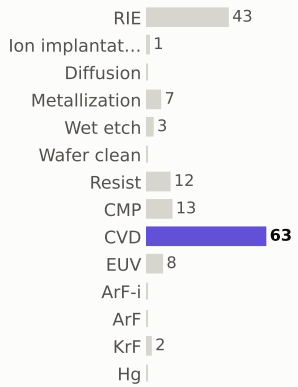
y cut at 0.049 µm



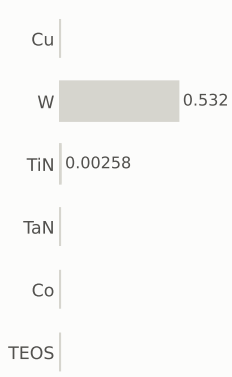
mask for this step

no mask for this step

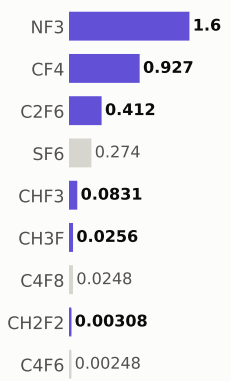
steps by process area



materials deposited (g)



gases emitted (g)



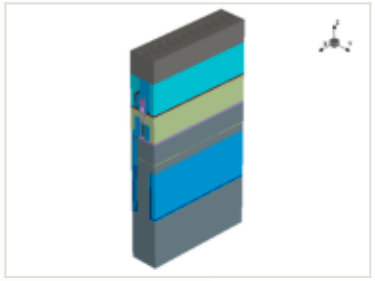
Step #1740: CVD

depo_fill · CVD

1.31	210.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

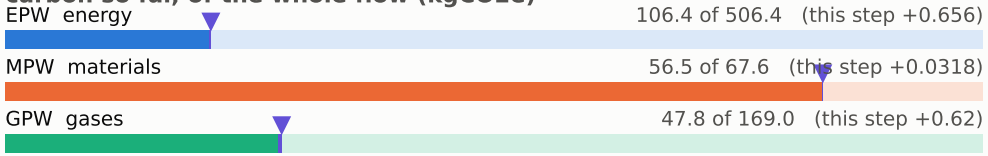


this step

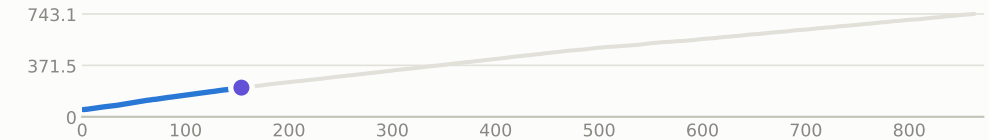
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

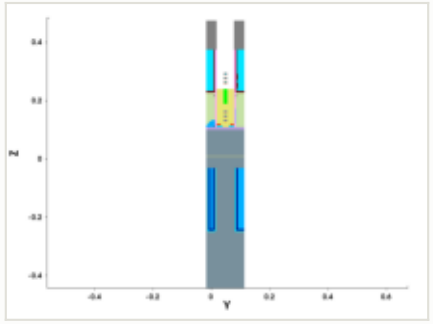
carbon so far, of the whole flow (kgCO2e)



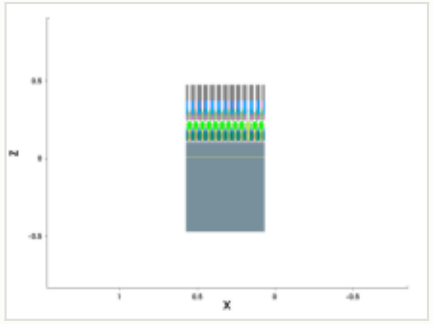
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



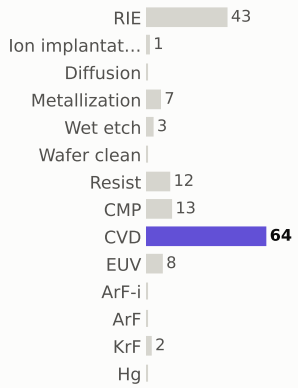
y cut at 0.049 µm



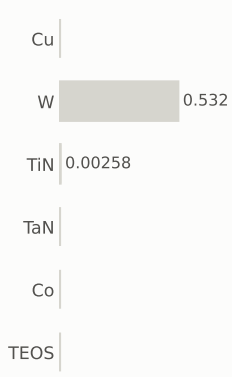
mask for this step

no mask for this step

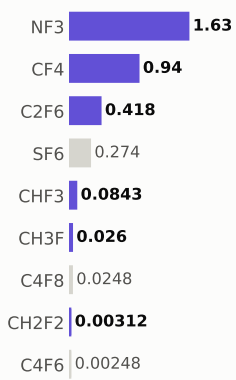
steps by process area



materials deposited (g)



gases emitted (g)



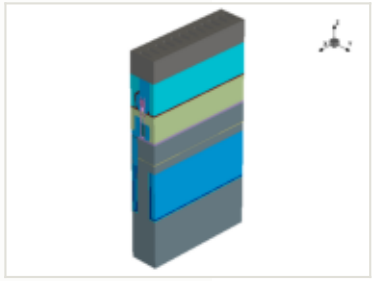
Step #1750: Wet_etch

etch_isotropic · Wet_etch

0.169	210.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

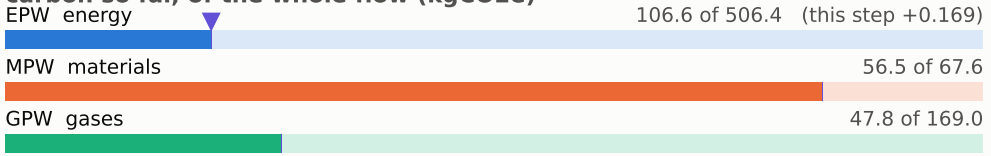


this step

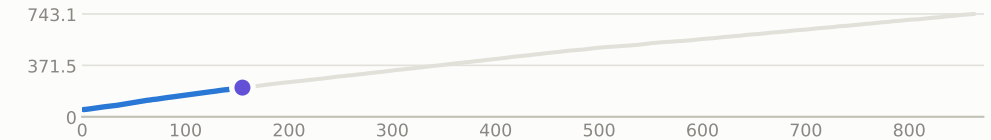
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

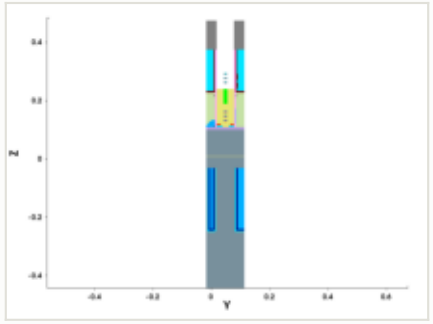
carbon so far, of the whole flow (kgCO2e)



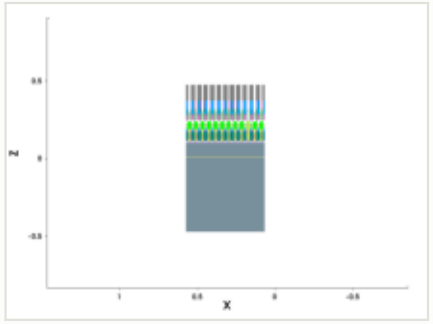
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



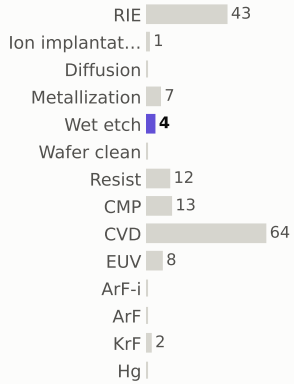
y cut at 0.049 µm



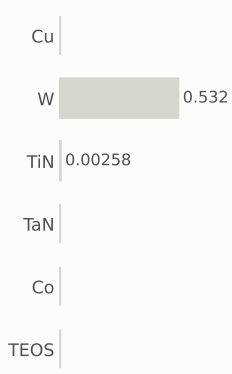
mask for this step

no mask for this step

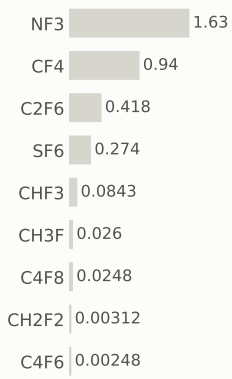
steps by process area



materials deposited (g)



gases emitted (g)



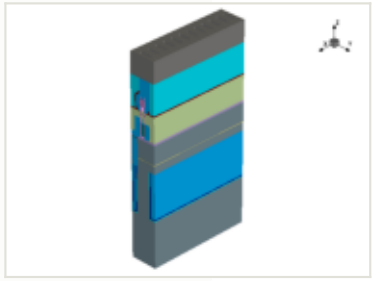
Step #1760: Resist

etch_strip · Resist

0.169	211.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

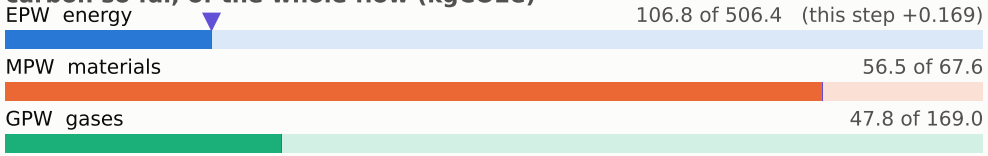


this step

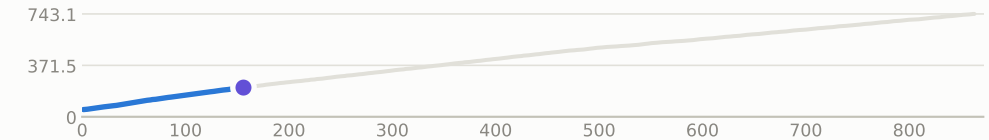
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

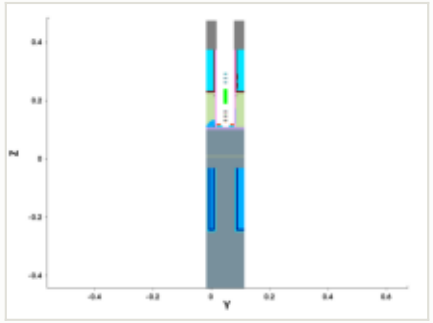
carbon so far, of the whole flow (kgCO2e)



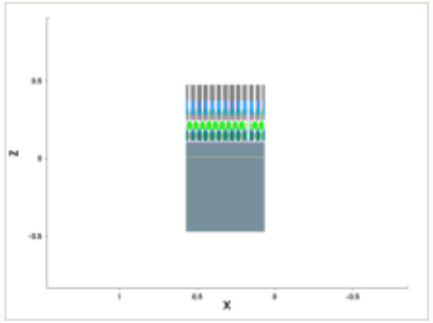
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



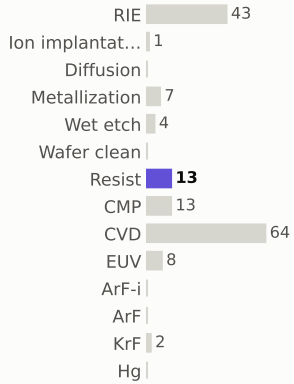
y cut at 0.049 µm



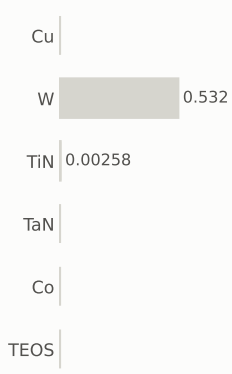
mask for this step

no mask for this step

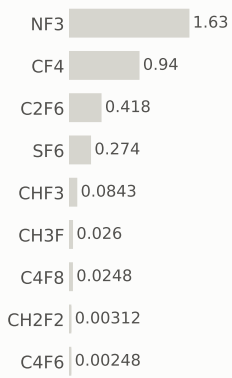
steps by process area



materials deposited (g)



gases emitted (g)



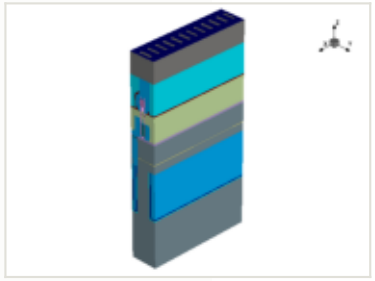
Step #1770: CVD

depo_isotropic · CVD

1.31	212.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

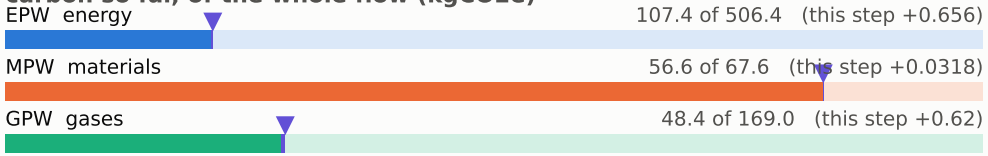
zoomed to fit



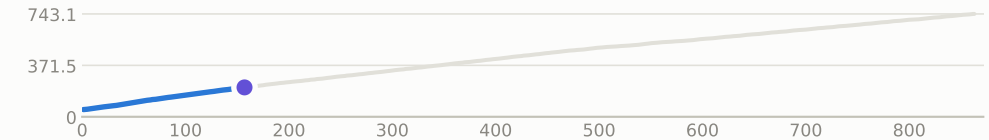
this step

process area	CVD	EPW	0.656 kgCO2e
material	TiN	EPW tool	0.469 kgCO2e
thickness	0.0005 µm	EPW overhead	0.187 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

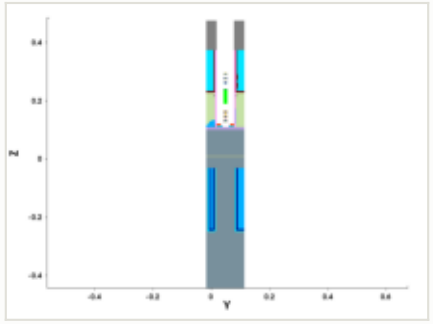
carbon so far, of the whole flow (kgCO2e)



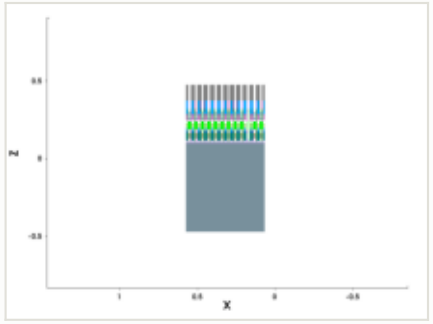
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



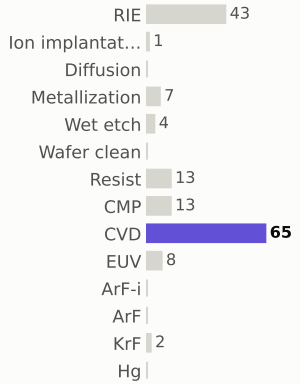
y cut at 0.049 µm



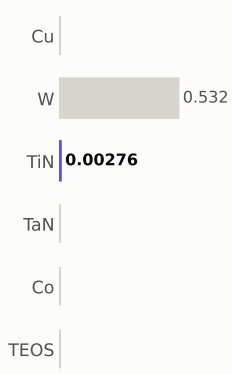
mask for this step

no mask for this step

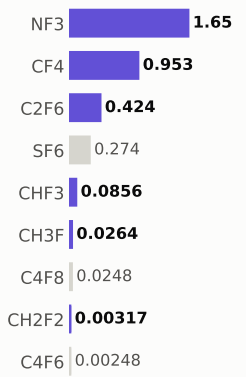
steps by process area



materials deposited (g)



gases emitted (g)



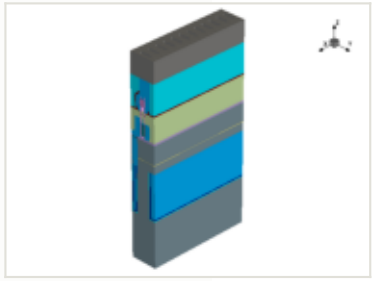
Step #1780: CMP

etch_cmp · CMP

0.169	212.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

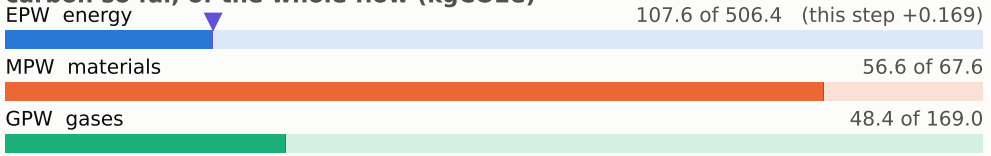


this step

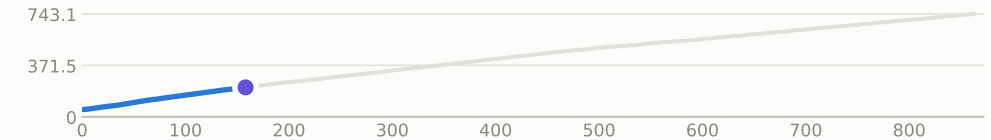
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

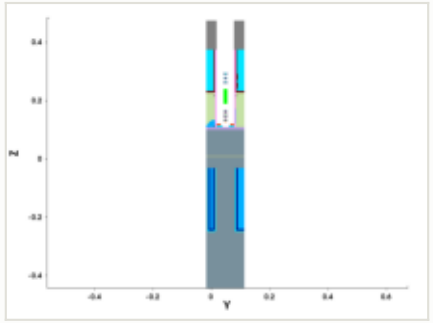
carbon so far, of the whole flow (kgCO2e)



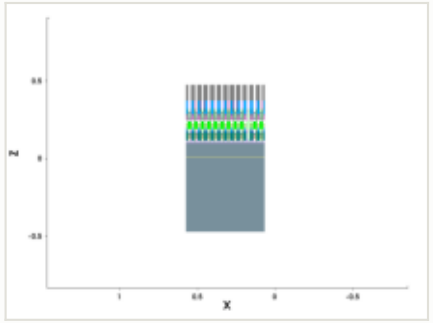
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



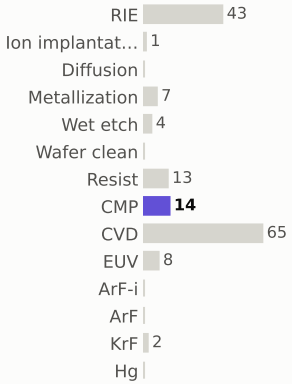
y cut at 0.049 µm



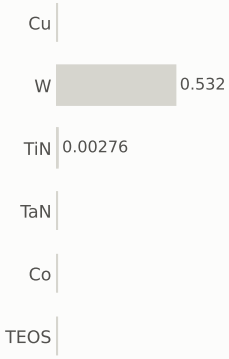
mask for this step

no mask for this step

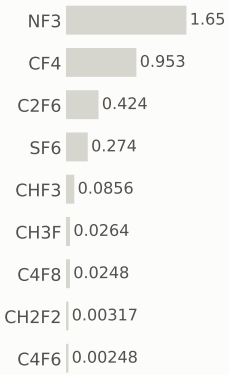
steps by process area



materials deposited (g)



gases emitted (g)



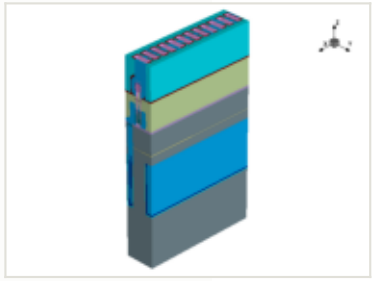
Step #1790: Resist

etch_strip · Resist

0.169	212.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

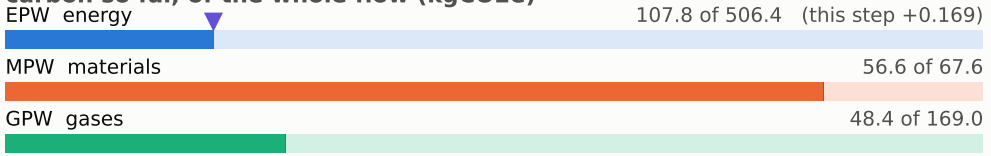


this step

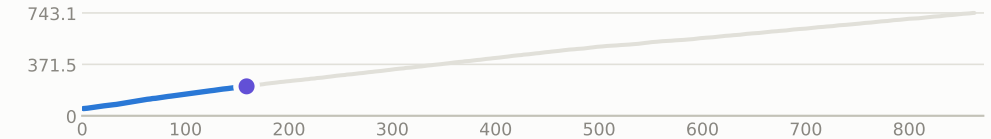
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

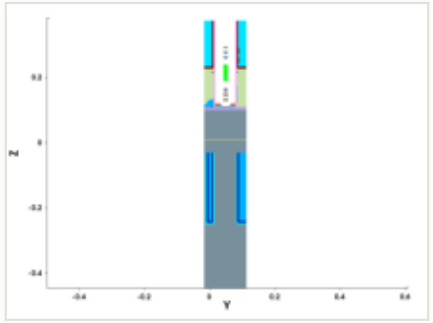
carbon so far, of the whole flow (kgCO2e)



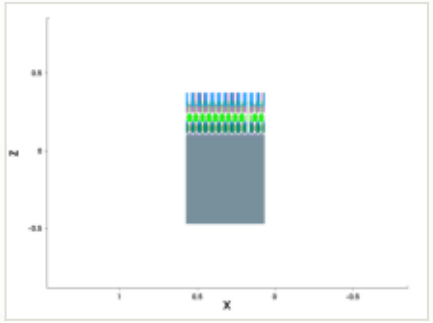
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



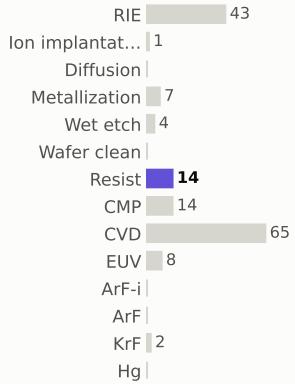
y cut at 0.049 µm



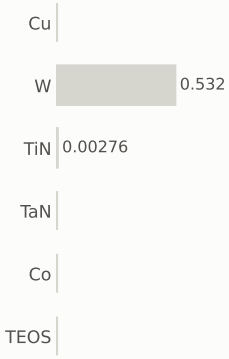
mask for this step

no mask for this step

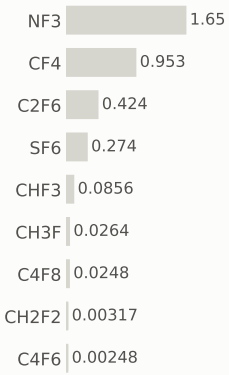
steps by process area



materials deposited (g)



gases emitted (g)



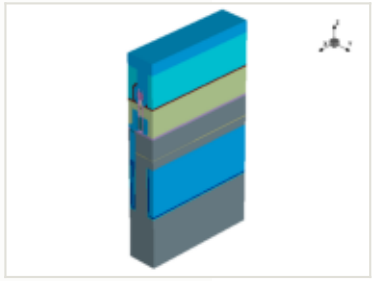
Step #1800: Metallization

depo_fill · Metallization

1.21	213.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

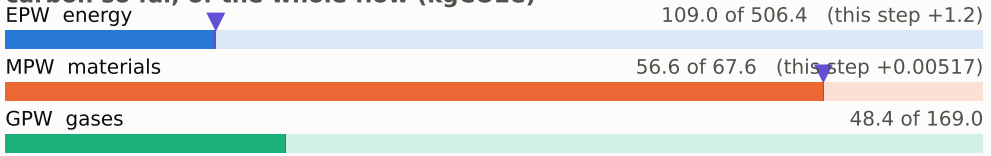


this step

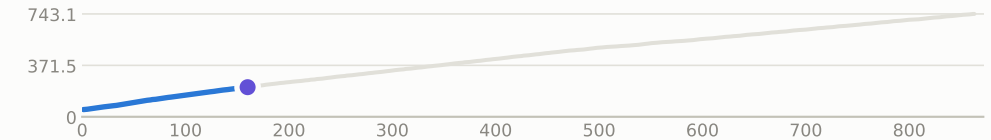
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

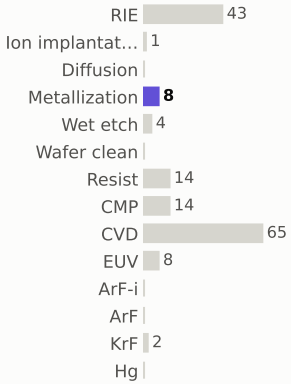
carbon so far, of the whole flow (kgCO2e)



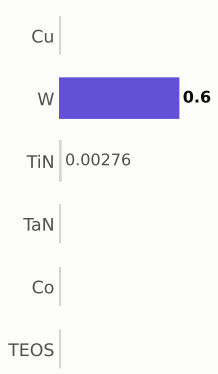
embodied carbon up to each step (kgCO2e per wafer)



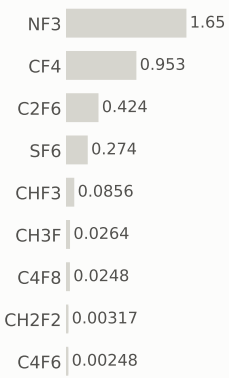
steps by process area



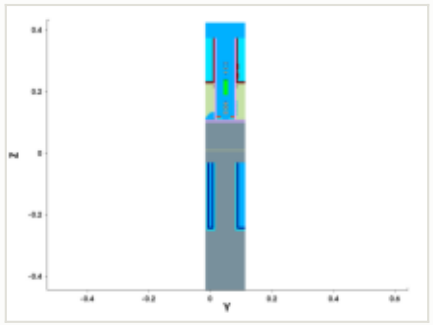
materials deposited (g)



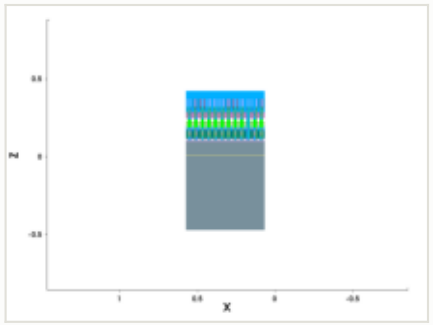
gases emitted (g)



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

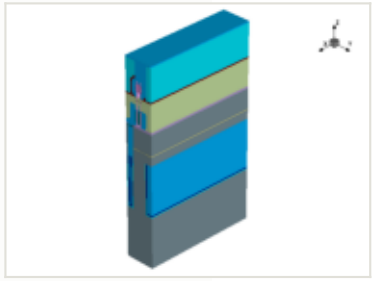
Step #1810: CMP

etch_cmp · CMP

0.169	214.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

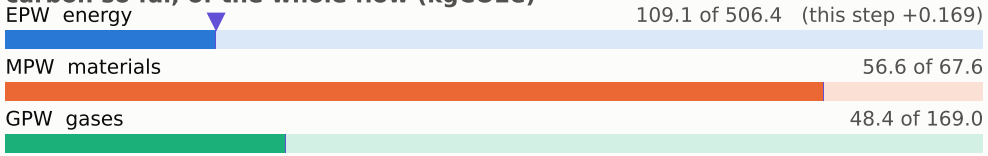


this step

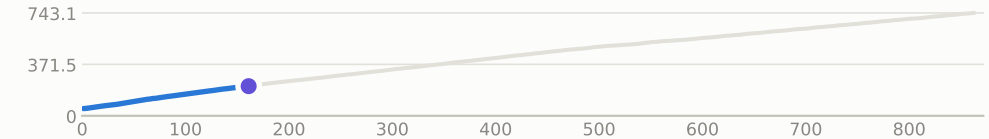
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

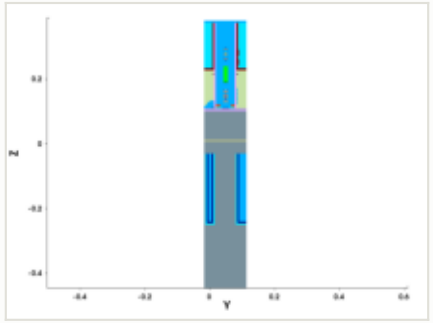
carbon so far, of the whole flow (kgCO2e)



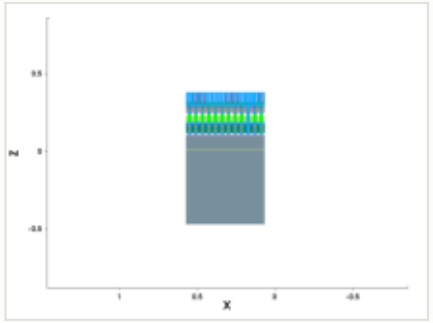
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



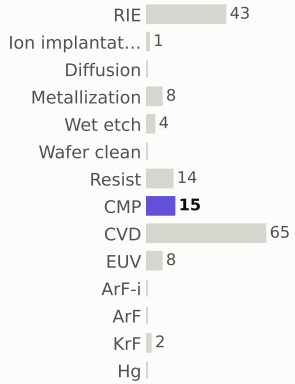
y cut at 0.049 µm



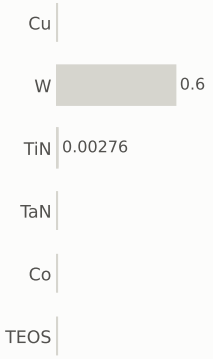
mask for this step

no mask for this step

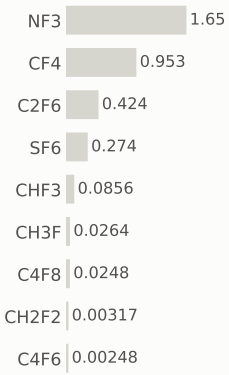
steps by process area



materials deposited (g)



gases emitted (g)



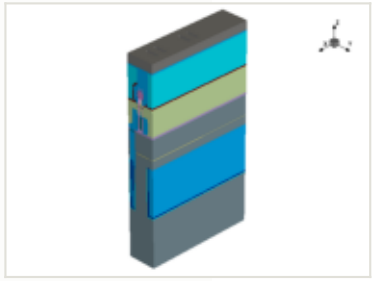
Step #1820: EUV

litho_patterning · EUV

1.96	216.0	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

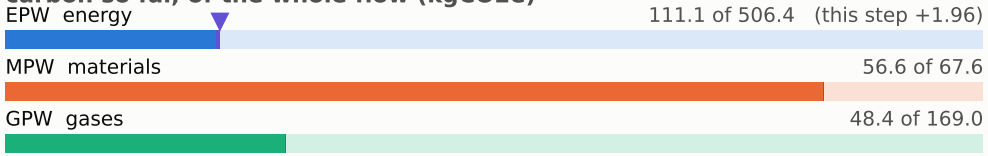


this step

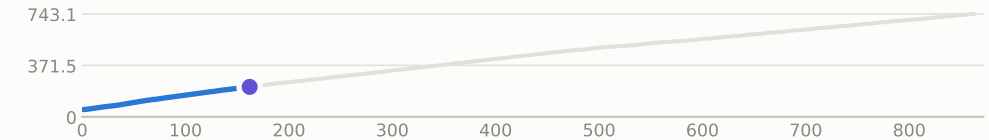
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

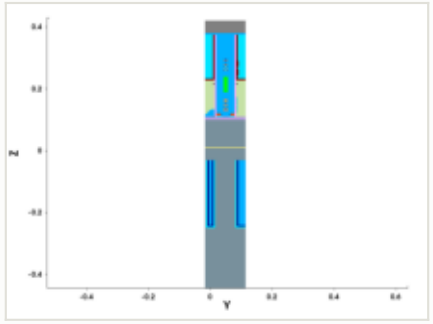
carbon so far, of the whole flow (kgCO2e)



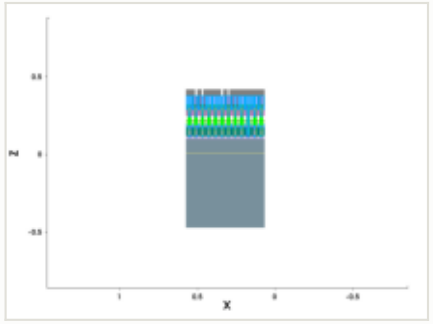
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

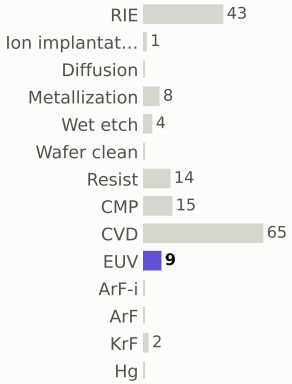


y cut at 0.049 µm

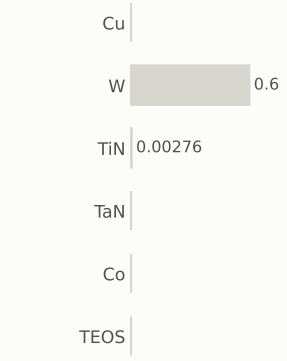


mask: layer 3:1 (GDSII)

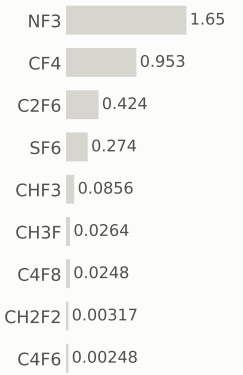
steps by process area



materials deposited (g)



gases emitted (g)



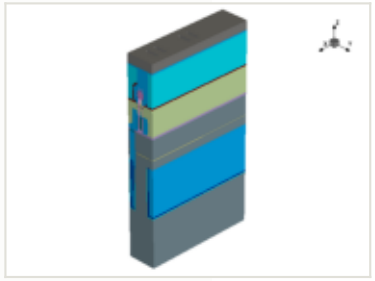
Step #1830: RIE

etch_anisotropic · RIE

1.04	217.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

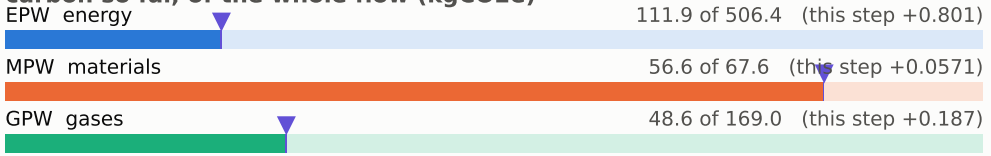


this step

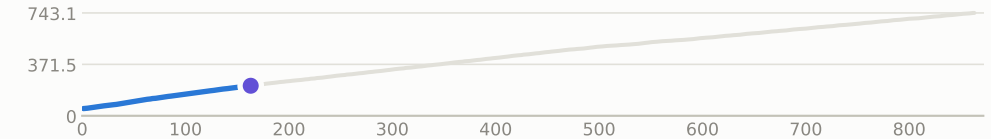
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

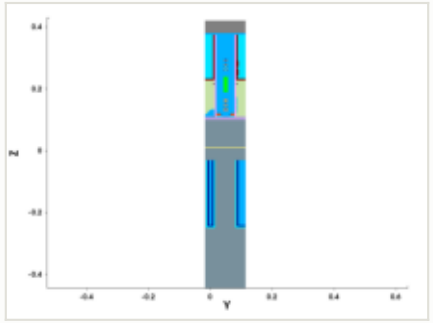
carbon so far, of the whole flow (kgCO2e)



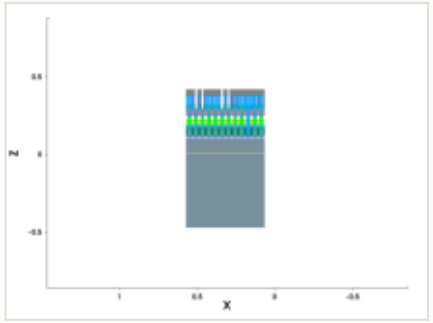
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



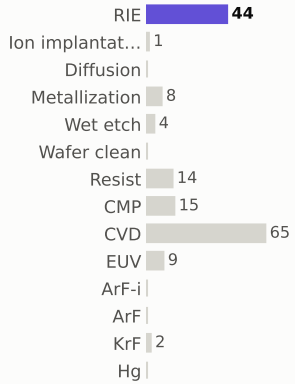
y cut at 0.049 µm



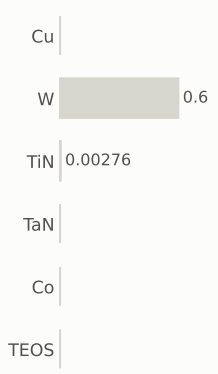
mask for this step

no mask for this step

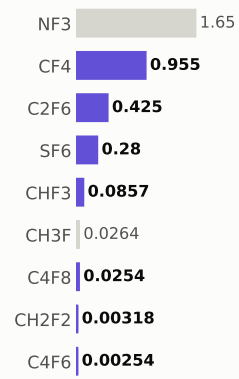
steps by process area



materials deposited (g)



gases emitted (g)



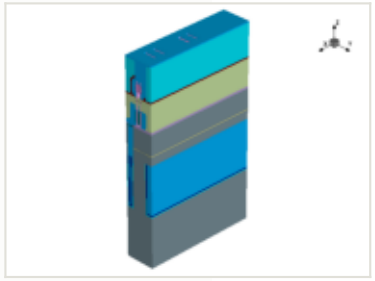
Step #1840: Resist

etch_strip · Resist

0.169	217.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

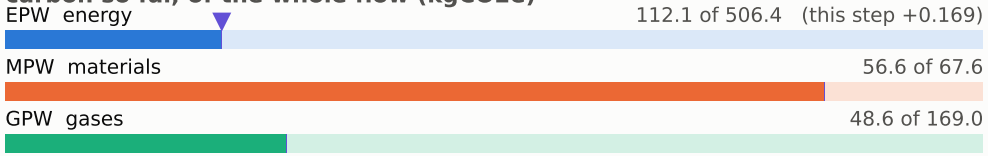


this step

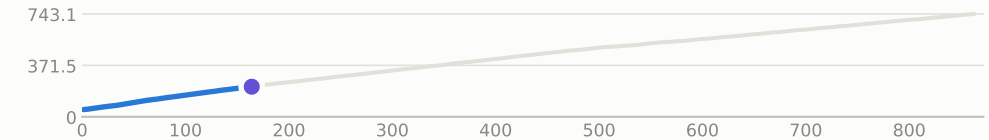
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

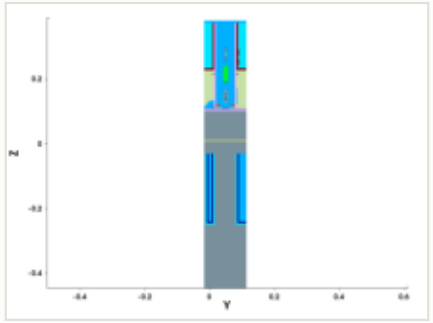
carbon so far, of the whole flow (kgCO2e)



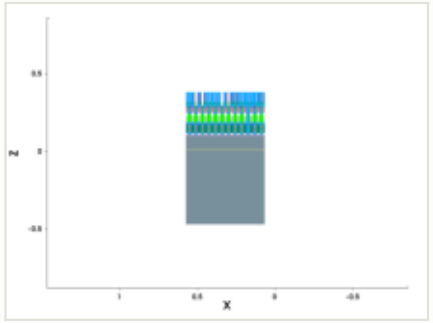
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



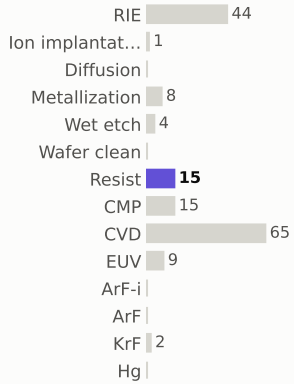
y cut at 0.049 µm



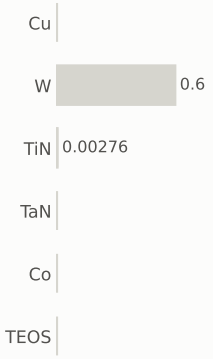
mask for this step

no mask for this step

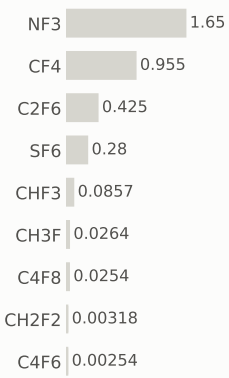
steps by process area



materials deposited (g)



gases emitted (g)



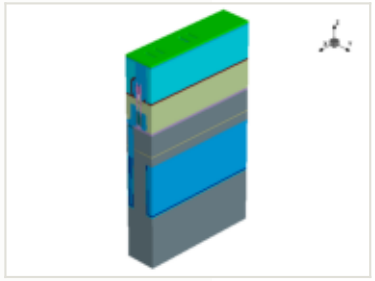
Step #1850: CVD

depo_anisotropic · CVD

1.31	218.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

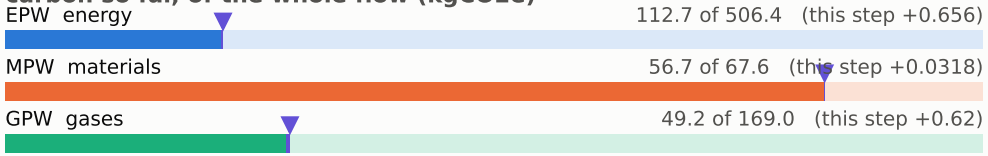


this step

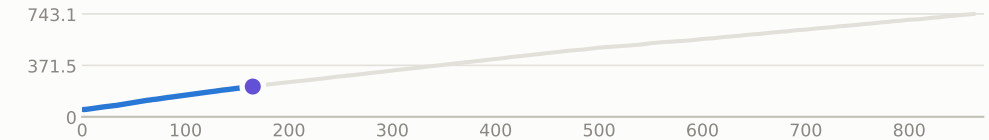
process area	CVD
material	—
thickness	0.008 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

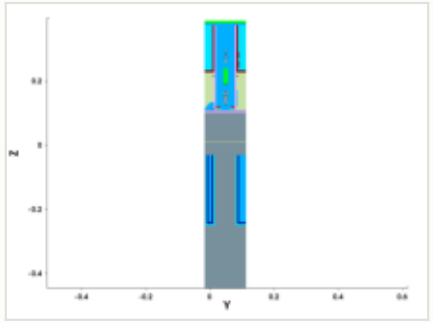
carbon so far, of the whole flow (kgCO2e)



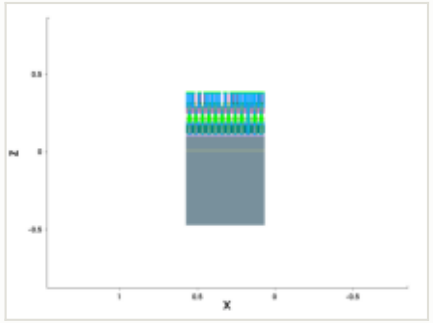
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



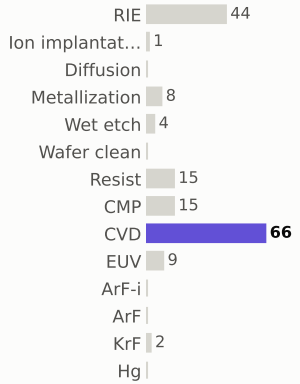
y cut at 0.049 µm



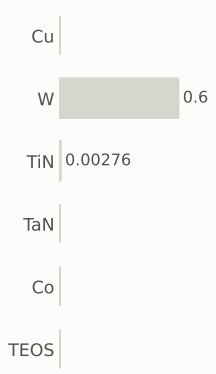
mask for this step

no mask for this step

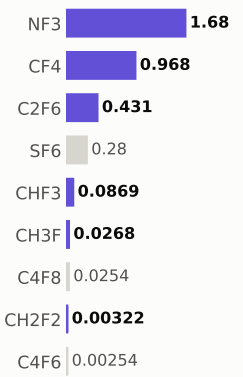
steps by process area



materials deposited (g)



gases emitted (g)



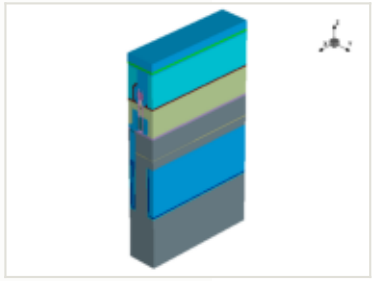
Step #1860: Metallization

depo_fill · Metallization

1.21	219.7	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

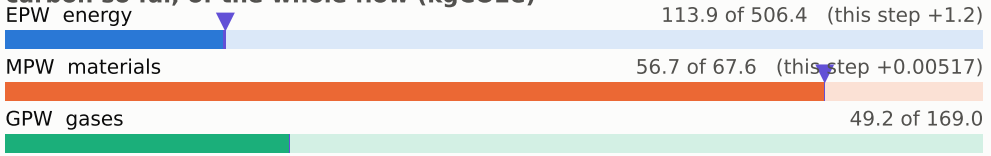


this step

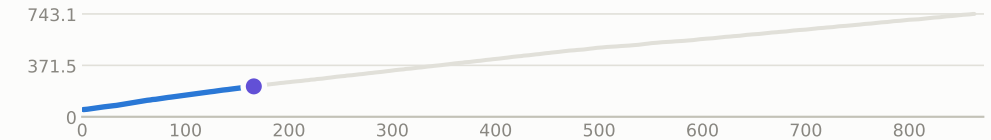
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

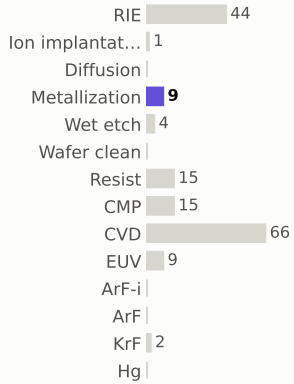
carbon so far, of the whole flow (kgCO2e)



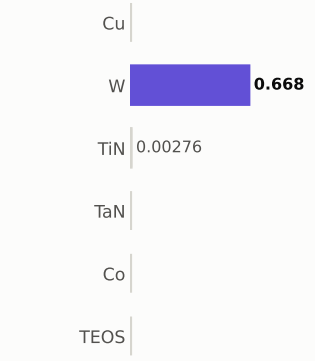
embodied carbon up to each step (kgCO2e per wafer)



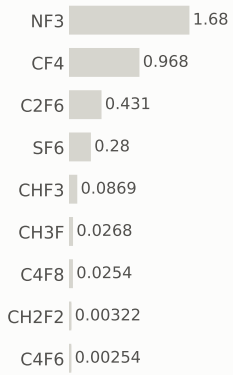
steps by process area



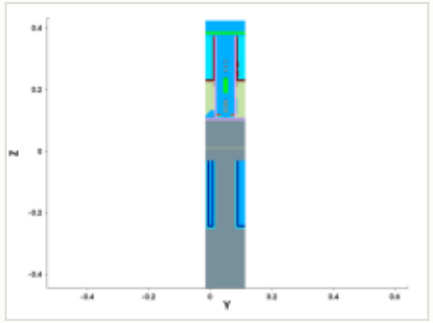
materials deposited (g)



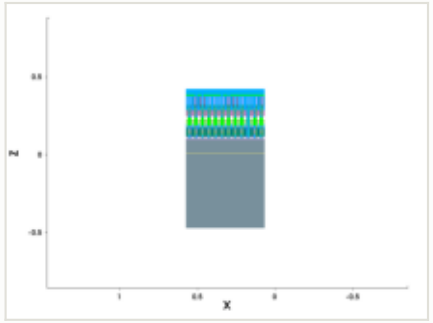
gases emitted (g)



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

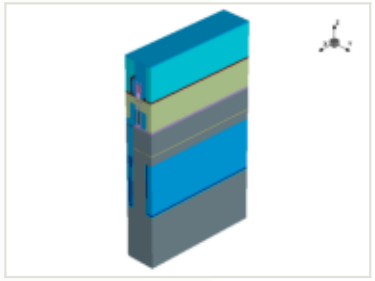
Step #1870: CMP

etch_cmp · CMP

0.169	219.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

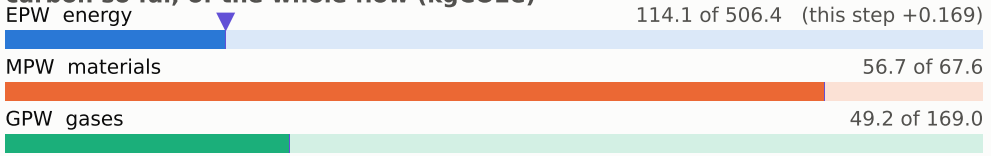


this step

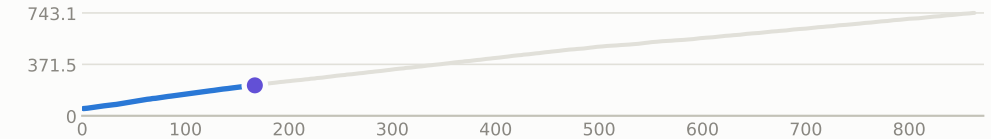
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

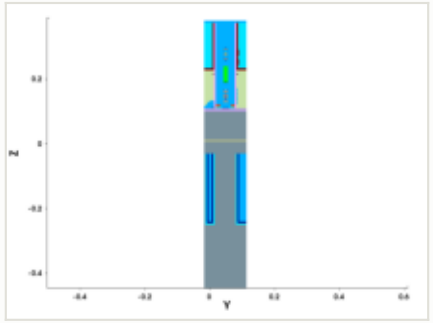
carbon so far, of the whole flow (kgCO2e)



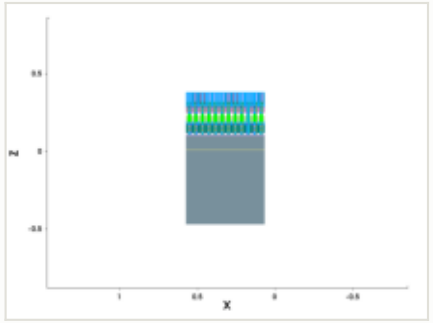
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



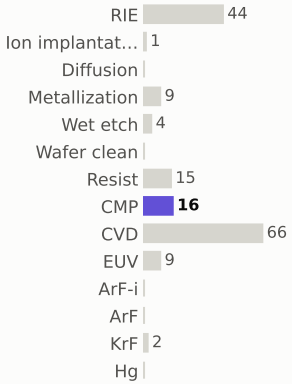
y cut at 0.049 µm



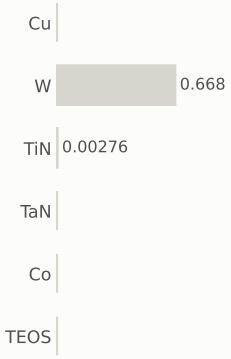
mask for this step

no mask for this step

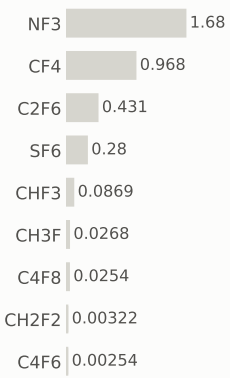
steps by process area



materials deposited (g)



gases emitted (g)



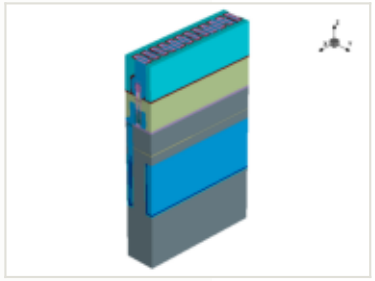
Step #1880: RIE

etch_height_dependent · RIE

1.04	221.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

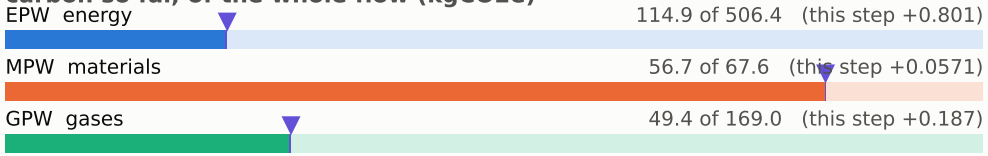


this step

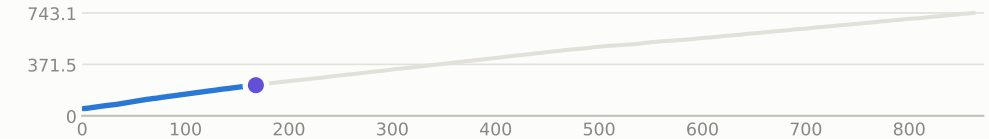
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

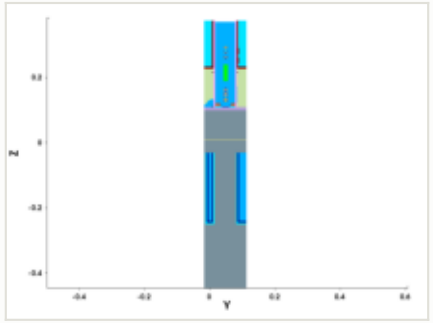
carbon so far, of the whole flow (kgCO2e)



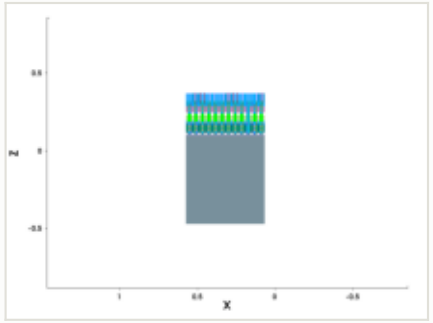
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



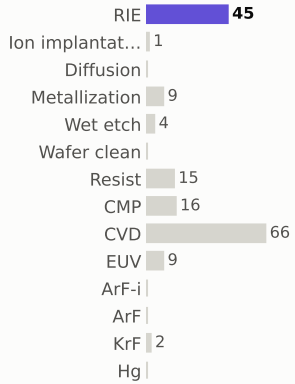
y cut at 0.049 µm



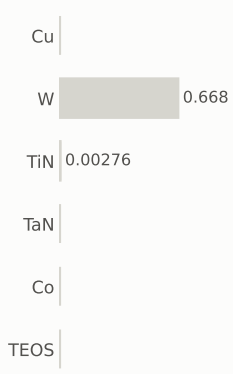
mask for this step

no mask for this step

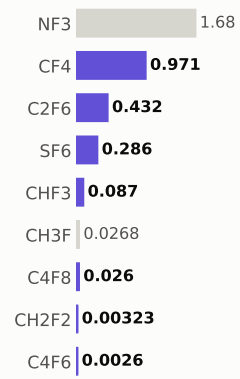
steps by process area



materials deposited (g)



gases emitted (g)



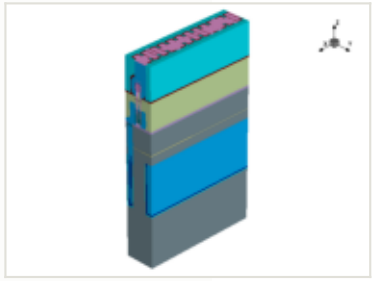
Step #1890: CVD

depo_fill · CVD

1.31	222.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

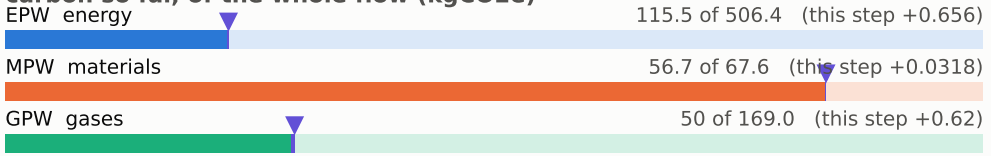


this step

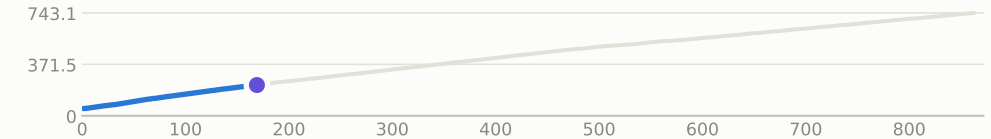
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

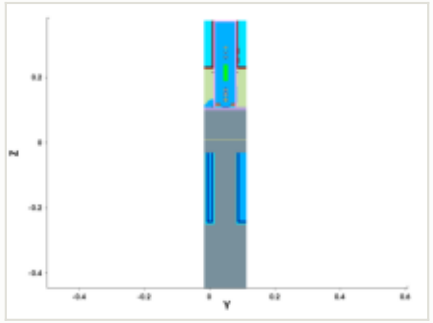
carbon so far, of the whole flow (kgCO2e)



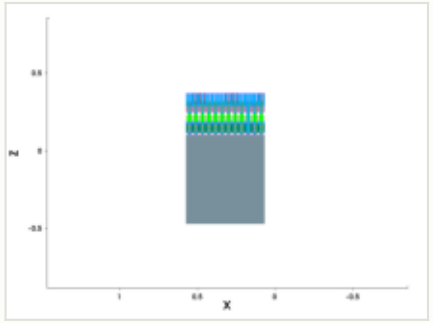
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



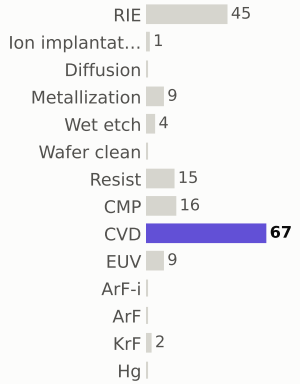
y cut at 0.049 µm



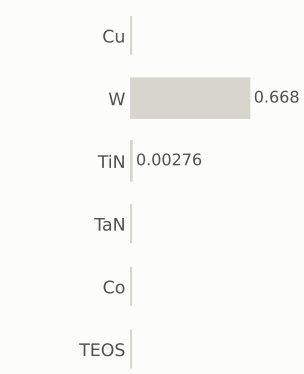
mask for this step

no mask for this step

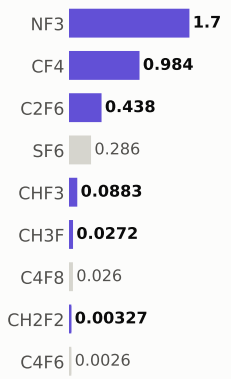
steps by process area



materials deposited (g)



gases emitted (g)



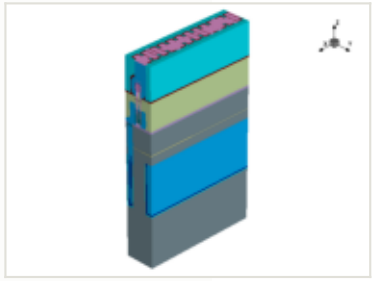
Step #1900: CMP

etch_cmp · CMP

0.169	222.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

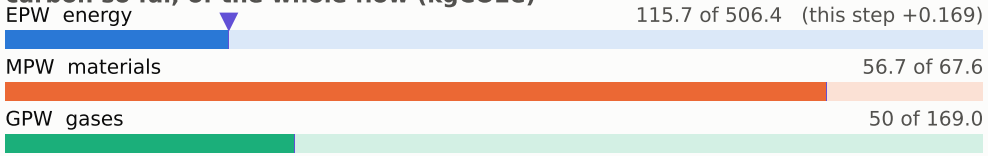


this step

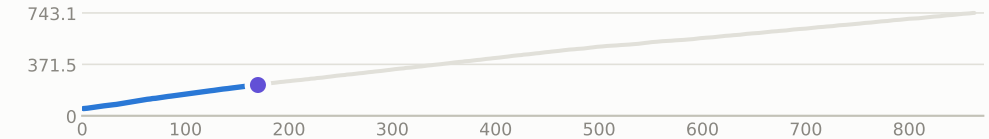
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

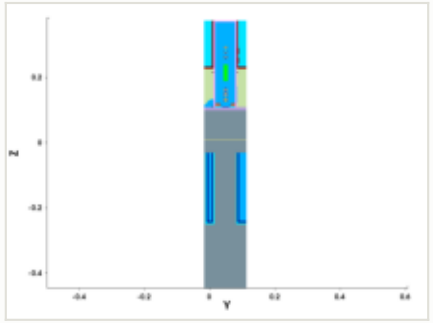
carbon so far, of the whole flow (kgCO2e)



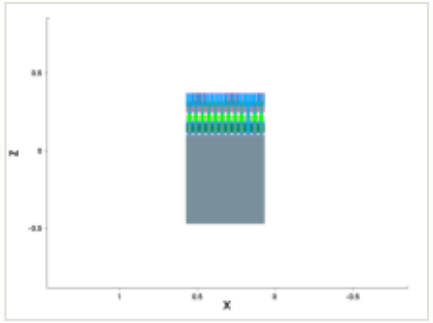
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



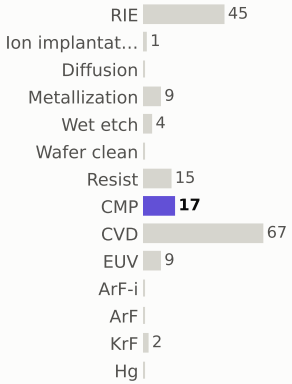
y cut at 0.049 µm



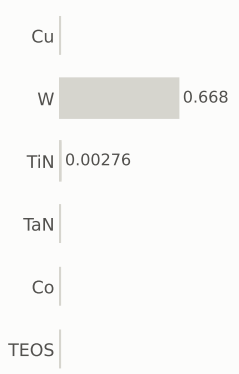
mask for this step

no mask for this step

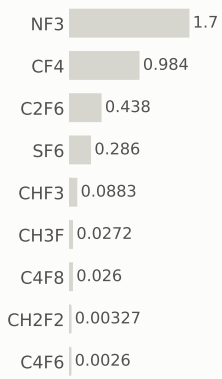
steps by process area



materials deposited (g)



gases emitted (g)



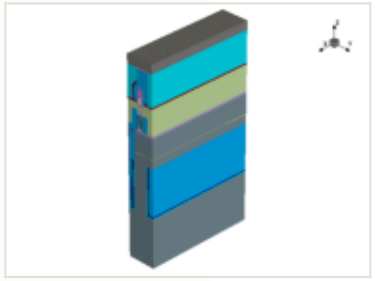
Step #1920: EUV

litho_patterning · EUV

1.96	224.4	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

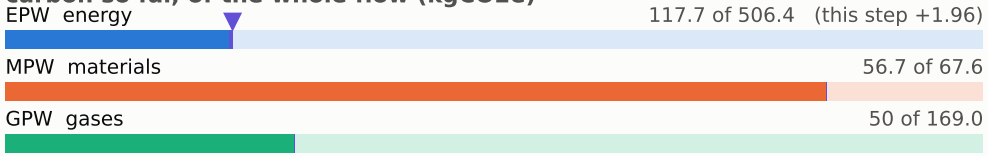


this step

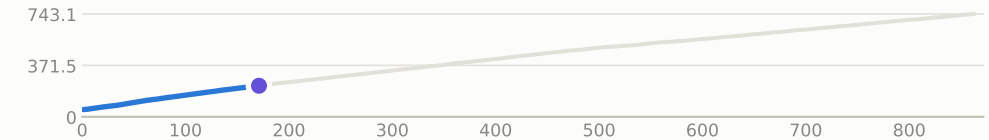
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

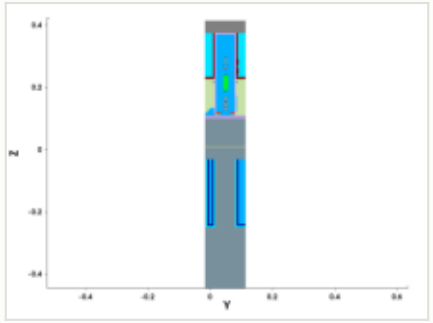
carbon so far, of the whole flow (kgCO2e)



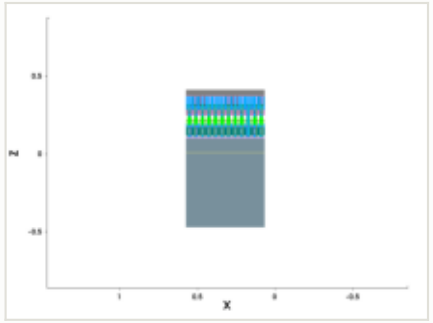
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

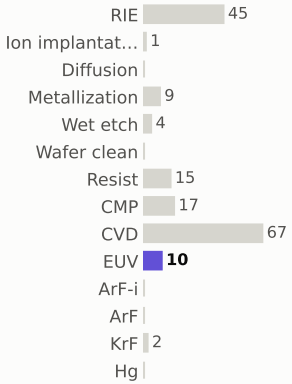


y cut at 0.049 µm

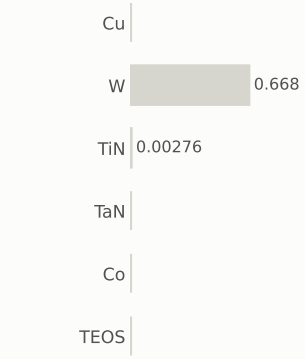


mask: layer 5:2 (GDSII)

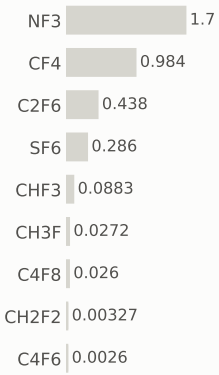
steps by process area



materials deposited (g)



gases emitted (g)



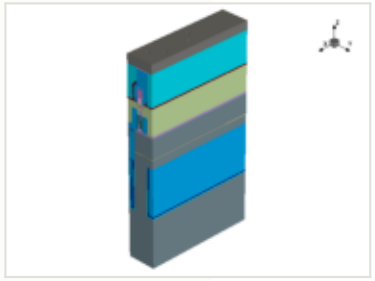
Step #1930: RIE

etch_anisotropic · RIE

1.04	225.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

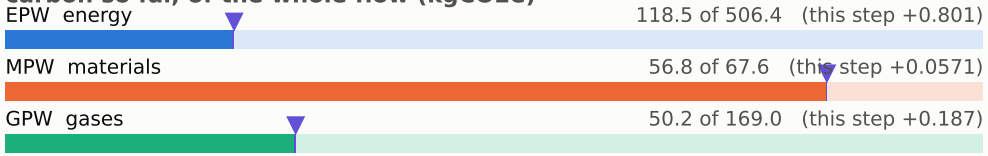


this step

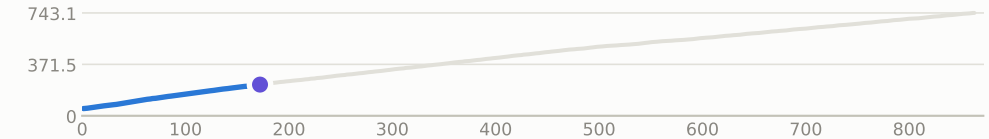
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

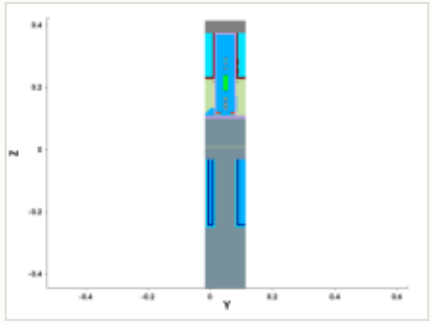
carbon so far, of the whole flow (kgCO2e)



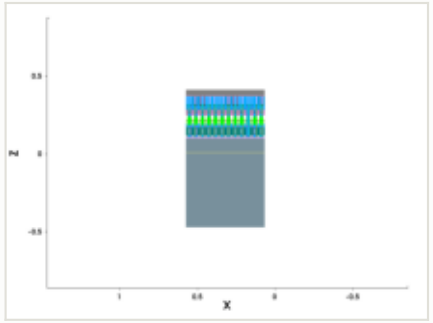
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



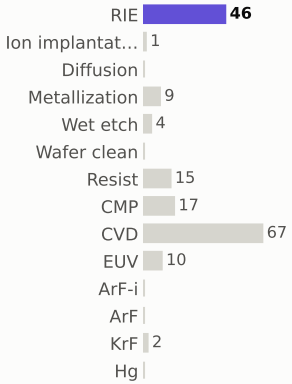
y cut at 0.049 µm



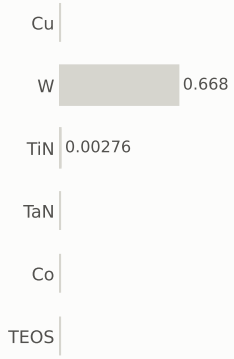
mask for this step

no mask for this step

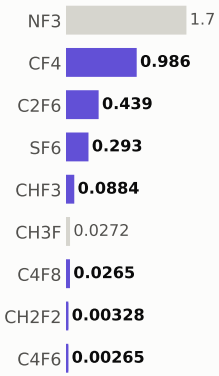
steps by process area



materials deposited (g)



gases emitted (g)



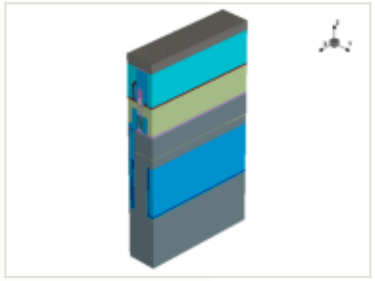
Step #1940: RIE

etch_anisotropic · RIE

1.04	226.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

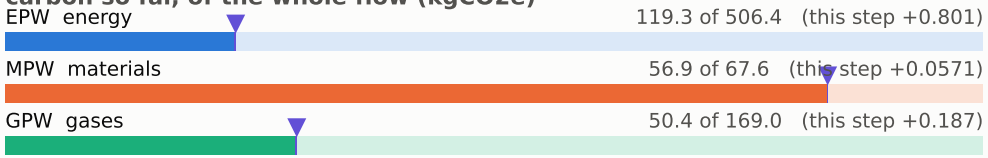
zoomed to fit



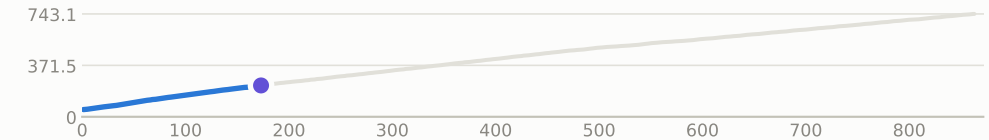
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

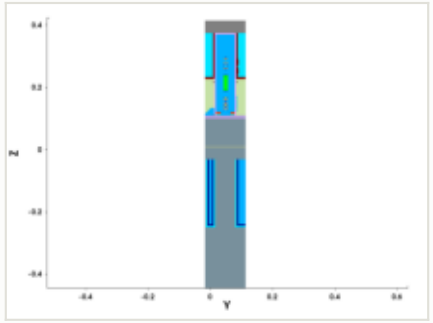
carbon so far, of the whole flow (kgCO2e)



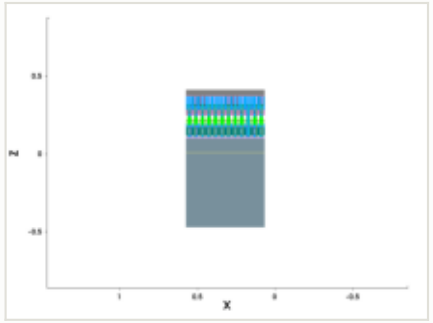
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



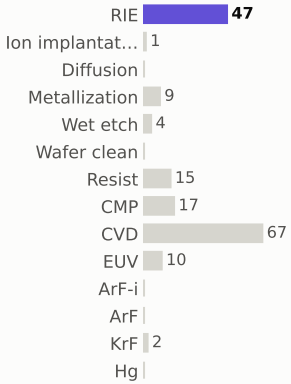
y cut at 0.049 µm



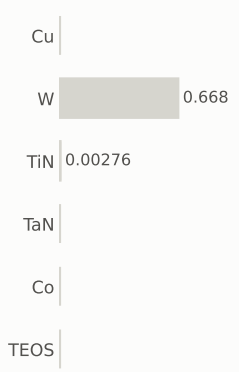
mask for this step

no mask for this step

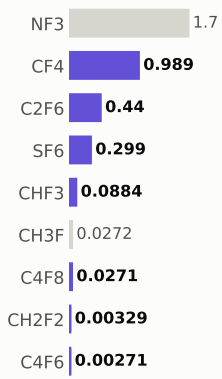
steps by process area



materials deposited (g)



gases emitted (g)



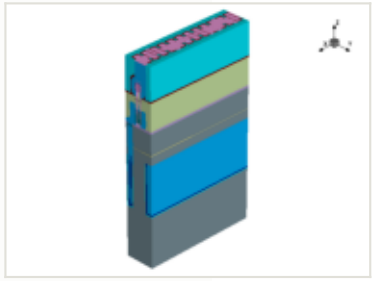
Step #1950: Resist

etch_strip · Resist

0.169	226.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

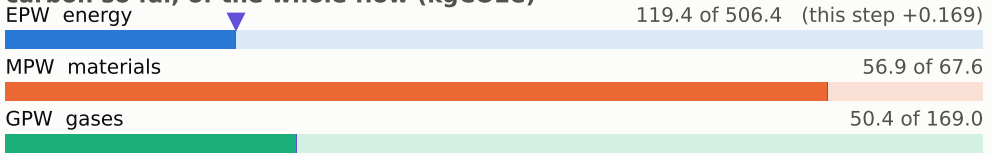


this step

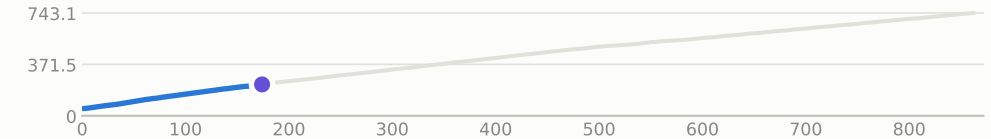
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

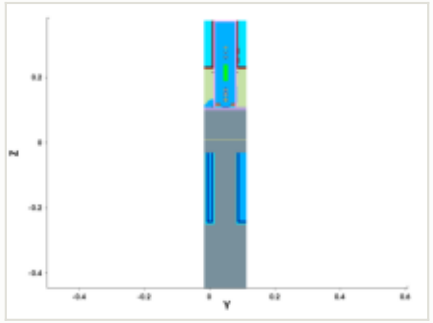
carbon so far, of the whole flow (kgCO2e)



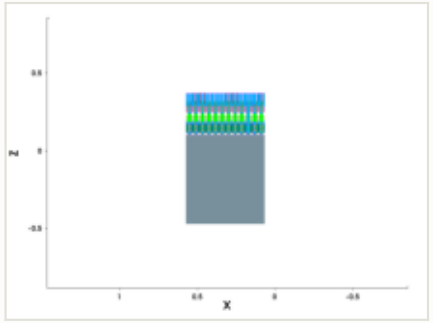
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



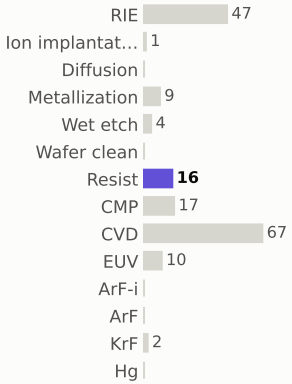
y cut at 0.049 µm



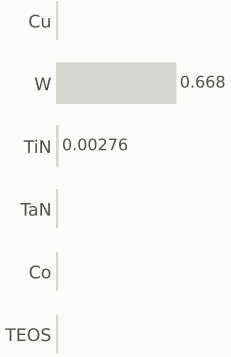
mask for this step

no mask for this step

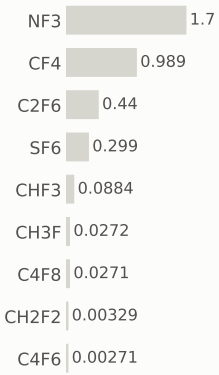
steps by process area



materials deposited (g)



gases emitted (g)



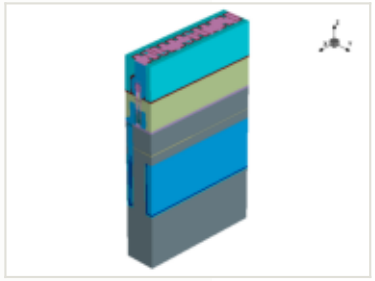
Step #1960: CVD

depo_fill · CVD

1.31	228.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

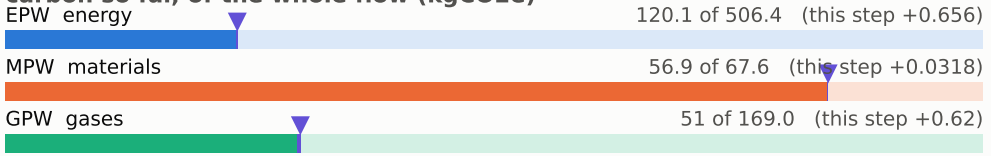


this step

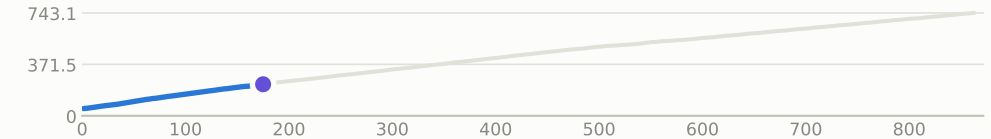
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

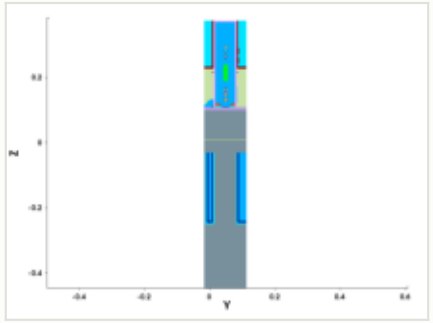
carbon so far, of the whole flow (kgCO2e)



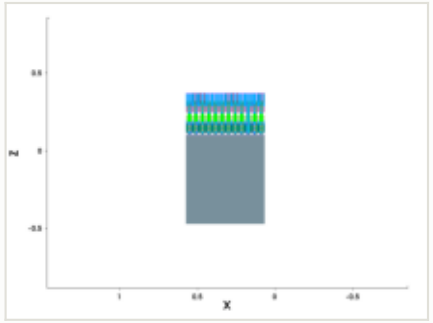
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



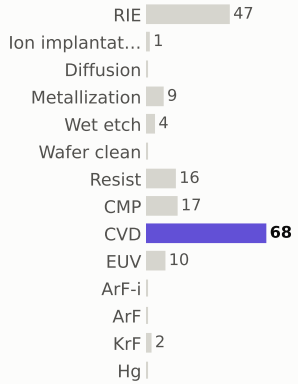
y cut at 0.049 µm



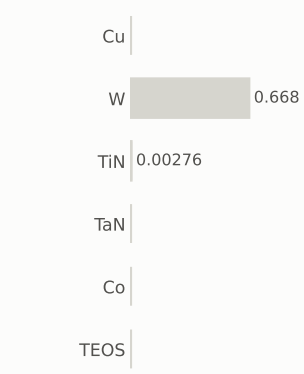
mask for this step

no mask for this step

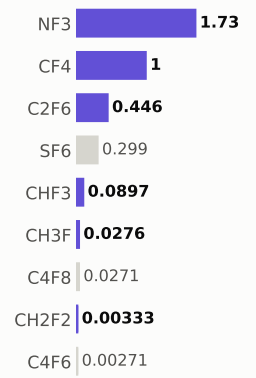
steps by process area



materials deposited (g)



gases emitted (g)



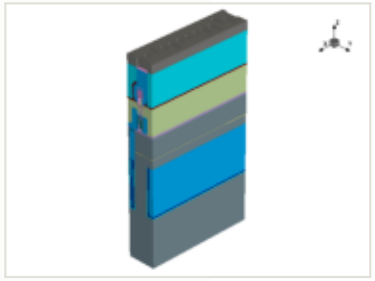
Step #1980: EUV

litho_patterning · EUV

1.96	229.9	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

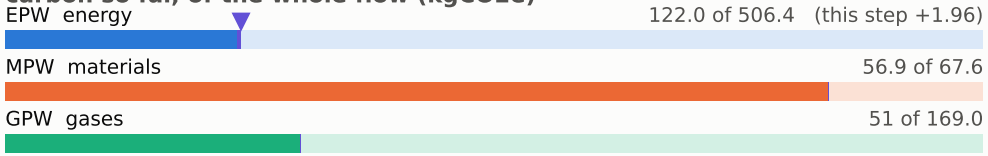


this step

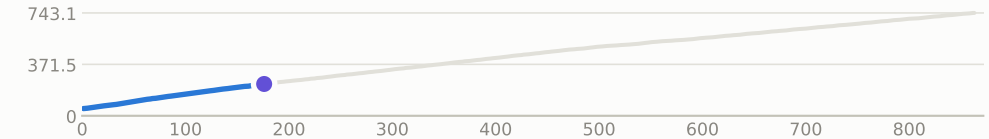
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

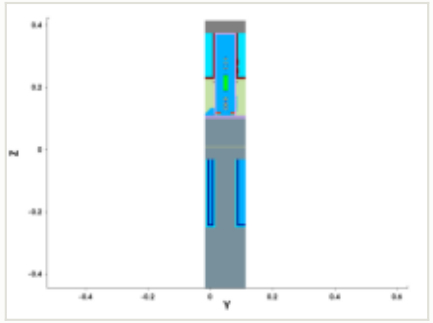
carbon so far, of the whole flow (kgCO2e)



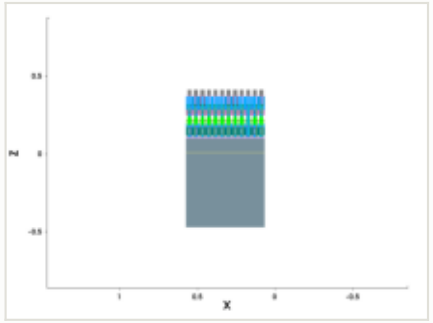
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

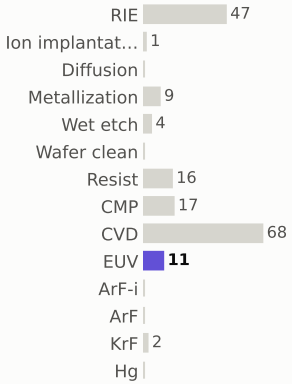


y cut at 0.049 µm

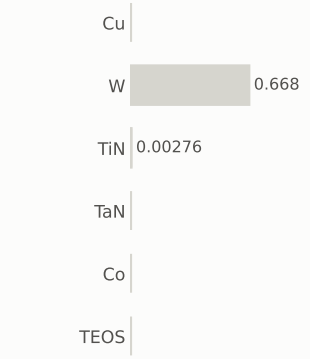


mask: layer 10:2 (GDSII)

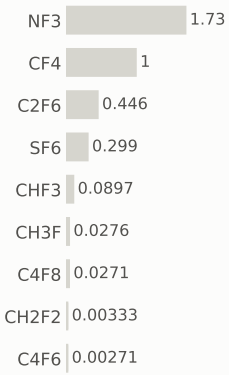
steps by process area



materials deposited (g)



gases emitted (g)



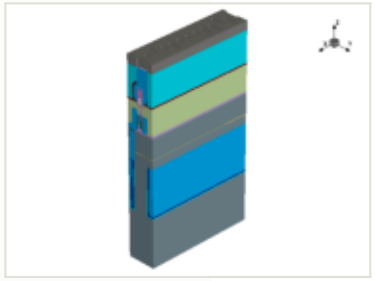
Step #1990: RIE

etch_anisotropic · RIE

1.04	231.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

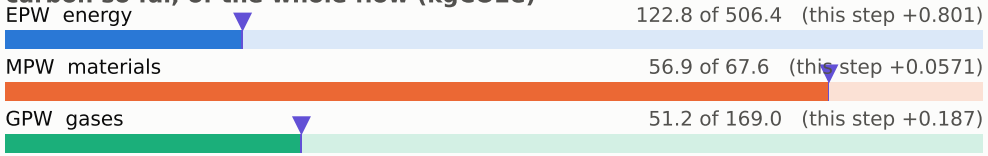


this step

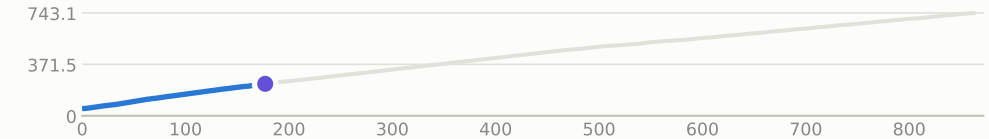
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

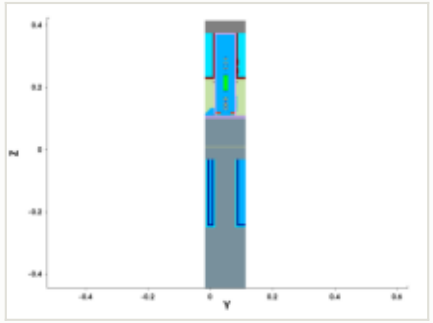
carbon so far, of the whole flow (kgCO2e)



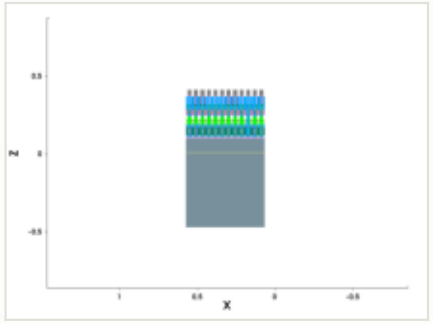
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



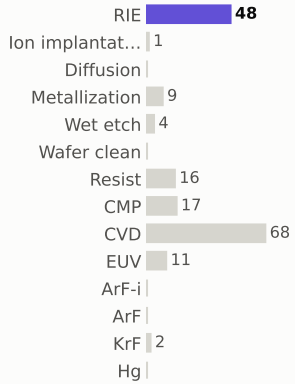
y cut at 0.049 µm



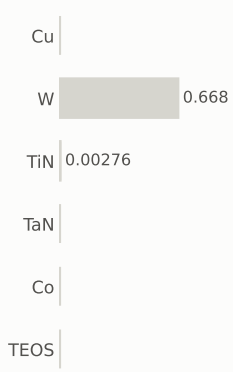
mask for this step

no mask for this step

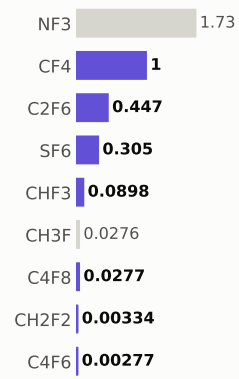
steps by process area



materials deposited (g)



gases emitted (g)



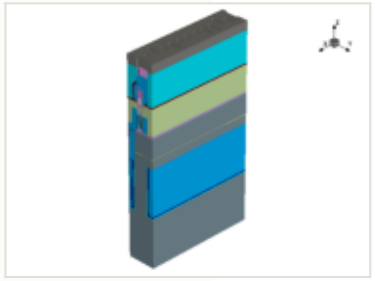
Step #2000: RIE

etch_anisotropic · RIE

1.04	232.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

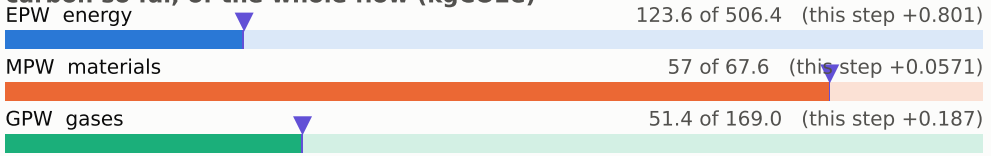


this step

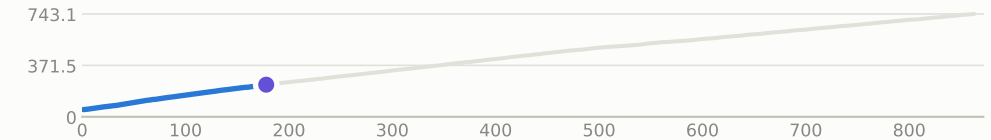
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

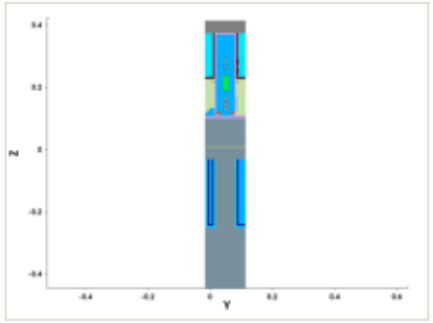
carbon so far, of the whole flow (kgCO2e)



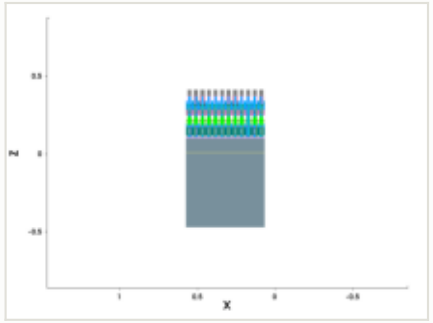
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



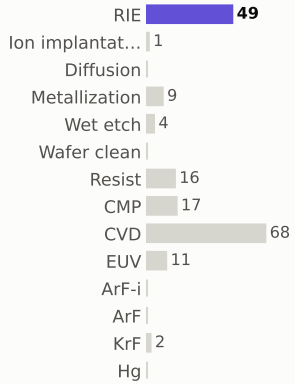
y cut at 0.049 µm



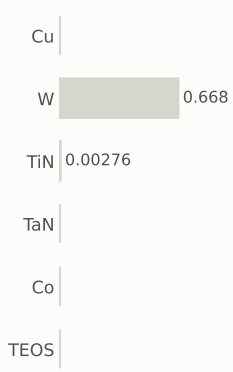
mask for this step

no mask for this step

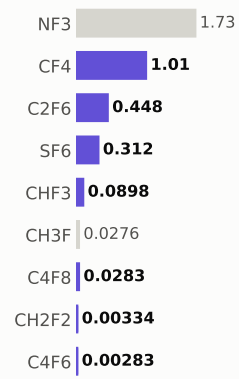
steps by process area



materials deposited (g)



gases emitted (g)



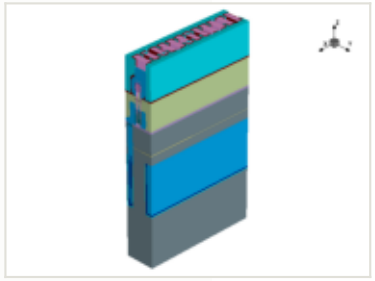
Step #2010: Resist

etch_strip · Resist

0.169	232.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

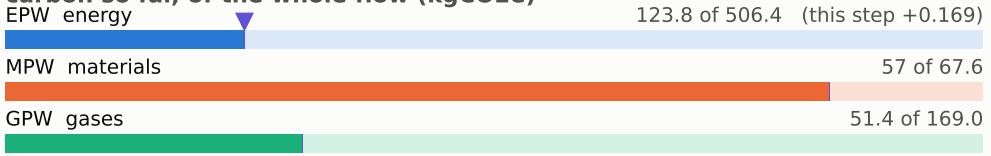


this step

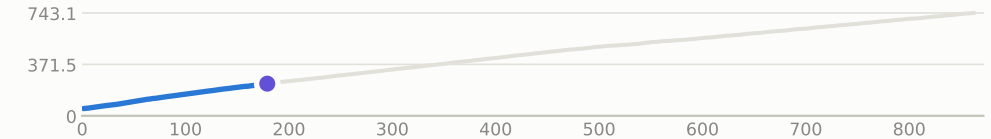
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

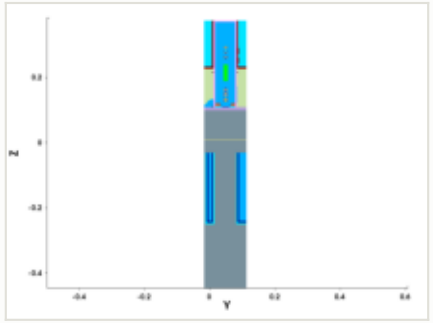
carbon so far, of the whole flow (kgCO2e)



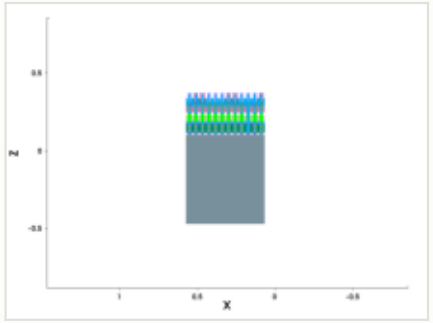
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



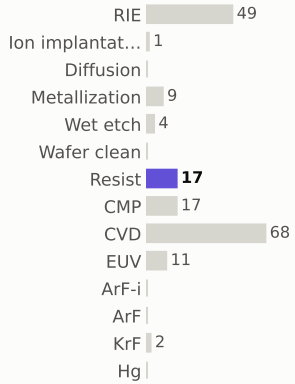
y cut at 0.049 µm



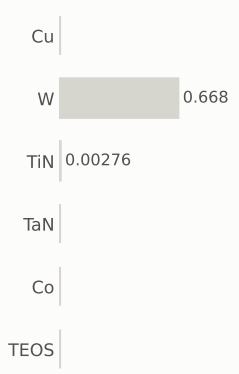
mask for this step

no mask for this step

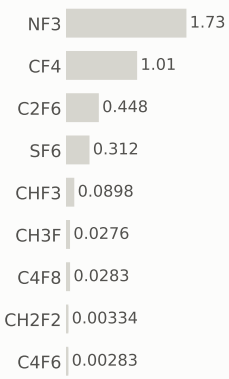
steps by process area



materials deposited (g)



gases emitted (g)



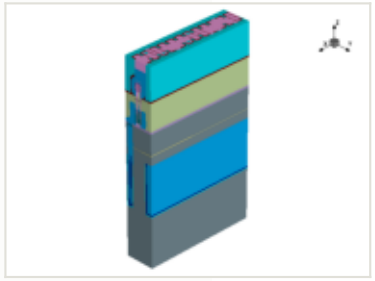
Step #2020: CVD

depo_fill · CVD

1.31	233.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

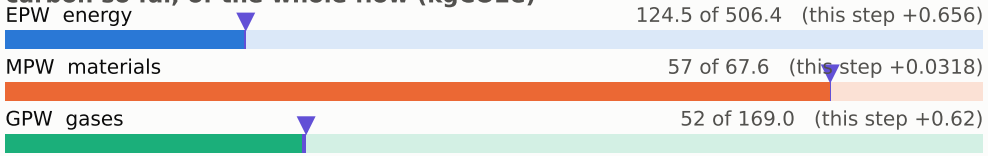


this step

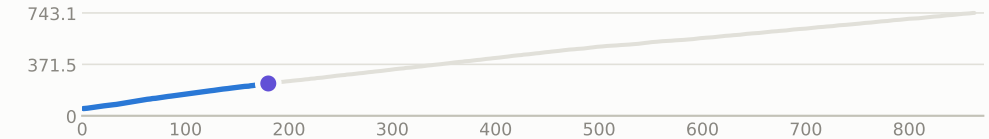
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

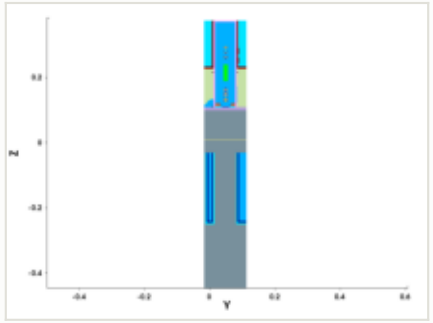
carbon so far, of the whole flow (kgCO2e)



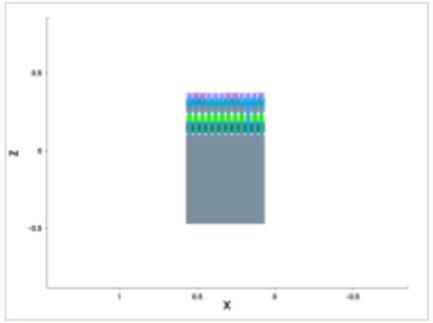
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



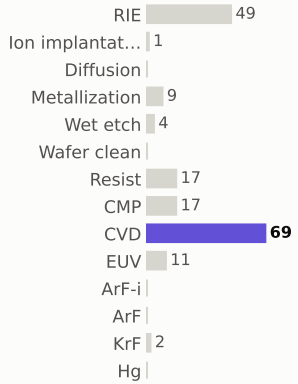
y cut at 0.049 µm



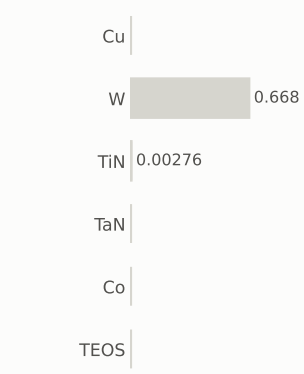
mask for this step

no mask for this step

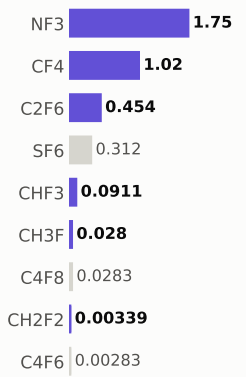
steps by process area



materials deposited (g)



gases emitted (g)



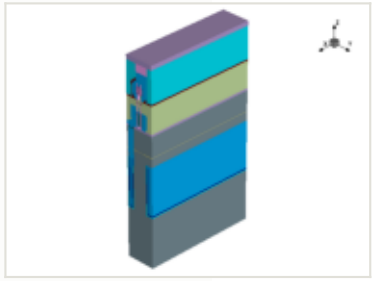
Step #2040: CVD

depo_fill · CVD

1.31	234.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

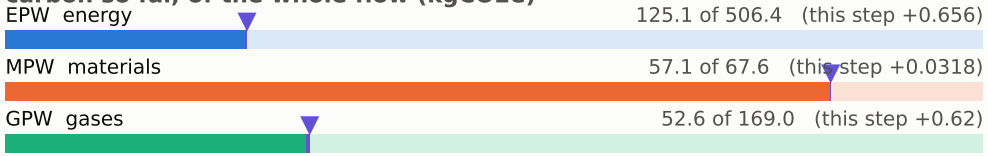


this step

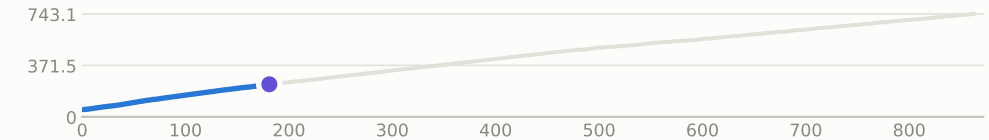
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

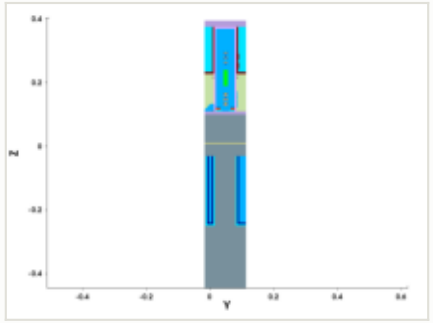
carbon so far, of the whole flow (kgCO2e)



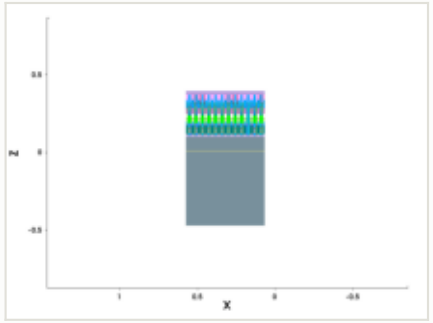
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



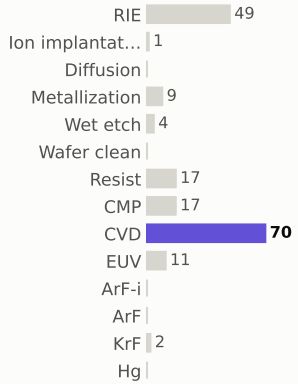
y cut at 0.049 µm



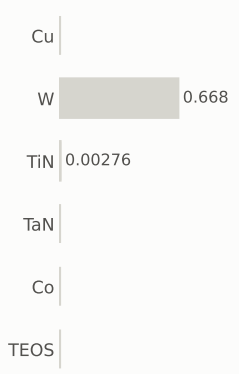
mask for this step

no mask for this step

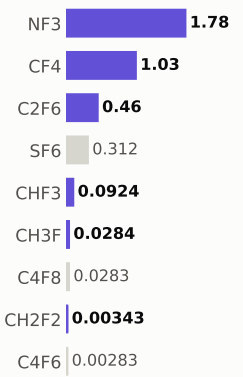
steps by process area



materials deposited (g)



gases emitted (g)



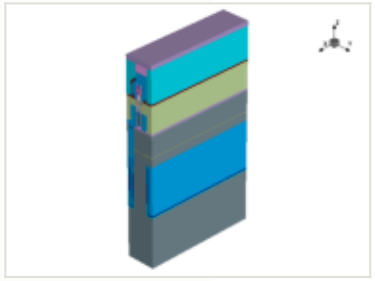
Step #2050: Diffusion

treatment · Diffusion

0.801	235.6	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

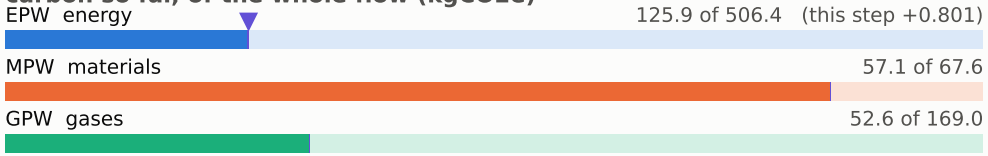


this step

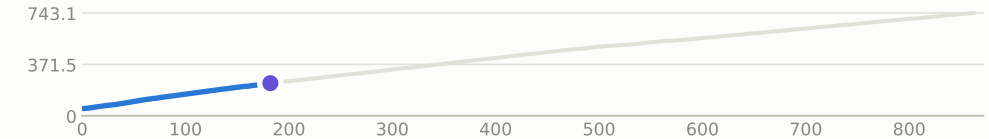
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

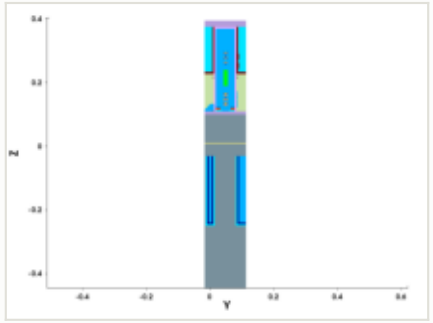
carbon so far, of the whole flow (kgCO2e)



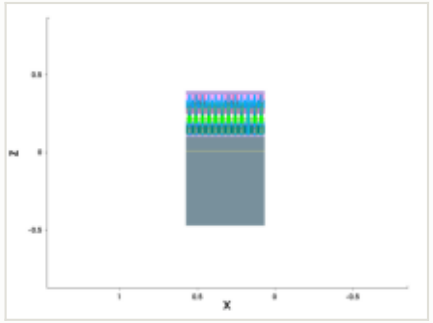
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



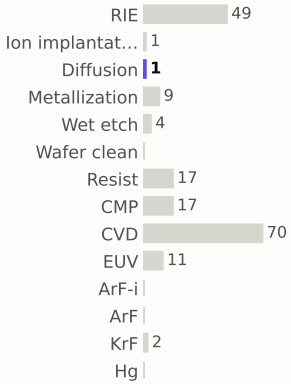
y cut at 0.049 µm



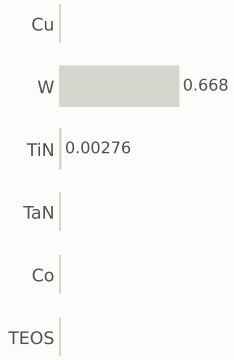
mask for this step

no mask for this step

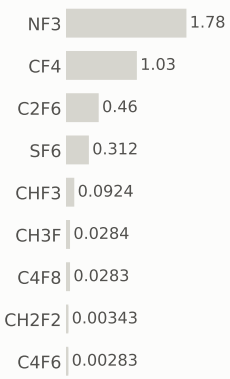
steps by process area



materials deposited (g)



gases emitted (g)



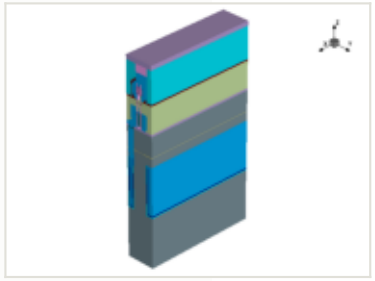
Step #2060: Wafer_clean

clean · Wafer_clean

0.169	235.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

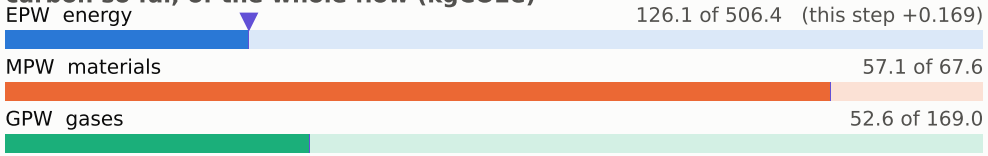


this step

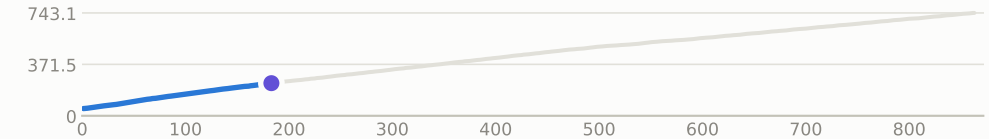
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

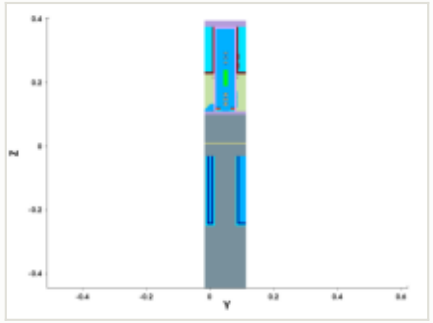
carbon so far, of the whole flow (kgCO2e)



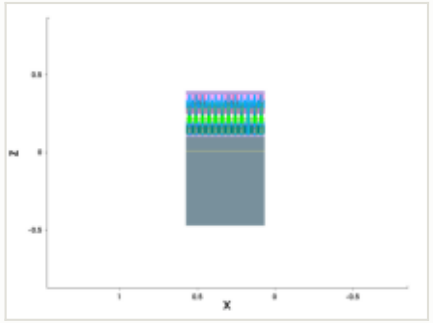
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



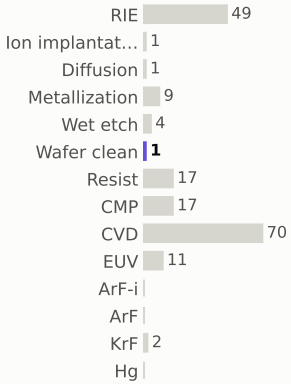
y cut at 0.049 µm



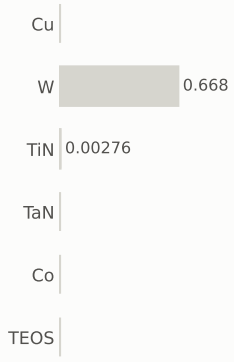
mask for this step

no mask for this step

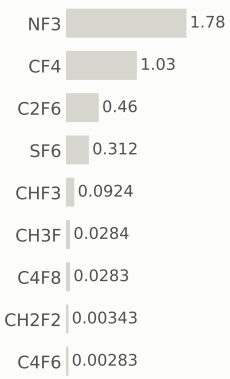
steps by process area



materials deposited (g)



gases emitted (g)



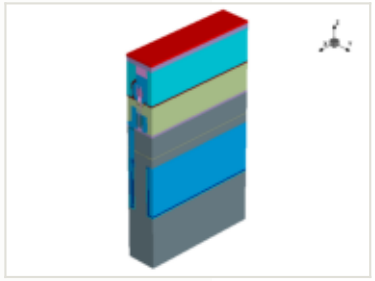
Step #2070: CVD

depo_fill · CVD

1.31	237.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

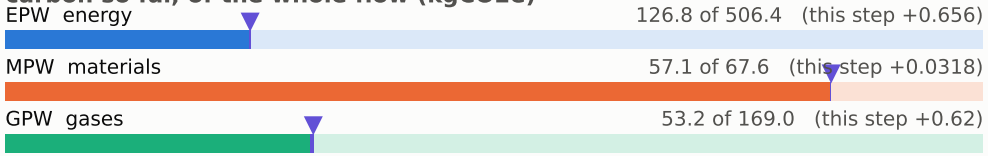


this step

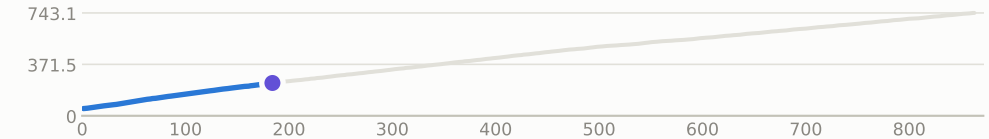
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

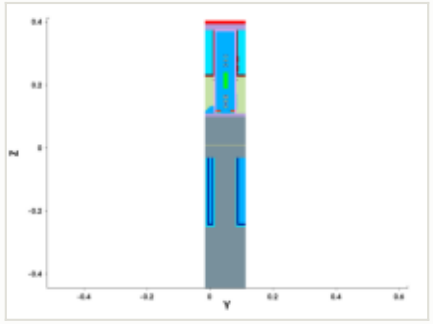
carbon so far, of the whole flow (kgCO2e)



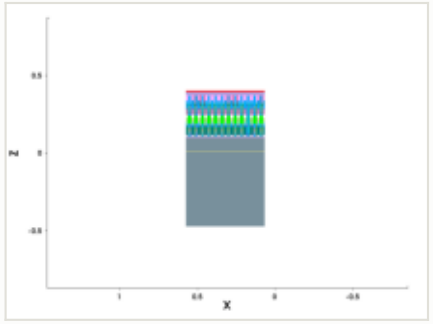
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



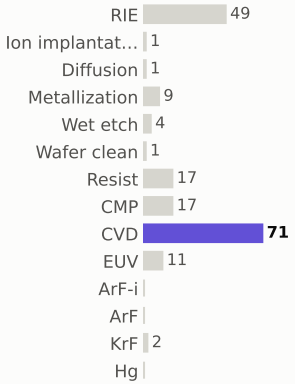
y cut at 0.049 µm



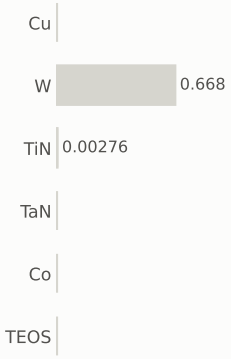
mask for this step

no mask for this step

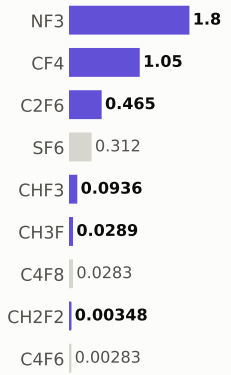
steps by process area



materials deposited (g)



gases emitted (g)



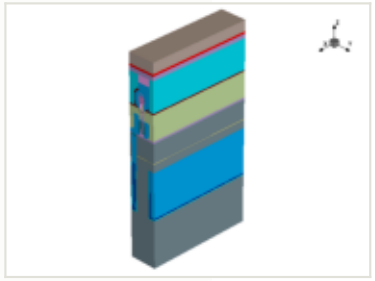
Step #2080: Resist

depo_fill · Resist

0.169	237.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

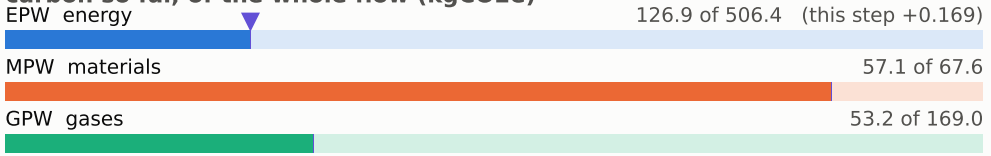


this step

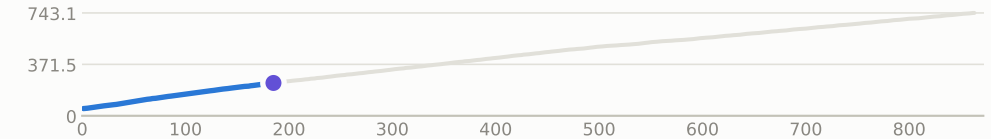
process area	Resist
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

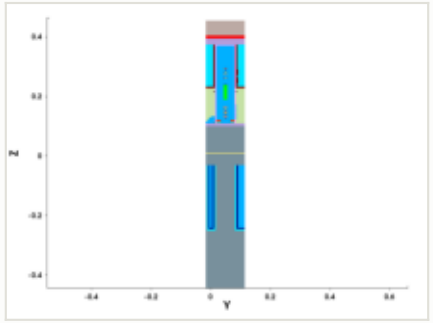
carbon so far, of the whole flow (kgCO2e)



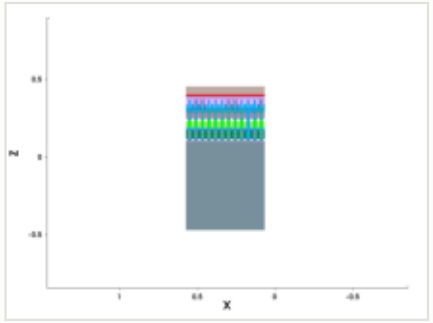
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



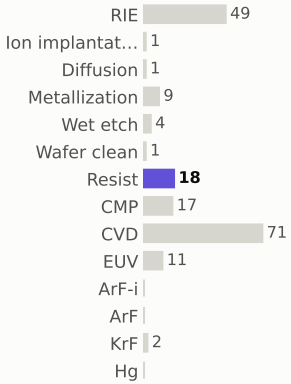
y cut at 0.049 µm



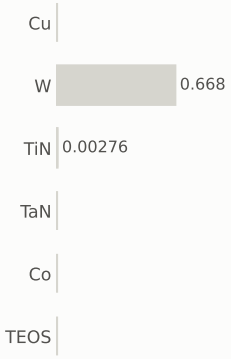
mask for this step

no mask for this step

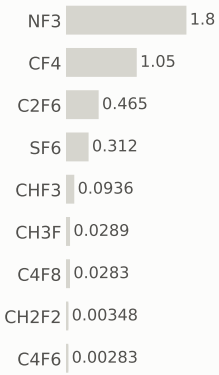
steps by process area



materials deposited (g)



gases emitted (g)



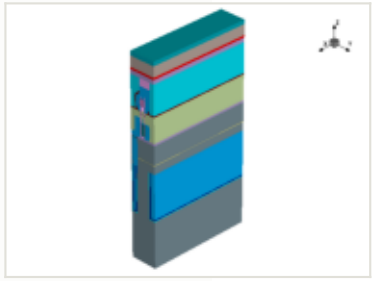
Step #2090: CVD

depo_fill · CVD

1.31	238.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

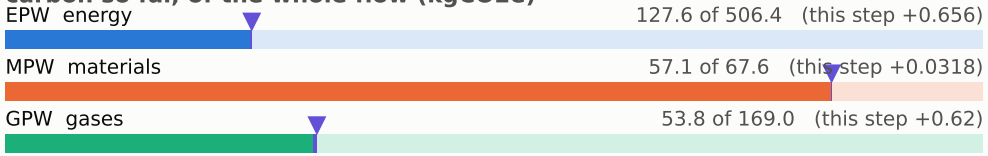
zoomed to fit



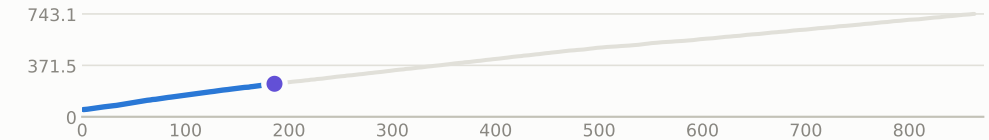
this step

process area	CVD	EPW	0.656 kgCO2e
material	TEOS	EPW tool	0.469 kgCO2e
thickness	0.02 µm	EPW overhead	0.187 kgCO2e
material GWP	2.85952 gCO2e/g	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

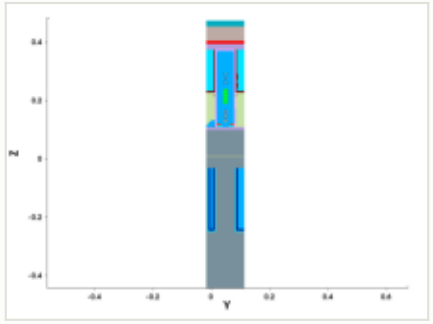
carbon so far, of the whole flow (kgCO2e)



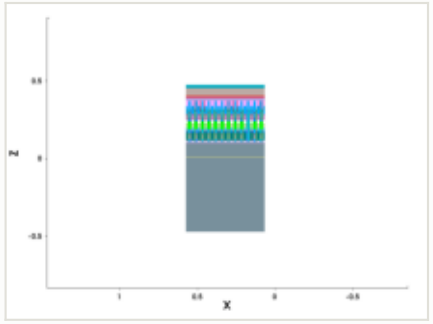
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



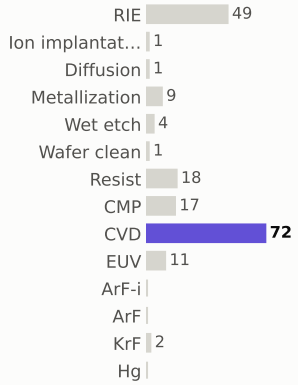
y cut at 0.049 µm



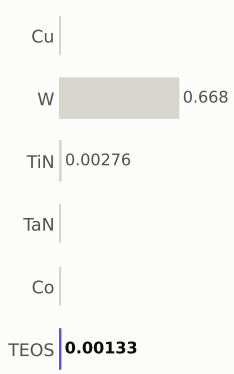
mask for this step

no mask for this step

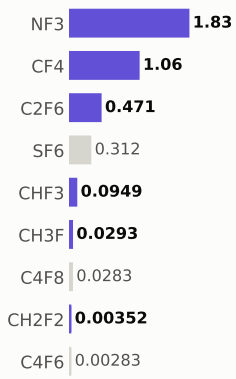
steps by process area



materials deposited (g)



gases emitted (g)



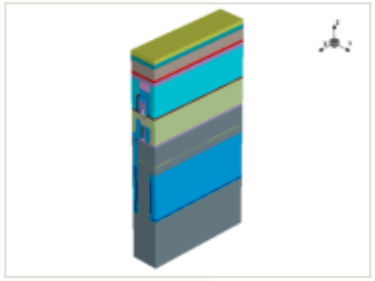
Step #2100: CVD

depo_fill · CVD

1.31	239.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

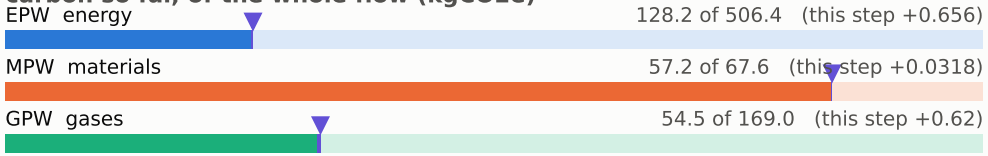


this step

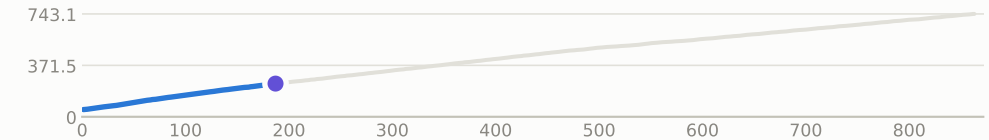
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

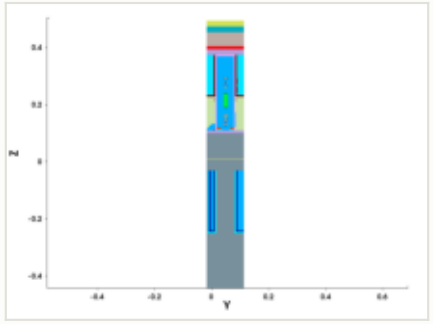
carbon so far, of the whole flow (kgCO2e)



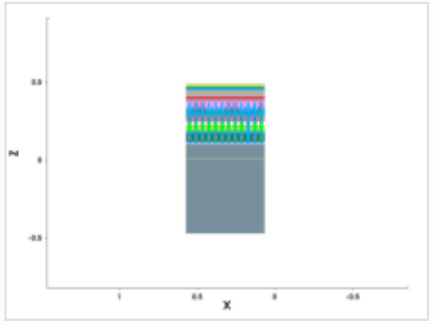
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



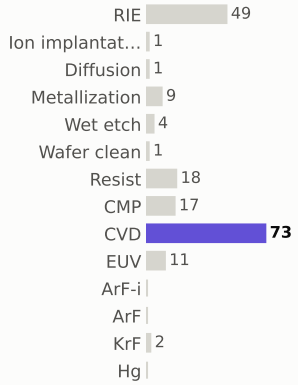
y cut at 0.049 µm



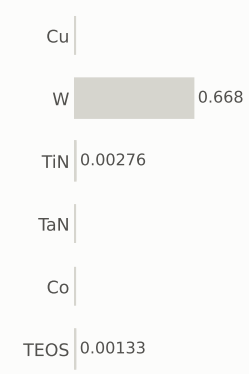
mask for this step

no mask for this step

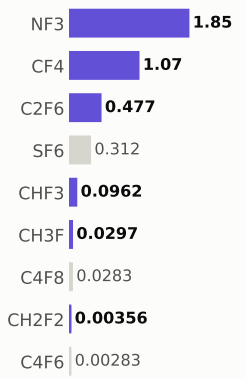
steps by process area



materials deposited (g)



gases emitted (g)



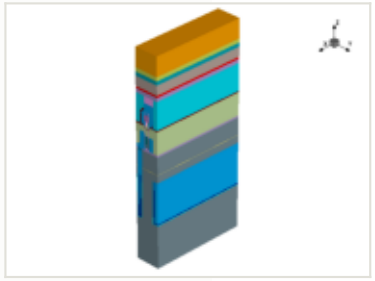
Step #2110: Resist

depo_fill · Resist

0.169	240.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

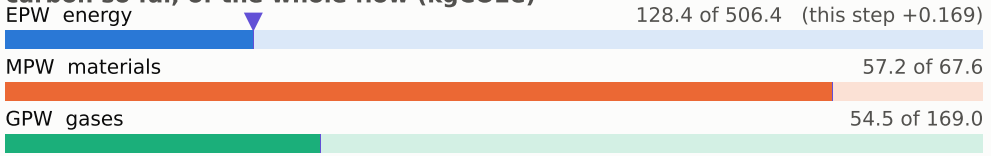


this step

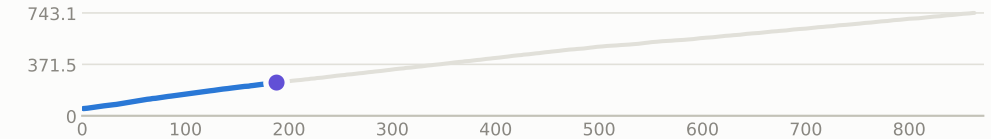
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

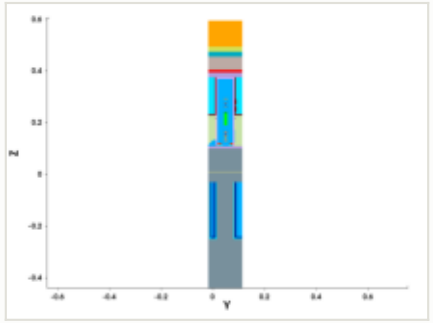
carbon so far, of the whole flow (kgCO2e)



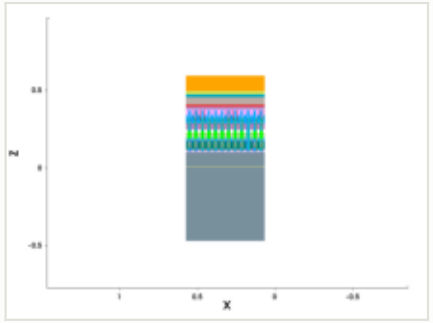
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



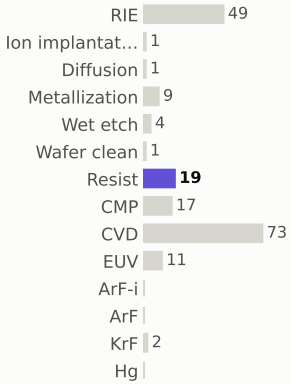
y cut at 0.049 µm



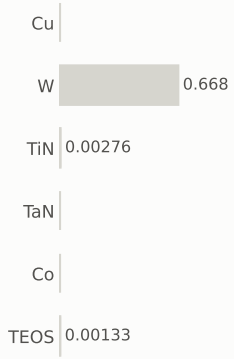
mask for this step

no mask for this step

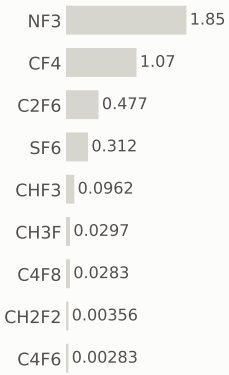
steps by process area



materials deposited (g)



gases emitted (g)



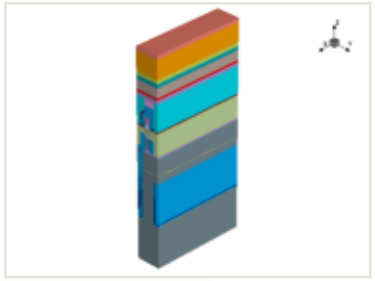
Step #2120: CVD

depo_fill · CVD

1.31	241.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

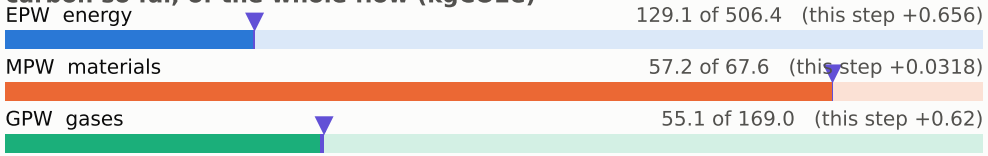
zoomed to fit



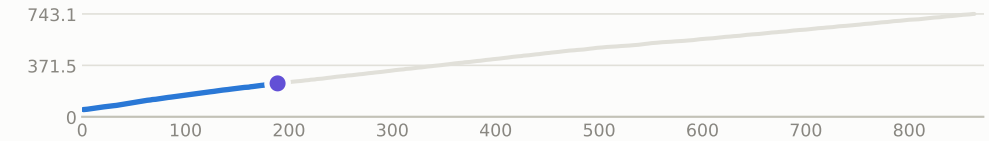
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.02 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

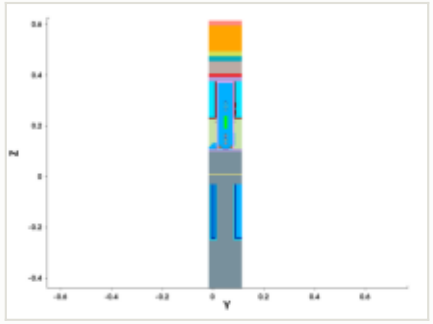
carbon so far, of the whole flow (kgCO2e)



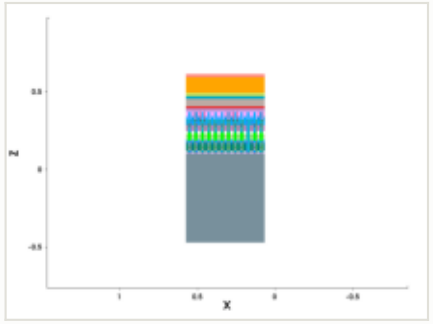
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



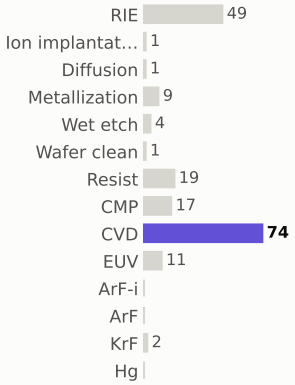
y cut at 0.049 µm



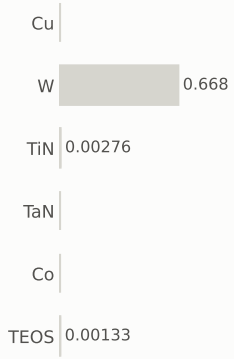
mask for this step

no mask for this step

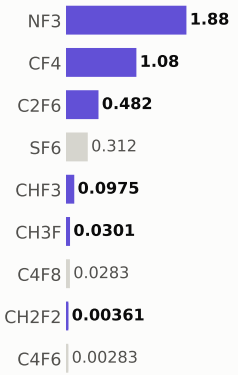
steps by process area



materials deposited (g)



gases emitted (g)



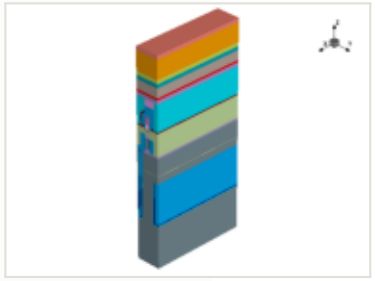
Step #2130: Wafer_clean

clean · Wafer_clean

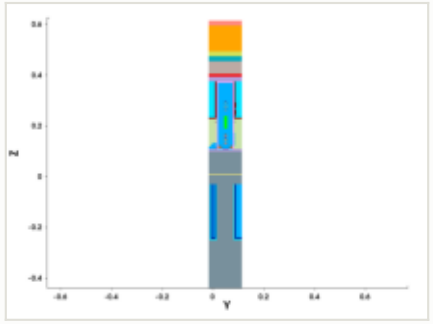
0.169	241.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

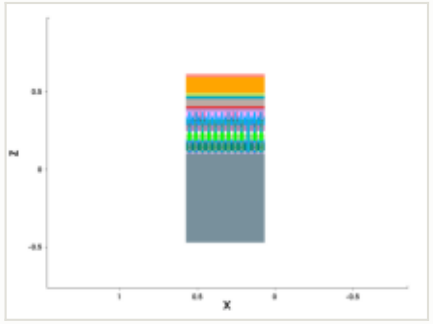
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

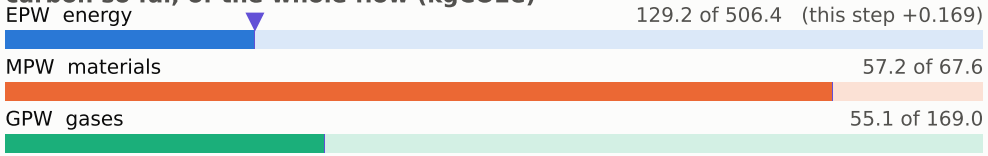
no mask for this step

this step

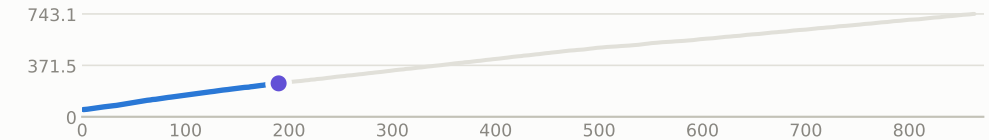
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

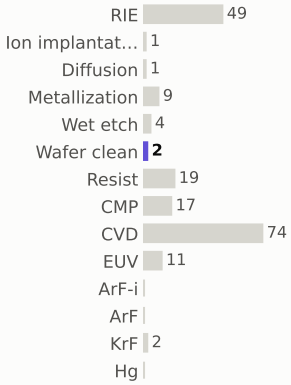
carbon so far, of the whole flow (kgCO2e)



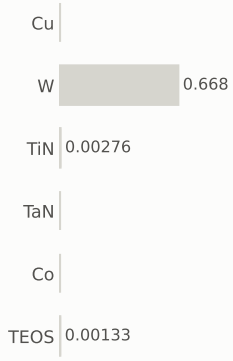
embodied carbon up to each step (kgCO2e per wafer)



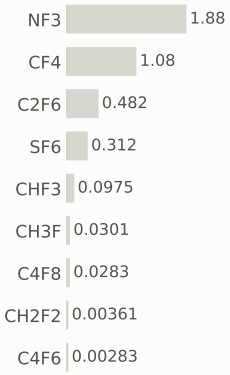
steps by process area



materials deposited (g)



gases emitted (g)



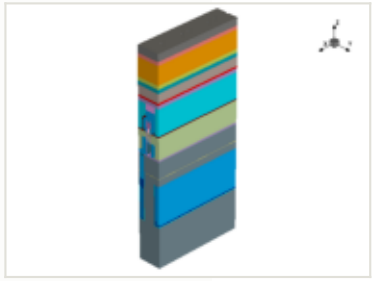
Step #2140: EUV

litho_patterning · EUV

1.96	243.5	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

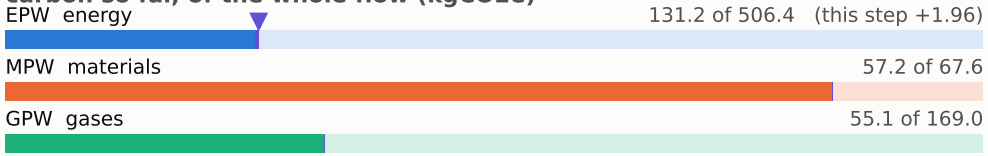


this step

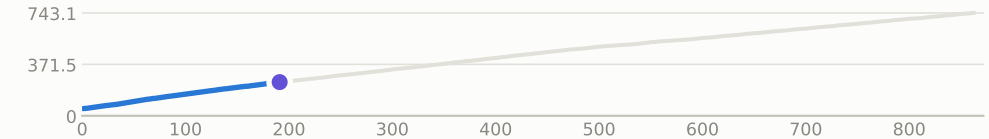
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

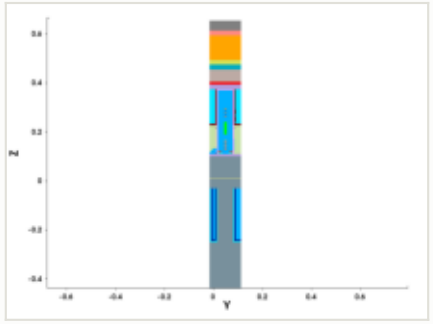
carbon so far, of the whole flow (kgCO2e)



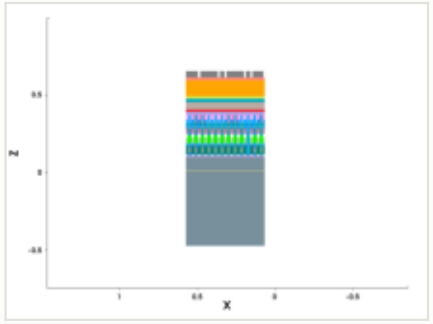
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

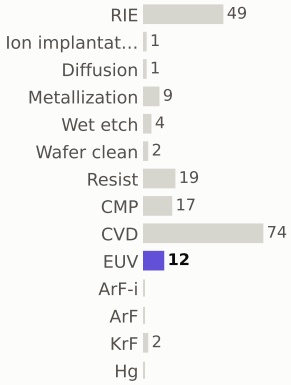


y cut at 0.049 µm

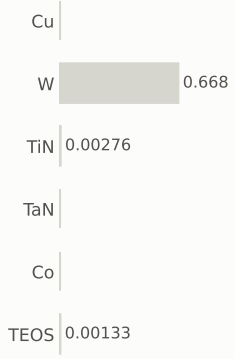


mask: layer 13:0 (GDSII)

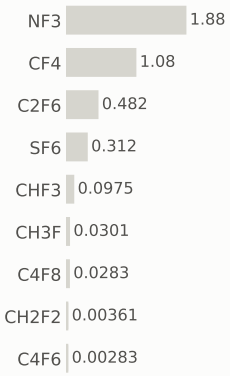
steps by process area



materials deposited (g)



gases emitted (g)



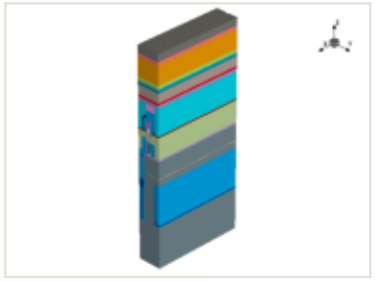
Step #2150: RIE

etch_anisotropic · RIE

1.04	244.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

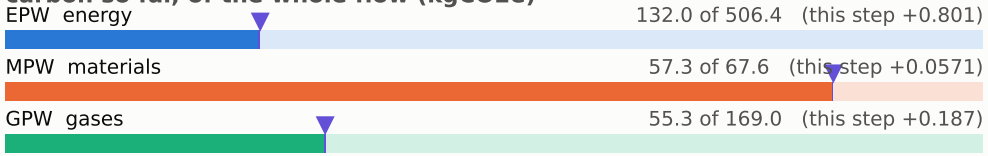
zoomed to fit



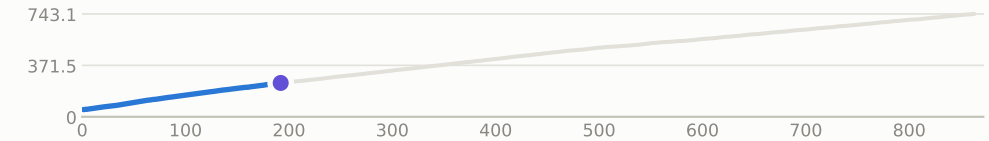
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

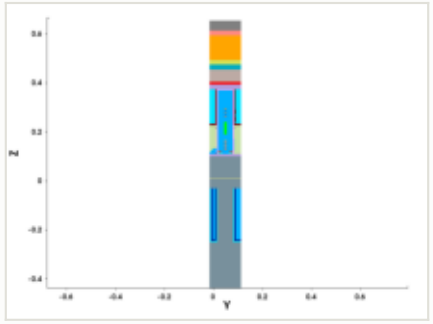
carbon so far, of the whole flow (kgCO2e)



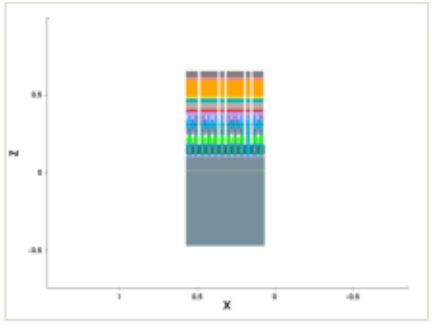
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



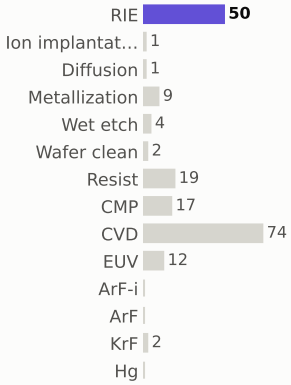
y cut at 0.049 µm



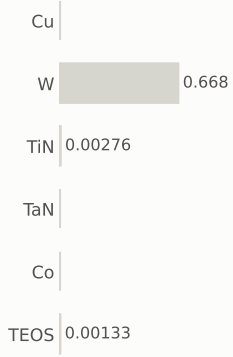
mask for this step

no mask for this step

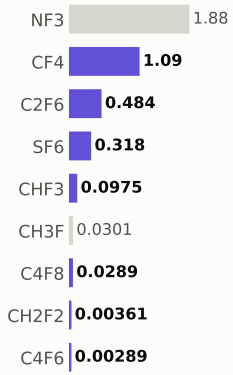
steps by process area



materials deposited (g)



gases emitted (g)



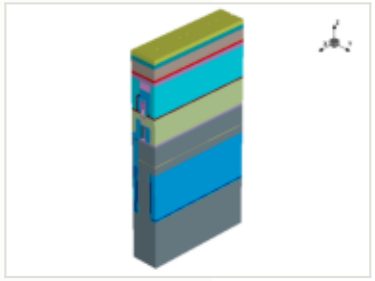
Step #2160: Wafer_clean

etch_strip · Wafer_clean

0.169	244.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

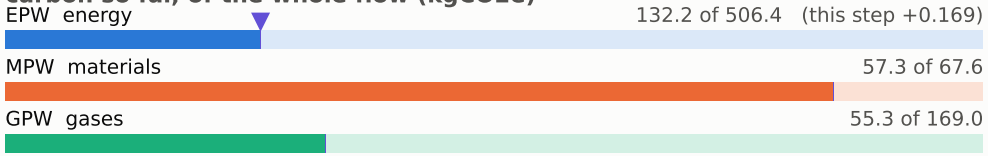


this step

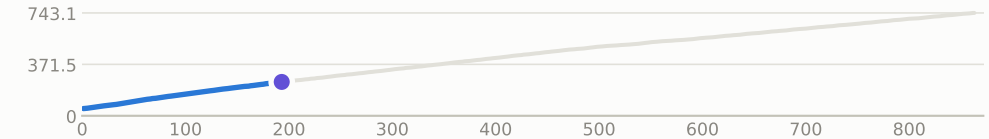
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

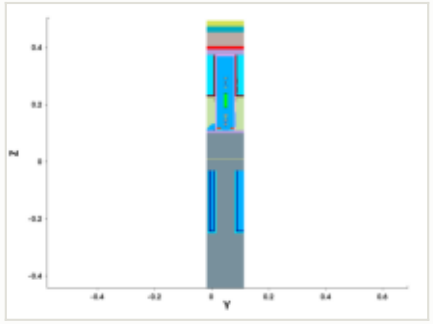
carbon so far, of the whole flow (kgCO2e)



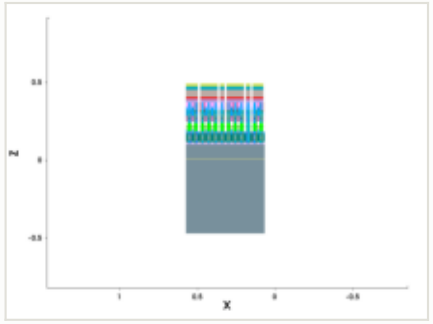
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



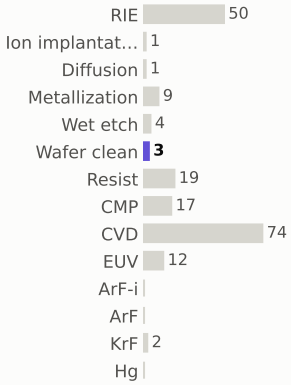
y cut at 0.049 µm



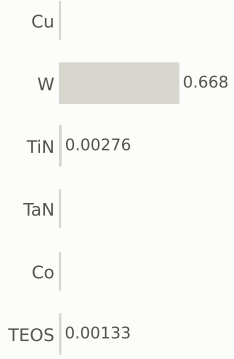
mask for this step

no mask for this step

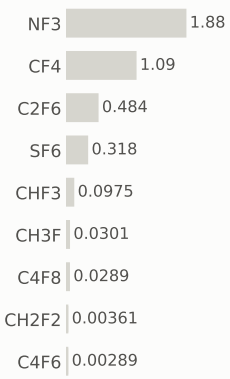
steps by process area



materials deposited (g)



gases emitted (g)



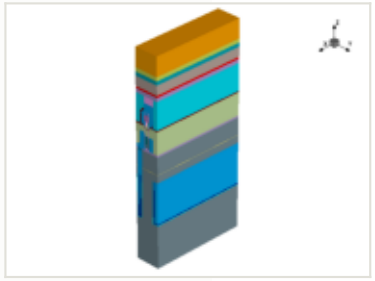
Step #2170: Resist

depo_fill · Resist

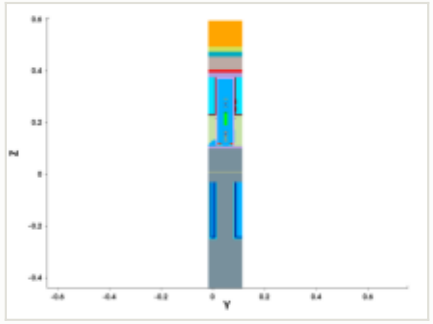
0.169	244.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

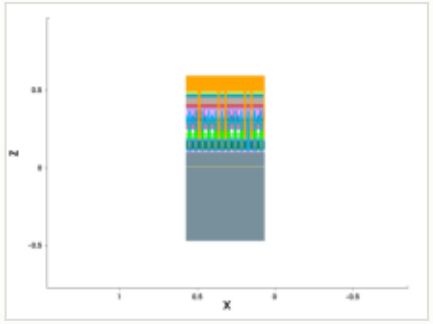
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



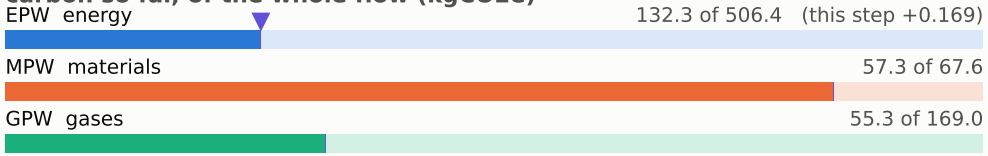
mask for this step

no mask for this step

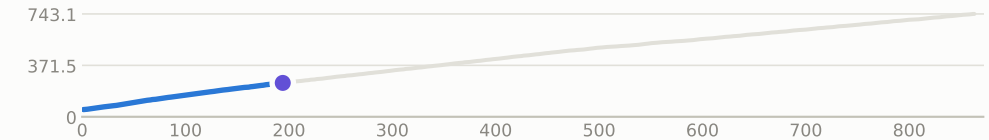
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

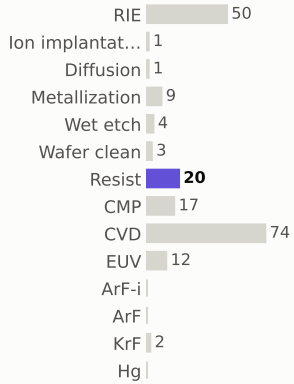
carbon so far, of the whole flow (kgCO2e)



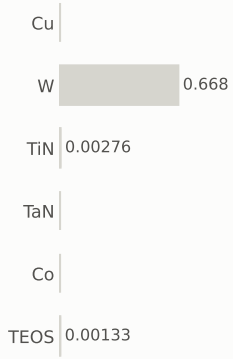
embodied carbon up to each step (kgCO2e per wafer)



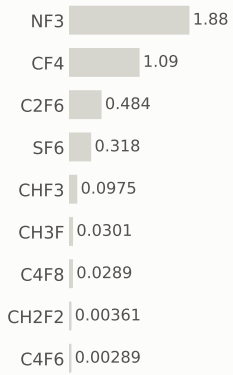
steps by process area



materials deposited (g)



gases emitted (g)



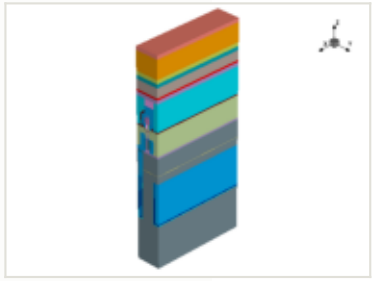
Step #2180: CVD

depo_fill · CVD

1.31	246.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

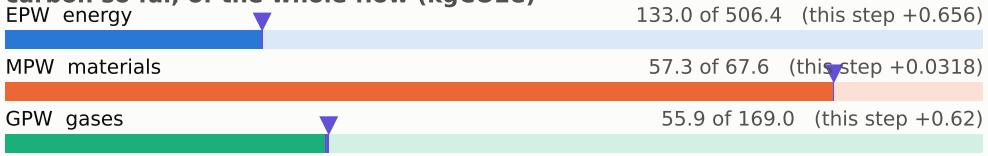
zoomed to fit



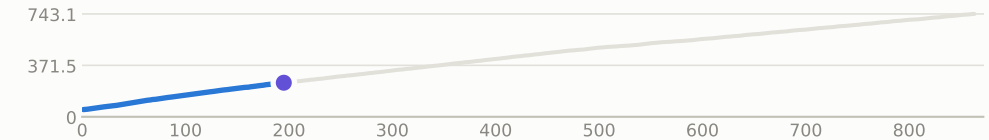
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.02 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

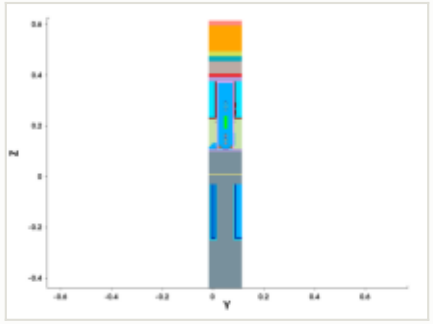
carbon so far, of the whole flow (kgCO2e)



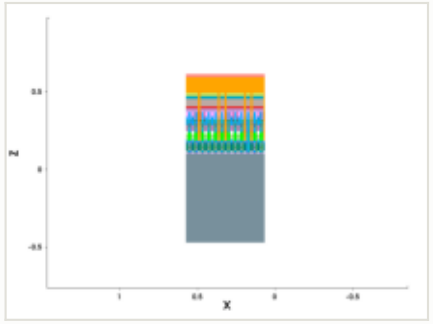
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



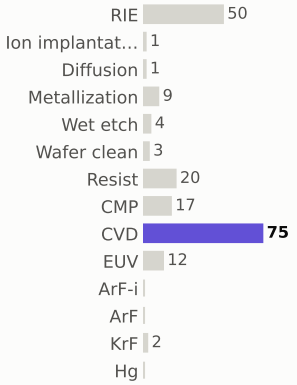
y cut at 0.049 µm



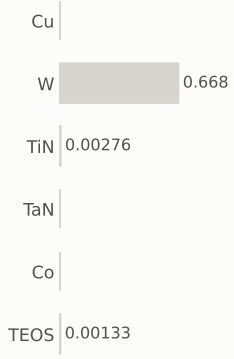
mask for this step

no mask for this step

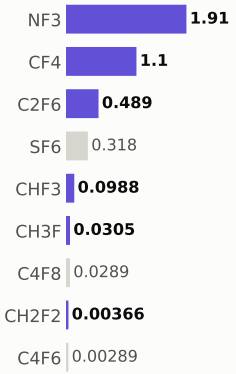
steps by process area



materials deposited (g)



gases emitted (g)



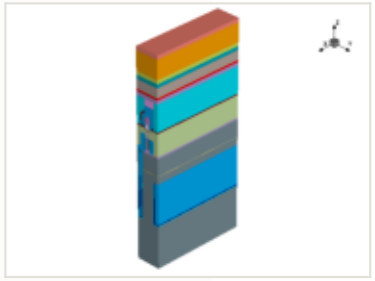
Step #2190: Wafer_clean

clean · Wafer_clean

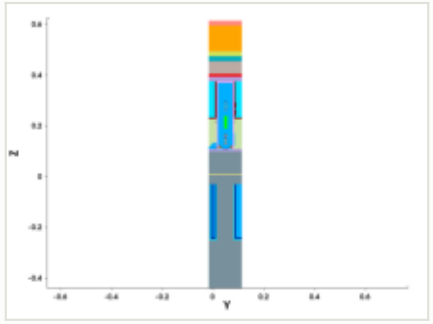
0.169	246.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

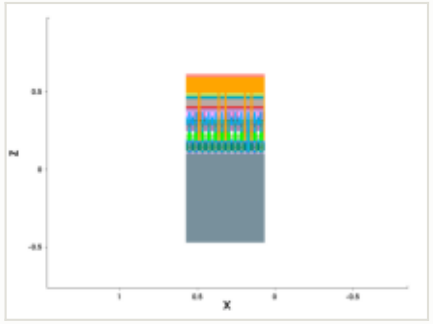
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

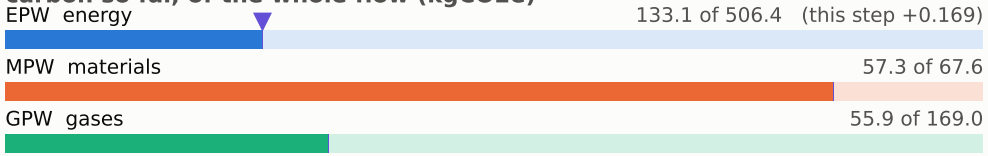
no mask for this step

this step

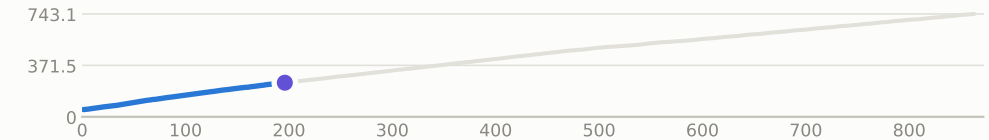
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

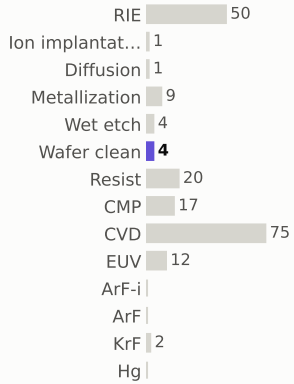
carbon so far, of the whole flow (kgCO2e)



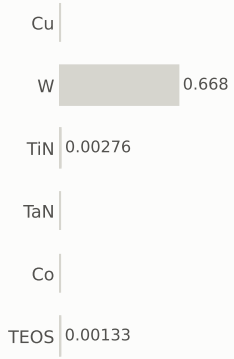
embodied carbon up to each step (kgCO2e per wafer)



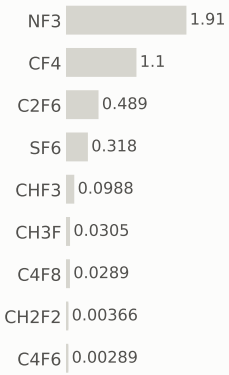
steps by process area



materials deposited (g)



gases emitted (g)



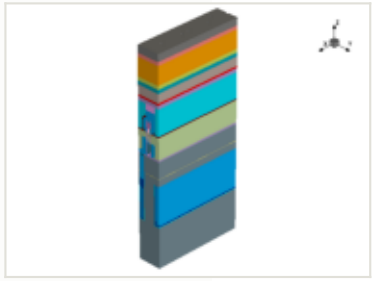
Step #2200: EUV

litho_patterning · EUV

1.96	248.3	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

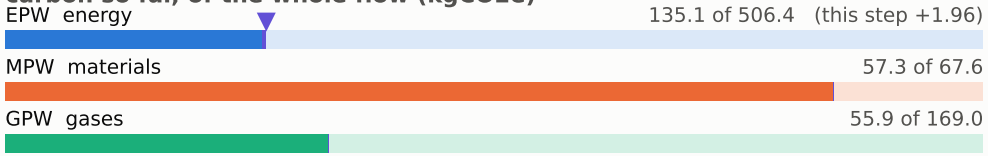


this step

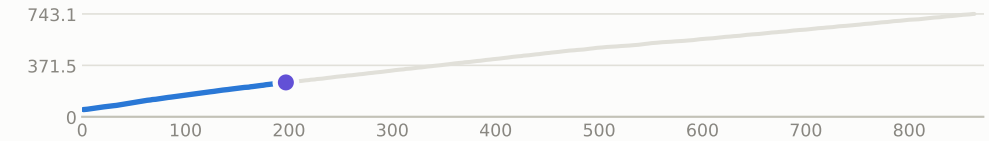
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

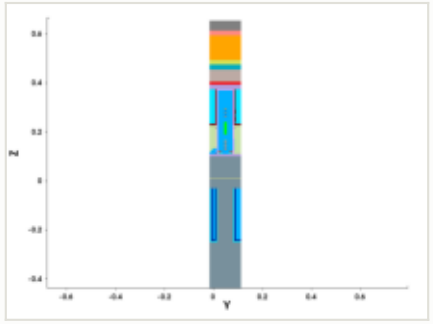
carbon so far, of the whole flow (kgCO2e)



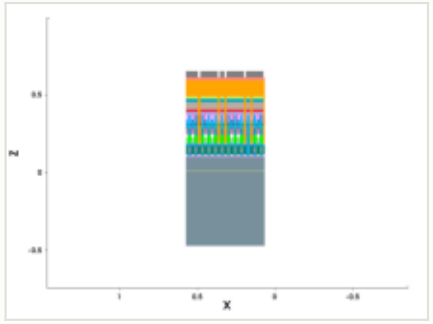
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

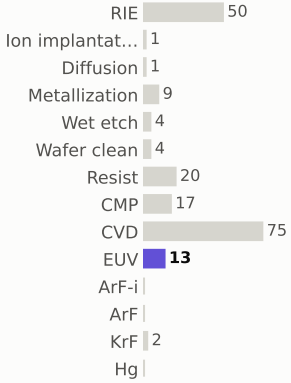


y cut at 0.049 µm

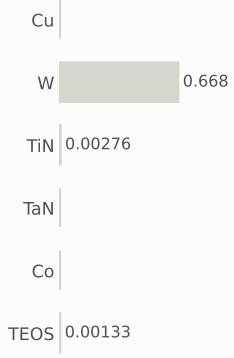


mask: layer 11:2 (GDSII)

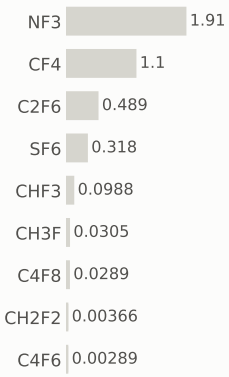
steps by process area



materials deposited (g)



gases emitted (g)



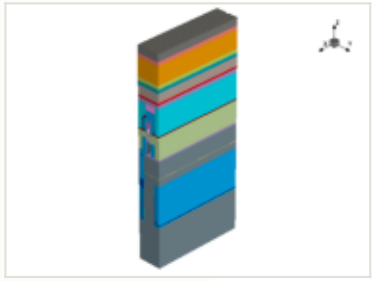
Step #2210: RIE

etch_anisotropic · RIE

1.04	249.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

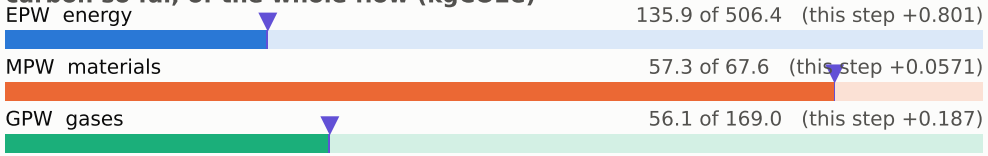
zoomed to fit



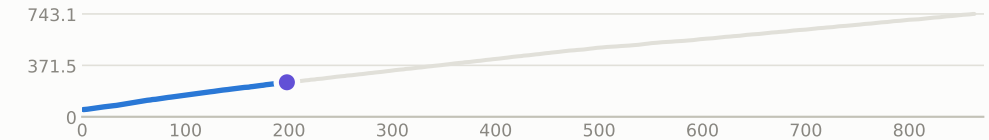
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

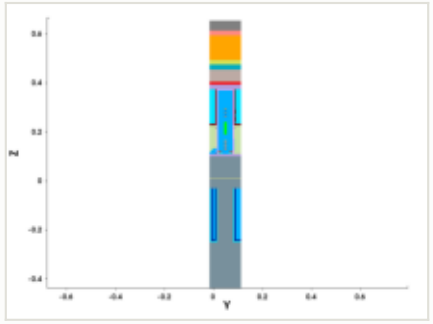
carbon so far, of the whole flow (kgCO2e)



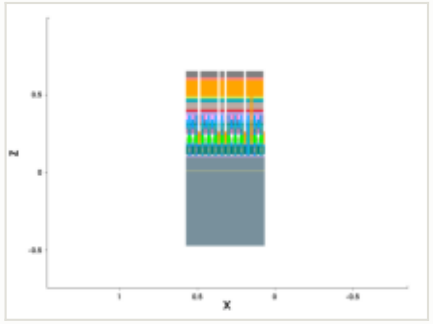
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



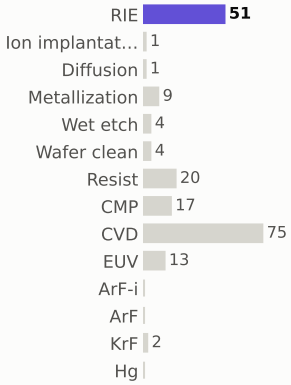
y cut at 0.049 µm



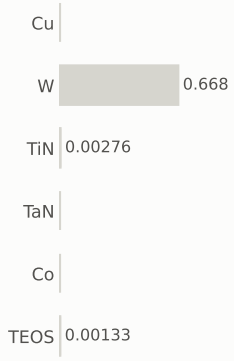
mask for this step

no mask for this step

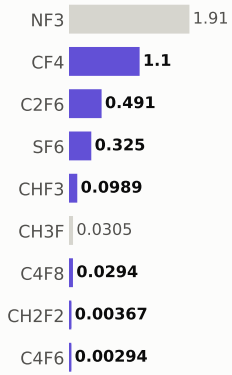
steps by process area



materials deposited (g)



gases emitted (g)



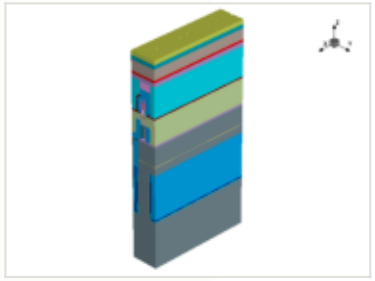
Step #2220: Wafer_clean

etch_strip · Wafer_clean

0.169	249.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

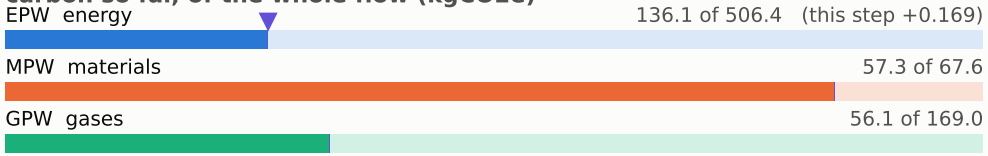


this step

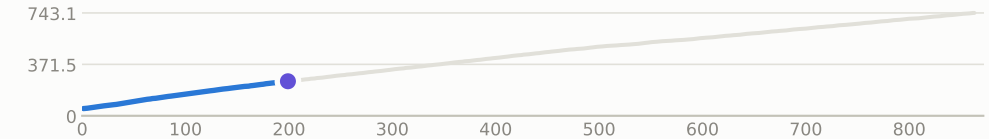
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

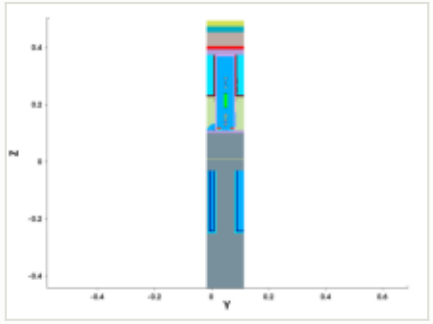
carbon so far, of the whole flow (kgCO2e)



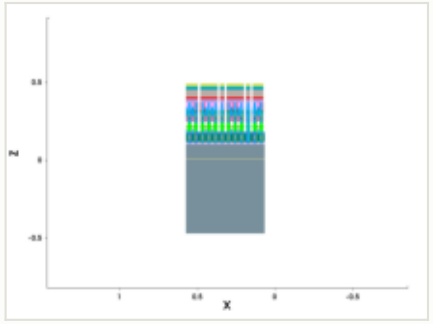
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



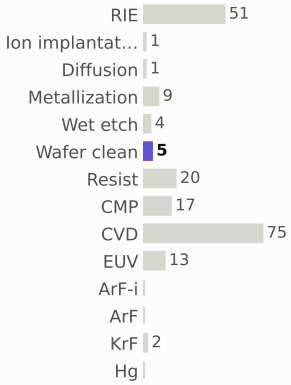
y cut at 0.049 µm



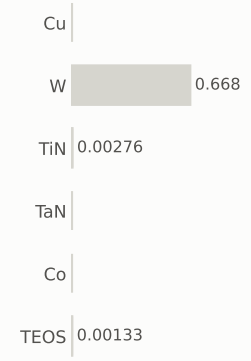
mask for this step

no mask for this step

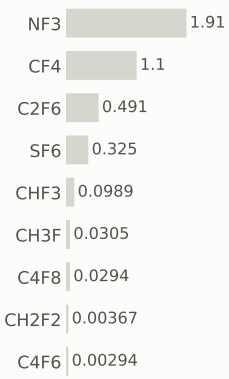
steps by process area



materials deposited (g)



gases emitted (g)



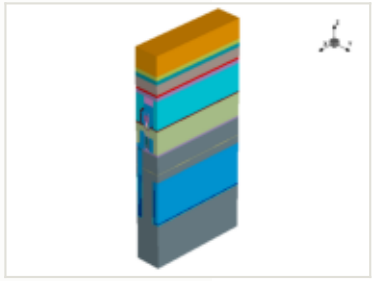
Step #2230: Resist

depo_fill · Resist

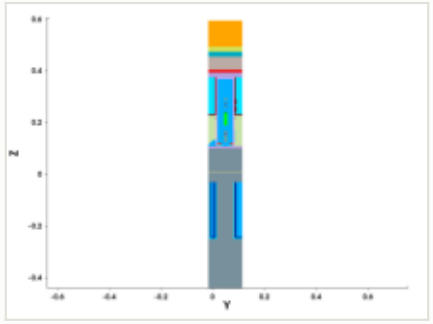
0.169	249.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

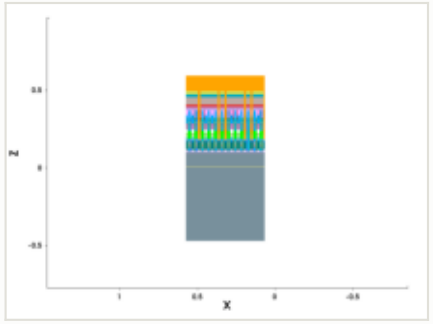
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



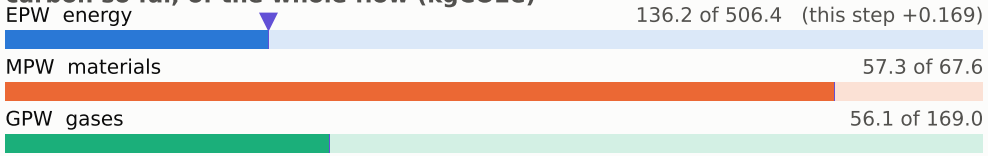
mask for this step

no mask for this step

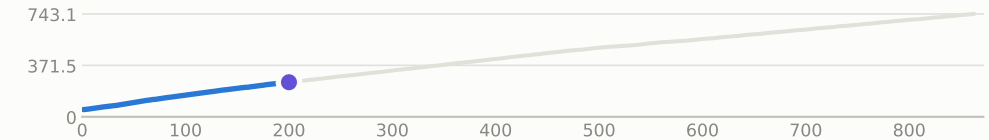
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

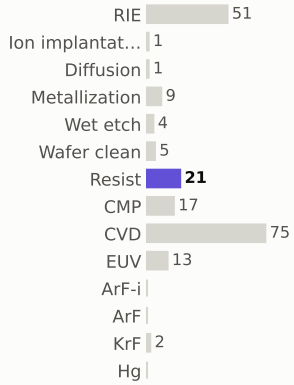
carbon so far, of the whole flow (kgCO2e)



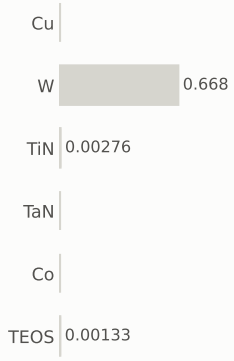
embodied carbon up to each step (kgCO2e per wafer)



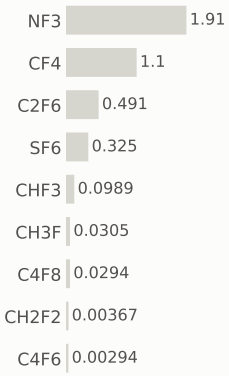
steps by process area



materials deposited (g)



gases emitted (g)



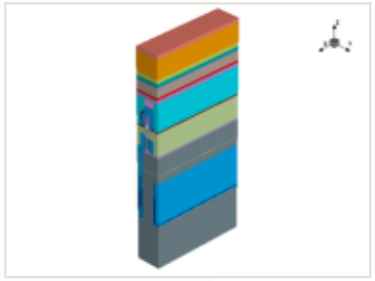
Step #2240: CVD

depo_fill · CVD

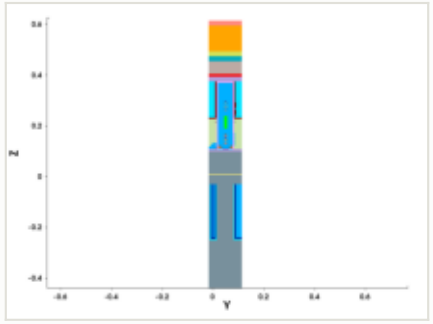
1.31	251.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

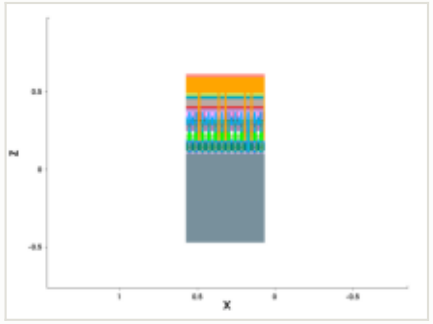
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

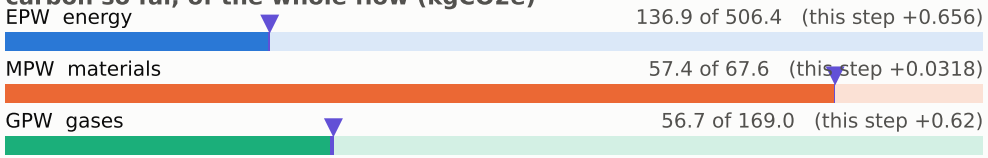
no mask for this step

this step

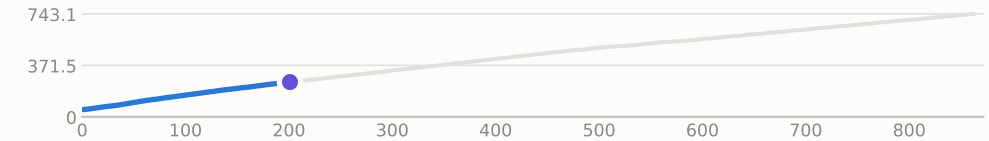
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

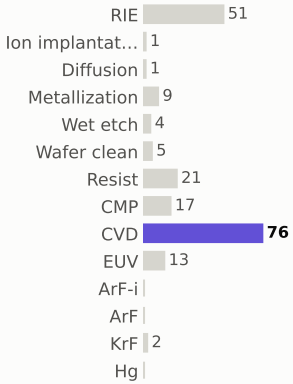
carbon so far, of the whole flow (kgCO2e)



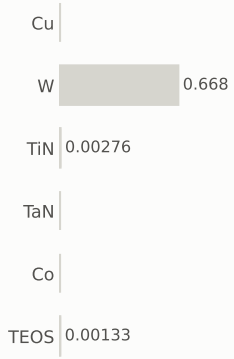
embodied carbon up to each step (kgCO2e per wafer)



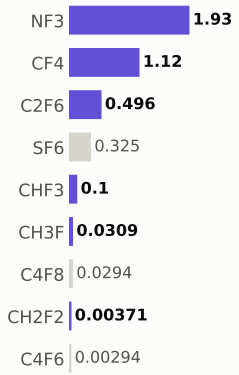
steps by process area



materials deposited (g)



gases emitted (g)



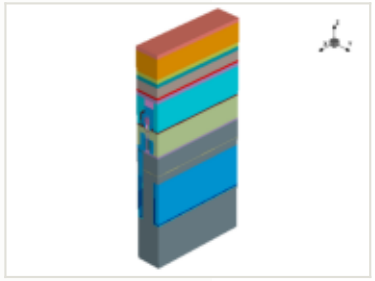
Step #2250: Wafer_clean

clean · Wafer_clean

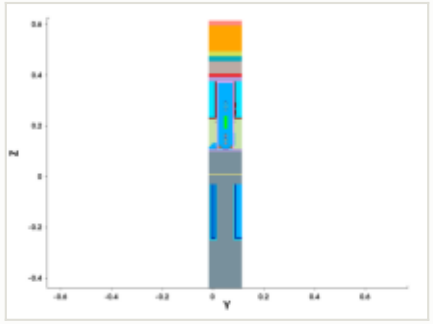
0.169	251.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

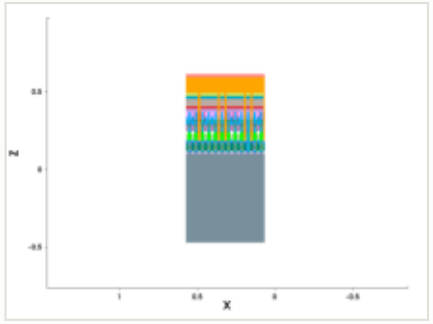
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

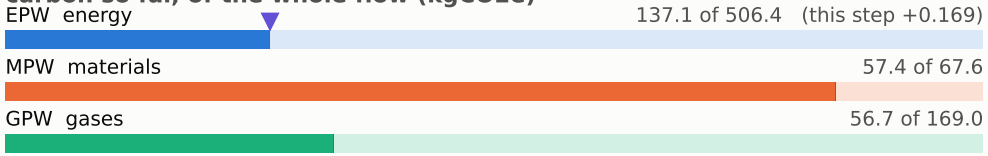
no mask for this step

this step

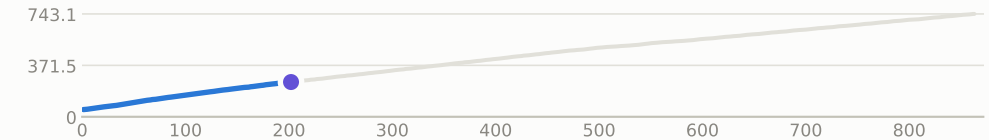
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

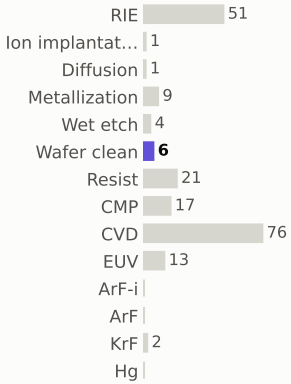
carbon so far, of the whole flow (kgCO2e)



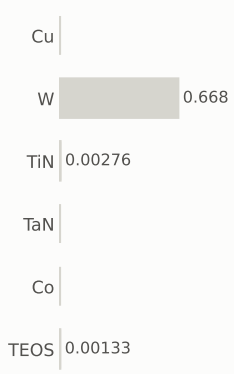
embodied carbon up to each step (kgCO2e per wafer)



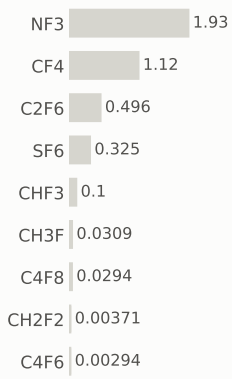
steps by process area



materials deposited (g)



gases emitted (g)



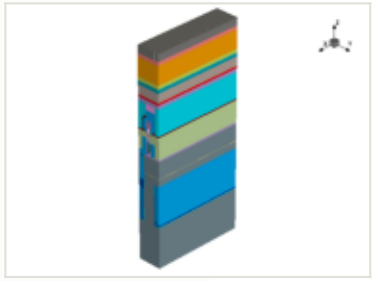
Step #2260: EUV

litho_patterning · EUV

1.96	253.1	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

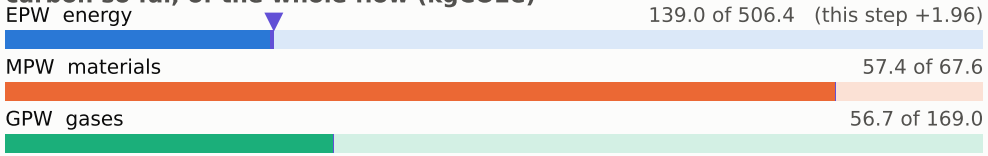


this step

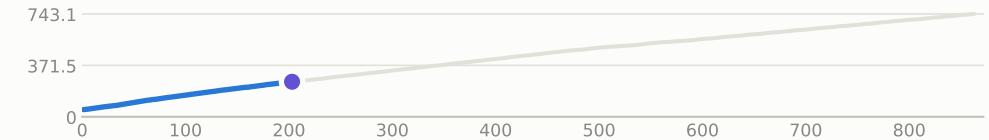
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

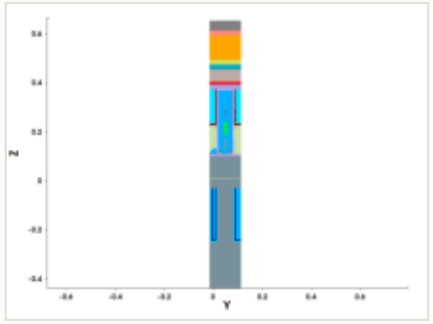
carbon so far, of the whole flow (kgCO2e)



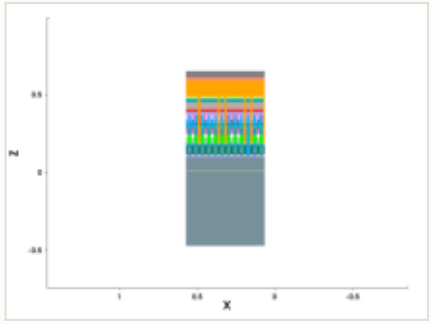
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

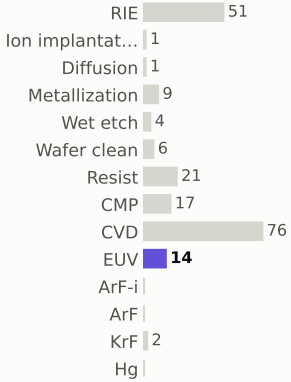


y cut at 0.049 µm

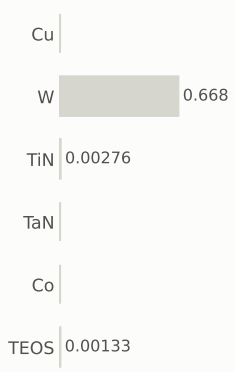


mask: layer 9:2 (GDSII)

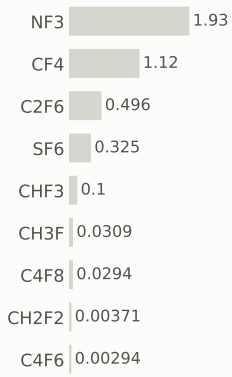
steps by process area



materials deposited (g)



gases emitted (g)



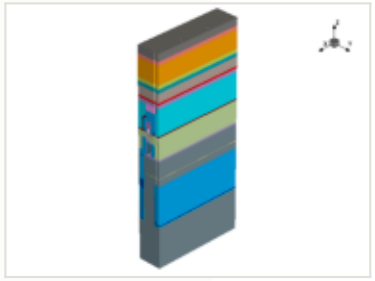
Step #2270: RIE

etch_anisotropic · RIE

1.04	254.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

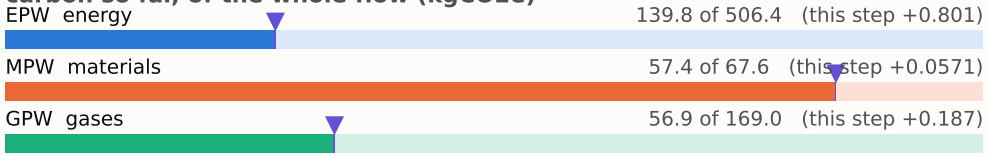
zoomed to fit



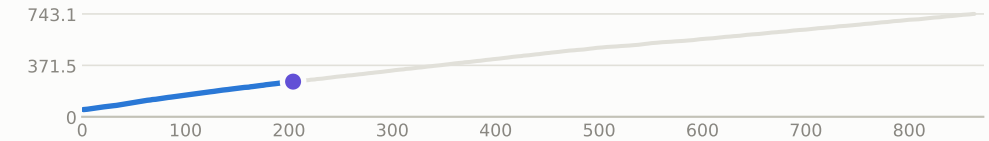
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

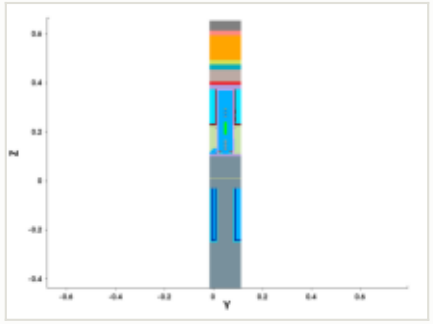
carbon so far, of the whole flow (kgCO2e)



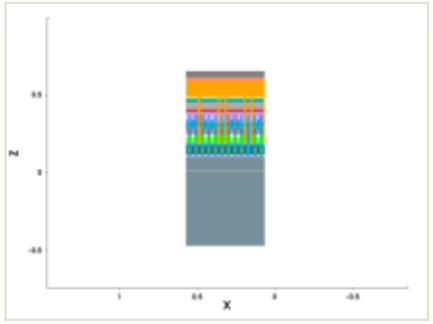
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



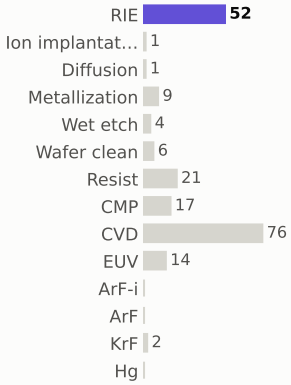
y cut at 0.049 µm



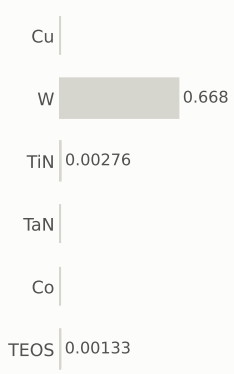
mask for this step

no mask for this step

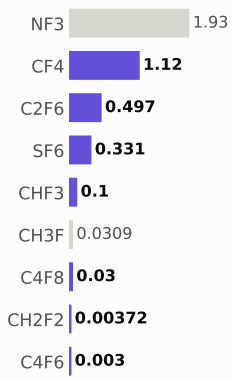
steps by process area



materials deposited (g)



gases emitted (g)



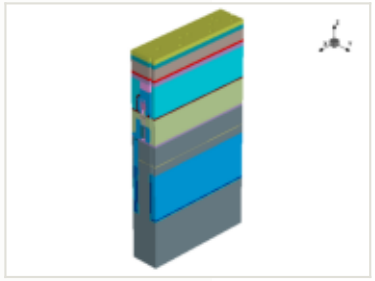
Step #2280: Wafer_clean

etch_strip · Wafer_clean

0.169	254.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

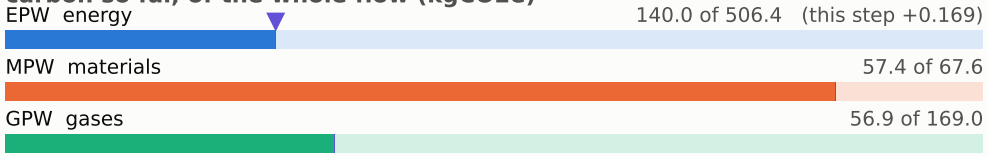


this step

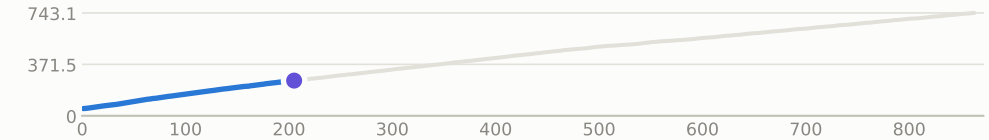
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

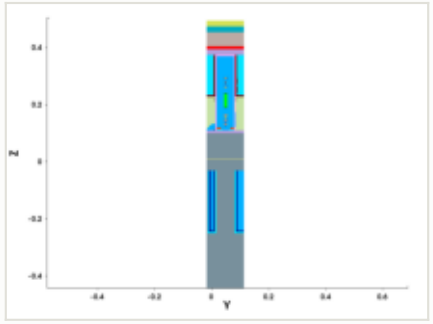
carbon so far, of the whole flow (kgCO2e)



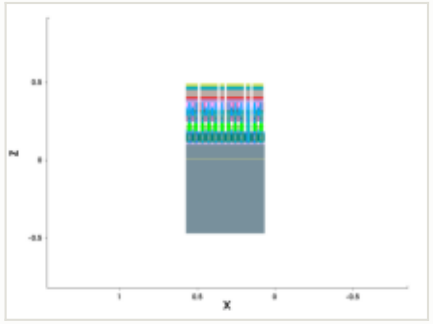
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



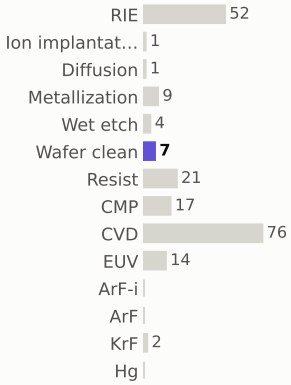
y cut at 0.049 µm



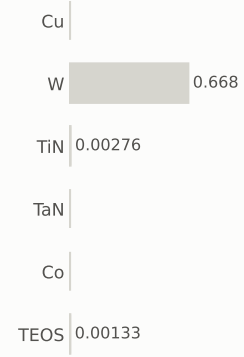
mask for this step

no mask for this step

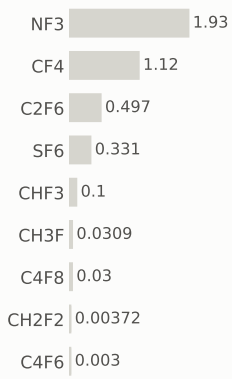
steps by process area



materials deposited (g)



gases emitted (g)



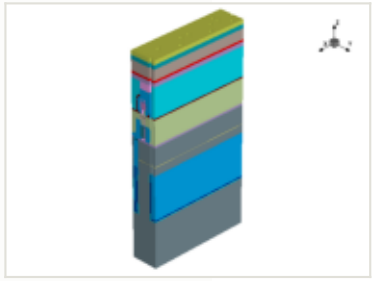
Step #2290: RIE

etch_noop · RIE

1.04	255.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



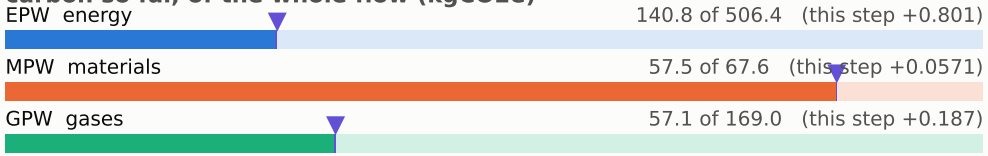
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

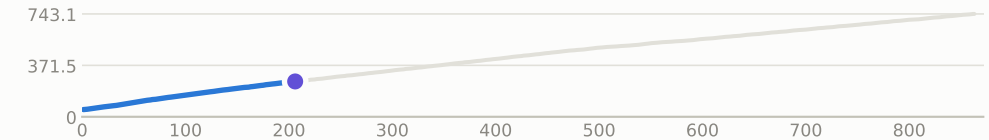
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

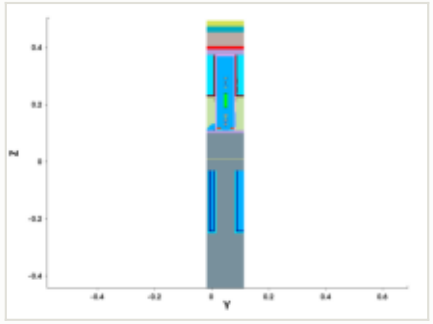
carbon so far, of the whole flow (kgCO2e)



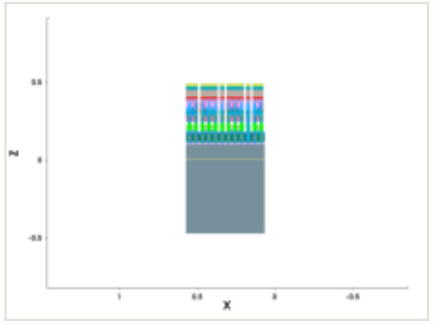
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



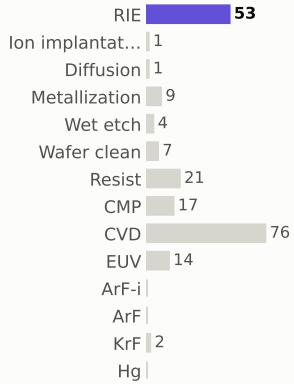
y cut at 0.049 µm



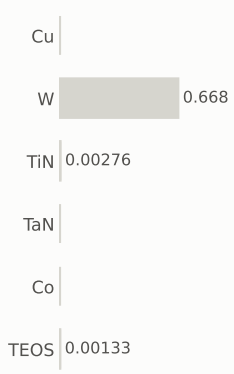
mask for this step

no mask for this step

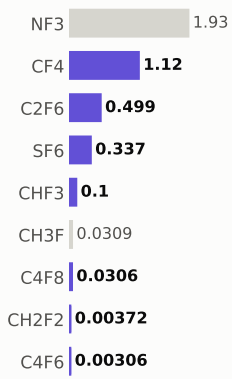
steps by process area



materials deposited (g)



gases emitted (g)



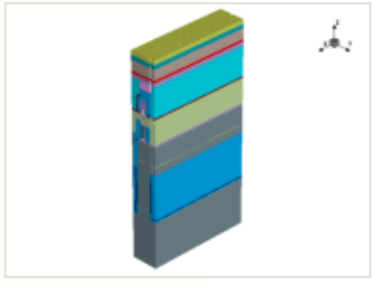
Step #2300: Wafer_clean

clean · Wafer_clean

0.169	255.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



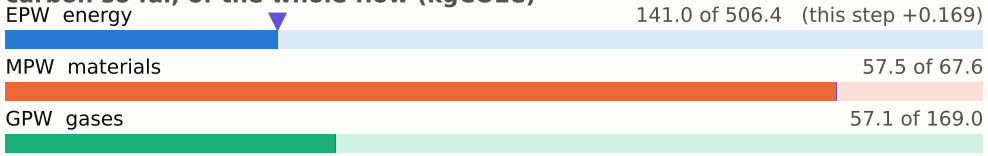
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

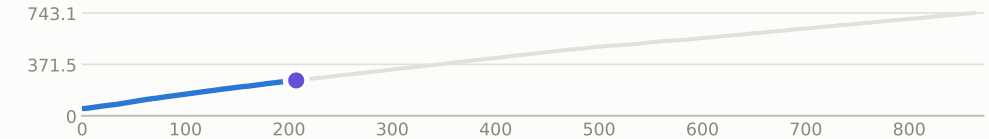
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

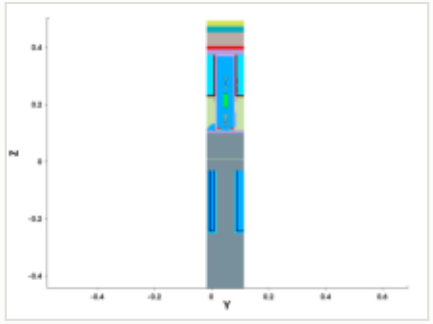
carbon so far, of the whole flow (kgCO2e)



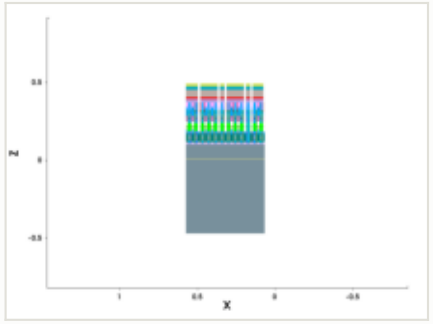
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



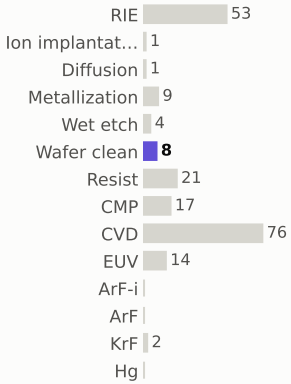
y cut at 0.049 µm



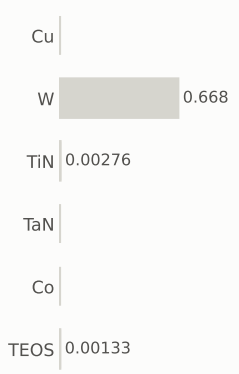
mask for this step

no mask for this step

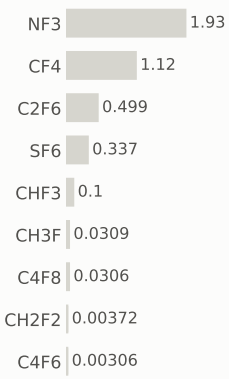
steps by process area



materials deposited (g)



gases emitted (g)



Step #2310: Metallization

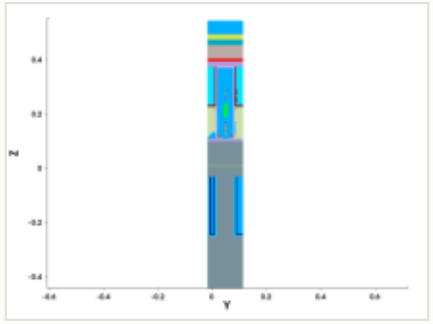
depo_fill · Metallization

1.21	256.7	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

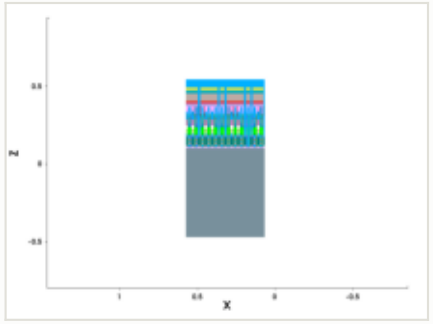
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

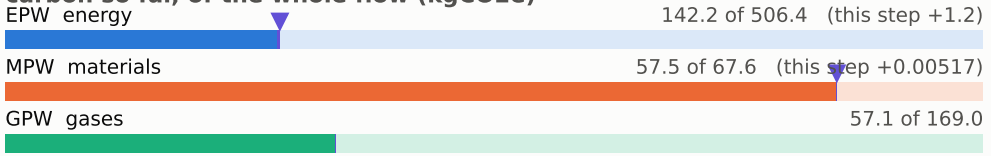
no mask for this step

this step

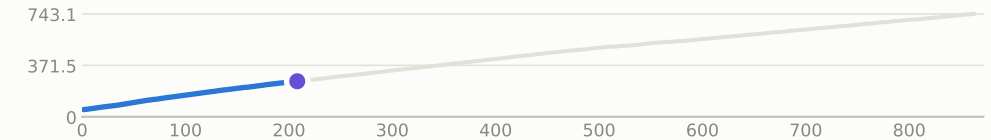
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

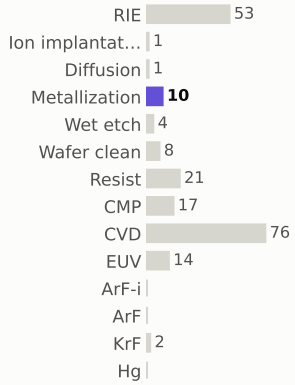
carbon so far, of the whole flow (kgCO2e)



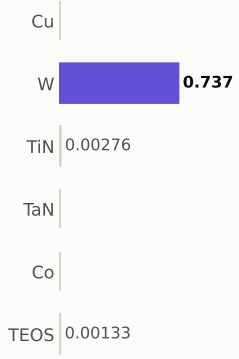
embodied carbon up to each step (kgCO2e per wafer)



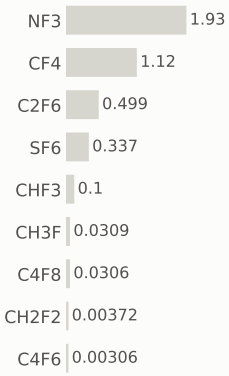
steps by process area



materials deposited (g)



gases emitted (g)



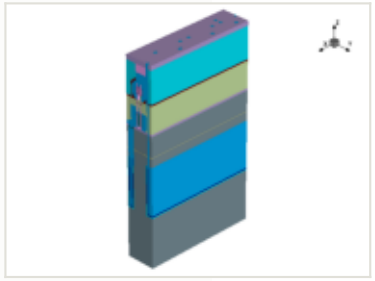
Step #2320: CMP

etch_cmp · CMP

0.169	256.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

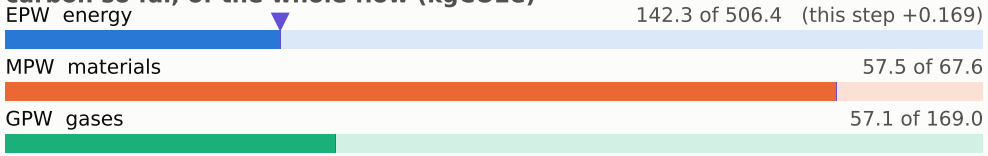


this step

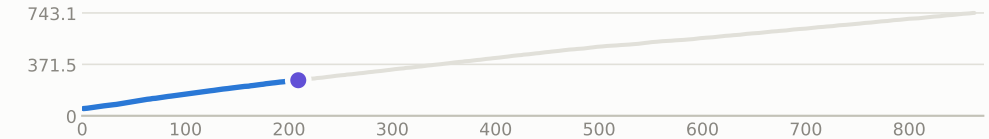
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

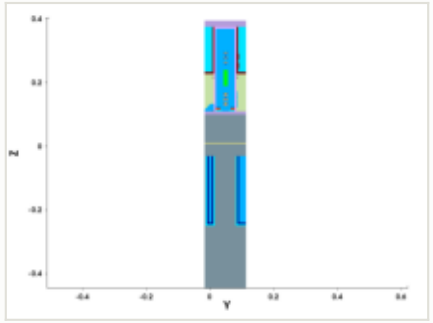
carbon so far, of the whole flow (kgCO2e)



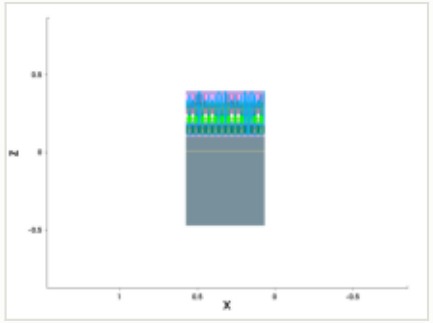
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



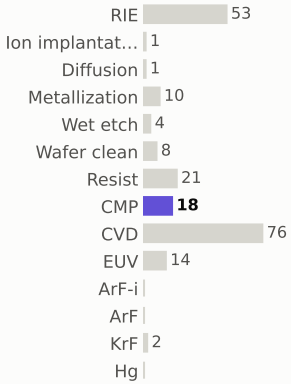
y cut at 0.049 µm



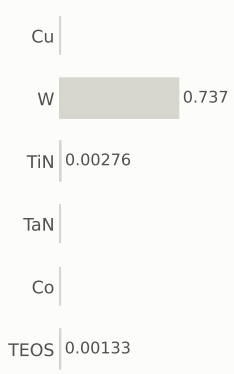
mask for this step

no mask for this step

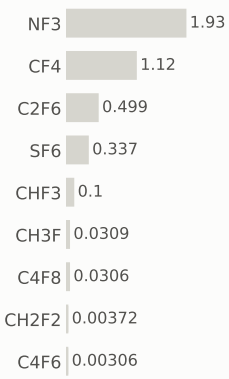
steps by process area



materials deposited (g)



gases emitted (g)



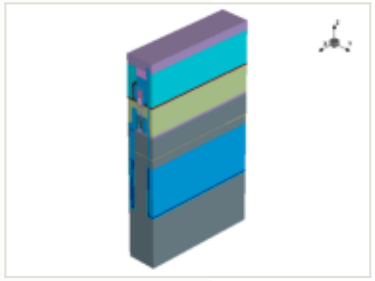
Step #2340: CVD

depo_fill · CVD

1.31	258.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

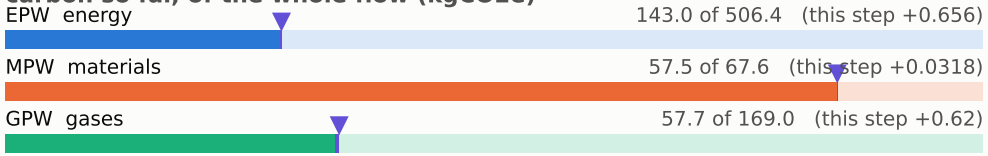


this step

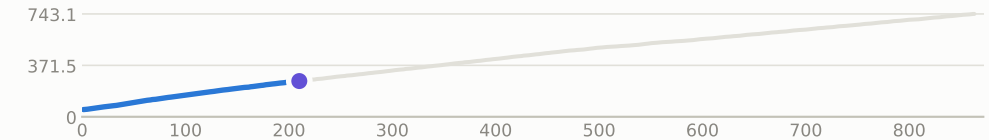
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

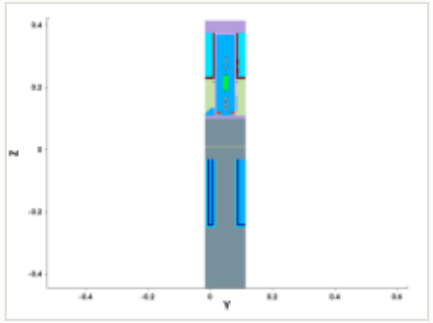
carbon so far, of the whole flow (kgCO2e)



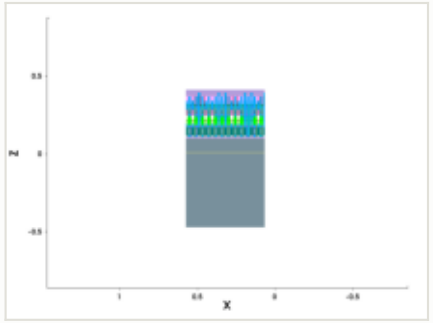
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



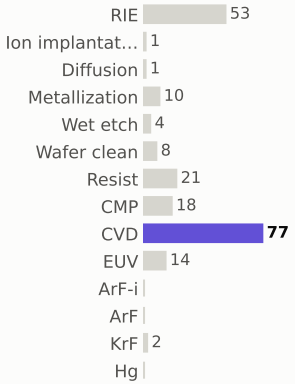
y cut at 0.049 µm



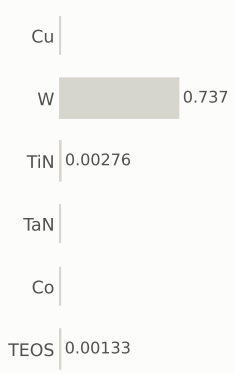
mask for this step

no mask for this step

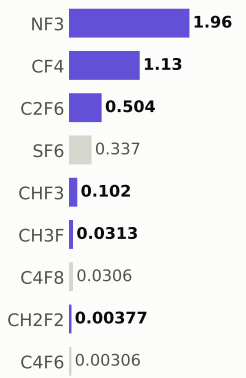
steps by process area



materials deposited (g)



gases emitted (g)



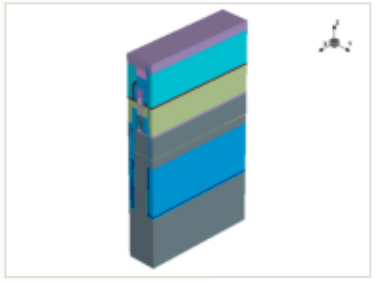
Step #2350: Diffusion

treatment · Diffusion

0.801	259.0	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

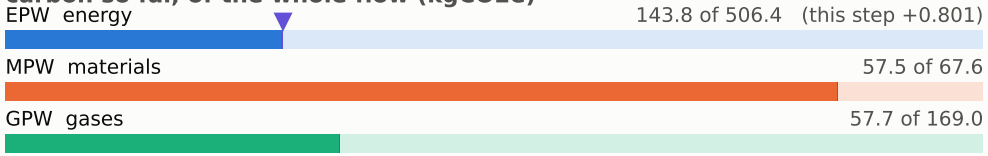


this step

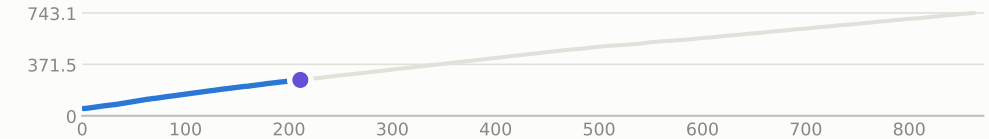
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

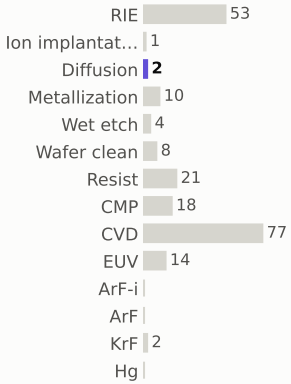
carbon so far, of the whole flow (kgCO2e)



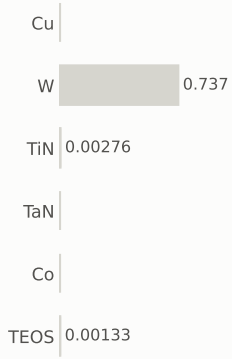
embodied carbon up to each step (kgCO2e per wafer)



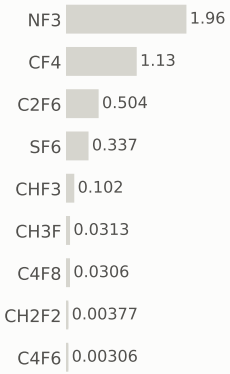
steps by process area



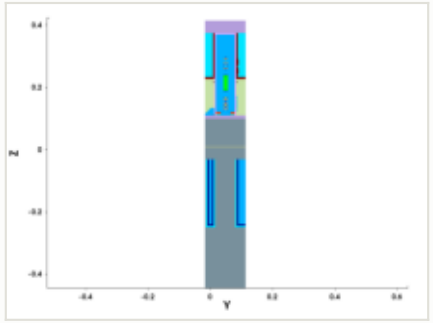
materials deposited (g)



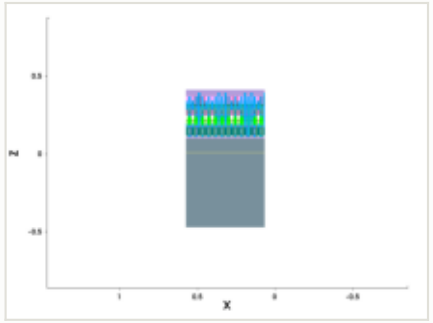
gases emitted (g)



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

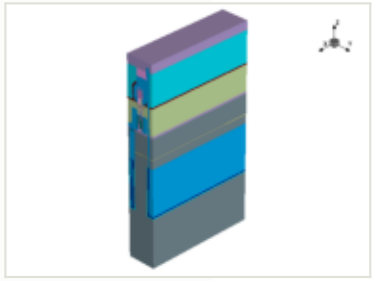
Step #2360: Wafer_clean

clean · Wafer_clean

0.169	259.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



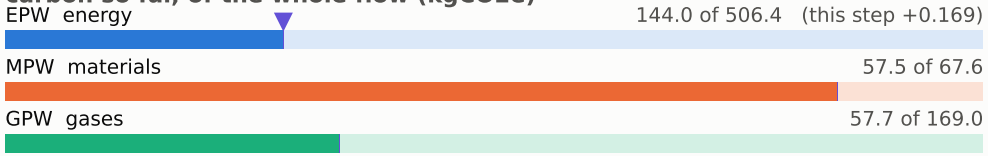
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

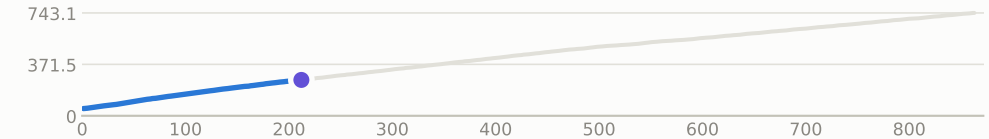
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

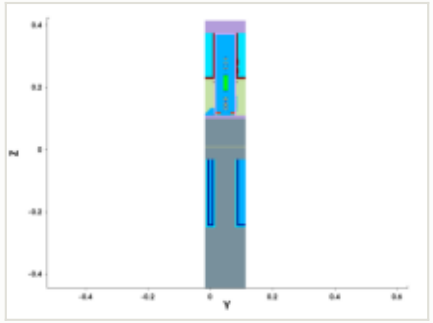
carbon so far, of the whole flow (kgCO2e)



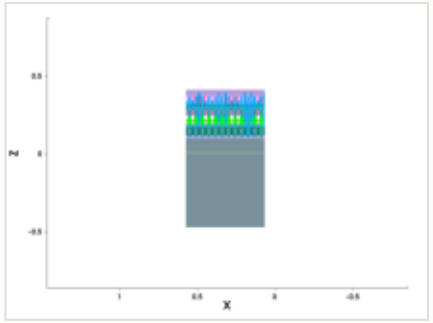
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



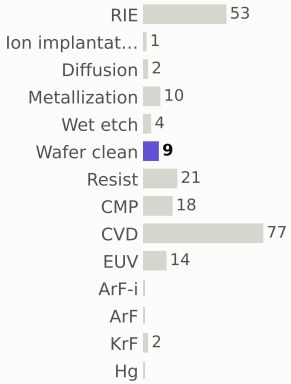
y cut at 0.049 µm



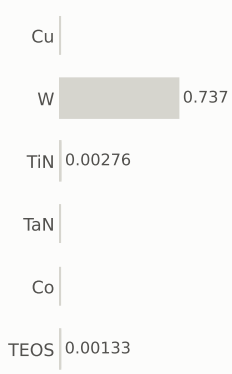
mask for this step

no mask for this step

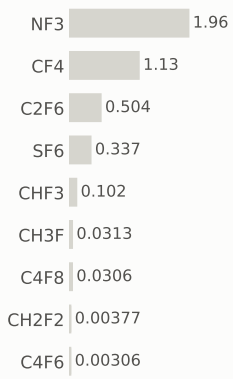
steps by process area



materials deposited (g)



gases emitted (g)



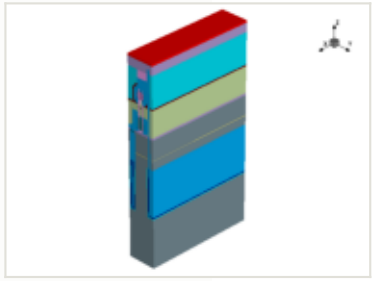
Step #2370: CVD

depo_fill · CVD

1.31	260.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

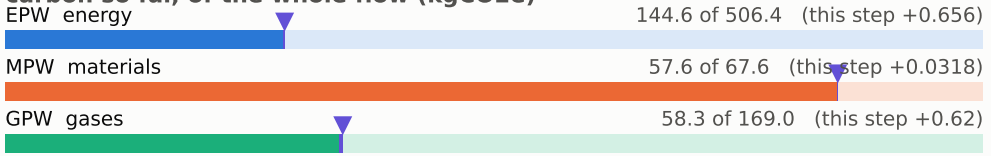


this step

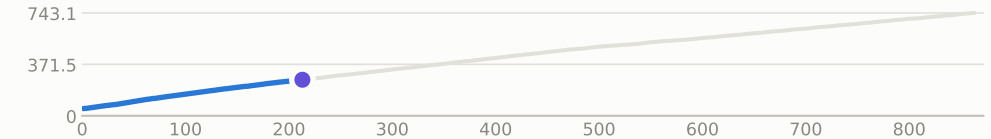
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

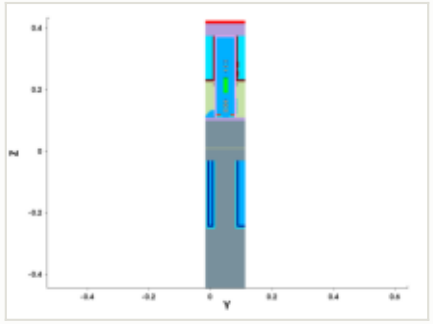
carbon so far, of the whole flow (kgCO2e)



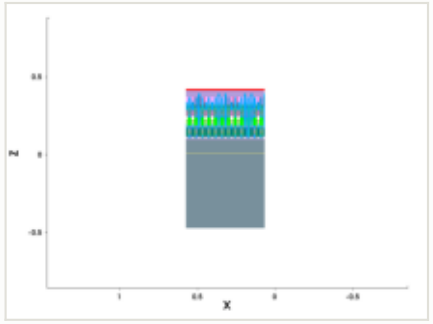
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



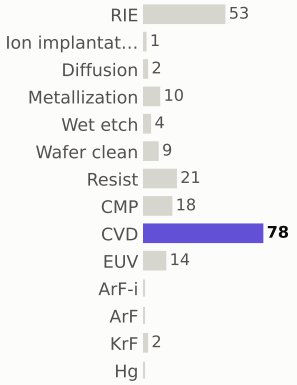
y cut at 0.049 µm



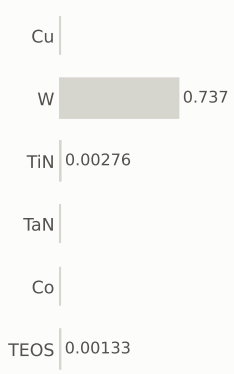
mask for this step

no mask for this step

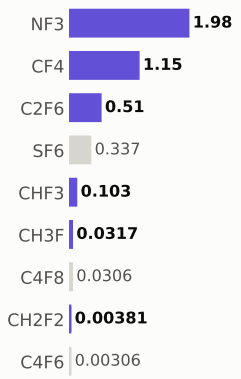
steps by process area



materials deposited (g)



gases emitted (g)



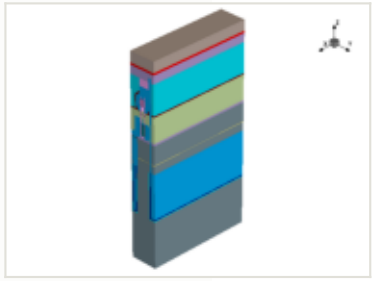
Step #2380: Resist

depo_fill · Resist

0.169	260.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

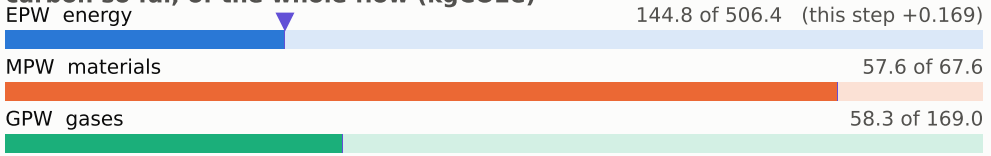


this step

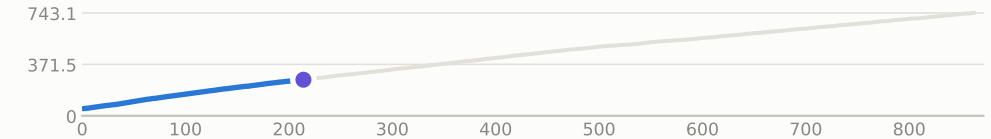
process area	Resist
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

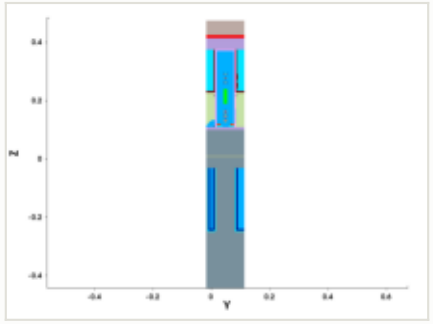
carbon so far, of the whole flow (kgCO2e)



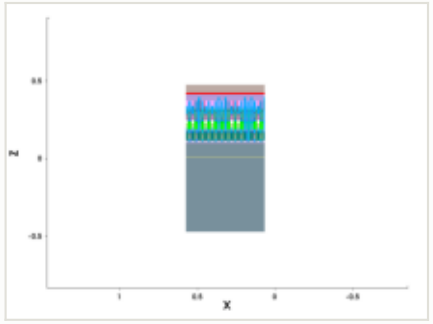
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



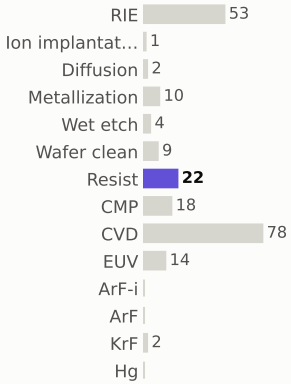
y cut at 0.049 µm



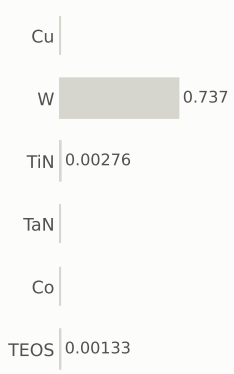
mask for this step

no mask for this step

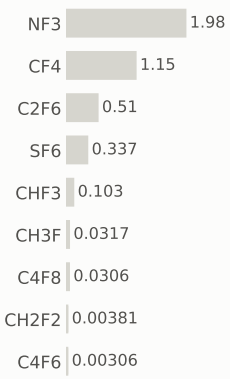
steps by process area



materials deposited (g)



gases emitted (g)



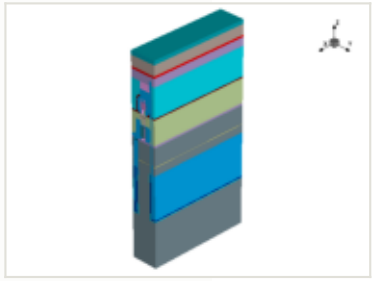
Step #2390: CVD

depo_fill · CVD

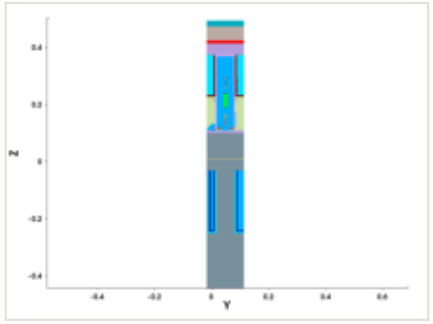
1.31	262.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

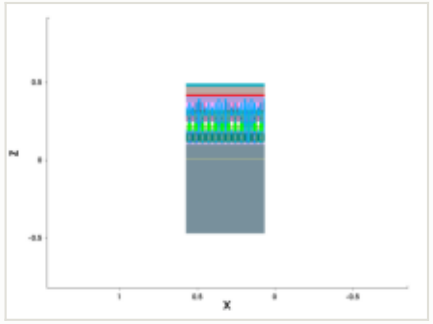
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

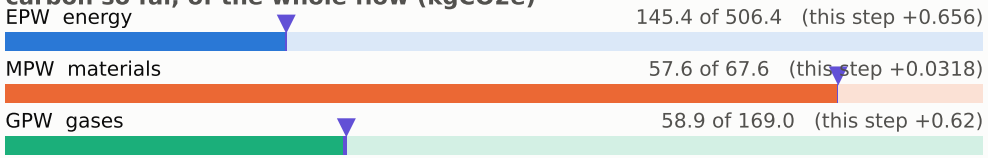
no mask for this step

this step

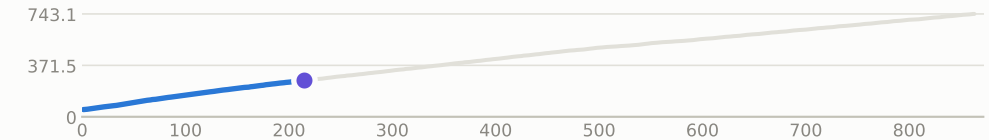
process area	CVD
material	TEOS
thickness	0.02 µm
material GWP	2.85952 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

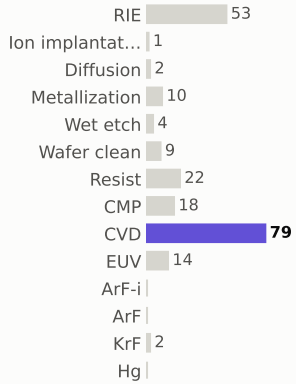
carbon so far, of the whole flow (kgCO2e)



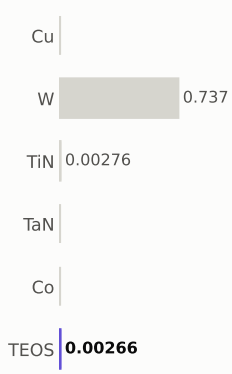
embodied carbon up to each step (kgCO2e per wafer)



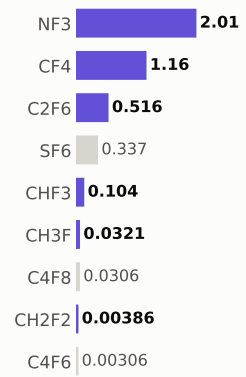
steps by process area



materials deposited (g)



gases emitted (g)



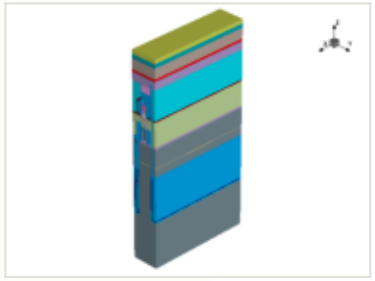
Step #2400: CVD

depo_fill · CVD

1.31	263.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

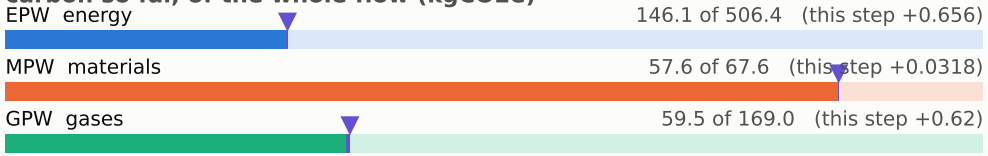


this step

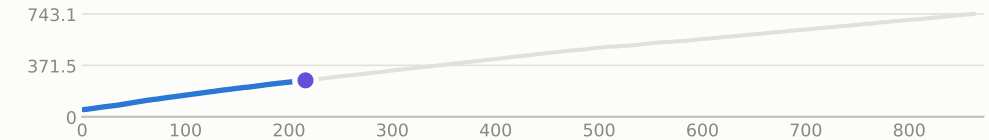
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

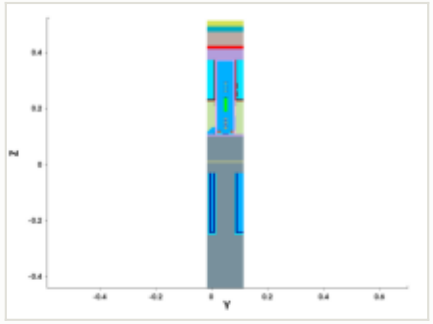
carbon so far, of the whole flow (kgCO2e)



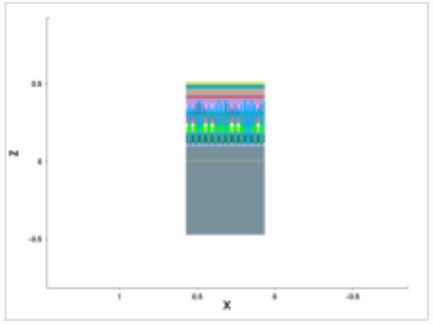
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



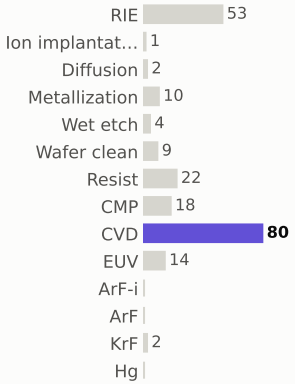
y cut at 0.049 µm



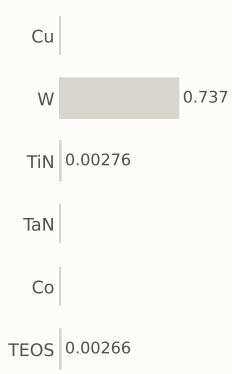
mask for this step

no mask for this step

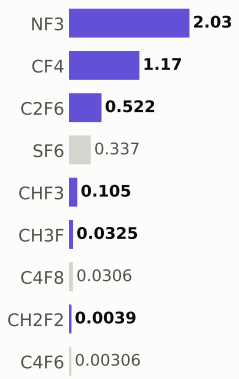
steps by process area



materials deposited (g)



gases emitted (g)



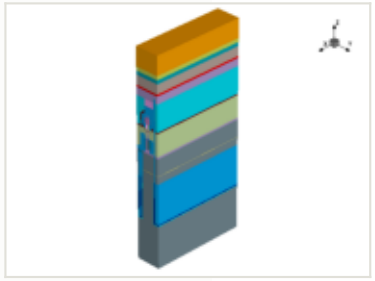
Step #2410: Resist

depo_fill · Resist

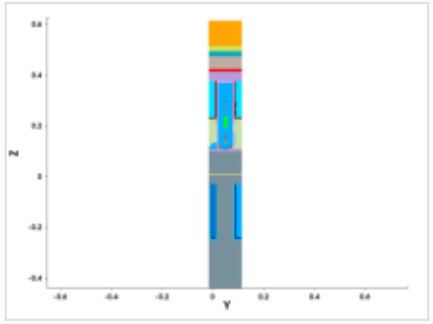
0.169	263.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

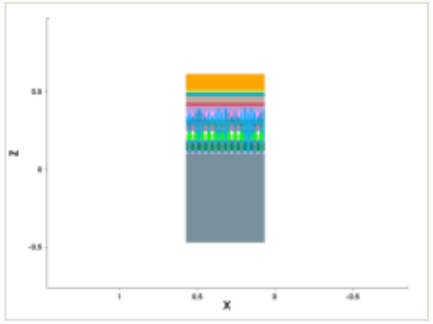
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



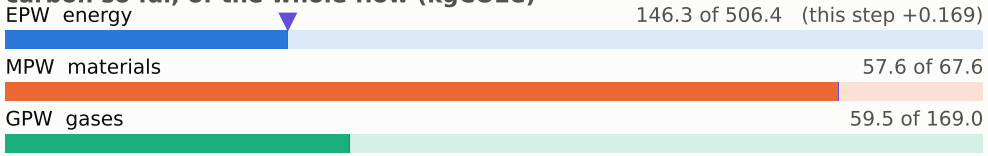
mask for this step

no mask for this step

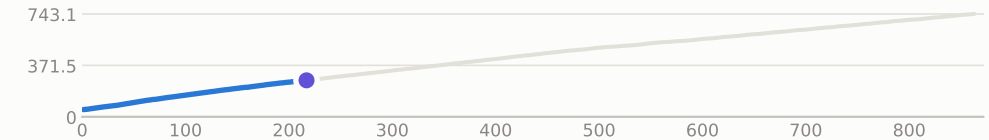
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

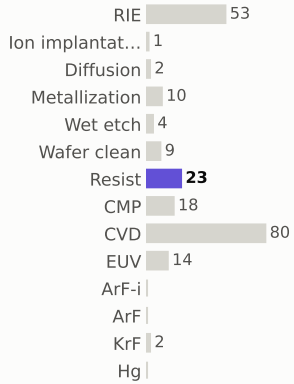
carbon so far, of the whole flow (kgCO2e)



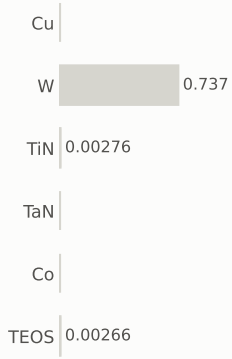
embodied carbon up to each step (kgCO2e per wafer)



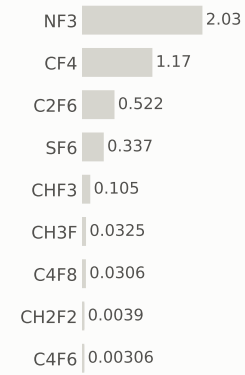
steps by process area



materials deposited (g)



gases emitted (g)



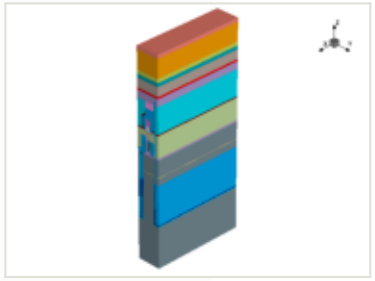
Step #2420: CVD

depo_fill · CVD

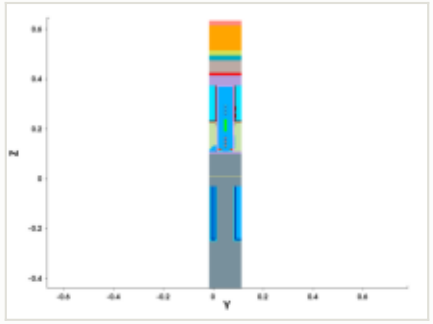
1.31	264.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

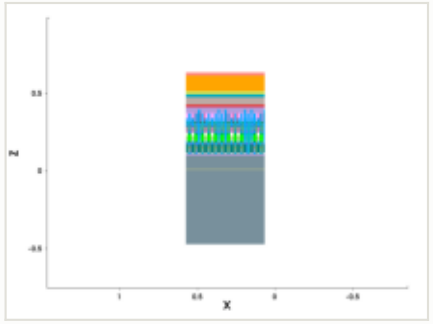
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

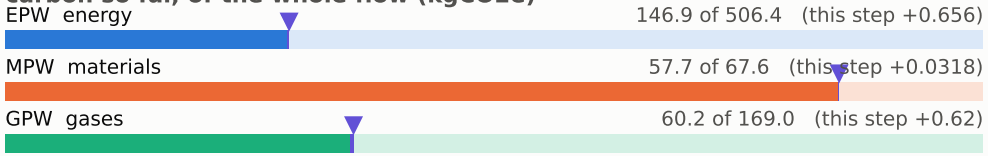
no mask for this step

this step

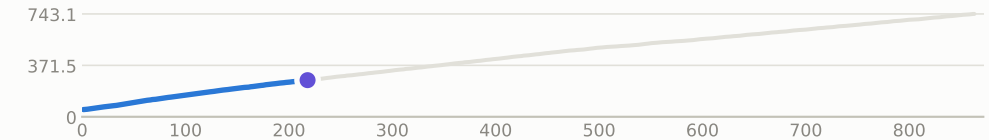
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

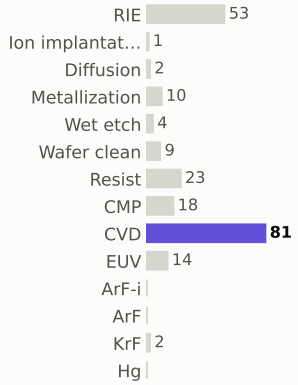
carbon so far, of the whole flow (kgCO2e)



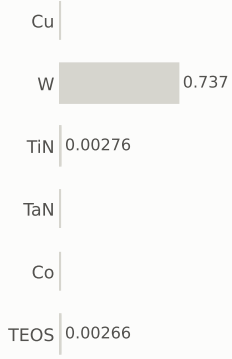
embodied carbon up to each step (kgCO2e per wafer)



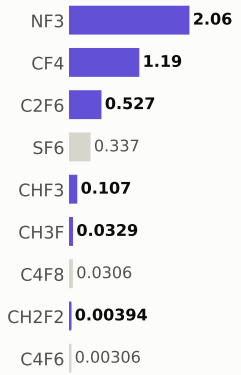
steps by process area



materials deposited (g)



gases emitted (g)



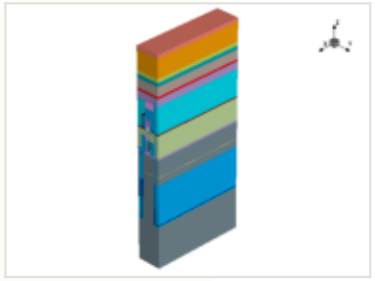
Step #2430: Wafer_clean

clean · Wafer_clean

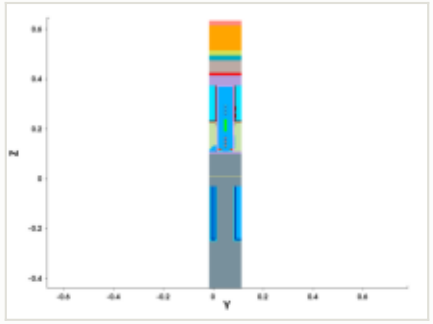
0.169	264.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

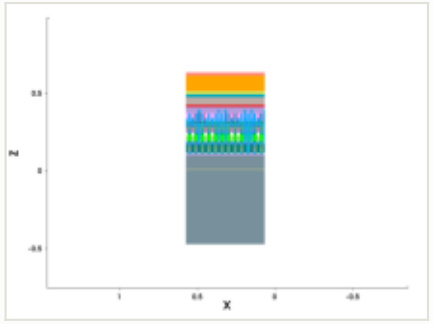
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

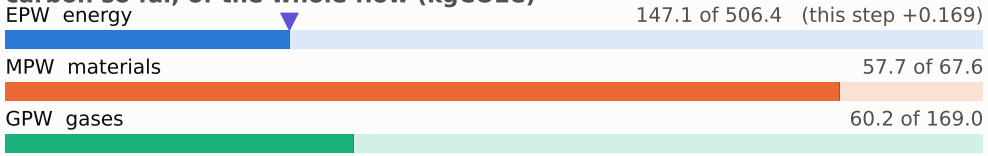
no mask for this step

this step

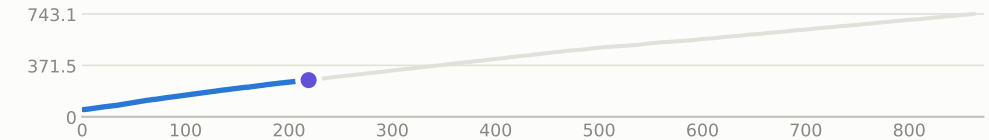
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

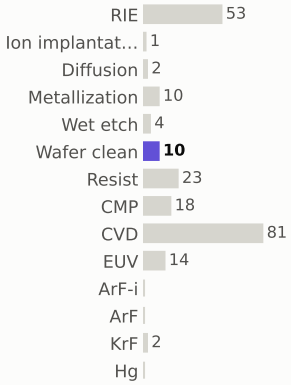
carbon so far, of the whole flow (kgCO2e)



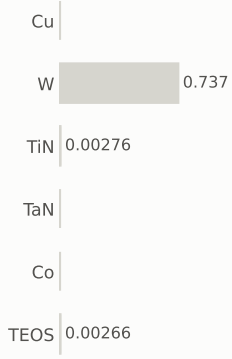
embodied carbon up to each step (kgCO2e per wafer)



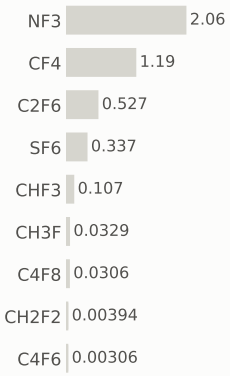
steps by process area



materials deposited (g)



gases emitted (g)



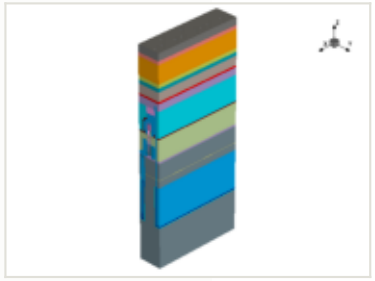
Step #2440: EUV

litho_patterning · EUV

1.96	266.9	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

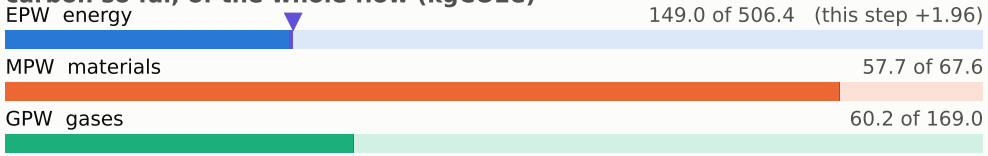


this step

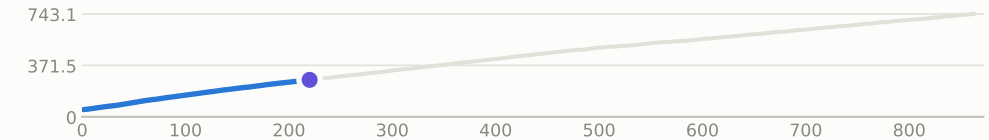
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

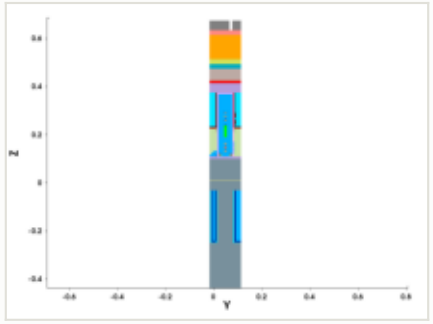
carbon so far, of the whole flow (kgCO2e)



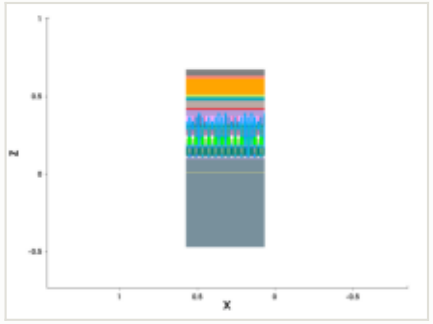
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

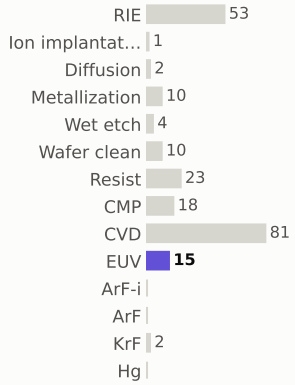


y cut at 0.049 µm

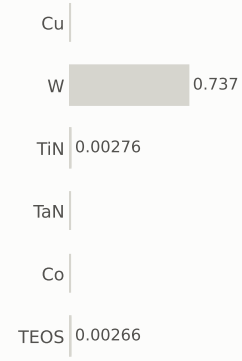


mask: layer 12:0, 12:2 (GDSII)

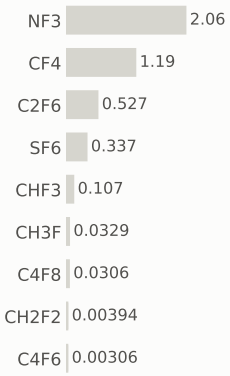
steps by process area



materials deposited (g)



gases emitted (g)



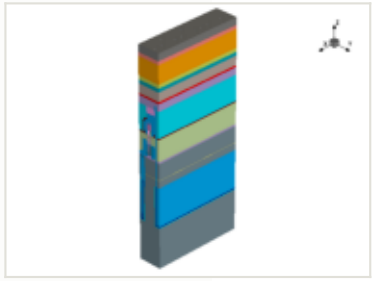
Step #2450: RIE

etch_anisotropic · RIE

1.04	267.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

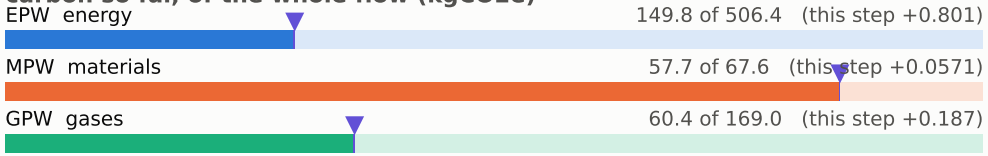


this step

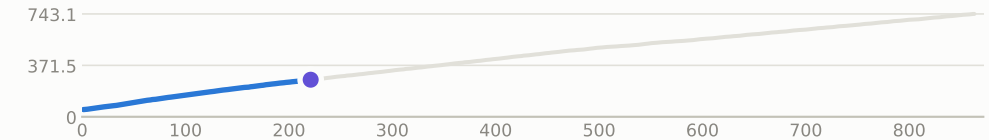
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

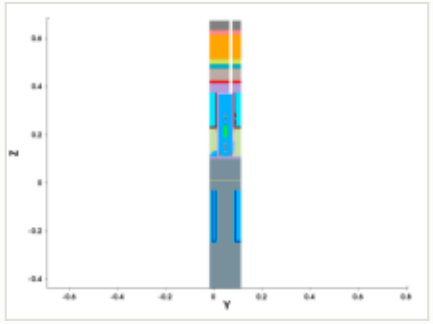
carbon so far, of the whole flow (kgCO2e)



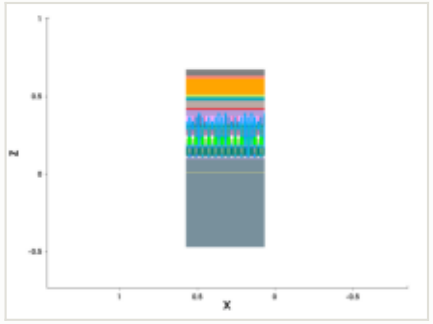
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



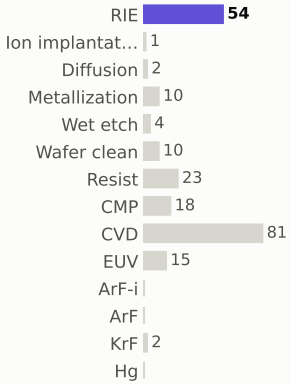
y cut at 0.049 µm



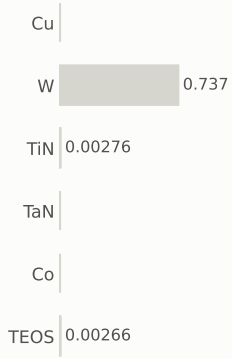
mask for this step

no mask for this step

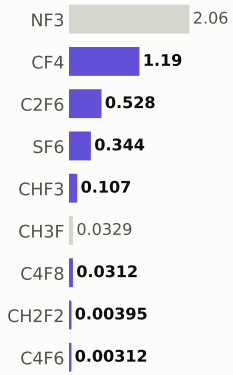
steps by process area



materials deposited (g)



gases emitted (g)



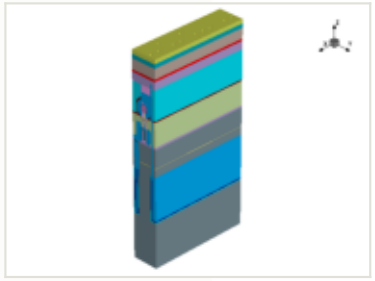
Step #2460: Wafer_clean

etch_strip · Wafer_clean

0.169	268.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

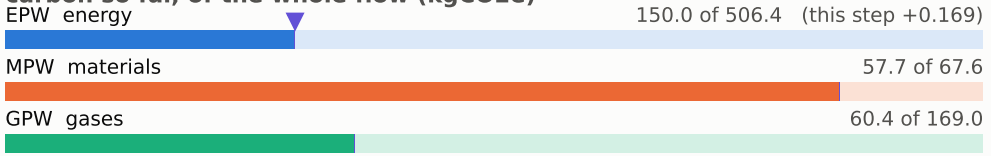


this step

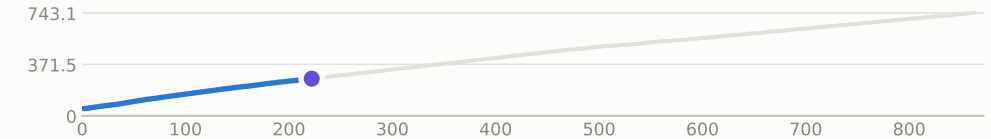
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

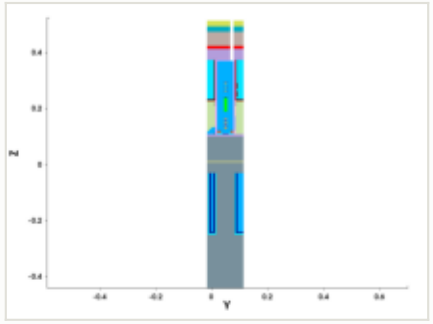
carbon so far, of the whole flow (kgCO2e)



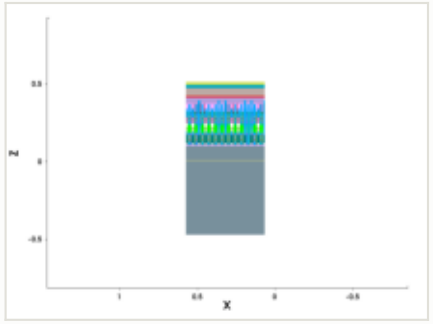
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



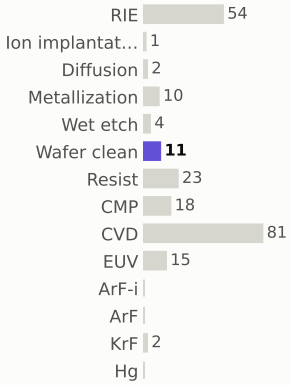
y cut at 0.049 µm



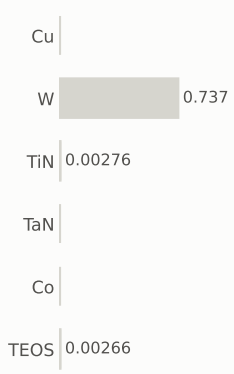
mask for this step

no mask for this step

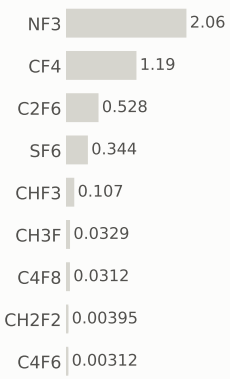
steps by process area



materials deposited (g)



gases emitted (g)



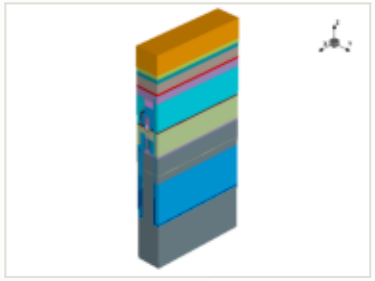
Step #2470: Resist

depo_fill · Resist

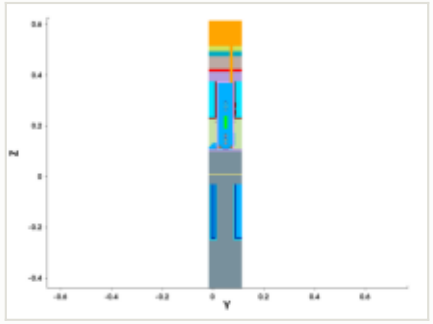
0.169	268.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

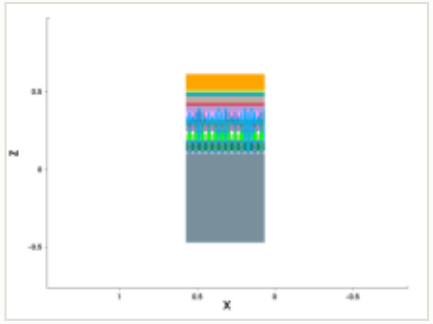
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



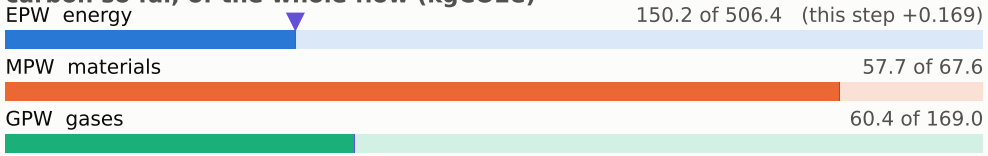
mask for this step

no mask for this step

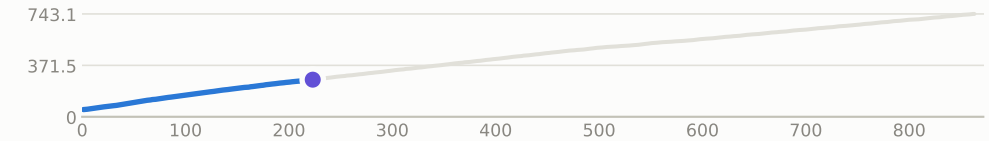
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

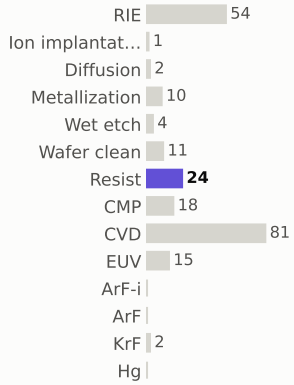
carbon so far, of the whole flow (kgCO2e)



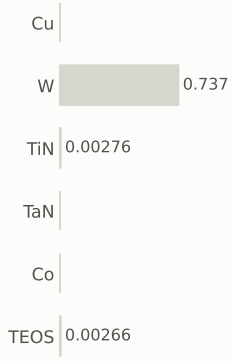
embodied carbon up to each step (kgCO2e per wafer)



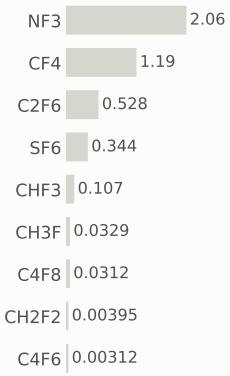
steps by process area



materials deposited (g)



gases emitted (g)



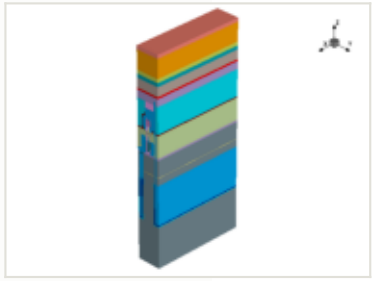
Step #2480: CVD

depo_fill · CVD

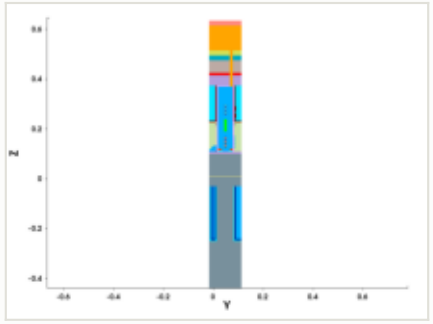
1.31	269.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

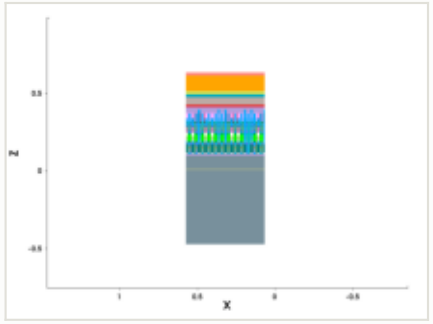
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

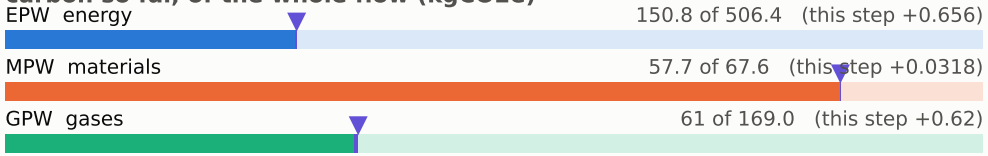
no mask for this step

this step

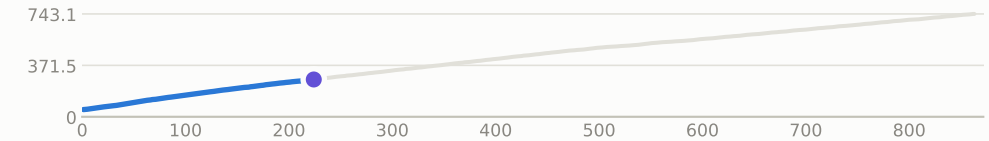
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

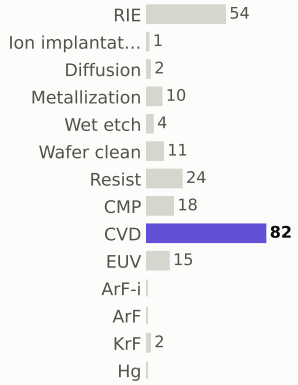
carbon so far, of the whole flow (kgCO2e)



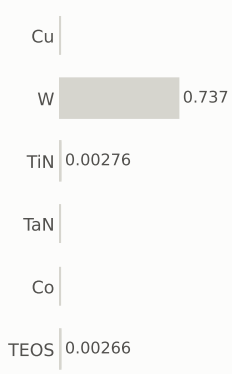
embodied carbon up to each step (kgCO2e per wafer)



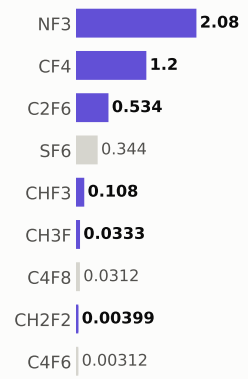
steps by process area



materials deposited (g)



gases emitted (g)



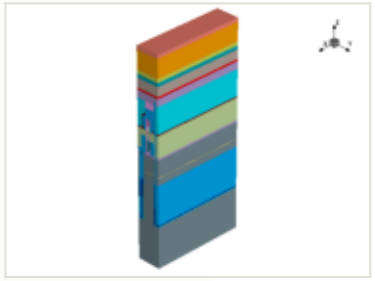
Step #2490: Wafer_clean

clean · Wafer_clean

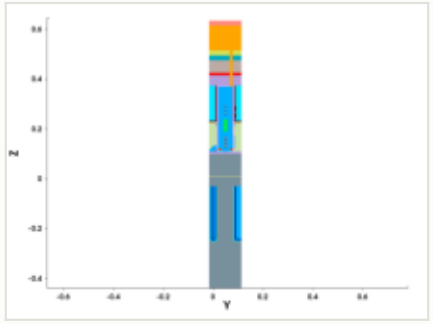
0.169	269.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

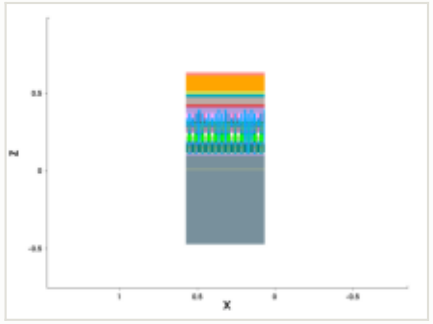
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

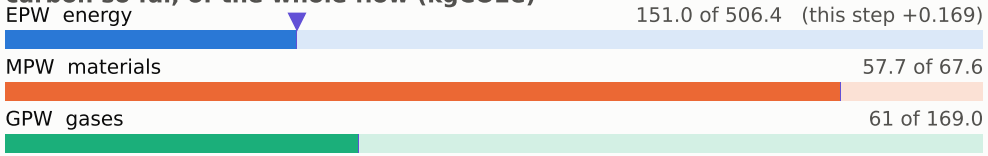
no mask for this step

this step

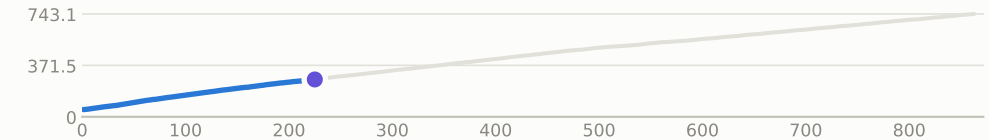
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

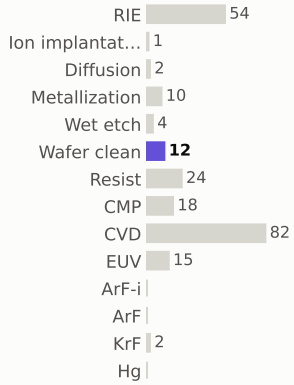
carbon so far, of the whole flow (kgCO2e)



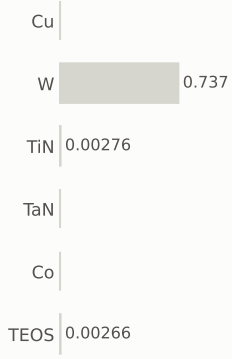
embodied carbon up to each step (kgCO2e per wafer)



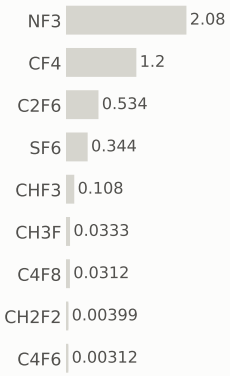
steps by process area



materials deposited (g)



gases emitted (g)



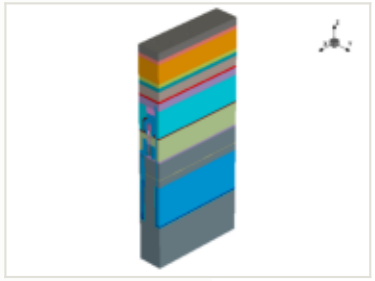
Step #2500: EUV

litho_patterning · EUV

1.96	271.7	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

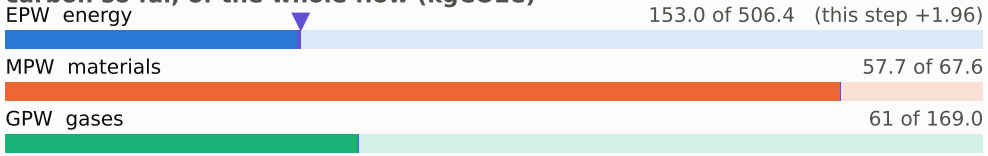


this step

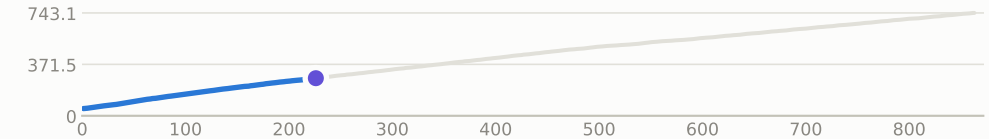
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

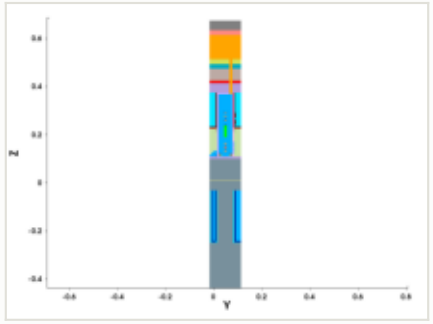
carbon so far, of the whole flow (kgCO2e)



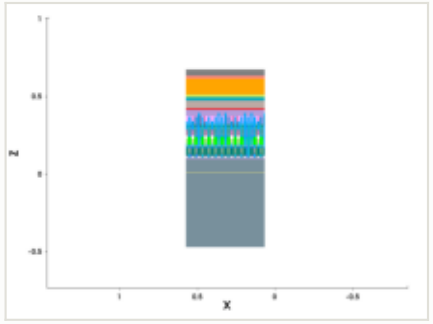
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

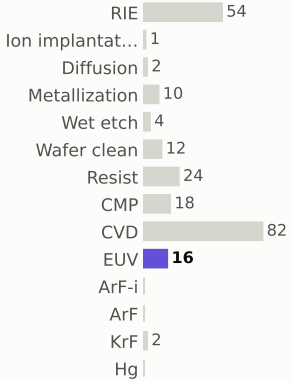


y cut at 0.049 µm

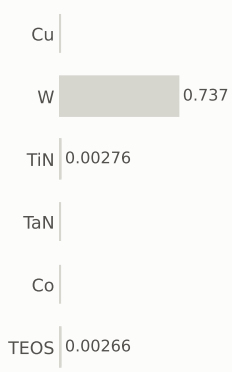


mask: layer 12:1 (GDSII)

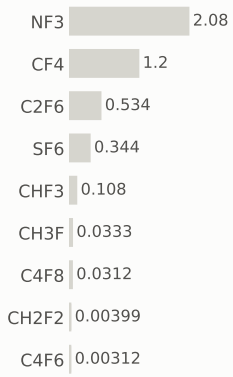
steps by process area



materials deposited (g)



gases emitted (g)



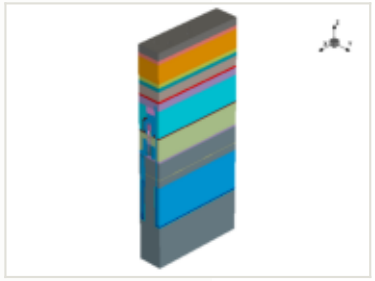
Step #2510: RIE

etch_anisotropic · RIE

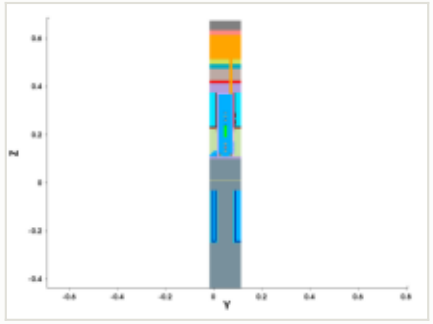
1.04	272.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

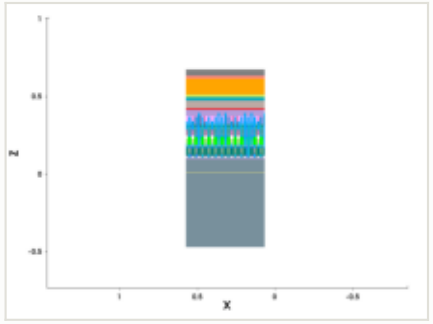
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

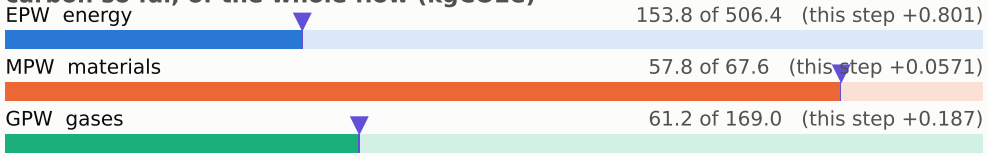
no mask for this step

this step

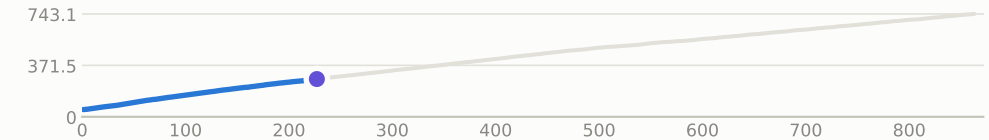
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

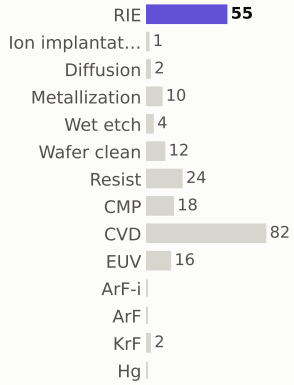
carbon so far, of the whole flow (kgCO2e)



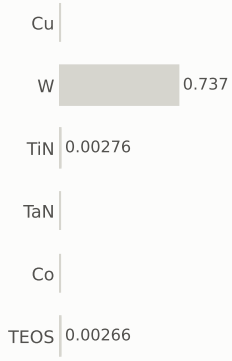
embodied carbon up to each step (kgCO2e per wafer)



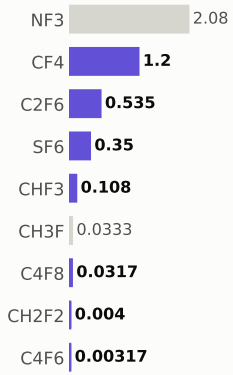
steps by process area



materials deposited (g)



gases emitted (g)



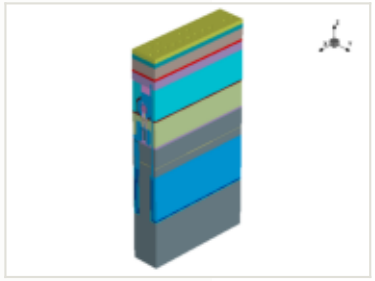
Step #2520: Wafer_clean

etch_strip · Wafer_clean

0.169	272.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

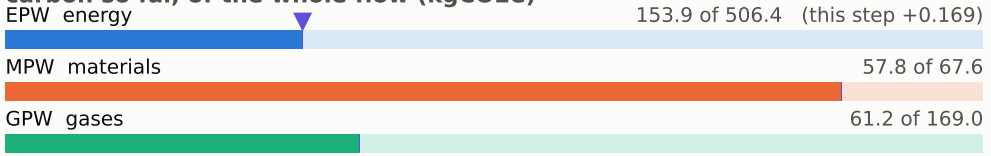


this step

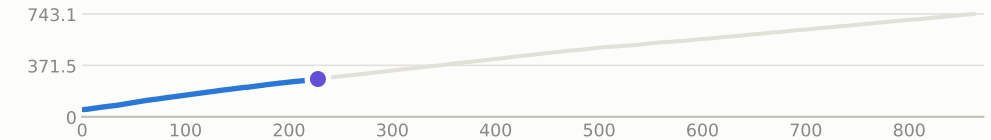
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

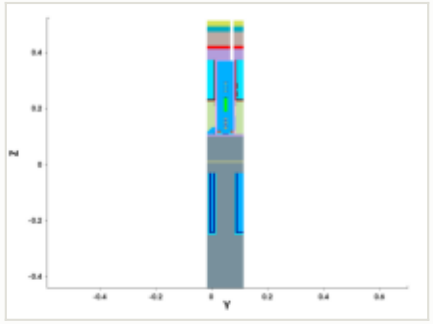
carbon so far, of the whole flow (kgCO2e)



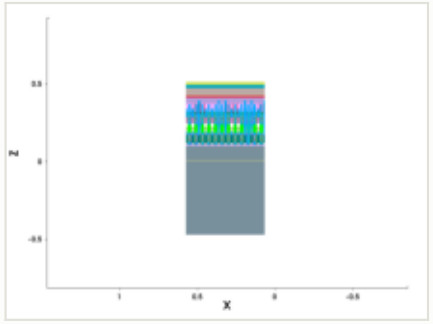
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



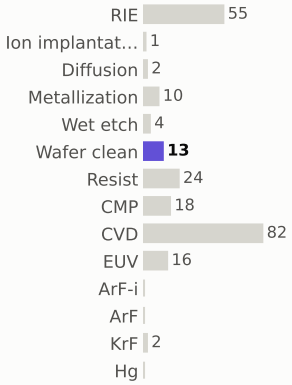
y cut at 0.049 µm



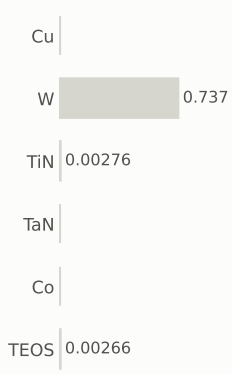
mask for this step

no mask for this step

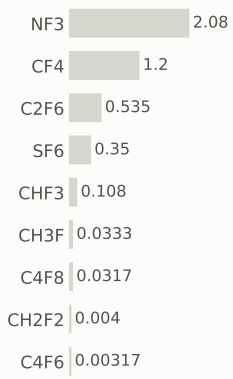
steps by process area



materials deposited (g)



gases emitted (g)



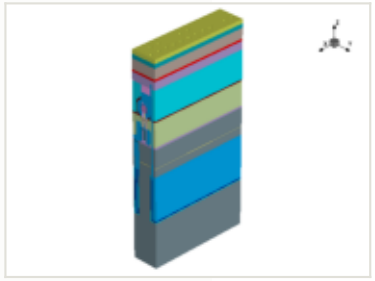
Step #2530: RIE

etch_noop · RIE

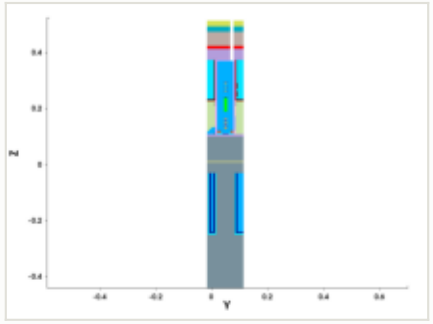
1.04	273.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

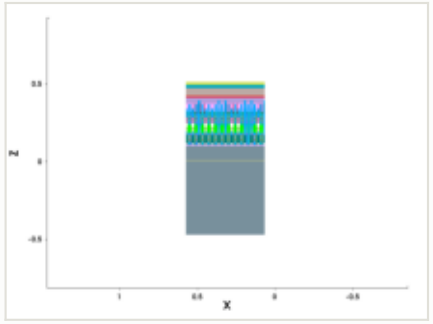
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

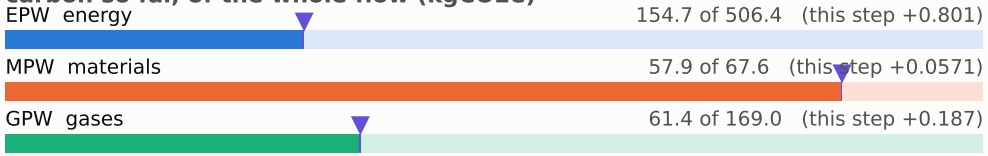
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

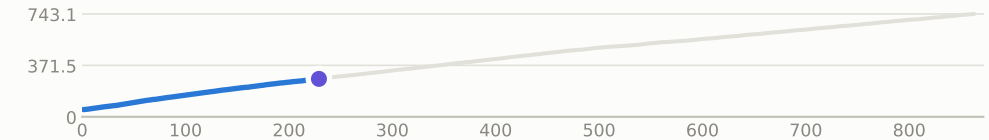
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

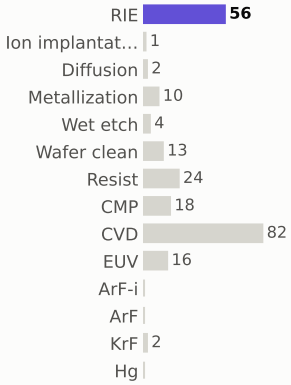
carbon so far, of the whole flow (kgCO2e)



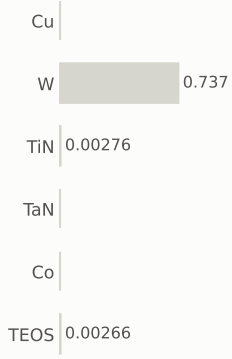
embodied carbon up to each step (kgCO2e per wafer)



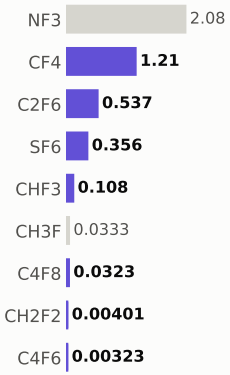
steps by process area



materials deposited (g)



gases emitted (g)



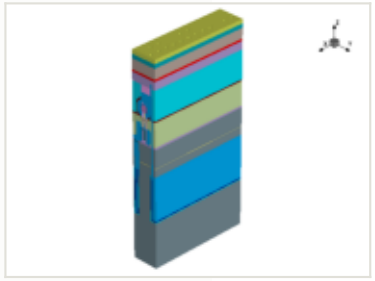
Step #2540: Wafer_clean

clean · Wafer_clean

0.169	274.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



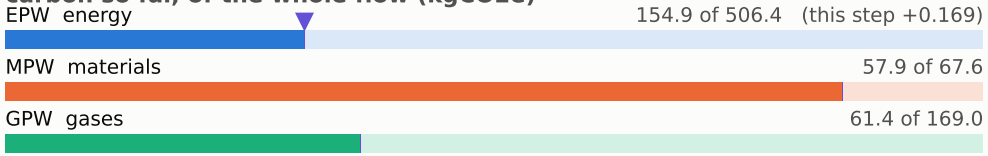
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

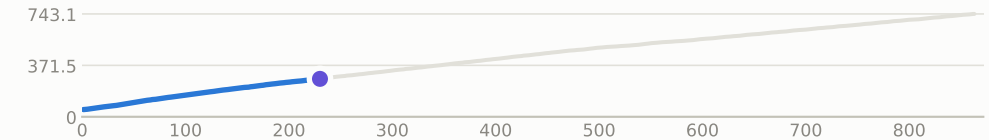
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

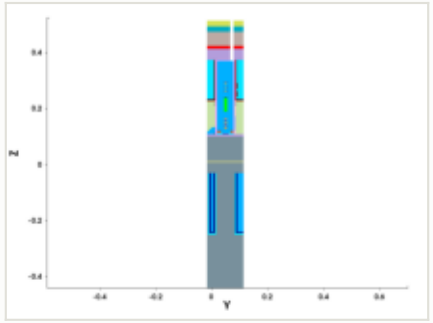
carbon so far, of the whole flow (kgCO2e)



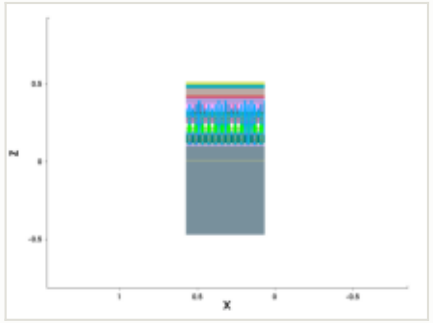
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



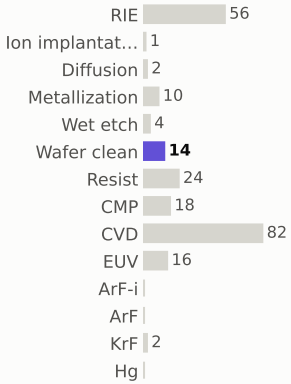
y cut at 0.049 µm



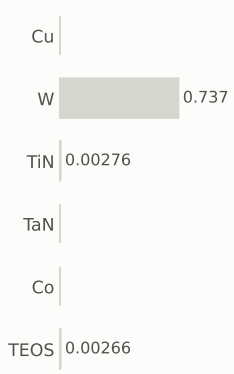
mask for this step

no mask for this step

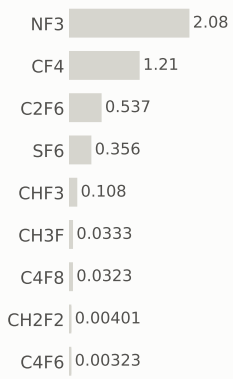
steps by process area



materials deposited (g)



gases emitted (g)



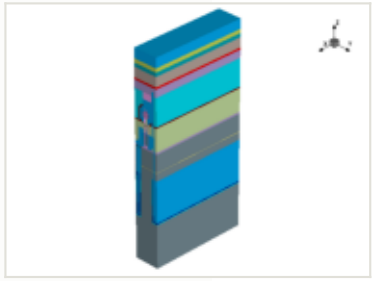
Step #2550: Metallization

depo_fill · Metallization

1.21	275.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

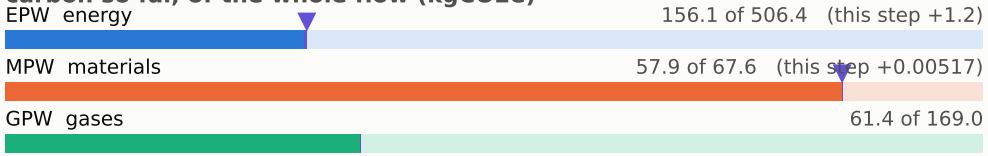


this step

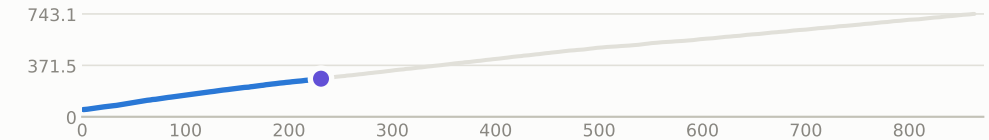
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

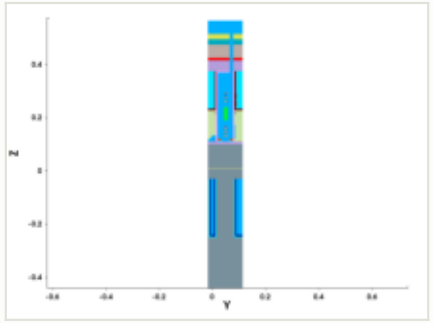
carbon so far, of the whole flow (kgCO2e)



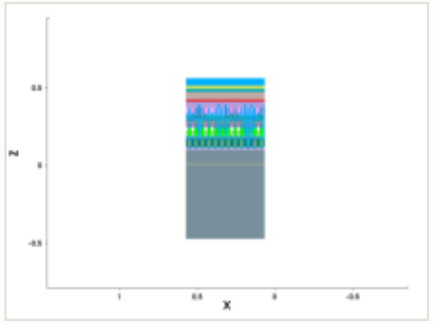
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



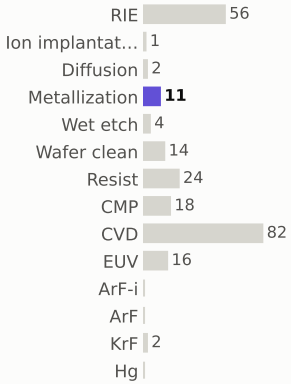
y cut at 0.049 µm



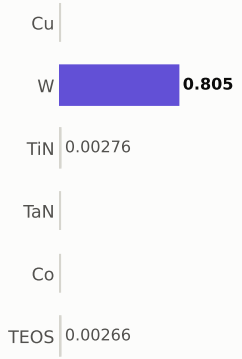
mask for this step

no mask for this step

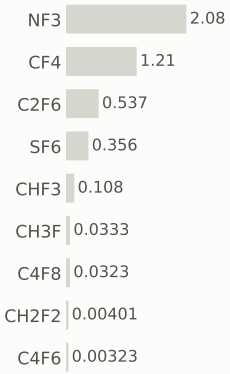
steps by process area



materials deposited (g)



gases emitted (g)



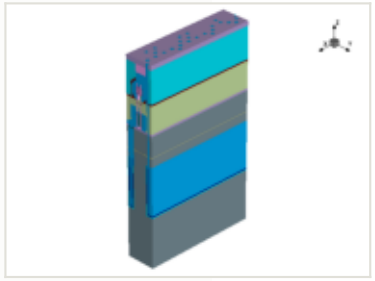
Step #2560: CMP

etch_cmp · CMP

0.169	275.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

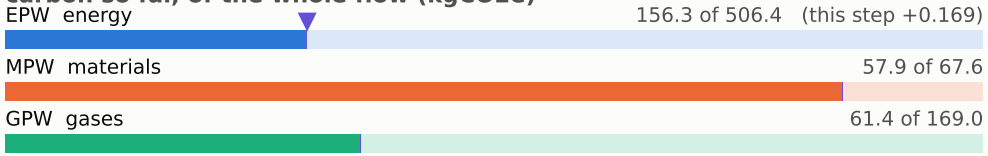


this step

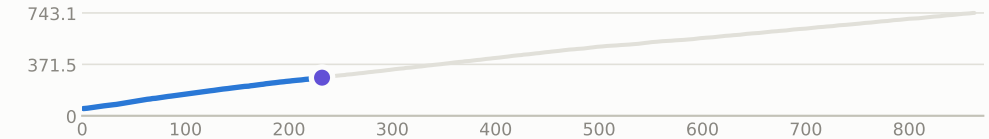
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

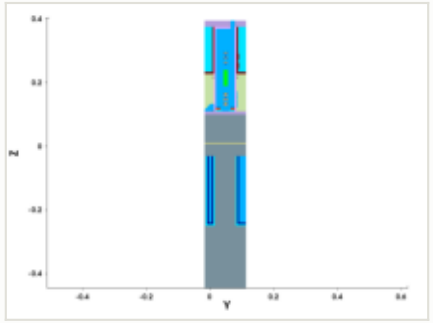
carbon so far, of the whole flow (kgCO2e)



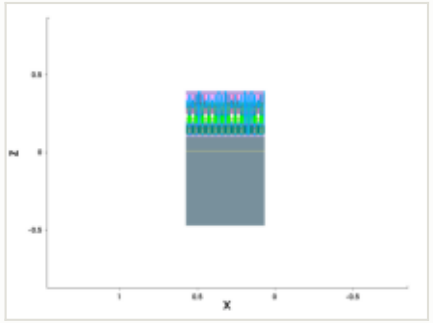
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



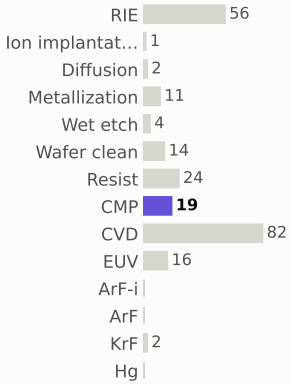
y cut at 0.049 µm



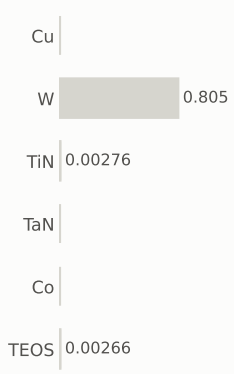
mask for this step

no mask for this step

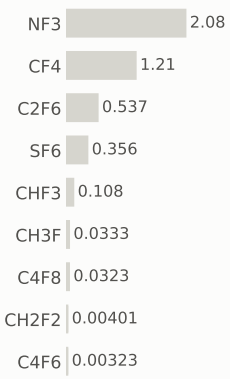
steps by process area



materials deposited (g)



gases emitted (g)



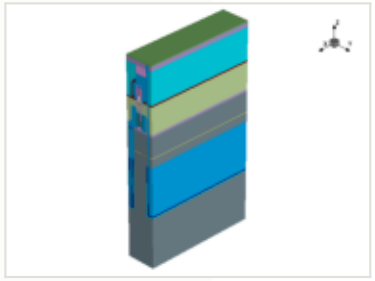
Step #2580: CVD

depo_fill · CVD

1.31	276.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

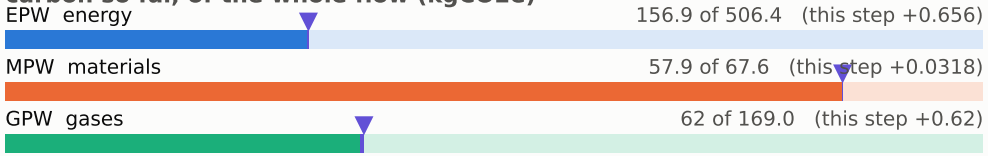


this step

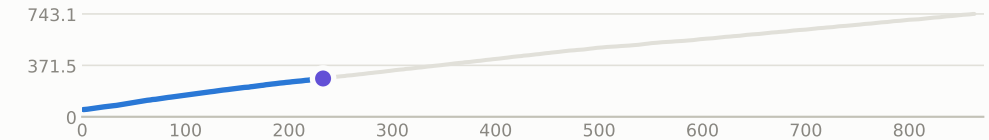
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

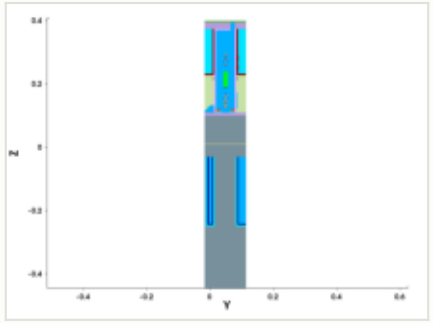
carbon so far, of the whole flow (kgCO2e)



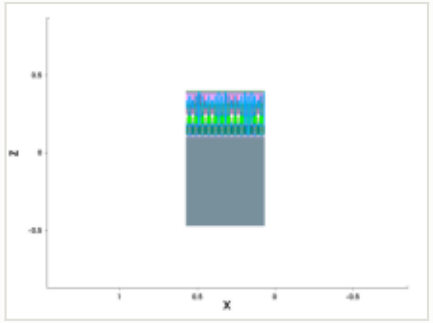
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



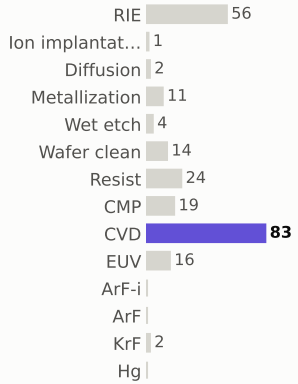
y cut at 0.049 µm



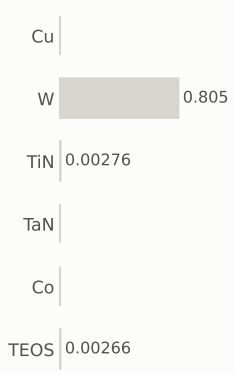
mask for this step

no mask for this step

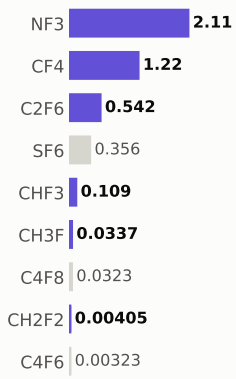
steps by process area



materials deposited (g)



gases emitted (g)



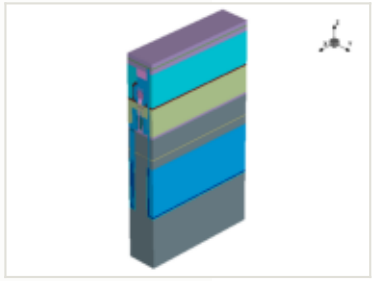
Step #2590: CVD

depo_fill · CVD

1.31	278.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

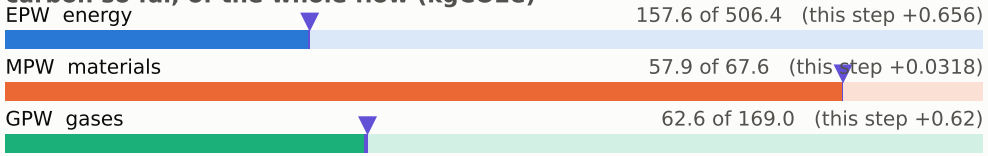


this step

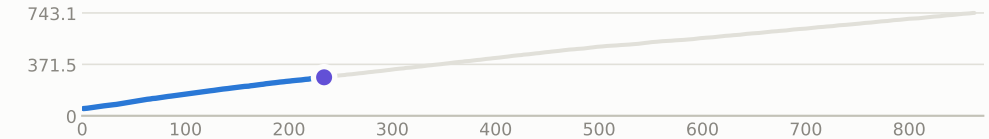
process area	CVD
material	—
thickness	0.018 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

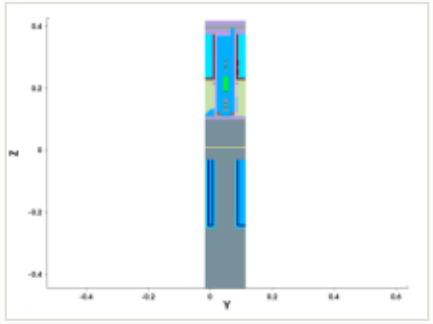
carbon so far, of the whole flow (kgCO2e)



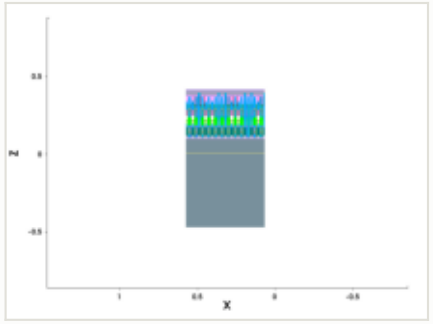
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



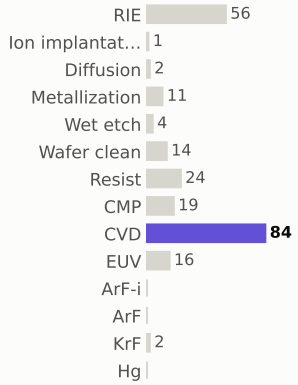
y cut at 0.049 µm



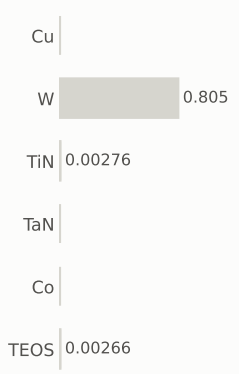
mask for this step

no mask for this step

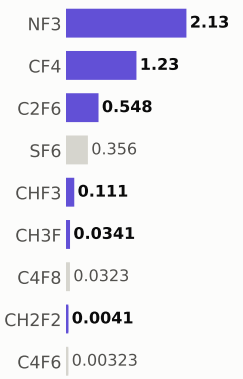
steps by process area



materials deposited (g)



gases emitted (g)



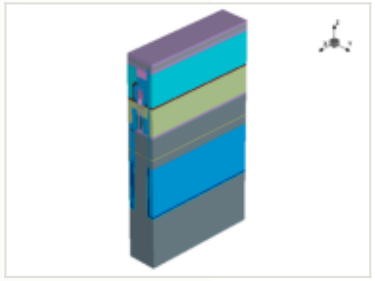
Step #2600: Diffusion

treatment · Diffusion

0.801	278.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

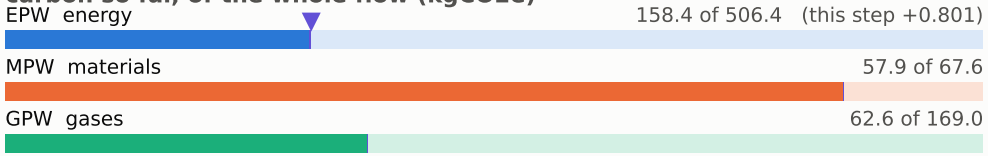


this step

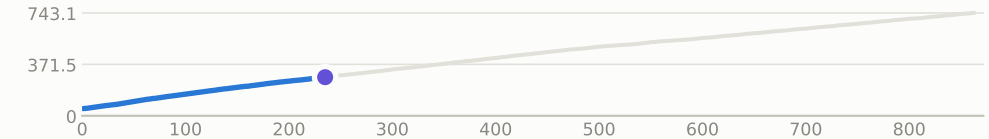
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

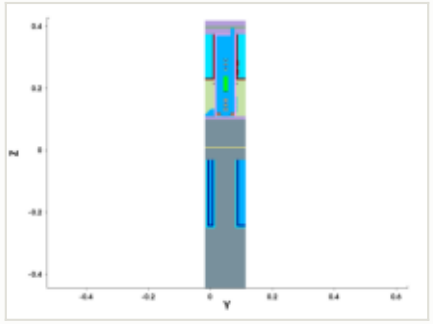
carbon so far, of the whole flow (kgCO2e)



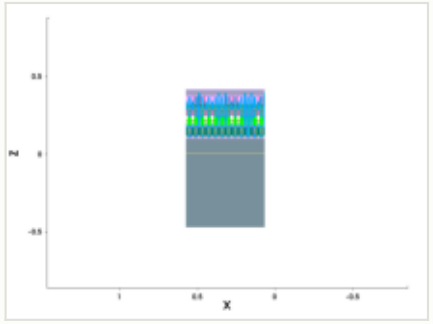
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



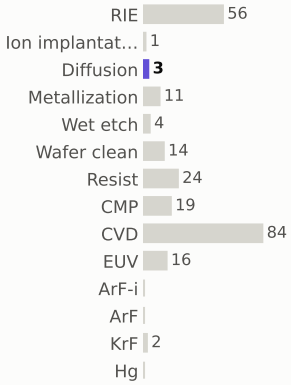
y cut at 0.049 µm



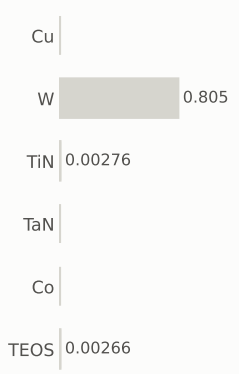
mask for this step

no mask for this step

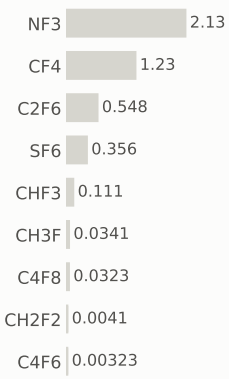
steps by process area



materials deposited (g)



gases emitted (g)



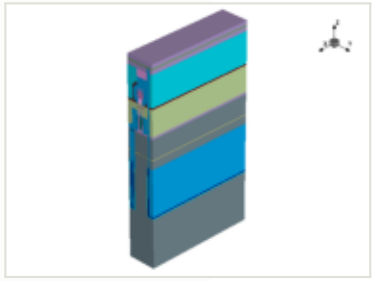
Step #2610: Wafer_clean

clean · Wafer_clean

0.169	279.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



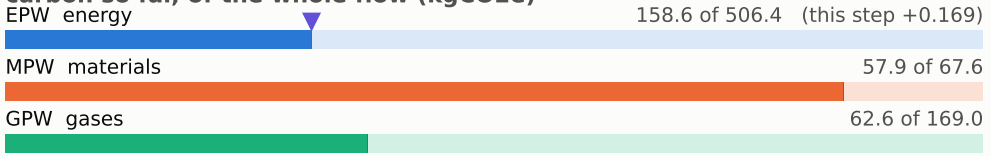
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

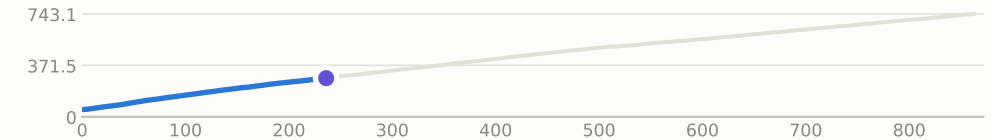
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

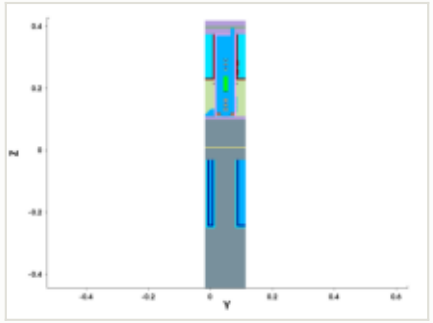
carbon so far, of the whole flow (kgCO2e)



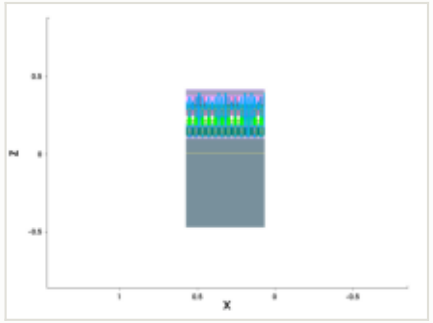
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



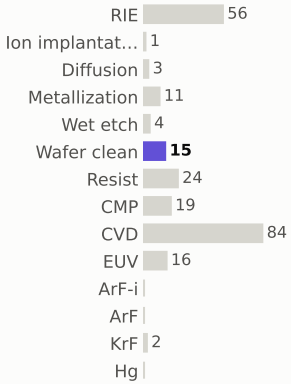
y cut at 0.049 µm



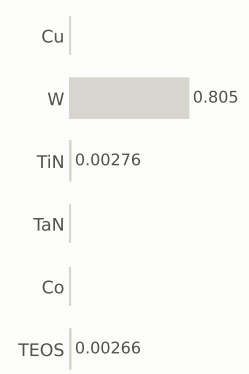
mask for this step

no mask for this step

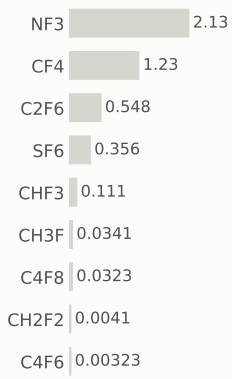
steps by process area



materials deposited (g)



gases emitted (g)



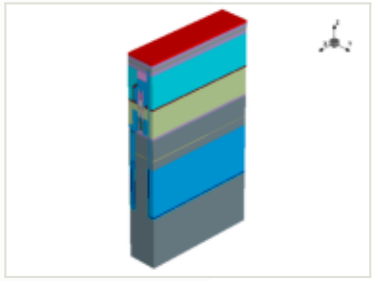
Step #2620: CVD

depo_fill · CVD

1.31	280.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

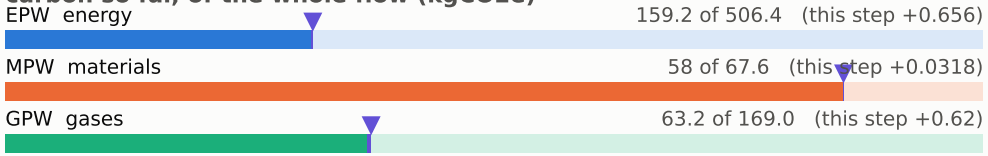


this step

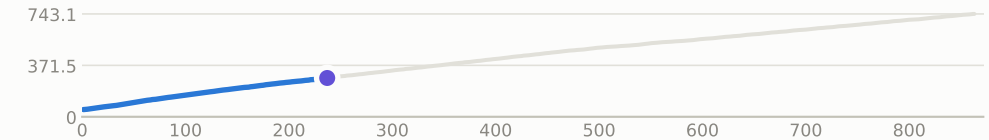
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

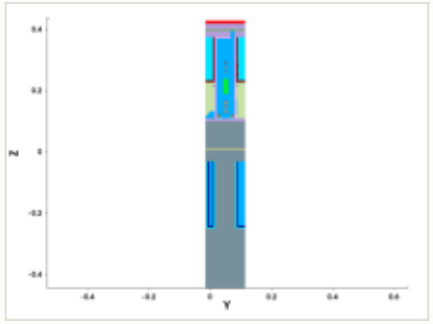
carbon so far, of the whole flow (kgCO2e)



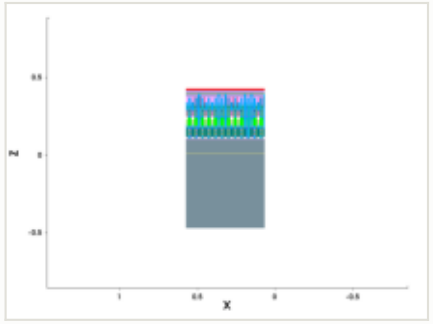
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



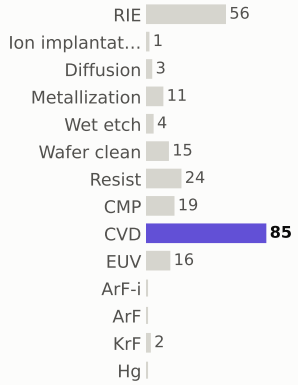
y cut at 0.049 µm



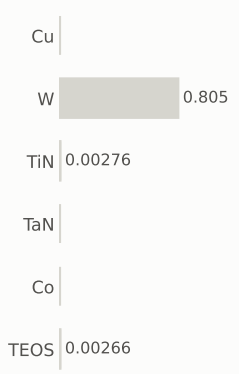
mask for this step

no mask for this step

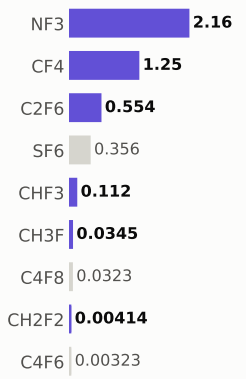
steps by process area



materials deposited (g)



gases emitted (g)



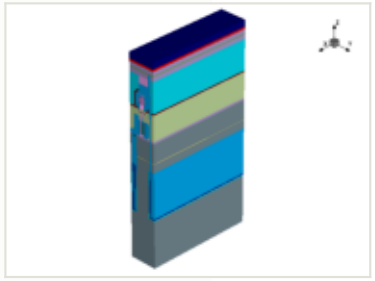
Step #2630: CVD

depo_fill · CVD

1.31	281.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

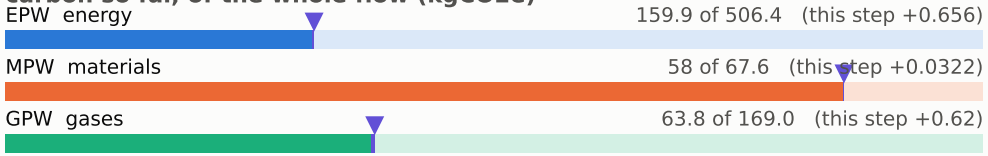


this step

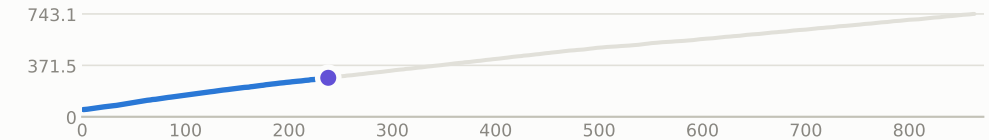
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

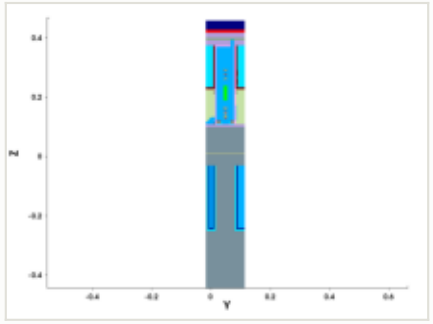
carbon so far, of the whole flow (kgCO2e)



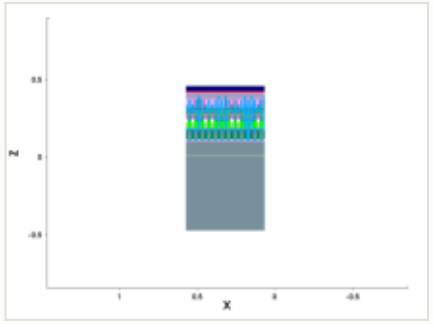
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



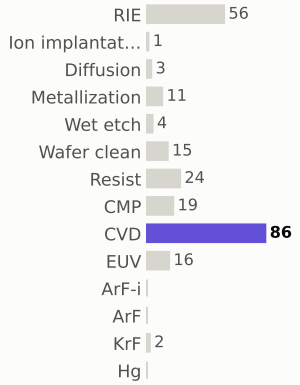
y cut at 0.049 µm



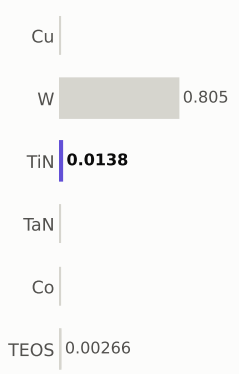
mask for this step

no mask for this step

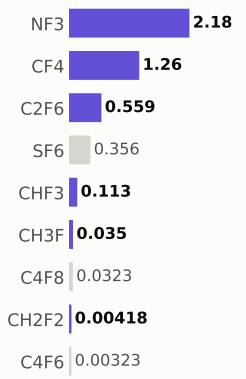
steps by process area



materials deposited (g)



gases emitted (g)



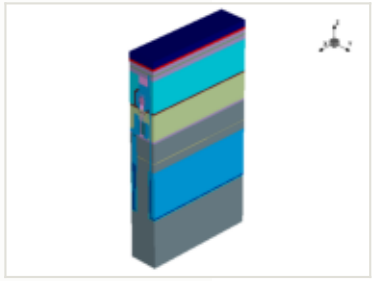
Step #2640: Wafer_clean

clean · Wafer_clean

0.169	281.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



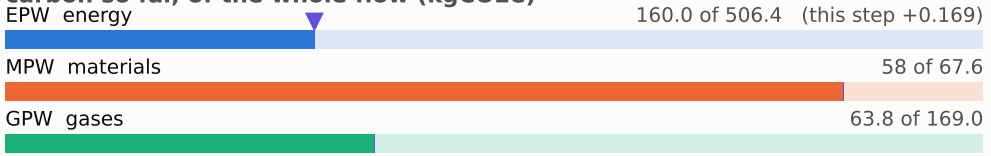
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

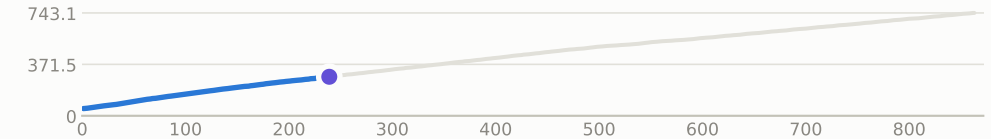
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

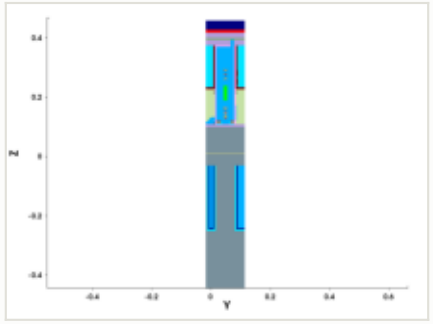
carbon so far, of the whole flow (kgCO2e)



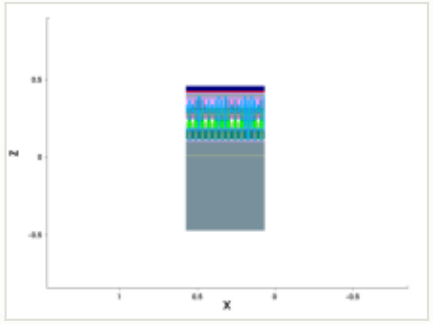
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



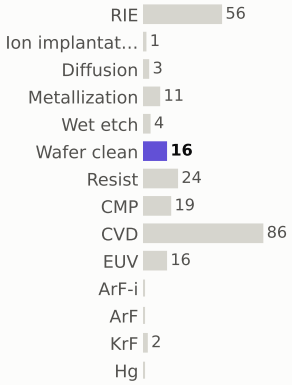
y cut at 0.049 µm



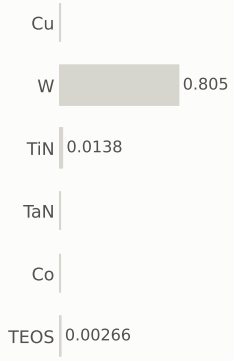
mask for this step

no mask for this step

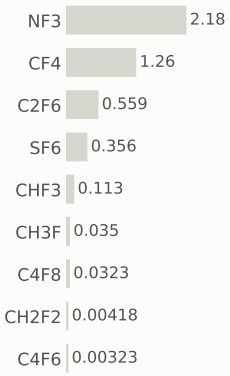
steps by process area



materials deposited (g)



gases emitted (g)



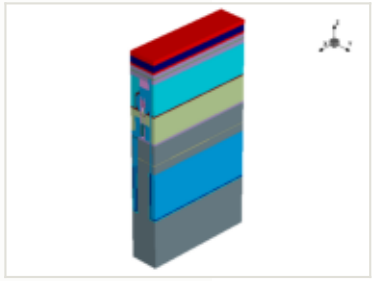
Step #2650: CVD

depo_fill · CVD

1.31	283.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

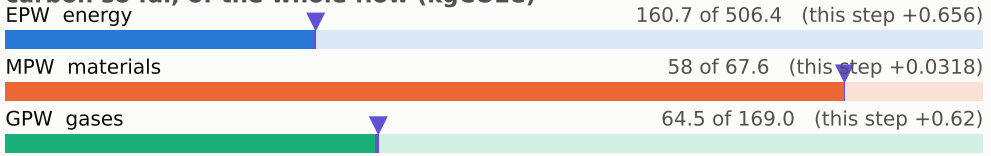


this step

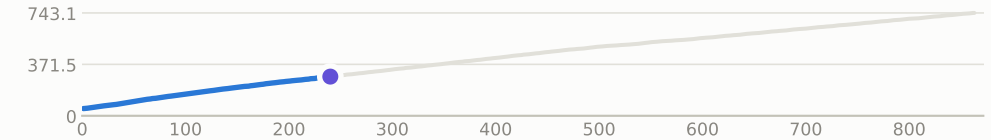
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

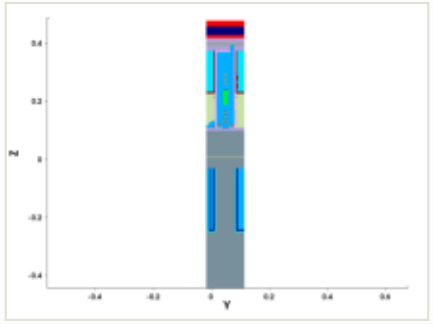
carbon so far, of the whole flow (kgCO2e)



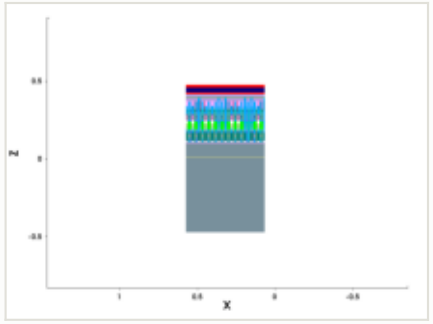
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



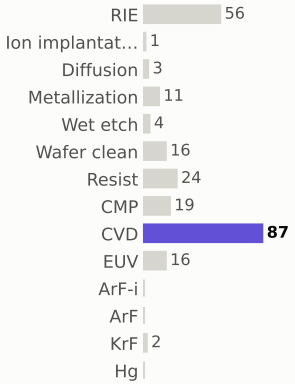
y cut at 0.049 µm



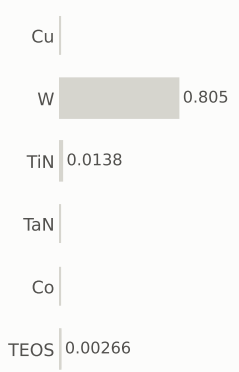
mask for this step

no mask for this step

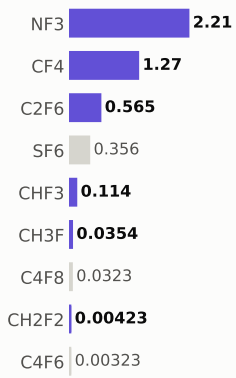
steps by process area



materials deposited (g)



gases emitted (g)



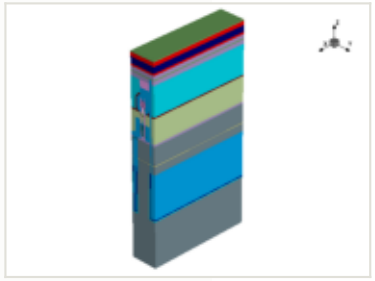
Step #2660: CVD

depo_fill · CVD

1.31	284.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

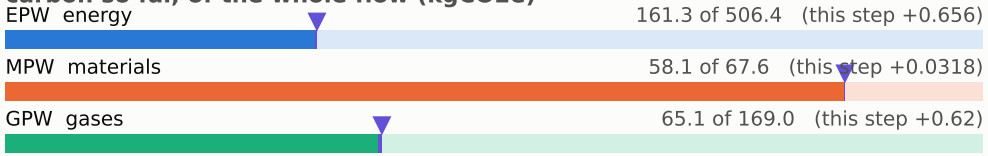


this step

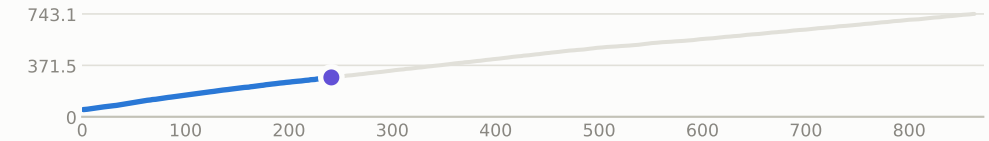
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

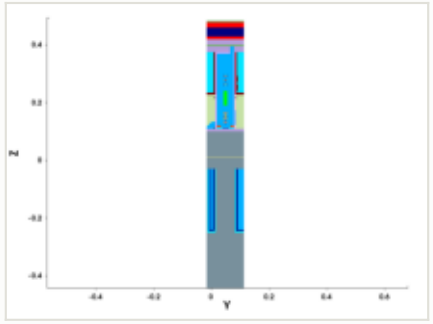
carbon so far, of the whole flow (kgCO2e)



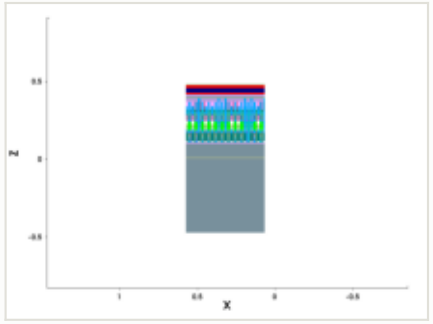
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



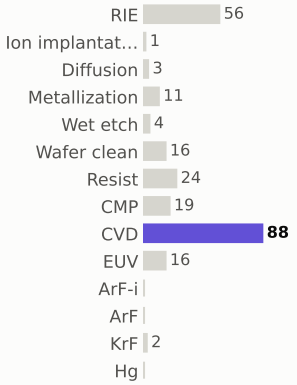
y cut at 0.049 µm



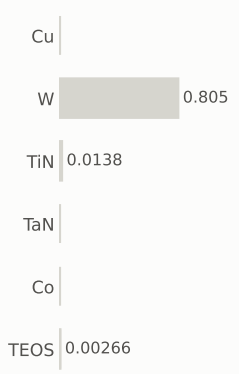
mask for this step

no mask for this step

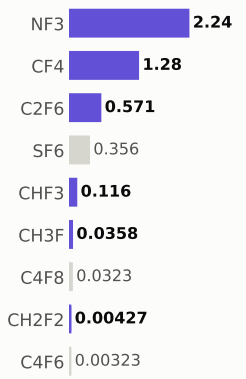
steps by process area



materials deposited (g)



gases emitted (g)



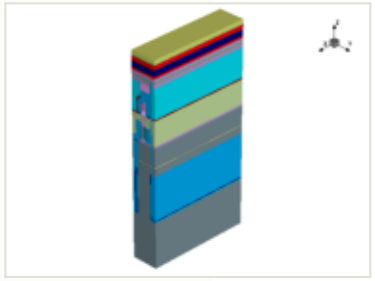
Step #2670: CVD

depo_fill · CVD

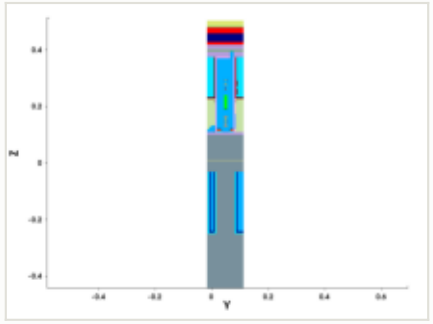
1.31	285.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

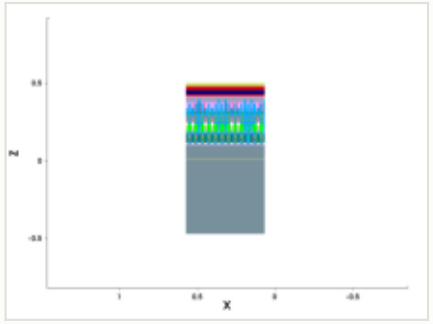
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

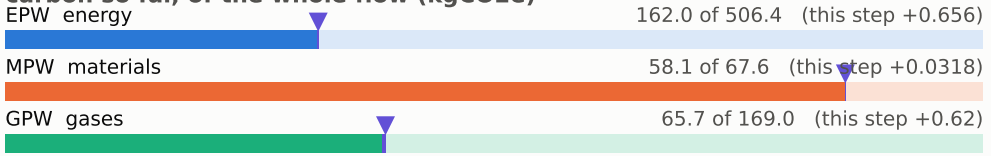
no mask for this step

this step

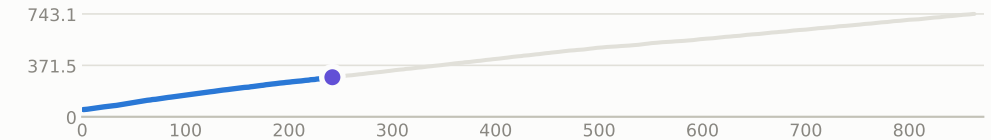
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

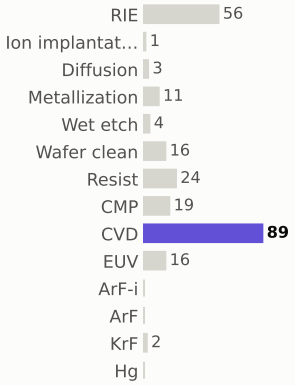
carbon so far, of the whole flow (kgCO2e)



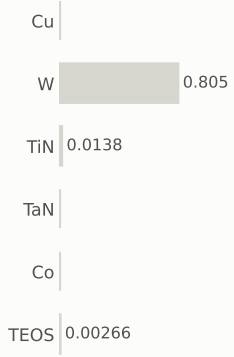
embodied carbon up to each step (kgCO2e per wafer)



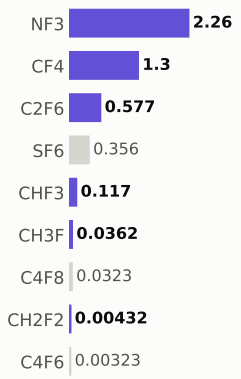
steps by process area



materials deposited (g)



gases emitted (g)



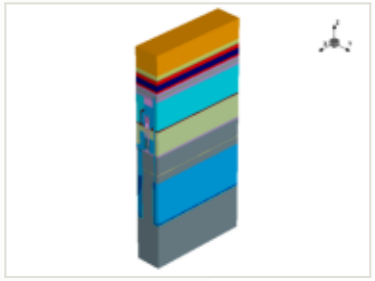
Step #2680: Resist

depo_fill · Resist

0.169	286.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

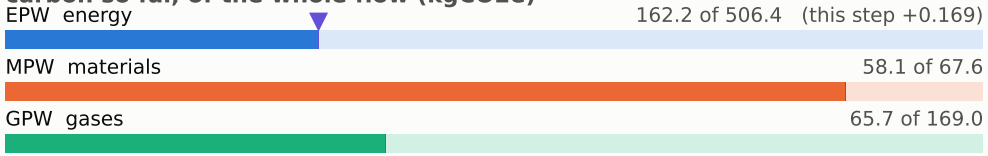


this step

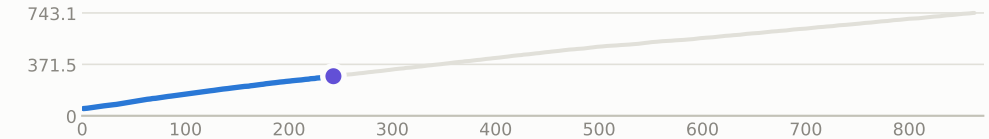
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

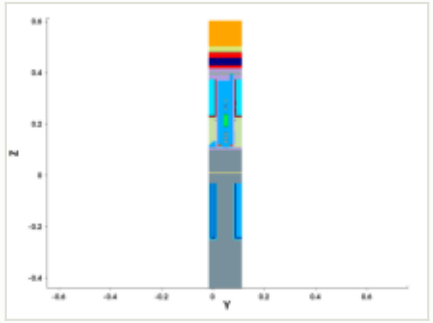
carbon so far, of the whole flow (kgCO2e)



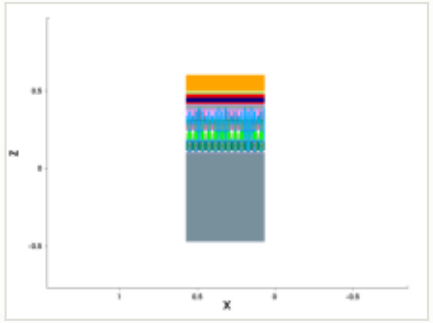
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



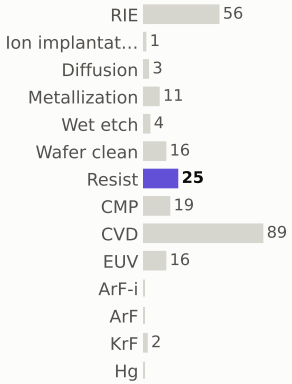
y cut at 0.049 µm



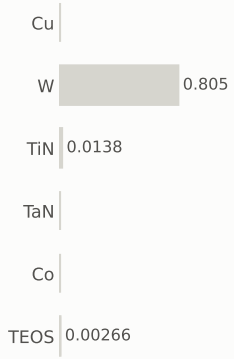
mask for this step

no mask for this step

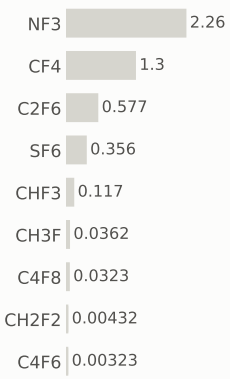
steps by process area



materials deposited (g)



gases emitted (g)



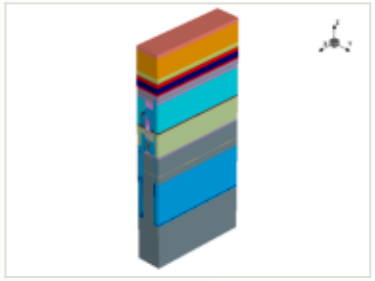
Step #2690: CVD

depo_fill · CVD

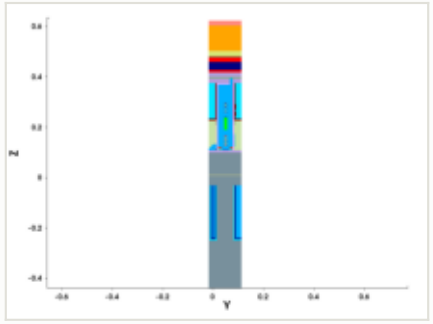
1.31	287.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

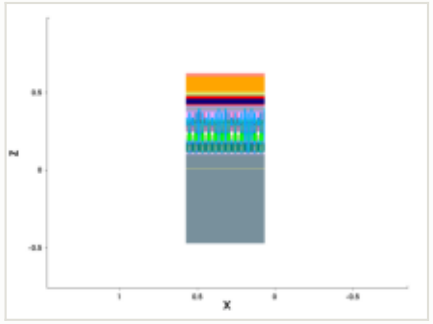
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

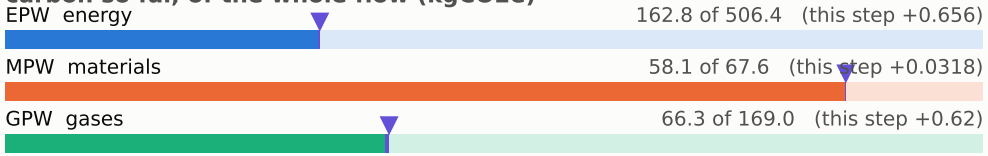
no mask for this step

this step

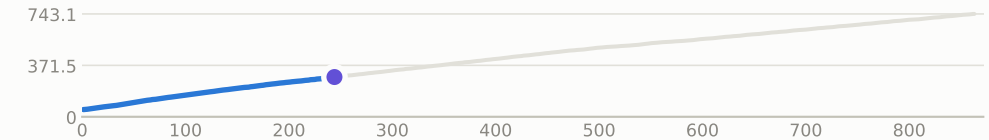
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

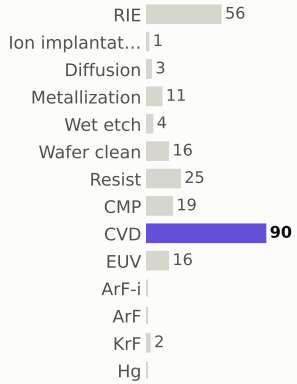
carbon so far, of the whole flow (kgCO2e)



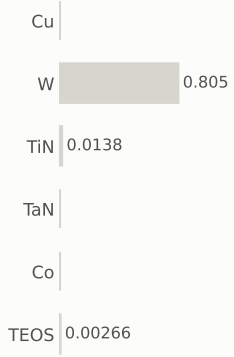
embodied carbon up to each step (kgCO2e per wafer)



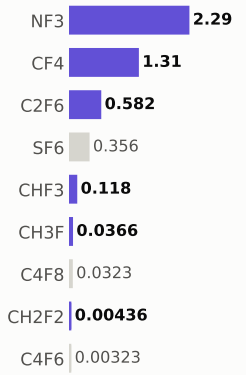
steps by process area



materials deposited (g)



gases emitted (g)



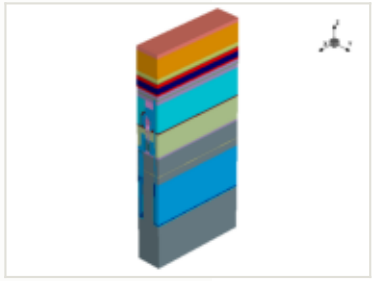
Step #2700: Wafer_clean

clean · Wafer_clean

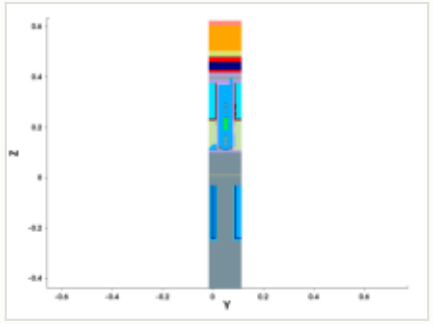
0.169	287.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

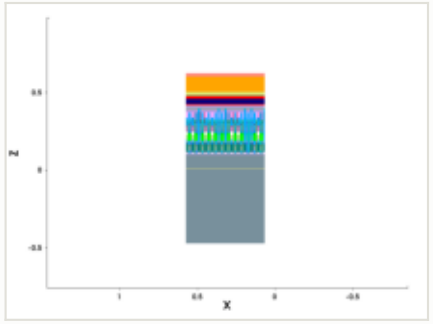
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

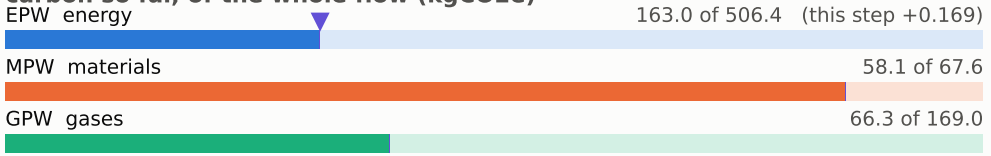
no mask for this step

this step

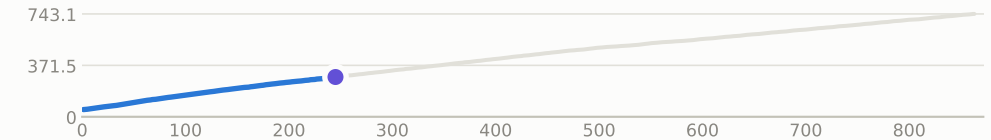
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

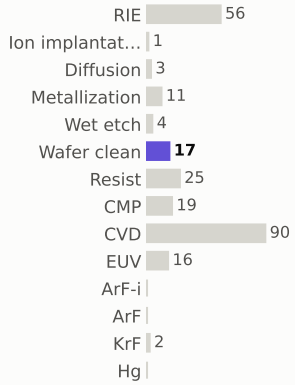
carbon so far, of the whole flow (kgCO2e)



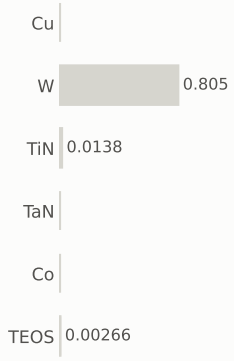
embodied carbon up to each step (kgCO2e per wafer)



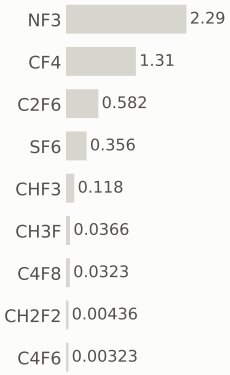
steps by process area



materials deposited (g)



gases emitted (g)



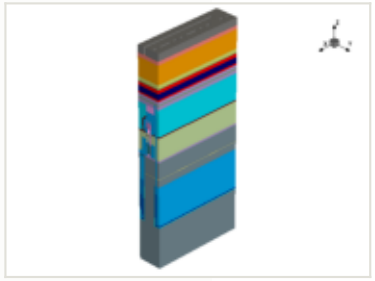
Step #2710: EUV

litho_patterning · EUV

1.96	289.4	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

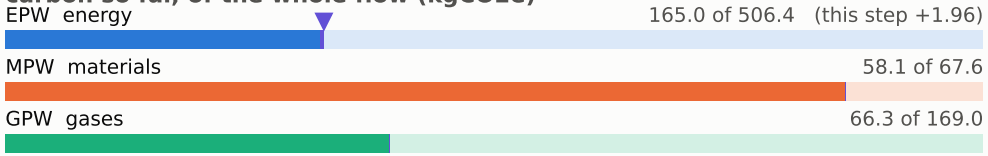


this step

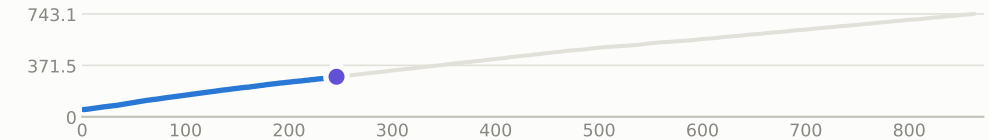
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

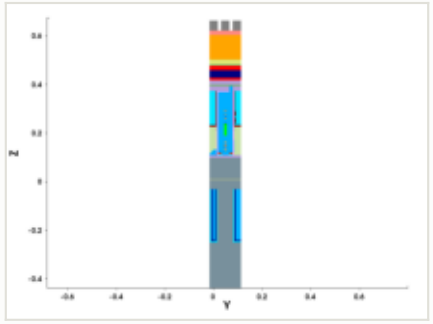
carbon so far, of the whole flow (kgCO2e)



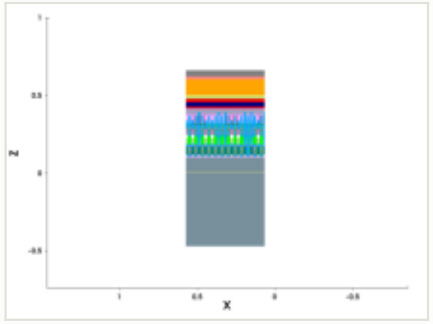
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

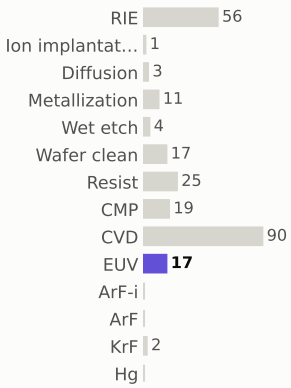


y cut at 0.049 µm

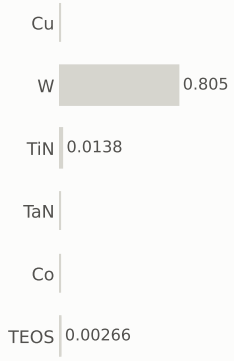


mask: layer 10201:0 (GDSII)

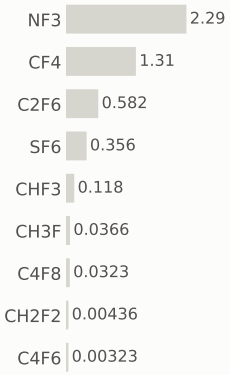
steps by process area



materials deposited (g)



gases emitted (g)



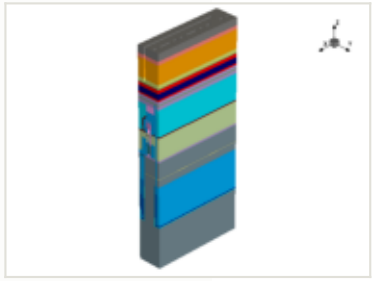
Step #2720: RIE

etch_anisotropic · RIE

1.04	290.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

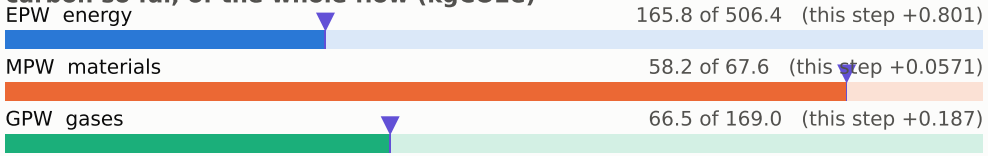


this step

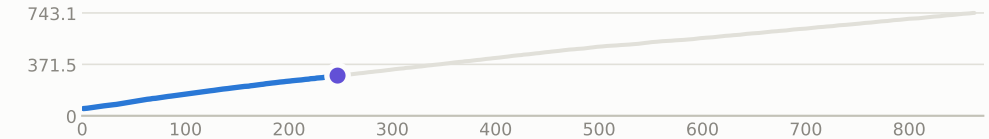
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

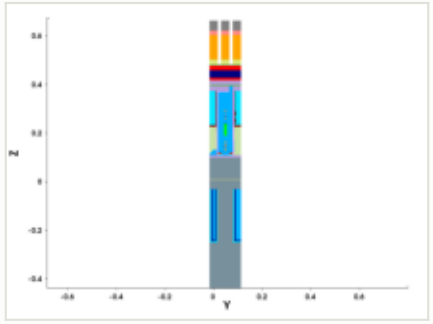
carbon so far, of the whole flow (kgCO2e)



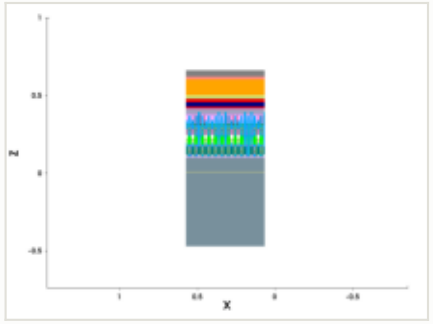
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



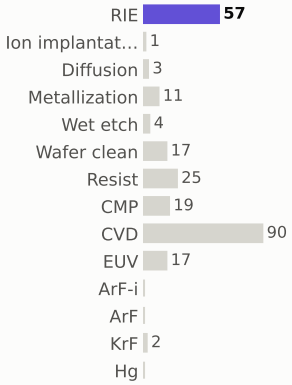
y cut at 0.049 µm



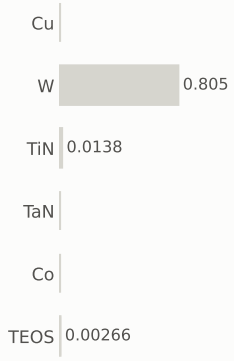
mask for this step

no mask for this step

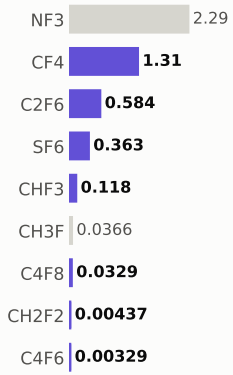
steps by process area



materials deposited (g)



gases emitted (g)



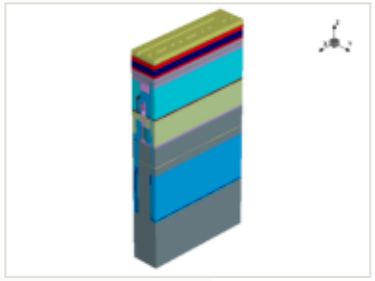
Step #2730: Wafer_clean

etch_strip · Wafer_clean

0.169	290.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

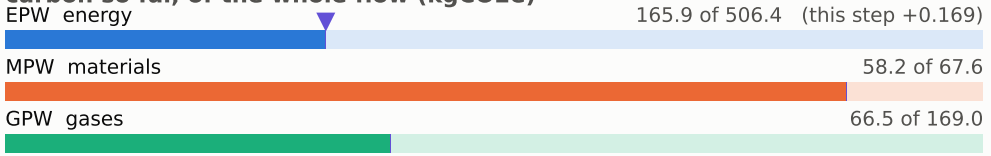


this step

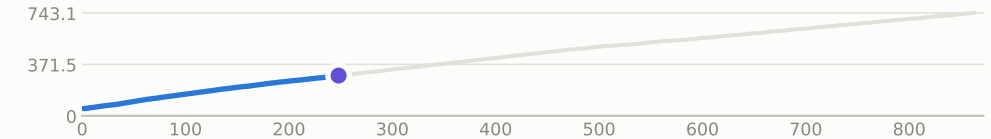
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

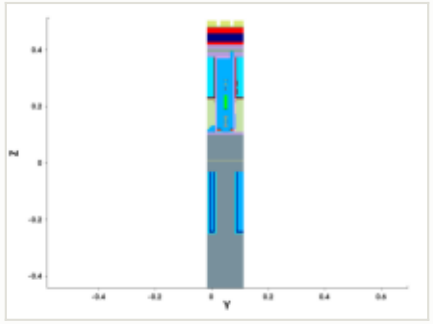
carbon so far, of the whole flow (kgCO2e)



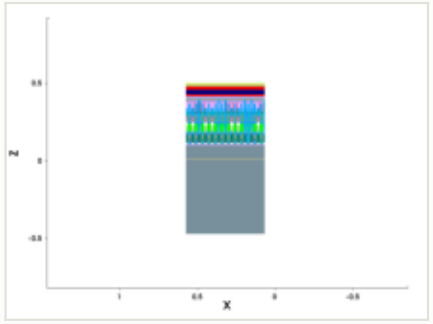
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



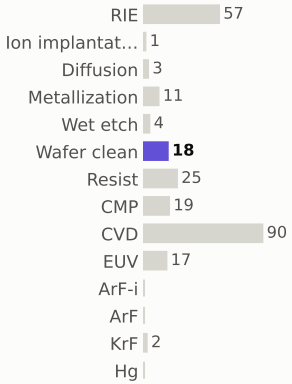
y cut at 0.049 µm



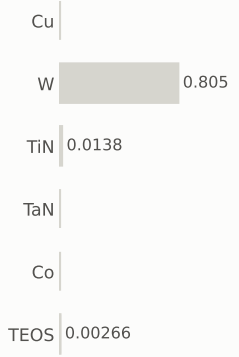
mask for this step

no mask for this step

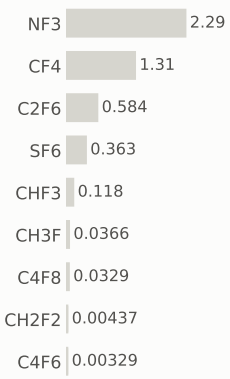
steps by process area



materials deposited (g)



gases emitted (g)



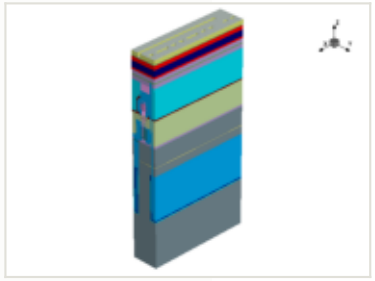
Step #2740: CVD

depo_isotropic · CVD

1.31	291.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

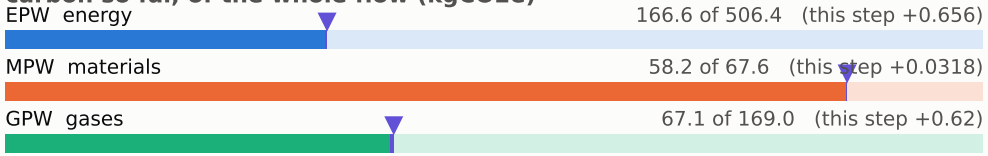


this step

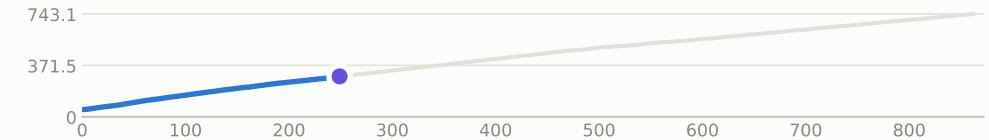
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

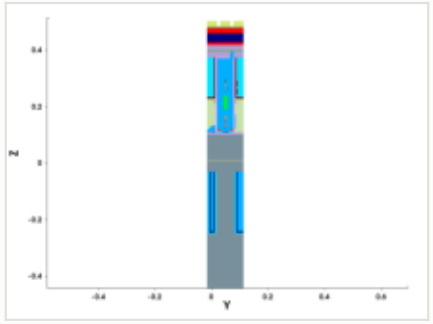
carbon so far, of the whole flow (kgCO2e)



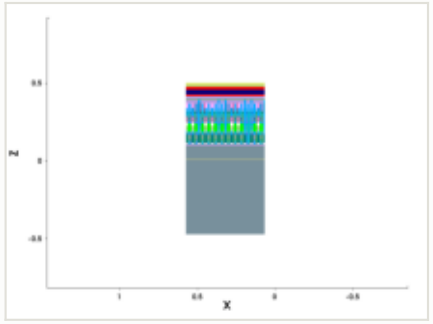
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



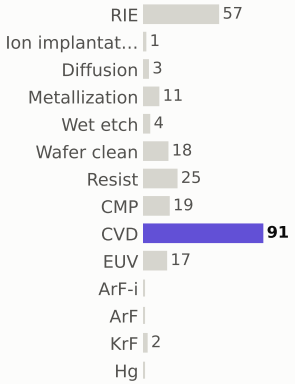
y cut at 0.049 µm



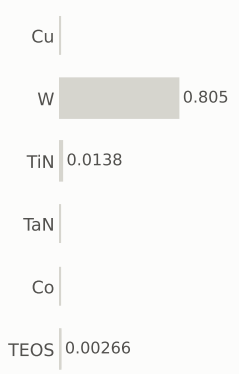
mask for this step

no mask for this step

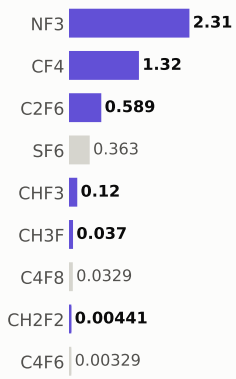
steps by process area



materials deposited (g)



gases emitted (g)



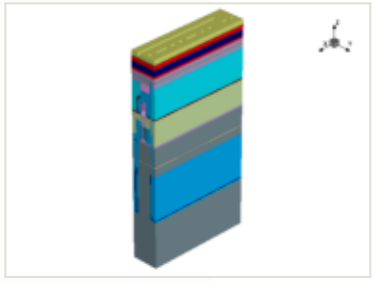
Step #2750: RIE

etch_anisotropic · RIE

1.04	293.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

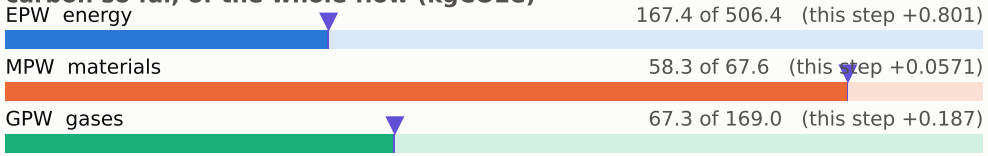


this step

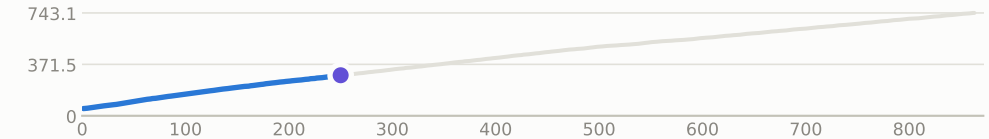
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

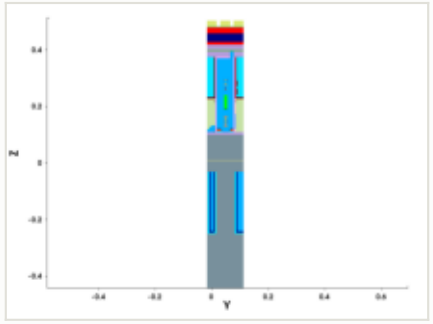
carbon so far, of the whole flow (kgCO2e)



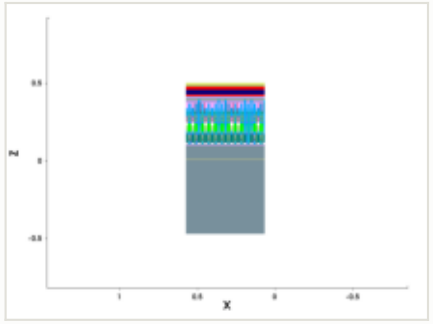
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



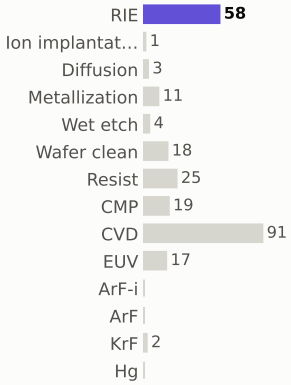
y cut at 0.049 µm



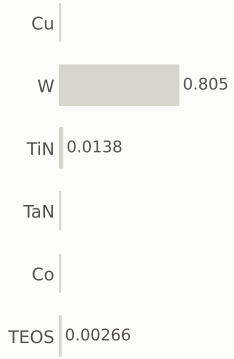
mask for this step

no mask for this step

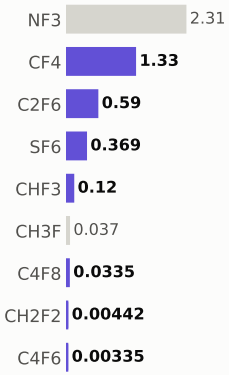
steps by process area



materials deposited (g)



gases emitted (g)



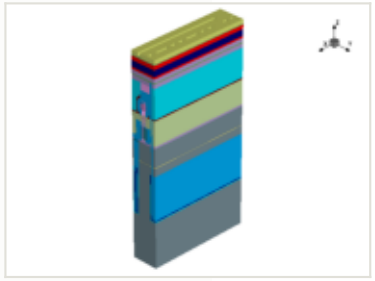
Step #2760: Wafer_clean

clean · Wafer_clean

0.169	293.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



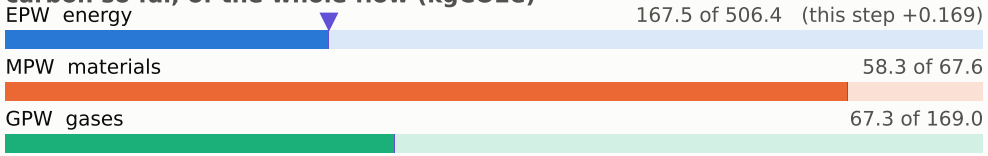
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

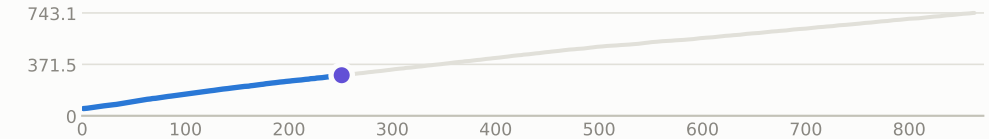
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

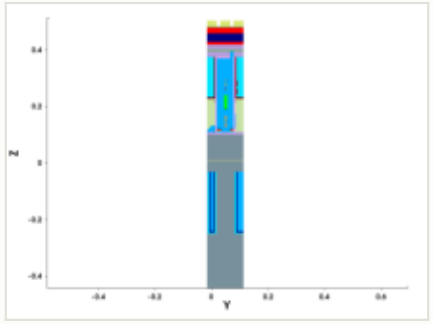
carbon so far, of the whole flow (kgCO2e)



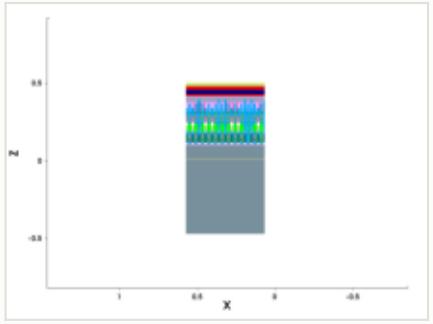
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



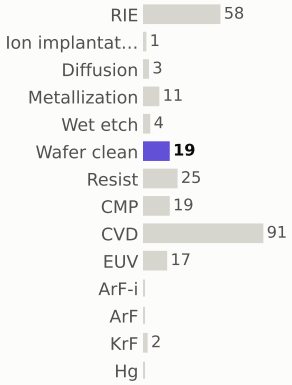
y cut at 0.049 µm



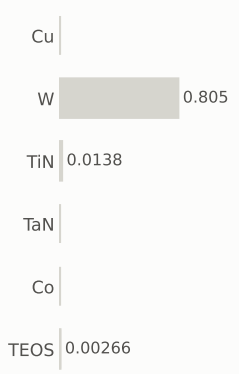
mask for this step

no mask for this step

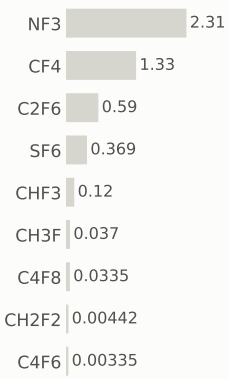
steps by process area



materials deposited (g)



gases emitted (g)



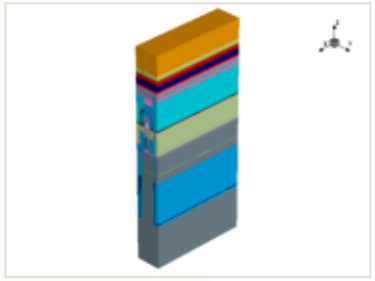
Step #2770: Resist

depo_fill · Resist

0.169	293.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

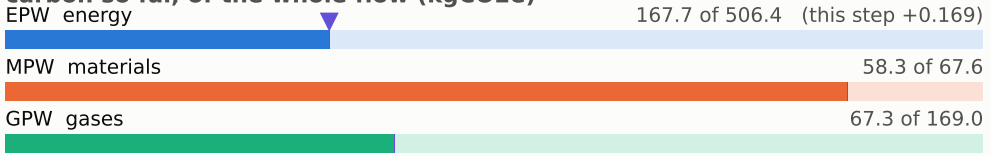


this step

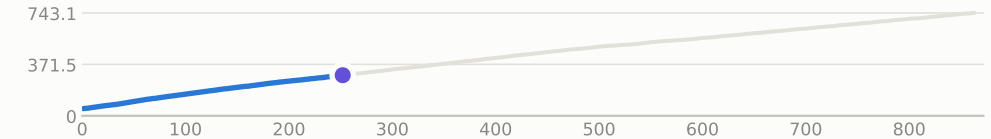
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

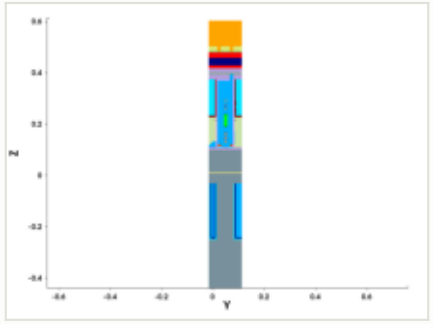
carbon so far, of the whole flow (kgCO2e)



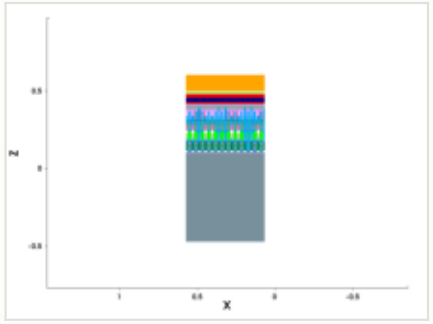
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



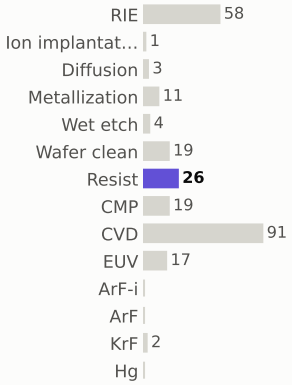
y cut at 0.049 µm



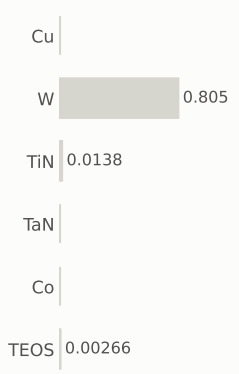
mask for this step

no mask for this step

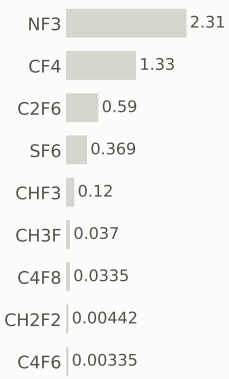
steps by process area



materials deposited (g)



gases emitted (g)



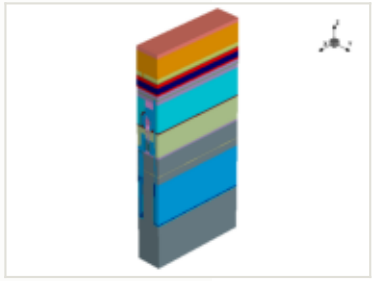
Step #2780: CVD

depo_fill · CVD

1.31	294.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

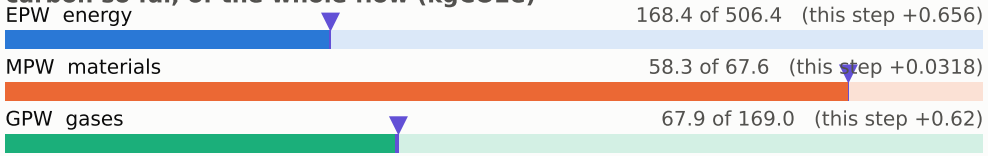


this step

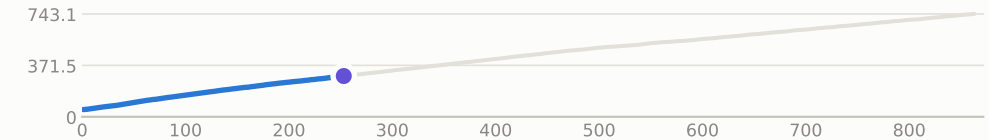
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

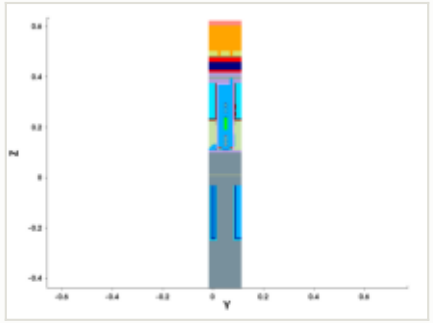
carbon so far, of the whole flow (kgCO2e)



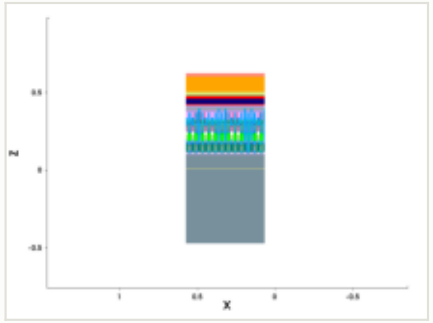
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



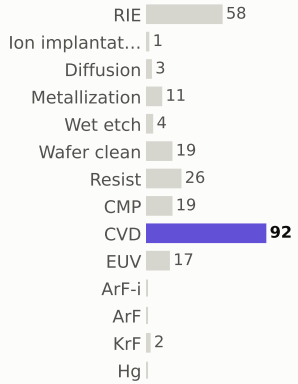
y cut at 0.049 µm



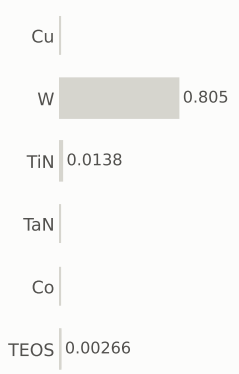
mask for this step

no mask for this step

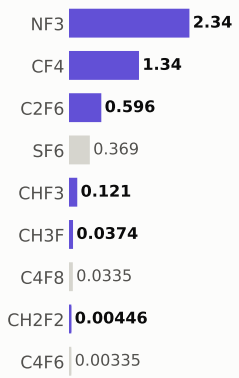
steps by process area



materials deposited (g)



gases emitted (g)



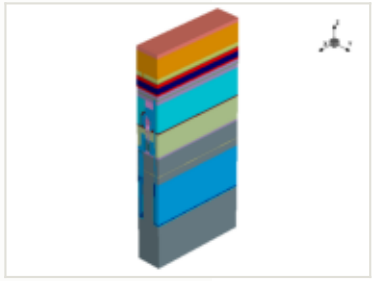
Step #2790: Wafer_clean

clean · Wafer_clean

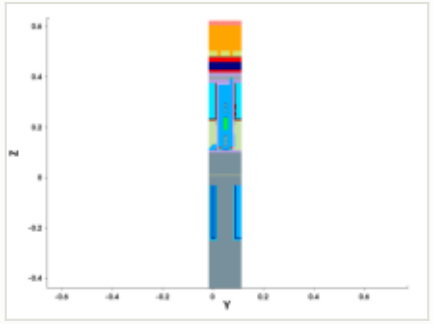
0.169	294.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

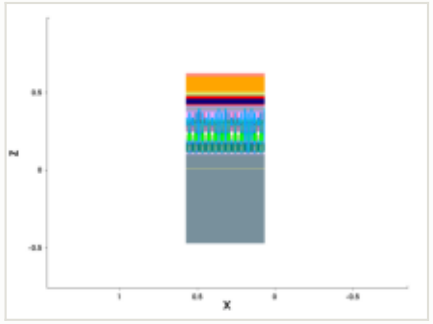
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

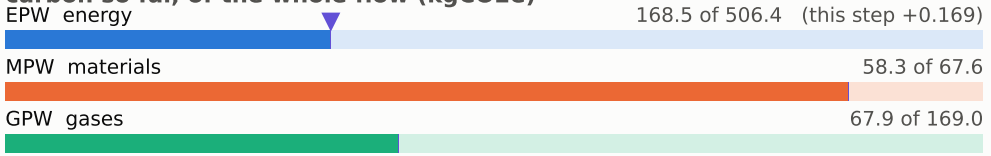
no mask for this step

this step

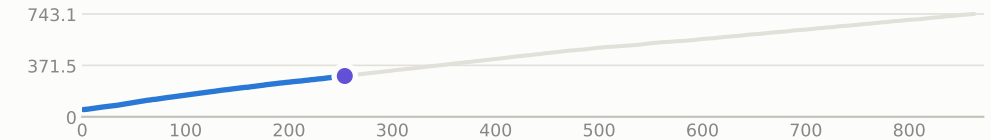
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

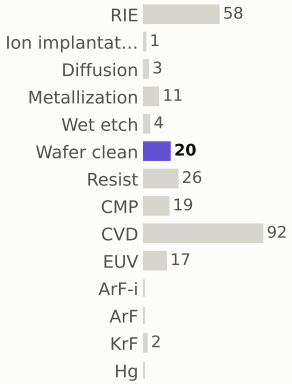
carbon so far, of the whole flow (kgCO2e)



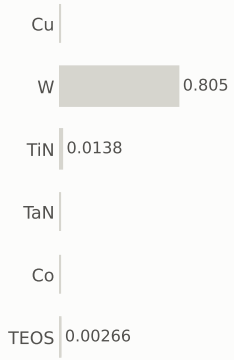
embodied carbon up to each step (kgCO2e per wafer)



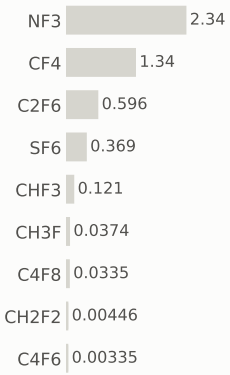
steps by process area



materials deposited (g)



gases emitted (g)



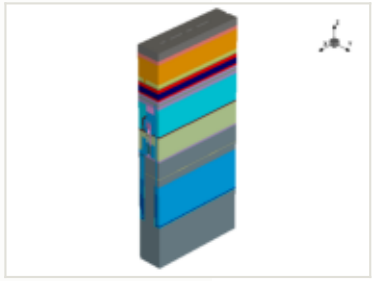
Step #2800: EUV

litho_patterning · EUV

1.96	296.7	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

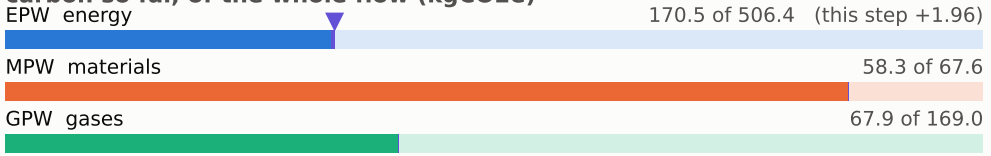


this step

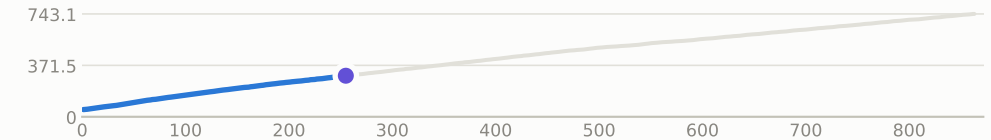
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

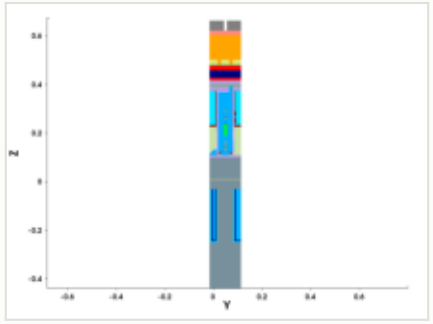
carbon so far, of the whole flow (kgCO2e)



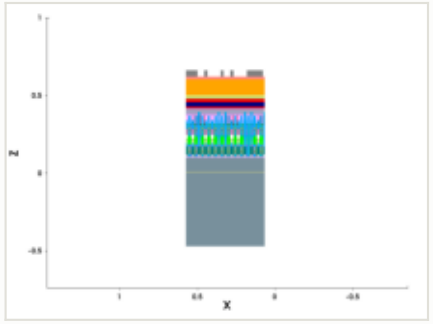
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

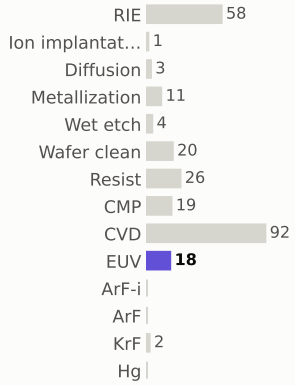


y cut at 0.049 µm

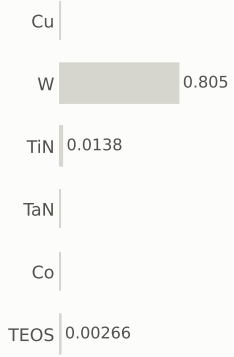


mask: layer 10202:0 (GDSII)

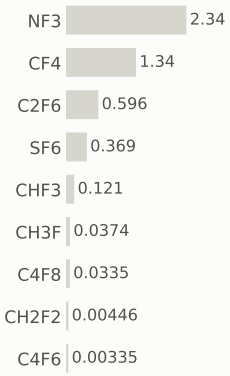
steps by process area



materials deposited (g)



gases emitted (g)



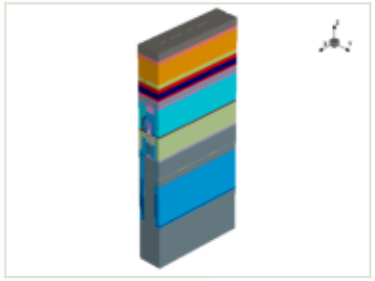
Step #2810: RIE

etch_anisotropic · RIE

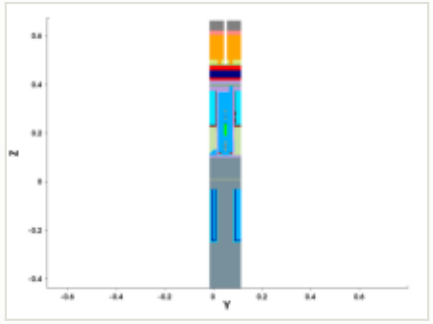
1.04	297.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

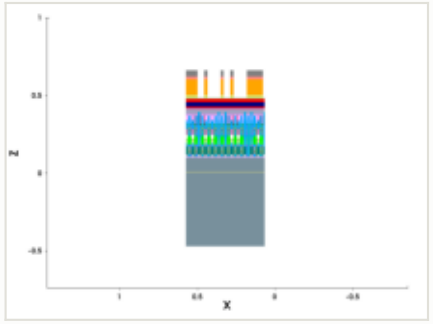
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

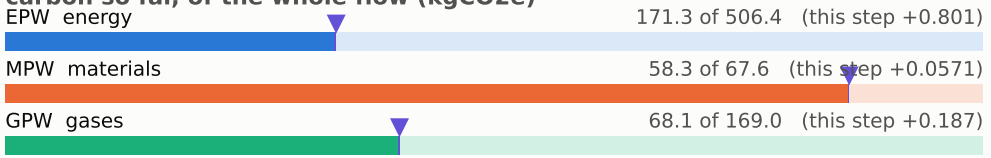
no mask for this step

this step

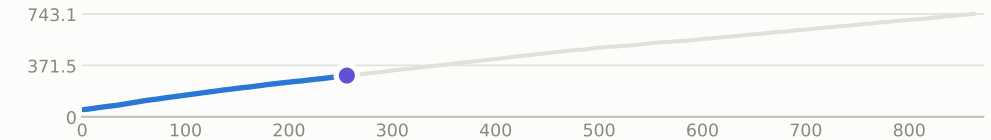
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

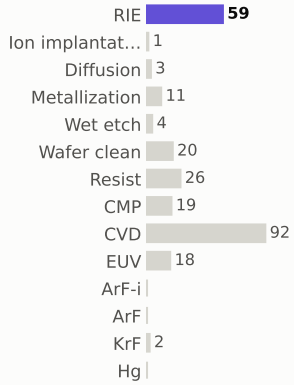
carbon so far, of the whole flow (kgCO2e)



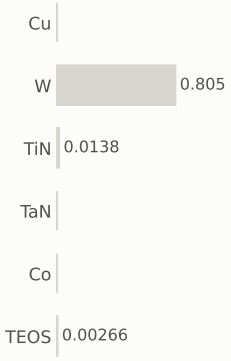
embodied carbon up to each step (kgCO2e per wafer)



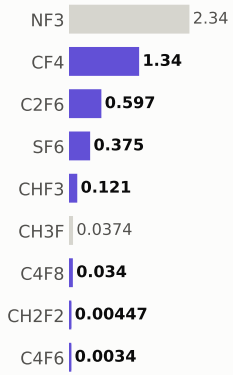
steps by process area



materials deposited (g)



gases emitted (g)



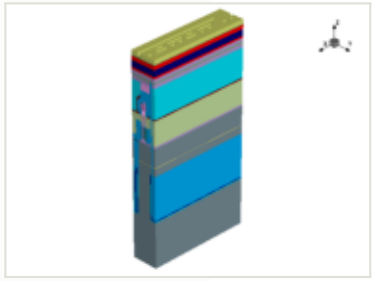
Step #2820: Wafer_clean

etch_strip · Wafer_clean

0.169	297.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

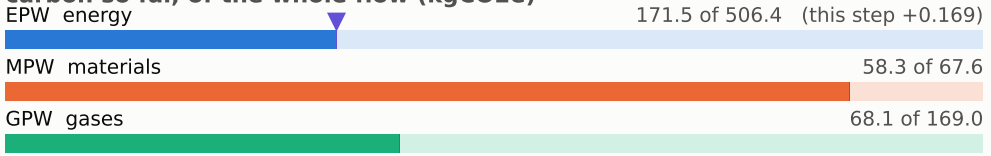


this step

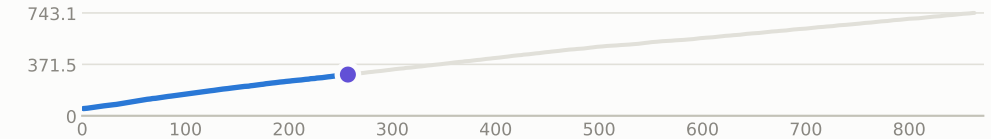
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

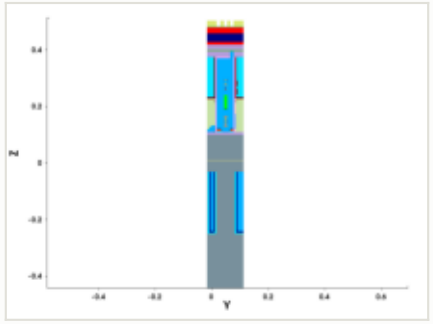
carbon so far, of the whole flow (kgCO2e)



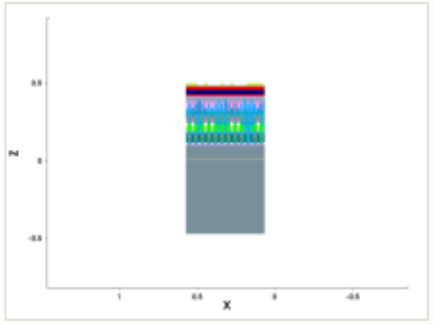
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



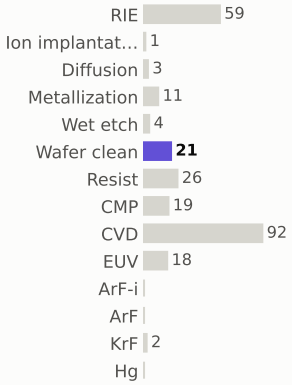
y cut at 0.049 µm



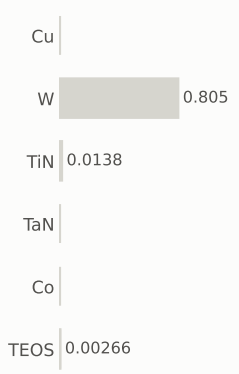
mask for this step

no mask for this step

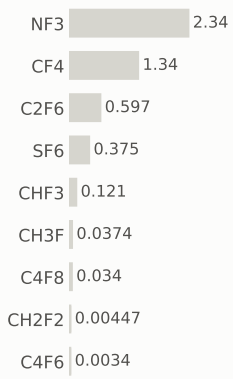
steps by process area



materials deposited (g)



gases emitted (g)



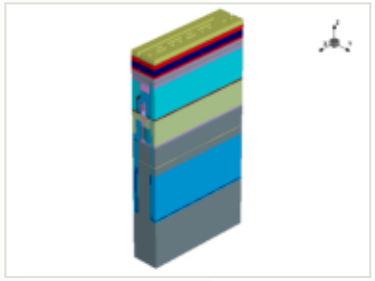
Step #2830: RIE

etch_anisotropic · RIE

1.04	299.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

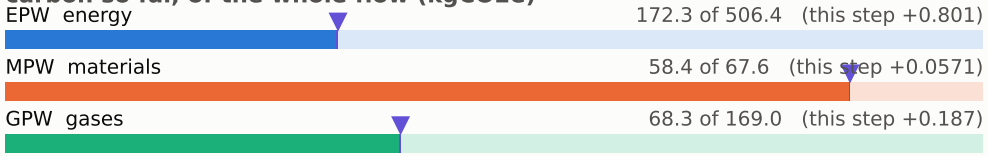


this step

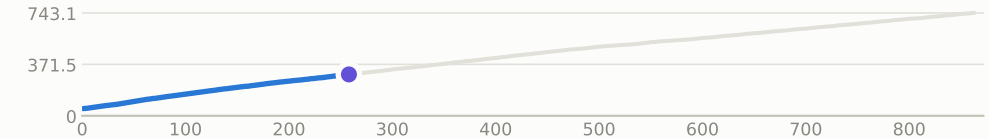
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

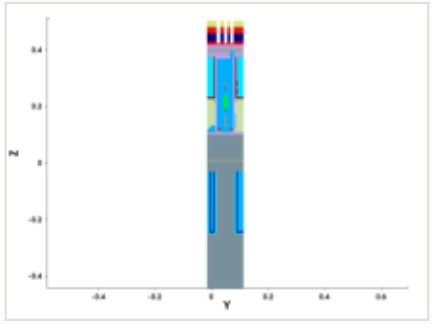
carbon so far, of the whole flow (kgCO2e)



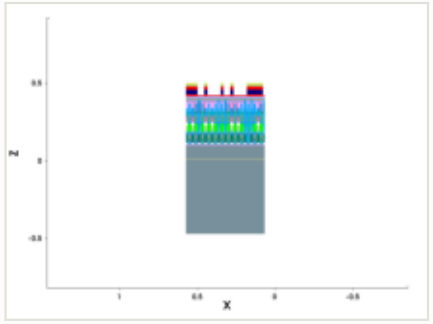
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



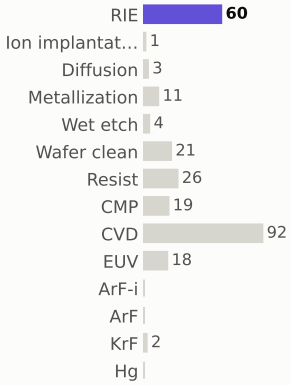
y cut at 0.049 µm



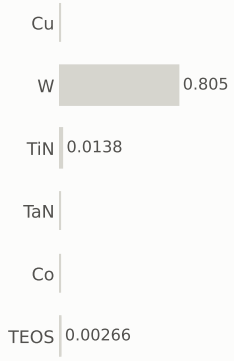
mask for this step

no mask for this step

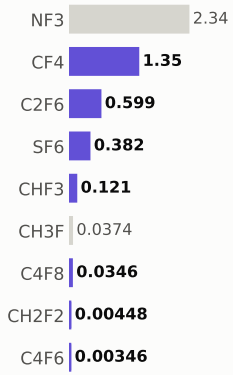
steps by process area



materials deposited (g)



gases emitted (g)



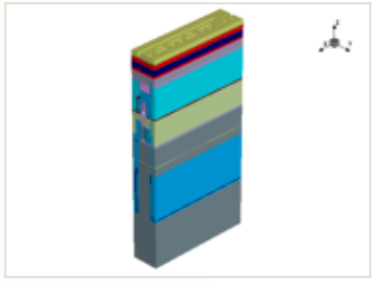
Step #2840: Wafer_clean

clean · Wafer_clean

0.169	299.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



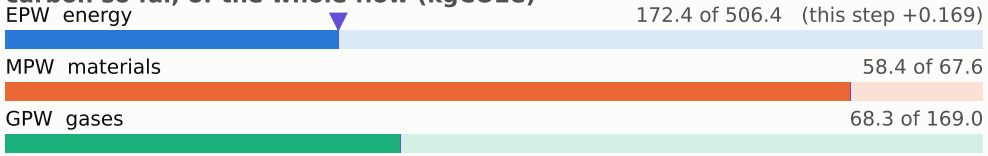
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

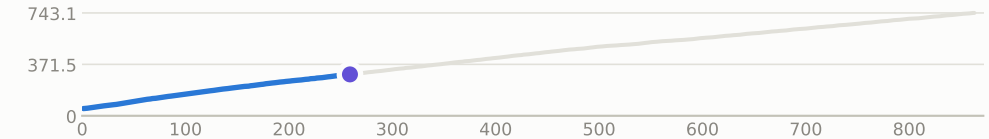
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

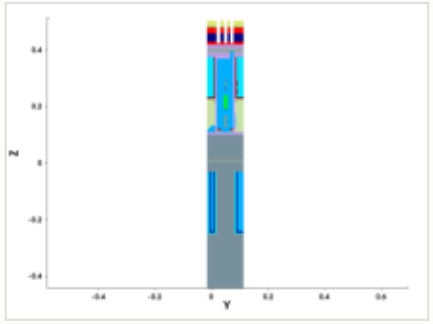
carbon so far, of the whole flow (kgCO2e)



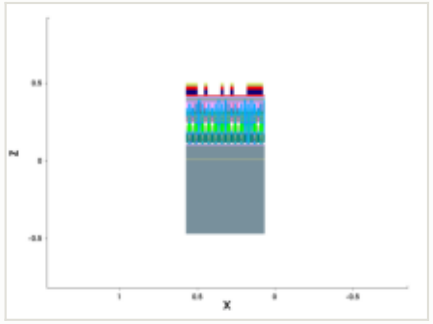
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



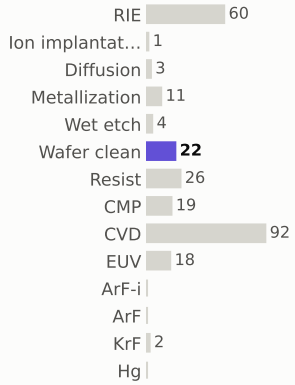
y cut at 0.049 µm



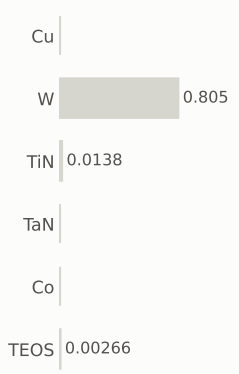
mask for this step

no mask for this step

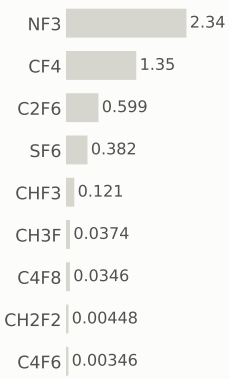
steps by process area



materials deposited (g)



gases emitted (g)



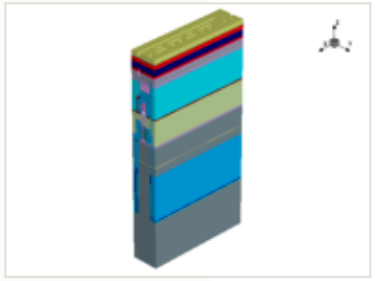
Step #2850: RIE

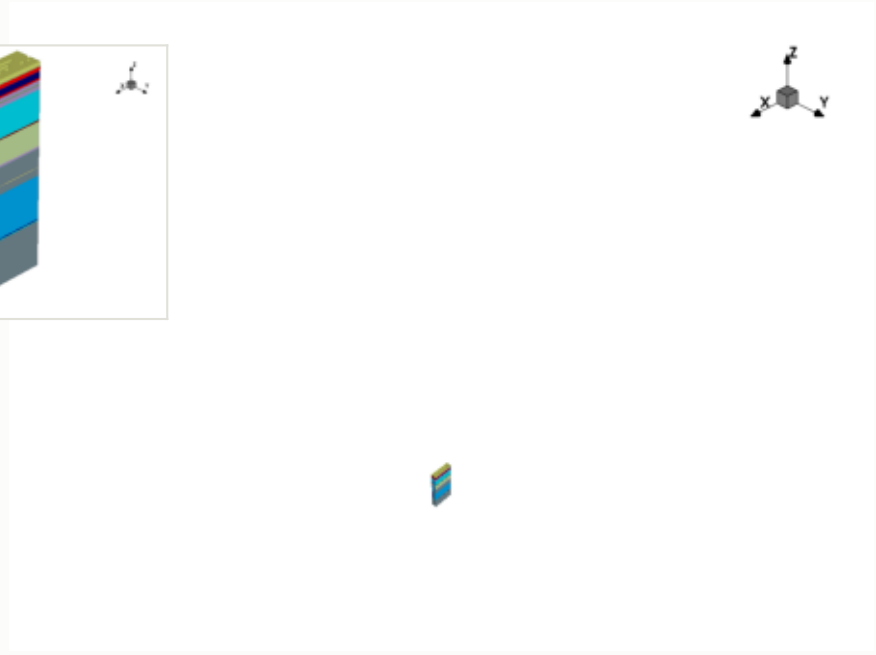
etch_anisotropic · RIE

1.04	300.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



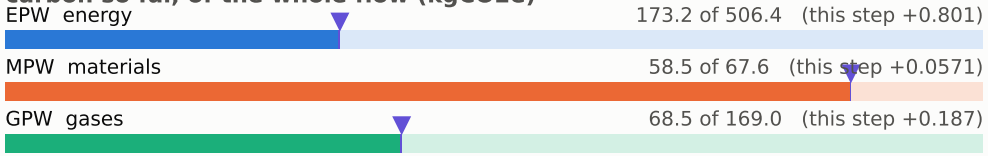


this step

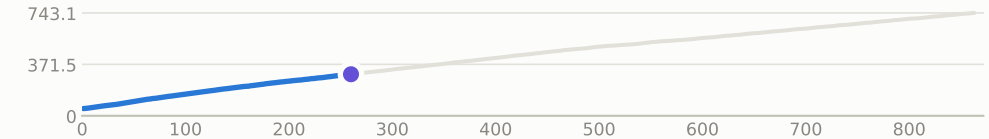
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

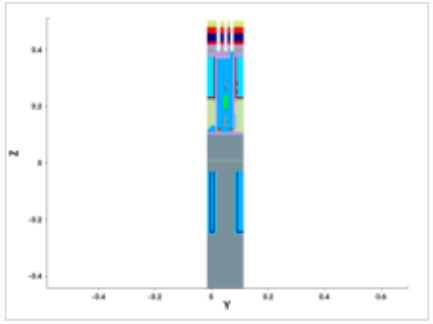
carbon so far, of the whole flow (kgCO2e)



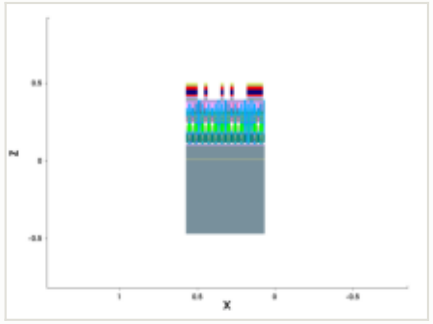
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



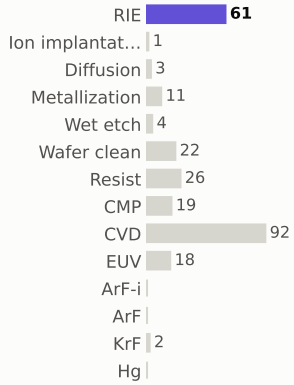
y cut at 0.049 µm



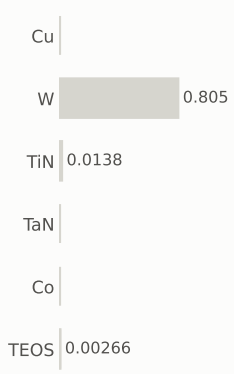
mask for this step

no mask for this step

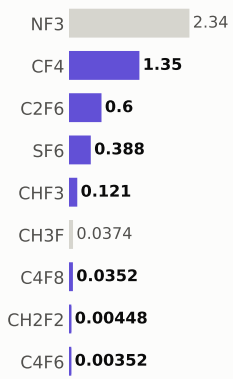
steps by process area



materials deposited (g)



gases emitted (g)



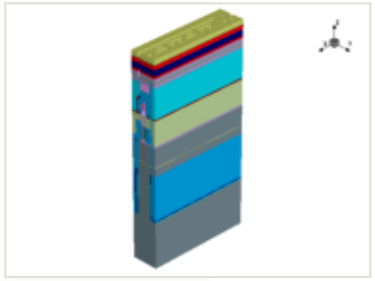
Step #2860: Wafer_clean

clean · Wafer_clean

0.169	300.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

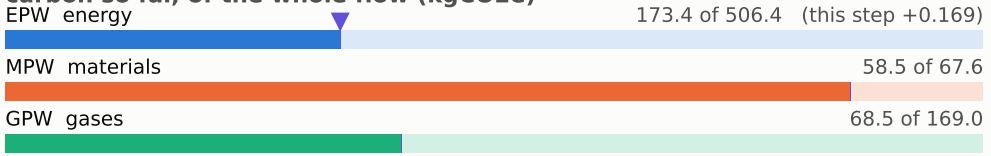


this step

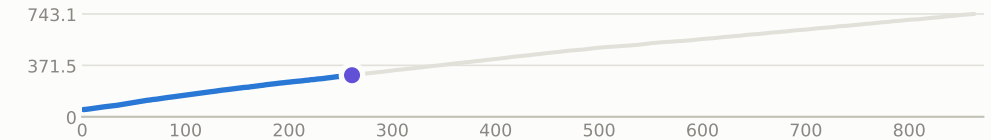
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

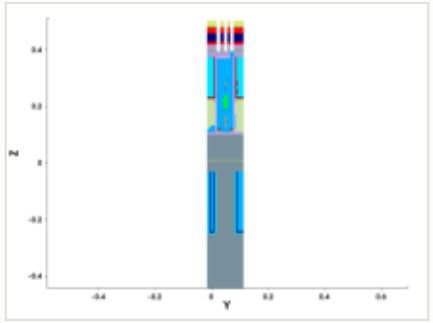
carbon so far, of the whole flow (kgCO2e)



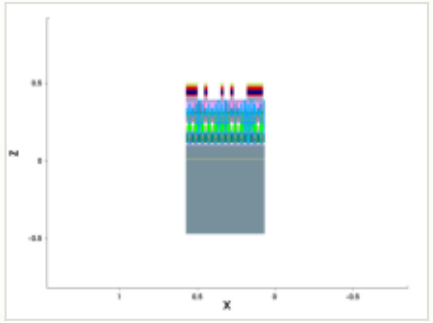
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



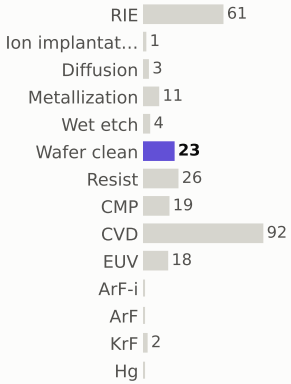
y cut at 0.049 µm



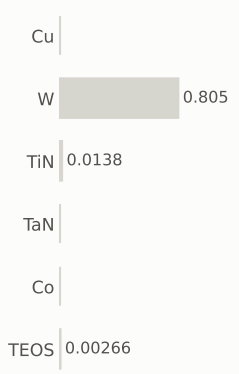
mask for this step

no mask for this step

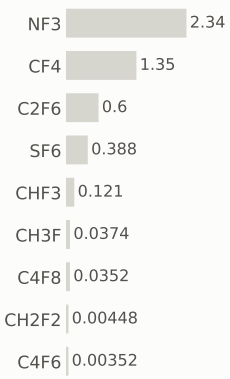
steps by process area



materials deposited (g)



gases emitted (g)



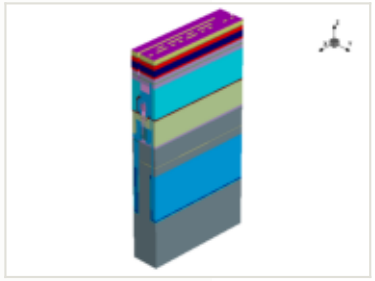
Step #2870: Metallization

depo_ellipsoidal · Metallization

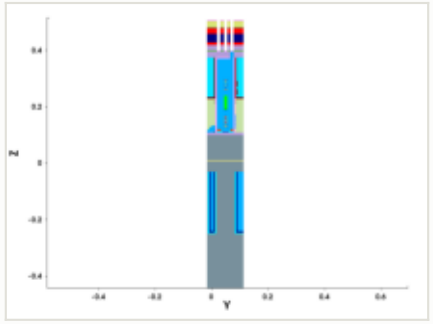
1.2	301.6	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

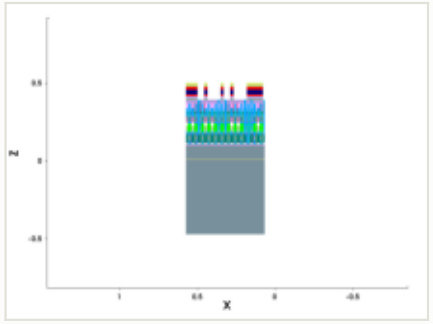
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

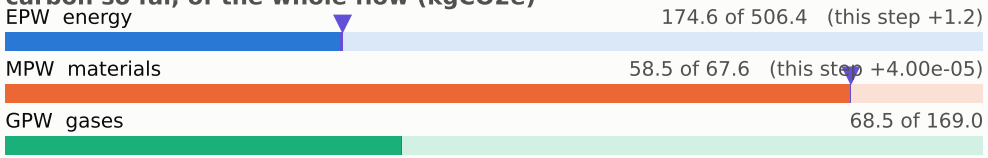
no mask for this step

this step

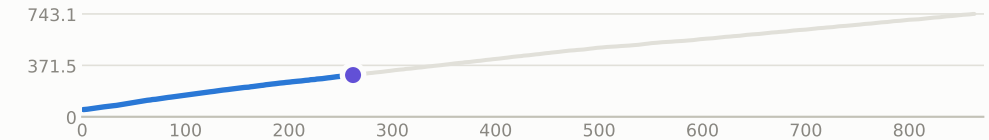
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

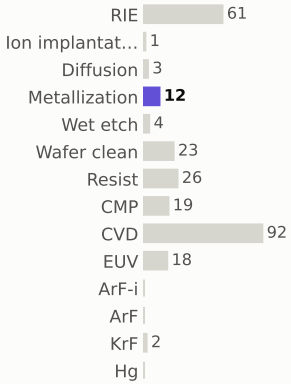
carbon so far, of the whole flow (kgCO2e)



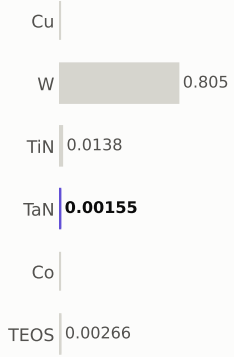
embodied carbon up to each step (kgCO2e per wafer)



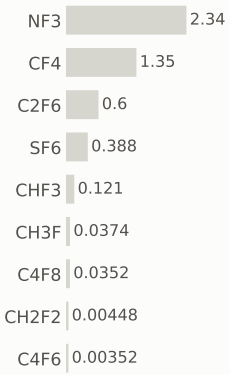
steps by process area



materials deposited (g)



gases emitted (g)



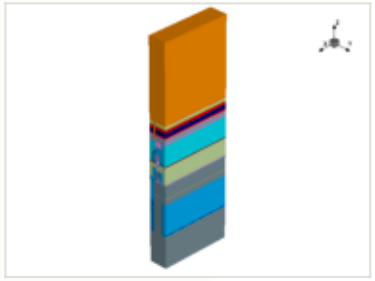
Step #2880: Metallization

depo_fill · Metallization

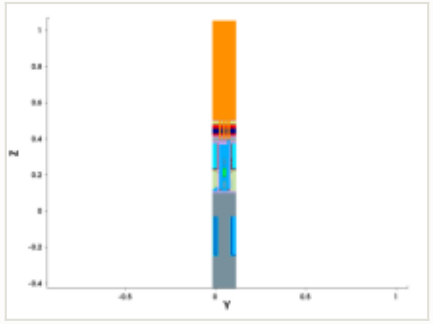
1.21	302.8	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

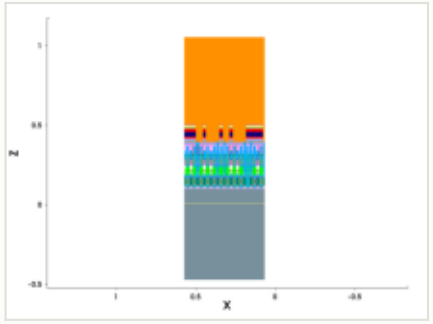
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



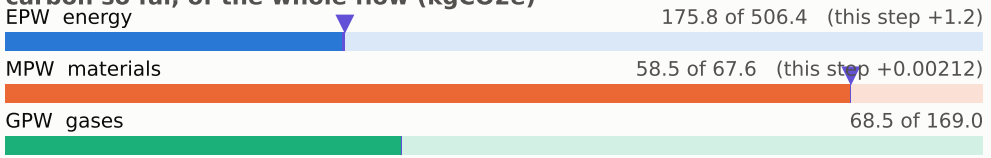
mask for this step

no mask for this step

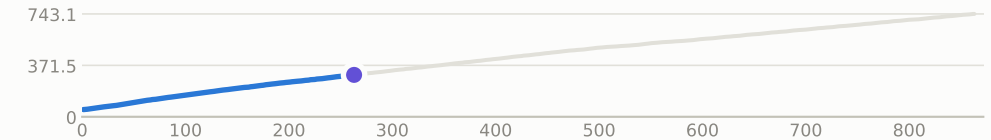
this step

process area	Metallization	EPW	1.2 kgCO2e
material	Cu	EPW tool	0.86 kgCO2e
thickness	0.55 µm	EPW overhead	0.344 kgCO2e
material GWP	6.09011 gCO2e/g	MPW	0.00212 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.21 kgCO2e**

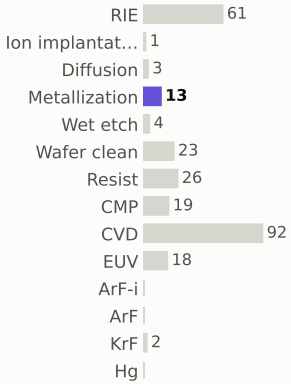
carbon so far, of the whole flow (kgCO2e)



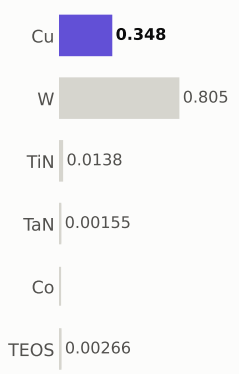
embodied carbon up to each step (kgCO2e per wafer)



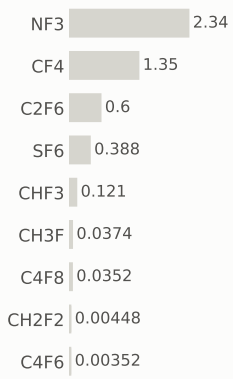
steps by process area



materials deposited (g)



gases emitted (g)



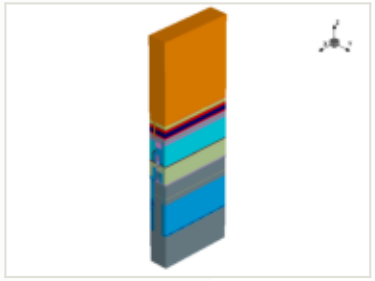
Step #2890: Diffusion

treatment · Diffusion

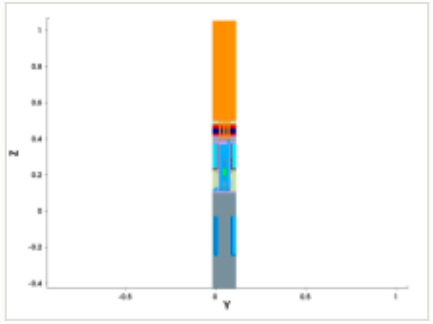
0.801	303.6	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

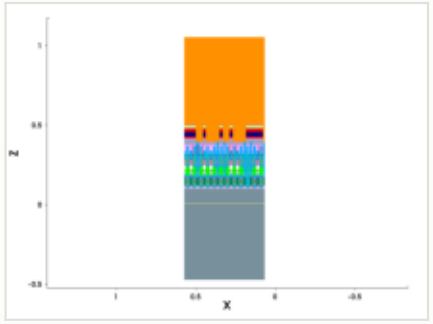
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

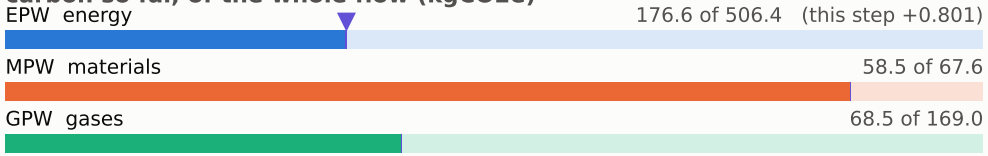
no mask for this step

this step

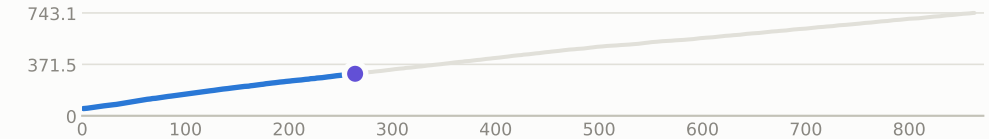
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

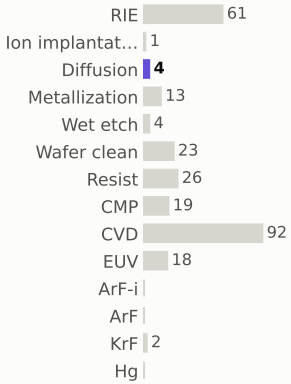
carbon so far, of the whole flow (kgCO2e)



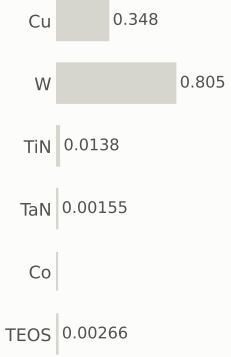
embodied carbon up to each step (kgCO2e per wafer)



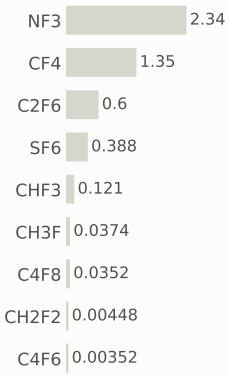
steps by process area



materials deposited (g)



gases emitted (g)



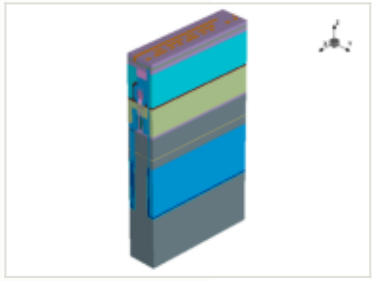
Step #2900: CMP

etch_cmp · CMP

0.169	303.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

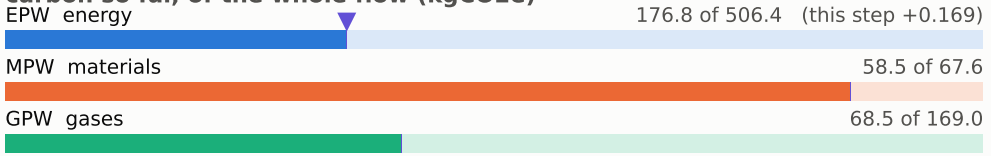


this step

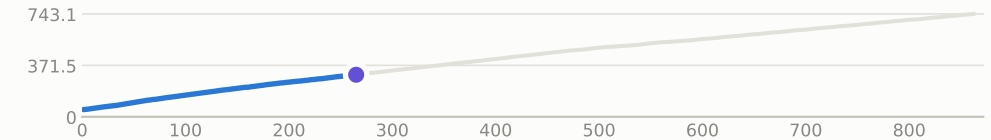
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

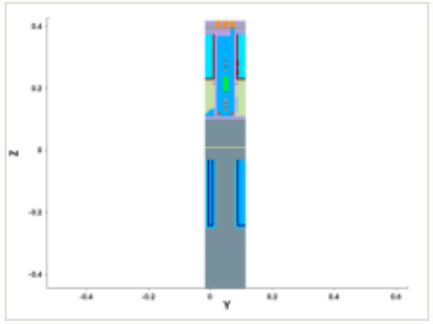
carbon so far, of the whole flow (kgCO2e)



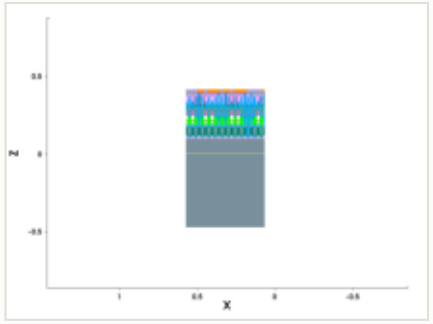
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



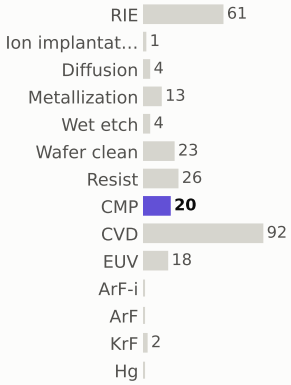
y cut at 0.049 µm



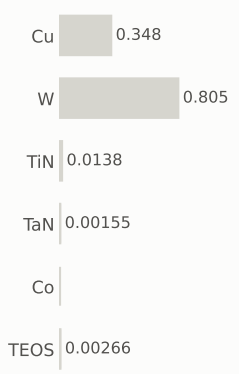
mask for this step

no mask for this step

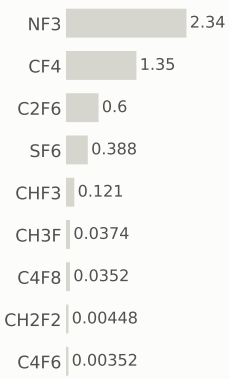
steps by process area



materials deposited (g)



gases emitted (g)



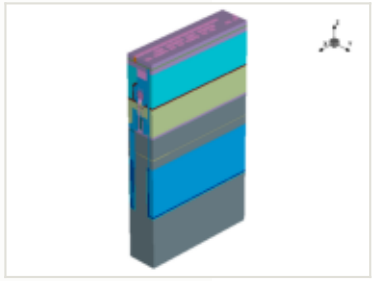
Step #2910: Metallization

depo_isotropic · Metallization

1.2	304.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

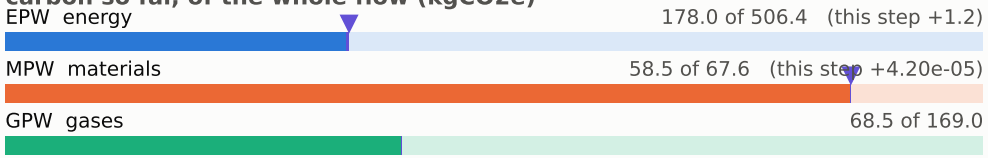


this step

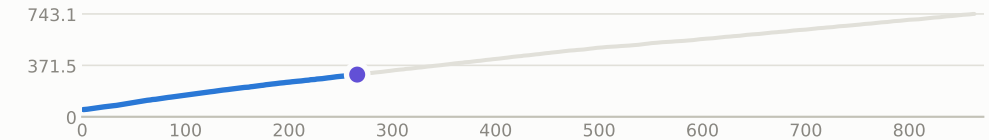
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

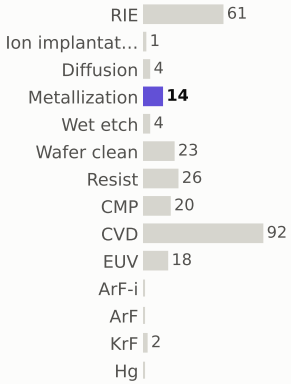
carbon so far, of the whole flow (kgCO2e)



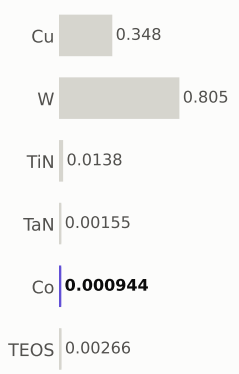
embodied carbon up to each step (kgCO2e per wafer)



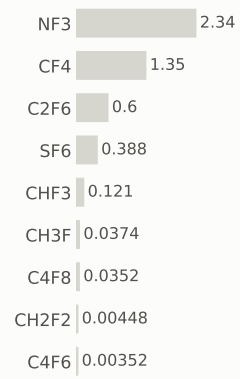
steps by process area



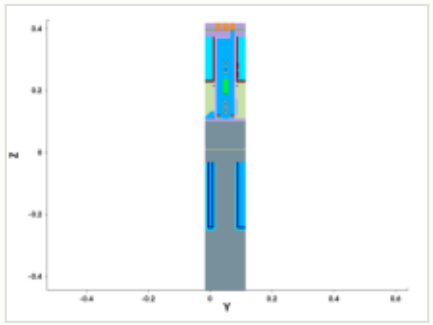
materials deposited (g)



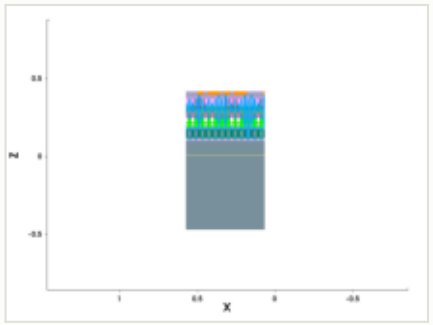
gases emitted (g)



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

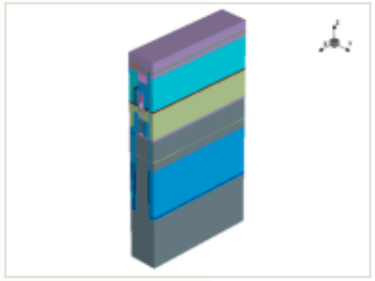
Step #2930: CVD

depo_fill · CVD

1.31	306.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

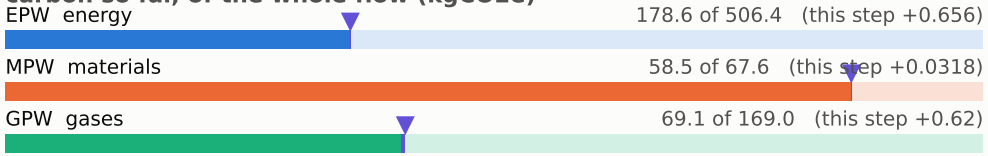


this step

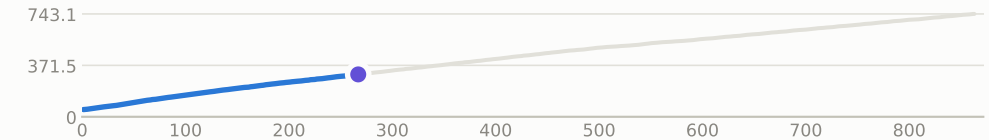
process area	CVD
material	—
thickness	0.024 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

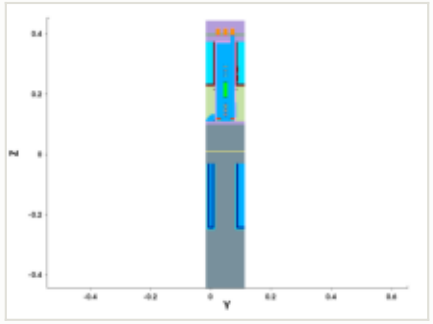
carbon so far, of the whole flow (kgCO2e)



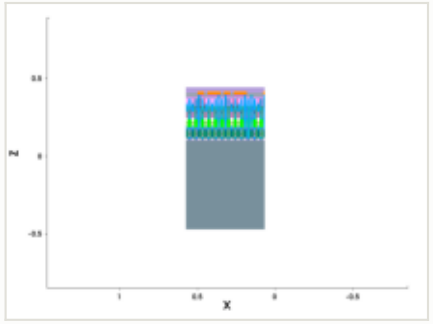
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



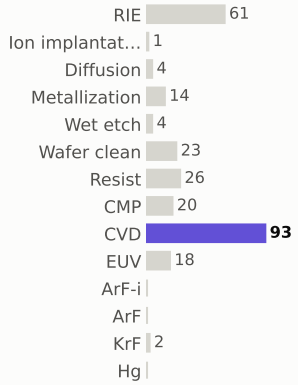
y cut at 0.049 µm



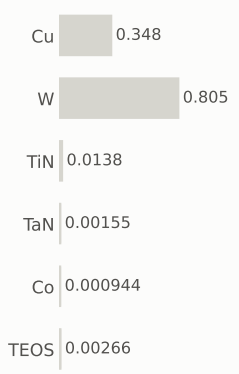
mask for this step

no mask for this step

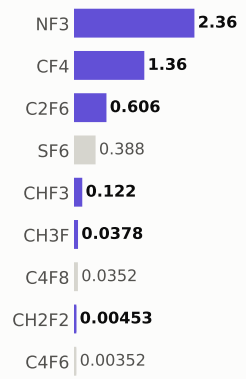
steps by process area



materials deposited (g)



gases emitted (g)



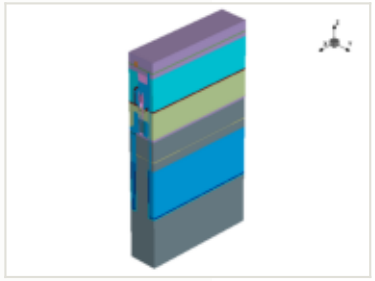
Step #2940: Diffusion

treatment · Diffusion

0.801	307.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

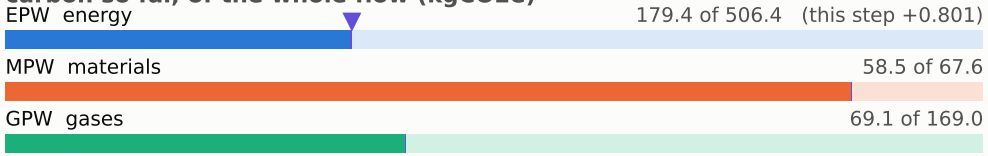


this step

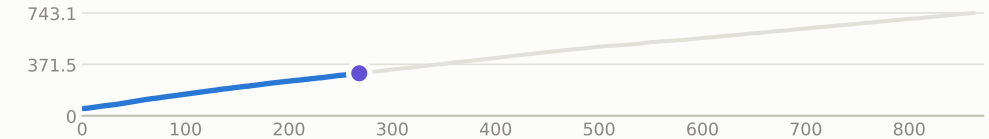
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

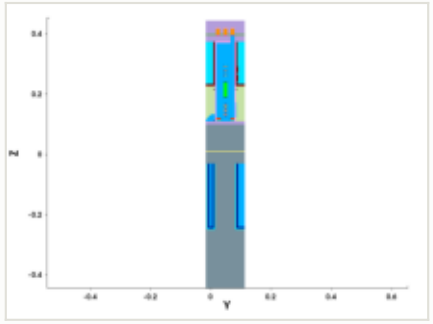
carbon so far, of the whole flow (kgCO2e)



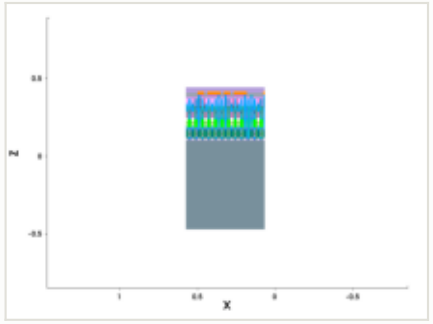
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



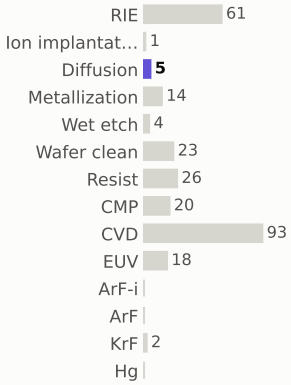
y cut at 0.049 µm



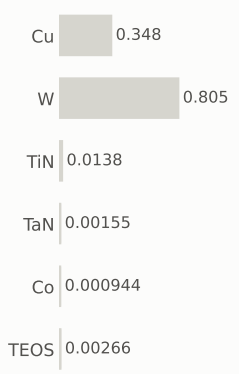
mask for this step

no mask for this step

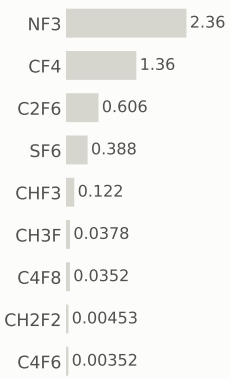
steps by process area



materials deposited (g)



gases emitted (g)



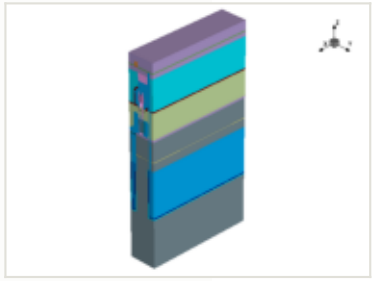
Step #2950: Wafer_clean

clean · Wafer_clean

0.169	307.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



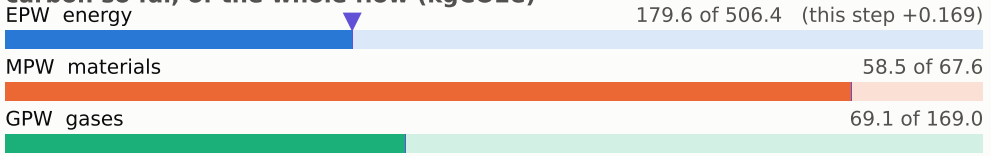
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

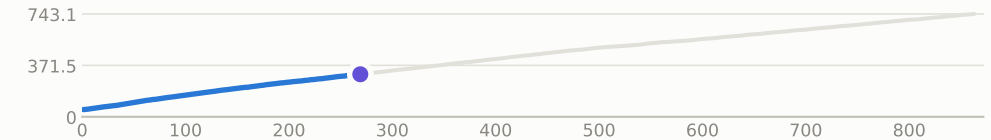
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

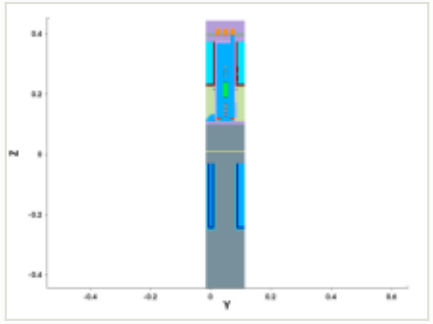
carbon so far, of the whole flow (kgCO2e)



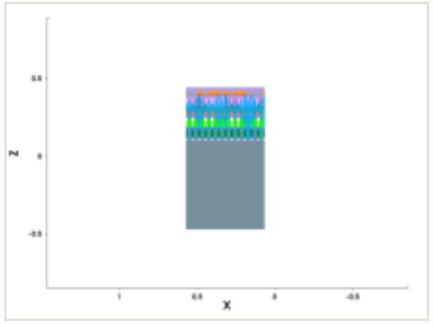
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



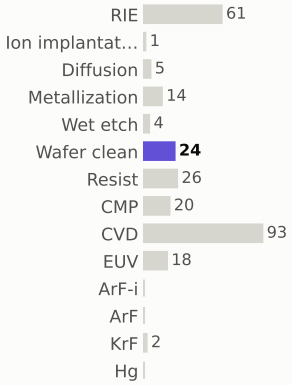
y cut at 0.049 µm



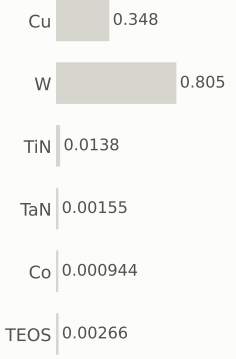
mask for this step

no mask for this step

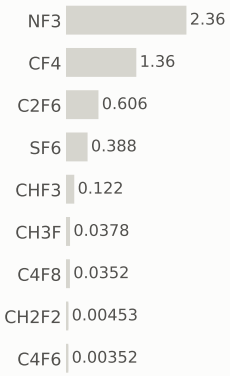
steps by process area



materials deposited (g)



gases emitted (g)



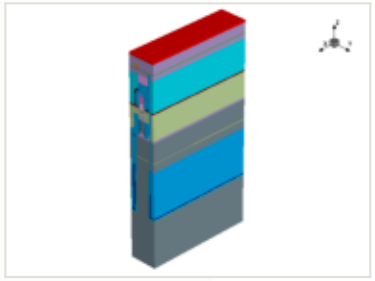
Step #2960: CVD

depo_fill · CVD

1.31	308.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

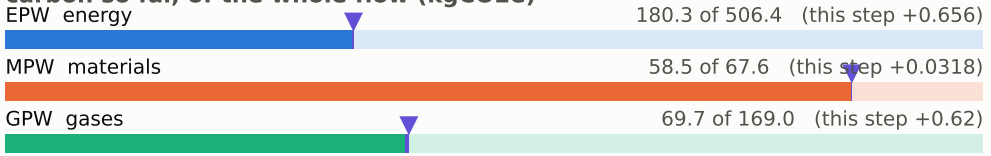


this step

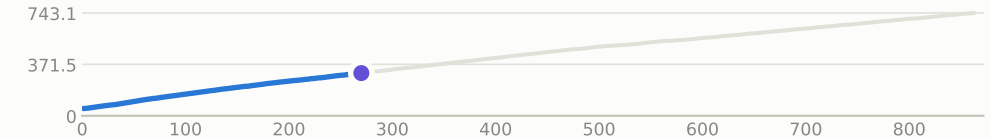
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

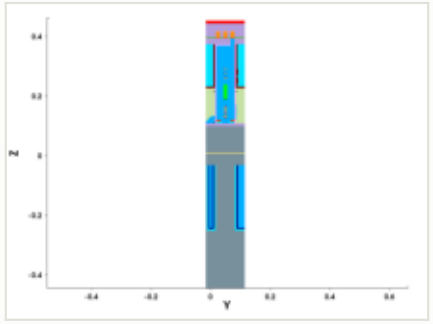
carbon so far, of the whole flow (kgCO2e)



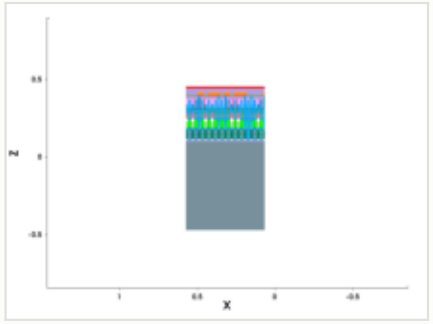
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



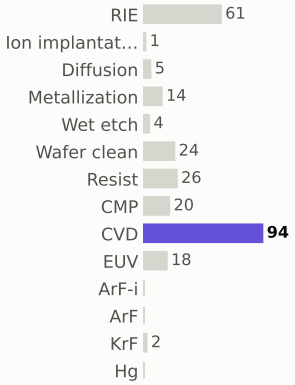
y cut at 0.049 µm



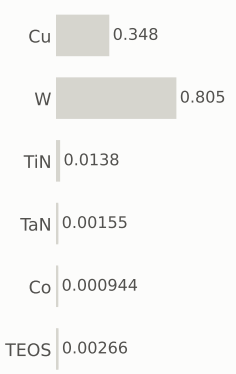
mask for this step

no mask for this step

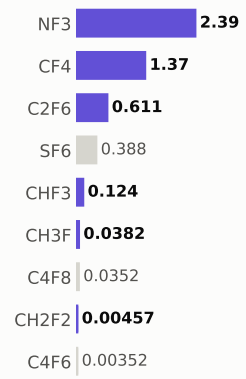
steps by process area



materials deposited (g)



gases emitted (g)



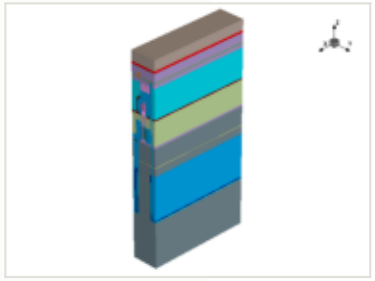
Step #2970: Resist

depo_fill · Resist

0.169	308.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

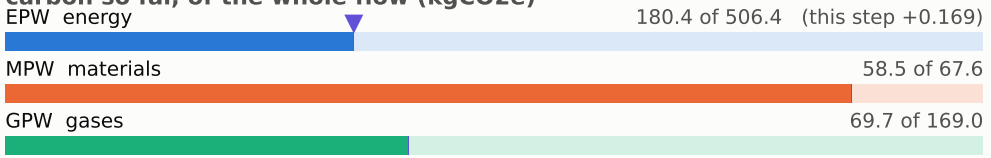


this step

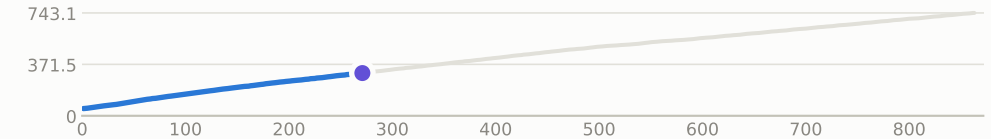
process area	Resist
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

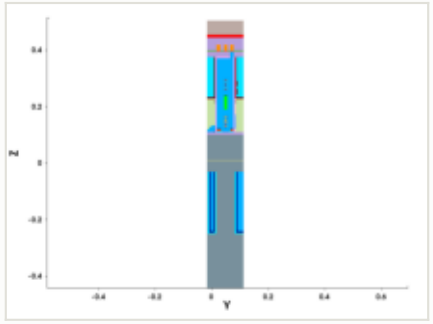
carbon so far, of the whole flow (kgCO2e)



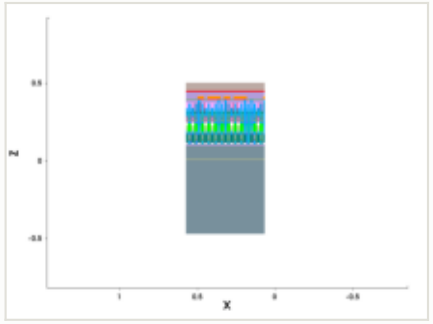
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



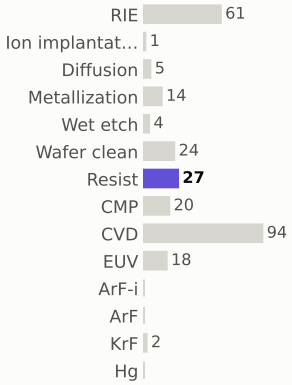
y cut at 0.049 µm



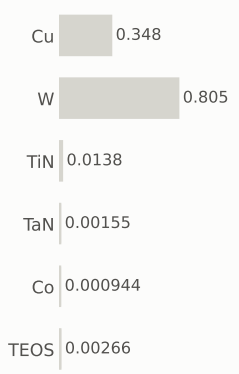
mask for this step

no mask for this step

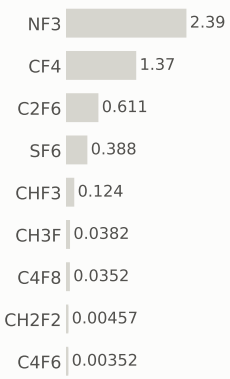
steps by process area



materials deposited (g)



gases emitted (g)



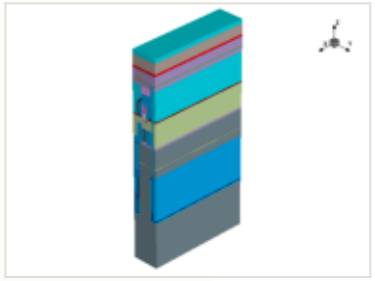
Step #2980: CVD

depo_fill · CVD

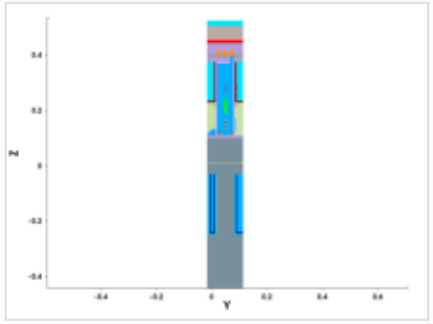
1.31	310.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

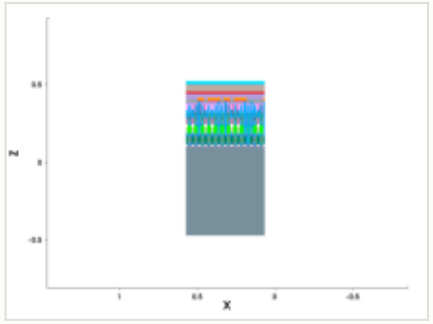
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

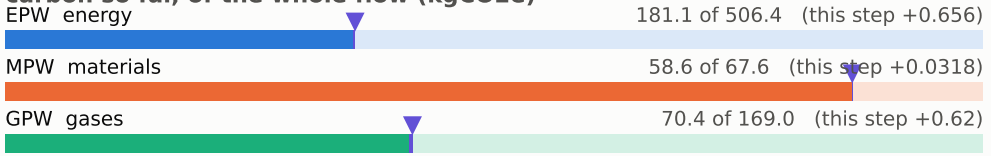
no mask for this step

this step

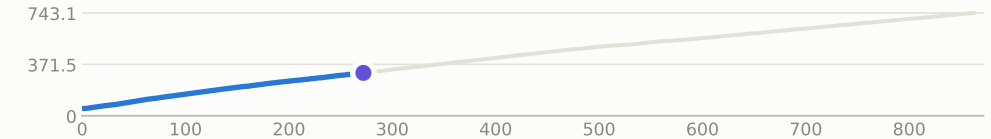
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

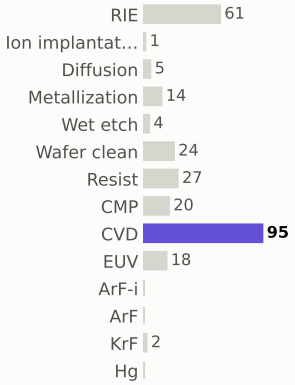
carbon so far, of the whole flow (kgCO2e)



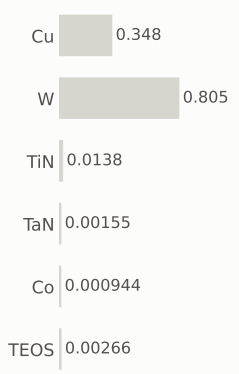
embodied carbon up to each step (kgCO2e per wafer)



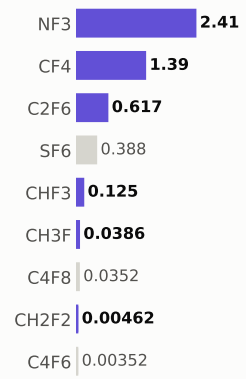
steps by process area



materials deposited (g)



gases emitted (g)



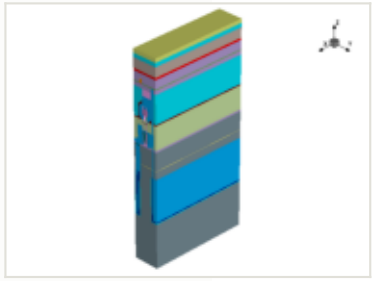
Step #2990: CVD

depo_fill · CVD

1.31	311.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

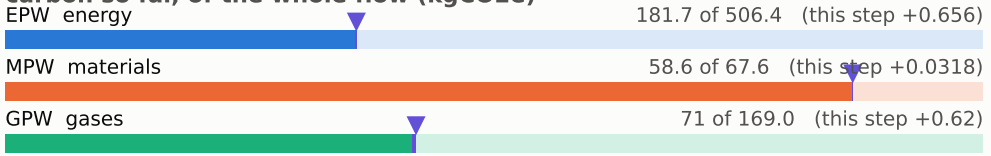


this step

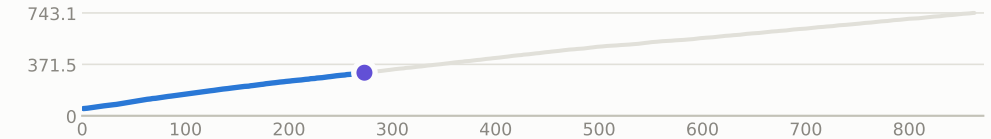
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

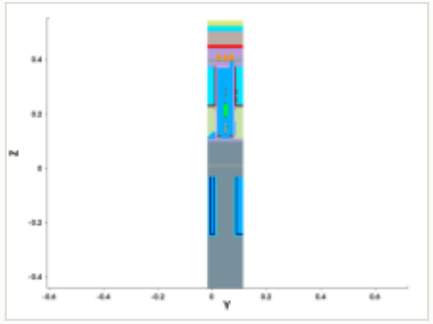
carbon so far, of the whole flow (kgCO2e)



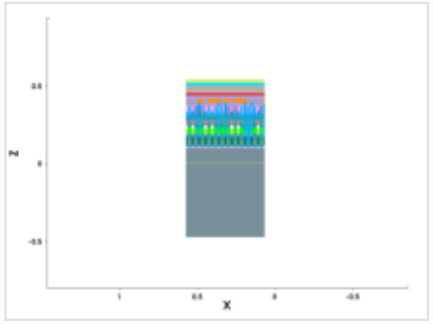
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



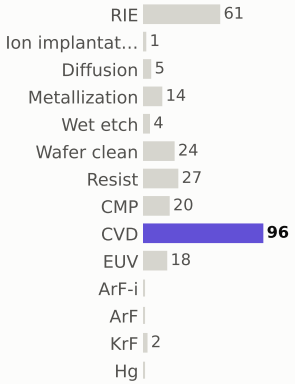
y cut at 0.049 µm



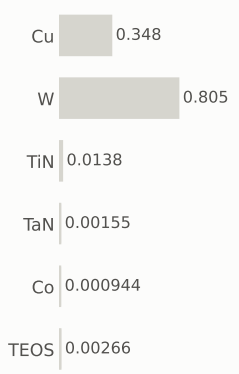
mask for this step

no mask for this step

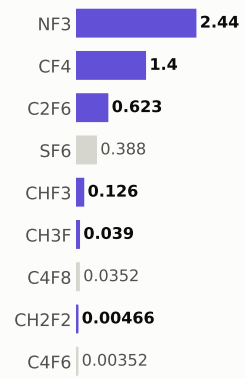
steps by process area



materials deposited (g)



gases emitted (g)



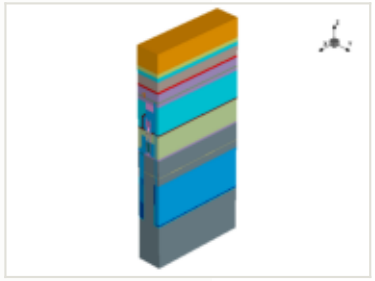
Step #3000: Resist

depo_fill · Resist

0.169	311.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

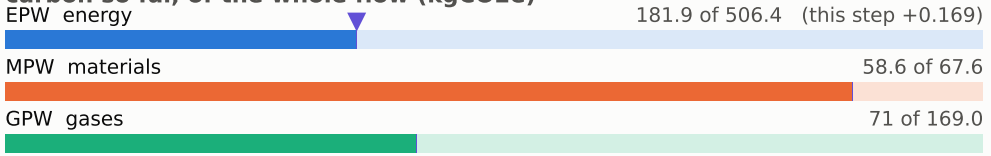


this step

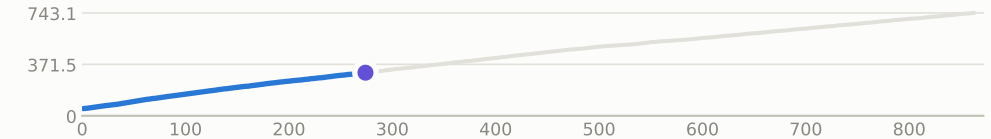
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

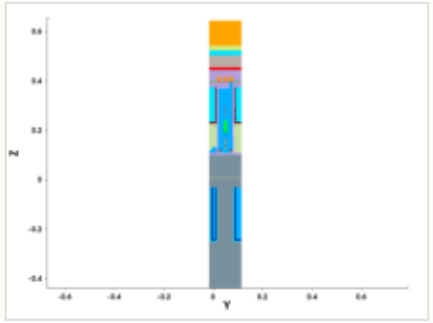
carbon so far, of the whole flow (kgCO2e)



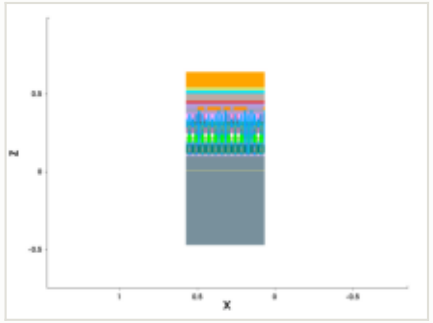
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



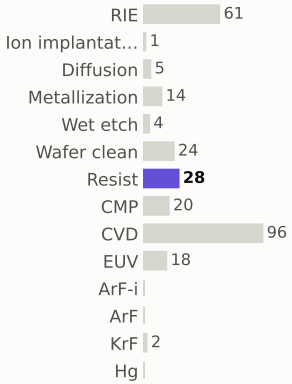
y cut at 0.049 µm



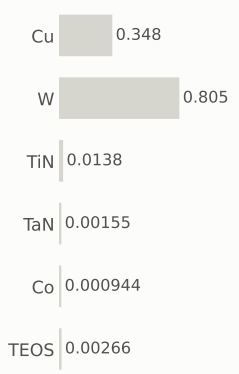
mask for this step

no mask for this step

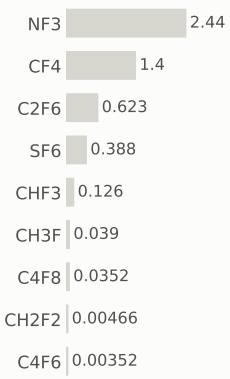
steps by process area



materials deposited (g)



gases emitted (g)



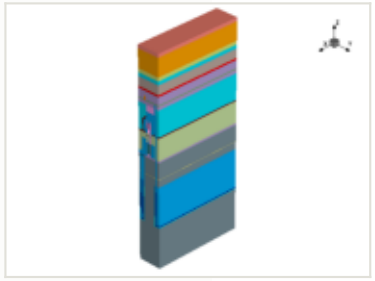
Step #3010: CVD

depo_fill · CVD

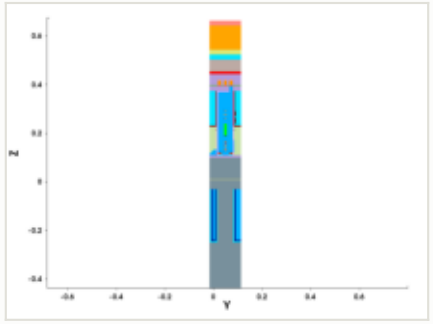
1.31	312.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

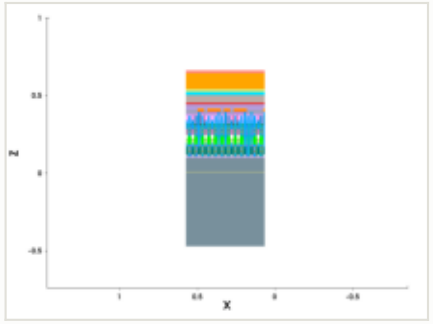
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

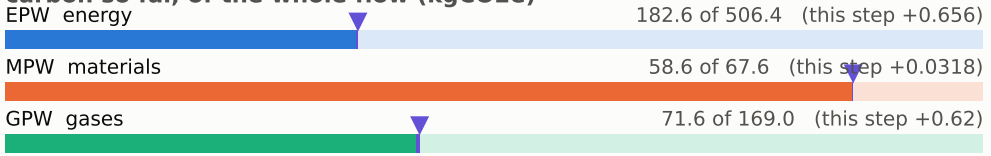
no mask for this step

this step

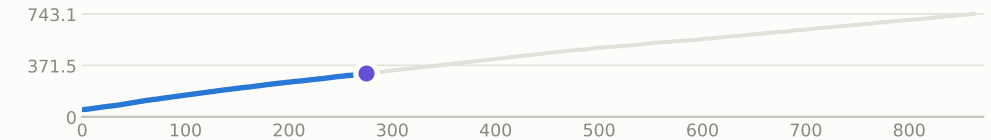
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

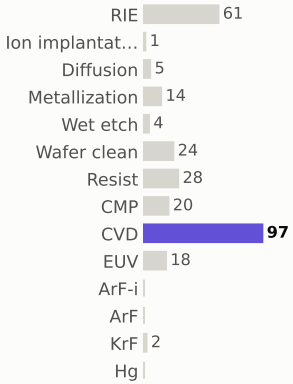
carbon so far, of the whole flow (kgCO2e)



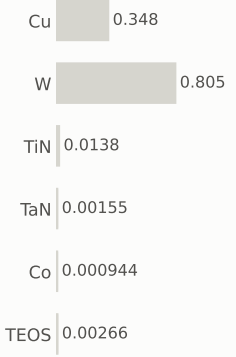
embodied carbon up to each step (kgCO2e per wafer)



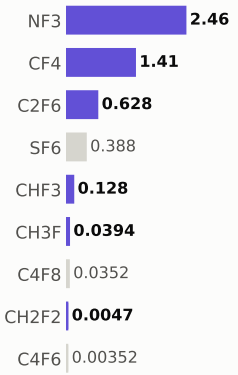
steps by process area



materials deposited (g)



gases emitted (g)



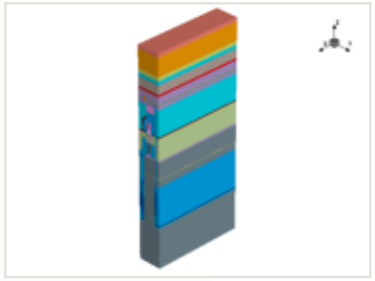
Step #3020: Wafer_clean

clean · Wafer_clean

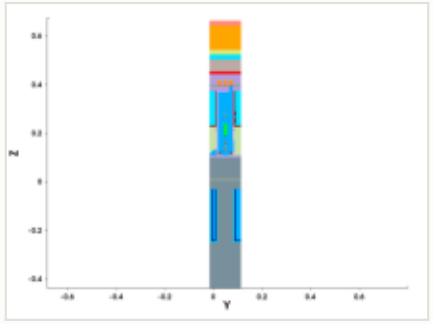
0.169	313.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

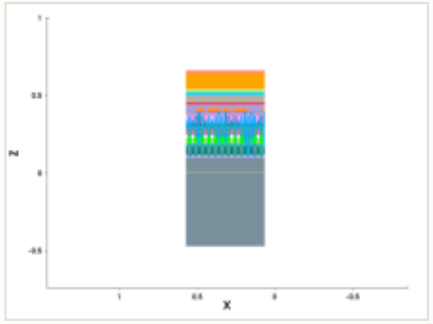
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

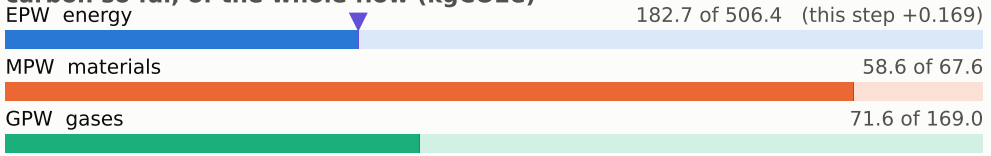
no mask for this step

this step

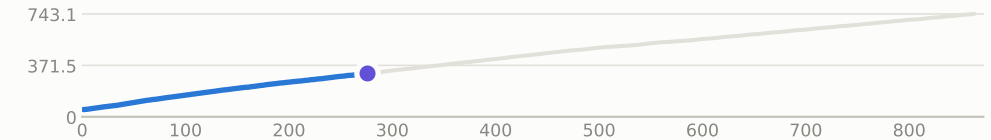
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

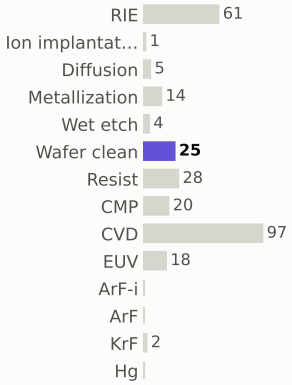
carbon so far, of the whole flow (kgCO2e)



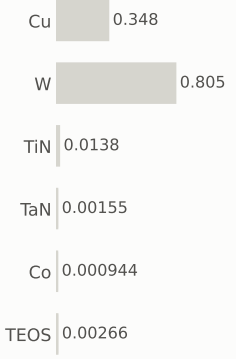
embodied carbon up to each step (kgCO2e per wafer)



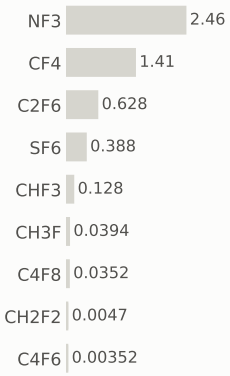
steps by process area



materials deposited (g)



gases emitted (g)



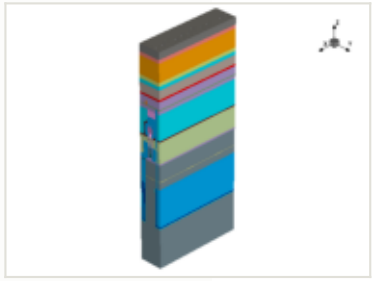
Step #3030: EUV

litho_patterning · EUV

1.96	314.9	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

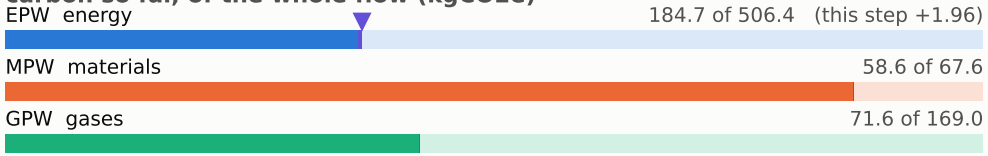


this step

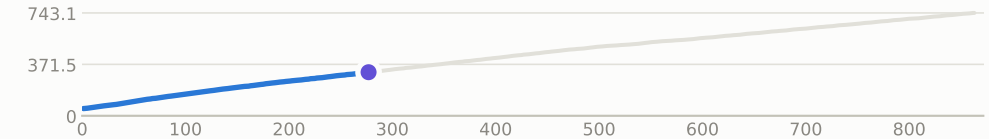
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

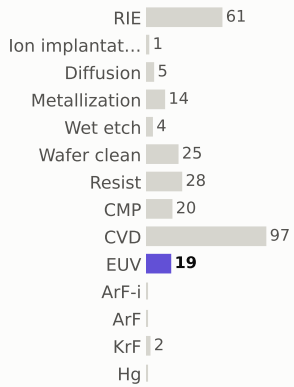
carbon so far, of the whole flow (kgCO2e)



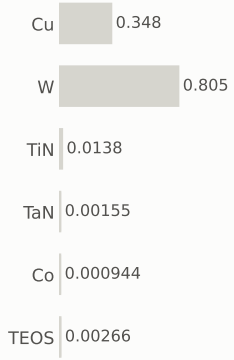
embodied carbon up to each step (kgCO2e per wafer)



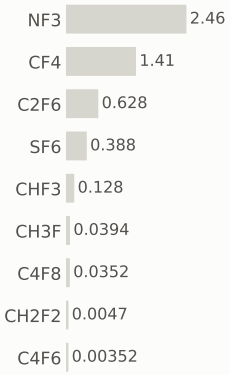
steps by process area



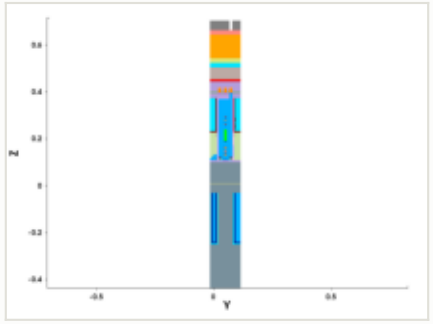
materials deposited (g)



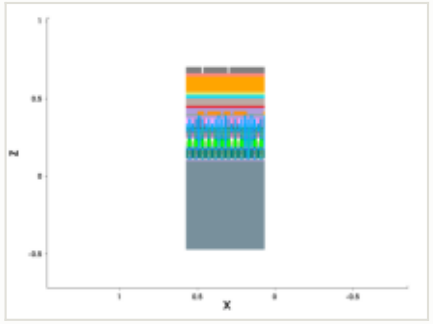
gases emitted (g)



x cut at 0.385 µm



y cut at 0.049 µm



mask: layer 10221:0 (GDSII)

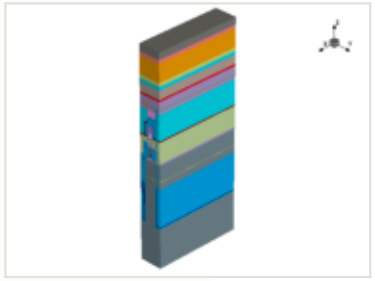
Step #3040: RIE

etch_anisotropic · RIE

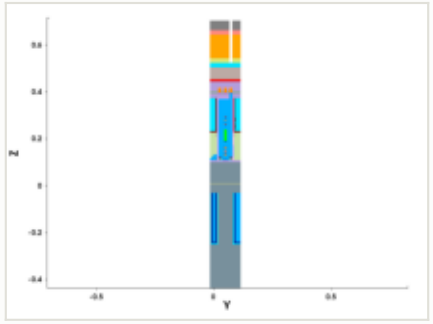
1.04	316.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

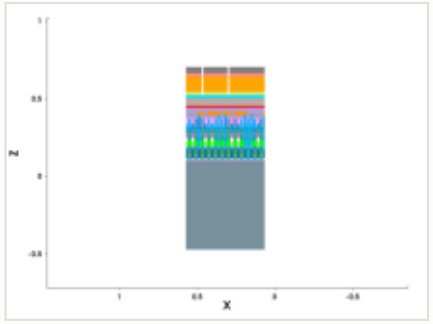
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

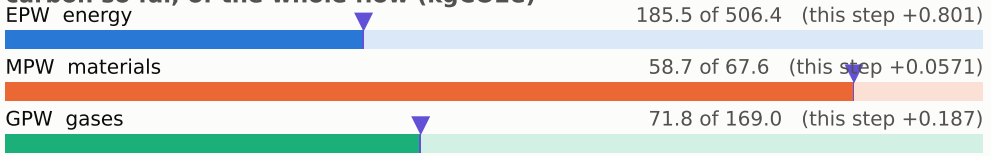
no mask for this step

this step

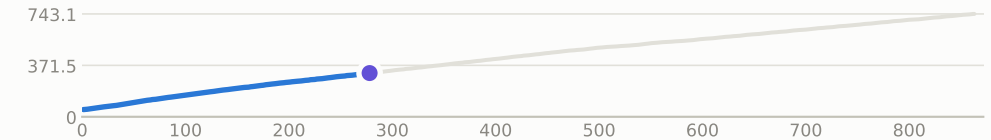
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

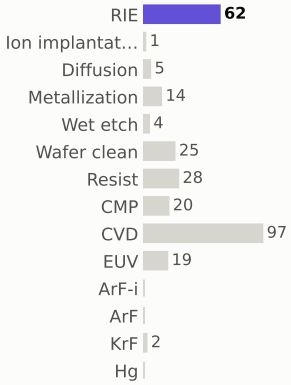
carbon so far, of the whole flow (kgCO2e)



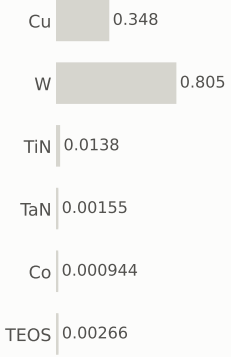
embodied carbon up to each step (kgCO2e per wafer)



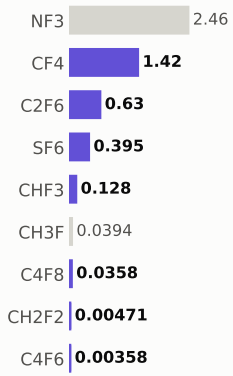
steps by process area



materials deposited (g)



gases emitted (g)



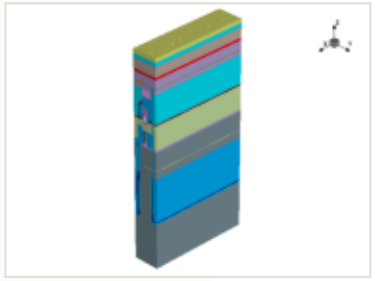
Step #3050: Wafer_clean

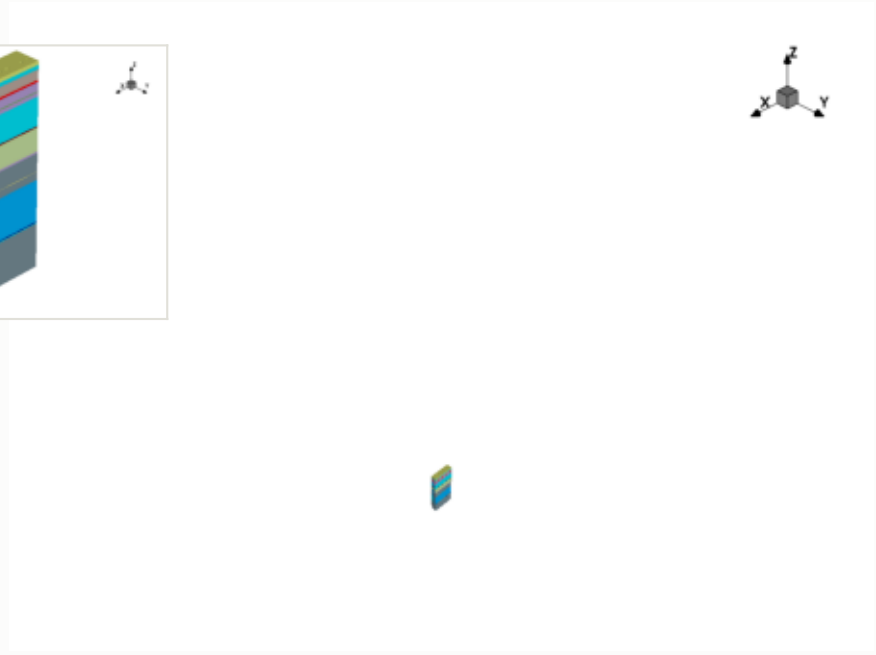
etch_strip · Wafer_clean

0.169	316.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



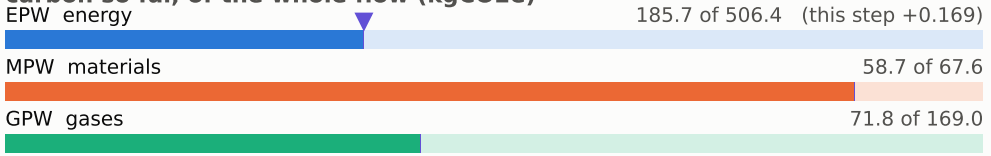


this step

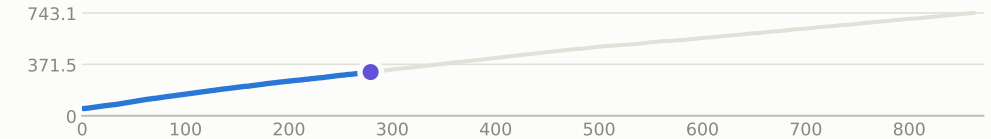
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

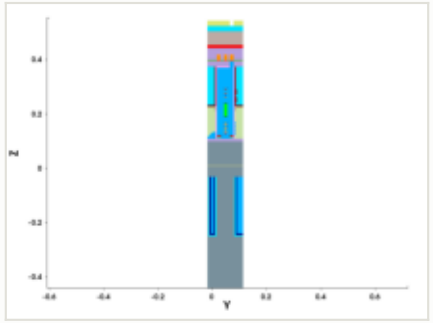
carbon so far, of the whole flow (kgCO2e)



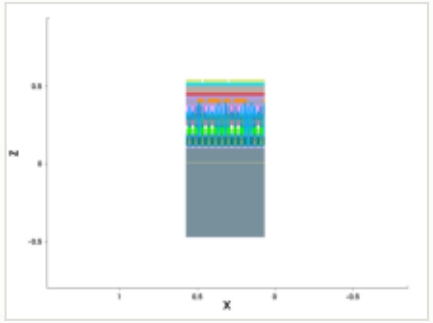
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



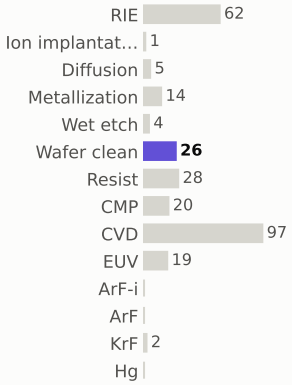
y cut at 0.049 µm



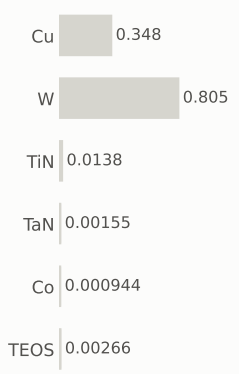
mask for this step

no mask for this step

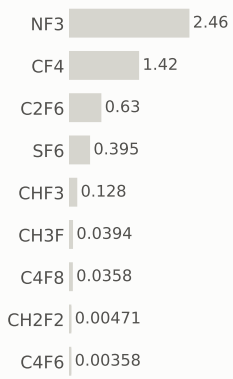
steps by process area



materials deposited (g)



gases emitted (g)



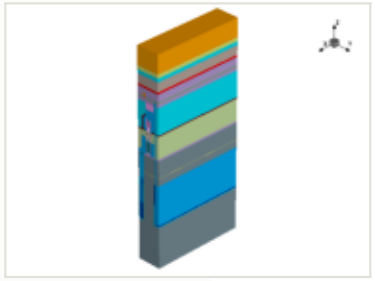
Step #3060: Resist

depo_fill · Resist

0.169	316.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

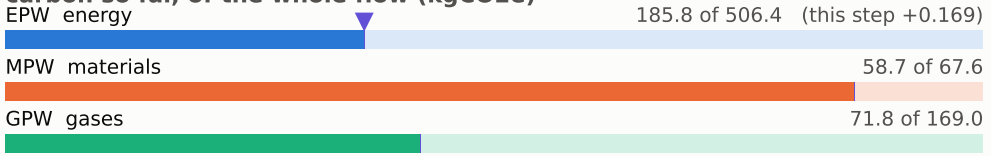


this step

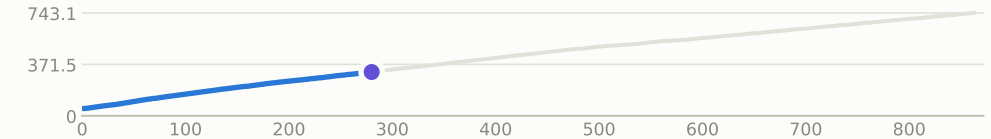
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

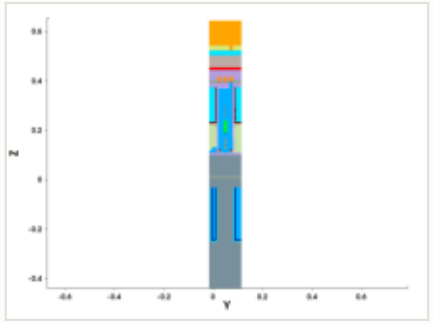
carbon so far, of the whole flow (kgCO2e)



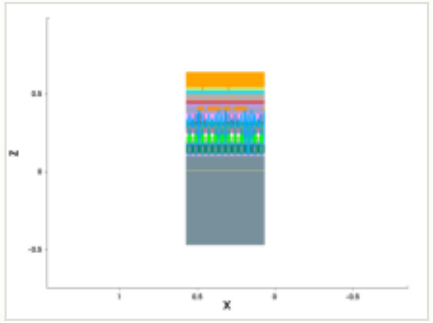
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



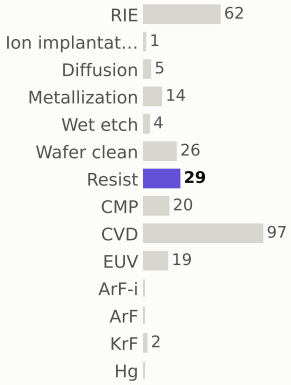
y cut at 0.049 µm



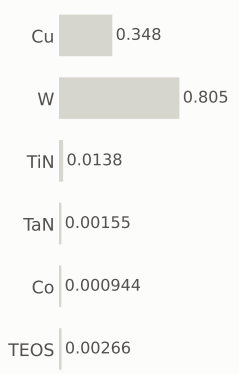
mask for this step

no mask for this step

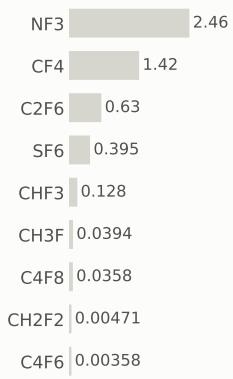
steps by process area



materials deposited (g)



gases emitted (g)



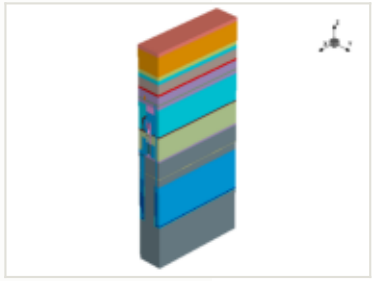
Step #3070: CVD

depo_fill · CVD

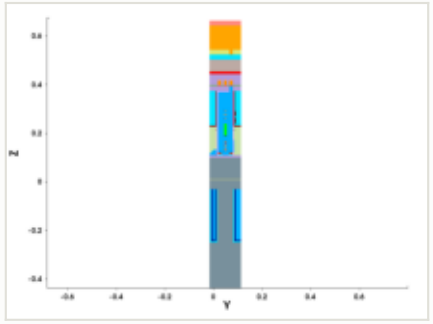
1.31	317.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

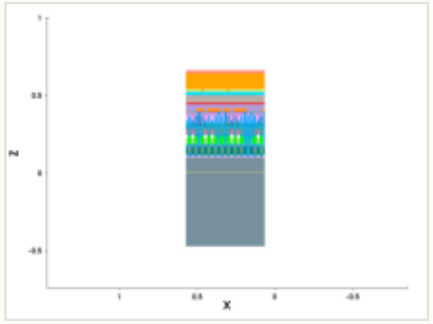
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

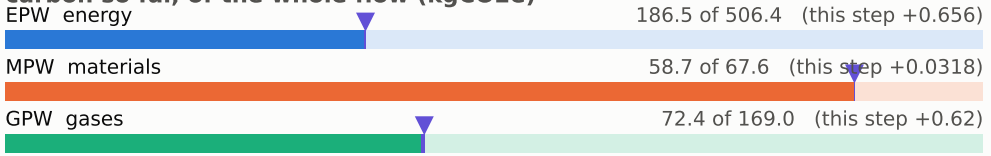
no mask for this step

this step

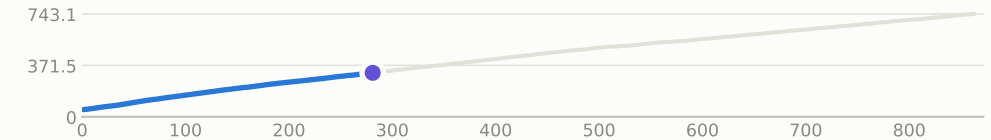
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

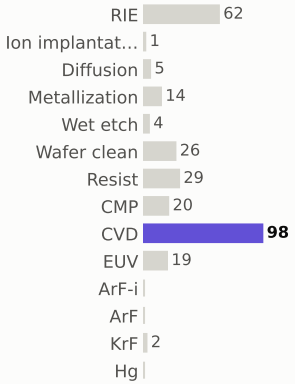
carbon so far, of the whole flow (kgCO2e)



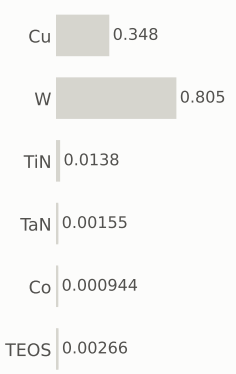
embodied carbon up to each step (kgCO2e per wafer)



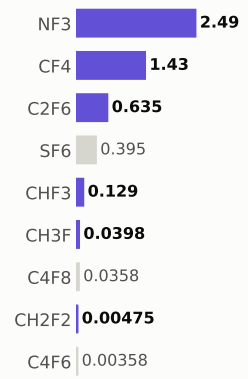
steps by process area



materials deposited (g)



gases emitted (g)



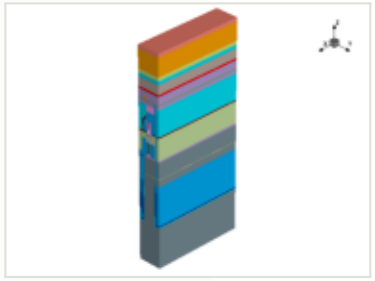
Step #3080: Wafer_clean

clean · Wafer_clean

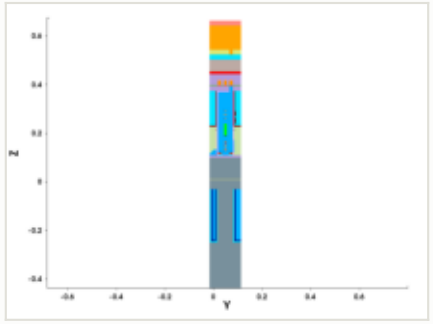
0.169	317.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

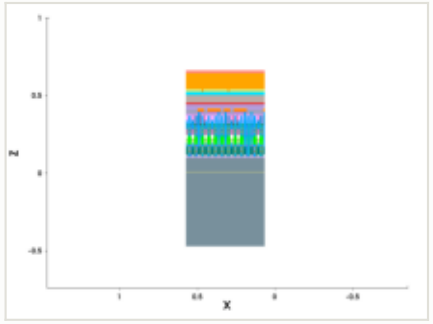
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

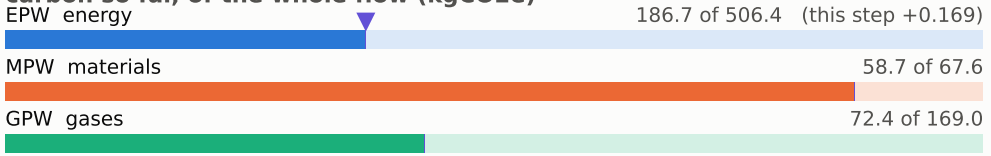
no mask for this step

this step

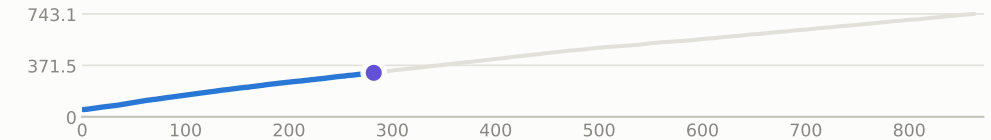
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

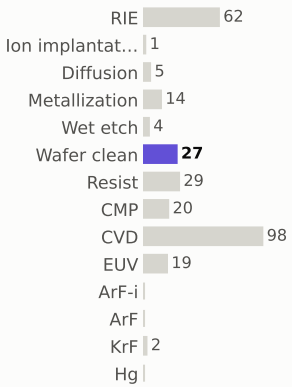
carbon so far, of the whole flow (kgCO2e)



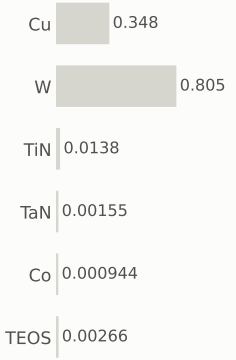
embodied carbon up to each step (kgCO2e per wafer)



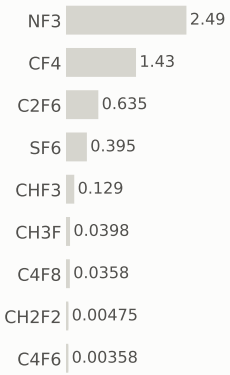
steps by process area



materials deposited (g)



gases emitted (g)



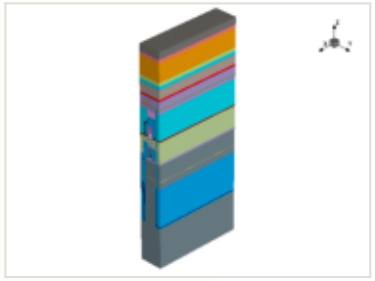
Step #3090: EUV

litho_patterning · EUV

1.96	319.7	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

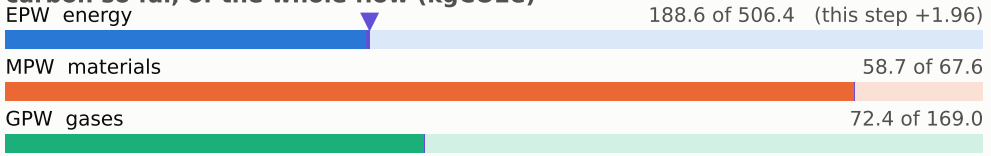


this step

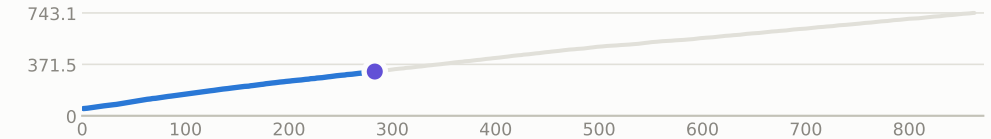
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

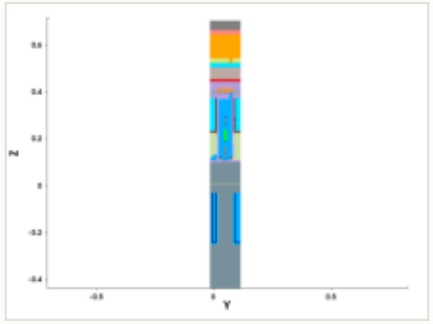
carbon so far, of the whole flow (kgCO2e)



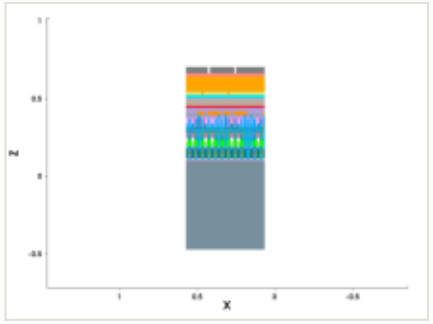
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

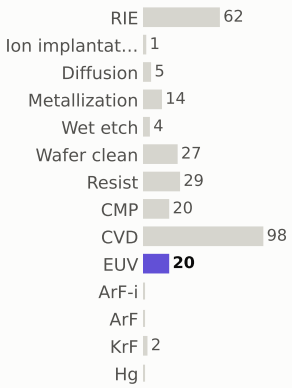


y cut at 0.049 µm

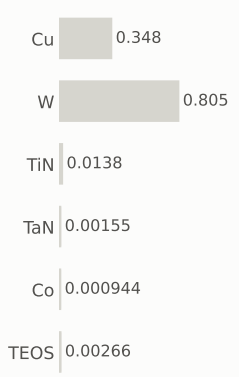


mask: layer 10222:0 (GDSII)

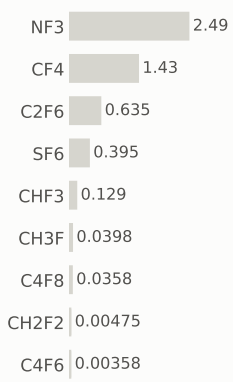
steps by process area



materials deposited (g)



gases emitted (g)



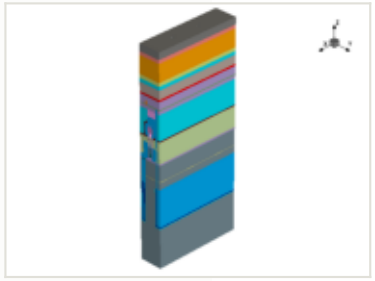
Step #3100: RIE

etch_anisotropic · RIE

1.04	320.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

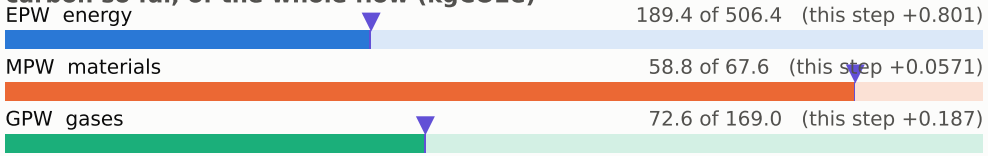


this step

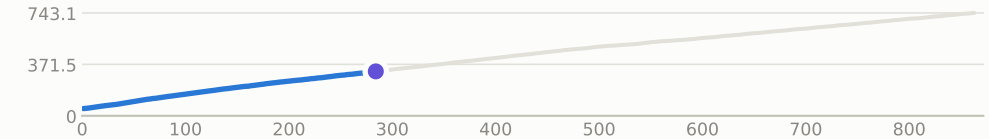
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

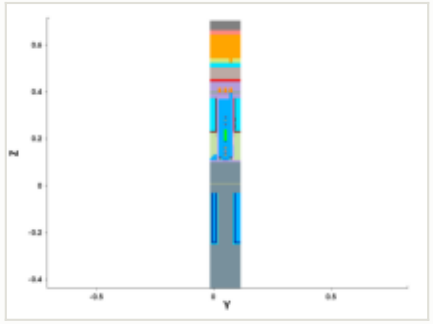
carbon so far, of the whole flow (kgCO2e)



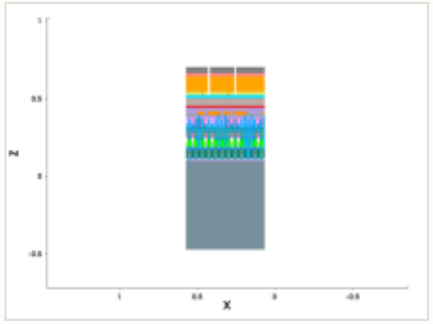
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



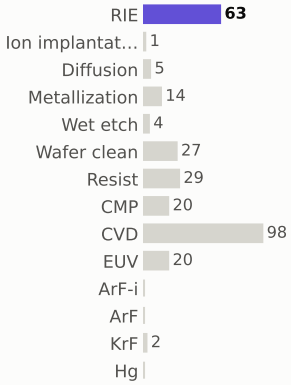
y cut at 0.049 µm



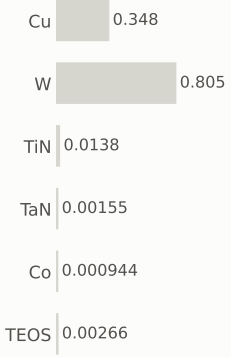
mask for this step

no mask for this step

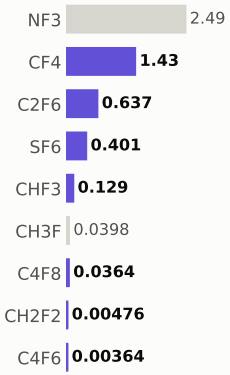
steps by process area



materials deposited (g)



gases emitted (g)



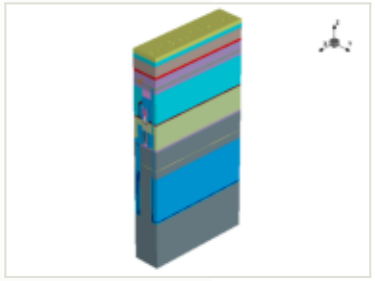
Step #3110: Wafer_clean

etch_strip · Wafer_clean

0.169	320.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

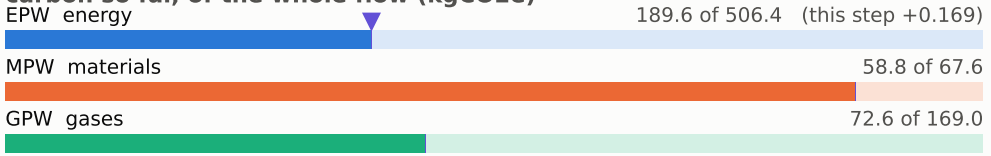


this step

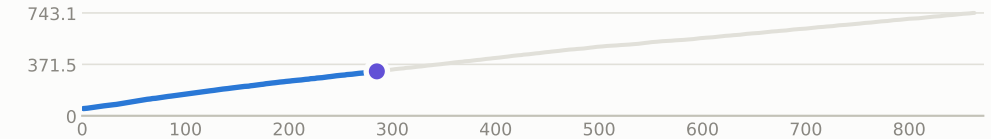
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

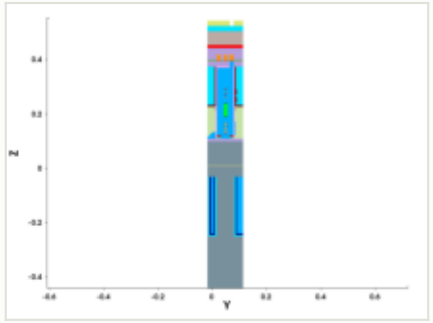
carbon so far, of the whole flow (kgCO2e)



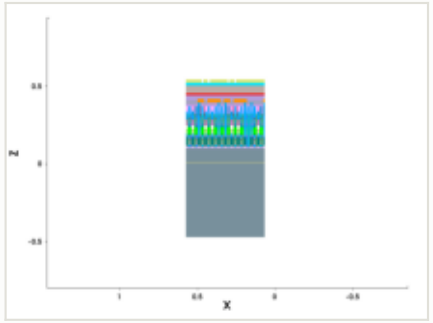
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



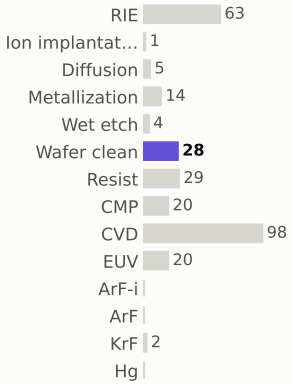
y cut at 0.049 µm



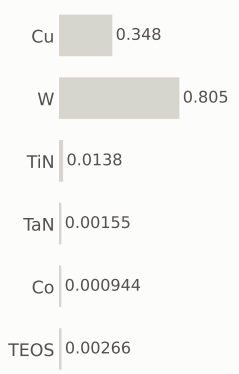
mask for this step

no mask for this step

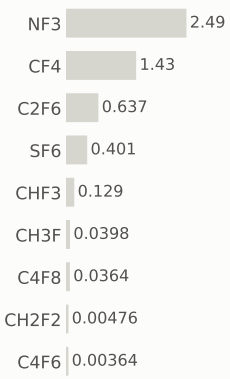
steps by process area



materials deposited (g)



gases emitted (g)



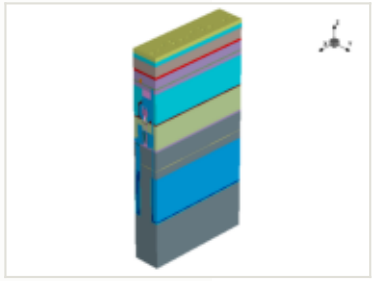
Step #3120: RIE

etch_anisotropic · RIE

1.04	322.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

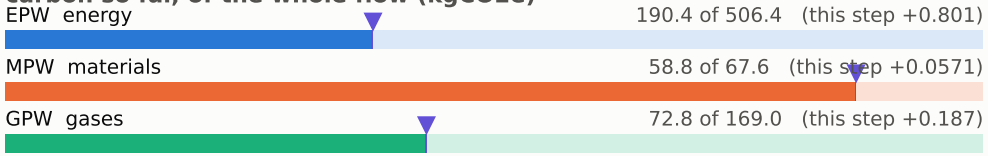


this step

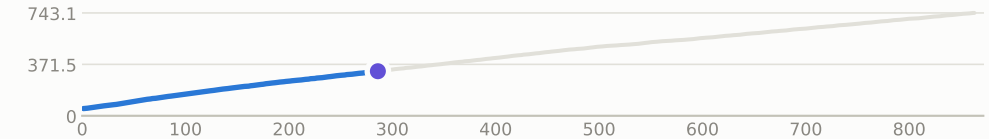
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

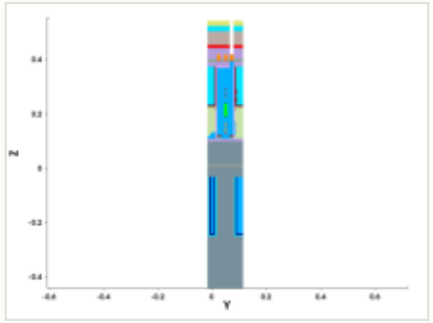
carbon so far, of the whole flow (kgCO2e)



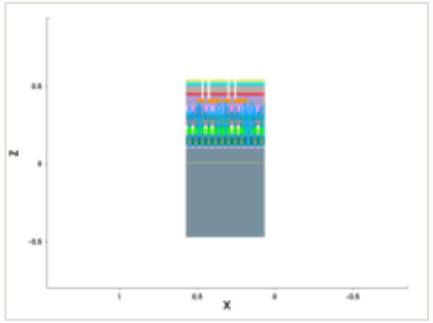
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



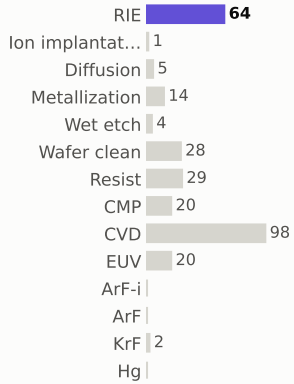
y cut at 0.049 µm



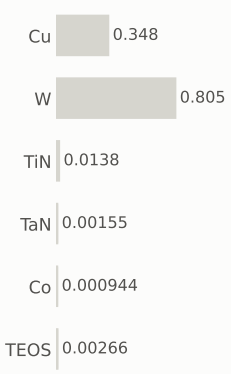
mask for this step

no mask for this step

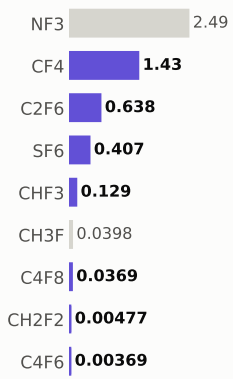
steps by process area



materials deposited (g)



gases emitted (g)



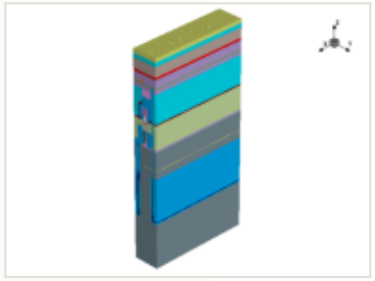
Step #3130: Wafer_clean

clean · Wafer_clean

0.169	322.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



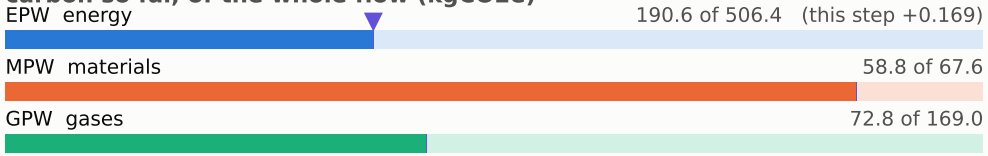
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

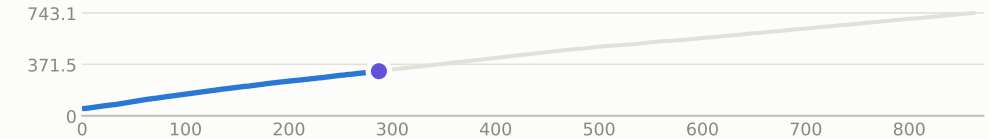
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

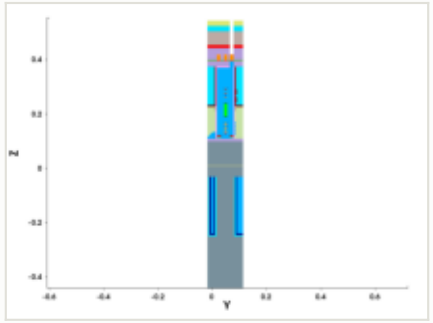
carbon so far, of the whole flow (kgCO2e)



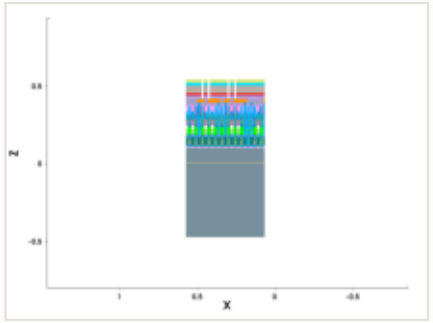
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



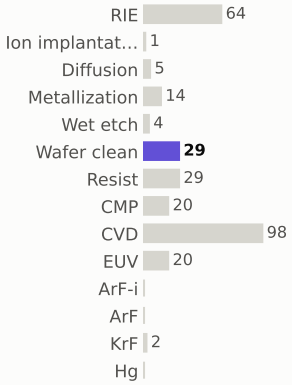
y cut at 0.049 µm



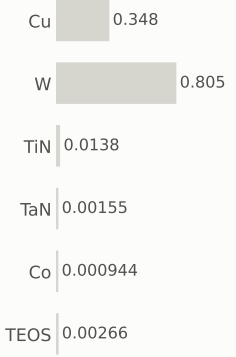
mask for this step

no mask for this step

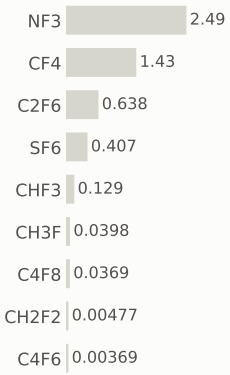
steps by process area



materials deposited (g)



gases emitted (g)



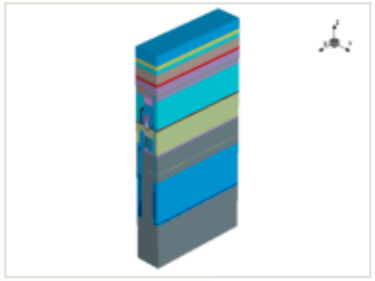
Step #3140: Metallization

depo_fill · Metallization

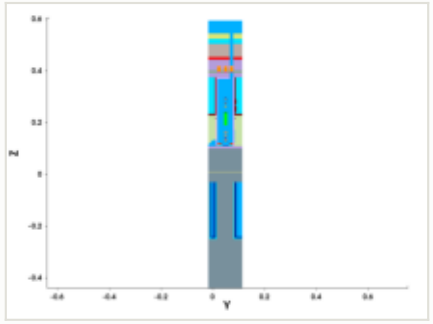
1.21	323.4	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

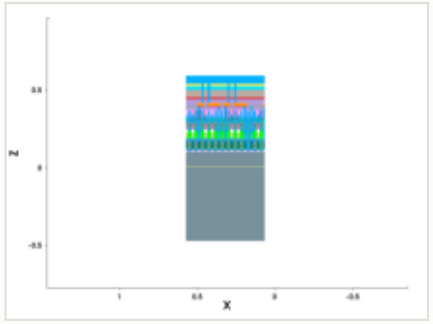
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

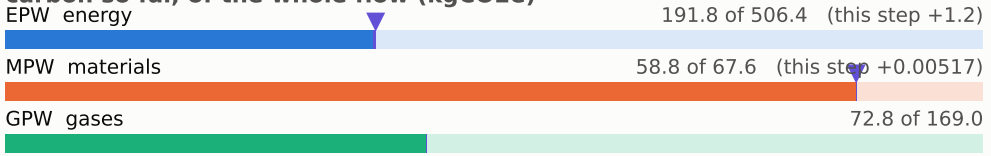
no mask for this step

this step

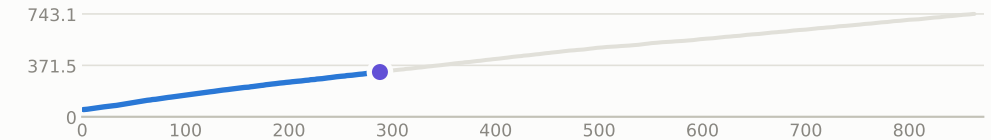
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

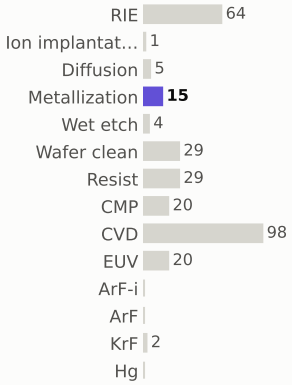
carbon so far, of the whole flow (kgCO2e)



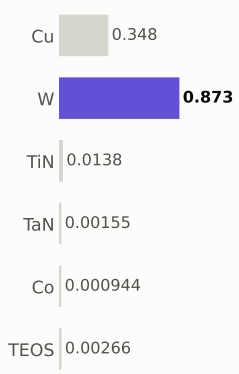
embodied carbon up to each step (kgCO2e per wafer)



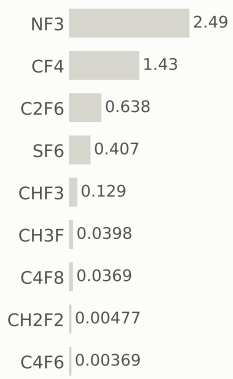
steps by process area



materials deposited (g)



gases emitted (g)



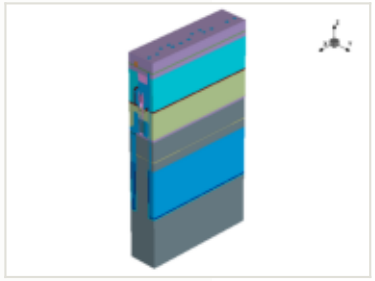
Step #3150: CMP

etch_cmp · CMP

0.169	323.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

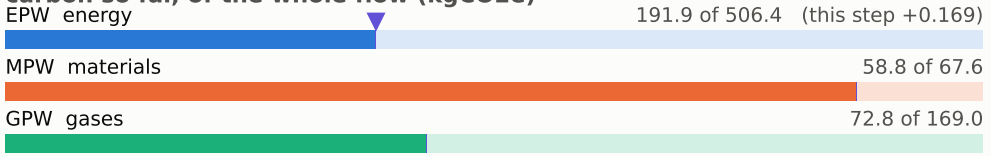


this step

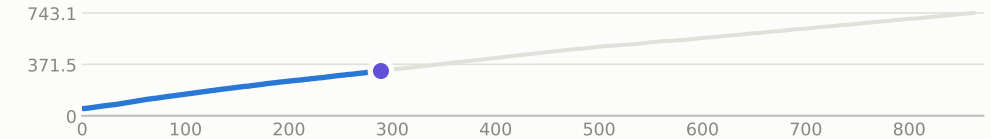
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

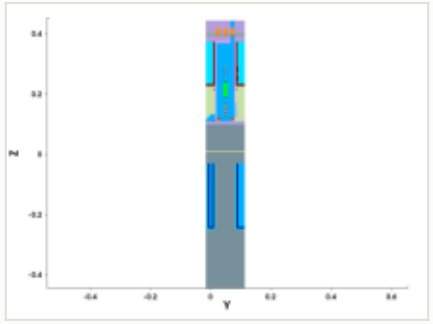
carbon so far, of the whole flow (kgCO2e)



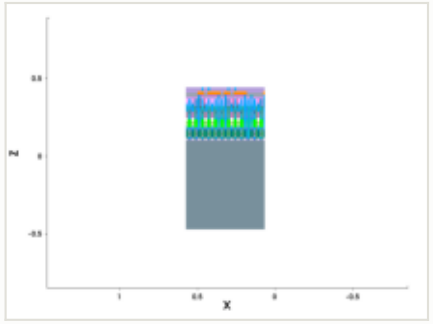
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



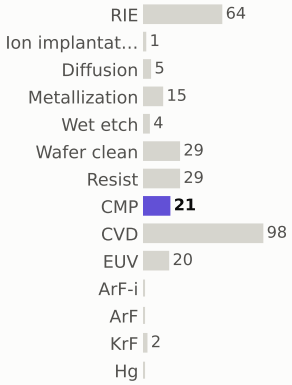
y cut at 0.049 µm



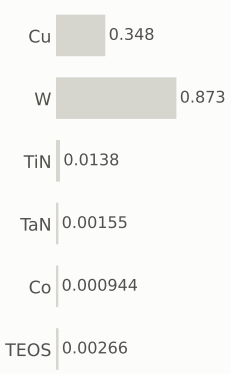
mask for this step

no mask for this step

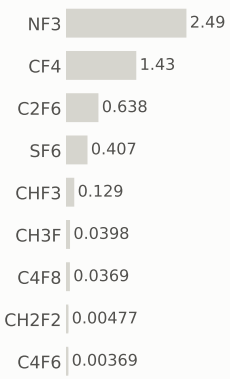
steps by process area



materials deposited (g)



gases emitted (g)



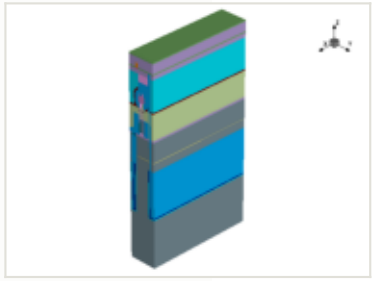
Step #3160: CVD

depo_fill · CVD

1.31	324.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

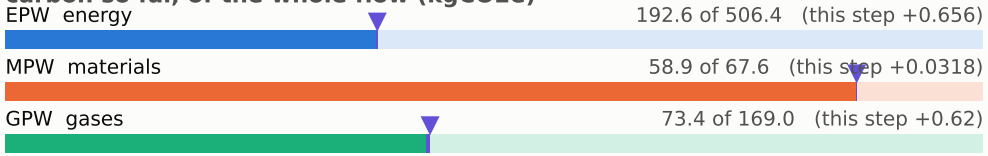


this step

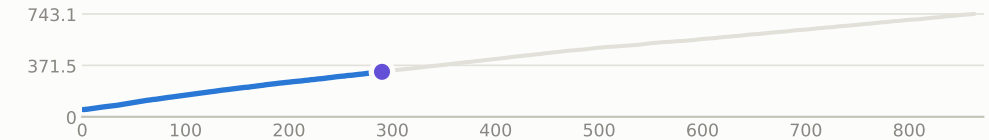
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

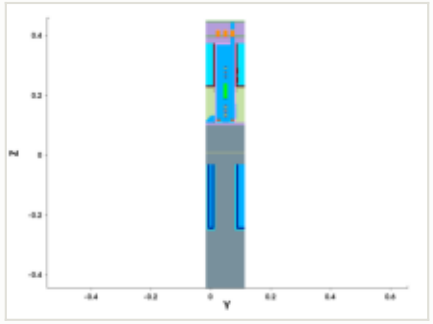
carbon so far, of the whole flow (kgCO2e)



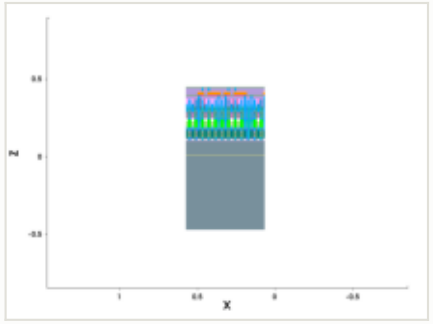
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



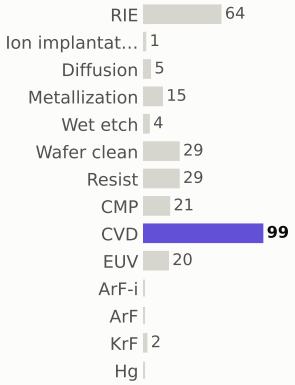
y cut at 0.049 µm



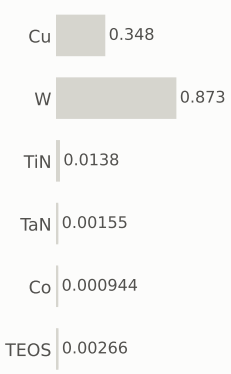
mask for this step

no mask for this step

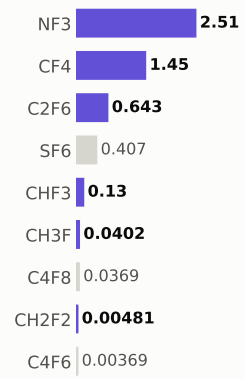
steps by process area



materials deposited (g)



gases emitted (g)



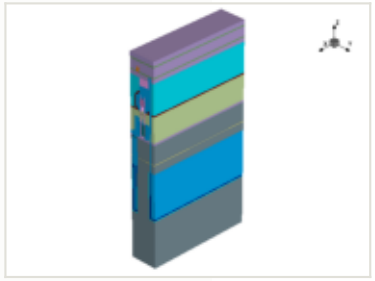
Step #3170: CVD

depo_fill · CVD

1.31	326.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

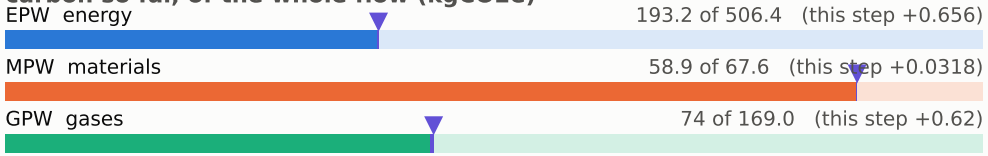


this step

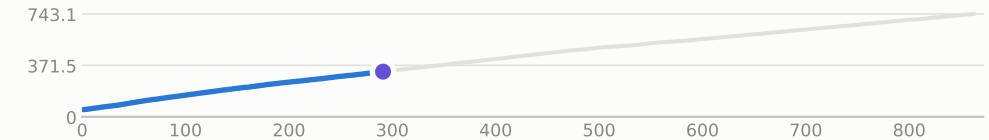
process area	CVD
material	—
thickness	0.022 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

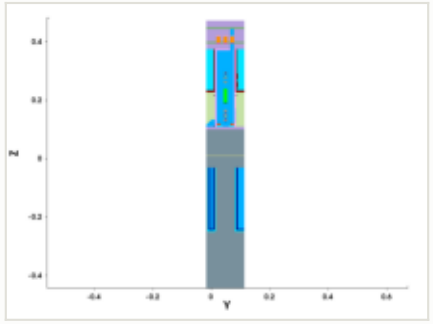
carbon so far, of the whole flow (kgCO2e)



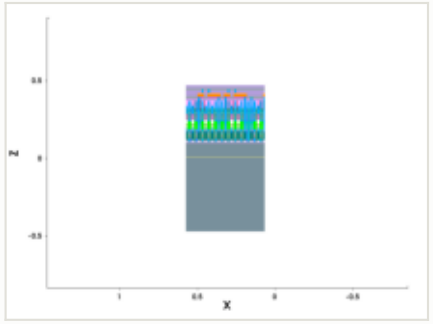
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



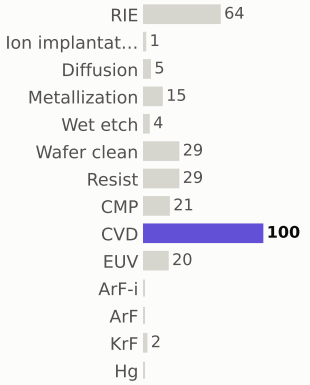
y cut at 0.049 µm



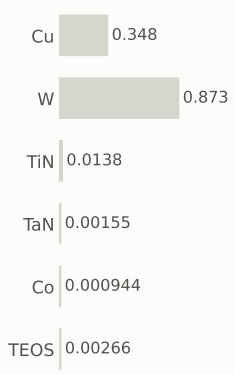
mask for this step

no mask for this step

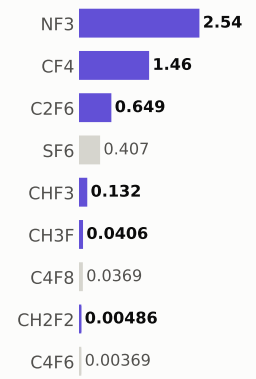
steps by process area



materials deposited (g)



gases emitted (g)



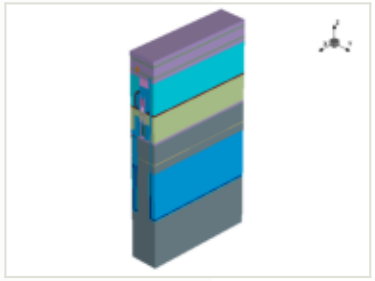
Step #3180: Diffusion

treatment · Diffusion

0.801	327.0	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

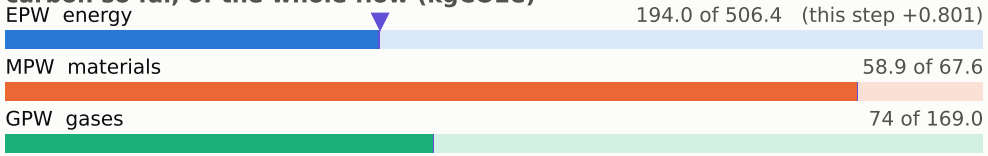


this step

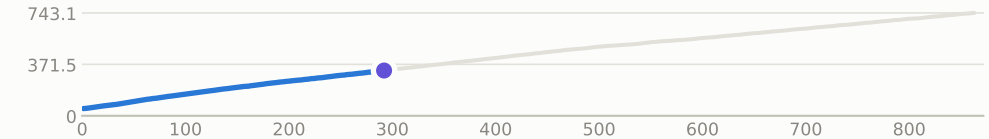
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

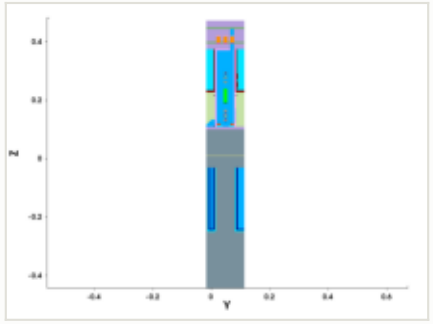
carbon so far, of the whole flow (kgCO2e)



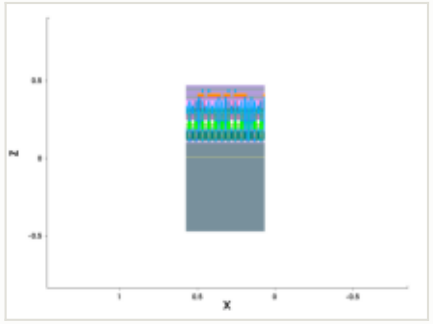
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



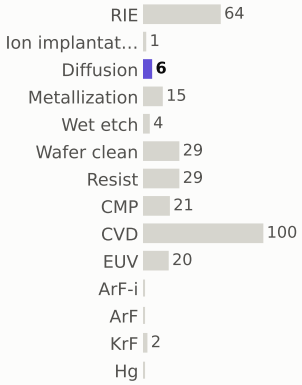
y cut at 0.049 µm



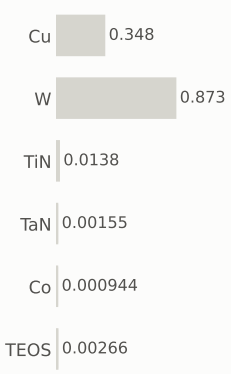
mask for this step

no mask for this step

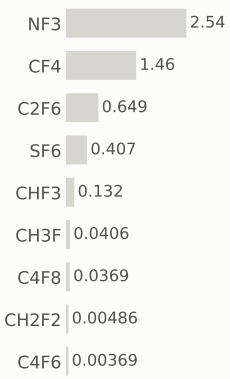
steps by process area



materials deposited (g)



gases emitted (g)



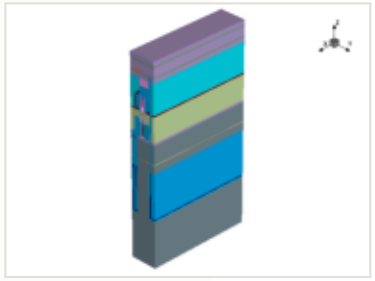
Step #3190: Wafer_clean

clean · Wafer_clean

0.169	327.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

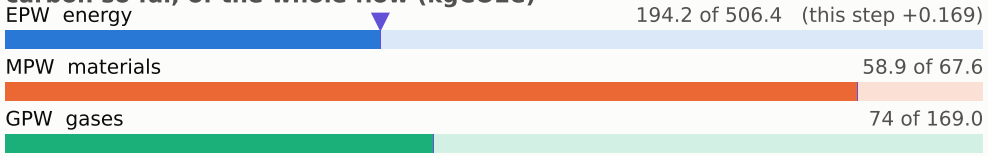


this step

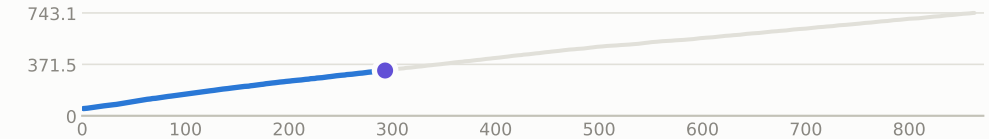
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

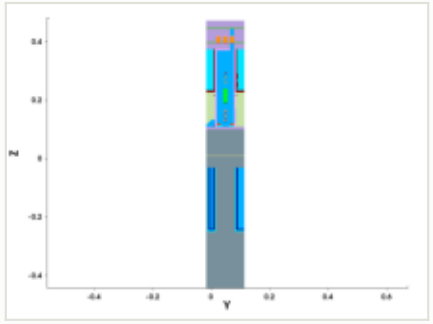
carbon so far, of the whole flow (kgCO2e)



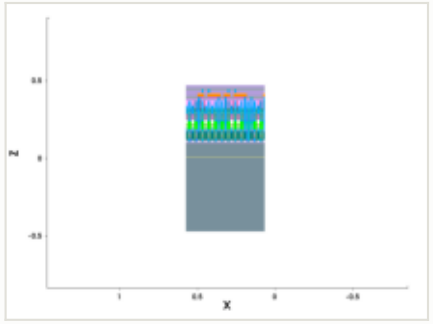
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



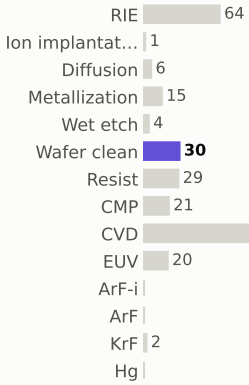
y cut at 0.049 µm



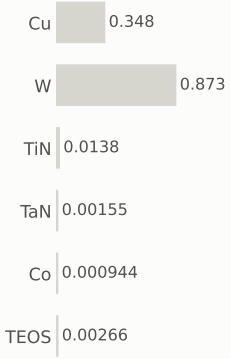
mask for this step

no mask for this step

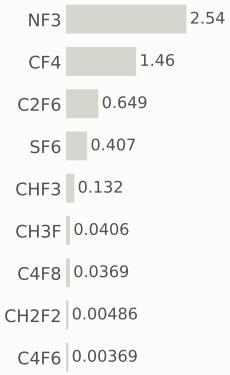
steps by process area



materials deposited (g)



gases emitted (g)



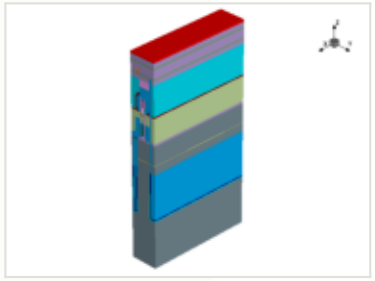
Step #3200: CVD

depo_fill · CVD

1.31	328.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

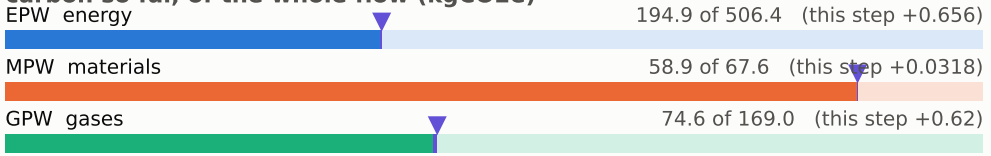


this step

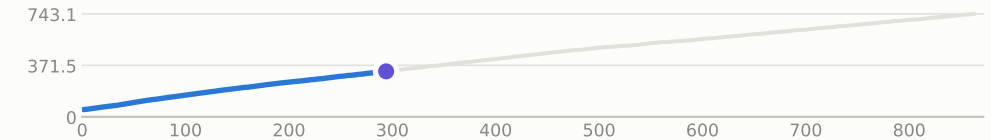
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

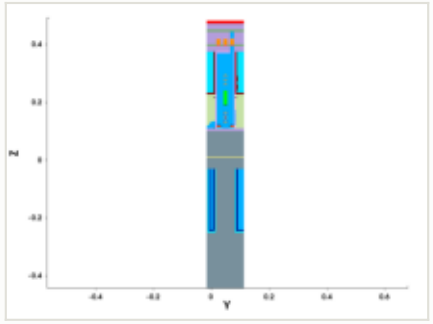
carbon so far, of the whole flow (kgCO2e)



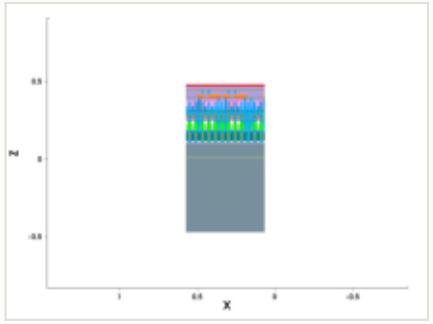
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



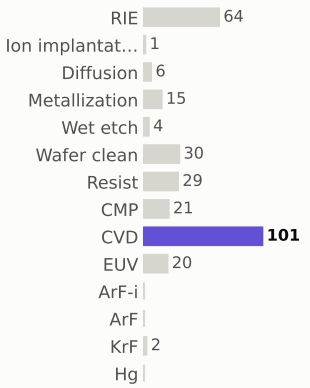
y cut at 0.049 µm



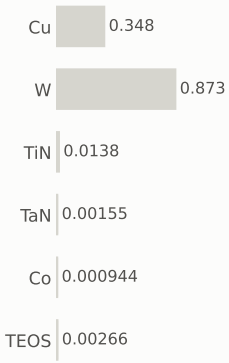
mask for this step

no mask for this step

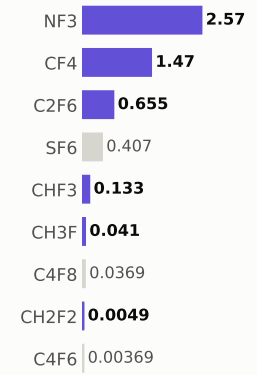
steps by process area



materials deposited (g)



gases emitted (g)



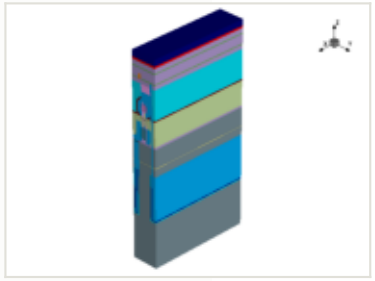
Step #3210: CVD

depo_fill · CVD

1.31	329.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

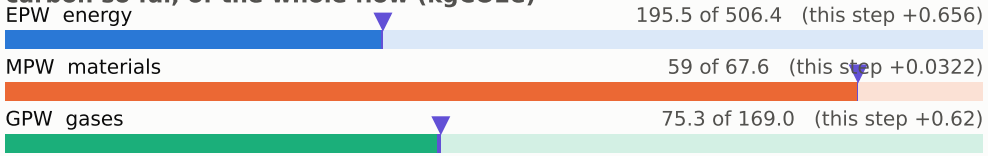
zoomed to fit



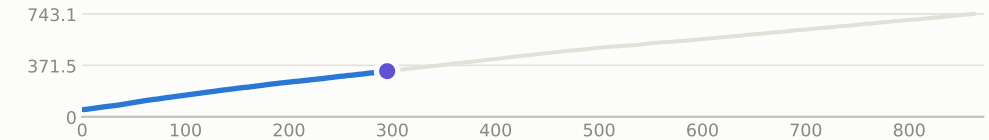
this step

process area	CVD	EPW	0.656 kgCO2e
material	TiN	EPW tool	0.469 kgCO2e
thickness	0.03 µm	EPW overhead	0.187 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	0.0322 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

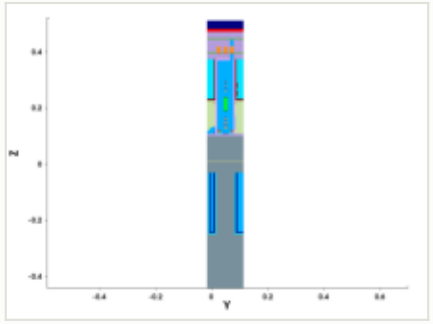
carbon so far, of the whole flow (kgCO2e)



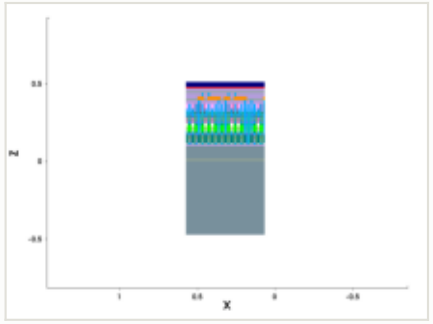
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



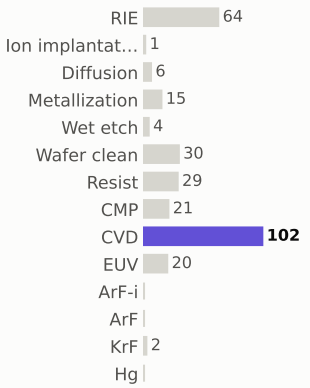
y cut at 0.049 µm



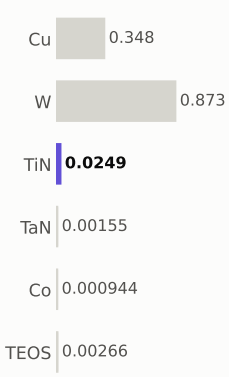
mask for this step

no mask for this step

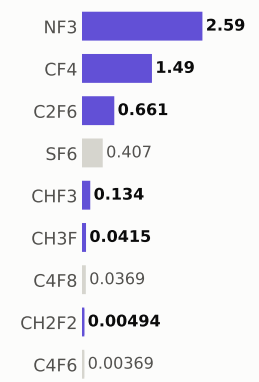
steps by process area



materials deposited (g)



gases emitted (g)



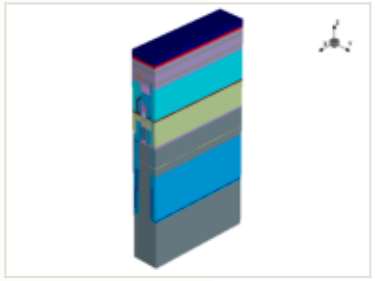
Step #3220: Wafer_clean

clean · Wafer_clean

0.169	329.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



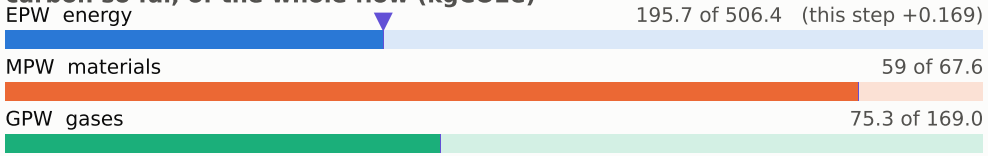
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

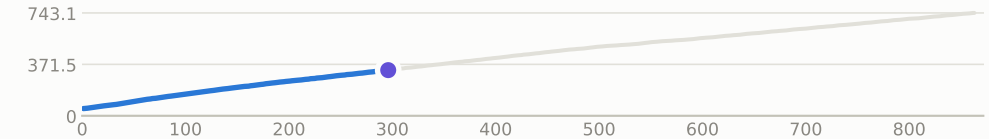
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

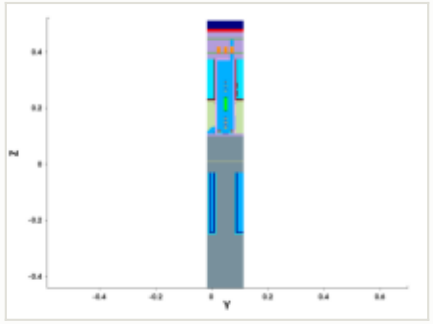
carbon so far, of the whole flow (kgCO2e)



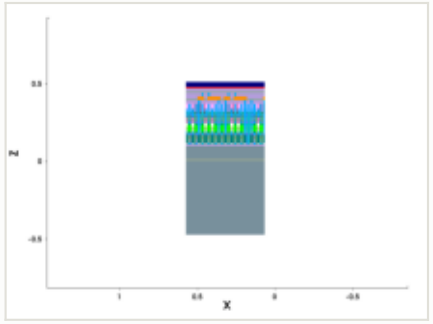
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



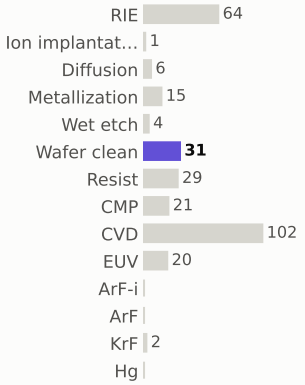
y cut at 0.049 µm



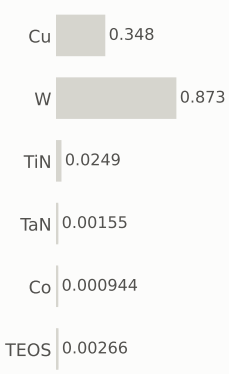
mask for this step

no mask for this step

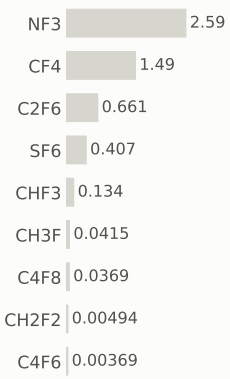
steps by process area



materials deposited (g)



gases emitted (g)



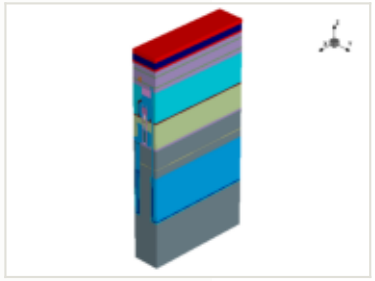
Step #3230: CVD

depo_fill · CVD

1.31	331.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

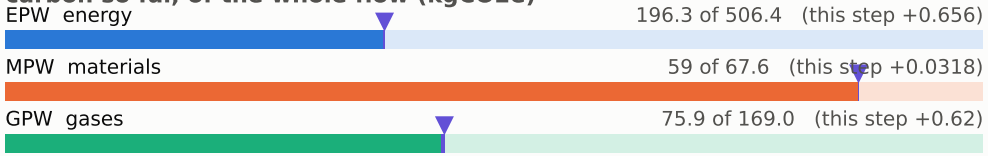
zoomed to fit



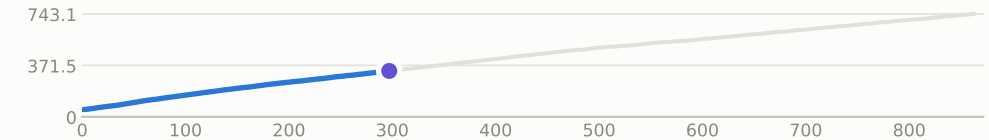
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.02 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

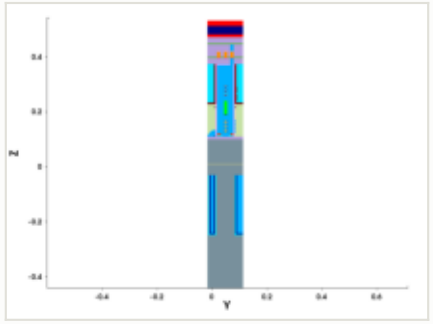
carbon so far, of the whole flow (kgCO2e)



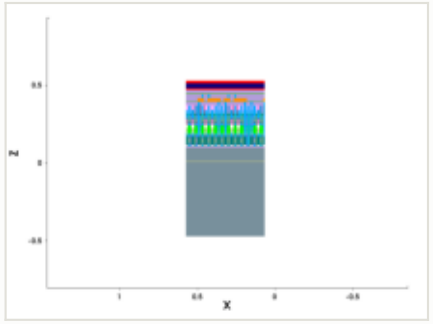
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



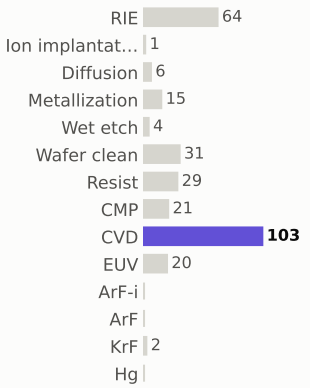
y cut at 0.049 µm



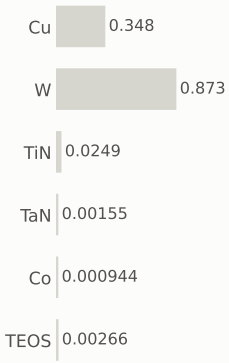
mask for this step

no mask for this step

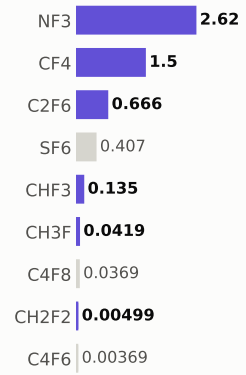
steps by process area



materials deposited (g)



gases emitted (g)



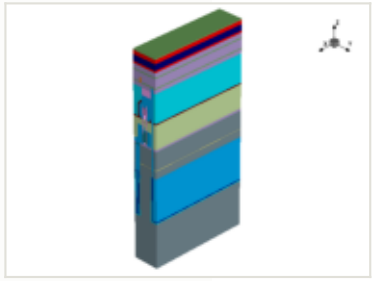
Step #3240: CVD

depo_fill · CVD

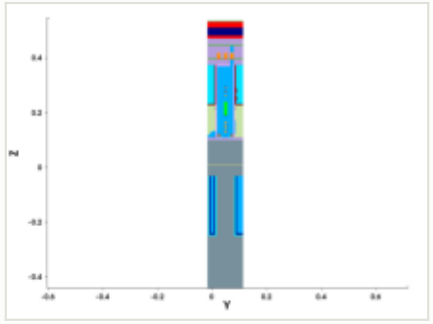
1.31	332.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

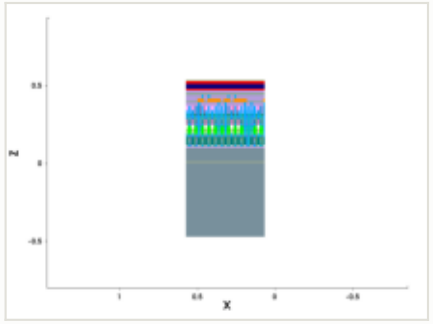
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

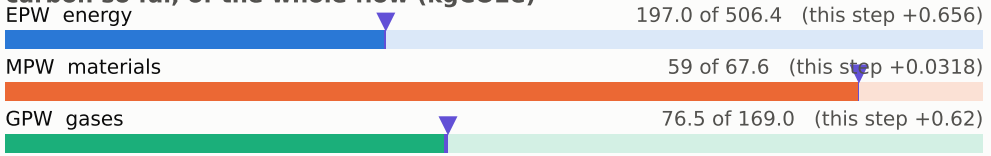
no mask for this step

this step

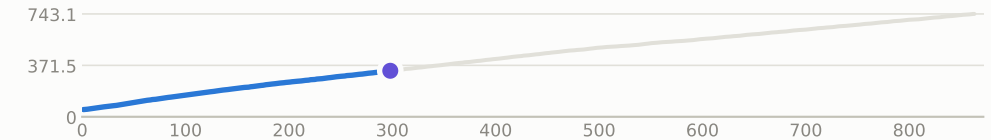
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

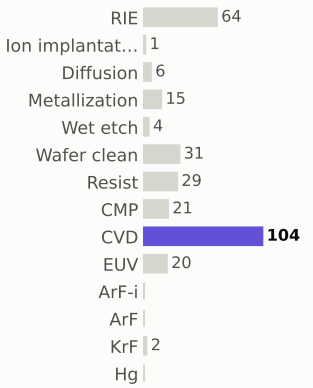
carbon so far, of the whole flow (kgCO2e)



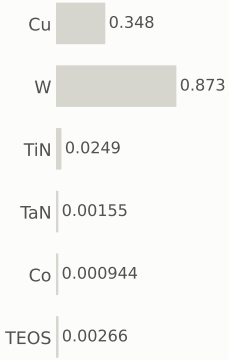
embodied carbon up to each step (kgCO2e per wafer)



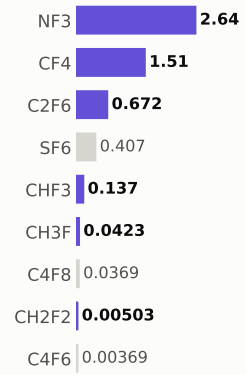
steps by process area



materials deposited (g)



gases emitted (g)



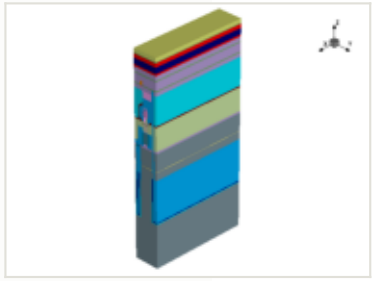
Step #3250: CVD

depo_fill · CVD

1.31	333.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

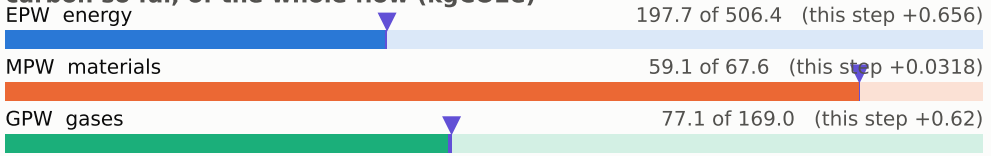


this step

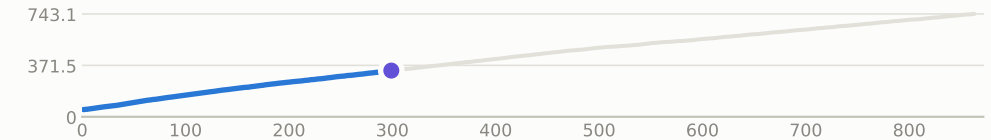
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

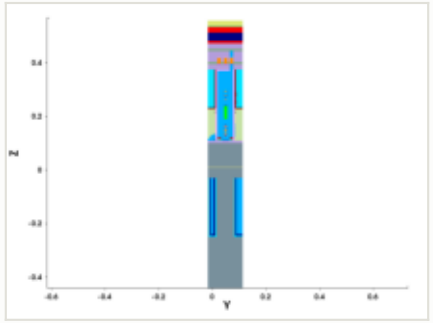
carbon so far, of the whole flow (kgCO2e)



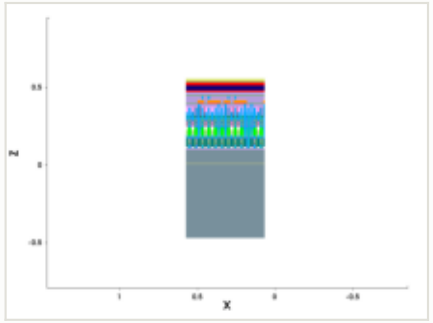
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



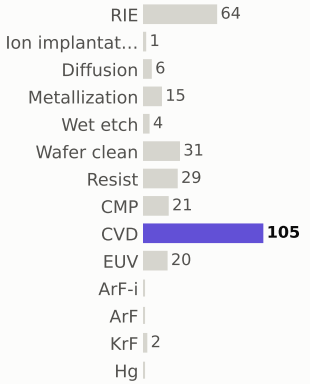
y cut at 0.049 µm



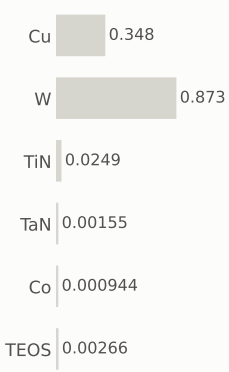
mask for this step

no mask for this step

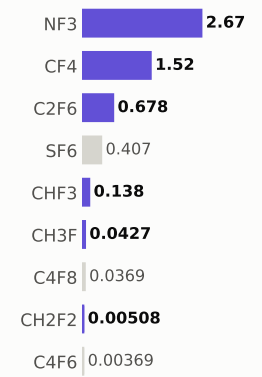
steps by process area



materials deposited (g)



gases emitted (g)



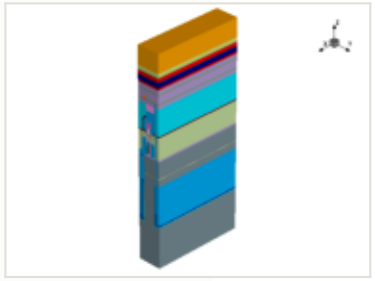
Step #3260: Resist

depo_fill · Resist

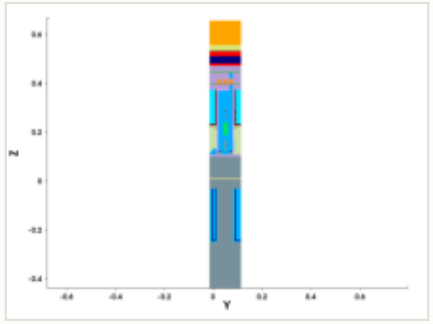
0.169	334.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

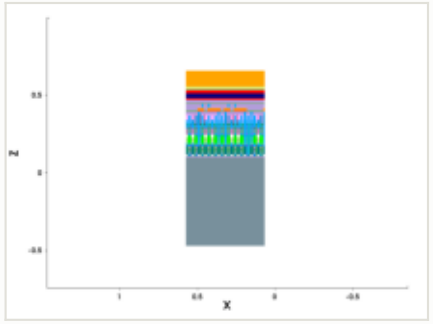
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



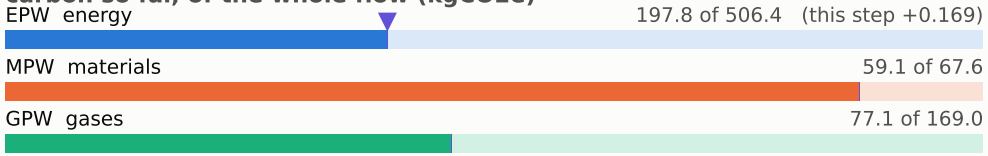
mask for this step

no mask for this step

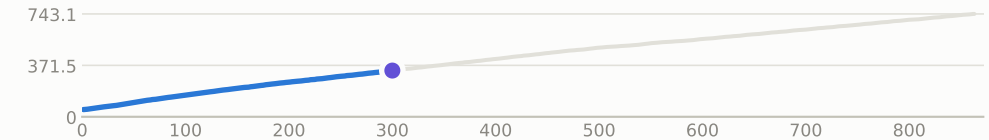
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

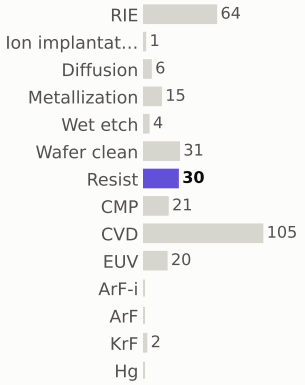
carbon so far, of the whole flow (kgCO2e)



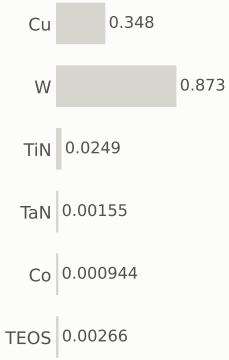
embodied carbon up to each step (kgCO2e per wafer)



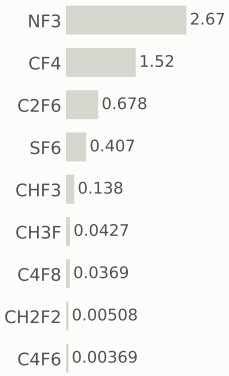
steps by process area



materials deposited (g)



gases emitted (g)



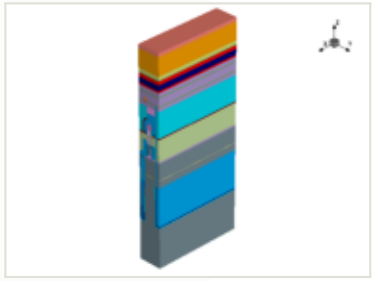
Step #3270: CVD

depo_fill · CVD

1.31	335.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

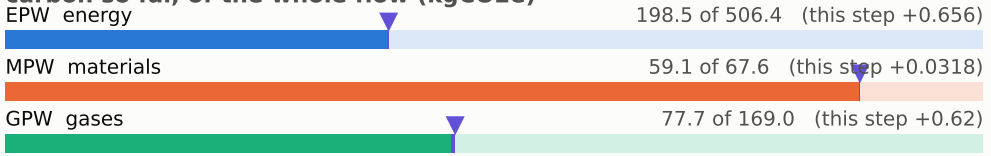


this step

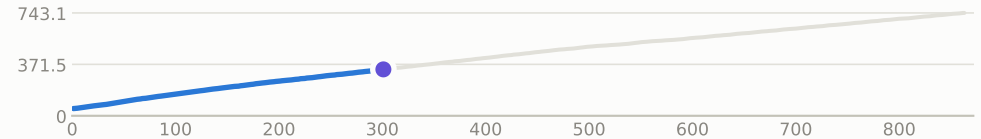
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

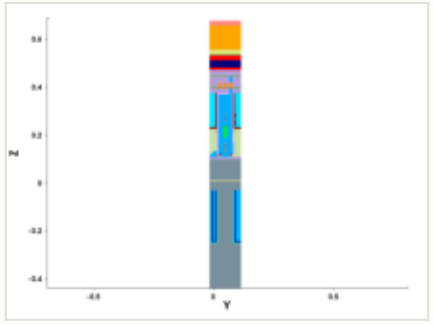
carbon so far, of the whole flow (kgCO2e)



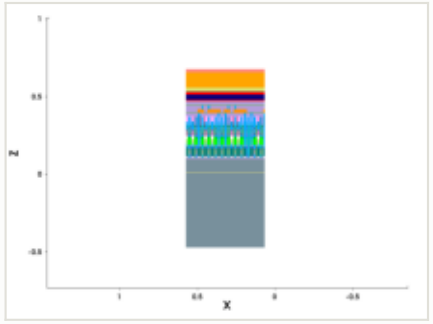
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



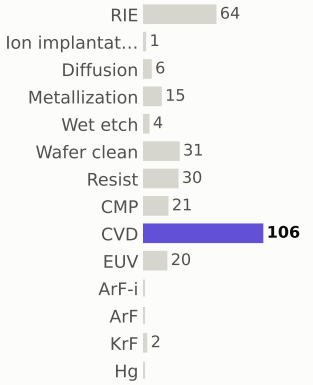
y cut at 0.049 µm



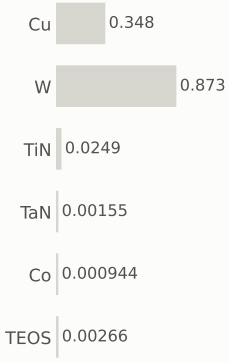
mask for this step

no mask for this step

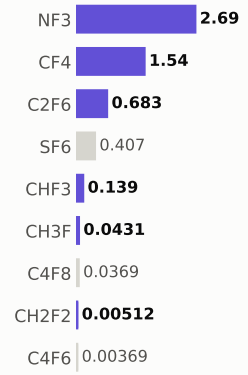
steps by process area



materials deposited (g)



gases emitted (g)



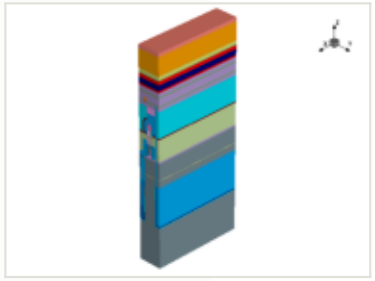
Step #3280: Wafer_clean

clean · Wafer_clean

0.169	335.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



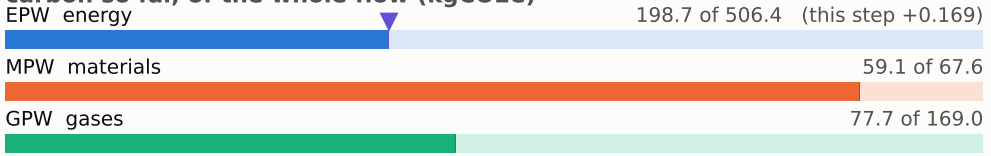
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

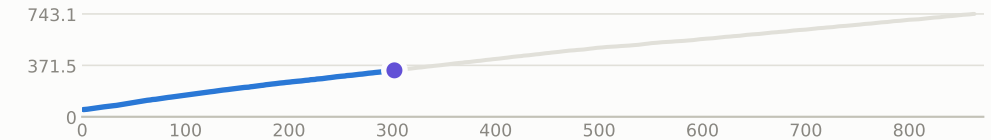
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

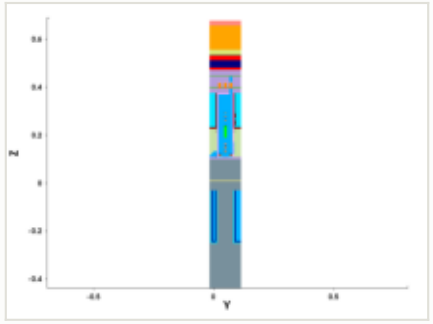
carbon so far, of the whole flow (kgCO2e)



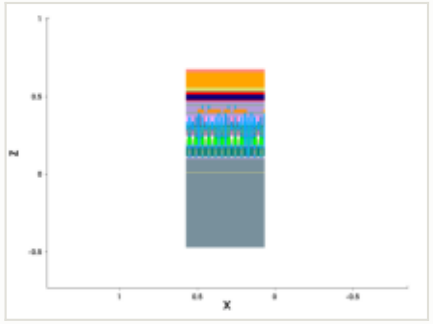
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



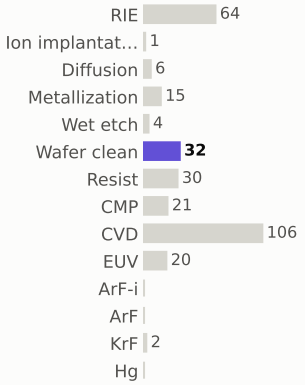
y cut at 0.049 µm



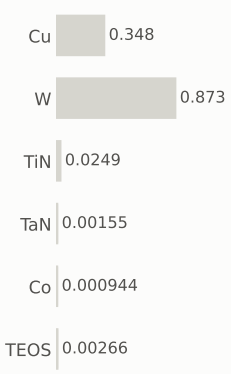
mask for this step

no mask for this step

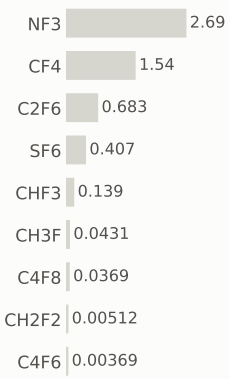
steps by process area



materials deposited (g)



gases emitted (g)



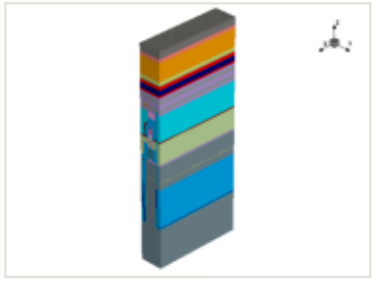
Step #3290: EUV

litho_patterning · EUV

1.96	337.4	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

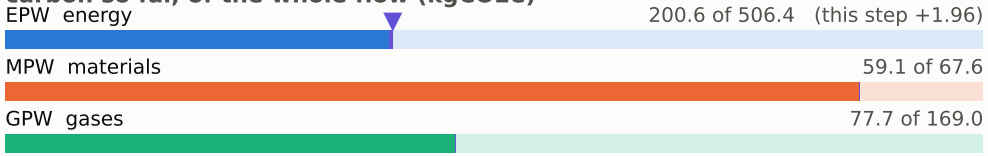


this step

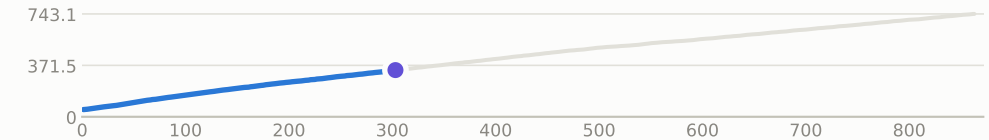
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

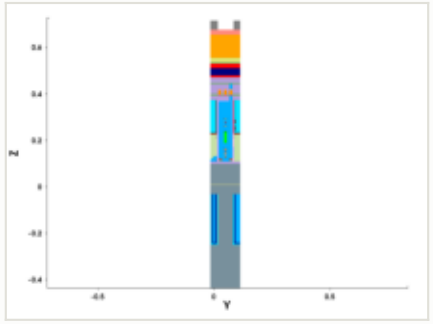
carbon so far, of the whole flow (kgCO2e)



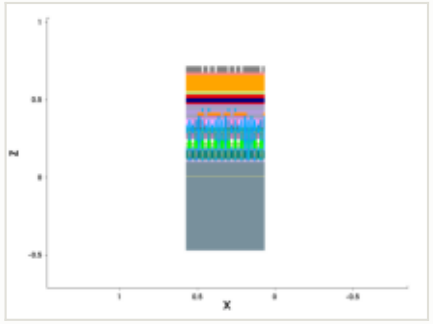
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

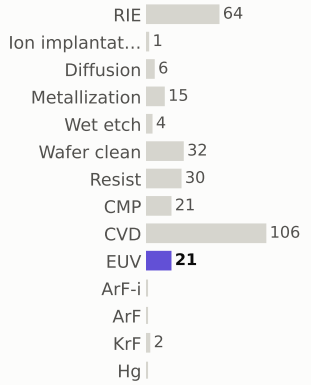


y cut at 0.049 µm

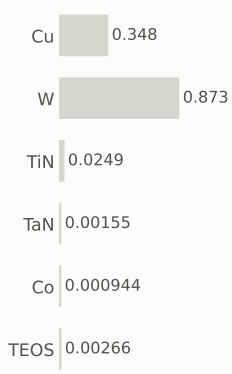


mask: layer 10251:0 (GDSII)

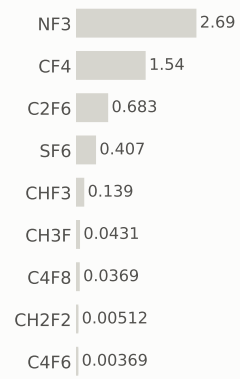
steps by process area



materials deposited (g)



gases emitted (g)



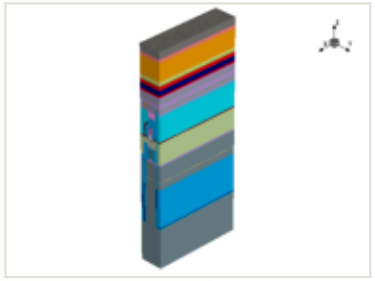
Step #3300: RIE

etch_anisotropic · RIE

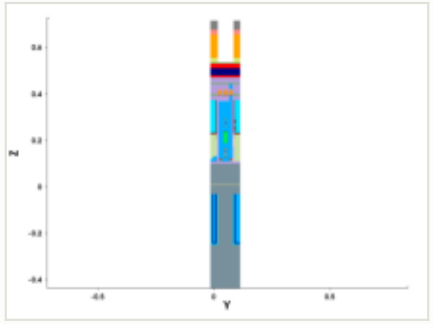
1.04	338.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

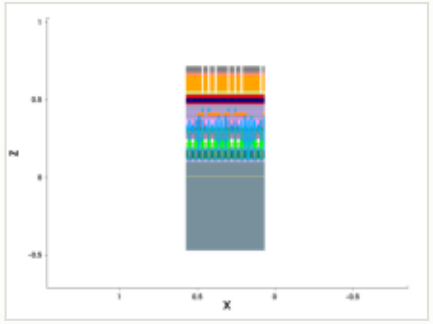
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

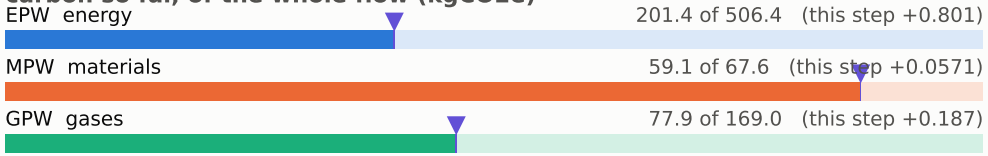
no mask for this step

this step

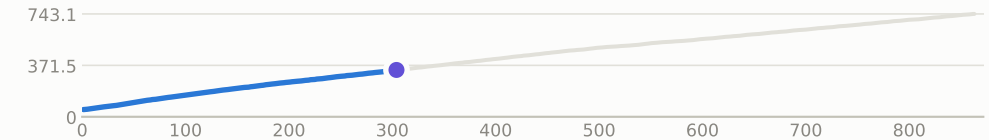
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

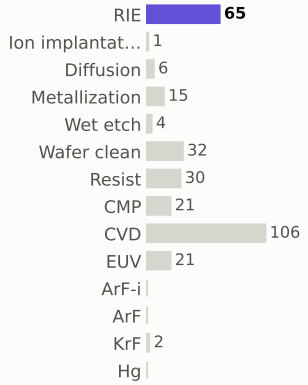
carbon so far, of the whole flow (kgCO2e)



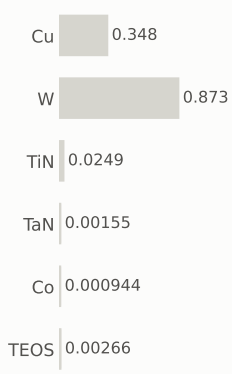
embodied carbon up to each step (kgCO2e per wafer)



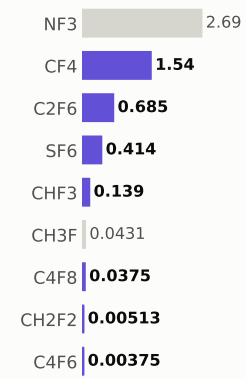
steps by process area



materials deposited (g)



gases emitted (g)



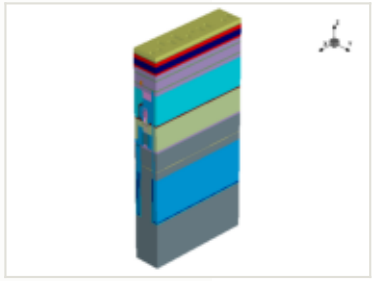
Step #3310: Wafer_clean

etch_strip · Wafer_clean

0.169	338.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

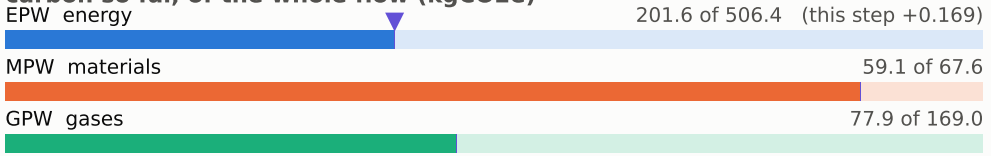


this step

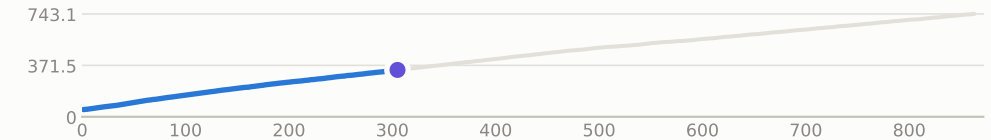
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

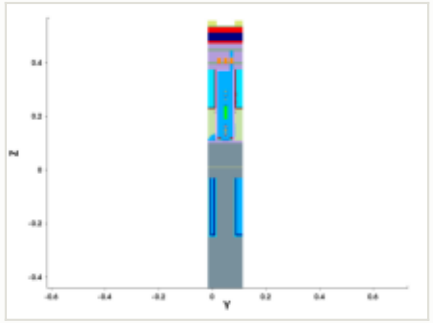
carbon so far, of the whole flow (kgCO2e)



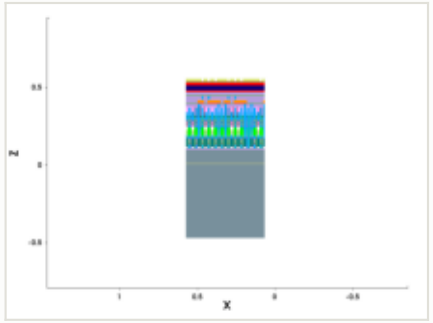
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



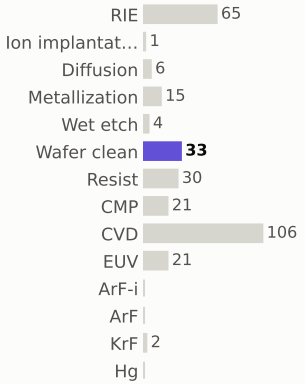
y cut at 0.049 µm



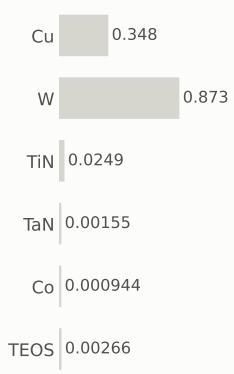
mask for this step

no mask for this step

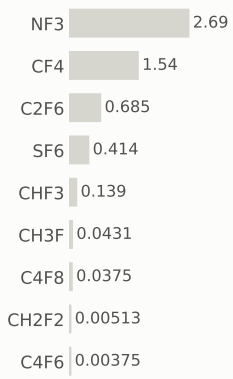
steps by process area



materials deposited (g)



gases emitted (g)



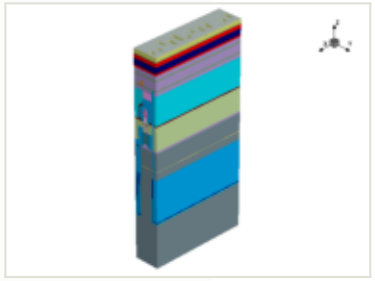
Step #3320: CVD

depo_isotropic · CVD

1.31	340.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

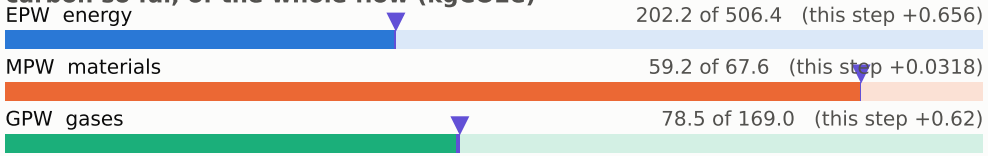


this step

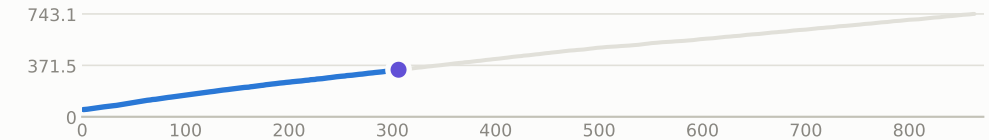
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

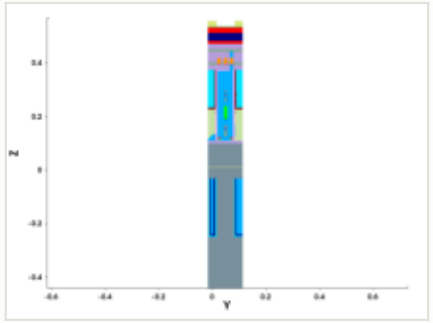
carbon so far, of the whole flow (kgCO2e)



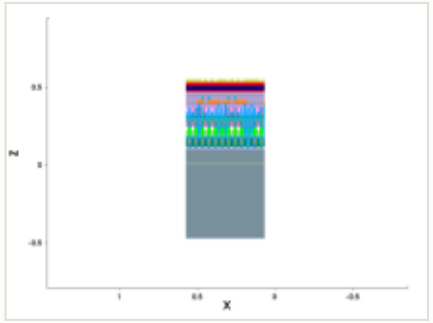
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



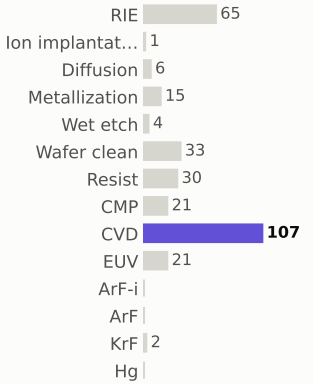
y cut at 0.049 µm



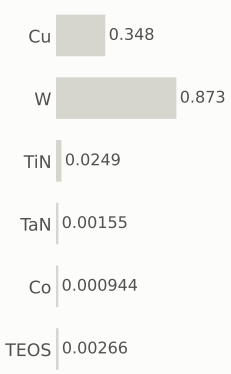
mask for this step

no mask for this step

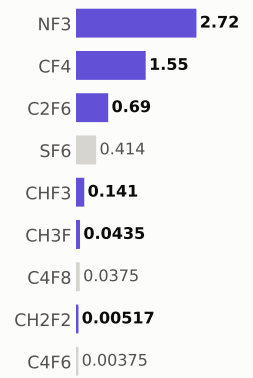
steps by process area



materials deposited (g)



gases emitted (g)



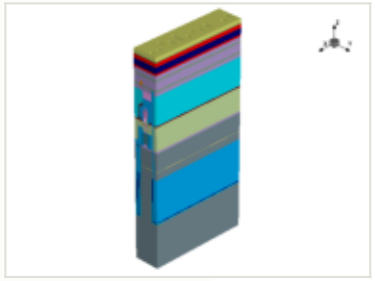
Step #3330: RIE

etch_anisotropic · RIE

1.04	341.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

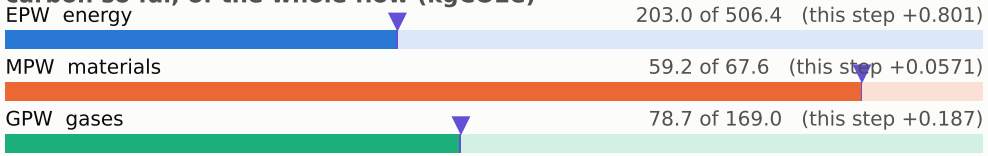


this step

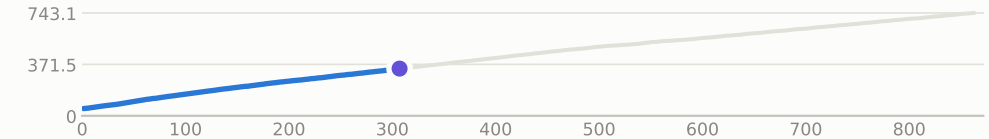
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

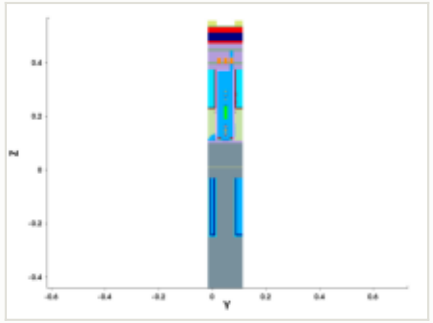
carbon so far, of the whole flow (kgCO2e)



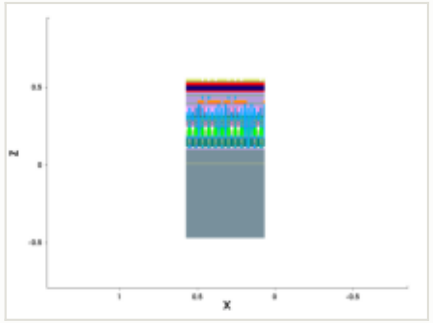
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



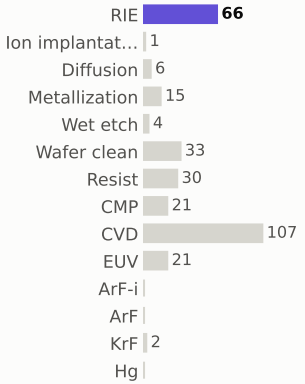
y cut at 0.049 µm



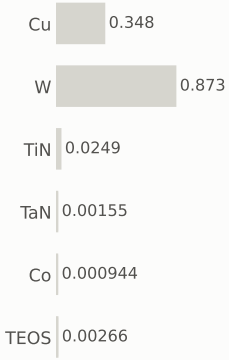
mask for this step

no mask for this step

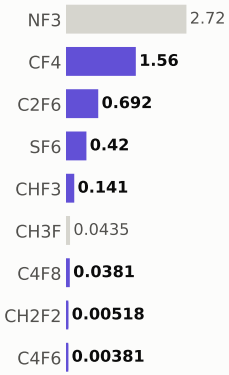
steps by process area



materials deposited (g)



gases emitted (g)



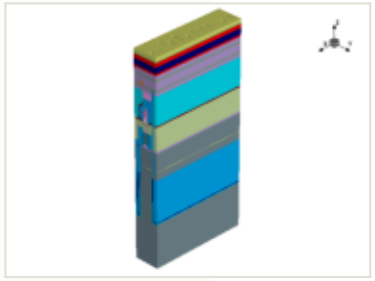
Step #3340: Wafer_clean

clean · Wafer_clean

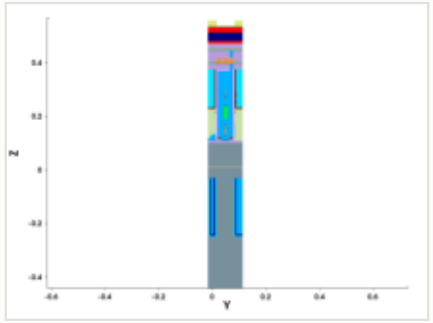
0.169	341.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

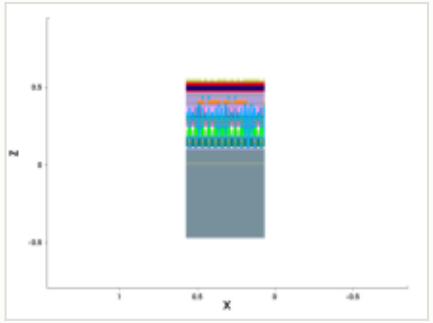
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

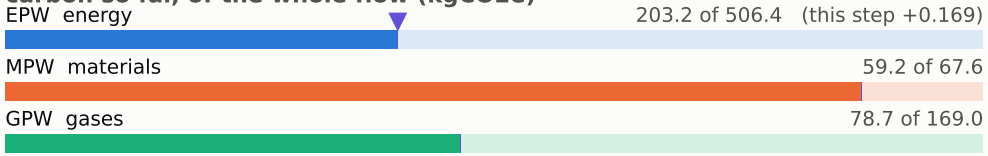
no mask for this step

this step

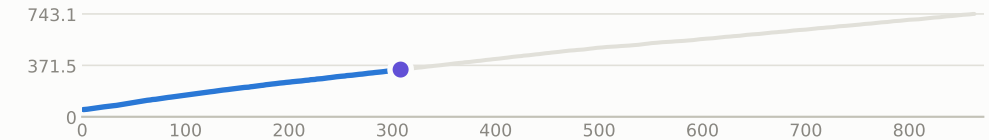
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

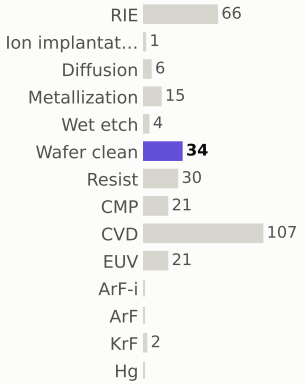
carbon so far, of the whole flow (kgCO2e)



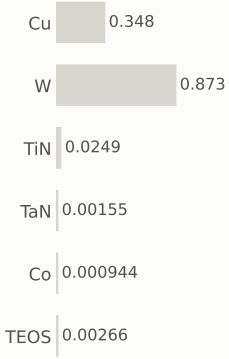
embodied carbon up to each step (kgCO2e per wafer)



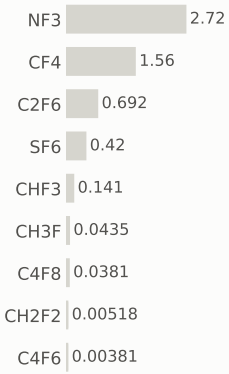
steps by process area



materials deposited (g)



gases emitted (g)



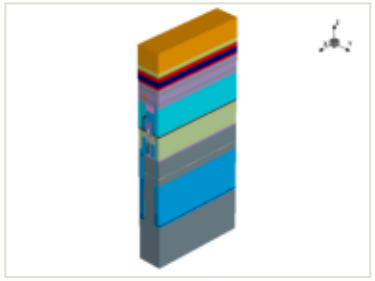
Step #3350: Resist

depo_fill · Resist

0.169	341.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

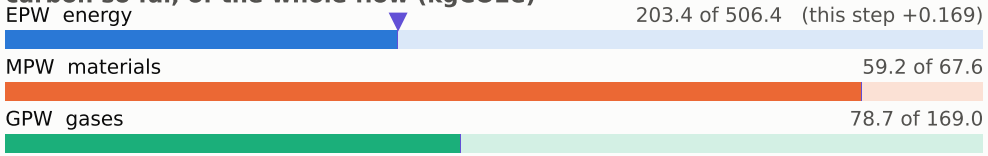


this step

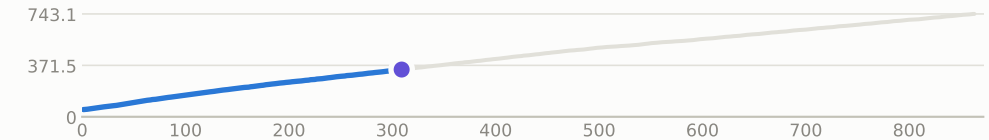
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

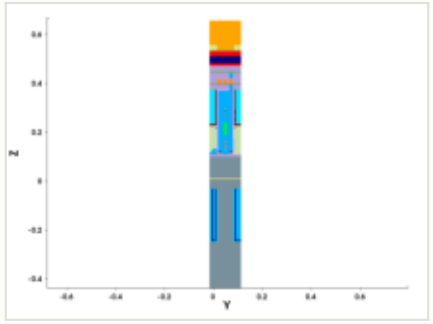
carbon so far, of the whole flow (kgCO2e)



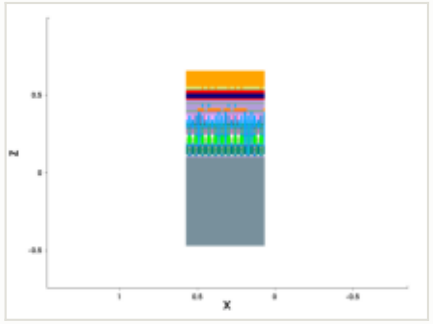
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



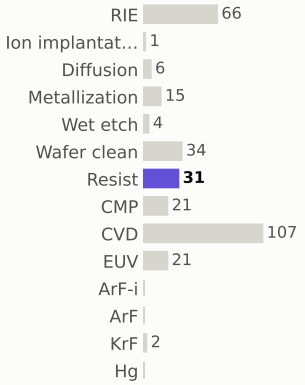
y cut at 0.049 µm



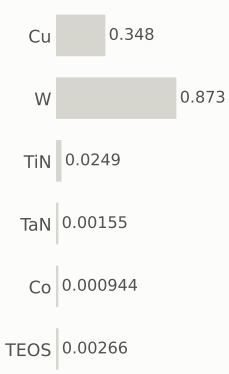
mask for this step

no mask for this step

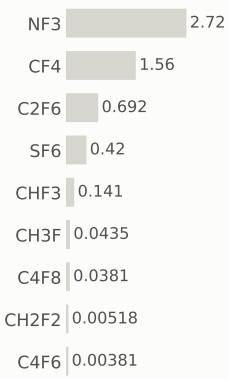
steps by process area



materials deposited (g)



gases emitted (g)



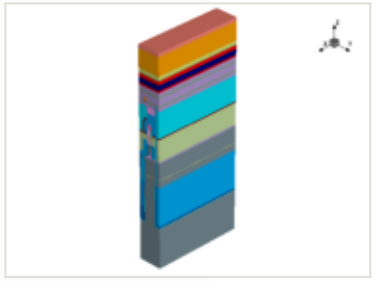
Step #3360: CVD

depo_fill · CVD

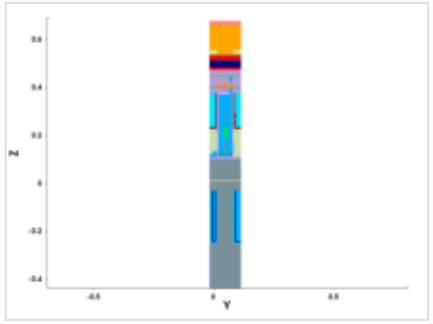
1.31	342.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

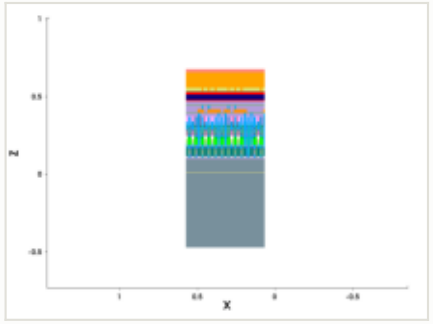
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

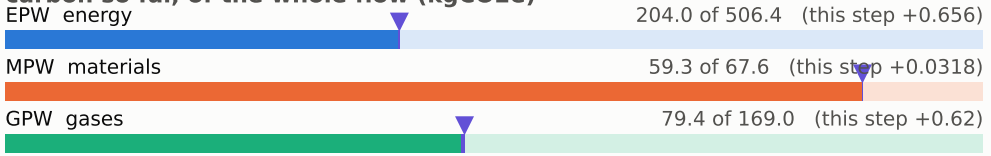
no mask for this step

this step

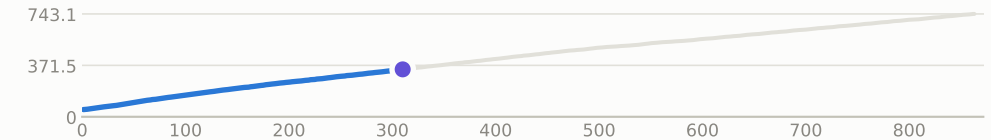
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

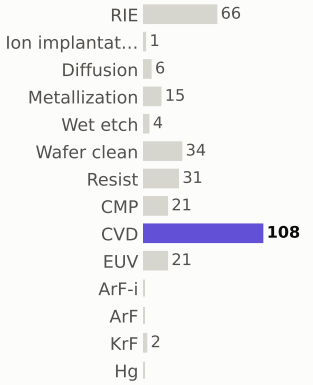
carbon so far, of the whole flow (kgCO2e)



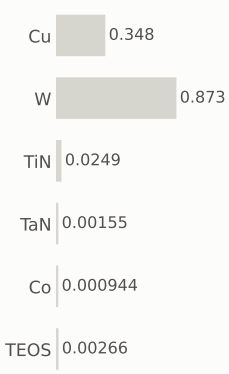
embodied carbon up to each step (kgCO2e per wafer)



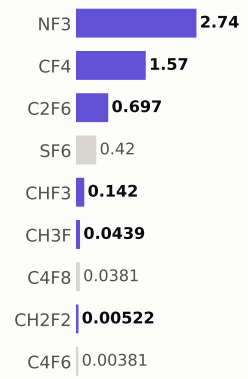
steps by process area



materials deposited (g)



gases emitted (g)



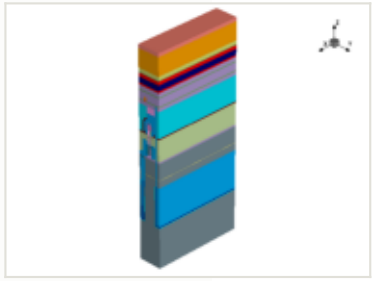
Step #3370: Wafer_clean

clean · Wafer_clean

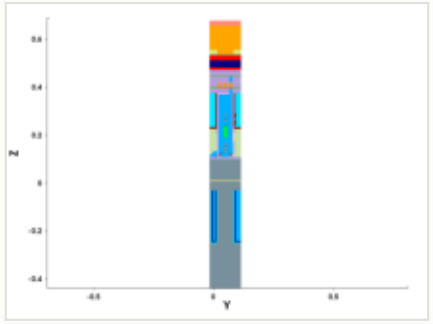
0.169	342.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

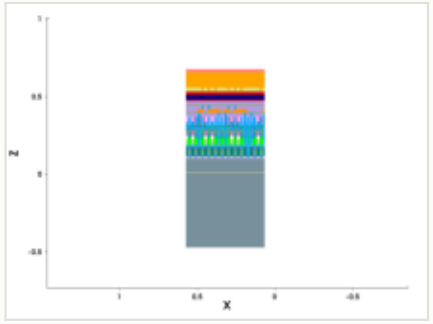
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

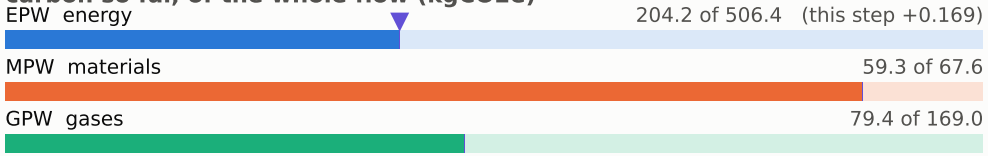
no mask for this step

this step

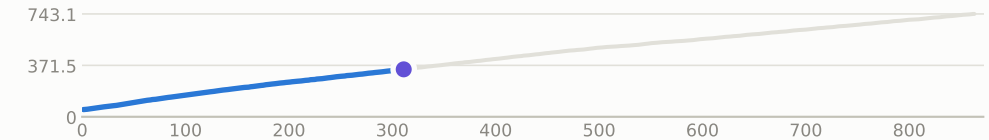
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

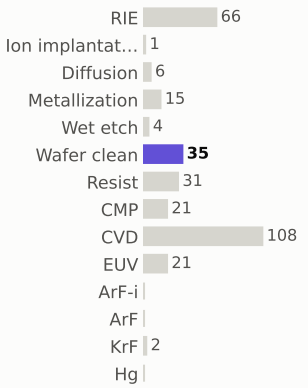
carbon so far, of the whole flow (kgCO2e)



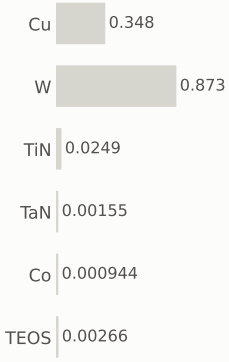
embodied carbon up to each step (kgCO2e per wafer)



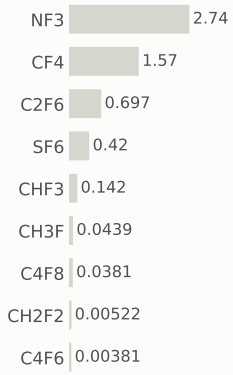
steps by process area



materials deposited (g)



gases emitted (g)



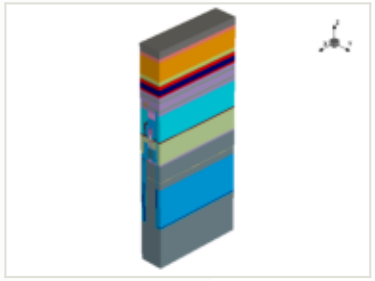
Step #3380: EUV

litho_patterning · EUV

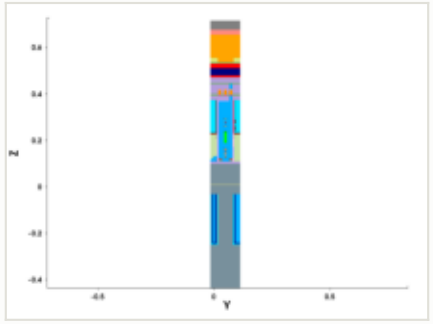
1.96	344.8	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

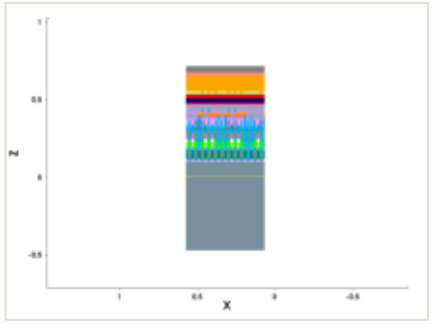
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask: layer 10252:0 (GDSII)

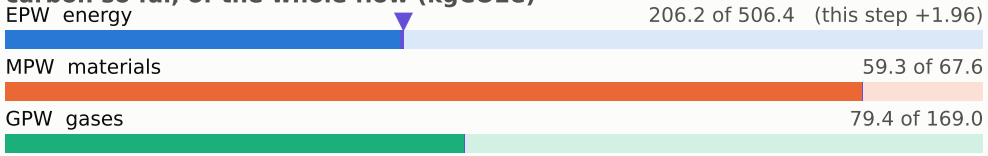
nothing in the clip window

this step

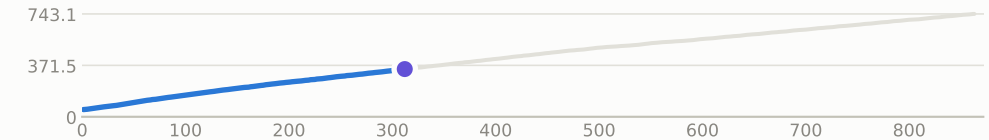
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

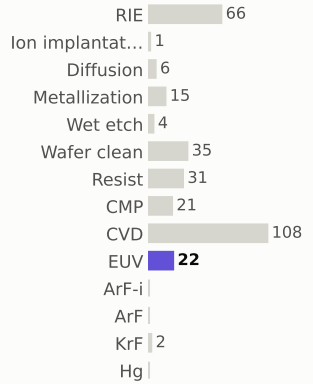
carbon so far, of the whole flow (kgCO2e)



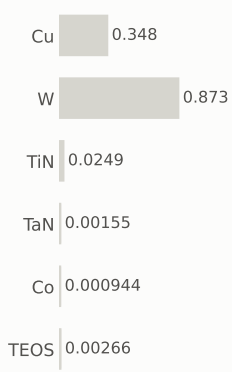
embodied carbon up to each step (kgCO2e per wafer)



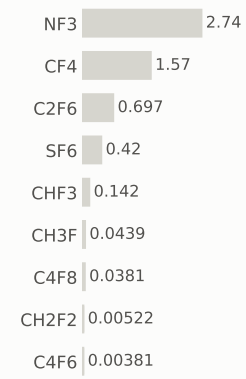
steps by process area



materials deposited (g)



gases emitted (g)



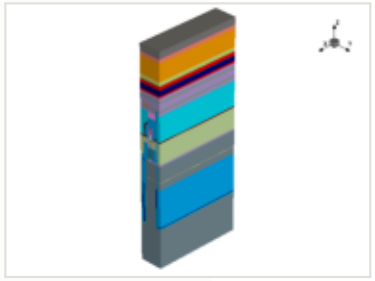
Step #3390: RIE

etch_anisotropic · RIE

1.04	345.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

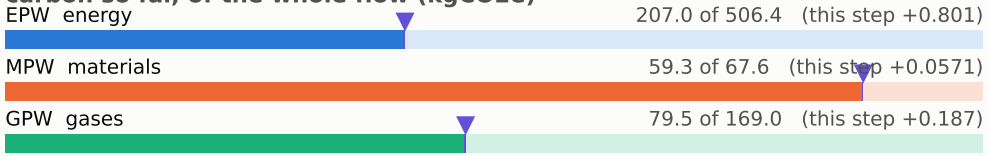


this step

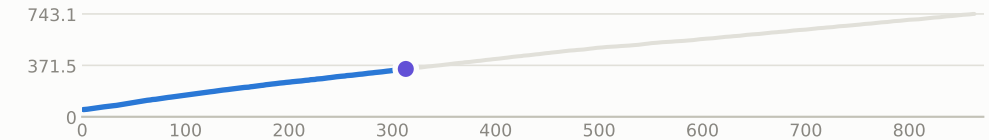
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

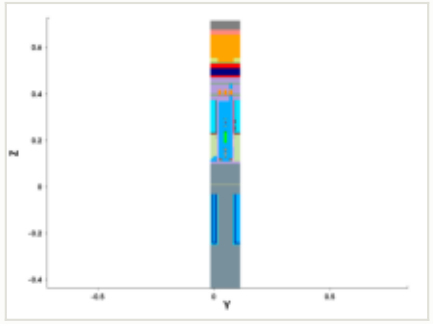
carbon so far, of the whole flow (kgCO2e)



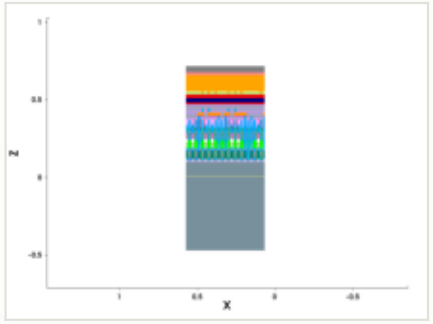
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



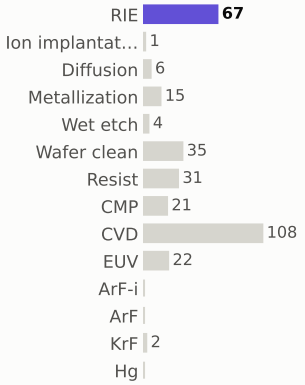
y cut at 0.049 µm



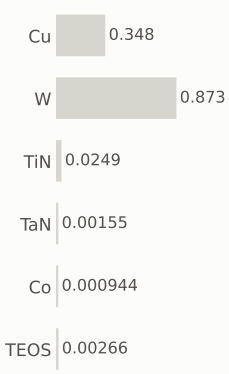
mask for this step

no mask for this step

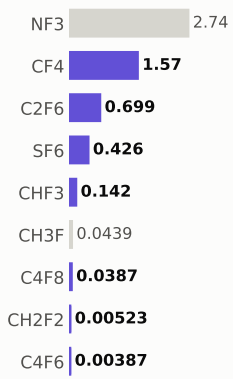
steps by process area



materials deposited (g)



gases emitted (g)



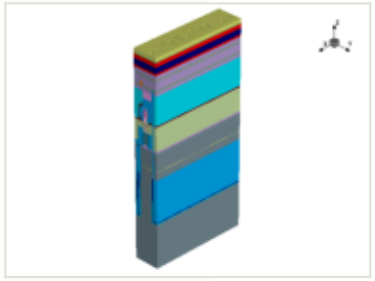
Step #3400: Wafer_clean

etch_strip · Wafer_clean

0.169	346.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

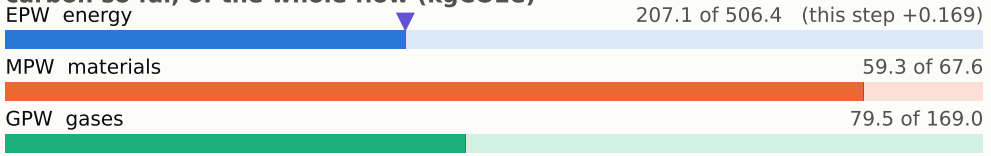


this step

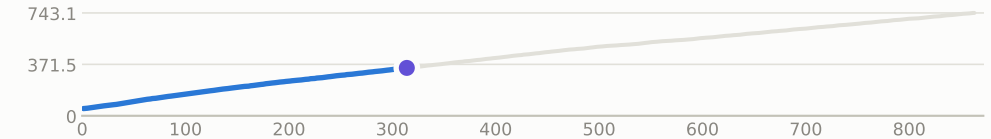
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

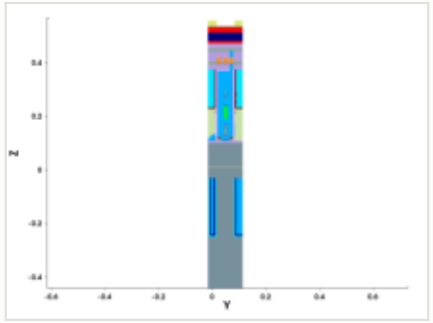
carbon so far, of the whole flow (kgCO2e)



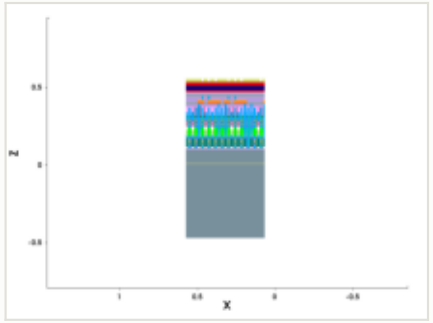
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



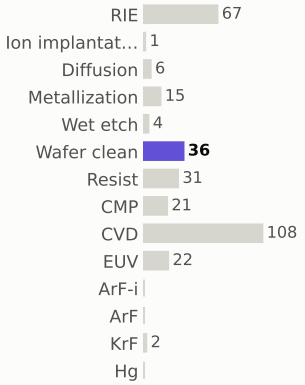
y cut at 0.049 µm



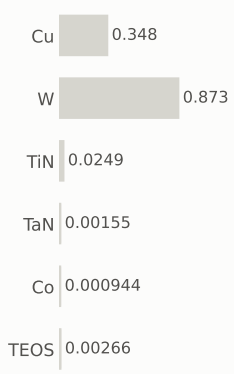
mask for this step

no mask for this step

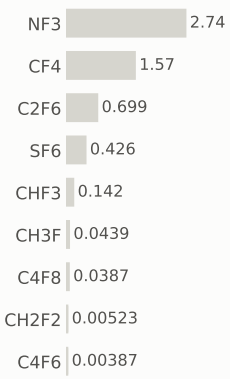
steps by process area



materials deposited (g)



gases emitted (g)



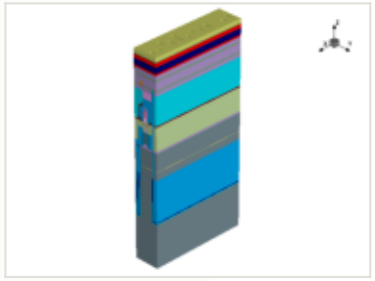
Step #3410: RIE

etch_anisotropic · RIE

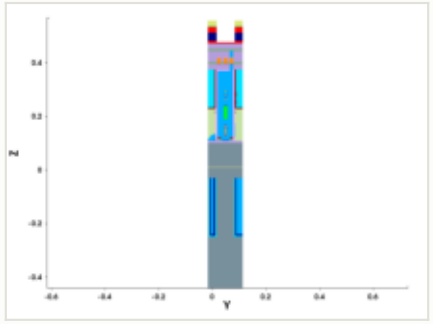
1.04	347.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

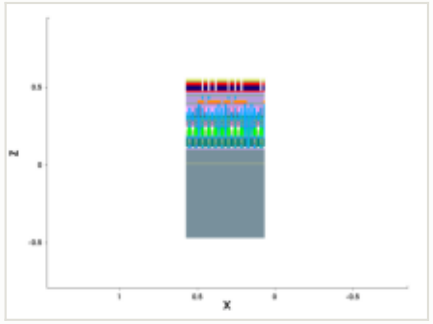
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

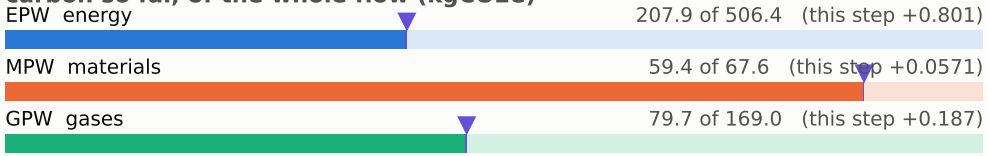
no mask for this step

this step

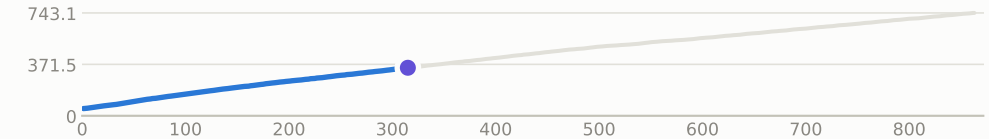
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

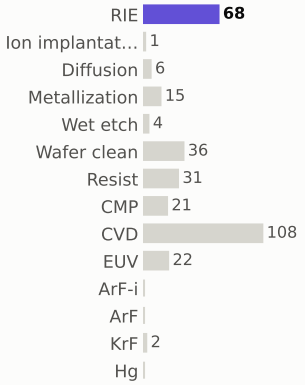
carbon so far, of the whole flow (kgCO2e)



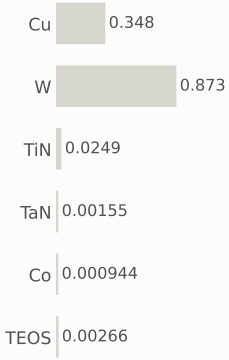
embodied carbon up to each step (kgCO2e per wafer)



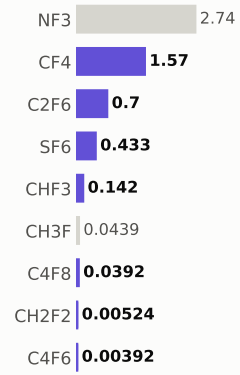
steps by process area



materials deposited (g)



gases emitted (g)



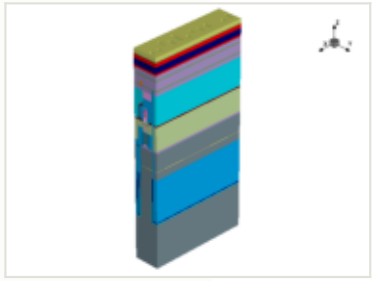
Step #3420: Wafer_clean

clean · Wafer_clean

0.169	347.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



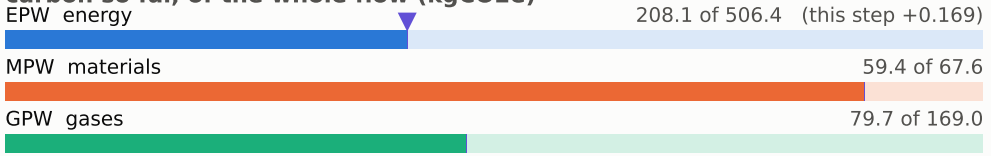
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

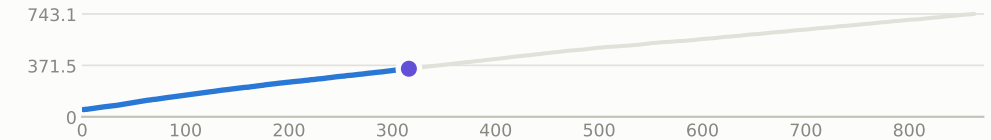
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

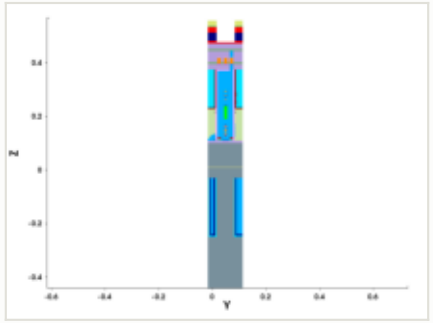
carbon so far, of the whole flow (kgCO2e)



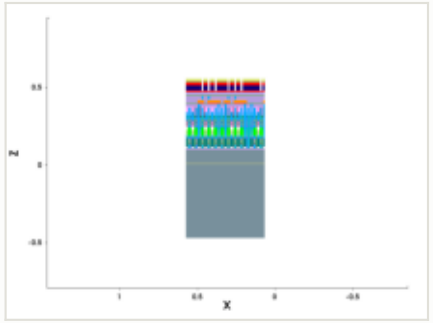
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



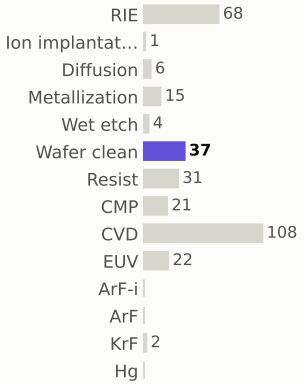
y cut at 0.049 µm



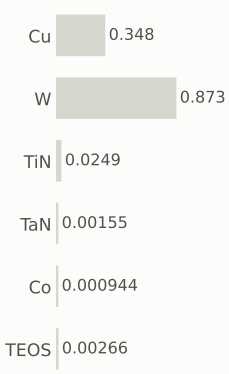
mask for this step

no mask for this step

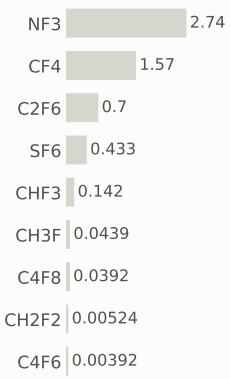
steps by process area



materials deposited (g)



gases emitted (g)



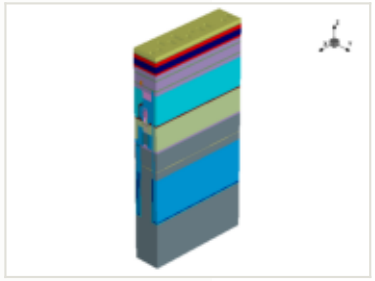
Step #3430: RIE

etch_anisotropic · RIE

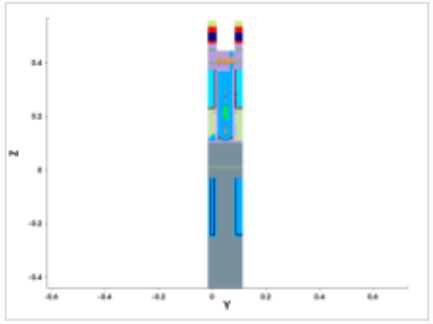
1.04	348.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

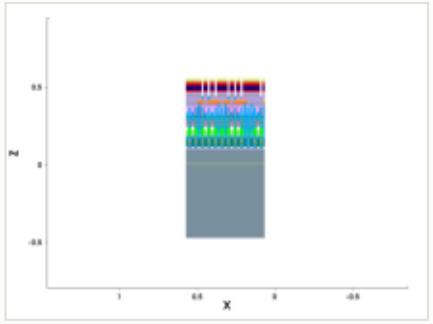
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

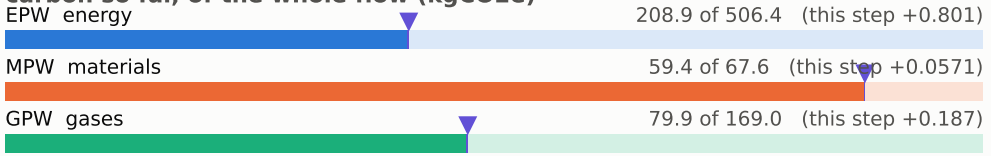
no mask for this step

this step

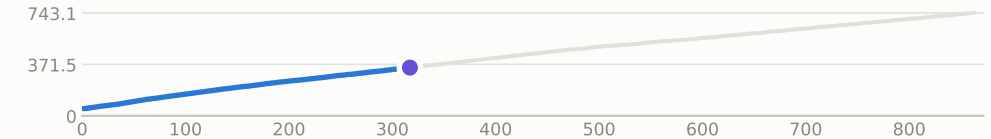
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

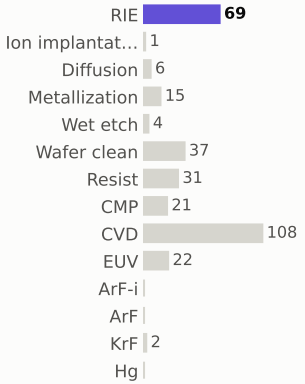
carbon so far, of the whole flow (kgCO2e)



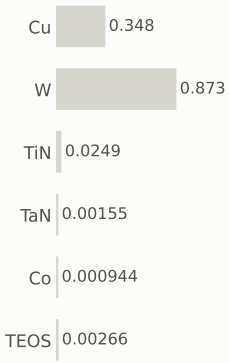
embodied carbon up to each step (kgCO2e per wafer)



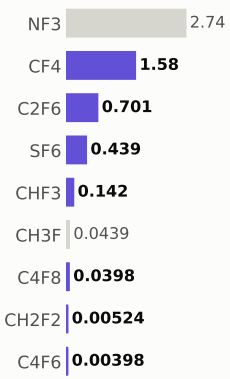
steps by process area



materials deposited (g)



gases emitted (g)



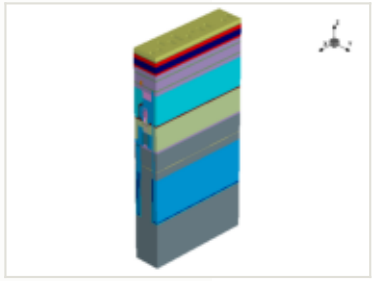
Step #3440: Wafer_clean

clean · Wafer_clean

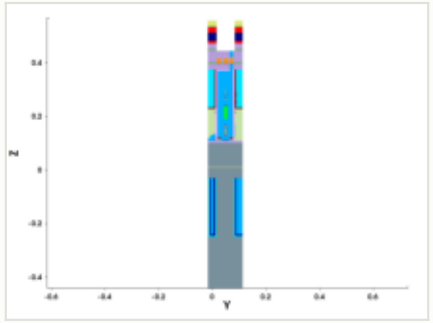
0.169	348.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

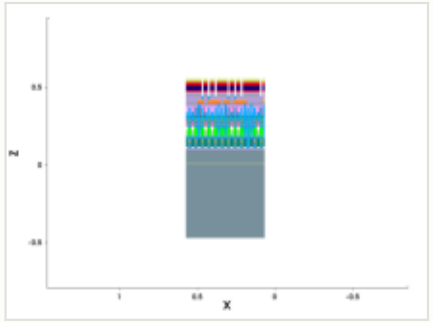
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

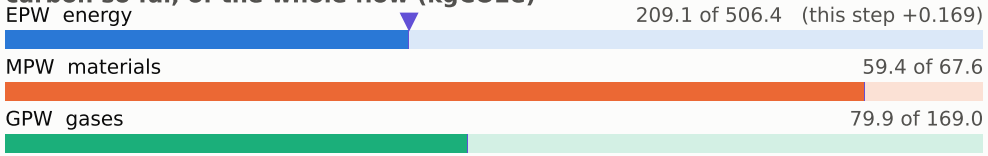
no mask for this step

this step

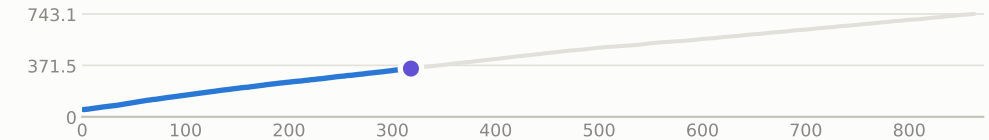
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

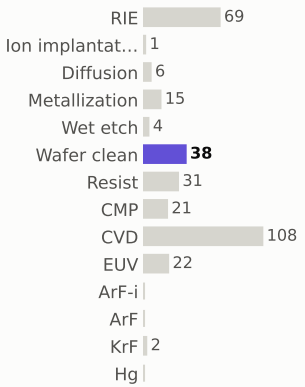
carbon so far, of the whole flow (kgCO2e)



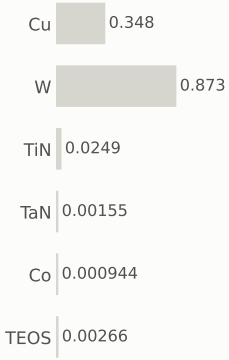
embodied carbon up to each step (kgCO2e per wafer)



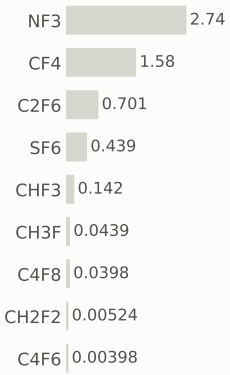
steps by process area



materials deposited (g)



gases emitted (g)



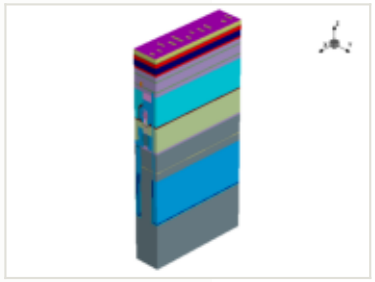
Step #3450: Metallization

depo_ellipsoidal · Metallization

1.2	349.6	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

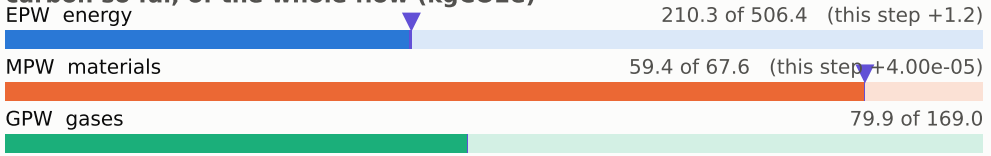
zoomed to fit



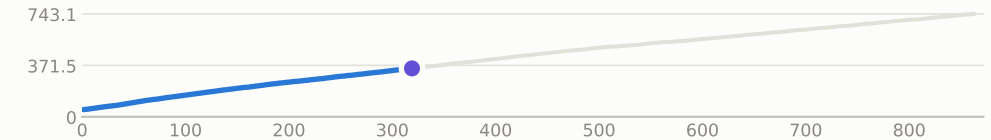
this step

process area	Metallization	EPW	1.2 kgCO2e
material	TaN	EPW tool	0.86 kgCO2e
thickness	0.0016 µm	EPW overhead	0.344 kgCO2e
material GWP	25.8159 gCO2e/g	MPW	4.00e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

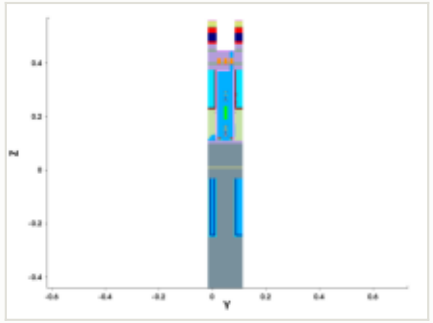
carbon so far, of the whole flow (kgCO2e)



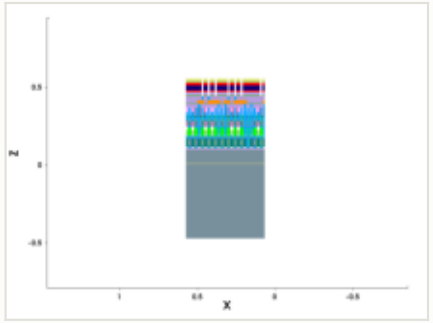
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



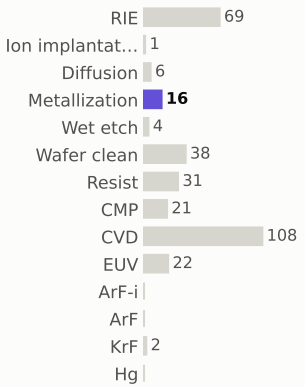
y cut at 0.049 µm



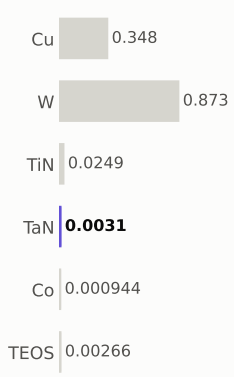
mask for this step

no mask for this step

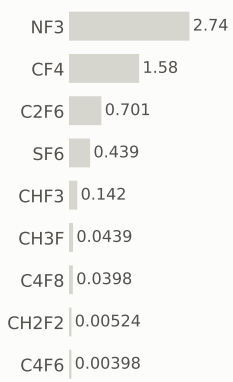
steps by process area



materials deposited (g)



gases emitted (g)



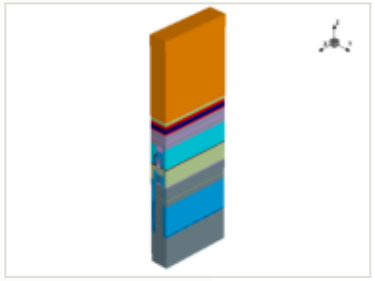
Step #3460: Metallization

depo_fill · Metallization

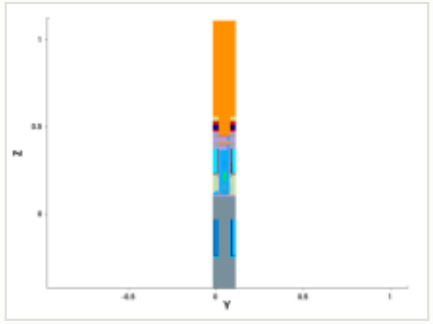
1.21	350.8	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

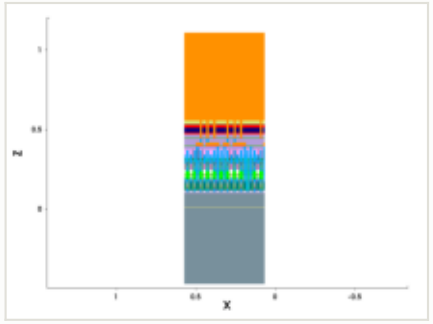
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

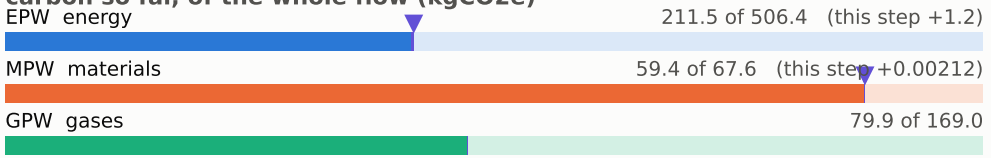
no mask for this step

this step

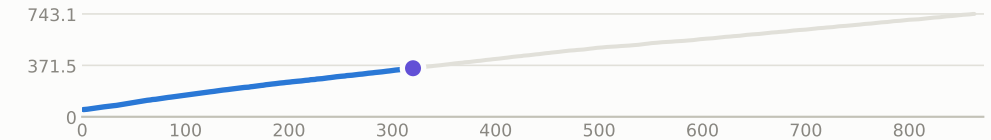
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

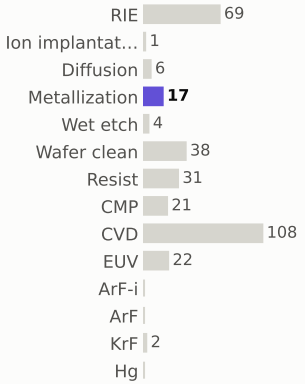
carbon so far, of the whole flow (kgCO2e)



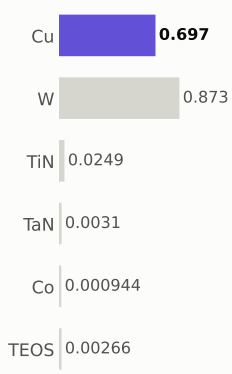
embodied carbon up to each step (kgCO2e per wafer)



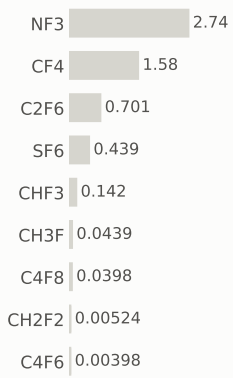
steps by process area



materials deposited (g)



gases emitted (g)



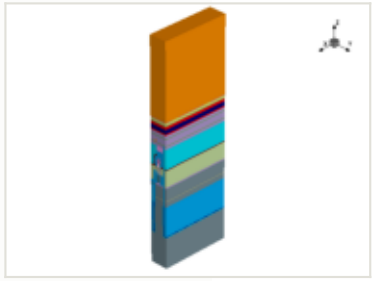
Step #3470: Diffusion

treatment · Diffusion

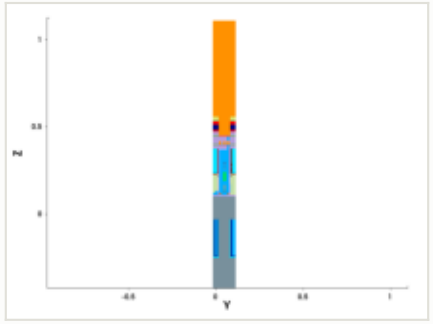
0.801	351.6	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

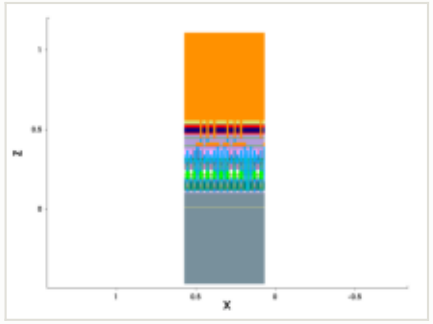
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

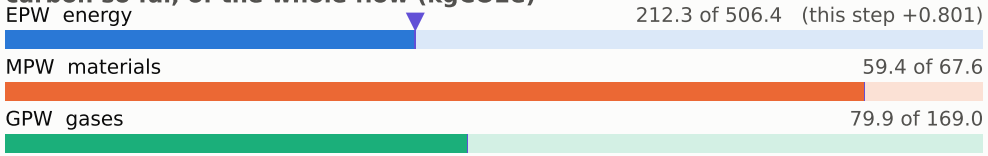
no mask for this step

this step

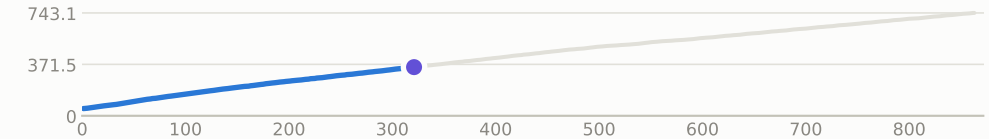
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

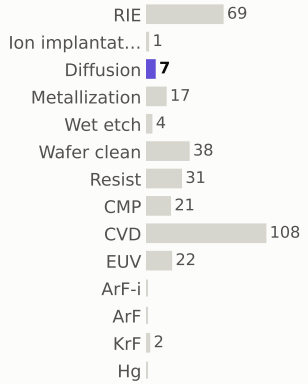
carbon so far, of the whole flow (kgCO2e)



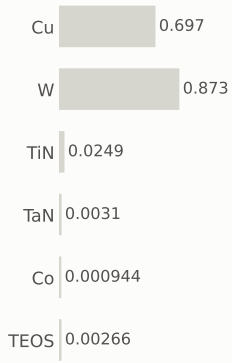
embodied carbon up to each step (kgCO2e per wafer)



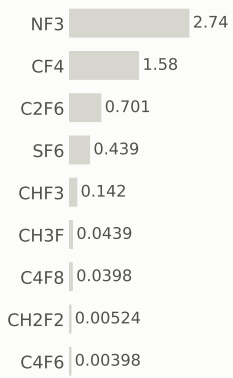
steps by process area



materials deposited (g)



gases emitted (g)



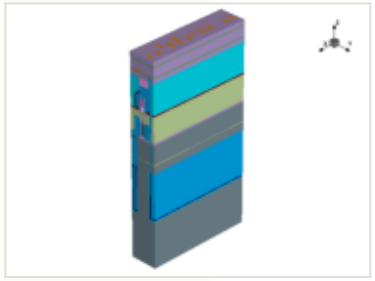
Step #3480: CMP

etch_cmp · CMP

0.169	351.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

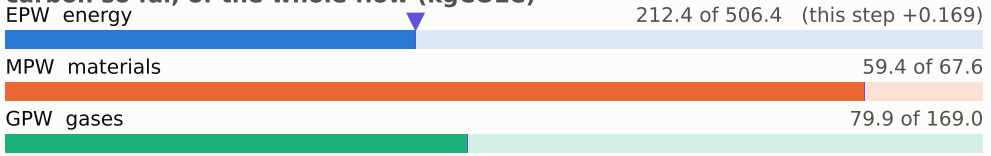


this step

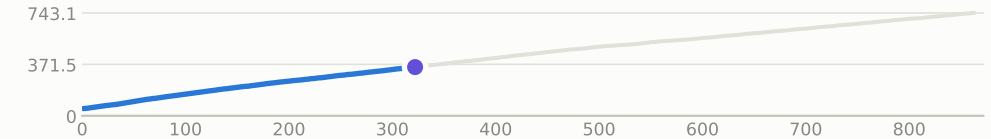
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

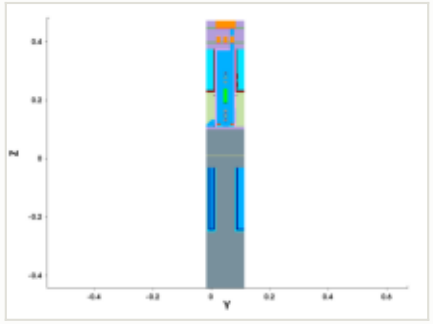
carbon so far, of the whole flow (kgCO2e)



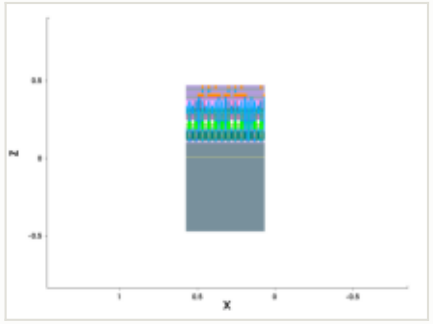
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



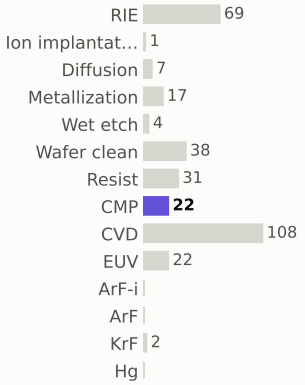
y cut at 0.049 µm



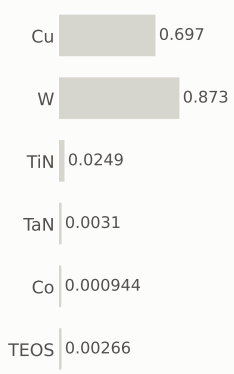
mask for this step

no mask for this step

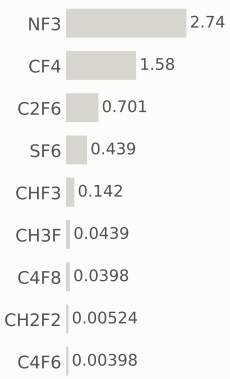
steps by process area



materials deposited (g)



gases emitted (g)



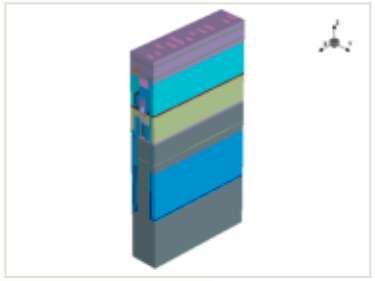
Step #3490: Metallization

depo_isotropic · Metallization

1.2	353.0	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

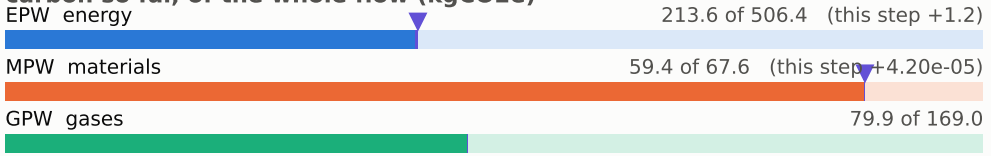
zoomed to fit



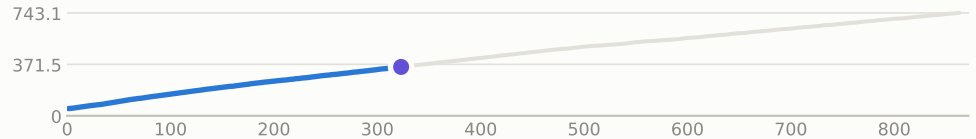
this step

process area	Metallization	EPW	1.2 kgCO2e
material	Co	EPW tool	0.86 kgCO2e
thickness	0.0015 µm	EPW overhead	0.344 kgCO2e
material GWP	44.5 gCO2e/g	MPW	4.20e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

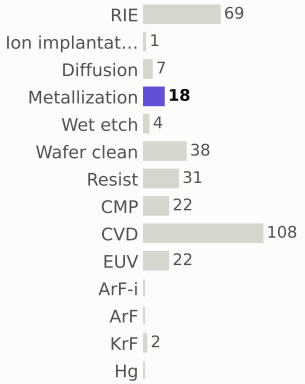
carbon so far, of the whole flow (kgCO2e)



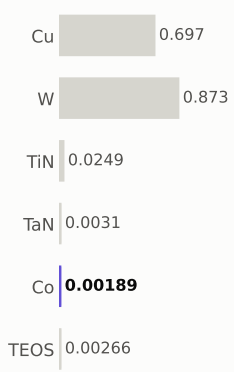
embodied carbon up to each step (kgCO2e per wafer)



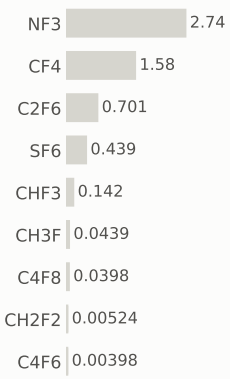
steps by process area



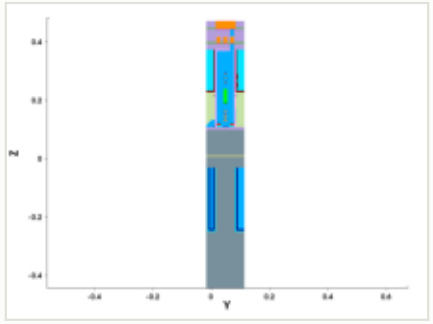
materials deposited (g)



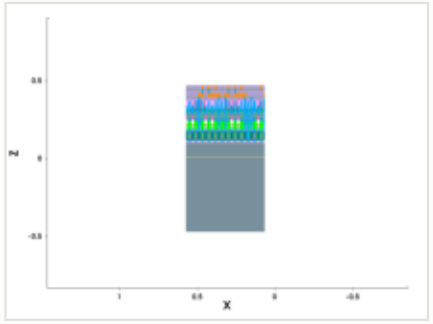
gases emitted (g)



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

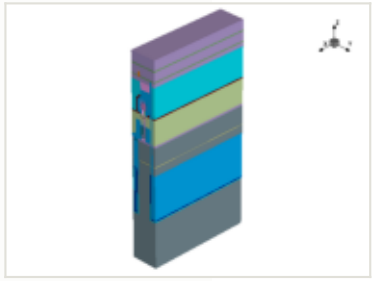
Step #3510: CVD

depo_fill · CVD

1.31	354.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

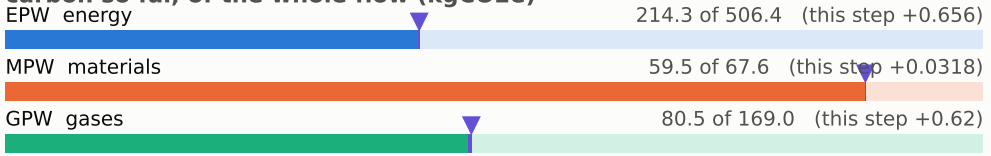


this step

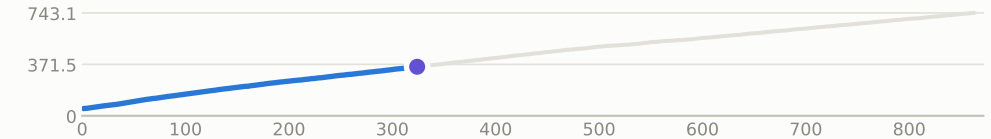
process area	CVD
material	—
thickness	0.024 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

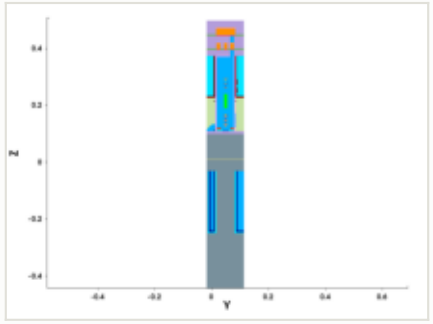
carbon so far, of the whole flow (kgCO2e)



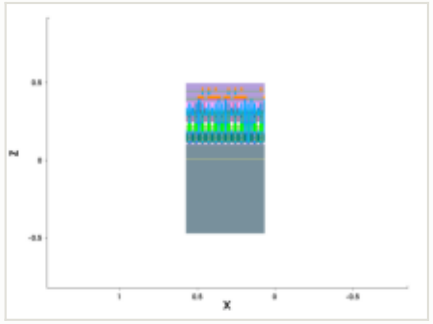
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



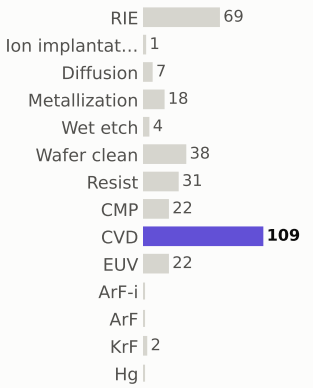
y cut at 0.049 µm



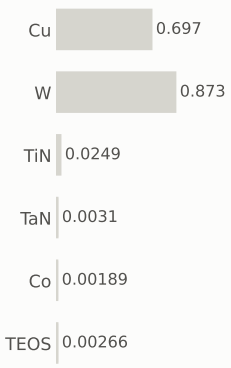
mask for this step

no mask for this step

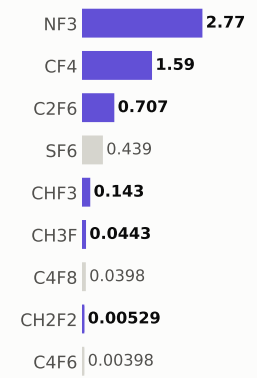
steps by process area



materials deposited (g)



gases emitted (g)



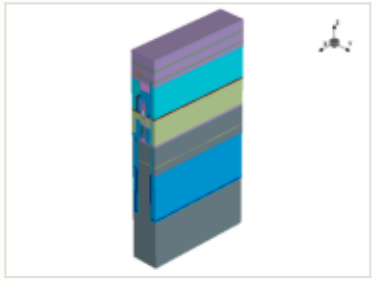
Step #3520: Diffusion

treatment · Diffusion

0.801	355.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

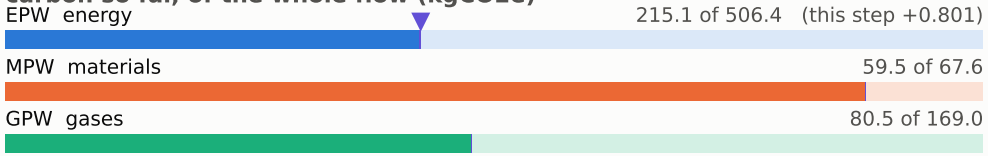


this step

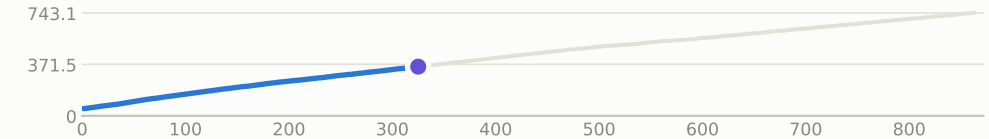
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

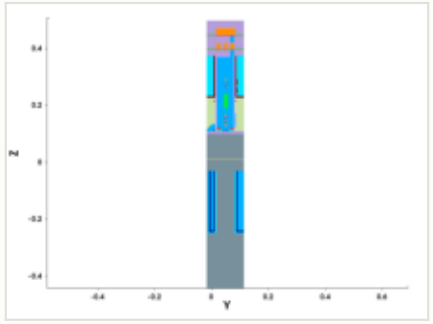
carbon so far, of the whole flow (kgCO2e)



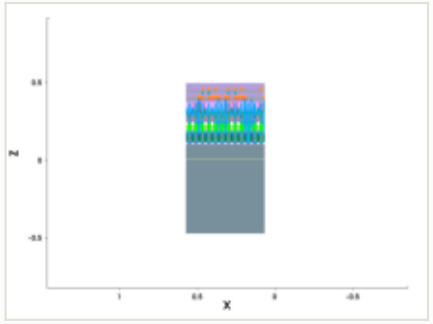
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



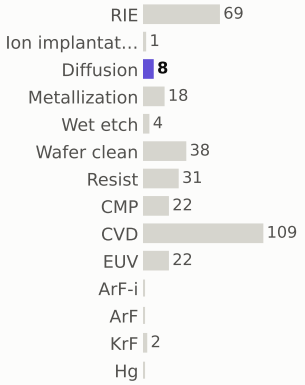
y cut at 0.049 µm



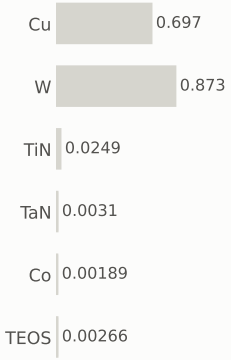
mask for this step

no mask for this step

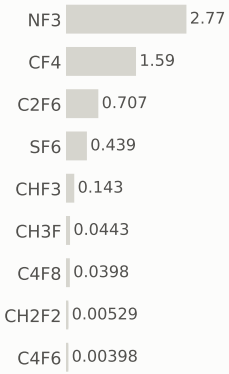
steps by process area



materials deposited (g)



gases emitted (g)



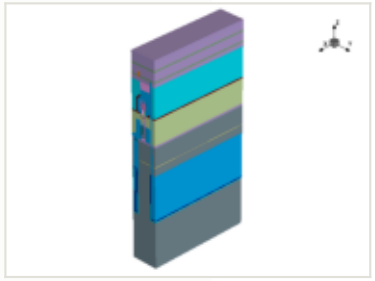
Step #3530: Wafer_clean

clean · Wafer_clean

0.169	355.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

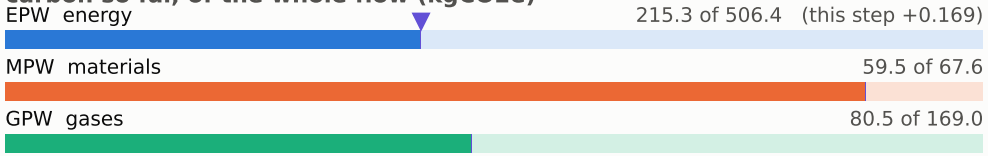


this step

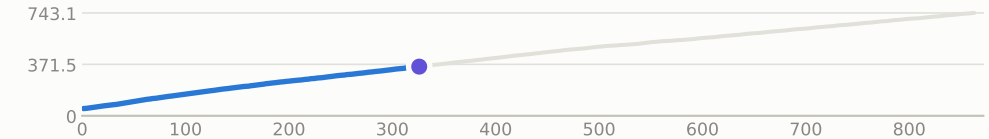
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

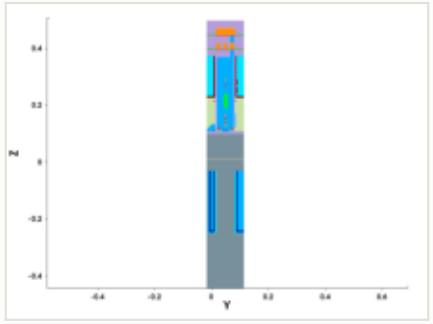
carbon so far, of the whole flow (kgCO2e)



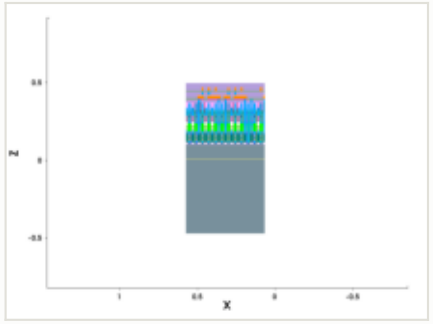
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



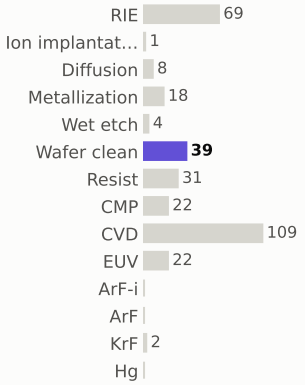
y cut at 0.049 µm



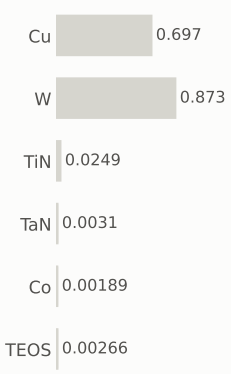
mask for this step

no mask for this step

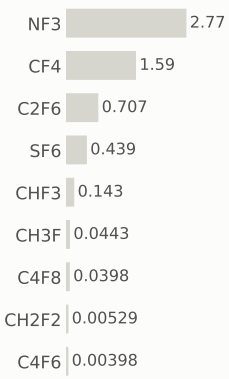
steps by process area



materials deposited (g)



gases emitted (g)



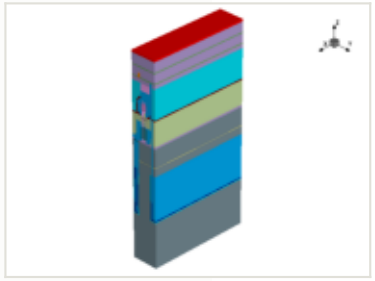
Step #3540: CVD

depo_fill · CVD

1.31	356.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

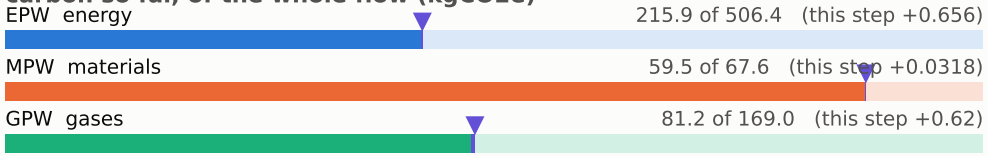


this step

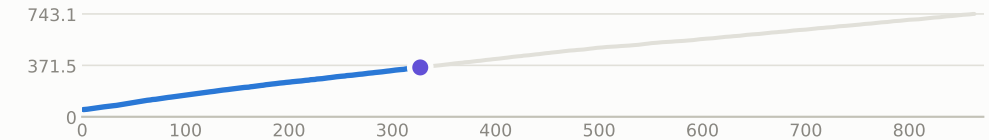
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

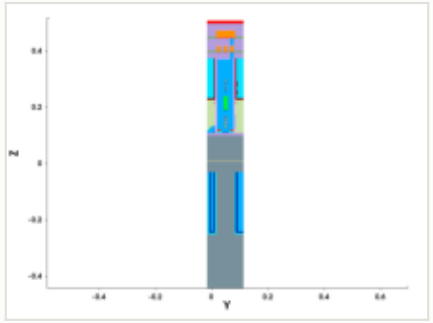
carbon so far, of the whole flow (kgCO2e)



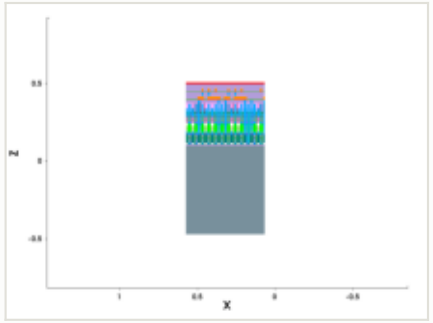
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



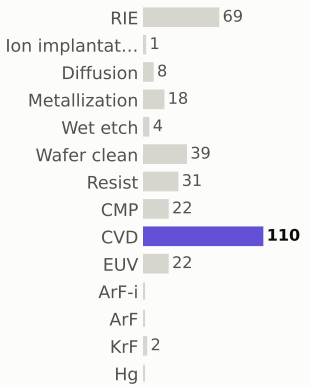
y cut at 0.049 µm



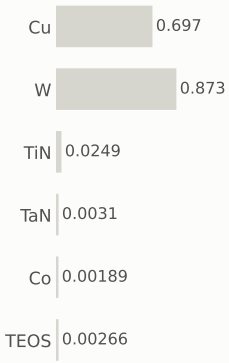
mask for this step

no mask for this step

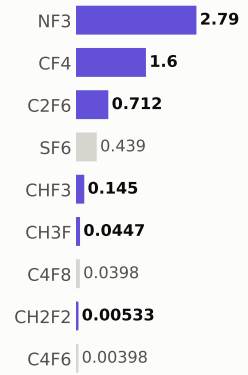
steps by process area



materials deposited (g)



gases emitted (g)



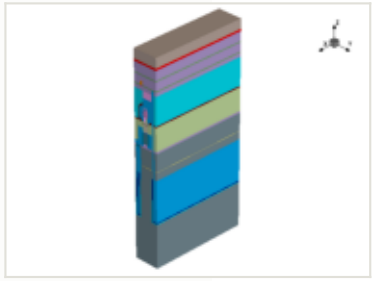
Step #3550: Resist

depo_fill · Resist

0.169	356.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

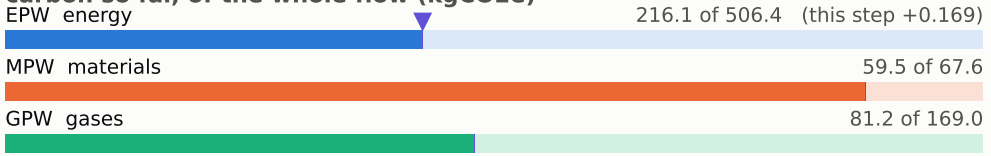


this step

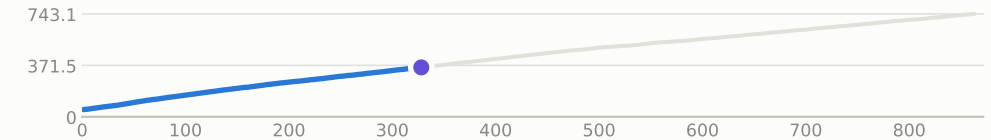
process area	Resist
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

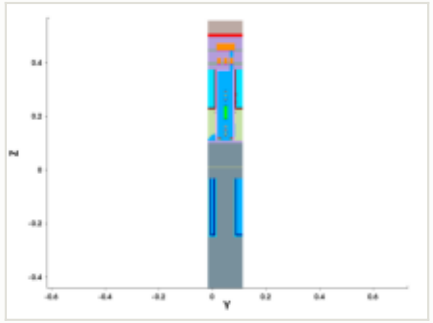
carbon so far, of the whole flow (kgCO2e)



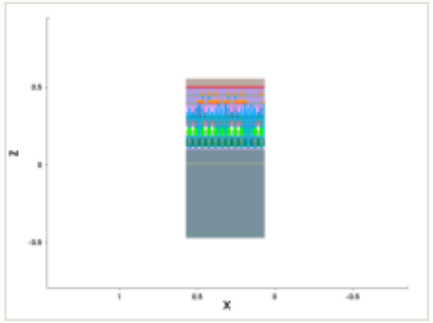
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



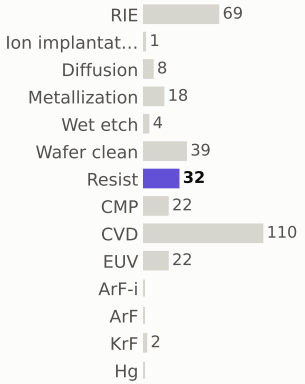
y cut at 0.049 µm



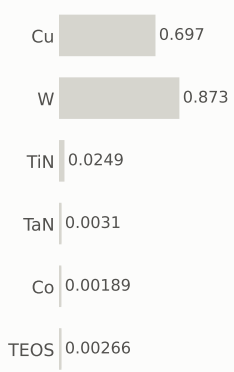
mask for this step

no mask for this step

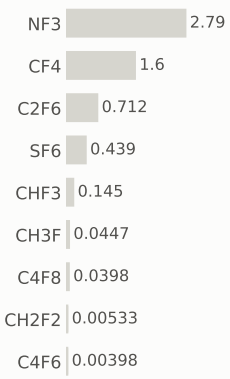
steps by process area



materials deposited (g)



gases emitted (g)



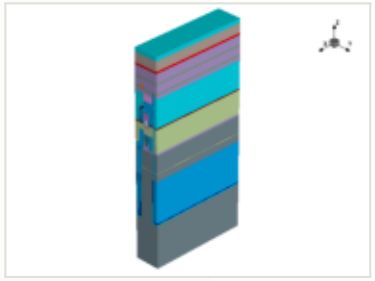
Step #3560: CVD

depo_fill · CVD

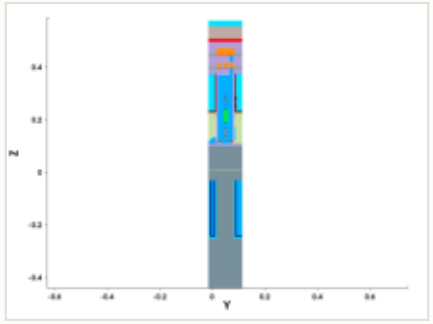
1.31	358.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

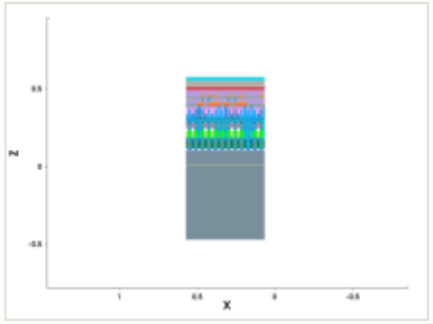
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



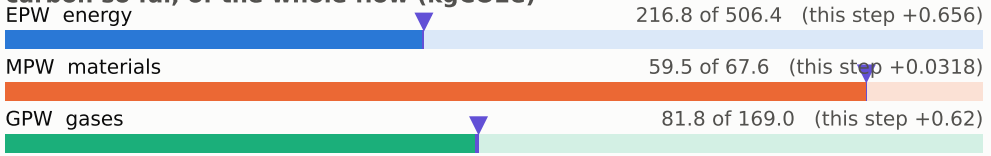
mask for this step

no mask for this step

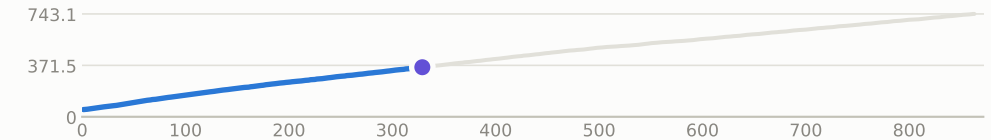
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.02 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

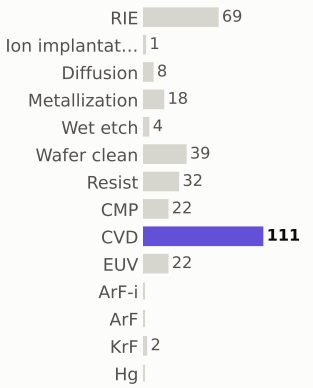
carbon so far, of the whole flow (kgCO2e)



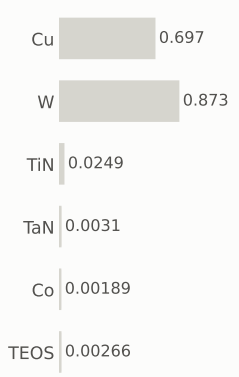
embodied carbon up to each step (kgCO2e per wafer)



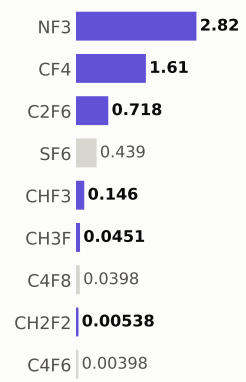
steps by process area



materials deposited (g)



gases emitted (g)



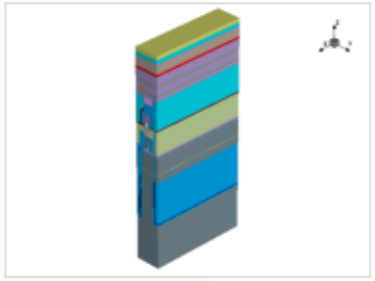
Step #3570: CVD

depo_fill · CVD

1.31	359.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

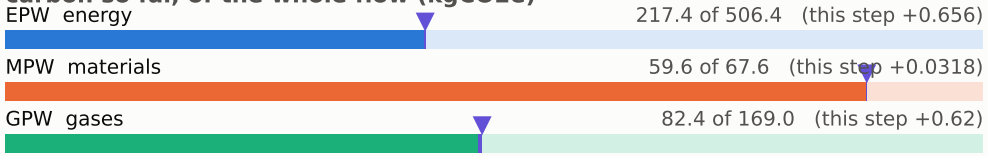


this step

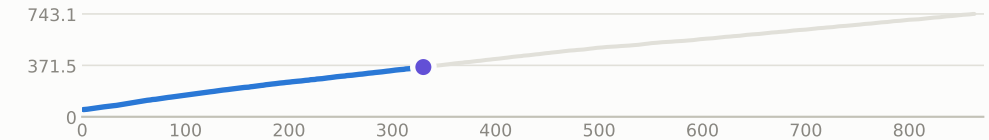
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

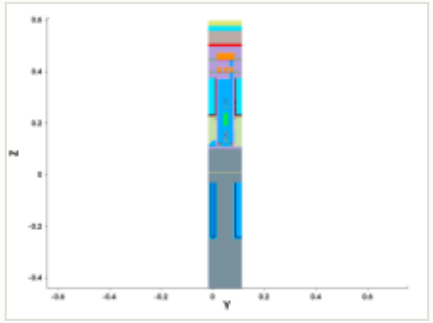
carbon so far, of the whole flow (kgCO2e)



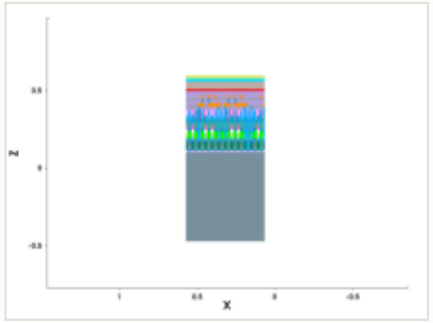
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



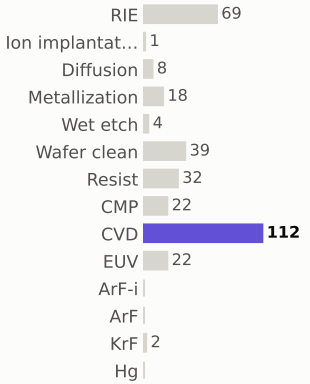
y cut at 0.049 µm



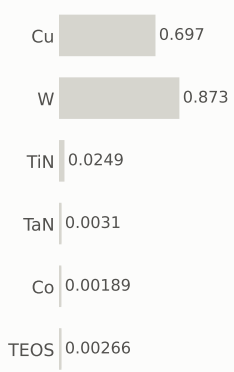
mask for this step

no mask for this step

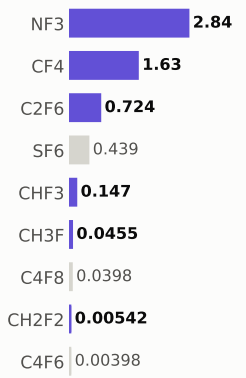
steps by process area



materials deposited (g)



gases emitted (g)



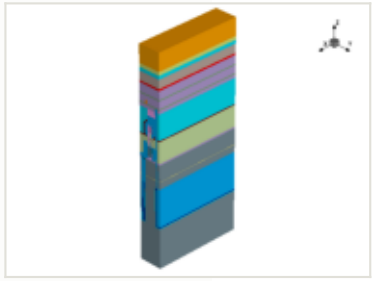
Step #3580: Resist

depo_fill · Resist

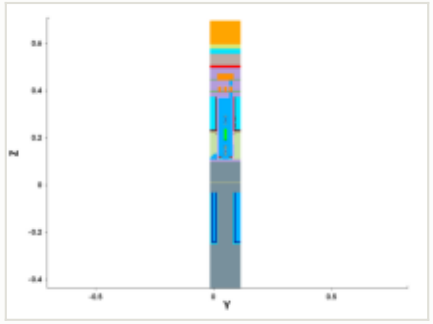
0.169	359.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

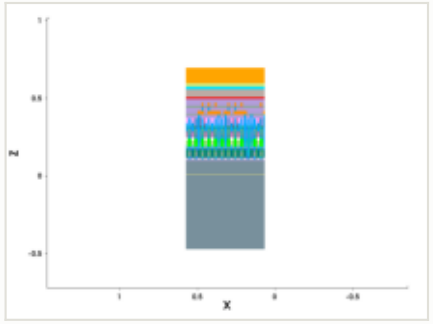
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



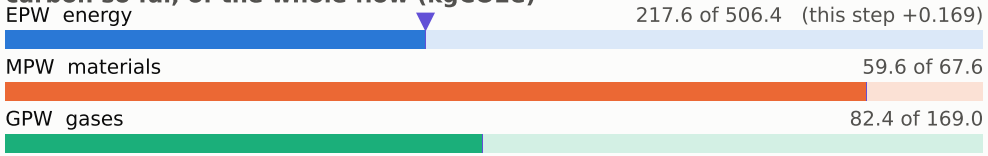
mask for this step

no mask for this step

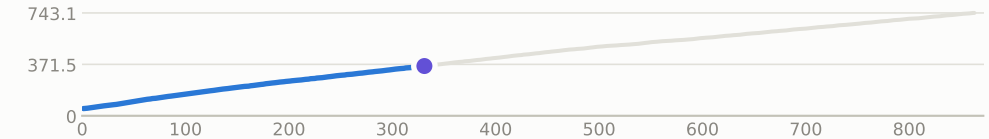
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

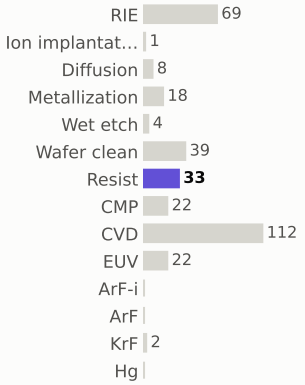
carbon so far, of the whole flow (kgCO2e)



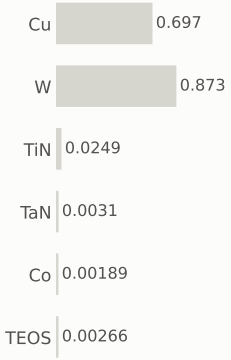
embodied carbon up to each step (kgCO2e per wafer)



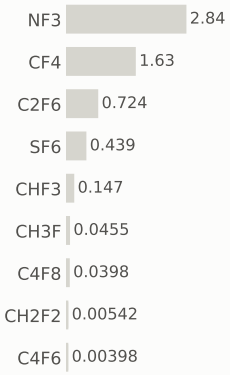
steps by process area



materials deposited (g)



gases emitted (g)



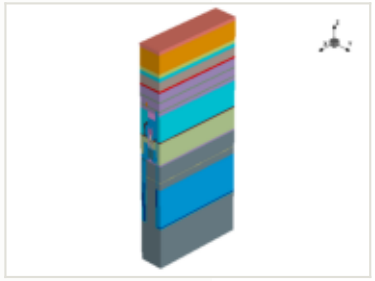
Step #3590: CVD

depo_fill · CVD

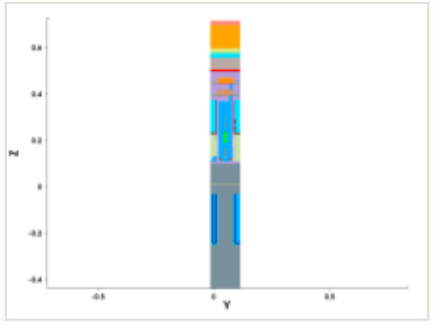
1.31	360.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

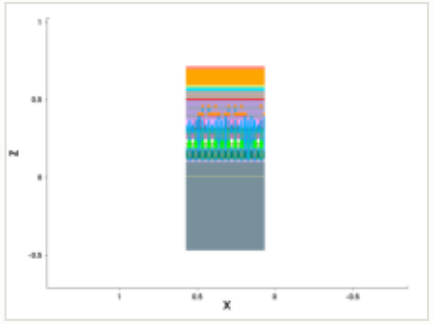
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

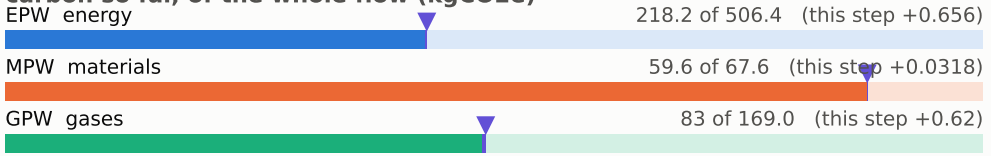
no mask for this step

this step

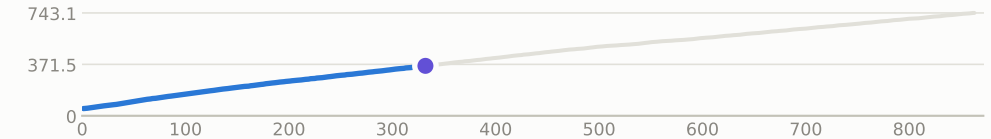
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

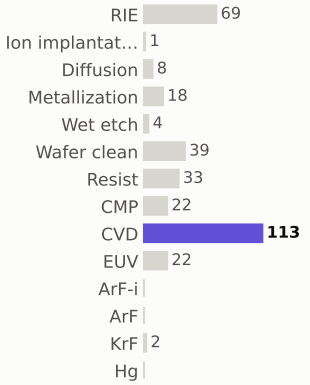
carbon so far, of the whole flow (kgCO2e)



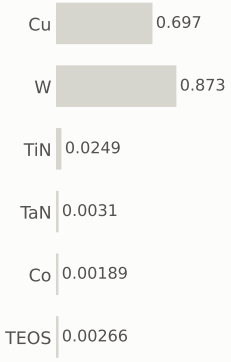
embodied carbon up to each step (kgCO2e per wafer)



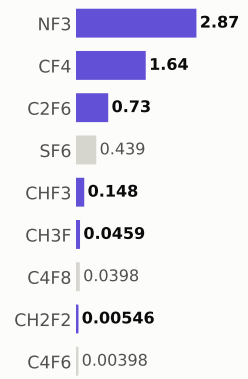
steps by process area



materials deposited (g)



gases emitted (g)



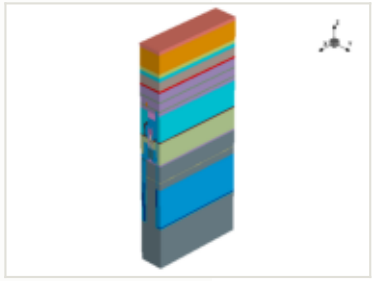
Step #3600: Wafer_clean

clean · Wafer_clean

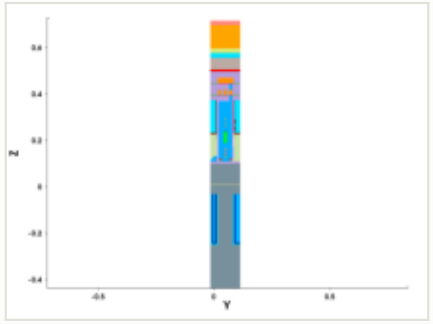
0.169	361.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

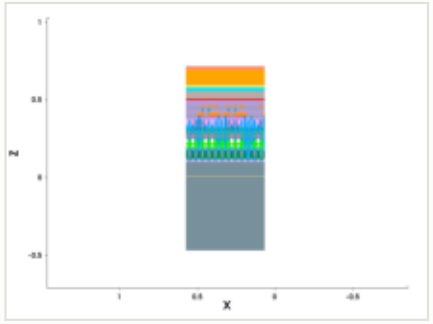
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

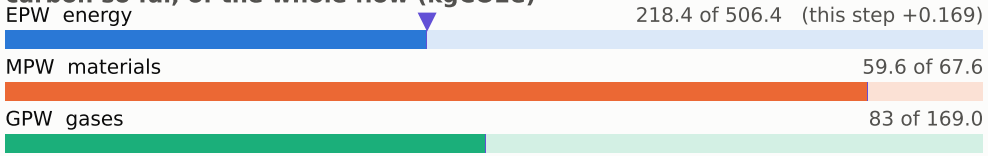
no mask for this step

this step

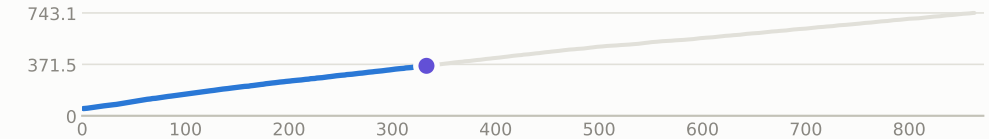
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

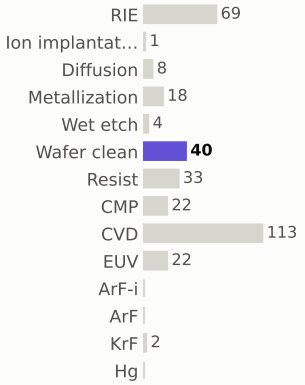
carbon so far, of the whole flow (kgCO2e)



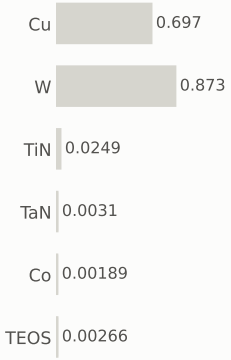
embodied carbon up to each step (kgCO2e per wafer)



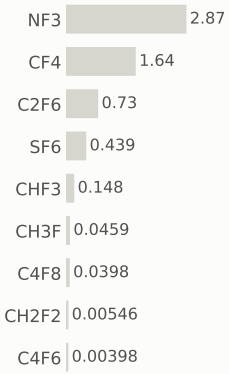
steps by process area



materials deposited (g)



gases emitted (g)



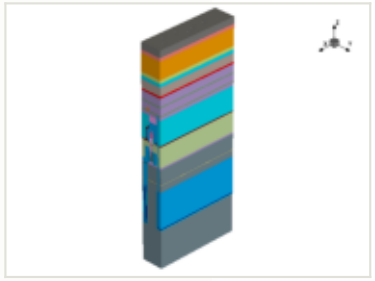
Step #3610: EUV

litho_patterning · EUV

1.96	363.0	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

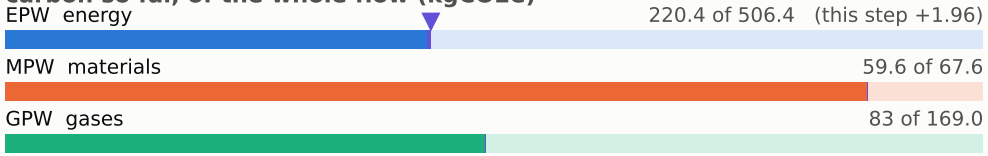


this step

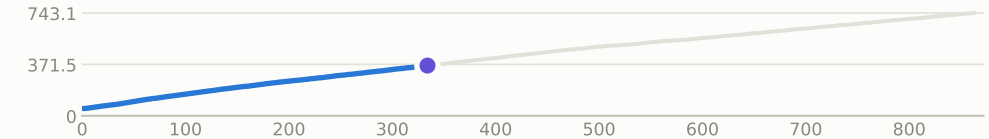
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

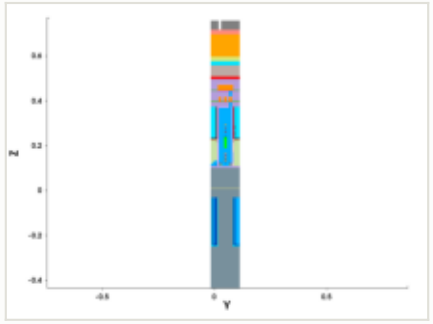
carbon so far, of the whole flow (kgCO2e)



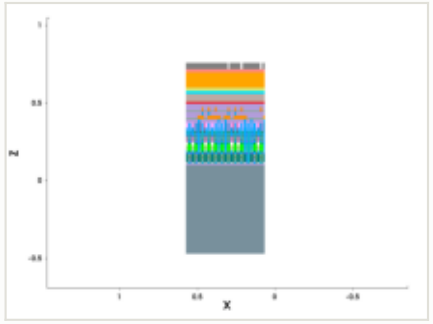
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

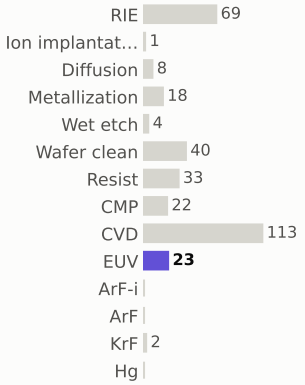


y cut at 0.049 µm

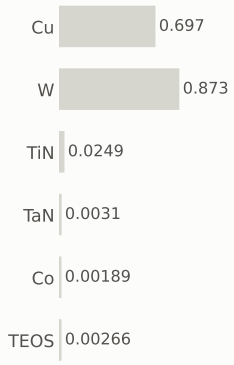


mask: layer 10271:0 (GDSII)

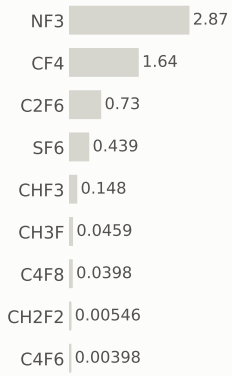
steps by process area



materials deposited (g)



gases emitted (g)



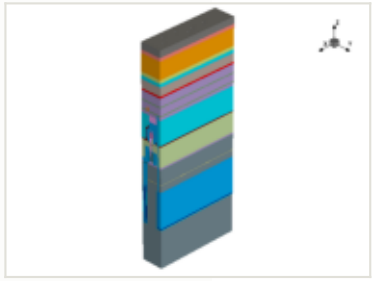
Step #3620: RIE

etch_anisotropic · RIE

1.04	364.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

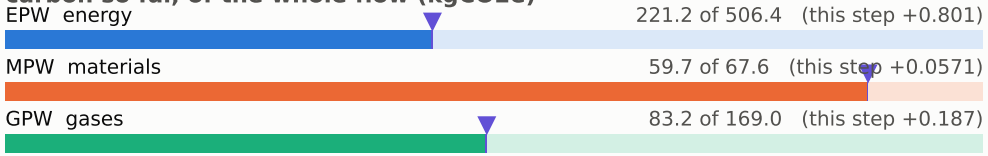


this step

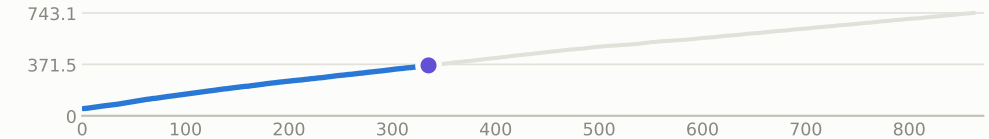
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

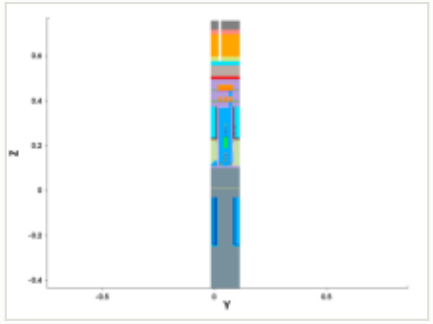
carbon so far, of the whole flow (kgCO2e)



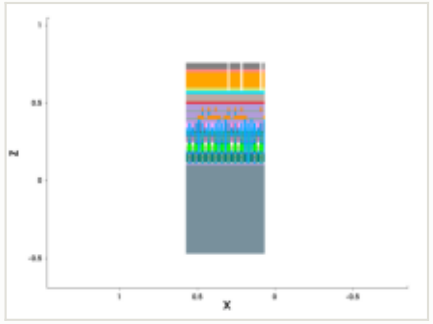
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



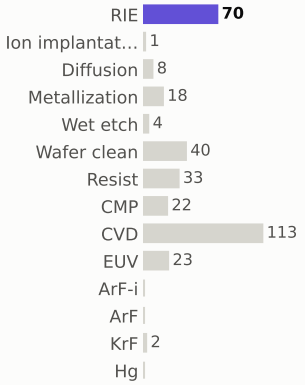
y cut at 0.049 µm



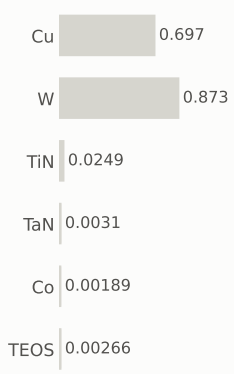
mask for this step

no mask for this step

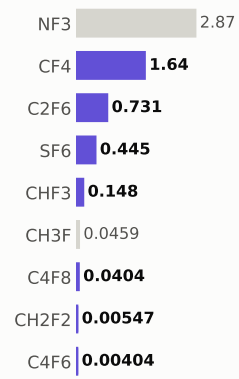
steps by process area



materials deposited (g)



gases emitted (g)



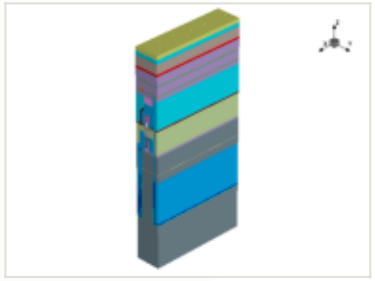
Step #3630: Wafer_clean

etch_strip · Wafer_clean

0.169	364.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

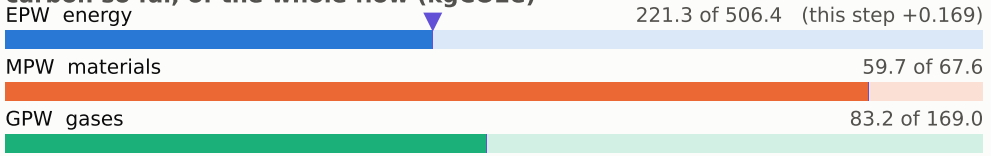


this step

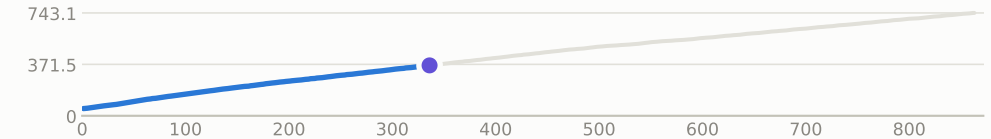
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

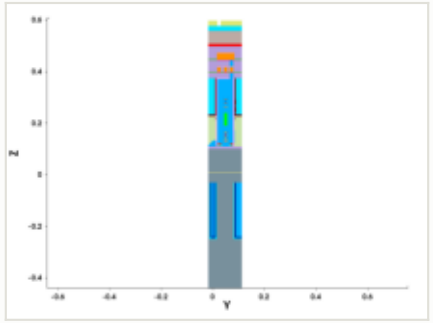
carbon so far, of the whole flow (kgCO2e)



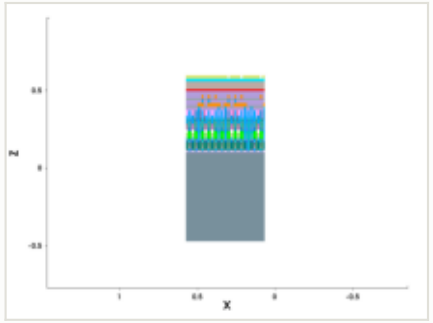
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



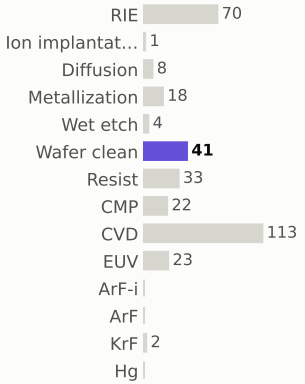
y cut at 0.049 µm



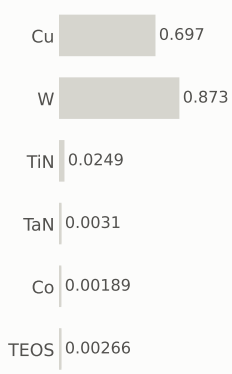
mask for this step

no mask for this step

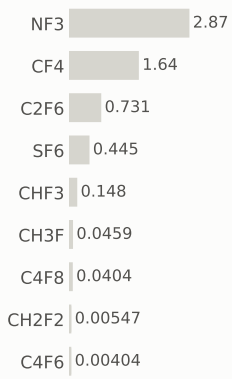
steps by process area



materials deposited (g)



gases emitted (g)



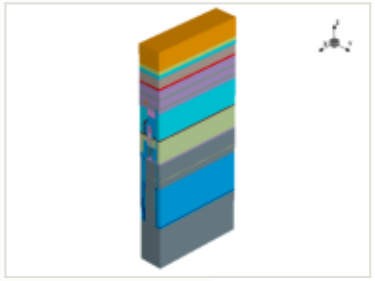
Step #3640: Resist

depo_fill · Resist

0.169	364.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

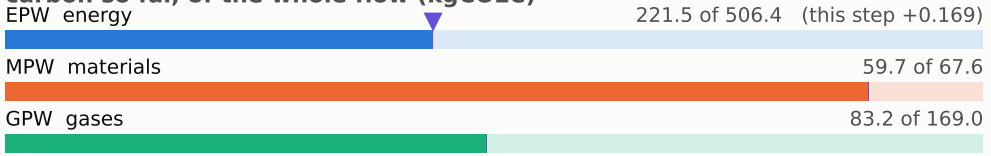


this step

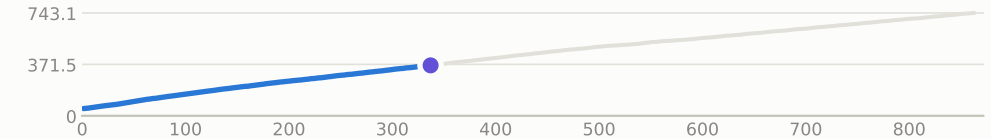
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

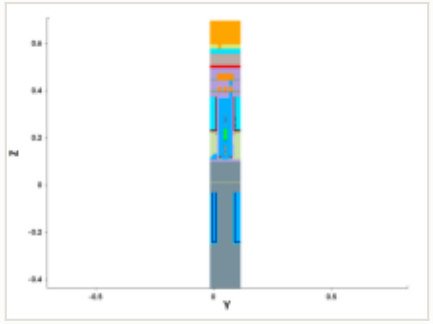
carbon so far, of the whole flow (kgCO2e)



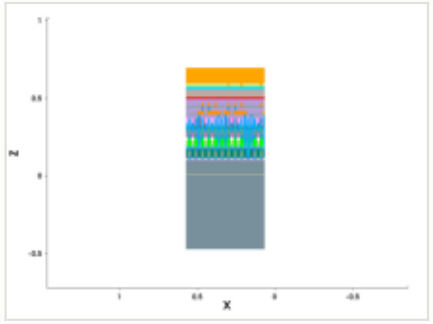
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



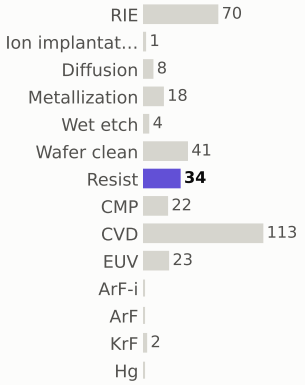
y cut at 0.049 µm



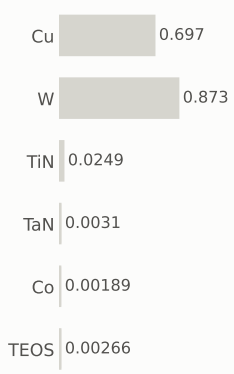
mask for this step

no mask for this step

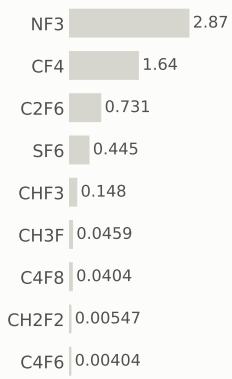
steps by process area



materials deposited (g)



gases emitted (g)



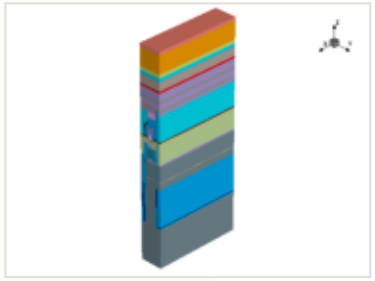
Step #3650: CVD

depo_fill · CVD

1.31	365.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

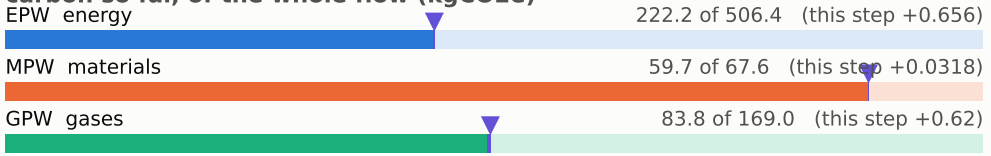


this step

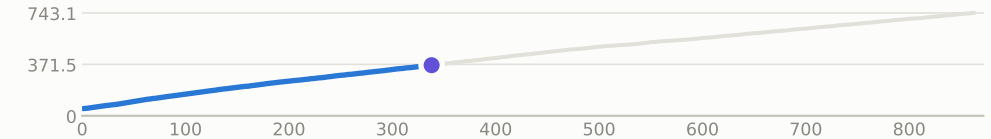
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

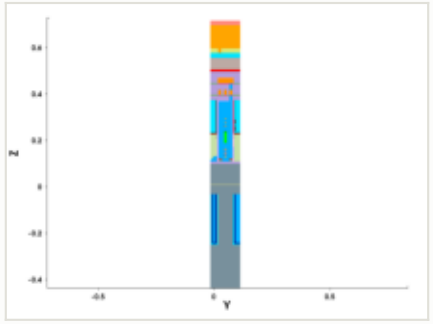
carbon so far, of the whole flow (kgCO2e)



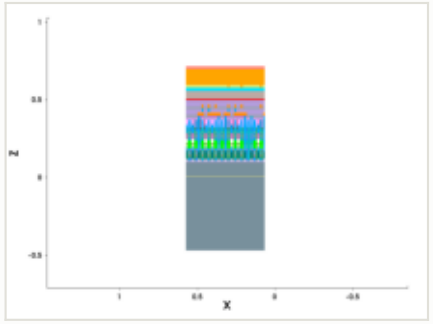
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



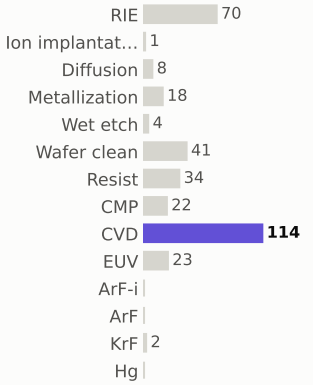
y cut at 0.049 µm



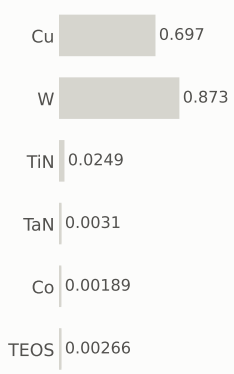
mask for this step

no mask for this step

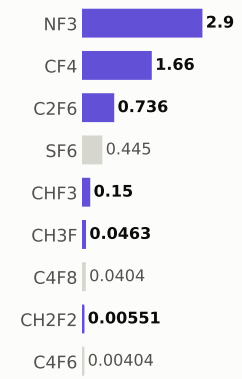
steps by process area



materials deposited (g)



gases emitted (g)



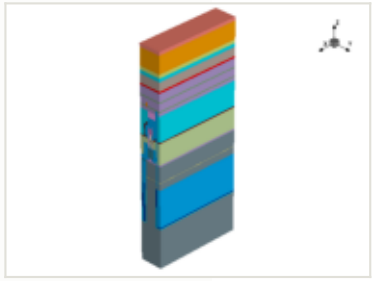
Step #3660: Wafer_clean

clean · Wafer_clean

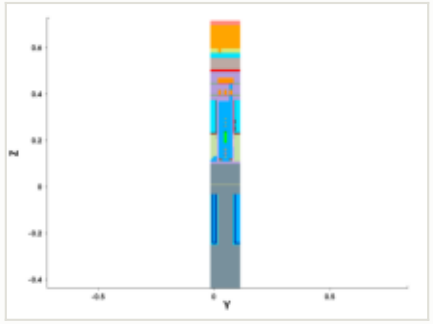
0.169	365.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

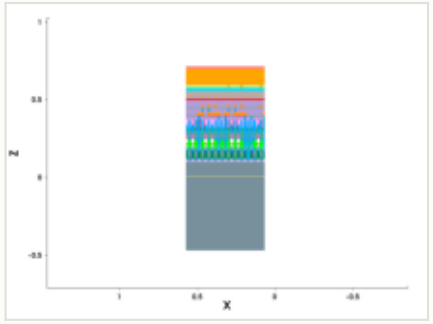
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

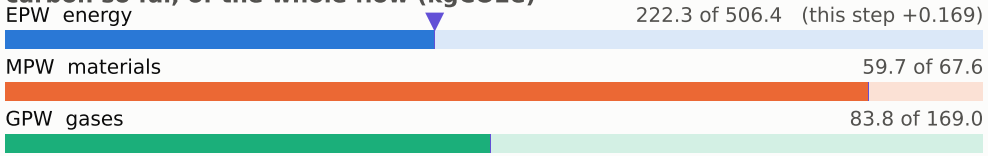
no mask for this step

this step

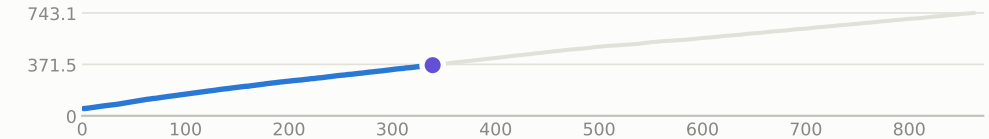
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

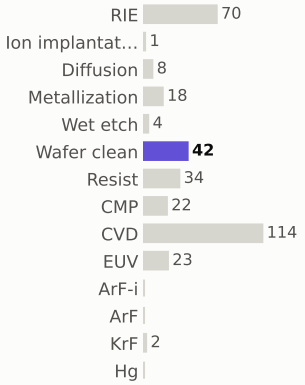
carbon so far, of the whole flow (kgCO2e)



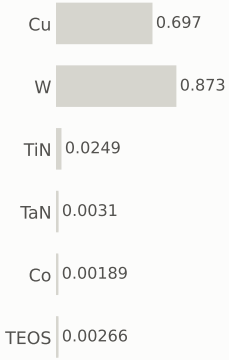
embodied carbon up to each step (kgCO2e per wafer)



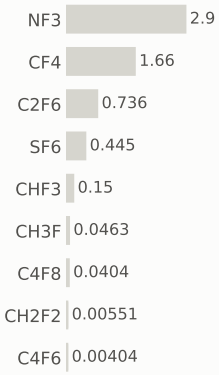
steps by process area



materials deposited (g)



gases emitted (g)



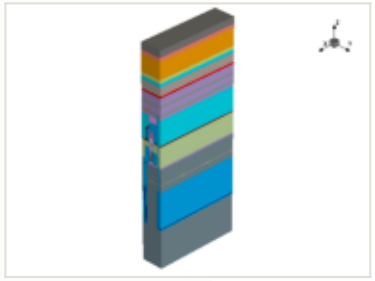
Step #3670: EUV

litho_patterning · EUV

1.96	367.8	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

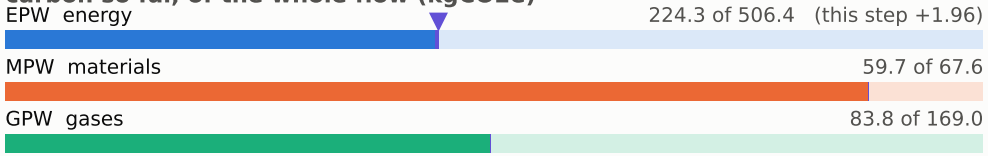


this step

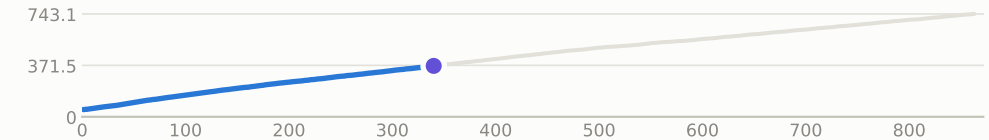
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

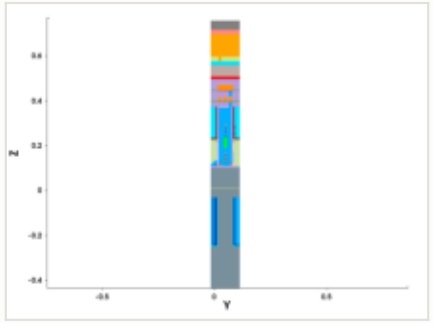
carbon so far, of the whole flow (kgCO2e)



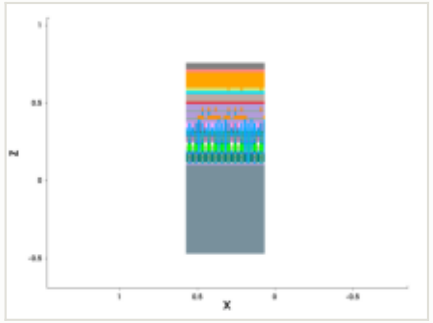
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



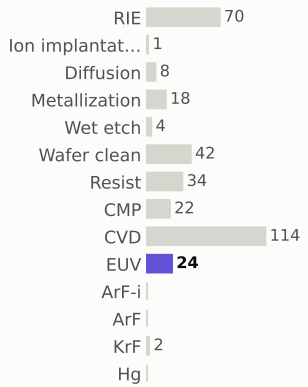
y cut at 0.049 µm



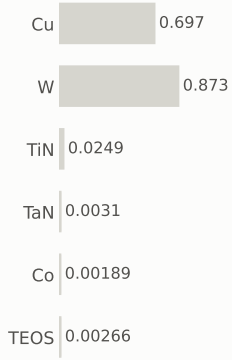
mask: layer 10272:0 (GDSII)

nothing in the clip window

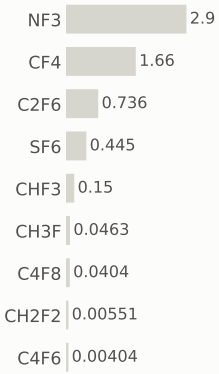
steps by process area



materials deposited (g)



gases emitted (g)



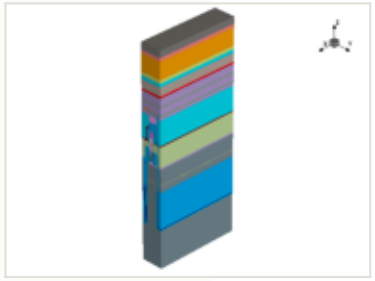
Step #3680: RIE

etch_anisotropic · RIE

1.04	368.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

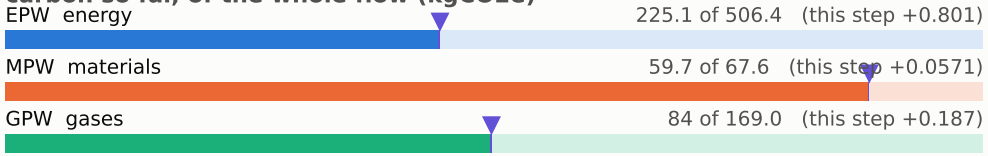


this step

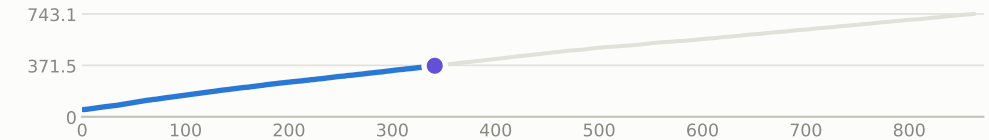
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

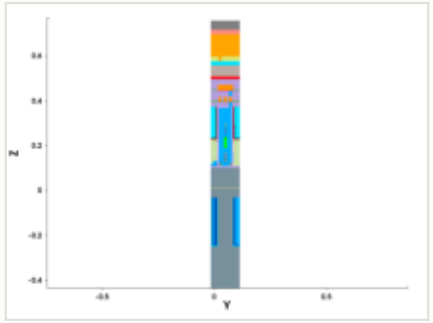
carbon so far, of the whole flow (kgCO2e)



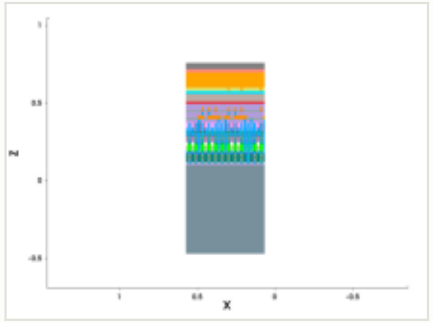
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



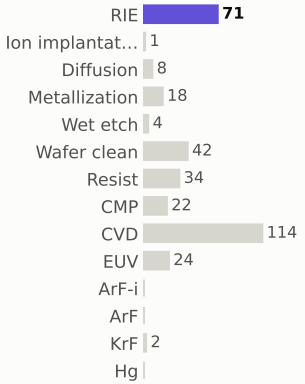
y cut at 0.049 µm



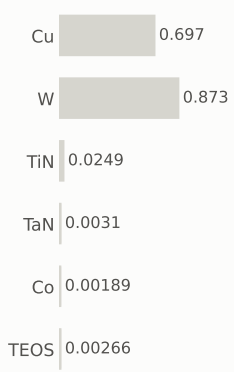
mask for this step

no mask for this step

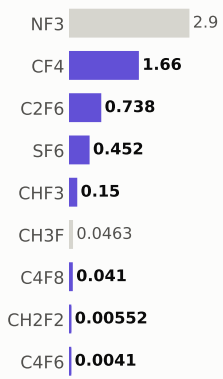
steps by process area



materials deposited (g)



gases emitted (g)



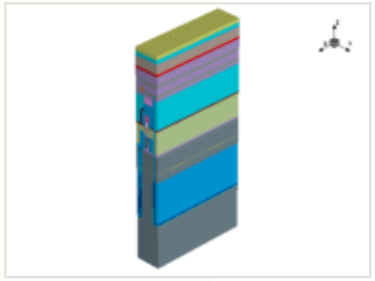
Step #3690: Wafer_clean

etch_strip · Wafer_clean

0.169	369.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

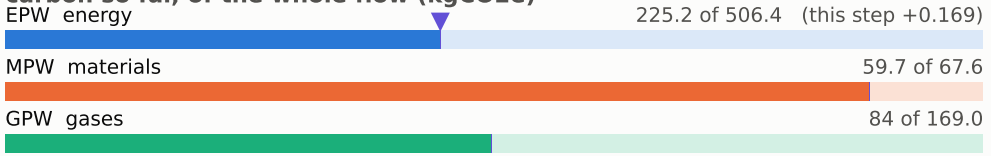


this step

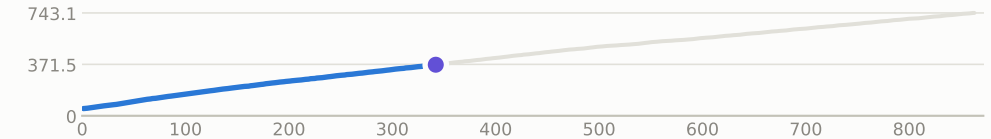
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

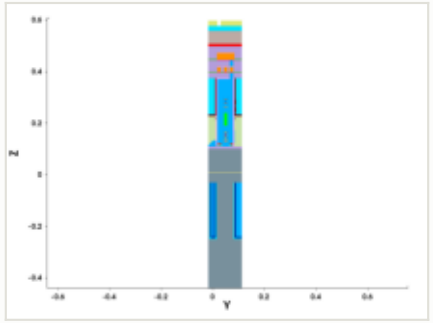
carbon so far, of the whole flow (kgCO2e)



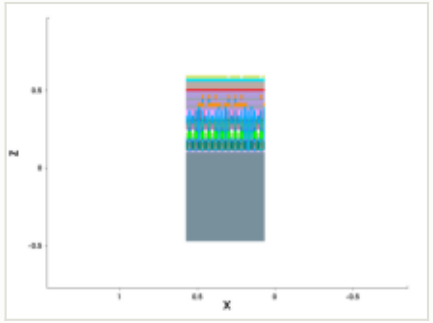
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



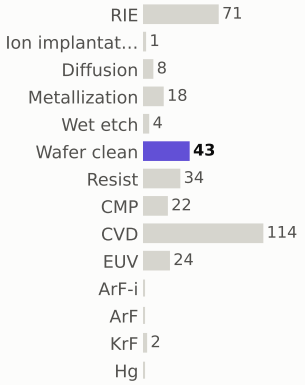
y cut at 0.049 µm



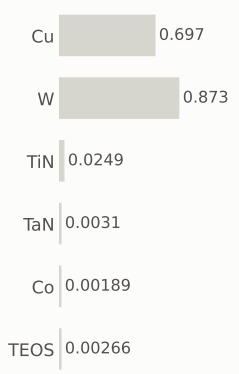
mask for this step

no mask for this step

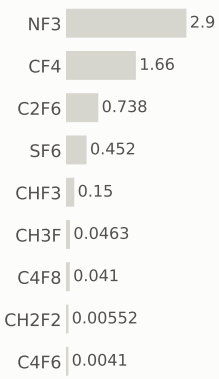
steps by process area



materials deposited (g)



gases emitted (g)



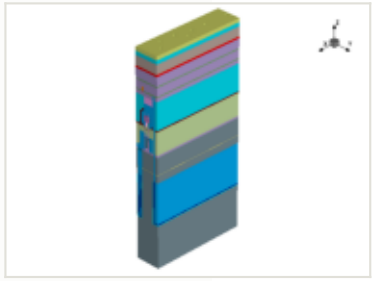
Step #3700: RIE

etch_anisotropic · RIE

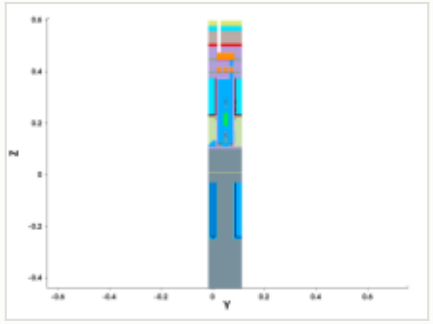
1.04	370.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

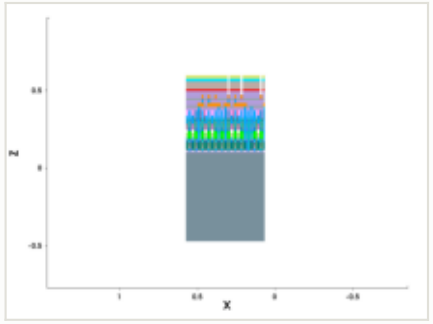
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

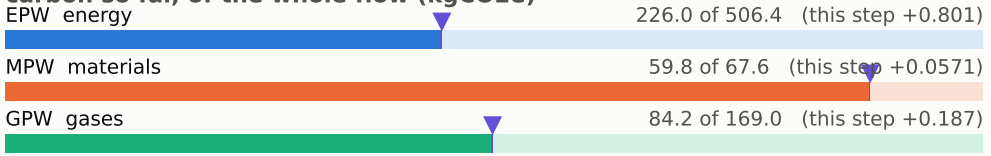
no mask for this step

this step

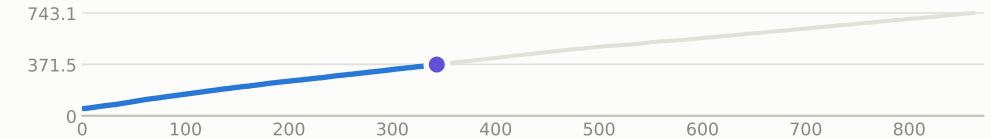
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

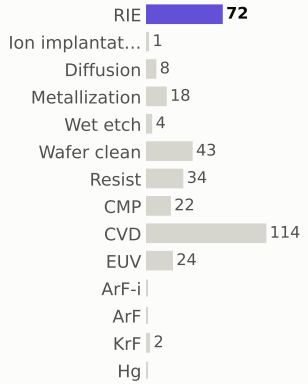
carbon so far, of the whole flow (kgCO2e)



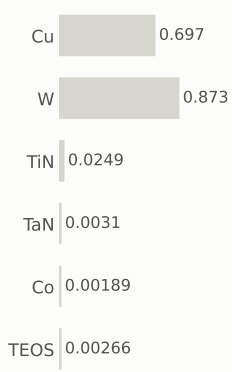
embodied carbon up to each step (kgCO2e per wafer)



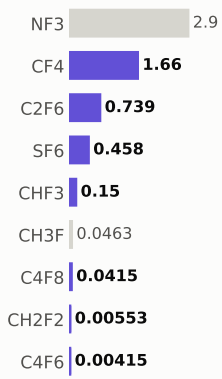
steps by process area



materials deposited (g)



gases emitted (g)



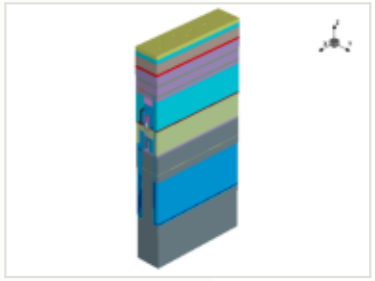
Step #3710: Wafer_clean

clean · Wafer_clean

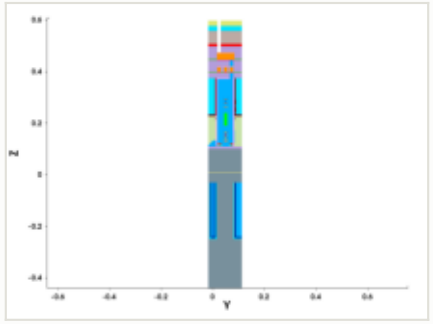
0.169	370.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

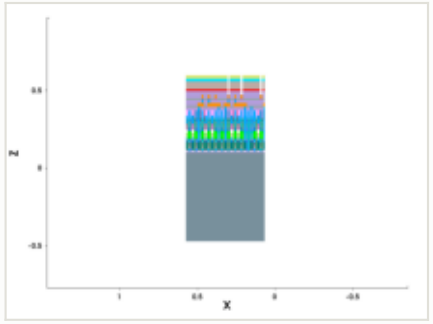
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

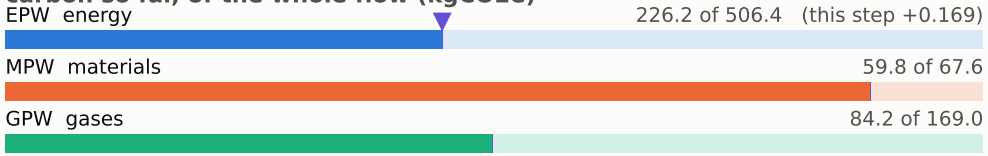
no mask for this step

this step

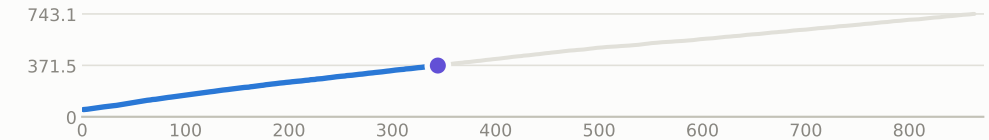
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

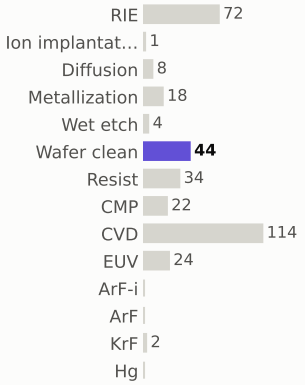
carbon so far, of the whole flow (kgCO2e)



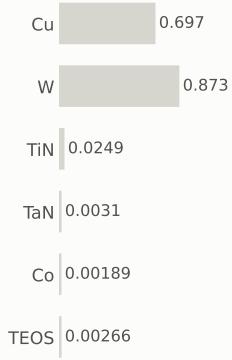
embodied carbon up to each step (kgCO2e per wafer)



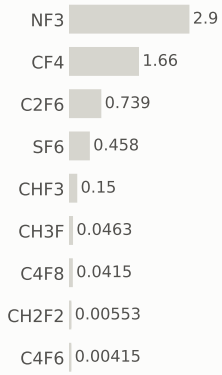
steps by process area



materials deposited (g)



gases emitted (g)



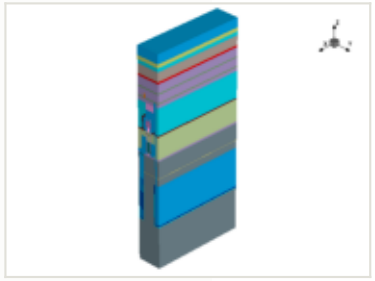
Step #3720: Metallization

depo_fill · Metallization

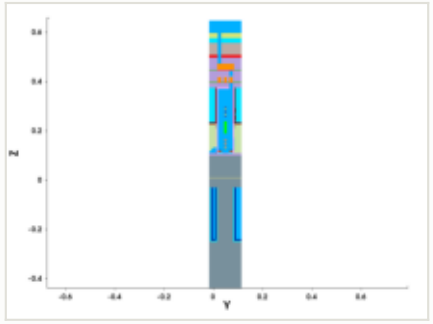
1.21	371.4	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

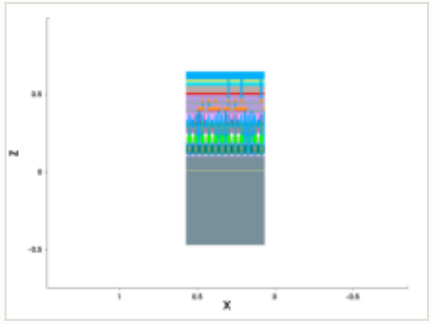
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

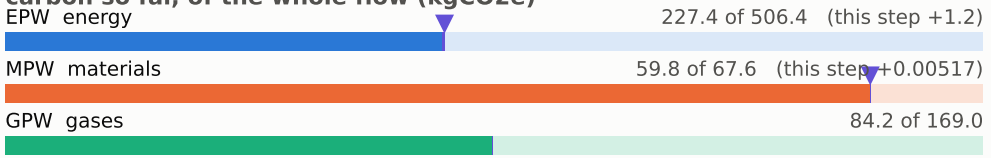
no mask for this step

this step

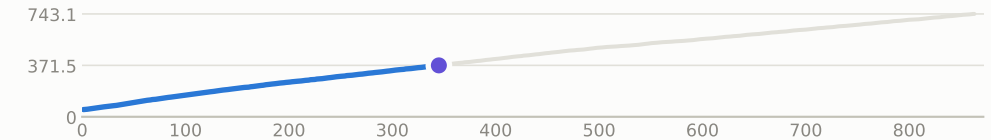
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

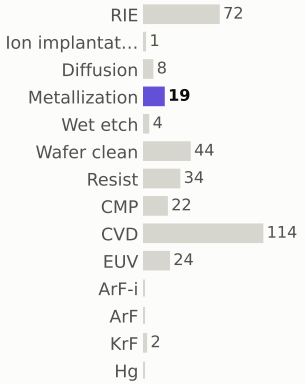
carbon so far, of the whole flow (kgCO2e)



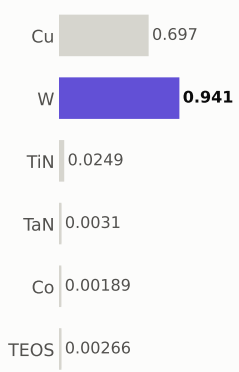
embodied carbon up to each step (kgCO2e per wafer)



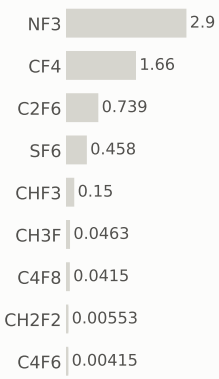
steps by process area



materials deposited (g)



gases emitted (g)



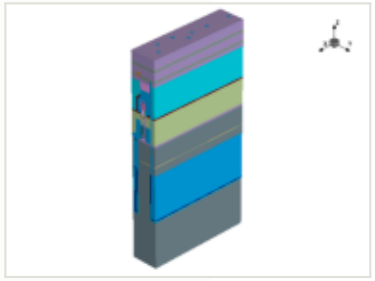
Step #3730: CMP

etch_cmp · CMP

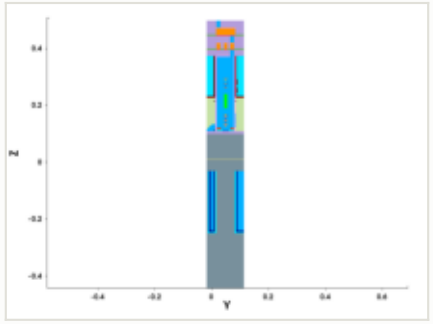
0.169	371.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

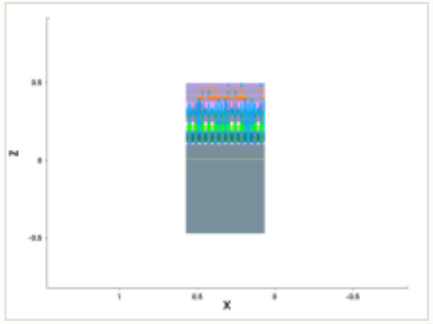
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

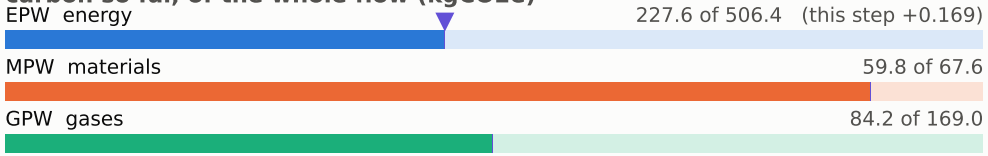
no mask for this step

this step

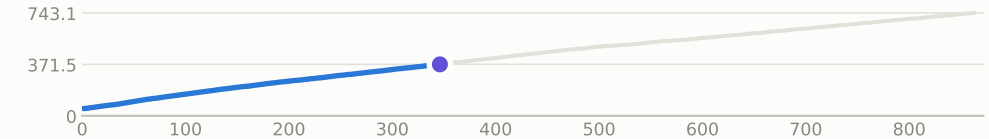
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

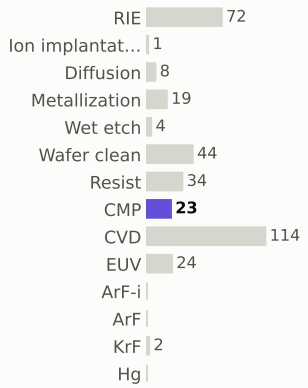
carbon so far, of the whole flow (kgCO2e)



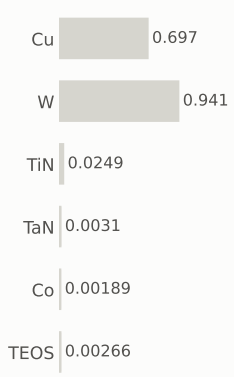
embodied carbon up to each step (kgCO2e per wafer)



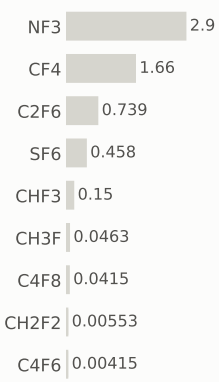
steps by process area



materials deposited (g)



gases emitted (g)



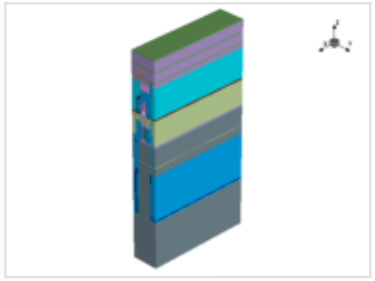
Step #3740: CVD

depo_fill · CVD

1.31	372.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

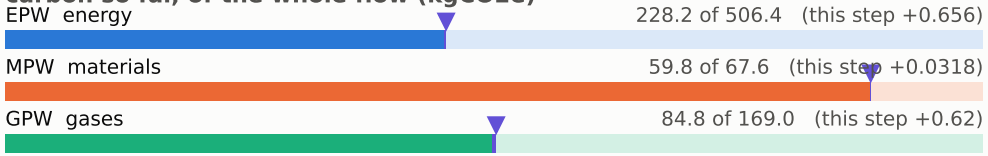


this step

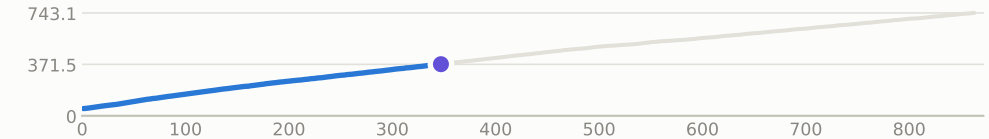
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

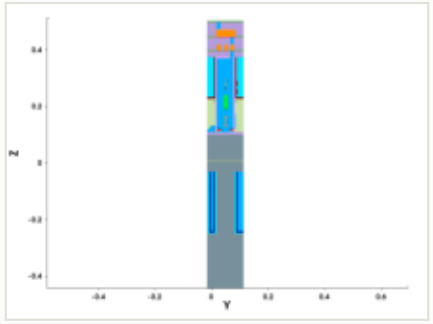
carbon so far, of the whole flow (kgCO2e)



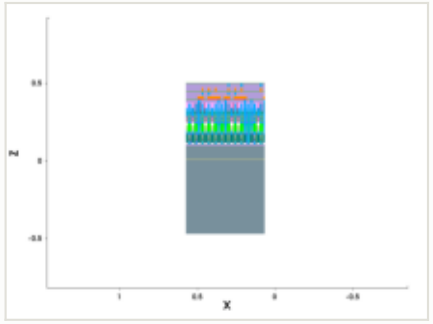
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



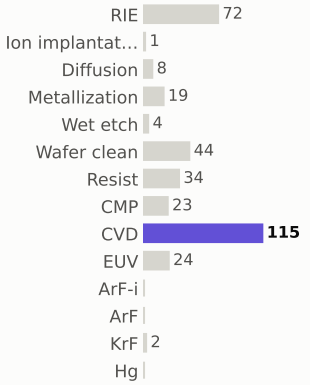
y cut at 0.049 µm



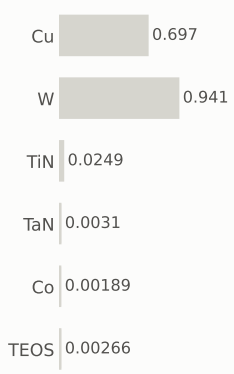
mask for this step

no mask for this step

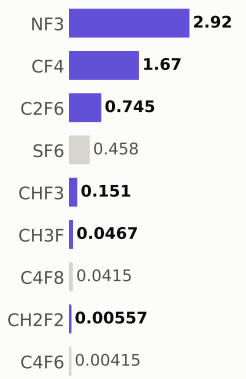
steps by process area



materials deposited (g)



gases emitted (g)



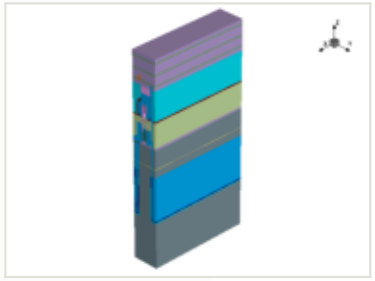
Step #3750: CVD

depo_fill · CVD

1.31	374.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

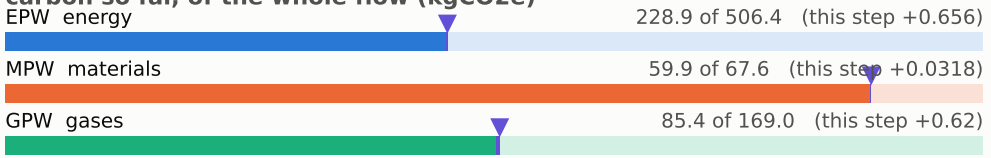


this step

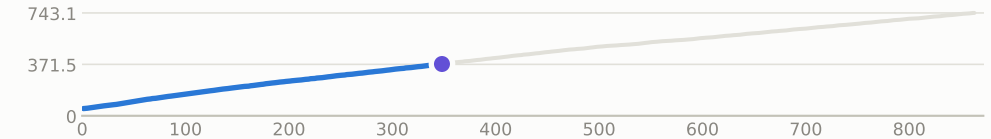
process area	CVD
material	—
thickness	0.018 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

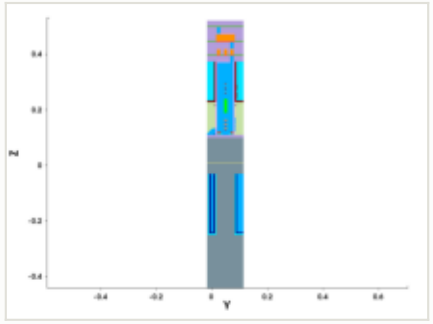
carbon so far, of the whole flow (kgCO2e)



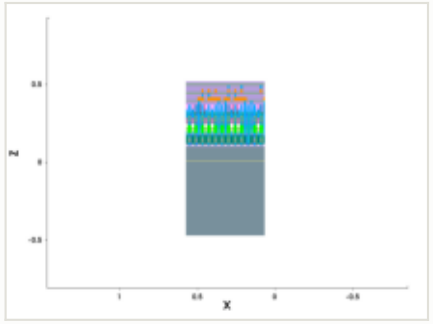
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



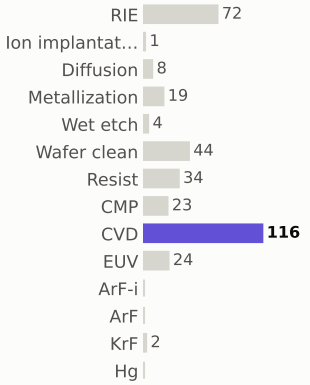
y cut at 0.049 µm



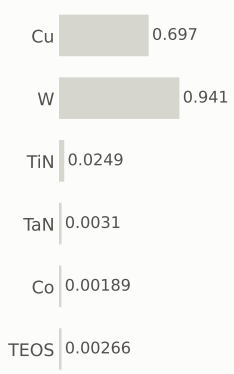
mask for this step

no mask for this step

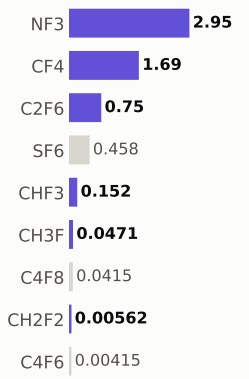
steps by process area



materials deposited (g)



gases emitted (g)



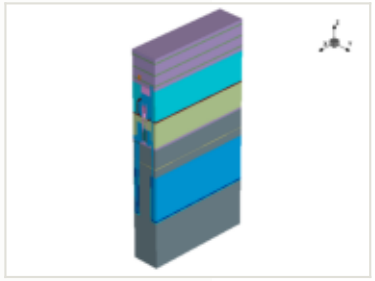
Step #3760: Diffusion

treatment · Diffusion

0.801	375.0	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

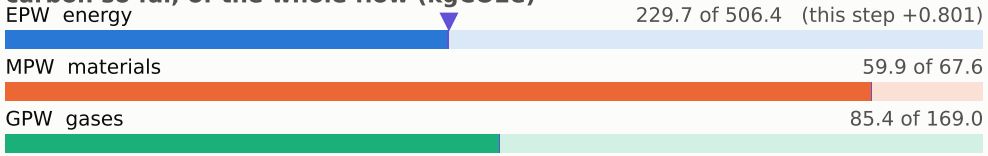


this step

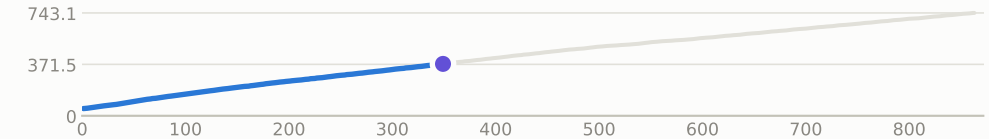
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

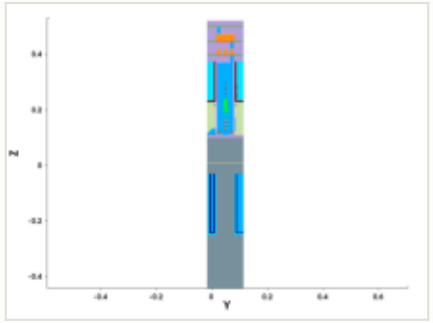
carbon so far, of the whole flow (kgCO2e)



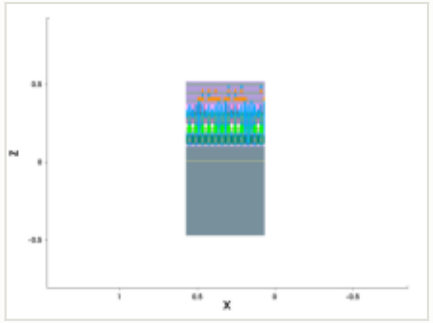
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



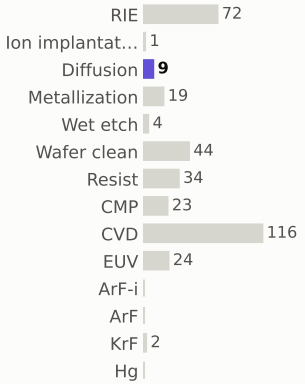
y cut at 0.049 µm



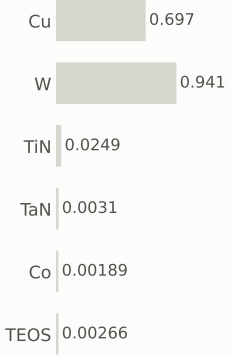
mask for this step

no mask for this step

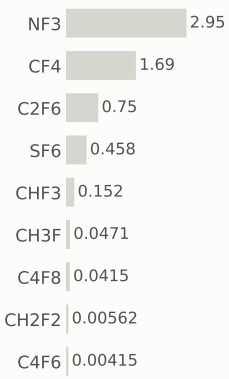
steps by process area



materials deposited (g)



gases emitted (g)



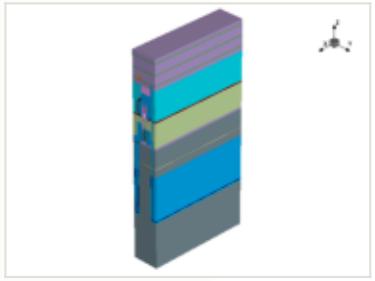
Step #3770: Wafer_clean

clean · Wafer_clean

0.169	375.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



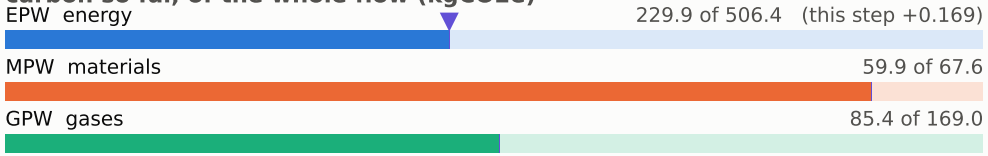
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

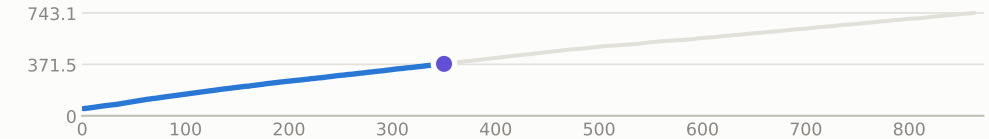
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

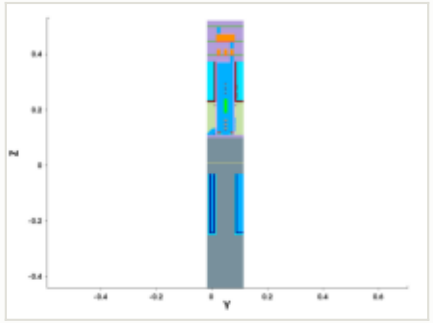
carbon so far, of the whole flow (kgCO2e)



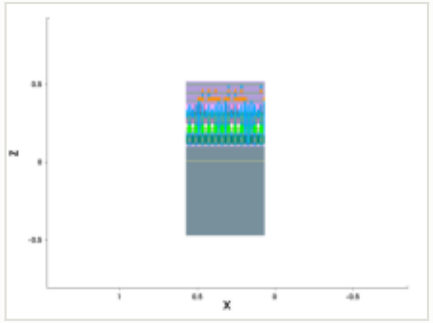
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



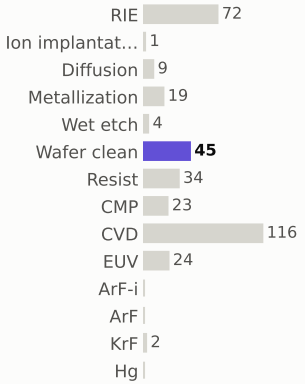
y cut at 0.049 µm



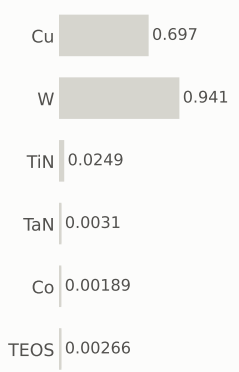
mask for this step

no mask for this step

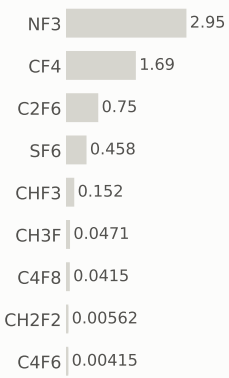
steps by process area



materials deposited (g)



gases emitted (g)



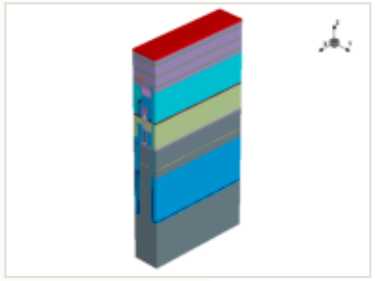
Step #3780: CVD

depo_fill · CVD

1.31	376.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

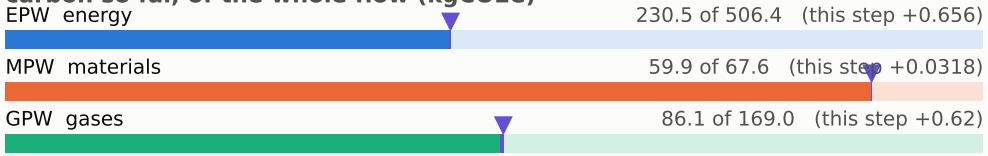


this step

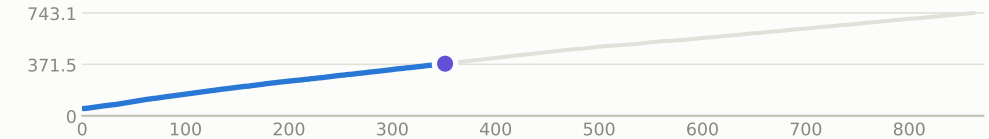
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

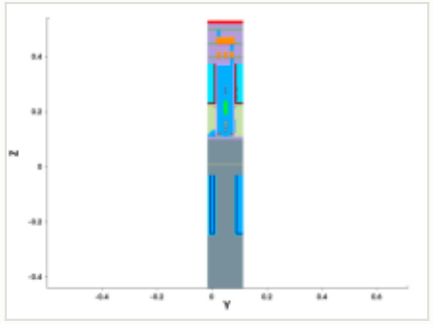
carbon so far, of the whole flow (kgCO2e)



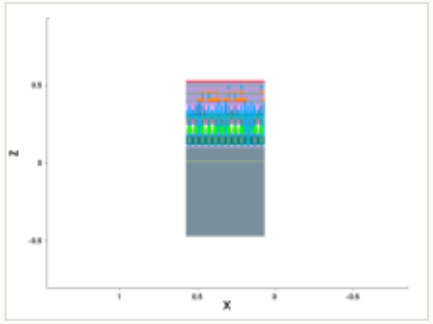
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



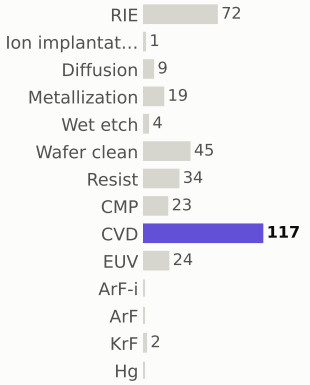
y cut at 0.049 µm



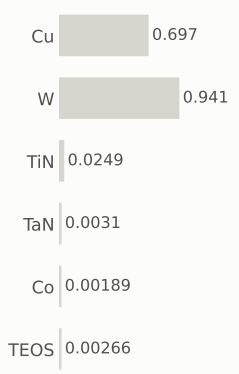
mask for this step

no mask for this step

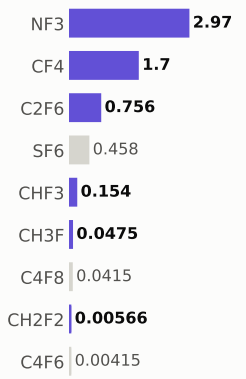
steps by process area



materials deposited (g)



gases emitted (g)



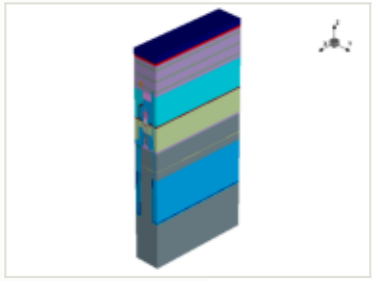
Step #3790: CVD

depo_fill · CVD

1.31	377.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

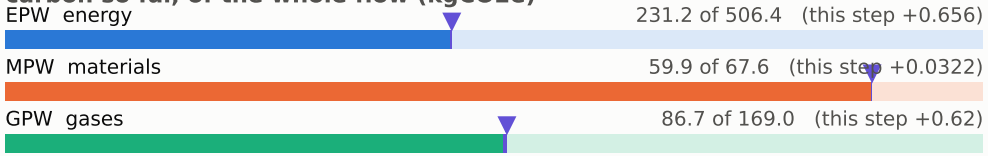


this step

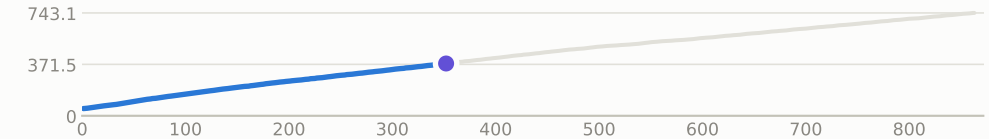
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

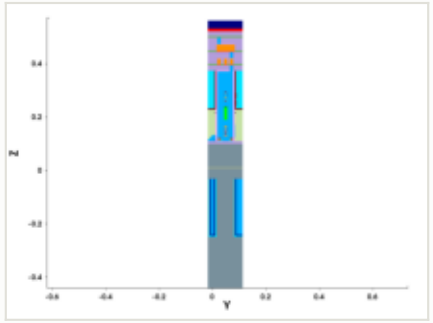
carbon so far, of the whole flow (kgCO2e)



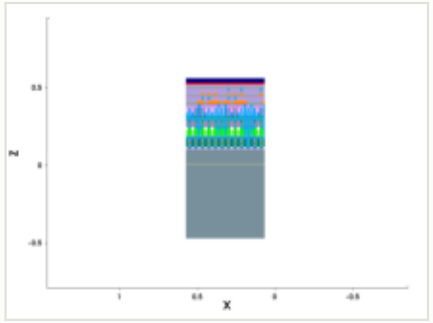
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



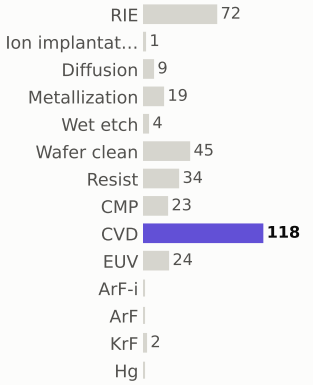
y cut at 0.049 µm



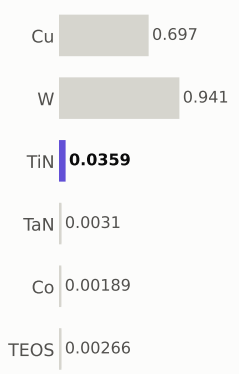
mask for this step

no mask for this step

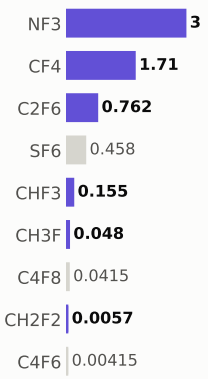
steps by process area



materials deposited (g)



gases emitted (g)



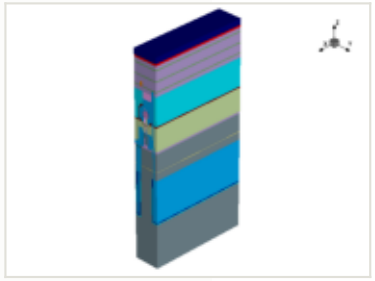
Step #3800: Wafer_clean

clean · Wafer_clean

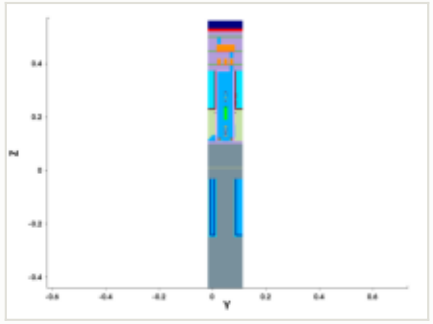
0.169	378.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

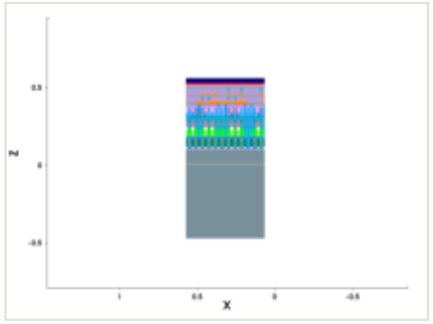
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

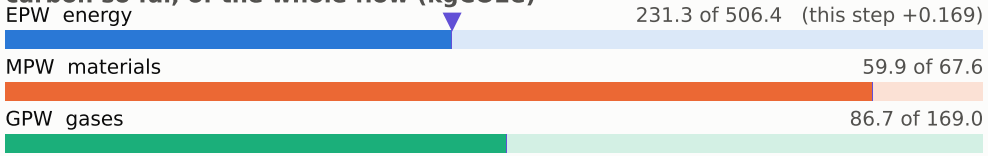
no mask for this step

this step

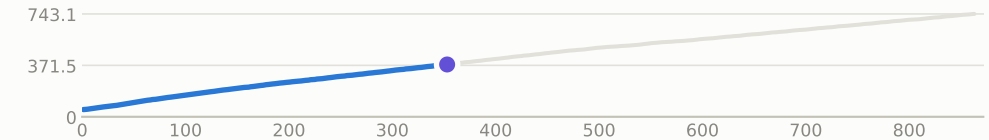
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

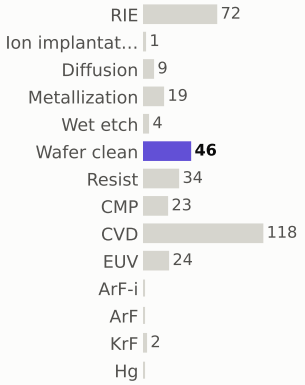
carbon so far, of the whole flow (kgCO2e)



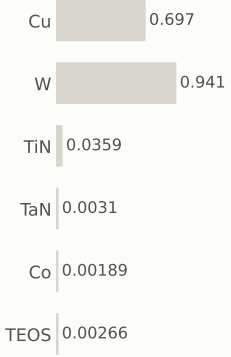
embodied carbon up to each step (kgCO2e per wafer)



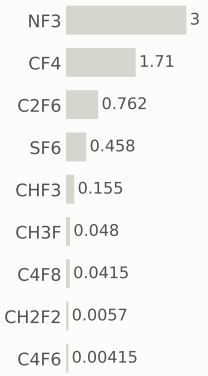
steps by process area



materials deposited (g)



gases emitted (g)



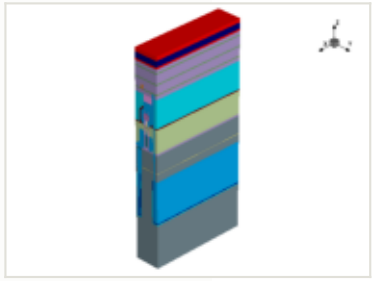
Step #3810: CVD

depo_fill · CVD

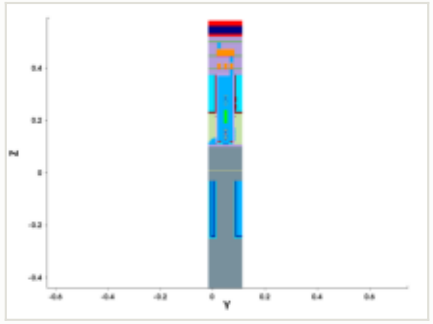
1.31	379.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

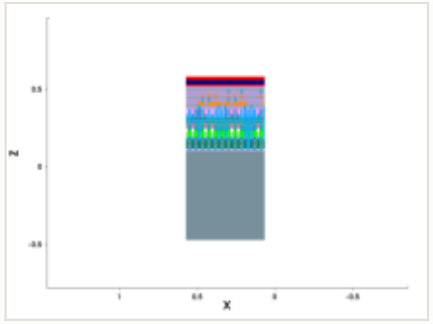
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

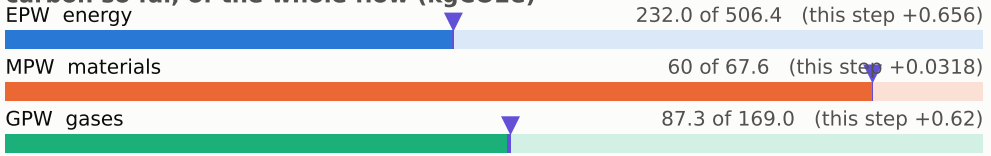
no mask for this step

this step

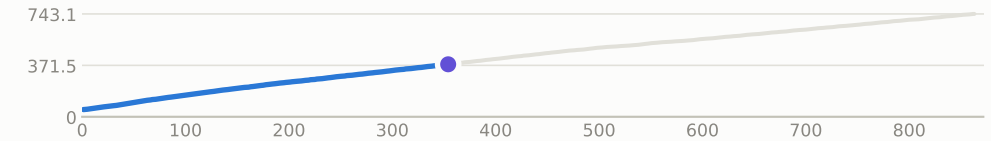
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

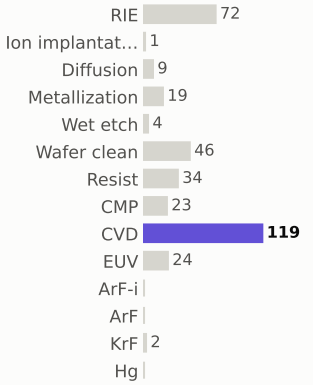
carbon so far, of the whole flow (kgCO2e)



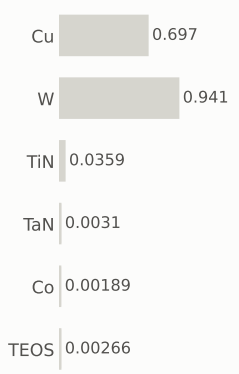
embodied carbon up to each step (kgCO2e per wafer)



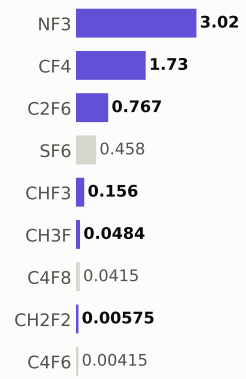
steps by process area



materials deposited (g)



gases emitted (g)



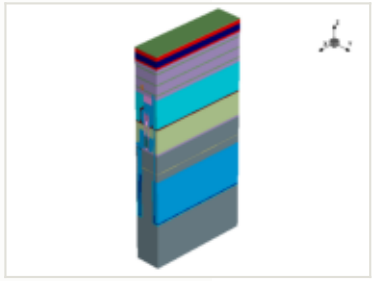
Step #3820: CVD

depo_fill · CVD

1.31	380.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

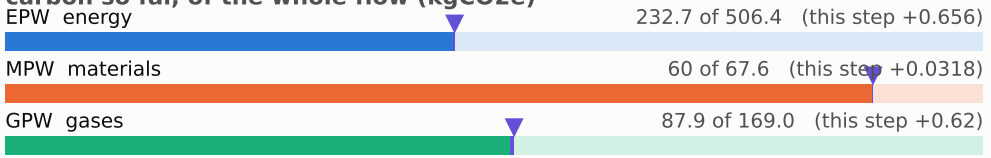


this step

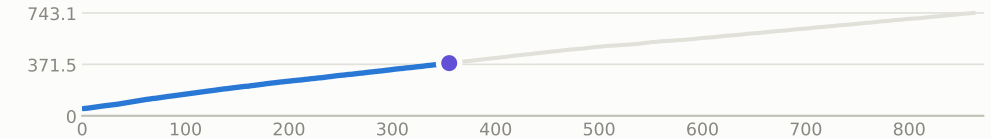
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

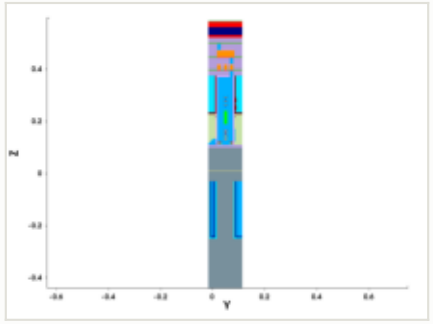
carbon so far, of the whole flow (kgCO2e)



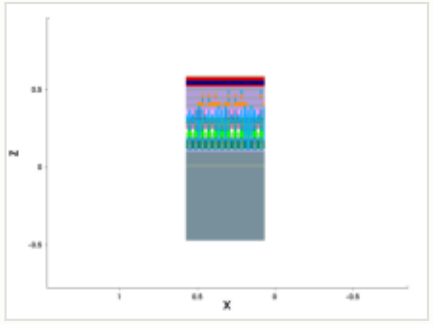
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



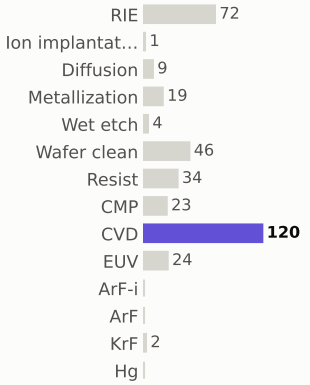
y cut at 0.049 µm



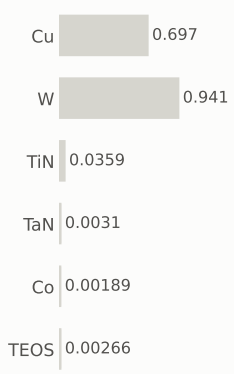
mask for this step

no mask for this step

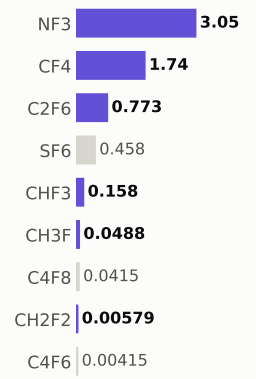
steps by process area



materials deposited (g)



gases emitted (g)



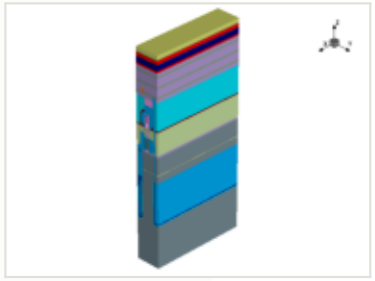
Step #3830: CVD

depo_fill · CVD

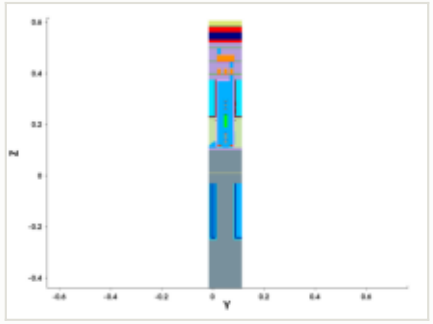
1.31	381.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

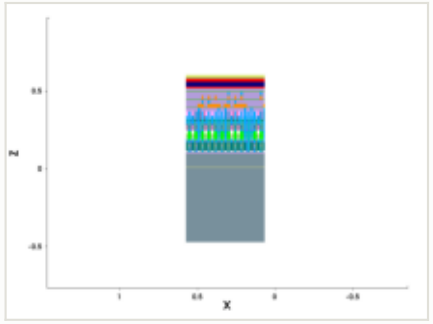
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

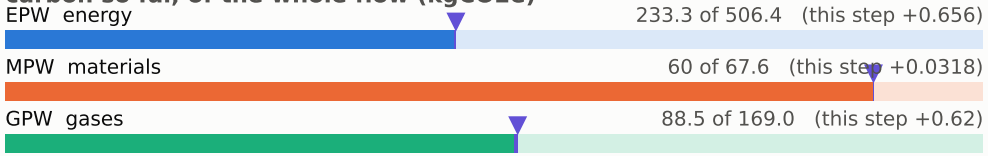
no mask for this step

this step

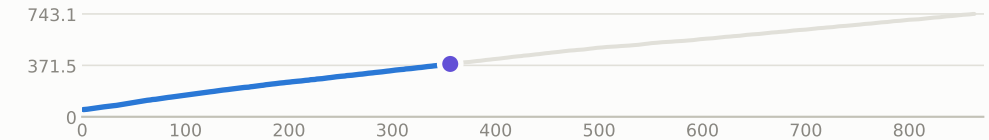
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

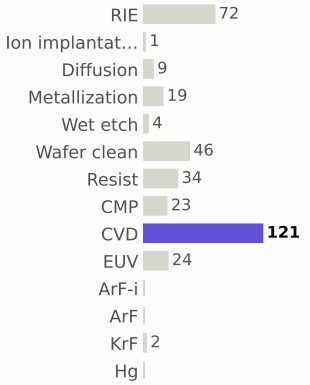
carbon so far, of the whole flow (kgCO2e)



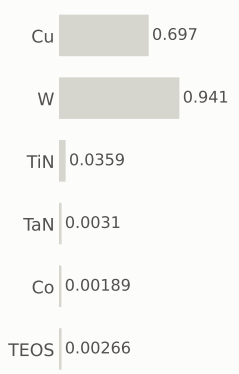
embodied carbon up to each step (kgCO2e per wafer)



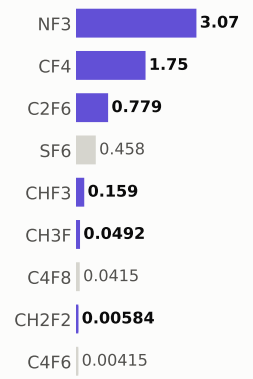
steps by process area



materials deposited (g)



gases emitted (g)



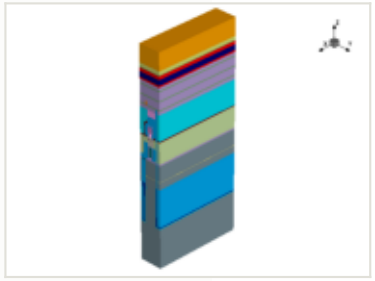
Step #3840: Resist

depo_fill · Resist

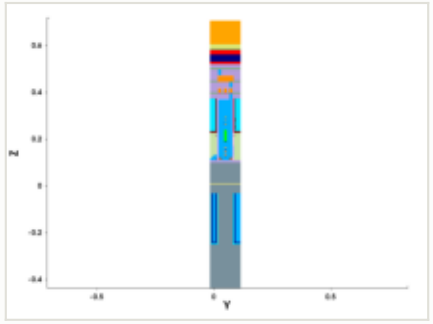
0.169	382.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

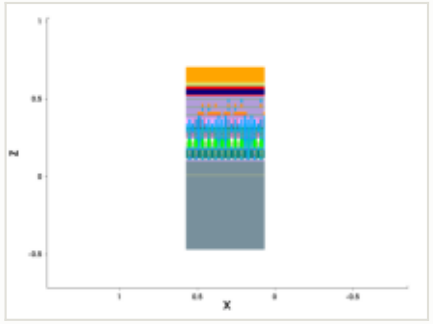
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



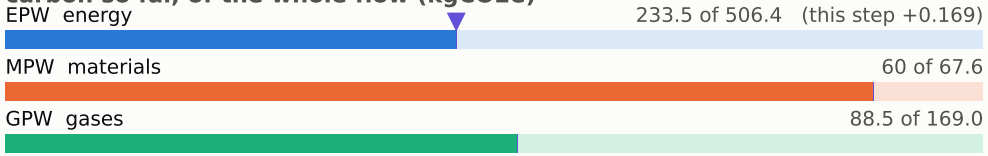
mask for this step

no mask for this step

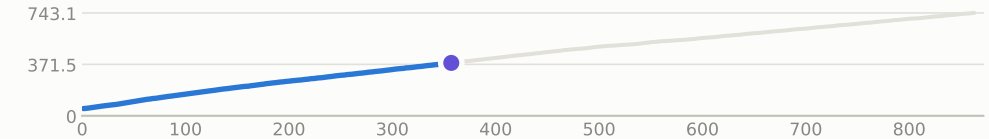
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

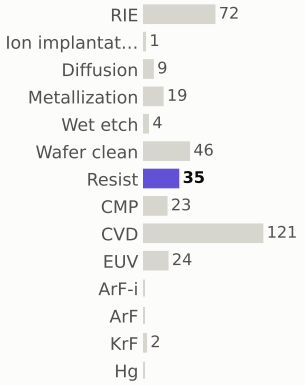
carbon so far, of the whole flow (kgCO2e)



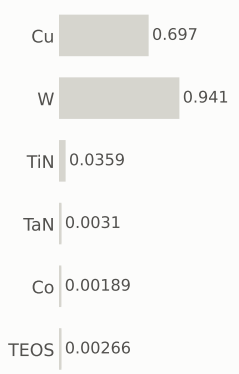
embodied carbon up to each step (kgCO2e per wafer)



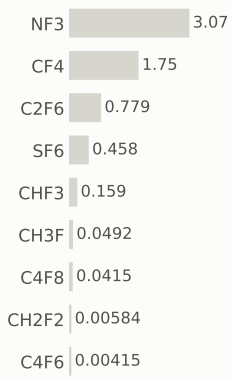
steps by process area



materials deposited (g)



gases emitted (g)



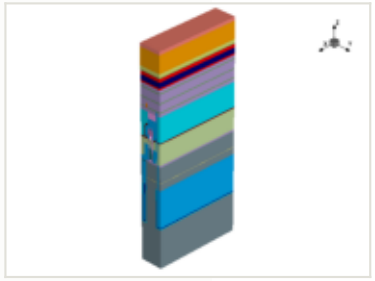
Step #3850: CVD

depo_fill · CVD

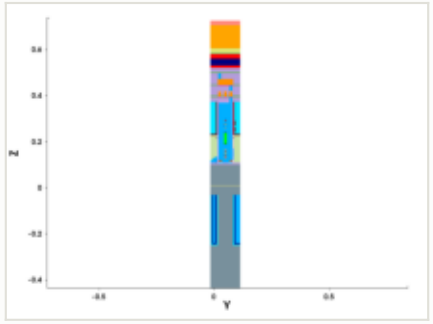
1.31	383.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

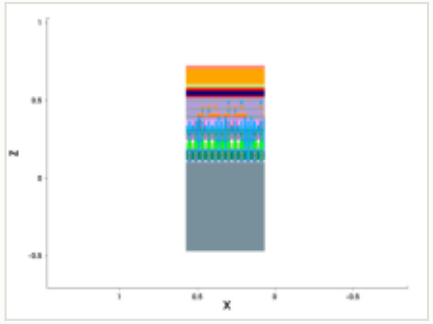
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

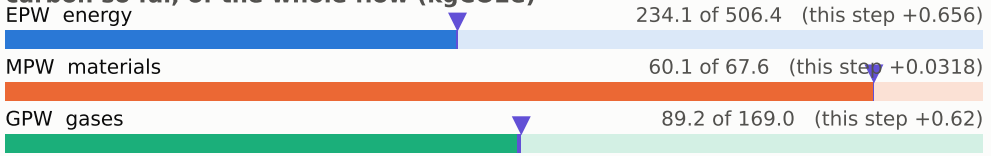
no mask for this step

this step

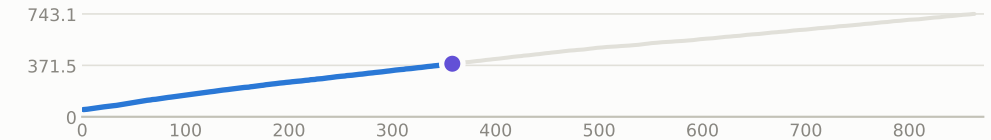
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

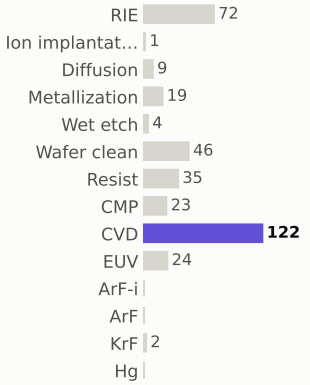
carbon so far, of the whole flow (kgCO2e)



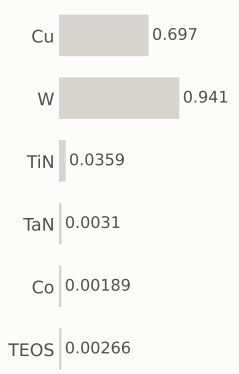
embodied carbon up to each step (kgCO2e per wafer)



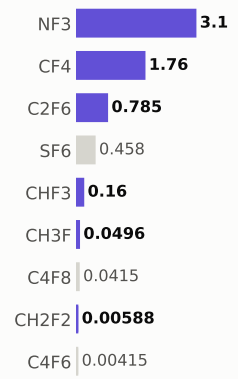
steps by process area



materials deposited (g)



gases emitted (g)



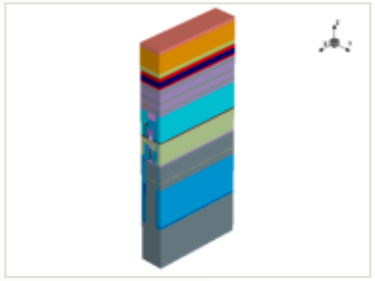
Step #3860: Wafer_clean

clean · Wafer_clean

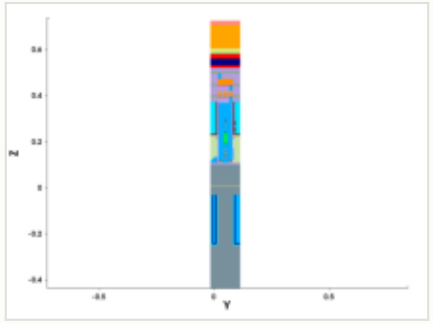
0.169	383.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

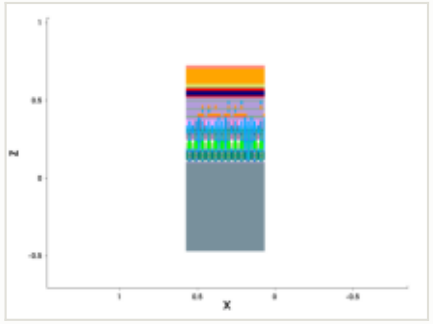
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

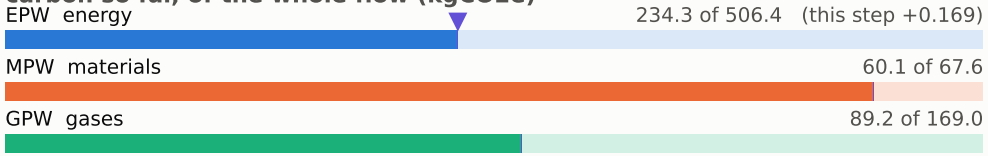
no mask for this step

this step

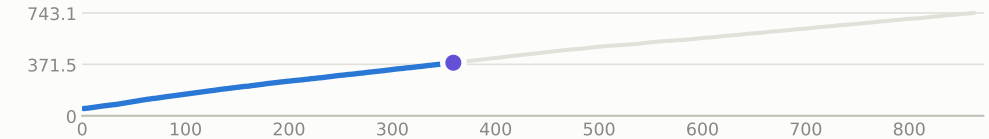
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

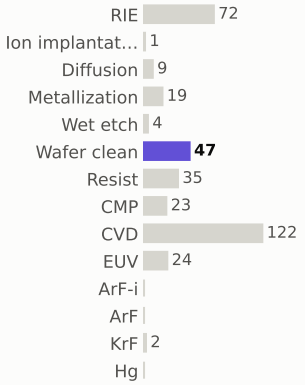
carbon so far, of the whole flow (kgCO2e)



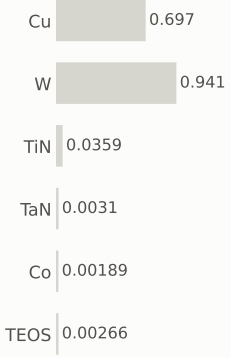
embodied carbon up to each step (kgCO2e per wafer)



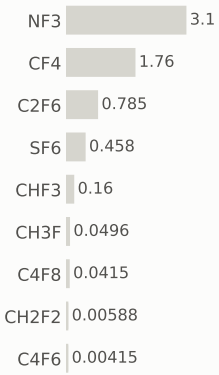
steps by process area



materials deposited (g)



gases emitted (g)



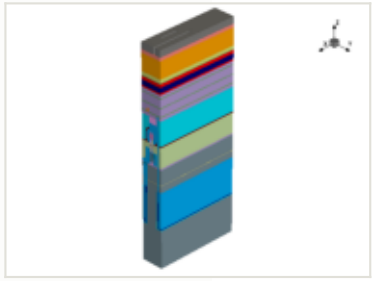
Step #3870: EUV

litho_patterning · EUV

1.96	385.5	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

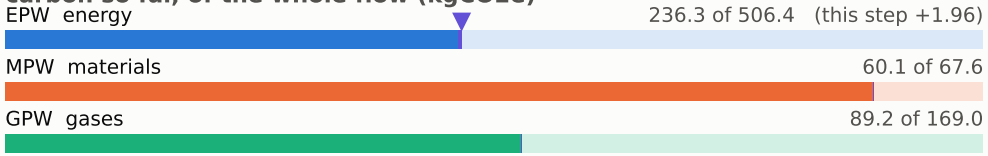


this step

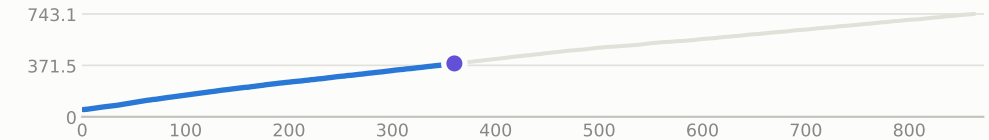
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

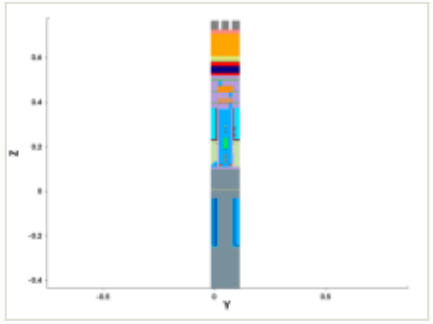
carbon so far, of the whole flow (kgCO2e)



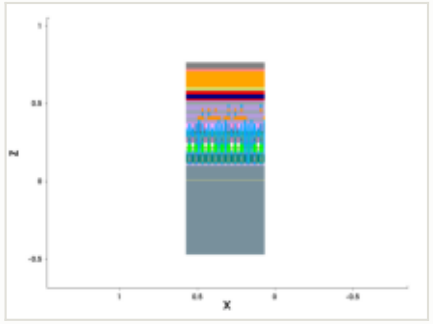
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

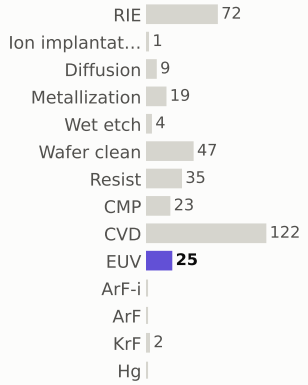


y cut at 0.049 µm

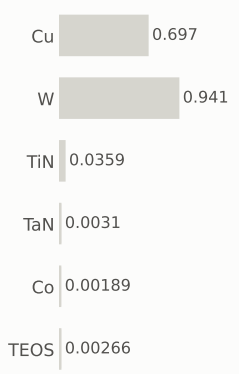


mask: layer 10301:0 (GDSII)

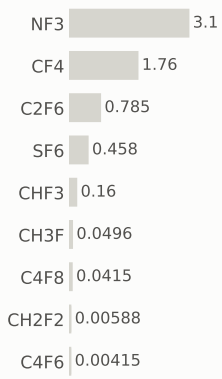
steps by process area



materials deposited (g)



gases emitted (g)



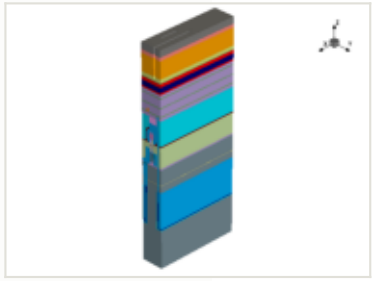
Step #3880: RIE

etch_anisotropic · RIE

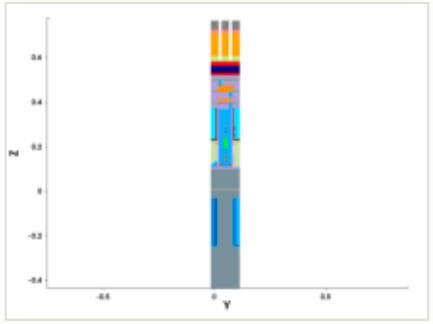
1.04	386.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

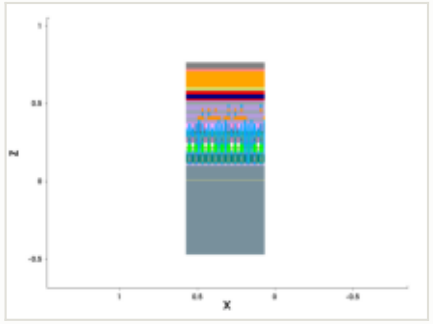
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

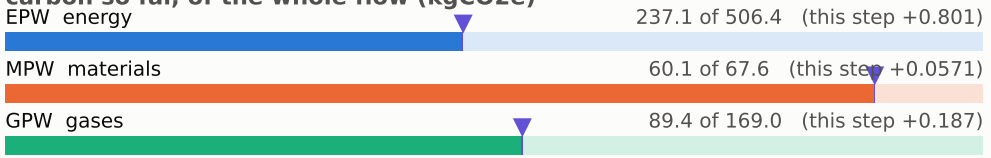
no mask for this step

this step

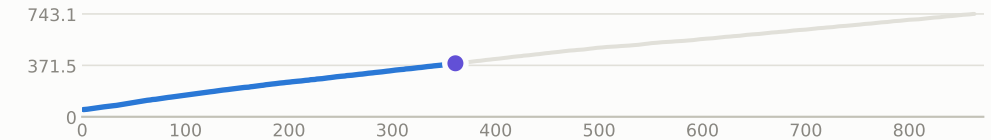
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

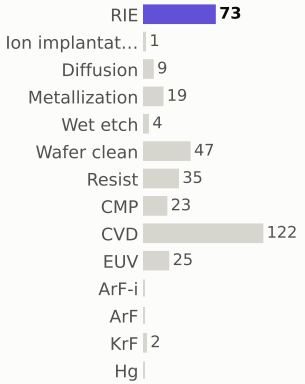
carbon so far, of the whole flow (kgCO2e)



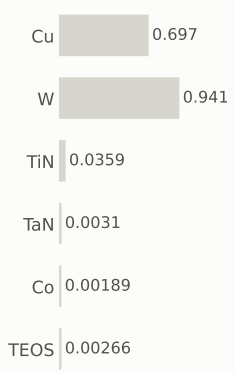
embodied carbon up to each step (kgCO2e per wafer)



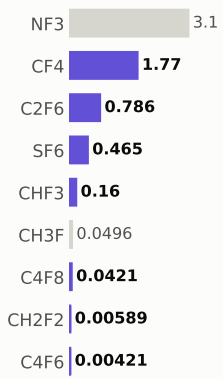
steps by process area



materials deposited (g)



gases emitted (g)



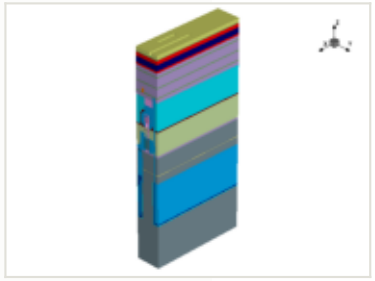
Step #3890: Wafer_clean

etch_strip · Wafer_clean

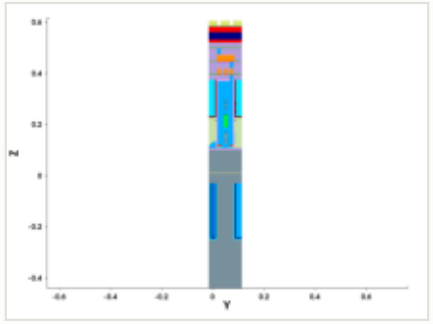
0.169	386.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

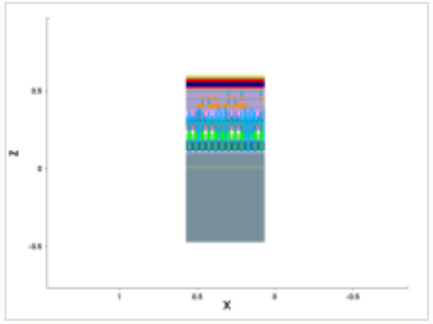
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



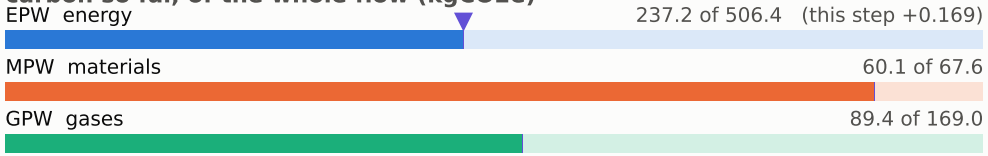
mask for this step

no mask for this step

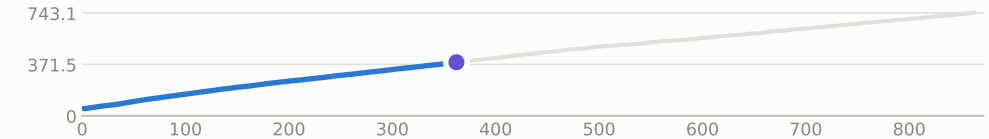
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

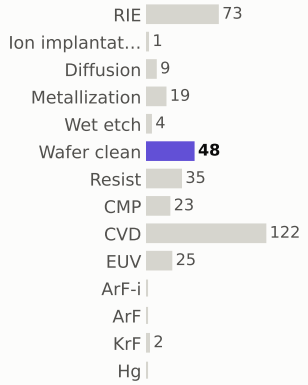
carbon so far, of the whole flow (kgCO2e)



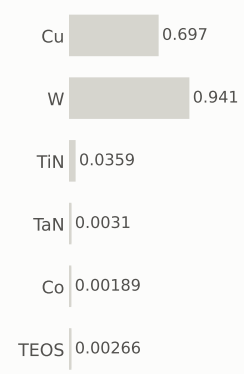
embodied carbon up to each step (kgCO2e per wafer)



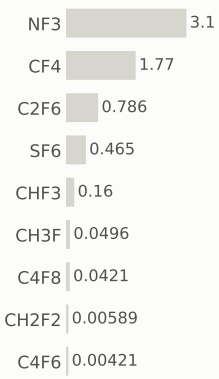
steps by process area



materials deposited (g)



gases emitted (g)



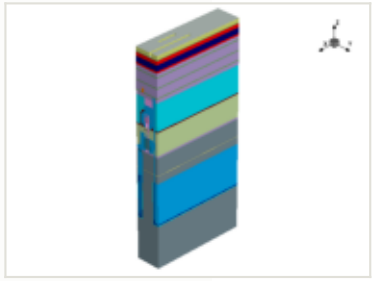
Step #3900: CVD

depo_isotropic · CVD

1.31	388.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

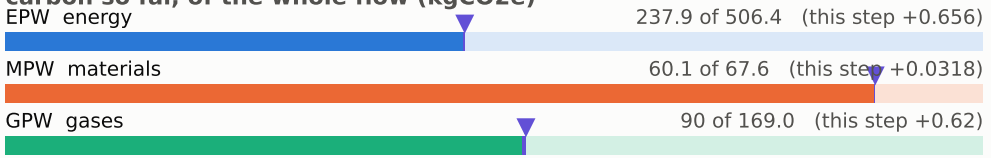


this step

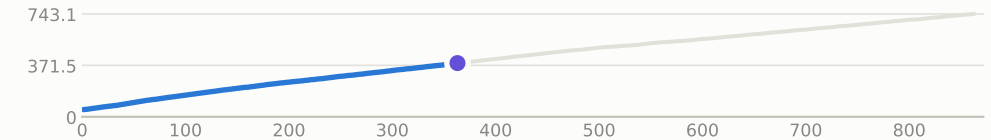
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

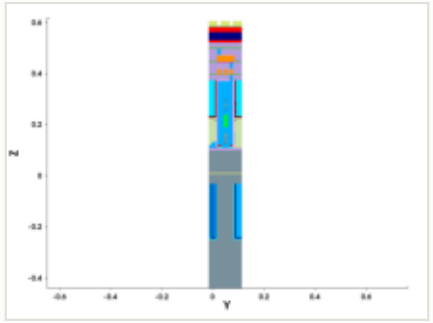
carbon so far, of the whole flow (kgCO2e)



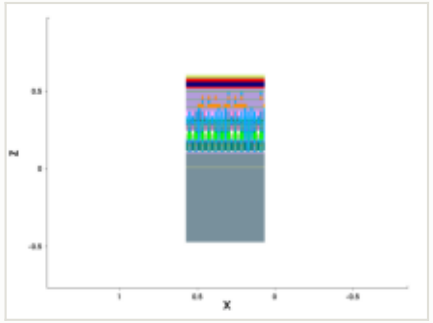
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



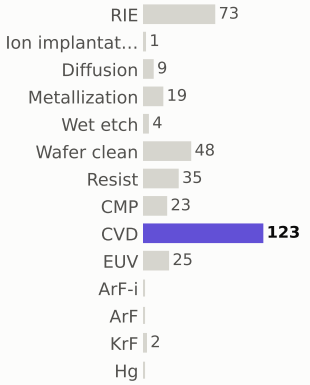
y cut at 0.049 µm



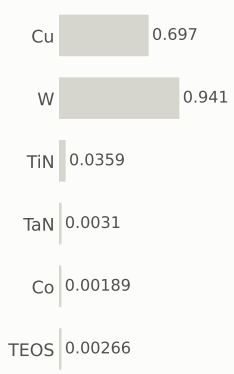
mask for this step

no mask for this step

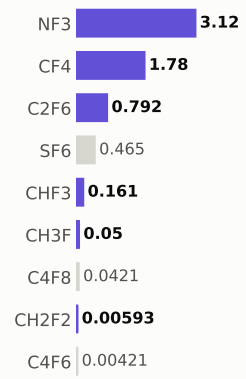
steps by process area



materials deposited (g)



gases emitted (g)



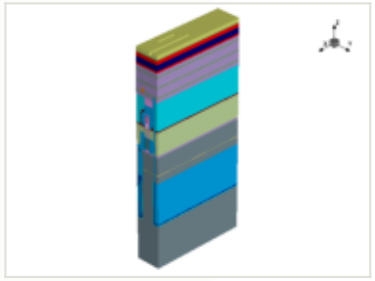
Step #3910: RIE

etch_anisotropic · RIE

1.04	389.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

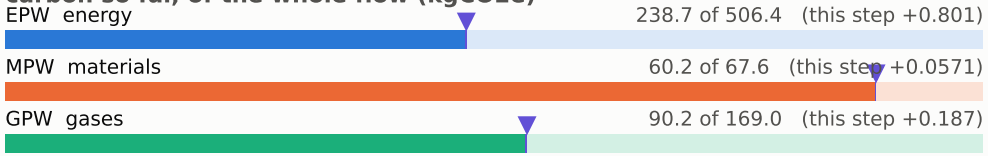


this step

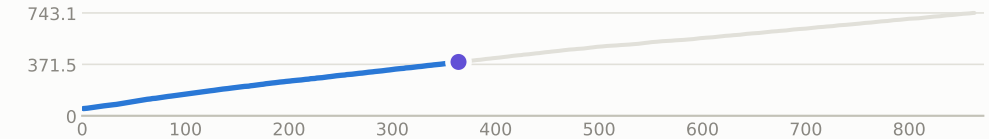
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

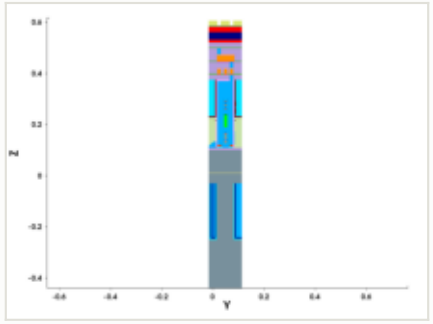
carbon so far, of the whole flow (kgCO2e)



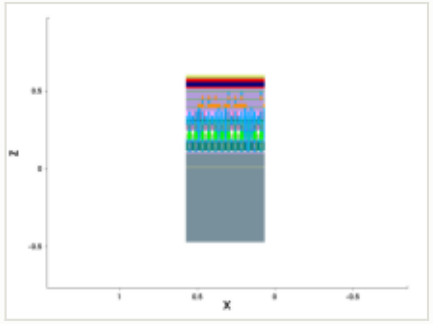
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



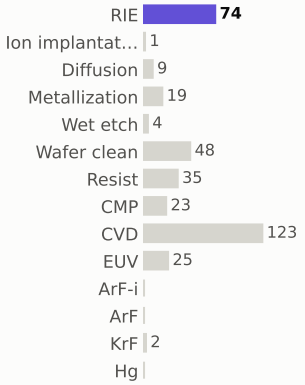
y cut at 0.049 µm



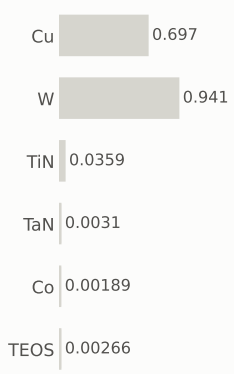
mask for this step

no mask for this step

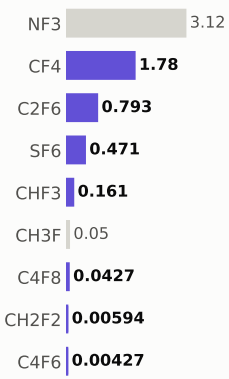
steps by process area



materials deposited (g)



gases emitted (g)



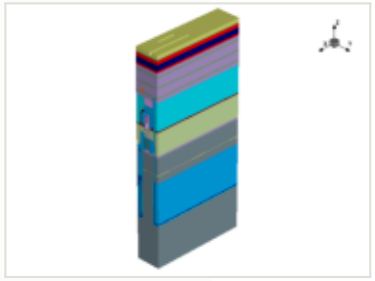
Step #3920: Wafer_clean

clean · Wafer_clean

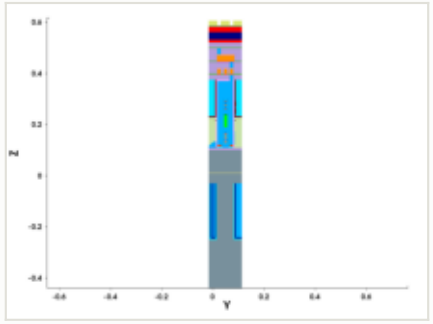
0.169	389.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

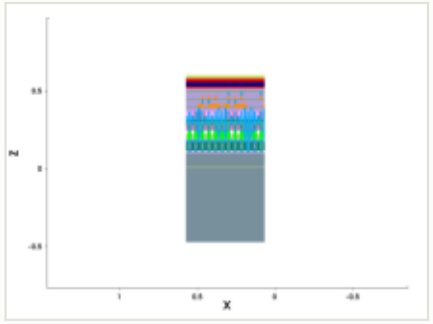
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

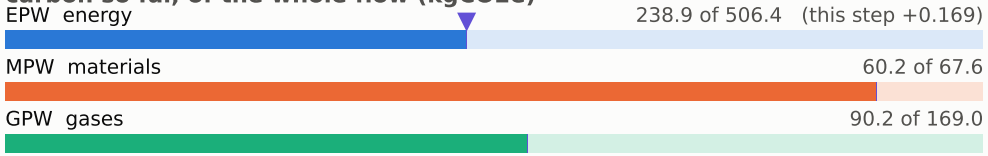
no mask for this step

this step

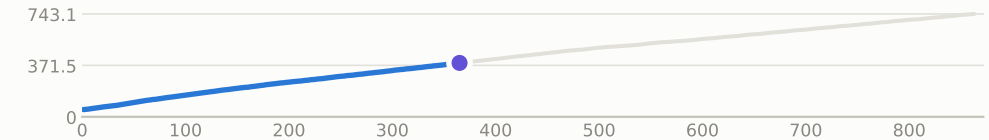
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

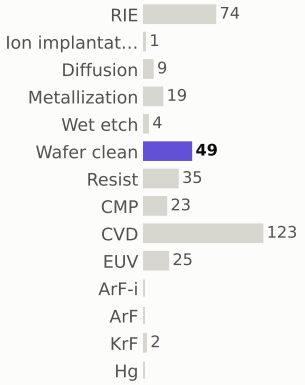
carbon so far, of the whole flow (kgCO2e)



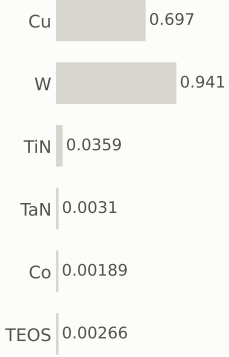
embodied carbon up to each step (kgCO2e per wafer)



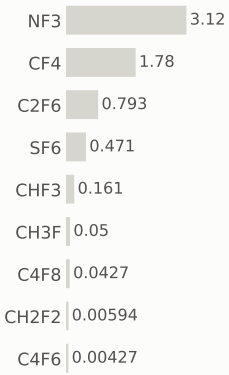
steps by process area



materials deposited (g)



gases emitted (g)



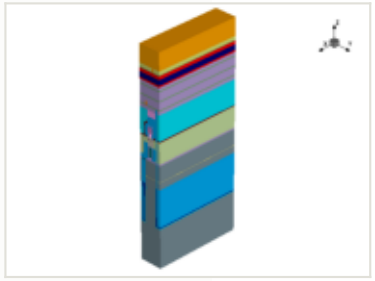
Step #3930: Resist

depo_fill · Resist

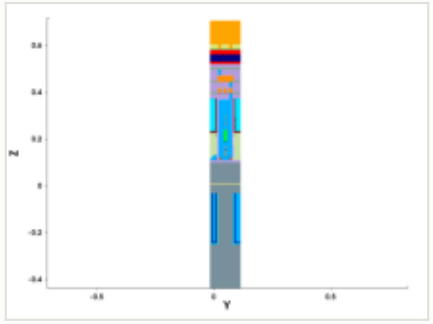
0.169	389.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

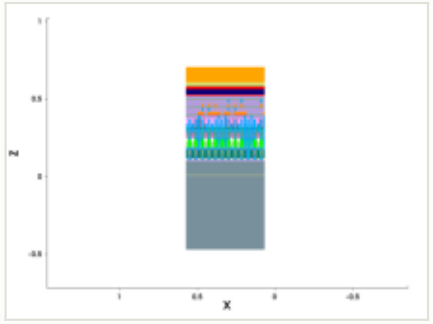
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



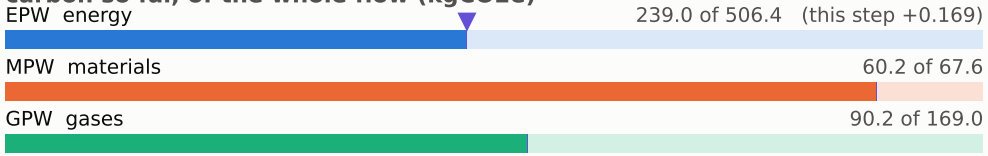
mask for this step

no mask for this step

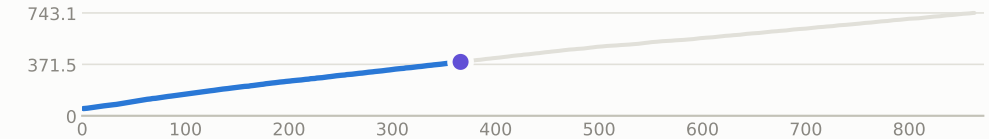
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

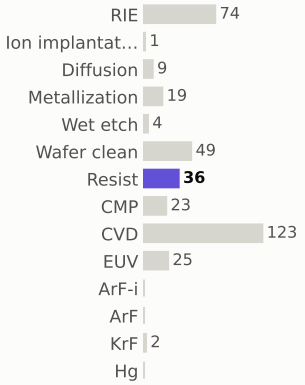
carbon so far, of the whole flow (kgCO2e)



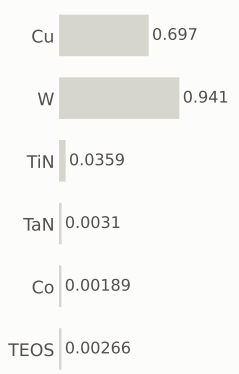
embodied carbon up to each step (kgCO2e per wafer)



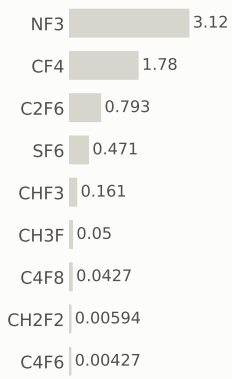
steps by process area



materials deposited (g)



gases emitted (g)



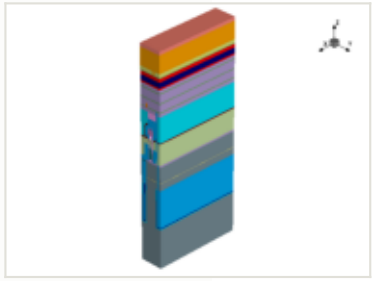
Step #3940: CVD

depo_fill · CVD

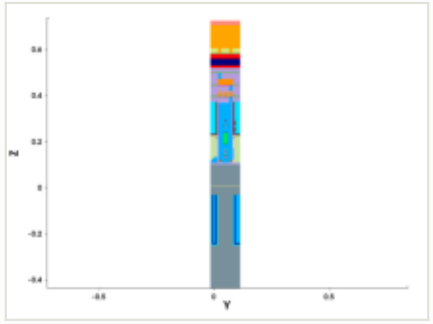
1.31	390.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

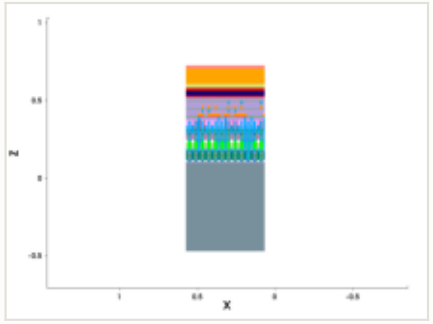
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

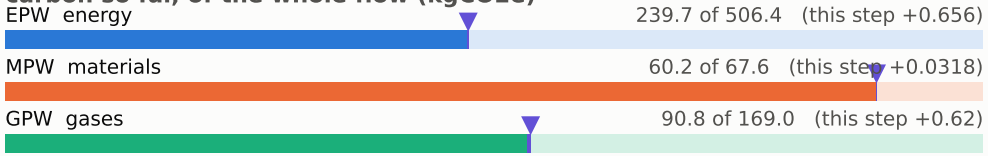
no mask for this step

this step

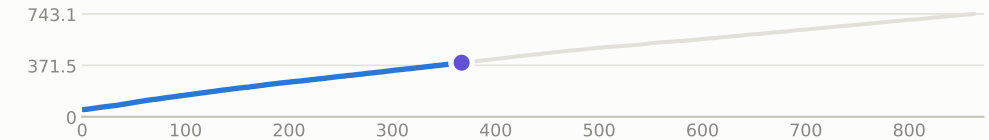
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

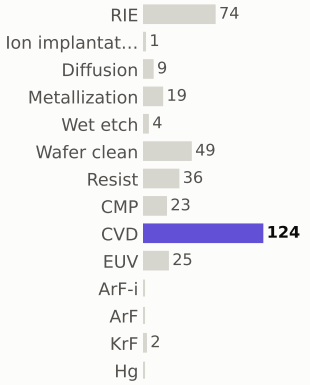
carbon so far, of the whole flow (kgCO2e)



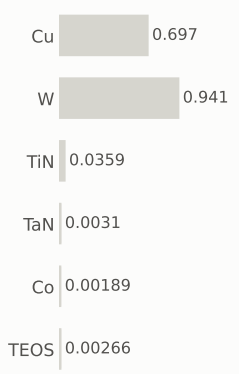
embodied carbon up to each step (kgCO2e per wafer)



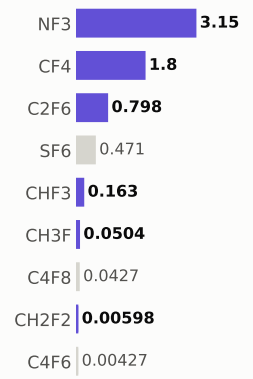
steps by process area



materials deposited (g)



gases emitted (g)



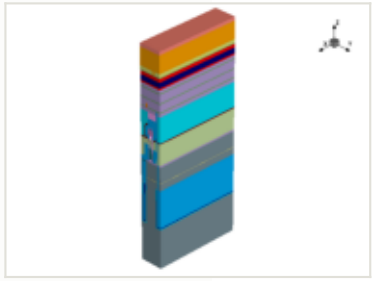
Step #3950: Wafer_clean

clean · Wafer_clean

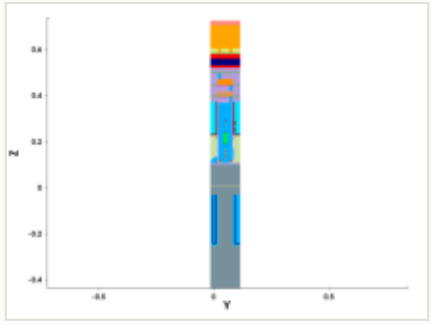
0.169	390.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

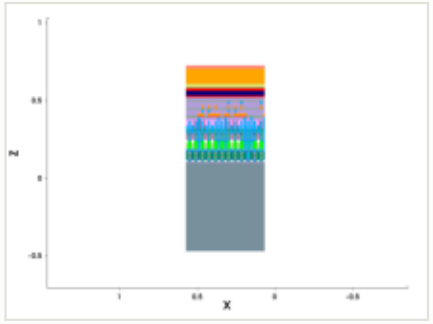
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

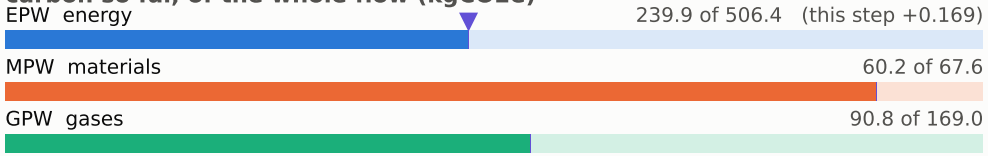
no mask for this step

this step

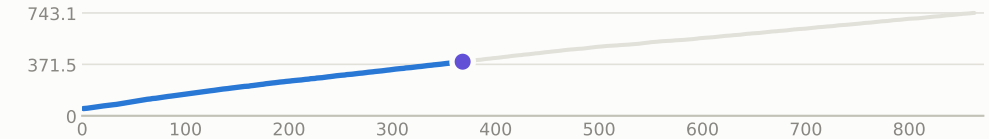
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

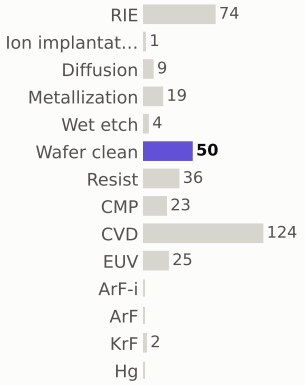
carbon so far, of the whole flow (kgCO2e)



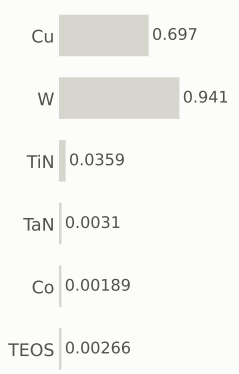
embodied carbon up to each step (kgCO2e per wafer)



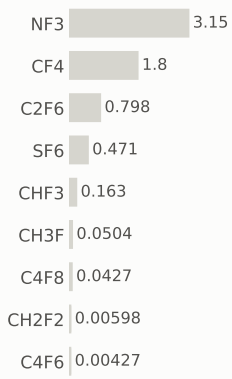
steps by process area



materials deposited (g)



gases emitted (g)



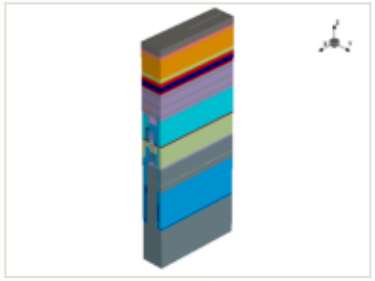
Step #3960: EUV

litho_patterning · EUV

1.96	392.8	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

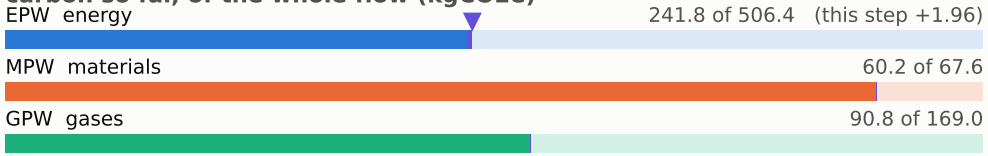


this step

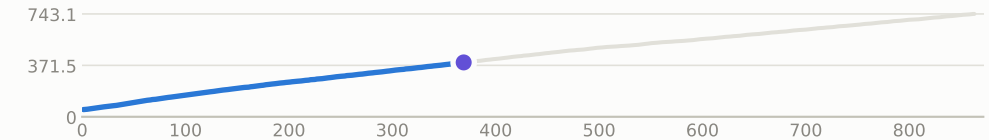
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

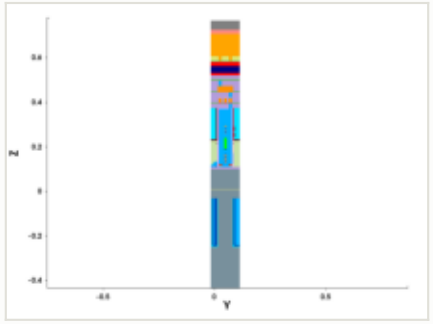
carbon so far, of the whole flow (kgCO2e)



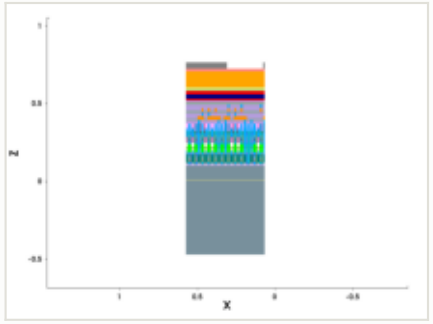
embodied carbon up to each step (kgCO2e per wafer)



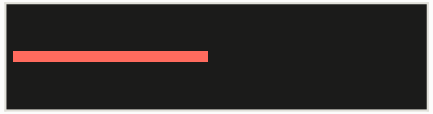
x cut at 0.385 µm



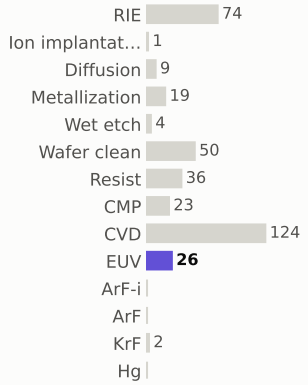
y cut at 0.049 µm



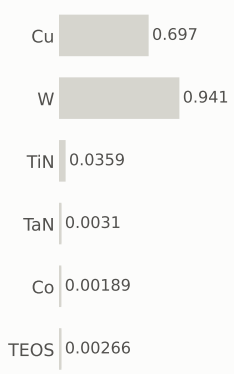
mask: layer 10302:0 (GDSII)



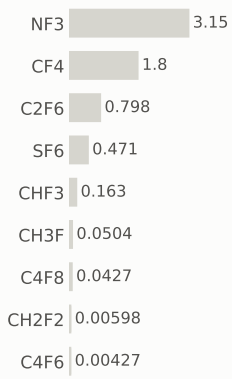
steps by process area



materials deposited (g)



gases emitted (g)



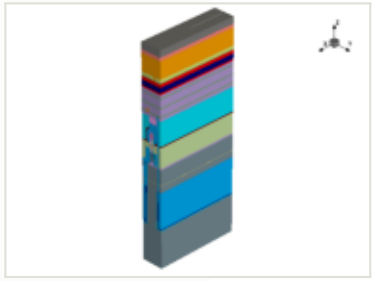
Step #3970: RIE

etch_anisotropic · RIE

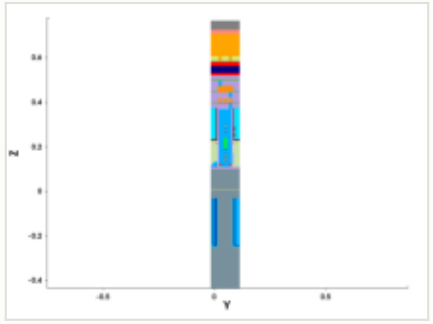
1.04	393.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

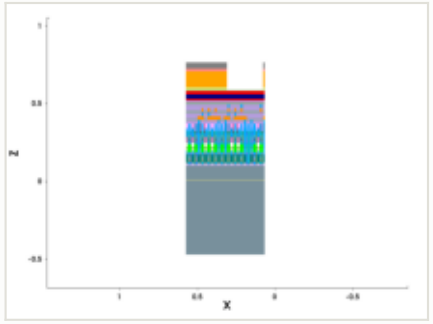
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

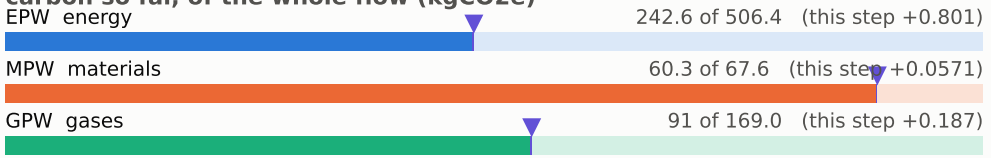
no mask for this step

this step

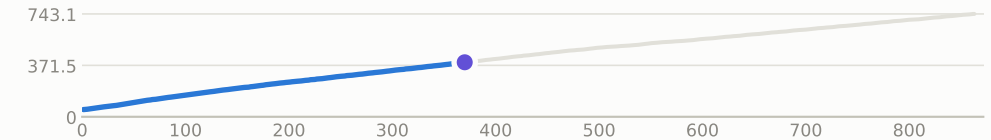
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

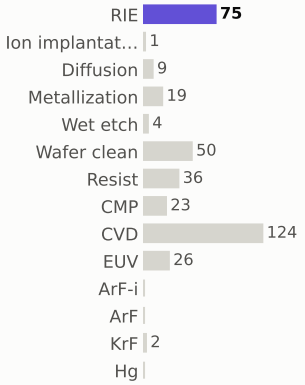
carbon so far, of the whole flow (kgCO2e)



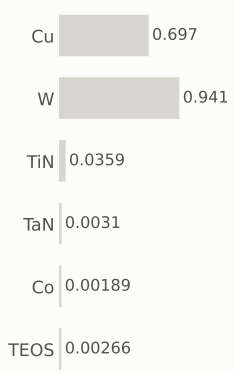
embodied carbon up to each step (kgCO2e per wafer)



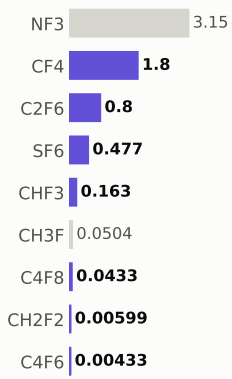
steps by process area



materials deposited (g)



gases emitted (g)



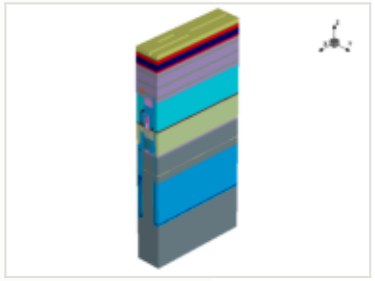
Step #3980: Wafer_clean

etch_strip · Wafer_clean

0.169	394.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

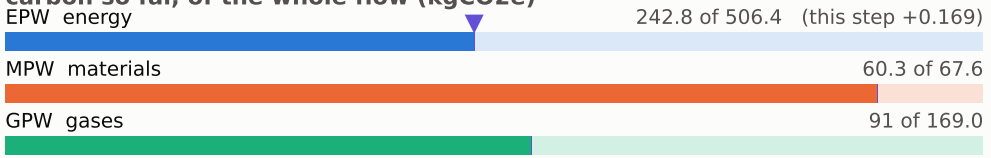


this step

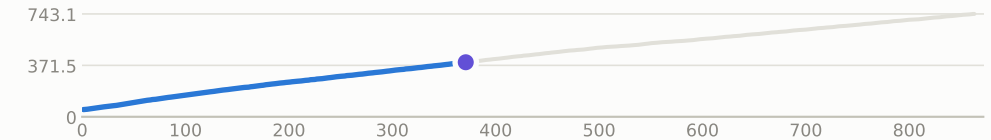
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

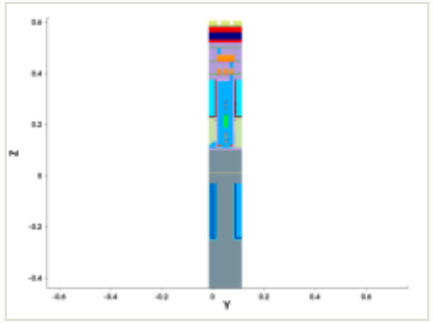
carbon so far, of the whole flow (kgCO2e)



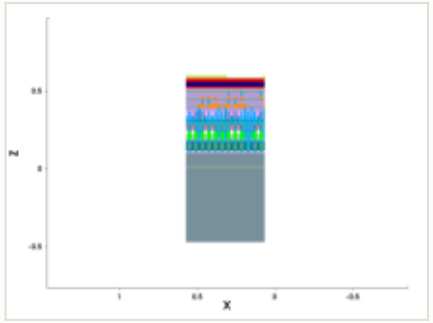
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



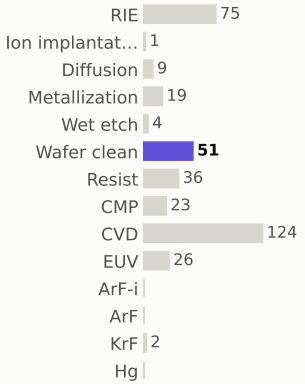
y cut at 0.049 µm



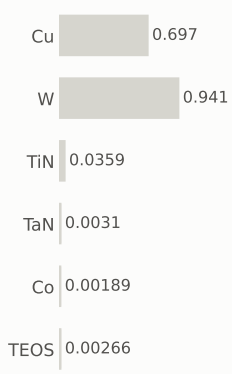
mask for this step

no mask for this step

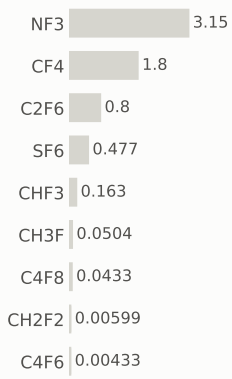
steps by process area



materials deposited (g)



gases emitted (g)



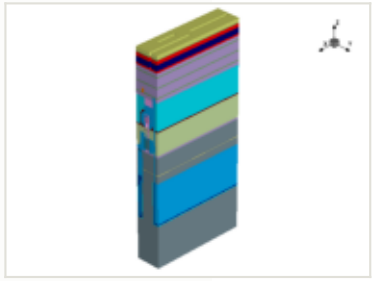
Step #3990: RIE

etch_anisotropic · RIE

1.04	395.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

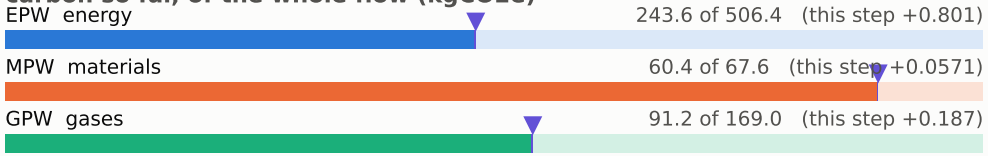


this step

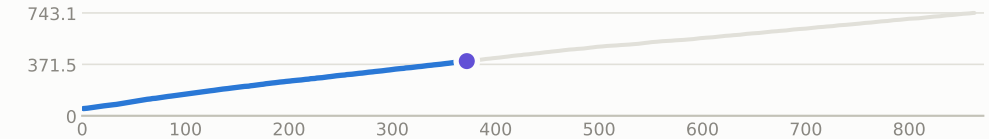
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

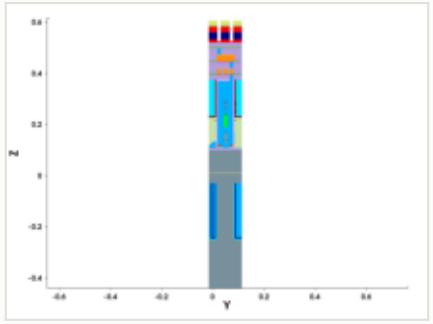
carbon so far, of the whole flow (kgCO2e)



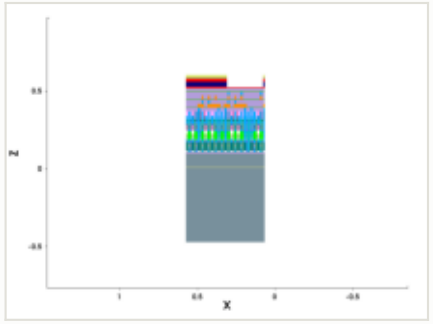
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



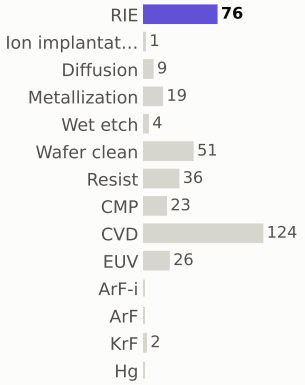
y cut at 0.049 µm



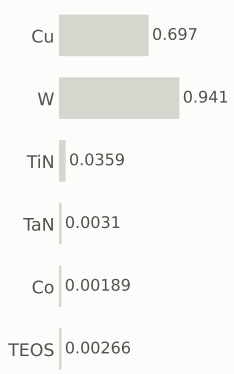
mask for this step

no mask for this step

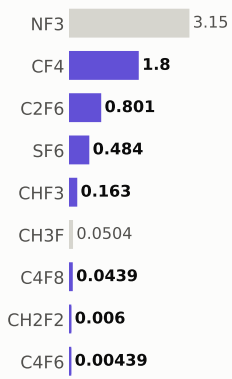
steps by process area



materials deposited (g)



gases emitted (g)



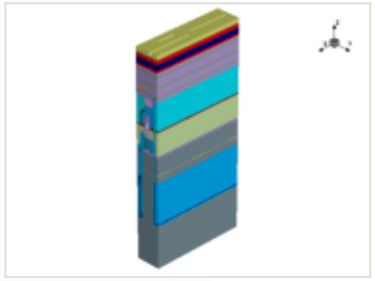
Step #4000: Wafer_clean

clean · Wafer_clean

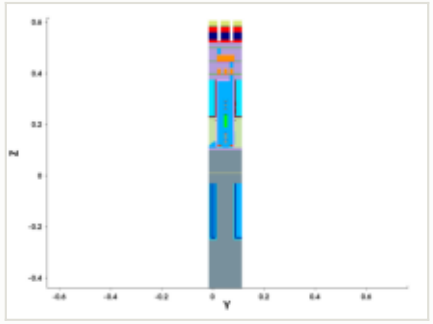
0.169	395.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

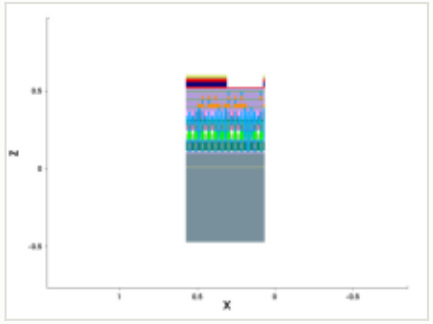
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

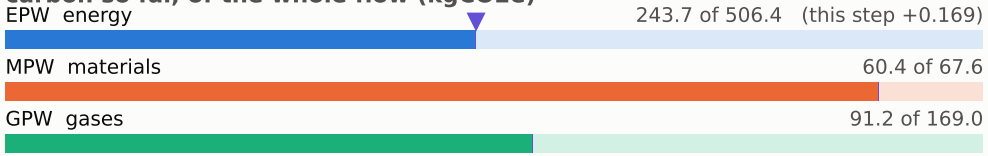
no mask for this step

this step

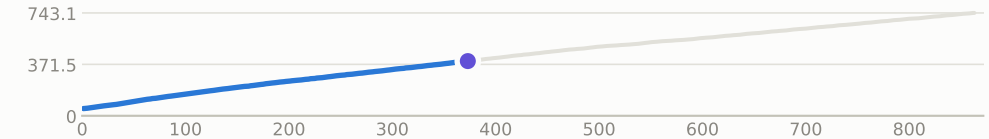
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

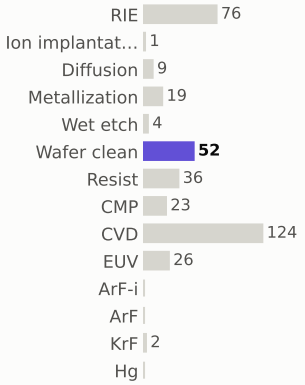
carbon so far, of the whole flow (kgCO2e)



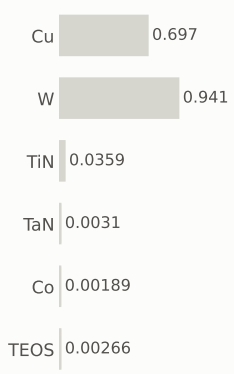
embodied carbon up to each step (kgCO2e per wafer)



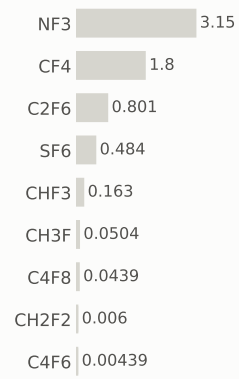
steps by process area



materials deposited (g)



gases emitted (g)



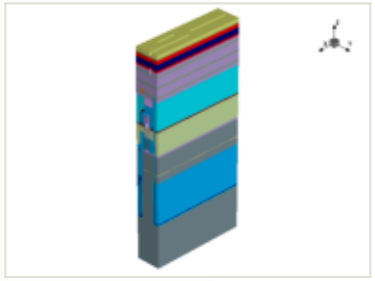
Step #4010: RIE

etch_anisotropic · RIE

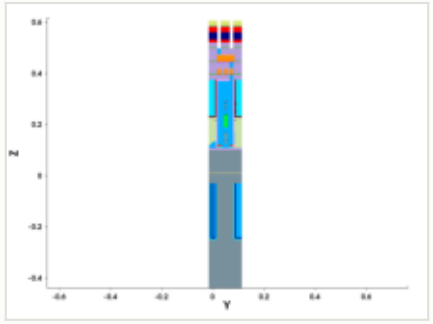
1.04	396.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

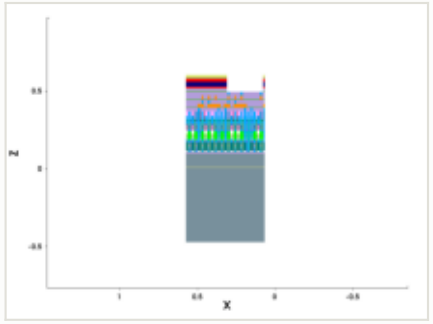
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

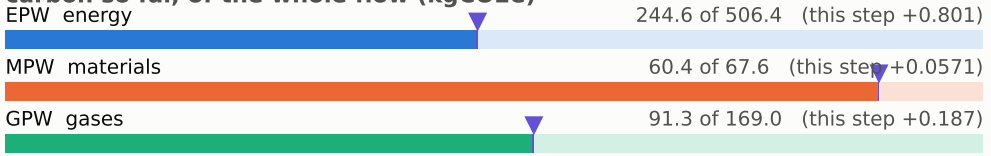
no mask for this step

this step

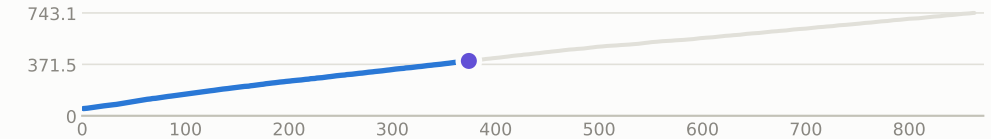
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

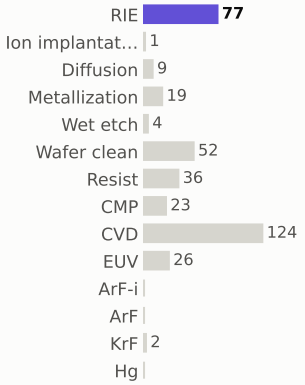
carbon so far, of the whole flow (kgCO2e)



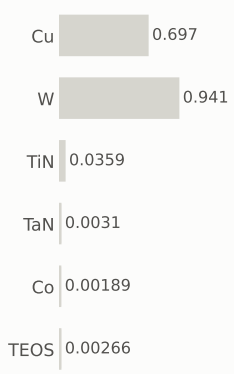
embodied carbon up to each step (kgCO2e per wafer)



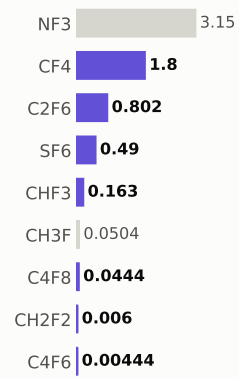
steps by process area



materials deposited (g)



gases emitted (g)



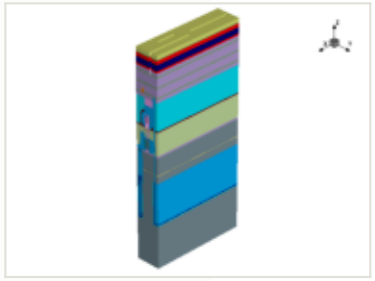
Step #4020: Wafer_clean

clean · Wafer_clean

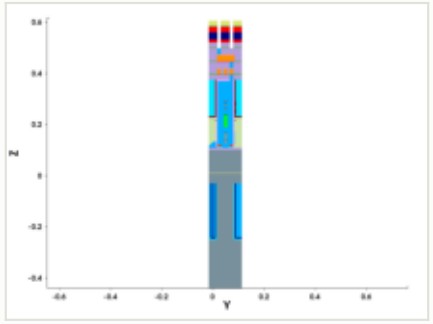
0.169	396.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

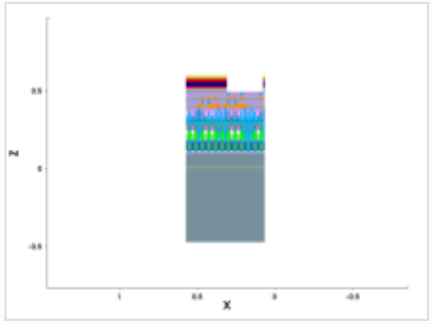
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

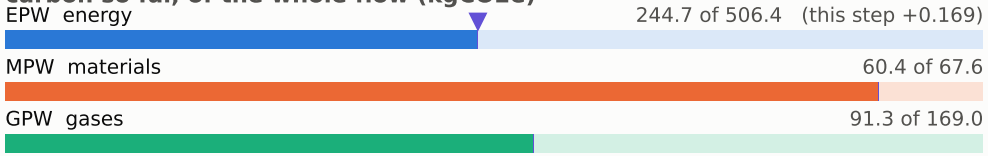
no mask for this step

this step

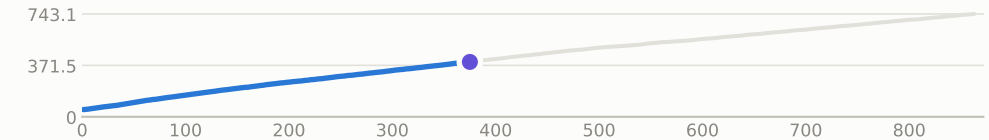
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

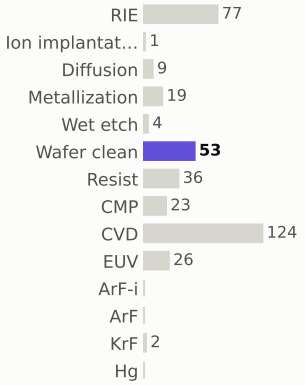
carbon so far, of the whole flow (kgCO2e)



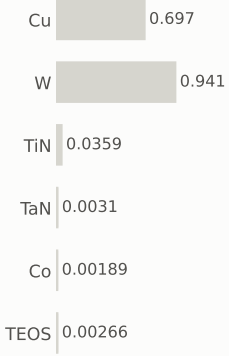
embodied carbon up to each step (kgCO2e per wafer)



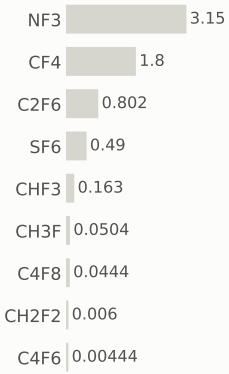
steps by process area



materials deposited (g)



gases emitted (g)



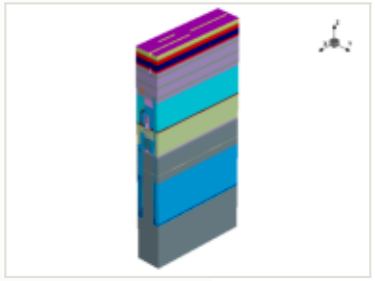
Step #4030: Metallization

depo_ellipsoidal · Metallization

1.2	397.7	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

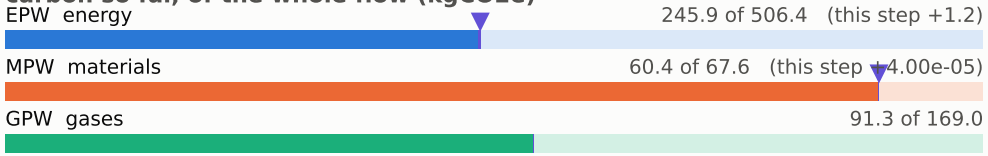


this step

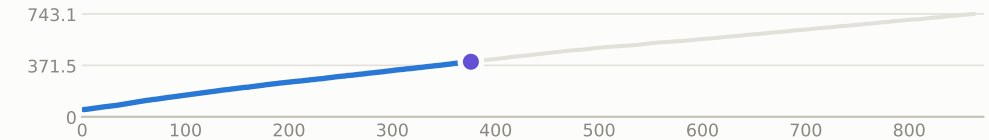
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

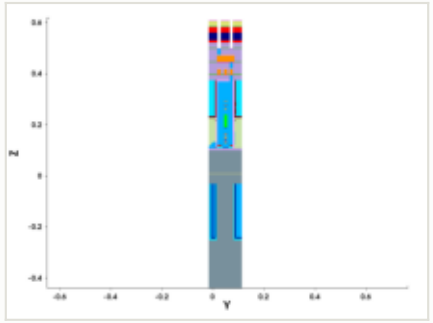
carbon so far, of the whole flow (kgCO2e)



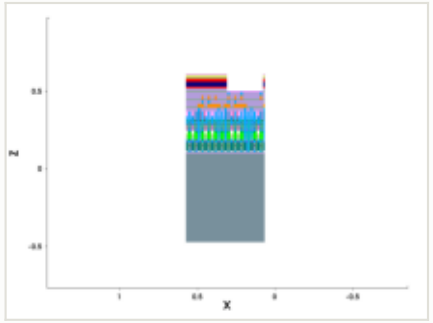
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



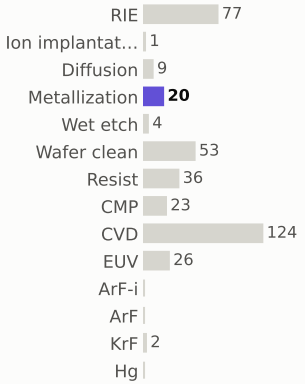
y cut at 0.049 µm



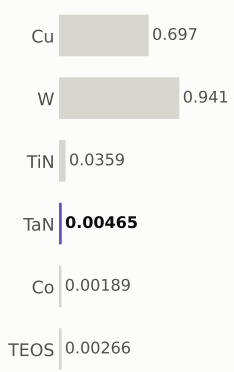
mask for this step

no mask for this step

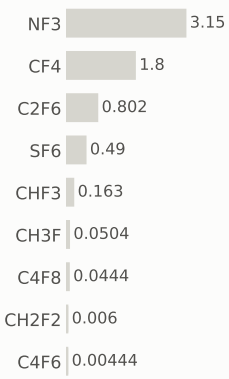
steps by process area



materials deposited (g)



gases emitted (g)



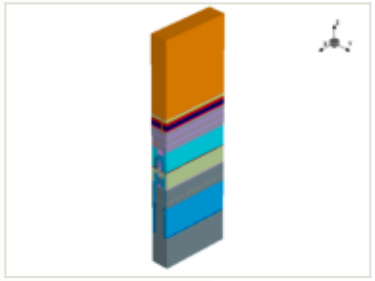
Step #4040: Metallization

depo_fill · Metallization

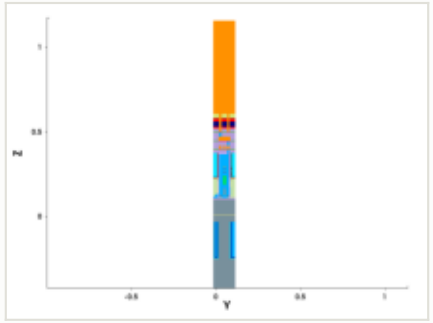
1.21	398.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

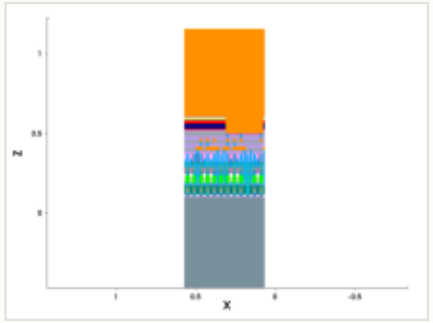
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

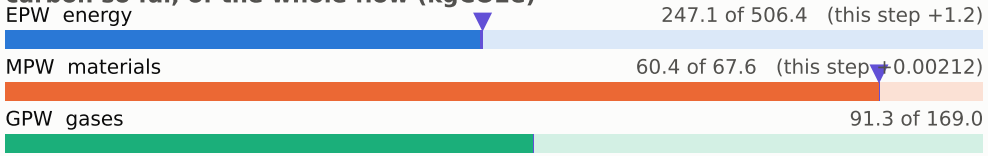
no mask for this step

this step

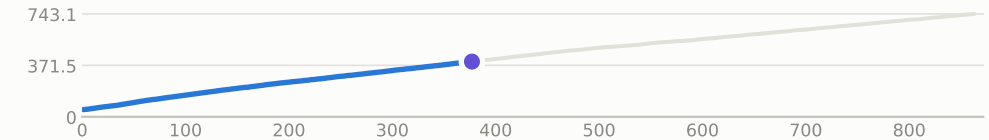
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

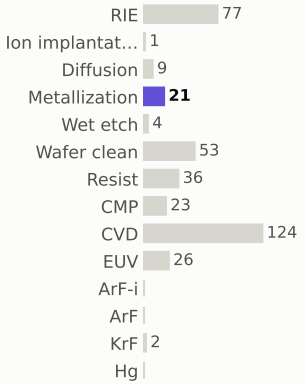
carbon so far, of the whole flow (kgCO2e)



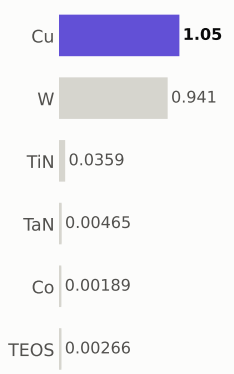
embodied carbon up to each step (kgCO2e per wafer)



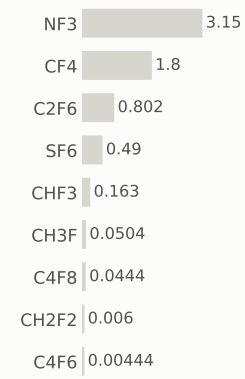
steps by process area



materials deposited (g)



gases emitted (g)



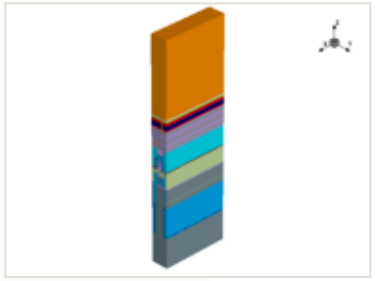
Step #4050: Diffusion

treatment · Diffusion

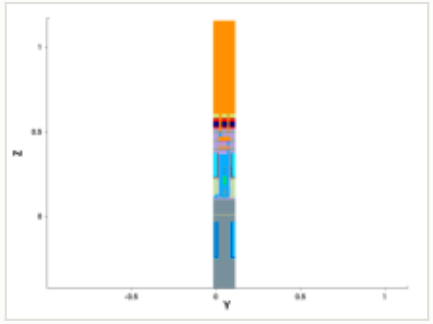
0.801	399.7	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

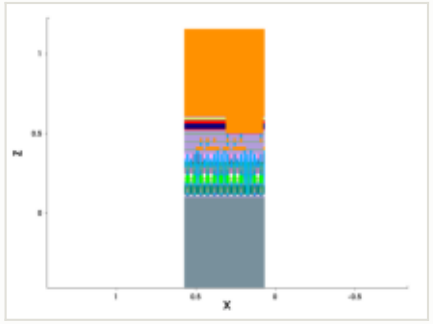
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

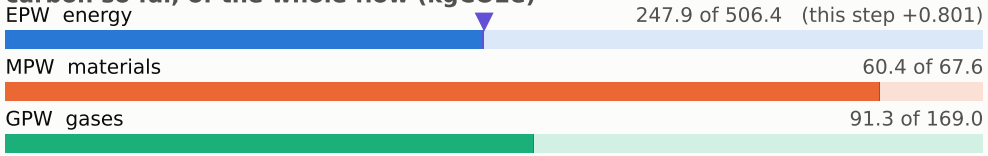
no mask for this step

this step

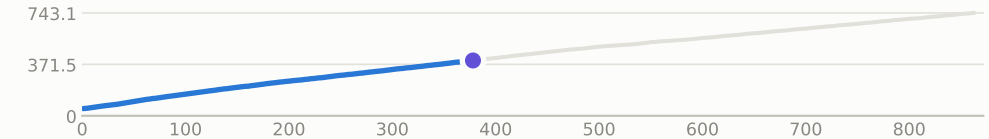
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

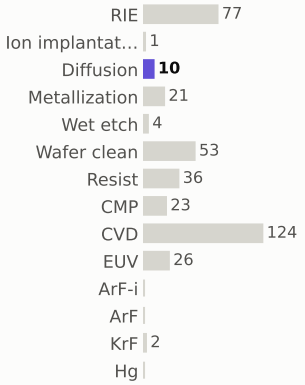
carbon so far, of the whole flow (kgCO2e)



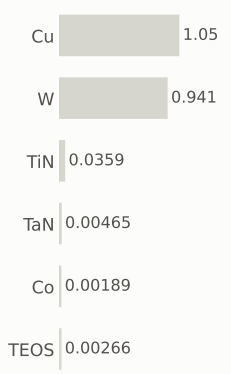
embodied carbon up to each step (kgCO2e per wafer)



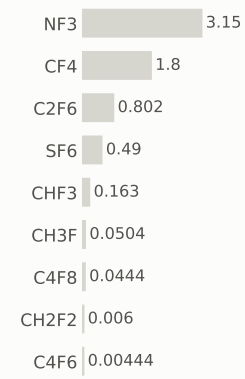
steps by process area



materials deposited (g)



gases emitted (g)



Step #4060: CMP

etch_cmp · CMP

0.169	399.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

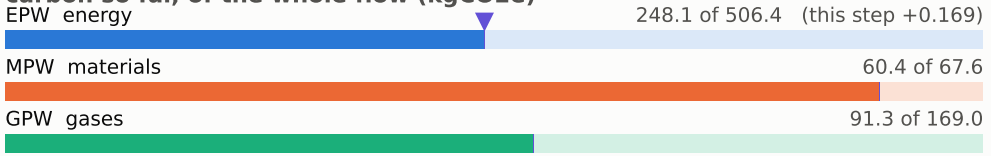
zoomed to fit

this step

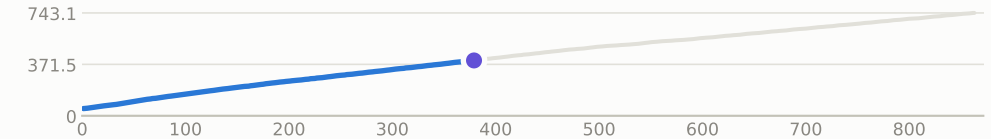
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

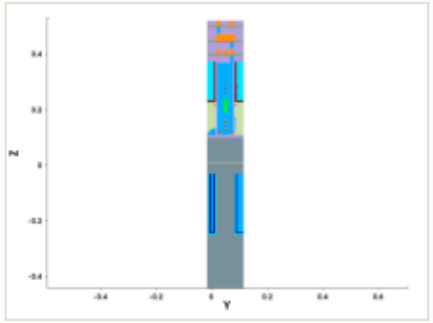
carbon so far, of the whole flow (kgCO2e)



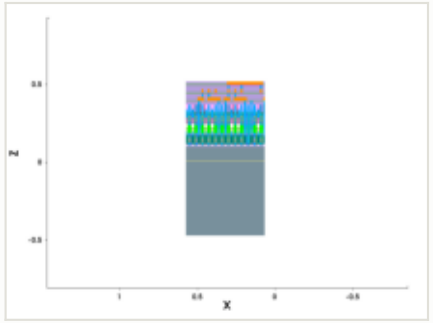
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



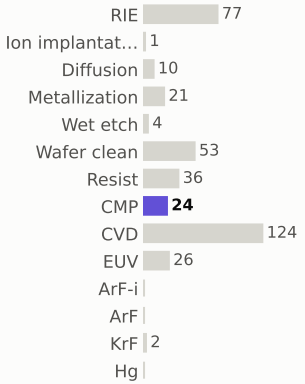
y cut at 0.049 µm



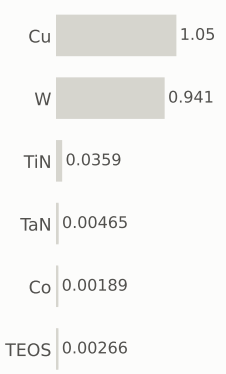
mask for this step

no mask for this step

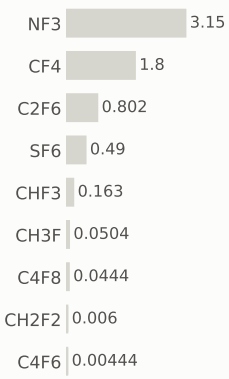
steps by process area



materials deposited (g)



gases emitted (g)



Step #4070: Metallization

depo_isotropic · Metallization

1.2	401.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

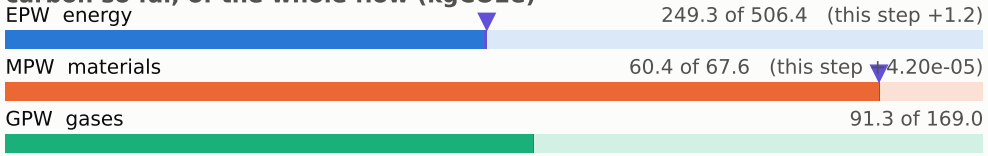
zoomed to fit

this step

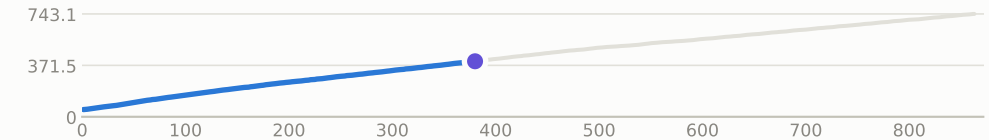
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

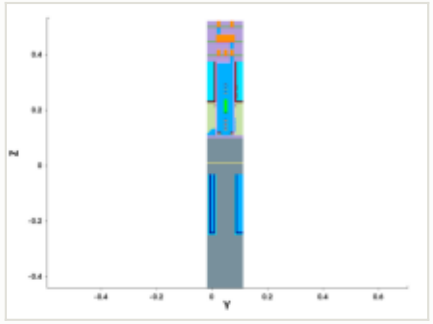
carbon so far, of the whole flow (kgCO2e)



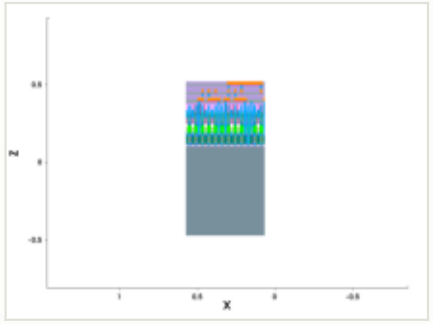
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



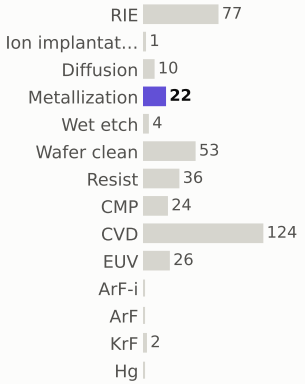
y cut at 0.049 µm



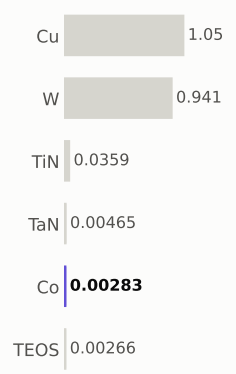
mask for this step

no mask for this step

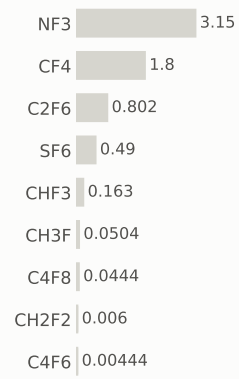
steps by process area



materials deposited (g)



gases emitted (g)



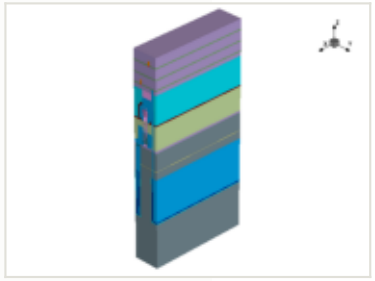
Step #4090: CVD

depo_fill · CVD

1.31	402.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

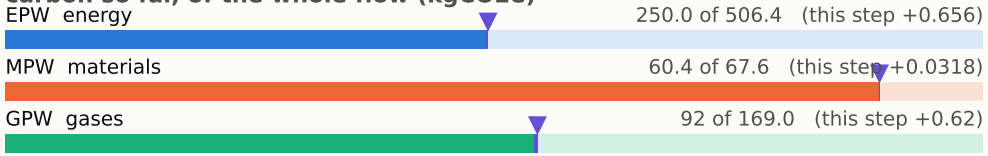


this step

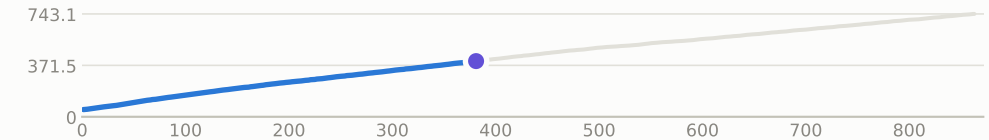
process area	CVD
material	—
thickness	0.028 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

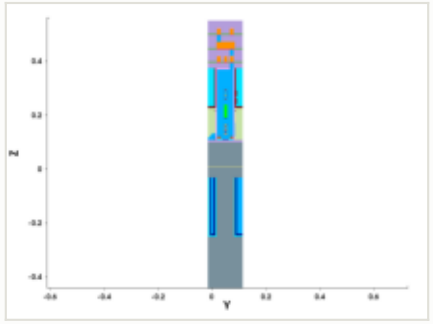
carbon so far, of the whole flow (kgCO2e)



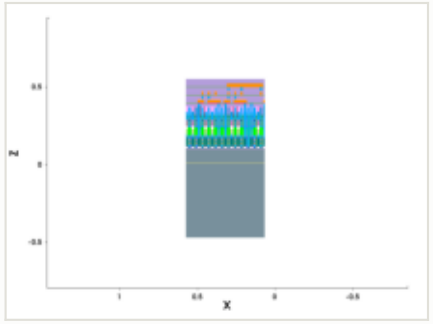
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



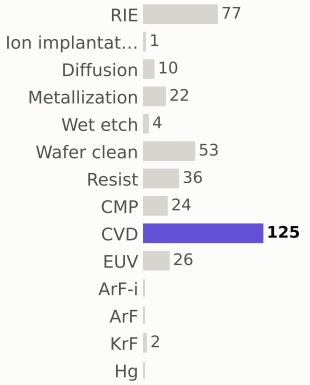
y cut at 0.049 µm



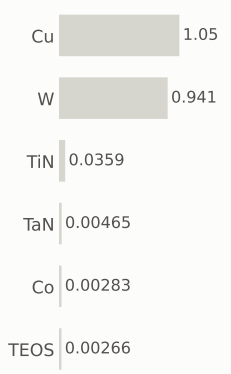
mask for this step

no mask for this step

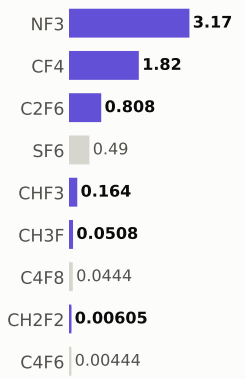
steps by process area



materials deposited (g)



gases emitted (g)



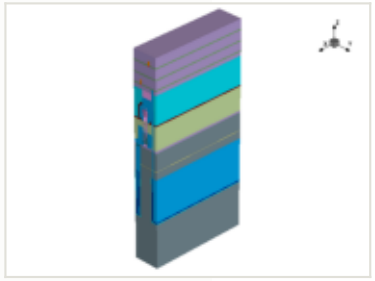
Step #4100: Diffusion

treatment · Diffusion

0.801	403.2	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

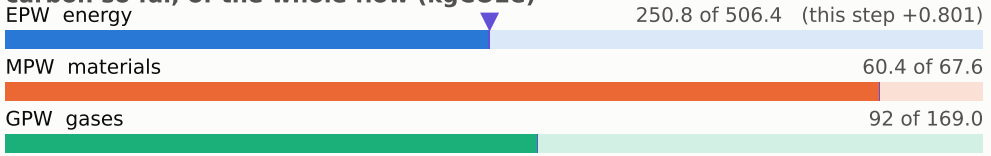


this step

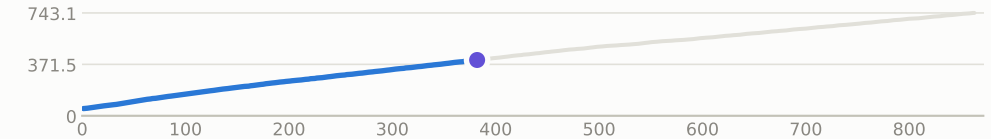
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

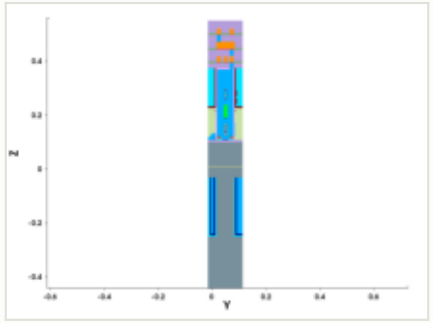
carbon so far, of the whole flow (kgCO2e)



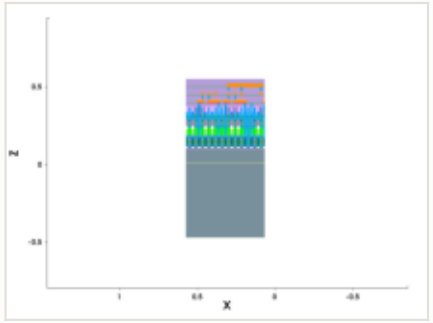
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



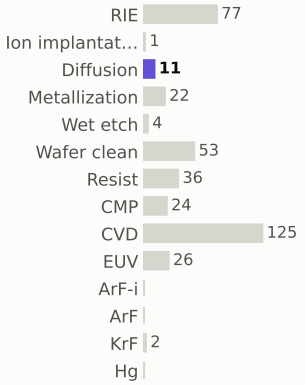
y cut at 0.049 µm



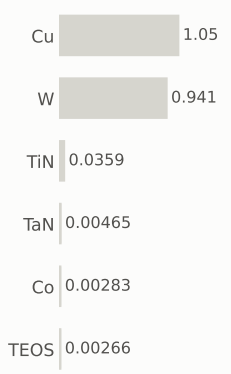
mask for this step

no mask for this step

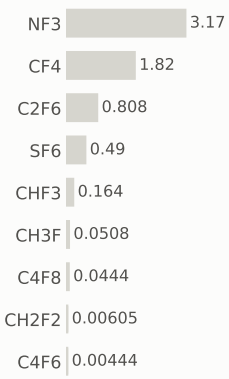
steps by process area



materials deposited (g)



gases emitted (g)



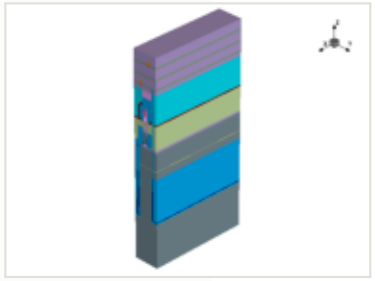
Step #4110: Wafer_clean

clean · Wafer_clean

0.169	403.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



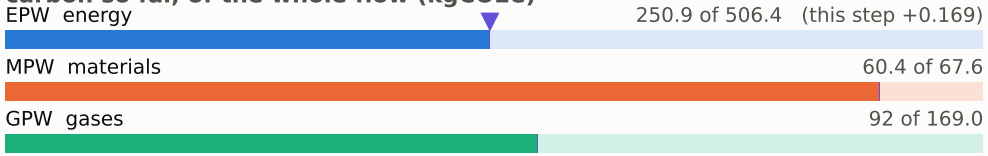
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

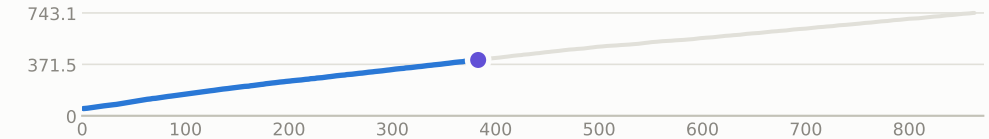
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

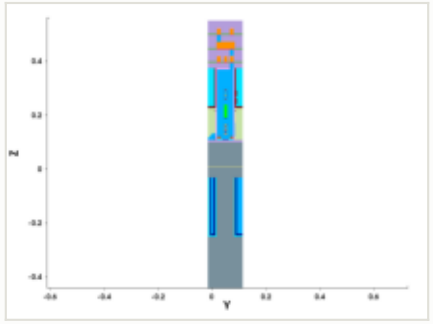
carbon so far, of the whole flow (kgCO2e)



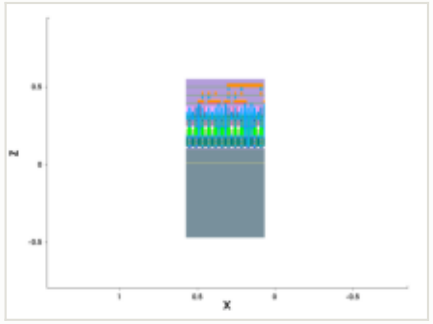
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



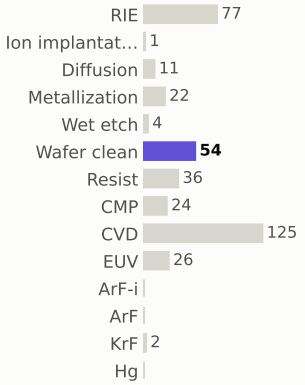
y cut at 0.049 µm



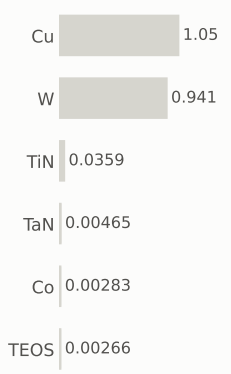
mask for this step

no mask for this step

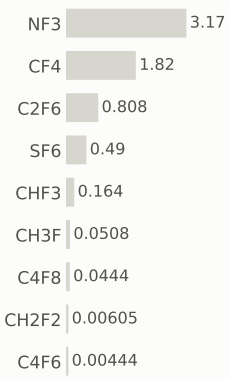
steps by process area



materials deposited (g)



gases emitted (g)



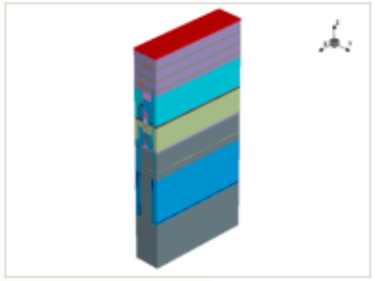
Step #4120: CVD

depo_fill · CVD

1.31	404.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

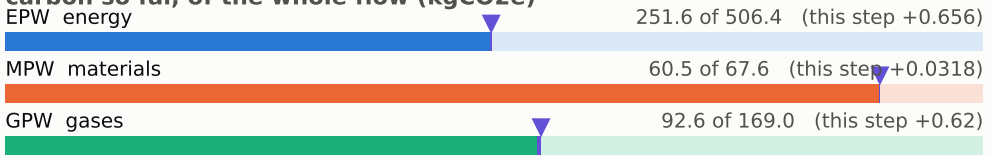


this step

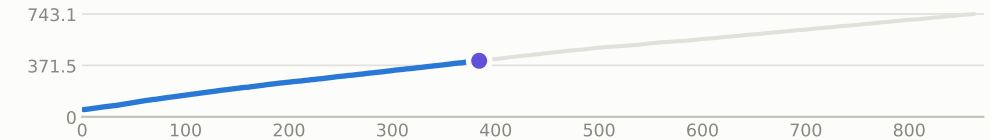
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

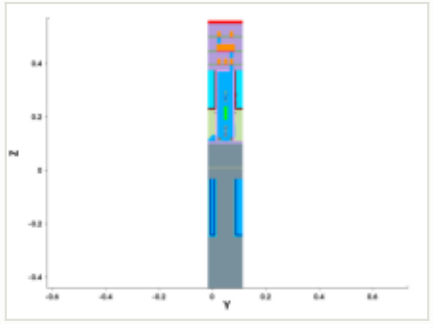
carbon so far, of the whole flow (kgCO2e)



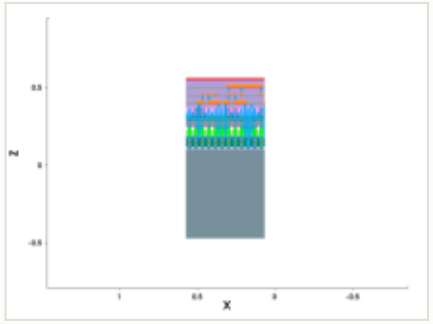
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



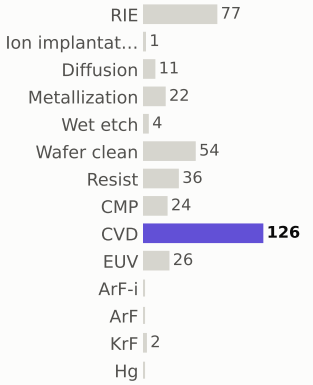
y cut at 0.049 µm



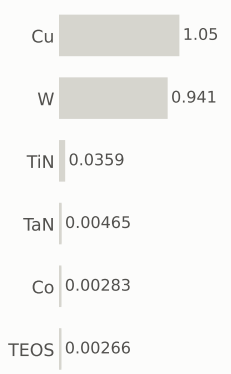
mask for this step

no mask for this step

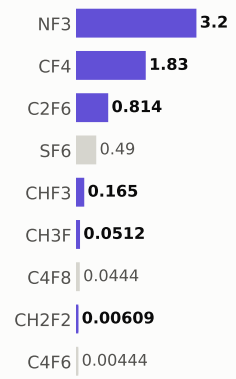
steps by process area



materials deposited (g)



gases emitted (g)



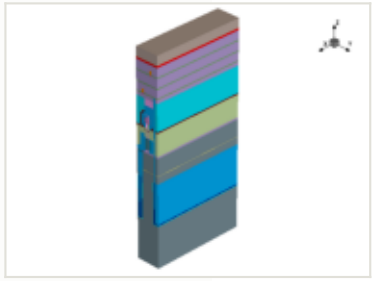
Step #4130: Resist

depo_fill · Resist

0.169	404.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

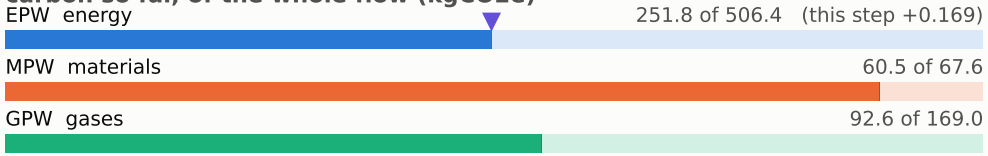


this step

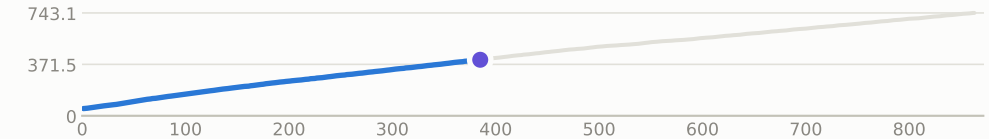
process area	Resist
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

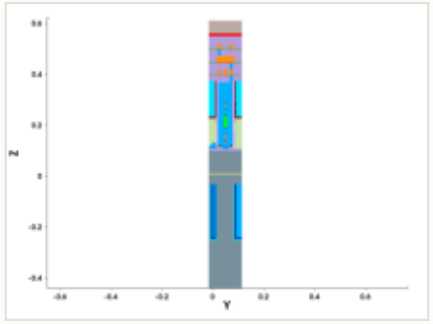
carbon so far, of the whole flow (kgCO2e)



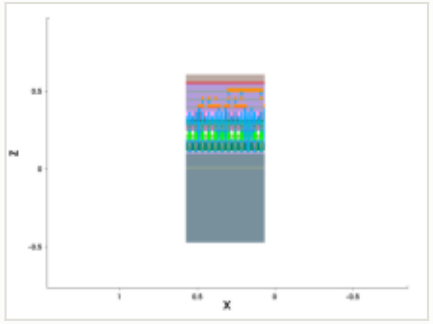
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



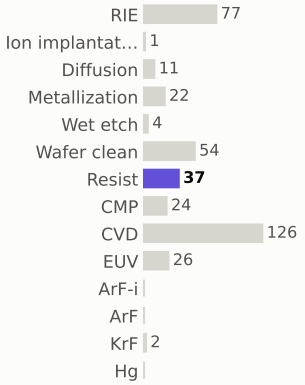
y cut at 0.049 µm



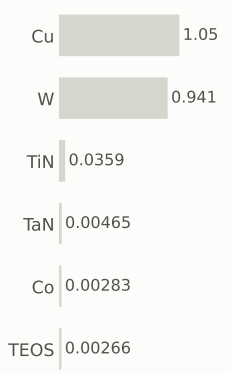
mask for this step

no mask for this step

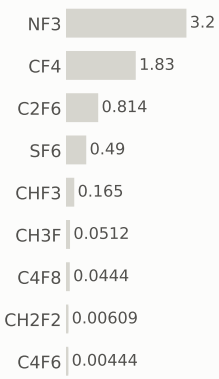
steps by process area



materials deposited (g)



gases emitted (g)



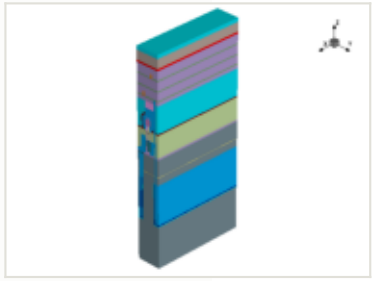
Step #4140: CVD

depo_fill · CVD

1.31	406.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

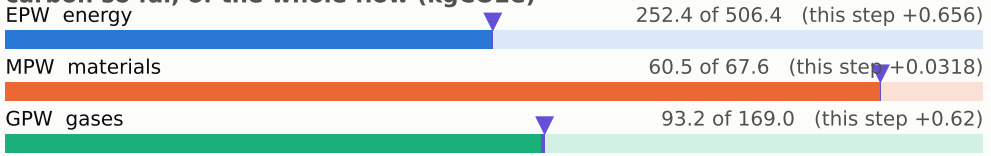


this step

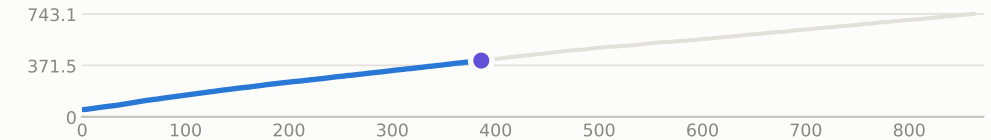
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

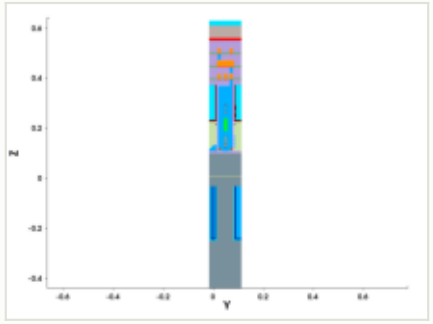
carbon so far, of the whole flow (kgCO2e)



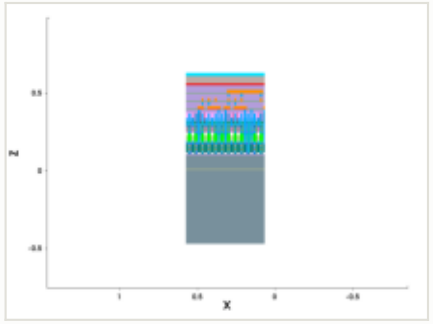
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



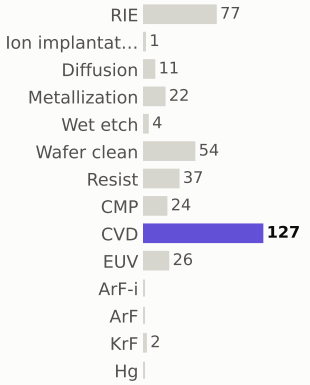
y cut at 0.049 µm



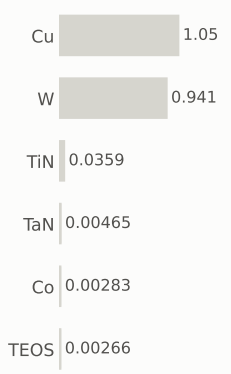
mask for this step

no mask for this step

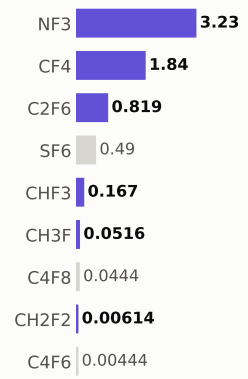
steps by process area



materials deposited (g)



gases emitted (g)



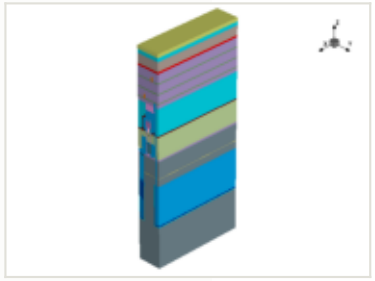
Step #4150: CVD

depo_fill · CVD

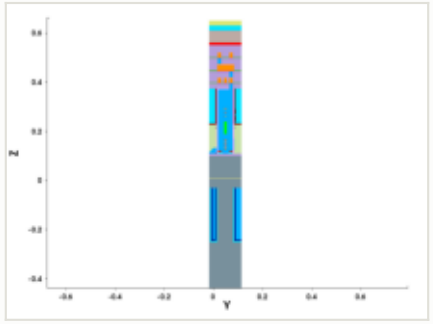
1.31	407.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

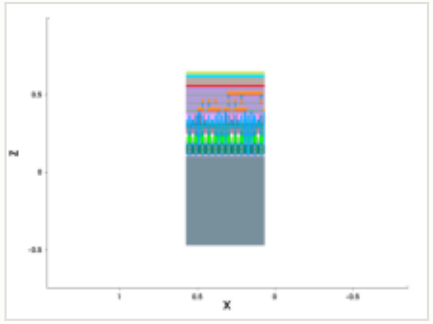
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

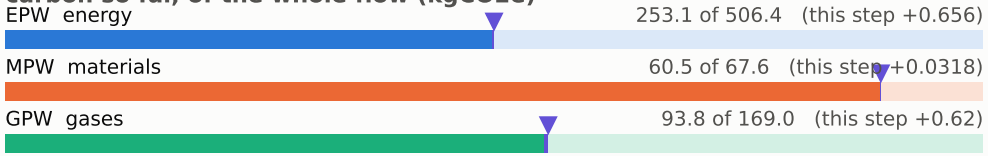
no mask for this step

this step

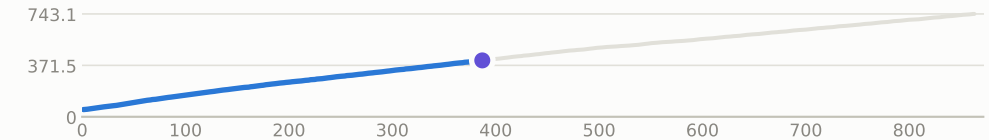
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

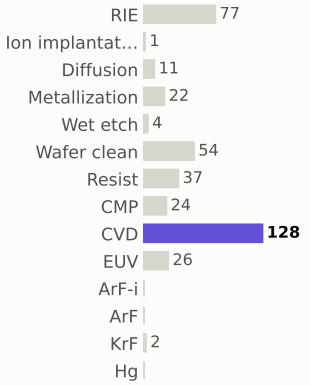
carbon so far, of the whole flow (kgCO2e)



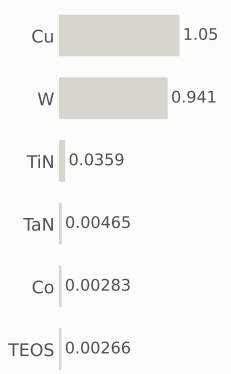
embodied carbon up to each step (kgCO2e per wafer)



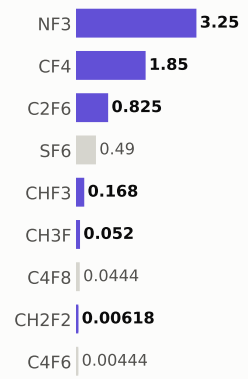
steps by process area



materials deposited (g)



gases emitted (g)



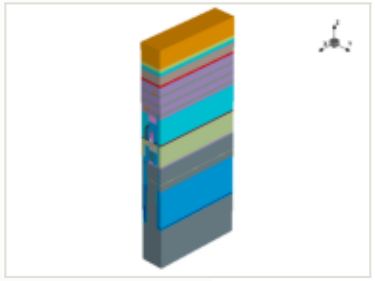
Step #4160: Resist

depo_fill · Resist

0.169	407.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

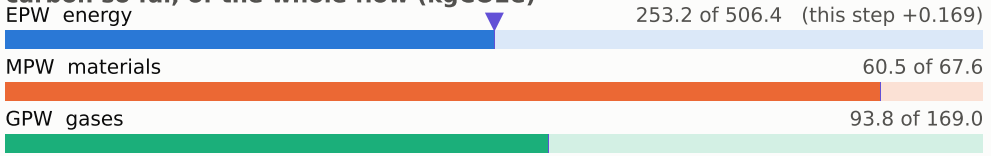


this step

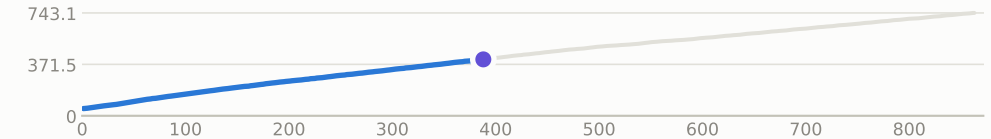
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

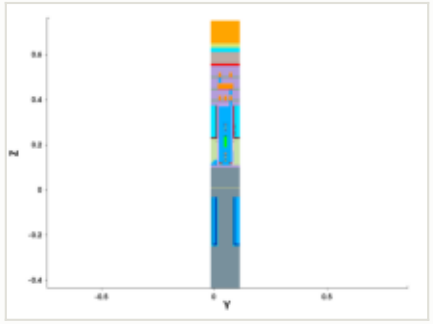
carbon so far, of the whole flow (kgCO2e)



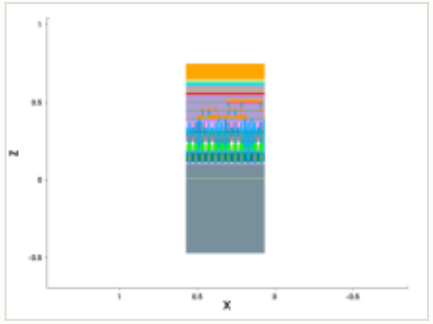
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



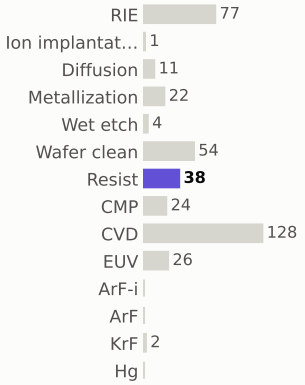
y cut at 0.049 µm



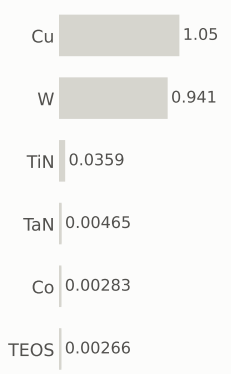
mask for this step

no mask for this step

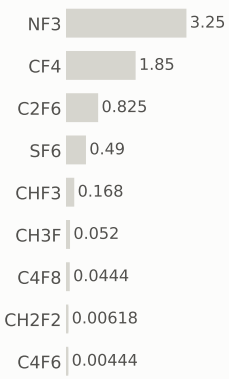
steps by process area



materials deposited (g)



gases emitted (g)



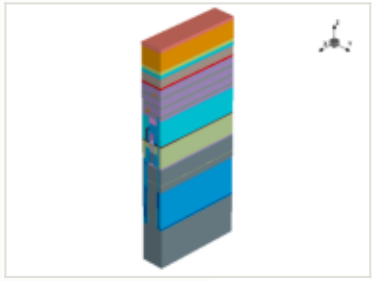
Step #4170: CVD

depo_fill · CVD

1.31	408.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

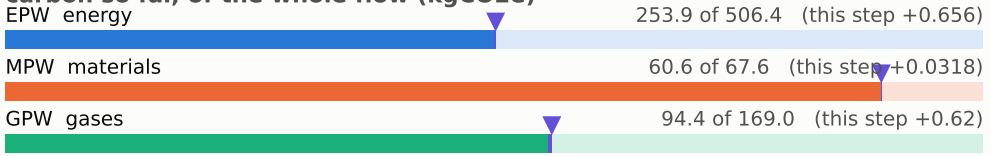


this step

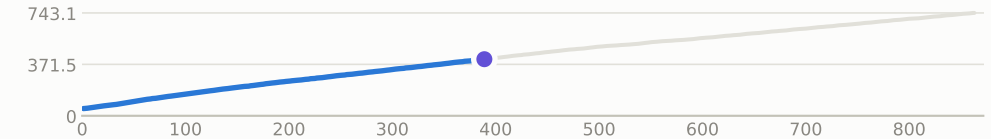
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

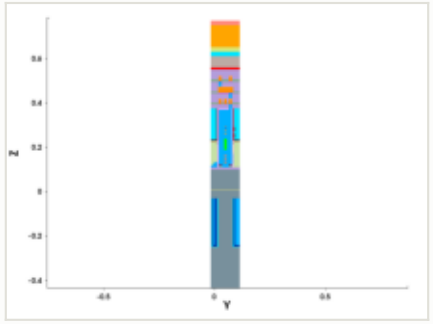
carbon so far, of the whole flow (kgCO2e)



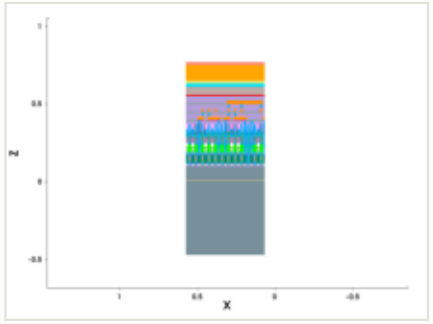
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



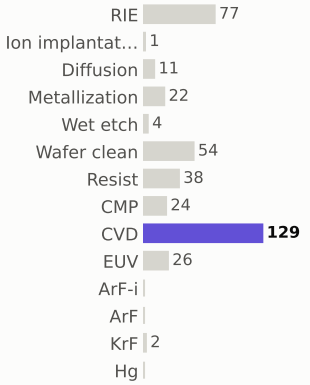
y cut at 0.049 µm



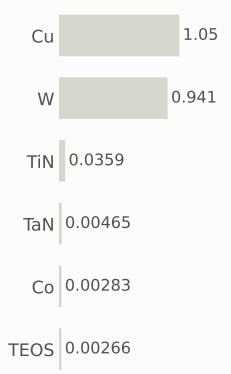
mask for this step

no mask for this step

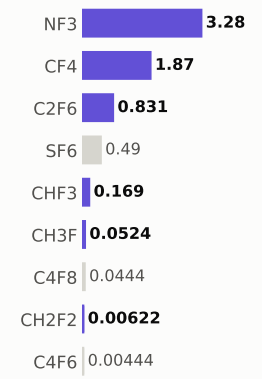
steps by process area



materials deposited (g)



gases emitted (g)



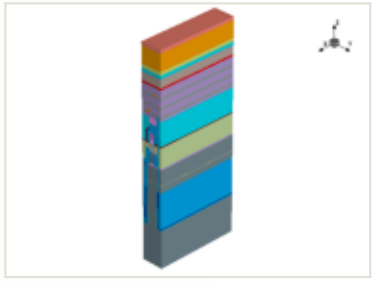
Step #4180: Wafer_clean

clean · Wafer_clean

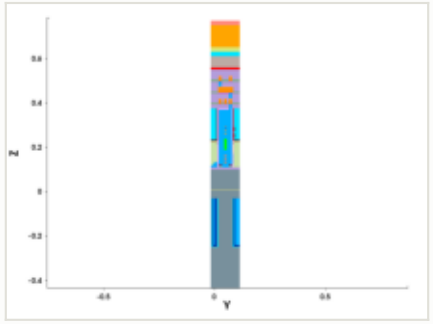
0.169	409.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

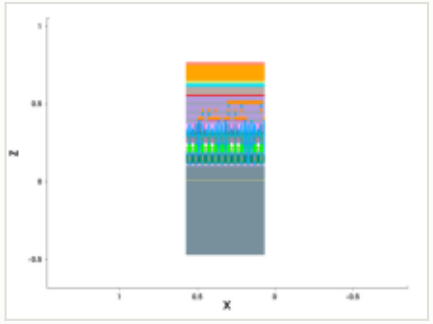
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

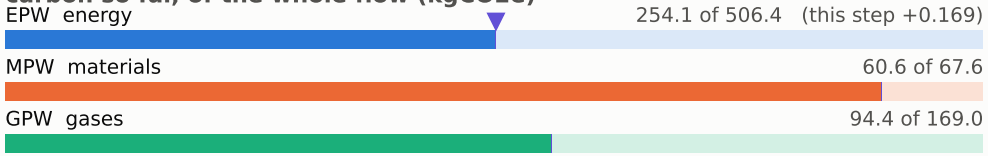
no mask for this step

this step

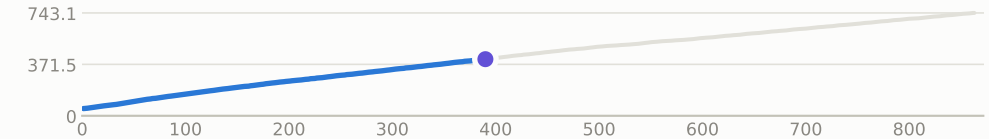
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

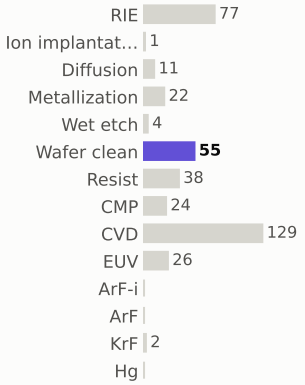
carbon so far, of the whole flow (kgCO2e)



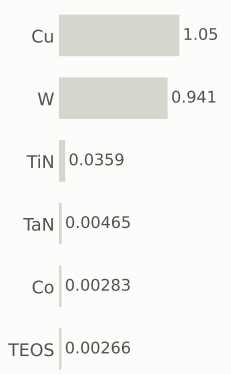
embodied carbon up to each step (kgCO2e per wafer)



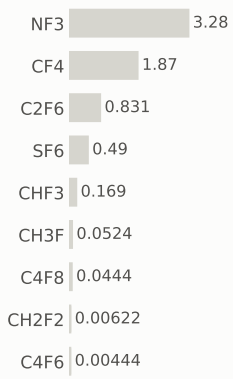
steps by process area



materials deposited (g)



gases emitted (g)



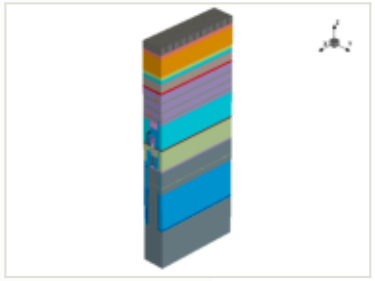
Step #4190: EUV

litho_patterning · EUV

1.96	411.0	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

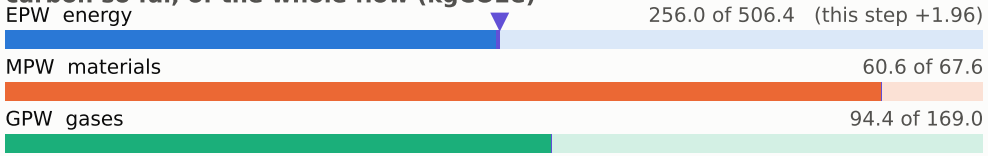


this step

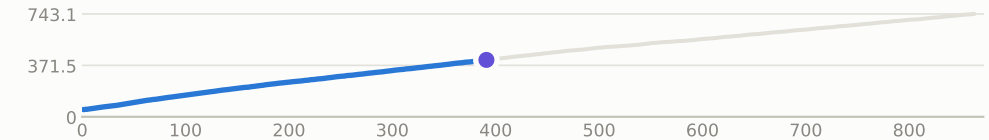
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

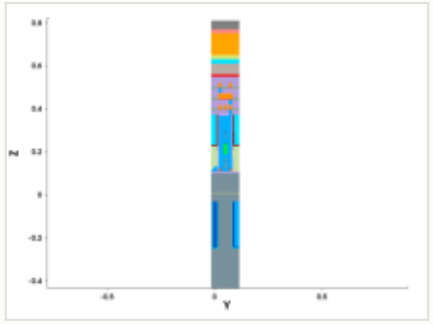
carbon so far, of the whole flow (kgCO2e)



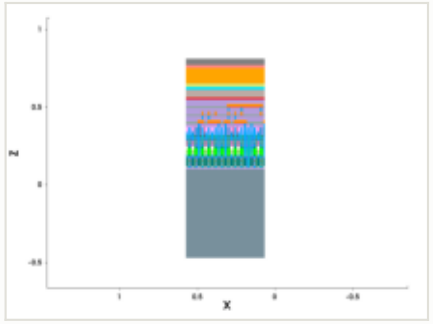
embodied carbon up to each step (kgCO2e per wafer)



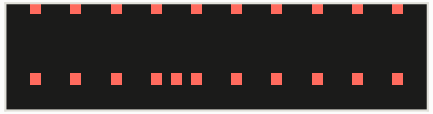
x cut at 0.385 µm



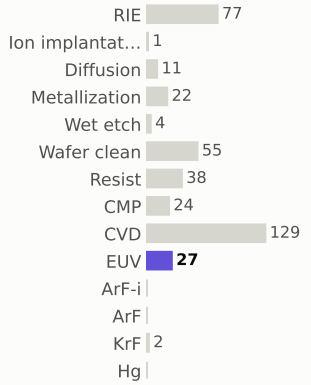
y cut at 0.049 µm



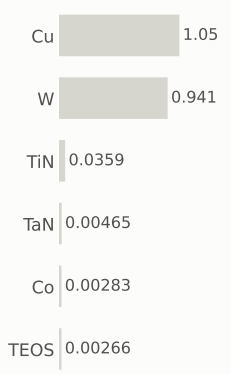
mask: layer 10321:0 (GDSII)



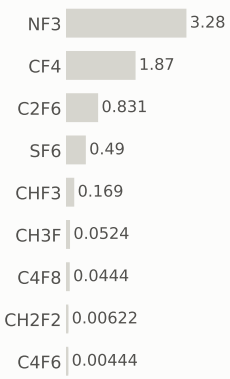
steps by process area



materials deposited (g)



gases emitted (g)



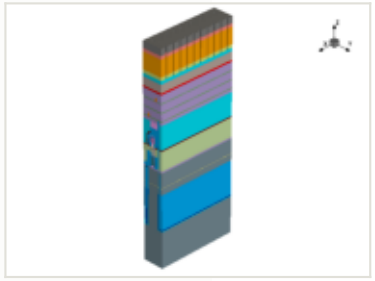
Step #4200: RIE

etch_anisotropic · RIE

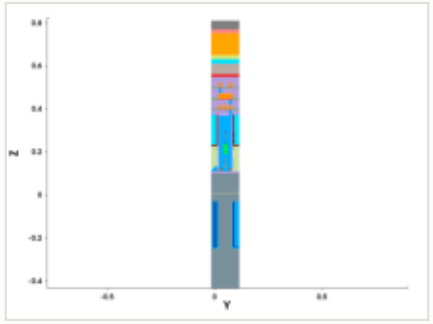
1.04	412.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

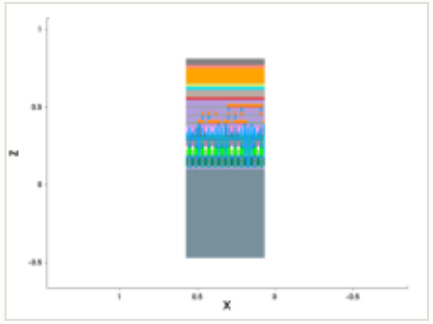
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

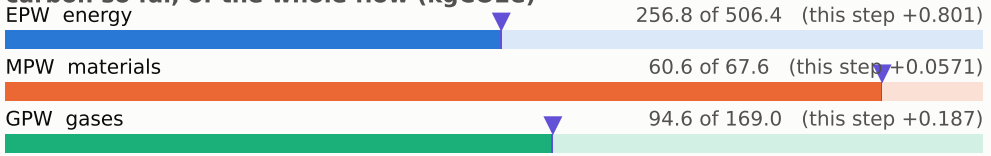
no mask for this step

this step

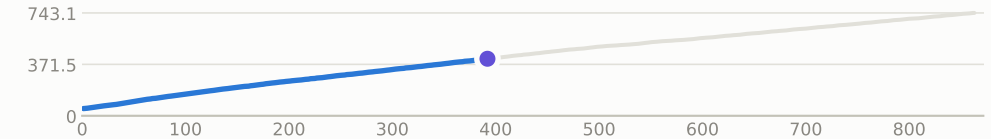
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

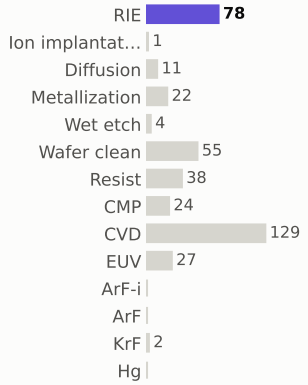
carbon so far, of the whole flow (kgCO2e)



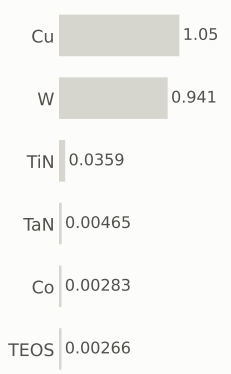
embodied carbon up to each step (kgCO2e per wafer)



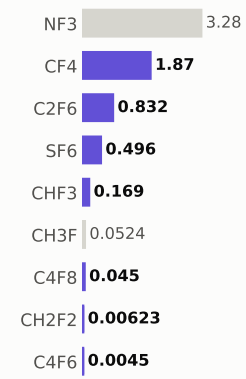
steps by process area



materials deposited (g)



gases emitted (g)



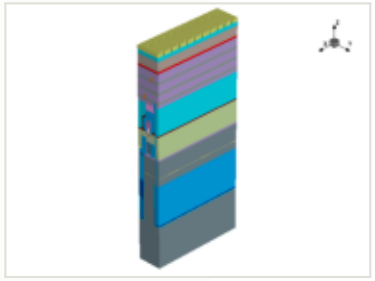
Step #4210: Wafer_clean

etch_strip · Wafer_clean

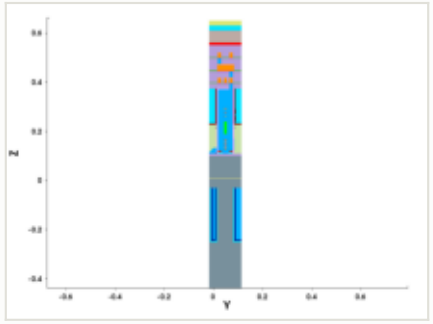
0.169	412.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

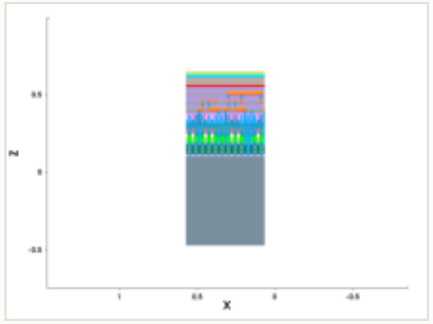
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



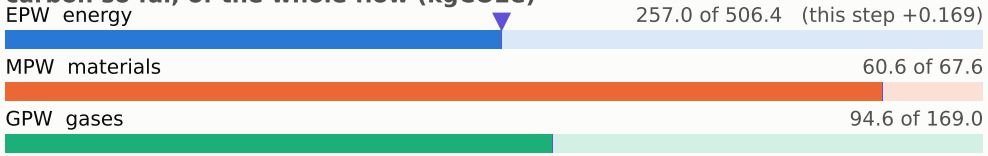
mask for this step

no mask for this step

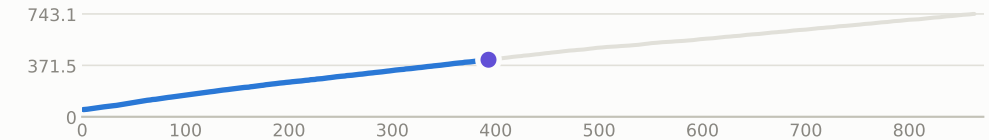
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

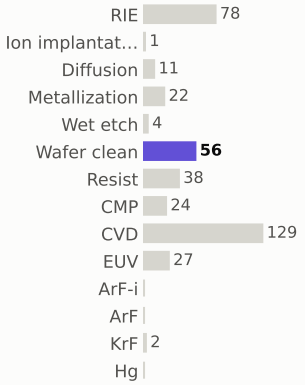
carbon so far, of the whole flow (kgCO2e)



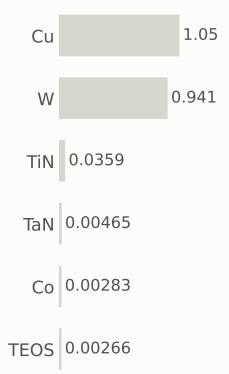
embodied carbon up to each step (kgCO2e per wafer)



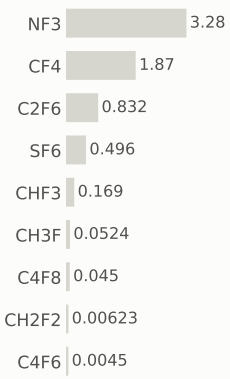
steps by process area



materials deposited (g)



gases emitted (g)



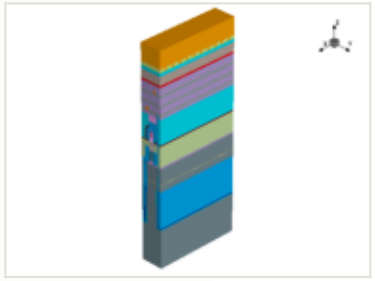
Step #4220: Resist

depo_fill · Resist

0.169	412.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

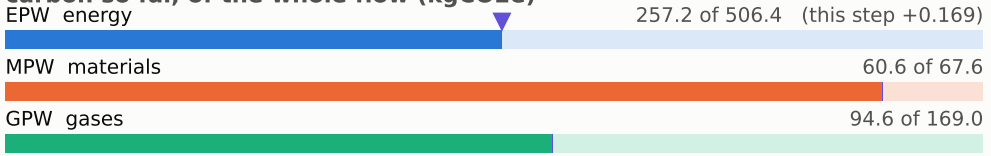


this step

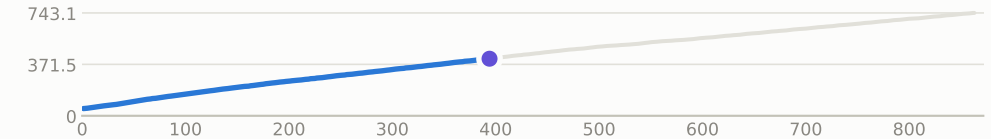
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

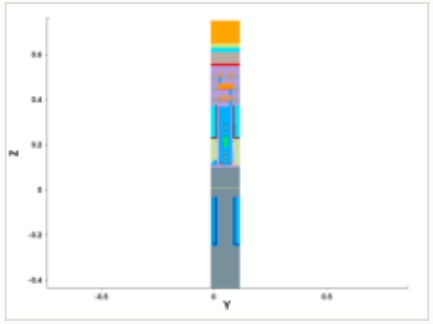
carbon so far, of the whole flow (kgCO2e)



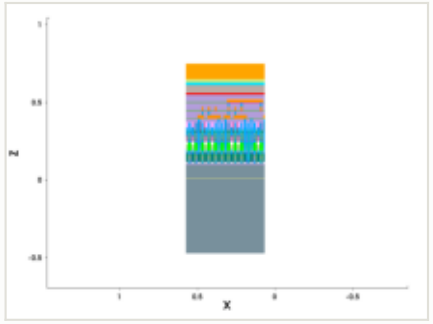
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



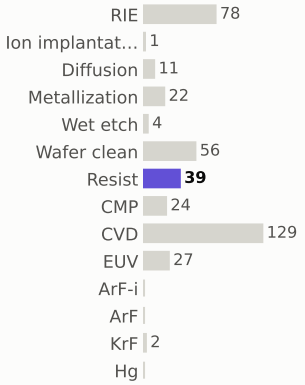
y cut at 0.049 µm



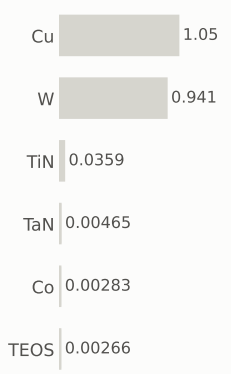
mask for this step

no mask for this step

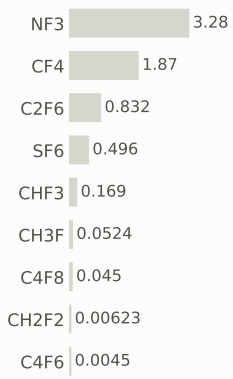
steps by process area



materials deposited (g)



gases emitted (g)



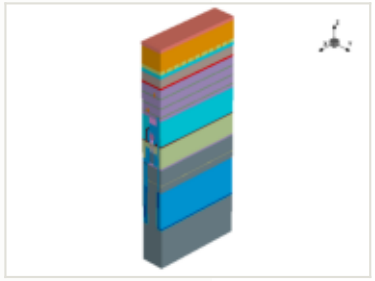
Step #4230: CVD

depo_fill · CVD

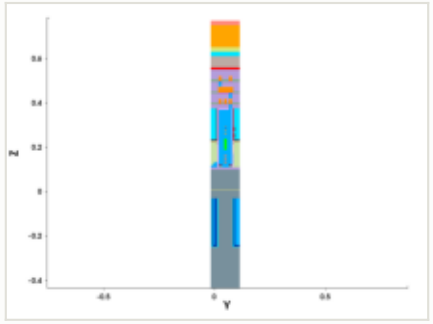
1.31	413.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

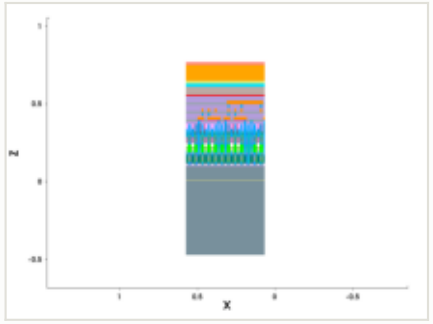
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

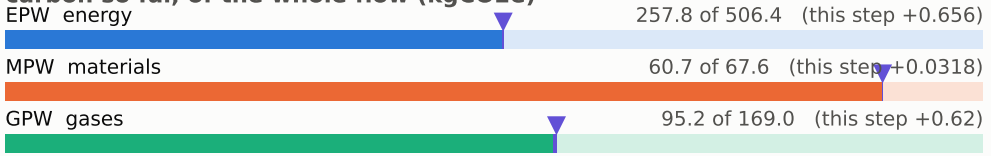
no mask for this step

this step

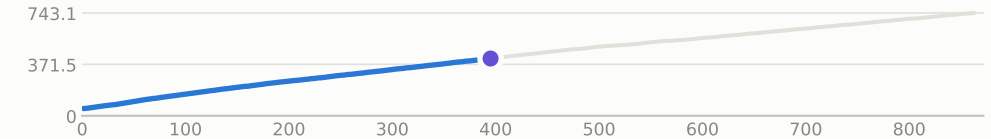
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

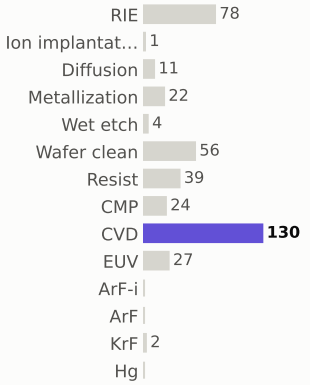
carbon so far, of the whole flow (kgCO2e)



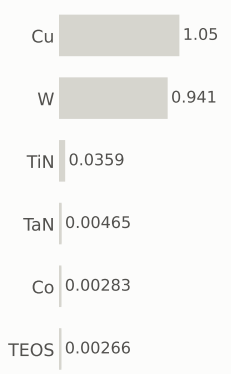
embodied carbon up to each step (kgCO2e per wafer)



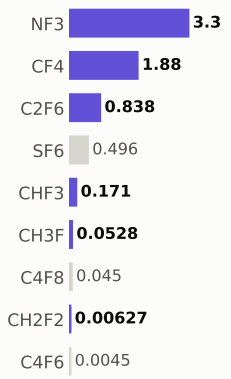
steps by process area



materials deposited (g)



gases emitted (g)



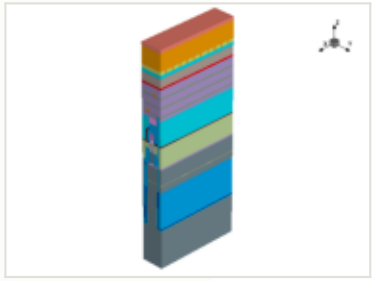
Step #4240: Wafer_clean

clean · Wafer_clean

0.169	413.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



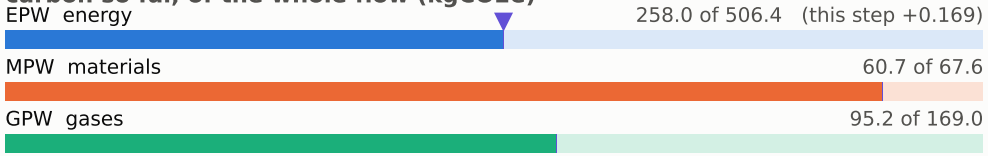
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

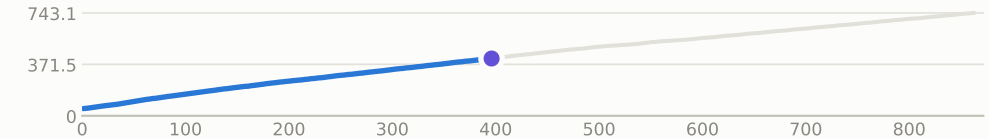
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

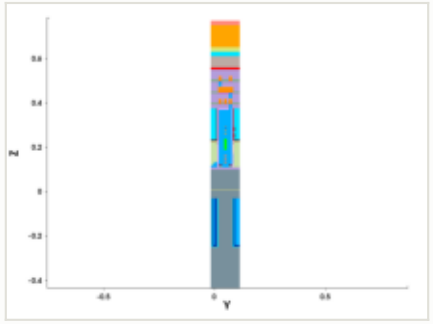
carbon so far, of the whole flow (kgCO2e)



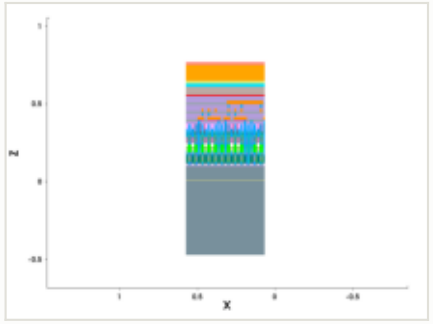
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



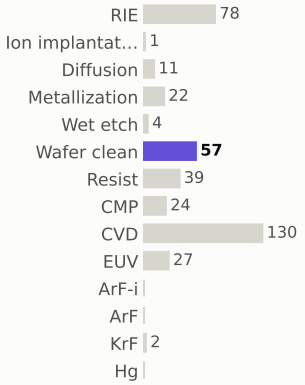
y cut at 0.049 µm



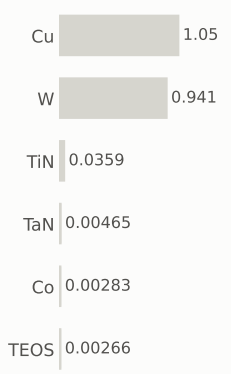
mask for this step

no mask for this step

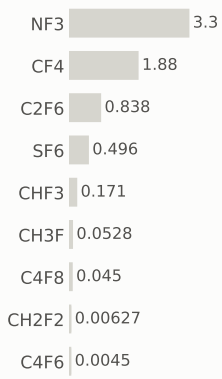
steps by process area



materials deposited (g)



gases emitted (g)



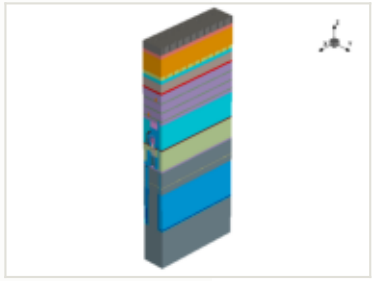
Step #4250: EUV

litho_patterning · EUV

1.96	415.8	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

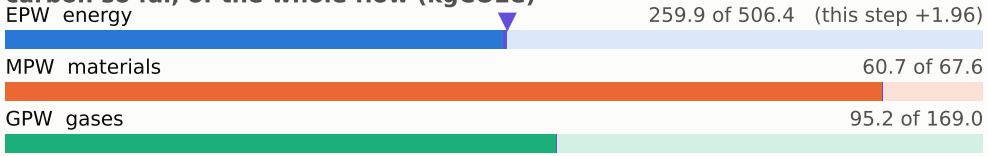


this step

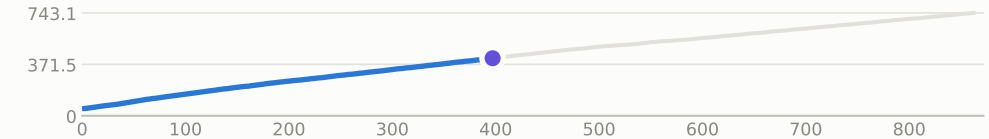
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

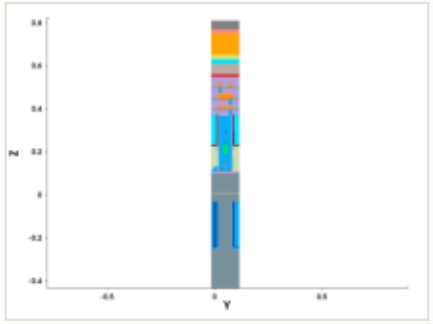
carbon so far, of the whole flow (kgCO2e)



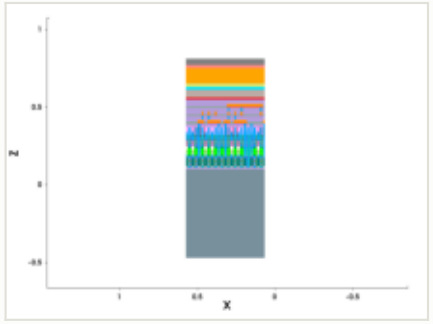
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

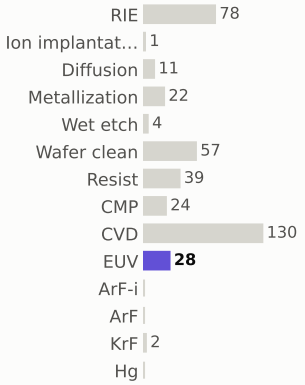


y cut at 0.049 µm

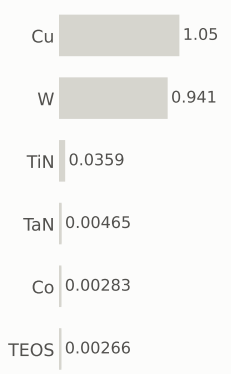


mask: layer 10322:0 (GDSII)

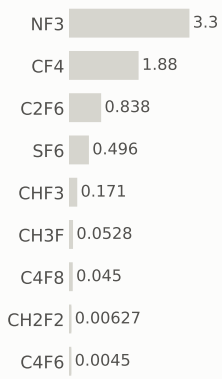
steps by process area



materials deposited (g)



gases emitted (g)



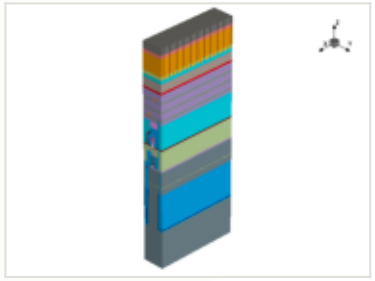
Step #4260: RIE

etch_anisotropic · RIE

1.04	416.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

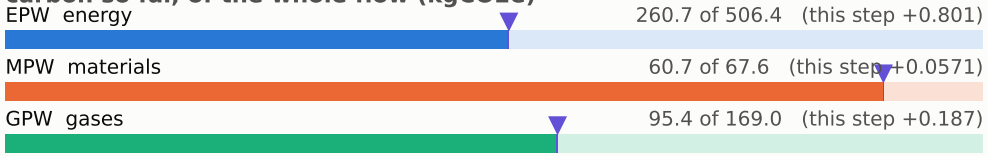


this step

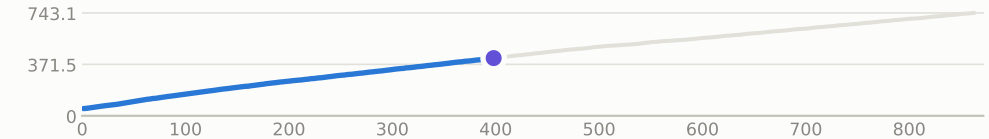
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

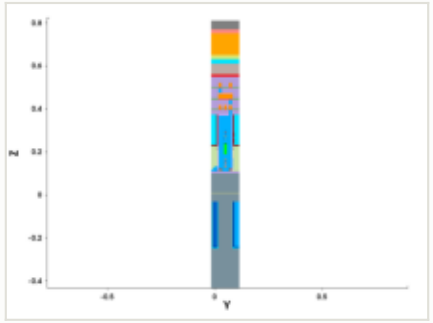
carbon so far, of the whole flow (kgCO2e)



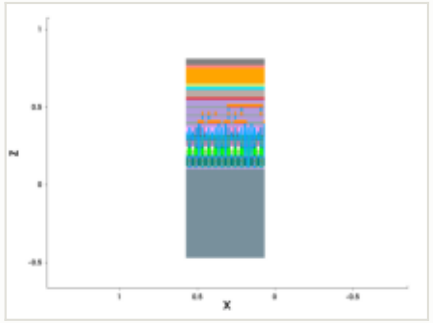
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



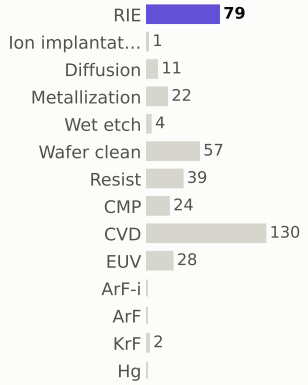
y cut at 0.049 µm



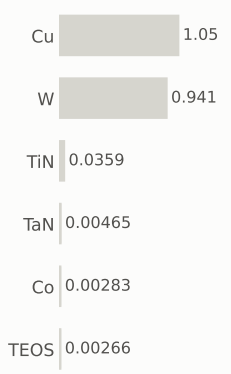
mask for this step

no mask for this step

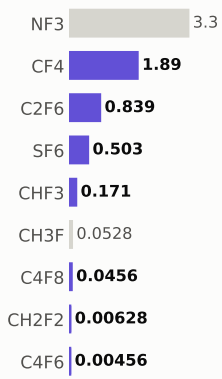
steps by process area



materials deposited (g)



gases emitted (g)



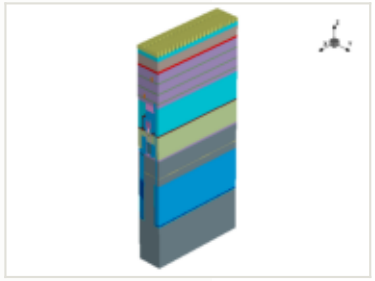
Step #4270: Wafer_clean

etch_strip · Wafer_clean

0.169	417.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

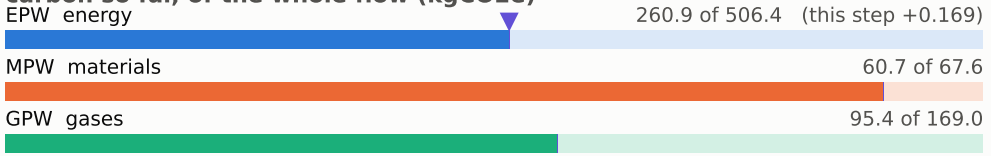


this step

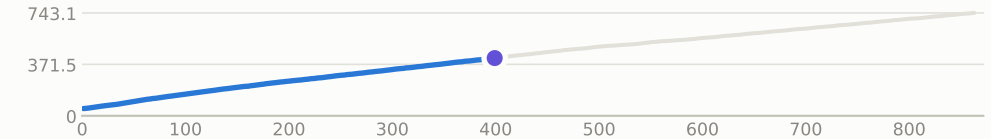
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

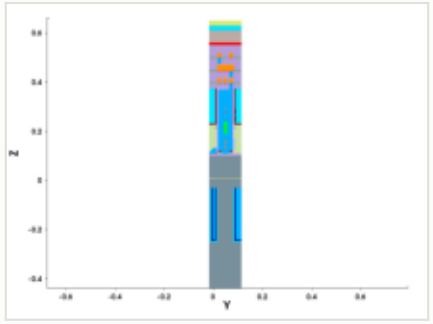
carbon so far, of the whole flow (kgCO2e)



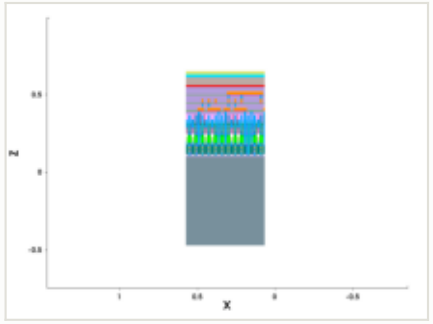
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



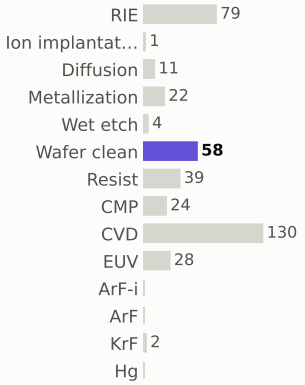
y cut at 0.049 µm



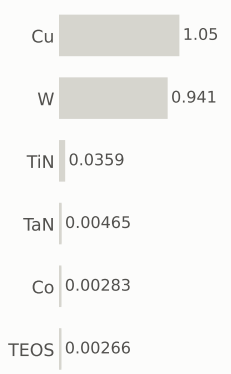
mask for this step

no mask for this step

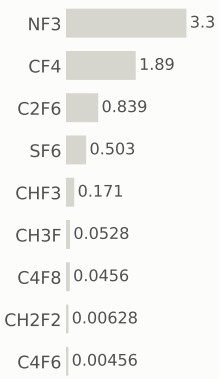
steps by process area



materials deposited (g)



gases emitted (g)



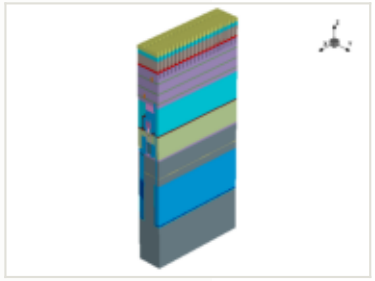
Step #4280: RIE

etch_anisotropic · RIE

1.04	418.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

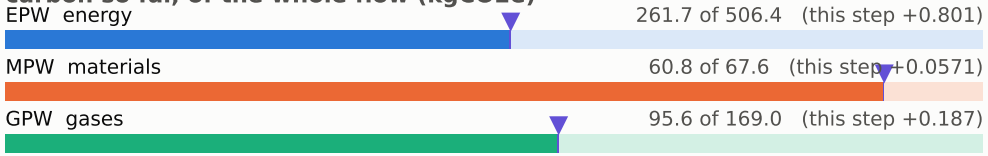


this step

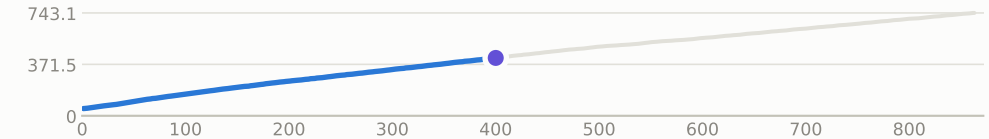
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

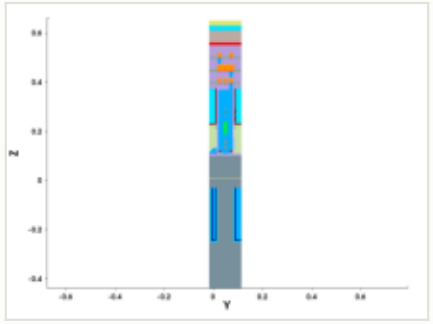
carbon so far, of the whole flow (kgCO2e)



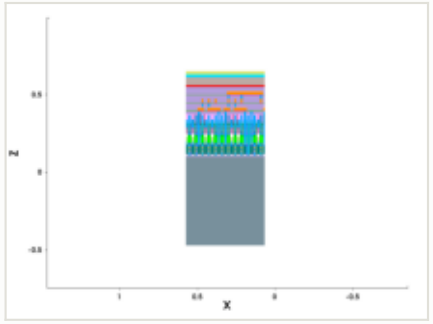
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



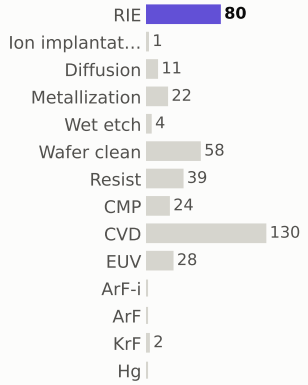
y cut at 0.049 µm



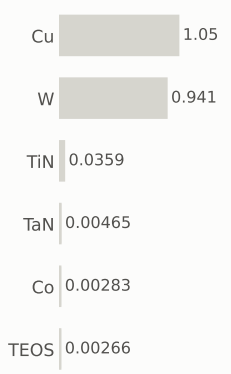
mask for this step

no mask for this step

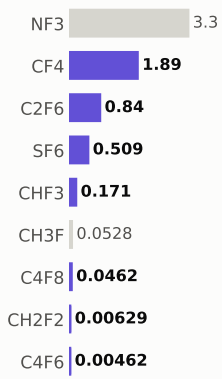
steps by process area



materials deposited (g)



gases emitted (g)



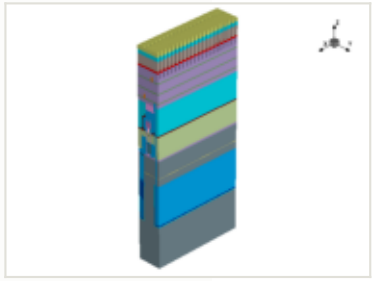
Step #4290: Wafer_clean

clean · Wafer_clean

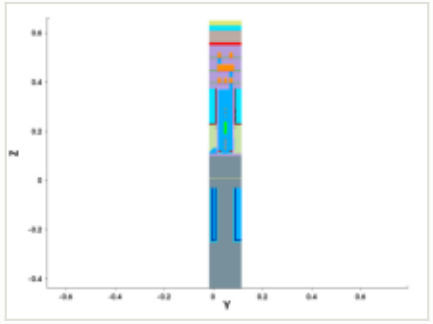
0.169	418.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

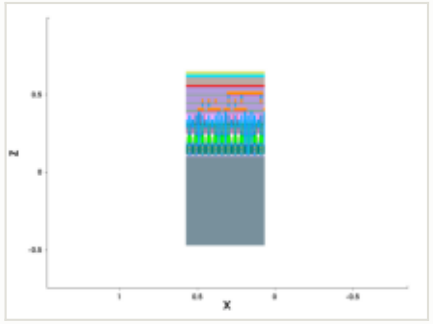
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

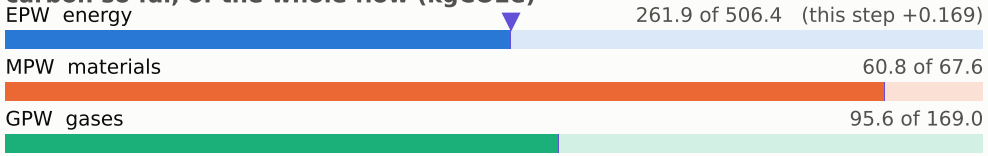
no mask for this step

this step

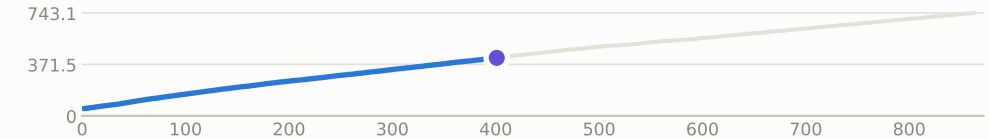
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

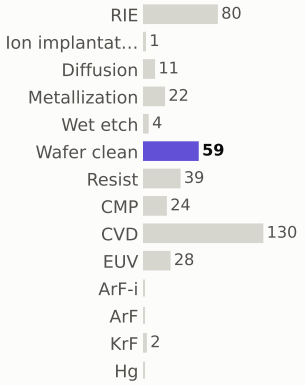
carbon so far, of the whole flow (kgCO2e)



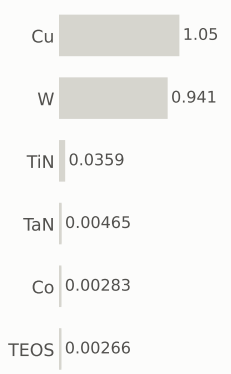
embodied carbon up to each step (kgCO2e per wafer)



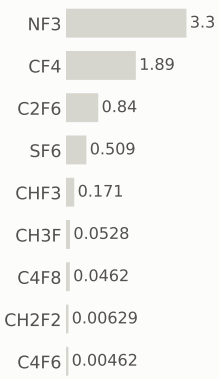
steps by process area



materials deposited (g)



gases emitted (g)



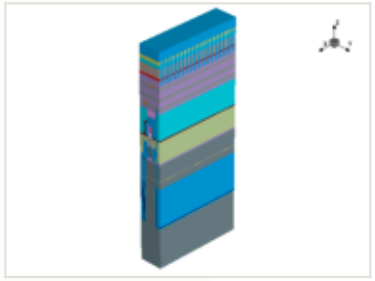
Step #4300: Metallization

depo_fill · Metallization

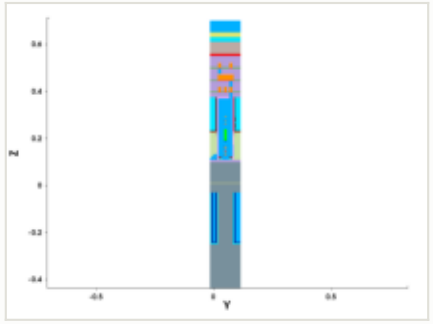
1.21	419.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

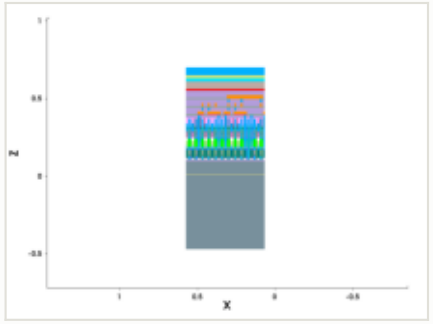
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

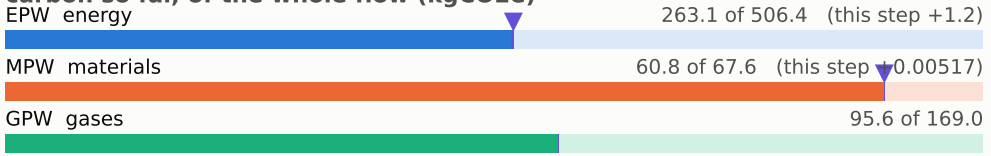
no mask for this step

this step

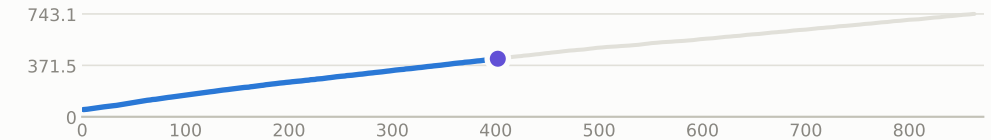
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

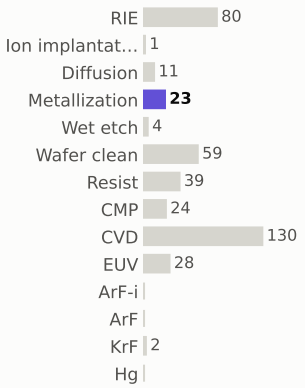
carbon so far, of the whole flow (kgCO2e)



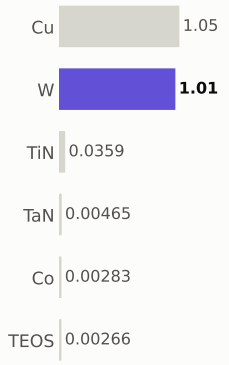
embodied carbon up to each step (kgCO2e per wafer)



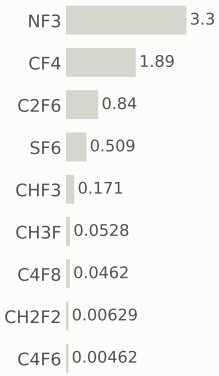
steps by process area



materials deposited (g)



gases emitted (g)



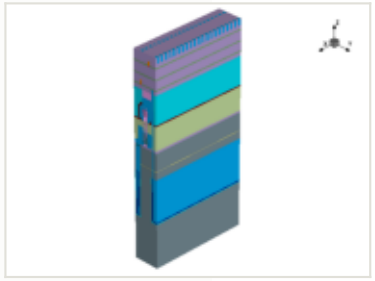
Step #4310: CMP

etch_cmp · CMP

0.169	419.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

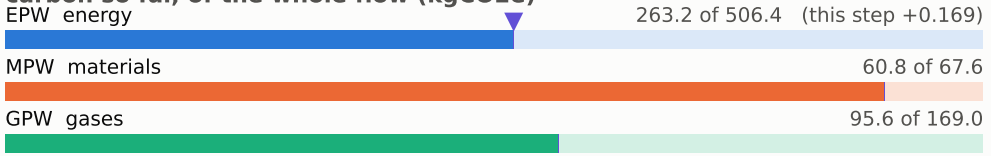


this step

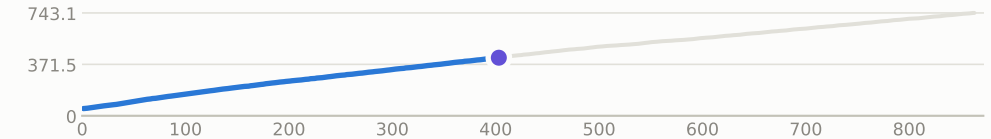
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

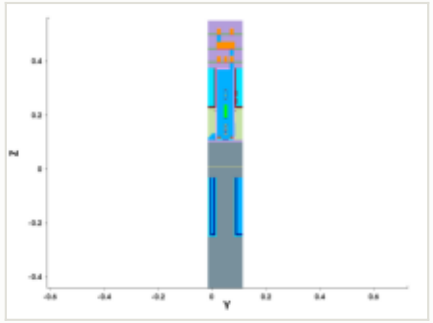
carbon so far, of the whole flow (kgCO2e)



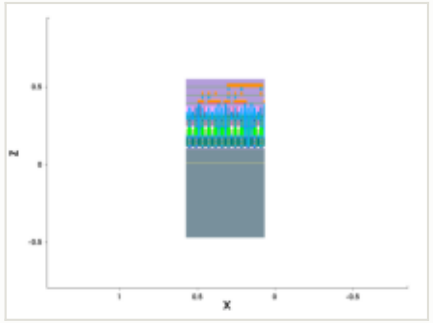
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



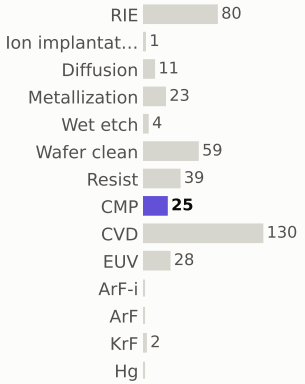
y cut at 0.049 µm



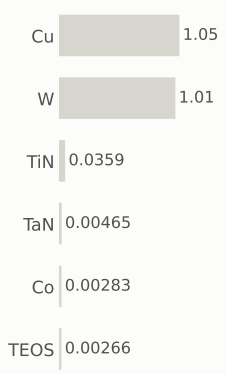
mask for this step

no mask for this step

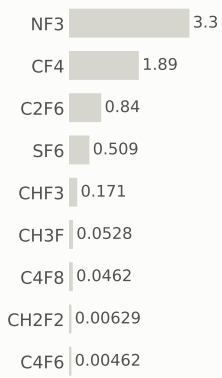
steps by process area



materials deposited (g)



gases emitted (g)



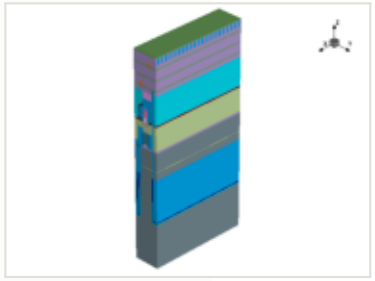
Step #4320: CVD

depo_fill · CVD

1.31	421.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

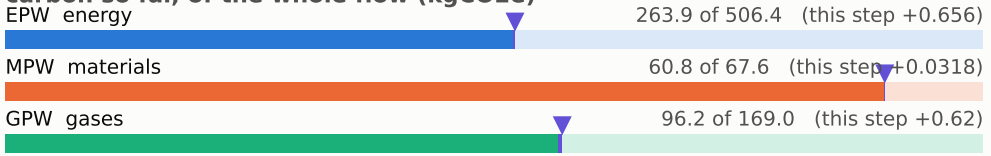


this step

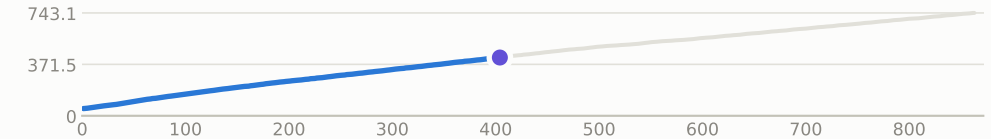
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

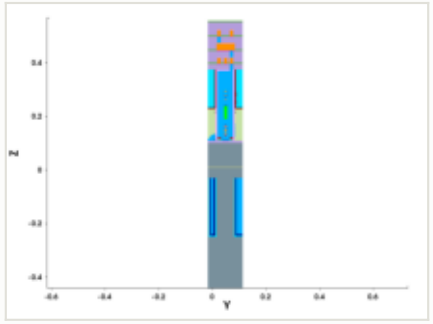
carbon so far, of the whole flow (kgCO2e)



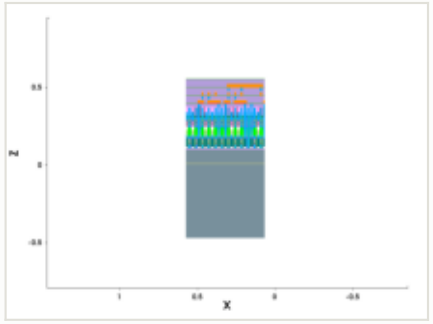
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



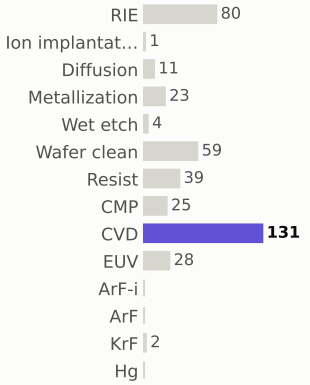
y cut at 0.049 µm



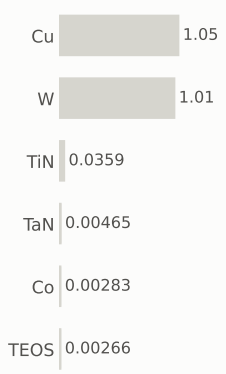
mask for this step

no mask for this step

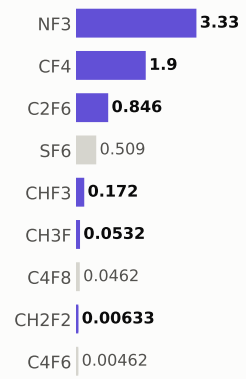
steps by process area



materials deposited (g)



gases emitted (g)



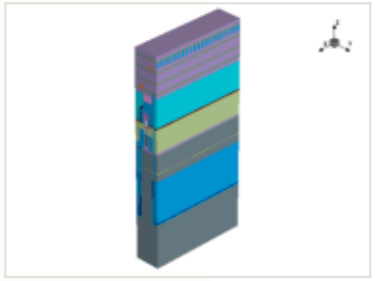
Step #4330: CVD

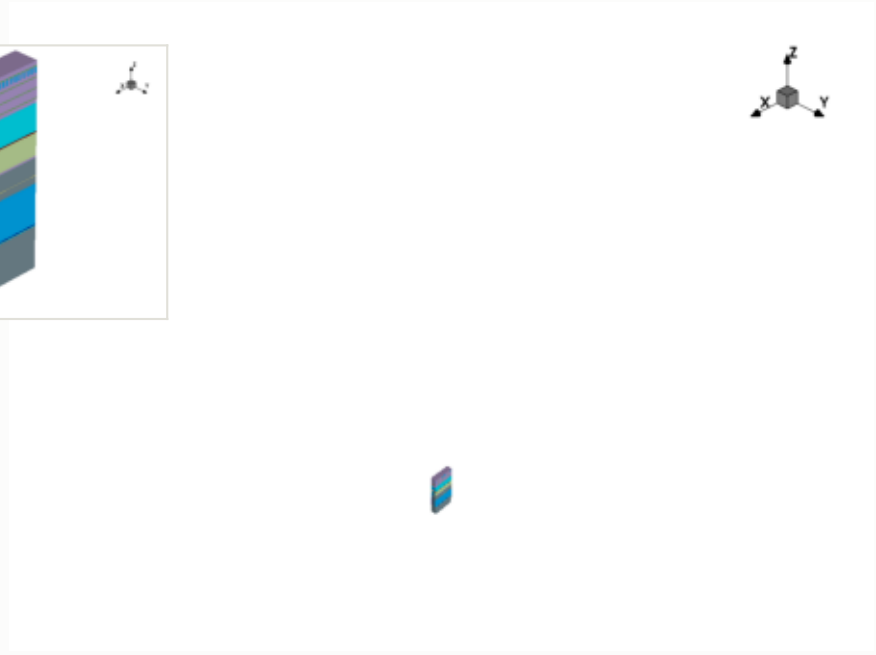
depo_fill · CVD

1.31	422.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



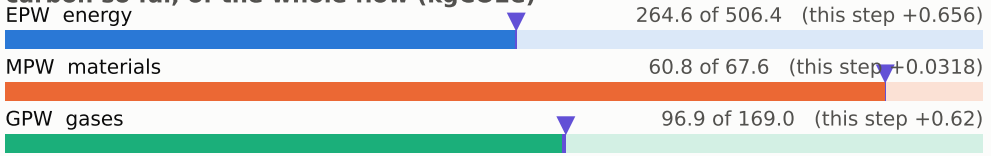


this step

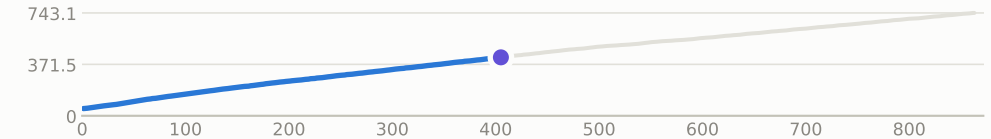
process area	CVD
material	—
thickness	0.022 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

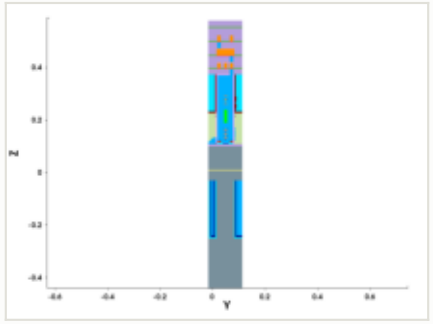
carbon so far, of the whole flow (kgCO2e)



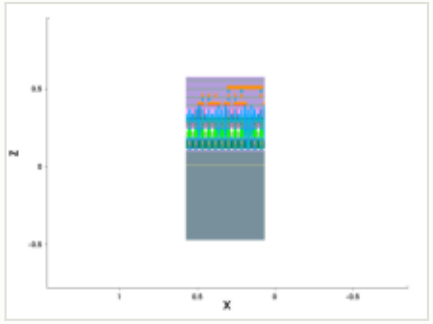
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



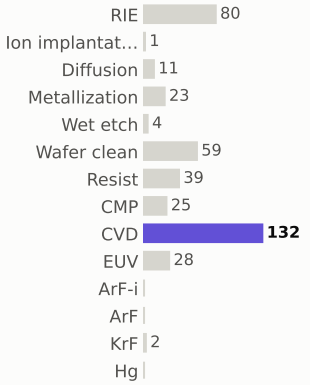
y cut at 0.049 µm



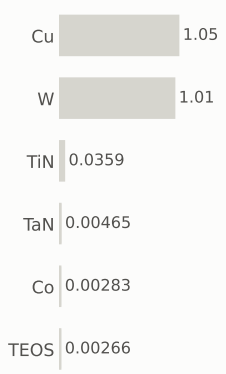
mask for this step

no mask for this step

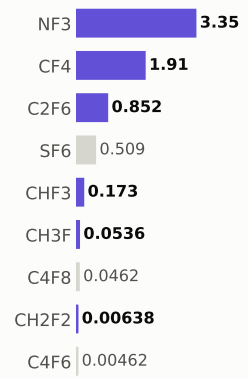
steps by process area



materials deposited (g)



gases emitted (g)



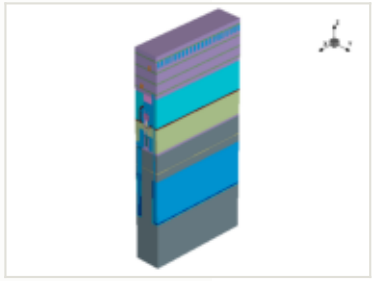
Step #4340: Diffusion

treatment · Diffusion

0.801	423.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



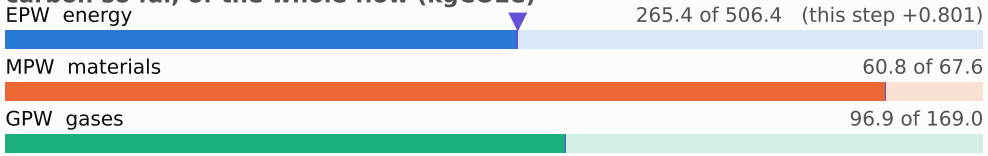
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

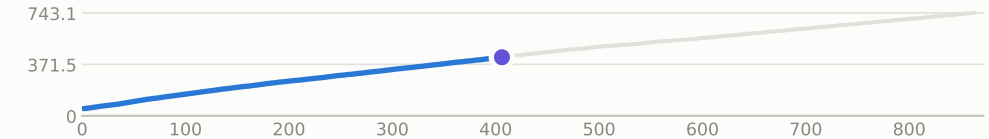
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

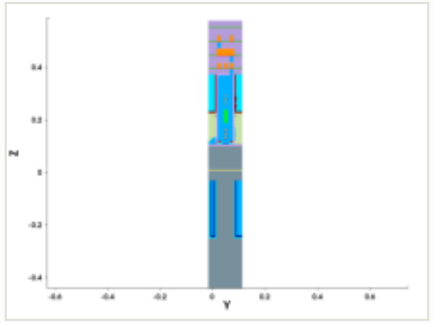
carbon so far, of the whole flow (kgCO2e)



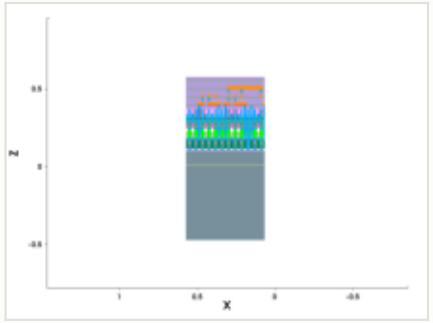
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



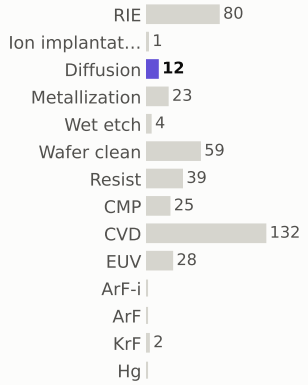
y cut at 0.049 µm



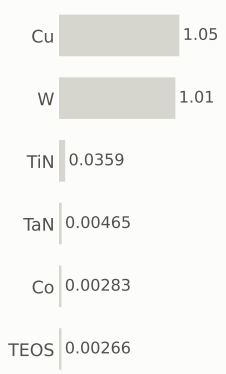
mask for this step

no mask for this step

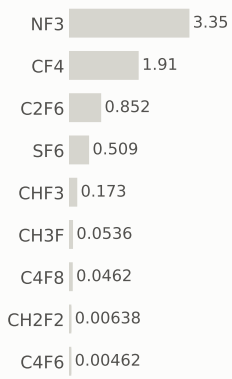
steps by process area



materials deposited (g)



gases emitted (g)



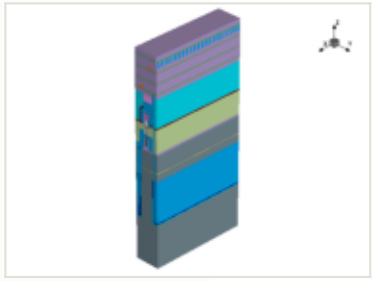
Step #4350: Wafer_clean

clean · Wafer_clean

0.169	423.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



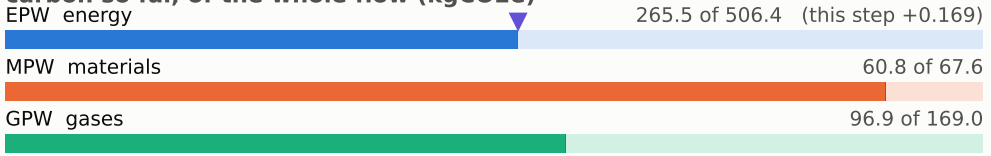
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

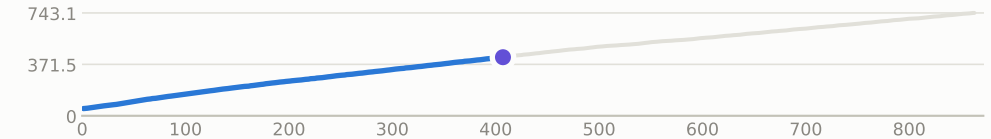
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

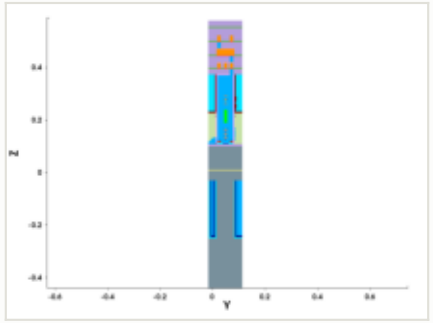
carbon so far, of the whole flow (kgCO2e)



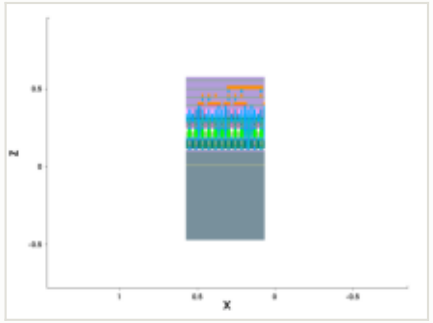
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



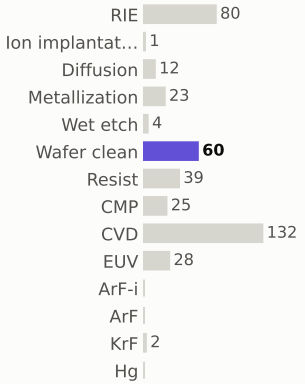
y cut at 0.049 µm



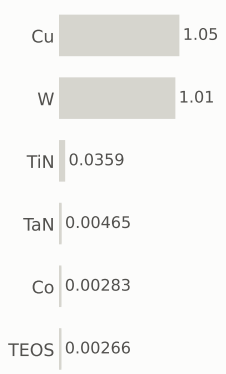
mask for this step

no mask for this step

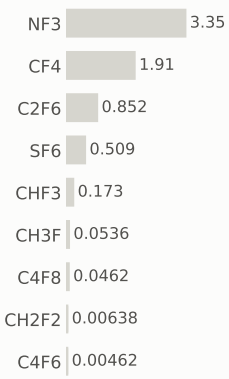
steps by process area



materials deposited (g)



gases emitted (g)



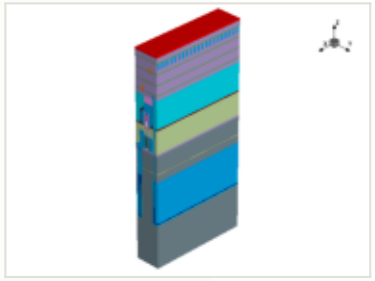
Step #4360: CVD

depo_fill · CVD

1.31	424.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

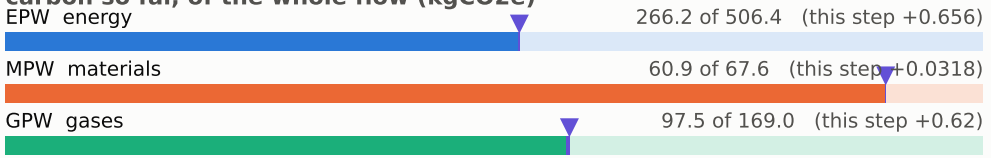


this step

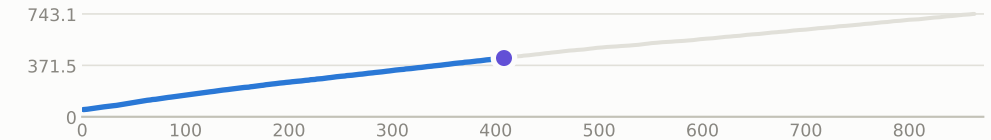
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

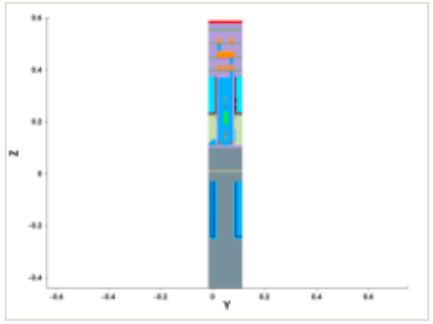
carbon so far, of the whole flow (kgCO2e)



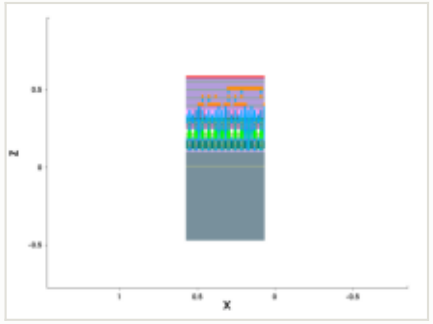
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



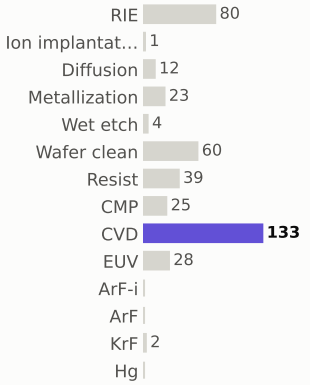
y cut at 0.049 µm



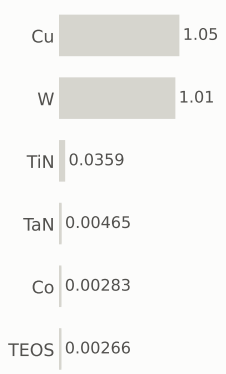
mask for this step

no mask for this step

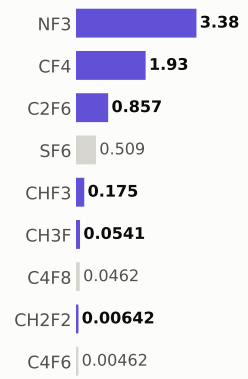
steps by process area



materials deposited (g)



gases emitted (g)



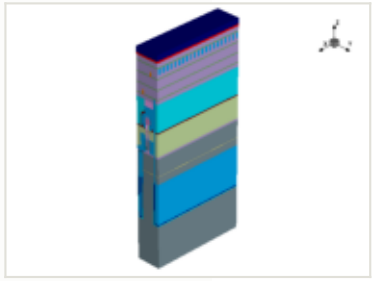
Step #4370: CVD

depo_fill · CVD

1.31	425.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



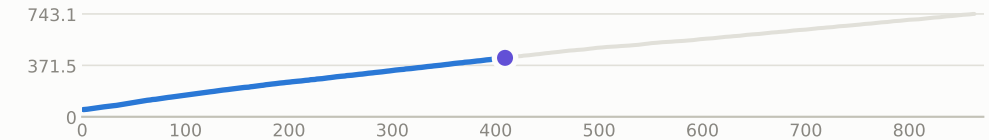
this step

process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

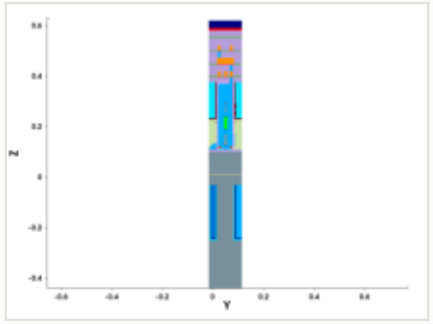
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

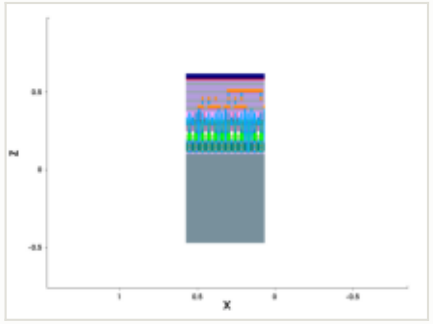
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



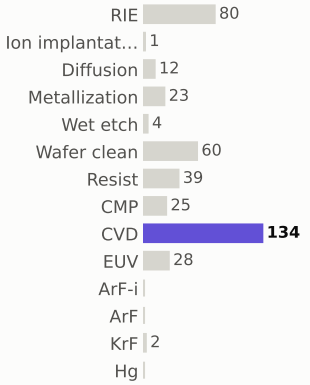
y cut at 0.049 µm



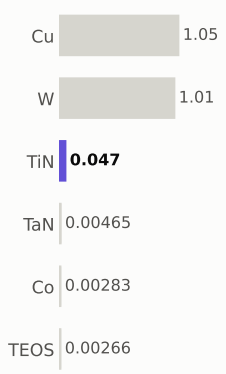
mask for this step

no mask for this step

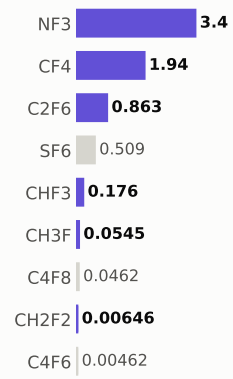
steps by process area



materials deposited (g)



gases emitted (g)



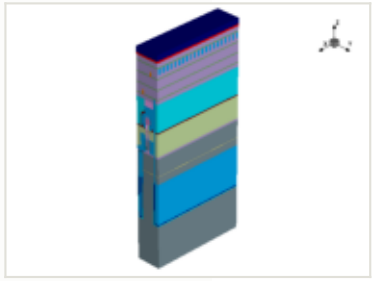
Step #4380: Wafer_clean

clean · Wafer_clean

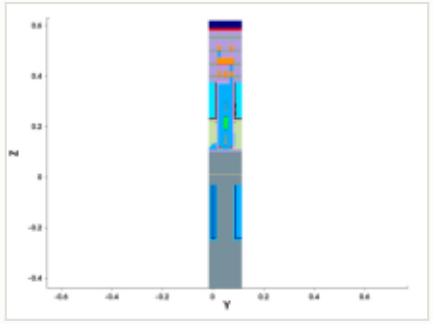
0.169	426.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

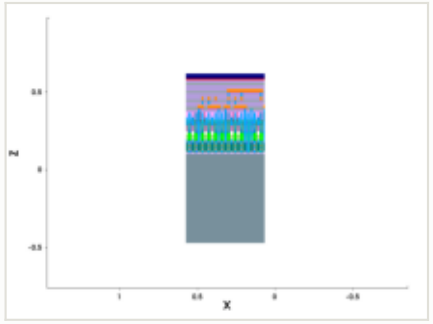
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

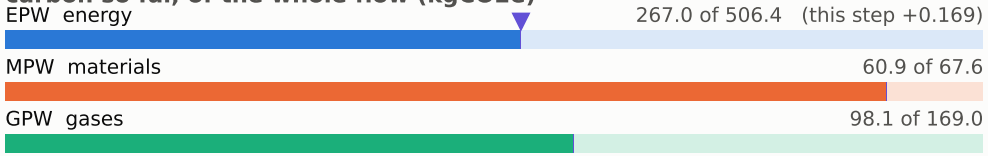
no mask for this step

this step

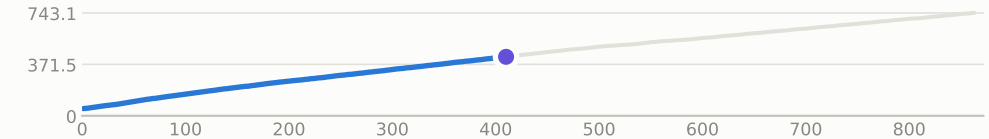
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

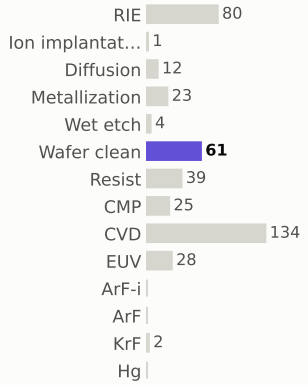
carbon so far, of the whole flow (kgCO2e)



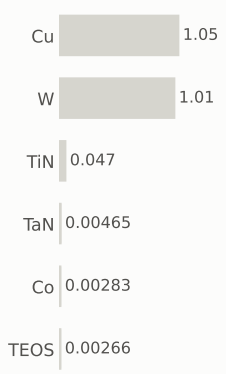
embodied carbon up to each step (kgCO2e per wafer)



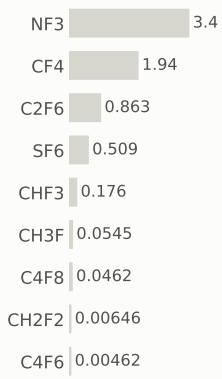
steps by process area



materials deposited (g)



gases emitted (g)



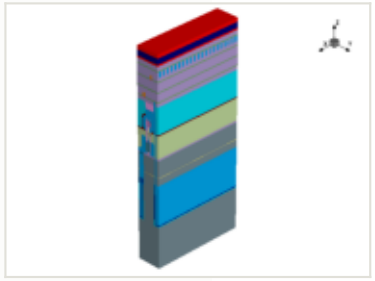
Step #4390: CVD

depo_fill · CVD

1.31	427.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

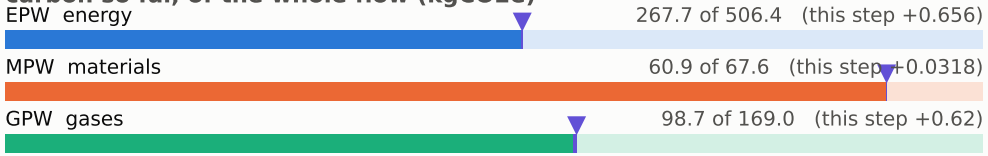


this step

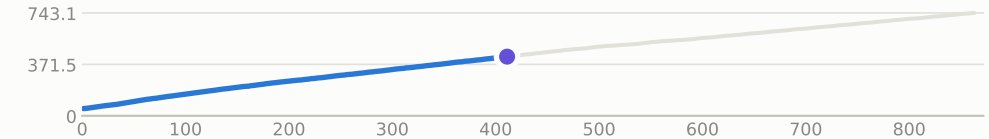
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

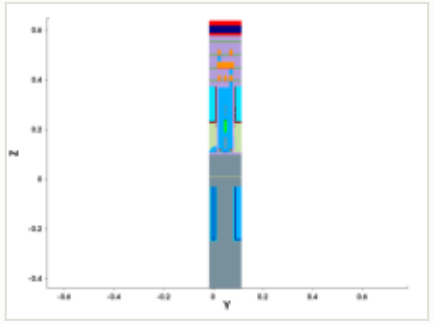
carbon so far, of the whole flow (kgCO2e)



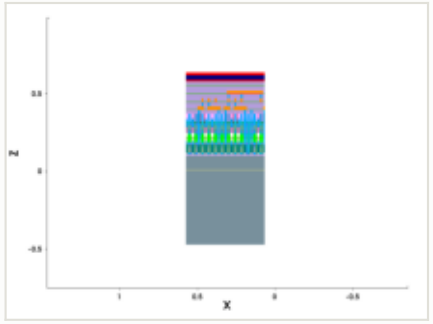
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



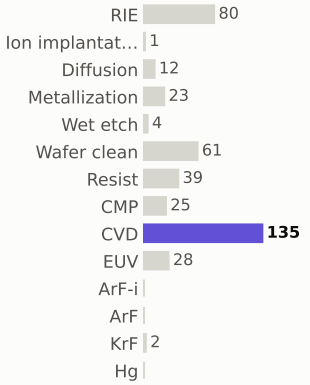
y cut at 0.049 µm



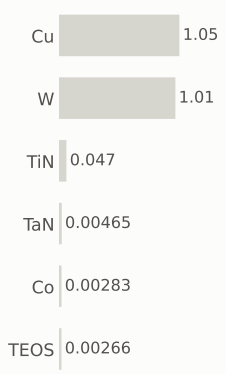
mask for this step

no mask for this step

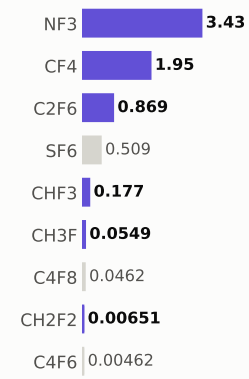
steps by process area



materials deposited (g)



gases emitted (g)



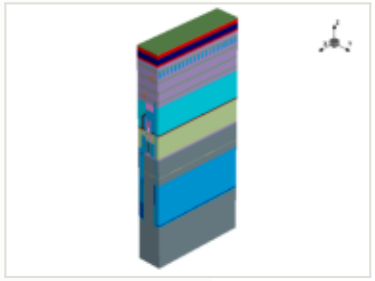
Step #4400: CVD

depo_fill · CVD

1.31	428.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

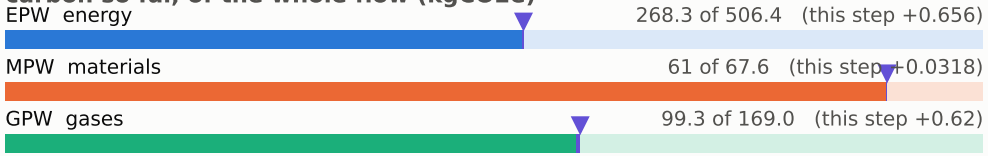


this step

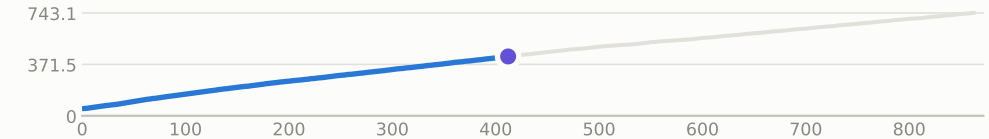
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

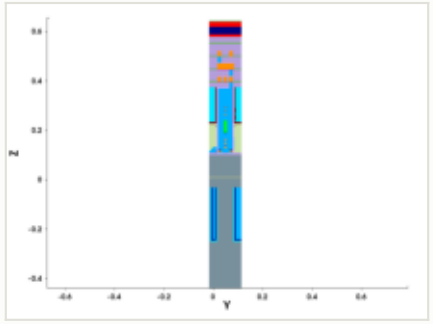
carbon so far, of the whole flow (kgCO2e)



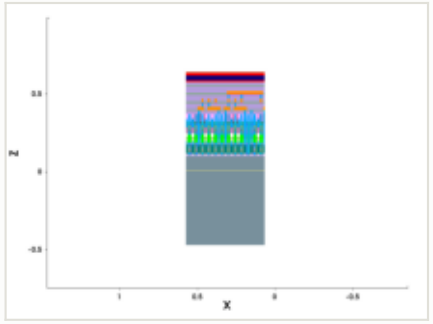
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



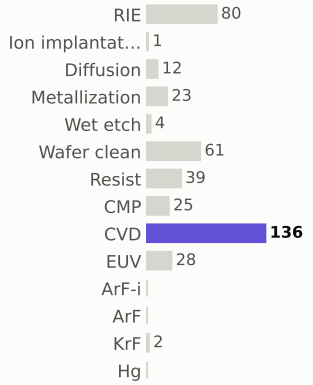
y cut at 0.049 µm



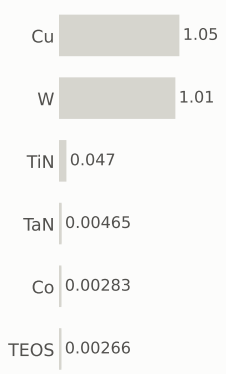
mask for this step

no mask for this step

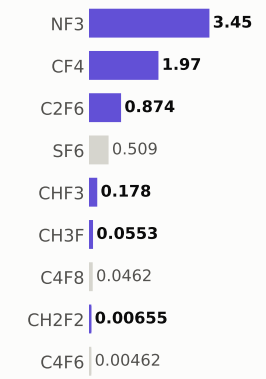
steps by process area



materials deposited (g)



gases emitted (g)



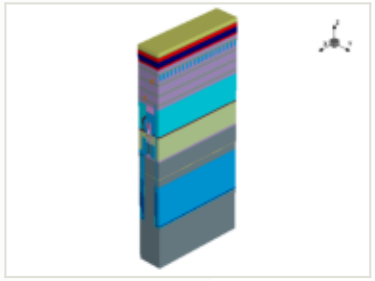
Step #4410: CVD

depo_fill · CVD

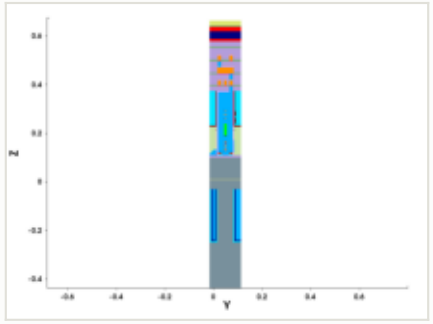
1.31	429.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

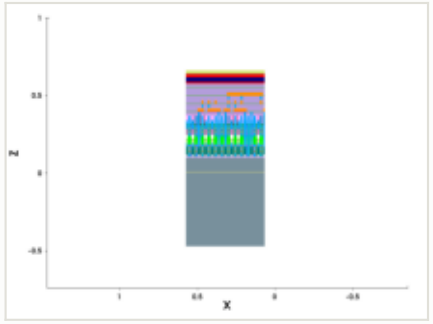
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

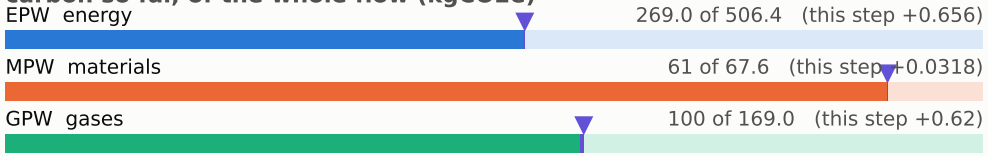
no mask for this step

this step

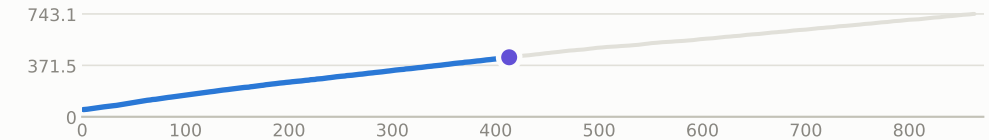
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

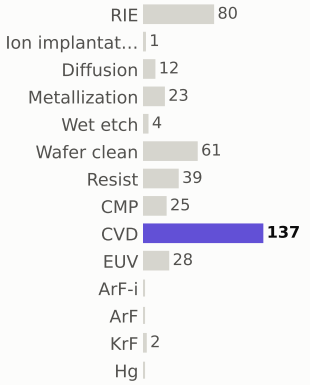
carbon so far, of the whole flow (kgCO2e)



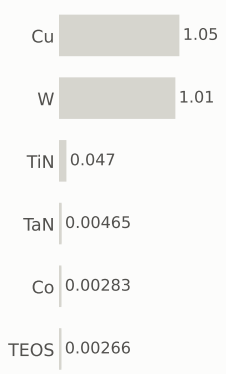
embodied carbon up to each step (kgCO2e per wafer)



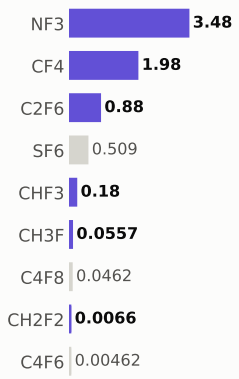
steps by process area



materials deposited (g)



gases emitted (g)



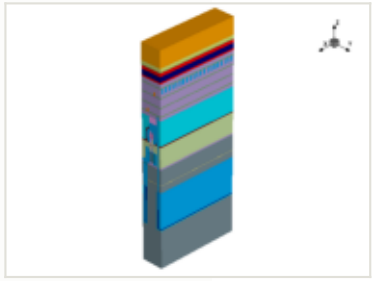
Step #4420: Resist

depo_fill · Resist

0.169	430.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

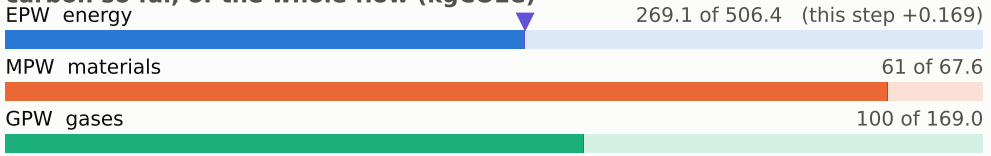


this step

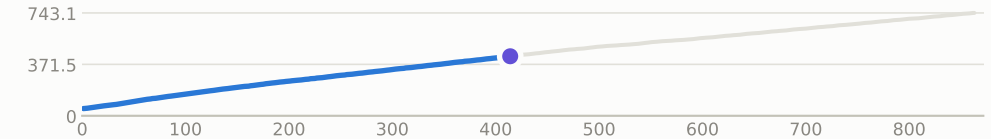
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

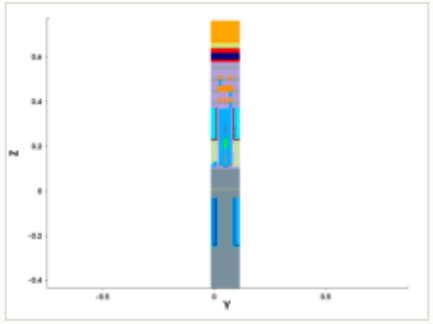
carbon so far, of the whole flow (kgCO2e)



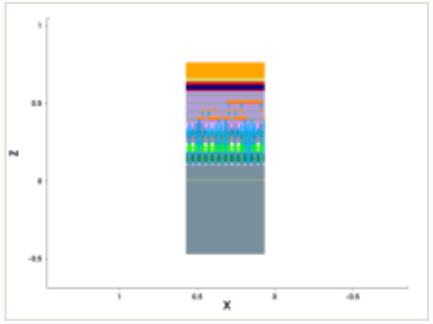
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



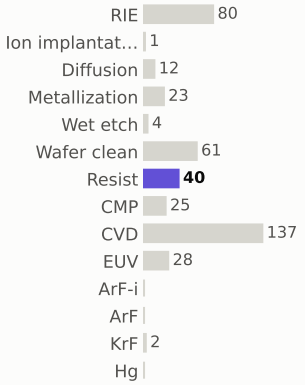
y cut at 0.049 µm



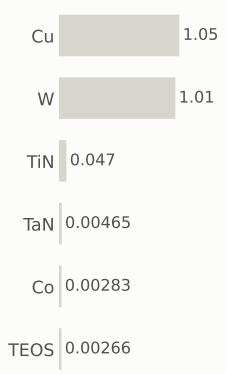
mask for this step

no mask for this step

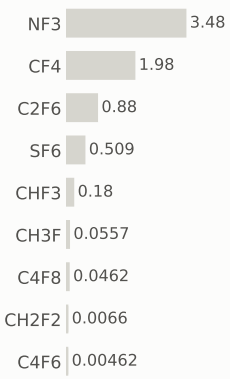
steps by process area



materials deposited (g)



gases emitted (g)



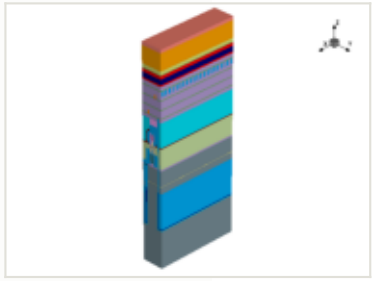
Step #4430: CVD

depo_fill · CVD

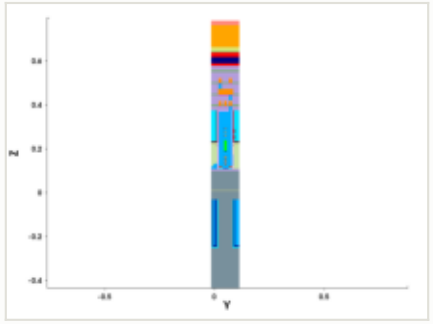
1.31	431.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

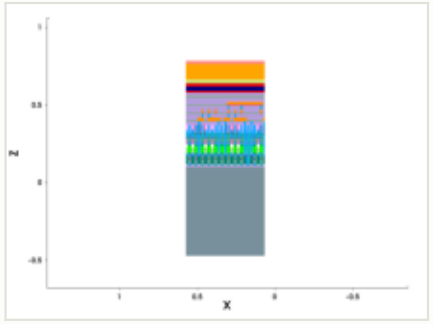
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

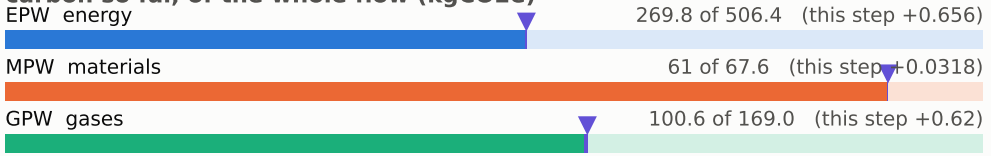
no mask for this step

this step

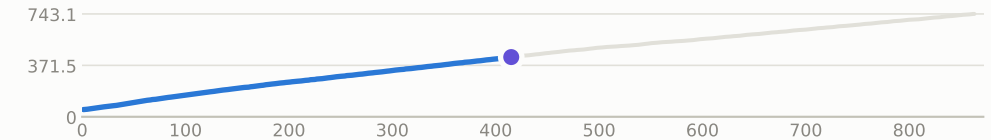
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

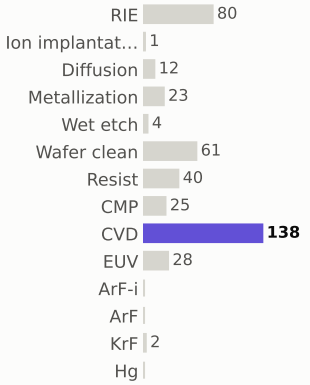
carbon so far, of the whole flow (kgCO2e)



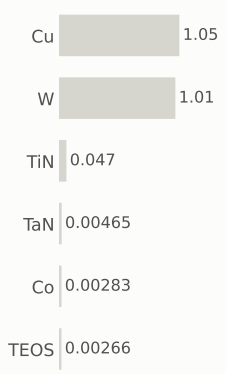
embodied carbon up to each step (kgCO2e per wafer)



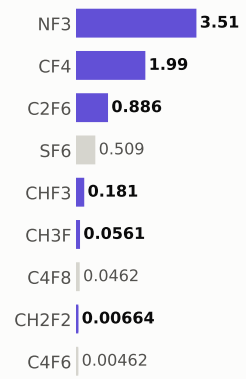
steps by process area



materials deposited (g)



gases emitted (g)



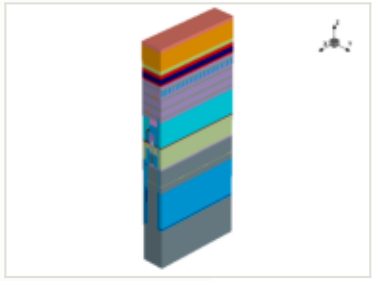
Step #4440: Wafer_clean

clean · Wafer_clean

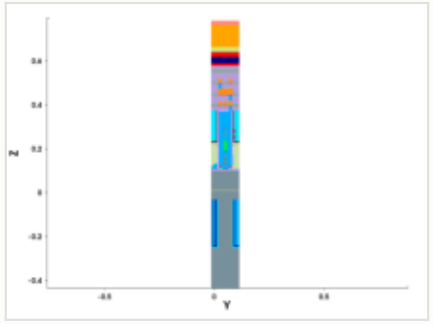
0.169	431.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

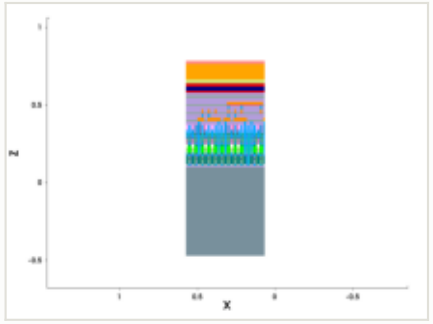
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

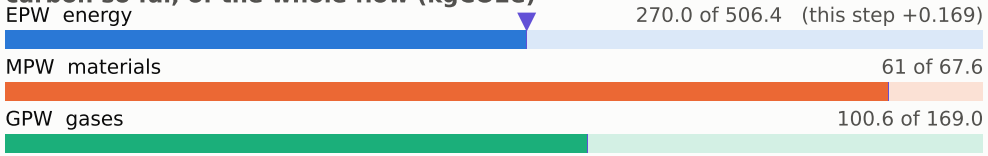
no mask for this step

this step

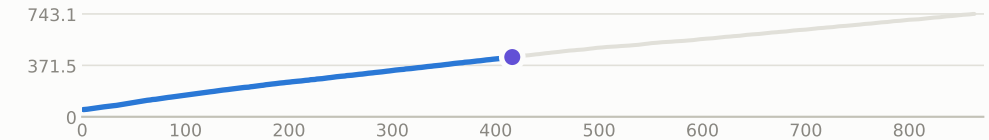
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

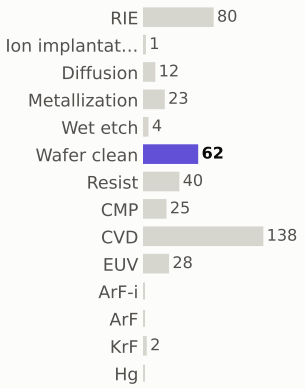
carbon so far, of the whole flow (kgCO2e)



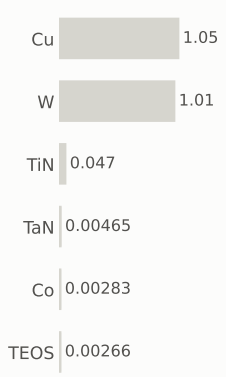
embodied carbon up to each step (kgCO2e per wafer)



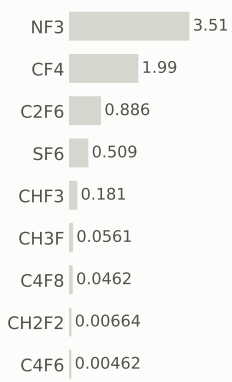
steps by process area



materials deposited (g)



gases emitted (g)



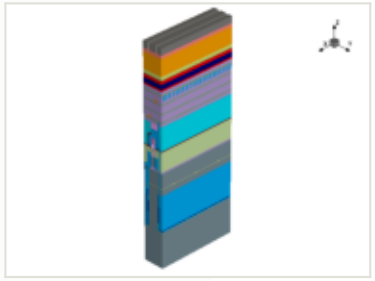
Step #4450: EUV

litho_patterning · EUV

1.96	433.5	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

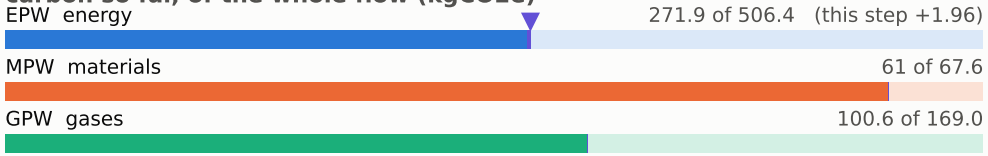


this step

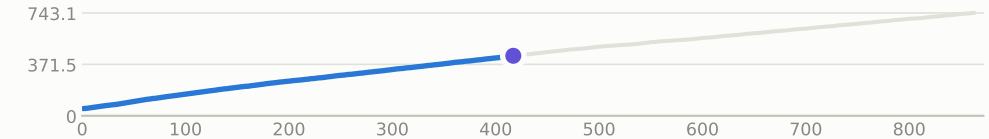
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

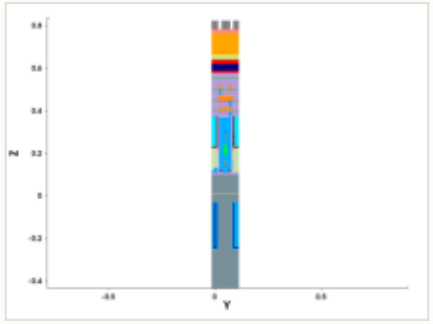
carbon so far, of the whole flow (kgCO2e)



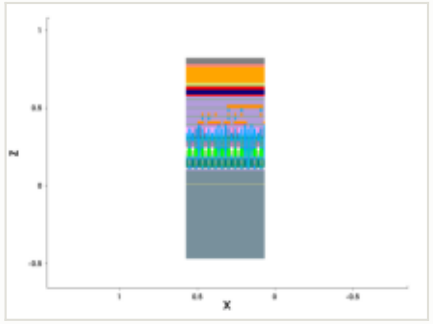
embodied carbon up to each step (kgCO2e per wafer)



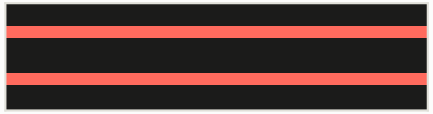
x cut at 0.385 µm



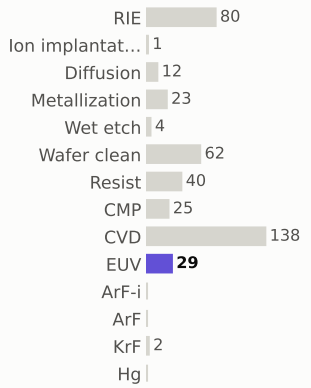
y cut at 0.049 µm



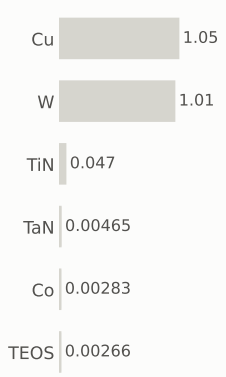
mask: layer 10351:0 (GDSII)



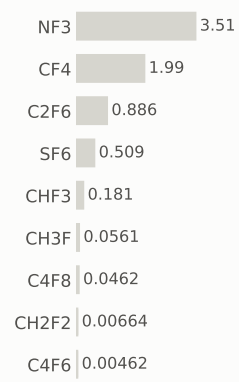
steps by process area



materials deposited (g)



gases emitted (g)



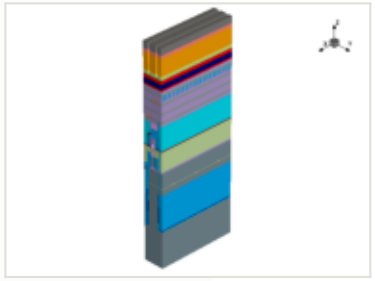
Step #4460: RIE

etch_anisotropic · RIE

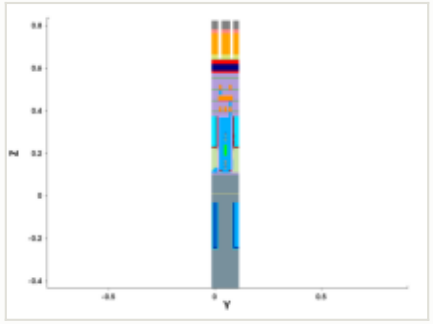
1.04	434.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

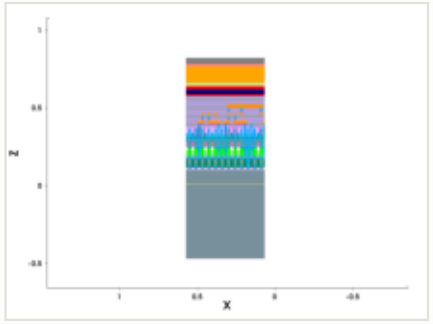
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

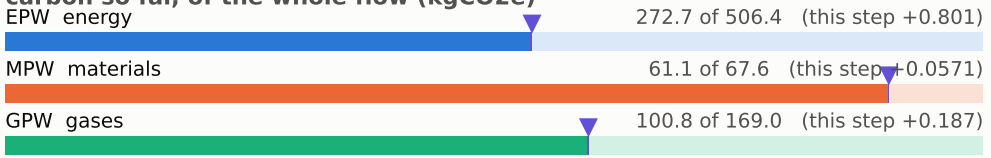
no mask for this step

this step

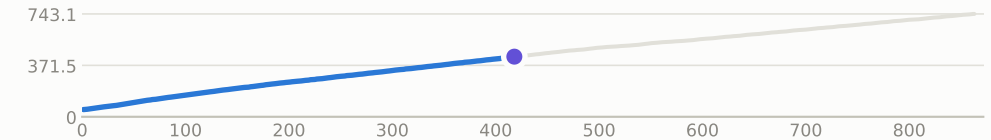
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

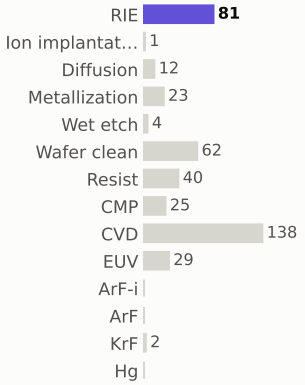
carbon so far, of the whole flow (kgCO2e)



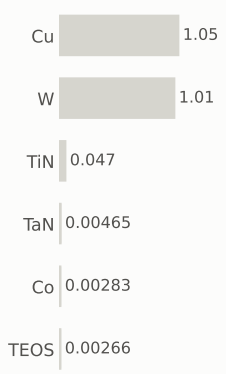
embodied carbon up to each step (kgCO2e per wafer)



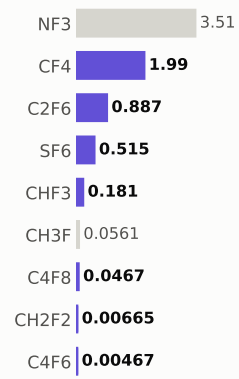
steps by process area



materials deposited (g)



gases emitted (g)



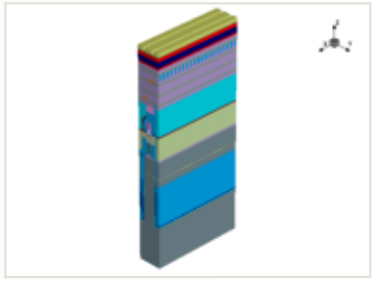
Step #4470: Wafer_clean

etch_strip · Wafer_clean

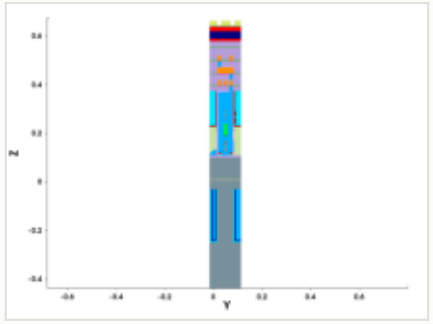
0.169	434.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

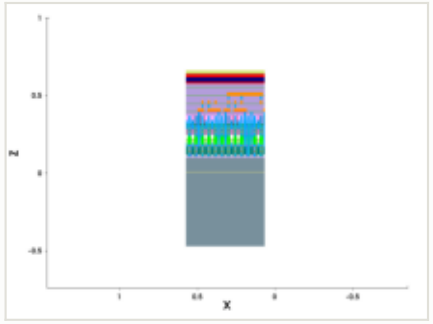
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



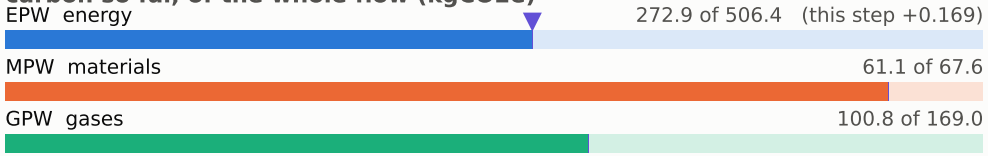
mask for this step

no mask for this step

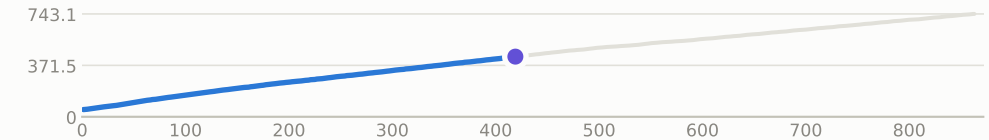
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

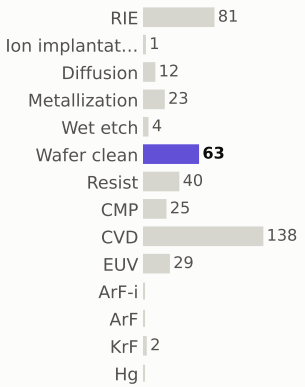
carbon so far, of the whole flow (kgCO2e)



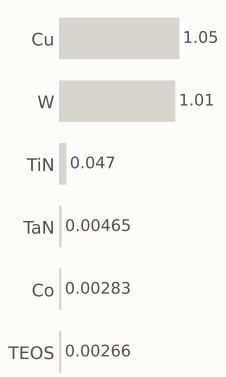
embodied carbon up to each step (kgCO2e per wafer)



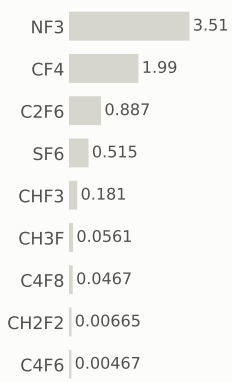
steps by process area



materials deposited (g)



gases emitted (g)



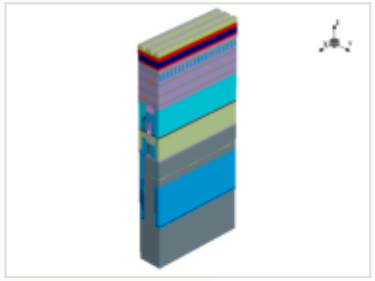
Step #4480: CVD

depo_isotropic · CVD

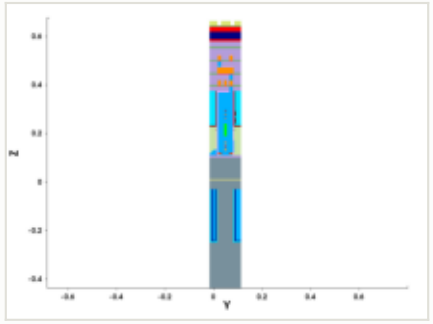
1.31	436.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

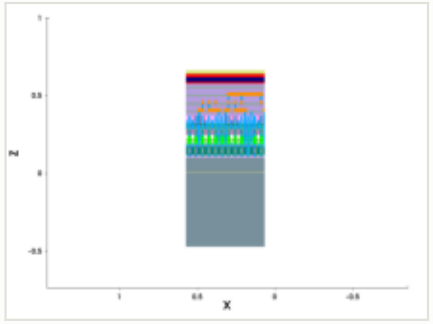
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

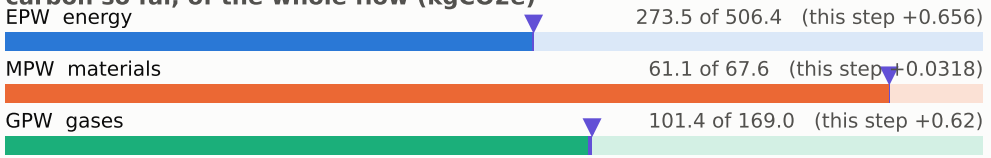
no mask for this step

this step

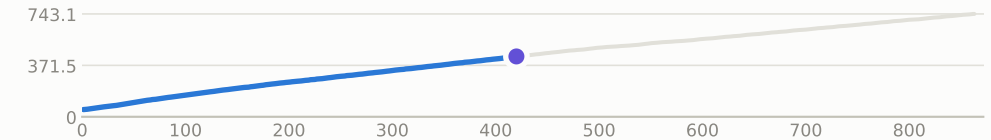
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

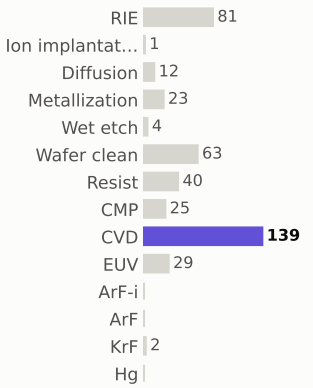
carbon so far, of the whole flow (kgCO2e)



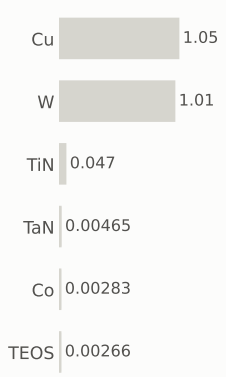
embodied carbon up to each step (kgCO2e per wafer)



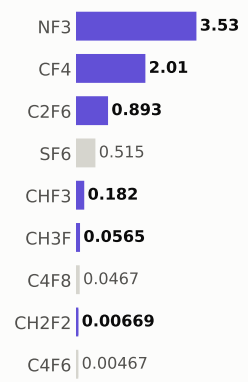
steps by process area



materials deposited (g)



gases emitted (g)



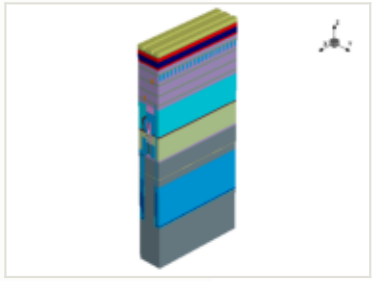
Step #4490: RIE

etch_anisotropic · RIE

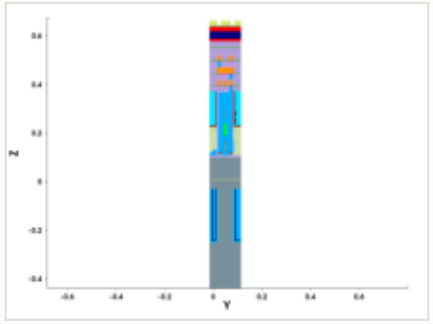
1.04	437.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

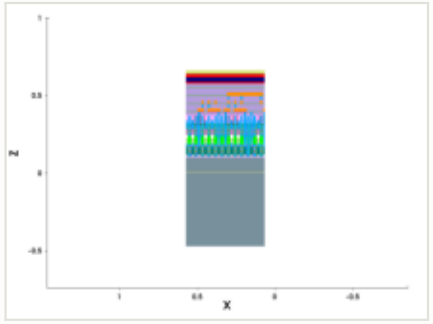
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

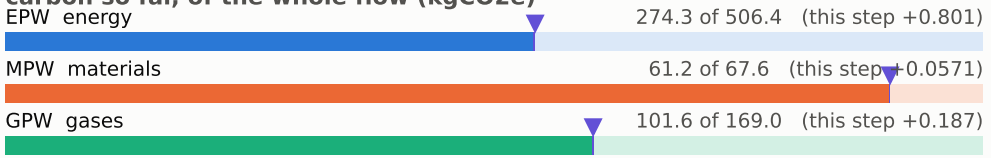
no mask for this step

this step

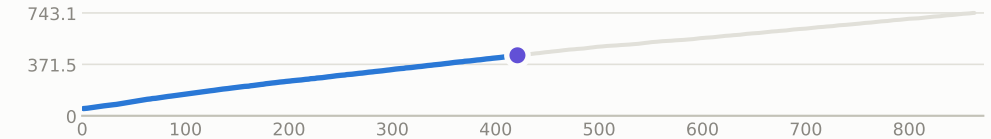
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

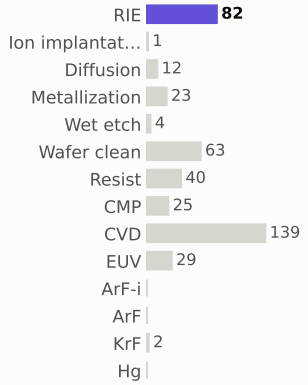
carbon so far, of the whole flow (kgCO2e)



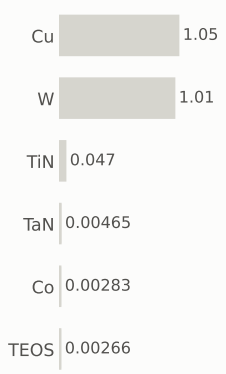
embodied carbon up to each step (kgCO2e per wafer)



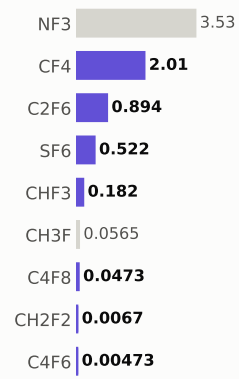
steps by process area



materials deposited (g)



gases emitted (g)



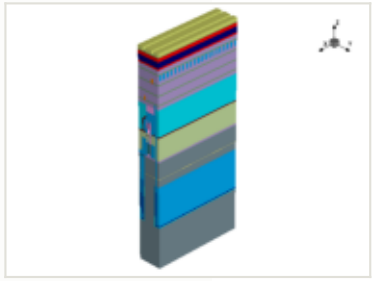
Step #4500: Wafer_clean

clean · Wafer_clean

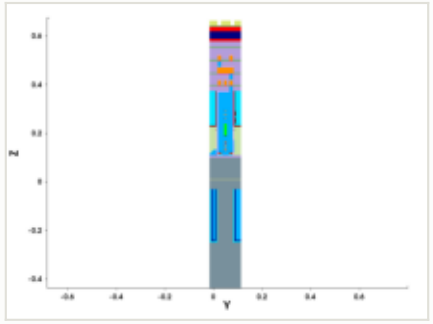
0.169	437.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

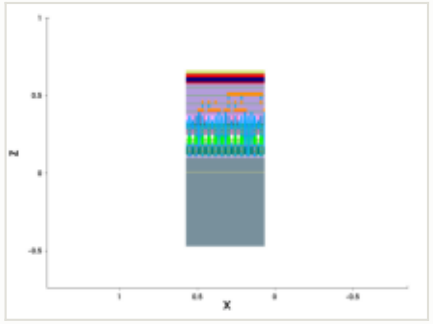
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

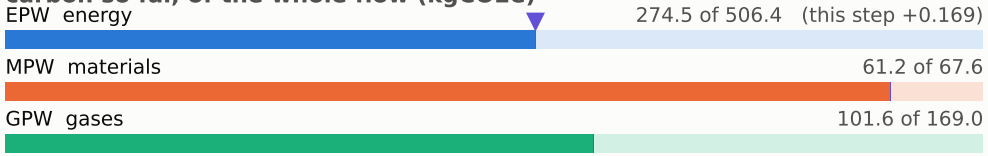
no mask for this step

this step

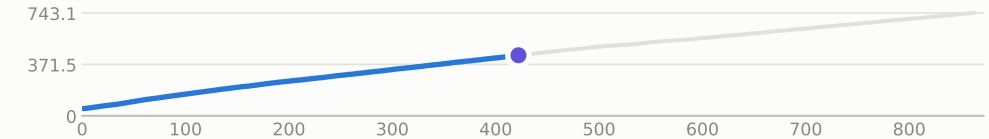
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

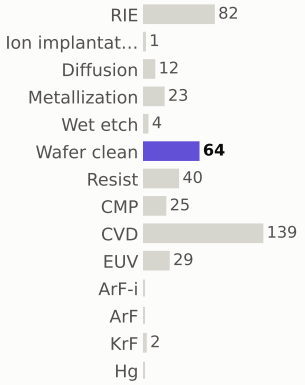
carbon so far, of the whole flow (kgCO2e)



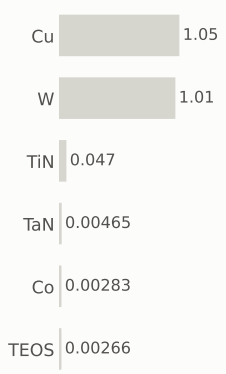
embodied carbon up to each step (kgCO2e per wafer)



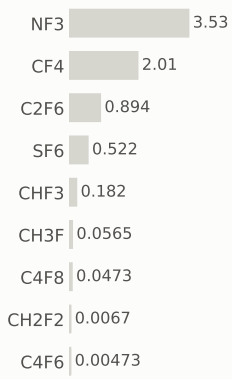
steps by process area



materials deposited (g)



gases emitted (g)



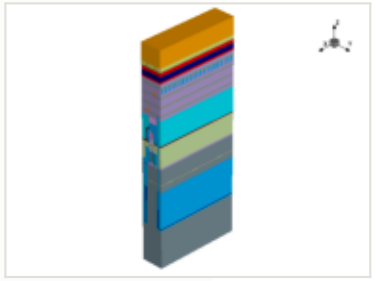
Step #4510: Resist

depo_fill · Resist

0.169	437.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

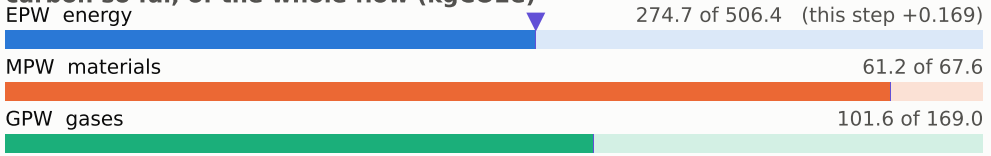


this step

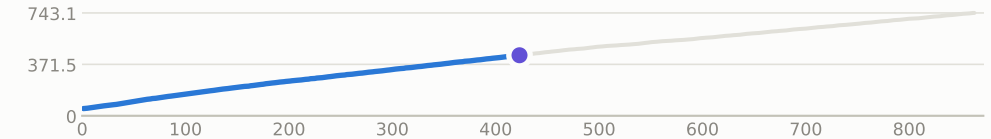
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

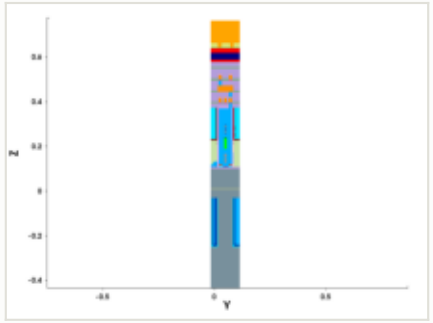
carbon so far, of the whole flow (kgCO2e)



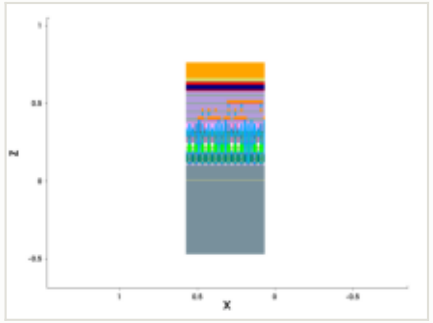
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



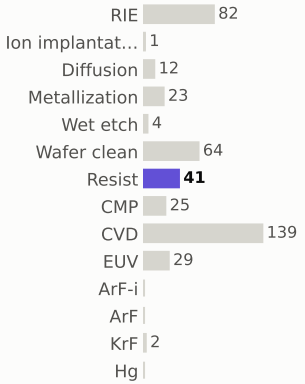
y cut at 0.049 µm



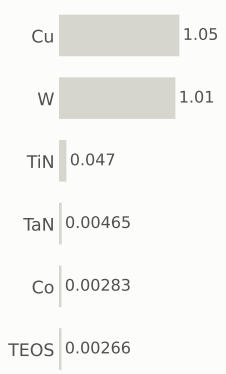
mask for this step

no mask for this step

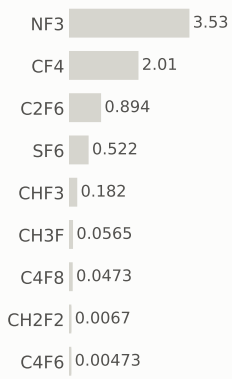
steps by process area



materials deposited (g)



gases emitted (g)



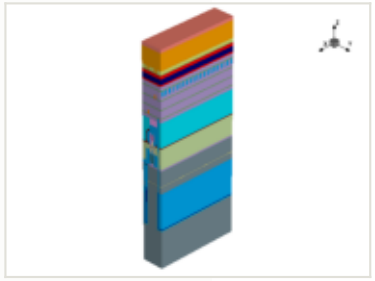
Step #4520: CVD

depo_fill · CVD

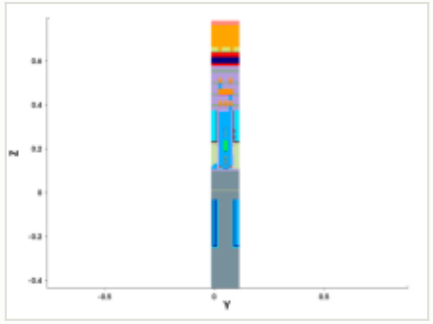
1.31	438.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

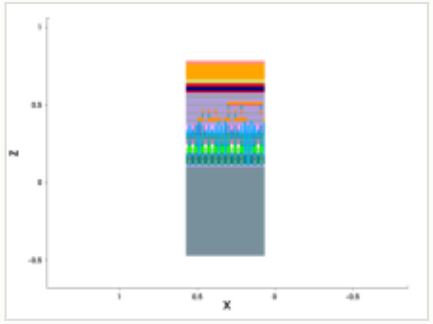
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

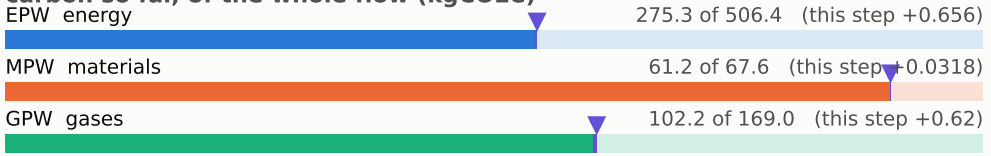
no mask for this step

this step

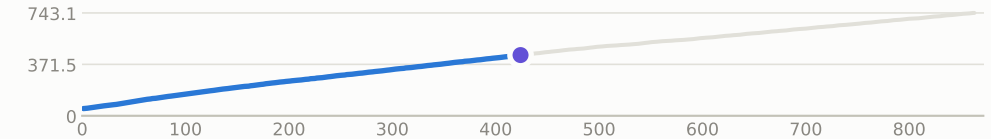
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

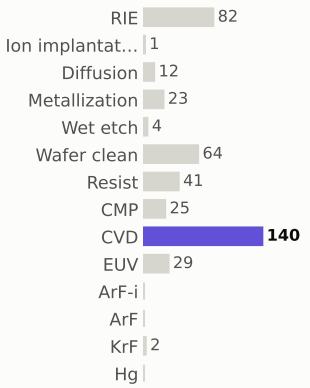
carbon so far, of the whole flow (kgCO2e)



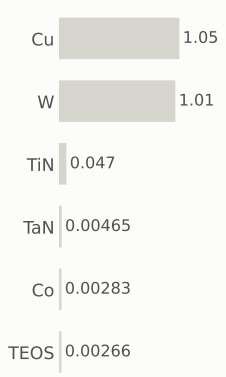
embodied carbon up to each step (kgCO2e per wafer)



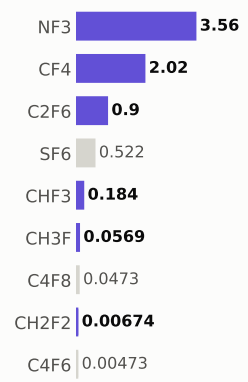
steps by process area



materials deposited (g)



gases emitted (g)



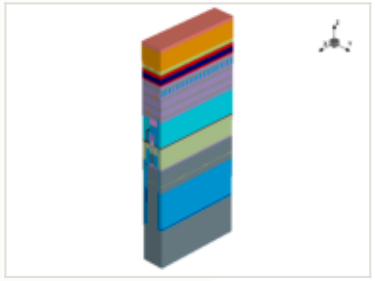
Step #4530: Wafer_clean

clean · Wafer_clean

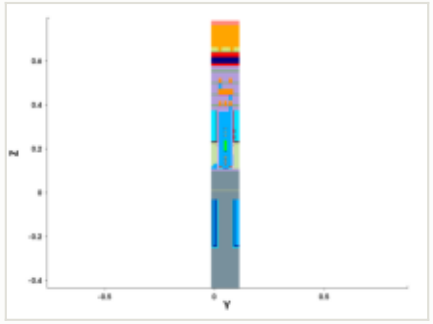
0.169	438.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

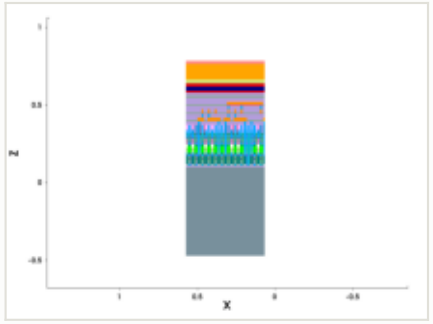
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

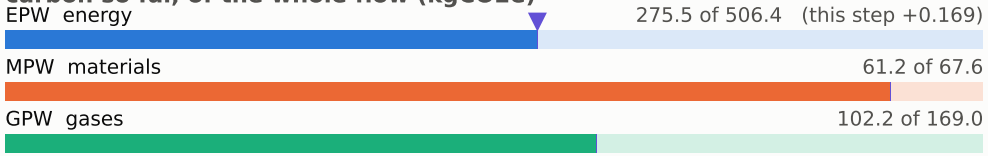
no mask for this step

this step

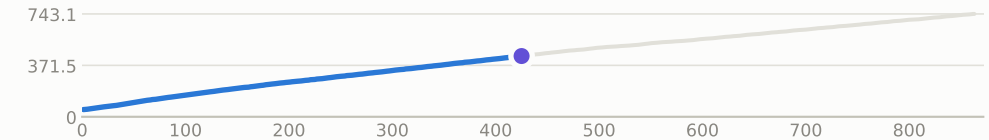
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

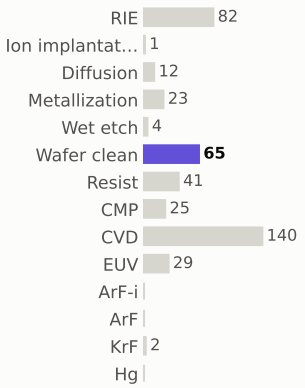
carbon so far, of the whole flow (kgCO2e)



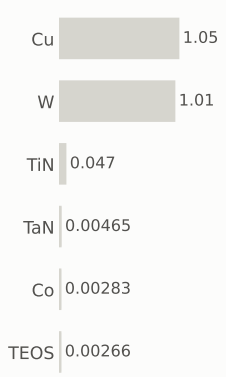
embodied carbon up to each step (kgCO2e per wafer)



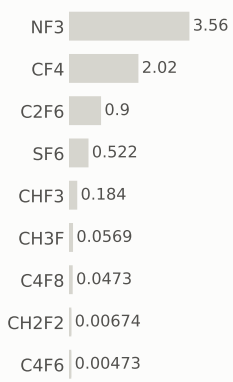
steps by process area



materials deposited (g)



gases emitted (g)



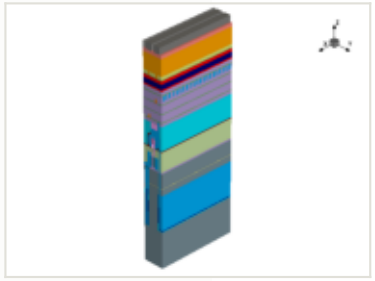
Step #4540: EUV

litho_patterning · EUV

1.96	440.9	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

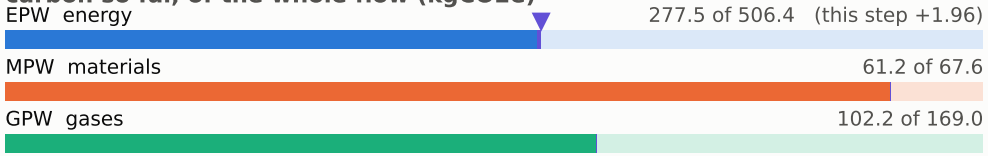


this step

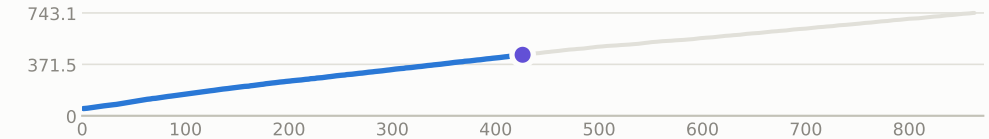
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

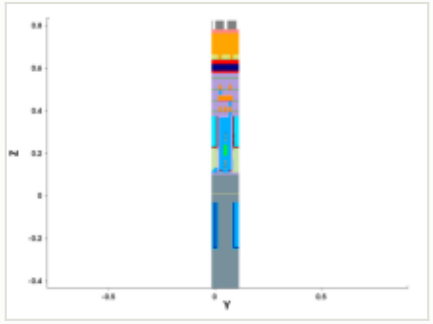
carbon so far, of the whole flow (kgCO2e)



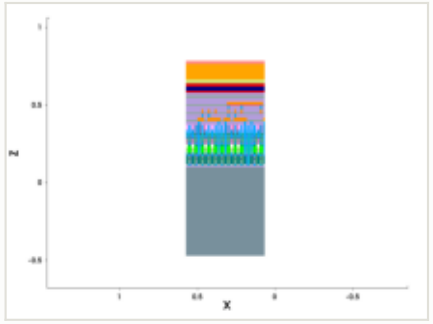
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

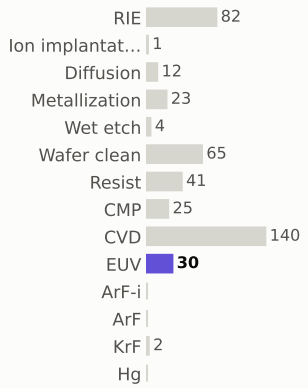


y cut at 0.049 µm

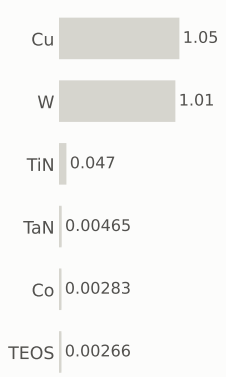


mask: layer 10352:0 (GDSII)

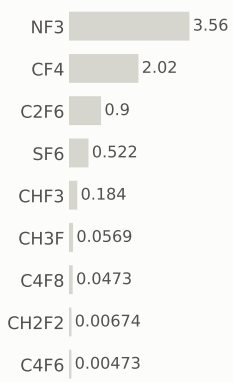
steps by process area



materials deposited (g)



gases emitted (g)



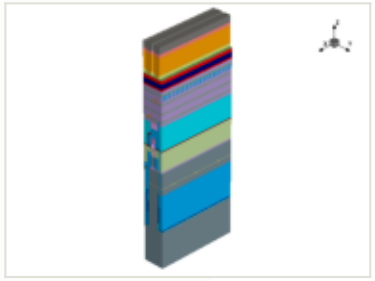
Step #4550: RIE

etch_anisotropic · RIE

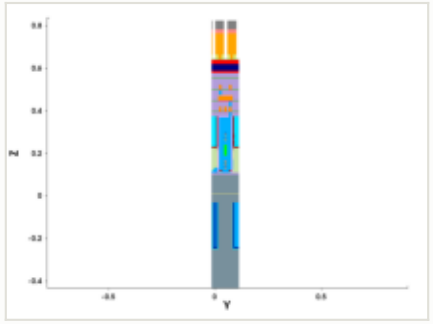
1.04	441.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

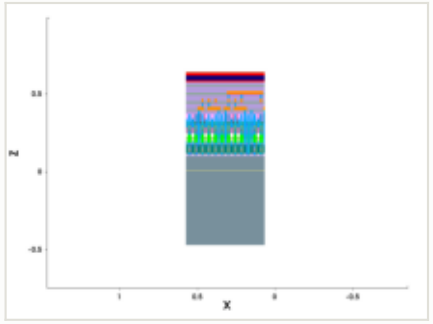
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

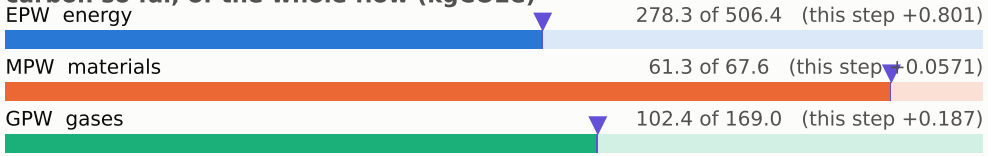
no mask for this step

this step

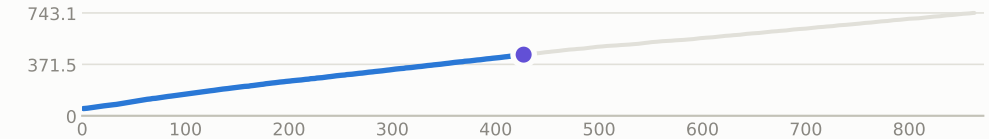
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

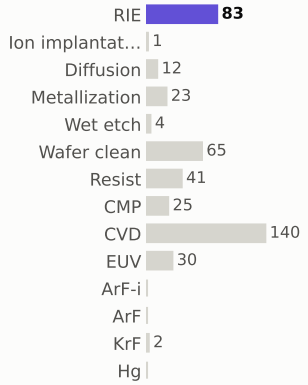
carbon so far, of the whole flow (kgCO2e)



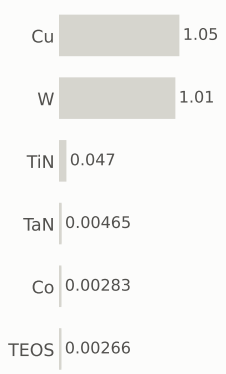
embodied carbon up to each step (kgCO2e per wafer)



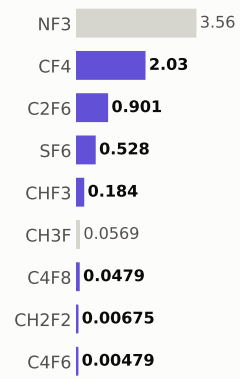
steps by process area



materials deposited (g)



gases emitted (g)



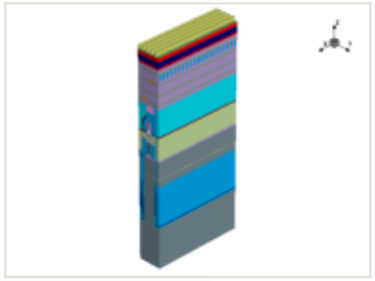
Step #4560: Wafer_clean

etch_strip · Wafer_clean

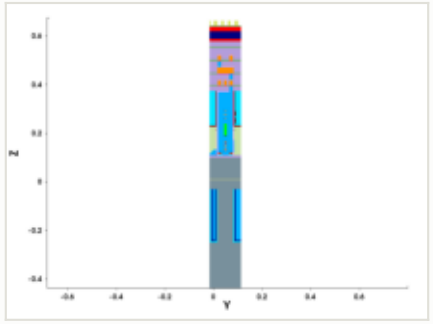
0.169	442.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

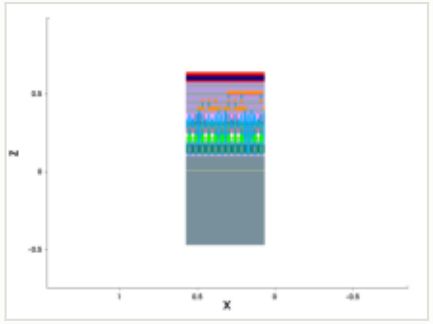
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



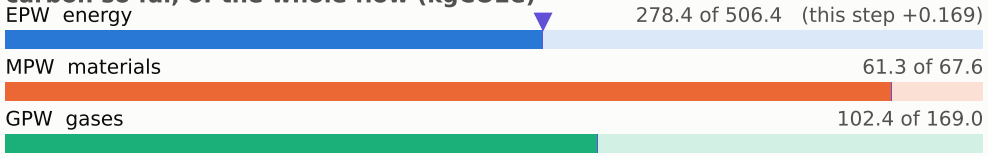
mask for this step

no mask for this step

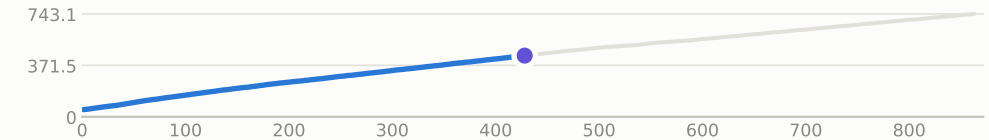
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

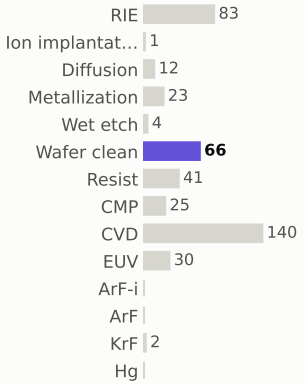
carbon so far, of the whole flow (kgCO2e)



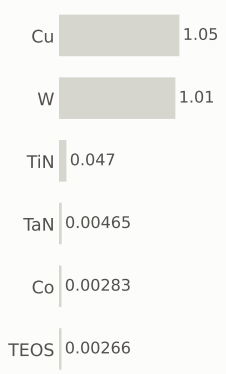
embodied carbon up to each step (kgCO2e per wafer)



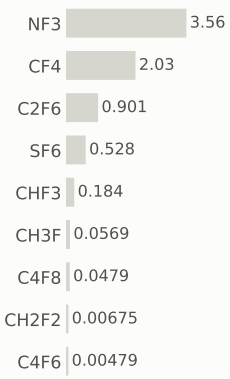
steps by process area



materials deposited (g)



gases emitted (g)



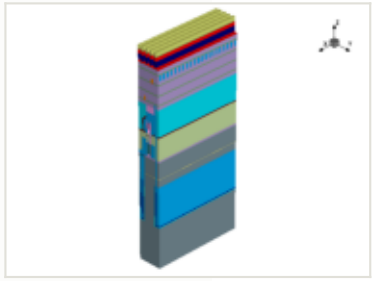
Step #4570: RIE

etch_anisotropic · RIE

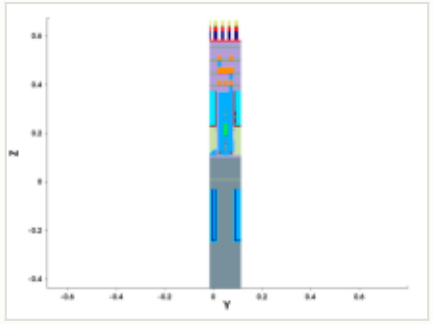
1.04	443.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

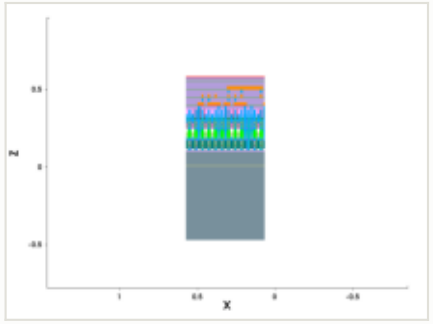
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

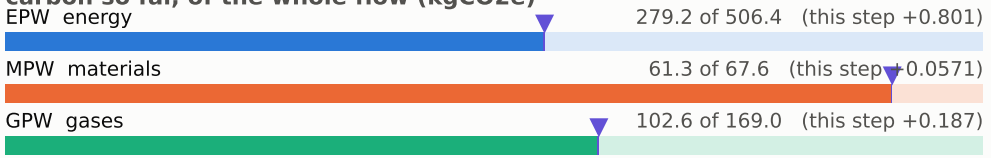
no mask for this step

this step

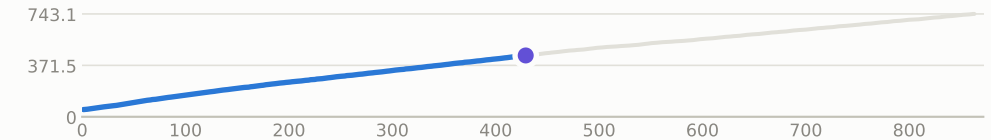
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

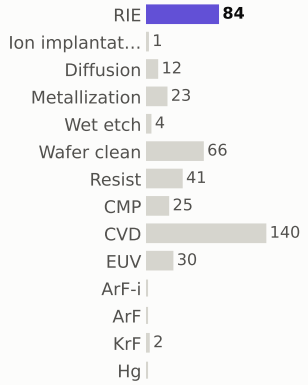
carbon so far, of the whole flow (kgCO2e)



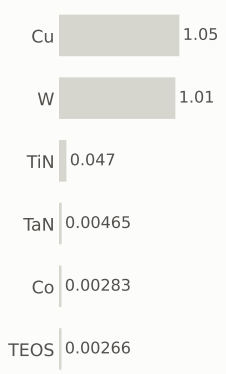
embodied carbon up to each step (kgCO2e per wafer)



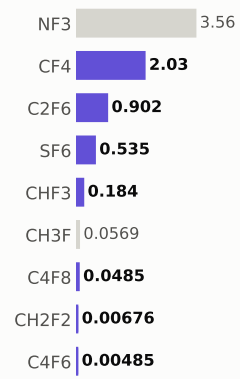
steps by process area



materials deposited (g)



gases emitted (g)



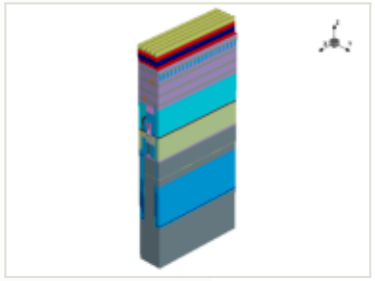
Step #4580: Wafer_clean

clean · Wafer_clean

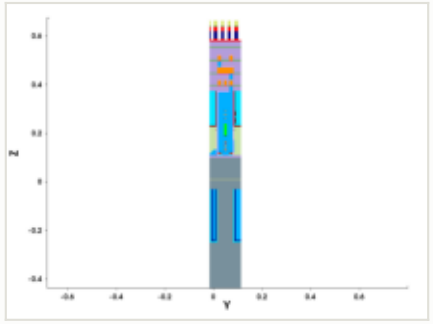
0.169	443.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

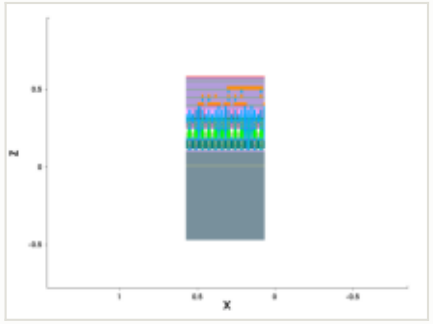
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

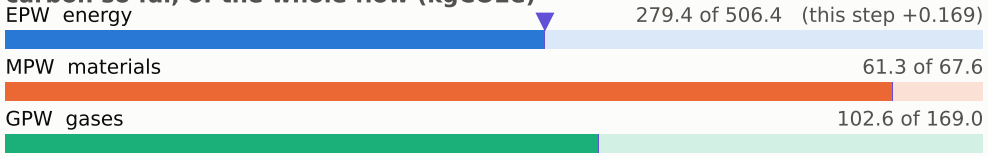
no mask for this step

this step

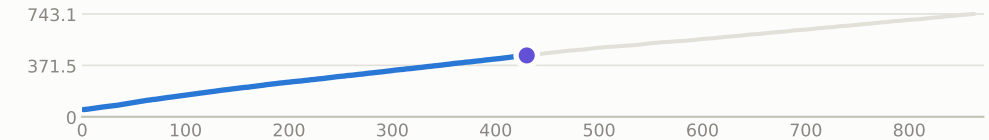
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

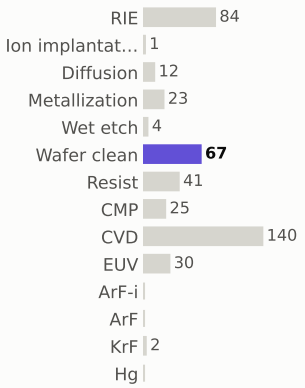
carbon so far, of the whole flow (kgCO2e)



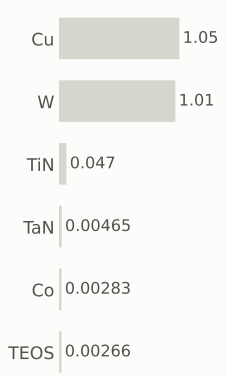
embodied carbon up to each step (kgCO2e per wafer)



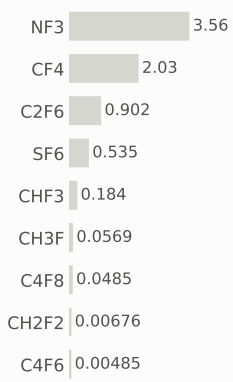
steps by process area



materials deposited (g)



gases emitted (g)



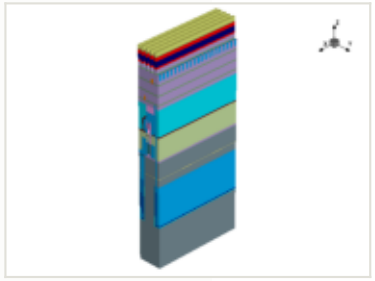
Step #4590: RIE

etch_anisotropic · RIE

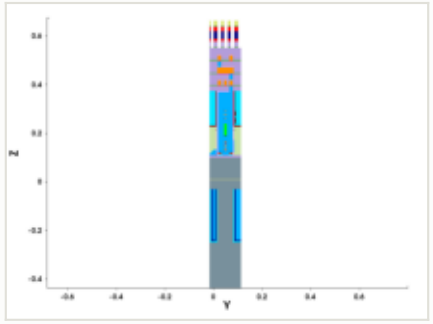
1.04	444.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

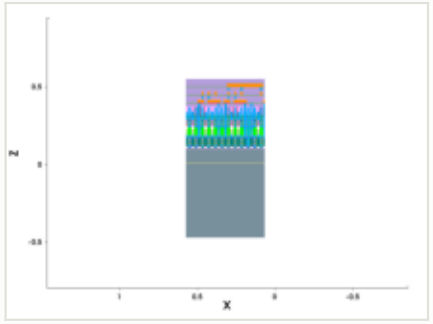
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

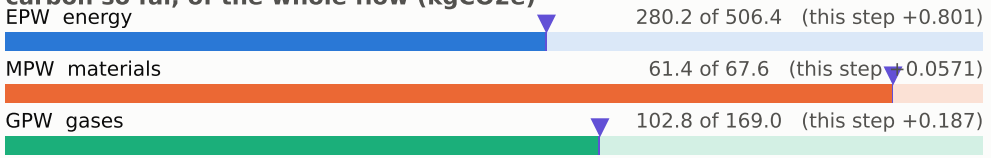
no mask for this step

this step

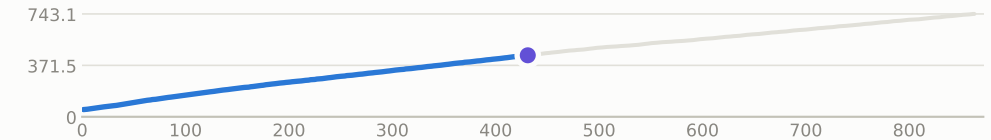
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

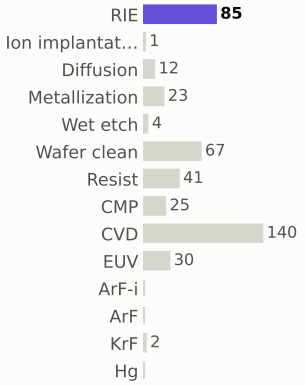
carbon so far, of the whole flow (kgCO2e)



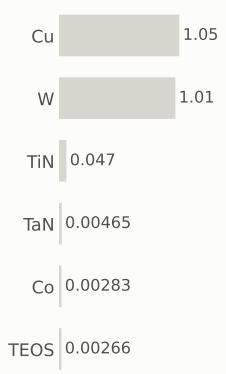
embodied carbon up to each step (kgCO2e per wafer)



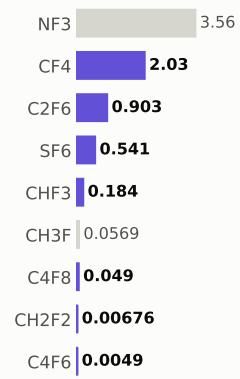
steps by process area



materials deposited (g)



gases emitted (g)



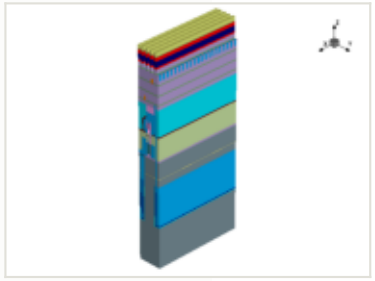
Step #4600: Wafer_clean

clean · Wafer_clean

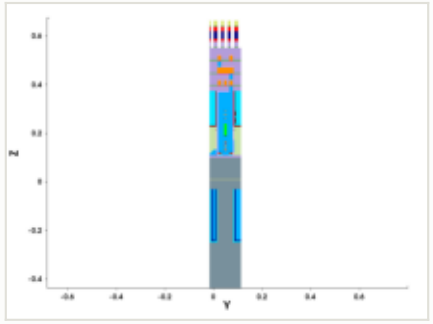
0.169	444.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

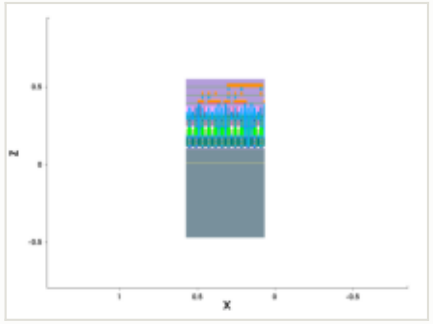
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

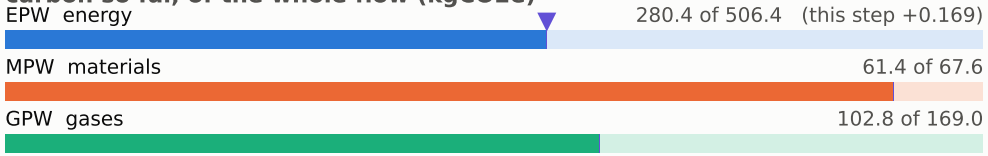
no mask for this step

this step

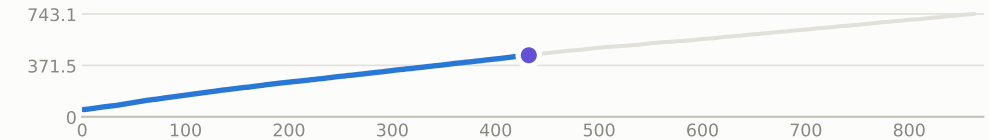
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

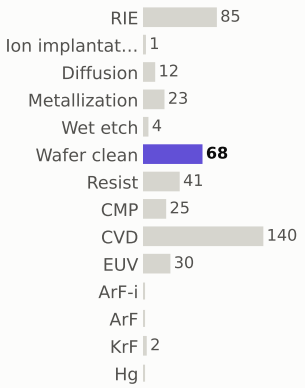
carbon so far, of the whole flow (kgCO2e)



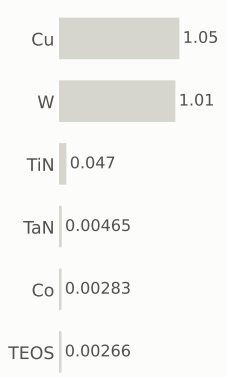
embodied carbon up to each step (kgCO2e per wafer)



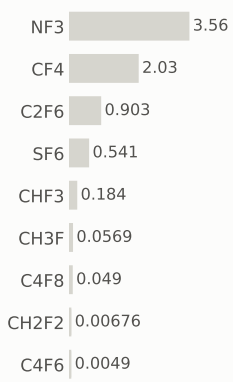
steps by process area



materials deposited (g)



gases emitted (g)



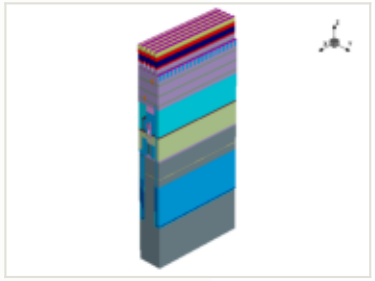
Step #4610: Metallization

depo_ellipsoidal · Metallization

1.2	445.7	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

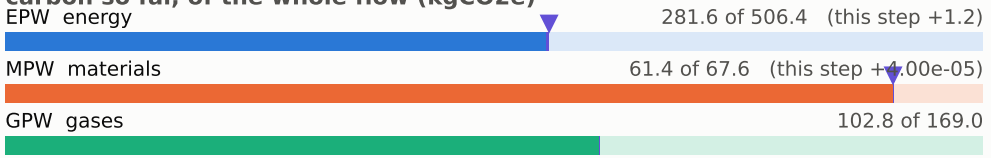


this step

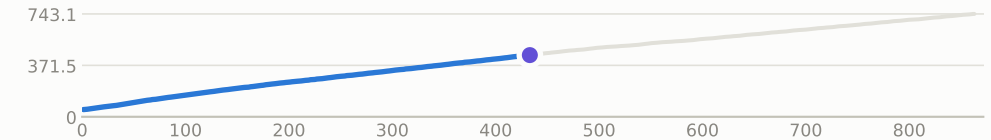
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

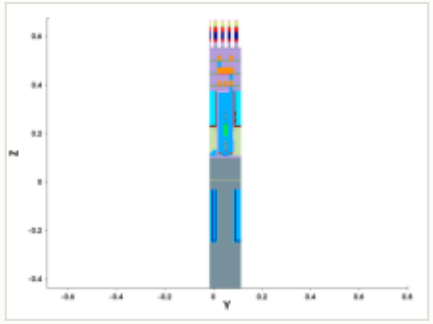
carbon so far, of the whole flow (kgCO2e)



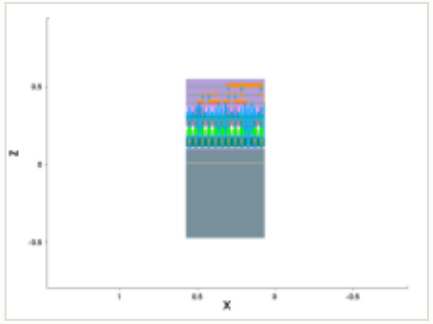
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



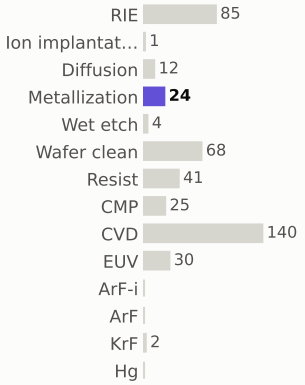
y cut at 0.049 µm



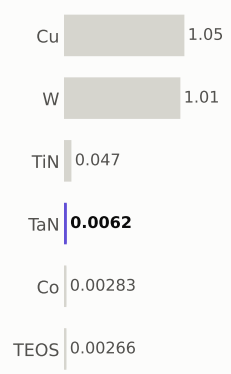
mask for this step

no mask for this step

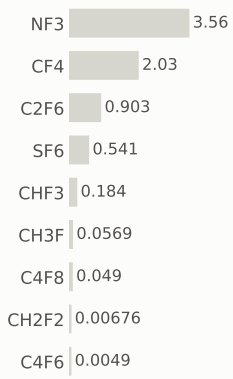
steps by process area



materials deposited (g)



gases emitted (g)



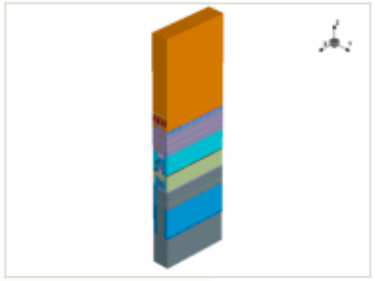
Step #4620: Metallization

depo_fill · Metallization

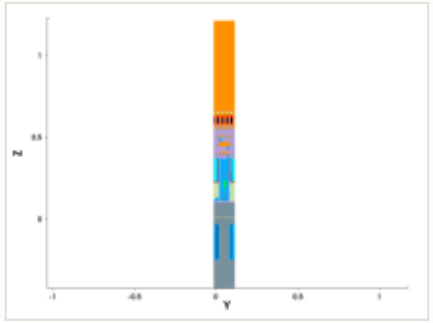
1.21	446.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

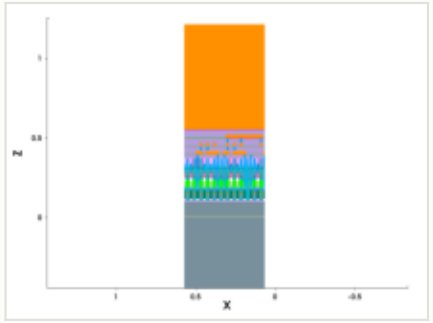
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

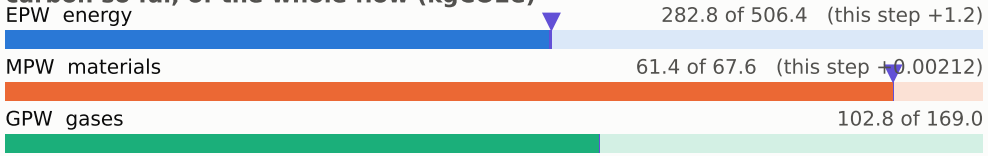
no mask for this step

this step

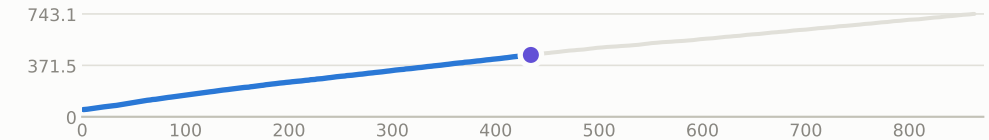
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

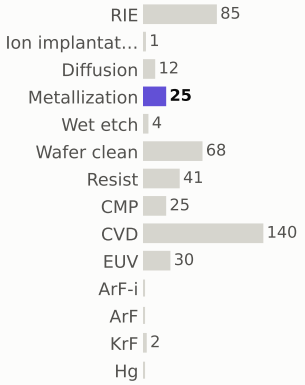
carbon so far, of the whole flow (kgCO2e)



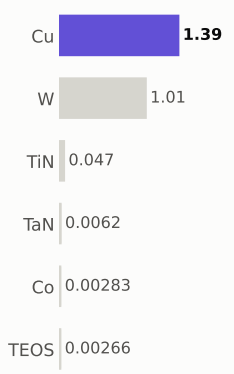
embodied carbon up to each step (kgCO2e per wafer)



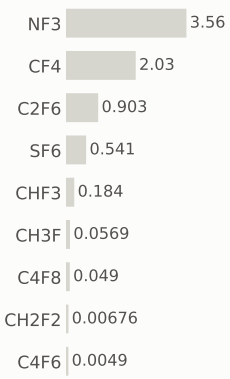
steps by process area



materials deposited (g)



gases emitted (g)



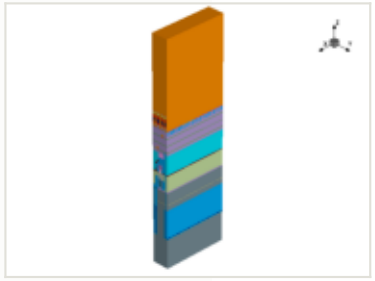
Step #4630: Diffusion

treatment · Diffusion

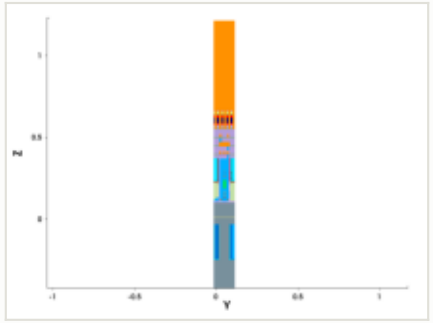
0.801	447.7	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

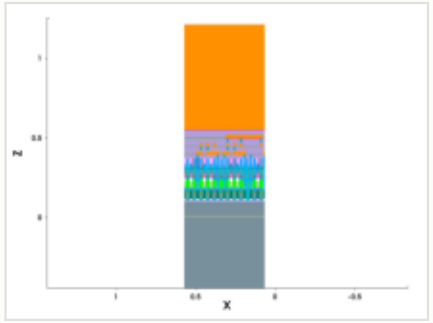
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

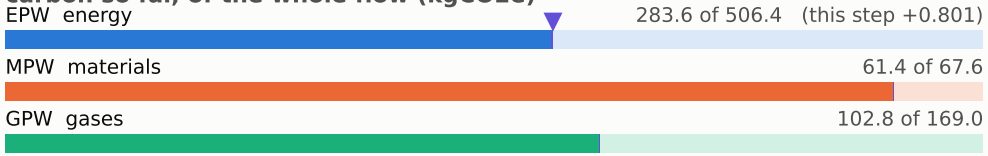
no mask for this step

this step

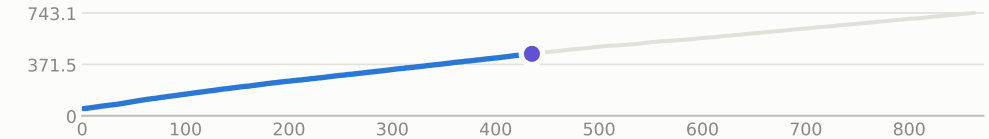
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

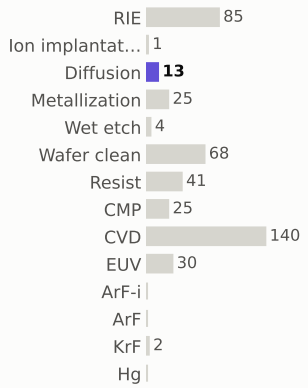
carbon so far, of the whole flow (kgCO2e)



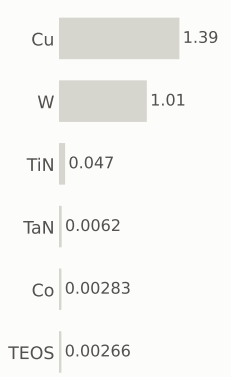
embodied carbon up to each step (kgCO2e per wafer)



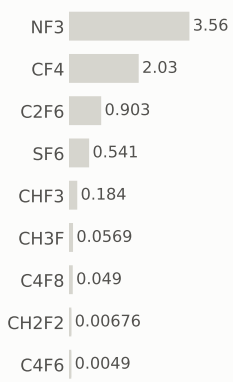
steps by process area



materials deposited (g)



gases emitted (g)



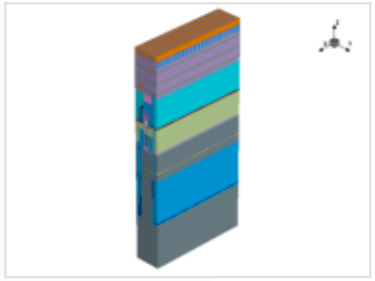
Step #4640: CMP

etch_cmp · CMP

0.169	447.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

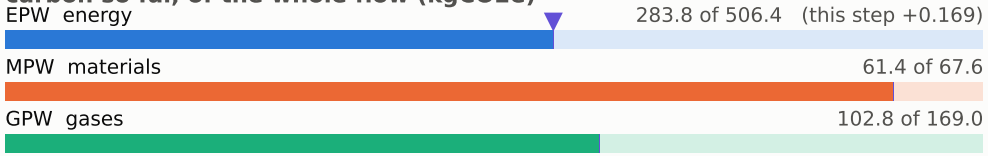


this step

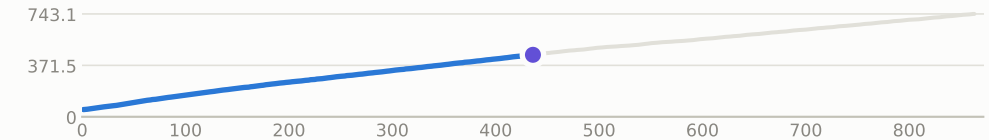
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

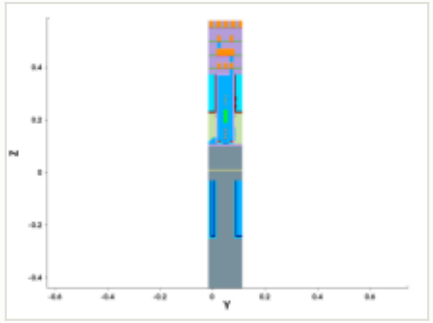
carbon so far, of the whole flow (kgCO2e)



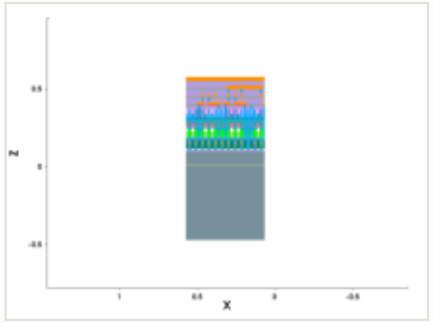
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



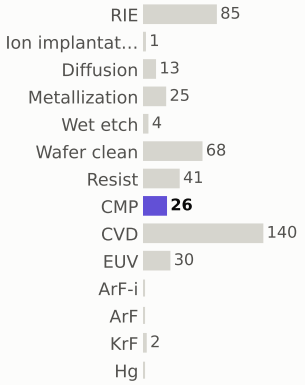
y cut at 0.049 µm



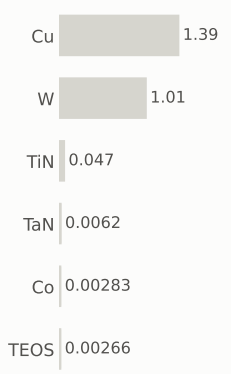
mask for this step

no mask for this step

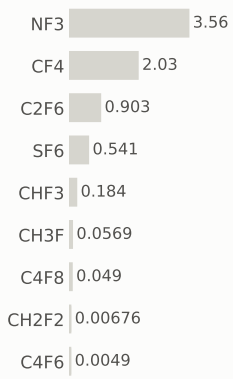
steps by process area



materials deposited (g)



gases emitted (g)



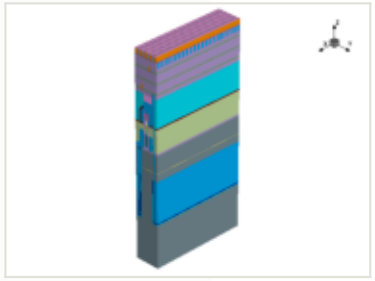
Step #4650: Metallization

depo_isotropic · Metallization

1.2	449.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

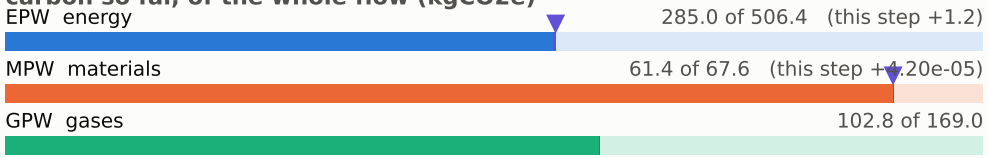


this step

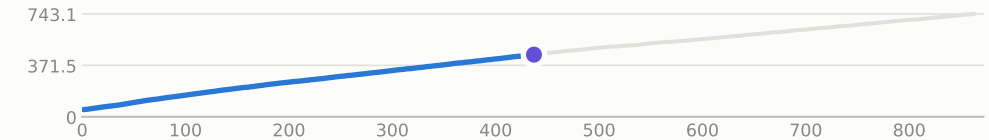
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

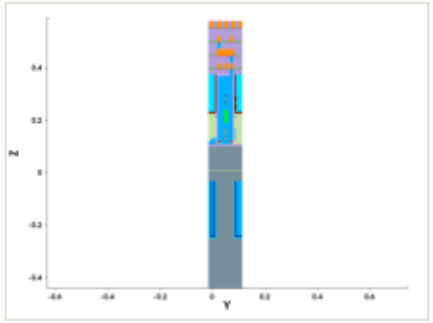
carbon so far, of the whole flow (kgCO2e)



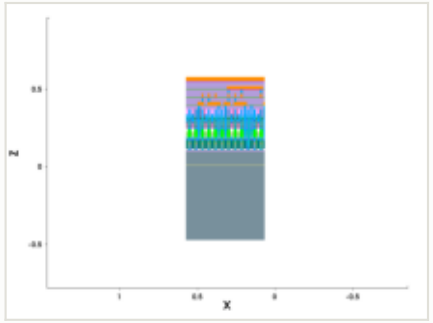
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



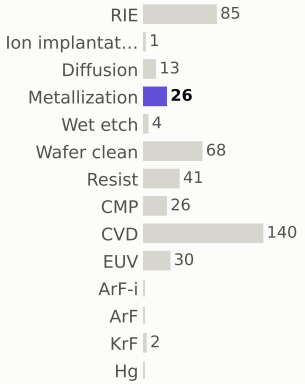
y cut at 0.049 µm



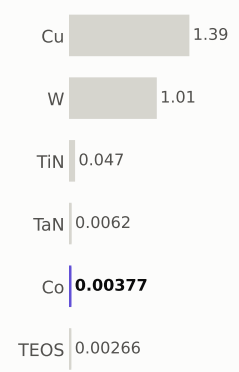
mask for this step

no mask for this step

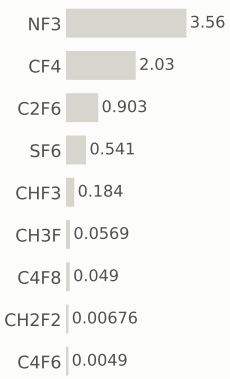
steps by process area



materials deposited (g)



gases emitted (g)



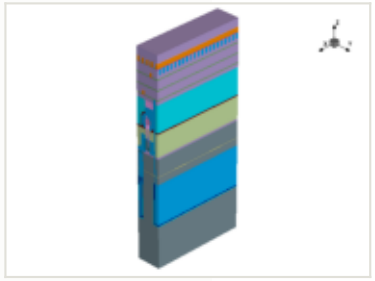
Step #4670: CVD

depo_fill · CVD

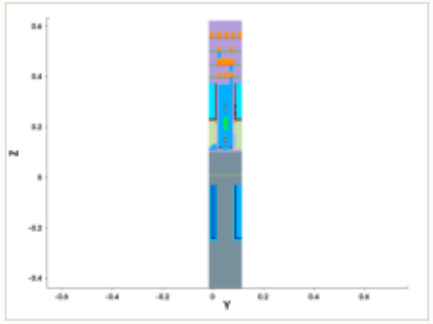
1.31	450.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

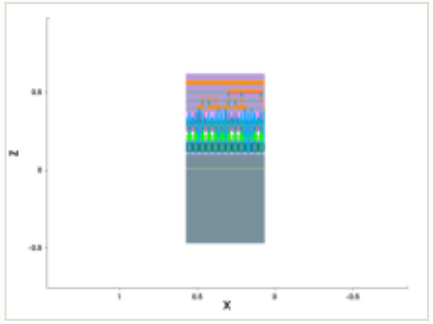
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

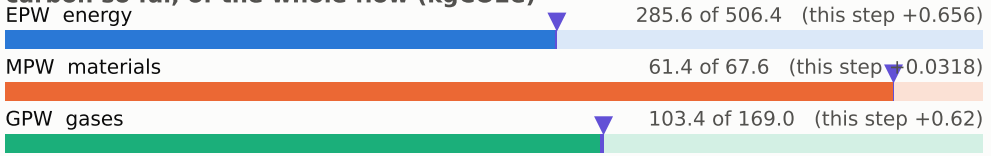
no mask for this step

this step

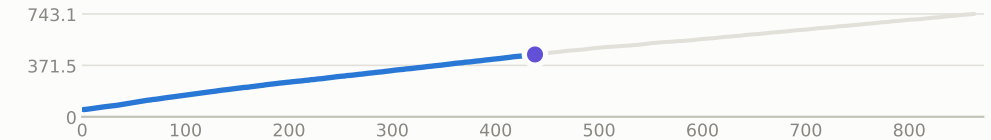
process area	CVD
material	—
thickness	0.042 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

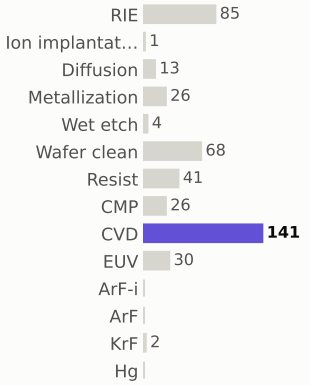
carbon so far, of the whole flow (kgCO2e)



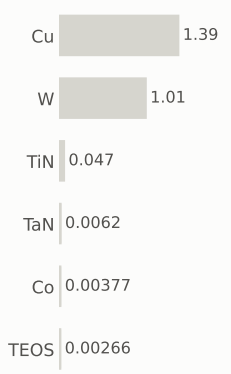
embodied carbon up to each step (kgCO2e per wafer)



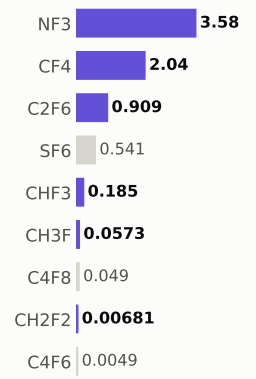
steps by process area



materials deposited (g)



gases emitted (g)



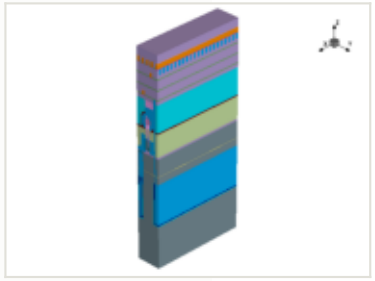
Step #4680: Diffusion

treatment · Diffusion

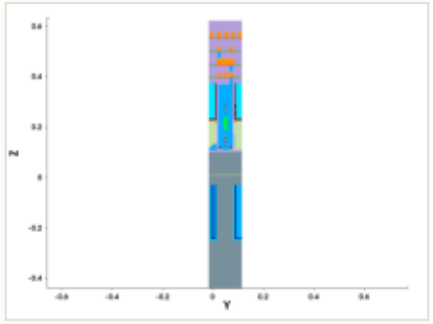
0.801	451.2	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

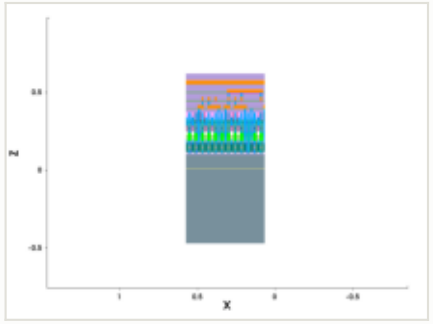
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

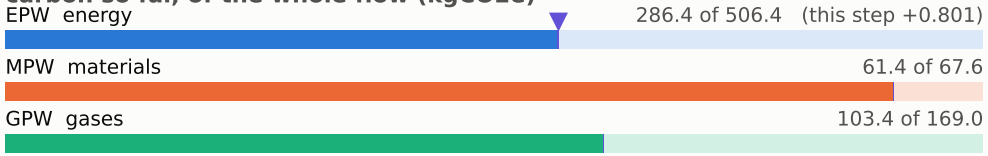
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

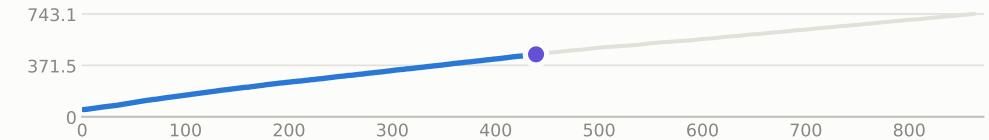
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

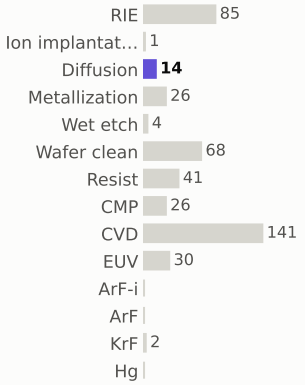
carbon so far, of the whole flow (kgCO2e)



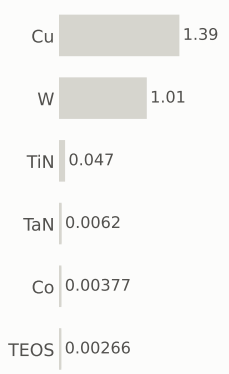
embodied carbon up to each step (kgCO2e per wafer)



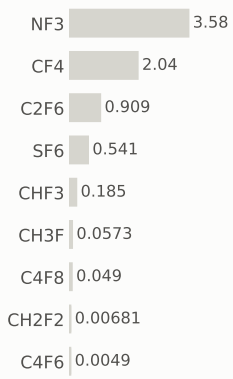
steps by process area



materials deposited (g)



gases emitted (g)



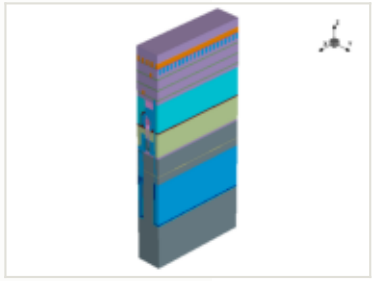
Step #4690: Wafer_clean

clean · Wafer_clean

0.169	451.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



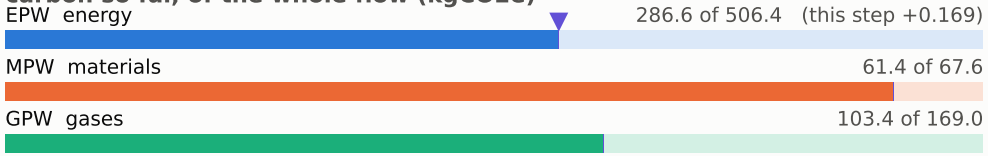
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

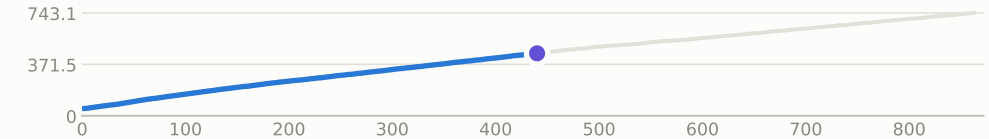
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

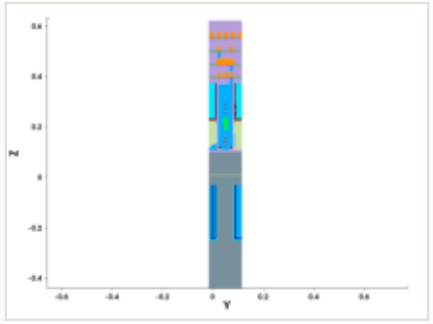
carbon so far, of the whole flow (kgCO2e)



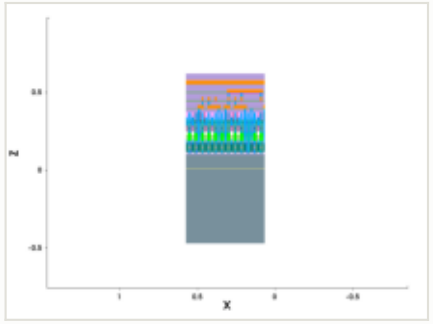
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



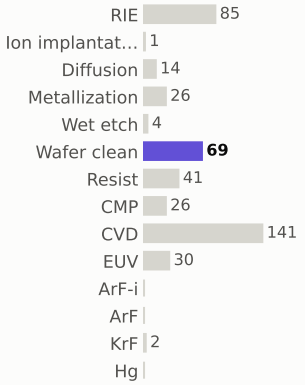
y cut at 0.049 µm



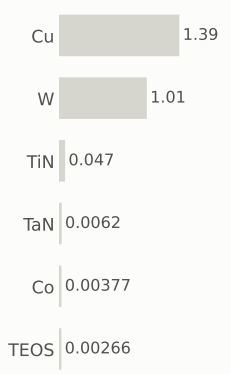
mask for this step

no mask for this step

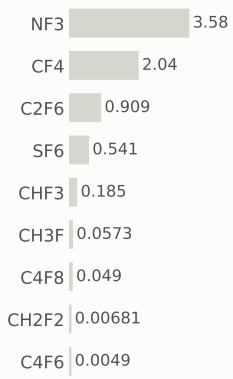
steps by process area



materials deposited (g)



gases emitted (g)



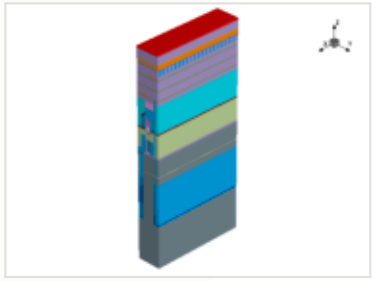
Step #4700: CVD

depo_fill · CVD

1.31	452.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

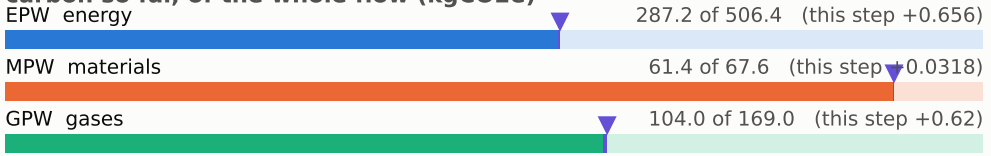


this step

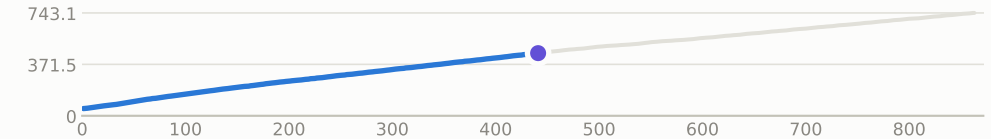
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

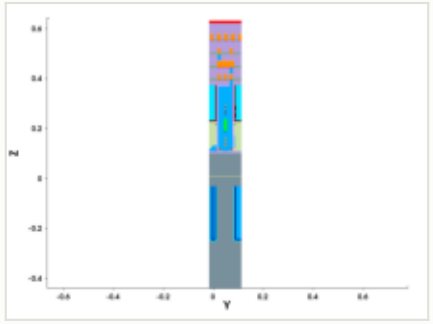
carbon so far, of the whole flow (kgCO2e)



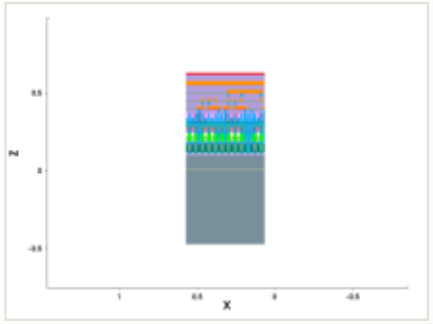
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



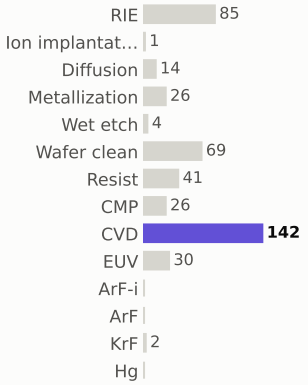
y cut at 0.049 µm



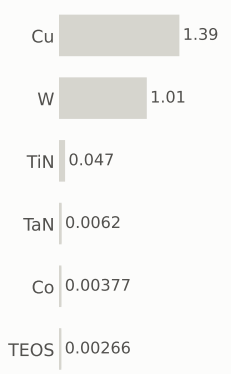
mask for this step

no mask for this step

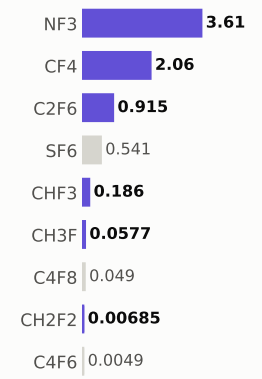
steps by process area



materials deposited (g)



gases emitted (g)



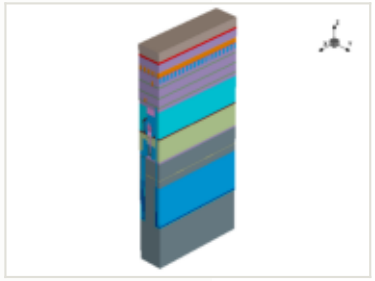
Step #4710: Resist

depo_fill · Resist

0.169	452.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

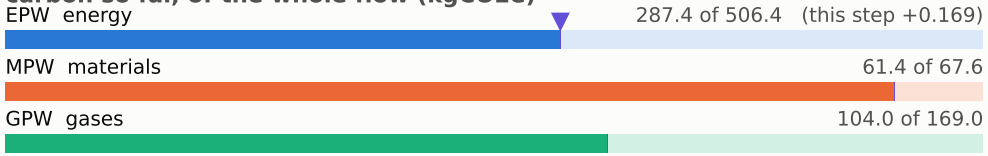


this step

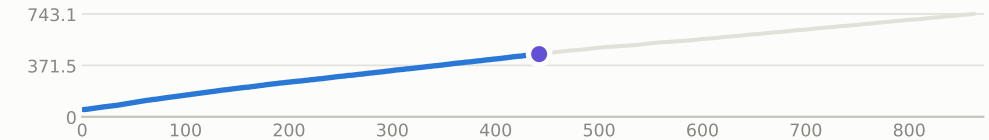
process area	Resist
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

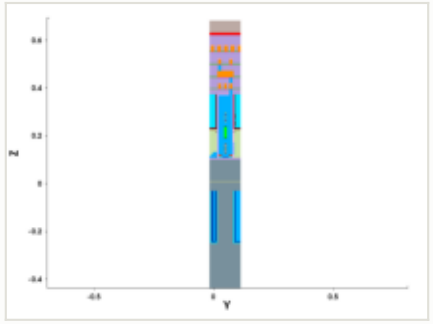
carbon so far, of the whole flow (kgCO2e)



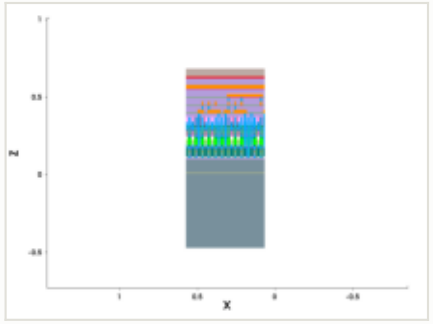
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



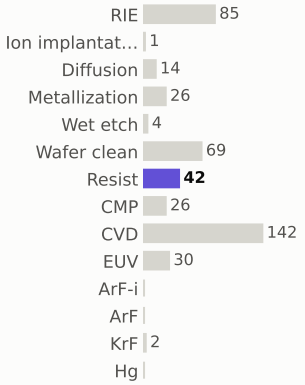
y cut at 0.049 µm



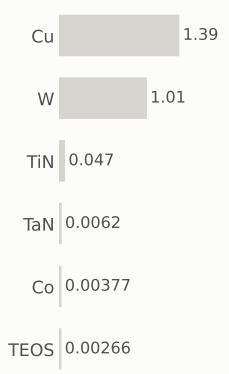
mask for this step

no mask for this step

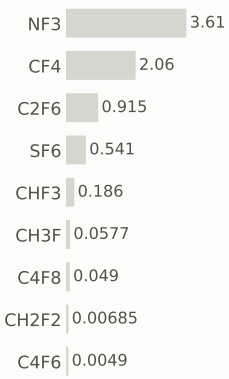
steps by process area



materials deposited (g)



gases emitted (g)



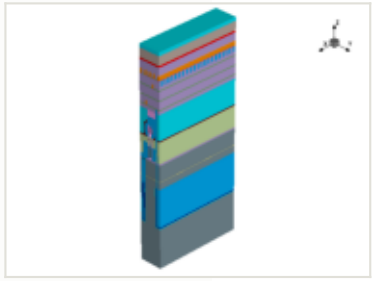
Step #4720: CVD

depo_fill · CVD

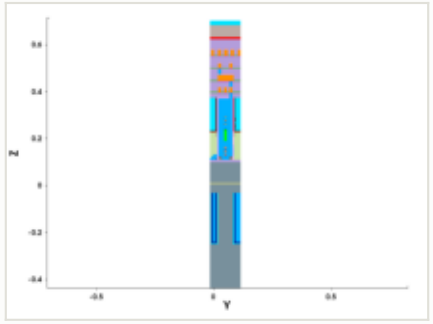
1.31	454.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

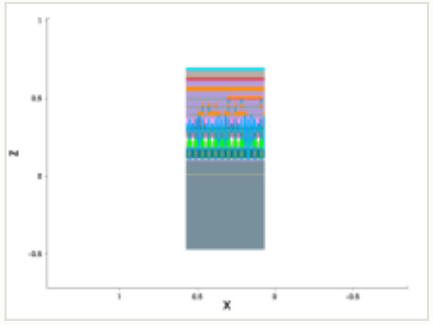
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

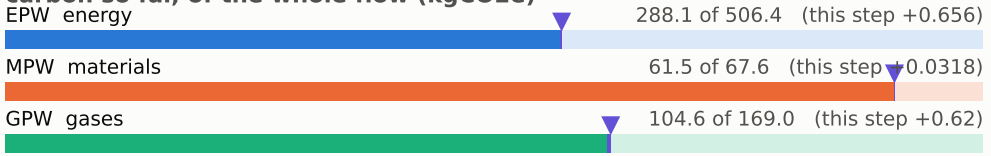
no mask for this step

this step

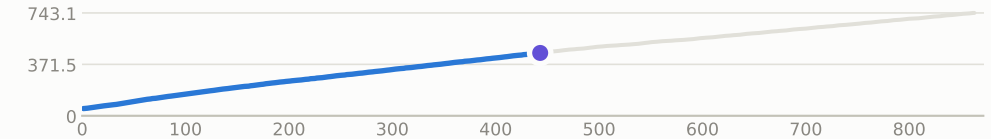
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

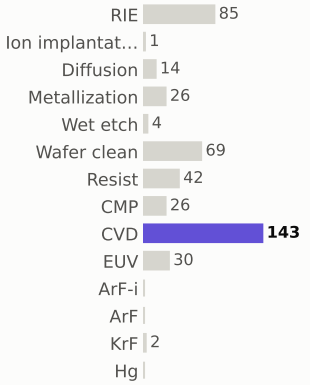
carbon so far, of the whole flow (kgCO2e)



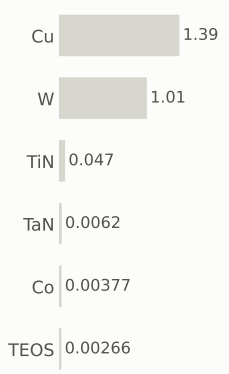
embodied carbon up to each step (kgCO2e per wafer)



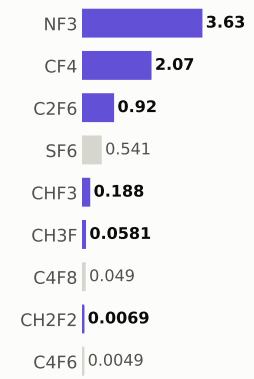
steps by process area



materials deposited (g)



gases emitted (g)



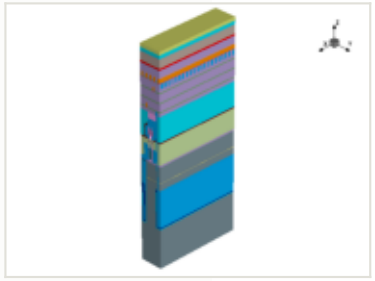
Step #4730: CVD

depo_fill · CVD

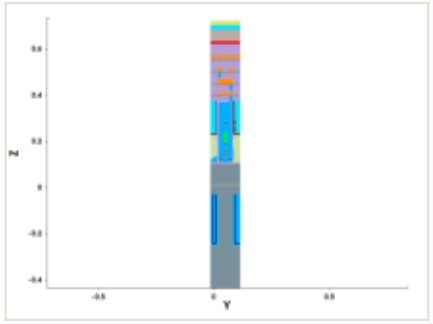
1.31	455.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

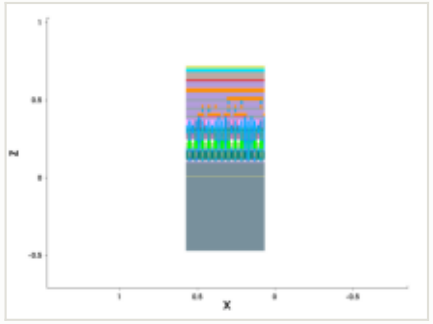
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

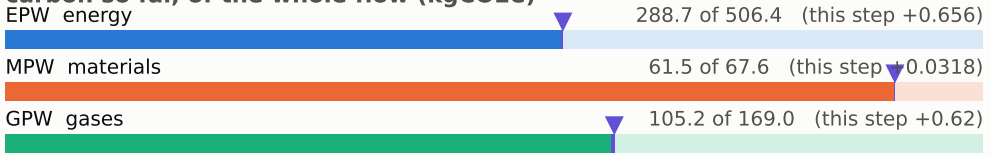
no mask for this step

this step

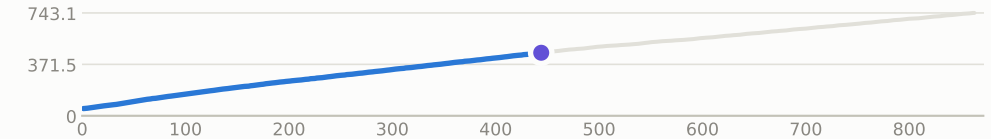
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

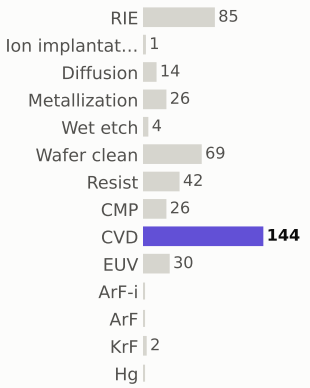
carbon so far, of the whole flow (kgCO2e)



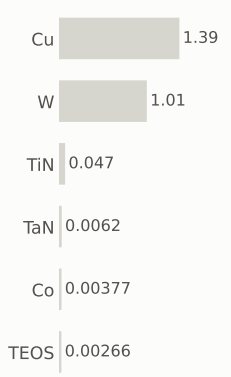
embodied carbon up to each step (kgCO2e per wafer)



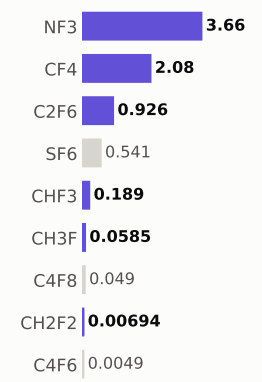
steps by process area



materials deposited (g)



gases emitted (g)



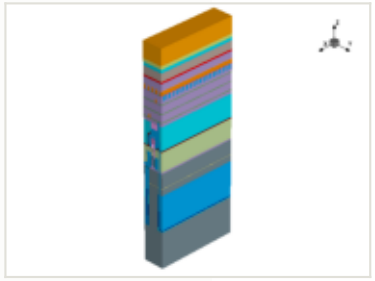
Step #4740: Resist

depo_fill · Resist

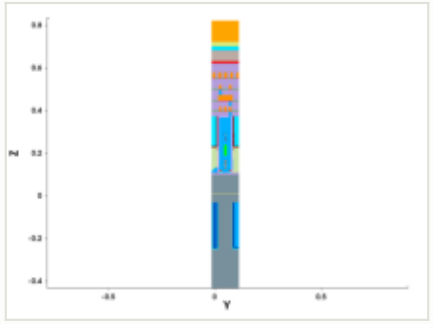
0.169	455.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

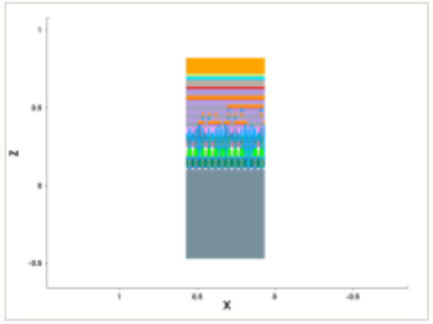
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

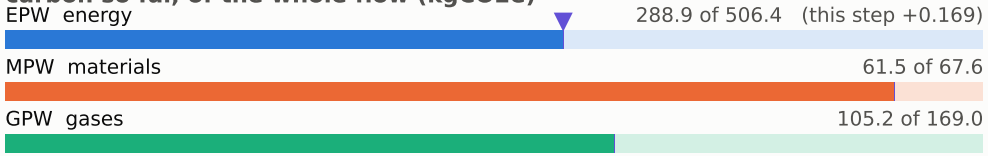
no mask for this step

this step

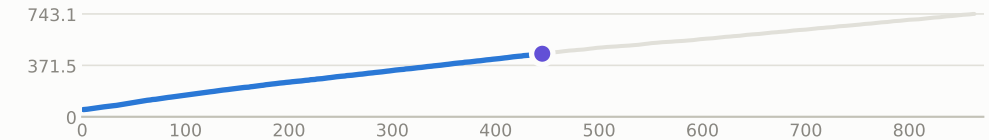
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

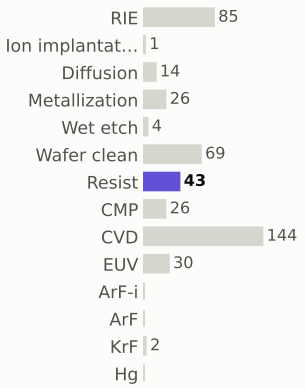
carbon so far, of the whole flow (kgCO2e)



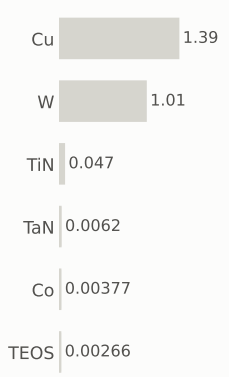
embodied carbon up to each step (kgCO2e per wafer)



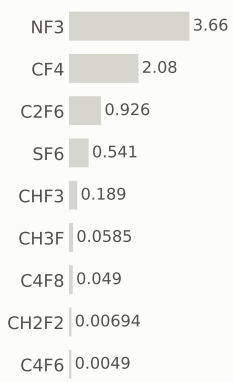
steps by process area



materials deposited (g)



gases emitted (g)



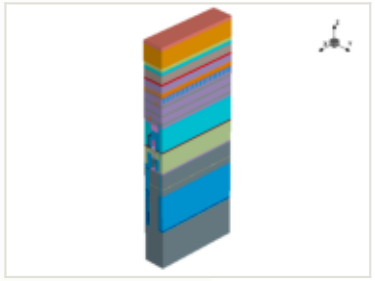
Step #4750: CVD

depo_fill · CVD

1.31	457.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

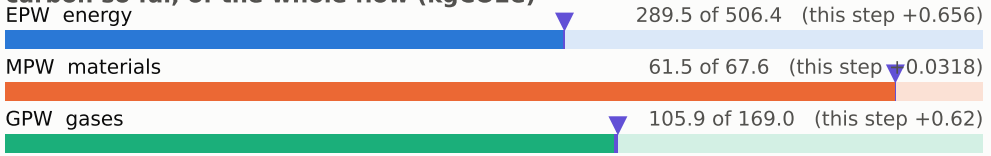


this step

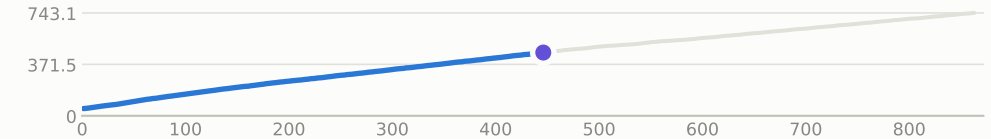
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

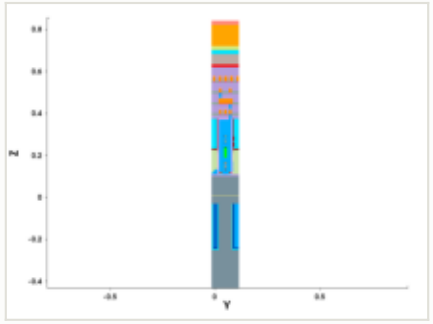
carbon so far, of the whole flow (kgCO2e)



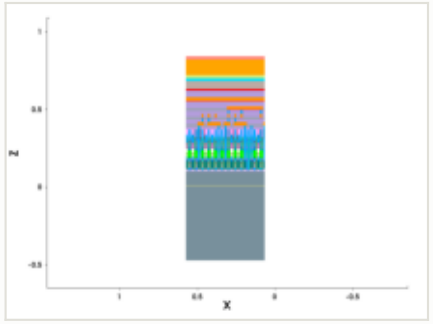
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



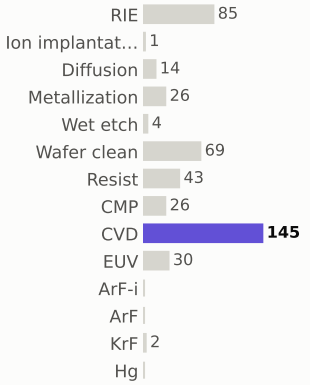
y cut at 0.049 µm



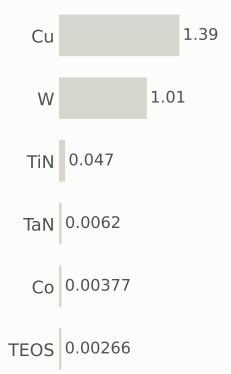
mask for this step

no mask for this step

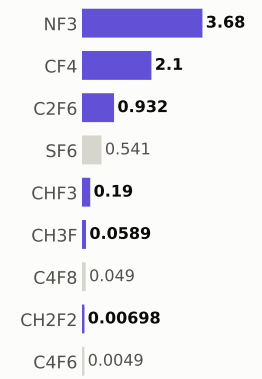
steps by process area



materials deposited (g)



gases emitted (g)



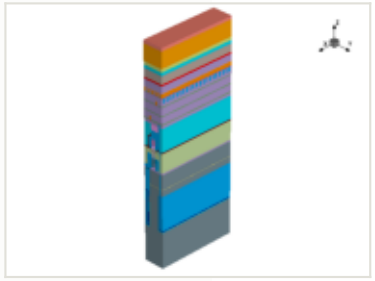
Step #4760: Wafer_clean

clean · Wafer_clean

0.169	457.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



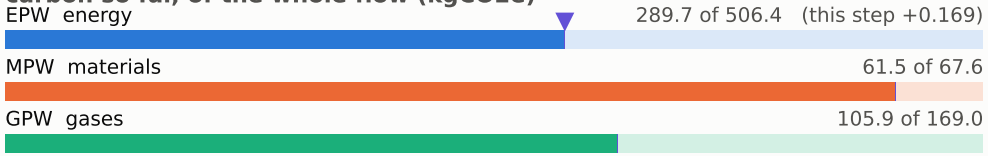
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

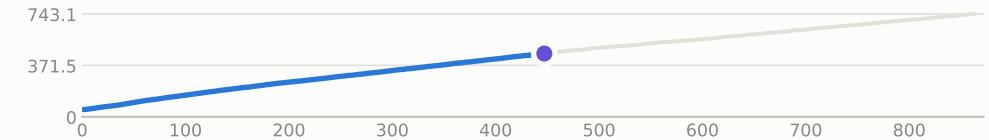
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

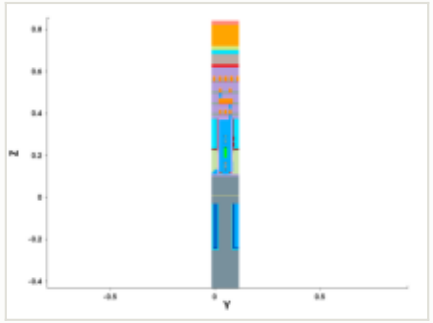
carbon so far, of the whole flow (kgCO2e)



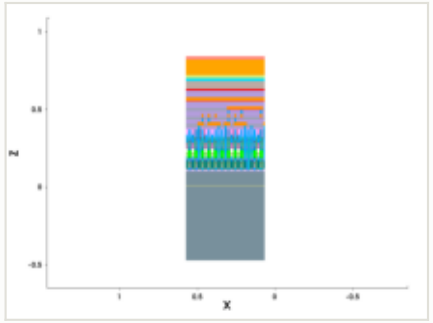
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



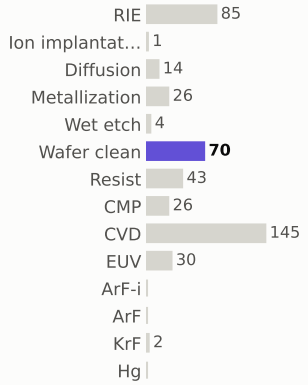
y cut at 0.049 µm



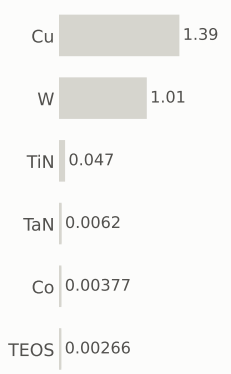
mask for this step

no mask for this step

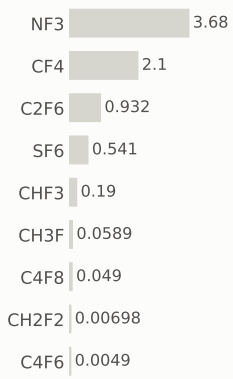
steps by process area



materials deposited (g)



gases emitted (g)



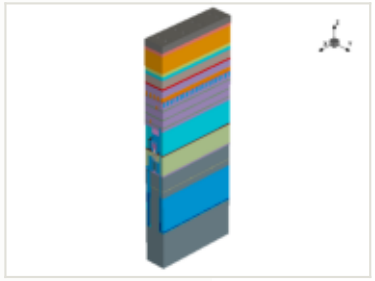
Step #4770: EUV

litho_patterning · EUV

1.96	459.1	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

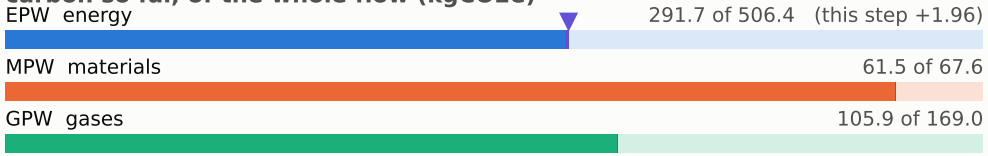


this step

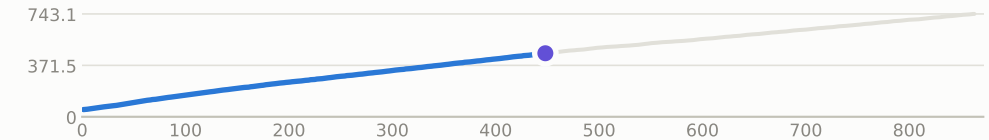
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

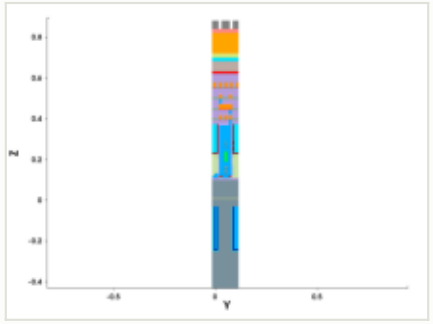
carbon so far, of the whole flow (kgCO2e)



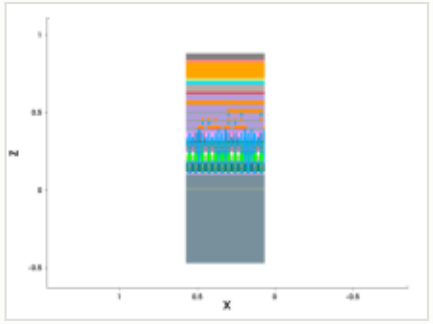
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

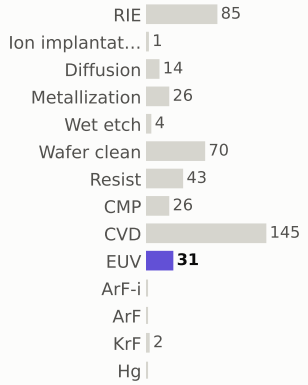


y cut at 0.049 µm

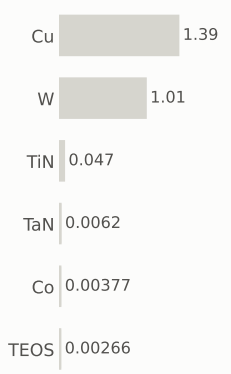


mask: layer 10371:0 (GDSII)

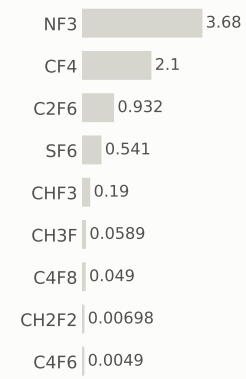
steps by process area



materials deposited (g)



gases emitted (g)



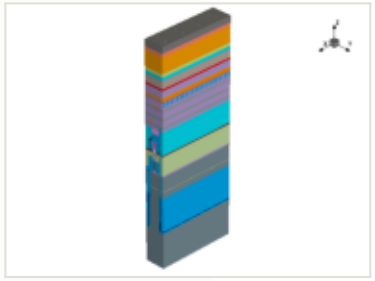
Step #4780: RIE

etch_anisotropic · RIE

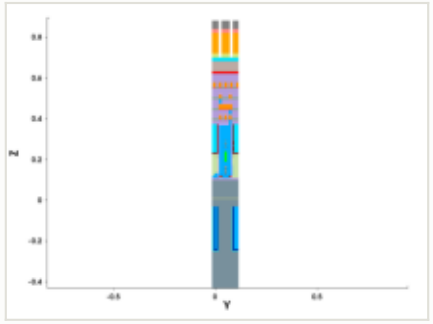
1.04	460.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

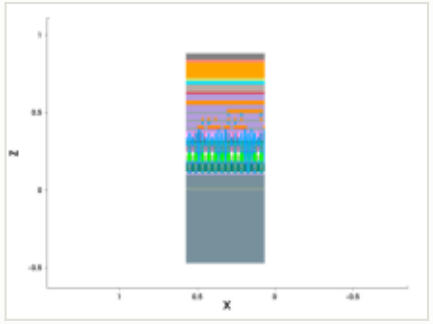
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

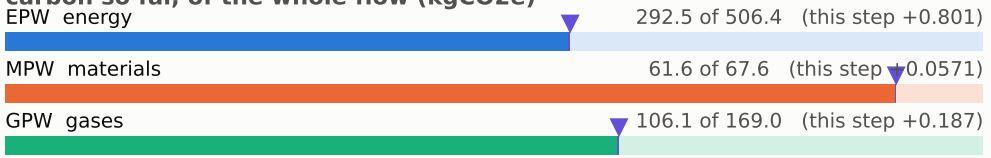
no mask for this step

this step

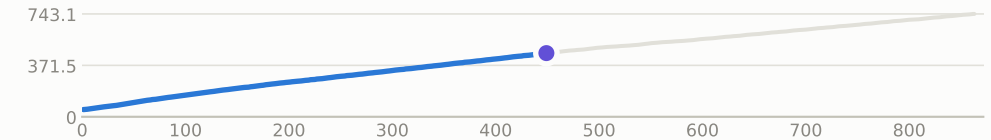
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

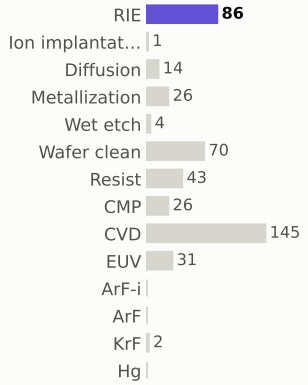
carbon so far, of the whole flow (kgCO2e)



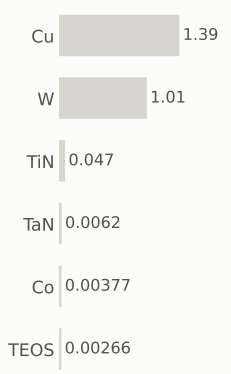
embodied carbon up to each step (kgCO2e per wafer)



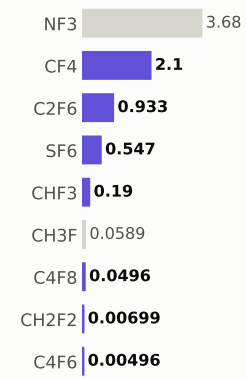
steps by process area



materials deposited (g)



gases emitted (g)



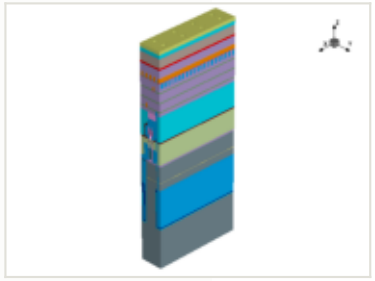
Step #4790: Wafer_clean

etch_strip · Wafer_clean

0.169	460.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

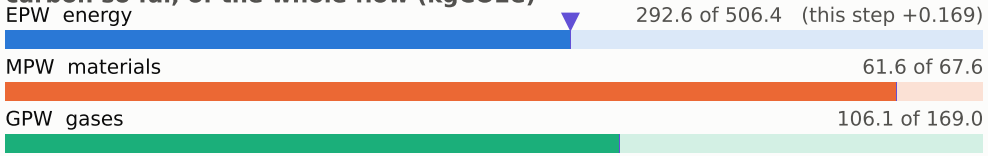


this step

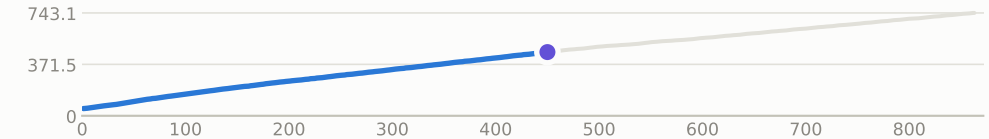
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

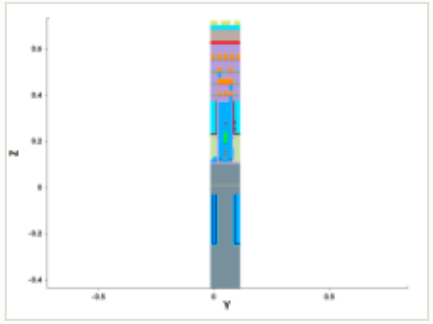
carbon so far, of the whole flow (kgCO2e)



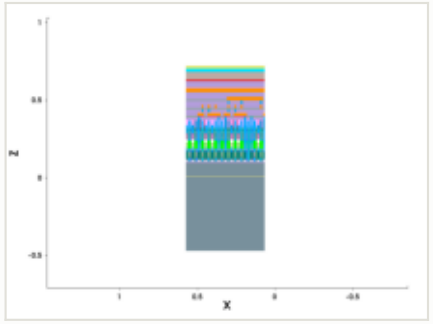
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



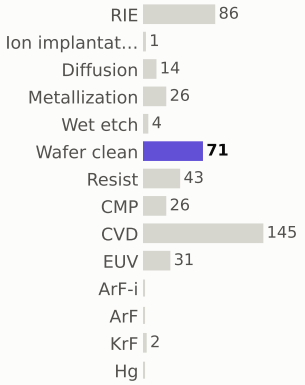
y cut at 0.049 µm



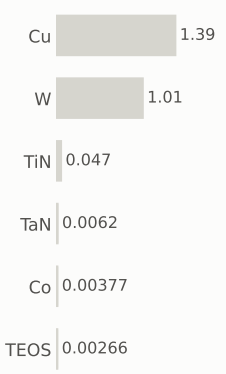
mask for this step

no mask for this step

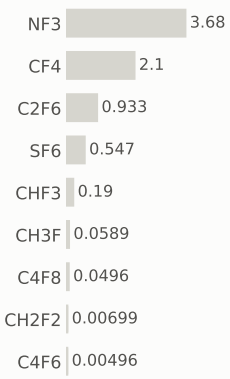
steps by process area



materials deposited (g)



gases emitted (g)



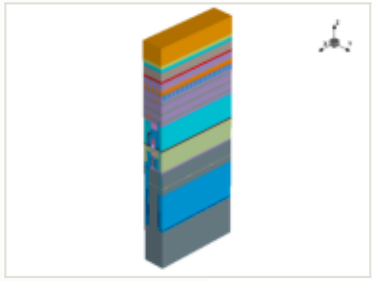
Step #4800: Resist

depo_fill · Resist

0.169	460.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

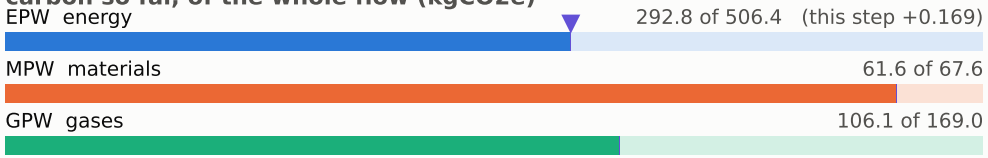


this step

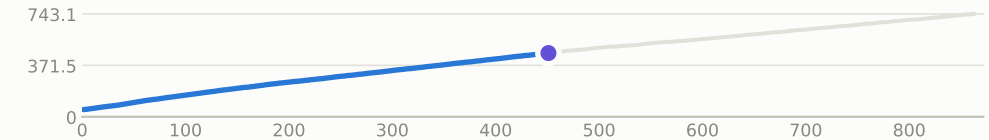
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

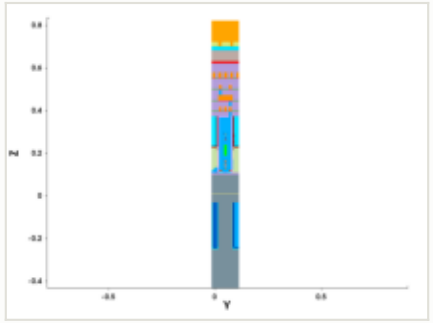
carbon so far, of the whole flow (kgCO2e)



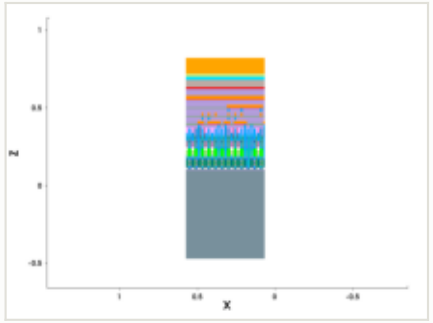
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



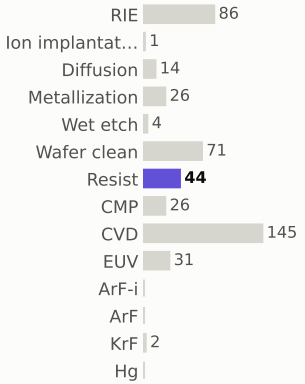
y cut at 0.049 µm



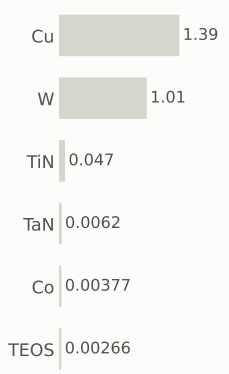
mask for this step

no mask for this step

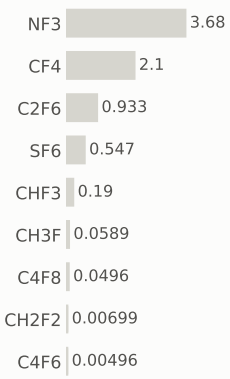
steps by process area



materials deposited (g)



gases emitted (g)



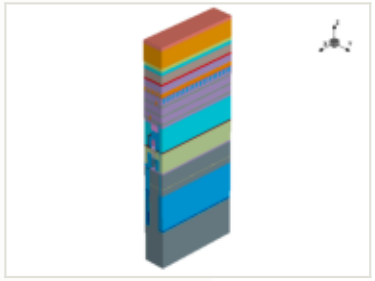
Step #4810: CVD

depo_fill · CVD

1.31	461.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

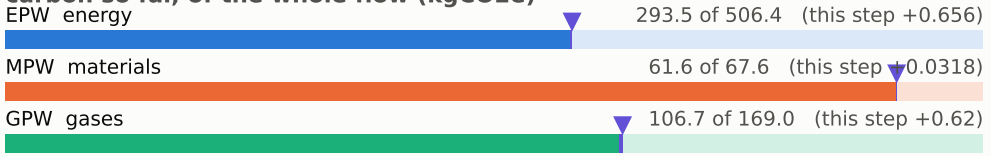


this step

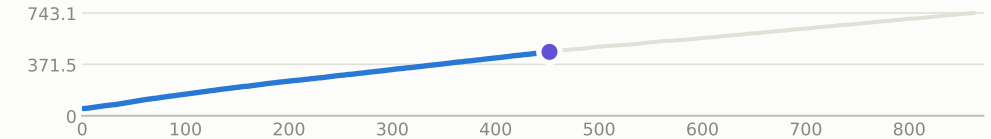
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

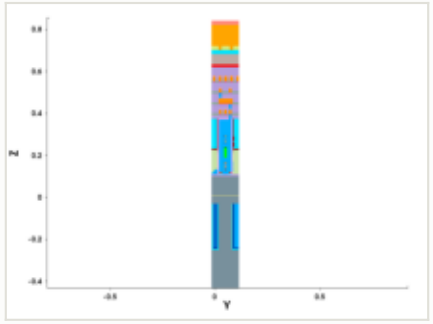
carbon so far, of the whole flow (kgCO2e)



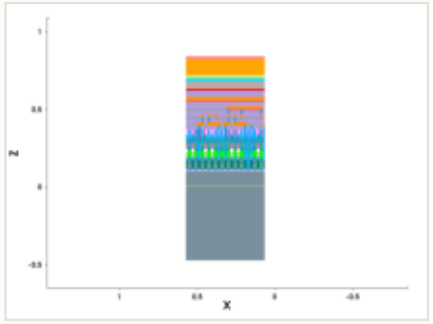
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



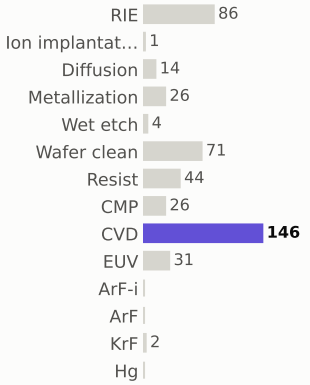
y cut at 0.049 µm



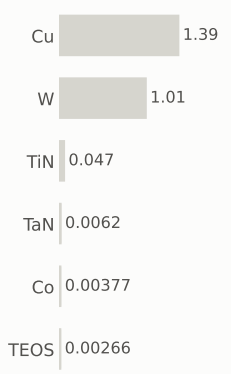
mask for this step

no mask for this step

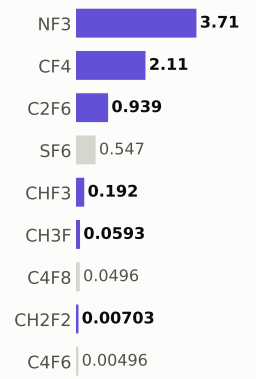
steps by process area



materials deposited (g)



gases emitted (g)



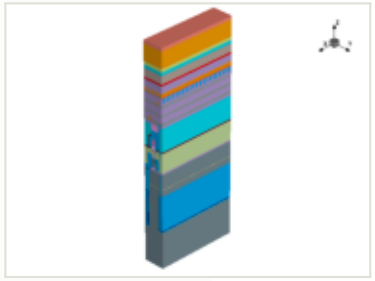
Step #4820: Wafer_clean

clean · Wafer_clean

0.169	461.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



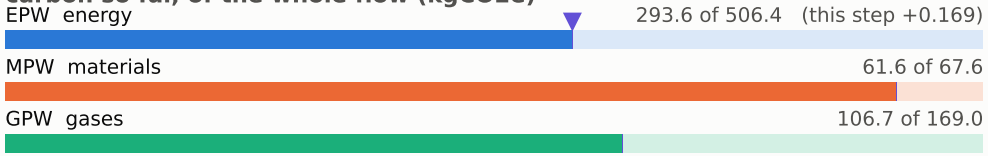
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

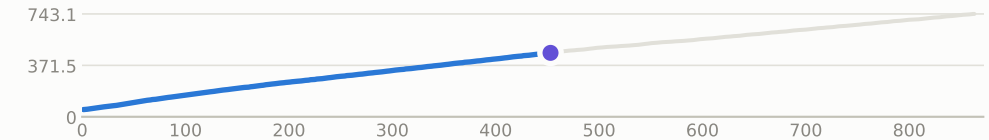
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

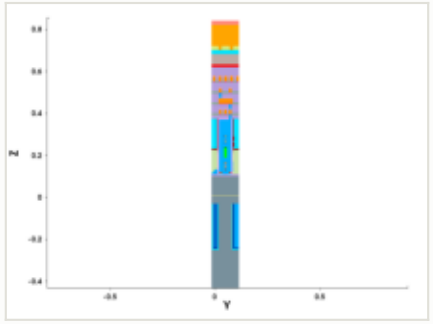
carbon so far, of the whole flow (kgCO2e)



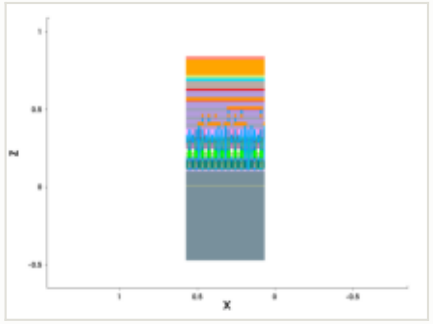
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



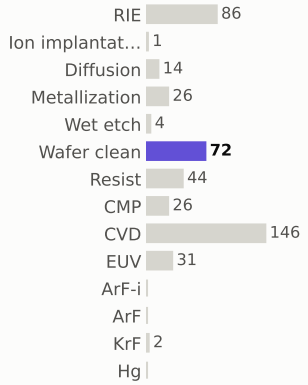
y cut at 0.049 µm



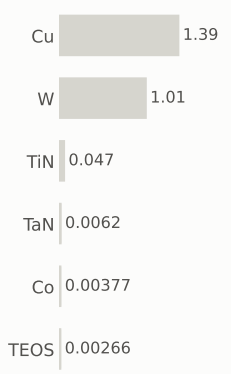
mask for this step

no mask for this step

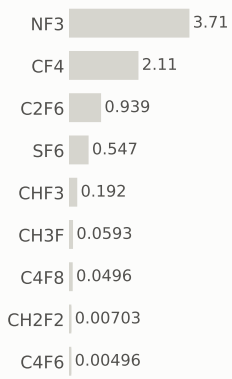
steps by process area



materials deposited (g)



gases emitted (g)



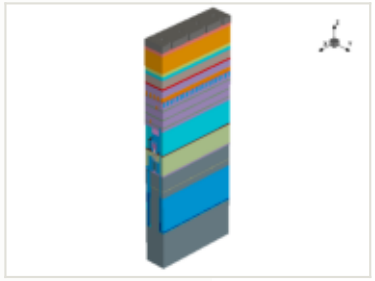
Step #4830: EUV

litho_patterning · EUV

1.96	463.9	0.26%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

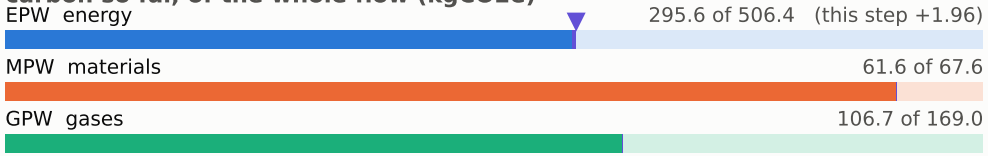


this step

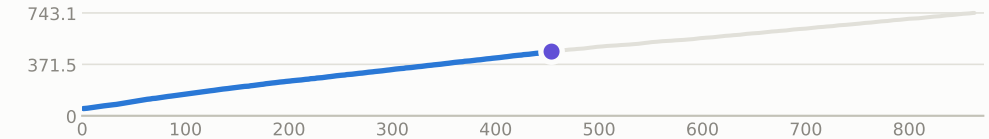
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

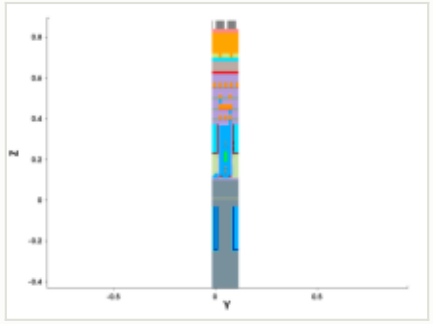
carbon so far, of the whole flow (kgCO2e)



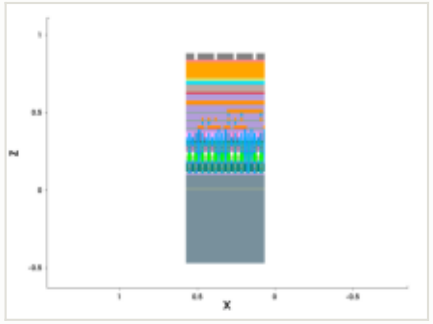
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

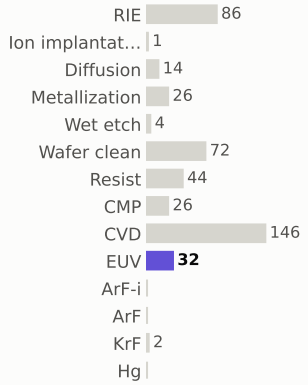


y cut at 0.049 µm

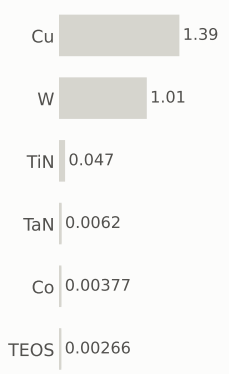


mask: layer 10372:0 (GDSII)

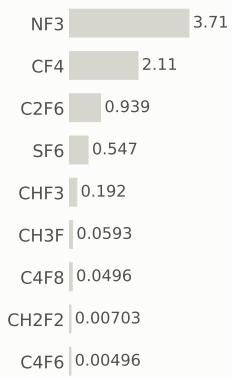
steps by process area



materials deposited (g)



gases emitted (g)



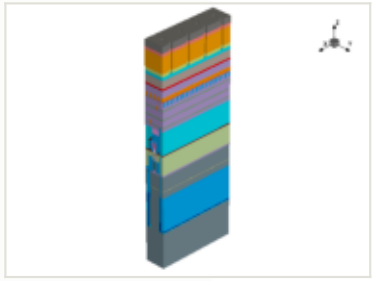
Step #4840: RIE

etch_anisotropic · RIE

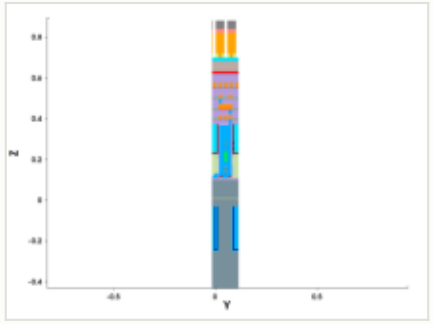
1.04	464.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

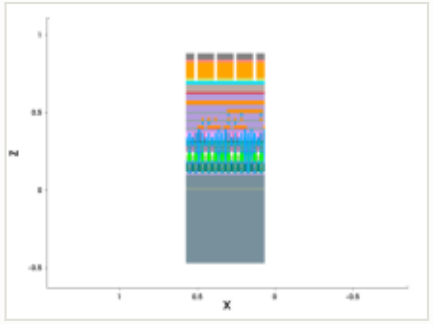
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

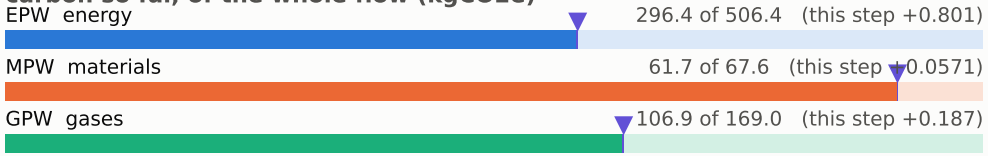
no mask for this step

this step

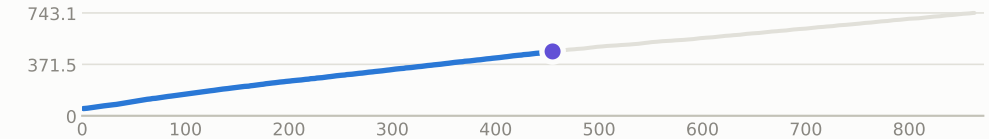
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

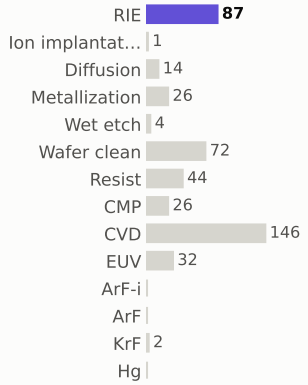
carbon so far, of the whole flow (kgCO2e)



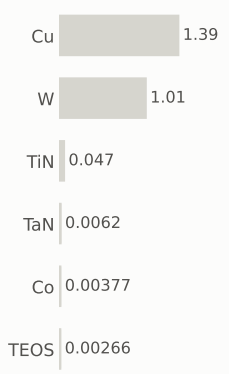
embodied carbon up to each step (kgCO2e per wafer)



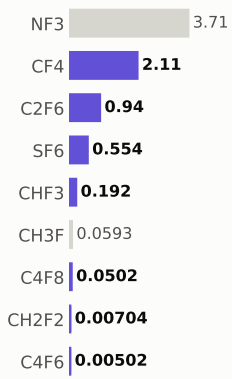
steps by process area



materials deposited (g)



gases emitted (g)



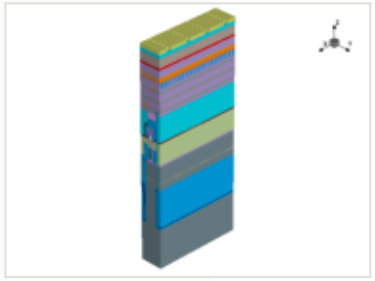
Step #4850: Wafer_clean

etch_strip · Wafer_clean

0.169	465.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

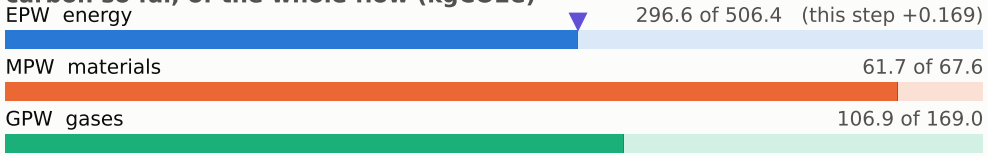


this step

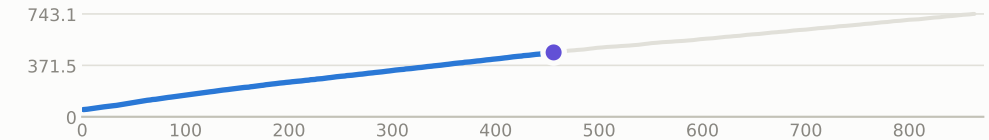
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

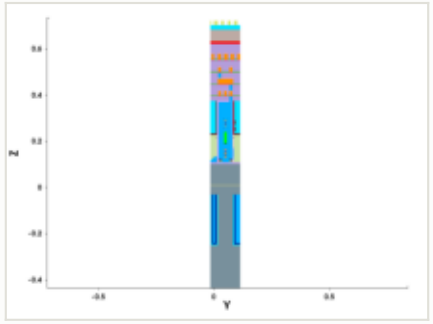
carbon so far, of the whole flow (kgCO2e)



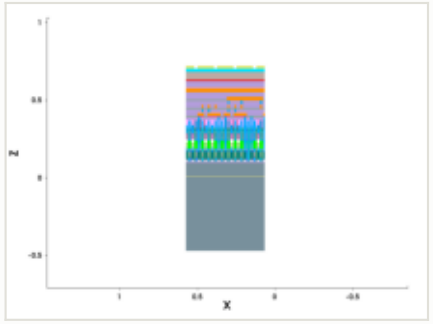
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



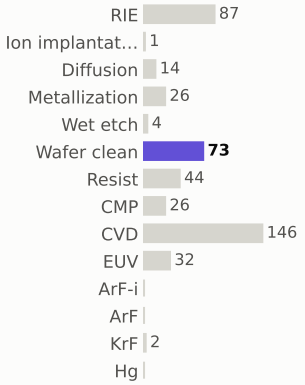
y cut at 0.049 µm



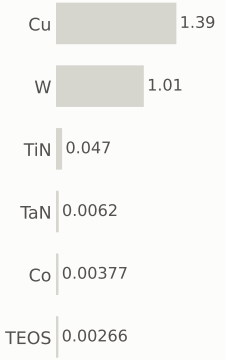
mask for this step

no mask for this step

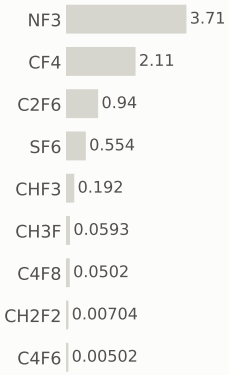
steps by process area



materials deposited (g)



gases emitted (g)



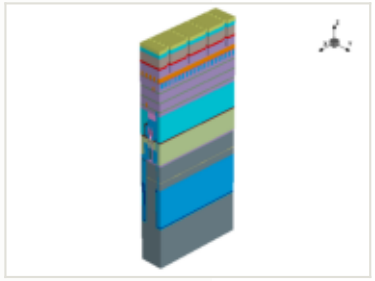
Step #4860: RIE

etch_anisotropic · RIE

1.04	466.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

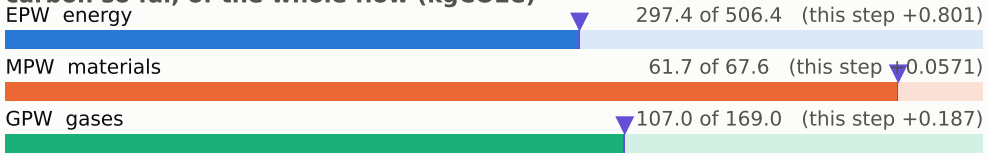


this step

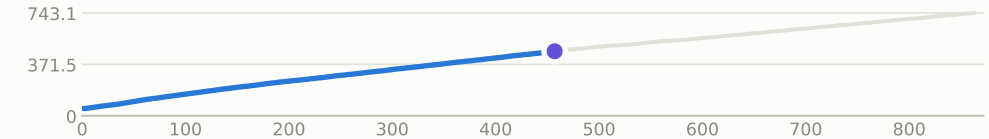
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

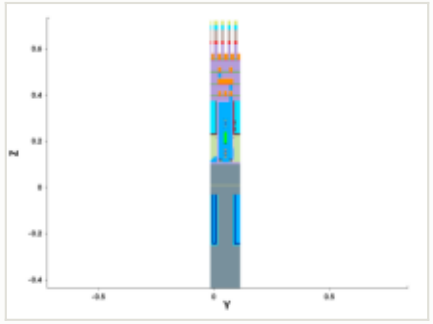
carbon so far, of the whole flow (kgCO2e)



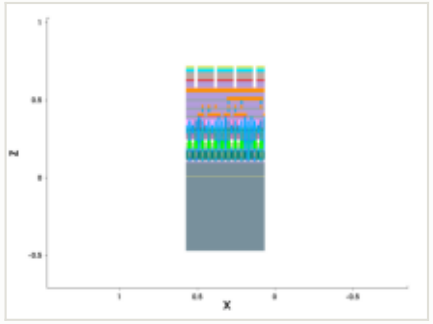
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



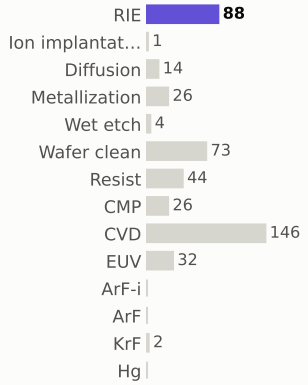
y cut at 0.049 µm



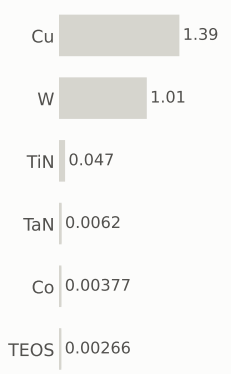
mask for this step

no mask for this step

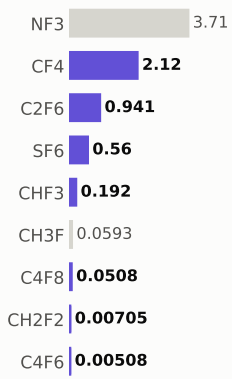
steps by process area



materials deposited (g)



gases emitted (g)



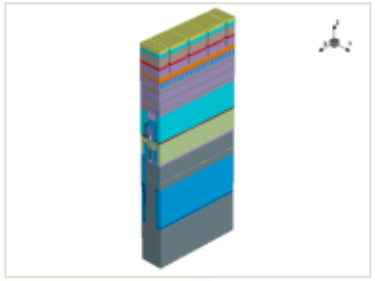
Step #4870: Wafer_clean

clean · Wafer_clean

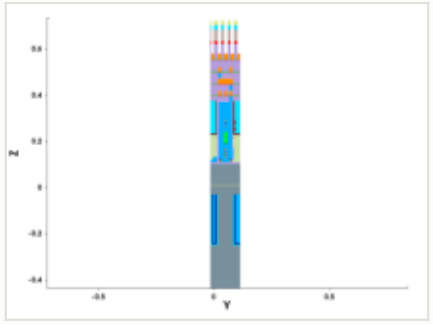
0.169	466.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

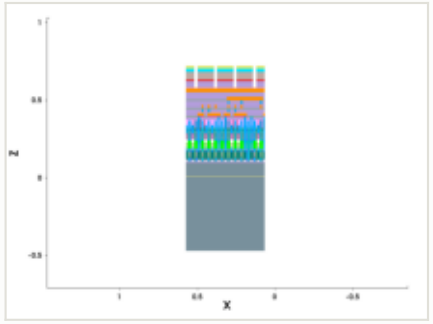
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

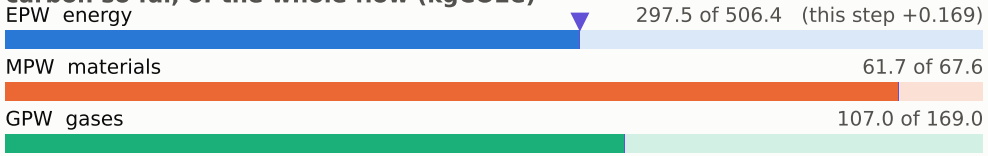
no mask for this step

this step

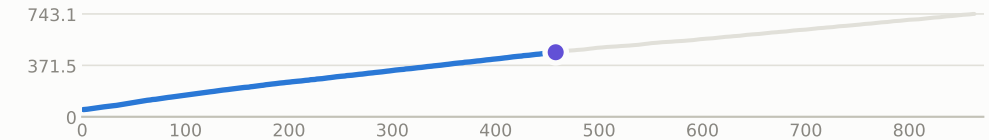
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

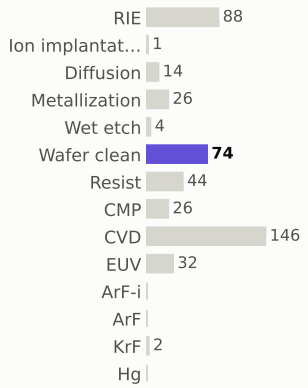
carbon so far, of the whole flow (kgCO2e)



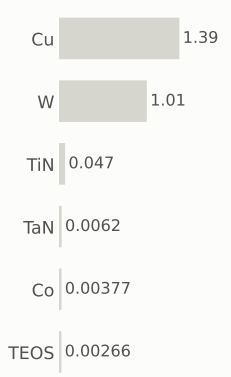
embodied carbon up to each step (kgCO2e per wafer)



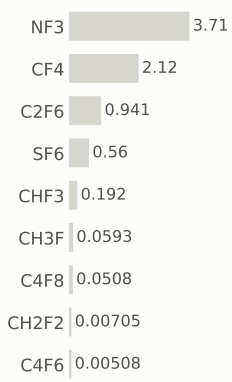
steps by process area



materials deposited (g)



gases emitted (g)



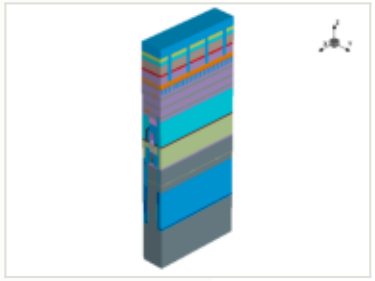
Step #4880: Metallization

depo_fill · Metallization

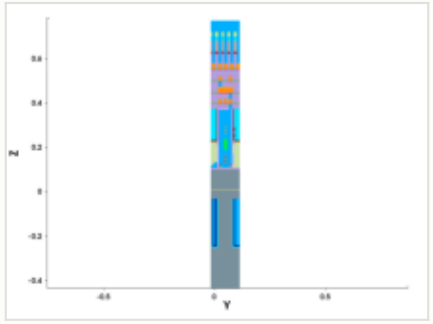
1.21	467.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

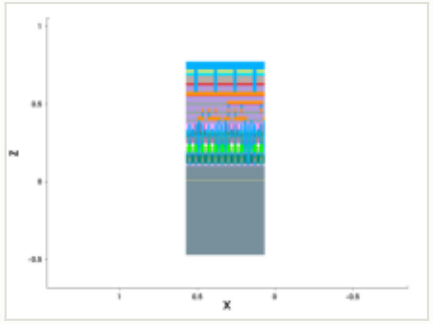
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

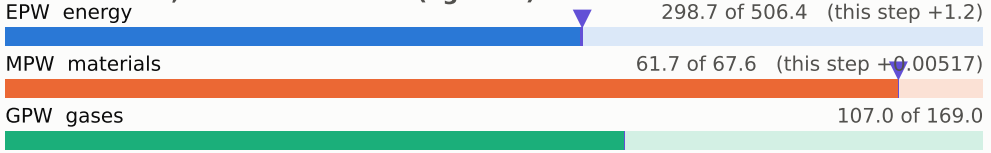
no mask for this step

this step

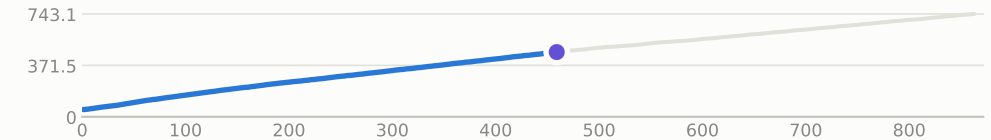
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

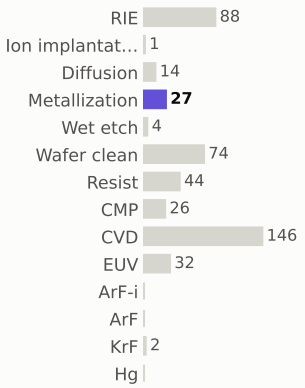
carbon so far, of the whole flow (kgCO2e)



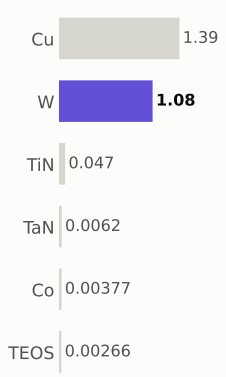
embodied carbon up to each step (kgCO2e per wafer)



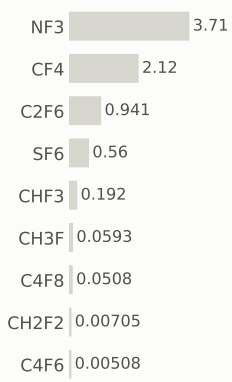
steps by process area



materials deposited (g)



gases emitted (g)



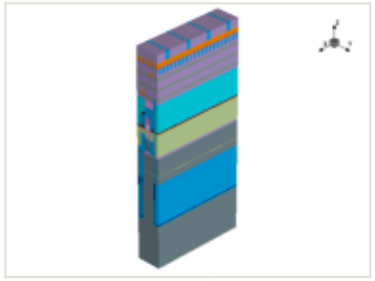
Step #4890: CMP

etch_cmp · CMP

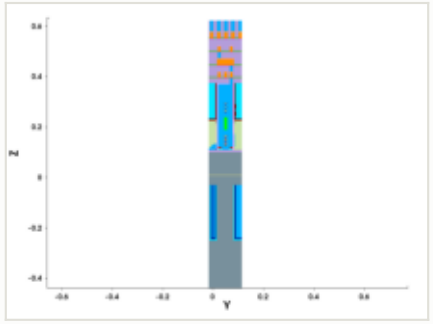
0.169	467.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

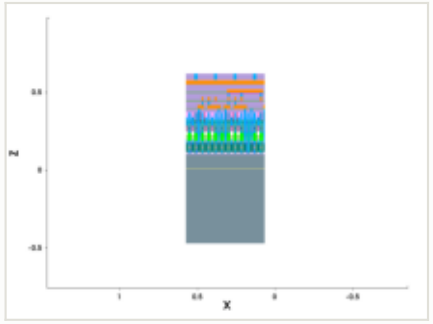
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



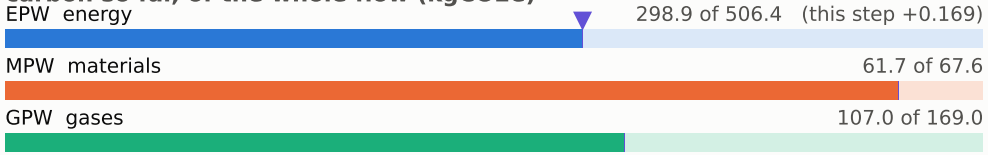
mask for this step

no mask for this step

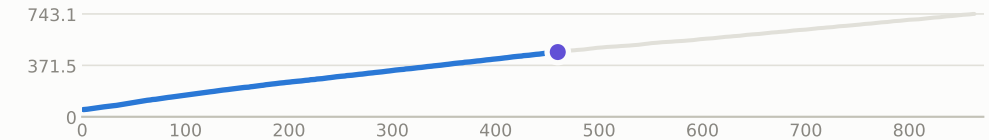
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

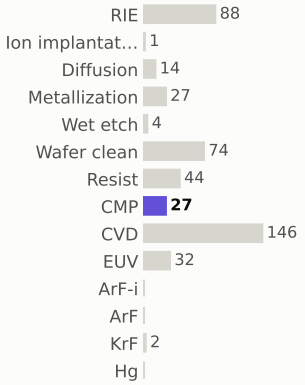
carbon so far, of the whole flow (kgCO2e)



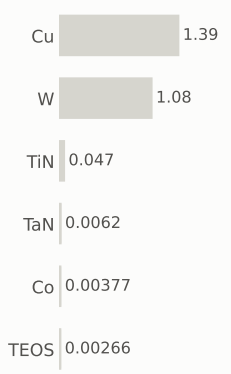
embodied carbon up to each step (kgCO2e per wafer)



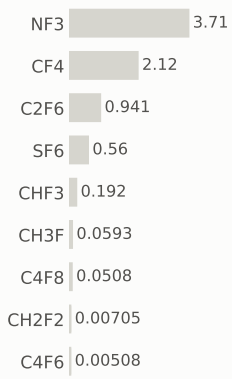
steps by process area



materials deposited (g)



gases emitted (g)



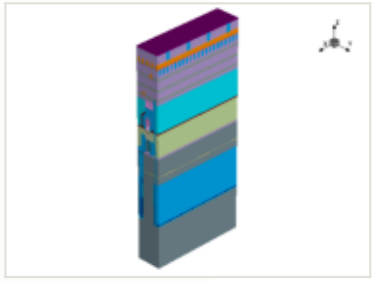
Step #4900: CVD

depo_fill · CVD

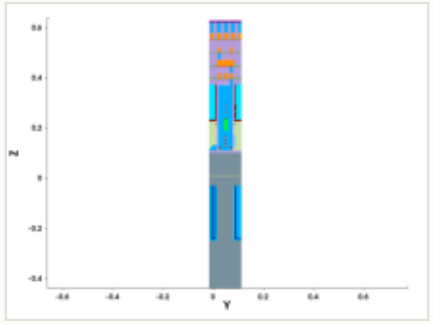
1.31	469.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

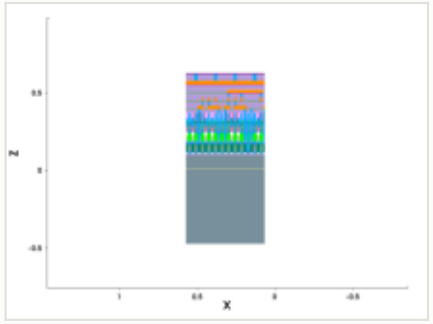
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

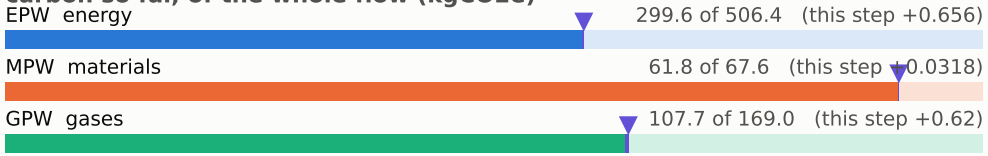
no mask for this step

this step

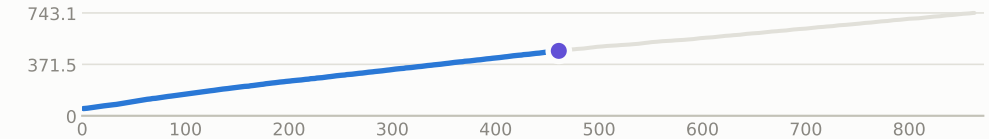
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

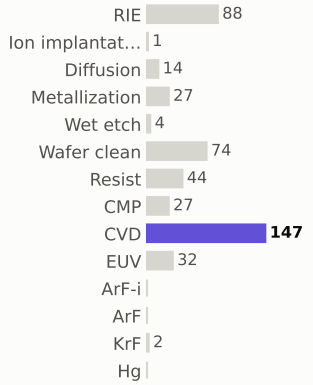
carbon so far, of the whole flow (kgCO2e)



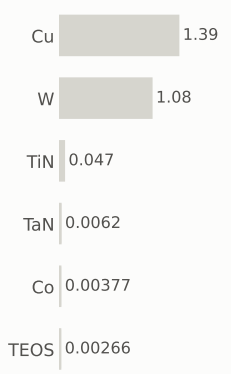
embodied carbon up to each step (kgCO2e per wafer)



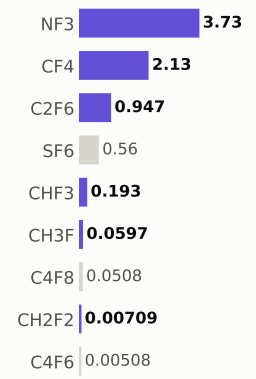
steps by process area



materials deposited (g)



gases emitted (g)



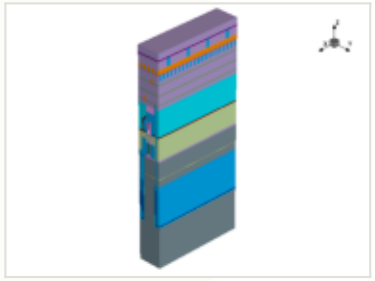
Step #4910: CVD

depo_fill · CVD

1.31	470.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

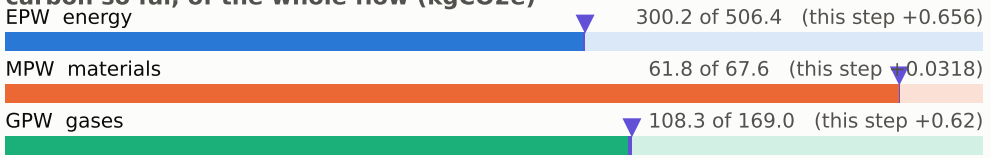


this step

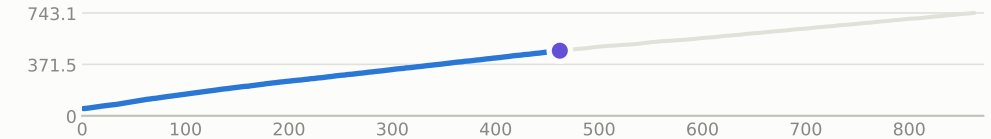
process area	CVD
material	—
thickness	0.036 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

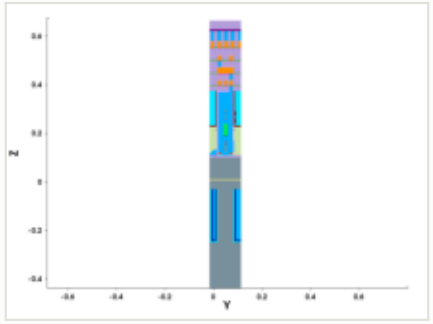
carbon so far, of the whole flow (kgCO2e)



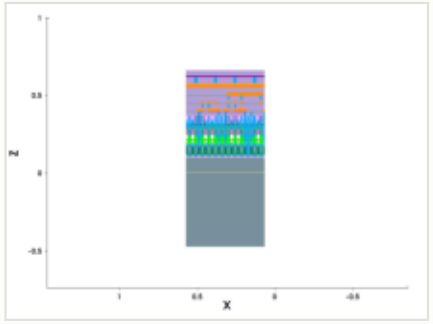
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



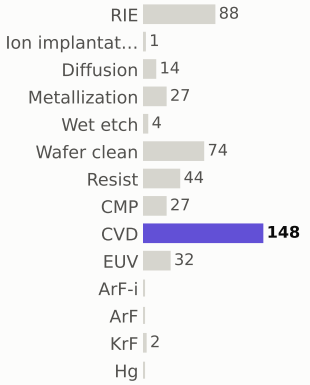
y cut at 0.049 µm



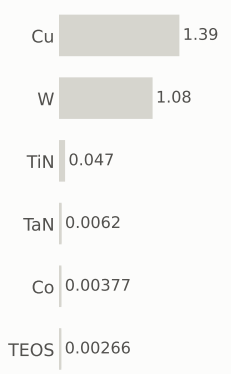
mask for this step

no mask for this step

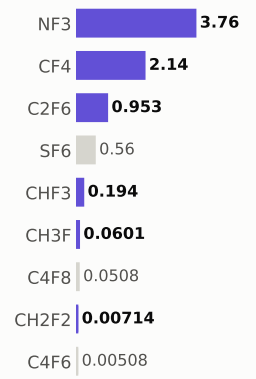
steps by process area



materials deposited (g)



gases emitted (g)



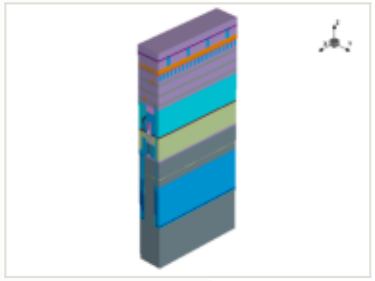
Step #4920: Diffusion

treatment · Diffusion

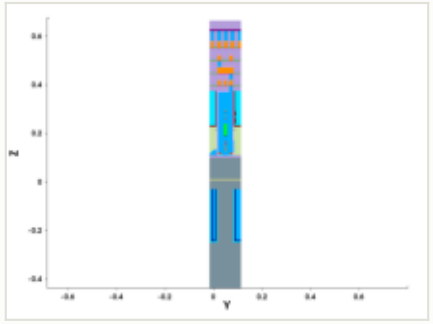
0.801	471.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

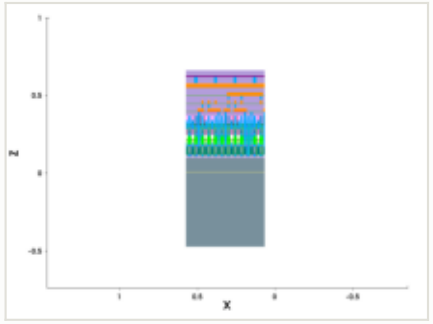
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

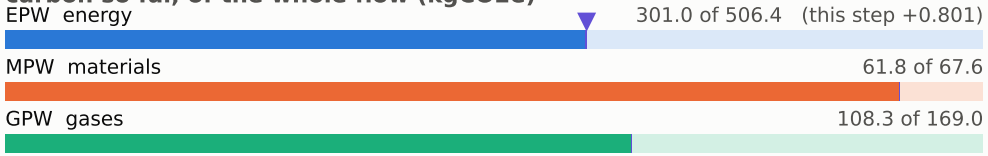
no mask for this step

this step

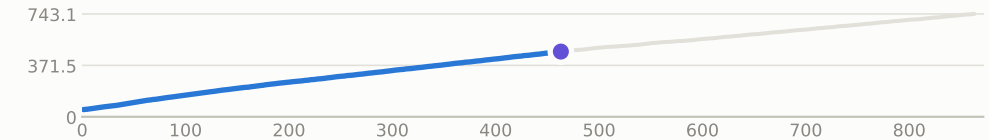
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

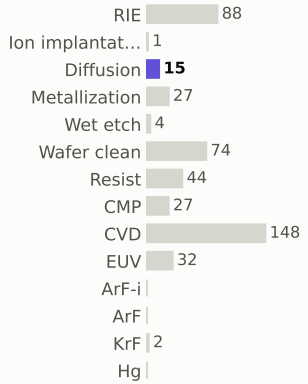
carbon so far, of the whole flow (kgCO2e)



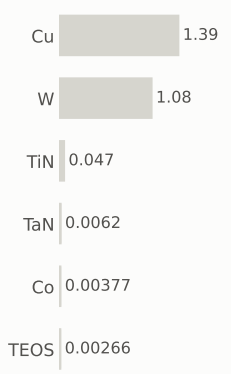
embodied carbon up to each step (kgCO2e per wafer)



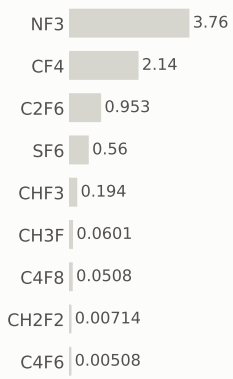
steps by process area



materials deposited (g)



gases emitted (g)



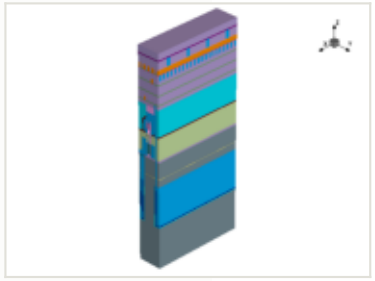
Step #4930: Wafer_clean

clean · Wafer_clean

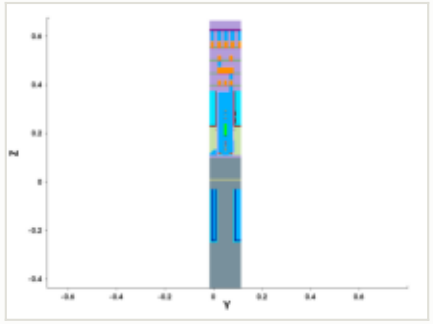
0.169	471.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

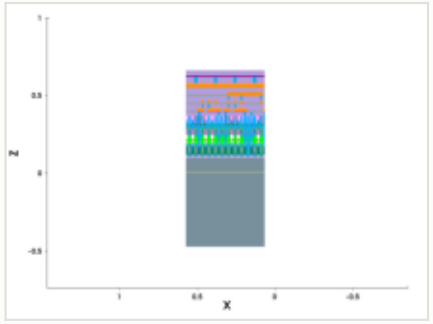
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

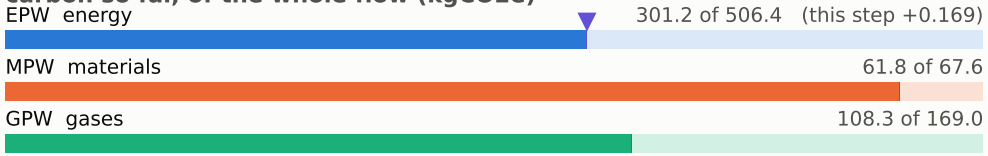
no mask for this step

this step

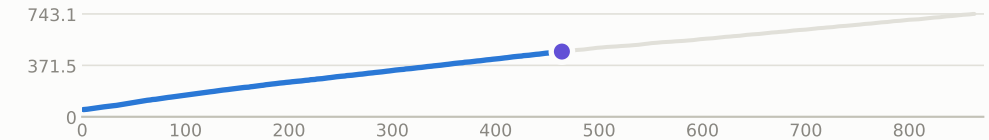
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

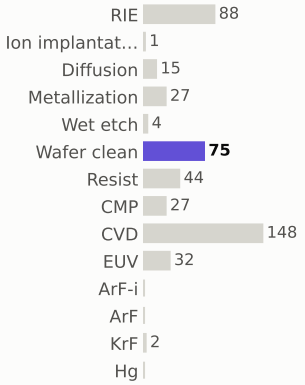
carbon so far, of the whole flow (kgCO2e)



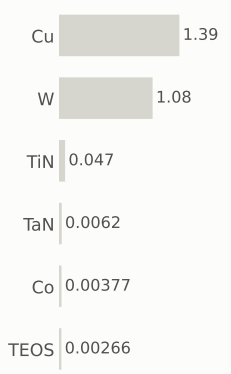
embodied carbon up to each step (kgCO2e per wafer)



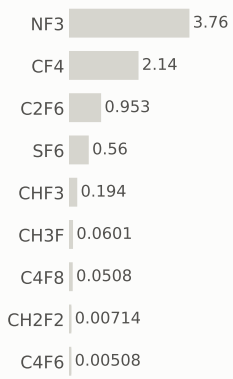
steps by process area



materials deposited (g)



gases emitted (g)



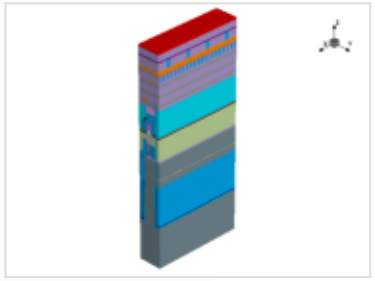
Step #4940: CVD

depo_fill · CVD

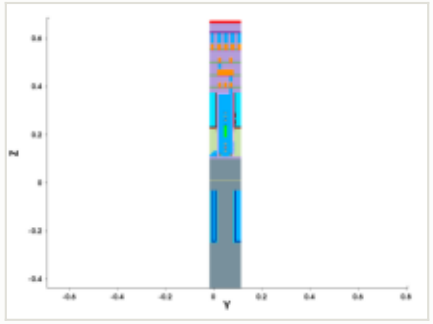
1.31	472.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

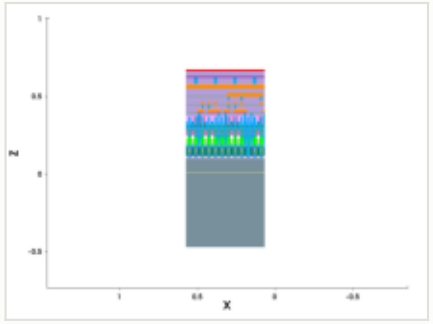
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

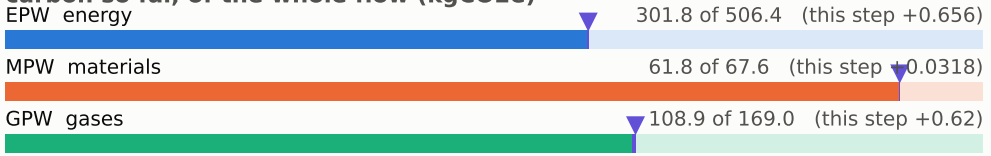
no mask for this step

this step

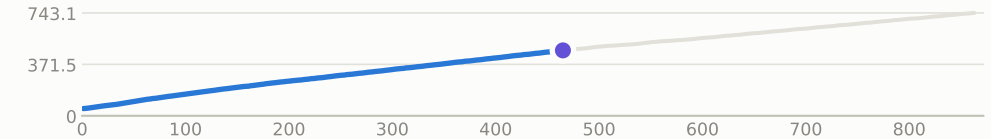
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

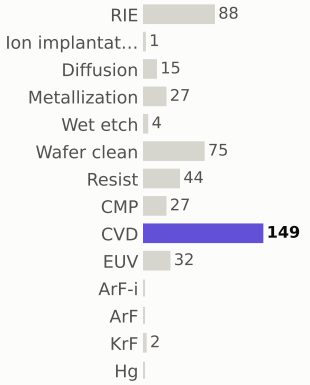
carbon so far, of the whole flow (kgCO2e)



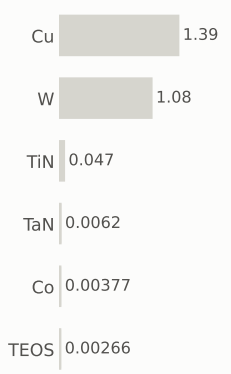
embodied carbon up to each step (kgCO2e per wafer)



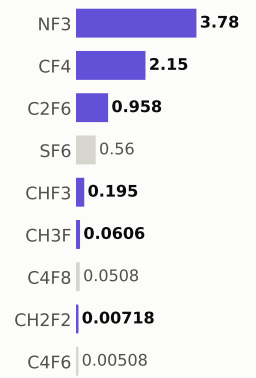
steps by process area



materials deposited (g)



gases emitted (g)



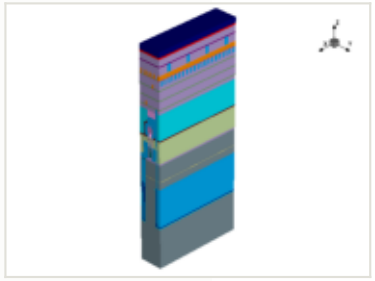
Step #4950: CVD

depo_fill · CVD

1.31	473.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

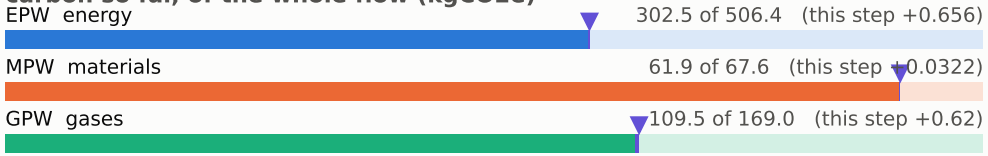
zoomed to fit



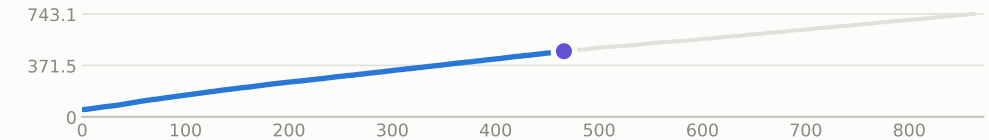
this step

process area	CVD	EPW	0.656 kgCO2e
material	TiN	EPW tool	0.469 kgCO2e
thickness	0.03 µm	EPW overhead	0.187 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	0.0322 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

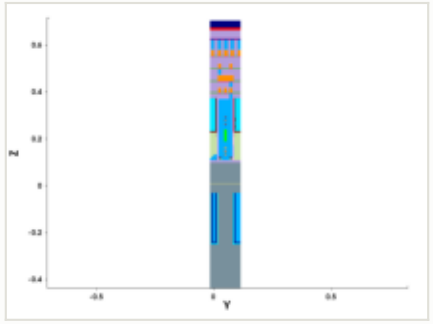
carbon so far, of the whole flow (kgCO2e)



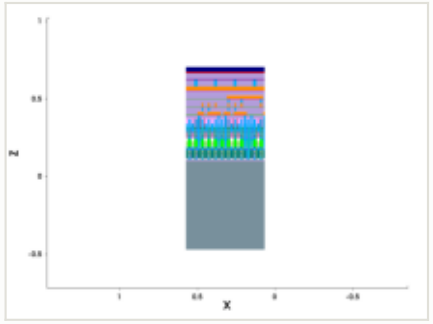
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



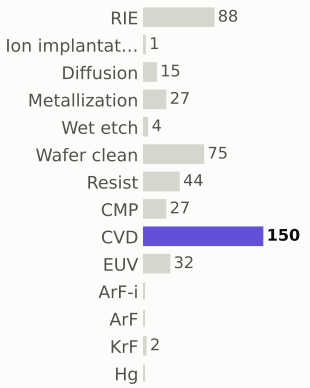
y cut at 0.049 µm



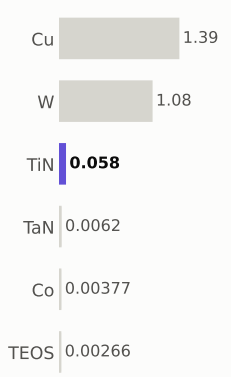
mask for this step

no mask for this step

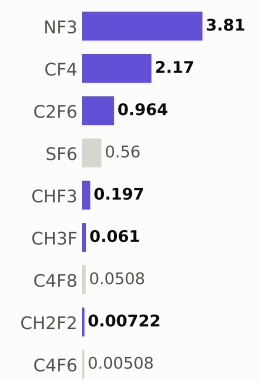
steps by process area



materials deposited (g)



gases emitted (g)



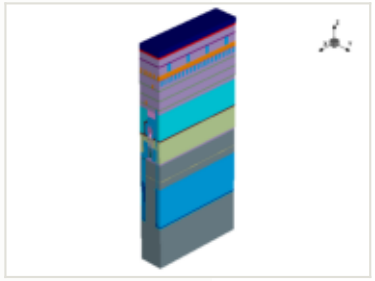
Step #4960: Wafer_clean

clean · Wafer_clean

0.169	474.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



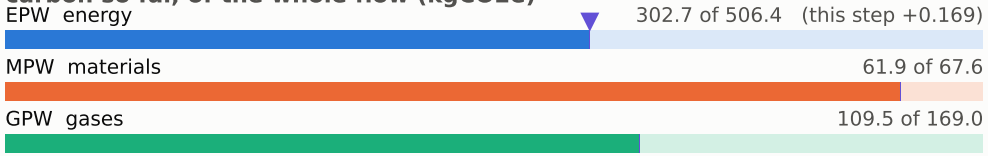
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

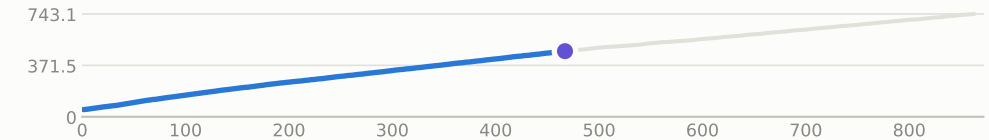
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

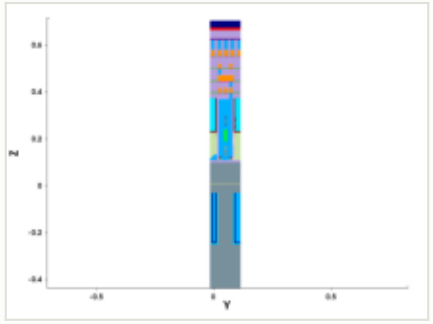
carbon so far, of the whole flow (kgCO2e)



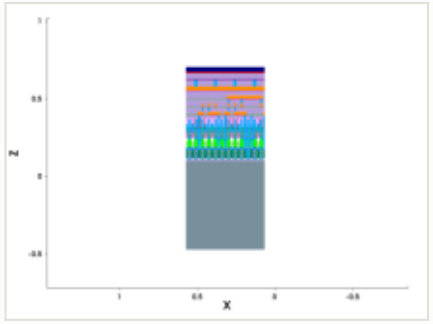
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



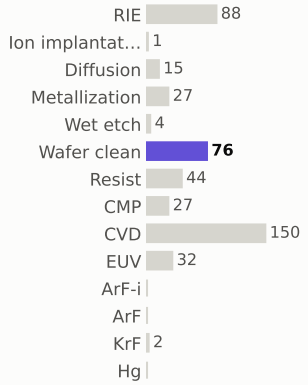
y cut at 0.049 µm



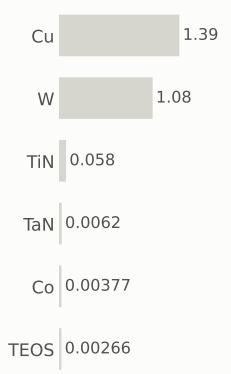
mask for this step

no mask for this step

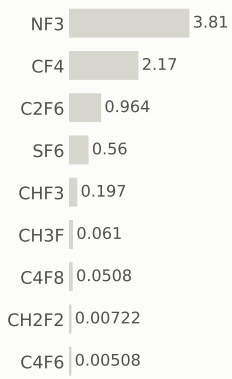
steps by process area



materials deposited (g)



gases emitted (g)



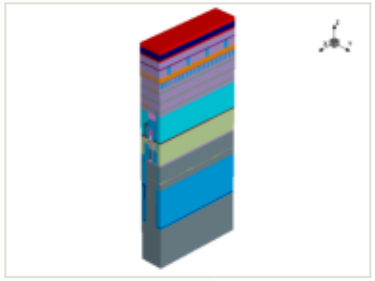
Step #4970: CVD

depo_fill · CVD

1.31	475.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

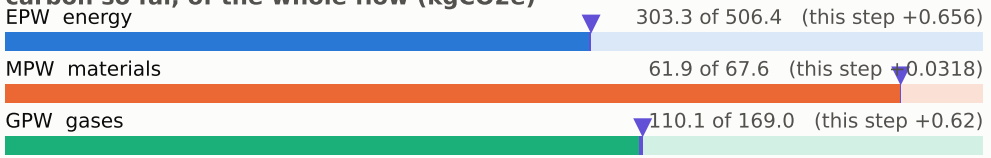


this step

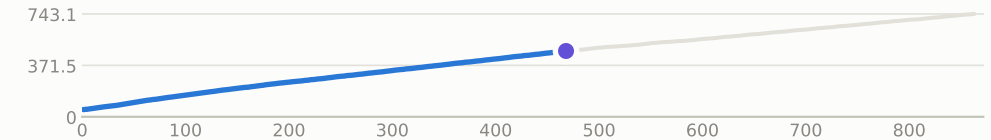
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

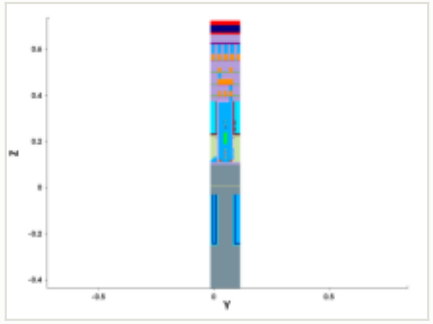
carbon so far, of the whole flow (kgCO2e)



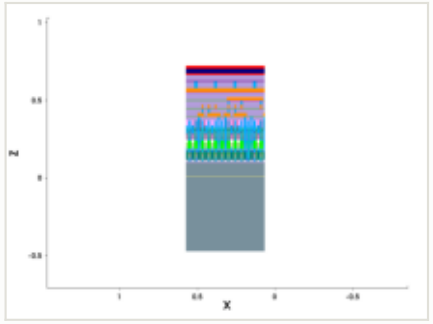
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



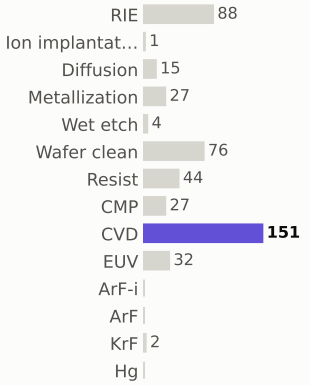
y cut at 0.049 µm



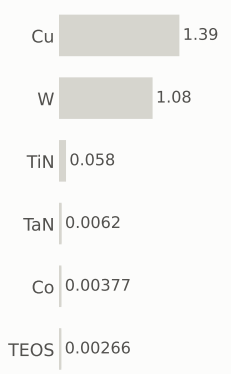
mask for this step

no mask for this step

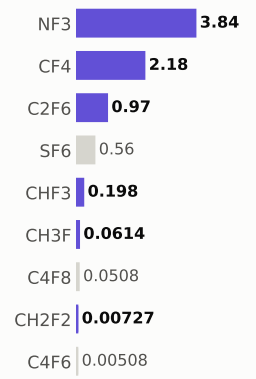
steps by process area



materials deposited (g)



gases emitted (g)



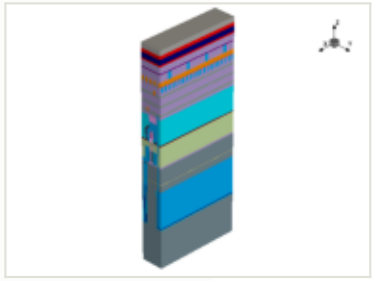
Step #4980: CVD

depo_fill · CVD

1.31	476.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

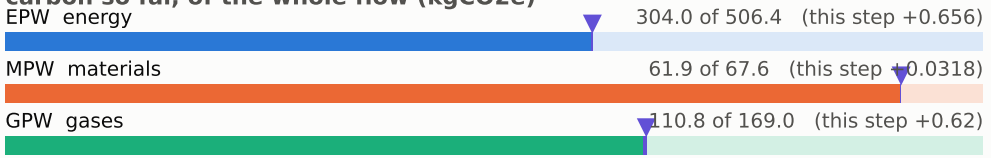


this step

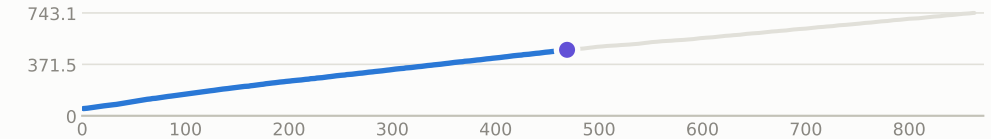
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

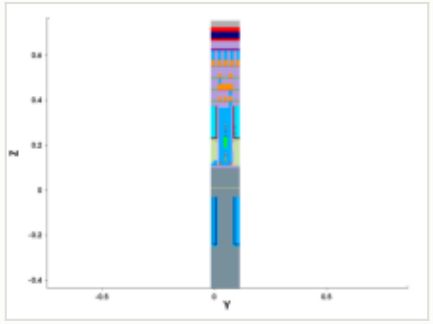
carbon so far, of the whole flow (kgCO2e)



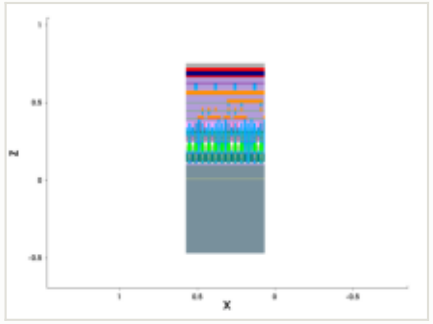
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



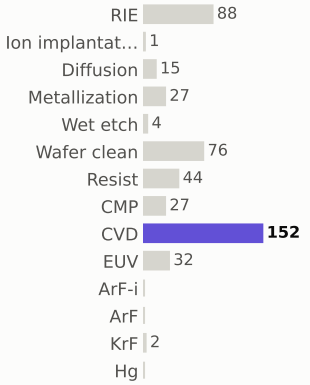
y cut at 0.049 µm



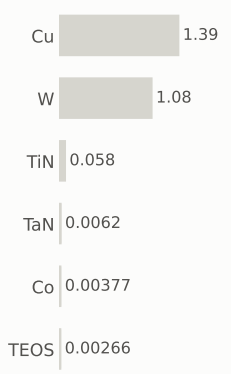
mask for this step

no mask for this step

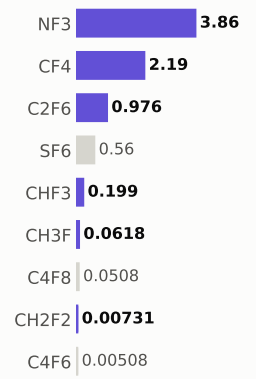
steps by process area



materials deposited (g)



gases emitted (g)



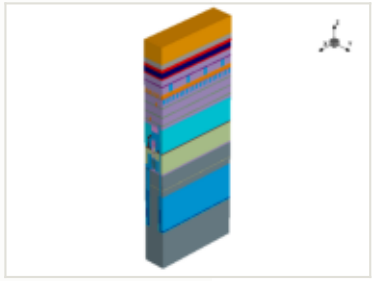
Step #4990: Resist

depo_fill · Resist

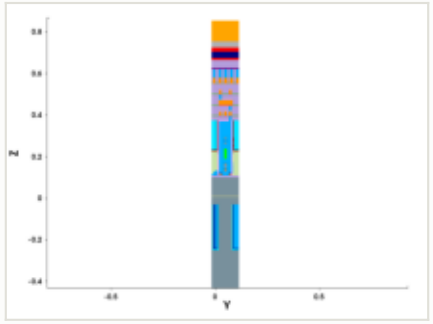
0.169	476.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

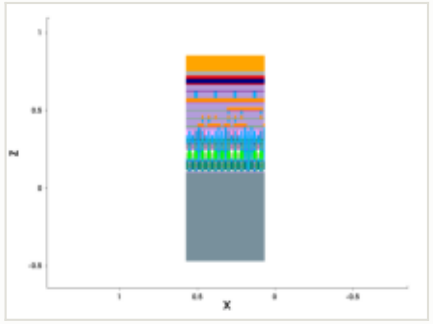
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



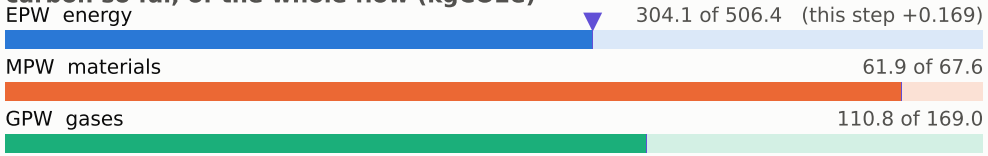
mask for this step

no mask for this step

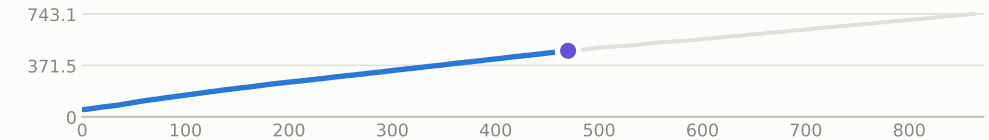
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

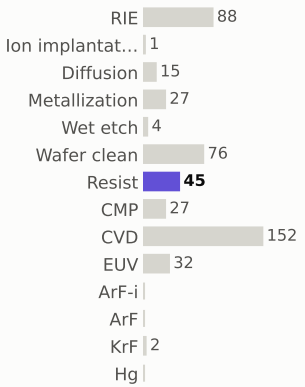
carbon so far, of the whole flow (kgCO2e)



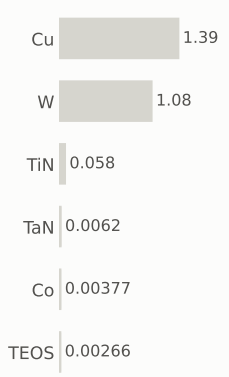
embodied carbon up to each step (kgCO2e per wafer)



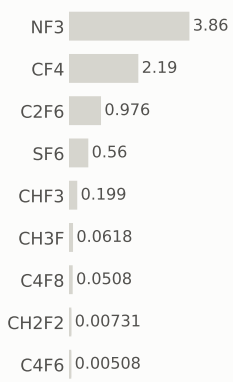
steps by process area



materials deposited (g)



gases emitted (g)



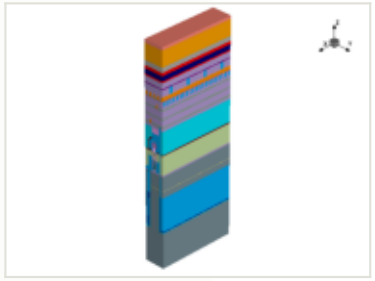
Step #5000: CVD

depo_fill · CVD

1.31	478.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

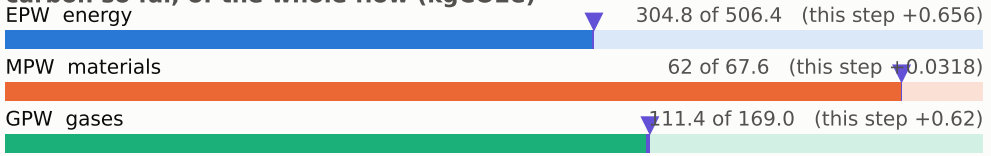


this step

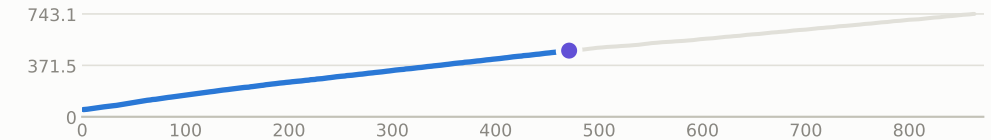
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

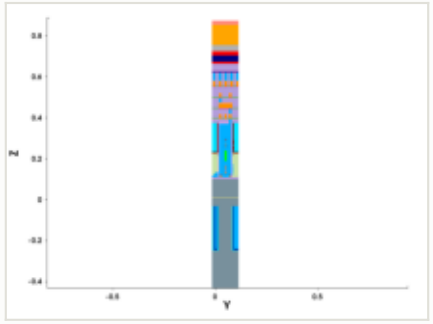
carbon so far, of the whole flow (kgCO2e)



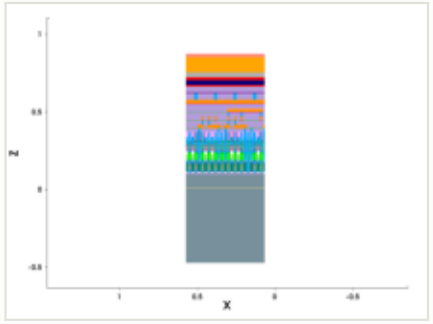
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



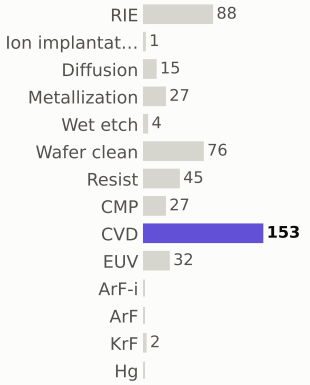
y cut at 0.049 µm



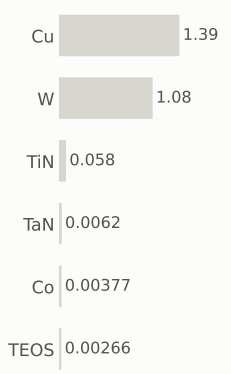
mask for this step

no mask for this step

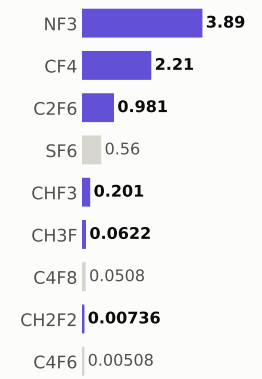
steps by process area



materials deposited (g)



gases emitted (g)



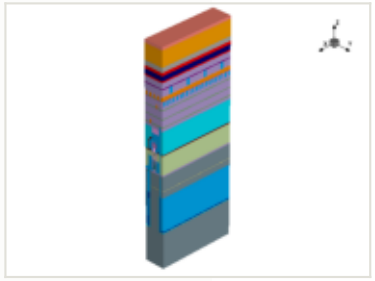
Step #5010: Wafer_clean

clean · Wafer_clean

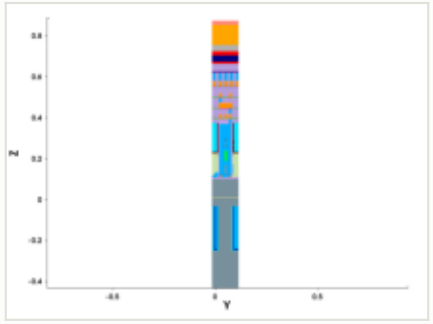
0.169	478.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

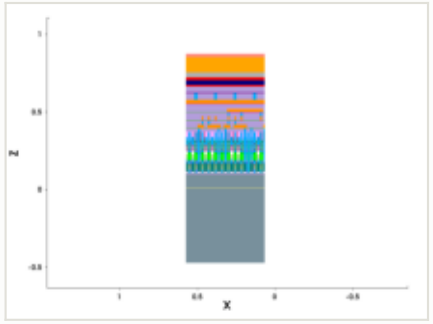
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

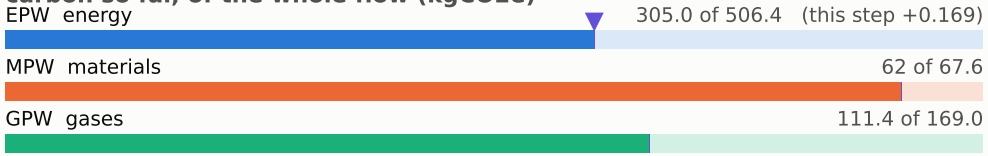
no mask for this step

this step

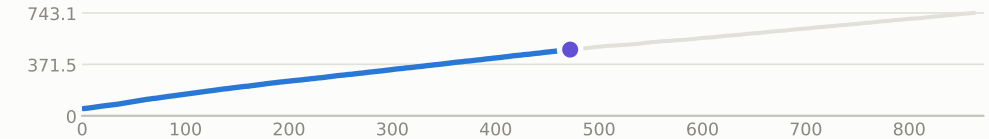
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

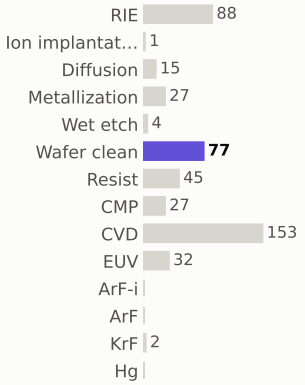
carbon so far, of the whole flow (kgCO2e)



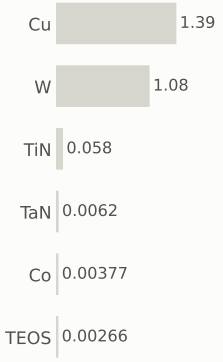
embodied carbon up to each step (kgCO2e per wafer)



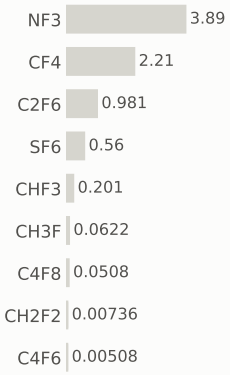
steps by process area



materials deposited (g)



gases emitted (g)



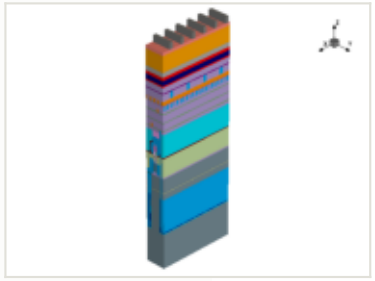
Step #5020: ArF-i

litho_patterning · ArF-i

0.217	478.5	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

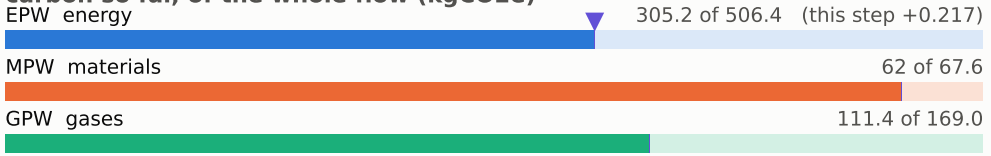


this step

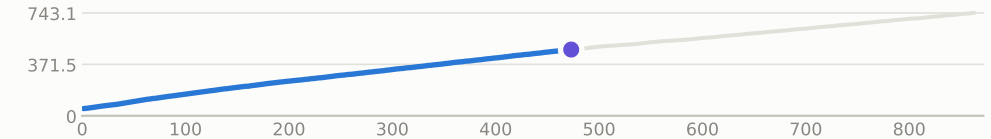
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

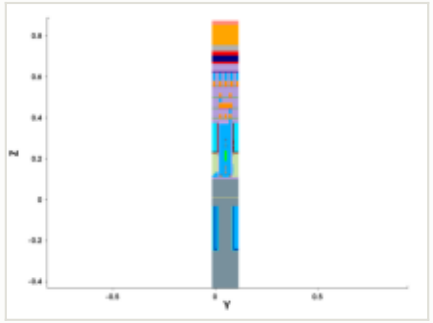
carbon so far, of the whole flow (kgCO2e)



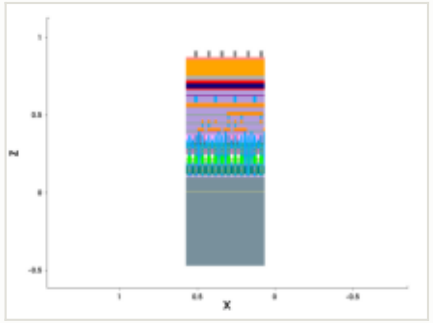
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

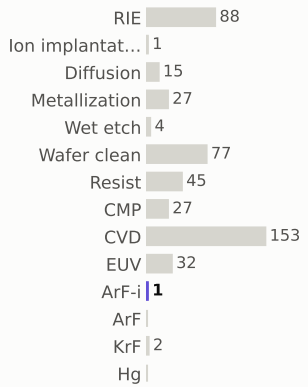


y cut at 0.049 µm

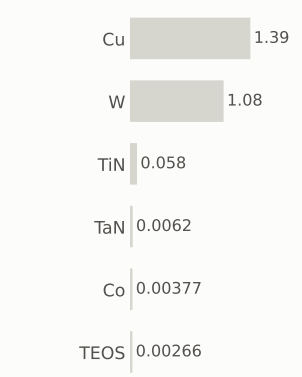


mask: layer 10401:0, positive (GDSII)

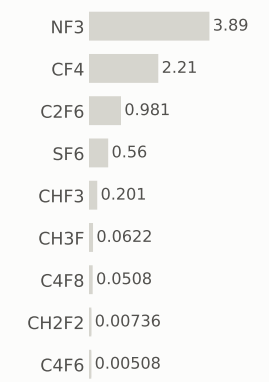
steps by process area



materials deposited (g)



gases emitted (g)



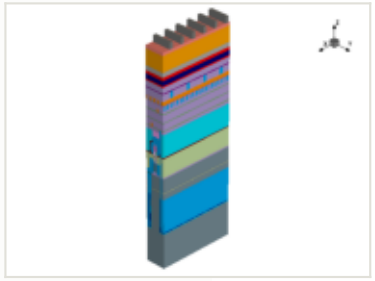
Step #5030: Ion_implantation

treatment · Ion_implantation

0.801	479.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



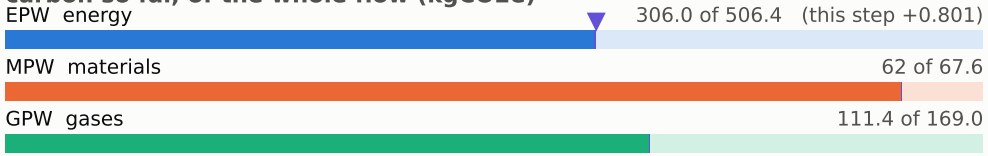
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

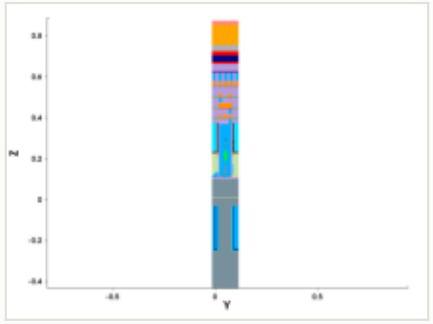
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

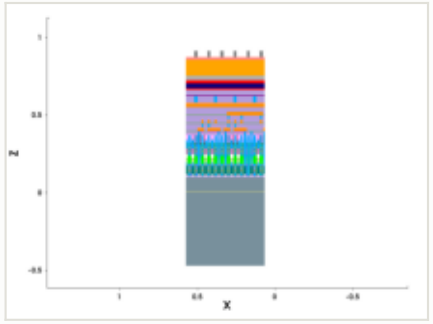


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.385 µm



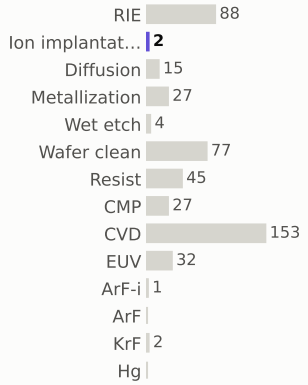
y cut at 0.049 µm



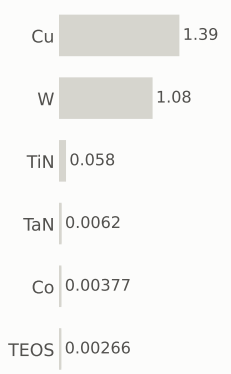
mask for this step

no mask for this step

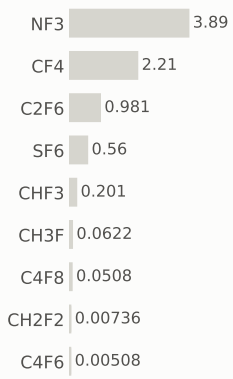
steps by process area



materials deposited (g)



gases emitted (g)



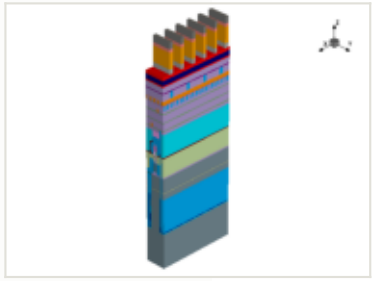
Step #5040: RIE

etch_anisotropic · RIE

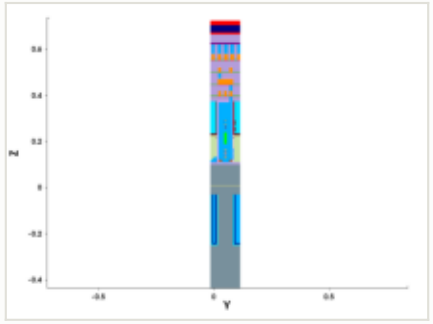
1.04	480.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

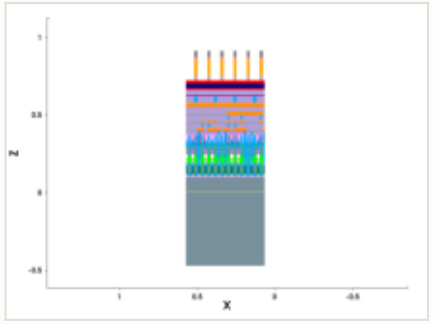
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

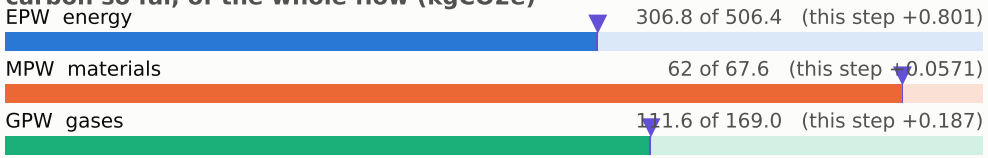
no mask for this step

this step

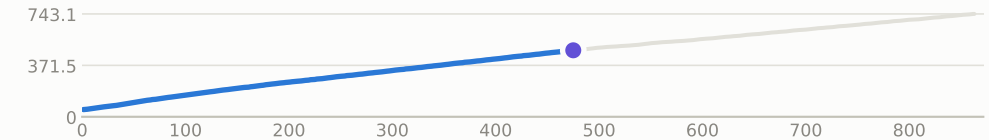
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

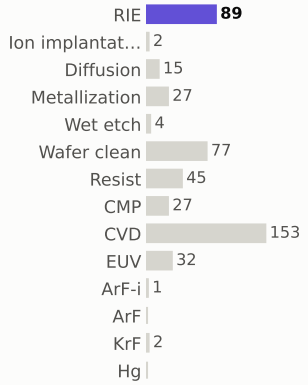
carbon so far, of the whole flow (kgCO2e)



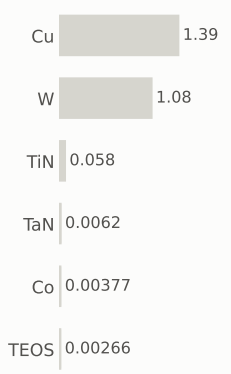
embodied carbon up to each step (kgCO2e per wafer)



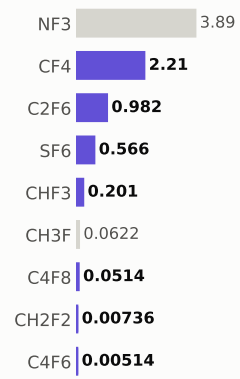
steps by process area



materials deposited (g)



gases emitted (g)



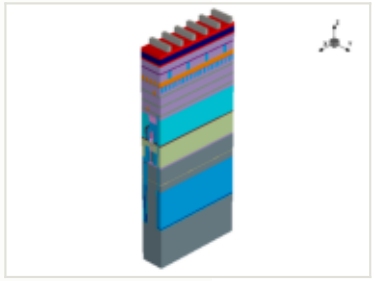
Step #5050: Wafer_clean

etch_strip · Wafer_clean

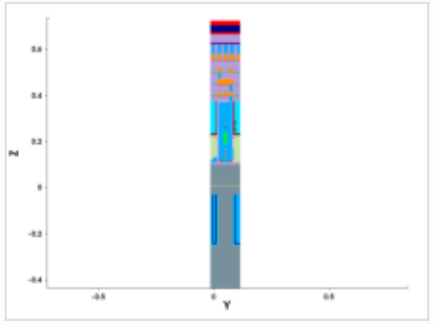
0.169	480.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

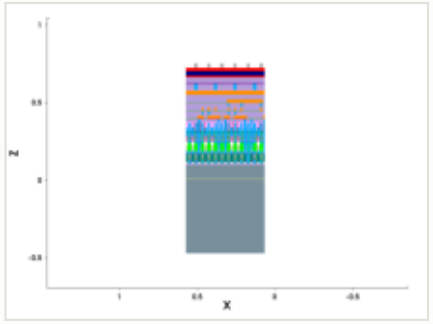
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



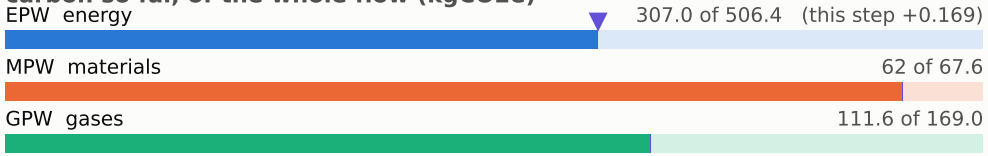
mask for this step

no mask for this step

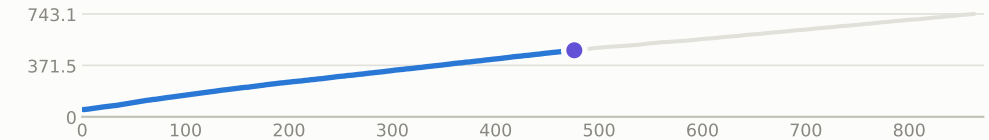
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

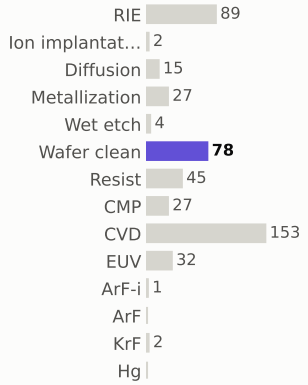
carbon so far, of the whole flow (kgCO2e)



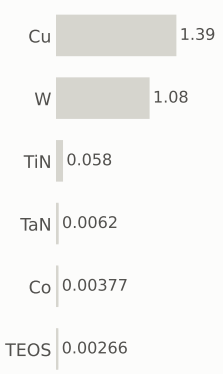
embodied carbon up to each step (kgCO2e per wafer)



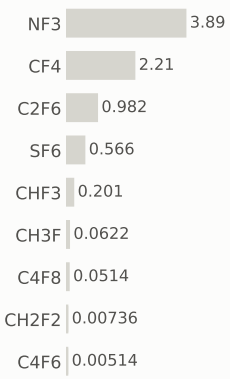
steps by process area



materials deposited (g)



gases emitted (g)



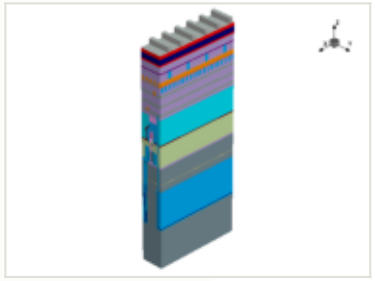
Step #5060: CVD

depo_isotropic · CVD

1.31	481.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

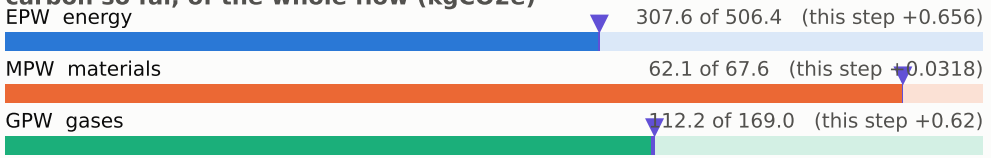


this step

process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

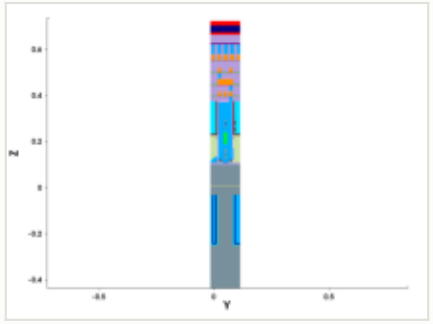
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

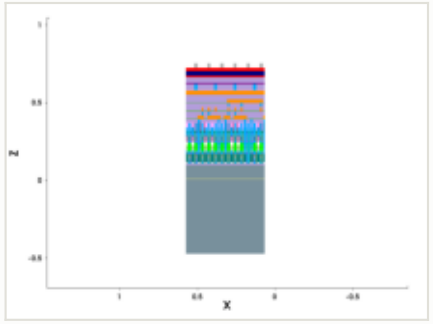


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.385 µm



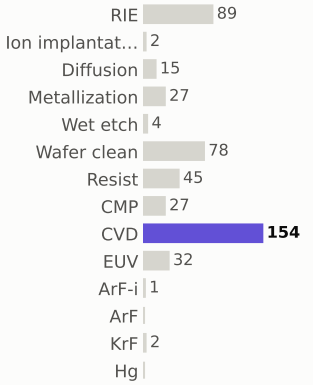
y cut at 0.049 µm



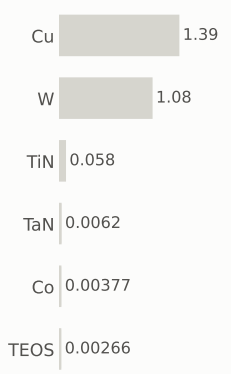
mask for this step

no mask for this step

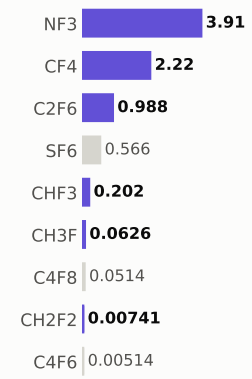
steps by process area



materials deposited (g)



gases emitted (g)



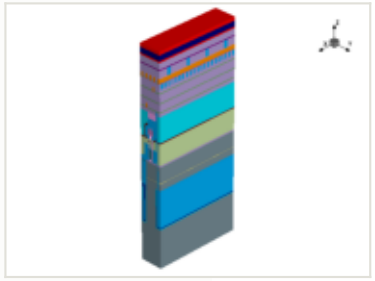
Step #5070: RIE

etch_anisotropic · RIE

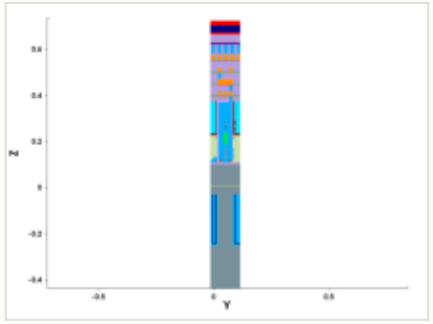
1.04	482.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

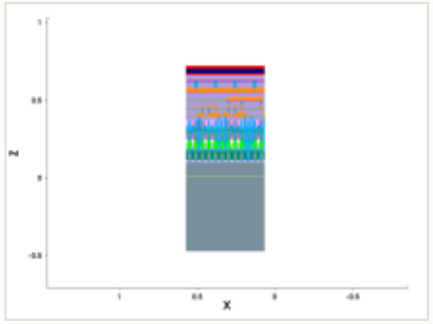
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

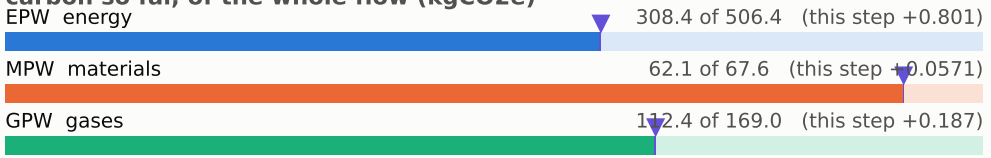
no mask for this step

this step

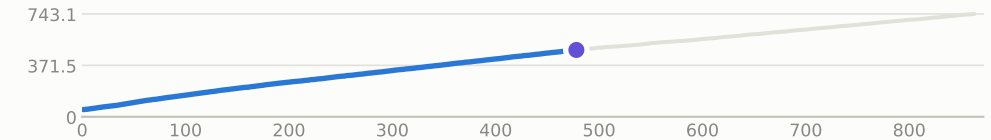
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

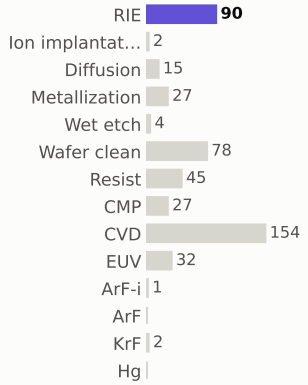
carbon so far, of the whole flow (kgCO2e)



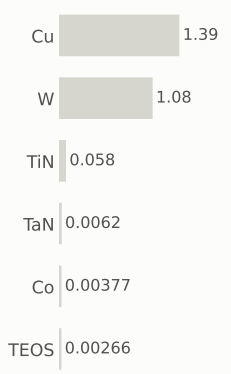
embodied carbon up to each step (kgCO2e per wafer)



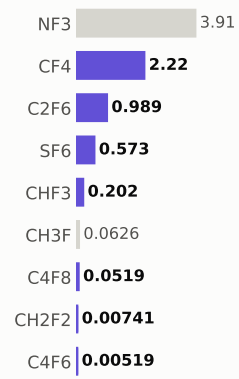
steps by process area



materials deposited (g)



gases emitted (g)



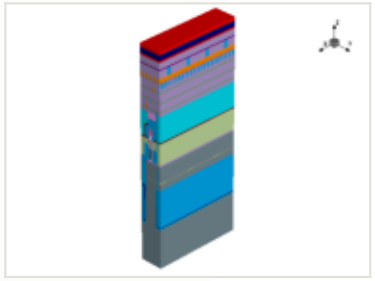
Step #5080: Wafer_clean

etch_strip · Wafer_clean

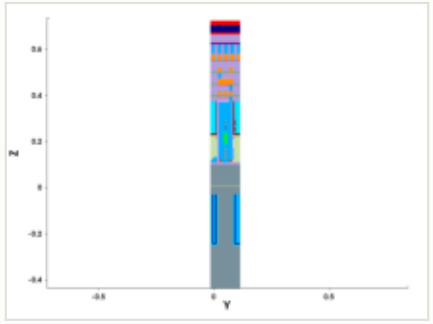
0.169	483.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

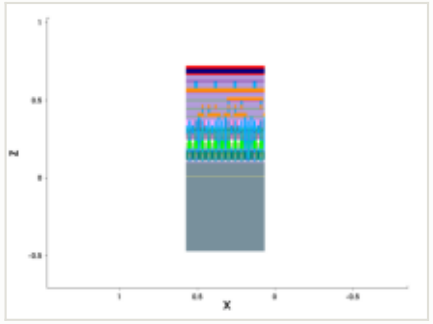
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



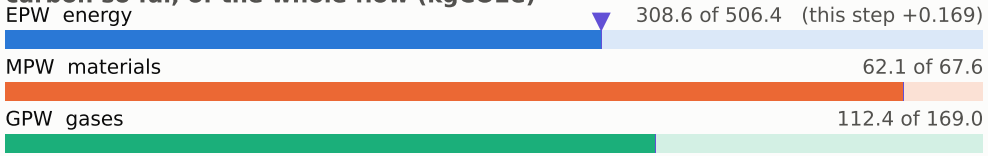
mask for this step

no mask for this step

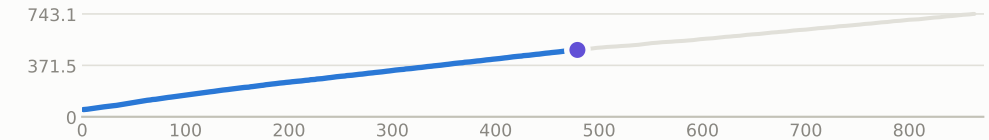
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

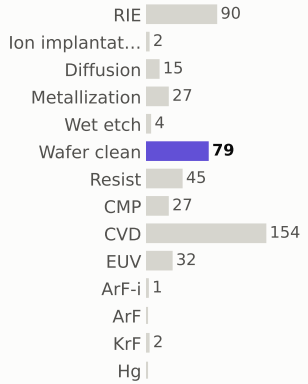
carbon so far, of the whole flow (kgCO2e)



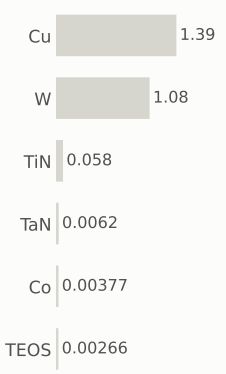
embodied carbon up to each step (kgCO2e per wafer)



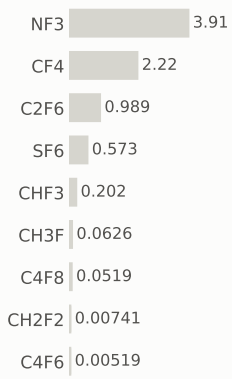
steps by process area



materials deposited (g)



gases emitted (g)



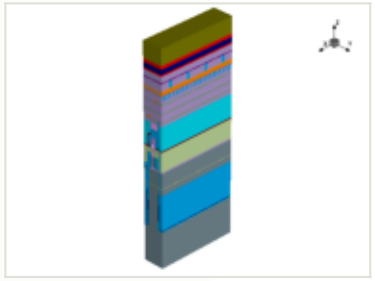
Step #5090: Resist

depo_fill · Resist

0.169	483.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

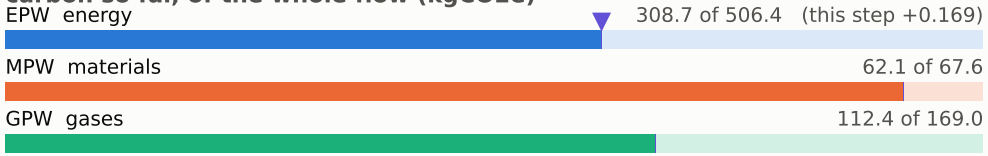


this step

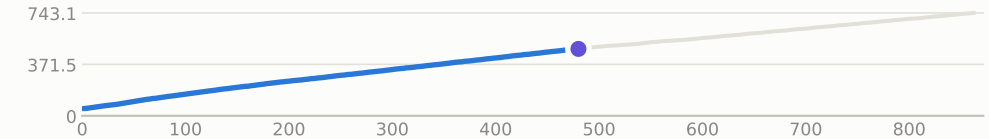
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

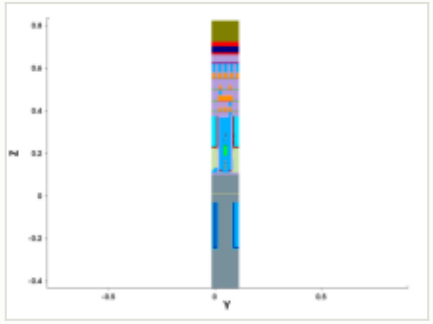
carbon so far, of the whole flow (kgCO2e)



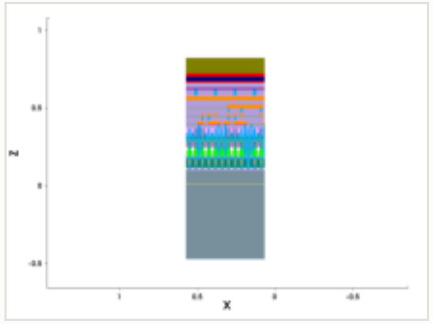
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



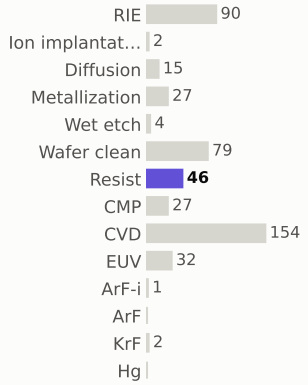
y cut at 0.049 µm



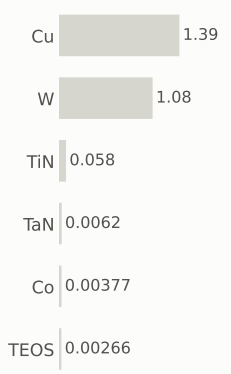
mask for this step

no mask for this step

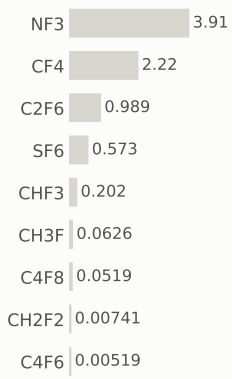
steps by process area



materials deposited (g)



gases emitted (g)



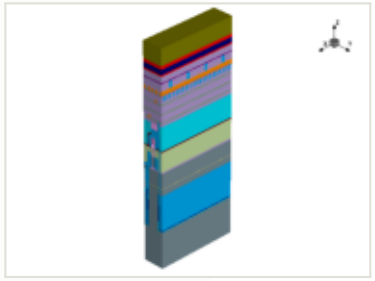
Step #5100: Wafer_clean

clean · Wafer_clean

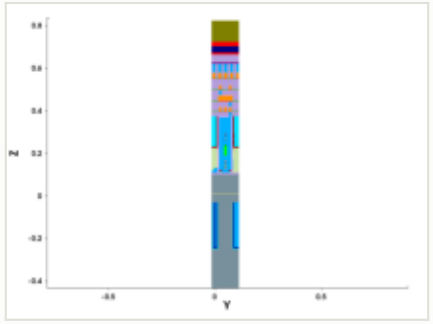
0.169	483.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

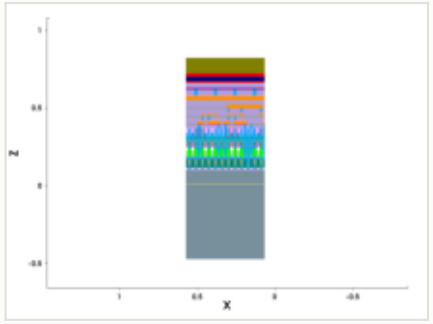
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

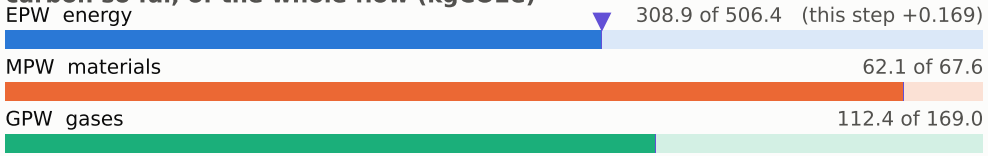
no mask for this step

this step

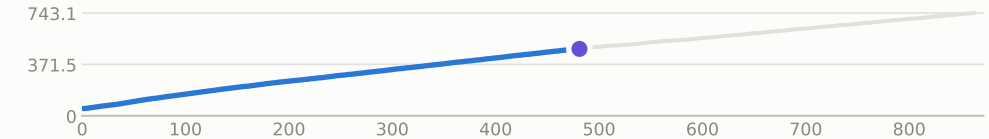
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

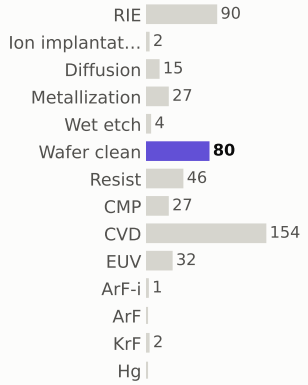
carbon so far, of the whole flow (kgCO2e)



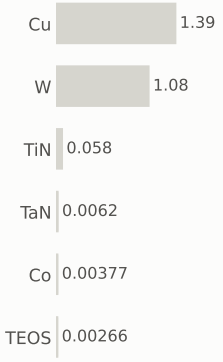
embodied carbon up to each step (kgCO2e per wafer)



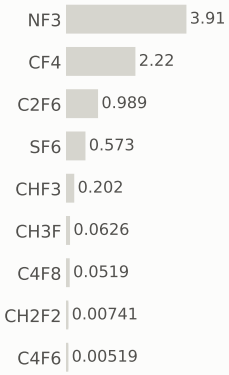
steps by process area



materials deposited (g)



gases emitted (g)



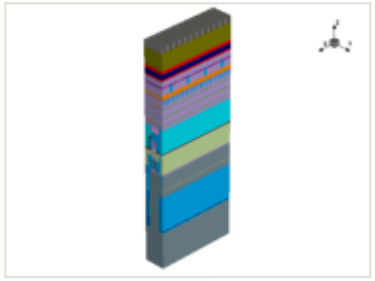
Step #5110: ArF-i

litho_patterning · ArF-i

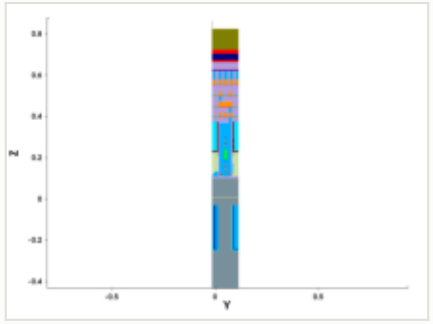
0.217	483.6	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

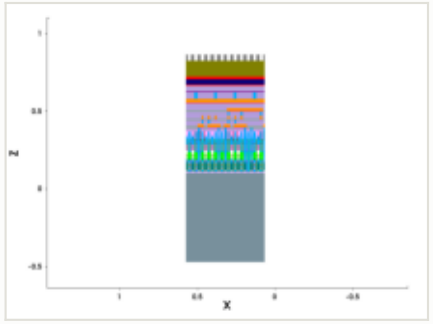
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



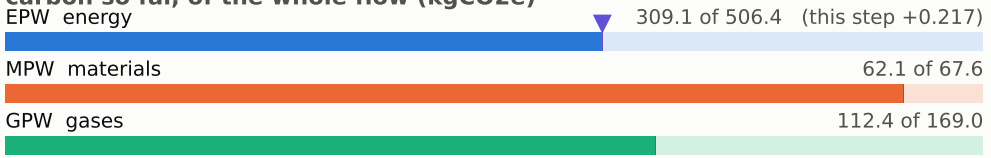
mask: layer 40:0 (GDSII)

this step

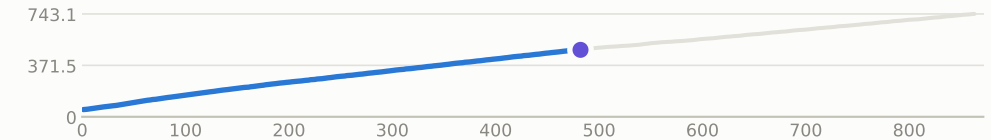
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

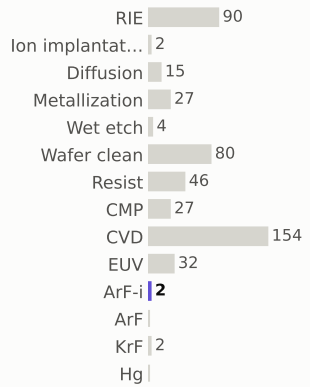
carbon so far, of the whole flow (kgCO2e)



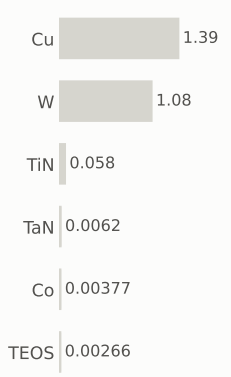
embodied carbon up to each step (kgCO2e per wafer)



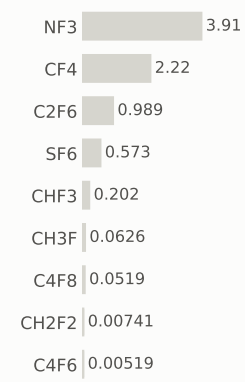
steps by process area



materials deposited (g)



gases emitted (g)



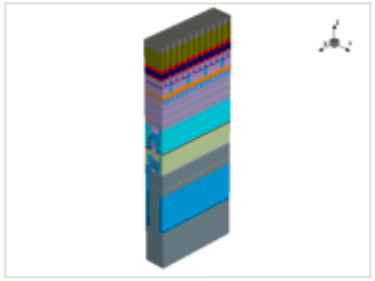
Step #5120: RIE

etch_anisotropic · RIE

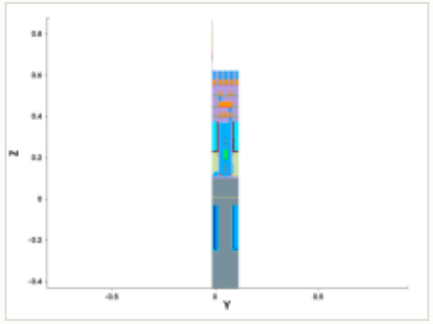
1.04	484.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

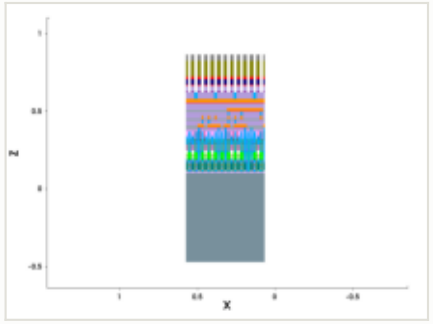
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

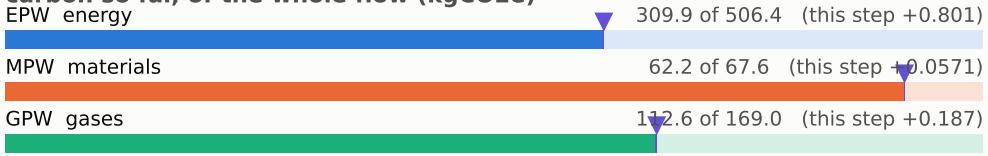
no mask for this step

this step

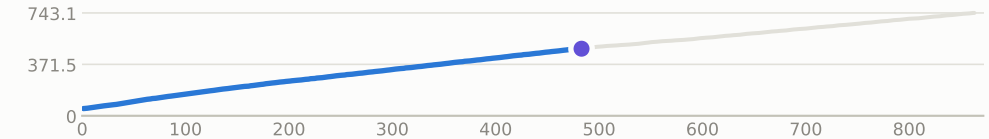
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

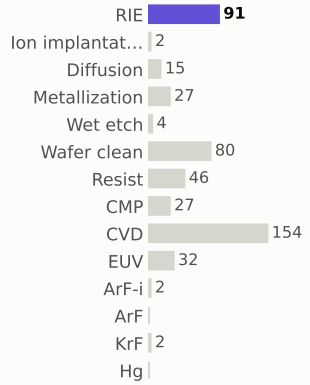
carbon so far, of the whole flow (kgCO2e)



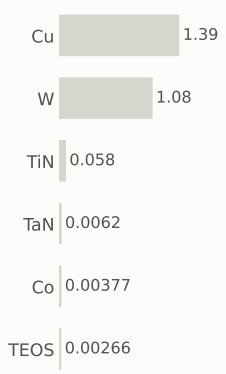
embodied carbon up to each step (kgCO2e per wafer)



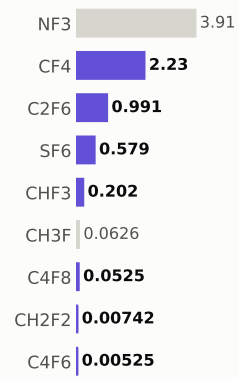
steps by process area



materials deposited (g)



gases emitted (g)



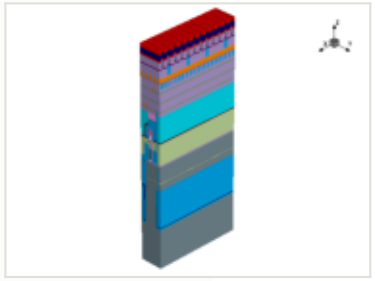
Step #5130: Wafer_clean

etch_strip · Wafer_clean

0.169	484.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

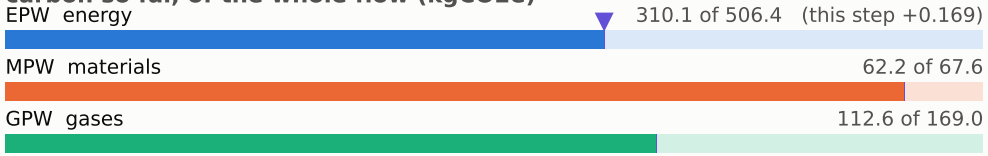


this step

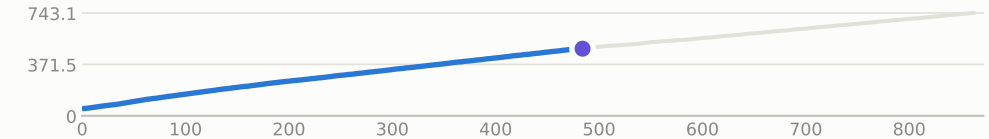
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

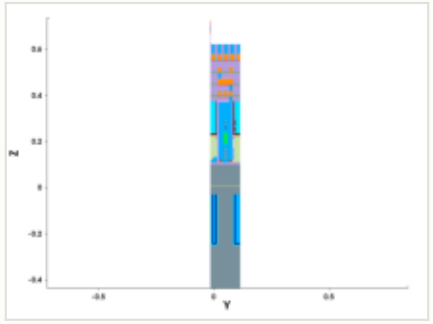
carbon so far, of the whole flow (kgCO2e)



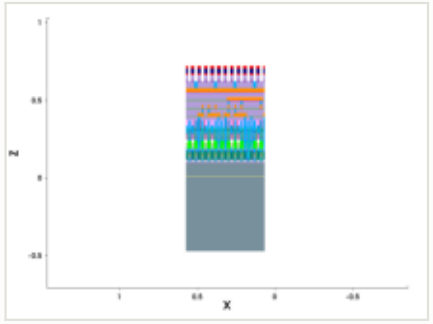
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



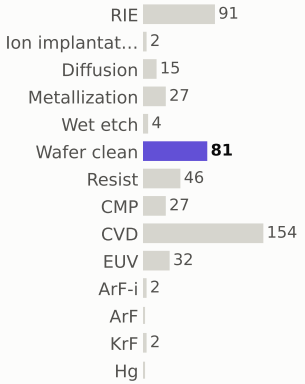
y cut at 0.049 µm



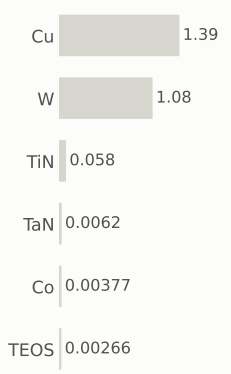
mask for this step

no mask for this step

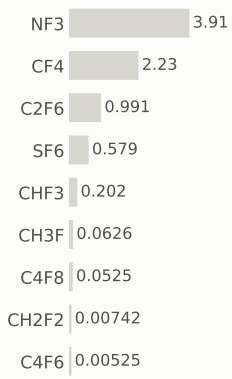
steps by process area



materials deposited (g)



gases emitted (g)



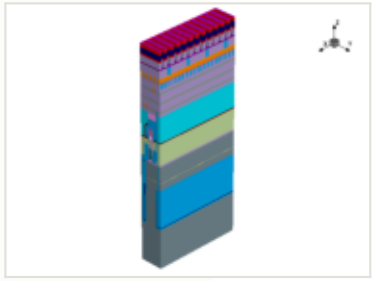
Step #5140: Metallization

depo_ellipsoidal · Metallization

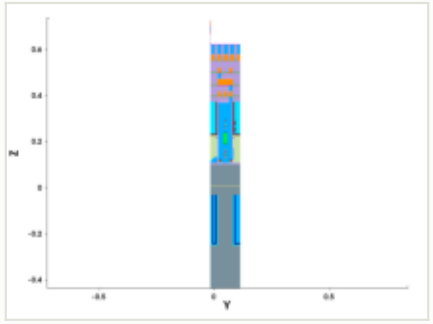
1.2	486.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

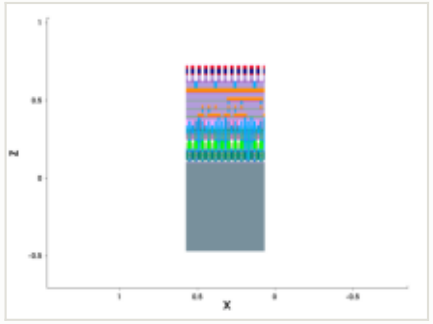
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

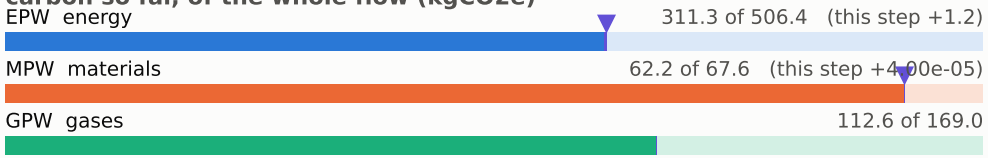
no mask for this step

this step

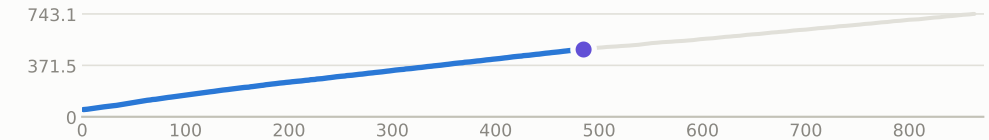
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

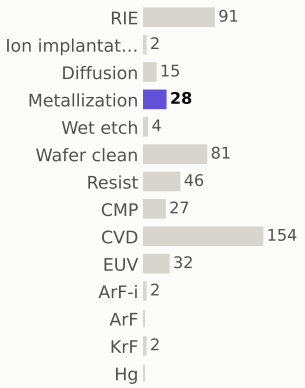
carbon so far, of the whole flow (kgCO2e)



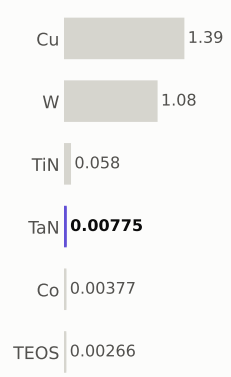
embodied carbon up to each step (kgCO2e per wafer)



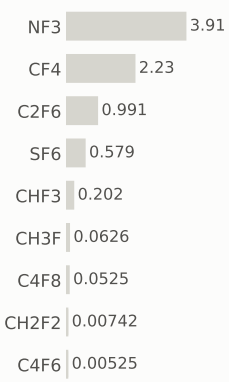
steps by process area



materials deposited (g)



gases emitted (g)



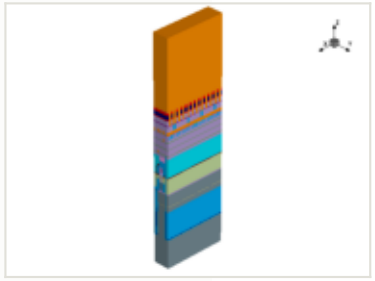
Step #5150: Metallization

depo_fill · Metallization

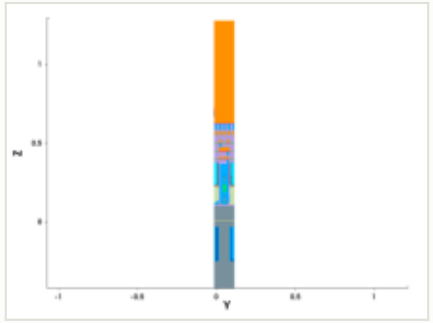
1.21	487.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

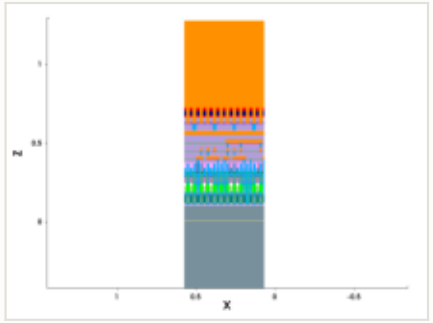
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

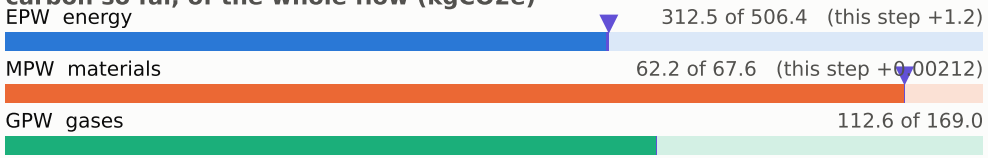
no mask for this step

this step

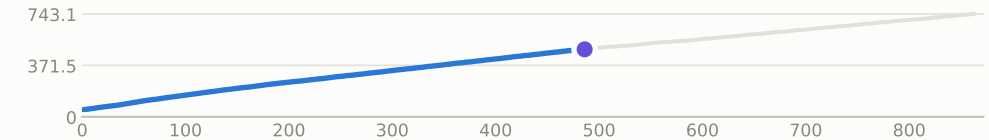
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

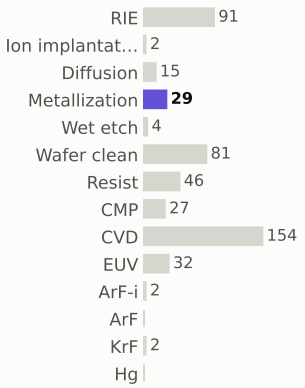
carbon so far, of the whole flow (kgCO2e)



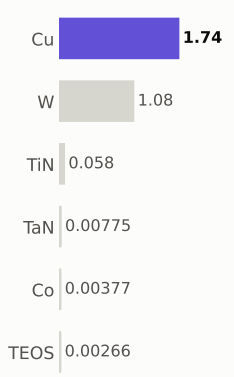
embodied carbon up to each step (kgCO2e per wafer)



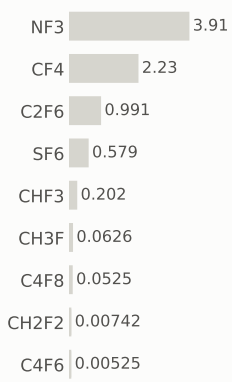
steps by process area



materials deposited (g)



gases emitted (g)



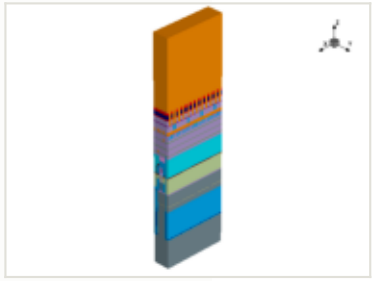
Step #5160: Diffusion

treatment · Diffusion

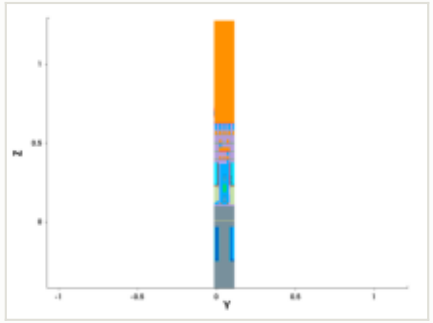
0.801	488.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

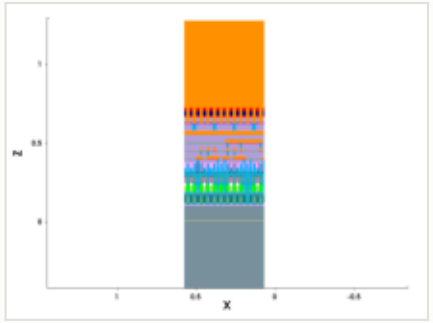
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

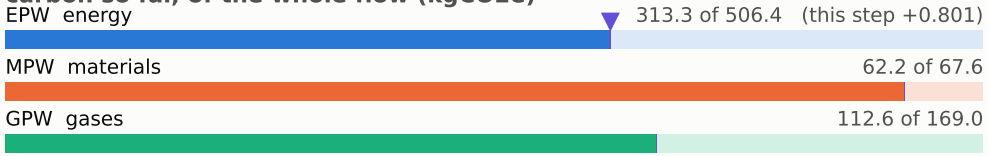
no mask for this step

this step

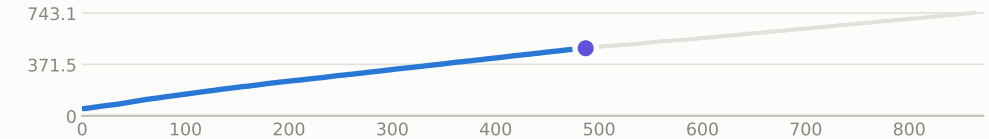
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

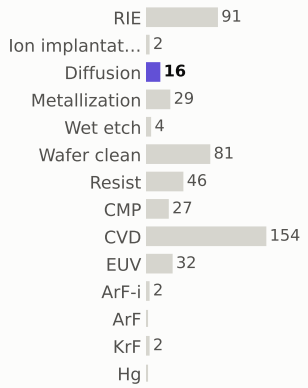
carbon so far, of the whole flow (kgCO2e)



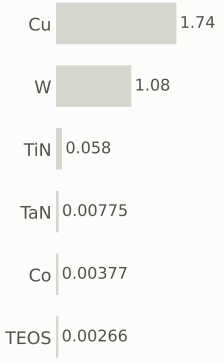
embodied carbon up to each step (kgCO2e per wafer)



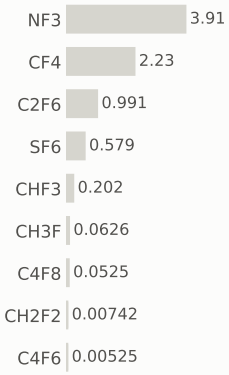
steps by process area



materials deposited (g)



gases emitted (g)



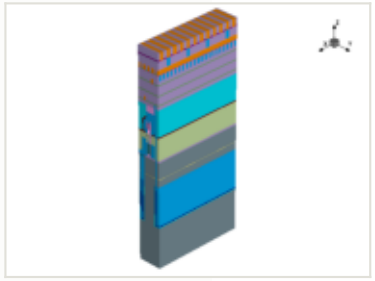
Step #5170: CMP

etch_cmp · CMP

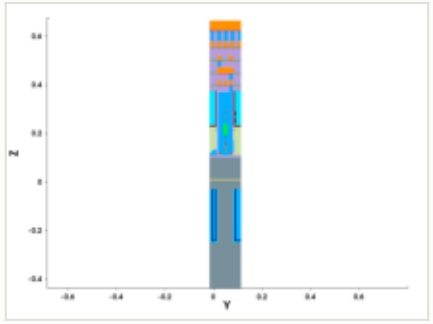
0.169	488.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

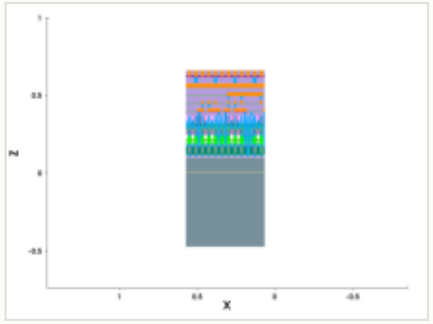
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

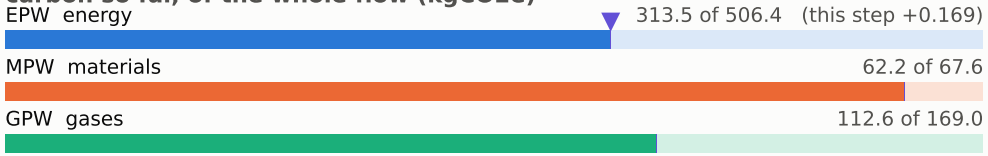
no mask for this step

this step

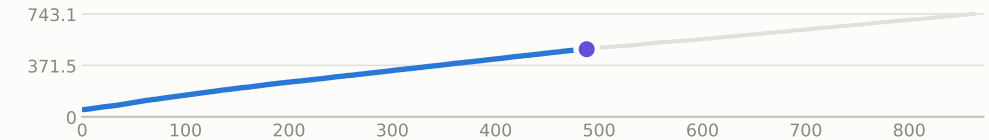
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

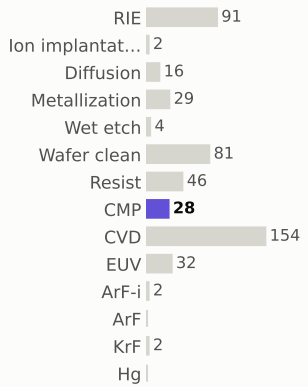
carbon so far, of the whole flow (kgCO2e)



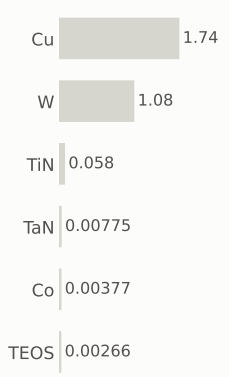
embodied carbon up to each step (kgCO2e per wafer)



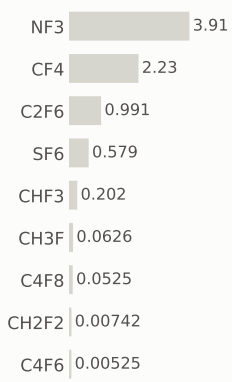
steps by process area



materials deposited (g)



gases emitted (g)



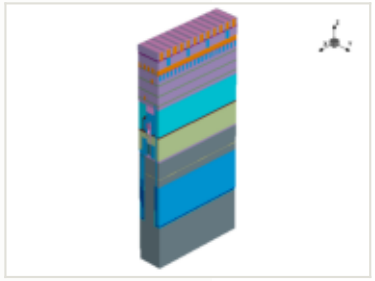
Step #5180: Metallization

depo_isotropic · Metallization

1.2	489.4	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

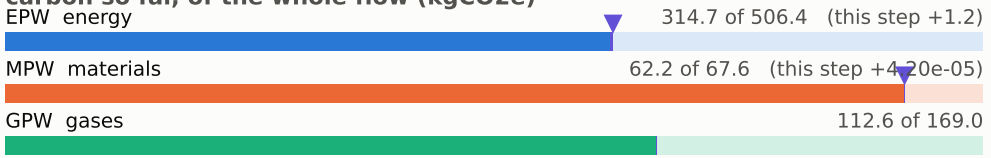


this step

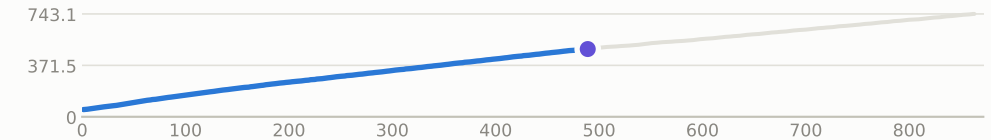
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

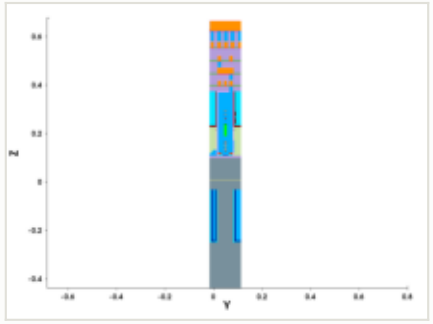
carbon so far, of the whole flow (kgCO2e)



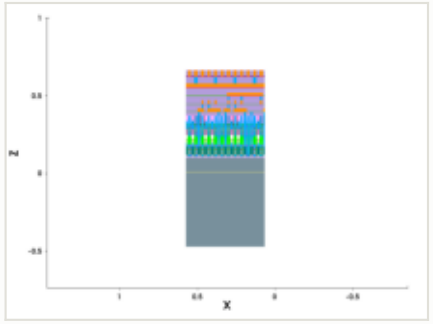
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



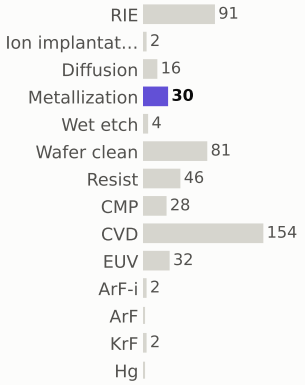
y cut at 0.049 µm



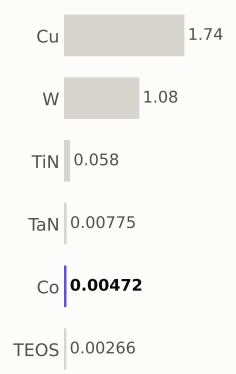
mask for this step

no mask for this step

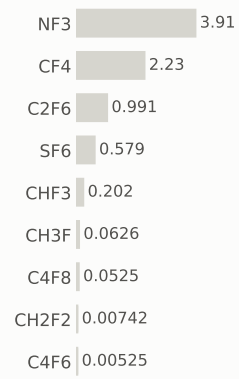
steps by process area



materials deposited (g)



gases emitted (g)



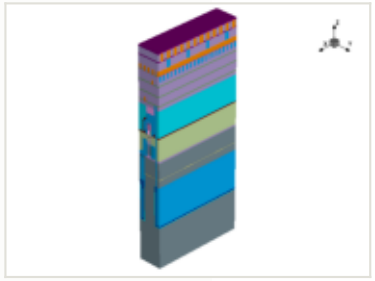
Step #5200: CVD

depo_fill · CVD

1.31	490.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

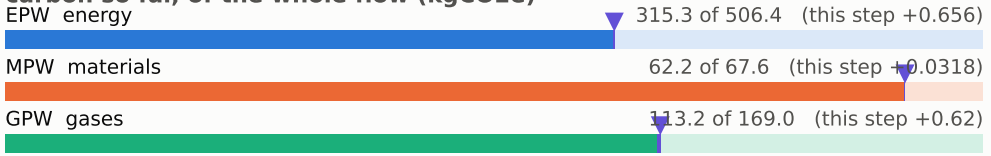


this step

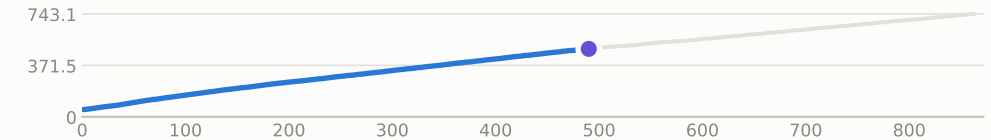
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

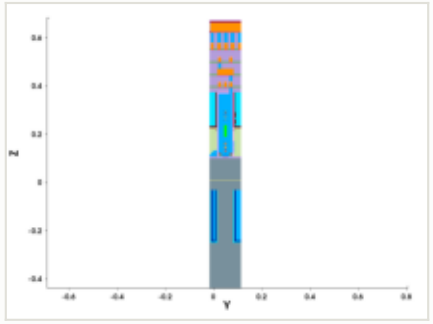
carbon so far, of the whole flow (kgCO2e)



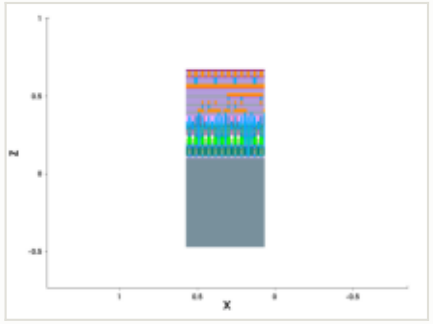
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



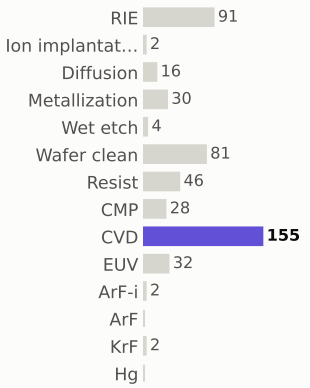
y cut at 0.049 µm



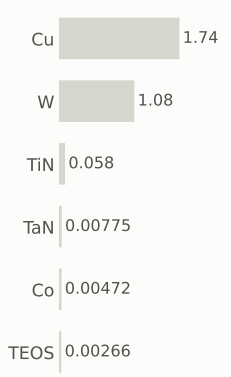
mask for this step

no mask for this step

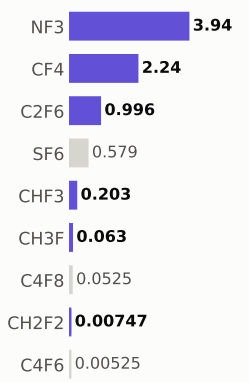
steps by process area



materials deposited (g)



gases emitted (g)



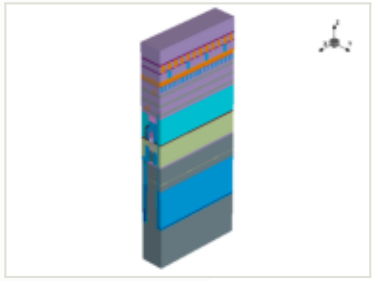
Step #5210: CVD

depo_fill · CVD

1.31	492.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

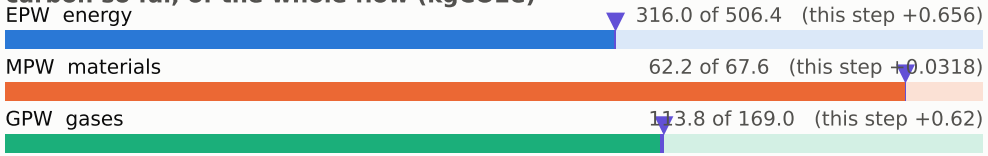


this step

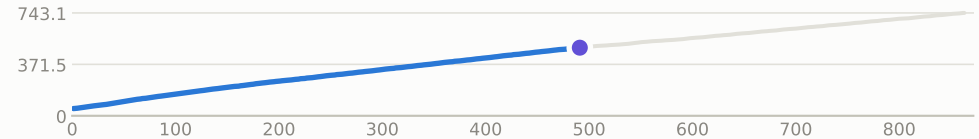
process area	CVD
material	—
thickness	0.078 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

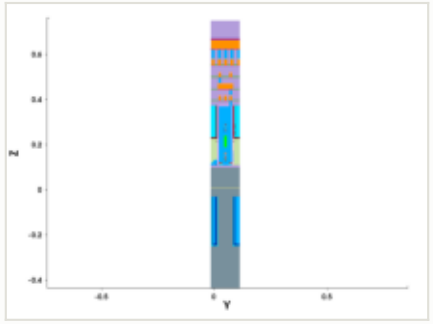
carbon so far, of the whole flow (kgCO2e)



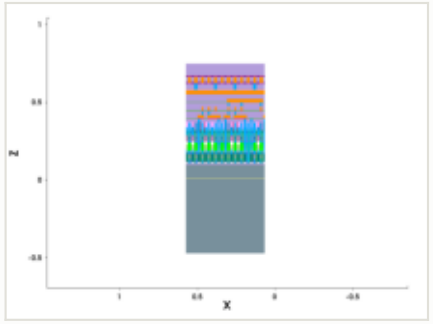
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



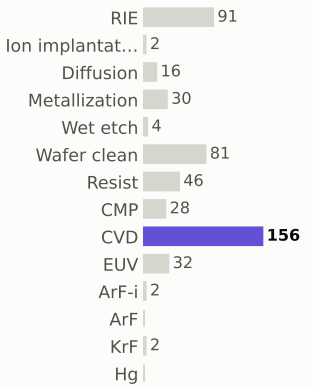
y cut at 0.049 µm



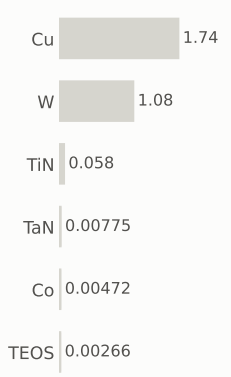
mask for this step

no mask for this step

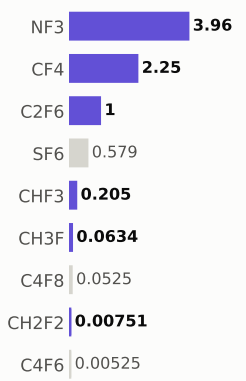
steps by process area



materials deposited (g)



gases emitted (g)



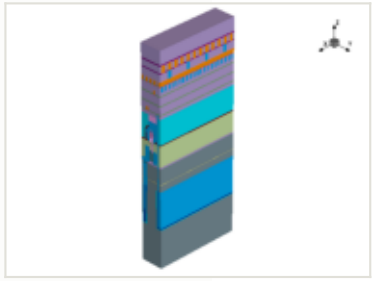
Step #5220: Diffusion

treatment · Diffusion

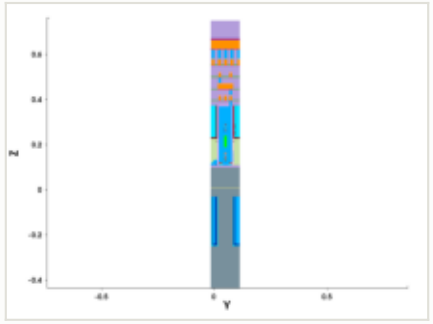
0.801	492.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

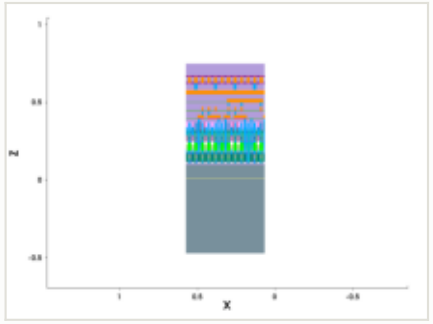
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

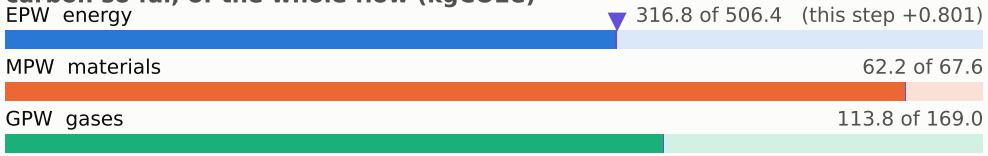
no mask for this step

this step

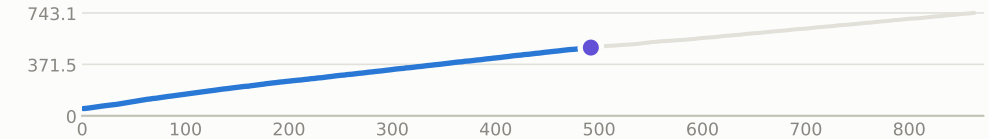
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

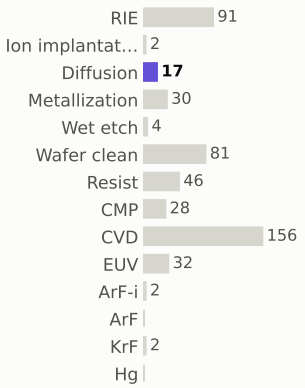
carbon so far, of the whole flow (kgCO2e)



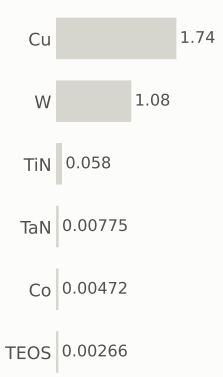
embodied carbon up to each step (kgCO2e per wafer)



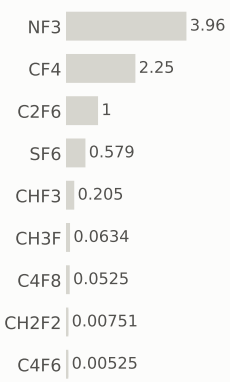
steps by process area



materials deposited (g)



gases emitted (g)



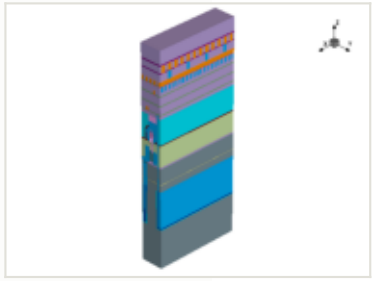
Step #5230: Wafer_clean

clean · Wafer_clean

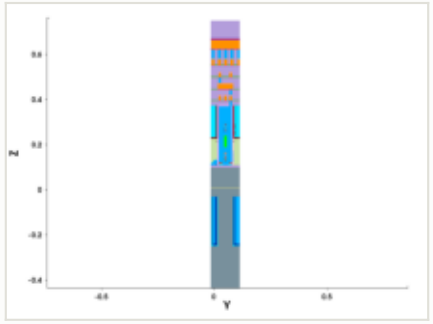
0.169	493.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

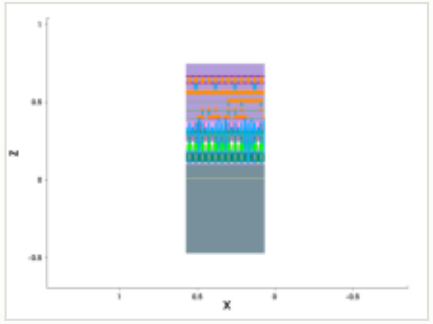
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

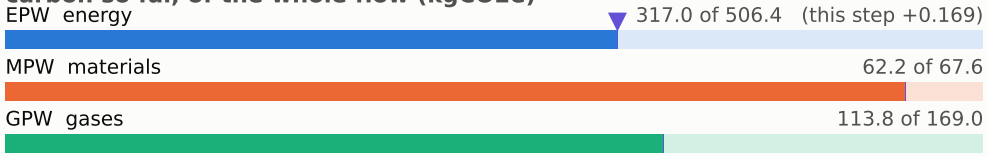
no mask for this step

this step

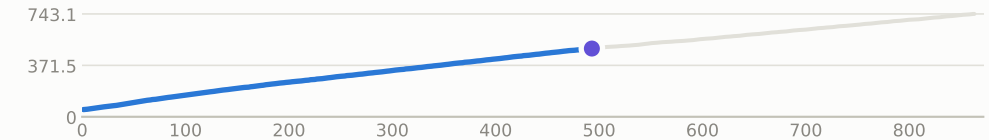
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

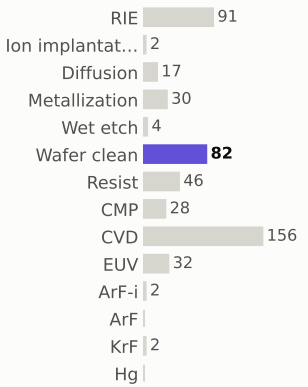
carbon so far, of the whole flow (kgCO2e)



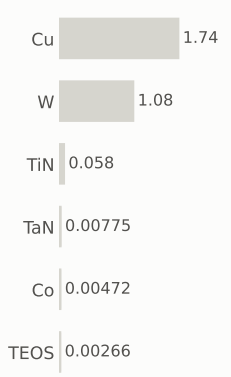
embodied carbon up to each step (kgCO2e per wafer)



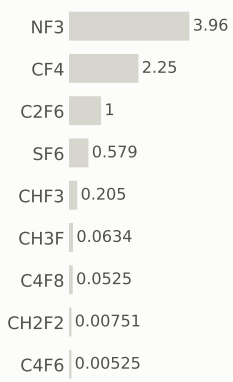
steps by process area



materials deposited (g)



gases emitted (g)



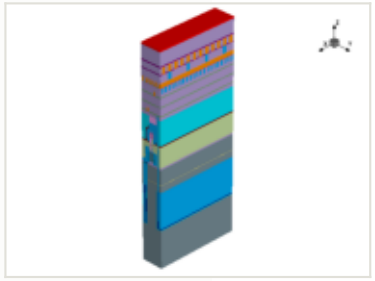
Step #5240: CVD

depo_fill · CVD

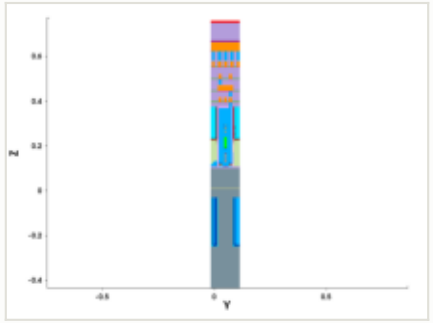
1.31	494.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

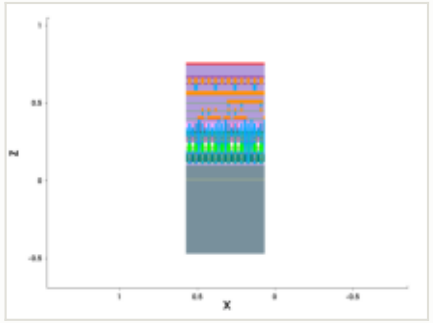
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

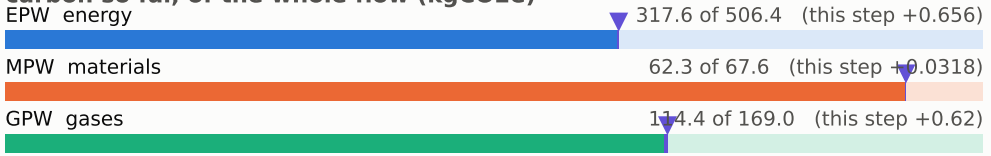
no mask for this step

this step

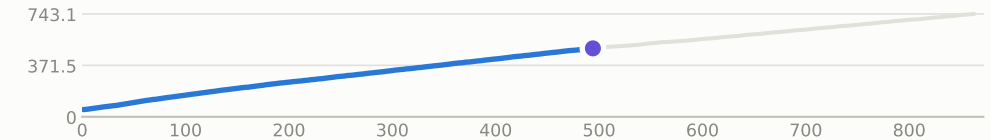
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

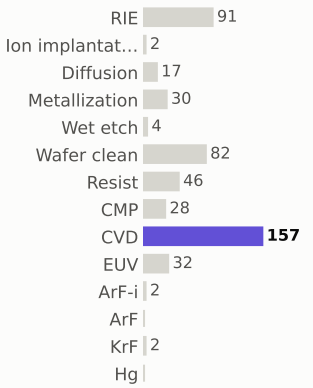
carbon so far, of the whole flow (kgCO2e)



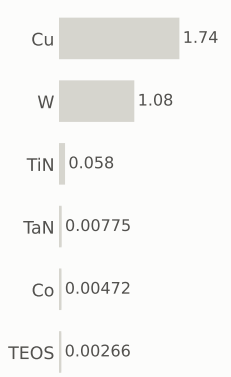
embodied carbon up to each step (kgCO2e per wafer)



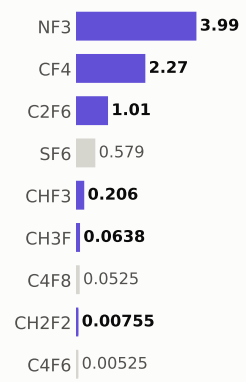
steps by process area



materials deposited (g)



gases emitted (g)



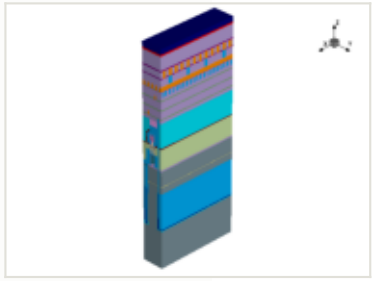
Step #5250: CVD

depo_fill · CVD

1.31	495.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

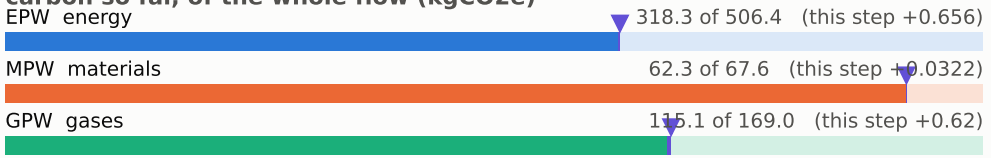


this step

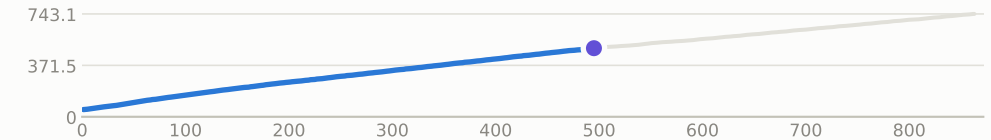
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

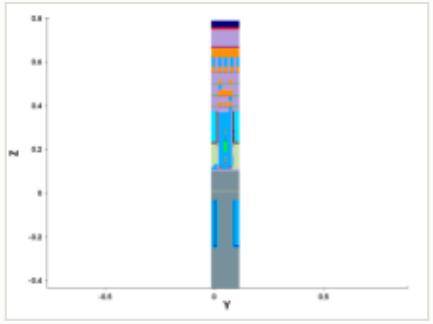
carbon so far, of the whole flow (kgCO2e)



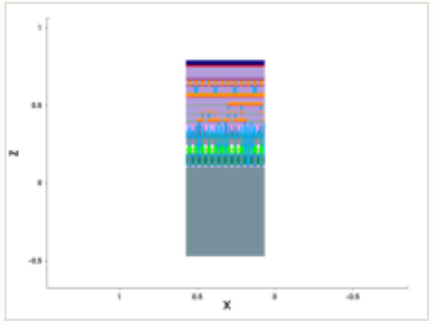
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



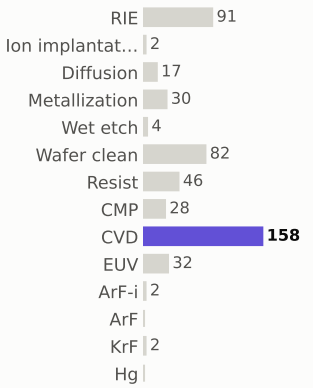
y cut at 0.049 µm



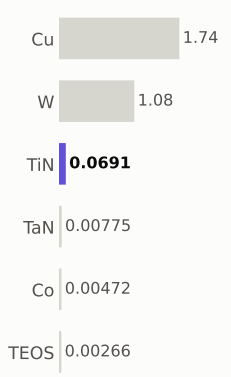
mask for this step

no mask for this step

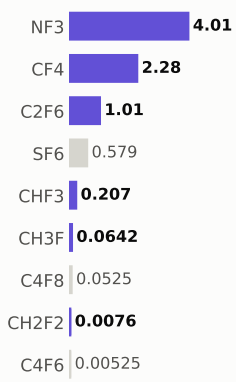
steps by process area



materials deposited (g)



gases emitted (g)



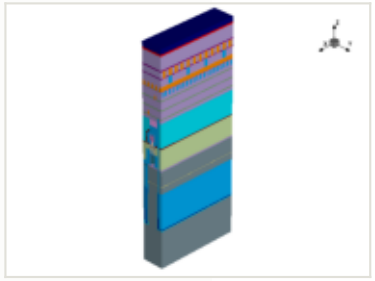
Step #5260: Wafer_clean

clean · Wafer_clean

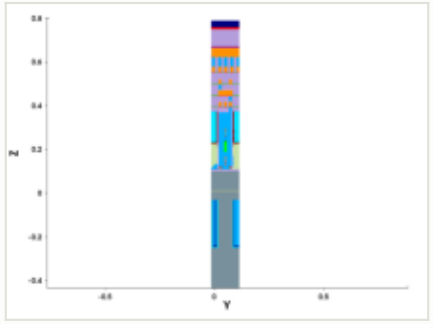
0.169	495.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

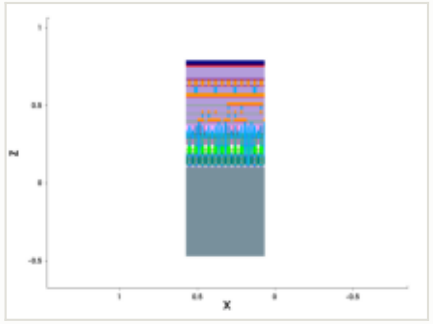
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

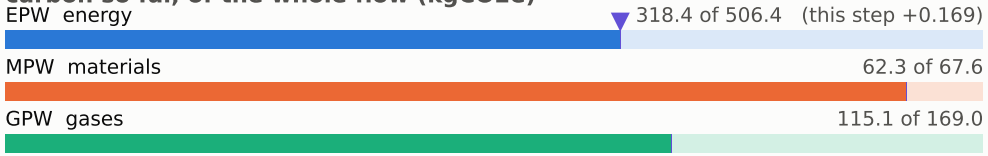
no mask for this step

this step

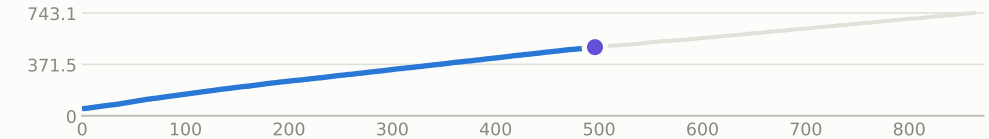
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

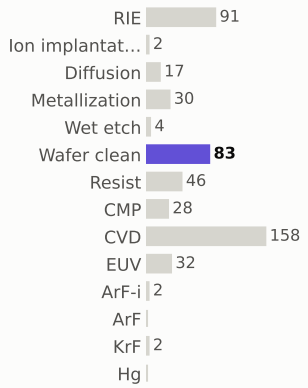
carbon so far, of the whole flow (kgCO2e)



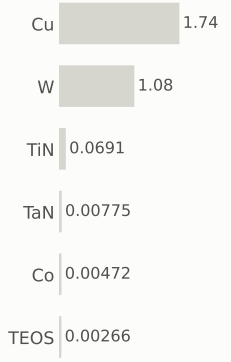
embodied carbon up to each step (kgCO2e per wafer)



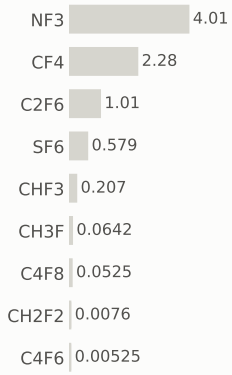
steps by process area



materials deposited (g)



gases emitted (g)



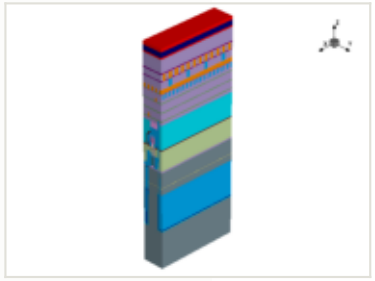
Step #5270: CVD

depo_fill · CVD

1.31	497.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

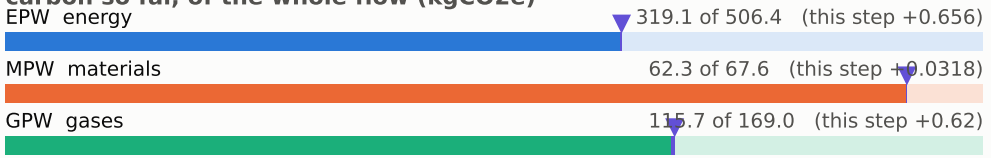


this step

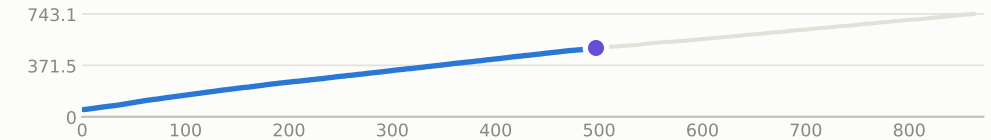
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

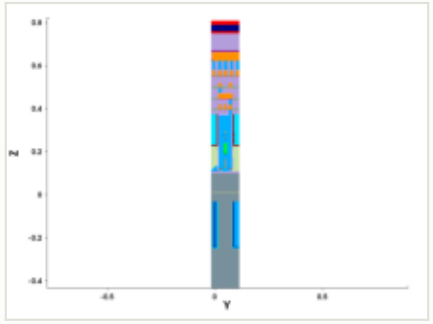
carbon so far, of the whole flow (kgCO2e)



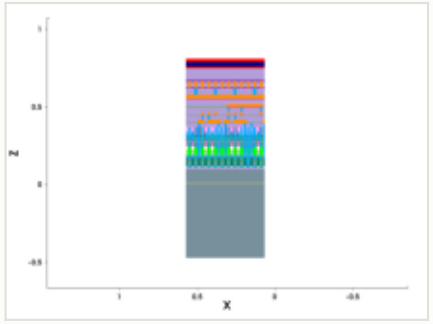
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



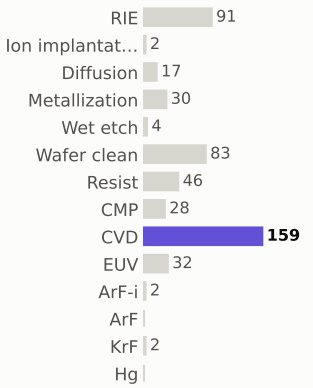
y cut at 0.049 µm



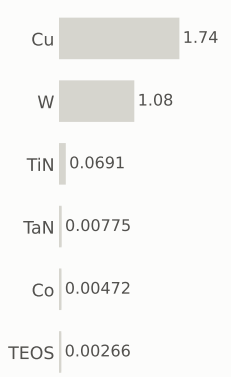
mask for this step

no mask for this step

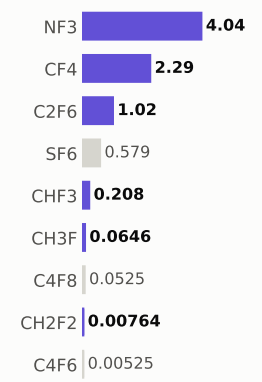
steps by process area



materials deposited (g)



gases emitted (g)



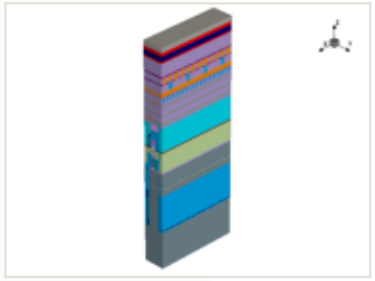
Step #5280: CVD

depo_fill · CVD

1.31	498.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

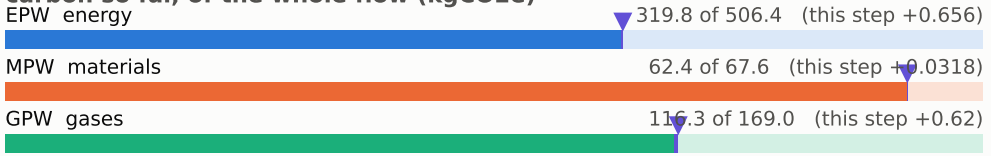


this step

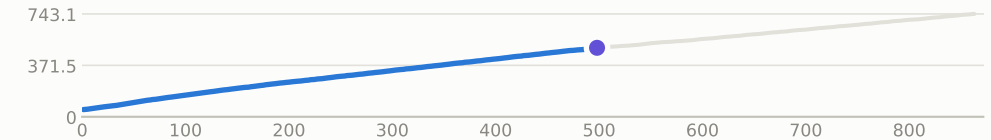
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

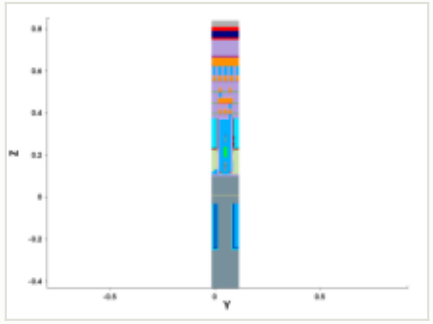
carbon so far, of the whole flow (kgCO2e)



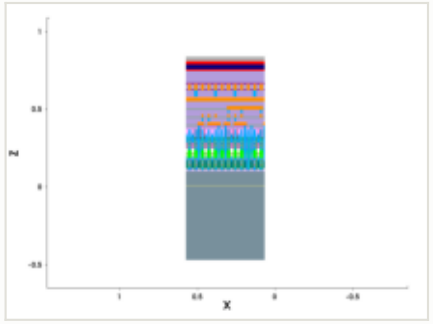
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



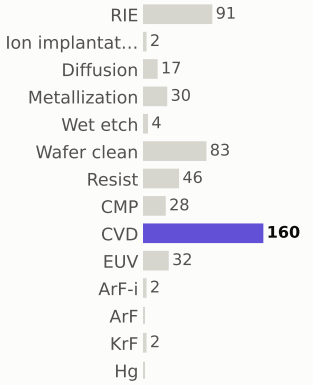
y cut at 0.049 µm



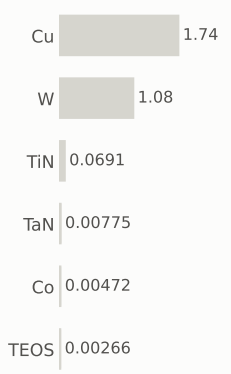
mask for this step

no mask for this step

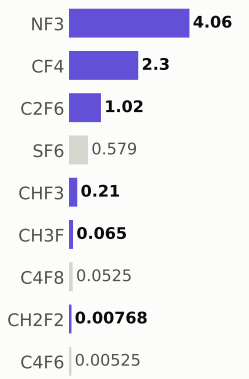
steps by process area



materials deposited (g)



gases emitted (g)



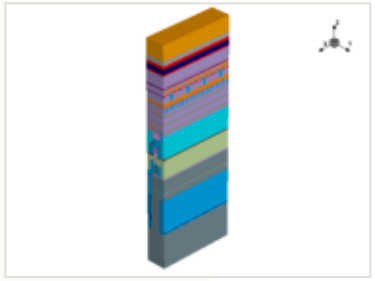
Step #5290: Resist

depo_fill · Resist

0.169	498.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

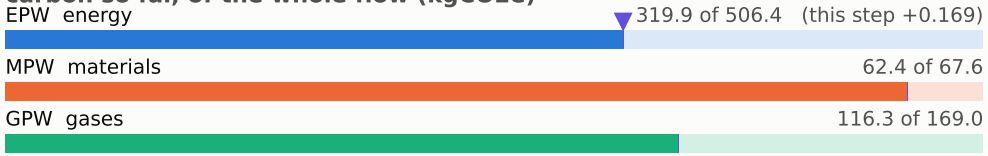


this step

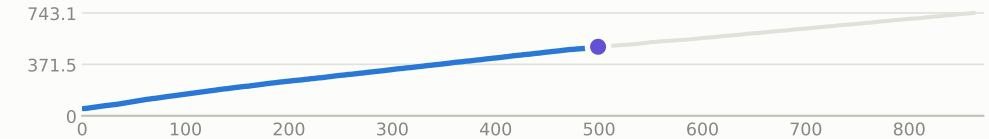
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

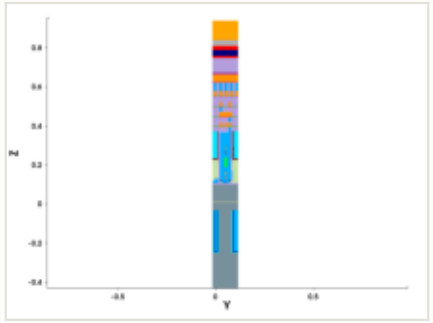
carbon so far, of the whole flow (kgCO2e)



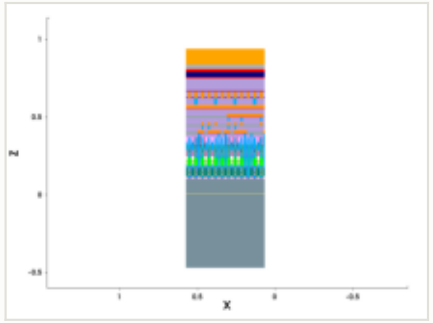
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



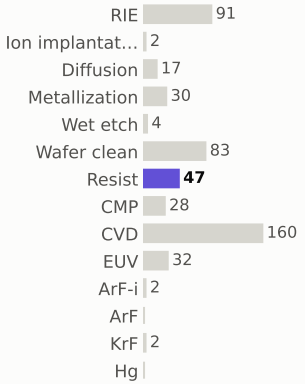
y cut at 0.049 µm



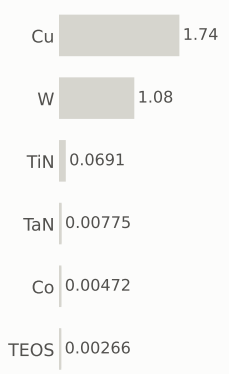
mask for this step

no mask for this step

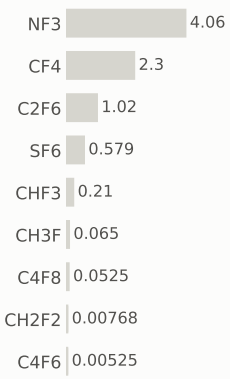
steps by process area



materials deposited (g)



gases emitted (g)



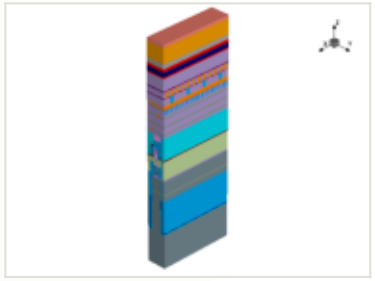
Step #5300: CVD

depo_fill · CVD

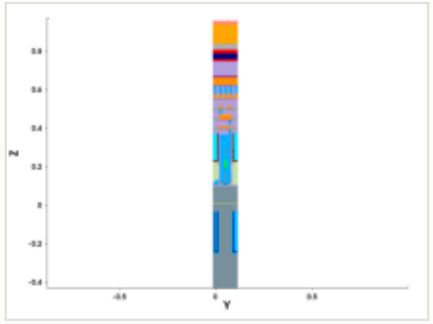
1.31	499.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

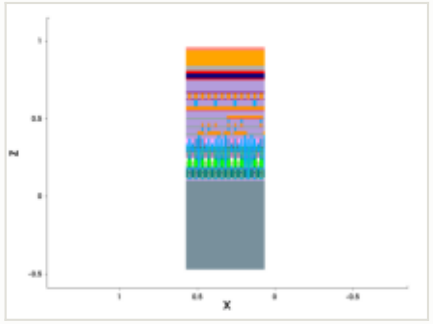
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

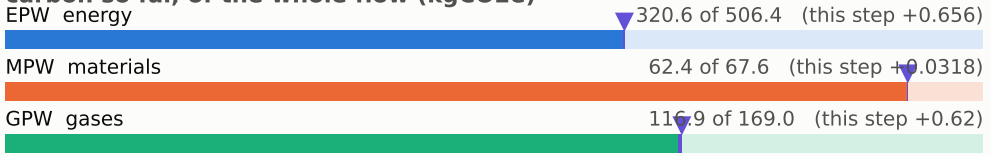
no mask for this step

this step

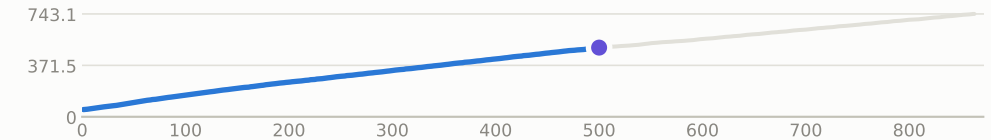
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

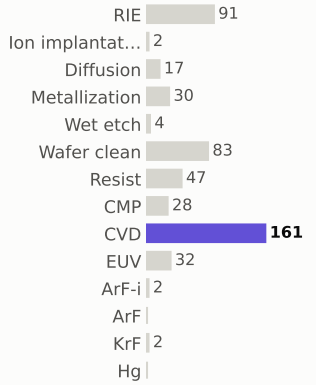
carbon so far, of the whole flow (kgCO2e)



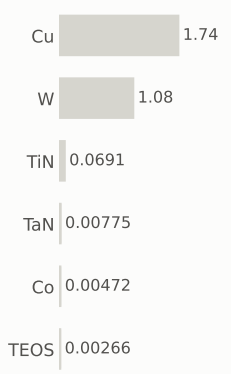
embodied carbon up to each step (kgCO2e per wafer)



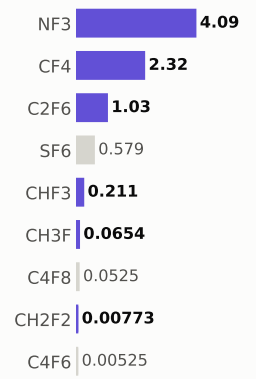
steps by process area



materials deposited (g)



gases emitted (g)



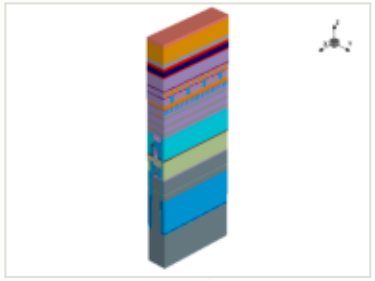
Step #5310: Wafer_clean

clean · Wafer_clean

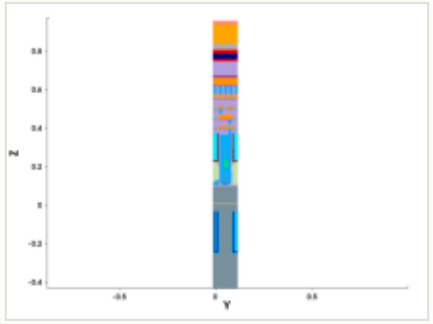
0.169	500.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

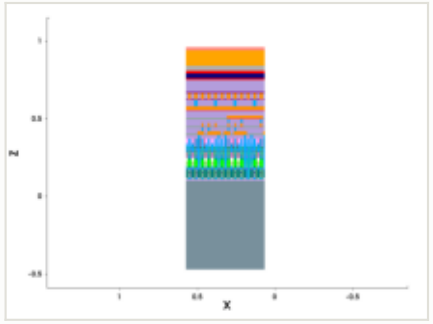
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

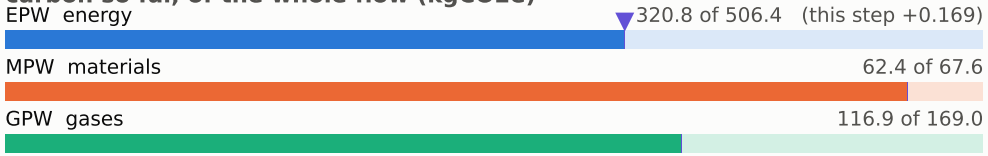
no mask for this step

this step

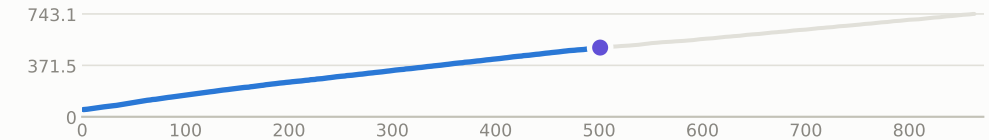
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

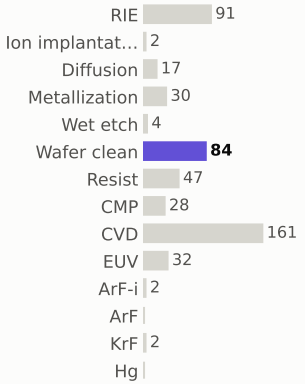
carbon so far, of the whole flow (kgCO2e)



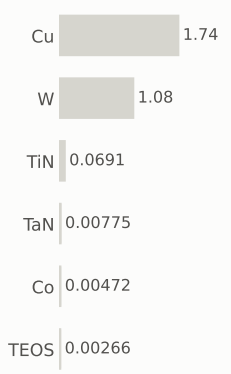
embodied carbon up to each step (kgCO2e per wafer)



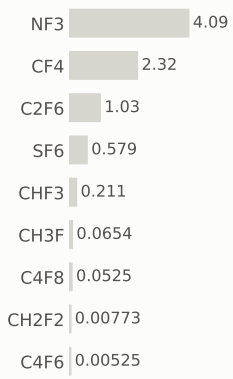
steps by process area



materials deposited (g)



gases emitted (g)



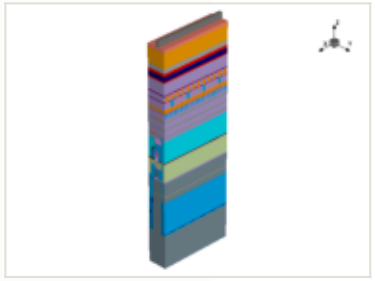
Step #5320: ArF-i

litho_patterning · ArF-i

0.217	500.3	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

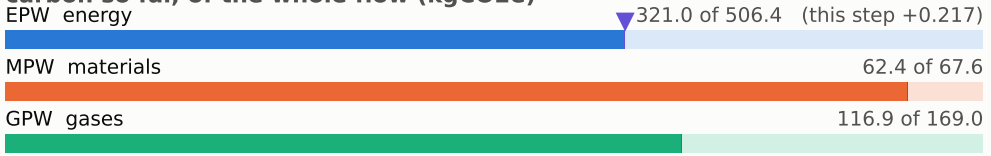


this step

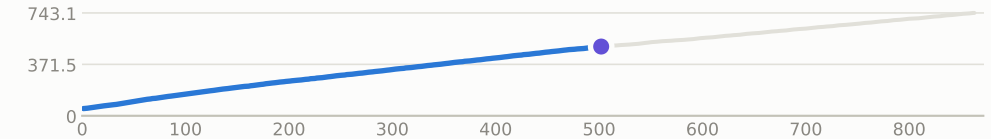
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

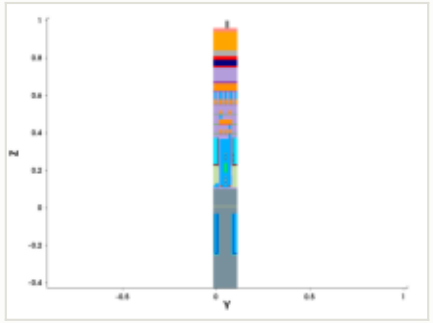
carbon so far, of the whole flow (kgCO2e)



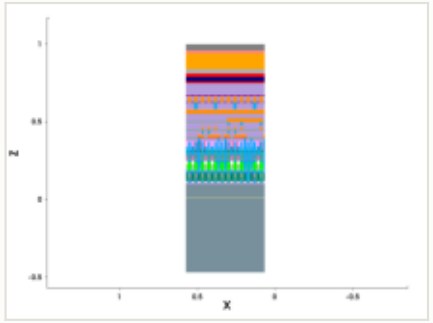
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

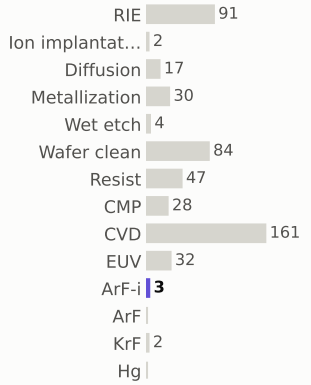


y cut at 0.049 µm

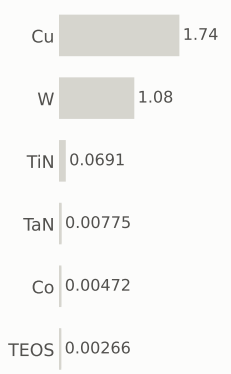


mask: layer 10451:0, positive (GDSII)

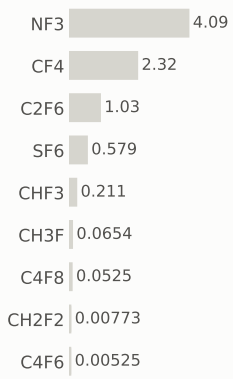
steps by process area



materials deposited (g)



gases emitted (g)



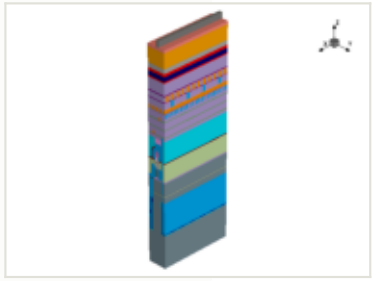
Step #5330: Ion_implantation

treatment · Ion_implantation

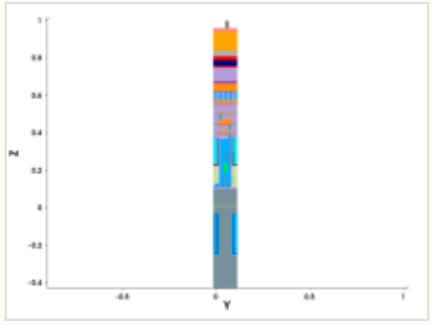
0.801	501.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

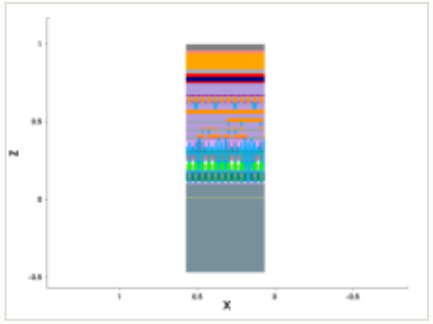
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

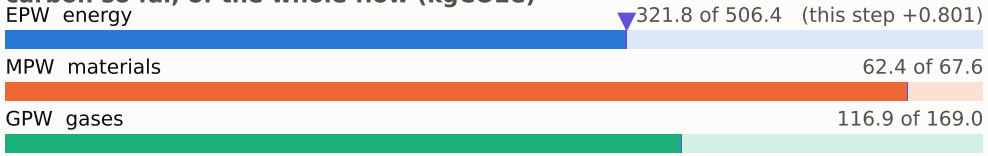
no mask for this step

this step

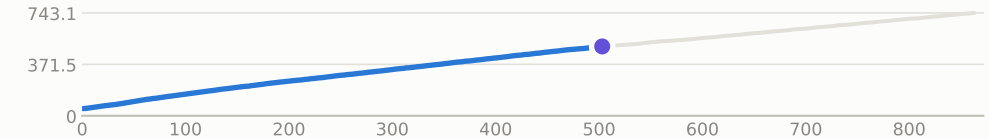
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

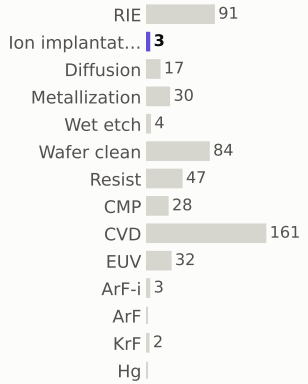
carbon so far, of the whole flow (kgCO2e)



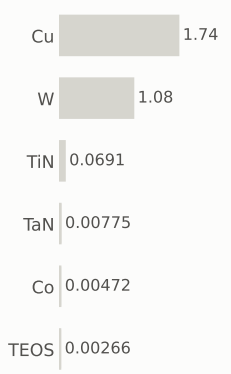
embodied carbon up to each step (kgCO2e per wafer)



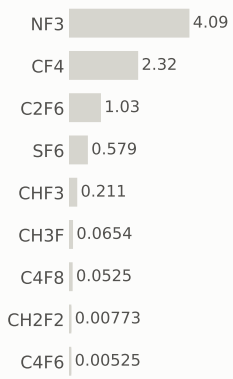
steps by process area



materials deposited (g)



gases emitted (g)



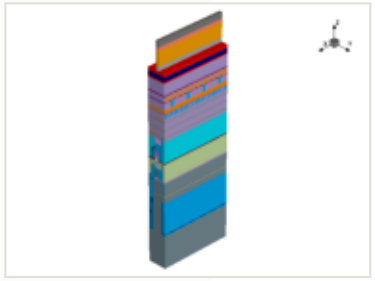
Step #5340: RIE

etch_anisotropic · RIE

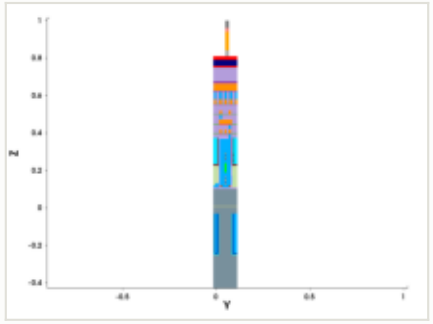
1.04	502.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

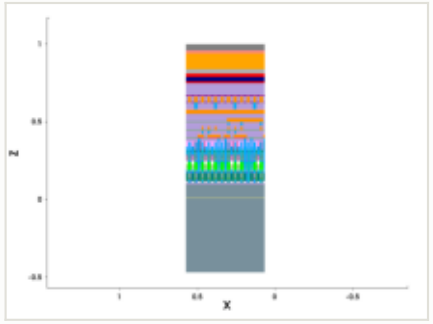
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

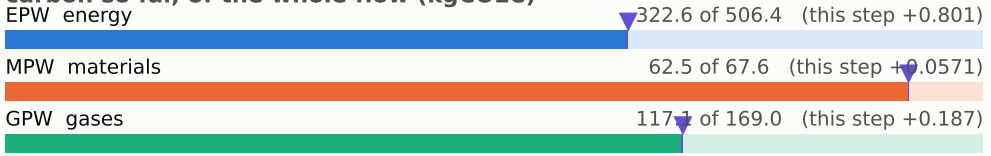
no mask for this step

this step

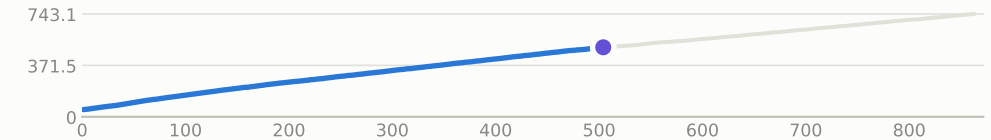
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

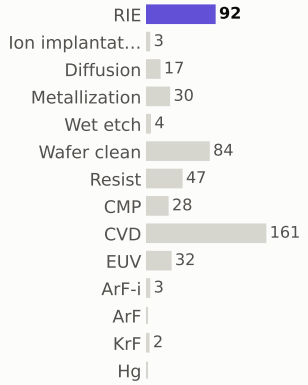
carbon so far, of the whole flow (kgCO2e)



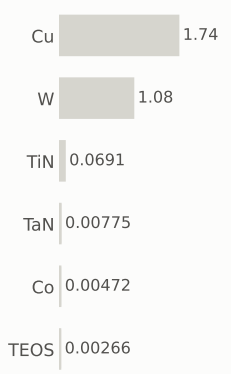
embodied carbon up to each step (kgCO2e per wafer)



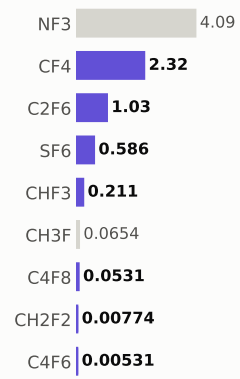
steps by process area



materials deposited (g)



gases emitted (g)



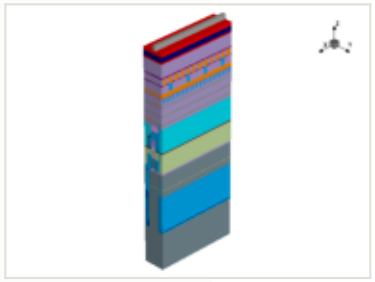
Step #5350: Wafer_clean

etch_strip · Wafer_clean

0.169	502.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

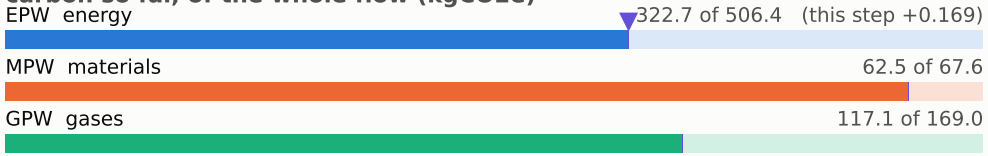


this step

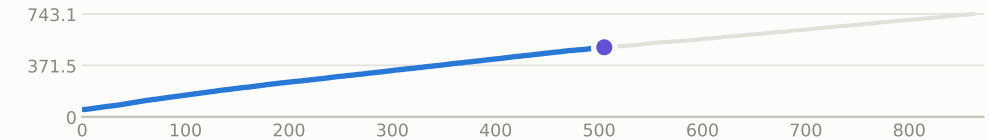
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

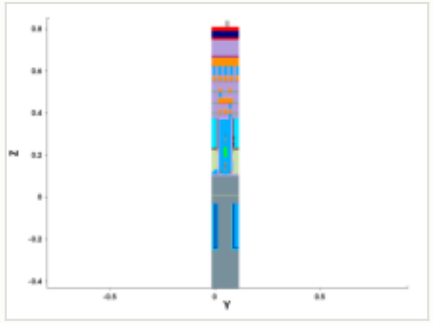
carbon so far, of the whole flow (kgCO2e)



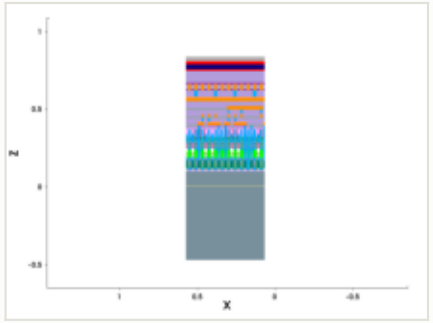
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



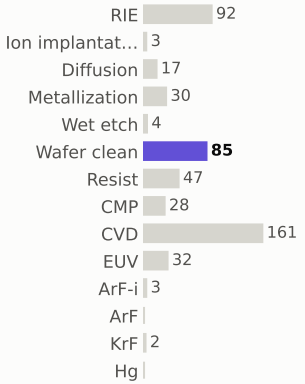
y cut at 0.049 µm



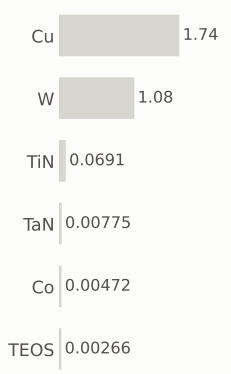
mask for this step

no mask for this step

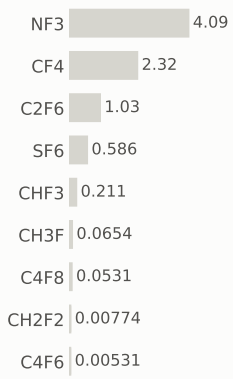
steps by process area



materials deposited (g)



gases emitted (g)



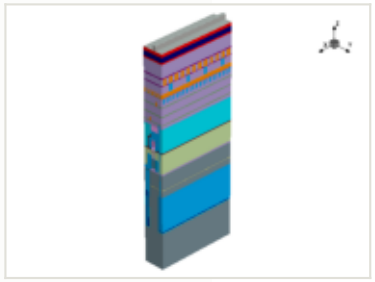
Step #5360: CVD

depo_isotropic · CVD

1.31	503.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

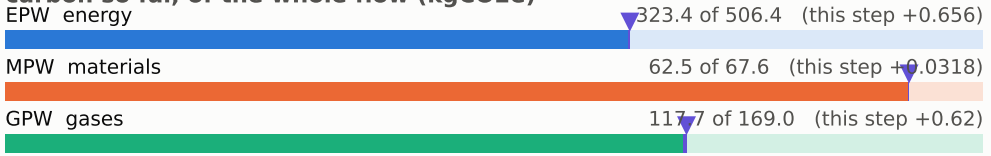


this step

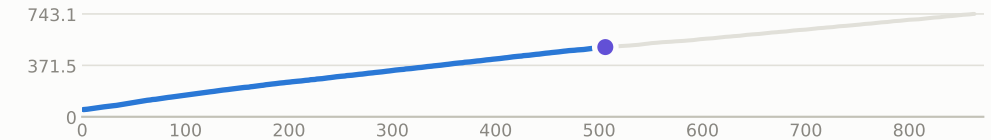
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

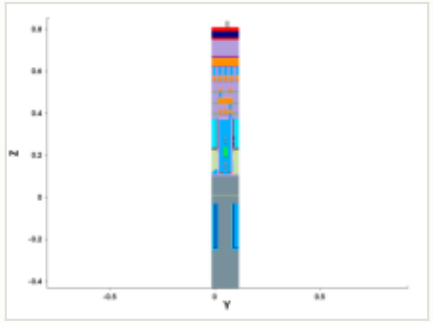
carbon so far, of the whole flow (kgCO2e)



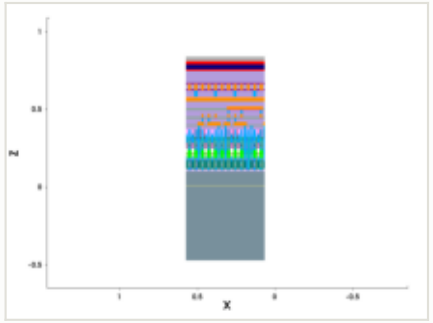
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



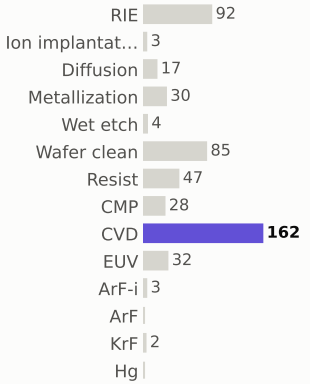
y cut at 0.049 µm



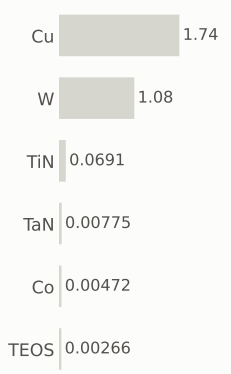
mask for this step

no mask for this step

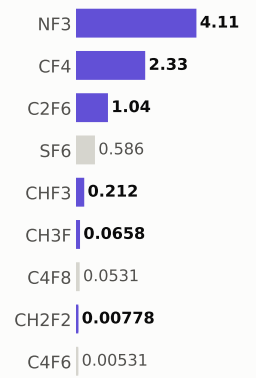
steps by process area



materials deposited (g)



gases emitted (g)



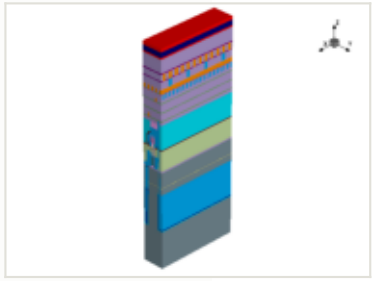
Step #5370: RIE

etch_anisotropic · RIE

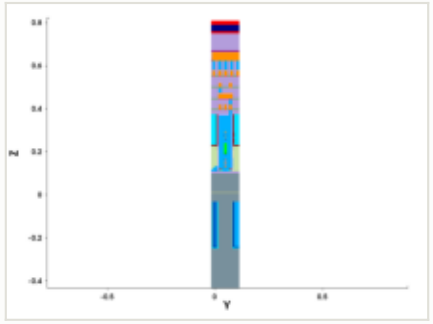
1.04	504.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

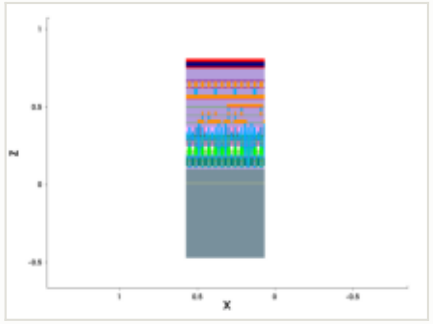
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

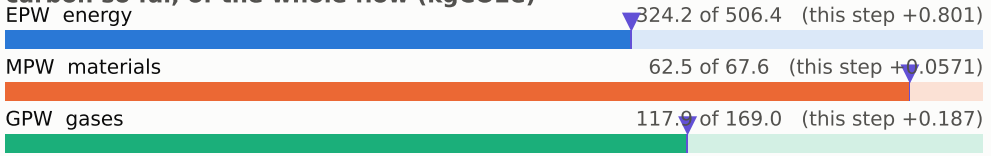
no mask for this step

this step

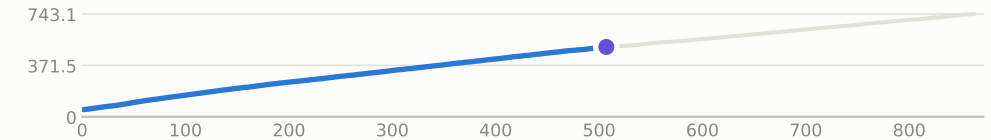
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

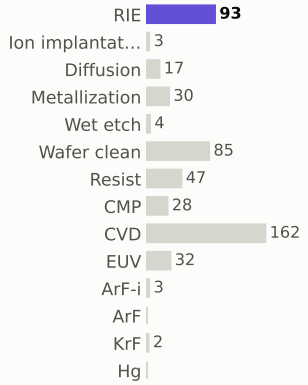
carbon so far, of the whole flow (kgCO2e)



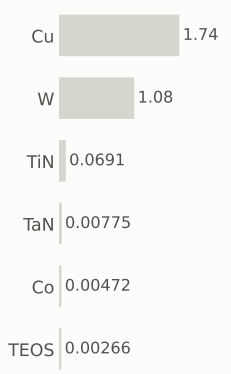
embodied carbon up to each step (kgCO2e per wafer)



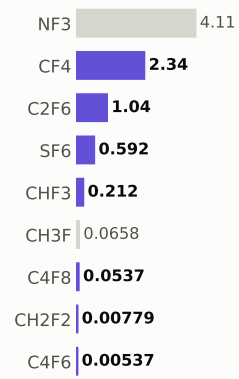
steps by process area



materials deposited (g)



gases emitted (g)



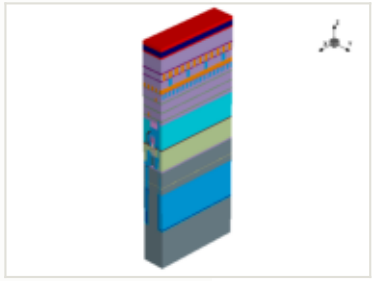
Step #5380: Wafer_clean

etch_strip · Wafer_clean

0.169	504.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

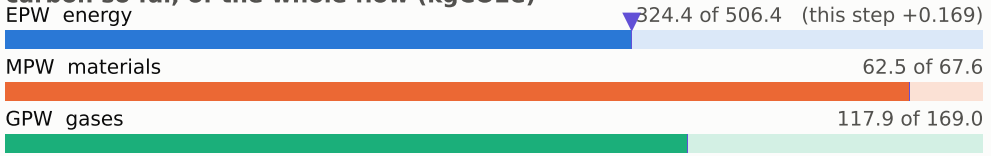


this step

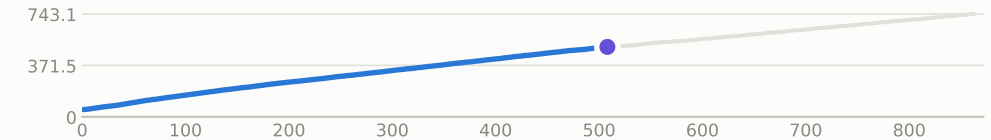
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

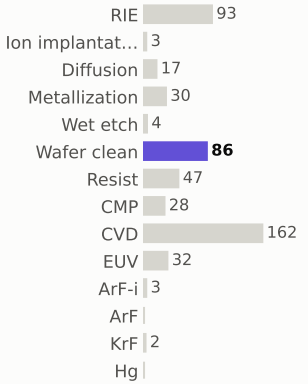
carbon so far, of the whole flow (kgCO2e)



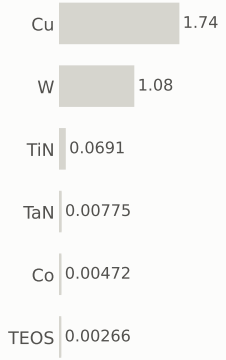
embodied carbon up to each step (kgCO2e per wafer)



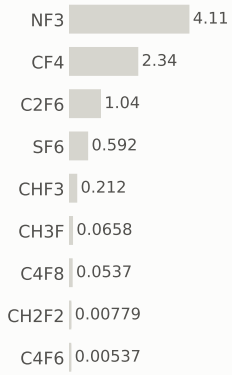
steps by process area



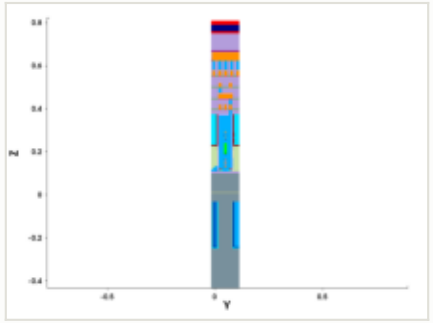
materials deposited (g)



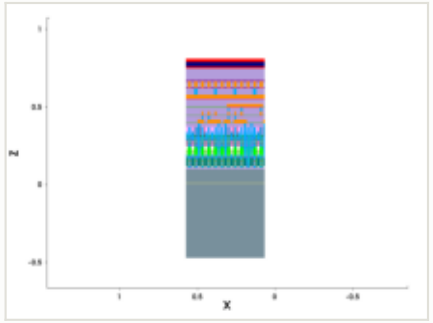
gases emitted (g)



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

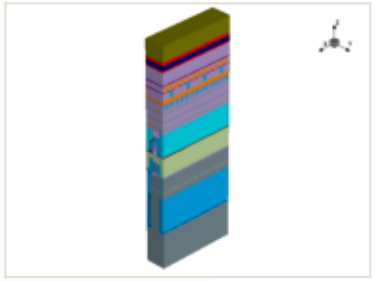
Step #5390: Resist

depo_fill · Resist

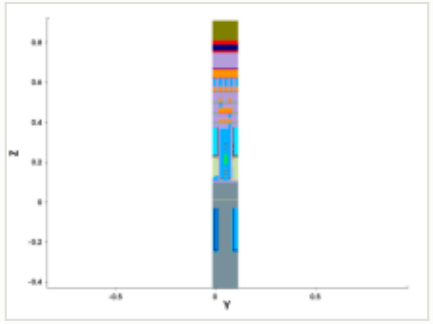
0.169	505.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

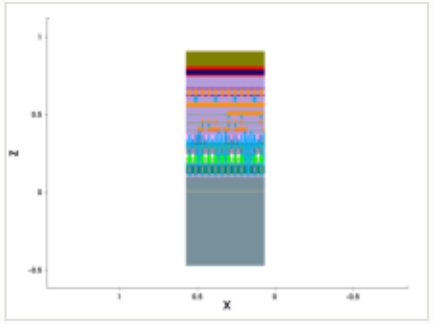
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



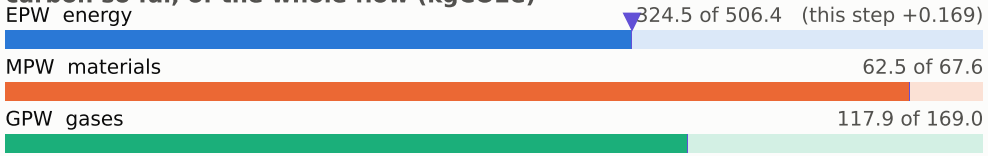
mask for this step

no mask for this step

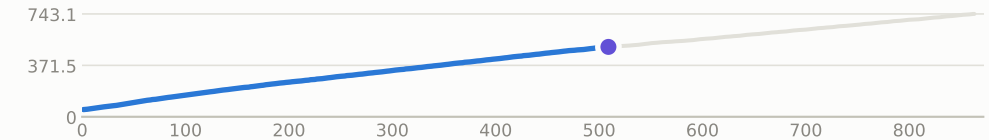
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

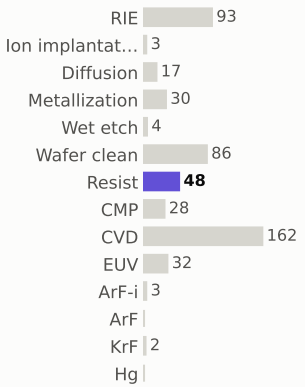
carbon so far, of the whole flow (kgCO2e)



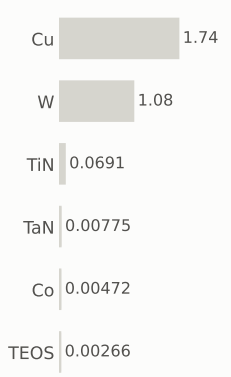
embodied carbon up to each step (kgCO2e per wafer)



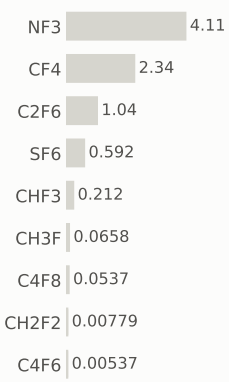
steps by process area



materials deposited (g)



gases emitted (g)



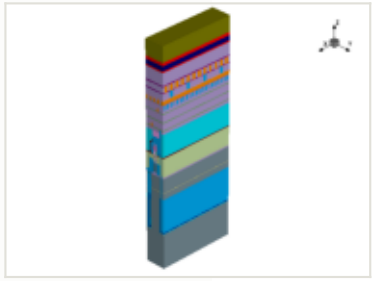
Step #5400: Wafer_clean

clean · Wafer_clean

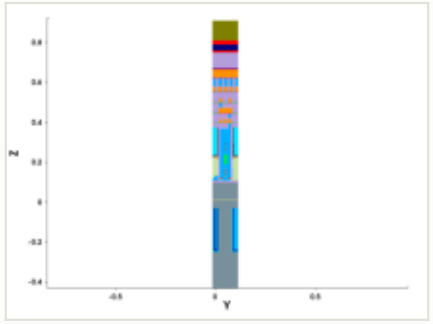
0.169	505.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

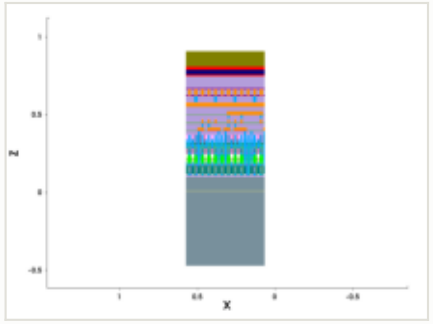
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

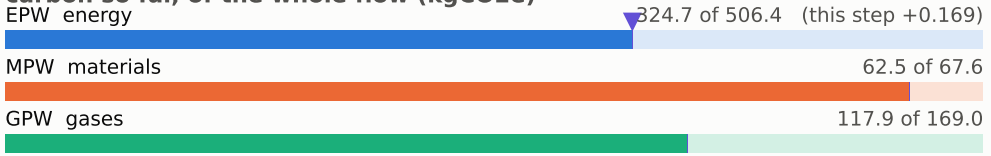
no mask for this step

this step

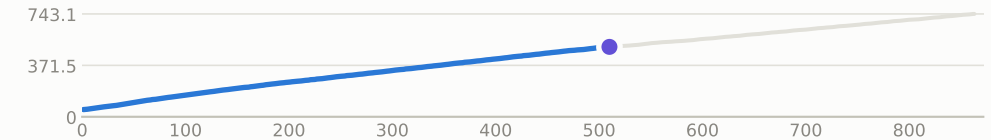
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

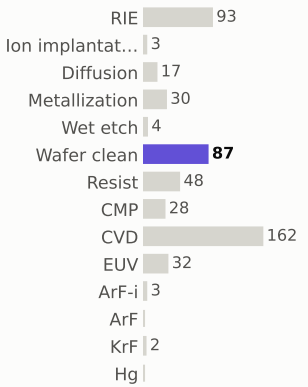
carbon so far, of the whole flow (kgCO2e)



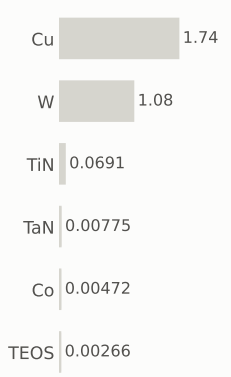
embodied carbon up to each step (kgCO2e per wafer)



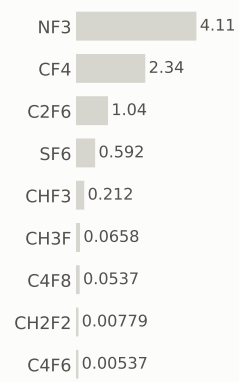
steps by process area



materials deposited (g)



gases emitted (g)



Step #5410: ArF-i

litho_patterning · ArF-i

0.217	505.4	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

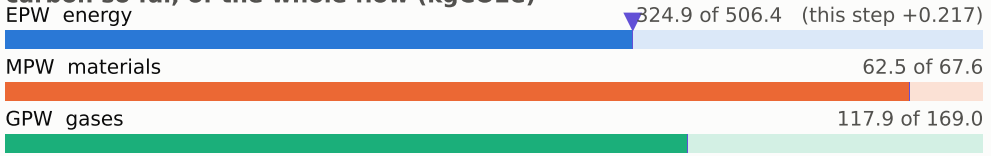
3D structure after this step

zoomed to fit

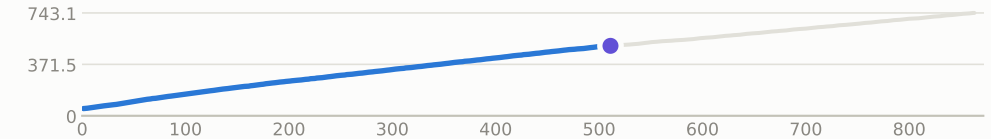
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

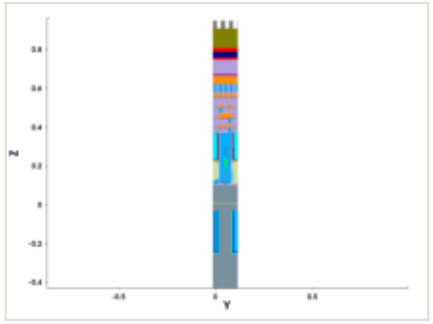
carbon so far, of the whole flow (kgCO2e)



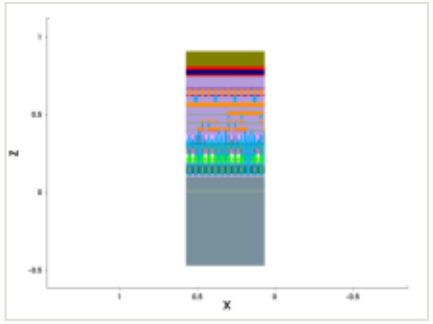
embodied carbon up to each step (kgCO2e per wafer)



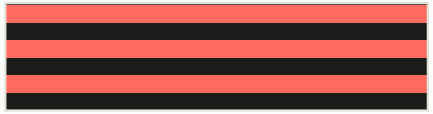
x cut at 0.385 µm



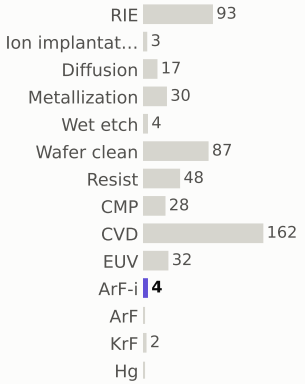
y cut at 0.049 µm



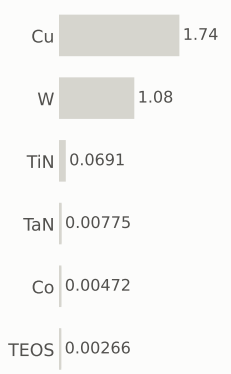
mask: layer 45:0 (GDSII)



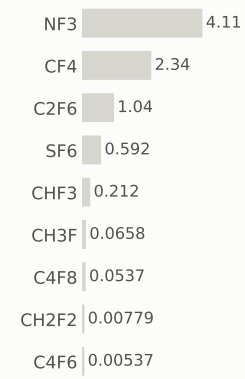
steps by process area



materials deposited (g)



gases emitted (g)



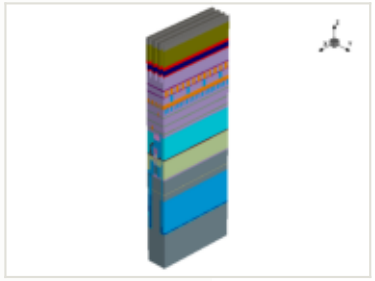
Step #5420: RIE

etch_anisotropic · RIE

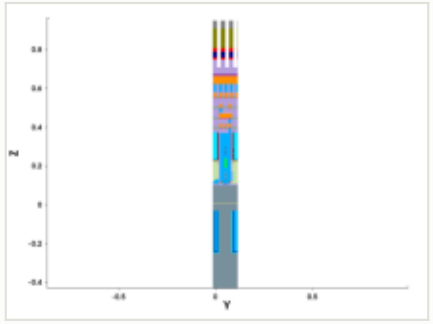
1.04	506.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

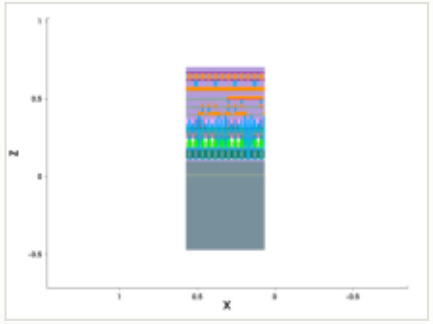
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

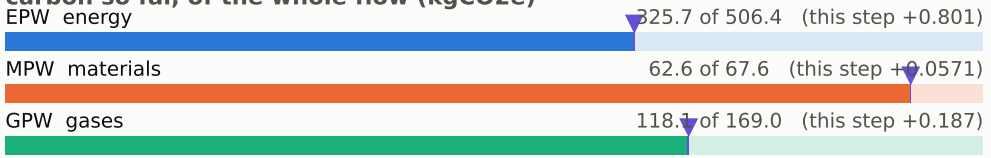
no mask for this step

this step

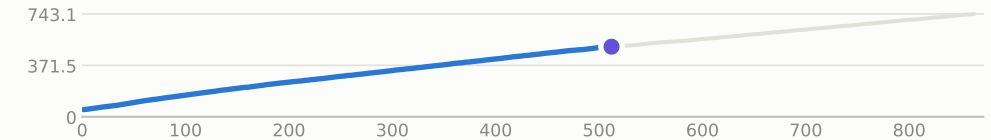
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

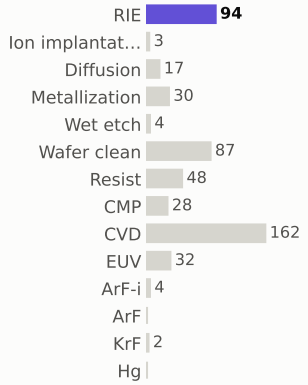
carbon so far, of the whole flow (kgCO2e)



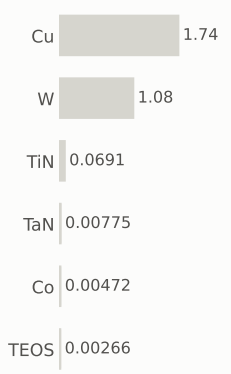
embodied carbon up to each step (kgCO2e per wafer)



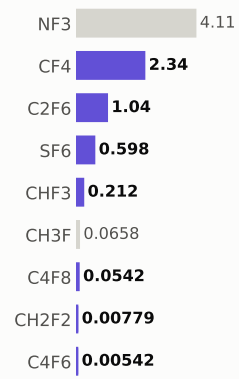
steps by process area



materials deposited (g)



gases emitted (g)



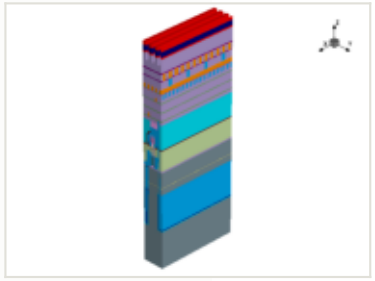
Step #5430: Wafer_clean

etch_strip · Wafer_clean

0.169	506.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

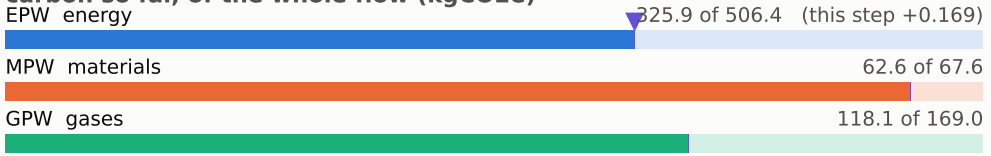


this step

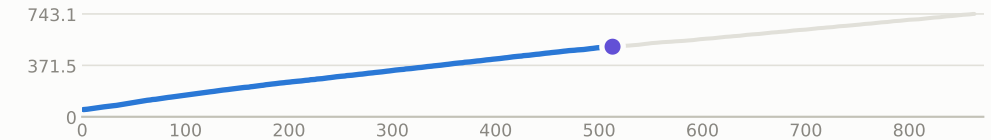
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

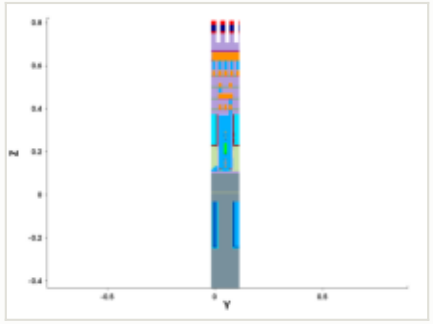
carbon so far, of the whole flow (kgCO2e)



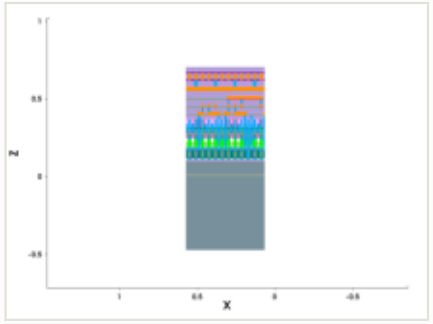
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



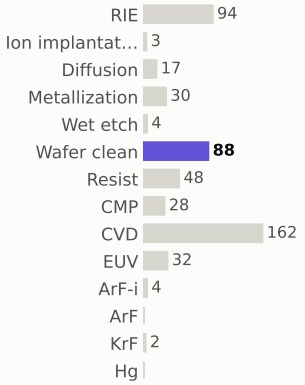
y cut at 0.049 µm



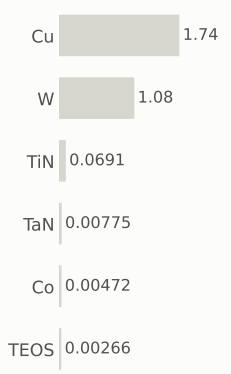
mask for this step

no mask for this step

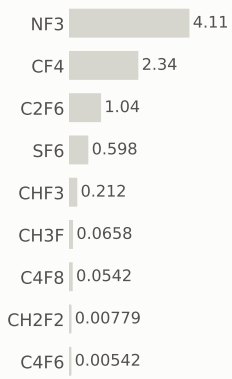
steps by process area



materials deposited (g)



gases emitted (g)



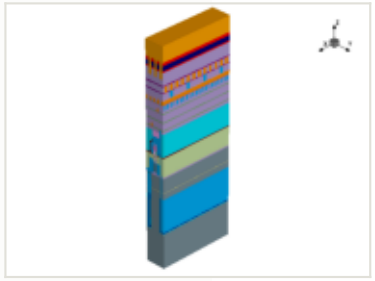
Step #5440: Resist

depo_fill · Resist

0.169	506.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

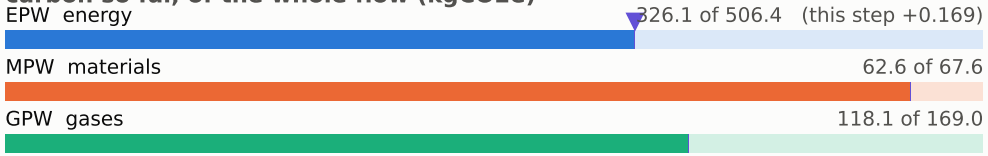


this step

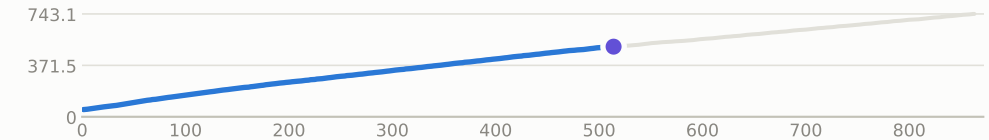
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

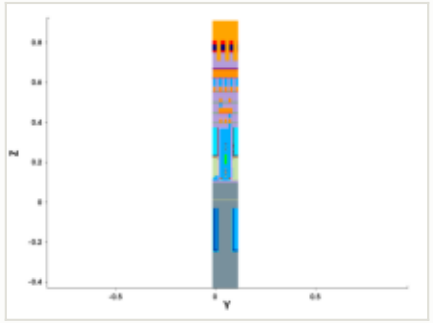
carbon so far, of the whole flow (kgCO2e)



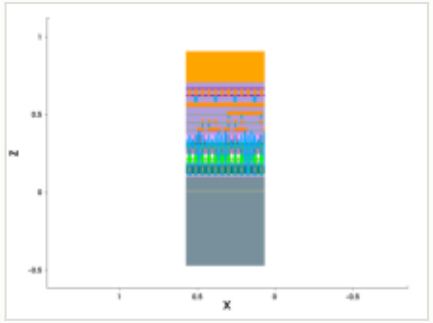
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



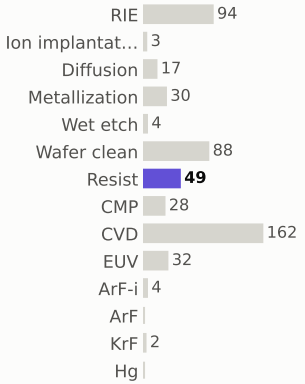
y cut at 0.049 µm



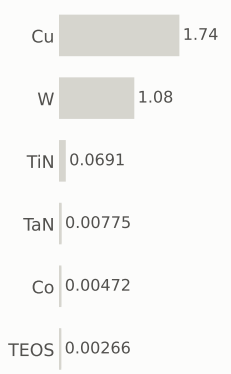
mask for this step

no mask for this step

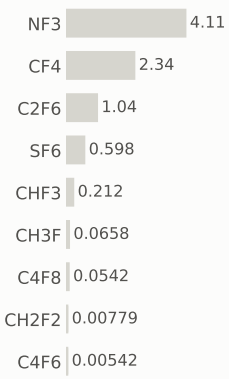
steps by process area



materials deposited (g)



gases emitted (g)



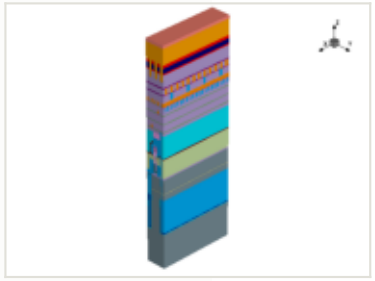
Step #5450: CVD

depo_fill · CVD

1.31	508.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

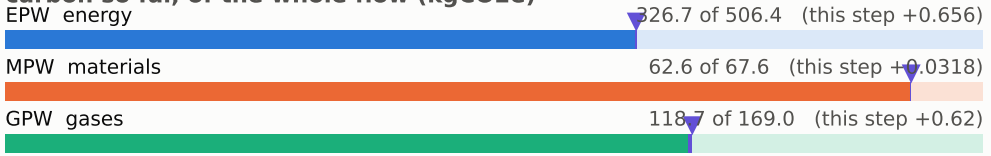


this step

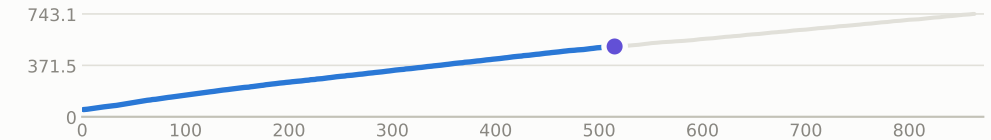
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

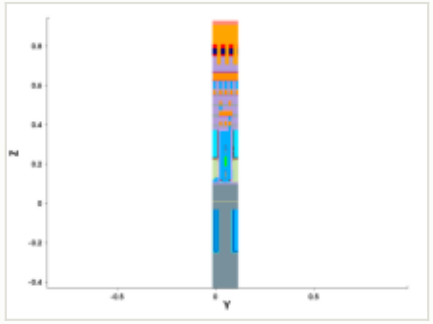
carbon so far, of the whole flow (kgCO2e)



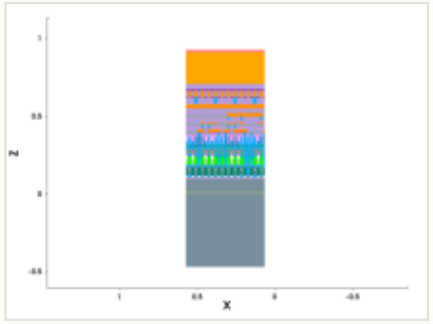
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



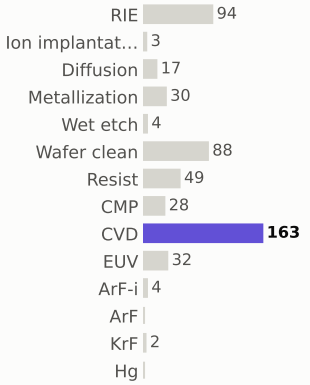
y cut at 0.049 µm



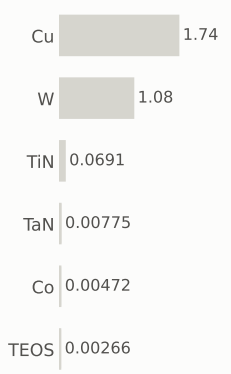
mask for this step

no mask for this step

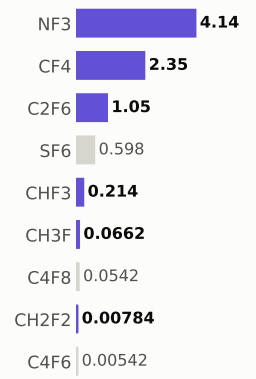
steps by process area



materials deposited (g)



gases emitted (g)



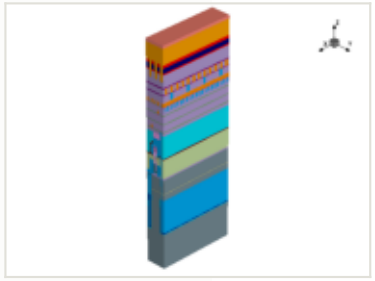
Step #5460: Wafer_clean

clean · Wafer_clean

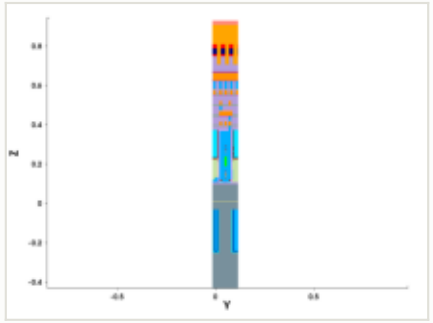
0.169	508.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

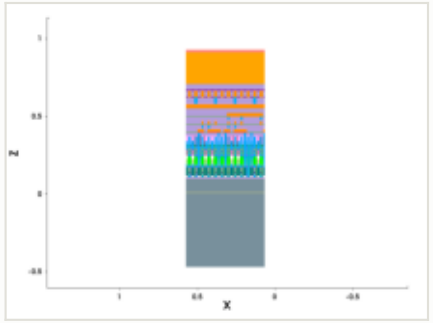
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

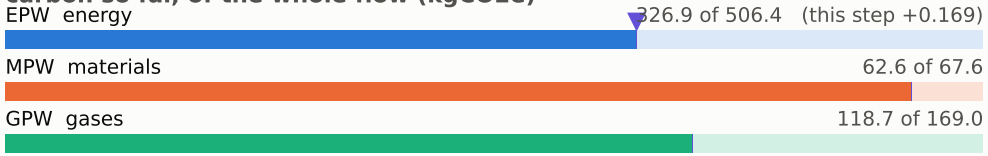
no mask for this step

this step

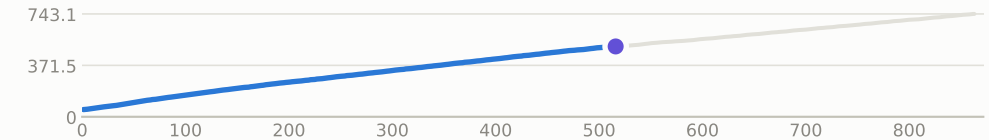
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

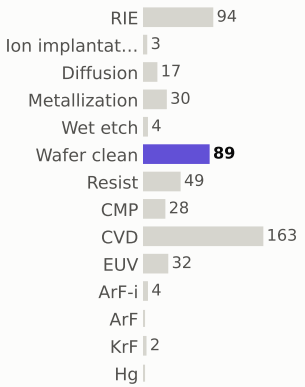
carbon so far, of the whole flow (kgCO2e)



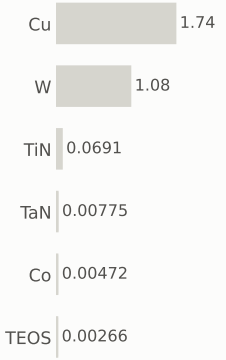
embodied carbon up to each step (kgCO2e per wafer)



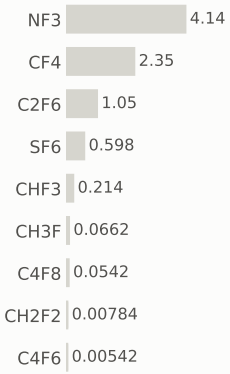
steps by process area



materials deposited (g)



gases emitted (g)



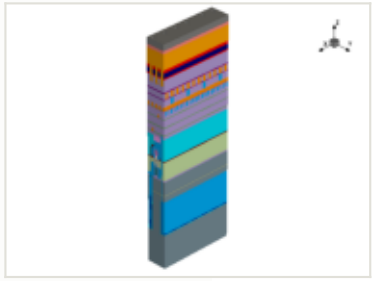
Step #5470: ArF-i

litho_patterning · ArF-i

0.217	508.4	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

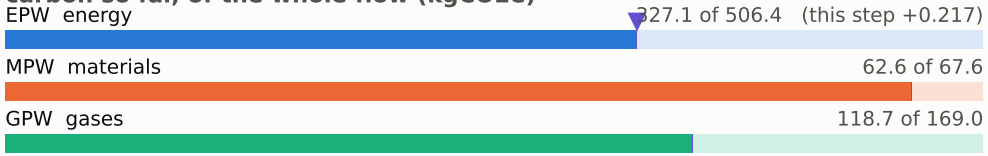


this step

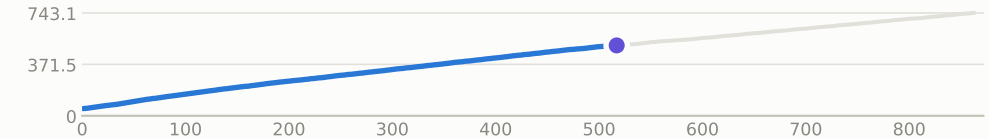
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

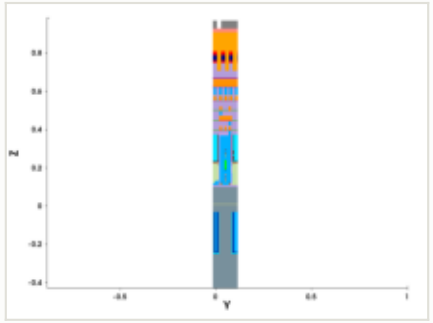
carbon so far, of the whole flow (kgCO2e)



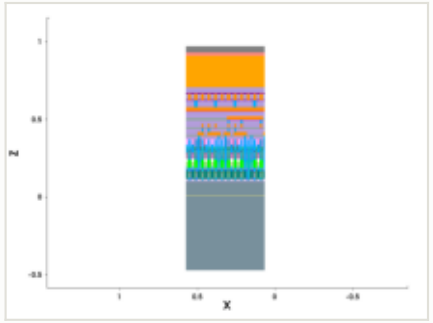
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

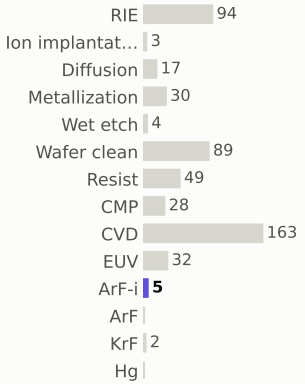


y cut at 0.049 µm

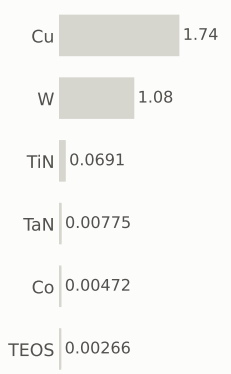


mask: layer 10421:0 (GDSII)

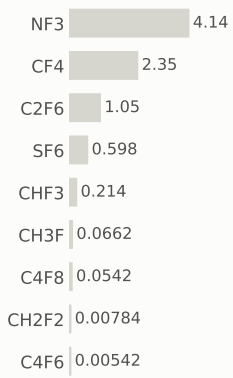
steps by process area



materials deposited (g)



gases emitted (g)



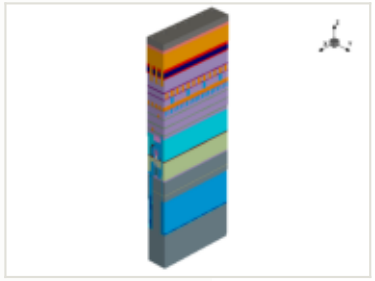
Step #5480: RIE

etch_anisotropic · RIE

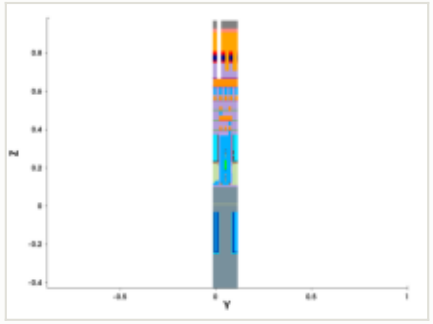
1.04	509.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

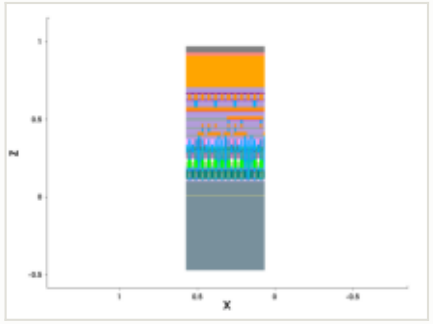
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

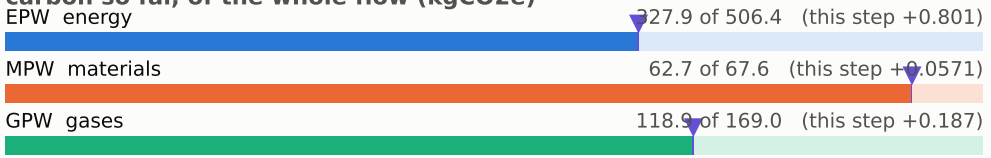
no mask for this step

this step

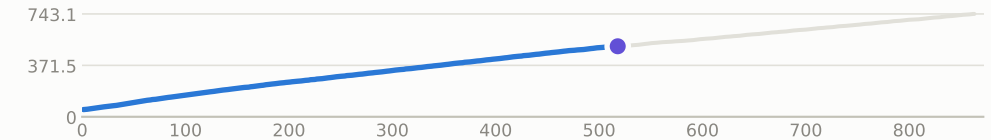
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

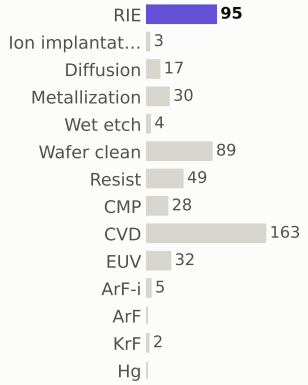
carbon so far, of the whole flow (kgCO2e)



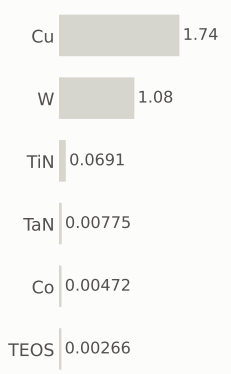
embodied carbon up to each step (kgCO2e per wafer)



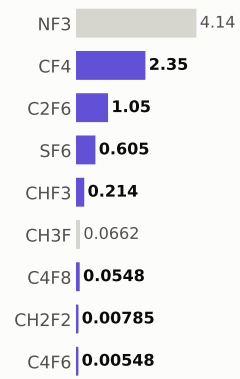
steps by process area



materials deposited (g)



gases emitted (g)



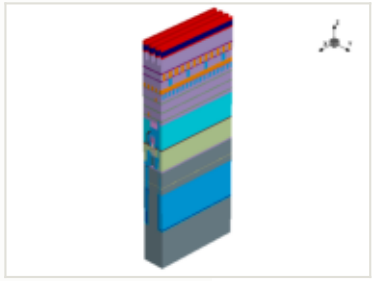
Step #5490: Wafer_clean

etch_strip · Wafer_clean

0.169	509.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

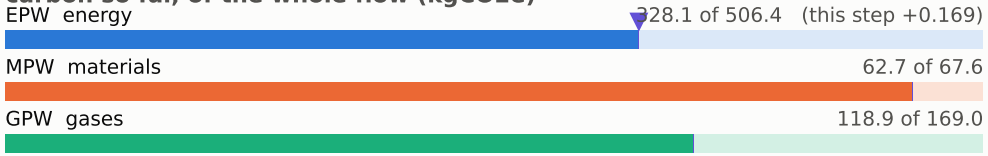


this step

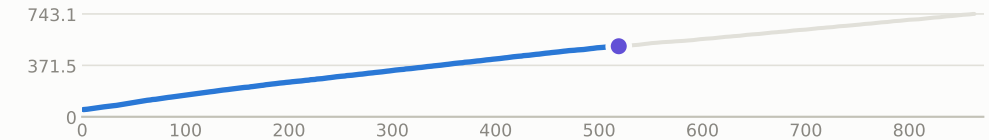
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

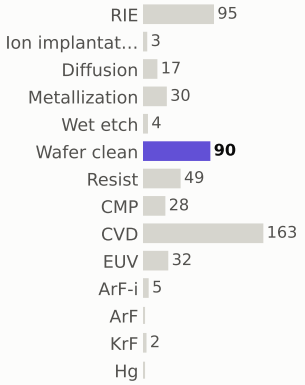
carbon so far, of the whole flow (kgCO2e)



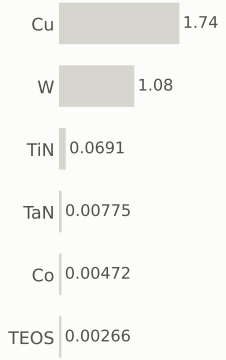
embodied carbon up to each step (kgCO2e per wafer)



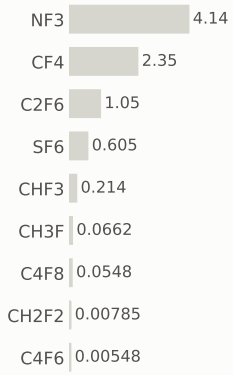
steps by process area



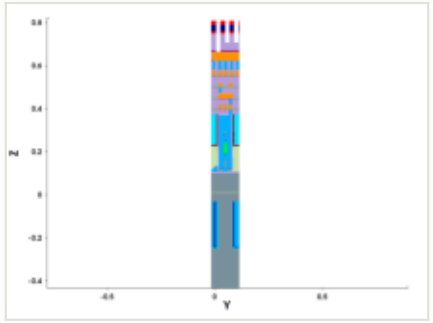
materials deposited (g)



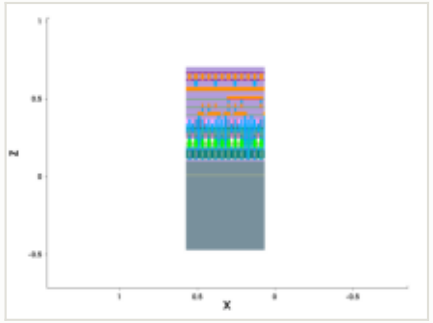
gases emitted (g)



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

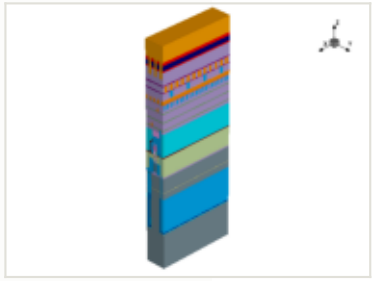
Step #5500: Resist

depo_fill · Resist

0.169	509.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

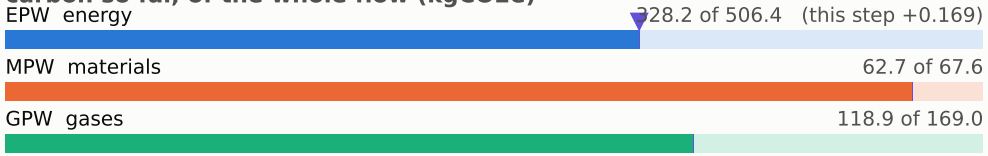


this step

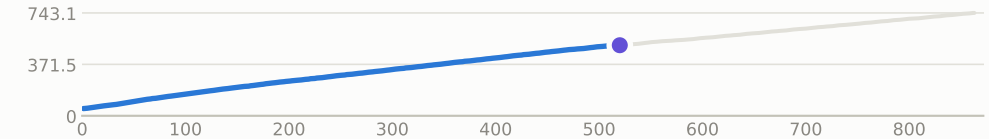
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

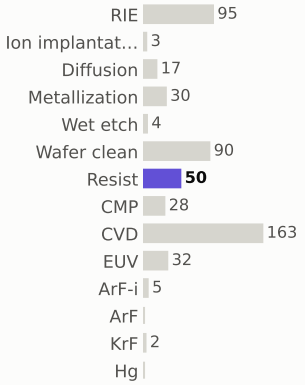
carbon so far, of the whole flow (kgCO2e)



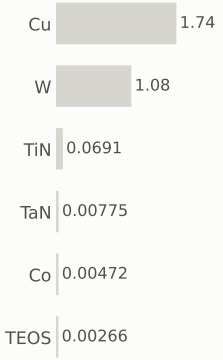
embodied carbon up to each step (kgCO2e per wafer)



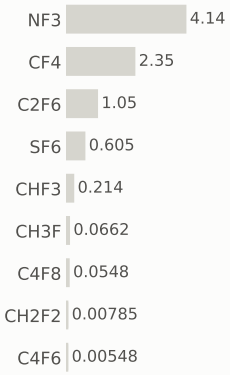
steps by process area



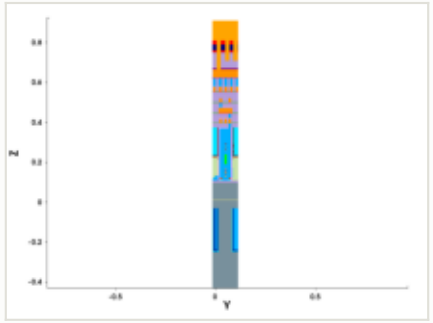
materials deposited (g)



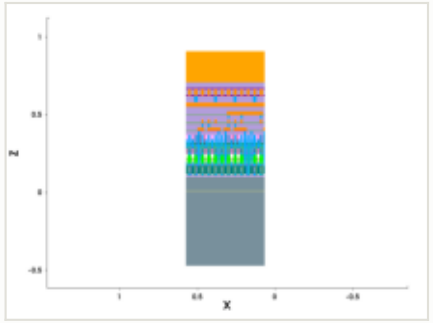
gases emitted (g)



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

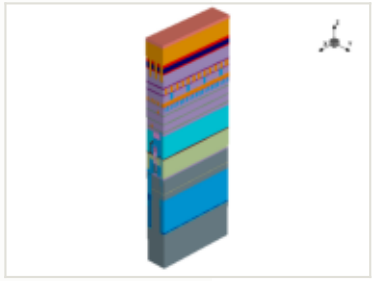
Step #5510: CVD

depo_fill · CVD

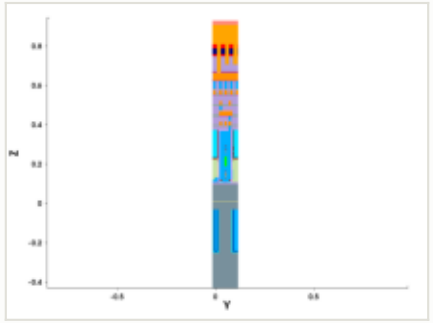
1.31	511.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

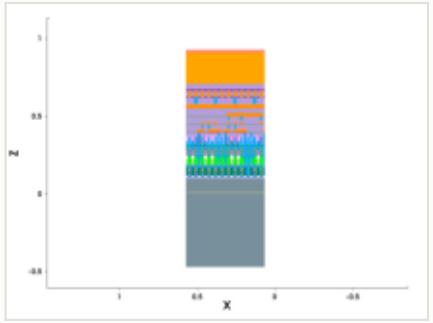
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

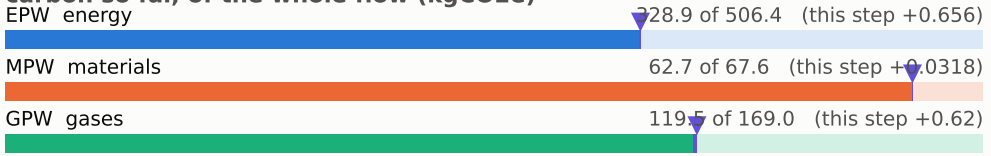
no mask for this step

this step

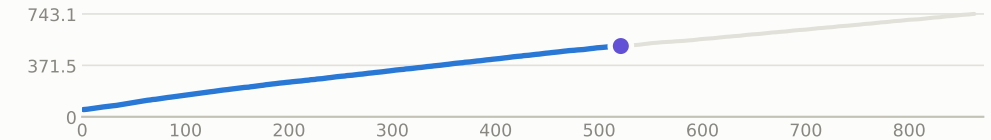
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

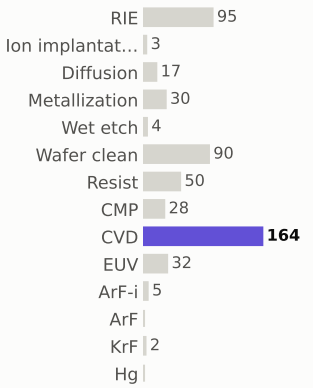
carbon so far, of the whole flow (kgCO2e)



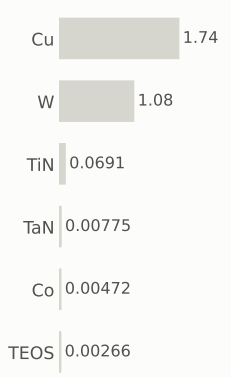
embodied carbon up to each step (kgCO2e per wafer)



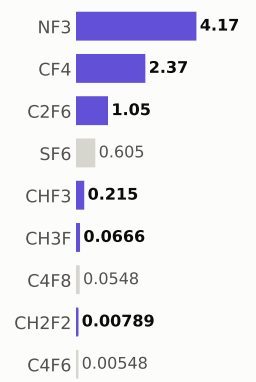
steps by process area



materials deposited (g)



gases emitted (g)



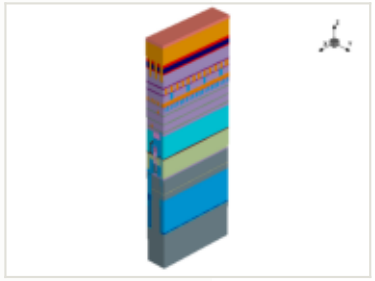
Step #5520: Wafer_clean

clean · Wafer_clean

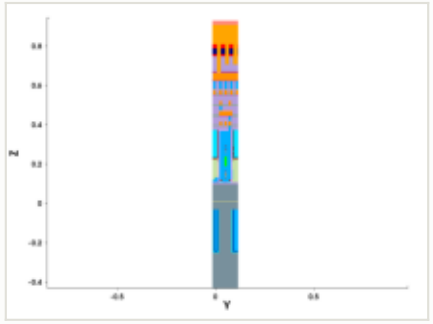
0.169	511.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

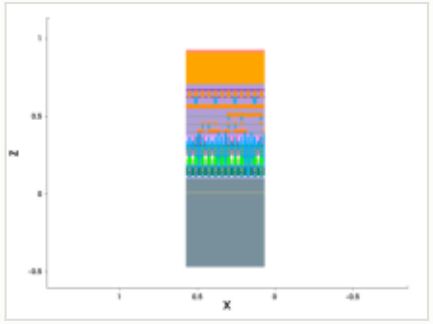
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

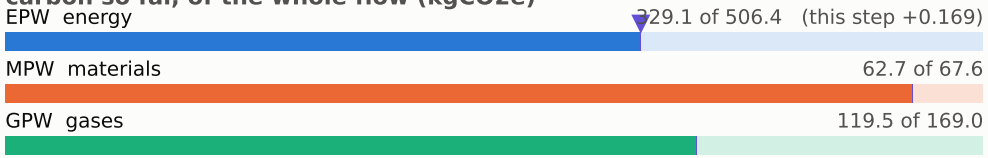
no mask for this step

this step

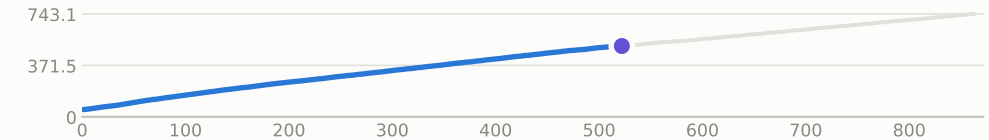
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

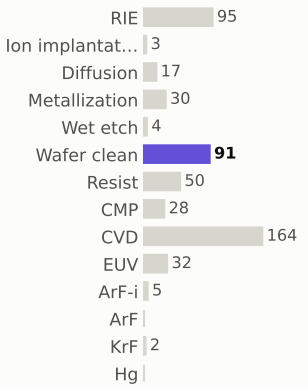
carbon so far, of the whole flow (kgCO2e)



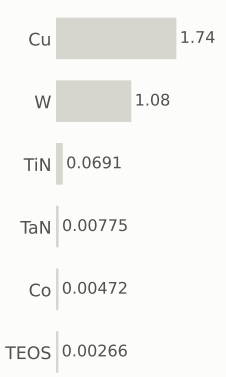
embodied carbon up to each step (kgCO2e per wafer)



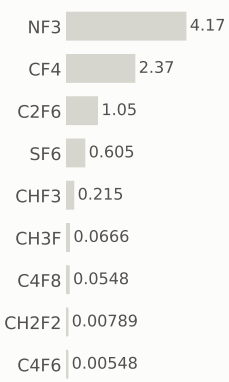
steps by process area



materials deposited (g)



gases emitted (g)



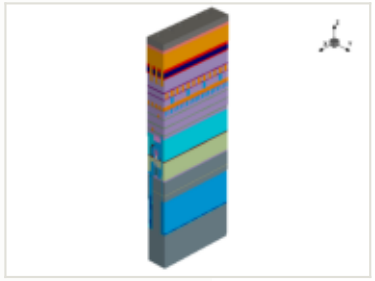
Step #5530: ArF-i

litho_patterning · ArF-i

0.217	511.5	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

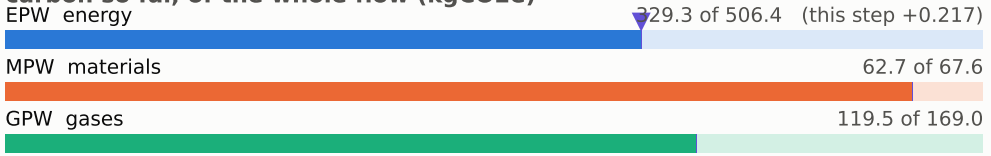


this step

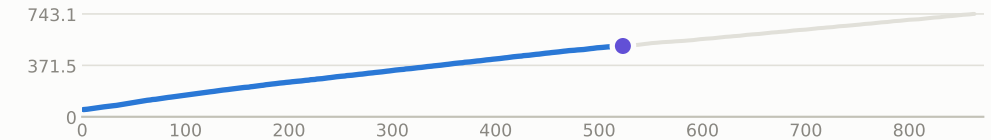
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

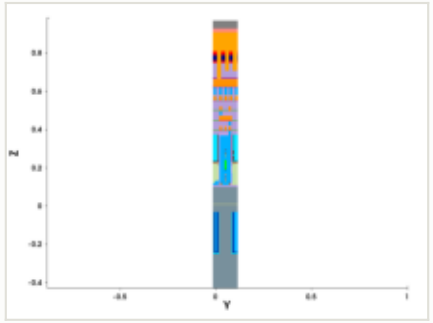
carbon so far, of the whole flow (kgCO2e)



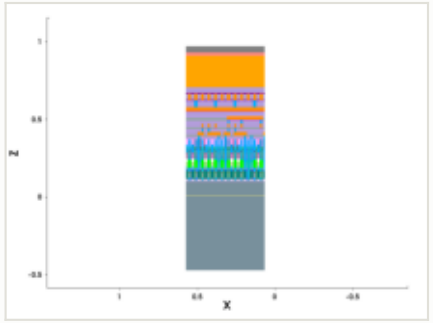
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

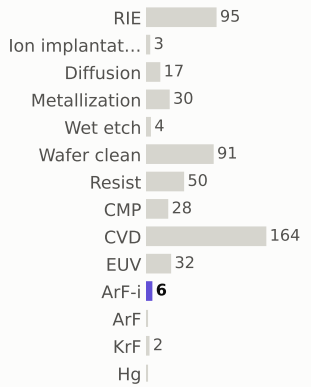


y cut at 0.049 µm

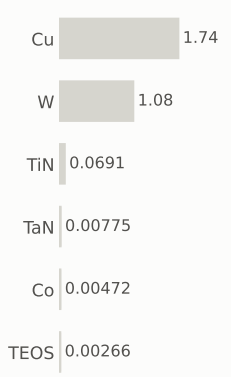


mask: layer 10422:0 (GDSII)

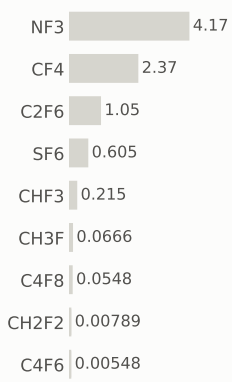
steps by process area



materials deposited (g)



gases emitted (g)



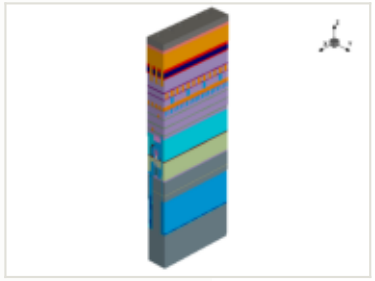
Step #5540: RIE

etch_anisotropic · RIE

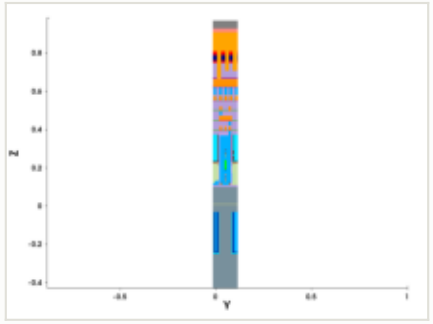
1.04	512.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

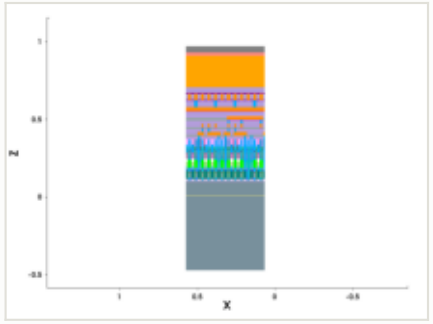
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

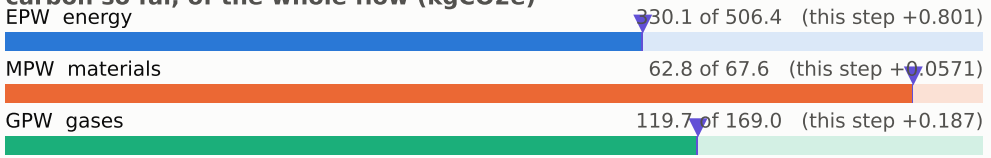
no mask for this step

this step

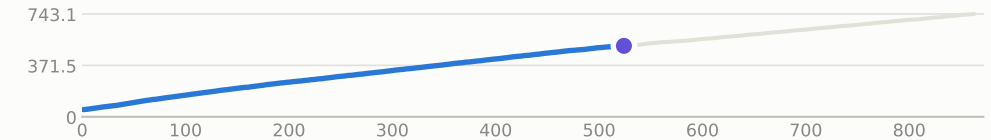
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

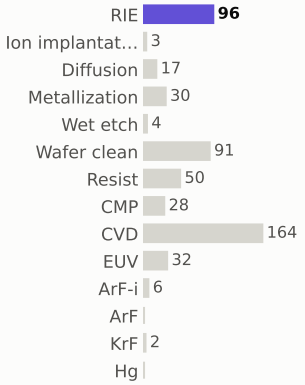
carbon so far, of the whole flow (kgCO2e)



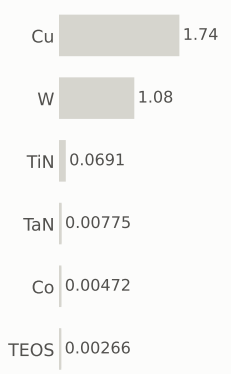
embodied carbon up to each step (kgCO2e per wafer)



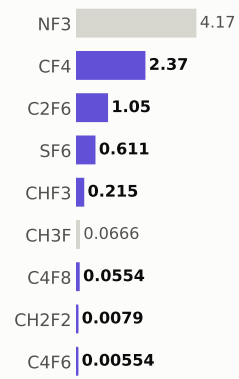
steps by process area



materials deposited (g)



gases emitted (g)



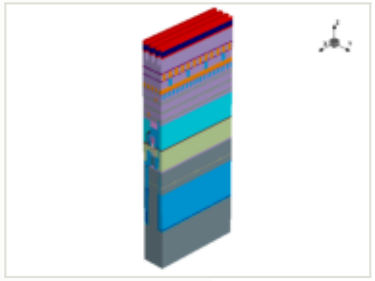
Step #5550: Wafer_clean

etch_strip · Wafer_clean

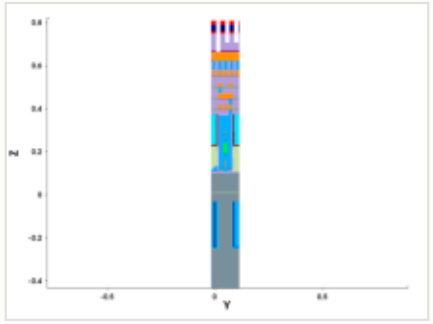
0.169	512.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

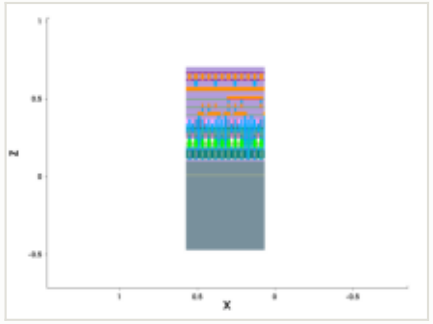
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



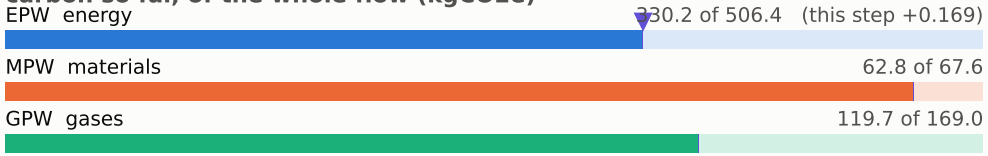
mask for this step

no mask for this step

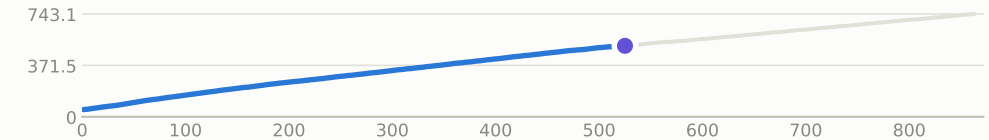
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

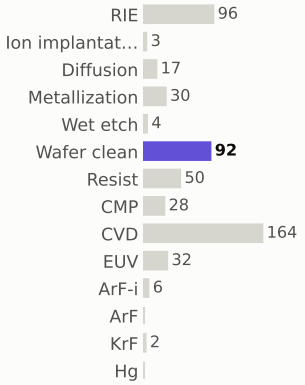
carbon so far, of the whole flow (kgCO2e)



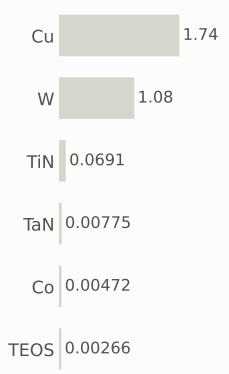
embodied carbon up to each step (kgCO2e per wafer)



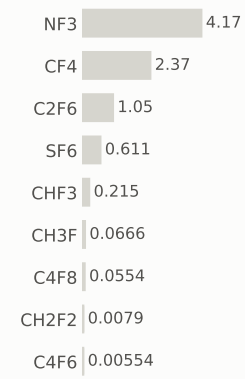
steps by process area



materials deposited (g)



gases emitted (g)



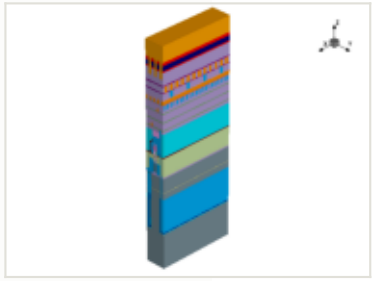
Step #5560: Resist

depo_fill · Resist

0.169	512.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

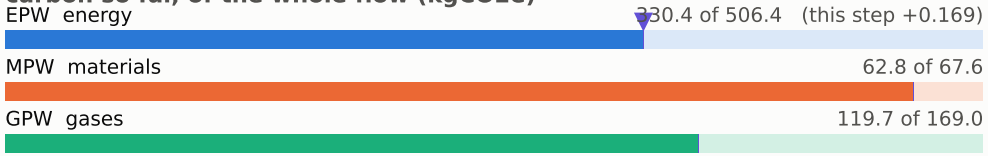


this step

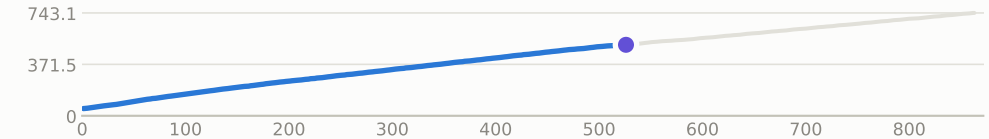
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

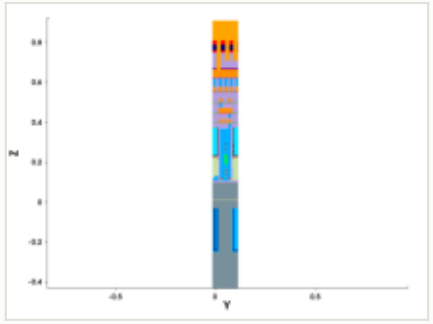
carbon so far, of the whole flow (kgCO2e)



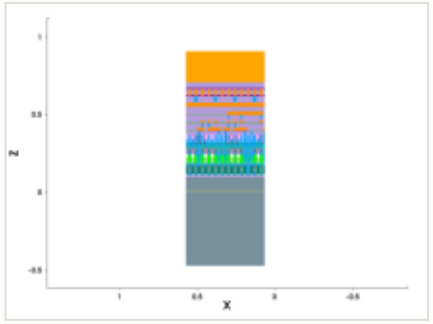
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



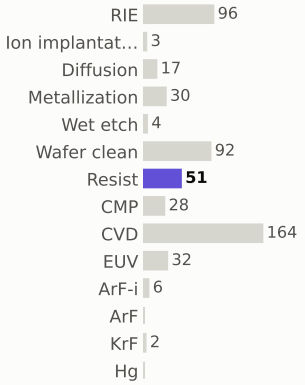
y cut at 0.049 µm



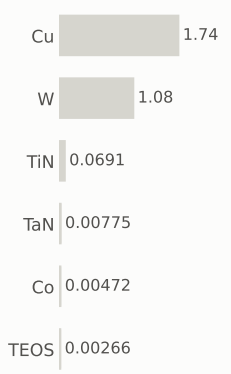
mask for this step

no mask for this step

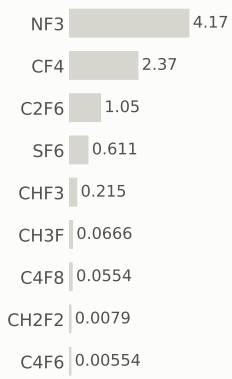
steps by process area



materials deposited (g)



gases emitted (g)



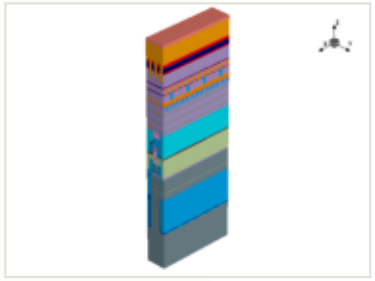
Step #5570: CVD

depo_fill · CVD

1.31	514.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

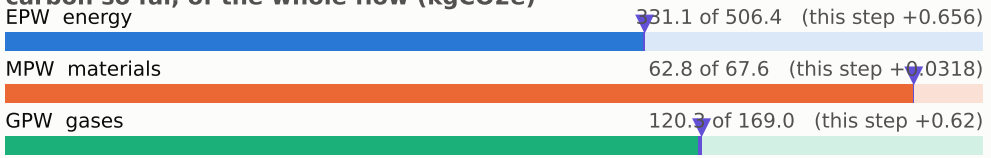


this step

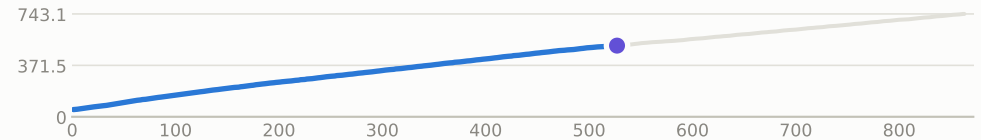
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

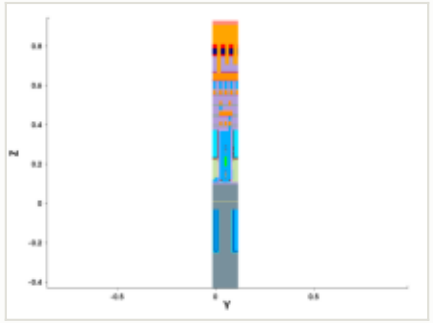
carbon so far, of the whole flow (kgCO2e)



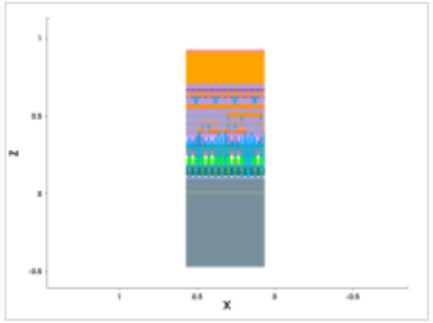
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



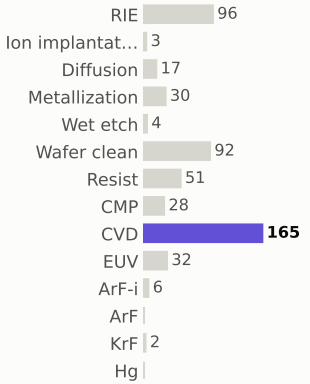
y cut at 0.049 µm



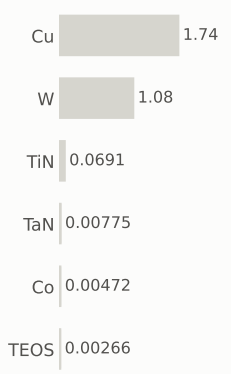
mask for this step

no mask for this step

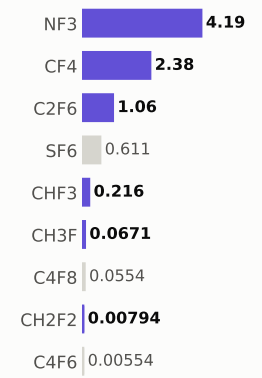
steps by process area



materials deposited (g)



gases emitted (g)



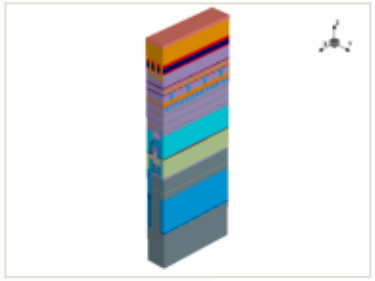
Step #5580: Wafer_clean

clean · Wafer_clean

0.169	514.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



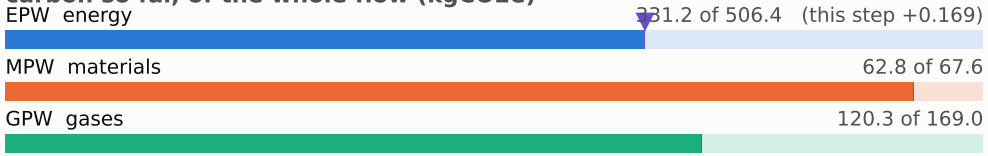
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

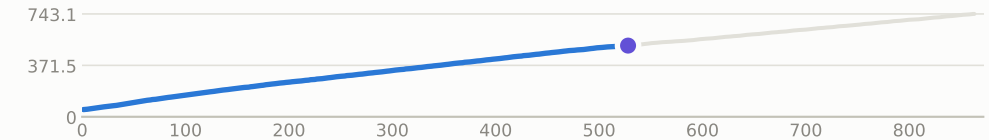
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

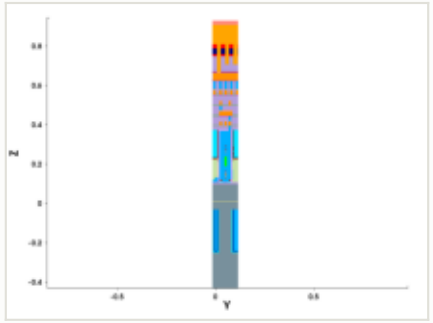
carbon so far, of the whole flow (kgCO2e)



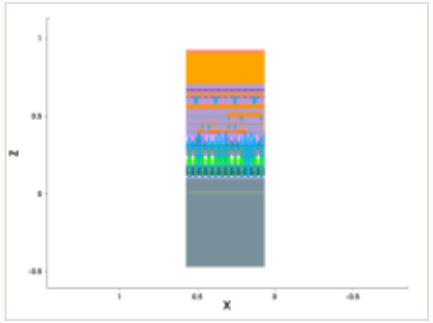
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



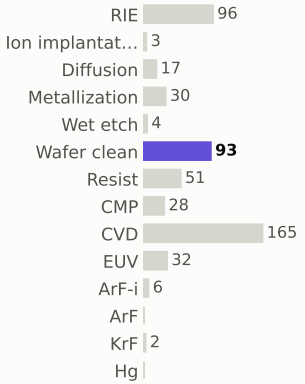
y cut at 0.049 µm



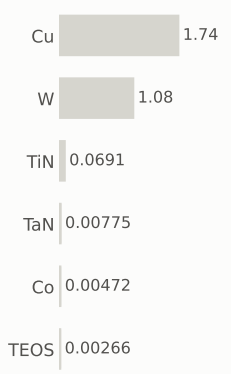
mask for this step

no mask for this step

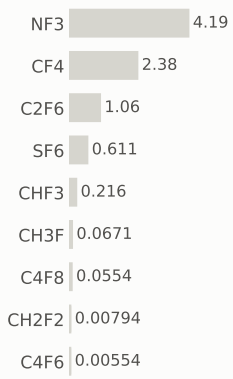
steps by process area



materials deposited (g)



gases emitted (g)



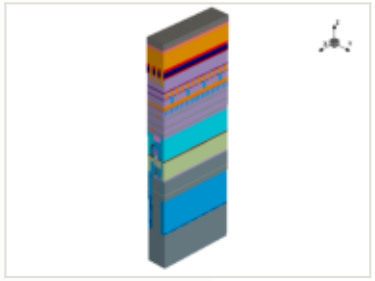
Step #5590: ArF-i

litho_patterning · ArF-i

0.217	514.6	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

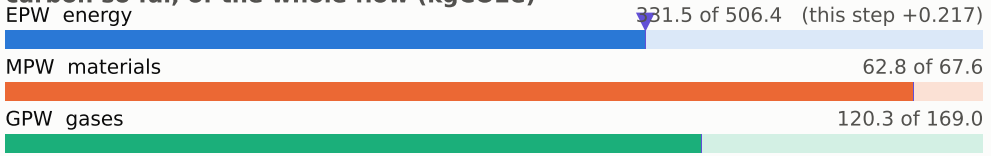


this step

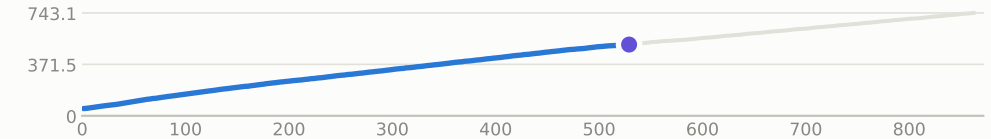
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

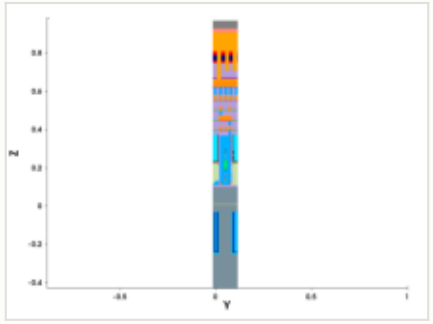
carbon so far, of the whole flow (kgCO2e)



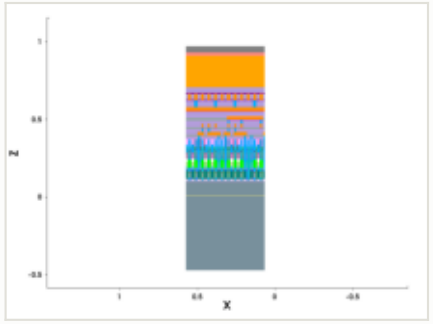
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

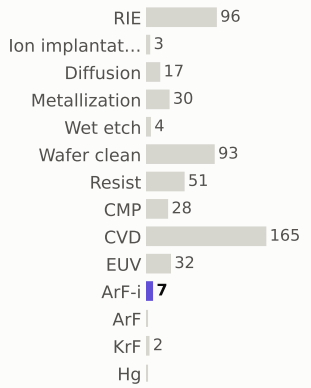


y cut at 0.049 µm

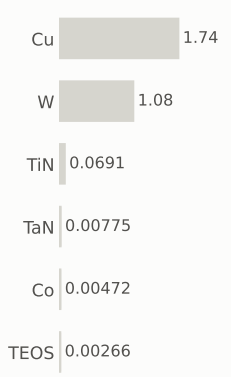


mask: layer 10423:0 (GDSII)

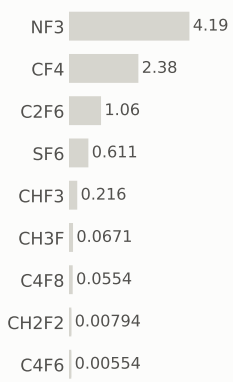
steps by process area



materials deposited (g)



gases emitted (g)



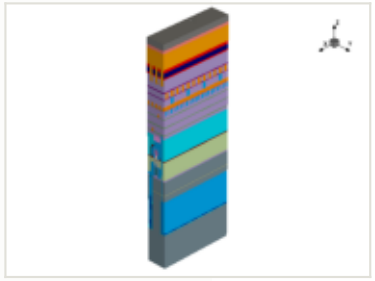
Step #5600: RIE

etch_anisotropic · RIE

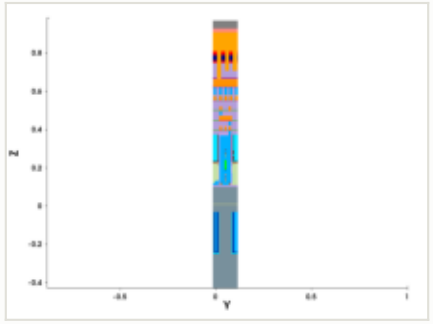
1.04	515.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

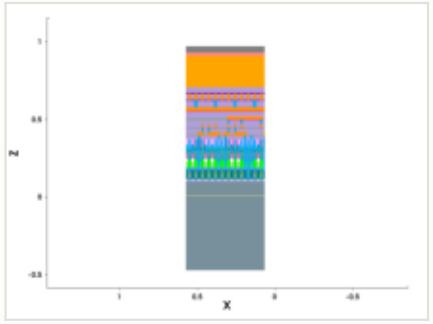
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

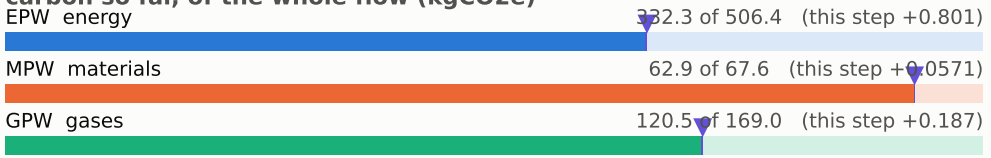
no mask for this step

this step

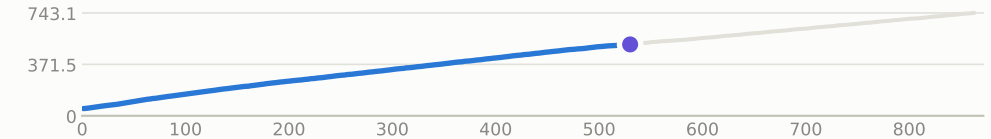
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

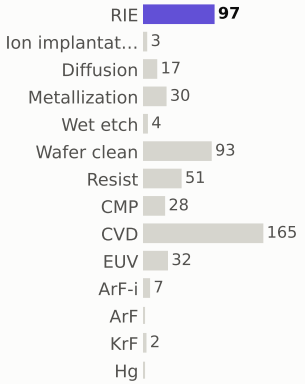
carbon so far, of the whole flow (kgCO2e)



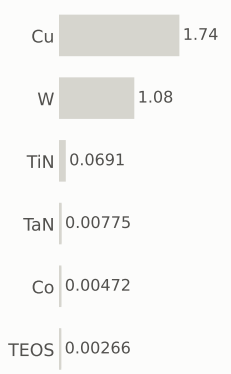
embodied carbon up to each step (kgCO2e per wafer)



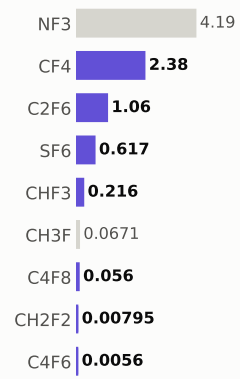
steps by process area



materials deposited (g)



gases emitted (g)



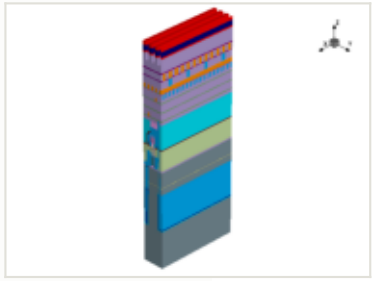
Step #5610: Wafer_clean

etch_strip · Wafer_clean

0.169	515.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

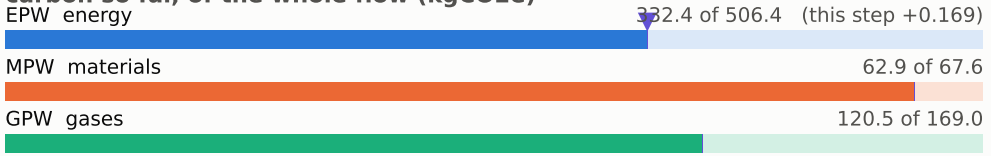


this step

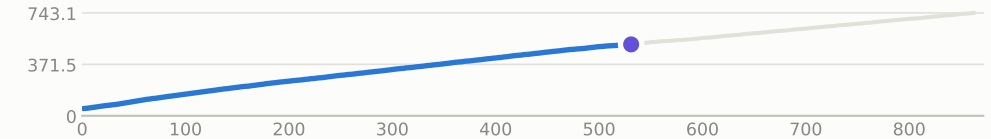
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

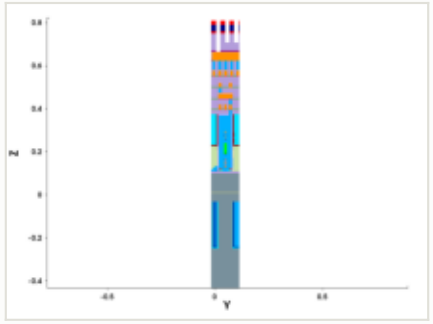
carbon so far, of the whole flow (kgCO2e)



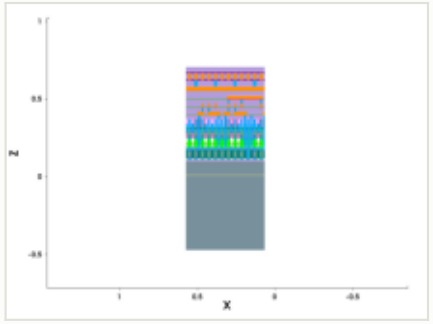
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



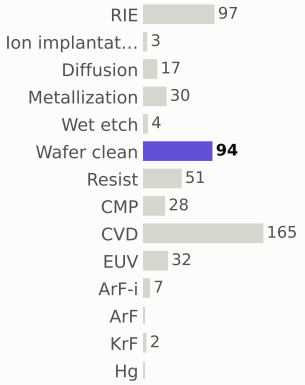
y cut at 0.049 µm



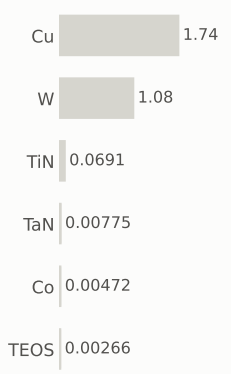
mask for this step

no mask for this step

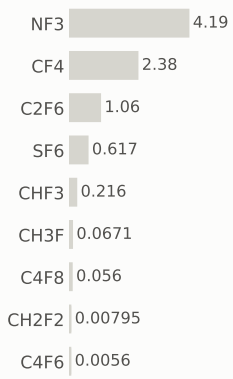
steps by process area



materials deposited (g)



gases emitted (g)



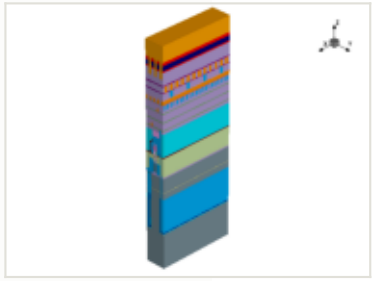
Step #5620: Resist

depo_fill · Resist

0.169	516.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

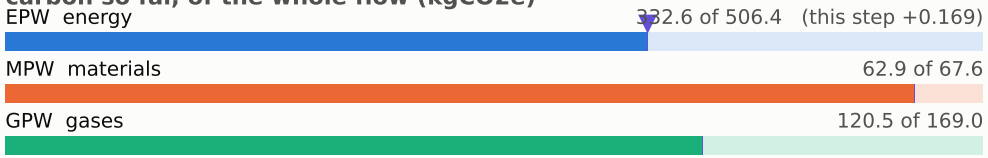


this step

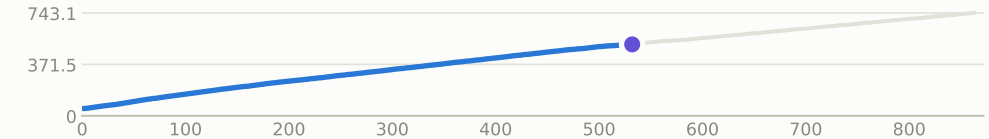
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

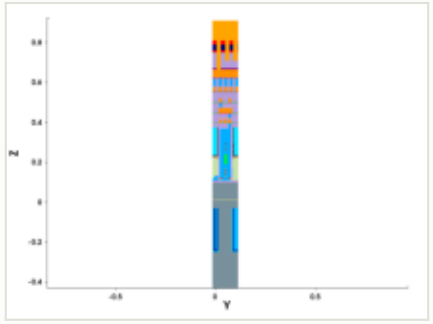
carbon so far, of the whole flow (kgCO2e)



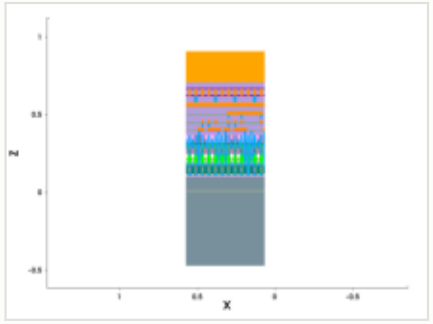
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



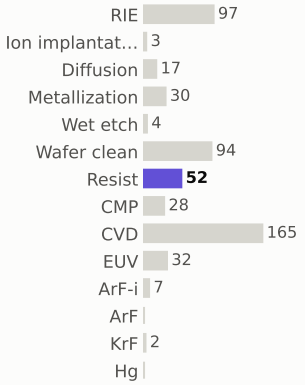
y cut at 0.049 µm



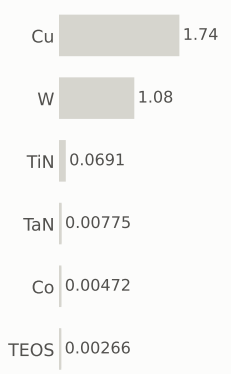
mask for this step

no mask for this step

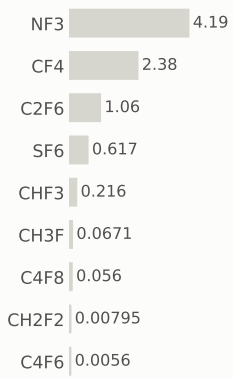
steps by process area



materials deposited (g)



gases emitted (g)



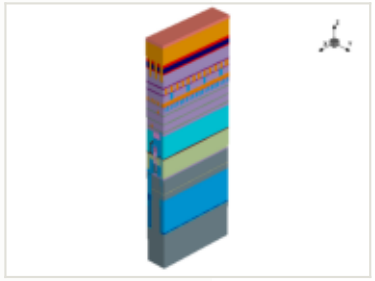
Step #5630: CVD

depo_fill · CVD

1.31	517.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

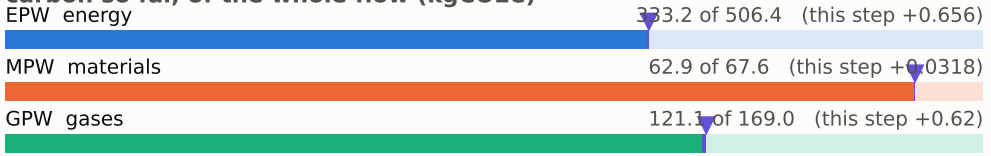


this step

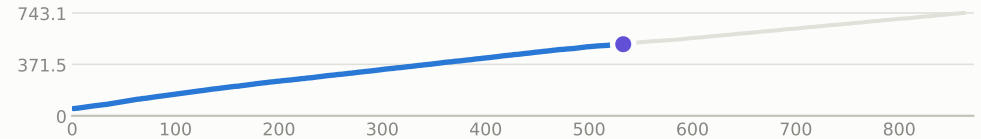
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

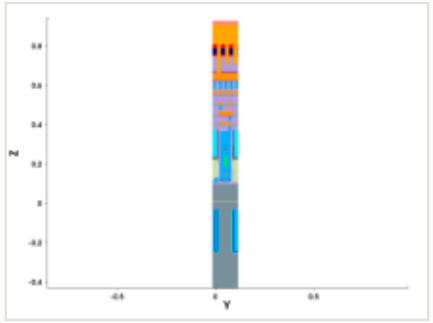
carbon so far, of the whole flow (kgCO2e)



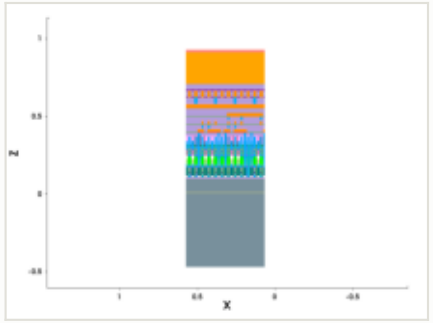
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



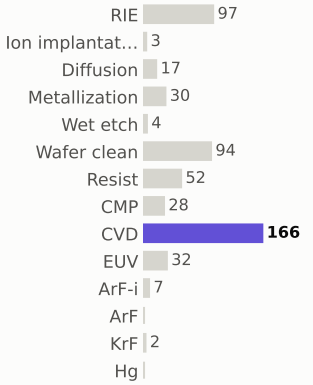
y cut at 0.049 µm



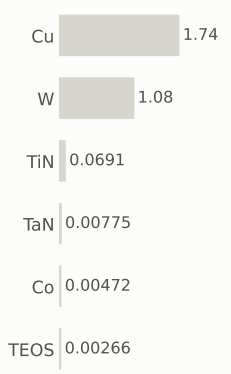
mask for this step

no mask for this step

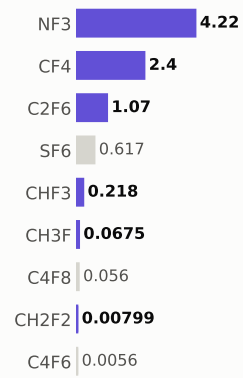
steps by process area



materials deposited (g)



gases emitted (g)



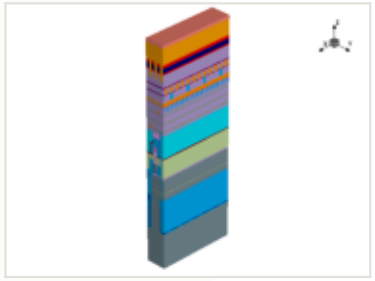
Step #5640: Wafer_clean

clean · Wafer_clean

0.169	517.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



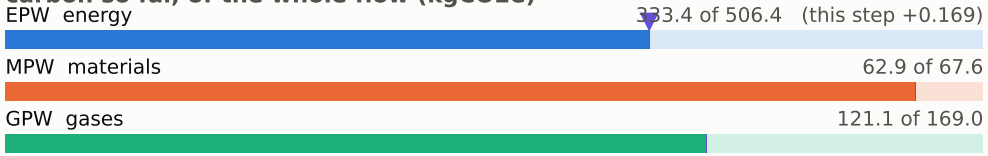
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

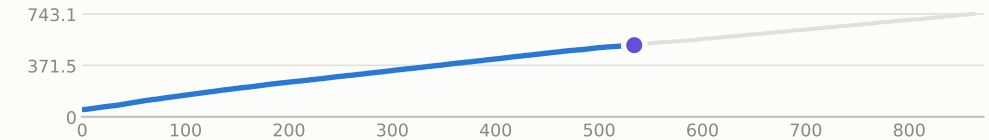
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

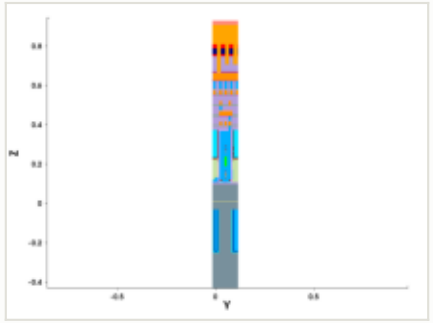
carbon so far, of the whole flow (kgCO2e)



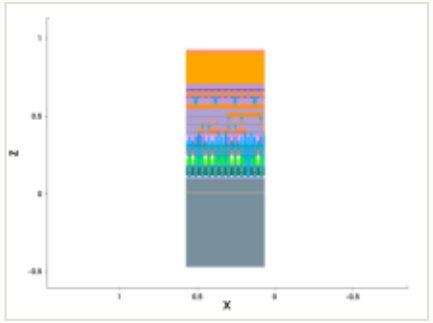
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



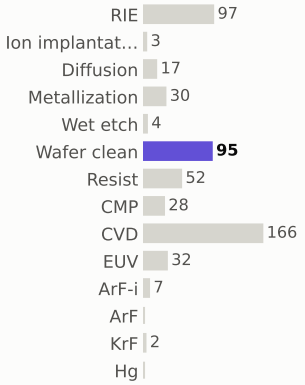
y cut at 0.049 µm



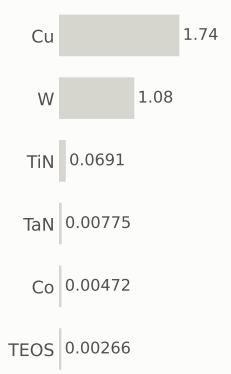
mask for this step

no mask for this step

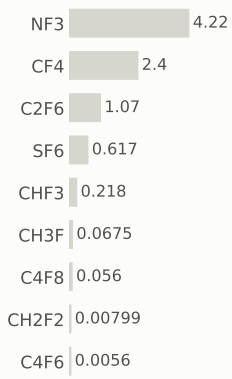
steps by process area



materials deposited (g)



gases emitted (g)



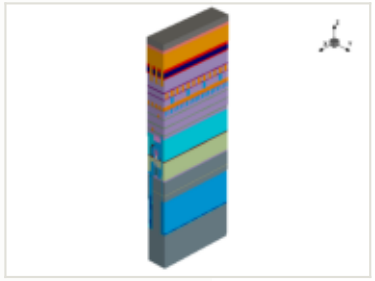
Step #5650: ArF-i

litho_patterning · ArF-i

0.217	517.7	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

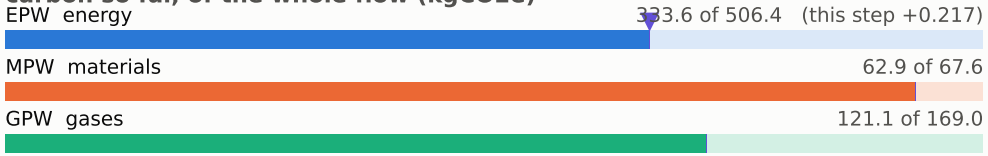


this step

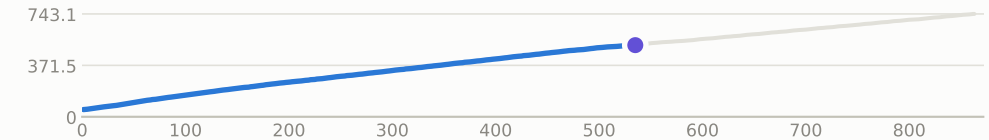
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

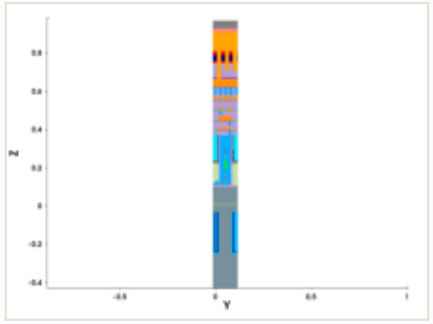
carbon so far, of the whole flow (kgCO2e)



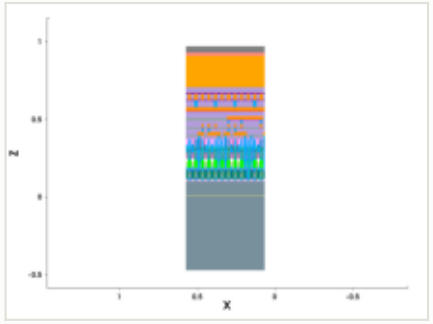
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



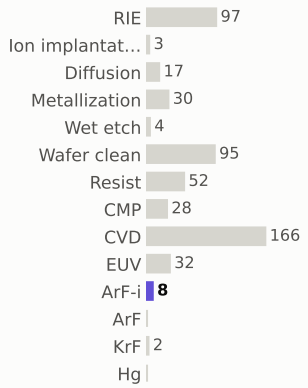
y cut at 0.049 µm



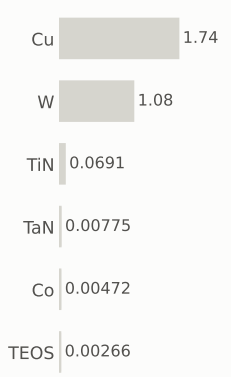
mask: layer 10424:0 (GDSII)

nothing in the clip window

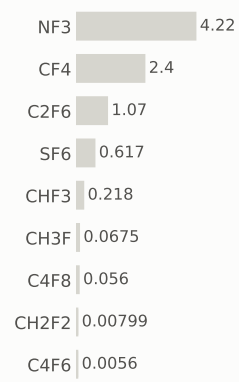
steps by process area



materials deposited (g)



gases emitted (g)



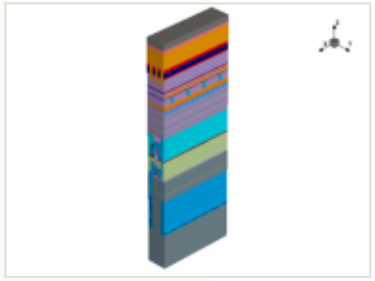
Step #5660: RIE

etch_anisotropic · RIE

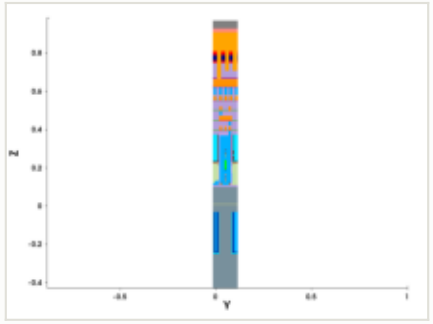
1.04	518.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

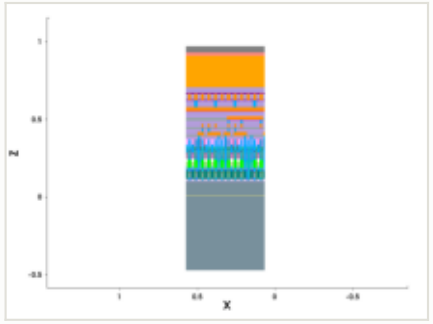
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

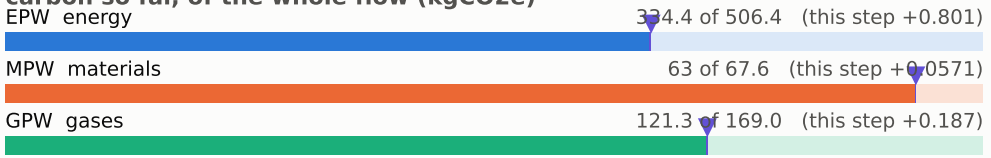
no mask for this step

this step

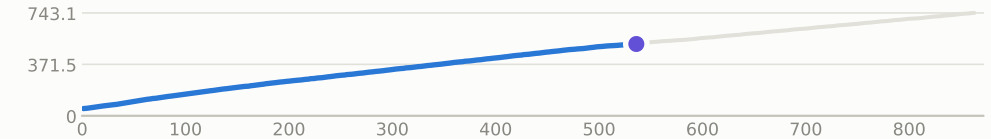
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

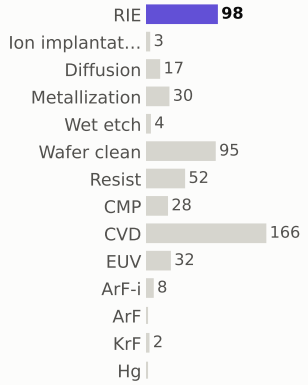
carbon so far, of the whole flow (kgCO2e)



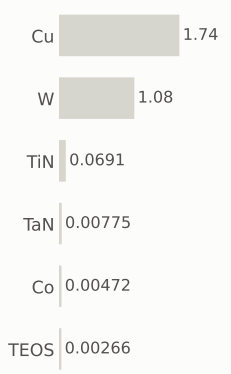
embodied carbon up to each step (kgCO2e per wafer)



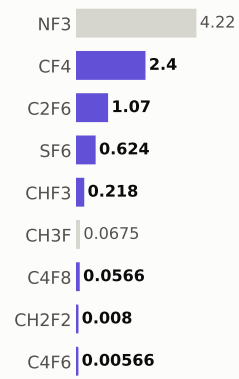
steps by process area



materials deposited (g)



gases emitted (g)



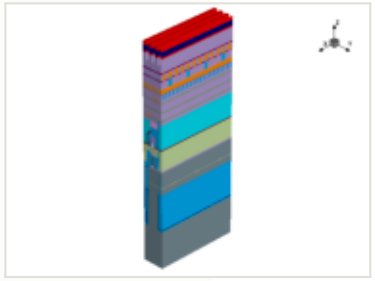
Step #5670: Wafer_clean

etch_strip · Wafer_clean

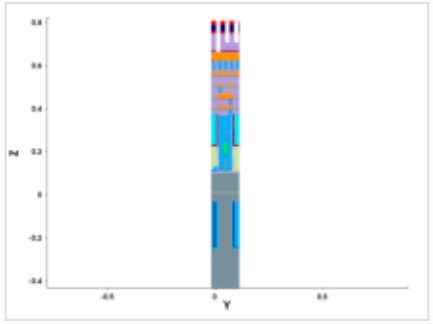
0.169	518.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

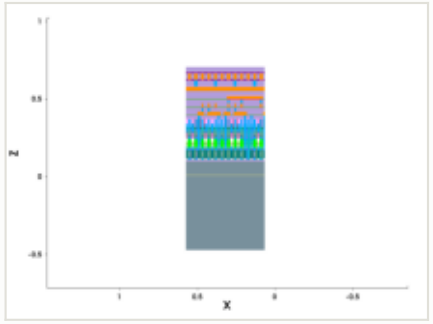
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



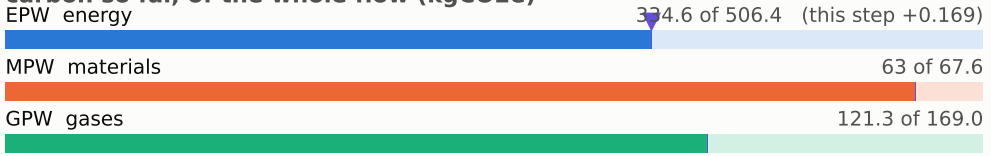
mask for this step

no mask for this step

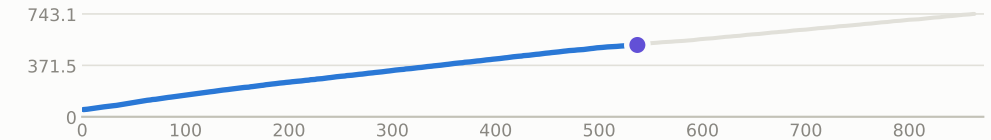
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

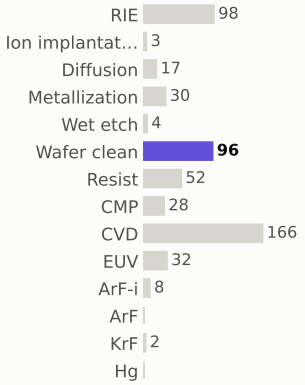
carbon so far, of the whole flow (kgCO2e)



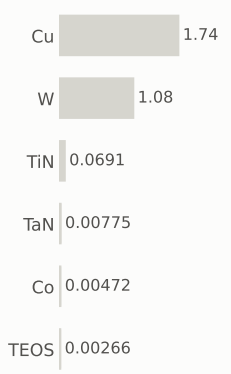
embodied carbon up to each step (kgCO2e per wafer)



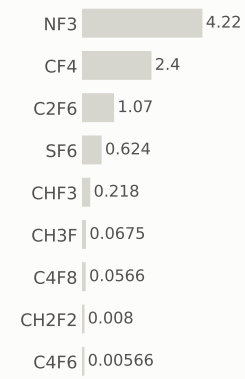
steps by process area



materials deposited (g)



gases emitted (g)



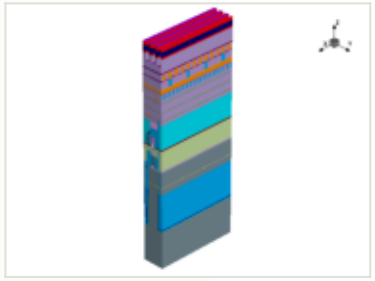
Step #5680: Metallization

depo_ellipsoidal · Metallization

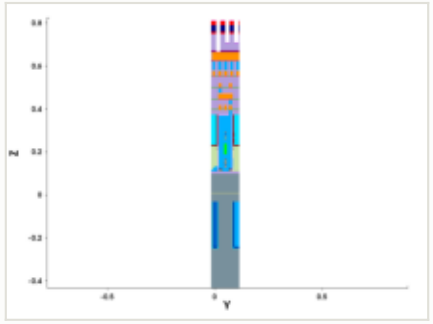
1.2	520.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

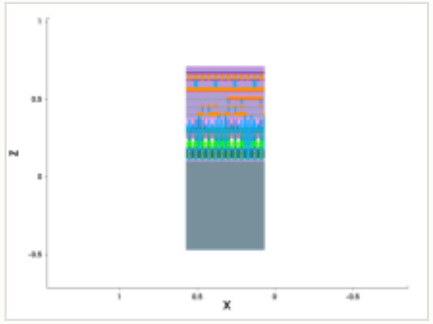
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

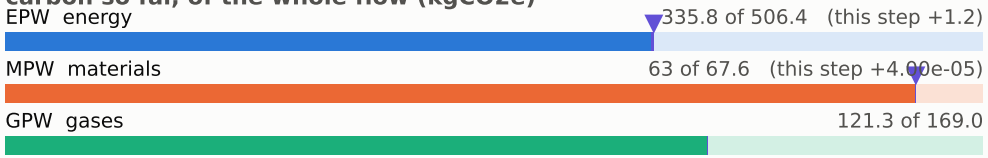
no mask for this step

this step

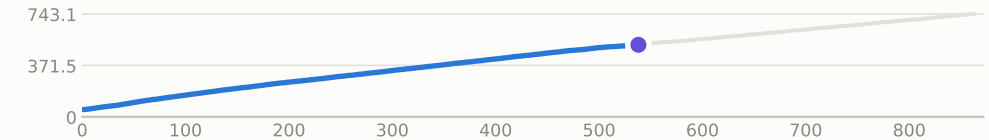
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

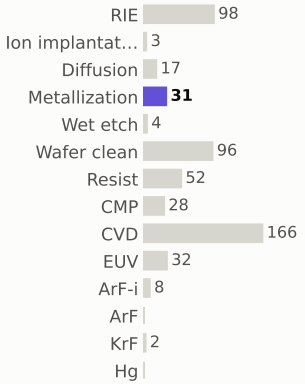
carbon so far, of the whole flow (kgCO2e)



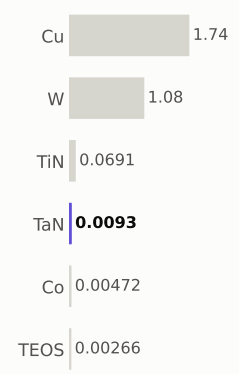
embodied carbon up to each step (kgCO2e per wafer)



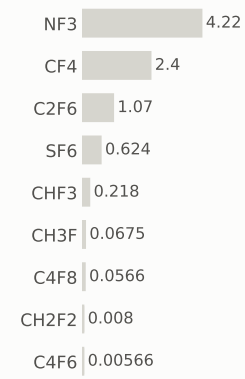
steps by process area



materials deposited (g)



gases emitted (g)



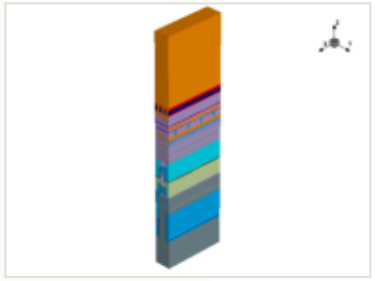
Step #5690: Metallization

depo_fill · Metallization

1.21	521.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

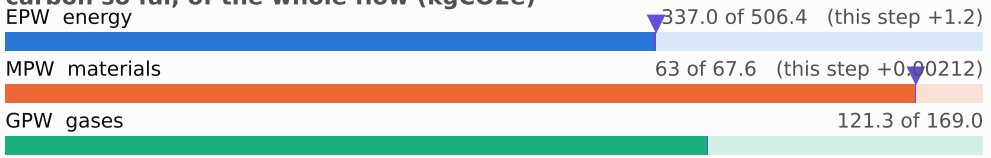


this step

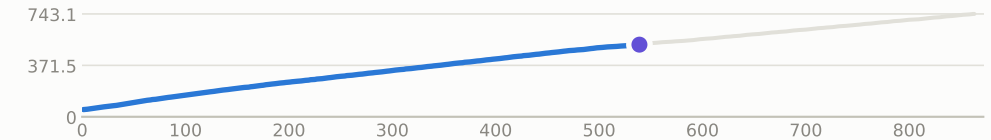
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

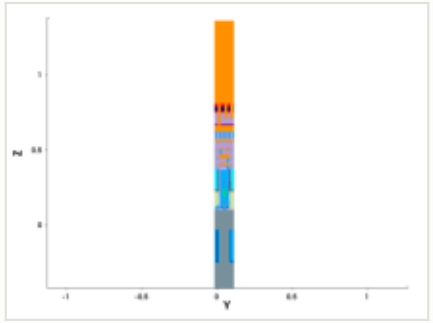
carbon so far, of the whole flow (kgCO2e)



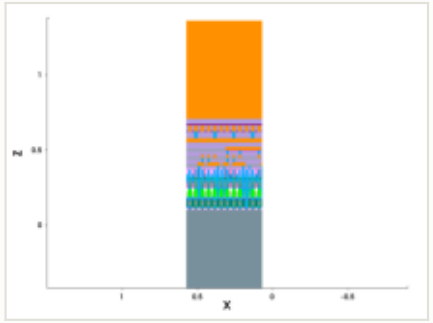
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



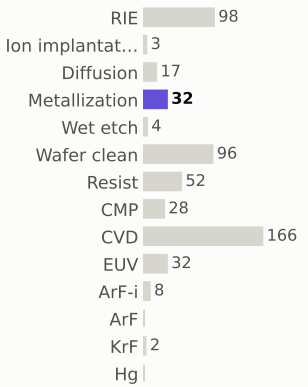
y cut at 0.049 µm



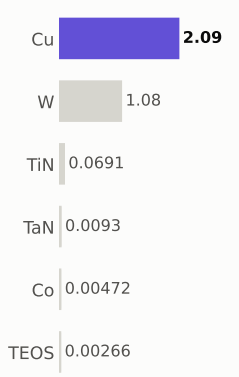
mask for this step

no mask for this step

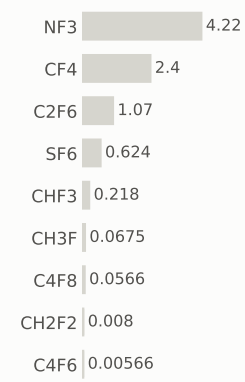
steps by process area



materials deposited (g)



gases emitted (g)



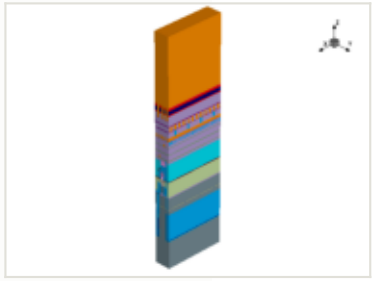
Step #5700: Diffusion

treatment · Diffusion

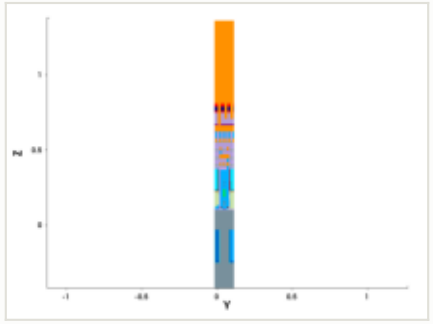
0.801	522.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

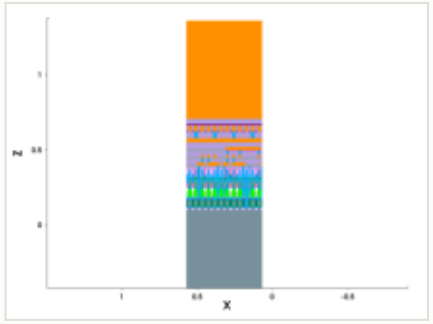
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

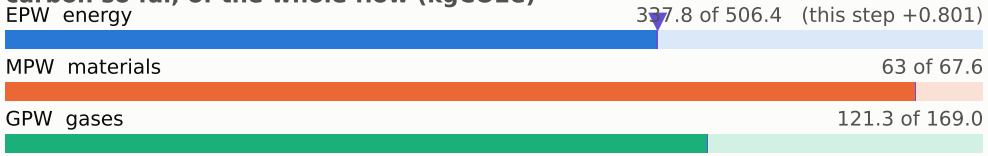
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

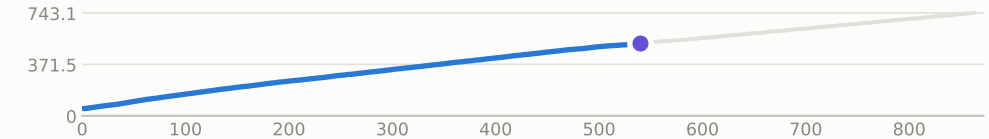
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

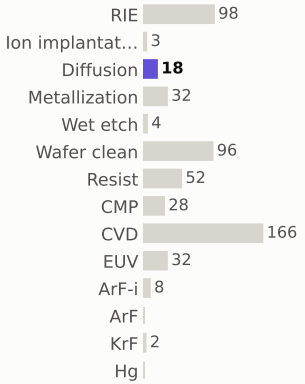
carbon so far, of the whole flow (kgCO2e)



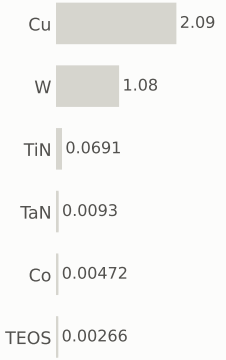
embodied carbon up to each step (kgCO2e per wafer)



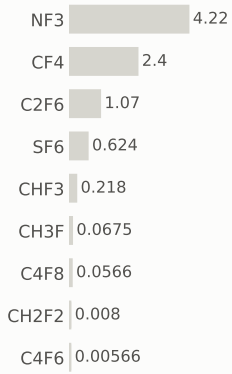
steps by process area



materials deposited (g)



gases emitted (g)



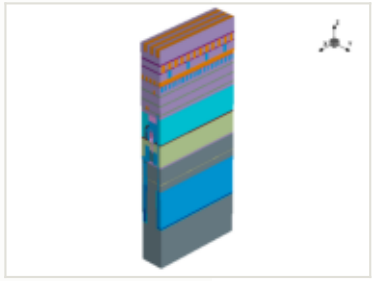
Step #5710: CMP

etch_cmp · CMP

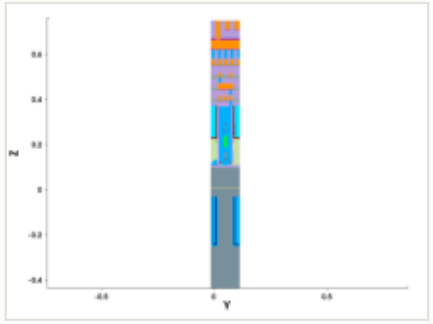
0.169	522.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

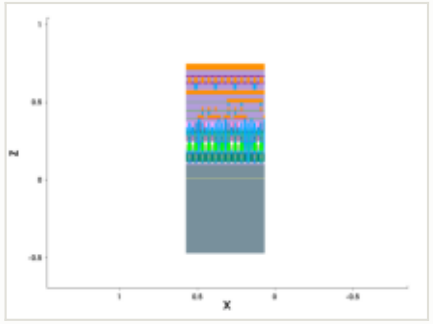
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

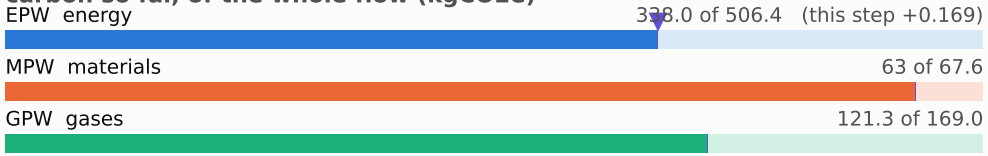
no mask for this step

this step

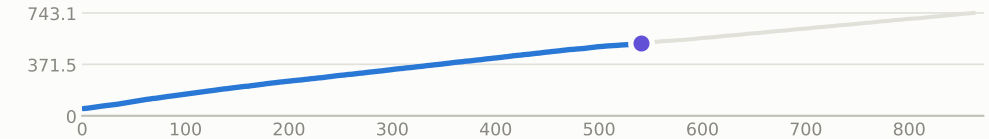
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

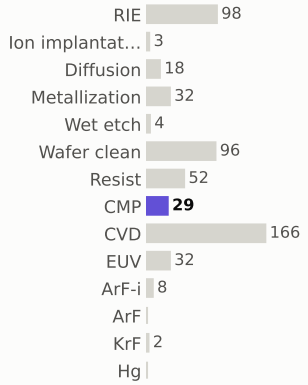
carbon so far, of the whole flow (kgCO2e)



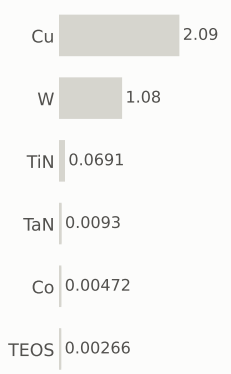
embodied carbon up to each step (kgCO2e per wafer)



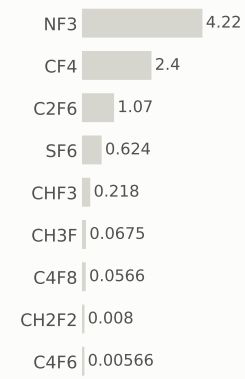
steps by process area



materials deposited (g)



gases emitted (g)



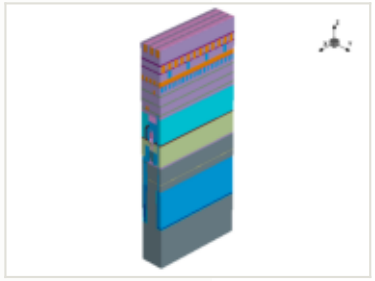
Step #5720: Metallization

depo_isotropic · Metallization

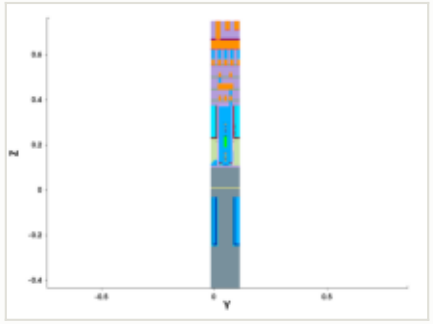
1.2	523.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

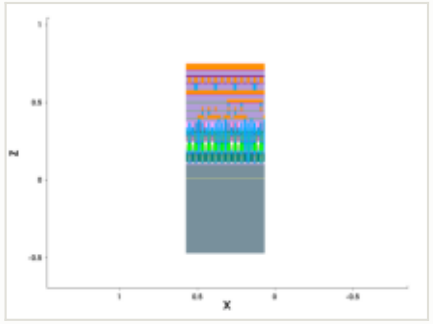
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

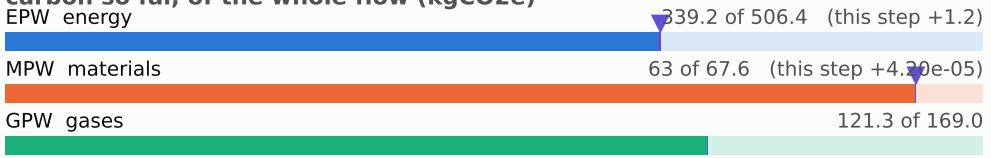
no mask for this step

this step

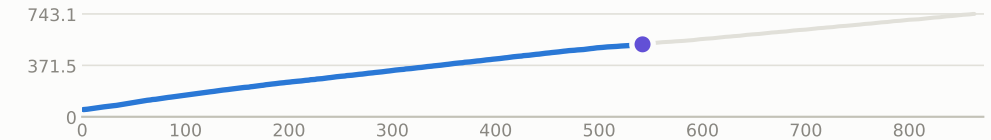
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

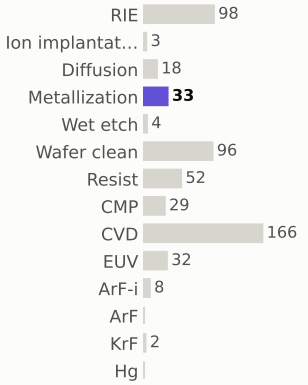
carbon so far, of the whole flow (kgCO2e)



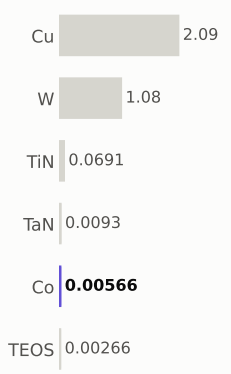
embodied carbon up to each step (kgCO2e per wafer)



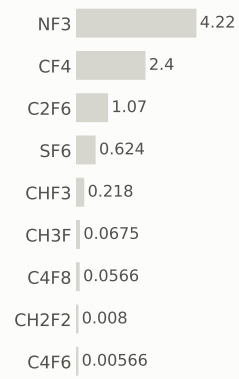
steps by process area



materials deposited (g)



gases emitted (g)



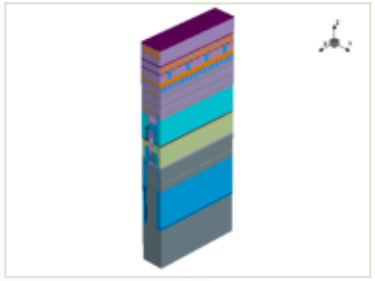
Step #5740: CVD

depo_fill · CVD

1.31	524.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

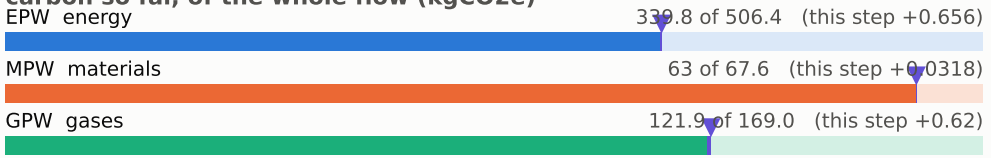


this step

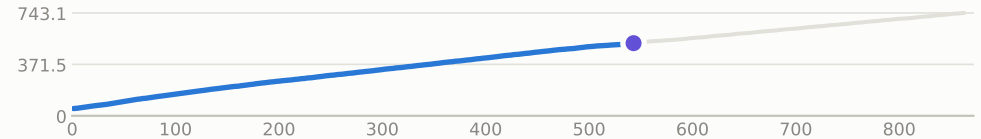
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

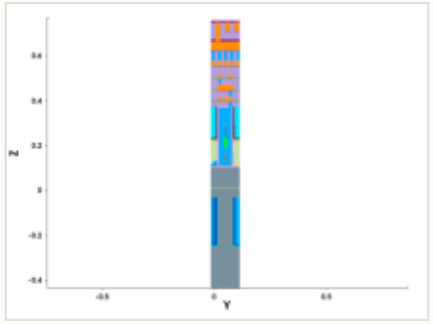
carbon so far, of the whole flow (kgCO2e)



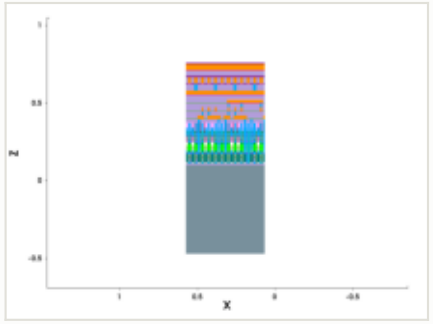
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



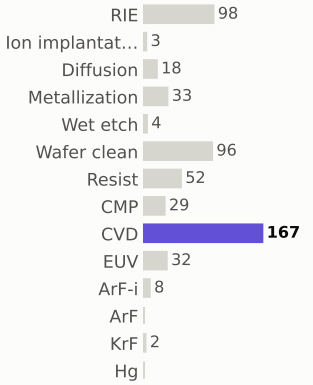
y cut at 0.049 µm



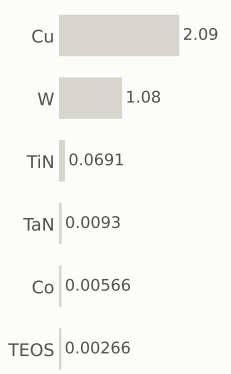
mask for this step

no mask for this step

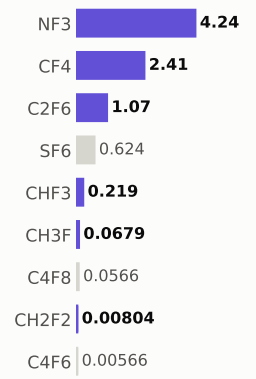
steps by process area



materials deposited (g)



gases emitted (g)



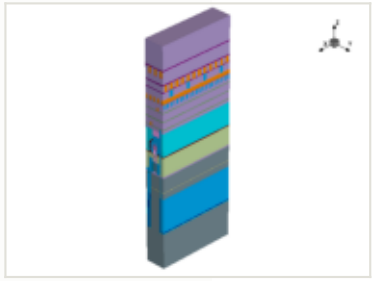
Step #5750: CVD

depo_fill · CVD

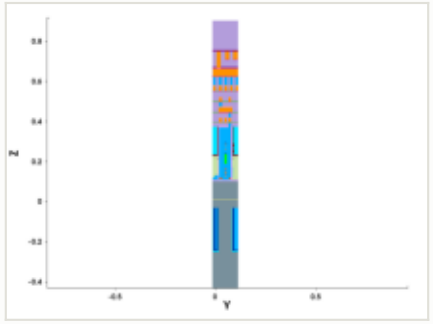
1.31	526.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

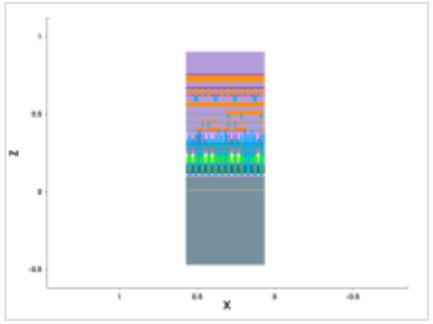
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

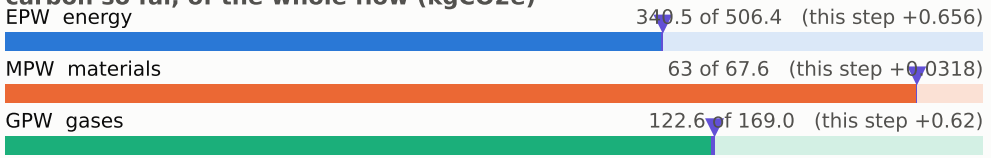
no mask for this step

this step

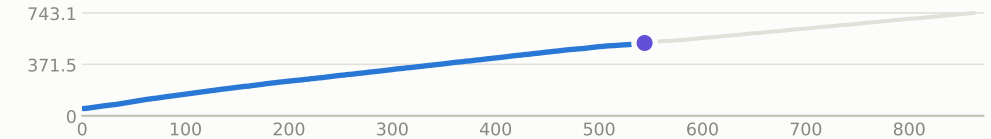
process area	CVD
material	—
thickness	0.146 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

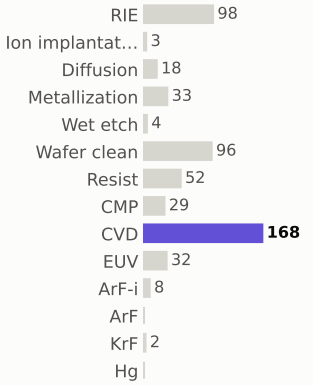
carbon so far, of the whole flow (kgCO2e)



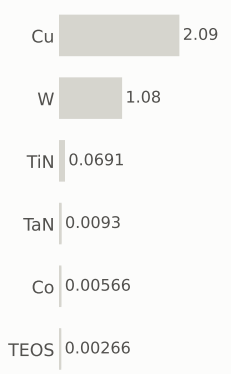
embodied carbon up to each step (kgCO2e per wafer)



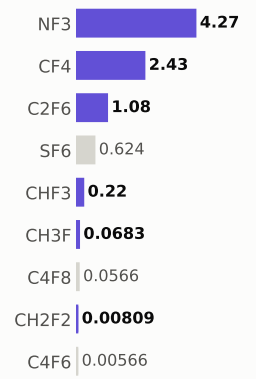
steps by process area



materials deposited (g)



gases emitted (g)



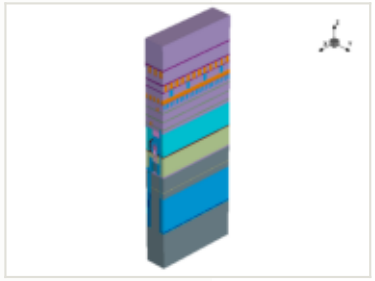
Step #5760: Diffusion

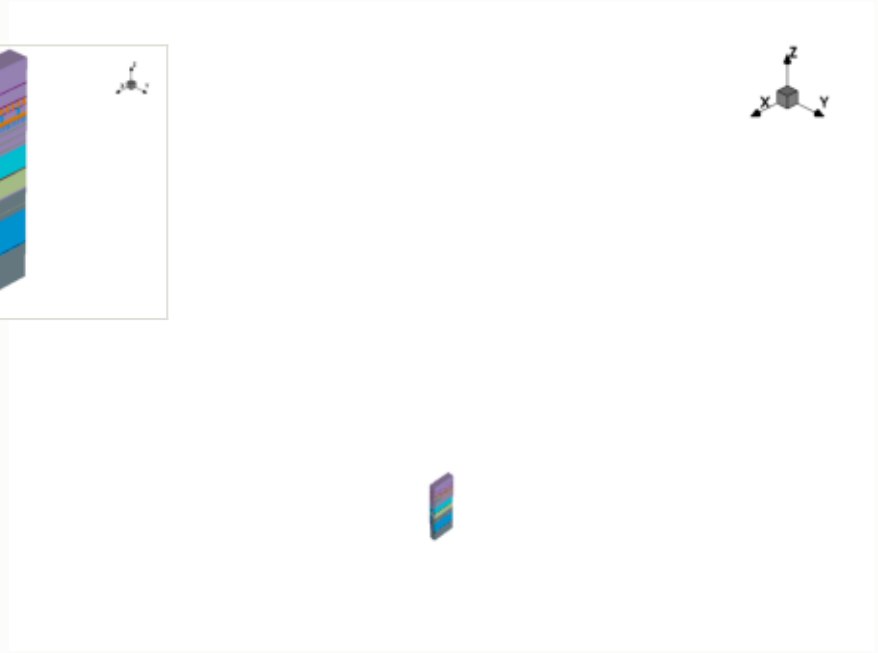
treatment · Diffusion

0.801	526.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

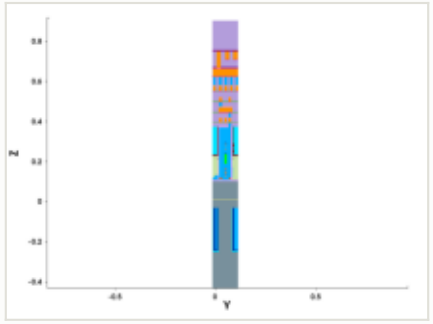
3D structure after this step

zoomed to fit

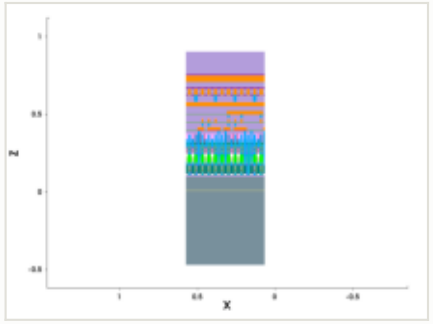




x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

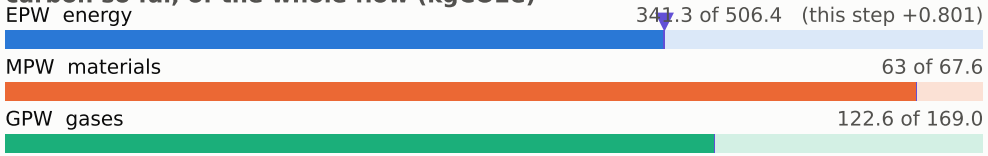
no mask for this step

this step

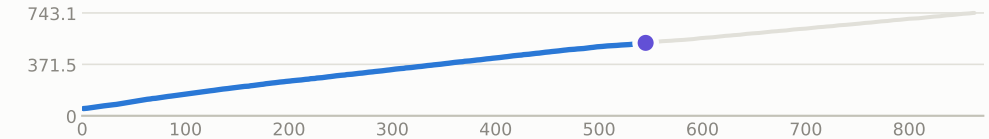
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

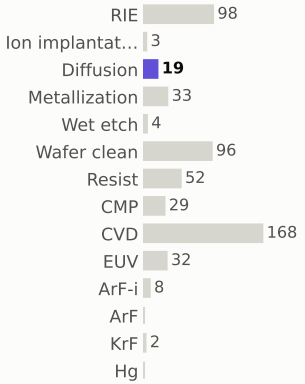
carbon so far, of the whole flow (kgCO2e)



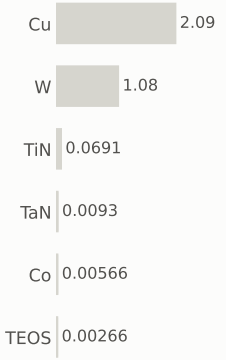
embodied carbon up to each step (kgCO2e per wafer)



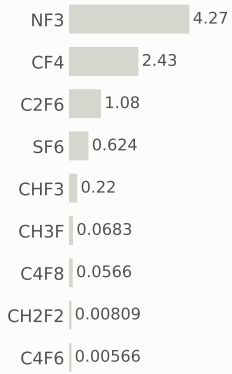
steps by process area



materials deposited (g)



gases emitted (g)



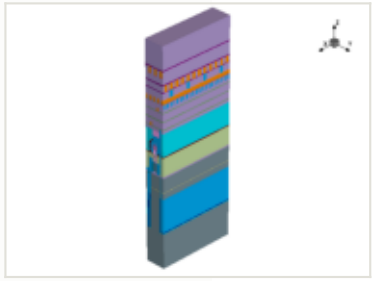
Step #5770: Wafer_clean

clean · Wafer_clean

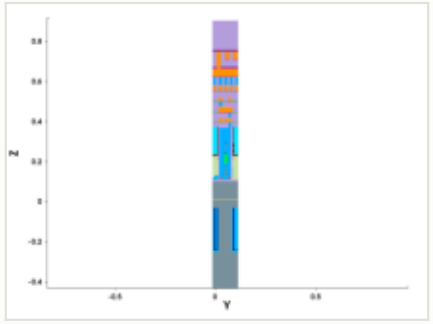
0.169	527.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

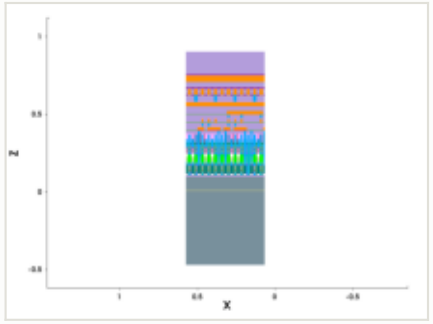
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

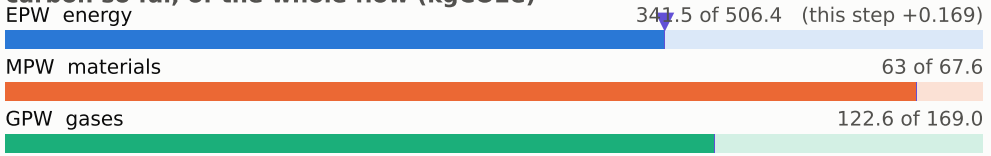
no mask for this step

this step

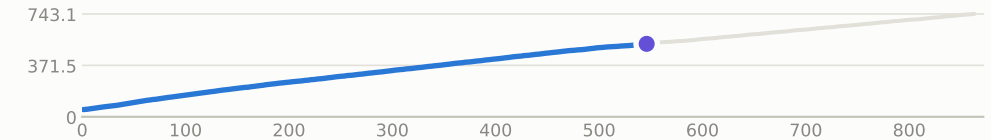
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

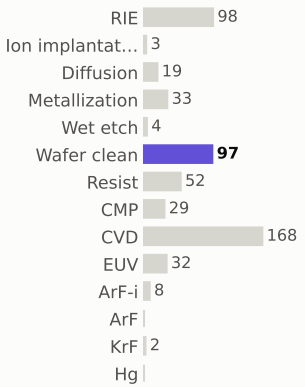
carbon so far, of the whole flow (kgCO2e)



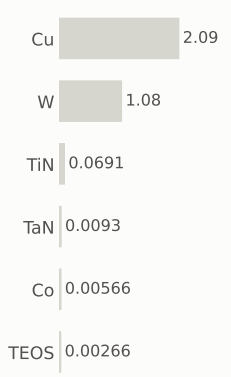
embodied carbon up to each step (kgCO2e per wafer)



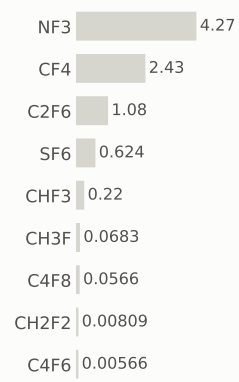
steps by process area



materials deposited (g)



gases emitted (g)



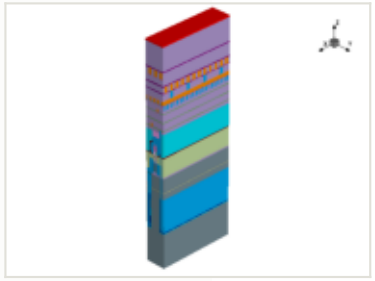
Step #5780: CVD

depo_fill · CVD

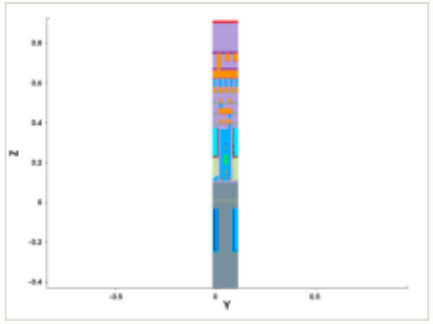
1.31	528.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

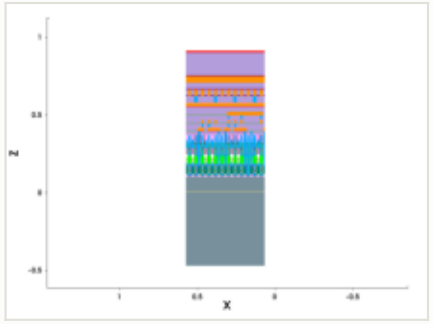
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

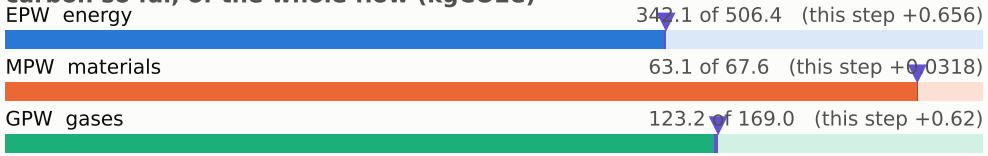
no mask for this step

this step

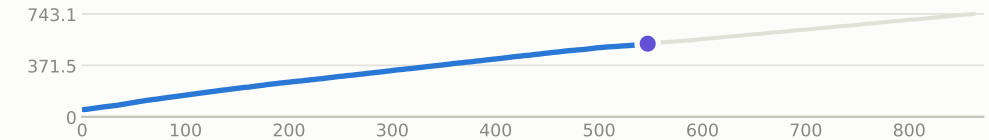
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

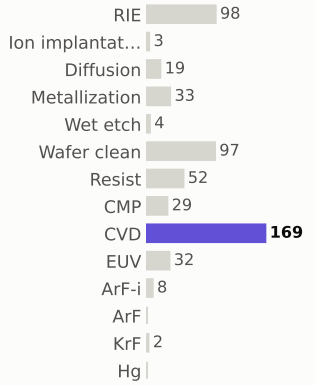
carbon so far, of the whole flow (kgCO2e)



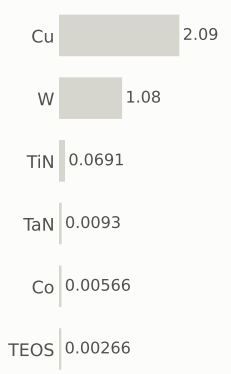
embodied carbon up to each step (kgCO2e per wafer)



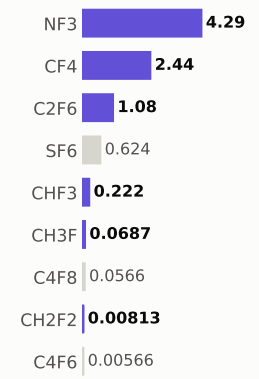
steps by process area



materials deposited (g)



gases emitted (g)



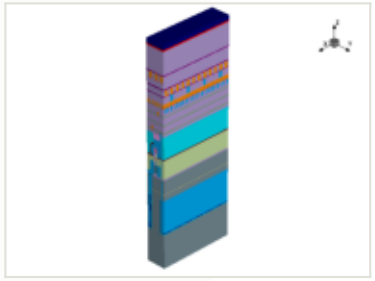
Step #5790: CVD

depo_fill · CVD

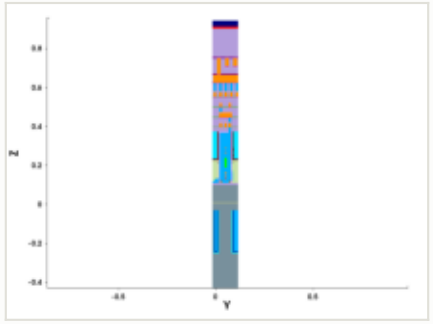
1.31	529.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

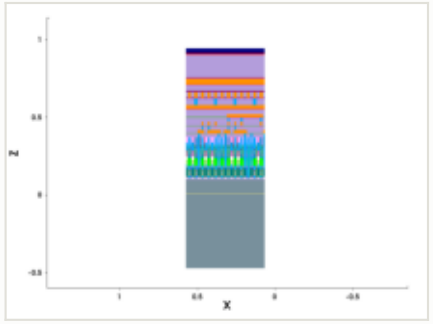
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

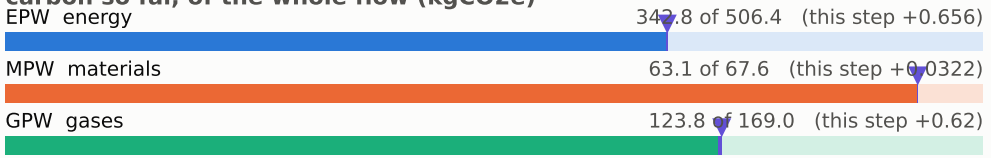
no mask for this step

this step

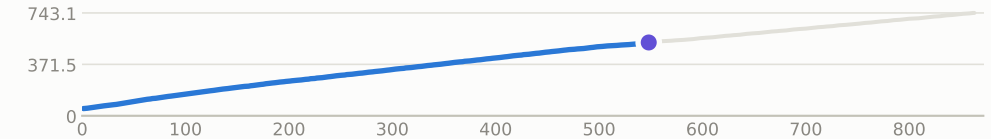
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

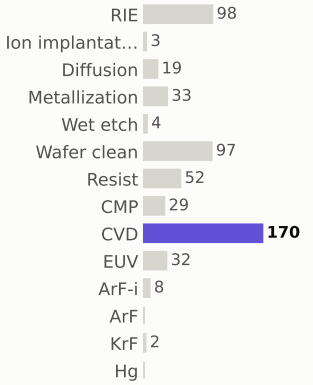
carbon so far, of the whole flow (kgCO2e)



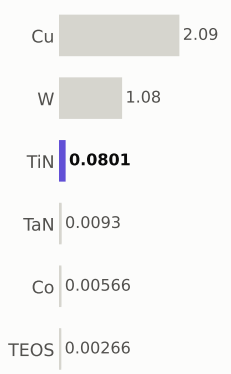
embodied carbon up to each step (kgCO2e per wafer)



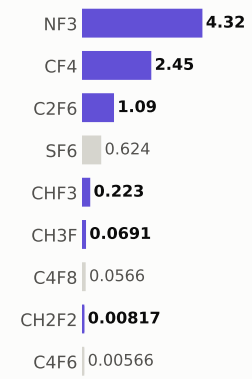
steps by process area



materials deposited (g)



gases emitted (g)



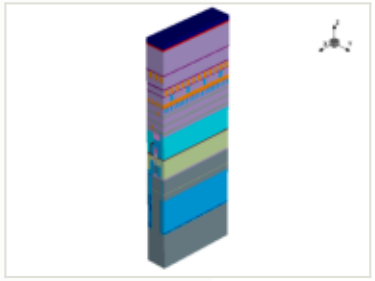
Step #5800: Wafer_clean

clean · Wafer_clean

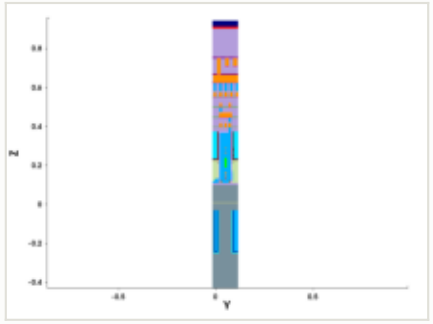
0.169	529.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

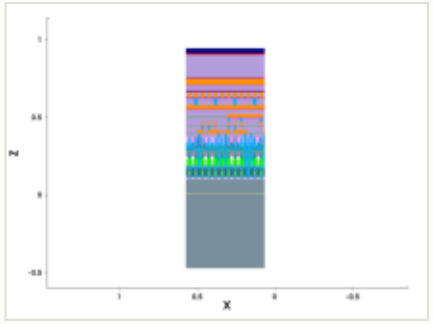
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

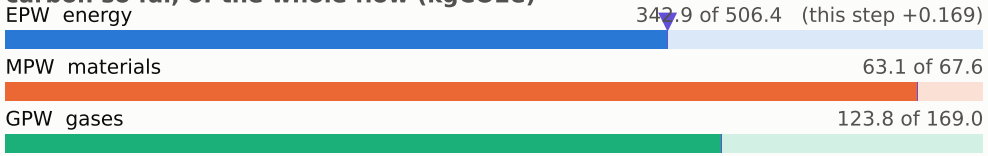
no mask for this step

this step

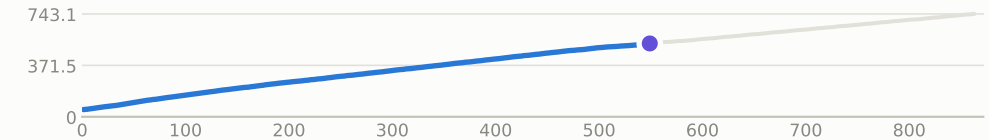
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

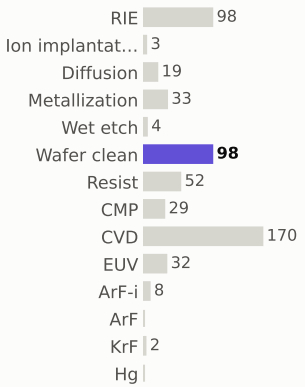
carbon so far, of the whole flow (kgCO2e)



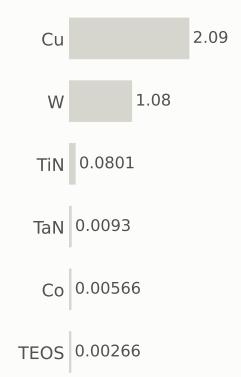
embodied carbon up to each step (kgCO2e per wafer)



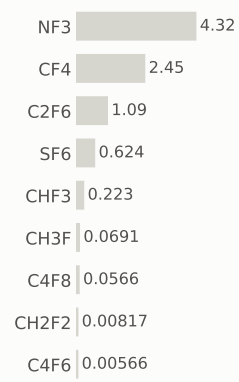
steps by process area



materials deposited (g)



gases emitted (g)



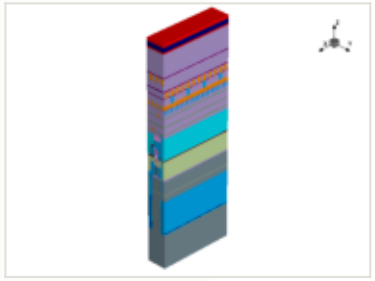
Step #5810: CVD

depo_fill · CVD

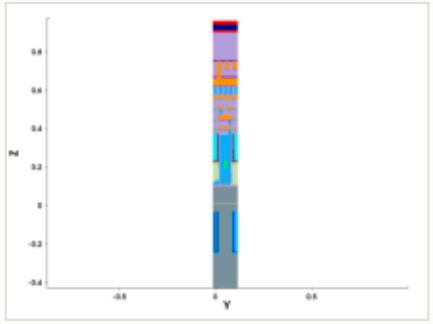
1.31	531.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

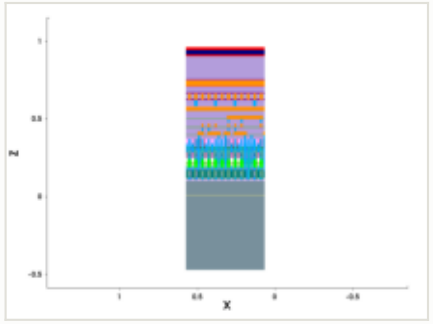
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

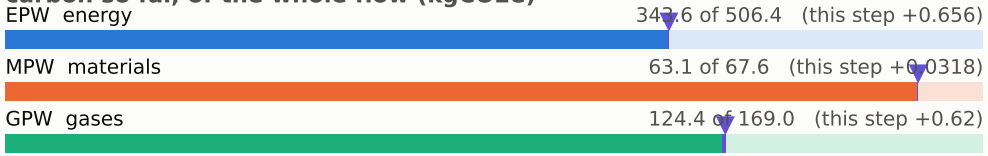
no mask for this step

this step

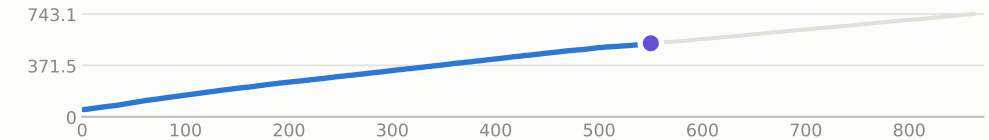
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

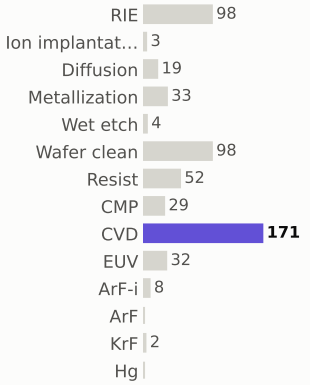
carbon so far, of the whole flow (kgCO2e)



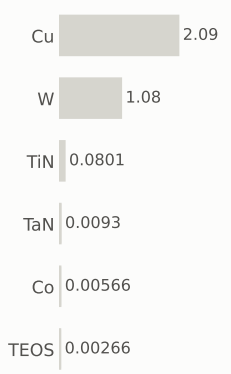
embodied carbon up to each step (kgCO2e per wafer)



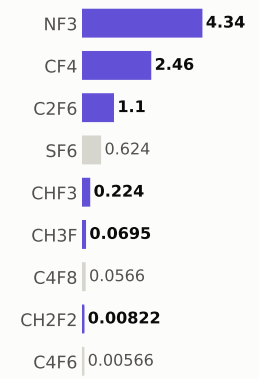
steps by process area



materials deposited (g)



gases emitted (g)



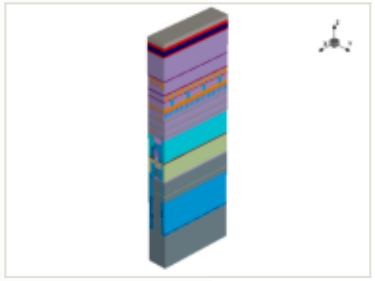
Step #5820: CVD

depo_fill · CVD

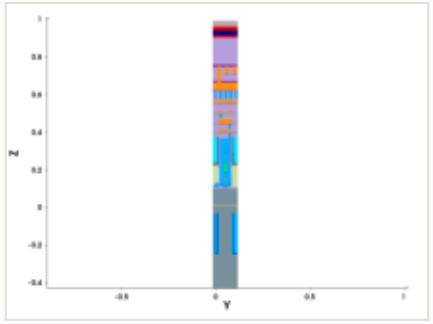
1.31	532.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

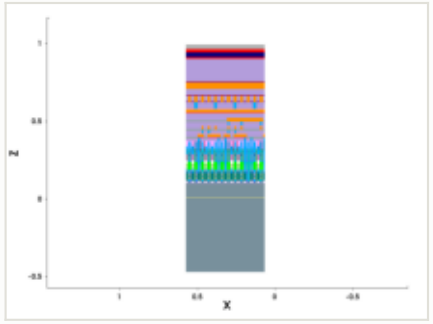
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

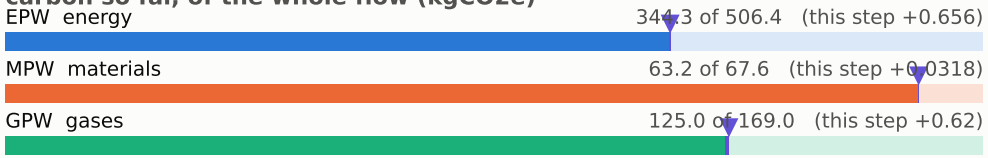
no mask for this step

this step

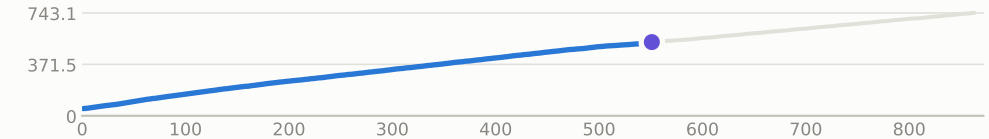
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

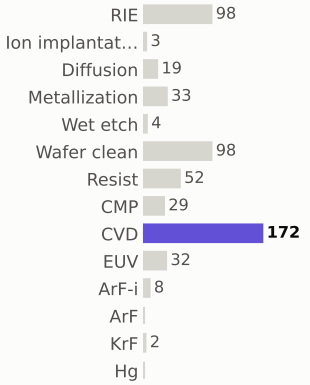
carbon so far, of the whole flow (kgCO2e)



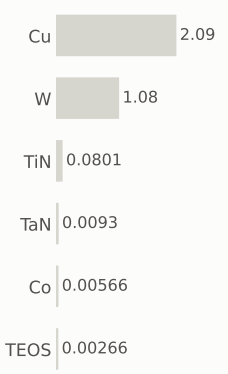
embodied carbon up to each step (kgCO2e per wafer)



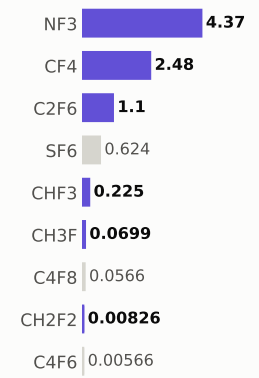
steps by process area



materials deposited (g)



gases emitted (g)



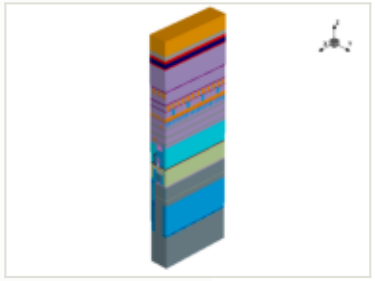
Step #5830: Resist

depo_fill · Resist

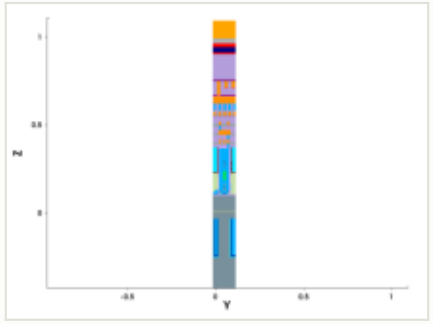
0.169	532.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

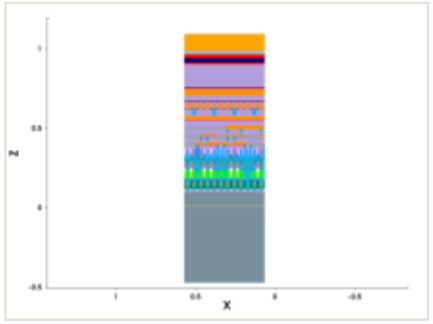
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



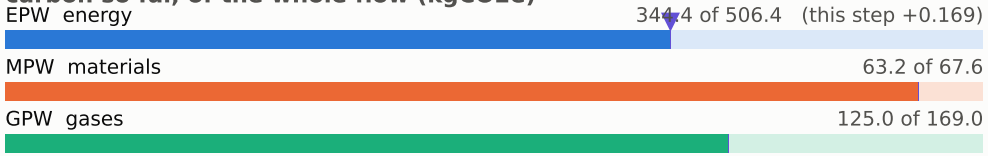
mask for this step

no mask for this step

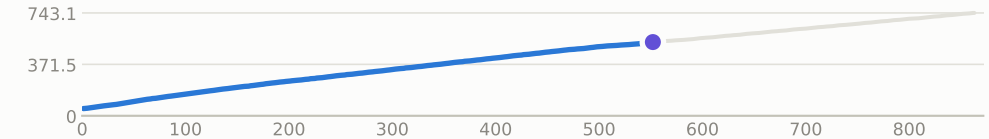
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

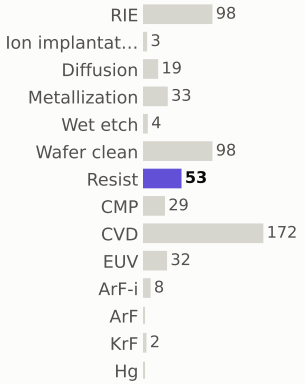
carbon so far, of the whole flow (kgCO2e)



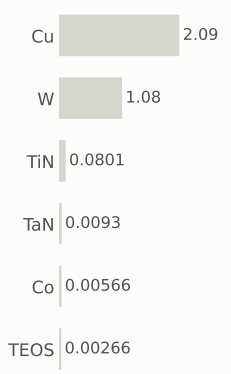
embodied carbon up to each step (kgCO2e per wafer)



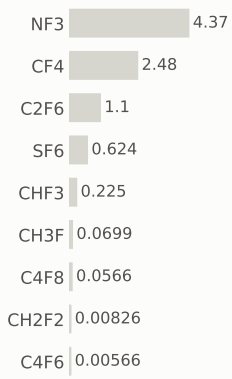
steps by process area



materials deposited (g)



gases emitted (g)



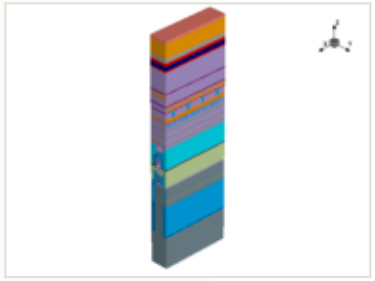
Step #5840: CVD

depo_fill · CVD

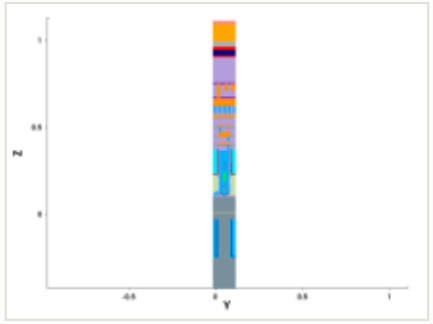
1.31	533.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

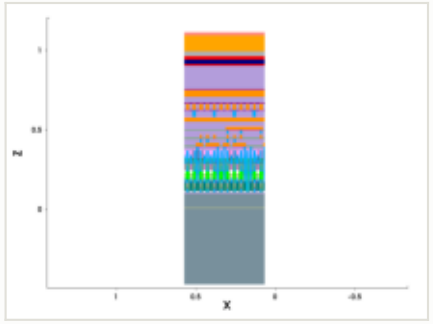
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

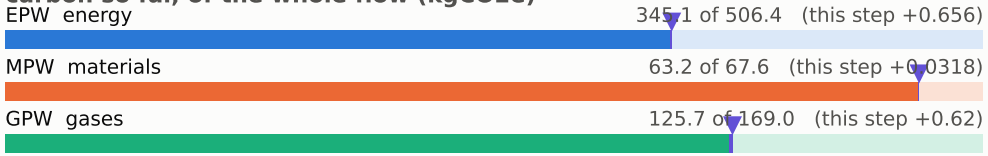
no mask for this step

this step

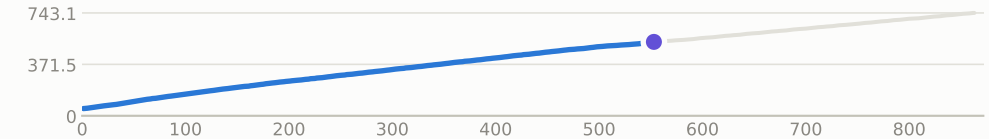
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

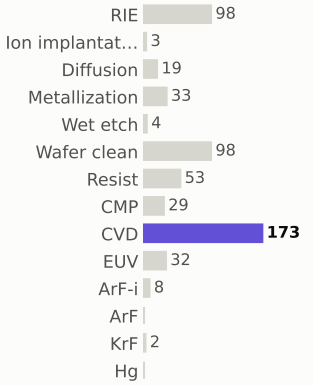
carbon so far, of the whole flow (kgCO2e)



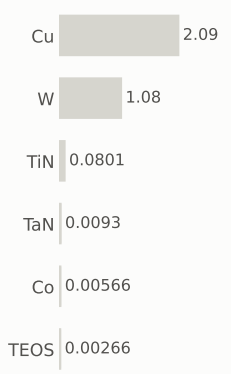
embodied carbon up to each step (kgCO2e per wafer)



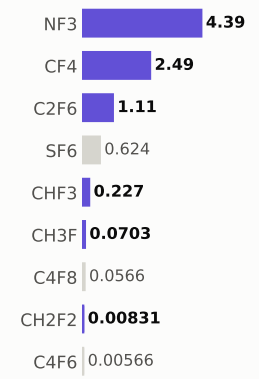
steps by process area



materials deposited (g)



gases emitted (g)



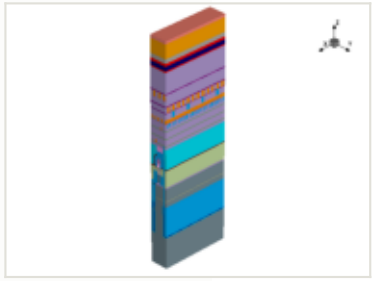
Step #5850: Wafer_clean

clean · Wafer_clean

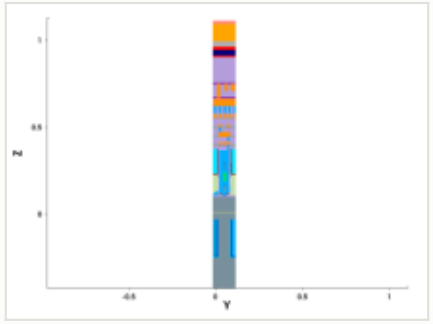
0.169	534.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

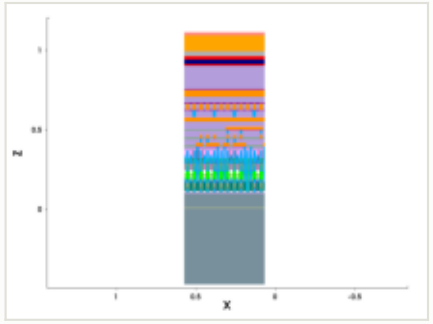
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

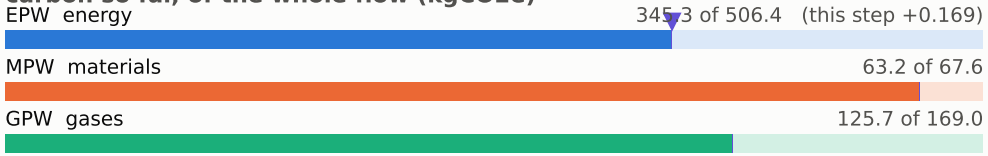
no mask for this step

this step

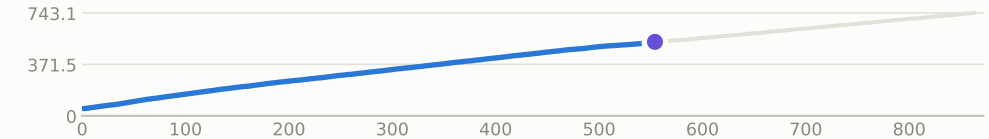
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

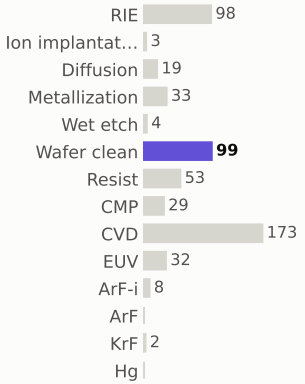
carbon so far, of the whole flow (kgCO2e)



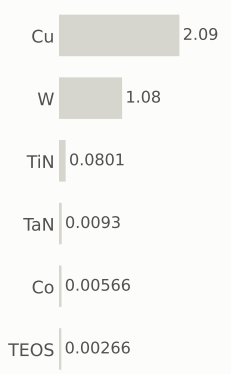
embodied carbon up to each step (kgCO2e per wafer)



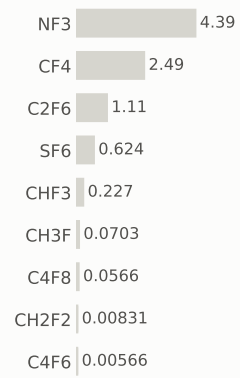
steps by process area



materials deposited (g)



gases emitted (g)



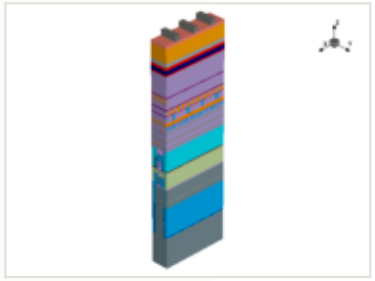
Step #5860: ArF-i

litho_patterning · ArF-i

0.217	534.3	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

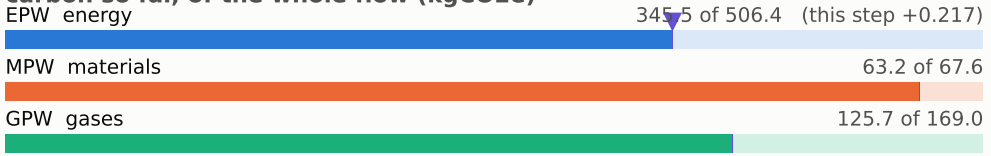


this step

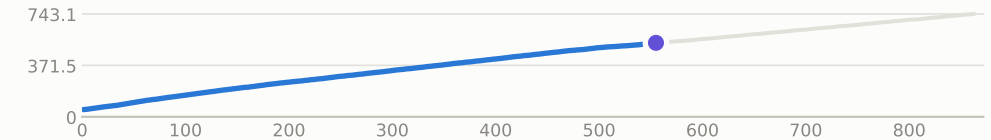
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

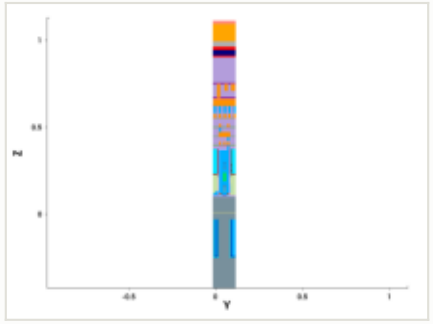
carbon so far, of the whole flow (kgCO2e)



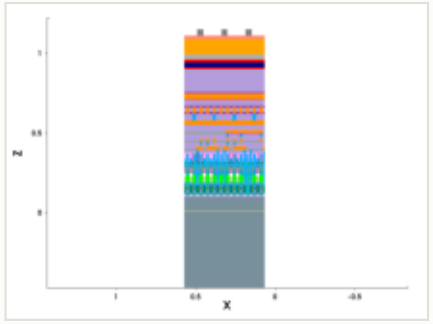
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

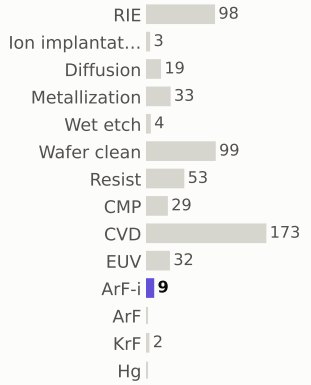


y cut at 0.049 µm

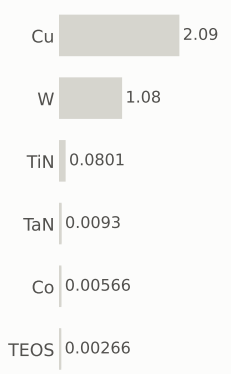


mask: layer 10501:0, positive (GDSII)

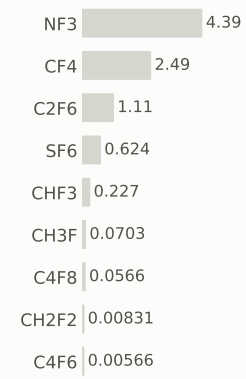
steps by process area



materials deposited (g)



gases emitted (g)



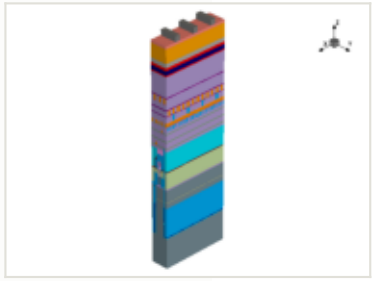
Step #5870: Ion_implantation

treatment · Ion_implantation

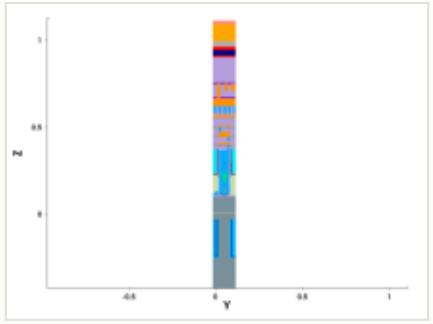
0.801	535.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

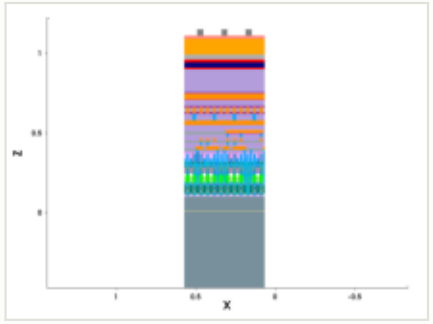
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

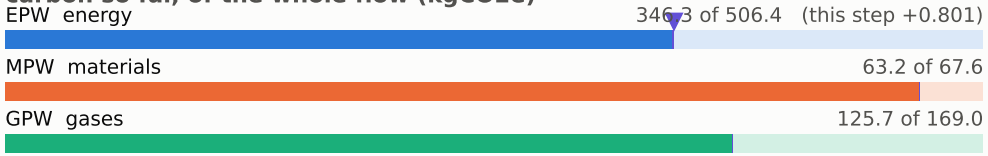
no mask for this step

this step

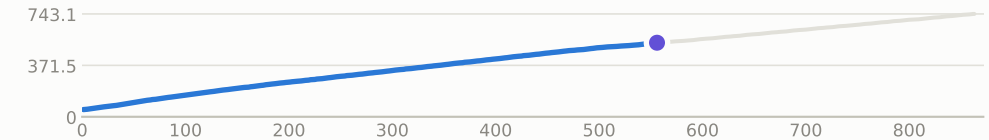
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

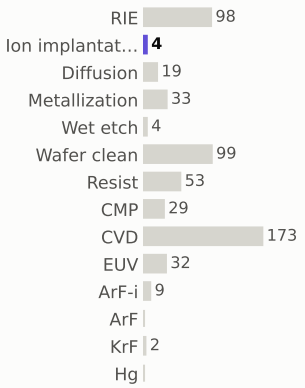
carbon so far, of the whole flow (kgCO2e)



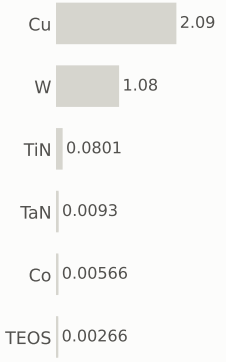
embodied carbon up to each step (kgCO2e per wafer)



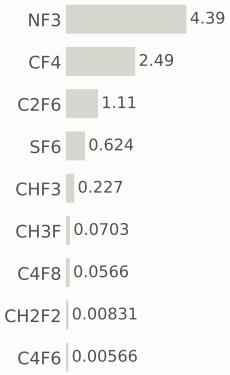
steps by process area



materials deposited (g)



gases emitted (g)



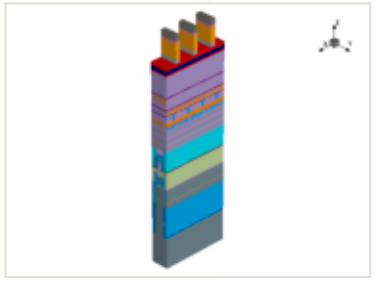
Step #5880: RIE

etch_anisotropic · RIE

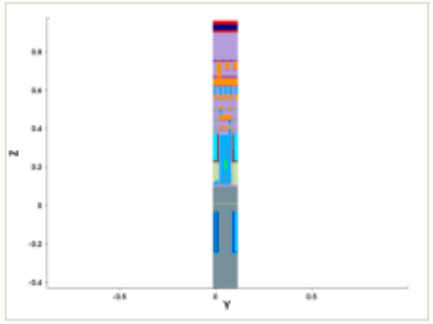
1.04	536.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

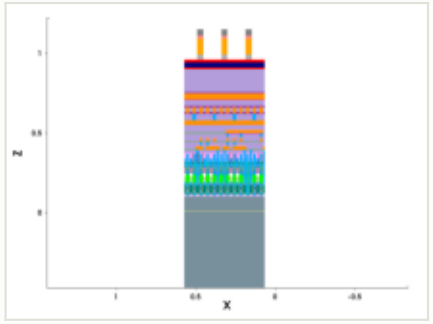
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

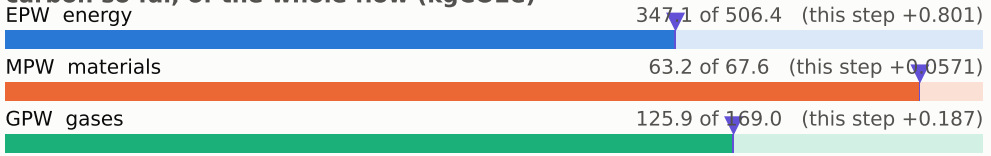
no mask for this step

this step

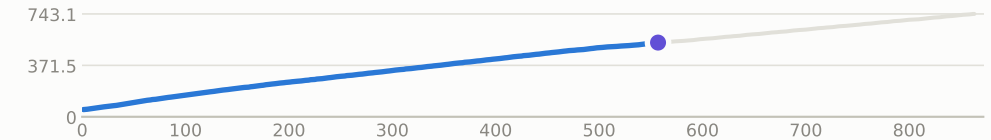
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

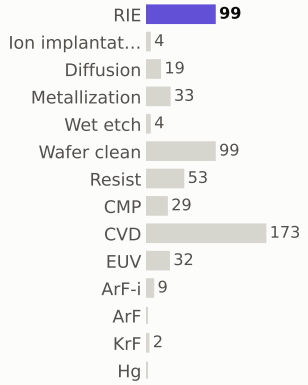
carbon so far, of the whole flow (kgCO2e)



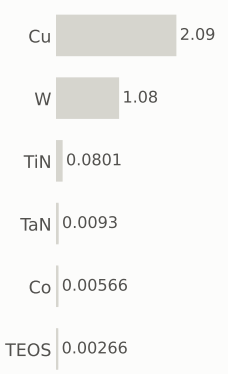
embodied carbon up to each step (kgCO2e per wafer)



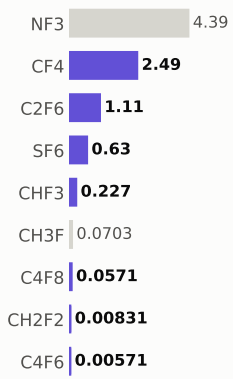
steps by process area



materials deposited (g)



gases emitted (g)



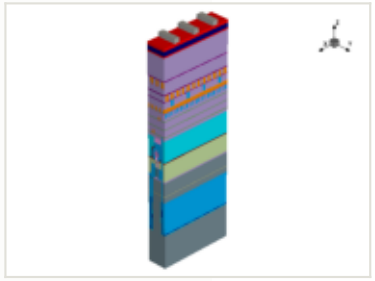
Step #5890: Wafer_clean

etch_strip · Wafer_clean

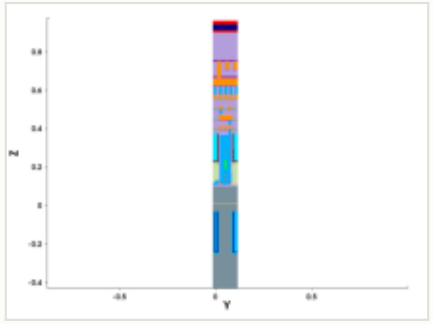
0.169	536.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

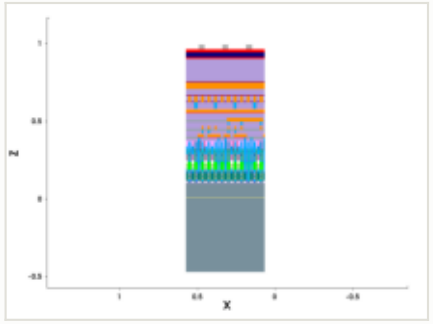
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



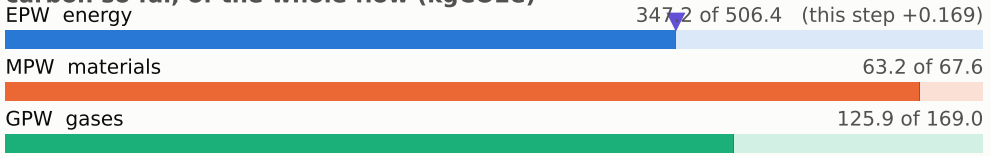
mask for this step

no mask for this step

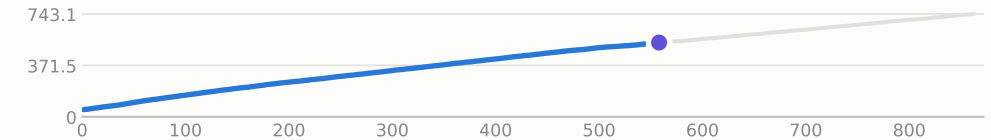
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

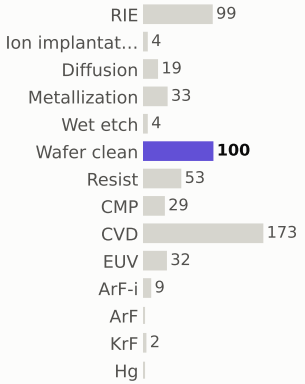
carbon so far, of the whole flow (kgCO2e)



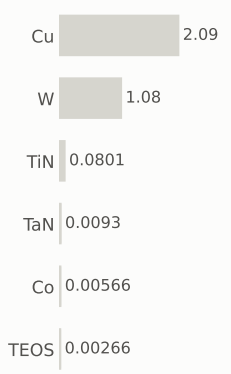
embodied carbon up to each step (kgCO2e per wafer)



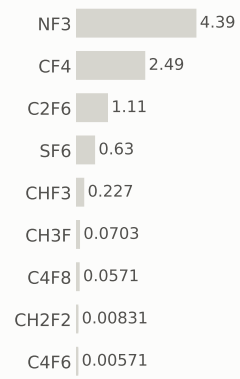
steps by process area



materials deposited (g)



gases emitted (g)



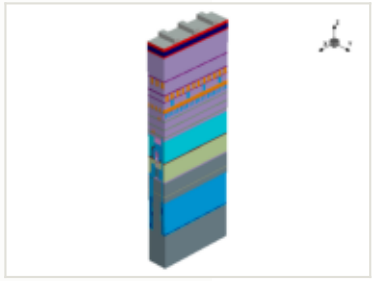
Step #5900: CVD

depo_isotropic · CVD

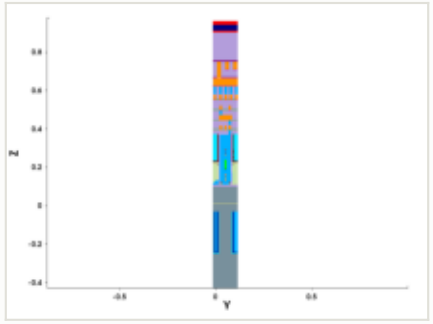
1.31	537.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

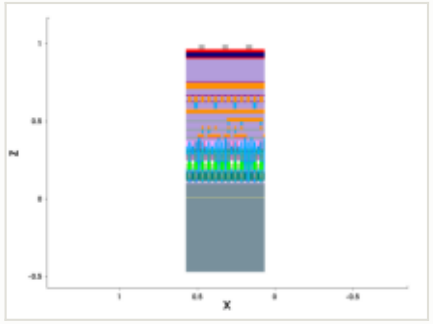
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

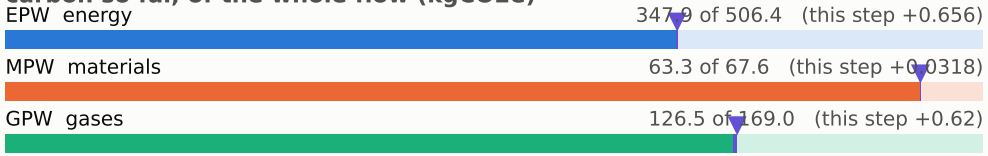
no mask for this step

this step

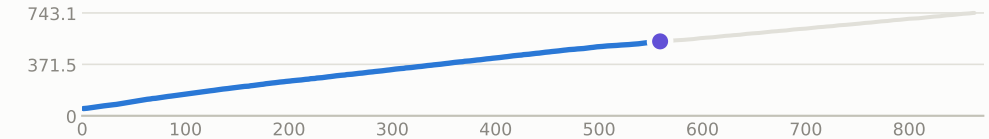
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

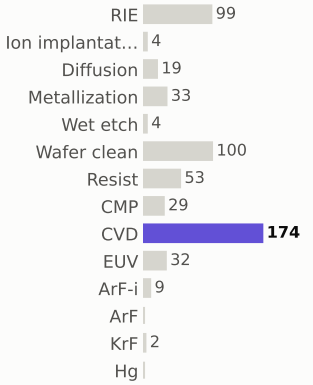
carbon so far, of the whole flow (kgCO2e)



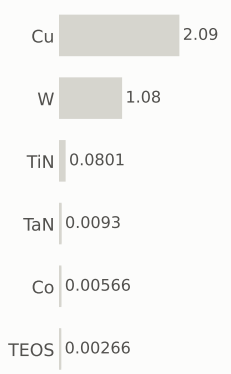
embodied carbon up to each step (kgCO2e per wafer)



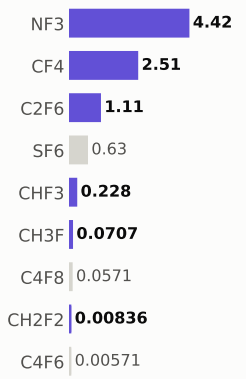
steps by process area



materials deposited (g)



gases emitted (g)



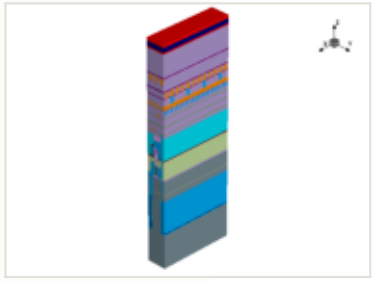
Step #5910: RIE

etch_anisotropic · RIE

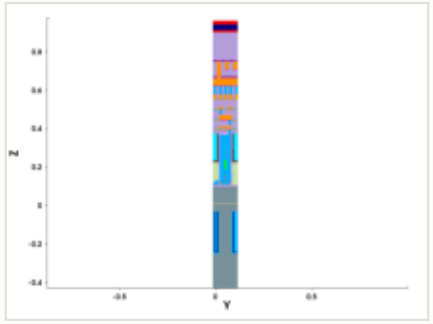
1.04	538.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

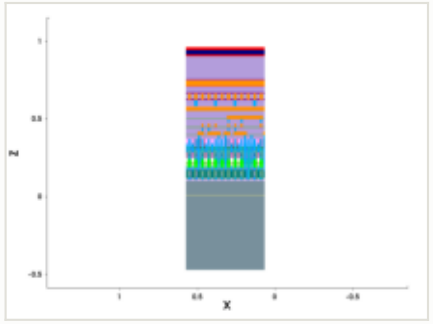
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

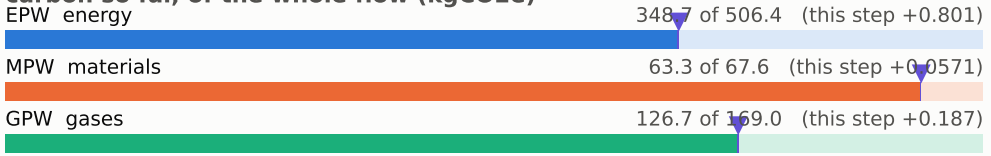
no mask for this step

this step

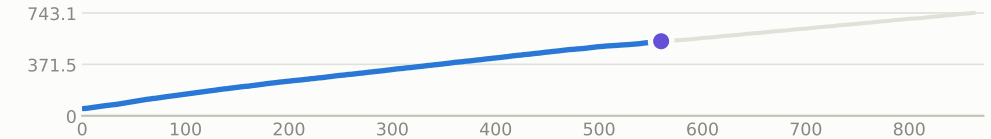
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

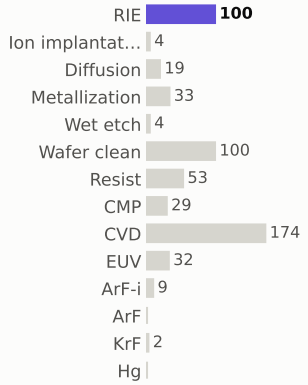
carbon so far, of the whole flow (kgCO2e)



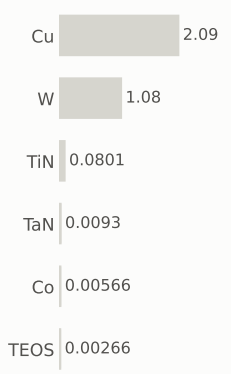
embodied carbon up to each step (kgCO2e per wafer)



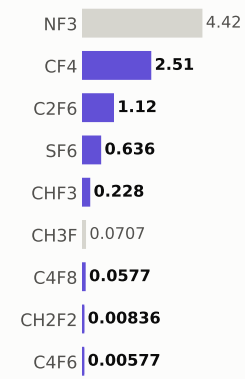
steps by process area



materials deposited (g)



gases emitted (g)



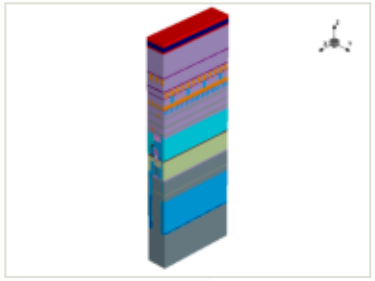
Step #5920: Wafer_clean

etch_strip · Wafer_clean

0.169	538.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

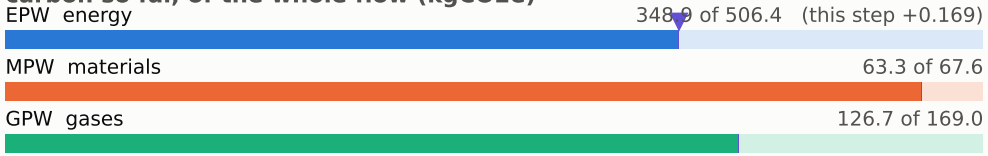


this step

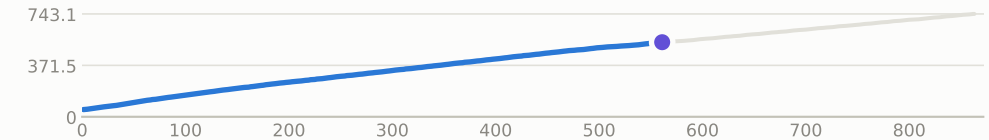
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

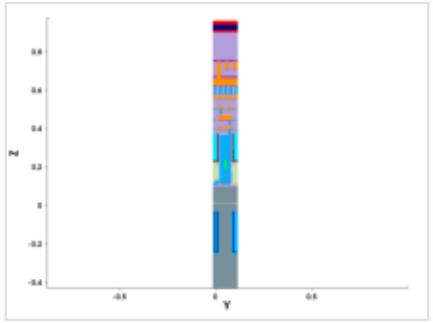
carbon so far, of the whole flow (kgCO2e)



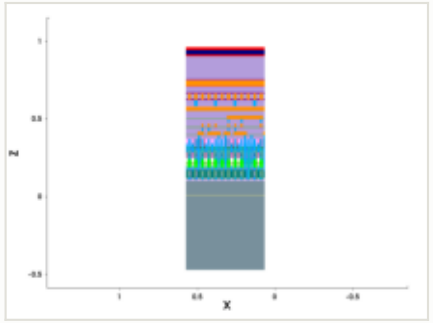
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



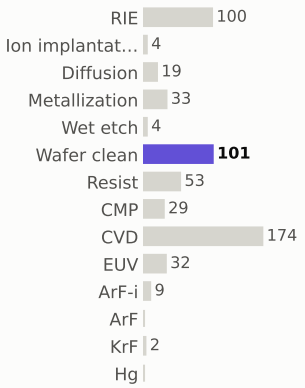
y cut at 0.049 µm



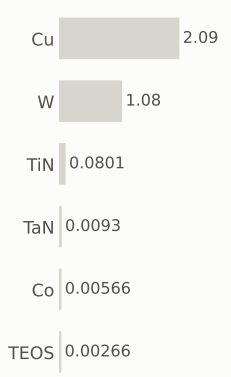
mask for this step

no mask for this step

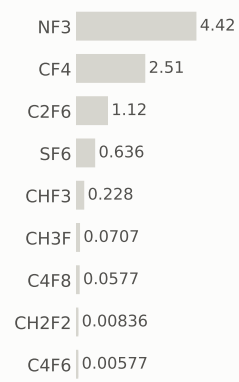
steps by process area



materials deposited (g)



gases emitted (g)



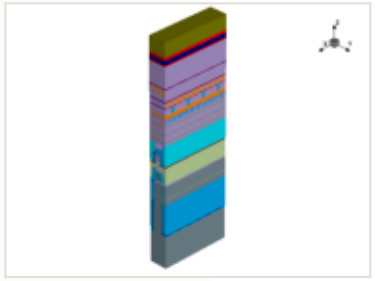
Step #5930: Resist

depo_fill · Resist

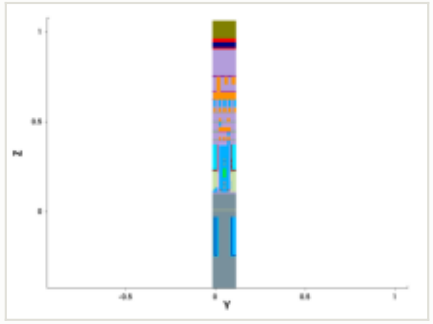
0.169	539.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

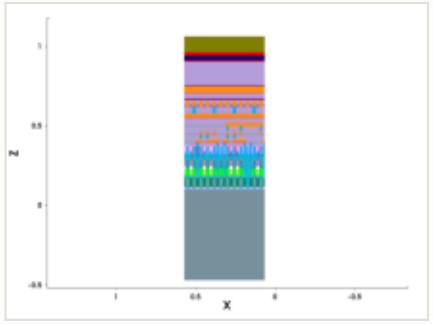
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



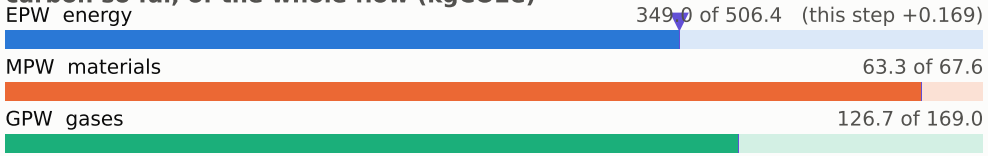
mask for this step

no mask for this step

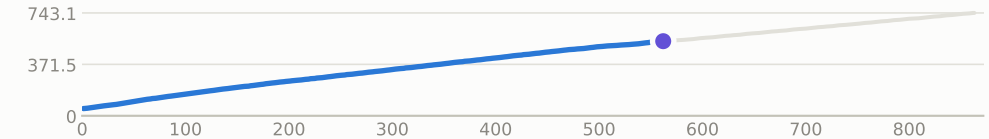
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

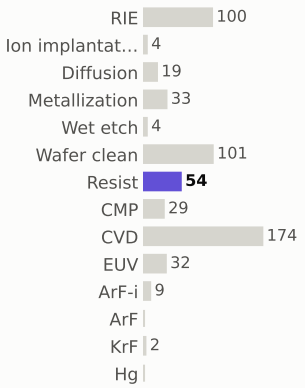
carbon so far, of the whole flow (kgCO2e)



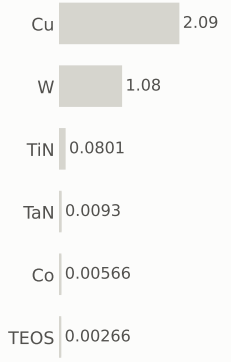
embodied carbon up to each step (kgCO2e per wafer)



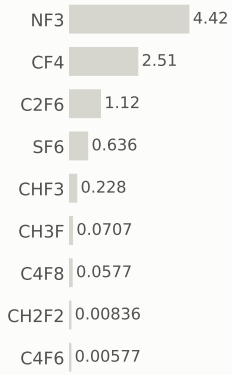
steps by process area



materials deposited (g)



gases emitted (g)



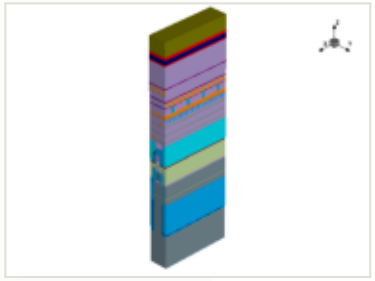
Step #5940: Wafer_clean

clean · Wafer_clean

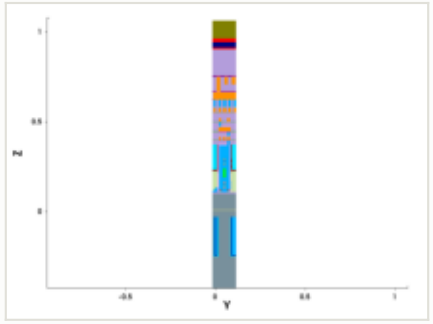
0.169	539.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

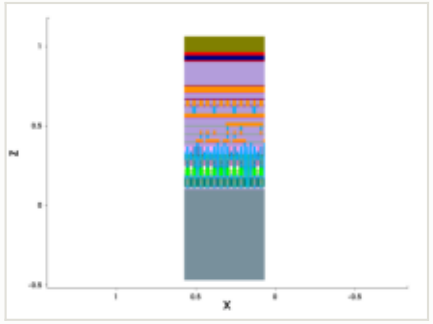
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

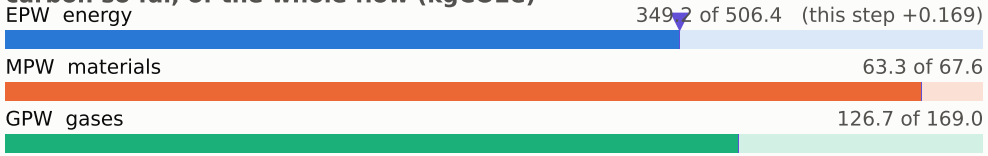
no mask for this step

this step

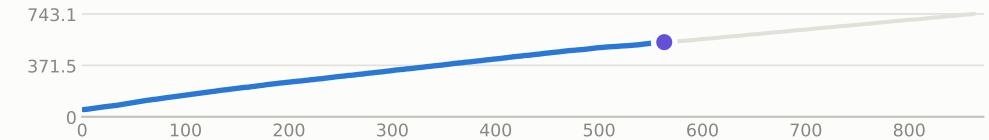
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

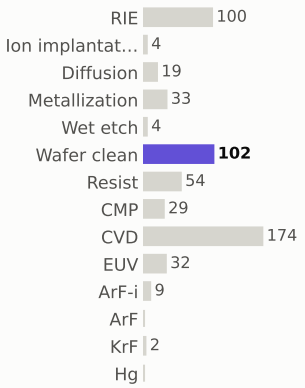
carbon so far, of the whole flow (kgCO2e)



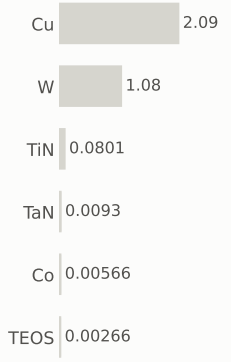
embodied carbon up to each step (kgCO2e per wafer)



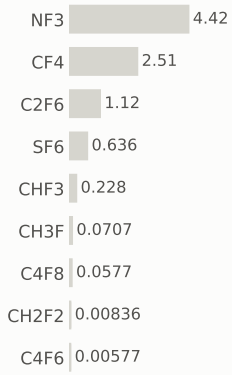
steps by process area



materials deposited (g)



gases emitted (g)



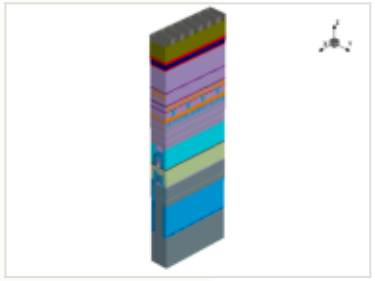
Step #5950: ArF-i

litho_patterning · ArF-i

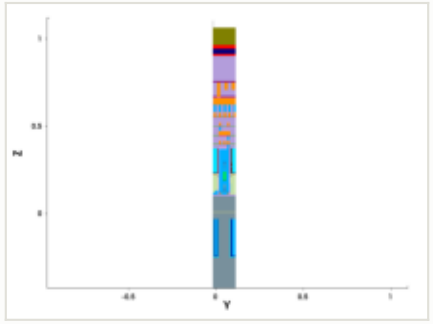
0.217	539.4	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

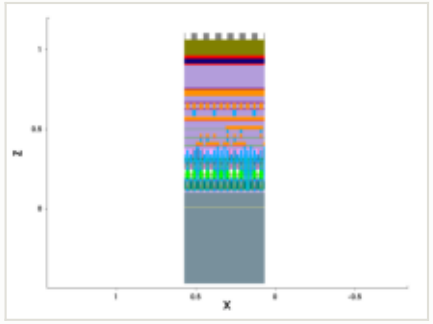
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



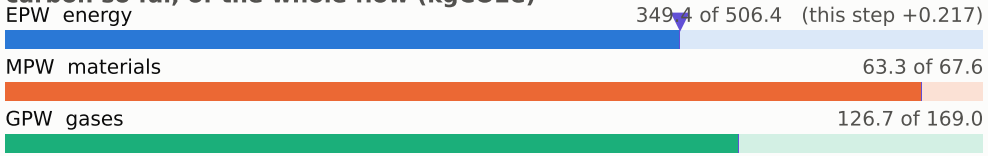
mask: layer 50:0 (GDSII)

this step

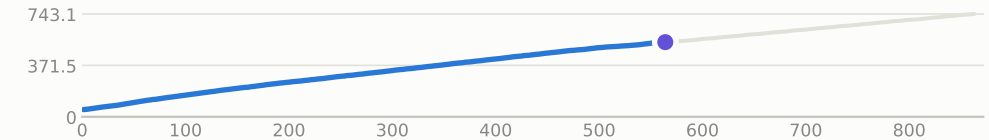
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

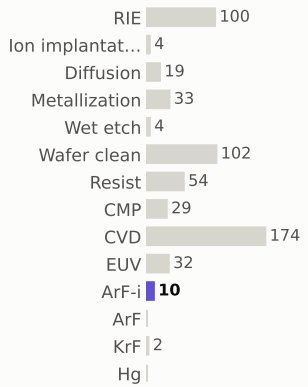
carbon so far, of the whole flow (kgCO2e)



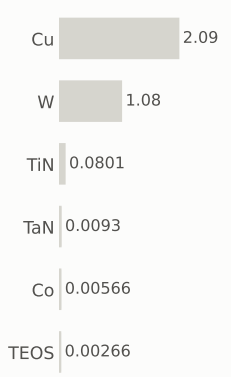
embodied carbon up to each step (kgCO2e per wafer)



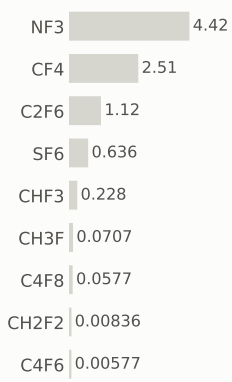
steps by process area



materials deposited (g)



gases emitted (g)



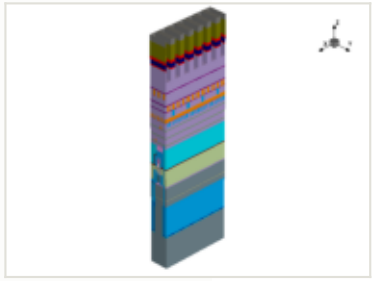
Step #5960: RIE

etch_anisotropic · RIE

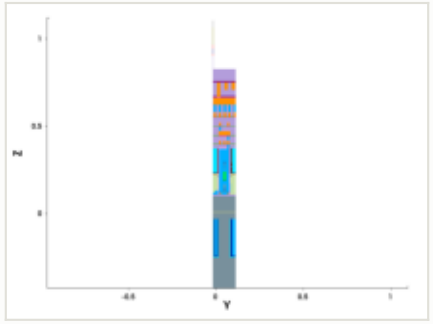
1.04	540.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

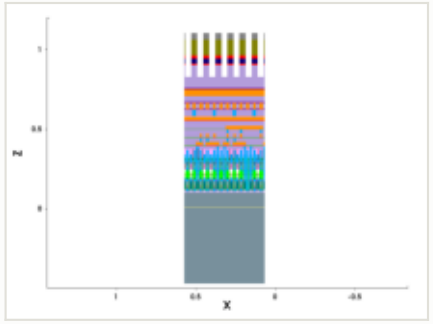
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

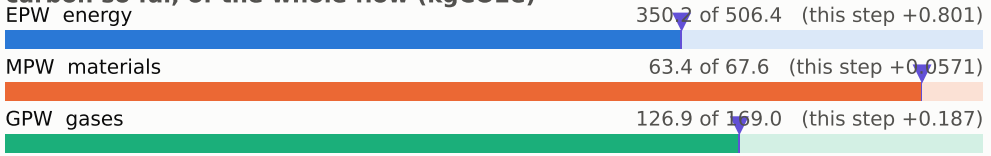
no mask for this step

this step

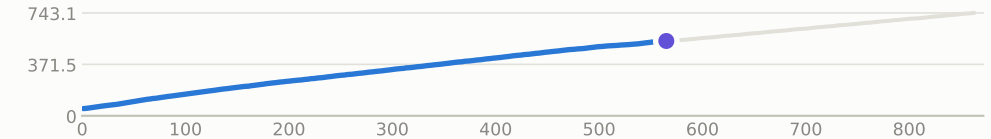
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

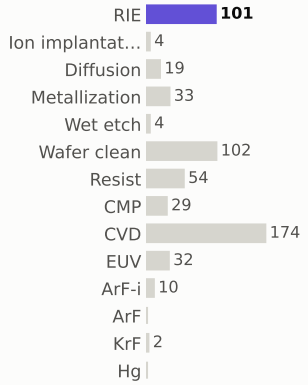
carbon so far, of the whole flow (kgCO2e)



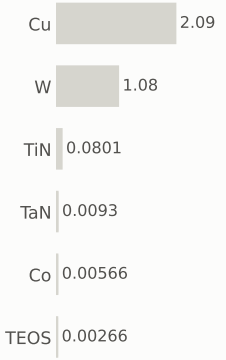
embodied carbon up to each step (kgCO2e per wafer)



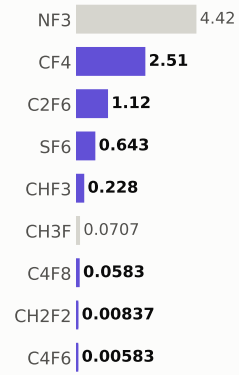
steps by process area



materials deposited (g)



gases emitted (g)



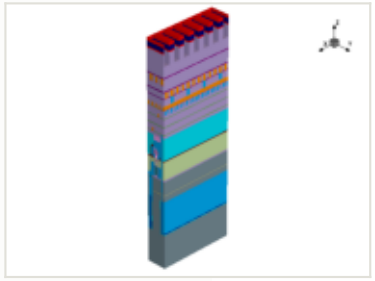
Step #5970: Wafer_clean

etch_strip · Wafer_clean

0.169	540.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

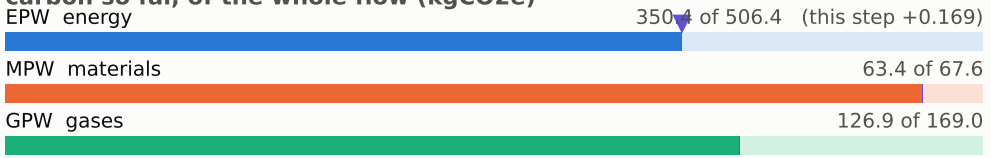


this step

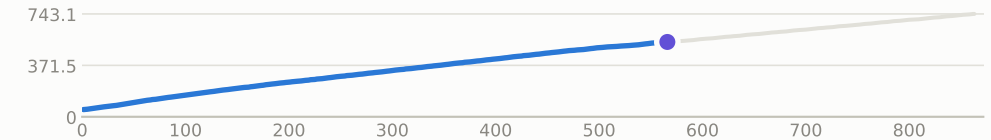
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

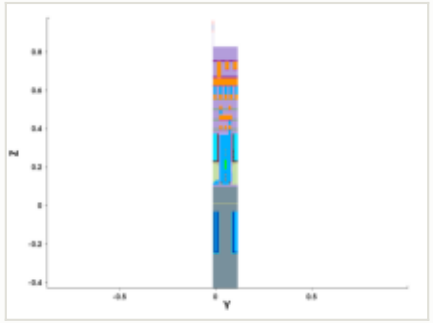
carbon so far, of the whole flow (kgCO2e)



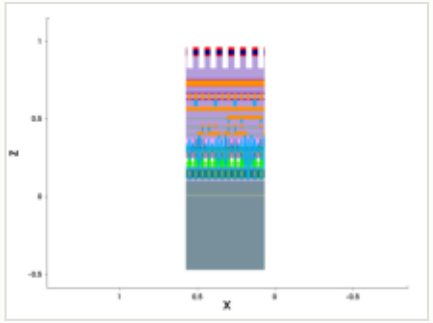
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



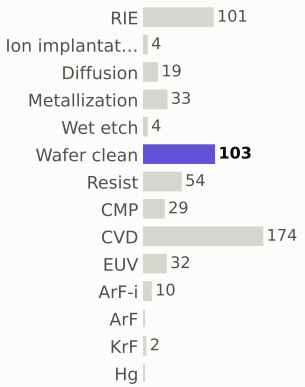
y cut at 0.049 µm



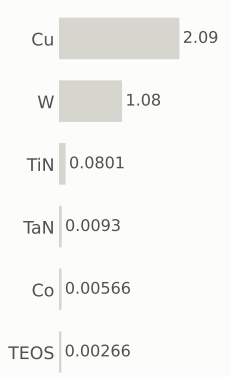
mask for this step

no mask for this step

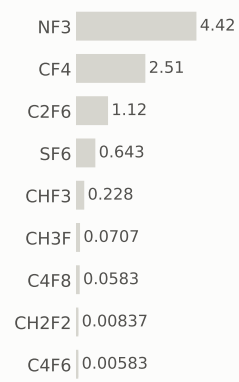
steps by process area



materials deposited (g)



gases emitted (g)



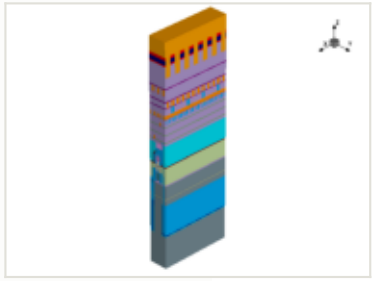
Step #5980: Resist

depo_fill · Resist

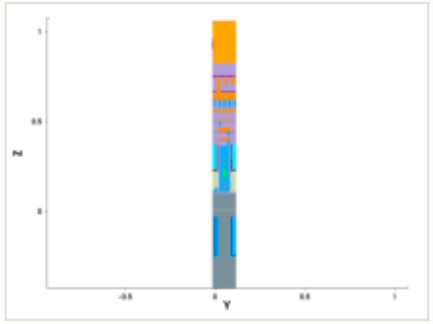
0.169	540.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

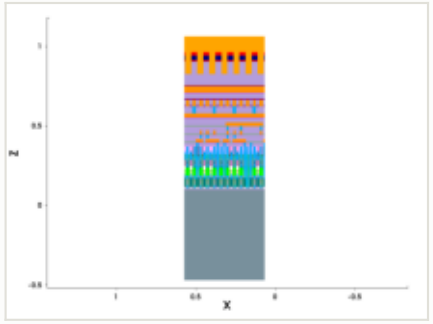
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



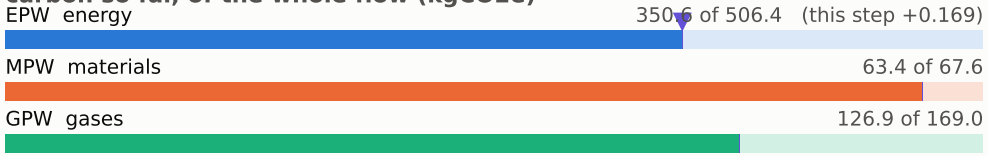
mask for this step

no mask for this step

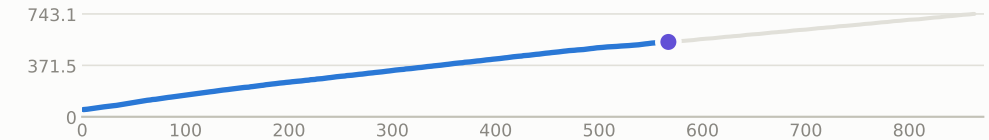
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

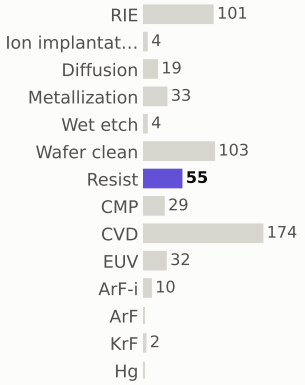
carbon so far, of the whole flow (kgCO2e)



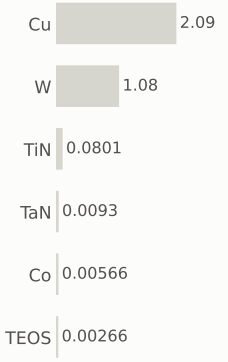
embodied carbon up to each step (kgCO2e per wafer)



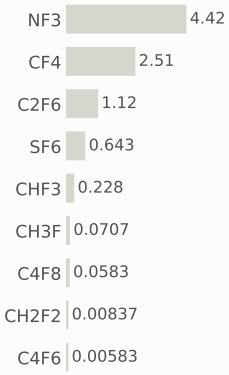
steps by process area



materials deposited (g)



gases emitted (g)



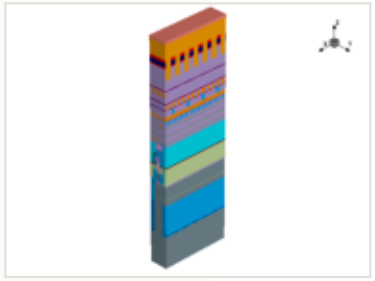
Step #5990: CVD

depo_fill · CVD

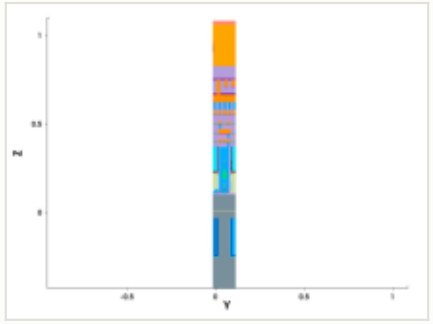
1.31	542.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

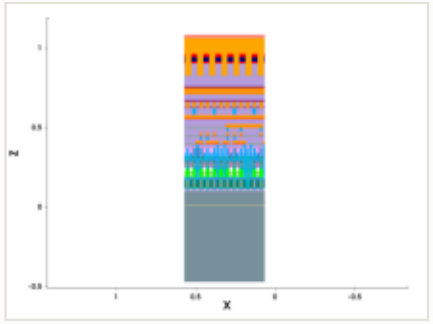
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

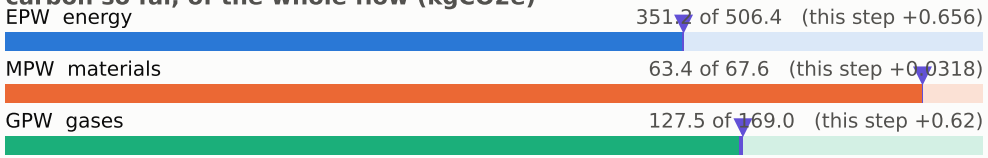
no mask for this step

this step

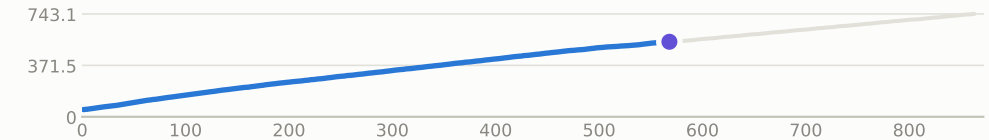
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

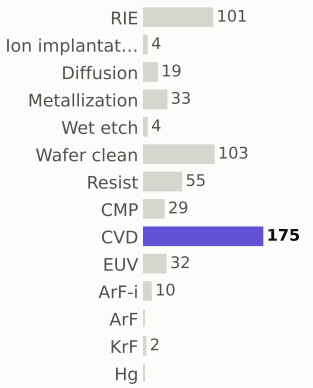
carbon so far, of the whole flow (kgCO2e)



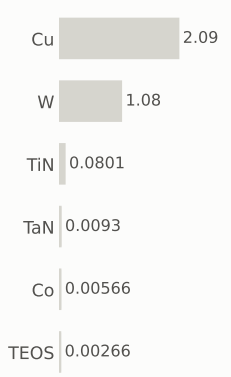
embodied carbon up to each step (kgCO2e per wafer)



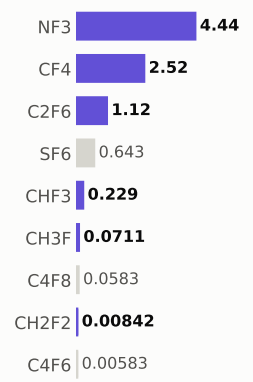
steps by process area



materials deposited (g)



gases emitted (g)



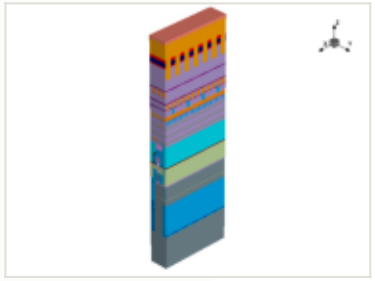
Step #6000: Wafer_clean

clean · Wafer_clean

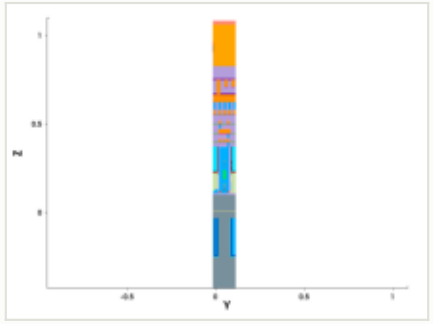
0.169	542.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

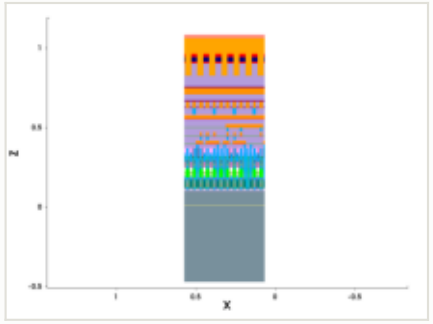
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

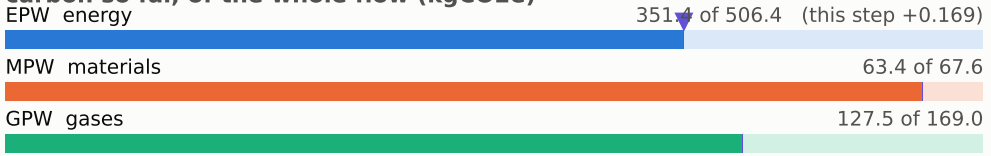
no mask for this step

this step

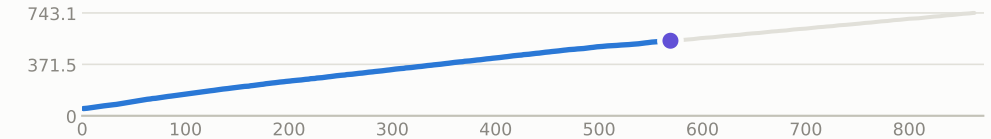
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

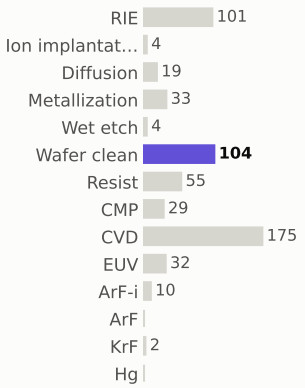
carbon so far, of the whole flow (kgCO2e)



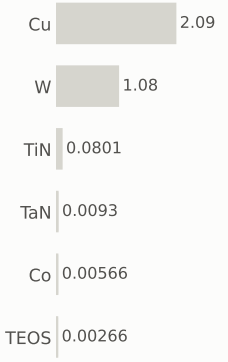
embodied carbon up to each step (kgCO2e per wafer)



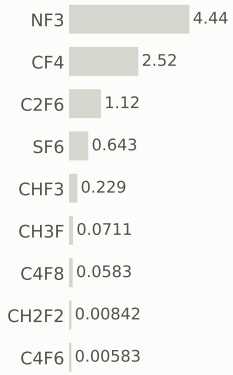
steps by process area



materials deposited (g)



gases emitted (g)



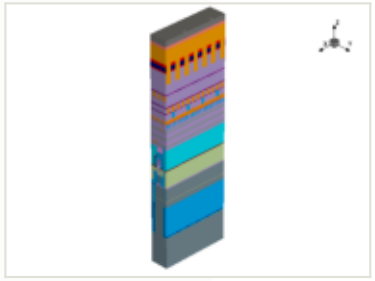
Step #6010: ArF-i

litho_patterning · ArF-i

0.217	542.5	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

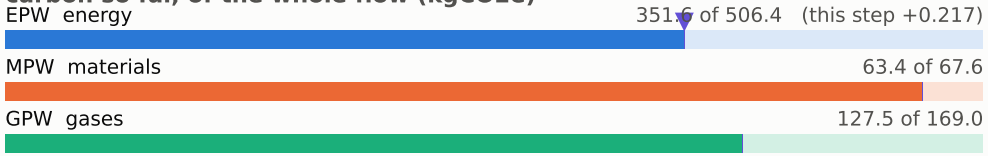


this step

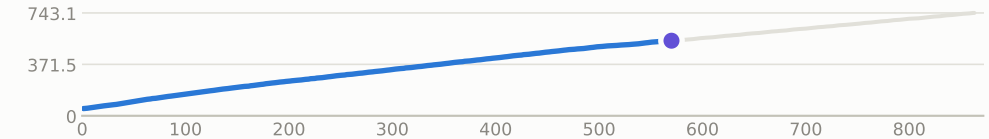
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

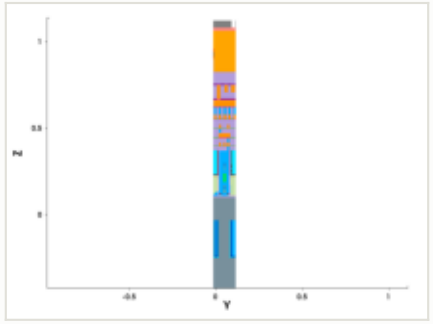
carbon so far, of the whole flow (kgCO2e)



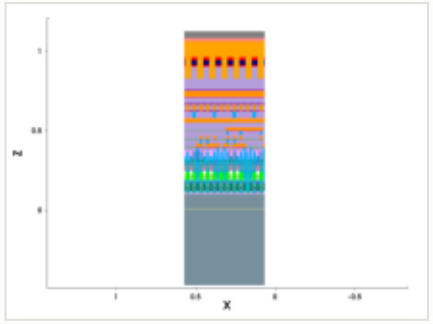
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

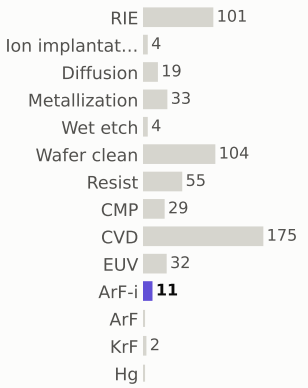


y cut at 0.049 µm

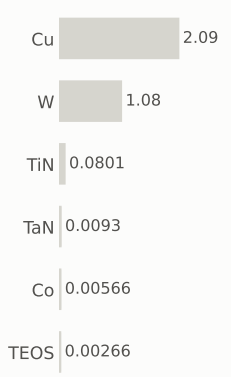


mask: layer 10471:0 (GDSII)

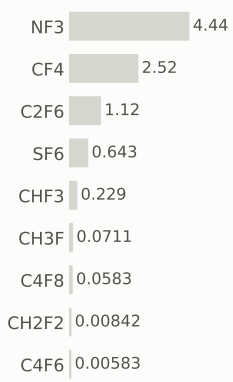
steps by process area



materials deposited (g)



gases emitted (g)



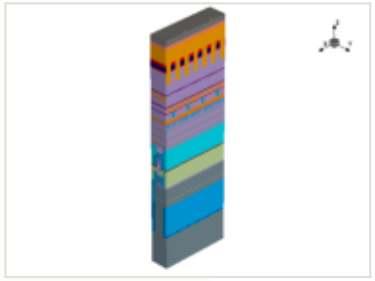
Step #6020: RIE

etch_anisotropic · RIE

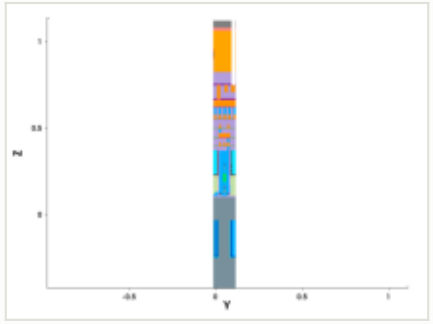
1.04	543.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

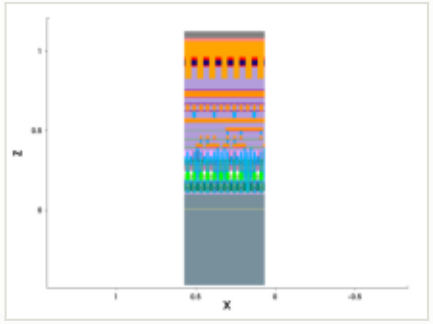
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

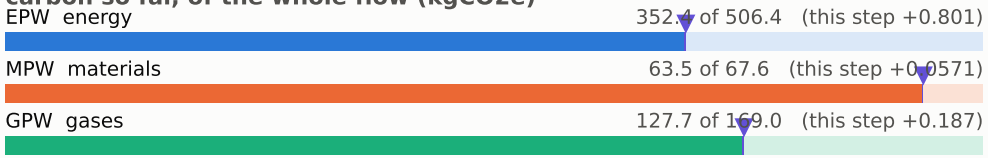
no mask for this step

this step

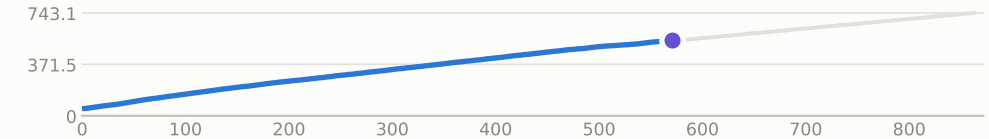
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

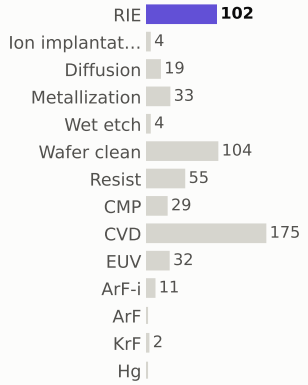
carbon so far, of the whole flow (kgCO2e)



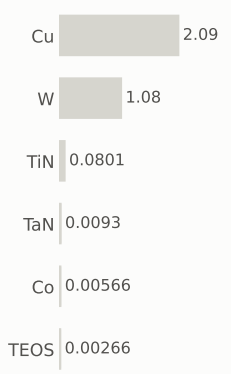
embodied carbon up to each step (kgCO2e per wafer)



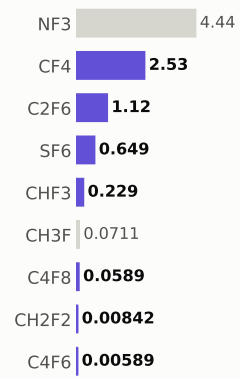
steps by process area



materials deposited (g)



gases emitted (g)



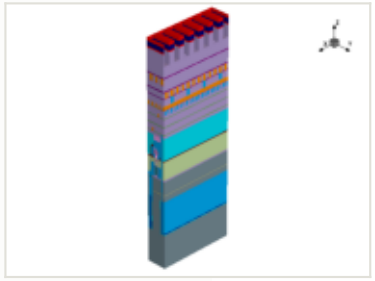
Step #6030: Wafer_clean

etch_strip · Wafer_clean

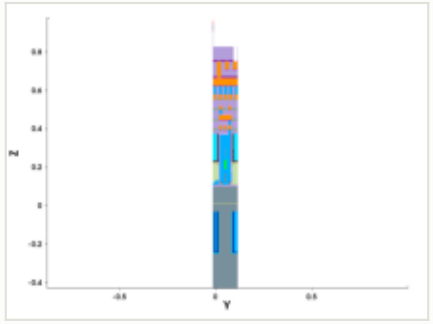
0.169	543.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

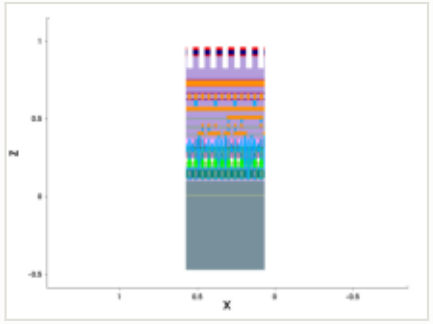
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



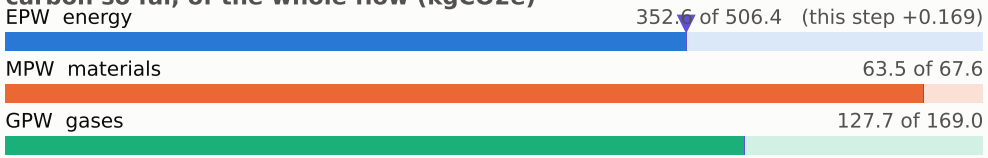
mask for this step

no mask for this step

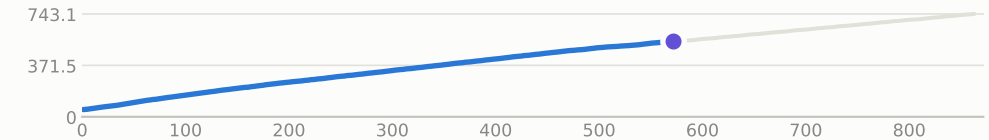
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

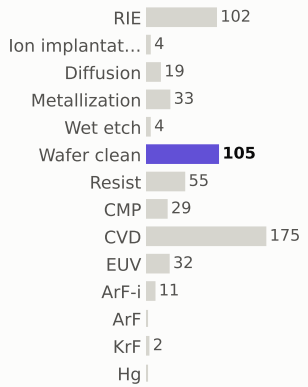
carbon so far, of the whole flow (kgCO2e)



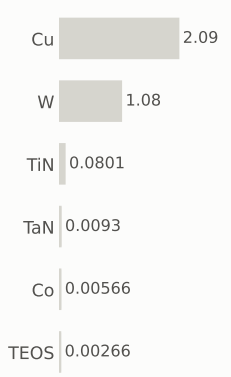
embodied carbon up to each step (kgCO2e per wafer)



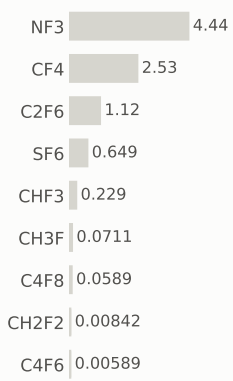
steps by process area



materials deposited (g)



gases emitted (g)



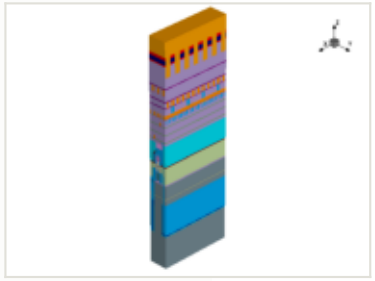
Step #6040: Resist

depo_fill · Resist

0.169	543.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

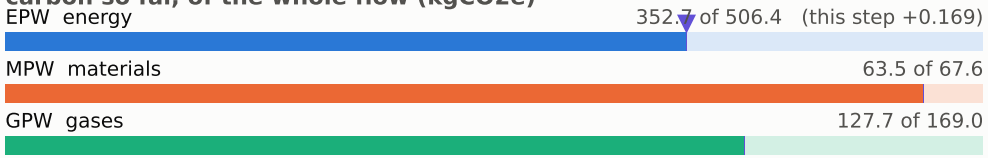


this step

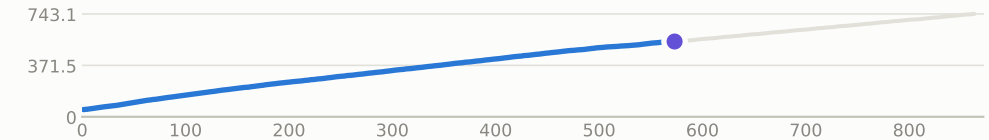
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

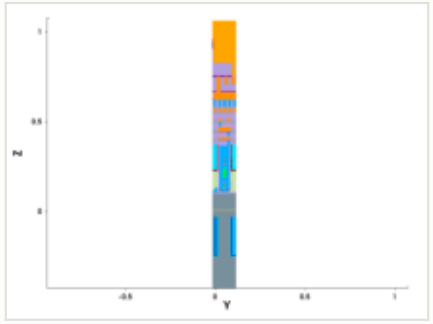
carbon so far, of the whole flow (kgCO2e)



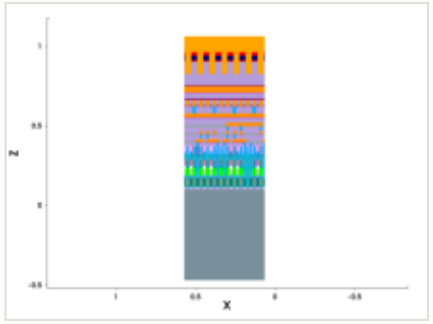
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



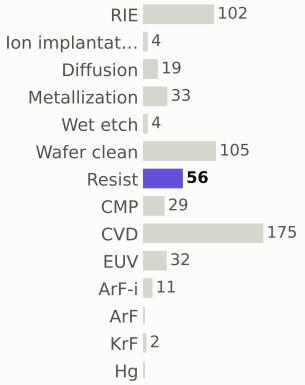
y cut at 0.049 µm



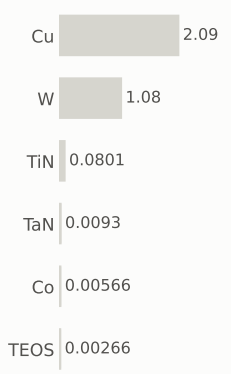
mask for this step

no mask for this step

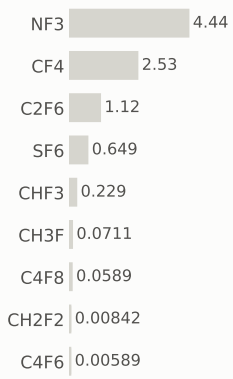
steps by process area



materials deposited (g)



gases emitted (g)



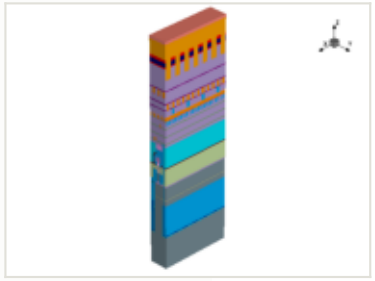
Step #6050: CVD

depo_fill · CVD

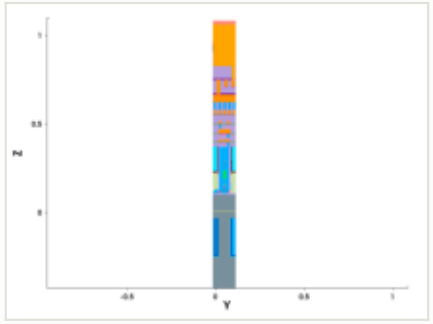
1.31	545.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

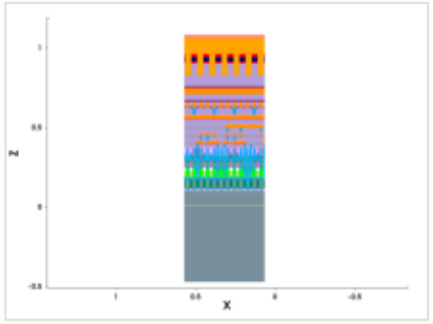
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

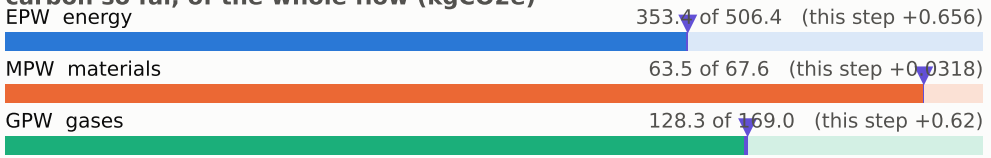
no mask for this step

this step

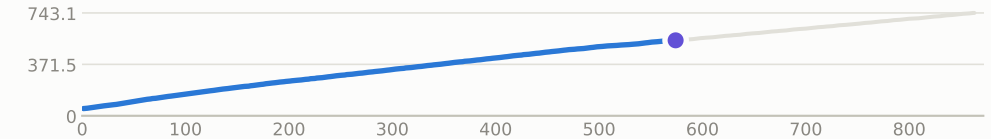
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

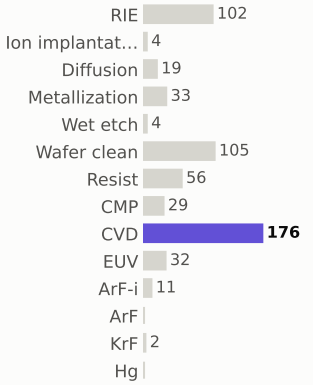
carbon so far, of the whole flow (kgCO2e)



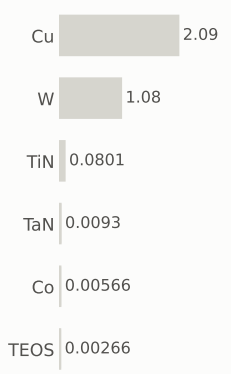
embodied carbon up to each step (kgCO2e per wafer)



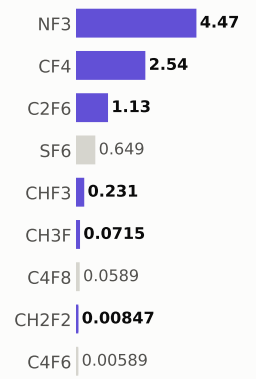
steps by process area



materials deposited (g)



gases emitted (g)



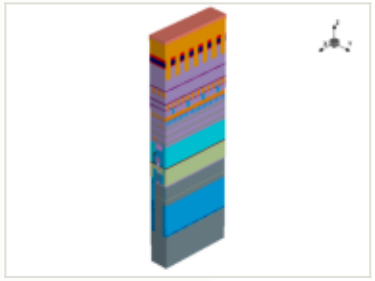
Step #6060: Wafer_clean

clean · Wafer_clean

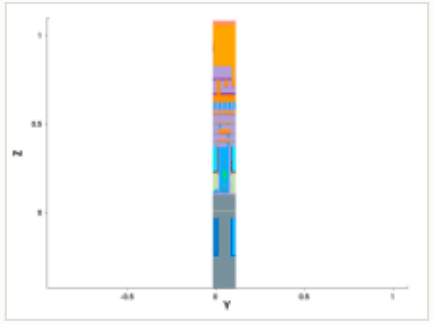
0.169	545.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

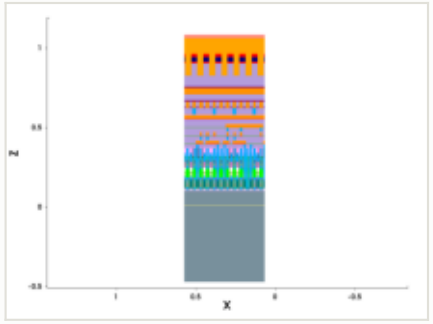
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

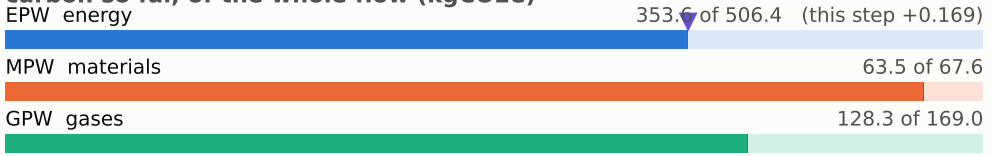
no mask for this step

this step

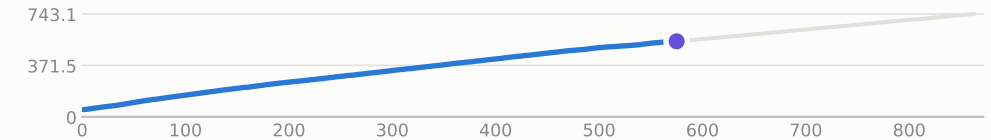
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

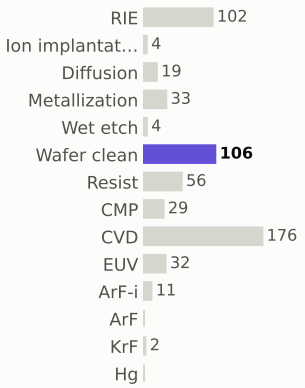
carbon so far, of the whole flow (kgCO2e)



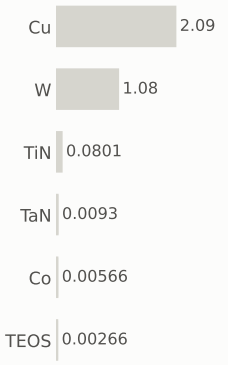
embodied carbon up to each step (kgCO2e per wafer)



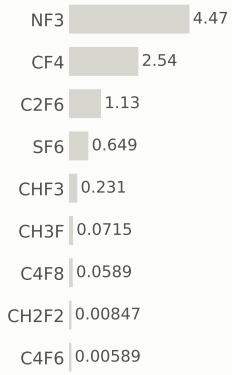
steps by process area



materials deposited (g)



gases emitted (g)



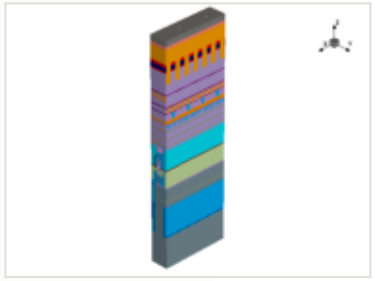
Step #6070: ArF-i

litho_patterning · ArF-i

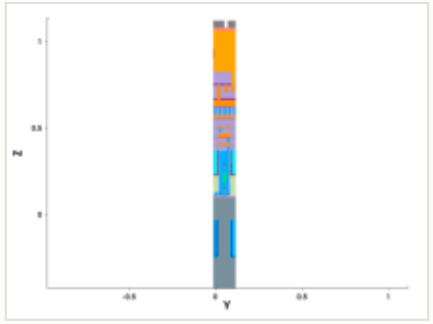
0.217	545.6	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

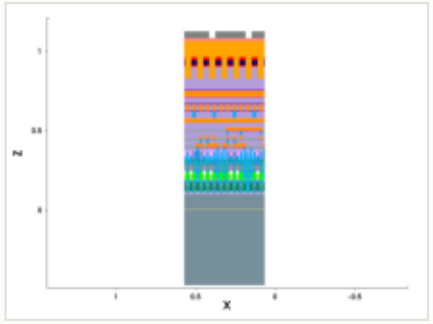
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



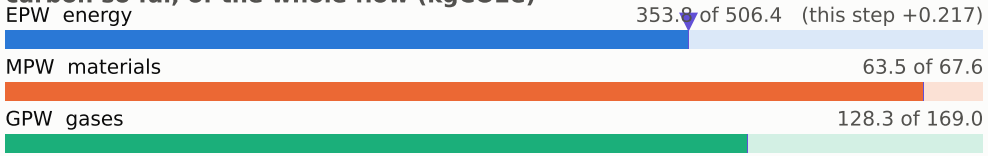
mask: layer 10472:0 (GDSII)

this step

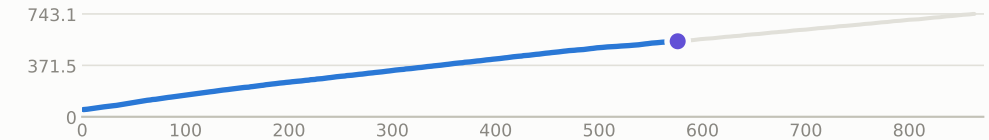
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

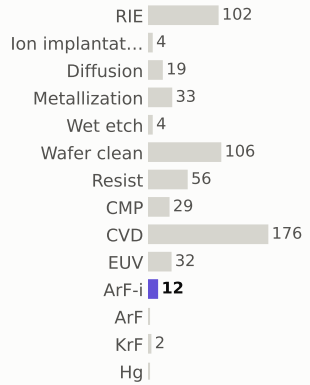
carbon so far, of the whole flow (kgCO2e)



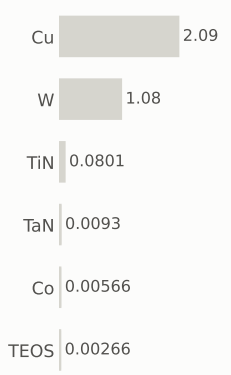
embodied carbon up to each step (kgCO2e per wafer)



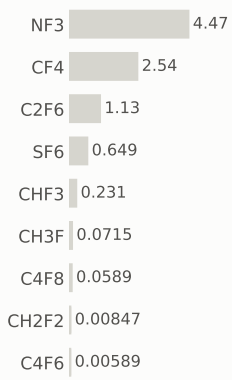
steps by process area



materials deposited (g)



gases emitted (g)



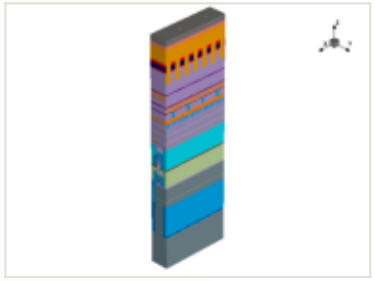
Step #6080: RIE

etch_anisotropic · RIE

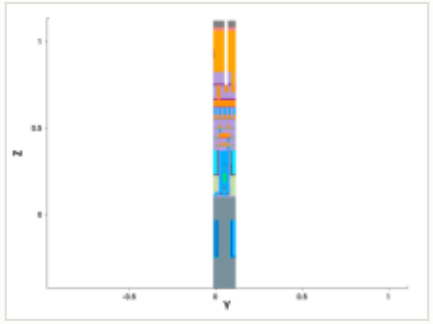
1.04	546.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

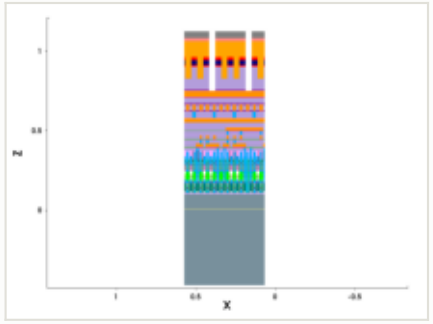
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

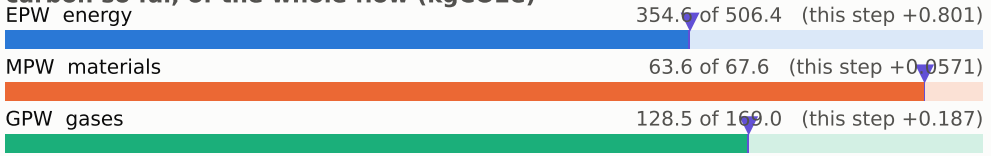
no mask for this step

this step

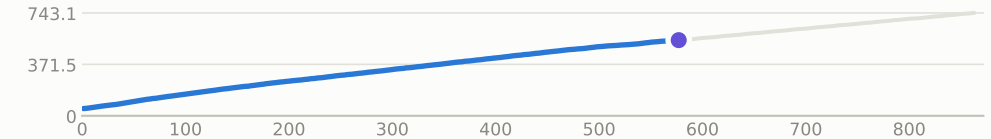
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

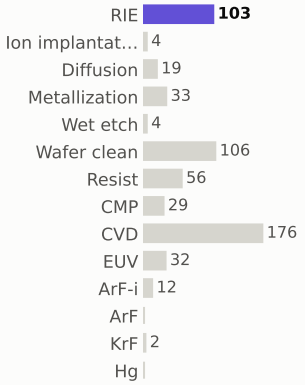
carbon so far, of the whole flow (kgCO2e)



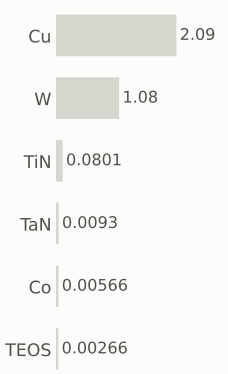
embodied carbon up to each step (kgCO2e per wafer)



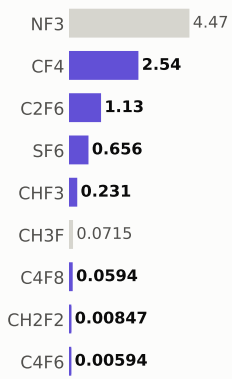
steps by process area



materials deposited (g)



gases emitted (g)



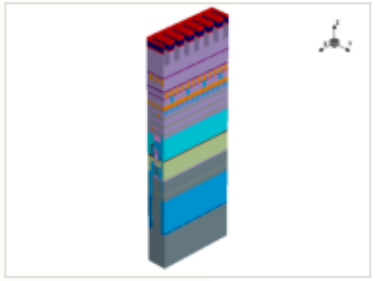
Step #6090: Wafer_clean

etch_strip · Wafer_clean

0.169	546.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

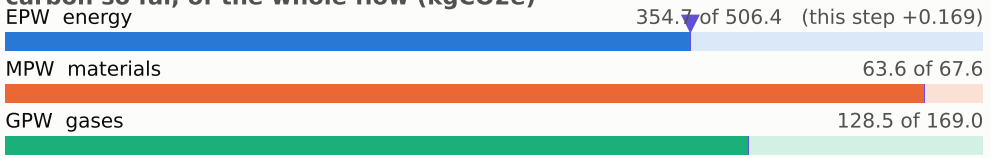


this step

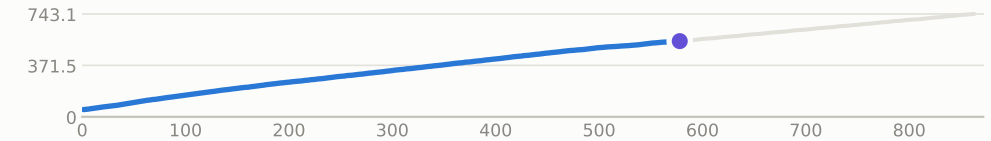
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

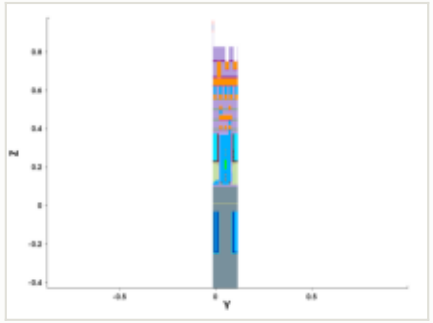
carbon so far, of the whole flow (kgCO2e)



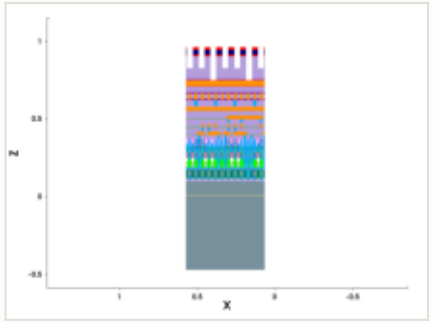
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



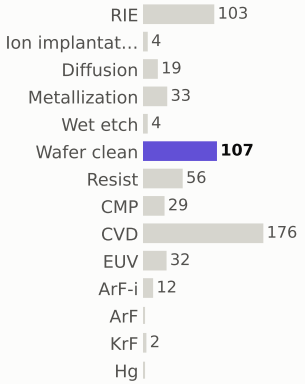
y cut at 0.049 µm



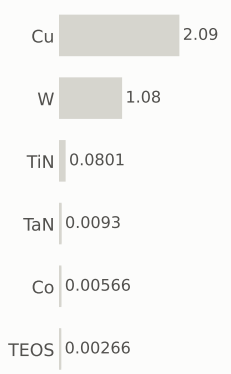
mask for this step

no mask for this step

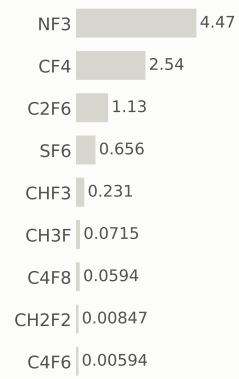
steps by process area



materials deposited (g)



gases emitted (g)



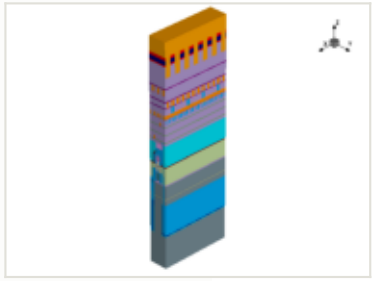
Step #6100: Resist

depo_fill · Resist

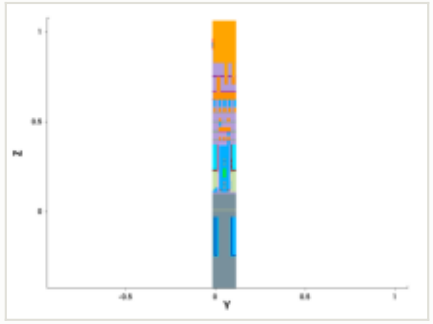
0.169	546.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

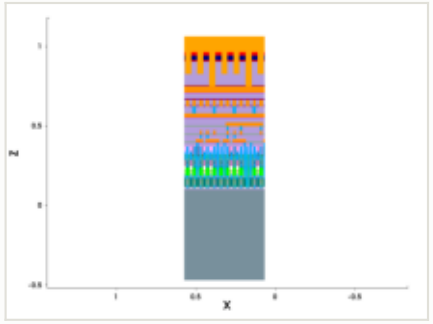
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



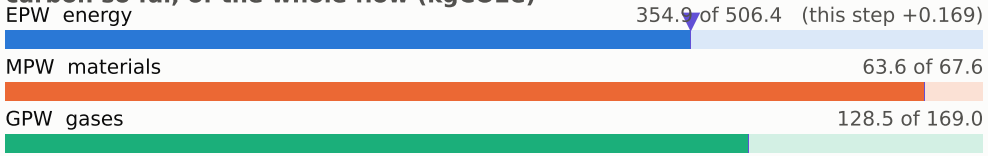
mask for this step

no mask for this step

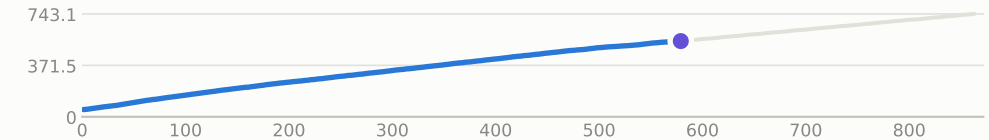
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

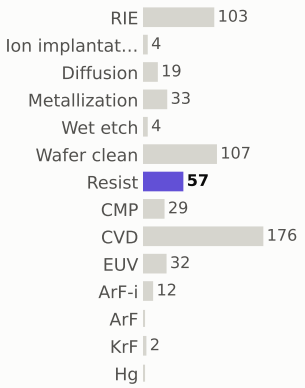
carbon so far, of the whole flow (kgCO2e)



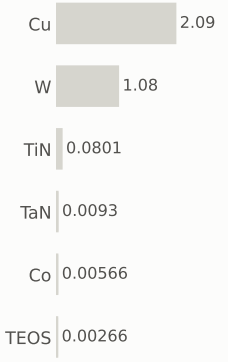
embodied carbon up to each step (kgCO2e per wafer)



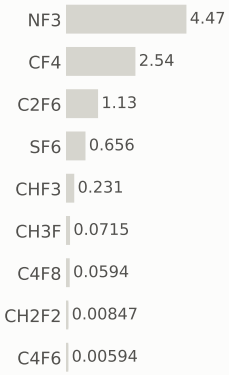
steps by process area



materials deposited (g)



gases emitted (g)



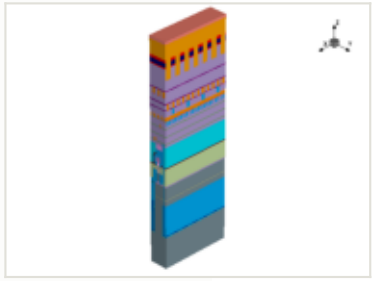
Step #6110: CVD

depo_fill · CVD

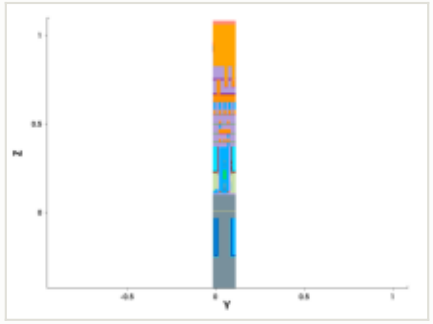
1.31	548.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

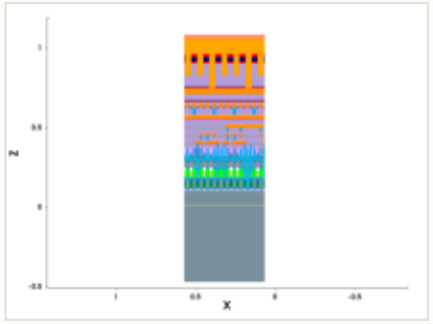
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

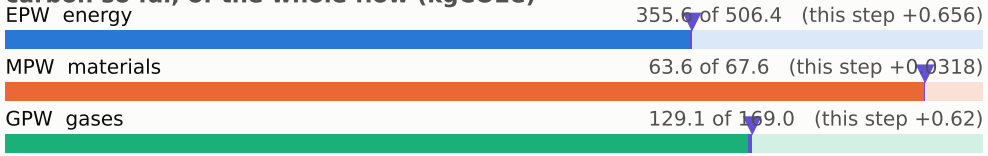
no mask for this step

this step

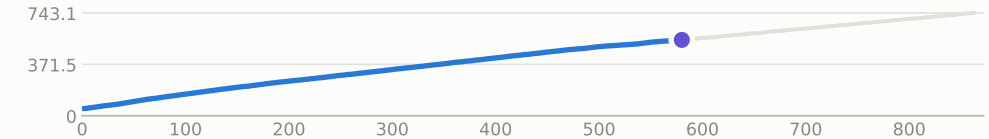
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

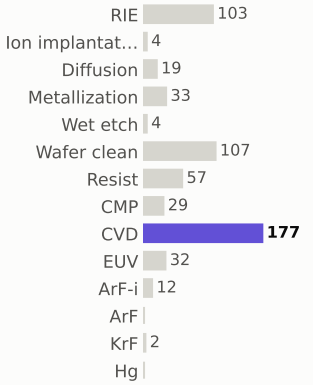
carbon so far, of the whole flow (kgCO2e)



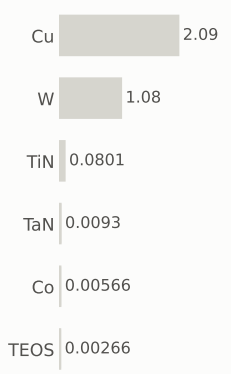
embodied carbon up to each step (kgCO2e per wafer)



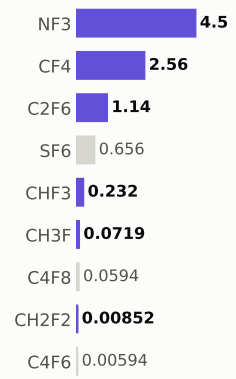
steps by process area



materials deposited (g)



gases emitted (g)



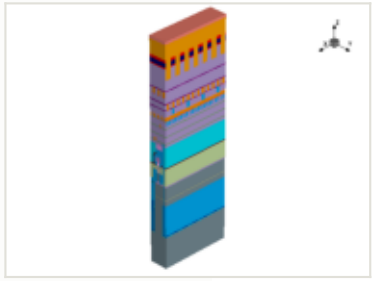
Step #6120: Wafer_clean

clean · Wafer_clean

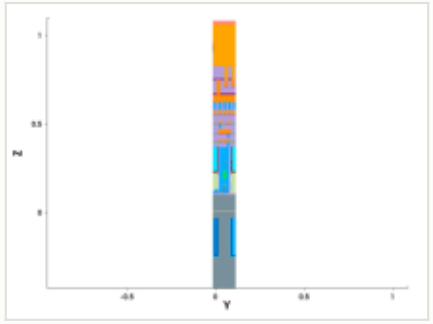
0.169	548.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

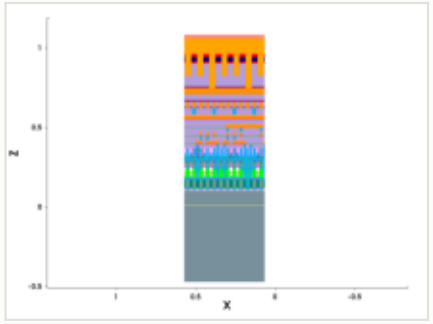
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

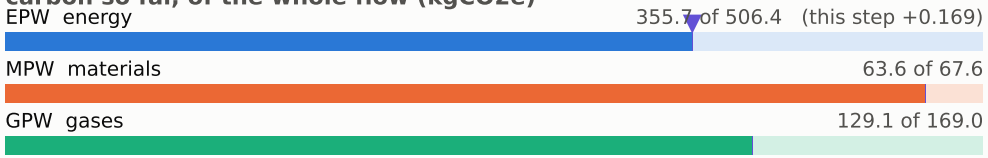
no mask for this step

this step

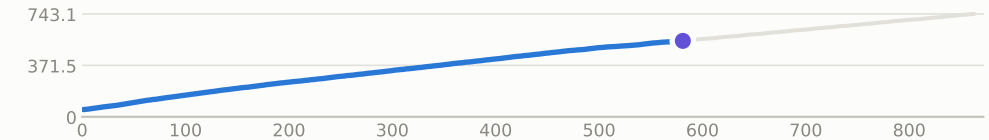
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

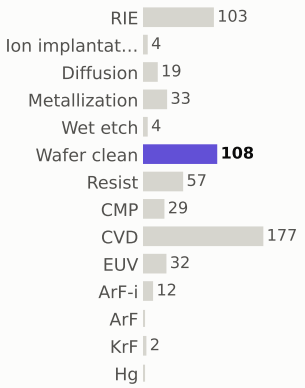
carbon so far, of the whole flow (kgCO2e)



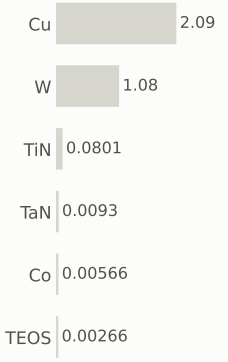
embodied carbon up to each step (kgCO2e per wafer)



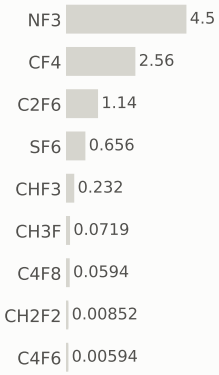
steps by process area



materials deposited (g)



gases emitted (g)



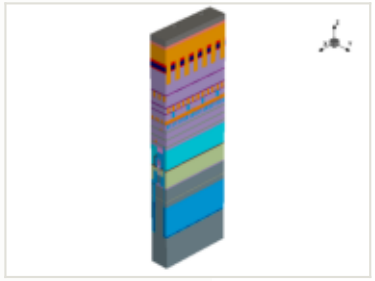
Step #6130: ArF-i

litho_patterning · ArF-i

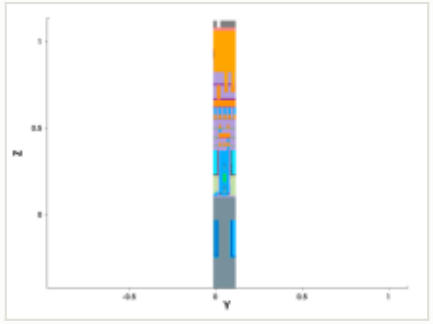
0.217	548.6	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

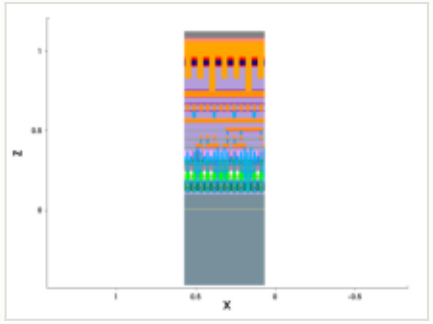
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



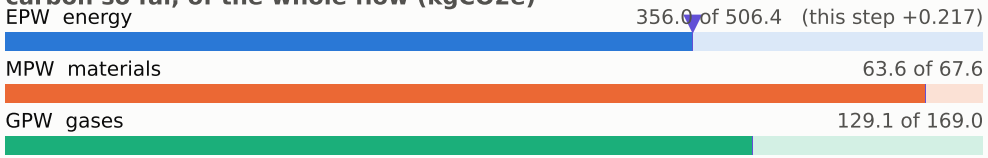
mask: layer 10473:0 (GDSII)

this step

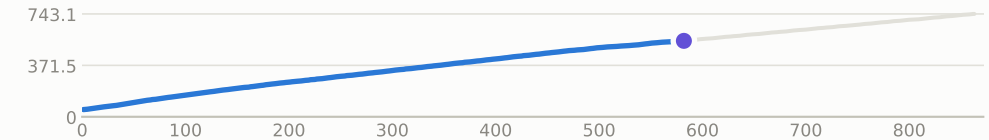
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

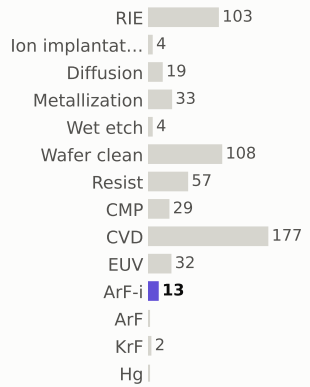
carbon so far, of the whole flow (kgCO2e)



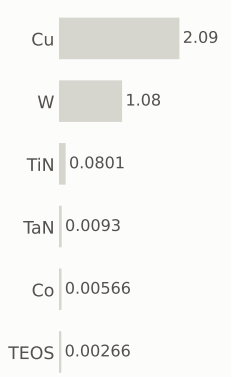
embodied carbon up to each step (kgCO2e per wafer)



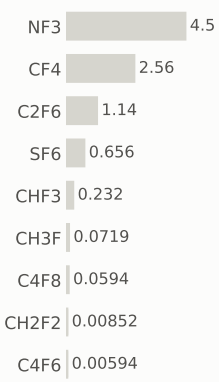
steps by process area



materials deposited (g)



gases emitted (g)



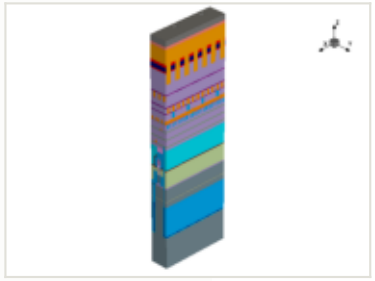
Step #6140: RIE

etch_anisotropic · RIE

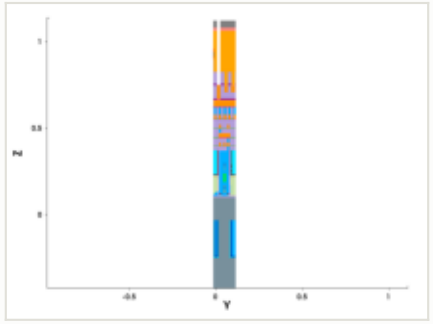
1.04	549.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

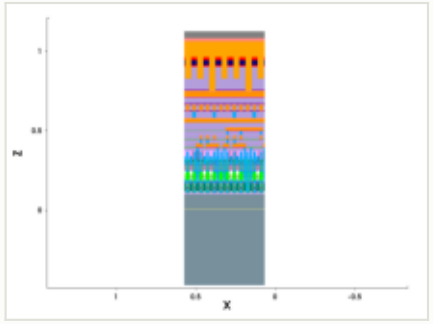
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

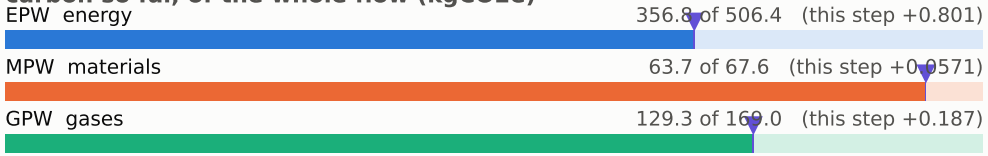
no mask for this step

this step

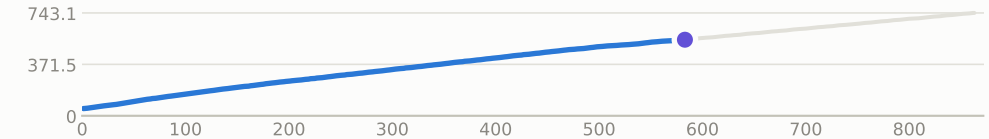
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

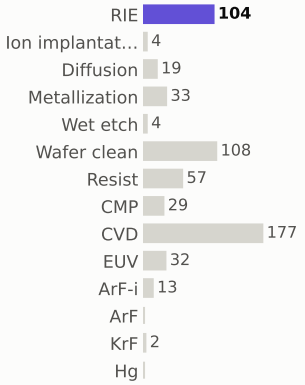
carbon so far, of the whole flow (kgCO2e)



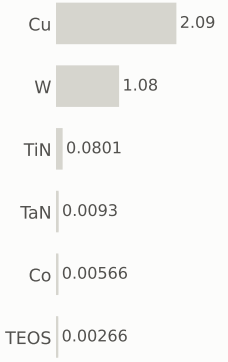
embodied carbon up to each step (kgCO2e per wafer)



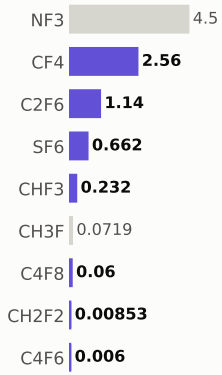
steps by process area



materials deposited (g)



gases emitted (g)



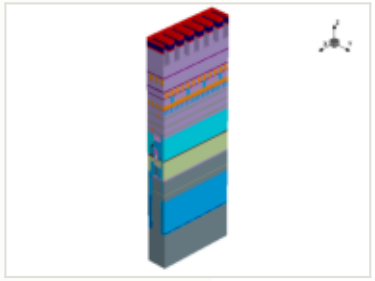
Step #6150: Wafer_clean

etch_strip · Wafer_clean

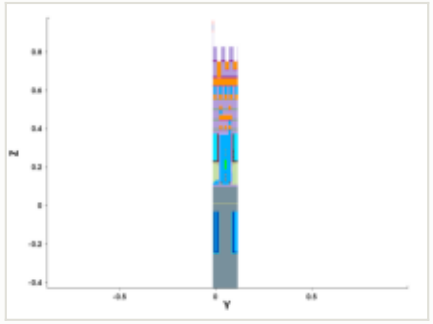
0.169	549.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

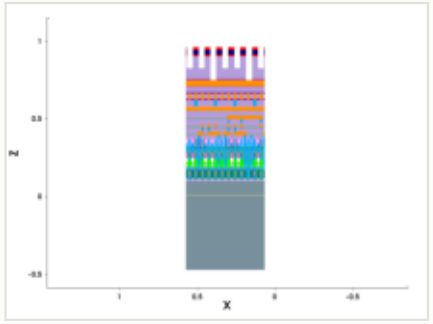
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



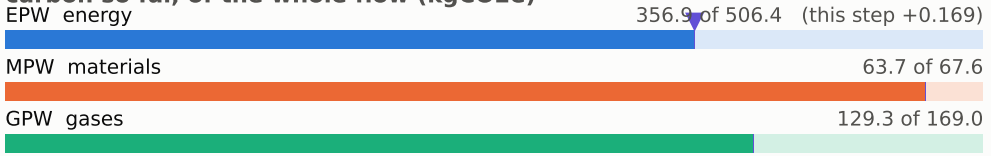
mask for this step

no mask for this step

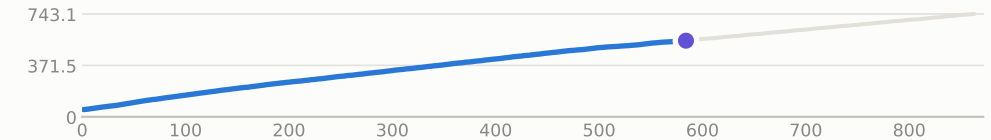
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

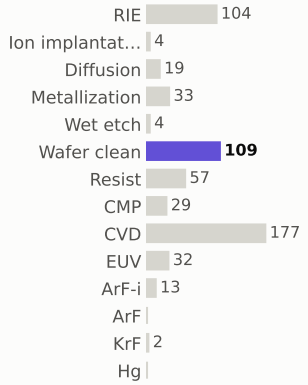
carbon so far, of the whole flow (kgCO2e)



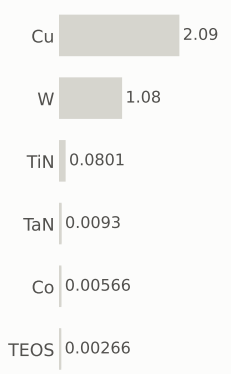
embodied carbon up to each step (kgCO2e per wafer)



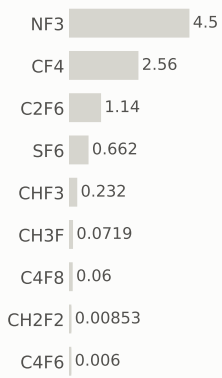
steps by process area



materials deposited (g)



gases emitted (g)



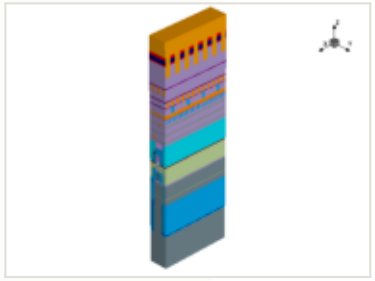
Step #6160: Resist

depo_fill · Resist

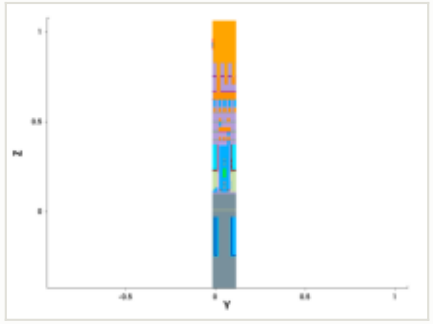
0.169	550.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

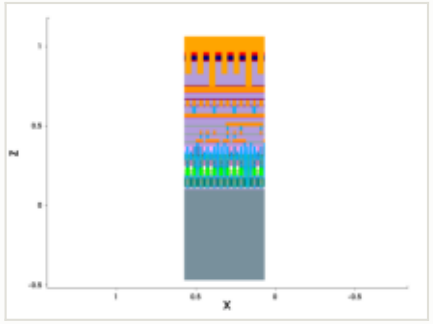
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



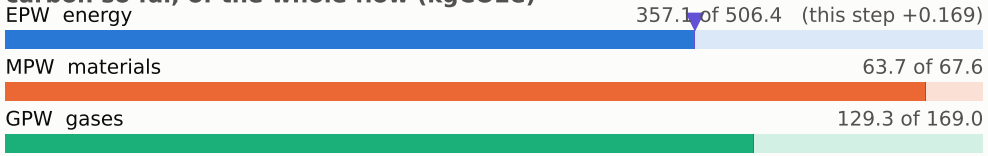
mask for this step

no mask for this step

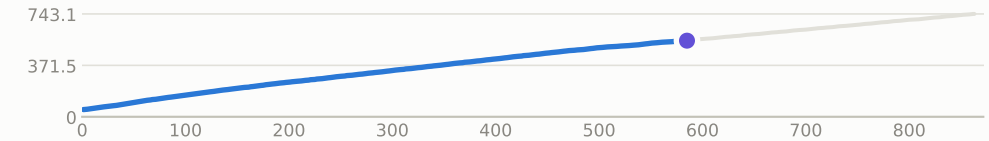
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

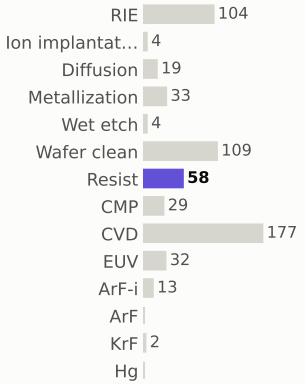
carbon so far, of the whole flow (kgCO2e)



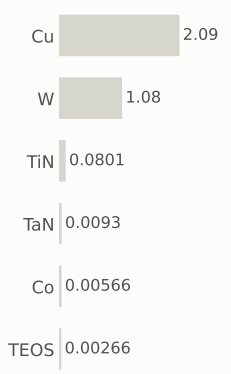
embodied carbon up to each step (kgCO2e per wafer)



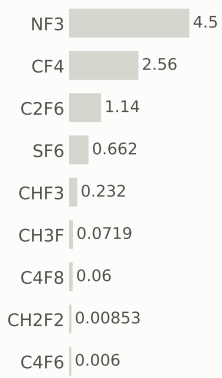
steps by process area



materials deposited (g)



gases emitted (g)



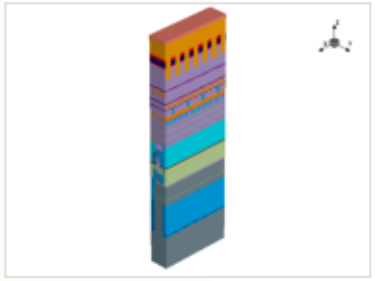
Step #6170: CVD

depo_fill · CVD

1.31	551.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

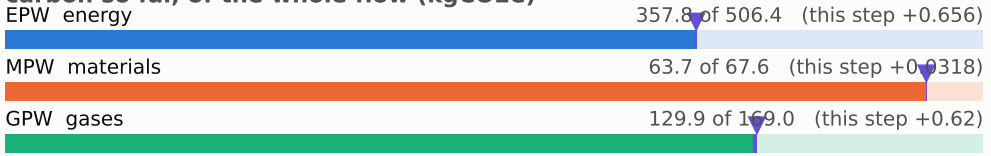


this step

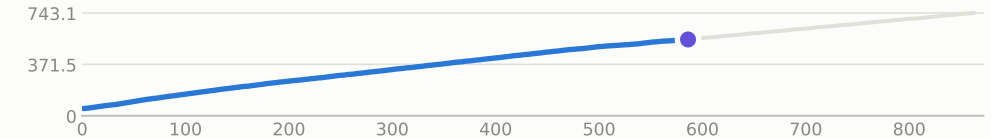
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

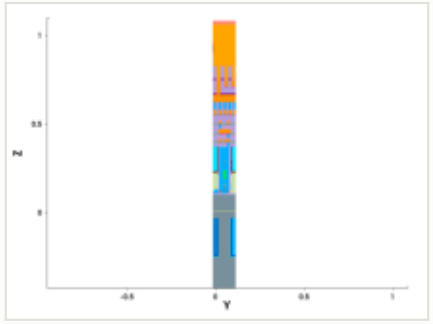
carbon so far, of the whole flow (kgCO2e)



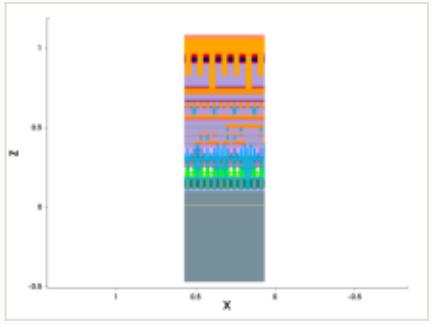
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



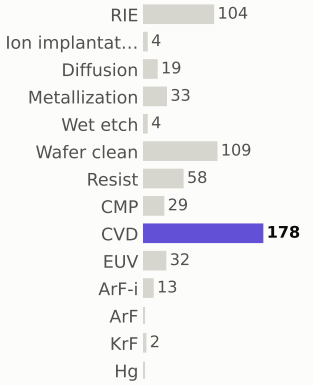
y cut at 0.049 µm



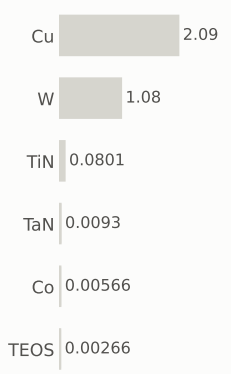
mask for this step

no mask for this step

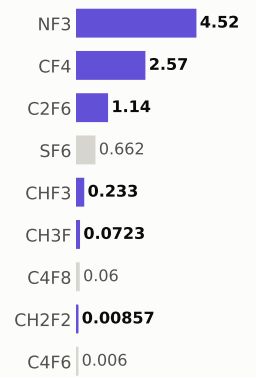
steps by process area



materials deposited (g)



gases emitted (g)



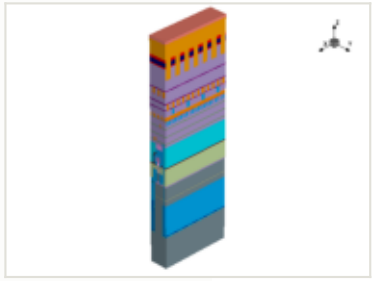
Step #6180: Wafer_clean

clean · Wafer_clean

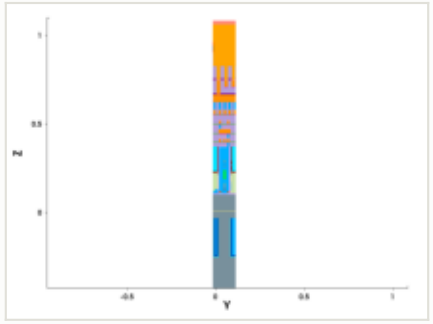
0.169	551.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

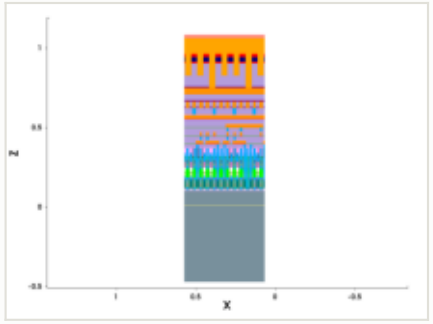
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

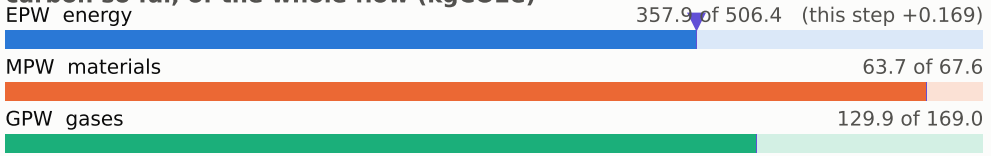
no mask for this step

this step

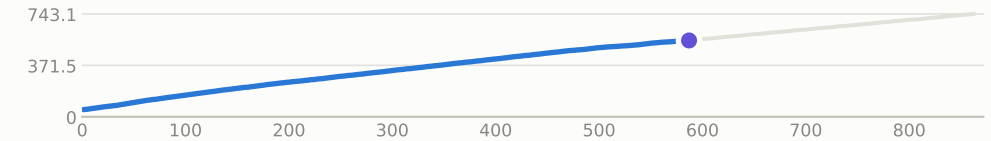
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

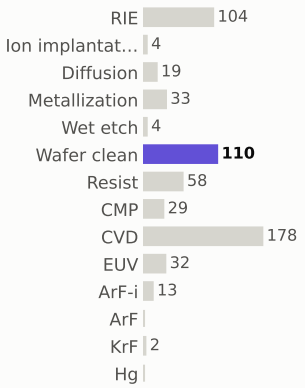
carbon so far, of the whole flow (kgCO2e)



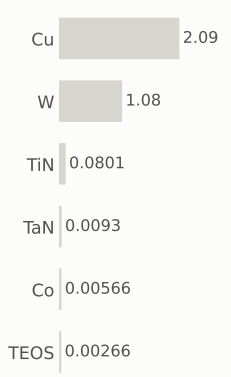
embodied carbon up to each step (kgCO2e per wafer)



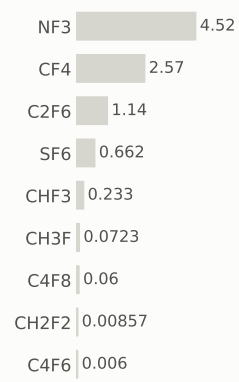
steps by process area



materials deposited (g)



gases emitted (g)



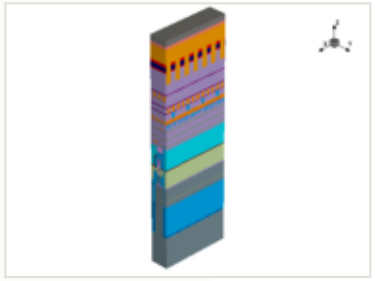
Step #6190: ArF-i

litho_patterning · ArF-i

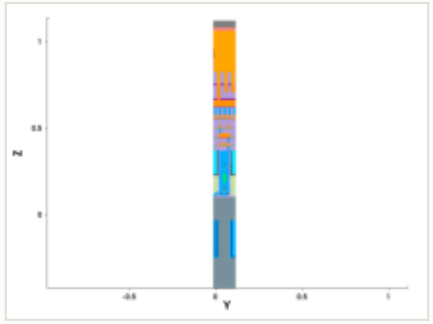
0.217	551.7	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

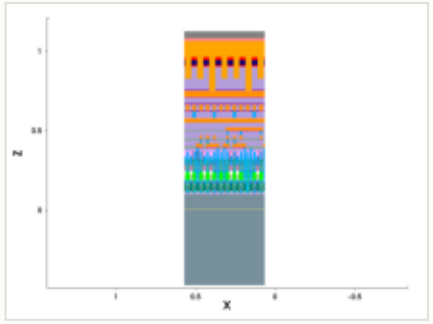
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask: layer 10474:0 (GDSII)

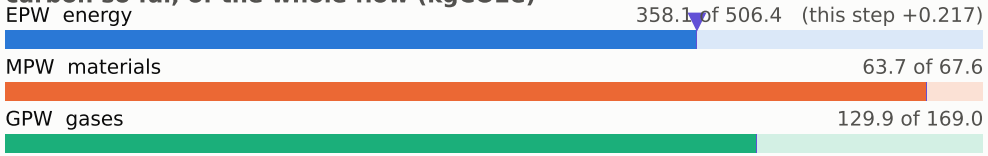
nothing in the clip window

this step

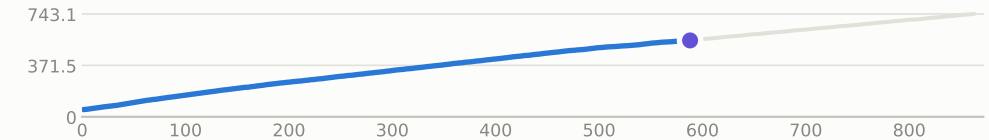
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

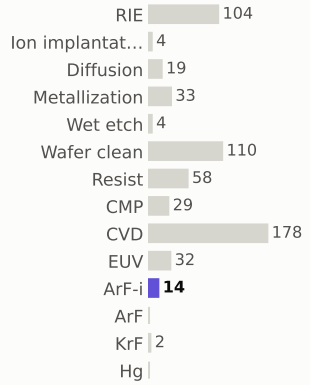
carbon so far, of the whole flow (kgCO2e)



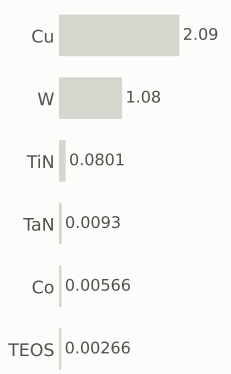
embodied carbon up to each step (kgCO2e per wafer)



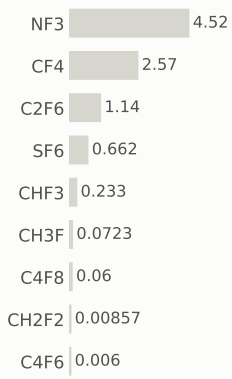
steps by process area



materials deposited (g)



gases emitted (g)



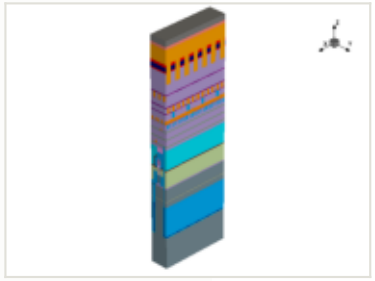
Step #6200: RIE

etch_anisotropic · RIE

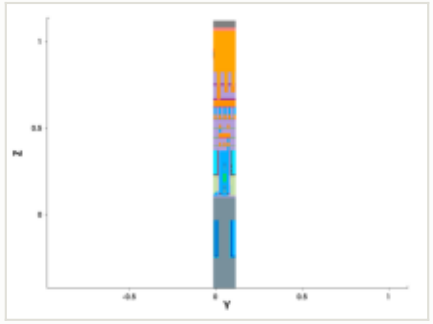
1.04	552.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

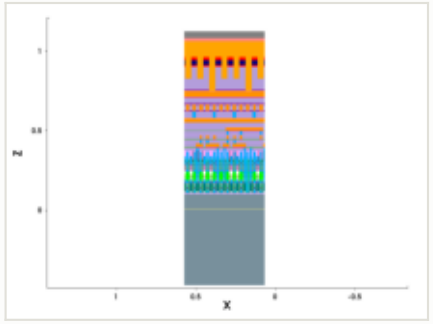
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

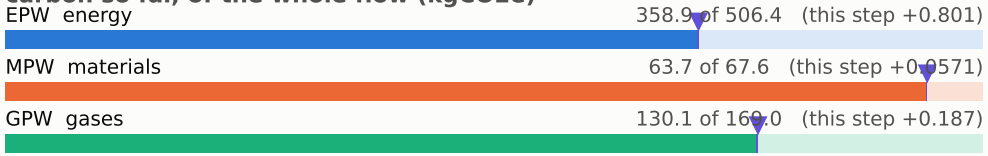
no mask for this step

this step

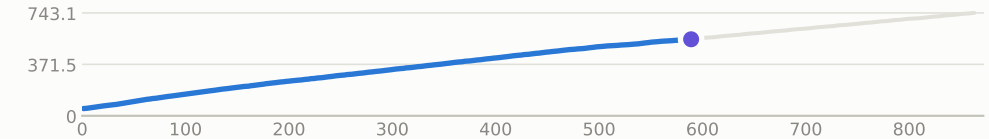
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

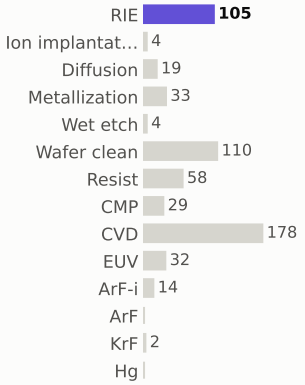
carbon so far, of the whole flow (kgCO2e)



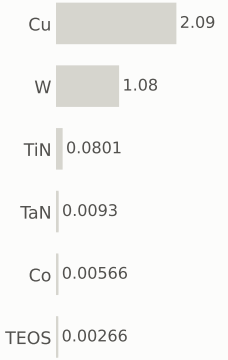
embodied carbon up to each step (kgCO2e per wafer)



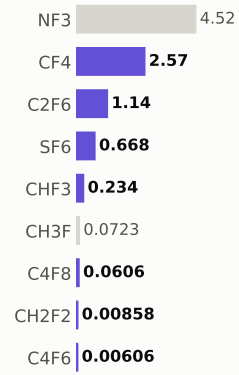
steps by process area



materials deposited (g)



gases emitted (g)



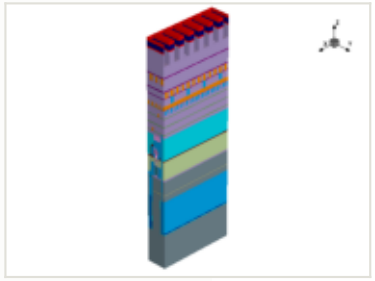
Step #6210: Wafer_clean

etch_strip · Wafer_clean

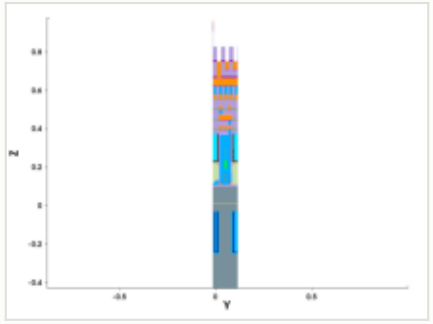
0.169	552.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

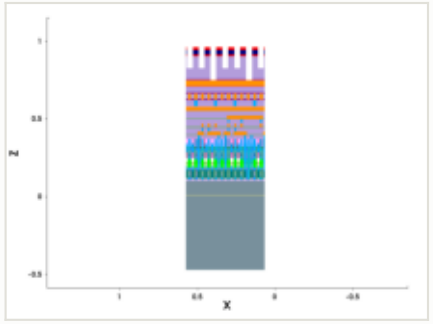
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



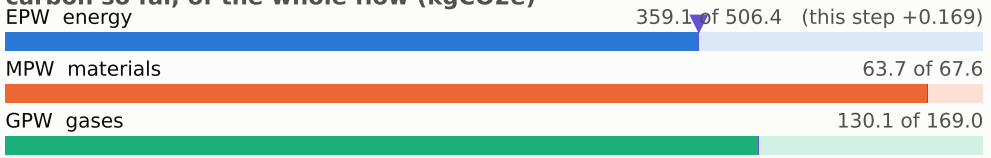
mask for this step

no mask for this step

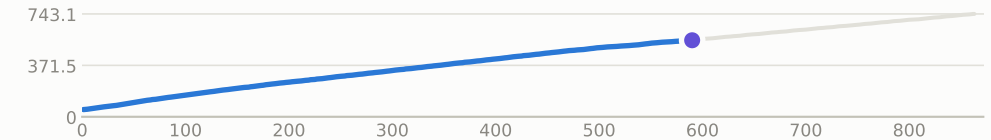
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

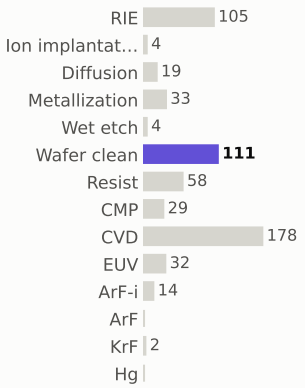
carbon so far, of the whole flow (kgCO2e)



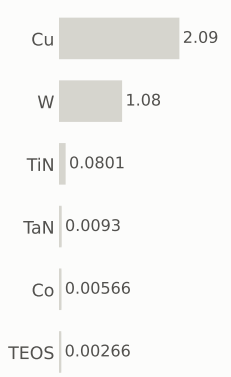
embodied carbon up to each step (kgCO2e per wafer)



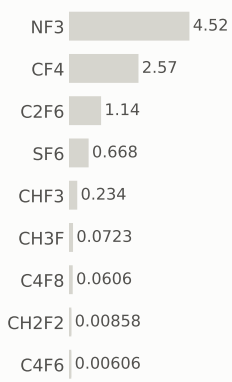
steps by process area



materials deposited (g)



gases emitted (g)



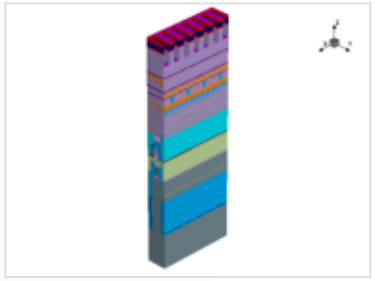
Step #6220: Metallization

depo_ellipsoidal · Metallization

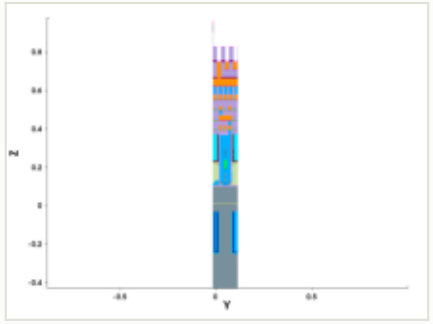
1.2	554.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

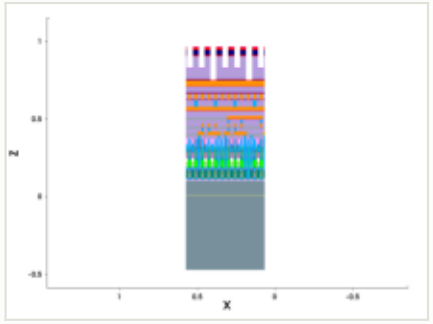
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

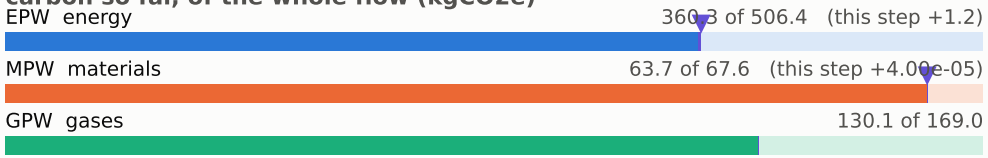
no mask for this step

this step

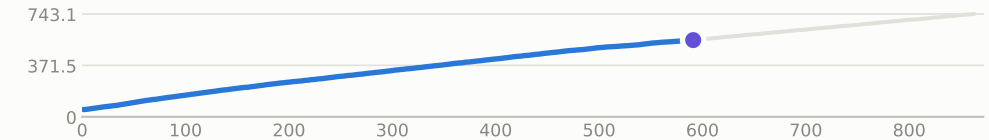
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

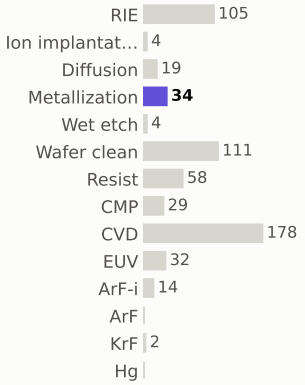
carbon so far, of the whole flow (kgCO2e)



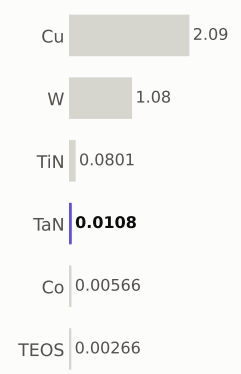
embodied carbon up to each step (kgCO2e per wafer)



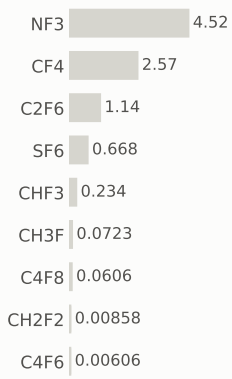
steps by process area



materials deposited (g)



gases emitted (g)



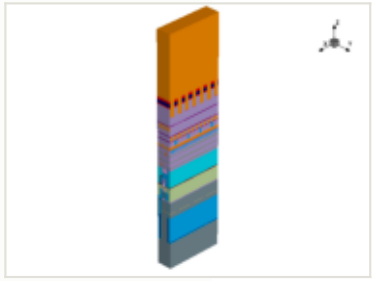
Step #6230: Metallization

depo_fill · Metallization

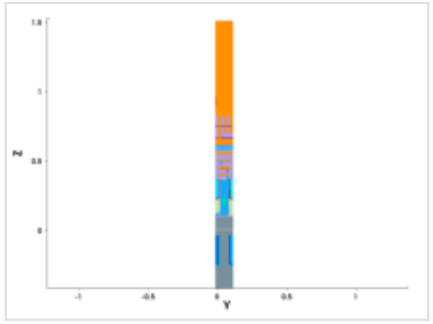
1.21	555.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

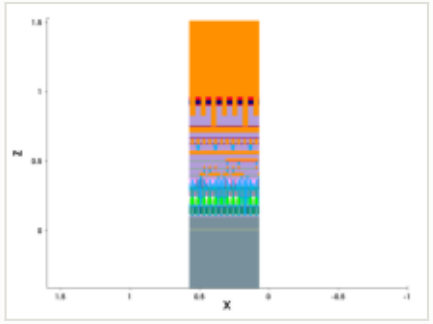
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

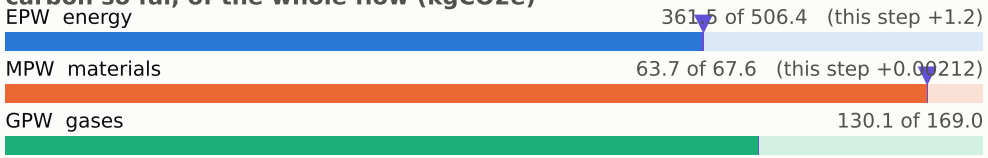
no mask for this step

this step

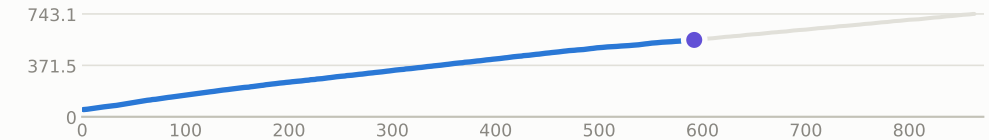
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

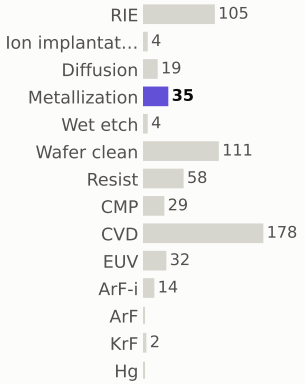
carbon so far, of the whole flow (kgCO2e)



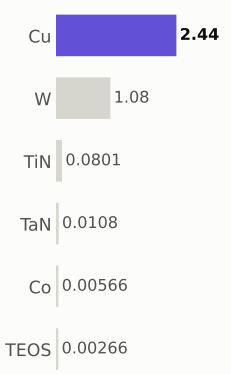
embodied carbon up to each step (kgCO2e per wafer)



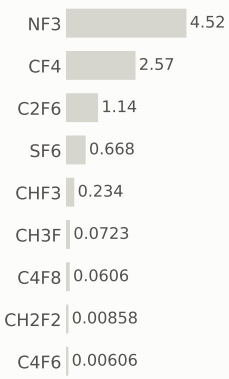
steps by process area



materials deposited (g)



gases emitted (g)



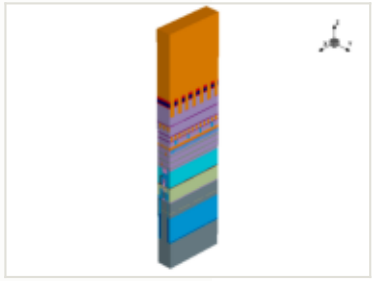
Step #6240: Diffusion

treatment · Diffusion

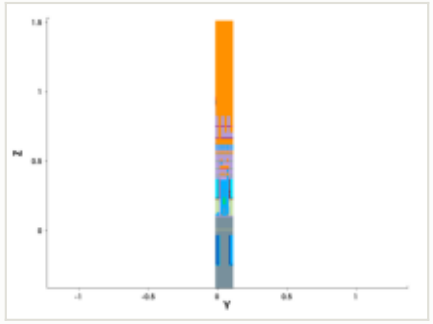
0.801	556.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

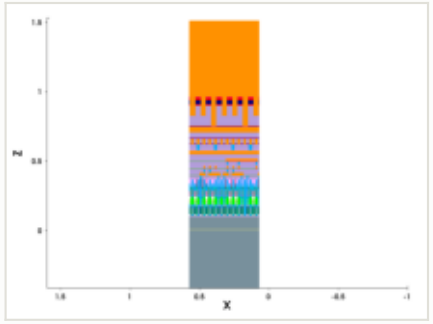
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

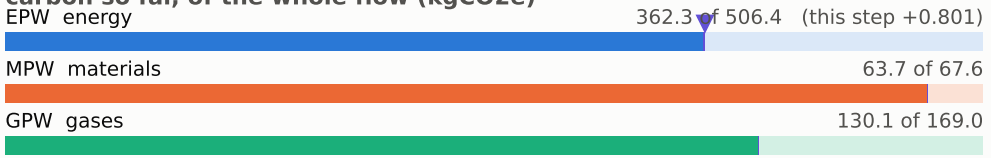
no mask for this step

this step

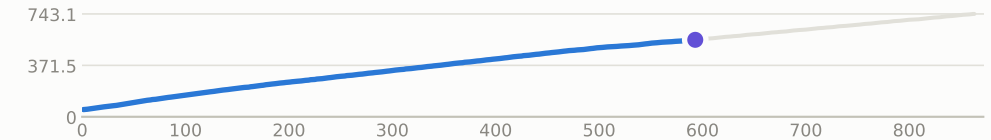
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

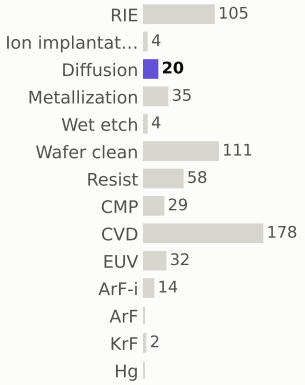
carbon so far, of the whole flow (kgCO2e)



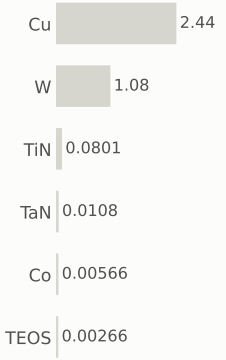
embodied carbon up to each step (kgCO2e per wafer)



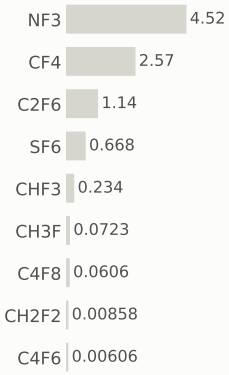
steps by process area



materials deposited (g)



gases emitted (g)



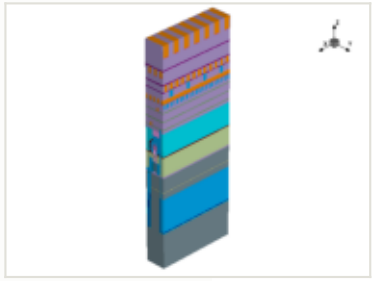
Step #6250: CMP

etch_cmp · CMP

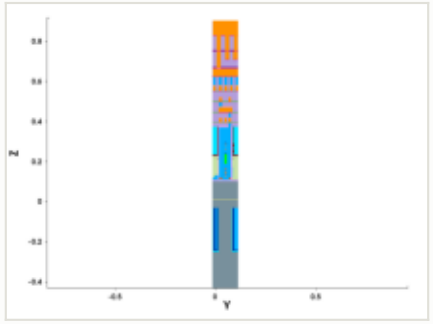
0.169	556.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

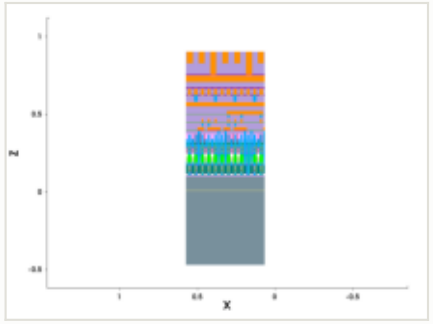
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



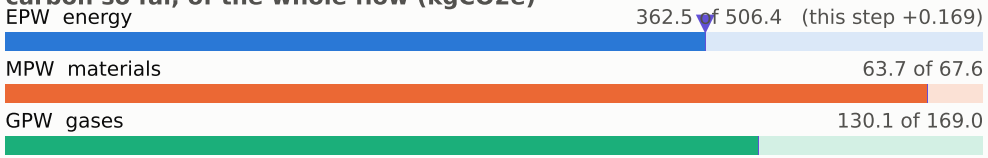
mask for this step

no mask for this step

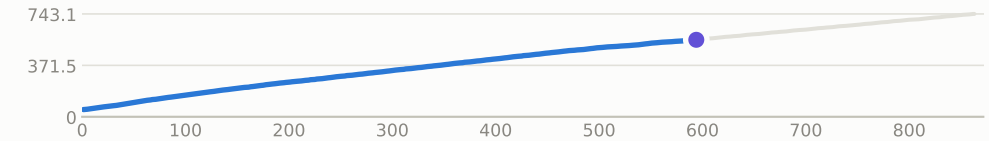
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

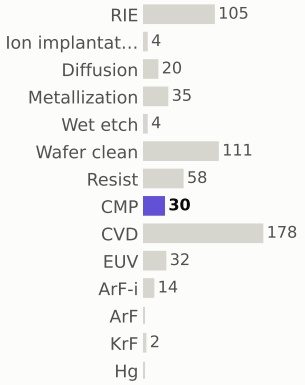
carbon so far, of the whole flow (kgCO2e)



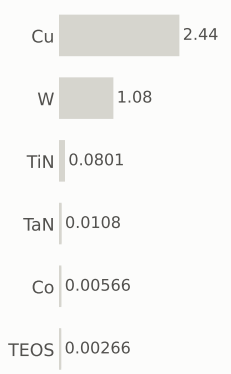
embodied carbon up to each step (kgCO2e per wafer)



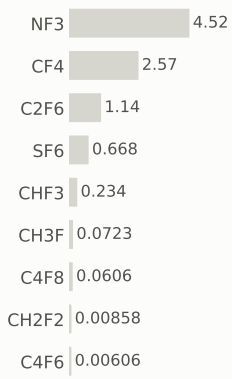
steps by process area



materials deposited (g)



gases emitted (g)



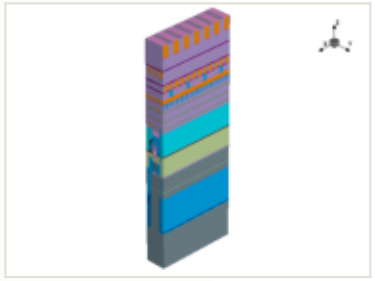
Step #6260: Metallization

depo_isotropic · Metallization

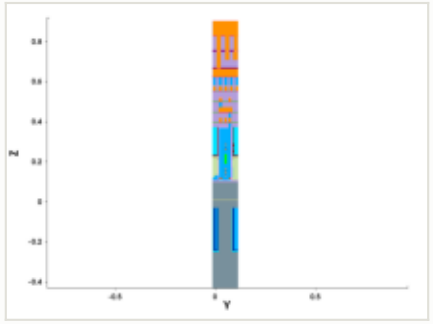
1.2	557.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

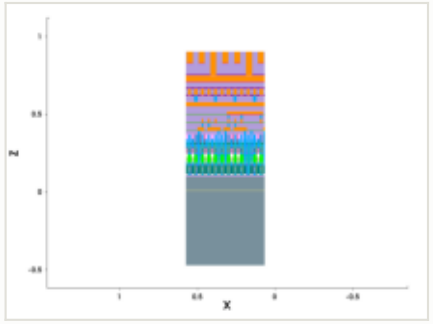
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

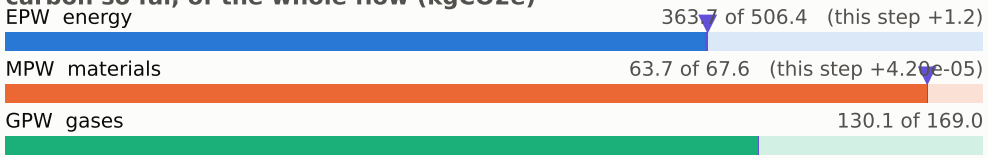
no mask for this step

this step

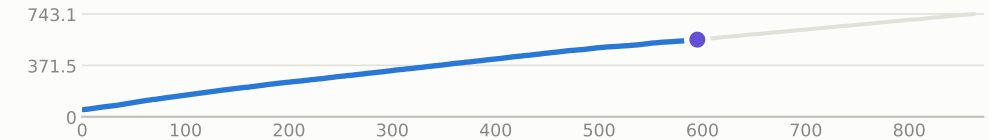
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

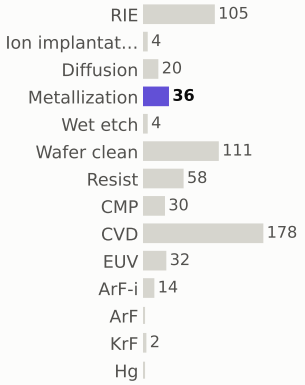
carbon so far, of the whole flow (kgCO2e)



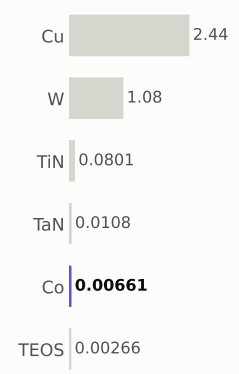
embodied carbon up to each step (kgCO2e per wafer)



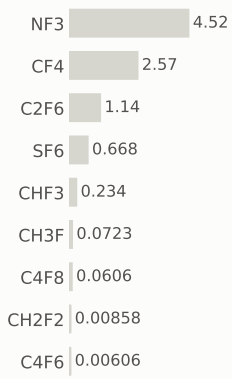
steps by process area



materials deposited (g)



gases emitted (g)



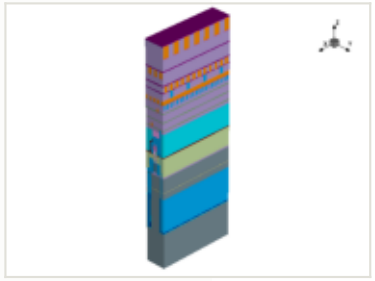
Step #6280: CVD

depo_fill · CVD

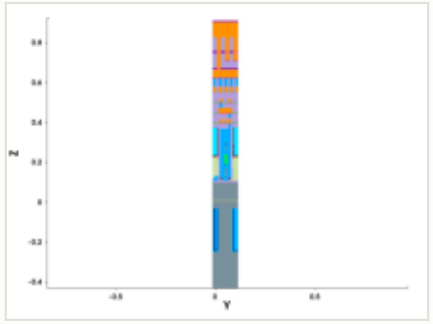
1.31	558.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

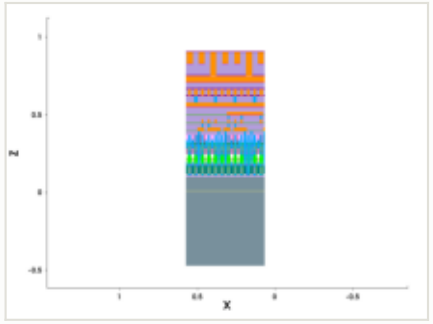
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

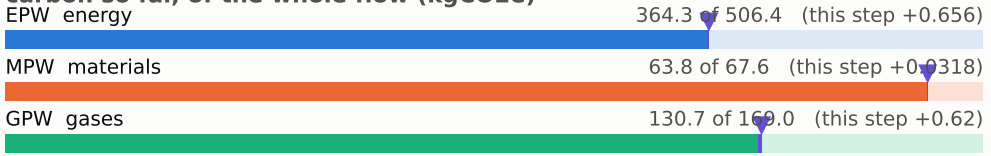
no mask for this step

this step

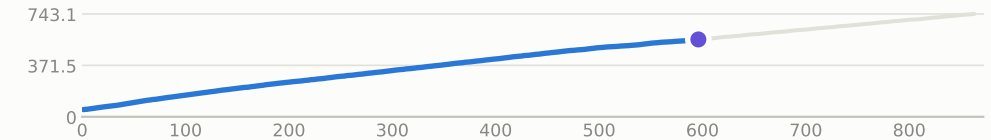
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

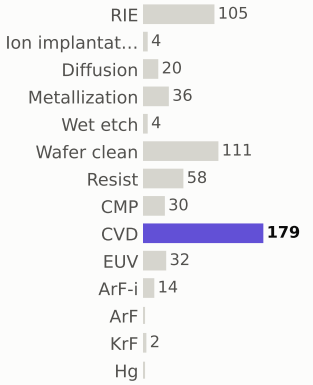
carbon so far, of the whole flow (kgCO2e)



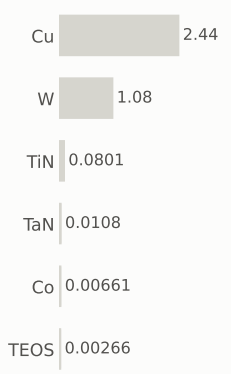
embodied carbon up to each step (kgCO2e per wafer)



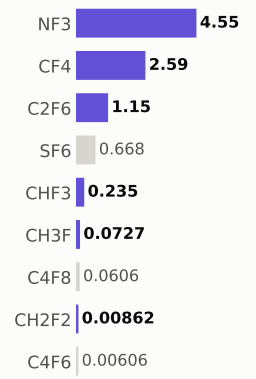
steps by process area



materials deposited (g)



gases emitted (g)



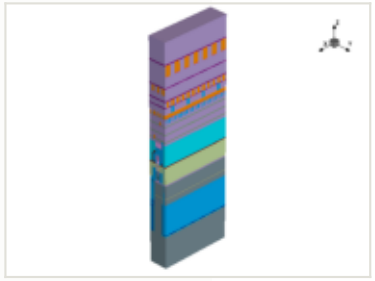
Step #6290: CVD

depo_fill · CVD

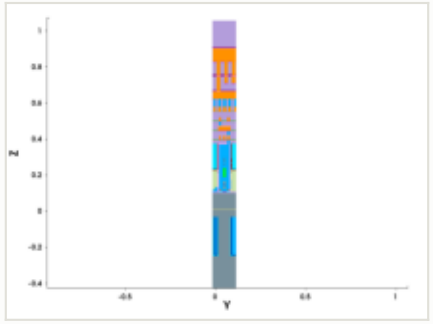
1.31	560.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

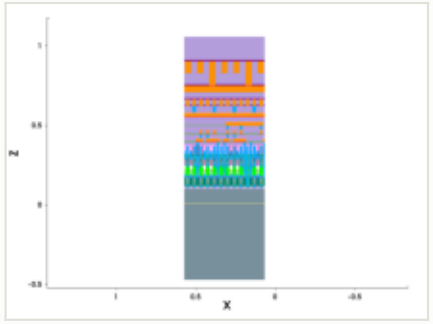
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

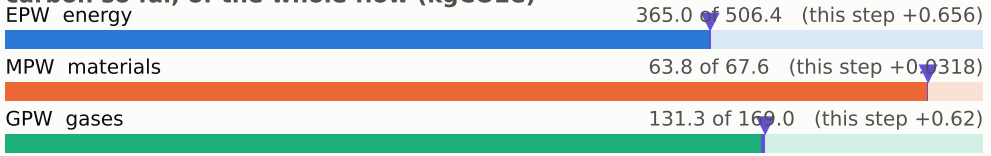
no mask for this step

this step

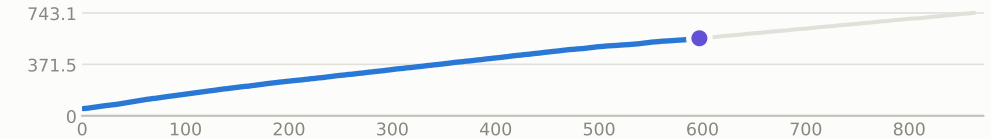
process area	CVD
material	—
thickness	0.146 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

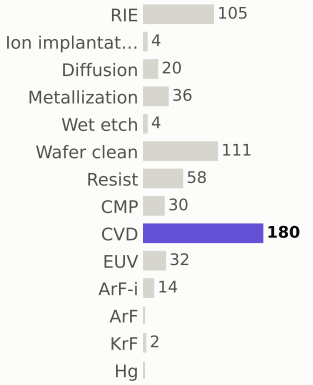
carbon so far, of the whole flow (kgCO2e)



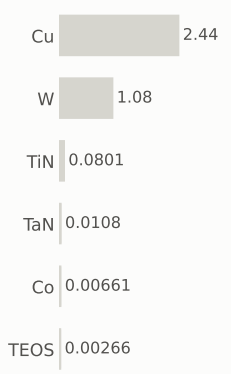
embodied carbon up to each step (kgCO2e per wafer)



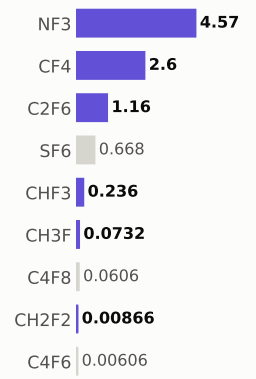
steps by process area



materials deposited (g)



gases emitted (g)



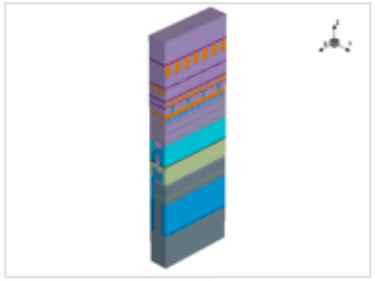
Step #6300: Diffusion

treatment · Diffusion

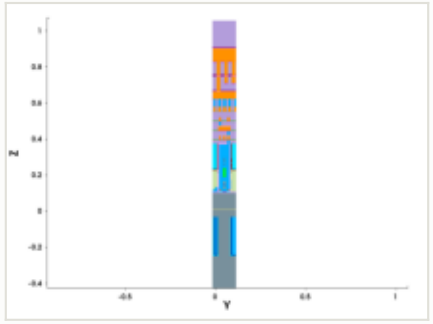
0.801	560.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

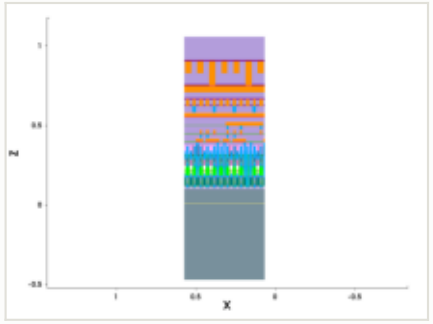
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

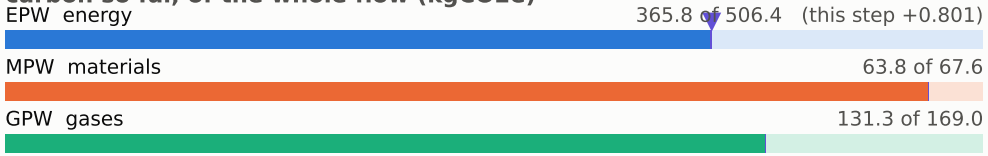
no mask for this step

this step

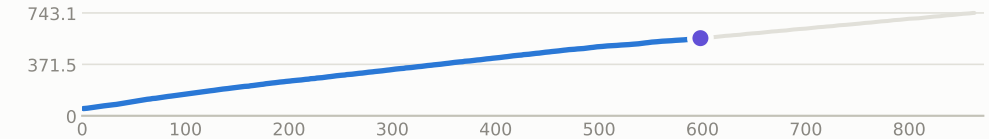
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

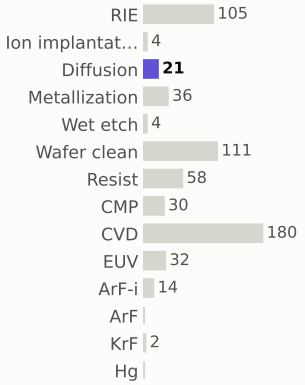
carbon so far, of the whole flow (kgCO2e)



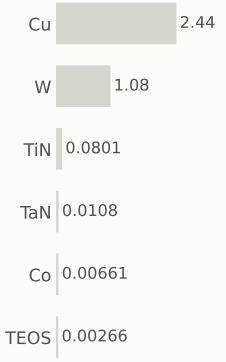
embodied carbon up to each step (kgCO2e per wafer)



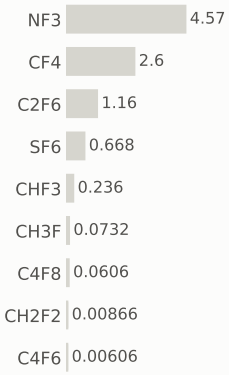
steps by process area



materials deposited (g)



gases emitted (g)



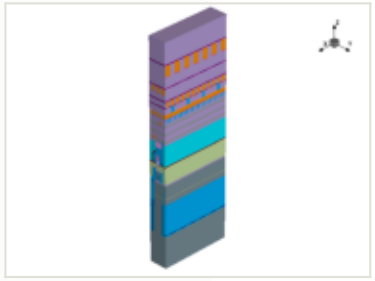
Step #6310: Wafer_clean

clean · Wafer_clean

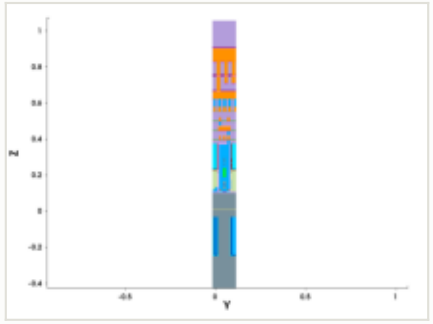
0.169	561.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

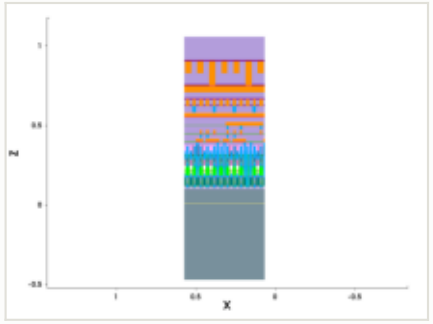
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

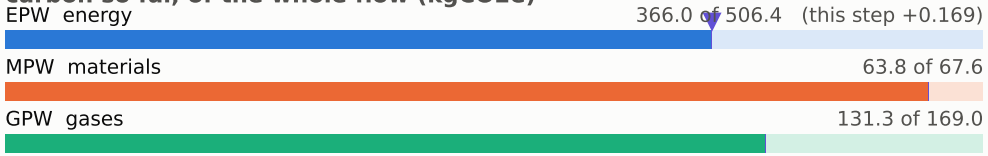
no mask for this step

this step

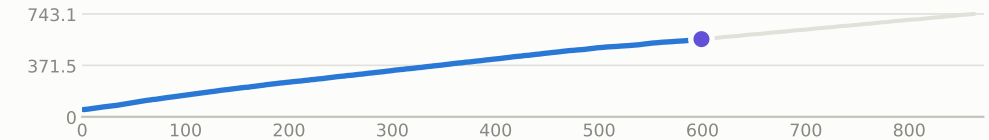
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

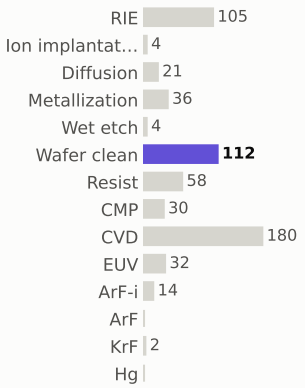
carbon so far, of the whole flow (kgCO2e)



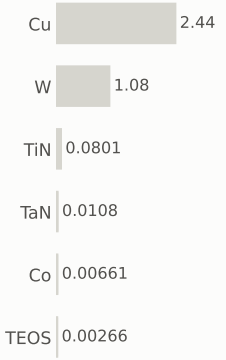
embodied carbon up to each step (kgCO2e per wafer)



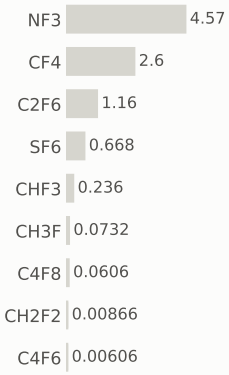
steps by process area



materials deposited (g)



gases emitted (g)



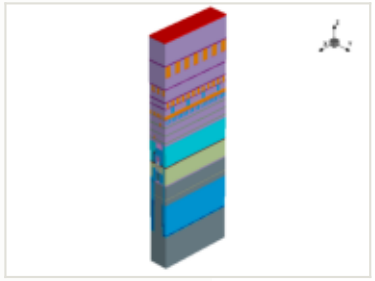
Step #6320: CVD

depo_fill · CVD

1.31	562.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

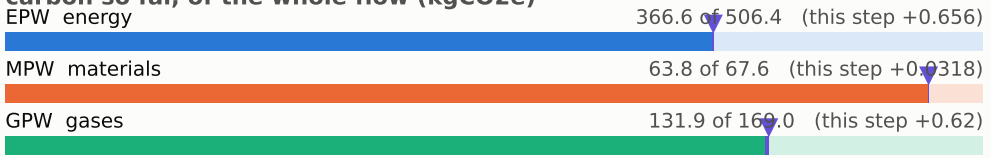


this step

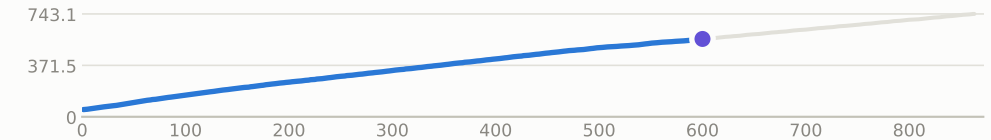
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

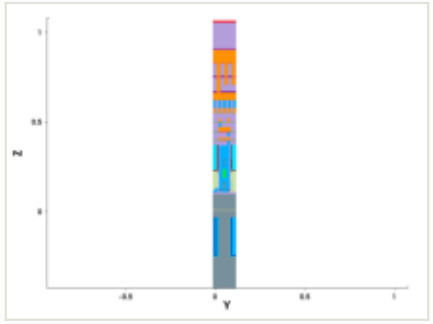
carbon so far, of the whole flow (kgCO2e)



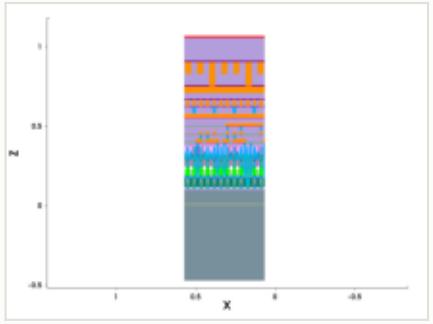
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



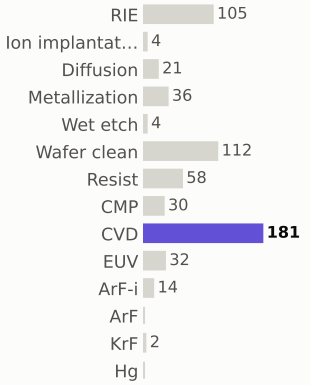
y cut at 0.049 µm



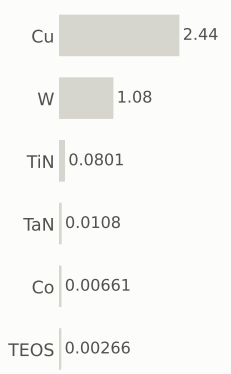
mask for this step

no mask for this step

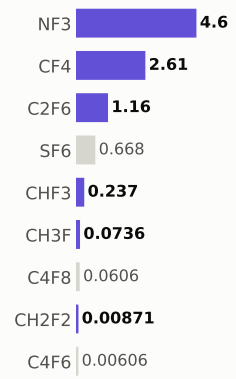
steps by process area



materials deposited (g)



gases emitted (g)



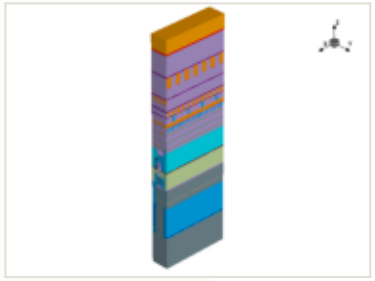
Step #6330: Resist

depo_fill · Resist

0.169	562.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

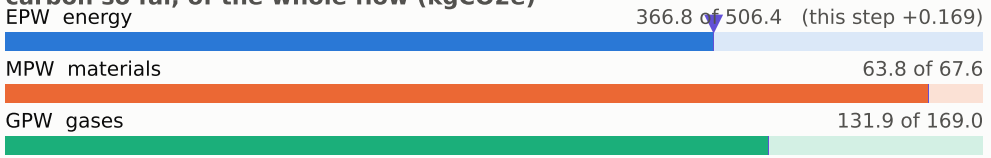


this step

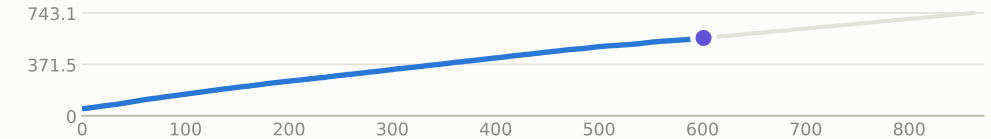
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

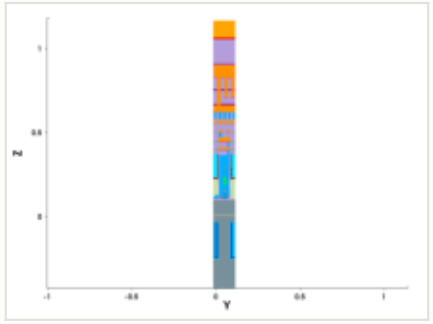
carbon so far, of the whole flow (kgCO2e)



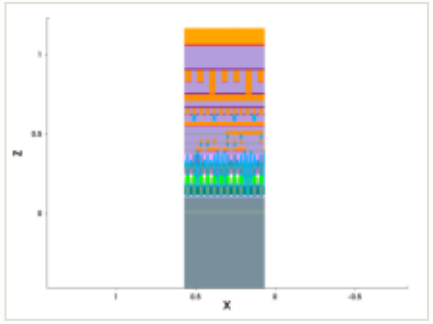
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



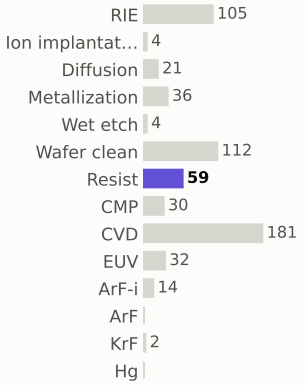
y cut at 0.049 µm



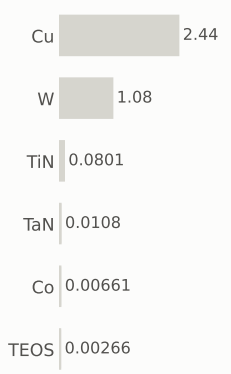
mask for this step

no mask for this step

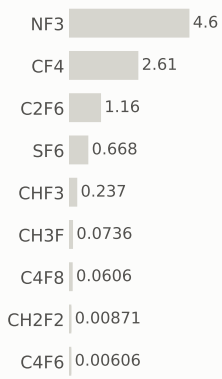
steps by process area



materials deposited (g)



gases emitted (g)



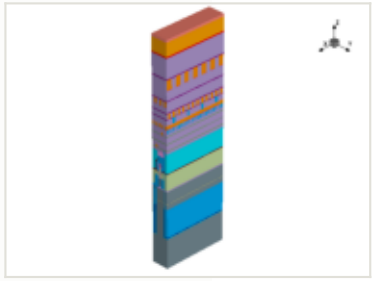
Step #6340: CVD

depo_fill · CVD

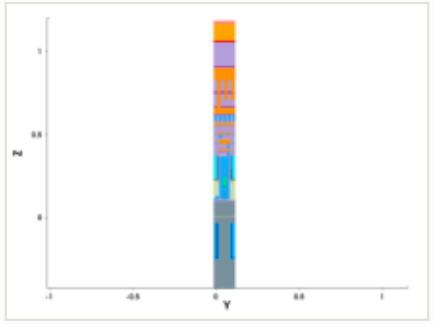
1.31	563.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

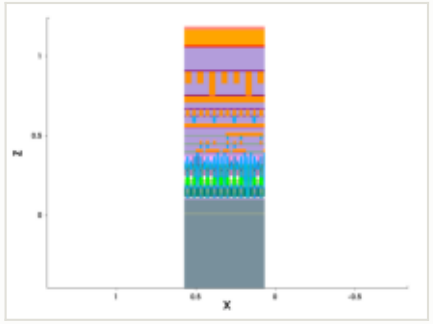
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

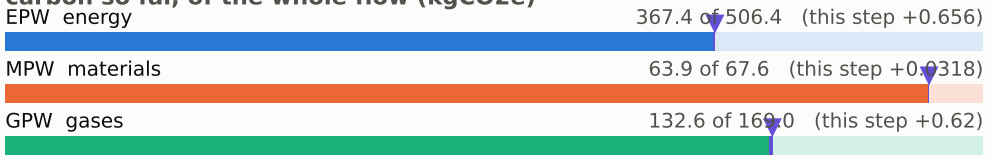
no mask for this step

this step

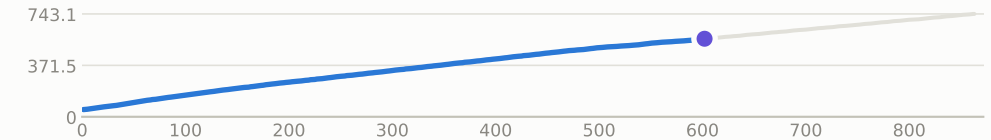
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

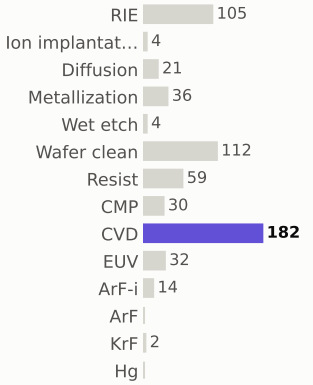
carbon so far, of the whole flow (kgCO2e)



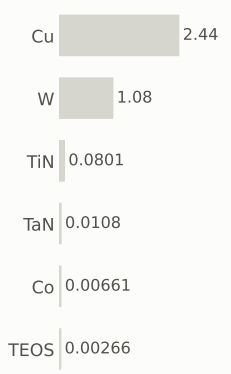
embodied carbon up to each step (kgCO2e per wafer)



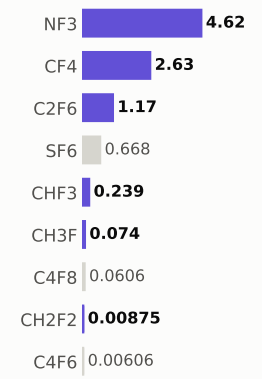
steps by process area



materials deposited (g)



gases emitted (g)



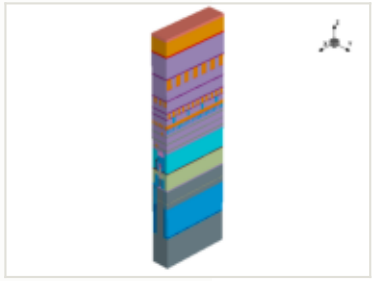
Step #6350: Wafer_clean

clean · Wafer_clean

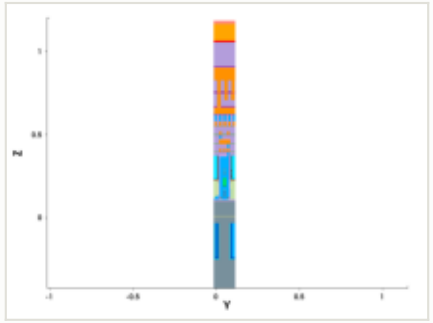
0.169	564.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

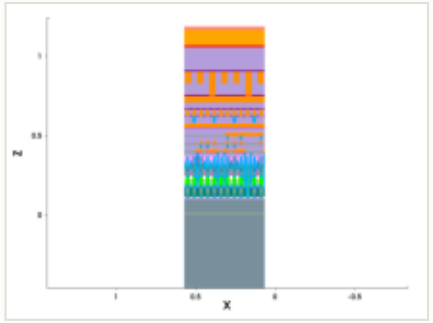
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

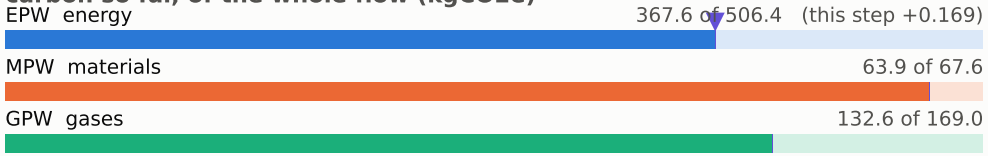
no mask for this step

this step

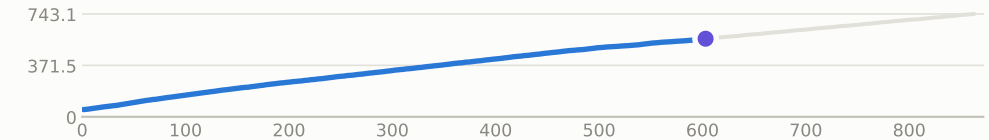
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

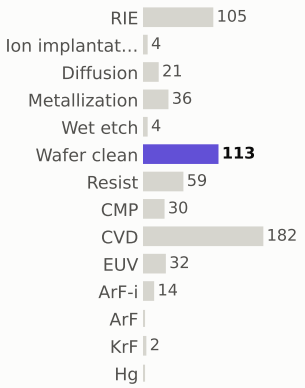
carbon so far, of the whole flow (kgCO2e)



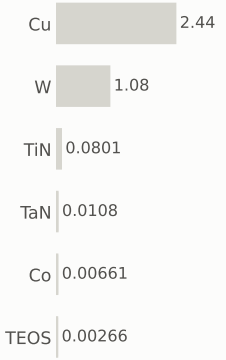
embodied carbon up to each step (kgCO2e per wafer)



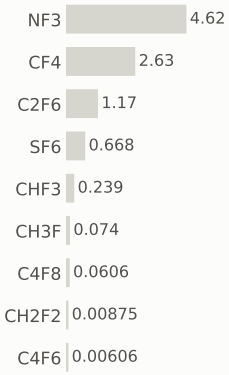
steps by process area



materials deposited (g)



gases emitted (g)



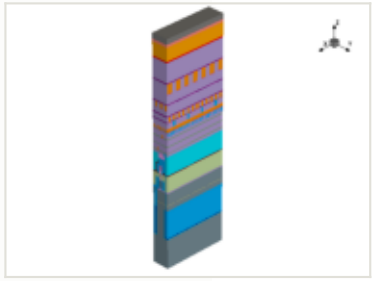
Step #6360: ArF-i

litho_patterning · ArF-i

0.217	564.3	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

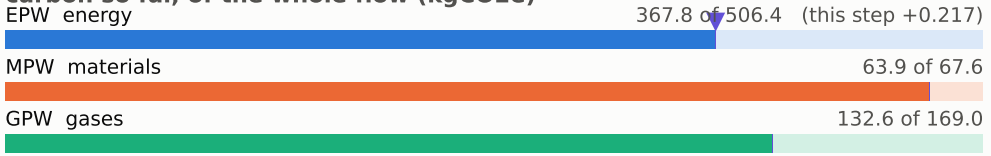


this step

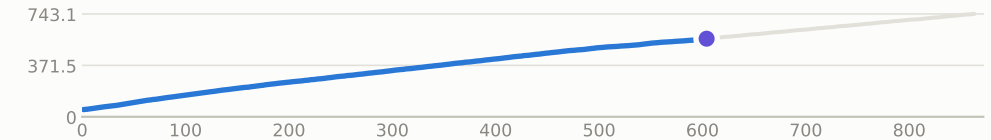
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

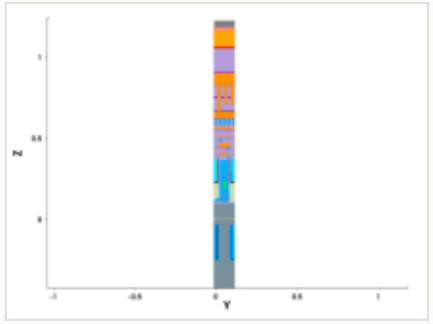
carbon so far, of the whole flow (kgCO2e)



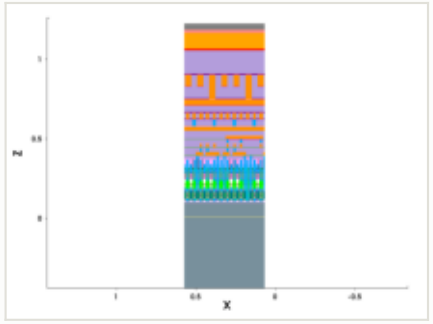
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

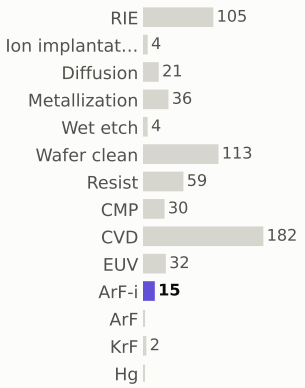


y cut at 0.049 µm

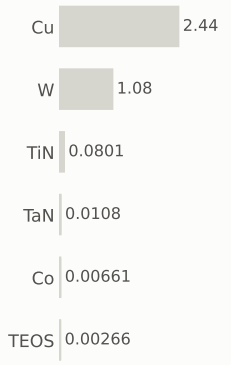


mask: layer 10521:0 (GDSII)

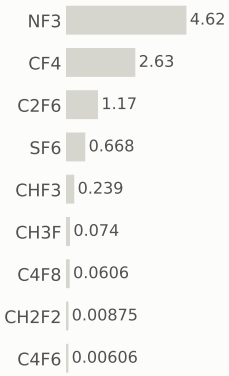
steps by process area



materials deposited (g)



gases emitted (g)



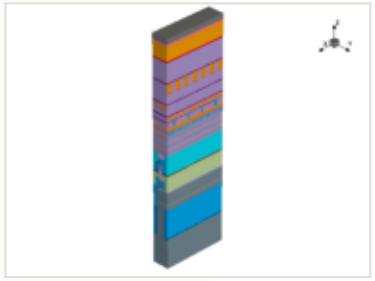
Step #6370: RIE

etch_anisotropic · RIE

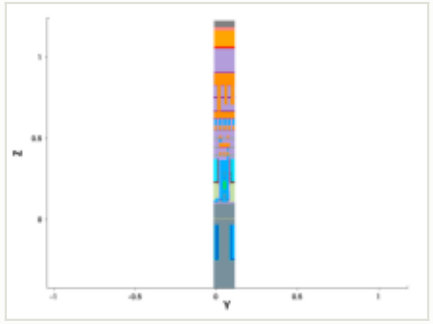
1.04	565.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

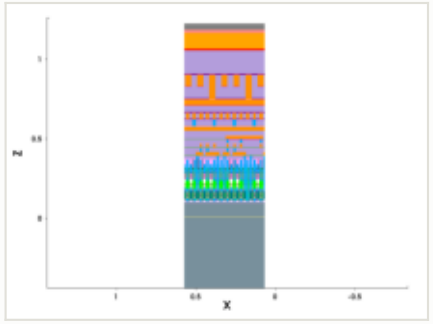
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

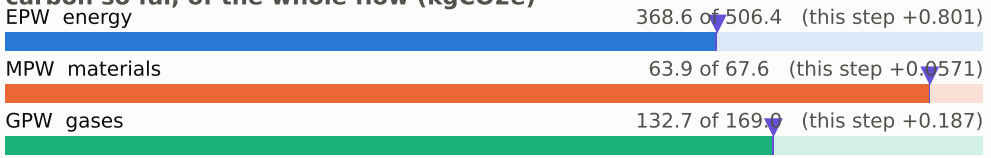
no mask for this step

this step

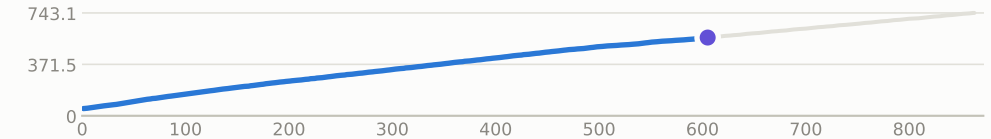
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

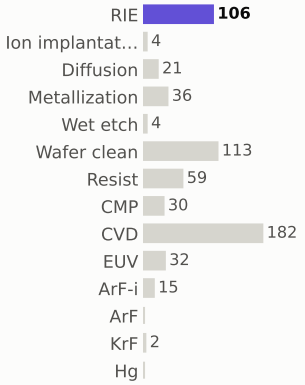
carbon so far, of the whole flow (kgCO2e)



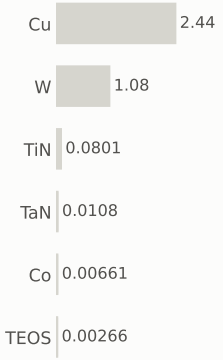
embodied carbon up to each step (kgCO2e per wafer)



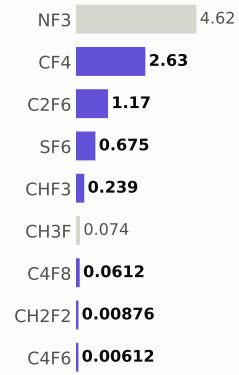
steps by process area



materials deposited (g)



gases emitted (g)



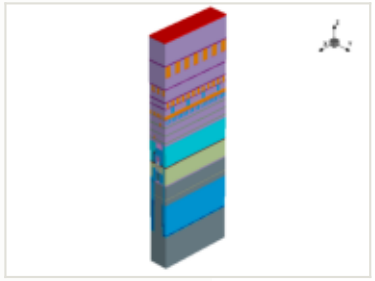
Step #6380: Wafer_clean

etch_strip · Wafer_clean

0.169	565.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

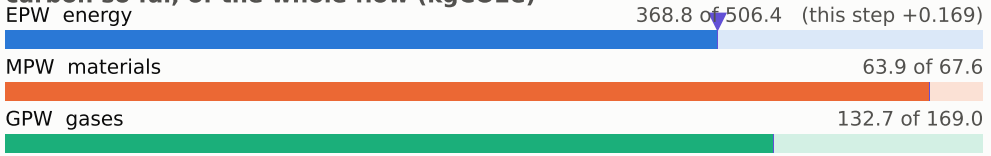


this step

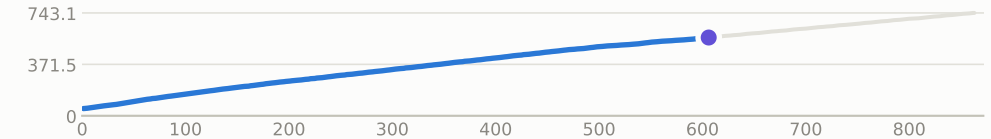
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

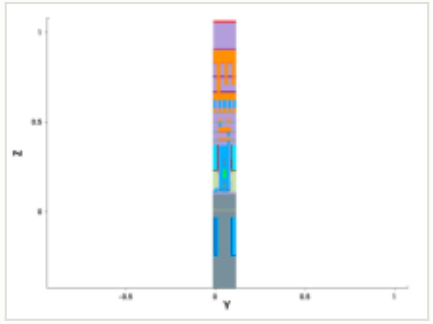
carbon so far, of the whole flow (kgCO2e)



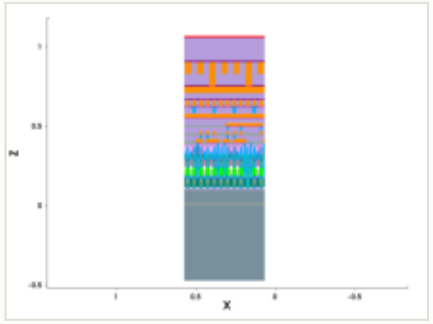
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



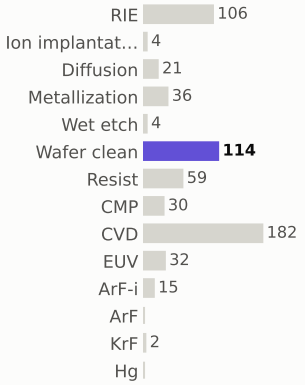
y cut at 0.049 µm



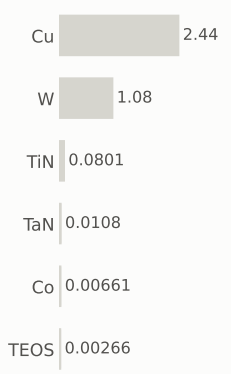
mask for this step

no mask for this step

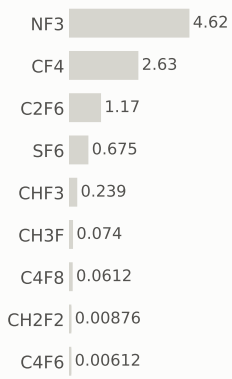
steps by process area



materials deposited (g)



gases emitted (g)



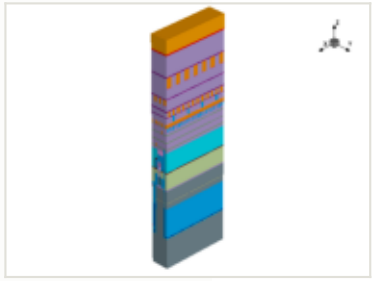
Step #6390: Resist

depo_fill · Resist

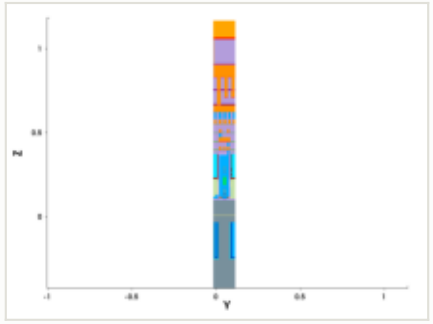
0.169	565.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

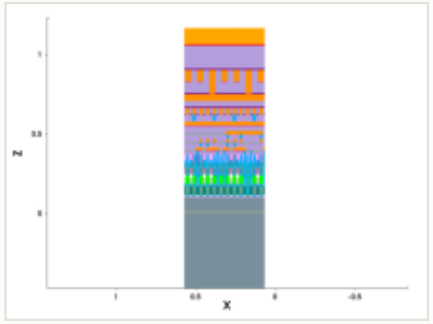
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



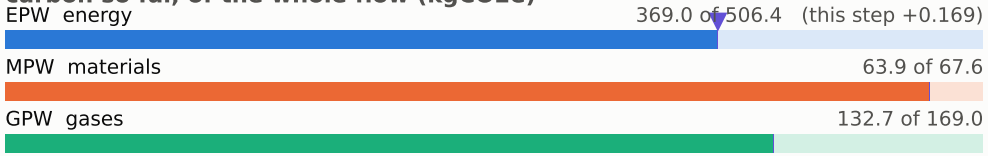
mask for this step

no mask for this step

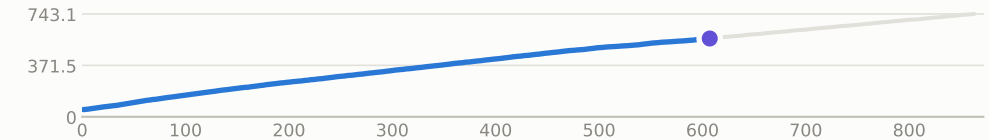
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

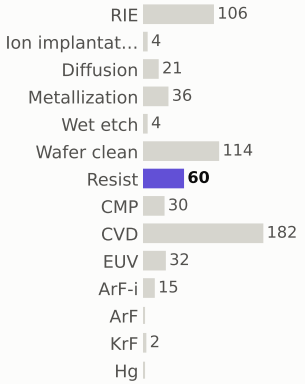
carbon so far, of the whole flow (kgCO2e)



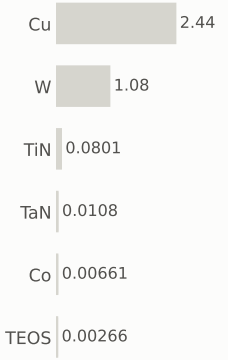
embodied carbon up to each step (kgCO2e per wafer)



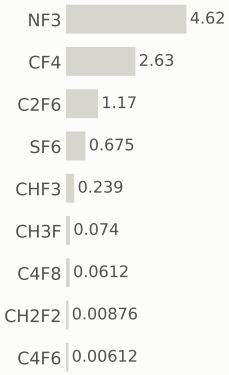
steps by process area



materials deposited (g)



gases emitted (g)



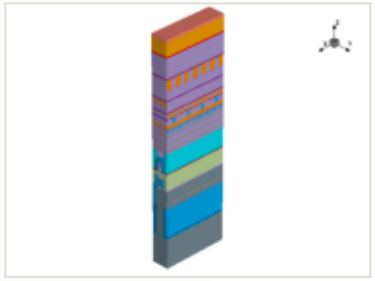
Step #6400: CVD

depo_fill · CVD

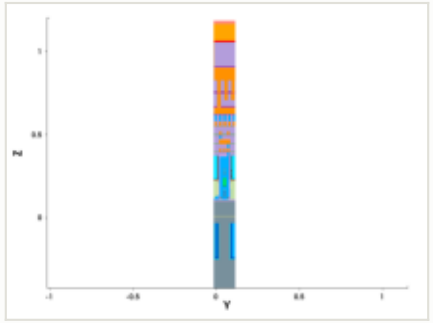
1.31	567.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

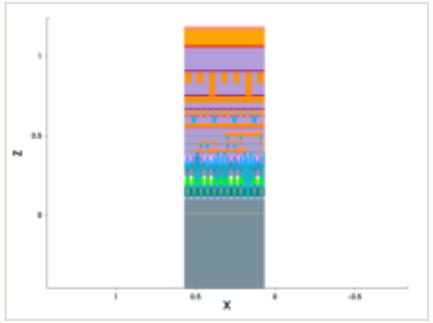
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

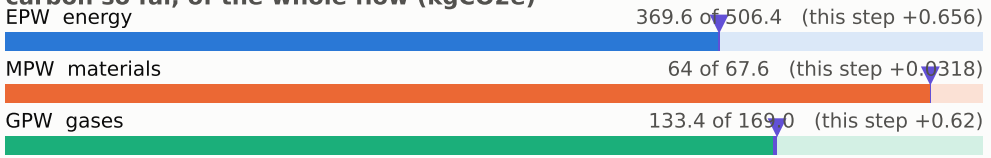
no mask for this step

this step

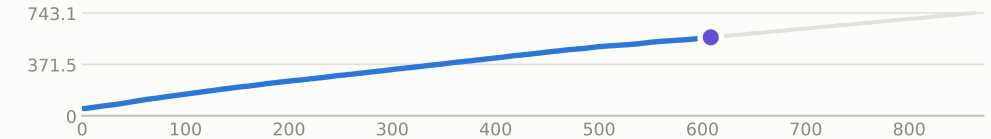
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

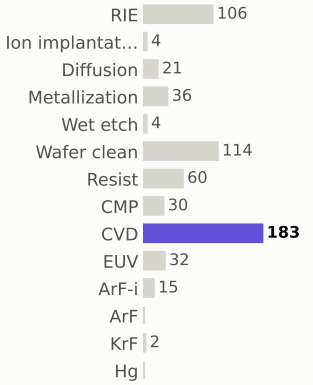
carbon so far, of the whole flow (kgCO2e)



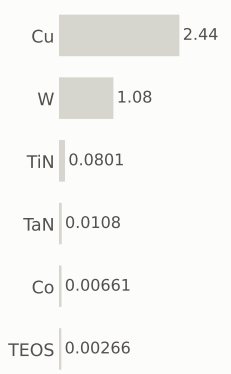
embodied carbon up to each step (kgCO2e per wafer)



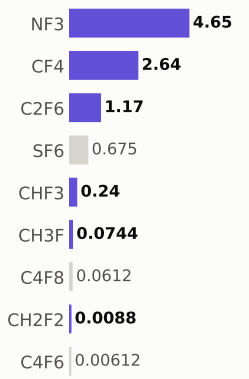
steps by process area



materials deposited (g)



gases emitted (g)



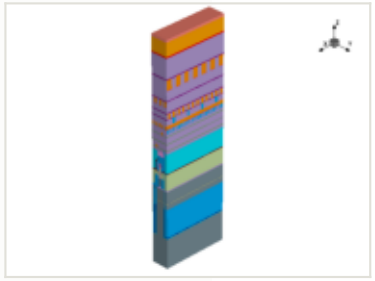
Step #6410: Wafer_clean

clean · Wafer_clean

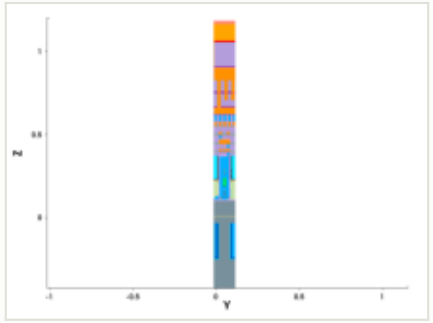
0.169	567.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

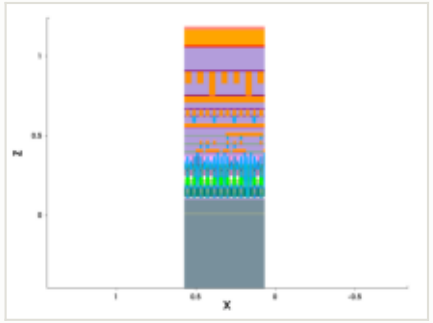
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

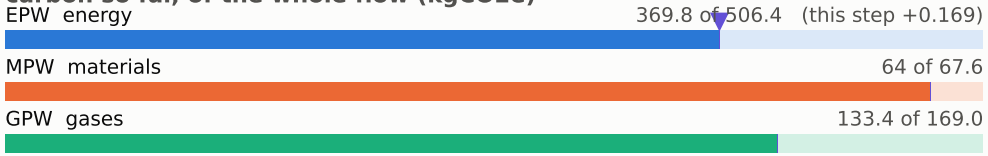
no mask for this step

this step

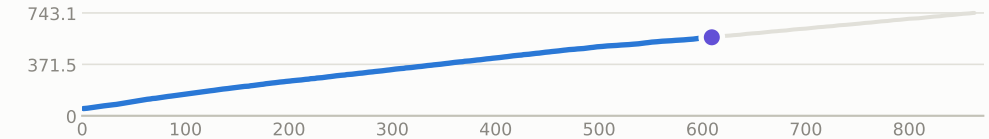
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

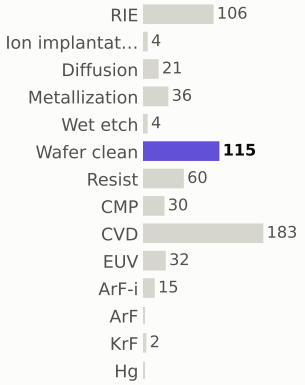
carbon so far, of the whole flow (kgCO2e)



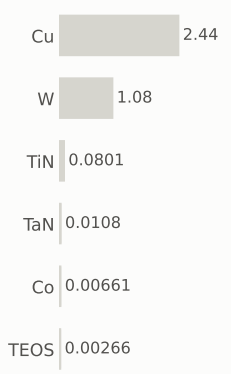
embodied carbon up to each step (kgCO2e per wafer)



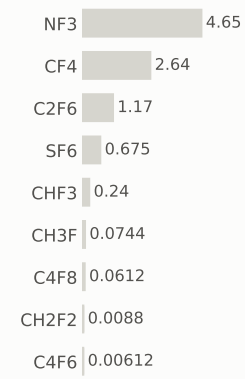
steps by process area



materials deposited (g)



gases emitted (g)



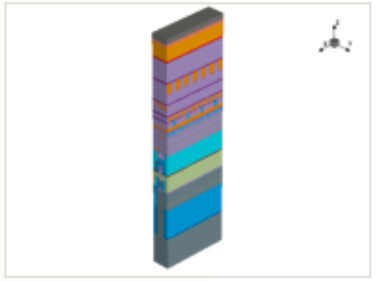
Step #6420: ArF-i

litho_patterning · ArF-i

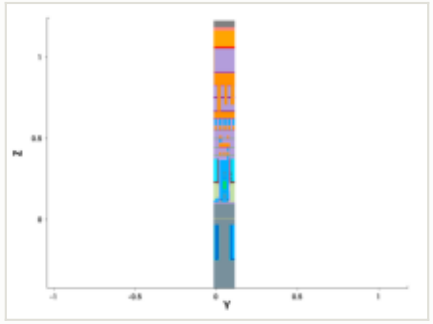
0.217	567.3	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

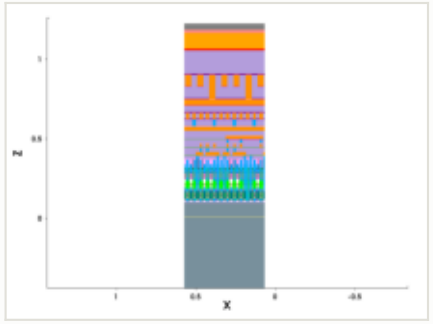
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask: layer 10522:0 (GDSII)

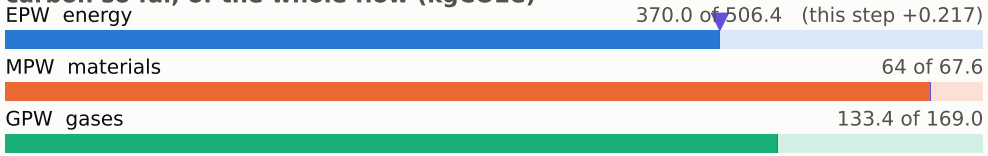
nothing in the clip window

this step

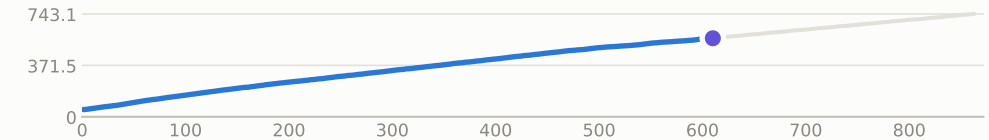
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

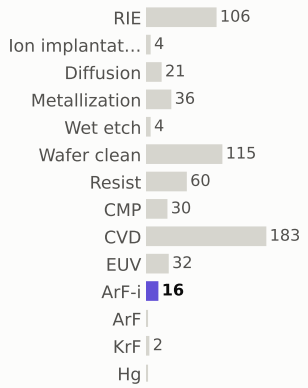
carbon so far, of the whole flow (kgCO2e)



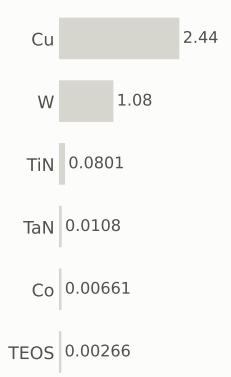
embodied carbon up to each step (kgCO2e per wafer)



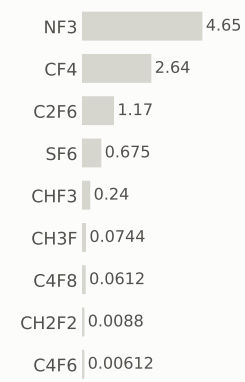
steps by process area



materials deposited (g)



gases emitted (g)



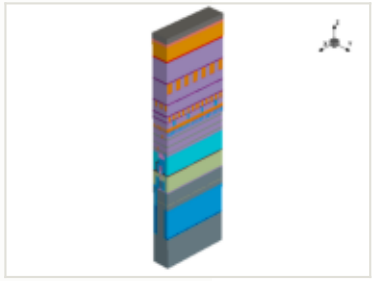
Step #6430: RIE

etch_anisotropic · RIE

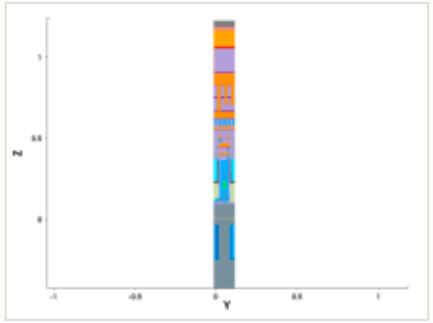
1.04	568.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

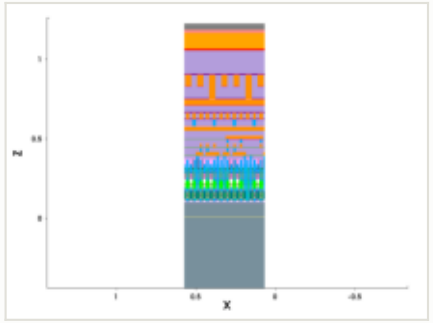
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

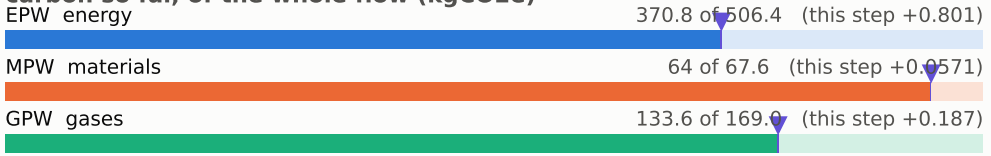
no mask for this step

this step

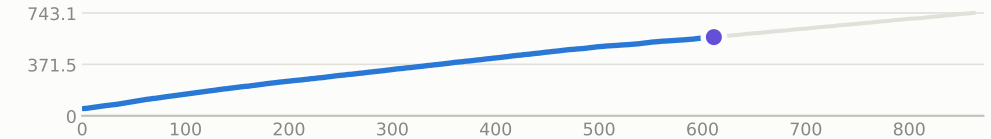
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

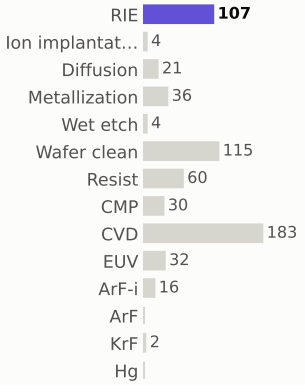
carbon so far, of the whole flow (kgCO2e)



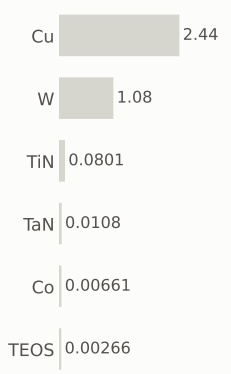
embodied carbon up to each step (kgCO2e per wafer)



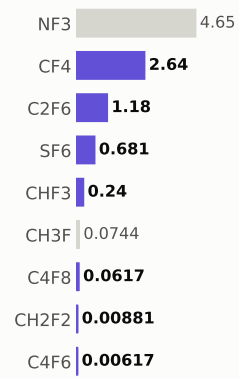
steps by process area



materials deposited (g)



gases emitted (g)



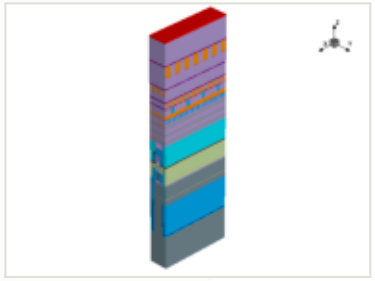
Step #6440: Wafer_clean

etch_strip · Wafer_clean

0.169	568.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

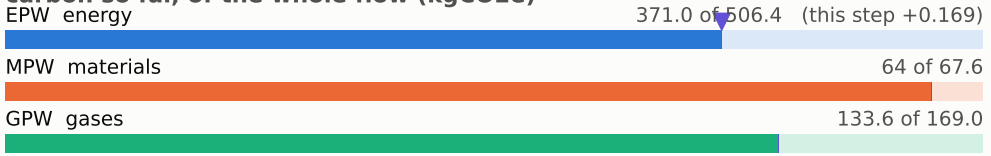


this step

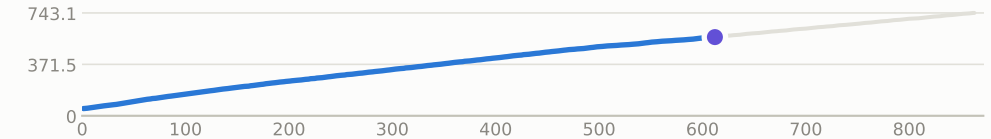
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

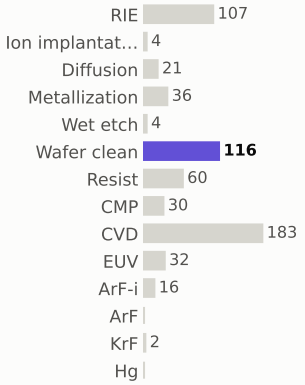
carbon so far, of the whole flow (kgCO2e)



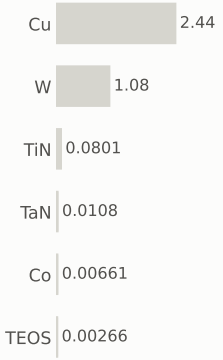
embodied carbon up to each step (kgCO2e per wafer)



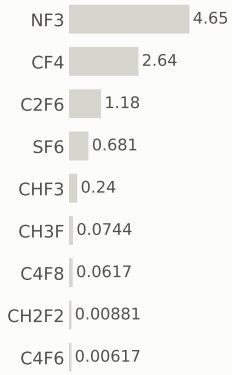
steps by process area



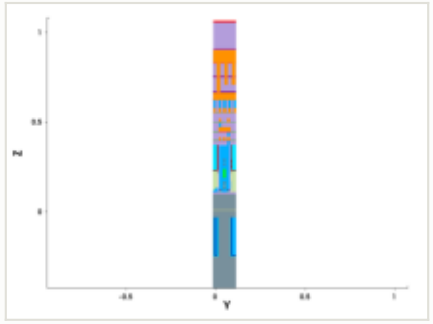
materials deposited (g)



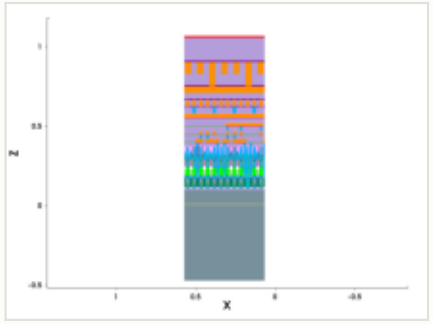
gases emitted (g)



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

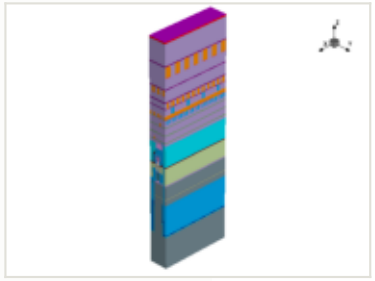
Step #6450: Metallization

depo_ellipsoidal · Metallization

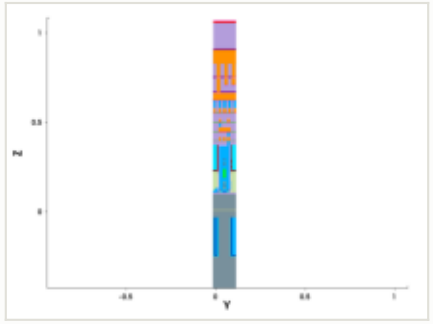
1.2	569.8	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

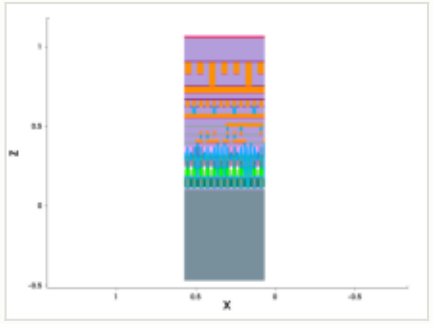
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

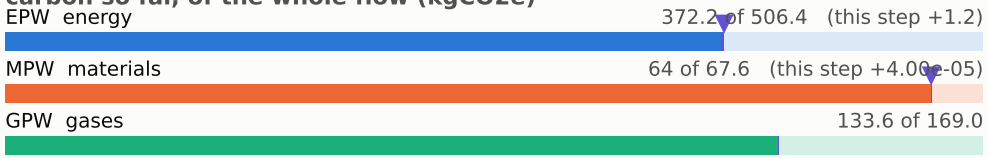
no mask for this step

this step

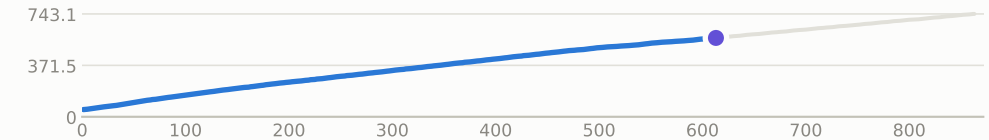
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

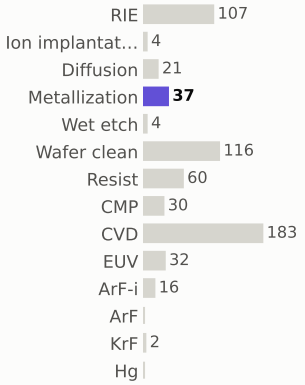
carbon so far, of the whole flow (kgCO2e)



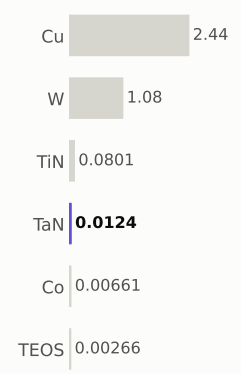
embodied carbon up to each step (kgCO2e per wafer)



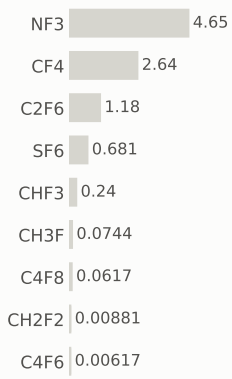
steps by process area



materials deposited (g)



gases emitted (g)



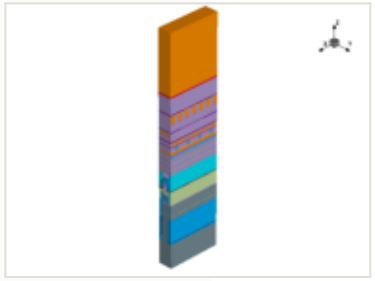
Step #6460: Metallization

depo_fill · Metallization

1.21	571.0	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

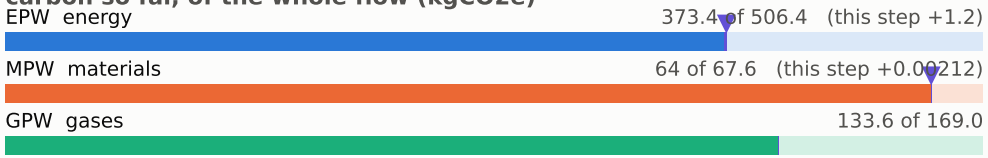


this step

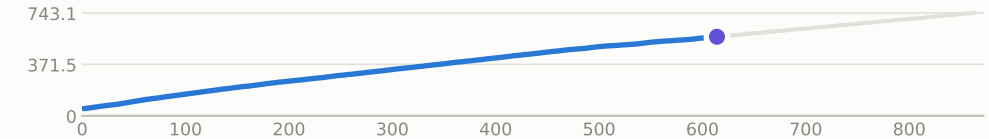
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

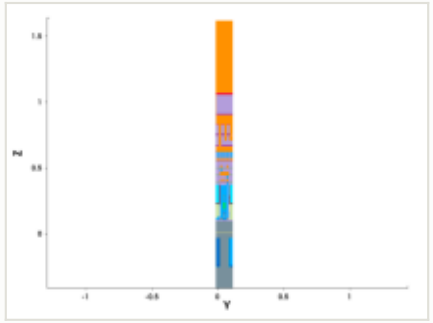
carbon so far, of the whole flow (kgCO2e)



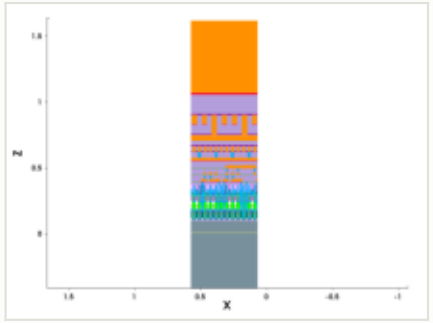
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



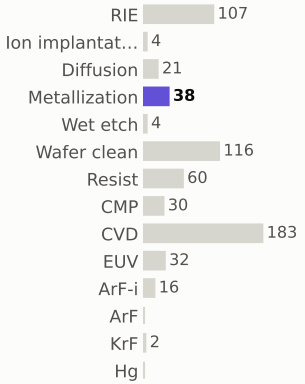
y cut at 0.049 µm



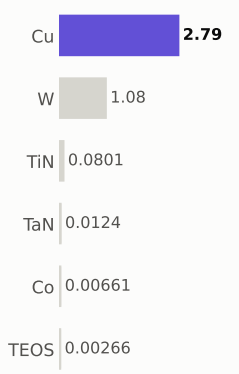
mask for this step

no mask for this step

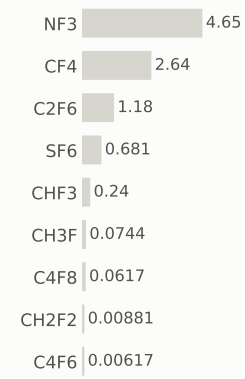
steps by process area



materials deposited (g)



gases emitted (g)



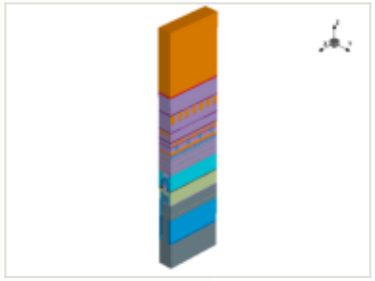
Step #6470: Diffusion

treatment · Diffusion

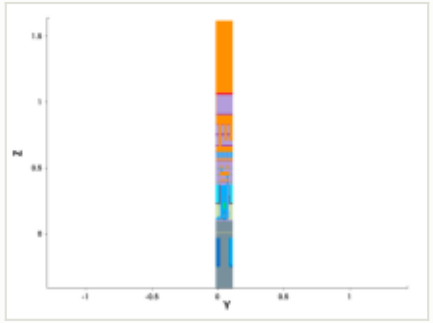
0.801	571.8	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

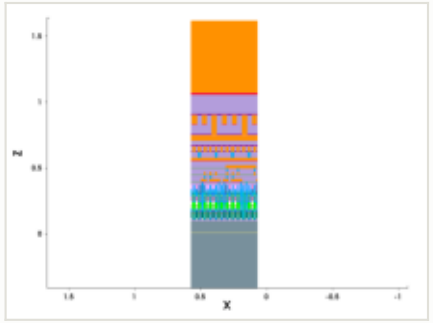
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

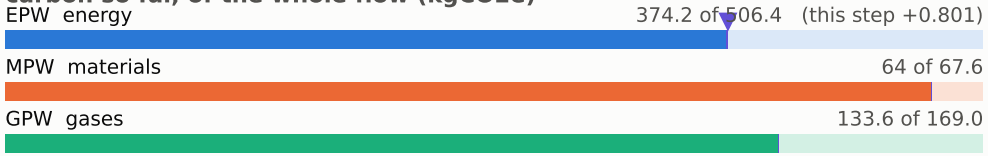
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

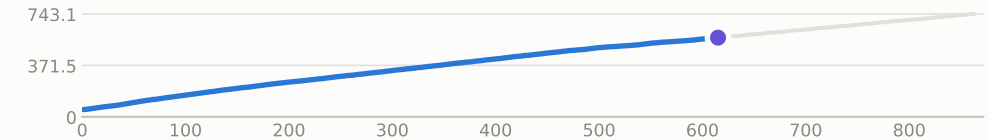
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

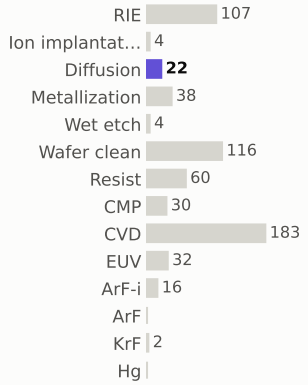
carbon so far, of the whole flow (kgCO2e)



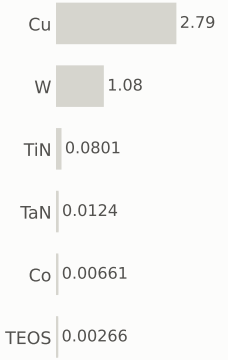
embodied carbon up to each step (kgCO2e per wafer)



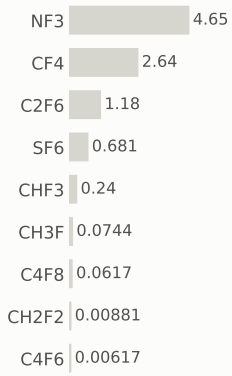
steps by process area



materials deposited (g)



gases emitted (g)



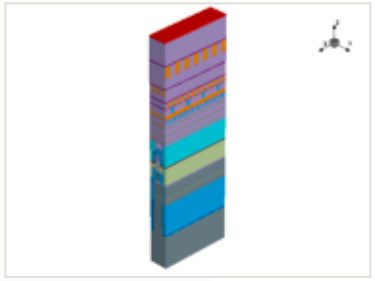
Step #6480: CMP

etch_cmp · CMP

0.169	571.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

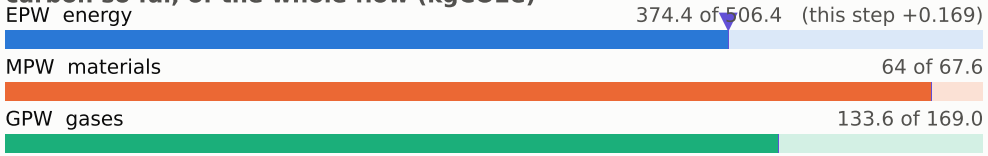


this step

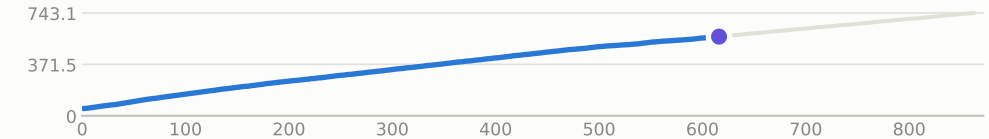
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

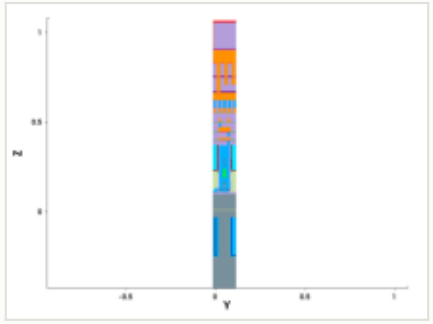
carbon so far, of the whole flow (kgCO2e)



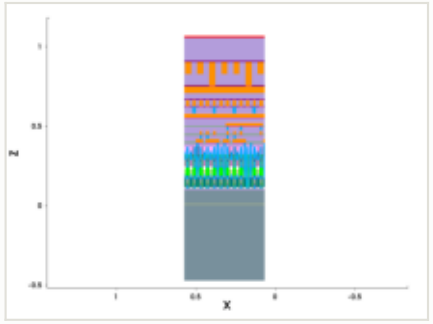
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



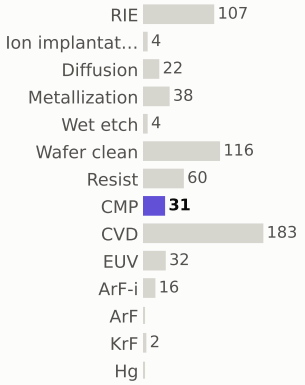
y cut at 0.049 µm



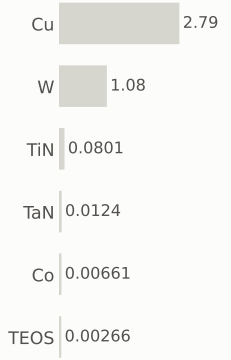
mask for this step

no mask for this step

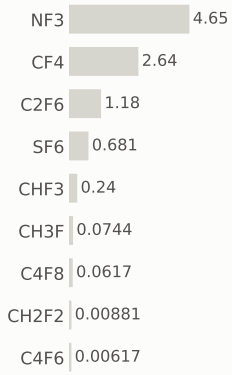
steps by process area



materials deposited (g)



gases emitted (g)



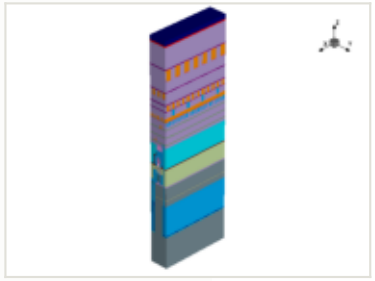
Step #6490: CVD

depo_fill · CVD

1.31	573.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

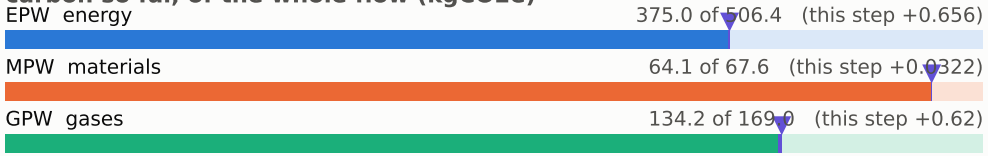


this step

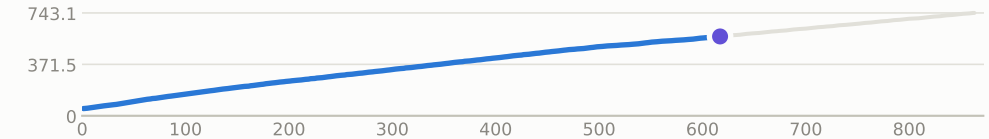
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

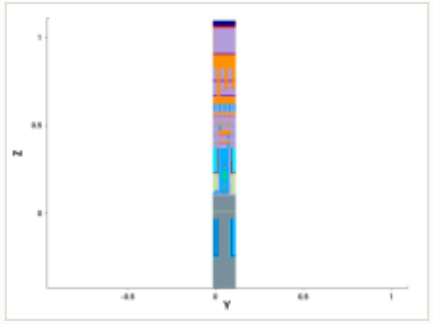
carbon so far, of the whole flow (kgCO2e)



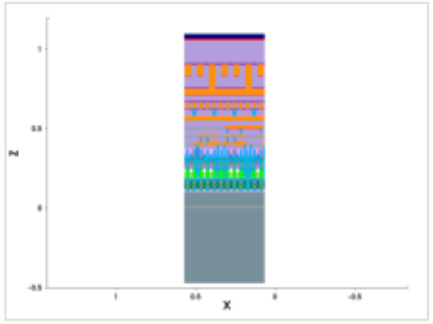
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



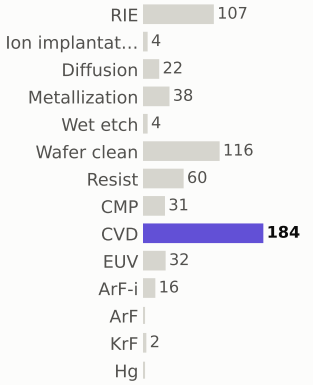
y cut at 0.049 µm



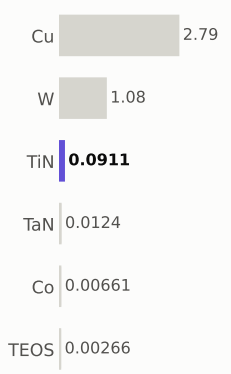
mask for this step

no mask for this step

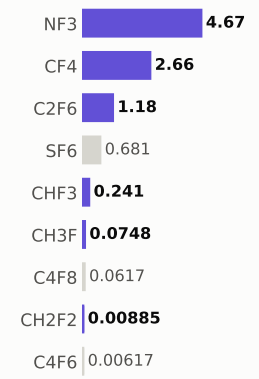
steps by process area



materials deposited (g)



gases emitted (g)



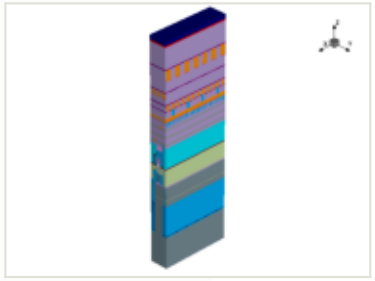
Step #6500: Wafer_clean

clean · Wafer_clean

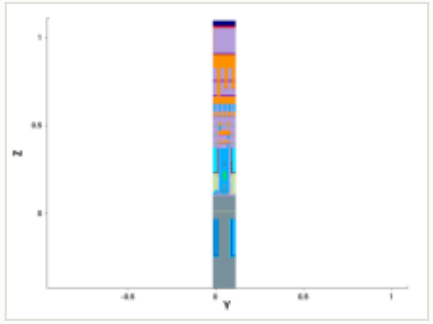
0.169	573.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

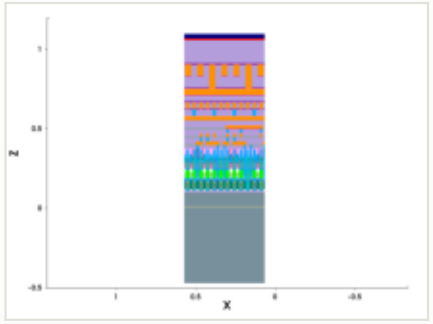
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

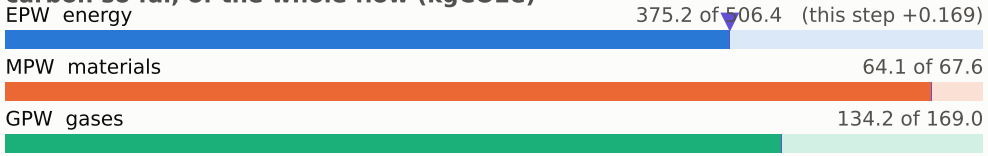
no mask for this step

this step

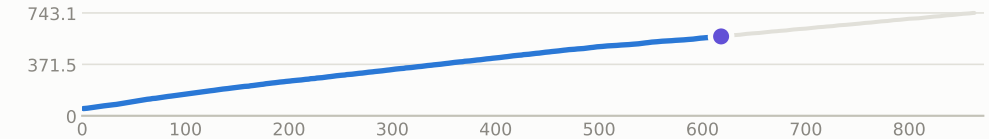
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

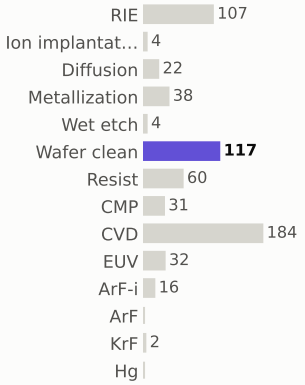
carbon so far, of the whole flow (kgCO2e)



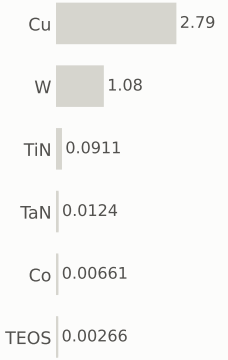
embodied carbon up to each step (kgCO2e per wafer)



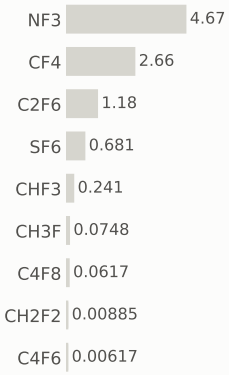
steps by process area



materials deposited (g)



gases emitted (g)



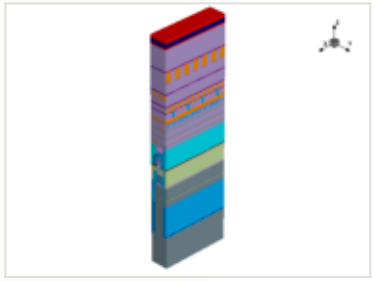
Step #6510: CVD

depo_fill · CVD

1.31	574.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

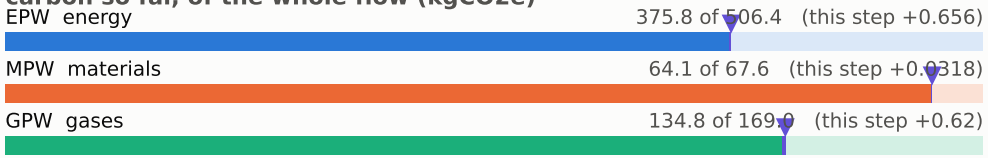


this step

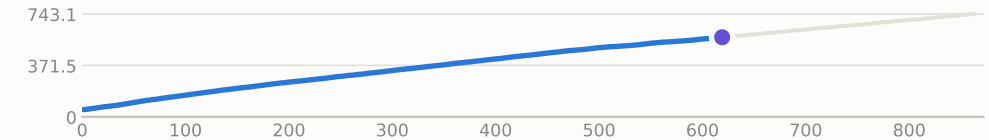
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

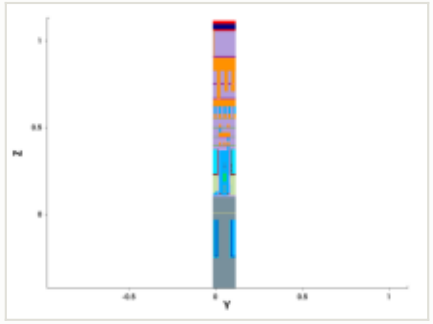
carbon so far, of the whole flow (kgCO2e)



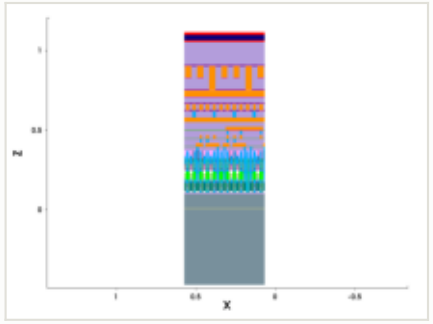
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



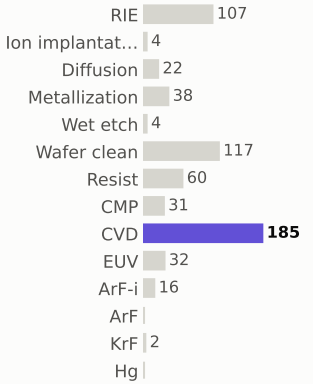
y cut at 0.049 µm



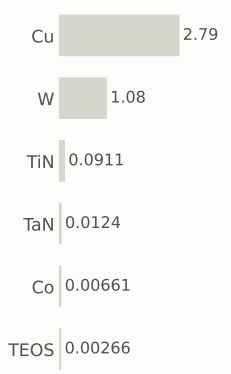
mask for this step

no mask for this step

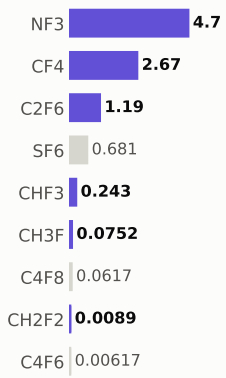
steps by process area



materials deposited (g)



gases emitted (g)



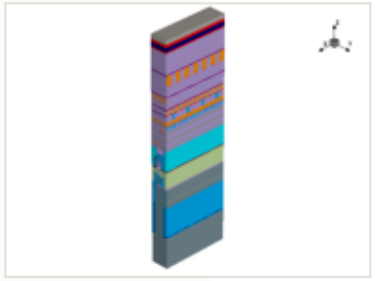
Step #6520: CVD

depo_fill · CVD

1.31	576.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

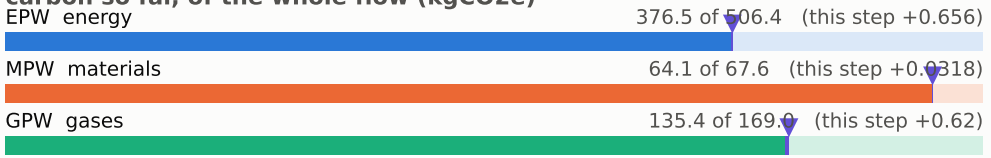


this step

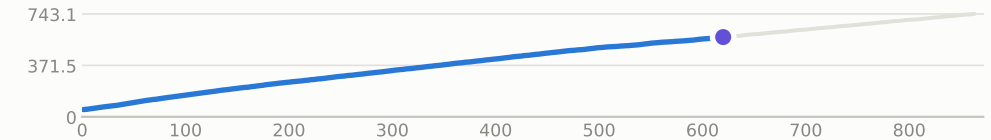
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

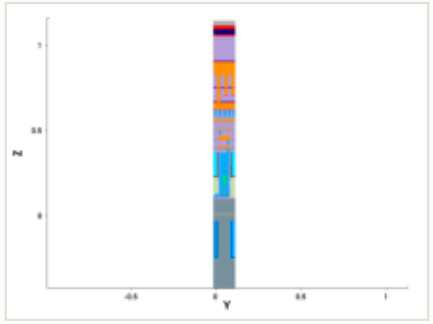
carbon so far, of the whole flow (kgCO2e)



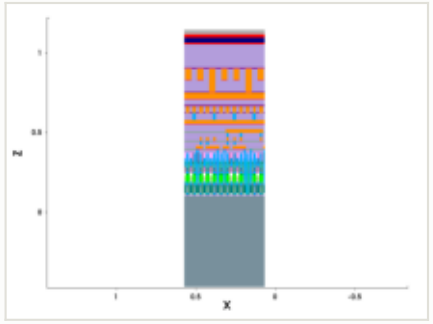
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



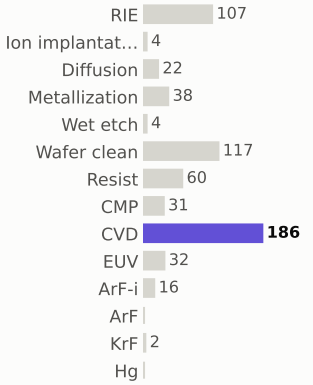
y cut at 0.049 µm



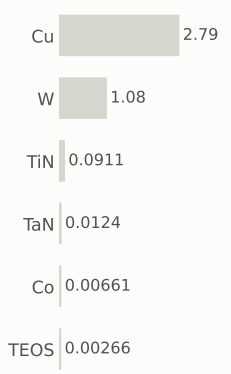
mask for this step

no mask for this step

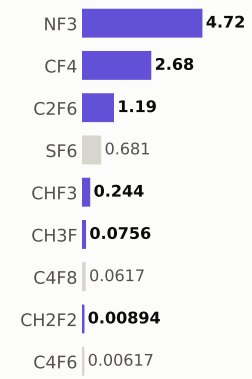
steps by process area



materials deposited (g)



gases emitted (g)



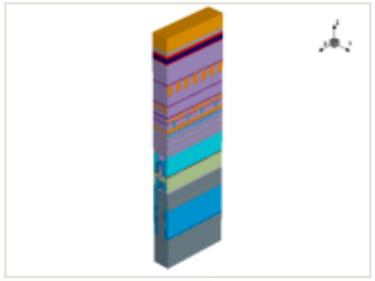
Step #6530: Resist

depo_fill · Resist

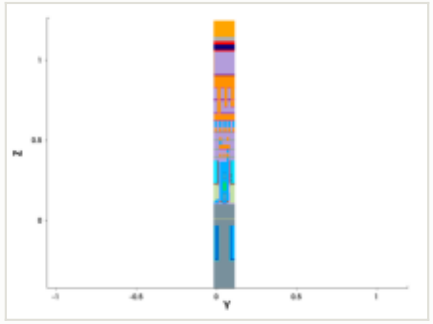
0.169	576.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

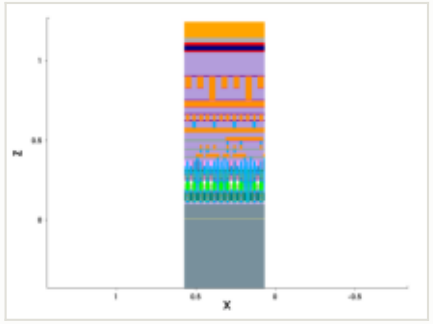
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



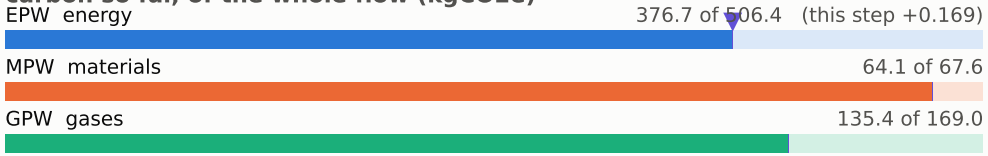
mask for this step

no mask for this step

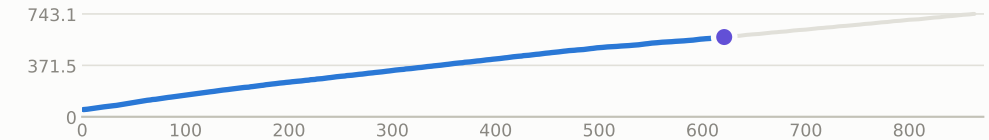
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

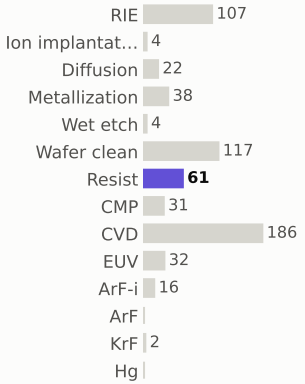
carbon so far, of the whole flow (kgCO2e)



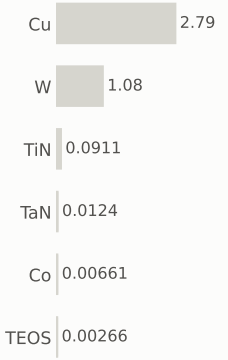
embodied carbon up to each step (kgCO2e per wafer)



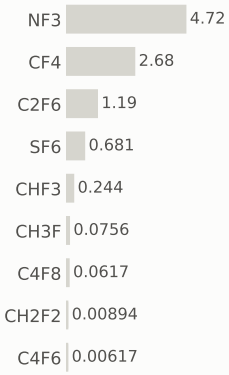
steps by process area



materials deposited (g)



gases emitted (g)



Step #6540: CVD

depo_fill · CVD

1.31	577.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

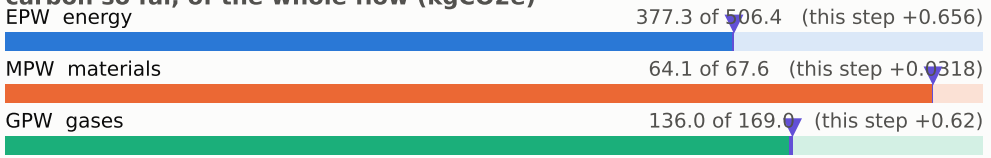
zoomed to fit

this step

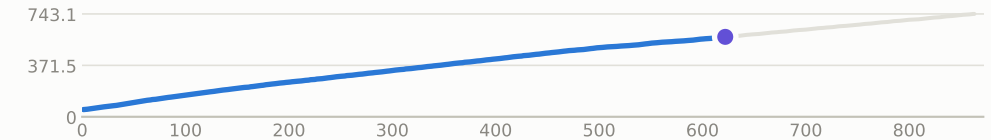
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

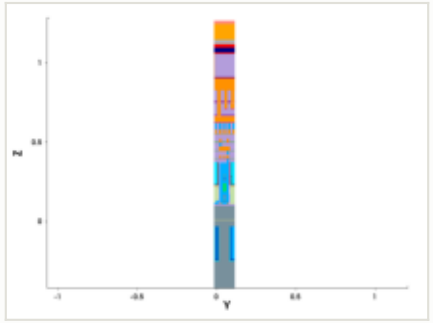
carbon so far, of the whole flow (kgCO2e)



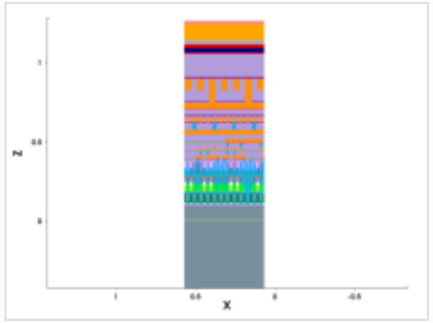
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



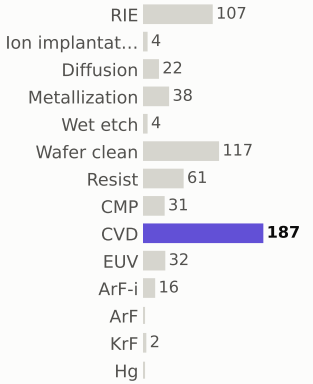
y cut at 0.049 µm



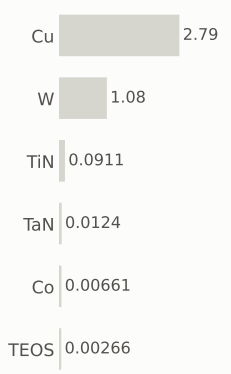
mask for this step

no mask for this step

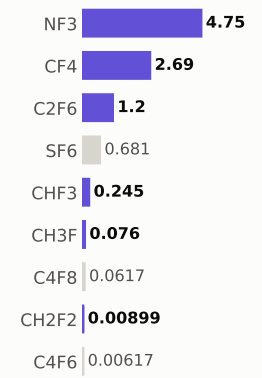
steps by process area



materials deposited (g)



gases emitted (g)



Step #6550: Wafer_clean

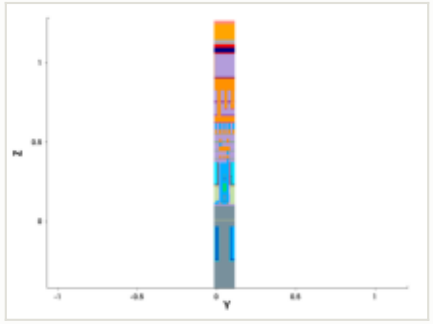
clean · Wafer_clean

0.169	577.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

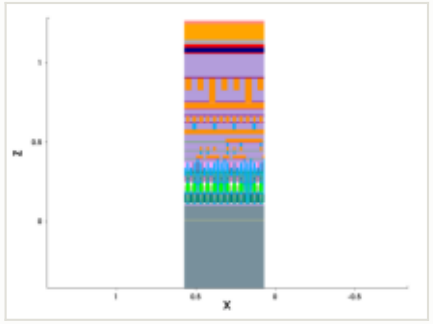
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

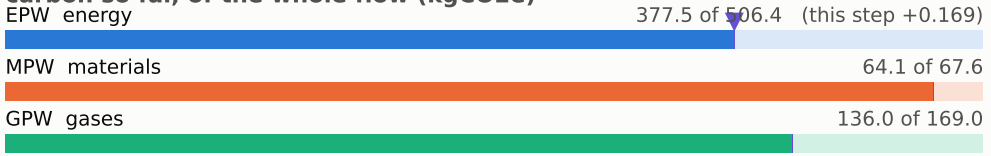
no mask for this step

this step

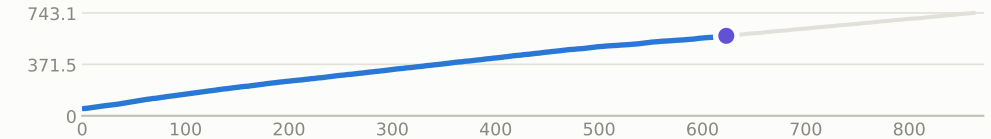
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

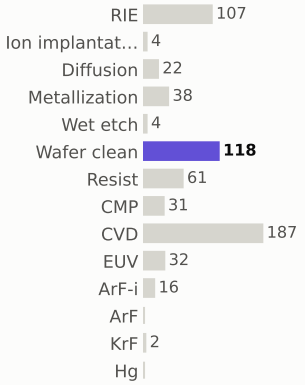
carbon so far, of the whole flow (kgCO2e)



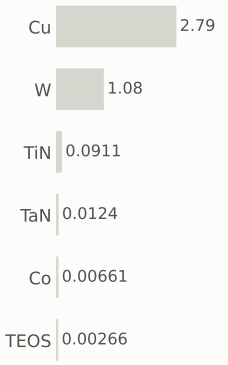
embodied carbon up to each step (kgCO2e per wafer)



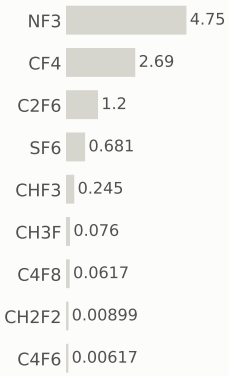
steps by process area



materials deposited (g)



gases emitted (g)



Step #6560: ArF-i

litho_patterning · ArF-i

0.217	577.9	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

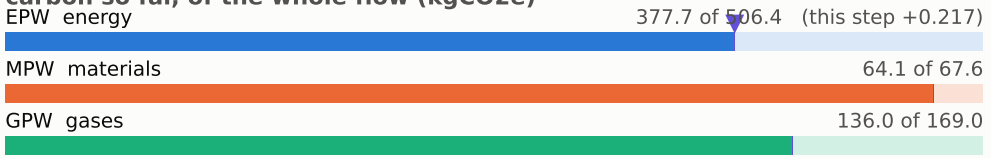
zoomed to fit

this step

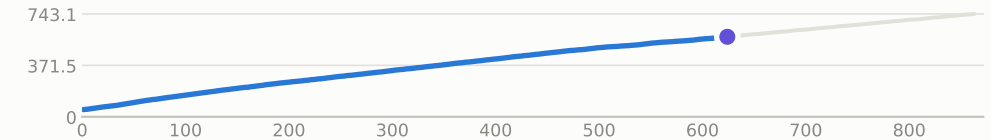
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

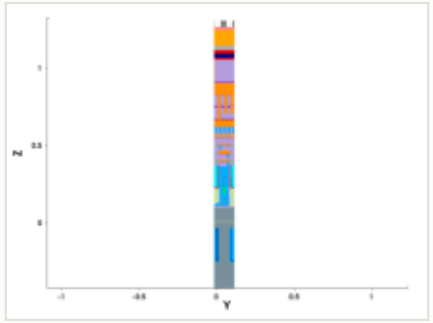
carbon so far, of the whole flow (kgCO2e)



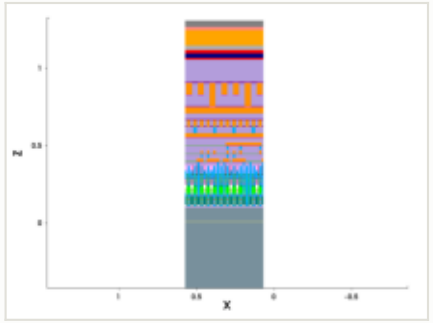
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

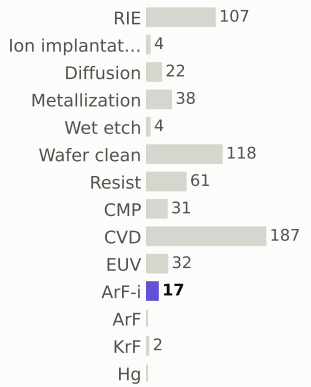


y cut at 0.049 µm

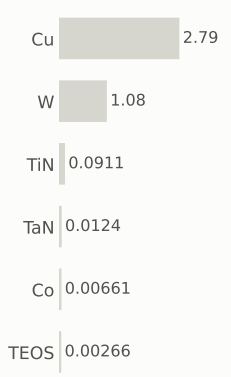


mask: layer 10551:0, positive (GDSII)

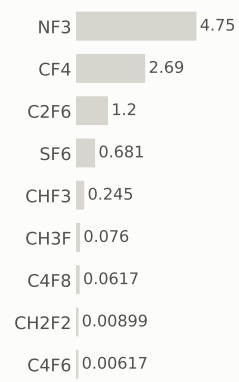
steps by process area



materials deposited (g)



gases emitted (g)



Step #6570: Ion_implantation

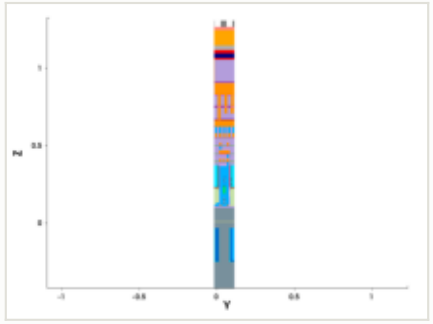
treatment · Ion_implantation

0.801	578.7	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

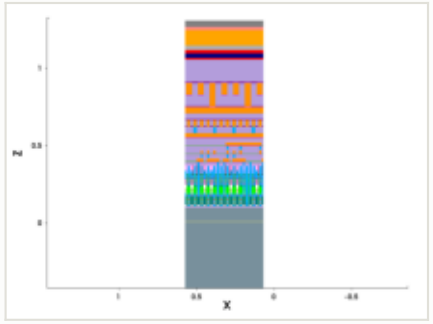
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

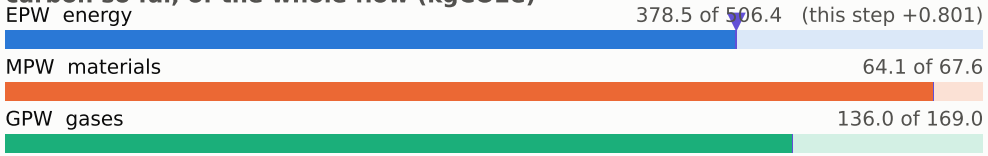
no mask for this step

this step

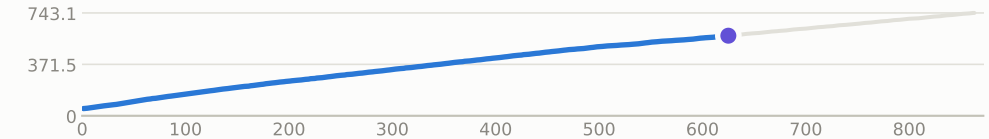
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

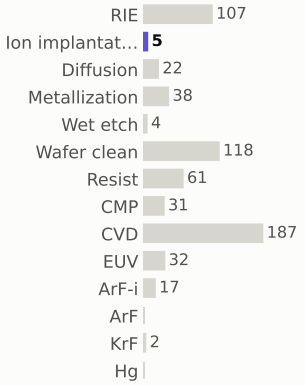
carbon so far, of the whole flow (kgCO2e)



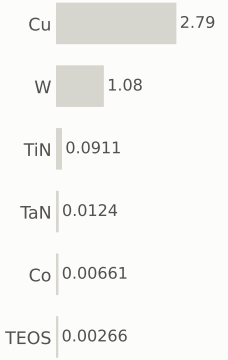
embodied carbon up to each step (kgCO2e per wafer)



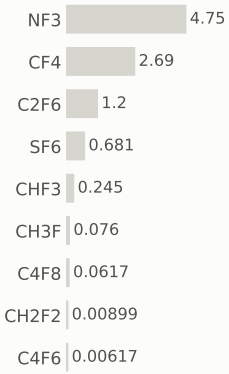
steps by process area



materials deposited (g)



gases emitted (g)



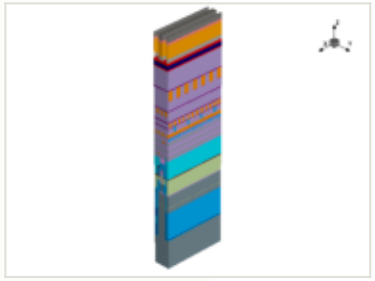
Step #6580: RIE

etch_anisotropic · RIE

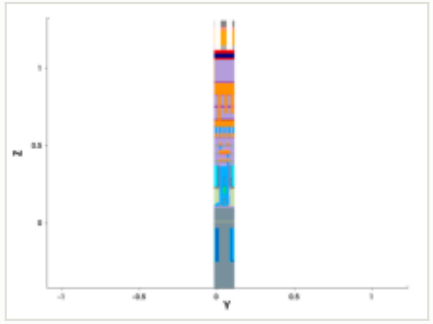
1.04	579.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

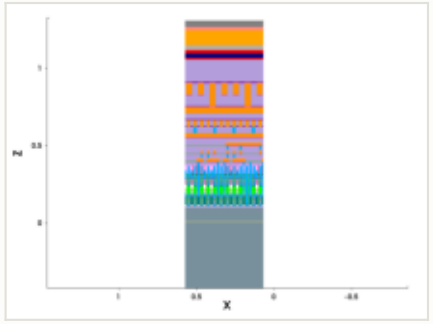
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

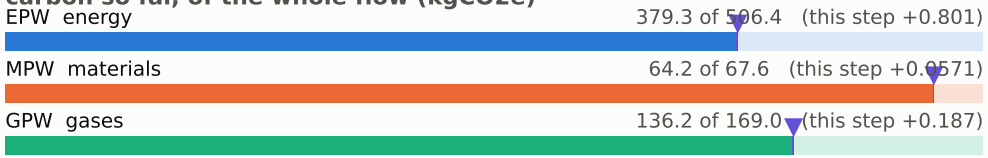
no mask for this step

this step

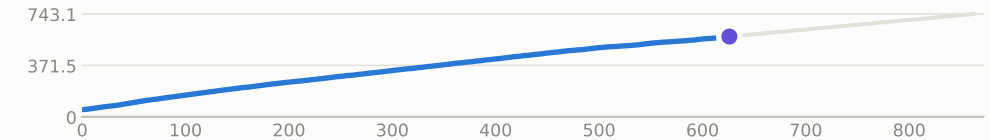
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

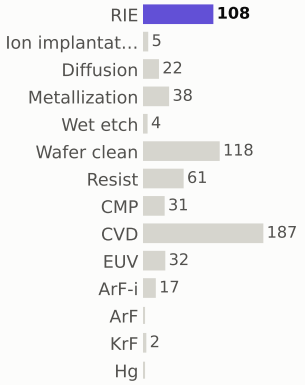
carbon so far, of the whole flow (kgCO2e)



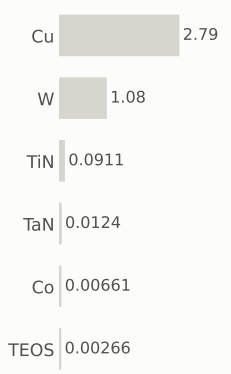
embodied carbon up to each step (kgCO2e per wafer)



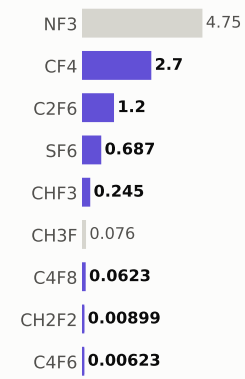
steps by process area



materials deposited (g)



gases emitted (g)



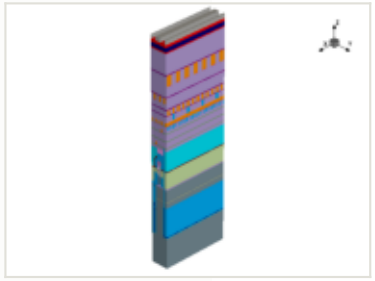
Step #6590: Wafer_clean

etch_strip · Wafer_clean

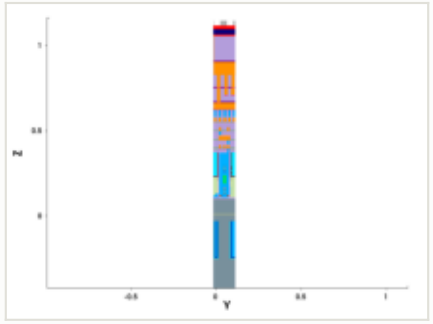
0.169	579.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

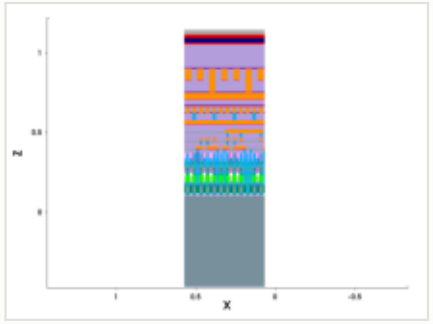
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



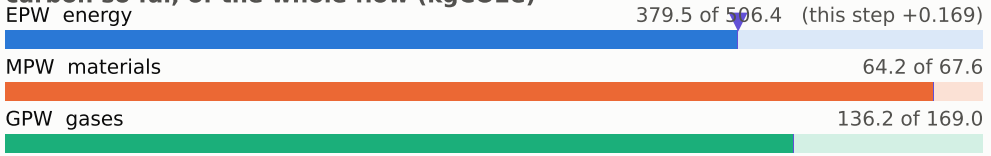
mask for this step

no mask for this step

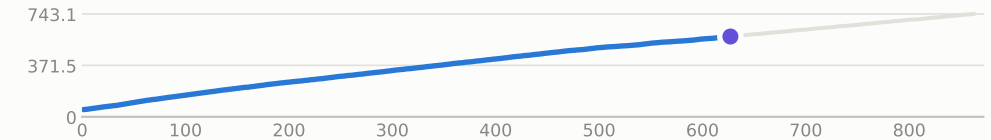
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

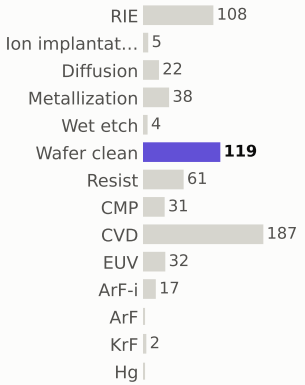
carbon so far, of the whole flow (kgCO2e)



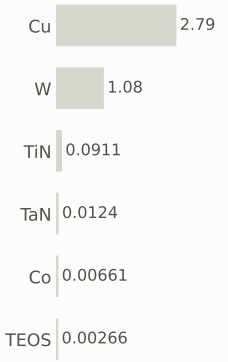
embodied carbon up to each step (kgCO2e per wafer)



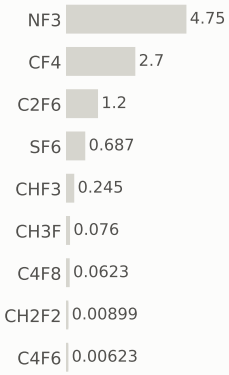
steps by process area



materials deposited (g)



gases emitted (g)



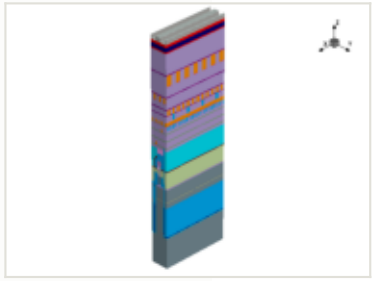
Step #6600: CVD

depo_isotropic · CVD

1.31	581.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

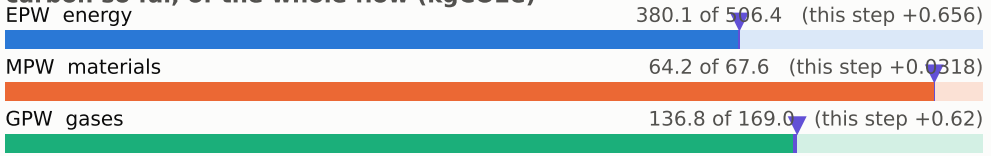


this step

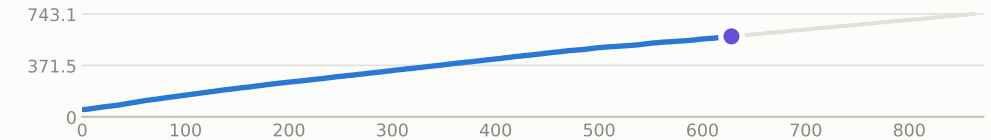
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

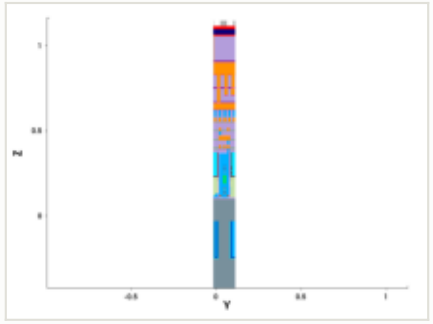
carbon so far, of the whole flow (kgCO2e)



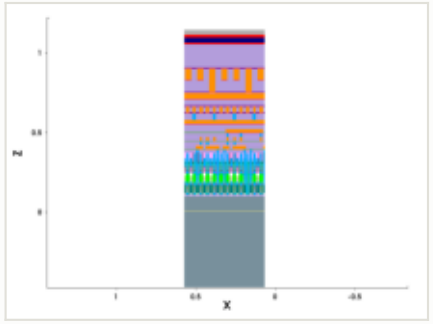
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



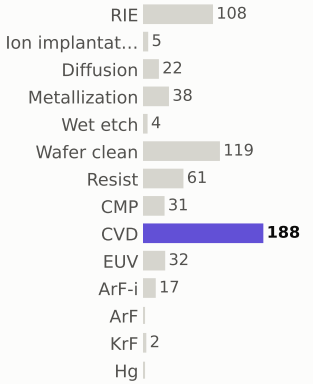
y cut at 0.049 µm



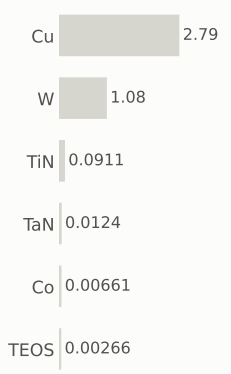
mask for this step

no mask for this step

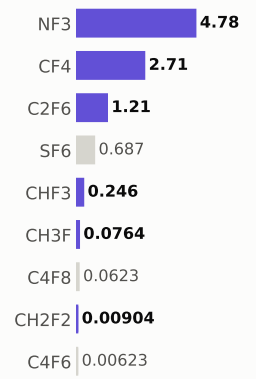
steps by process area



materials deposited (g)



gases emitted (g)



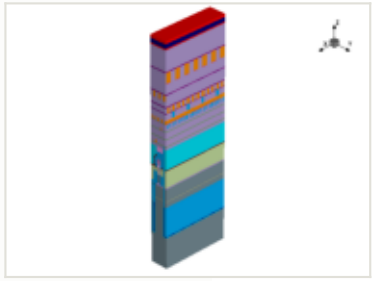
Step #6610: RIE

etch_anisotropic · RIE

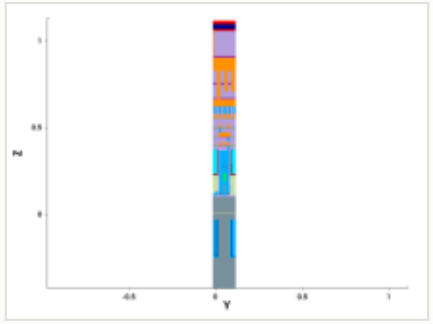
1.04	582.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

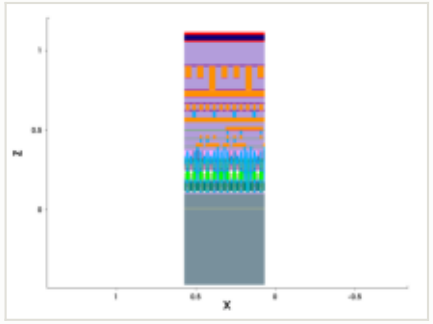
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

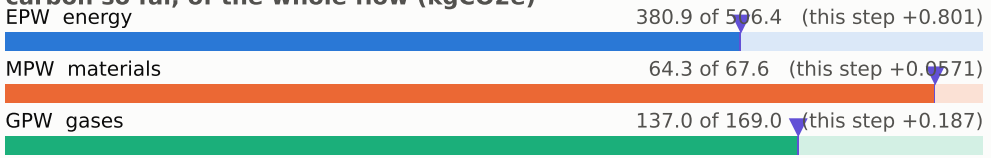
no mask for this step

this step

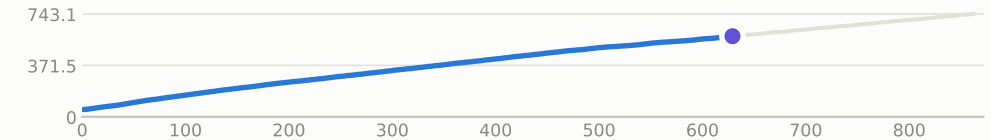
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

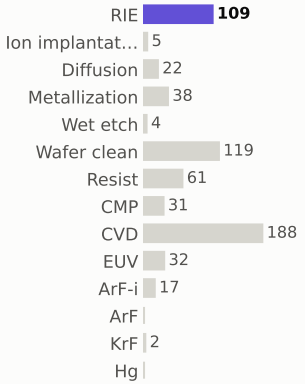
carbon so far, of the whole flow (kgCO2e)



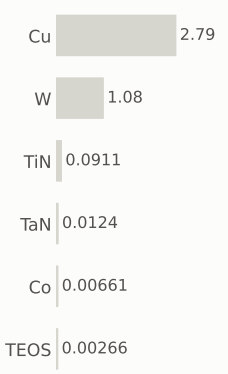
embodied carbon up to each step (kgCO2e per wafer)



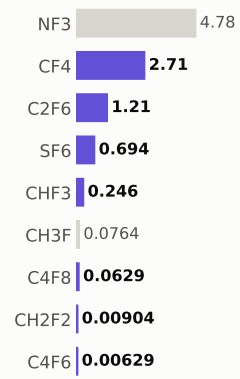
steps by process area



materials deposited (g)



gases emitted (g)



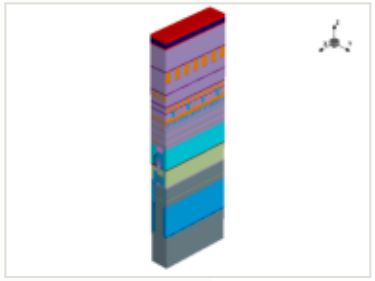
Step #6620: Wafer_clean

etch_strip · Wafer_clean

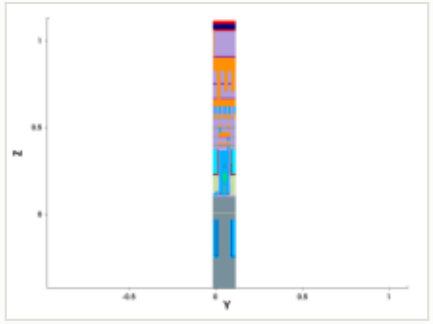
0.169	582.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

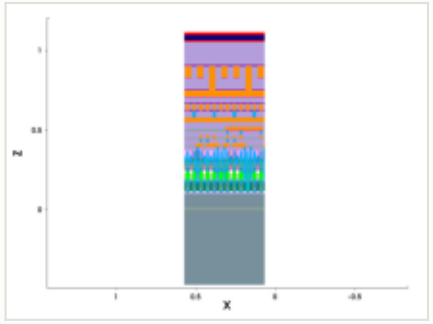
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



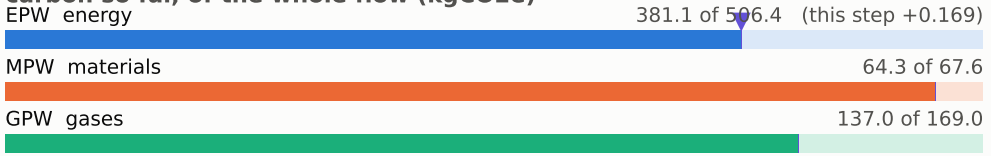
mask for this step

no mask for this step

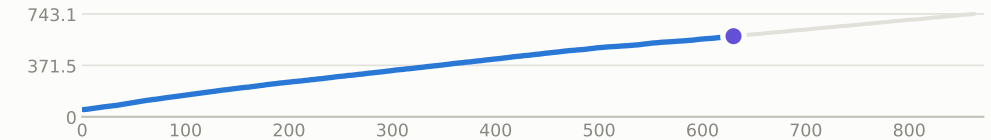
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

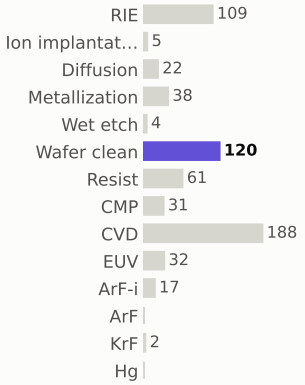
carbon so far, of the whole flow (kgCO2e)



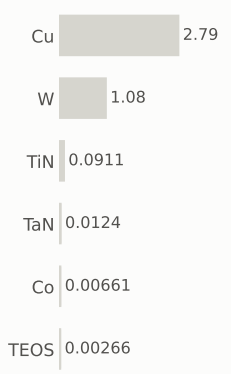
embodied carbon up to each step (kgCO2e per wafer)



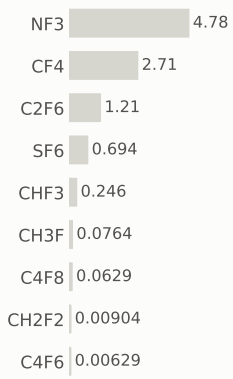
steps by process area



materials deposited (g)



gases emitted (g)



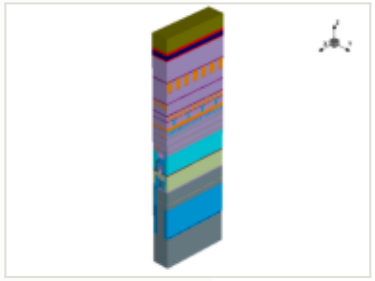
Step #6630: Resist

depo_fill · Resist

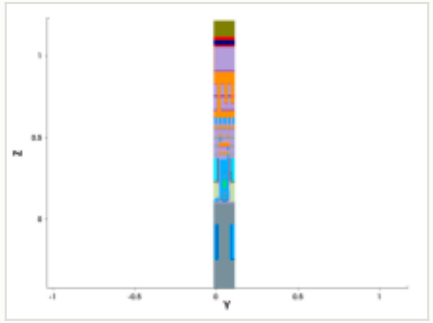
0.169	582.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

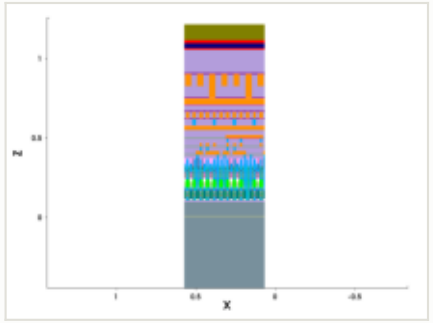
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



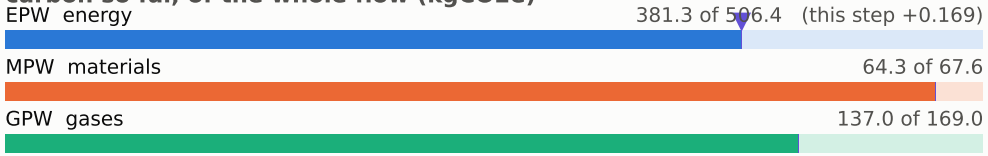
mask for this step

no mask for this step

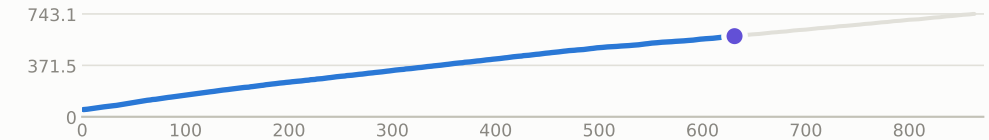
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

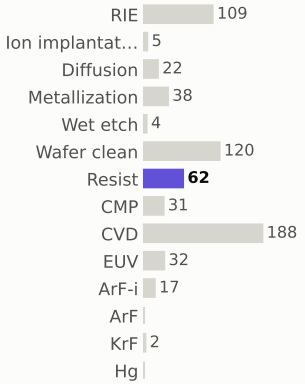
carbon so far, of the whole flow (kgCO2e)



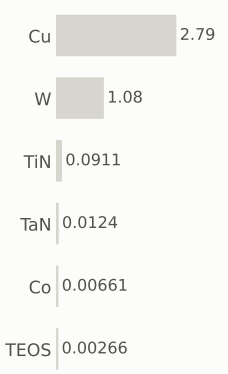
embodied carbon up to each step (kgCO2e per wafer)



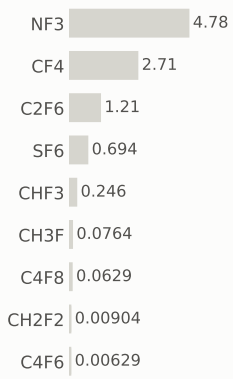
steps by process area



materials deposited (g)



gases emitted (g)



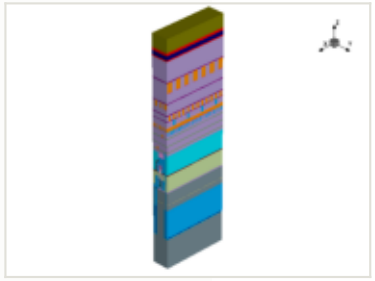
Step #6640: Wafer_clean

clean · Wafer_clean

0.169	582.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



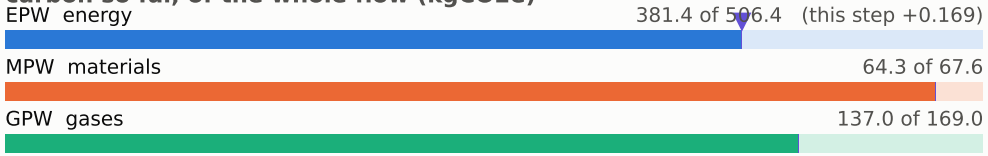
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

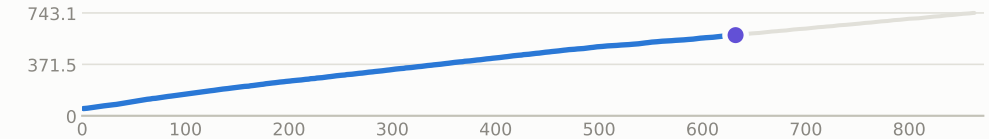
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

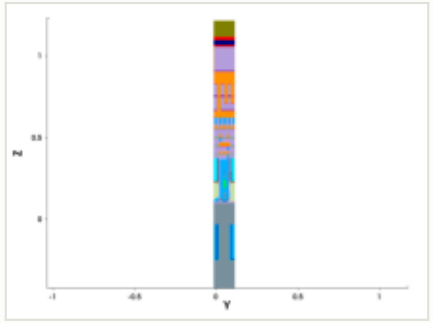
carbon so far, of the whole flow (kgCO2e)



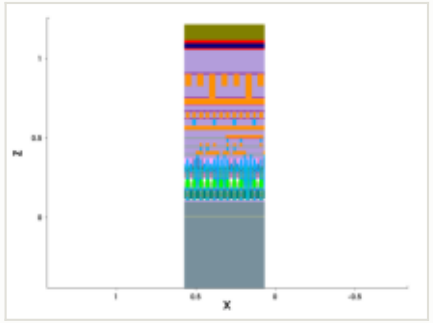
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



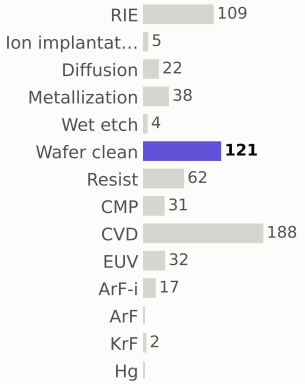
y cut at 0.049 µm



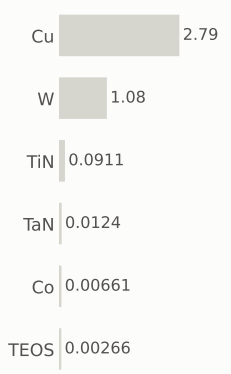
mask for this step

no mask for this step

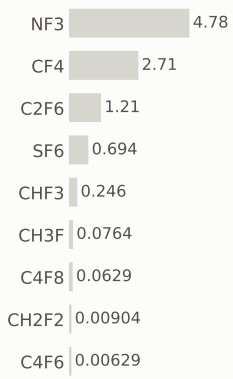
steps by process area



materials deposited (g)



gases emitted (g)



Step #6650: ArF-i

litho_patterning · ArF-i

0.217	583.0	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

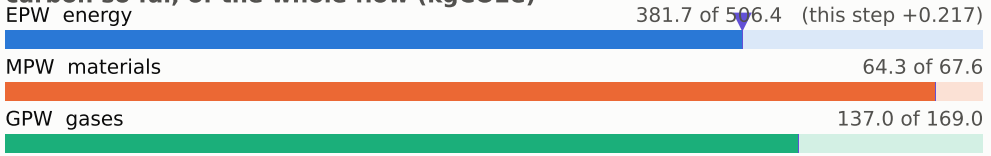
3D structure after this step

zoomed to fit

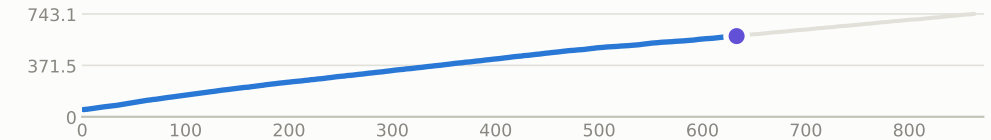
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

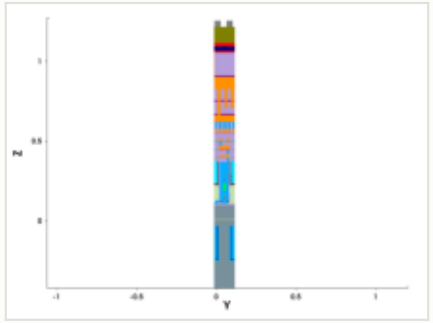
carbon so far, of the whole flow (kgCO2e)



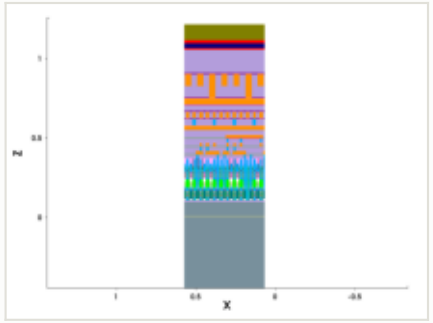
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

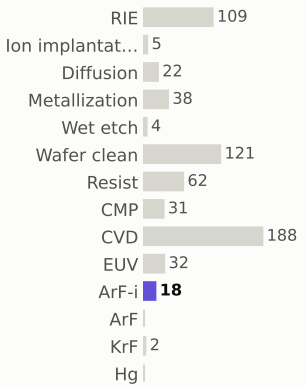


y cut at 0.049 µm

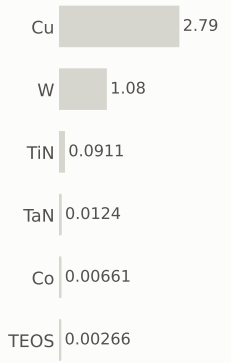


mask: layer 55:0 (GDSII)

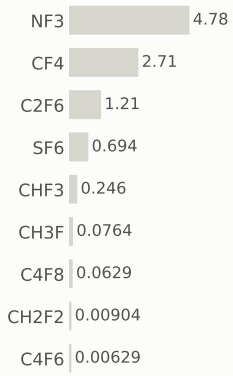
steps by process area



materials deposited (g)



gases emitted (g)



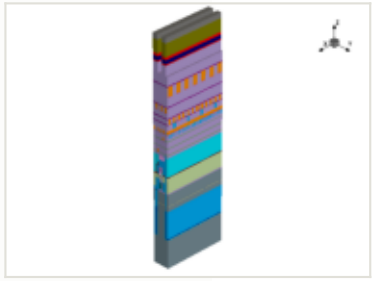
Step #6660: RIE

etch_anisotropic · RIE

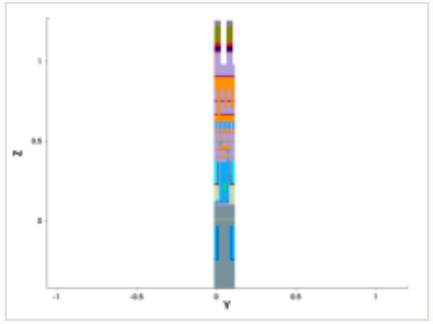
1.04	584.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

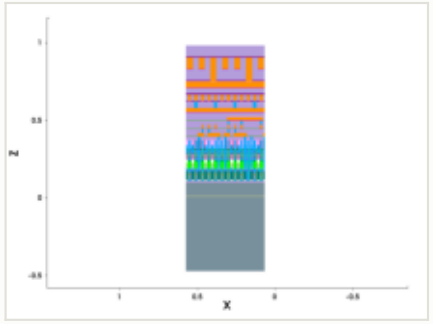
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

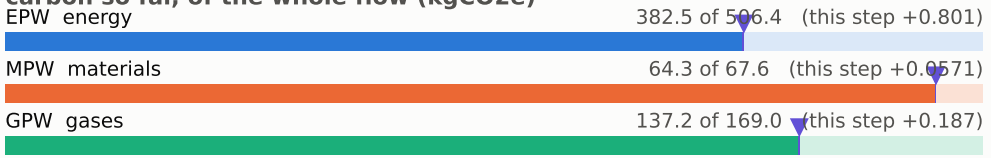
no mask for this step

this step

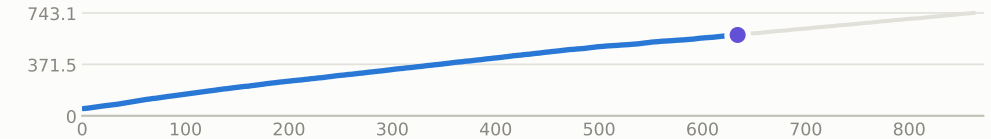
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

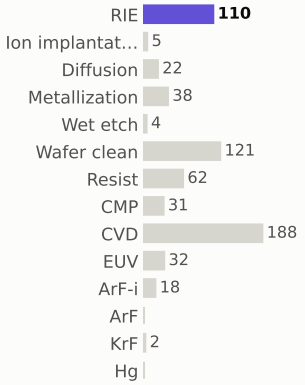
carbon so far, of the whole flow (kgCO2e)



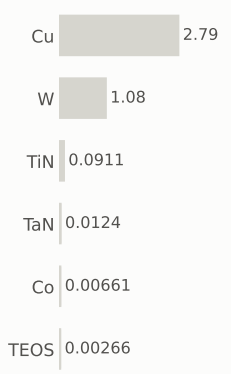
embodied carbon up to each step (kgCO2e per wafer)



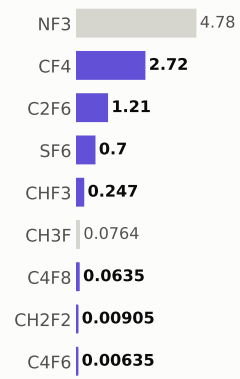
steps by process area



materials deposited (g)



gases emitted (g)



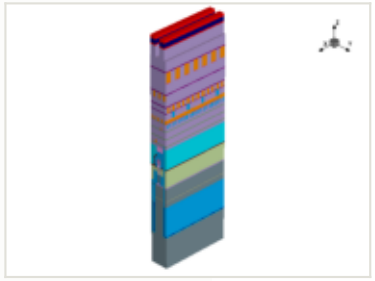
Step #6670: Wafer_clean

etch_strip · Wafer_clean

0.169	584.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

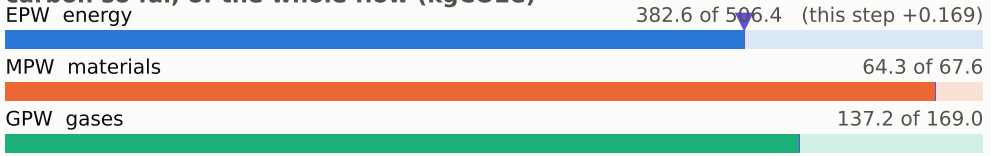


this step

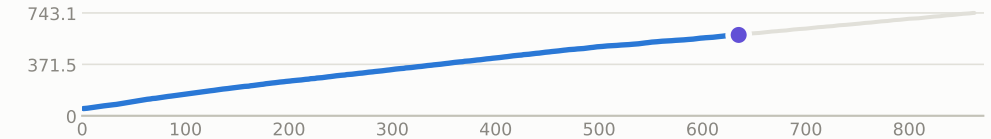
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

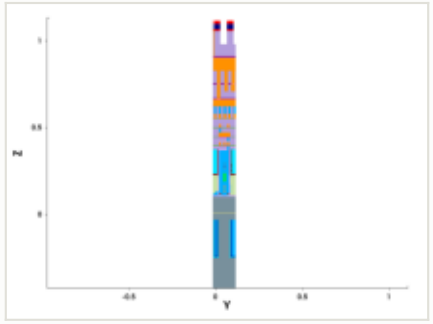
carbon so far, of the whole flow (kgCO2e)



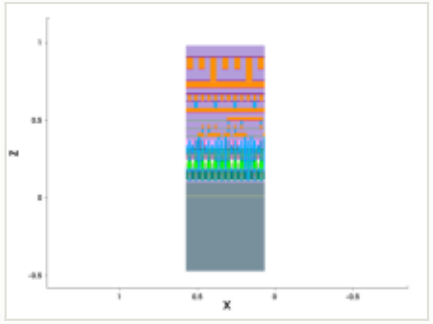
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



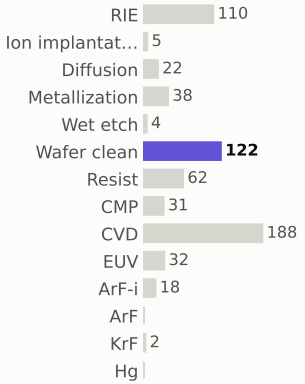
y cut at 0.049 µm



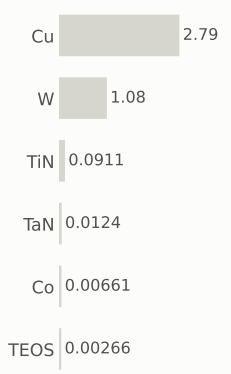
mask for this step

no mask for this step

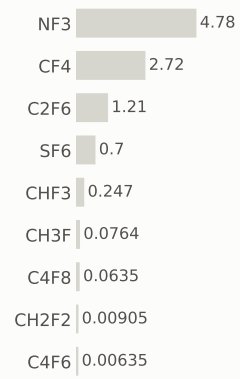
steps by process area



materials deposited (g)



gases emitted (g)



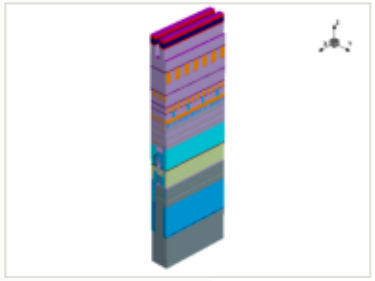
Step #6680: Metallization

depo_ellipsoidal · Metallization

1.2	585.4	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

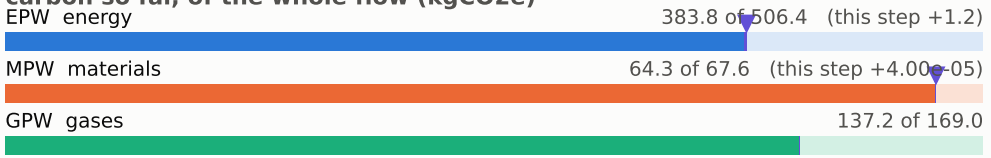


this step

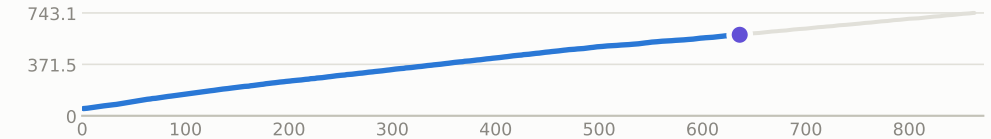
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

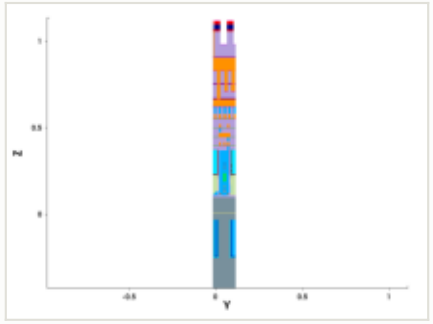
carbon so far, of the whole flow (kgCO2e)



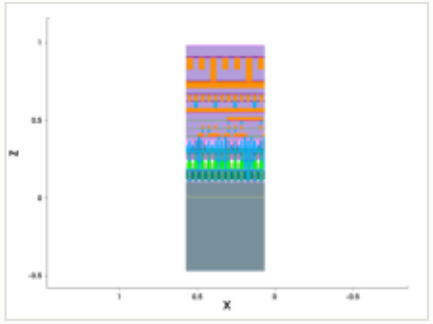
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



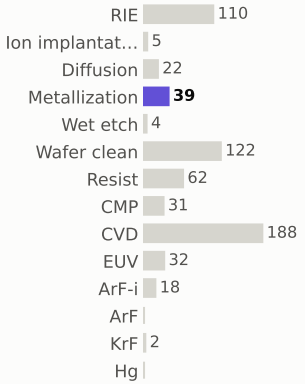
y cut at 0.049 µm



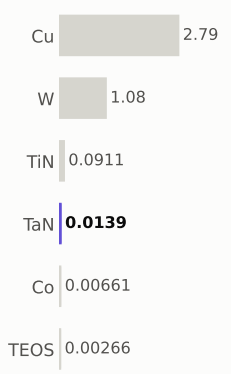
mask for this step

no mask for this step

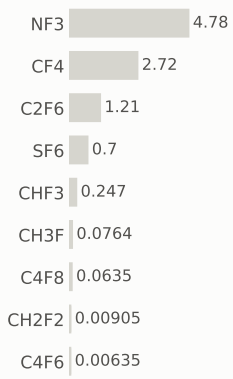
steps by process area



materials deposited (g)



gases emitted (g)



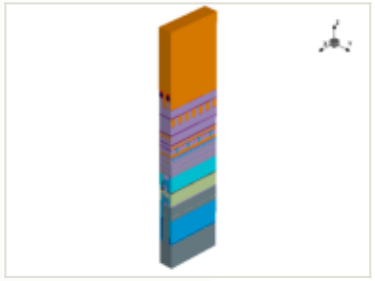
Step #6690: Metallization

depo_fill · Metallization

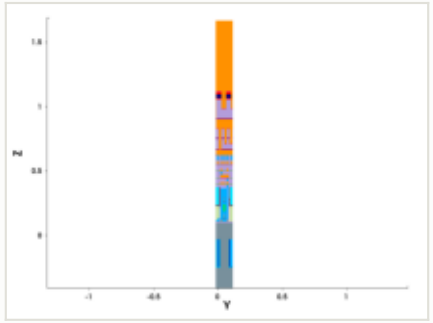
1.21	586.6	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

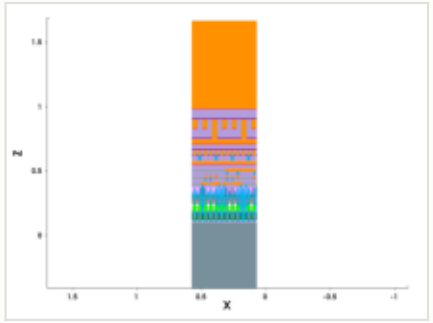
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

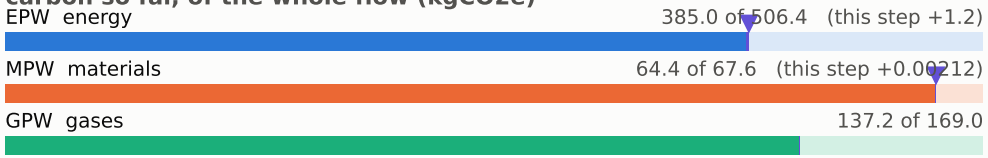
no mask for this step

this step

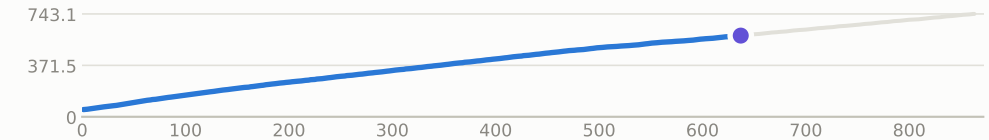
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

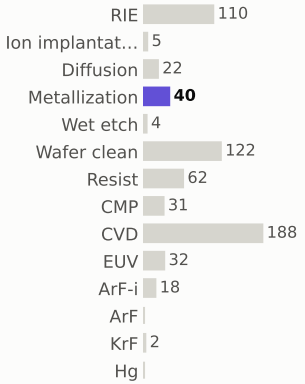
carbon so far, of the whole flow (kgCO2e)



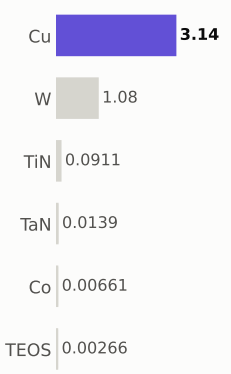
embodied carbon up to each step (kgCO2e per wafer)



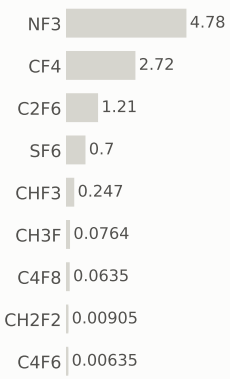
steps by process area



materials deposited (g)



gases emitted (g)



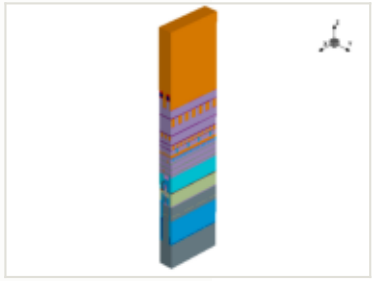
Step #6700: Diffusion

treatment · Diffusion

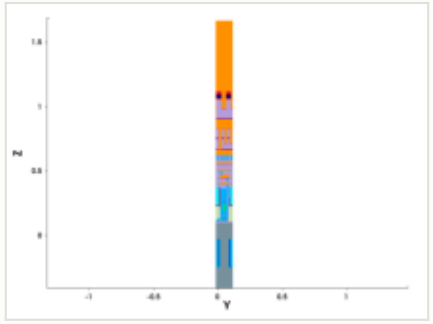
0.801	587.4	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

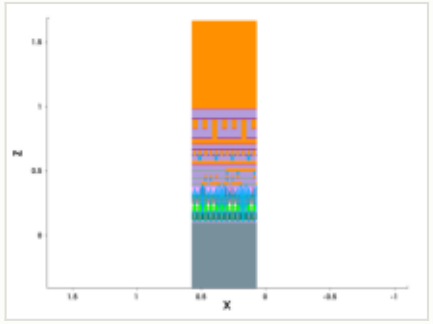
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

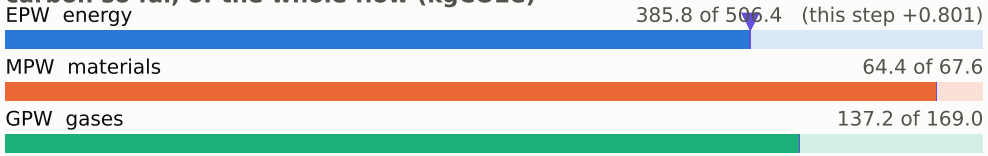
no mask for this step

this step

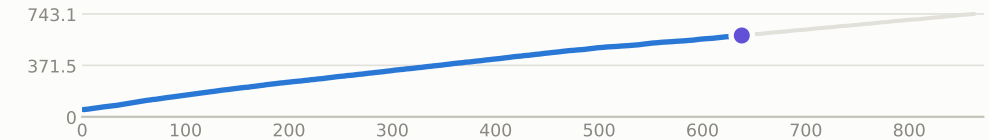
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

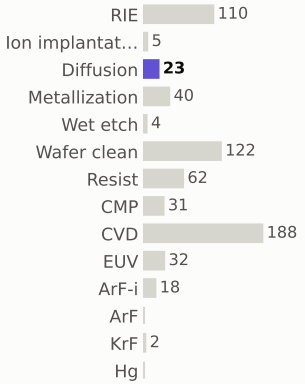
carbon so far, of the whole flow (kgCO2e)



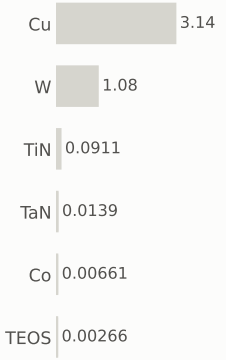
embodied carbon up to each step (kgCO2e per wafer)



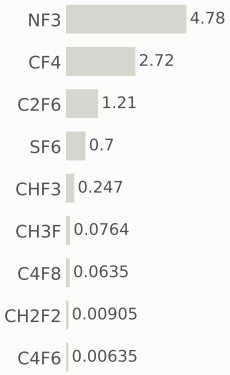
steps by process area



materials deposited (g)



gases emitted (g)



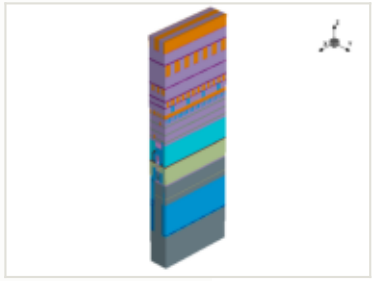
Step #6710: CMP

etch_cmp · CMP

0.169	587.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

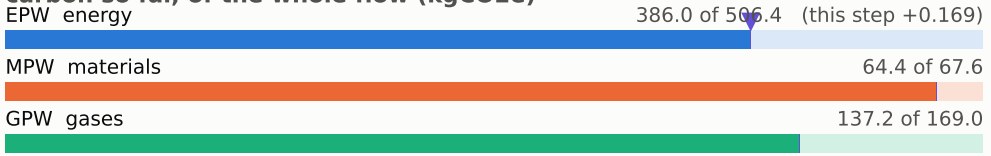


this step

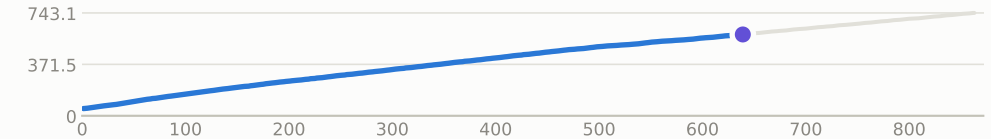
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

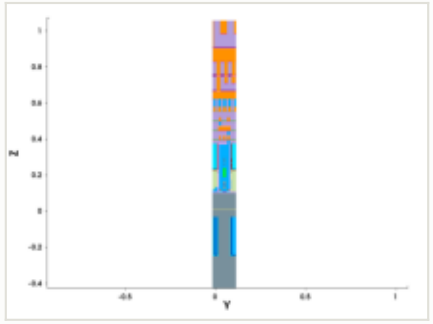
carbon so far, of the whole flow (kgCO2e)



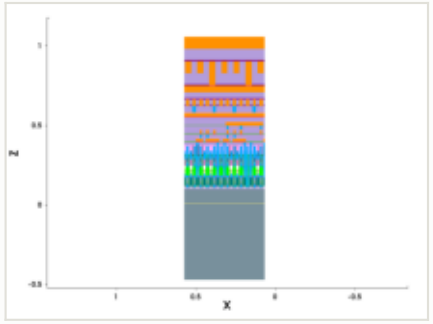
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



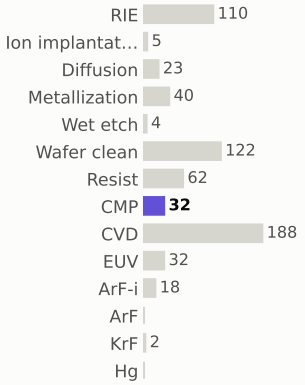
y cut at 0.049 µm



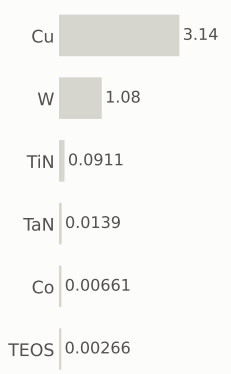
mask for this step

no mask for this step

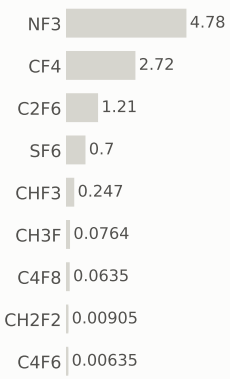
steps by process area



materials deposited (g)



gases emitted (g)



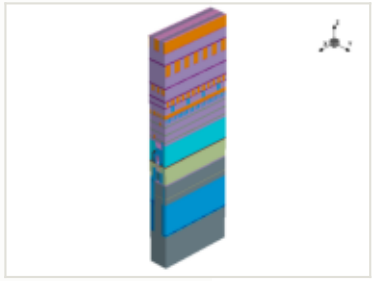
Step #6720: Metallization

depo_isotropic · Metallization

1.2	588.8	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

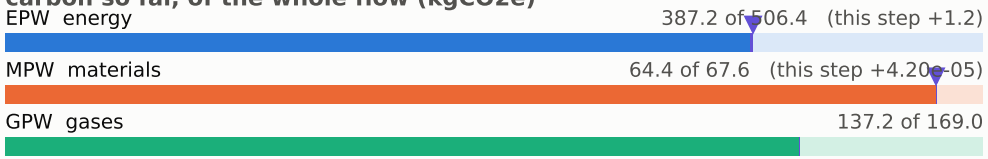


this step

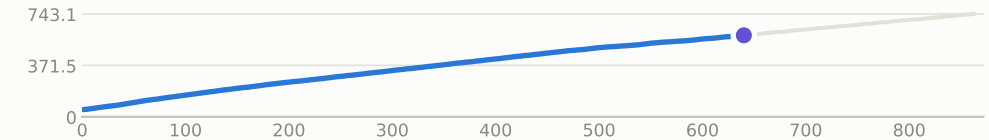
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

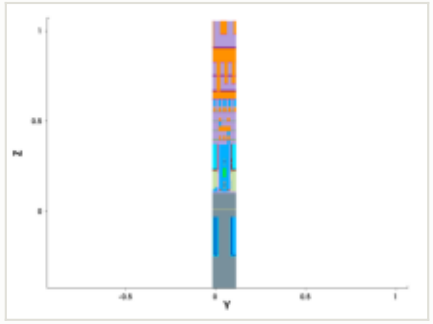
carbon so far, of the whole flow (kgCO2e)



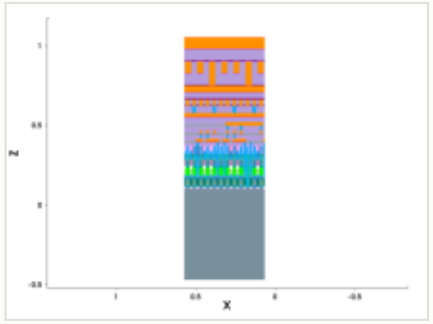
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



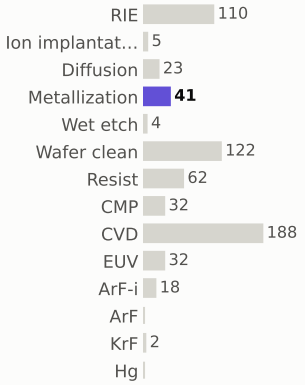
y cut at 0.049 µm



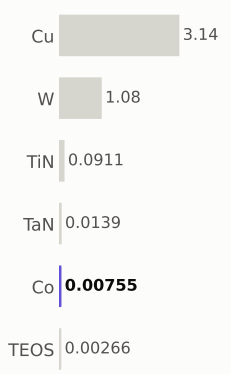
mask for this step

no mask for this step

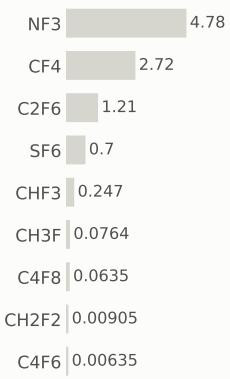
steps by process area



materials deposited (g)



gases emitted (g)



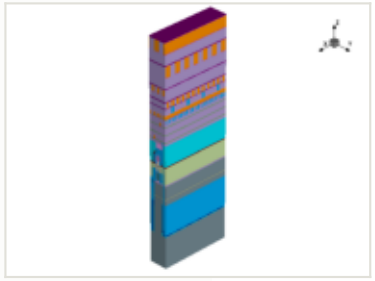
Step #6740: CVD

depo_fill · CVD

1.31	590.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

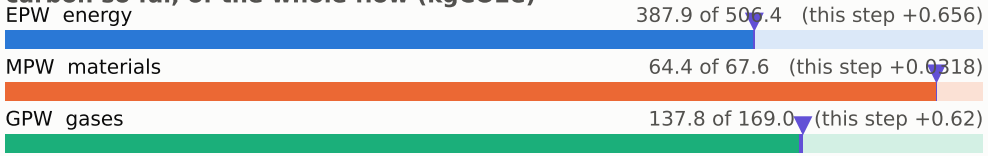


this step

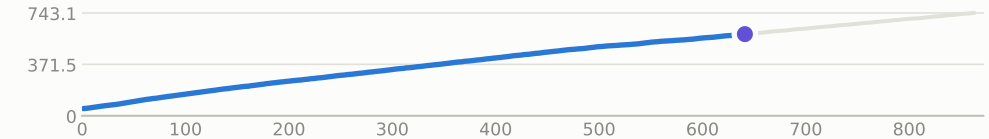
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

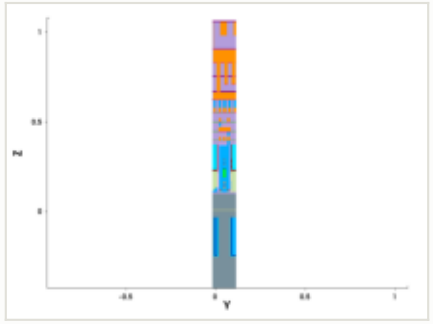
carbon so far, of the whole flow (kgCO2e)



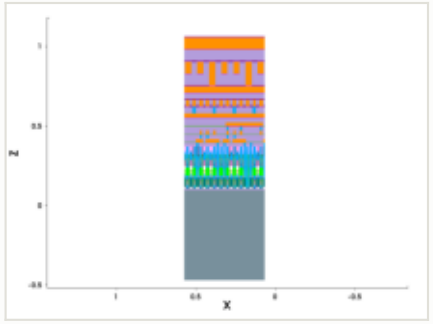
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



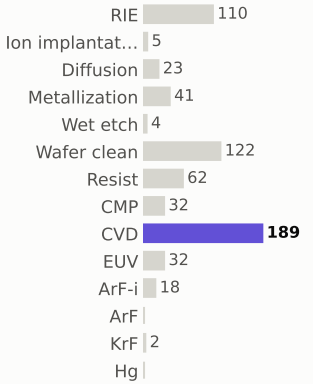
y cut at 0.049 µm



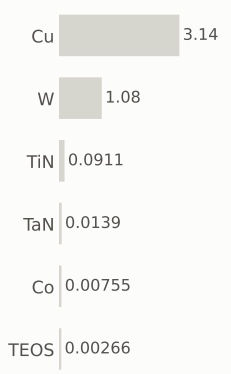
mask for this step

no mask for this step

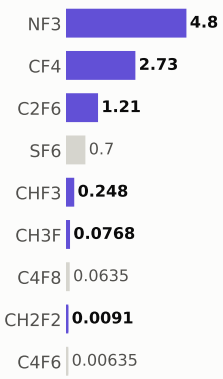
steps by process area



materials deposited (g)



gases emitted (g)



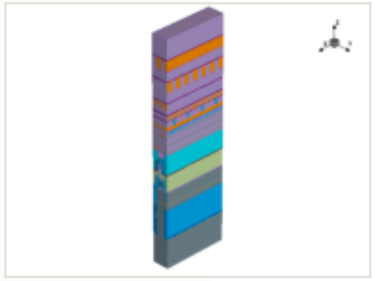
Step #6750: CVD

depo_fill · CVD

1.31	591.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

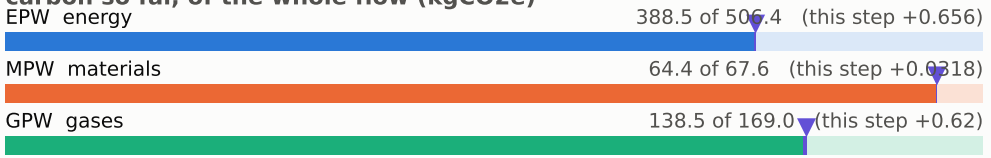


this step

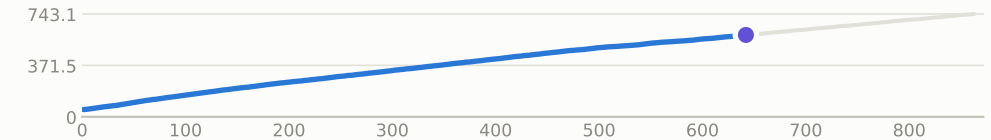
process area	CVD
material	—
thickness	0.146 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

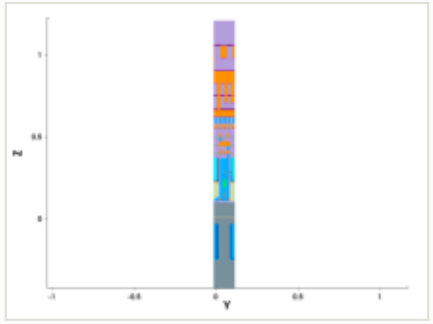
carbon so far, of the whole flow (kgCO2e)



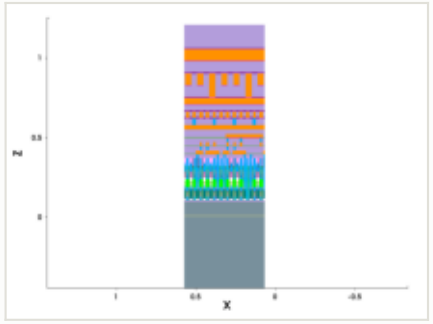
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



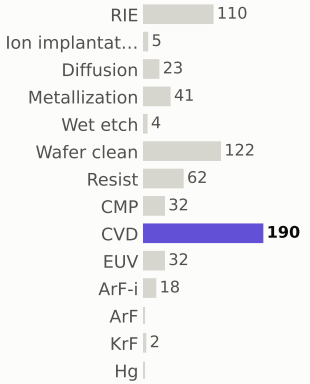
y cut at 0.049 µm



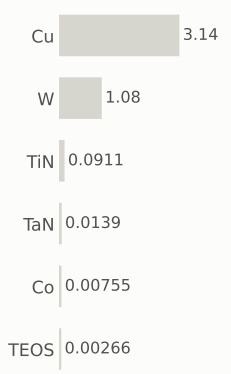
mask for this step

no mask for this step

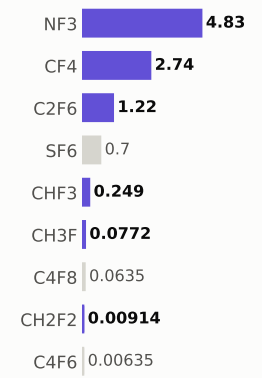
steps by process area



materials deposited (g)



gases emitted (g)



Step #6760: Diffusion

treatment · Diffusion

0.801	592.2	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

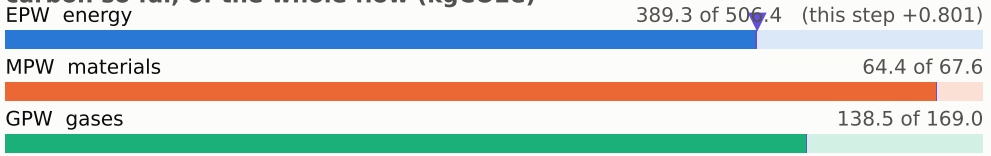
zoomed to fit

this step

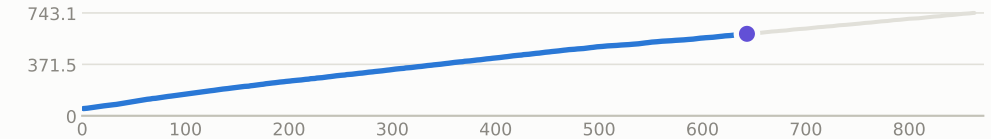
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

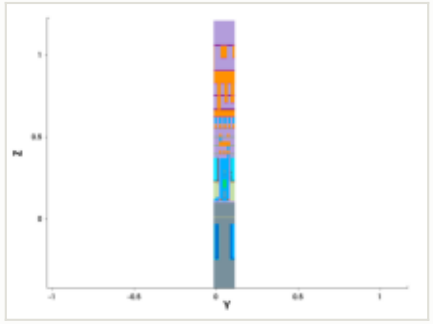
carbon so far, of the whole flow (kgCO2e)



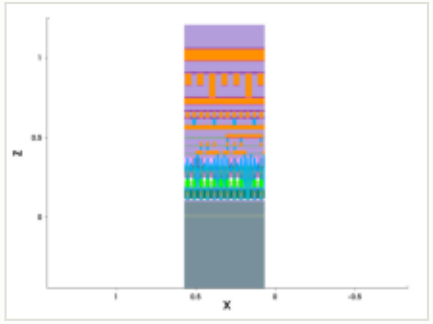
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



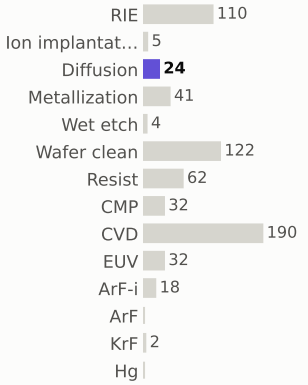
y cut at 0.049 µm



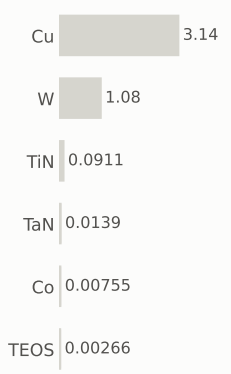
mask for this step

no mask for this step

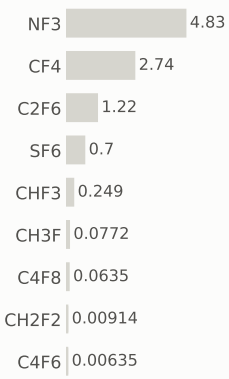
steps by process area



materials deposited (g)



gases emitted (g)



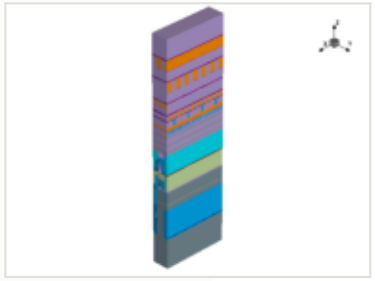
Step #6770: Wafer_clean

clean · Wafer_clean

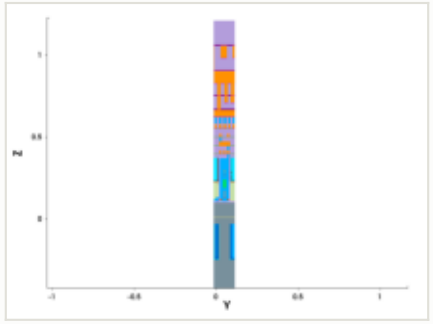
0.169	592.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

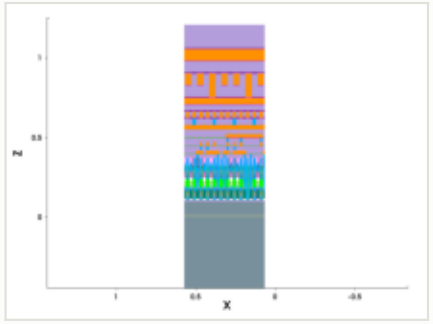
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

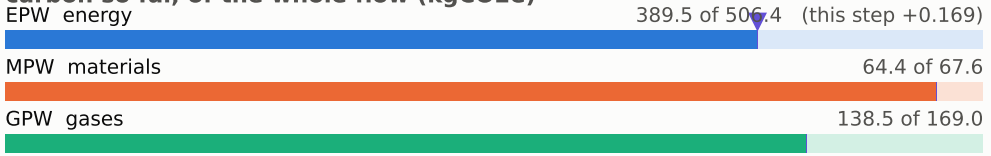
no mask for this step

this step

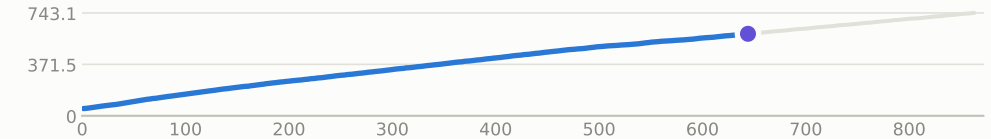
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

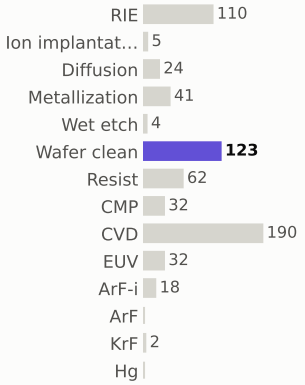
carbon so far, of the whole flow (kgCO2e)



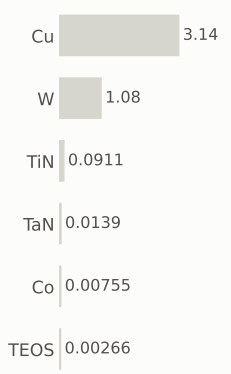
embodied carbon up to each step (kgCO2e per wafer)



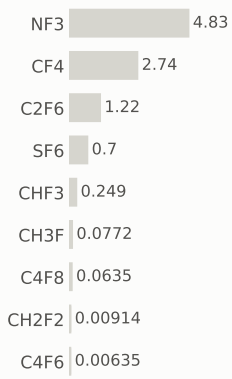
steps by process area



materials deposited (g)



gases emitted (g)



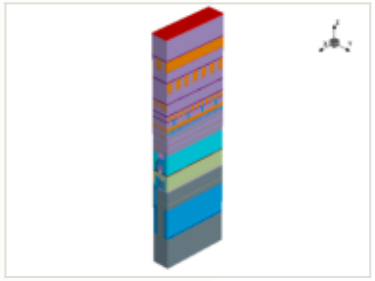
Step #6780: CVD

depo_fill · CVD

1.31	593.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

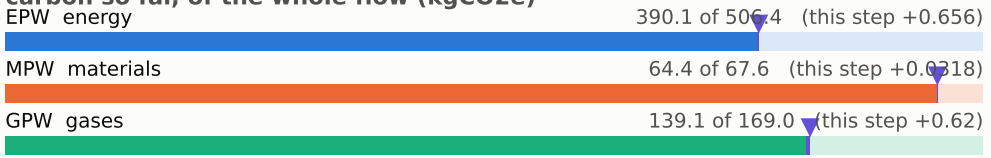


this step

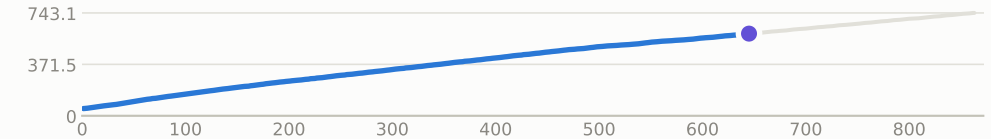
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

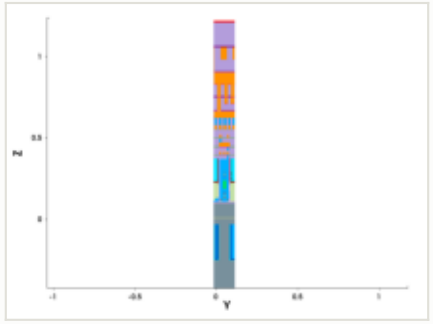
carbon so far, of the whole flow (kgCO2e)



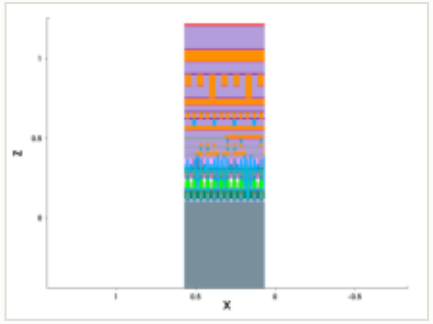
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



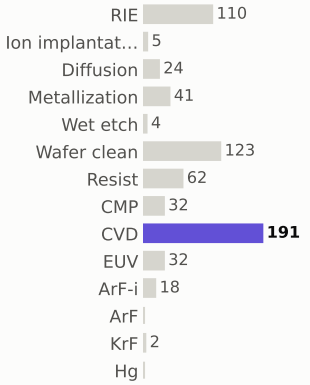
y cut at 0.049 µm



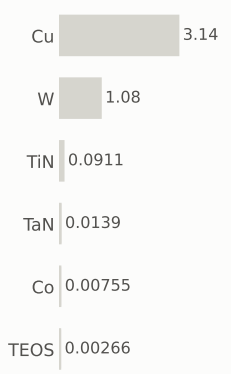
mask for this step

no mask for this step

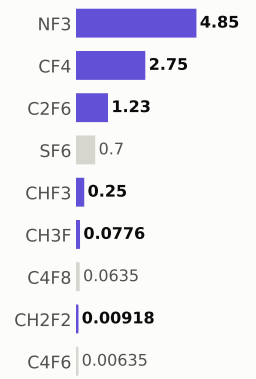
steps by process area



materials deposited (g)



gases emitted (g)



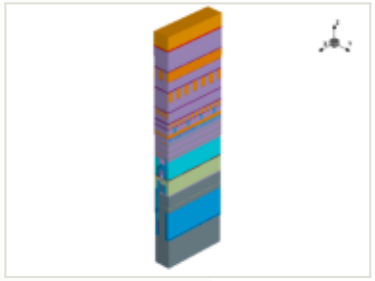
Step #6790: Resist

depo_fill · Resist

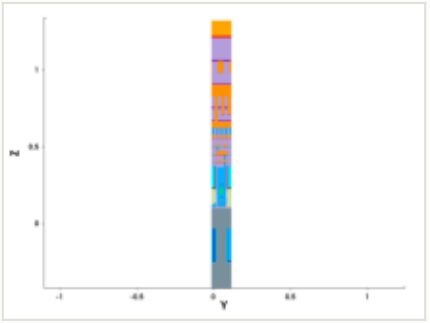
0.169	593.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

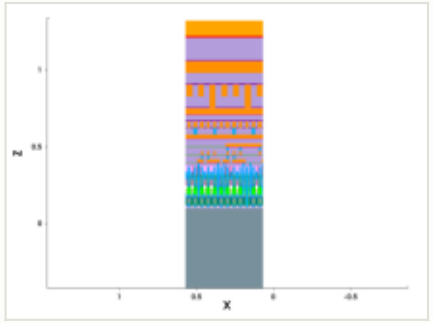
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



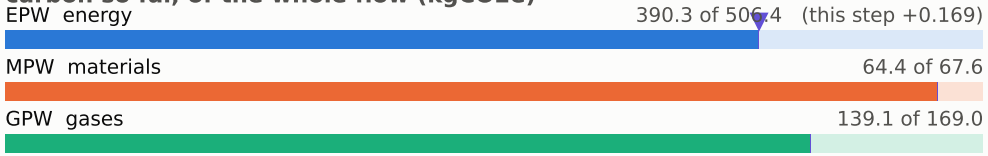
mask for this step

no mask for this step

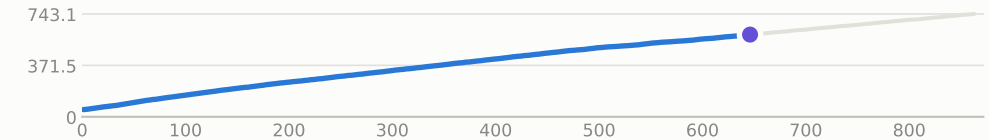
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

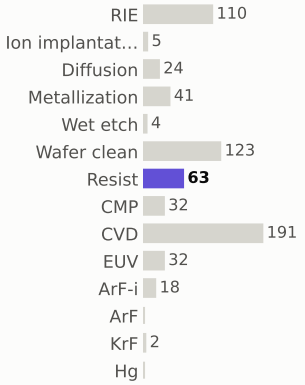
carbon so far, of the whole flow (kgCO2e)



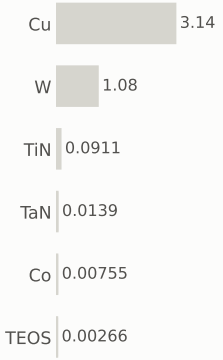
embodied carbon up to each step (kgCO2e per wafer)



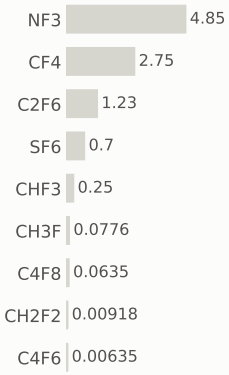
steps by process area



materials deposited (g)



gases emitted (g)



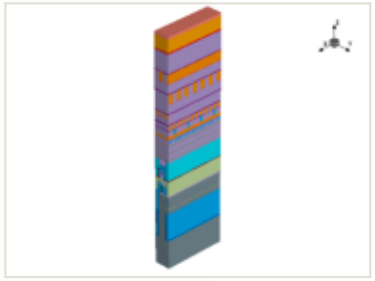
Step #6800: CVD

depo_fill · CVD

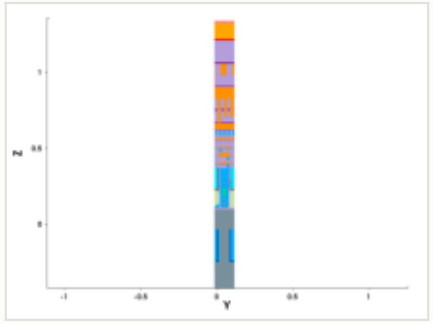
1.31	595.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

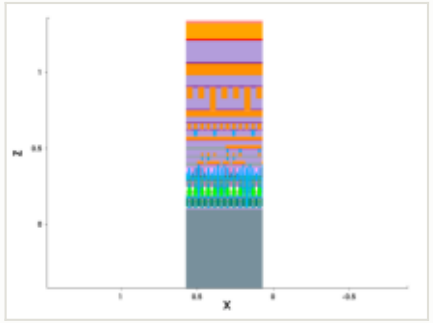
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

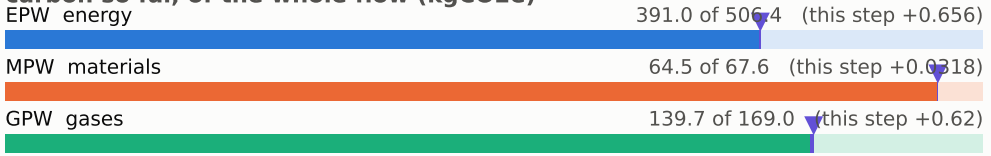
no mask for this step

this step

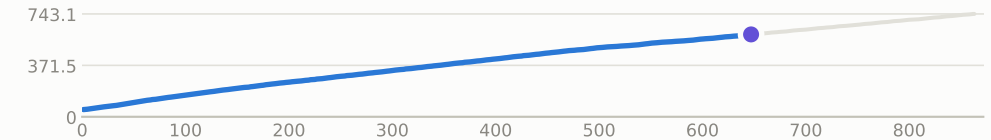
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

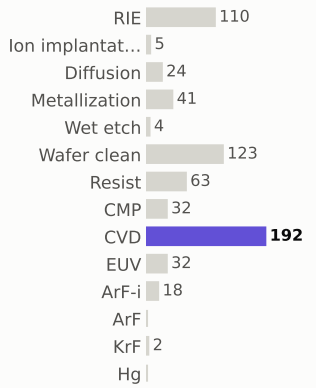
carbon so far, of the whole flow (kgCO2e)



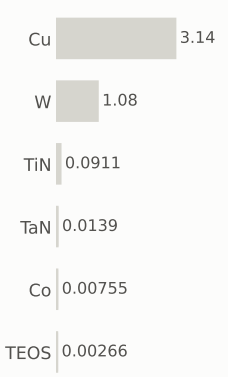
embodied carbon up to each step (kgCO2e per wafer)



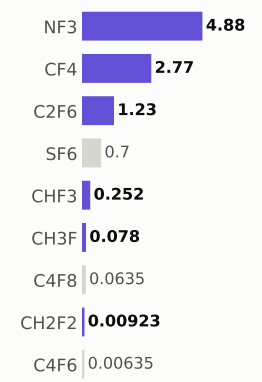
steps by process area



materials deposited (g)



gases emitted (g)



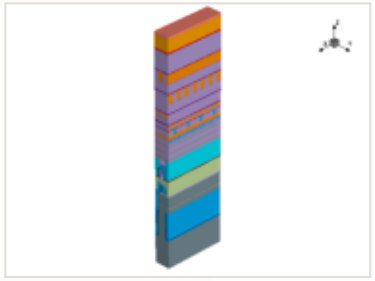
Step #6810: Wafer_clean

clean · Wafer_clean

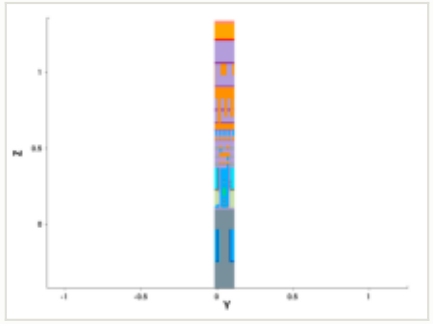
0.169	595.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

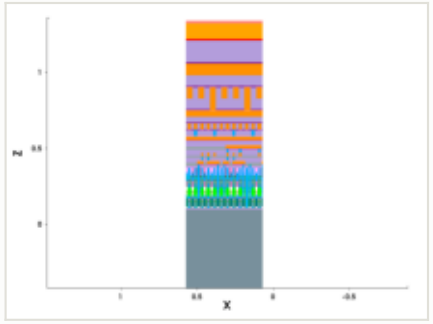
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

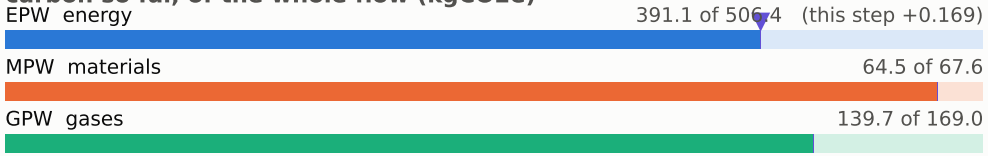
no mask for this step

this step

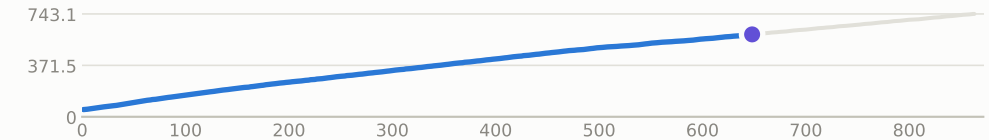
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

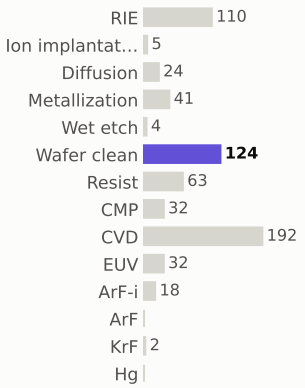
carbon so far, of the whole flow (kgCO2e)



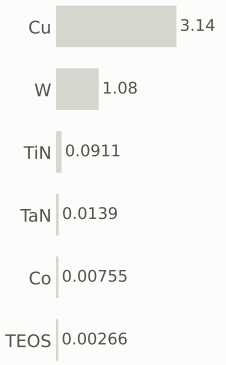
embodied carbon up to each step (kgCO2e per wafer)



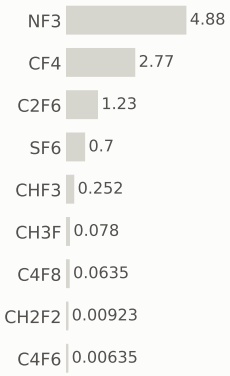
steps by process area



materials deposited (g)



gases emitted (g)



Step #6820: ArF-i

litho_patterning · ArF-i

0.217	595.5	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

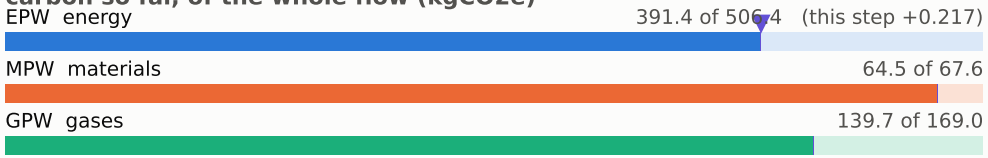
zoomed to fit

this step

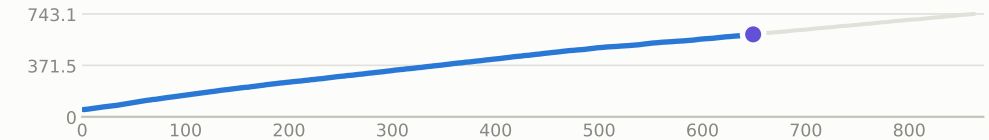
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

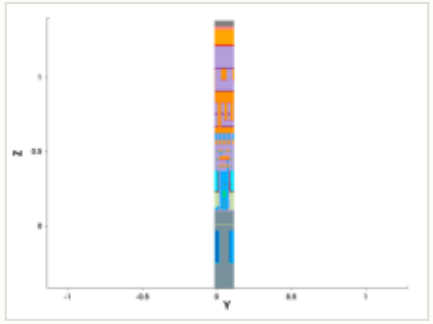
carbon so far, of the whole flow (kgCO2e)



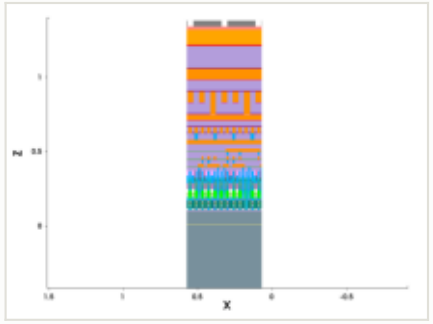
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

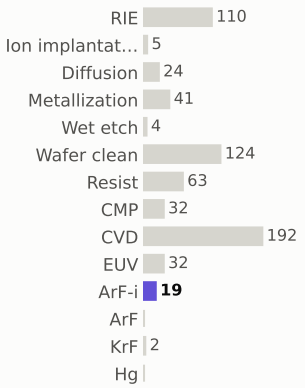


y cut at 0.049 µm

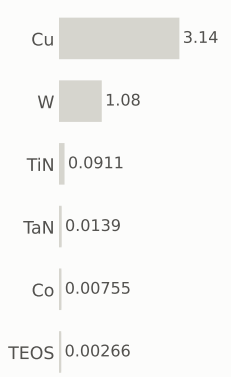


mask: layer 10571:0 (GDSII)

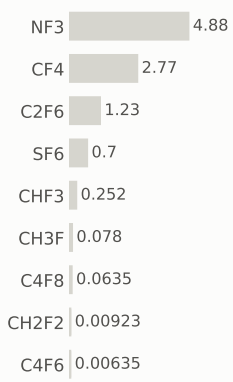
steps by process area



materials deposited (g)



gases emitted (g)



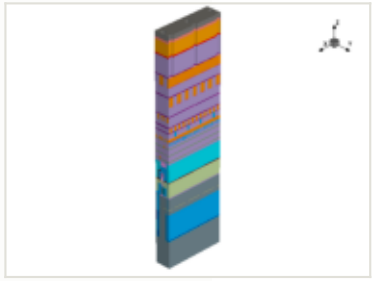
Step #6830: RIE

etch_anisotropic · RIE

1.04	596.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

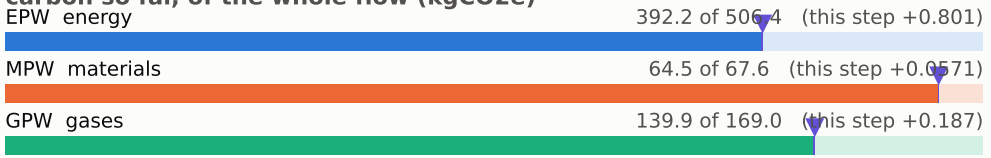


this step

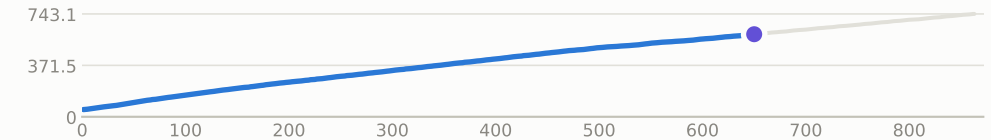
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

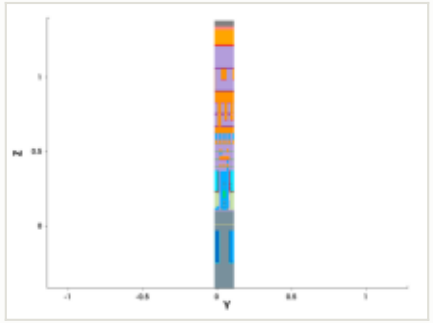
carbon so far, of the whole flow (kgCO2e)



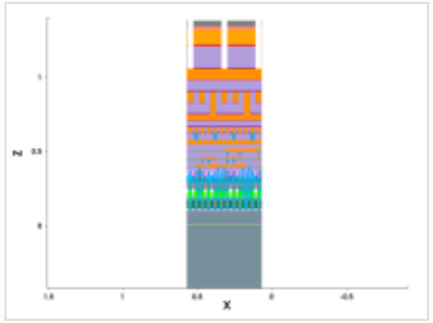
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



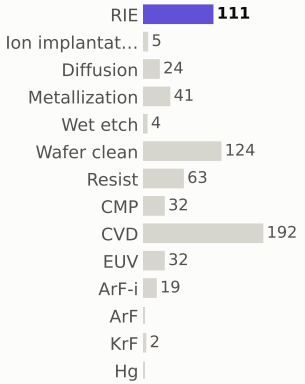
y cut at 0.049 µm



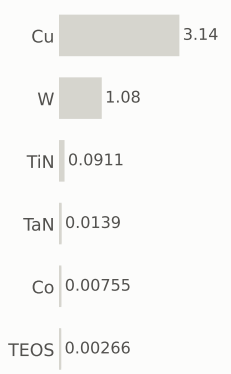
mask for this step

no mask for this step

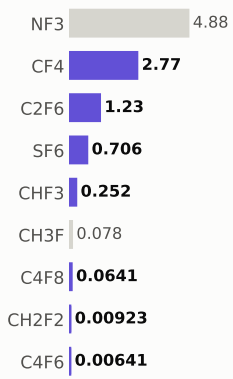
steps by process area



materials deposited (g)



gases emitted (g)



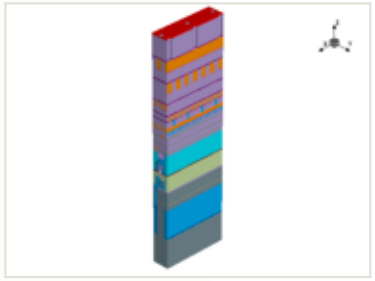
Step #6840: Wafer_clean

etch_strip · Wafer_clean

0.169	596.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

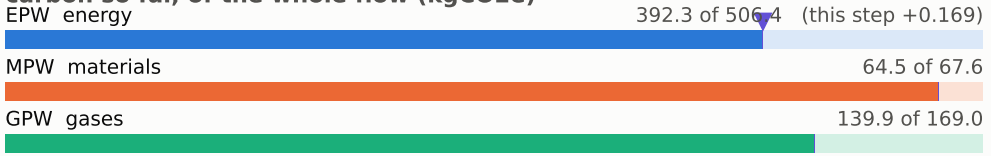


this step

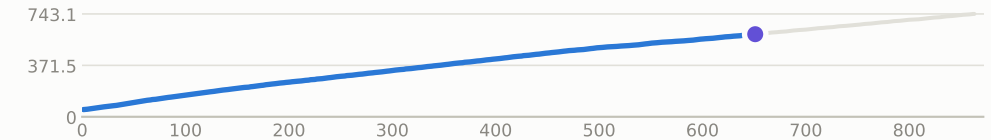
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

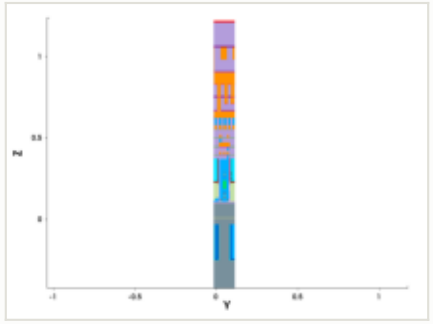
carbon so far, of the whole flow (kgCO2e)



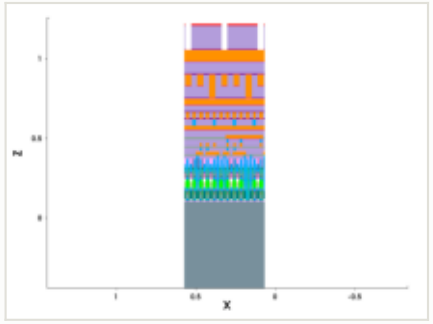
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



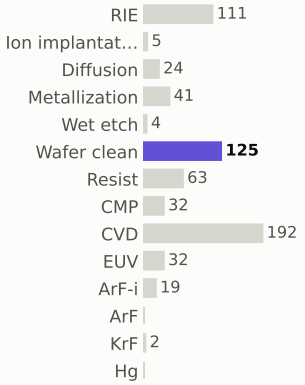
y cut at 0.049 µm



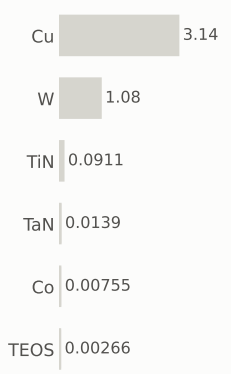
mask for this step

no mask for this step

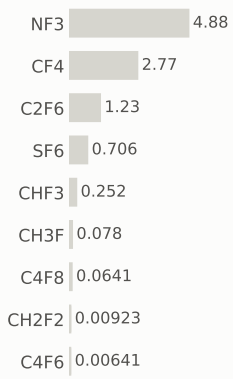
steps by process area



materials deposited (g)



gases emitted (g)



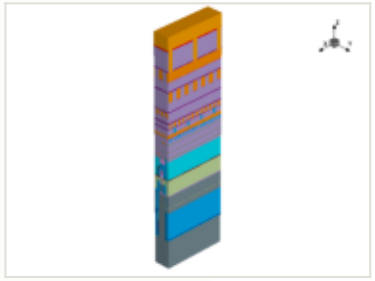
Step #6850: Resist

depo_fill · Resist

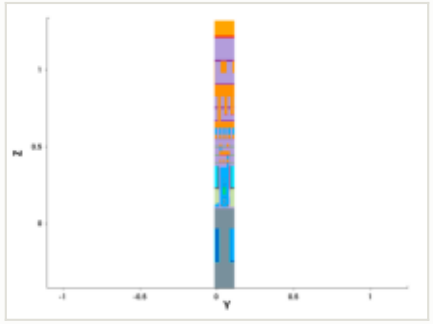
0.169	596.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

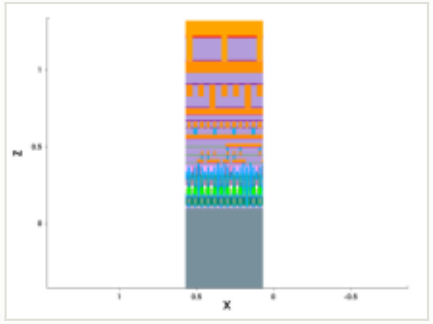
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



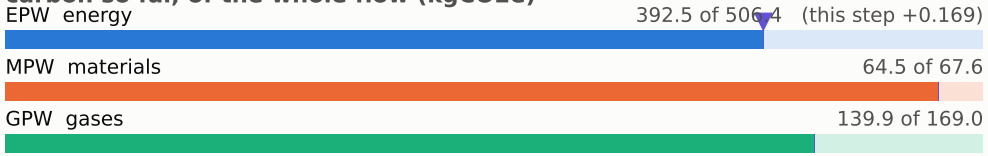
mask for this step

no mask for this step

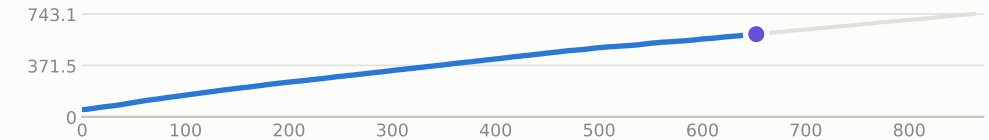
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

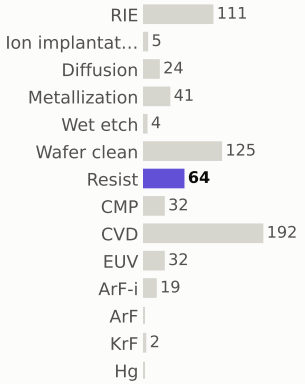
carbon so far, of the whole flow (kgCO2e)



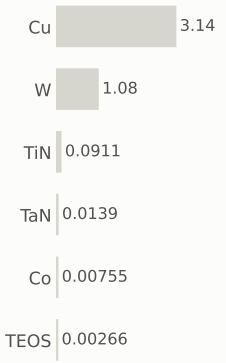
embodied carbon up to each step (kgCO2e per wafer)



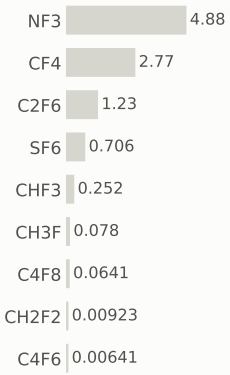
steps by process area



materials deposited (g)



gases emitted (g)



Step #6860: CVD

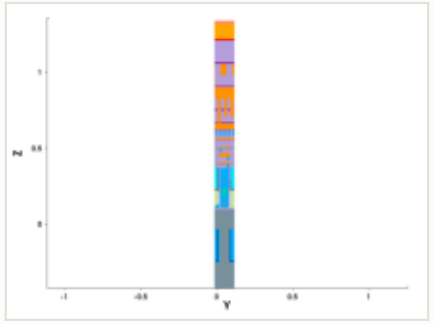
depo_fill · CVD

1.31	598.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

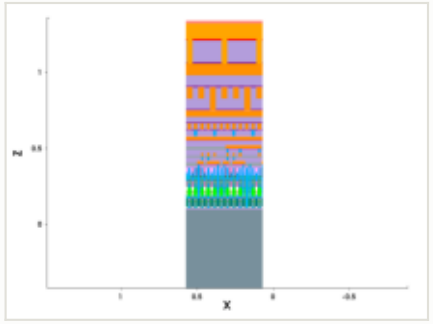
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

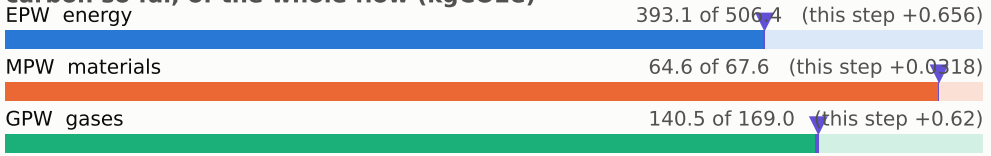
no mask for this step

this step

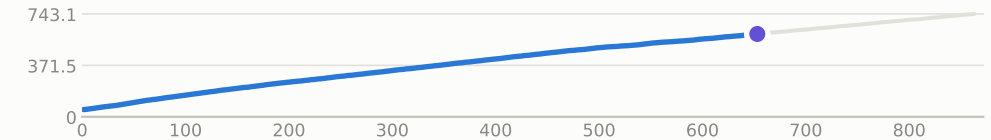
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

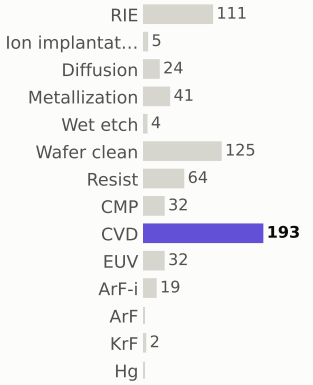
carbon so far, of the whole flow (kgCO2e)



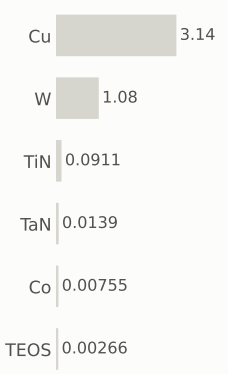
embodied carbon up to each step (kgCO2e per wafer)



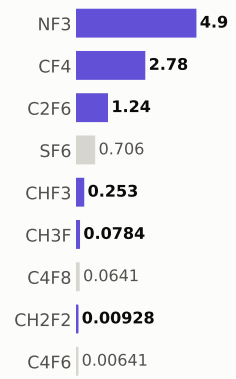
steps by process area



materials deposited (g)



gases emitted (g)



Step #6870: Wafer_clean

clean · Wafer_clean

0.169	598.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

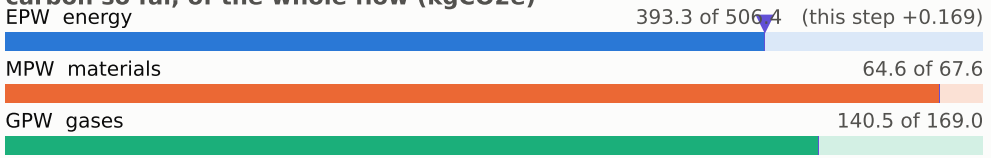
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

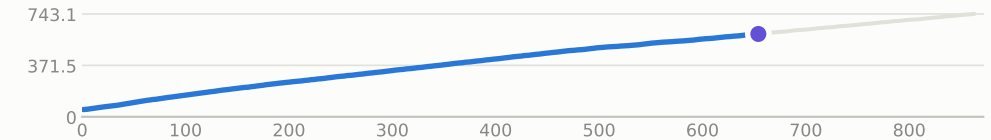
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

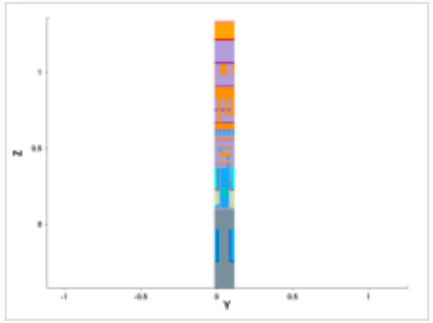
carbon so far, of the whole flow (kgCO2e)



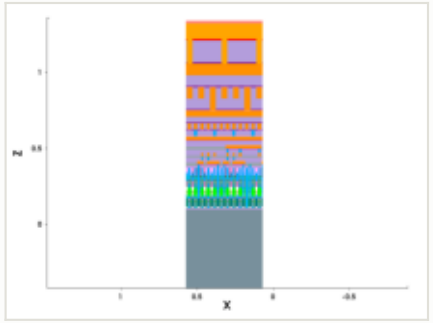
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



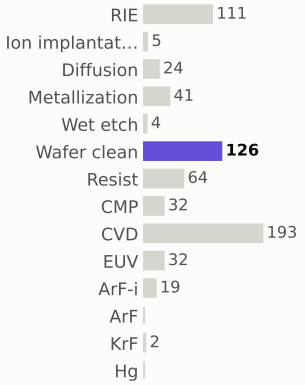
y cut at 0.049 µm



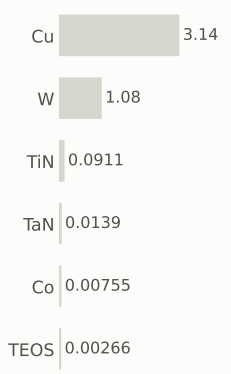
mask for this step

no mask for this step

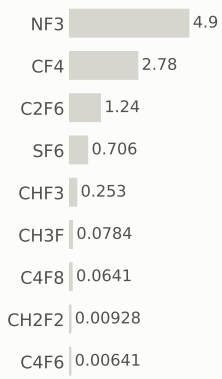
steps by process area



materials deposited (g)



gases emitted (g)



Step #6880: ArF-i

litho_patterning · ArF-i

0.217	598.6	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

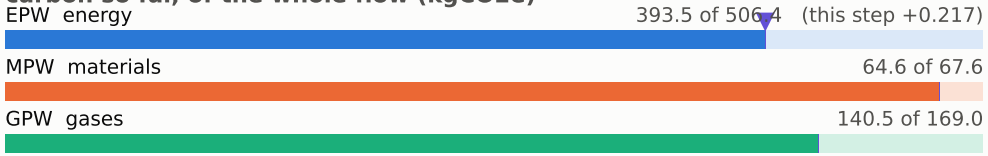
3D structure after this step

zoomed to fit

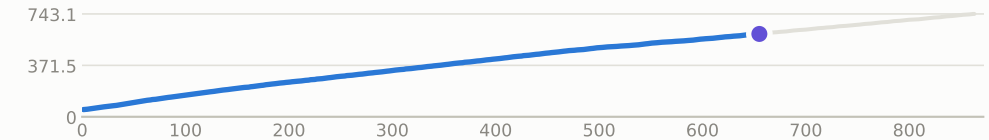
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

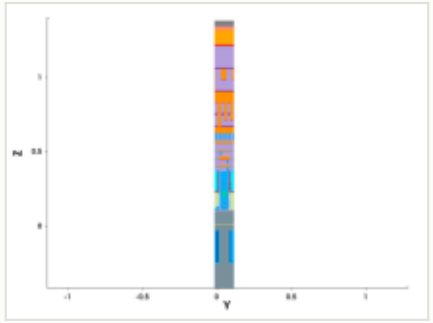
carbon so far, of the whole flow (kgCO2e)



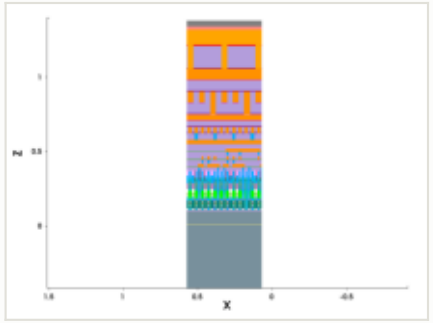
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



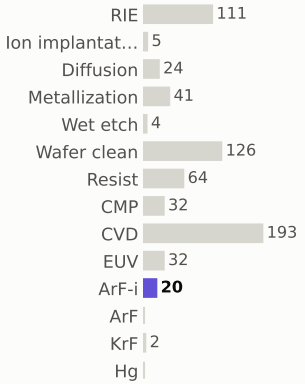
y cut at 0.049 µm



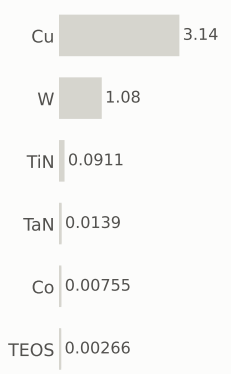
mask: layer 10572:0 (GDSII)

nothing in the clip window

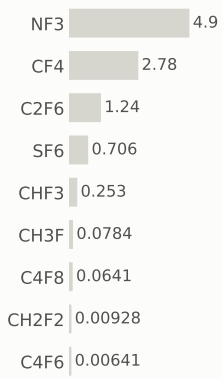
steps by process area



materials deposited (g)



gases emitted (g)



Step #6890: RIE

etch_anisotropic · RIE

1.04	599.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

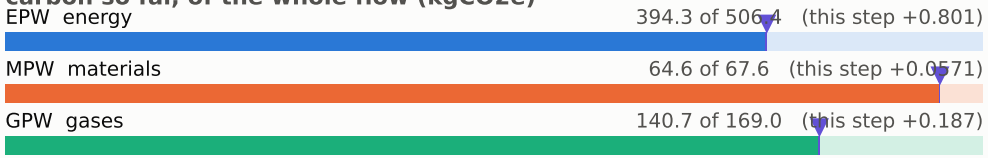
zoomed to fit

this step

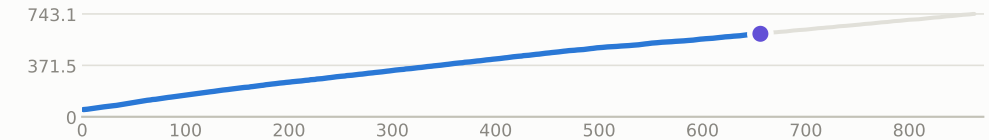
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

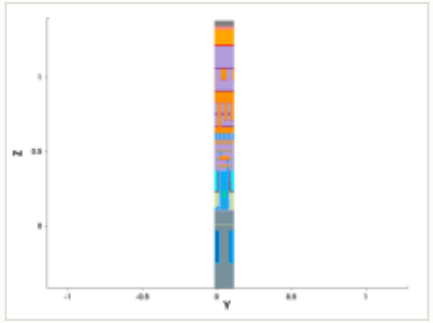
carbon so far, of the whole flow (kgCO2e)



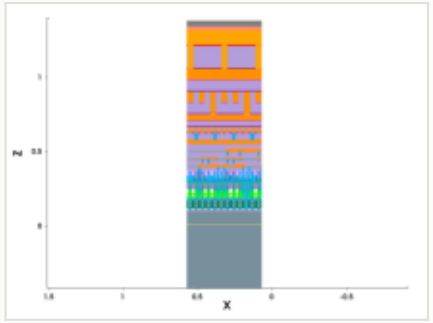
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



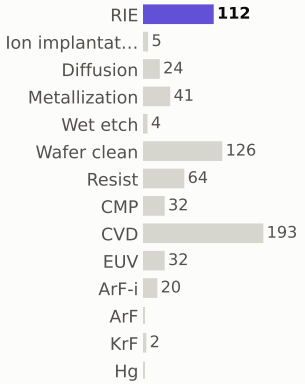
y cut at 0.049 µm



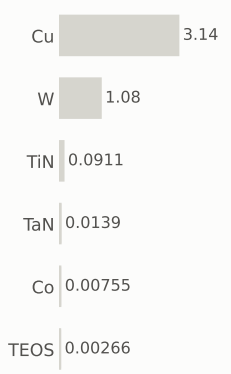
mask for this step

no mask for this step

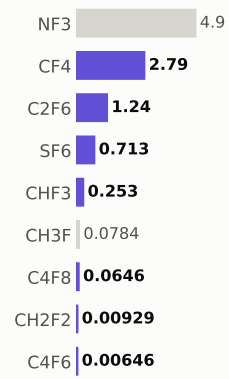
steps by process area



materials deposited (g)



gases emitted (g)



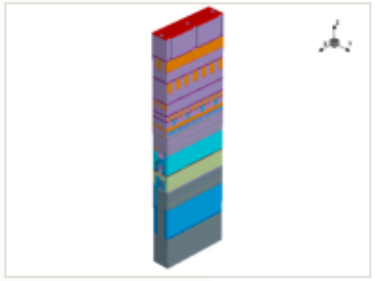
Step #6900: Wafer_clean

etch_strip · Wafer_clean

0.169	599.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

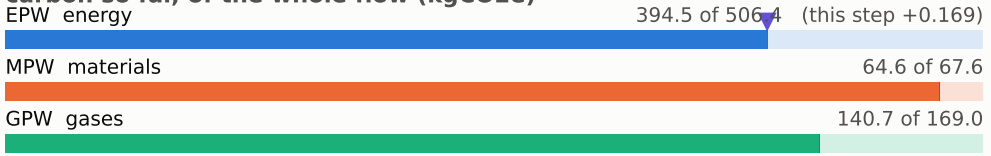


this step

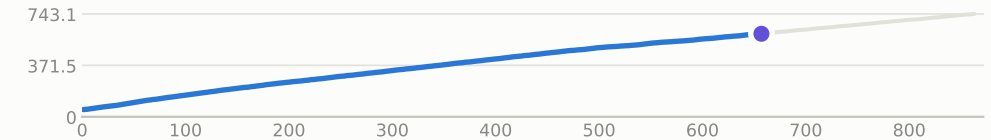
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

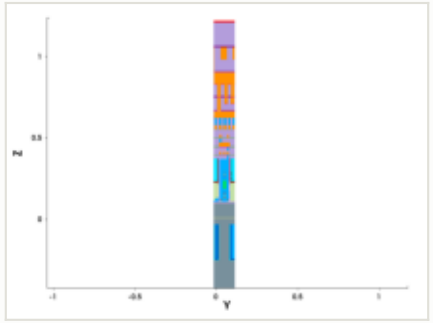
carbon so far, of the whole flow (kgCO2e)



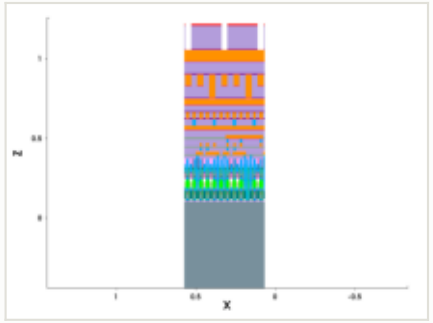
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



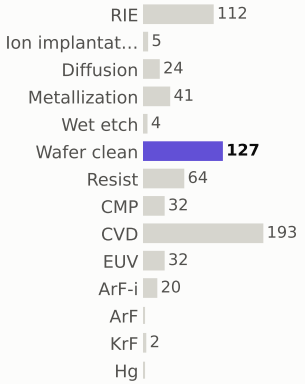
y cut at 0.049 µm



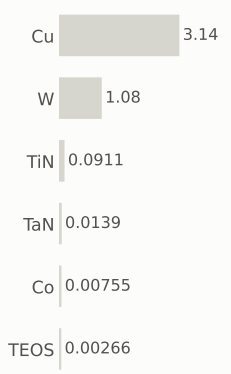
mask for this step

no mask for this step

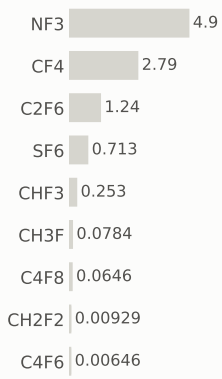
steps by process area



materials deposited (g)



gases emitted (g)



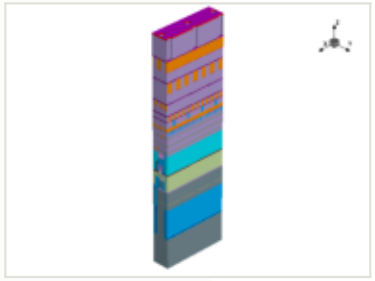
Step #6910: Metallization

depo_ellipsoidal · Metallization

1.2	601.0	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

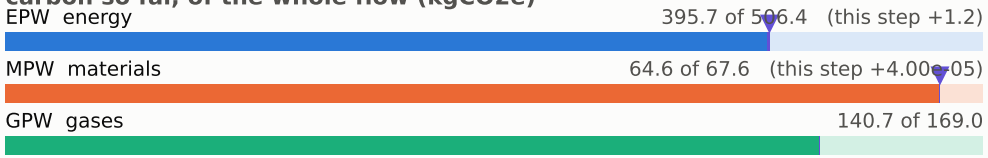


this step

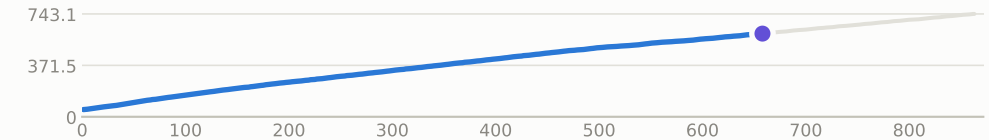
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

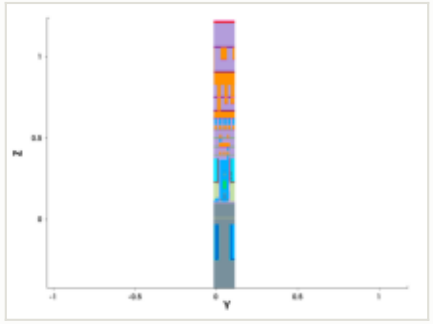
carbon so far, of the whole flow (kgCO2e)



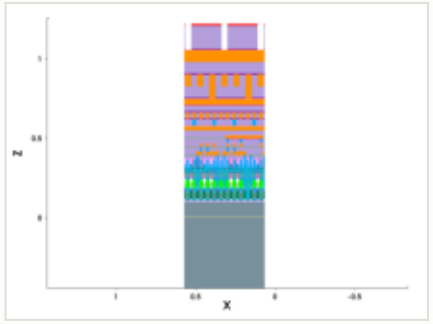
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



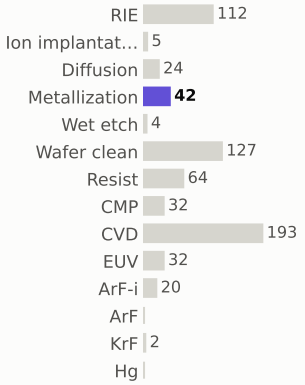
y cut at 0.049 µm



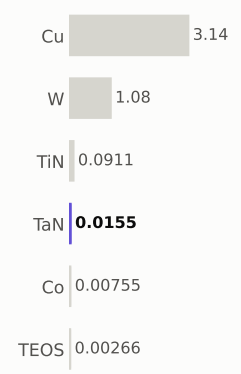
mask for this step

no mask for this step

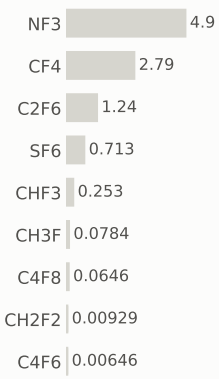
steps by process area



materials deposited (g)



gases emitted (g)



Step #6920: Metallization

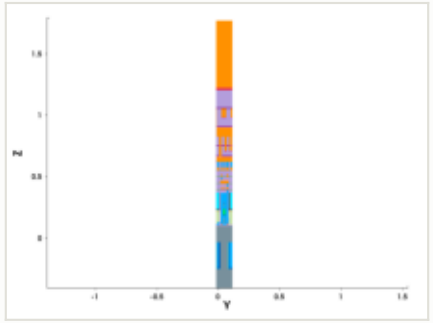
depo_fill · Metallization

1.21	602.2	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

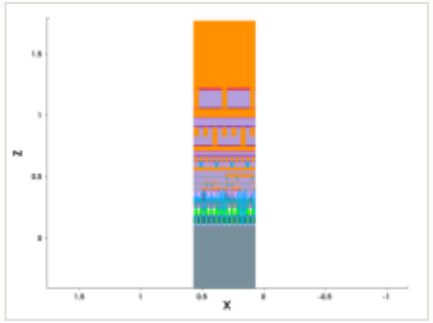
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

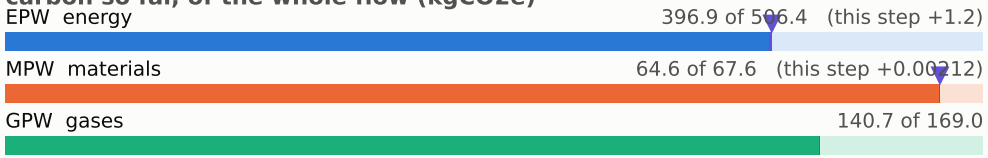
no mask for this step

this step

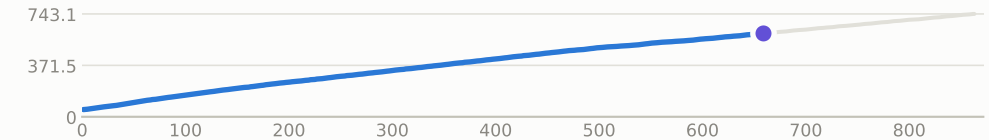
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

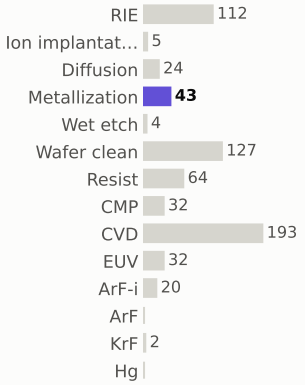
carbon so far, of the whole flow (kgCO2e)



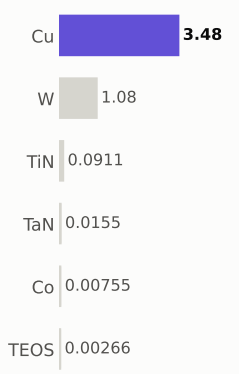
embodied carbon up to each step (kgCO2e per wafer)



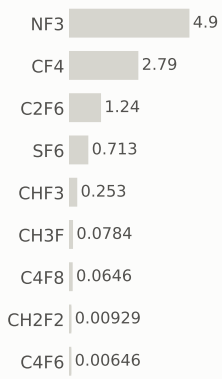
steps by process area



materials deposited (g)



gases emitted (g)



Step #6930: Diffusion

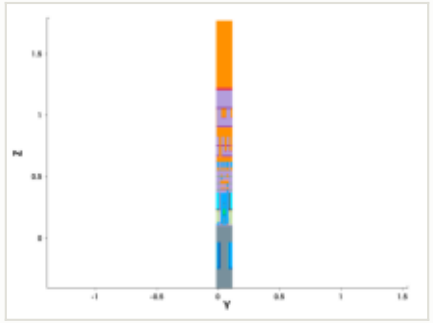
treatment · Diffusion

0.801	603.0	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

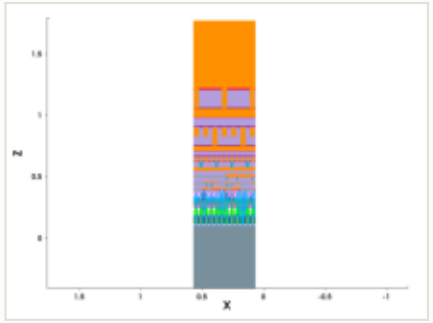
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

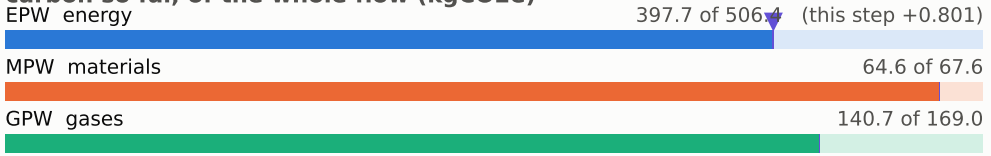
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

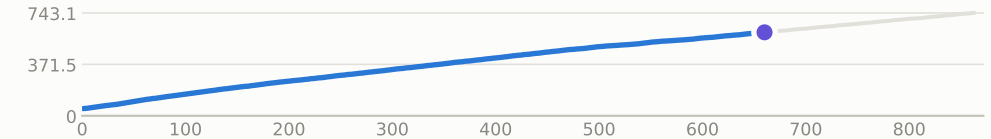
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

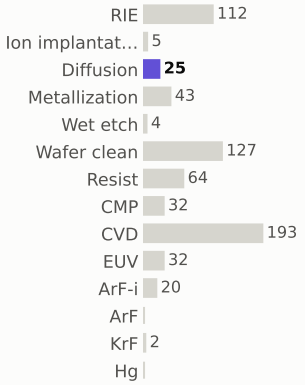
carbon so far, of the whole flow (kgCO2e)



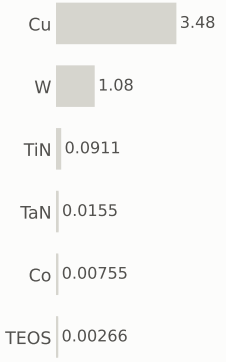
embodied carbon up to each step (kgCO2e per wafer)



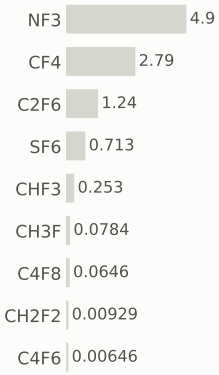
steps by process area



materials deposited (g)



gases emitted (g)



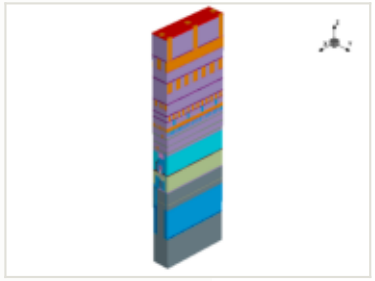
Step #6940: CMP

etch_cmp · CMP

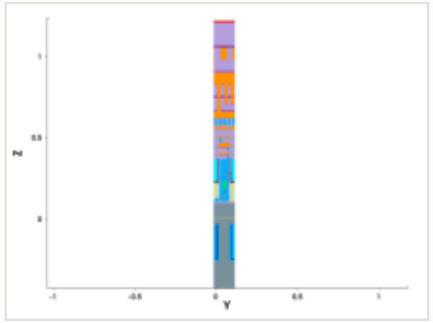
0.169	603.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

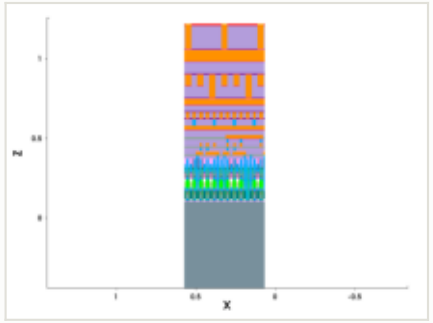
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

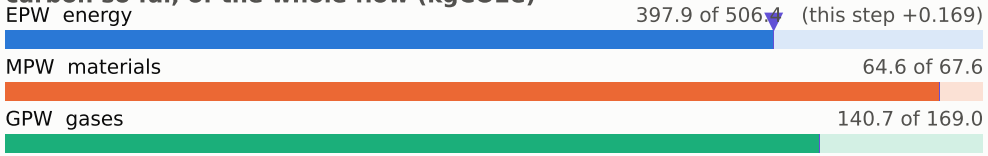
no mask for this step

this step

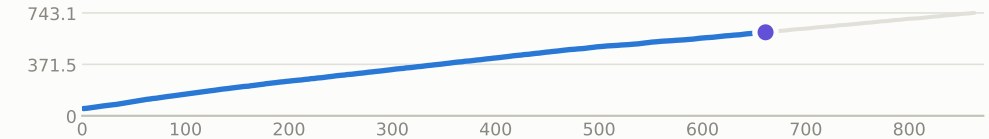
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

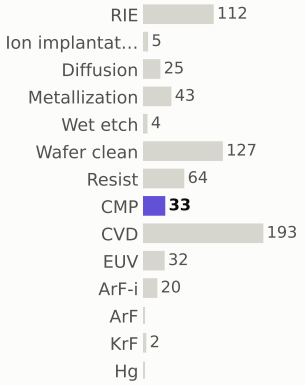
carbon so far, of the whole flow (kgCO2e)



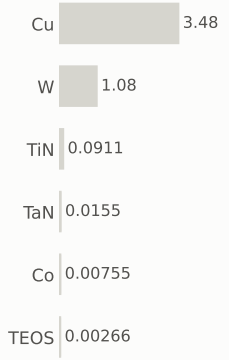
embodied carbon up to each step (kgCO2e per wafer)



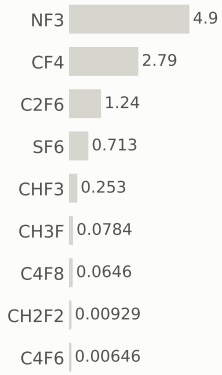
steps by process area



materials deposited (g)



gases emitted (g)



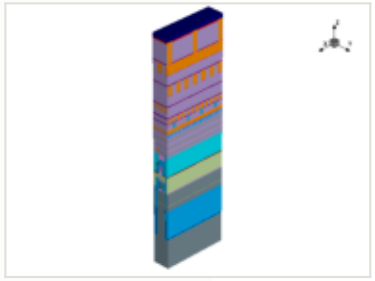
Step #6950: CVD

depo_fill · CVD

1.31	604.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

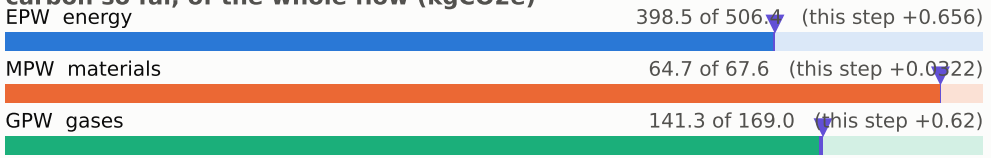


this step

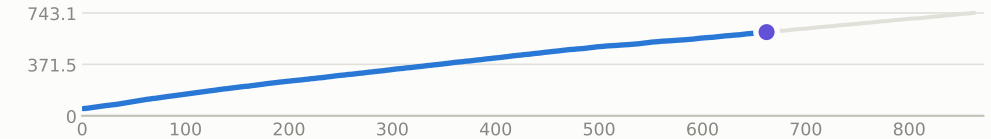
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

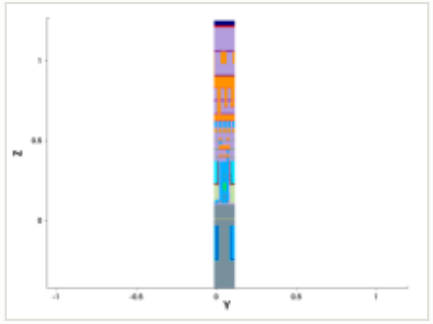
carbon so far, of the whole flow (kgCO2e)



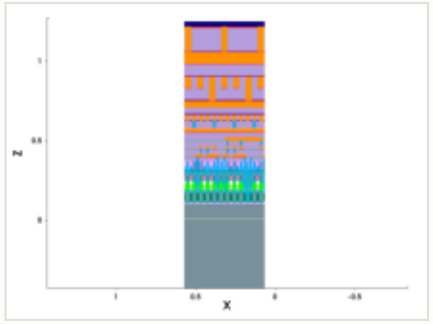
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



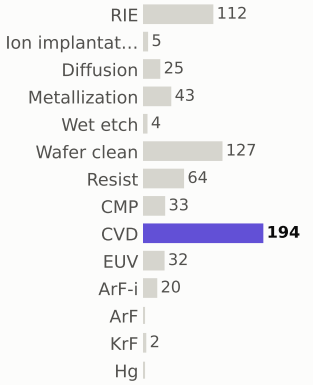
y cut at 0.049 µm



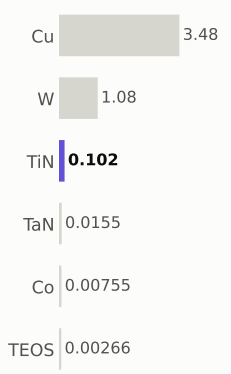
mask for this step

no mask for this step

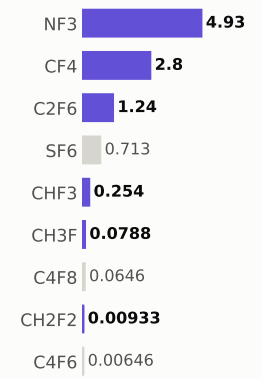
steps by process area



materials deposited (g)



gases emitted (g)



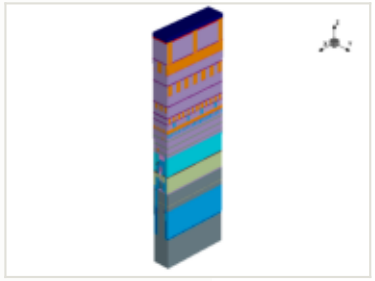
Step #6960: Wafer_clean

clean · Wafer_clean

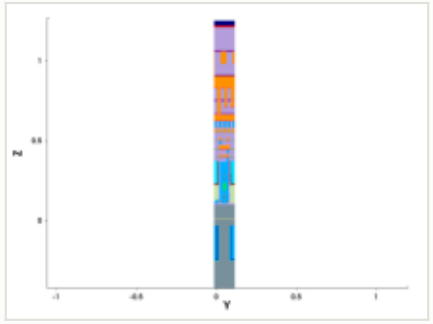
0.169	604.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

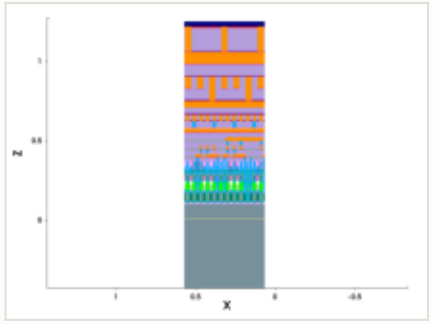
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

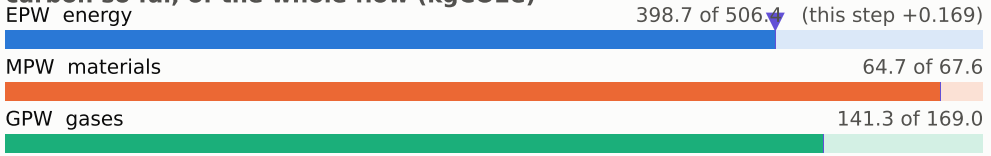
no mask for this step

this step

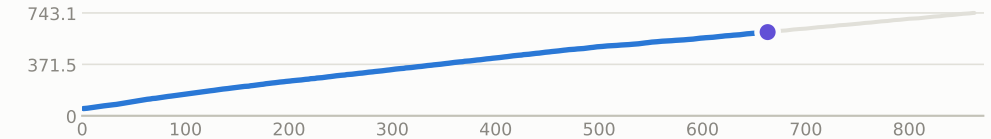
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

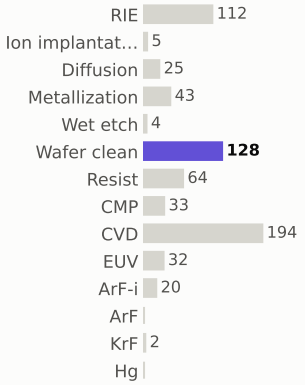
carbon so far, of the whole flow (kgCO2e)



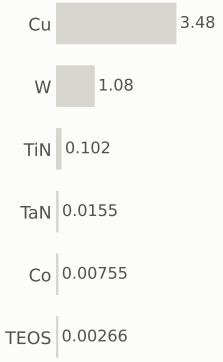
embodied carbon up to each step (kgCO2e per wafer)



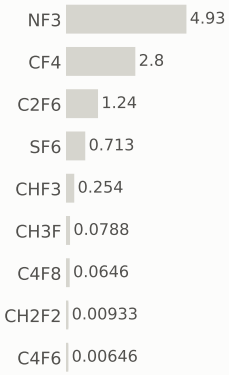
steps by process area



materials deposited (g)



gases emitted (g)



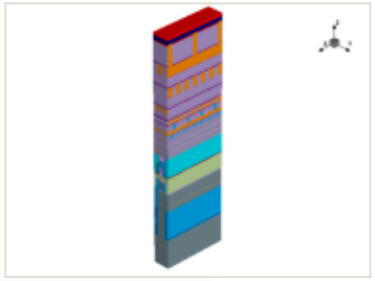
Step #6970: CVD

depo_fill · CVD

1.31	606.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

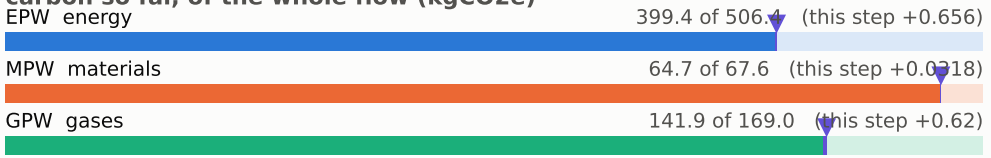


this step

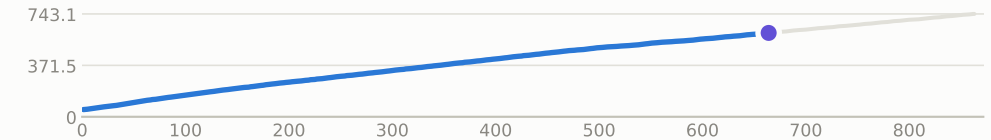
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

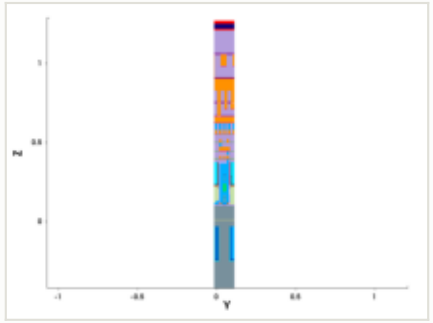
carbon so far, of the whole flow (kgCO2e)



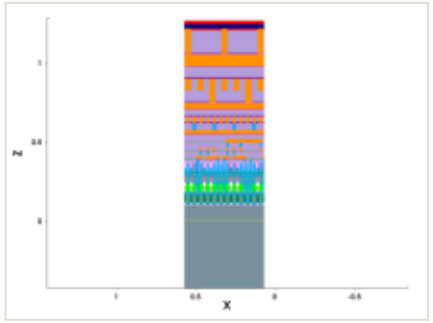
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



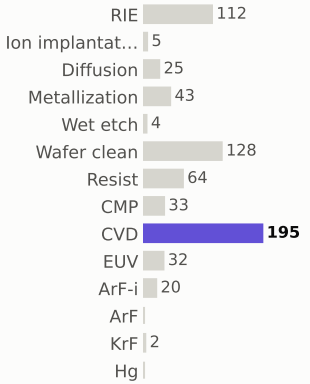
y cut at 0.049 µm



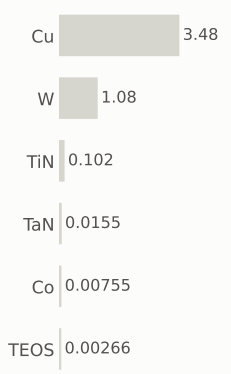
mask for this step

no mask for this step

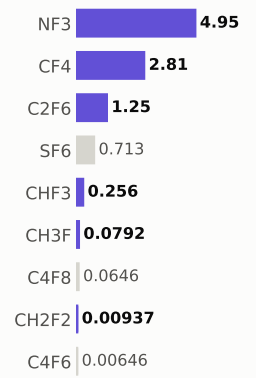
steps by process area



materials deposited (g)



gases emitted (g)



Step #6980: CVD

depo_fill · CVD

1.31	607.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

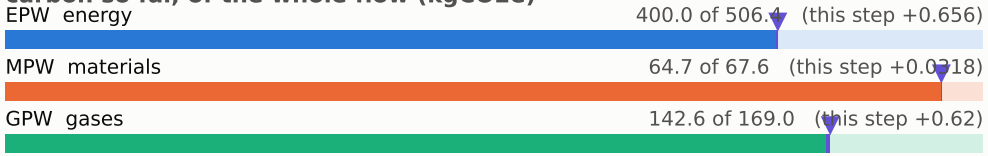
zoomed to fit

this step

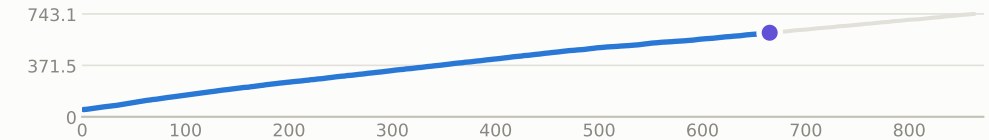
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

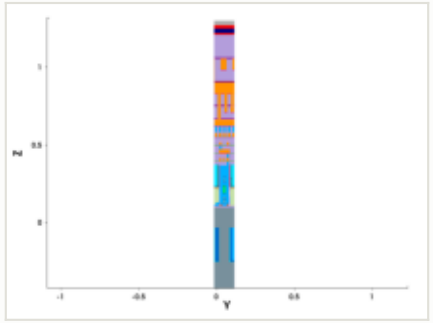
carbon so far, of the whole flow (kgCO2e)



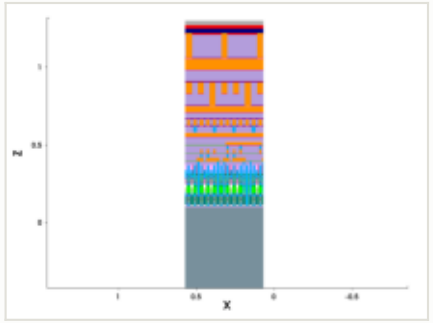
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



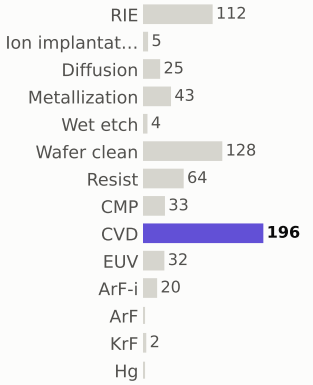
y cut at 0.049 µm



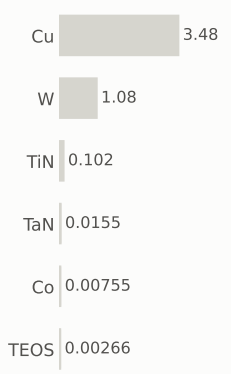
mask for this step

no mask for this step

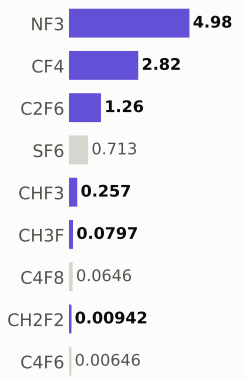
steps by process area



materials deposited (g)



gases emitted (g)



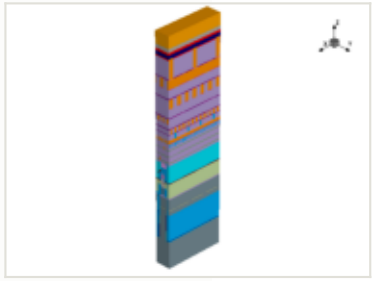
Step #6990: Resist

depo_fill · Resist

0.169	607.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

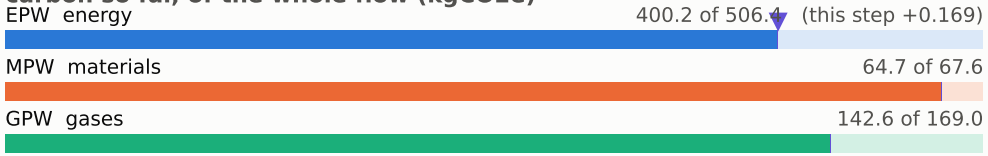
zoomed to fit



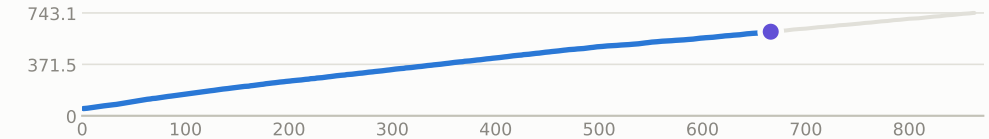
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

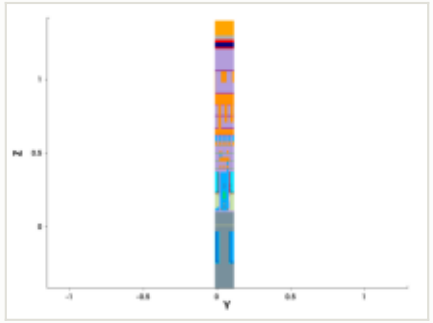
carbon so far, of the whole flow (kgCO2e)



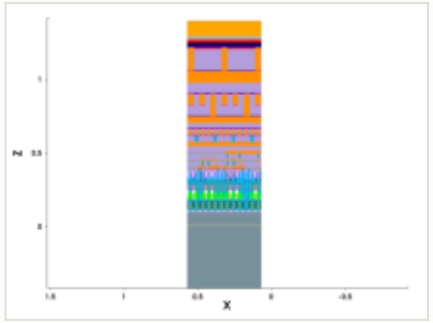
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



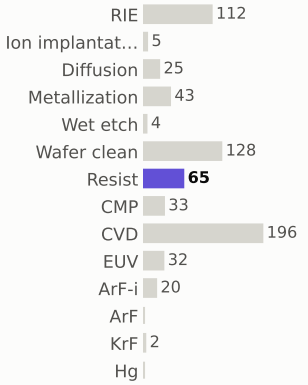
y cut at 0.049 µm



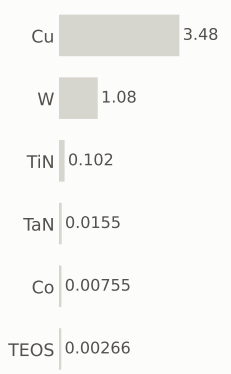
mask for this step

no mask for this step

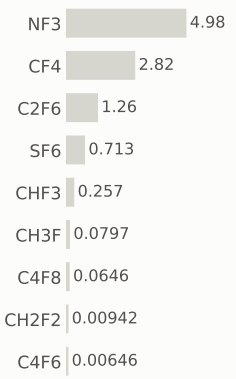
steps by process area



materials deposited (g)



gases emitted (g)



Step #7000: CVD

depo_fill · CVD

1.31	608.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

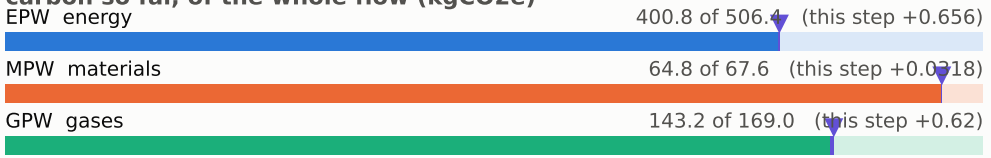
zoomed to fit

this step

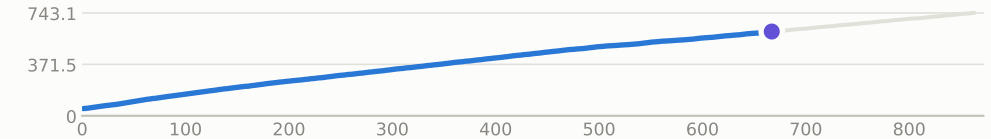
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

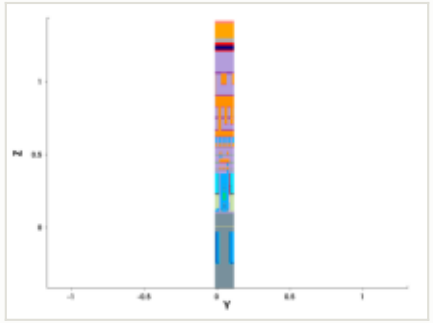
carbon so far, of the whole flow (kgCO2e)



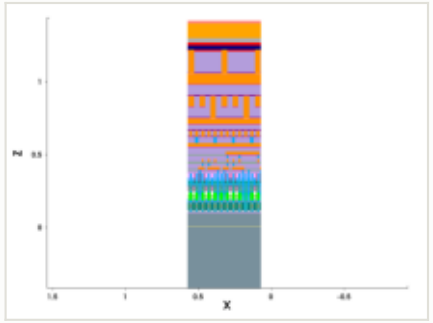
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



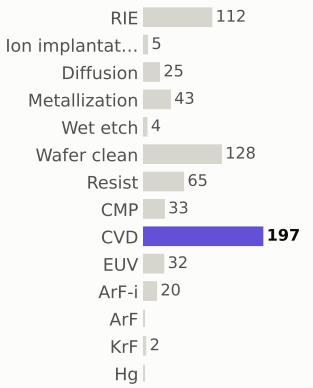
y cut at 0.049 µm



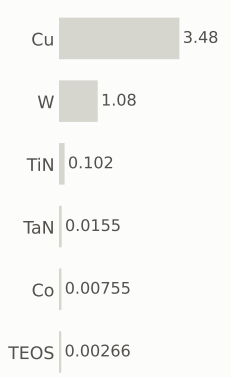
mask for this step

no mask for this step

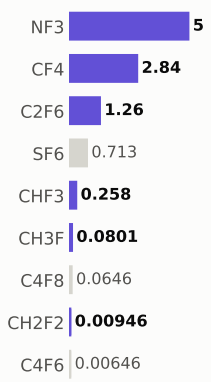
steps by process area



materials deposited (g)



gases emitted (g)



Step #7010: Wafer_clean

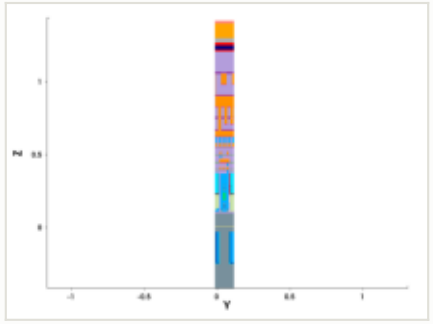
clean · Wafer_clean

0.169	608.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

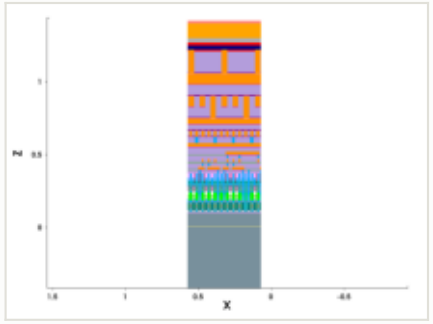
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

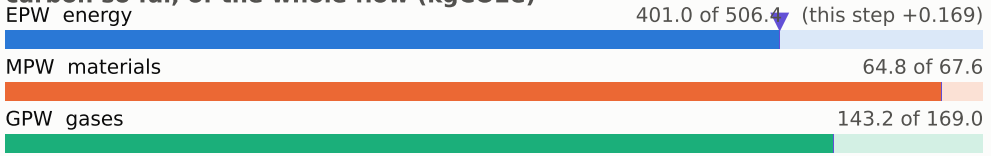
no mask for this step

this step

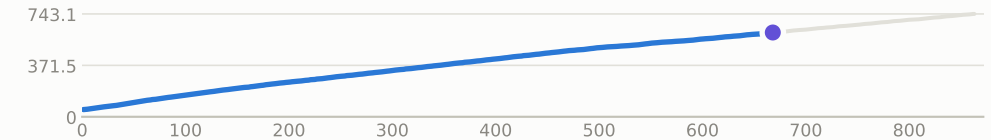
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

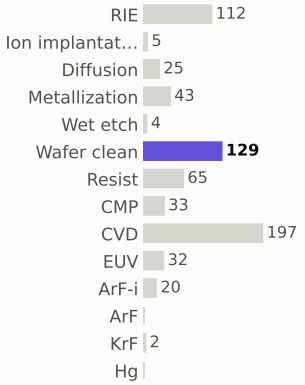
carbon so far, of the whole flow (kgCO2e)



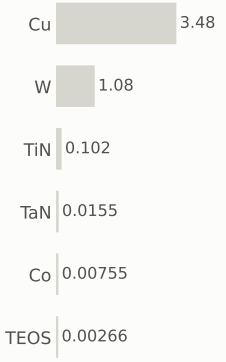
embodied carbon up to each step (kgCO2e per wafer)



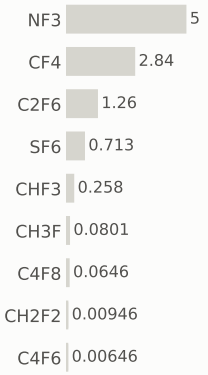
steps by process area



materials deposited (g)



gases emitted (g)



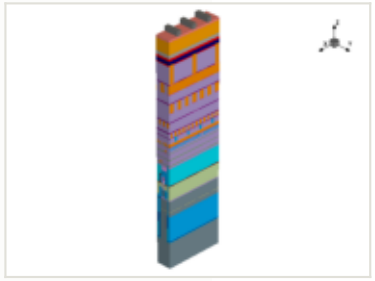
Step #7020: ArF-i

litho_patterning · ArF-i

0.217	609.2	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

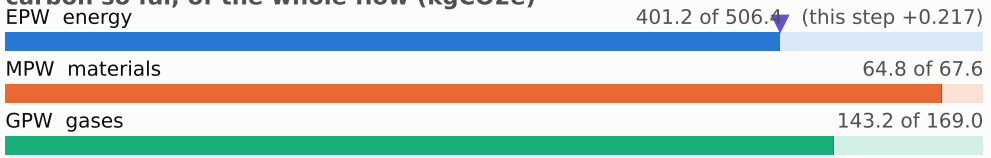


this step

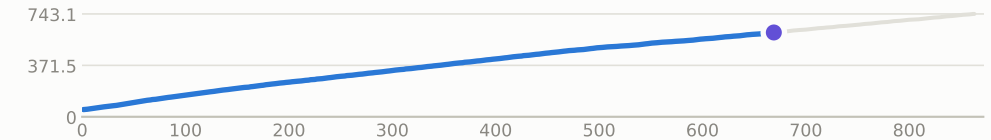
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

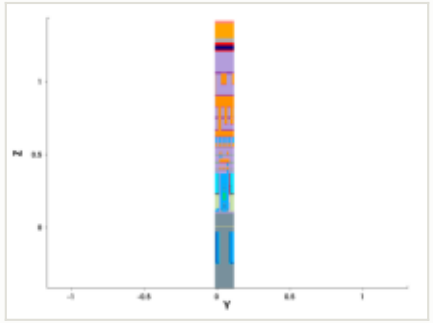
carbon so far, of the whole flow (kgCO2e)



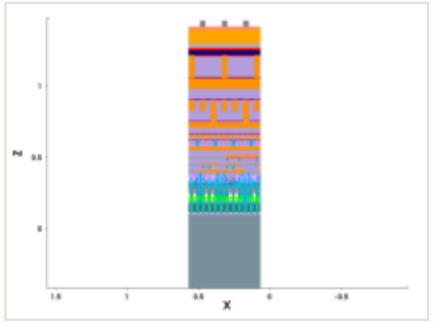
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

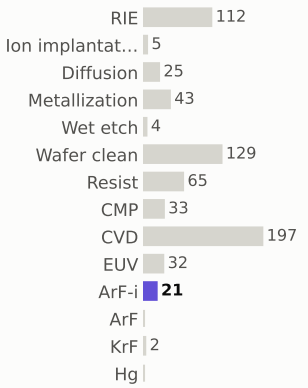


y cut at 0.049 µm

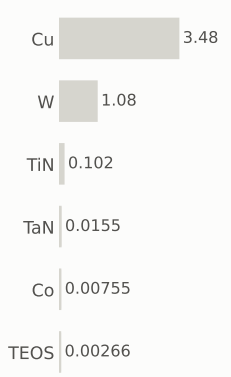


mask: layer 10601:0, positive (GDSII)

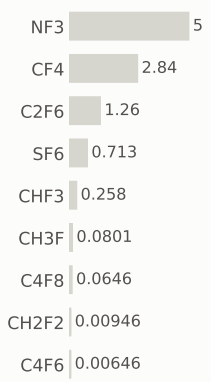
steps by process area



materials deposited (g)



gases emitted (g)



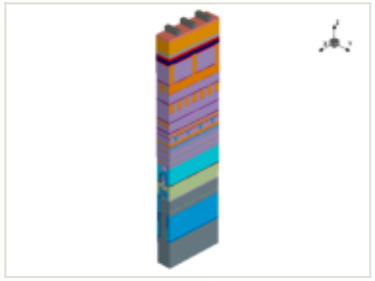
Step #7030: Ion_implantation

treatment · Ion_implantation

0.801	610.0	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

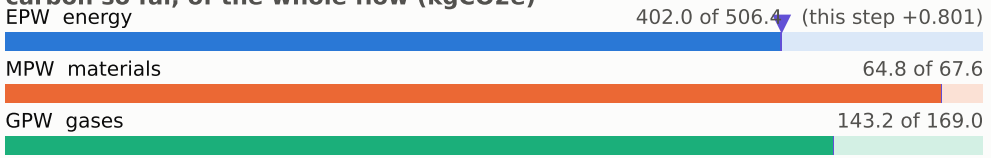


this step

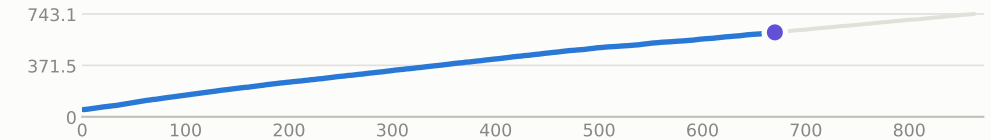
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

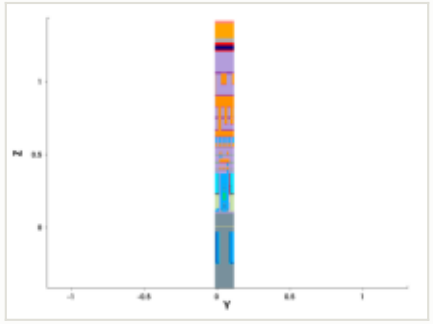
carbon so far, of the whole flow (kgCO2e)



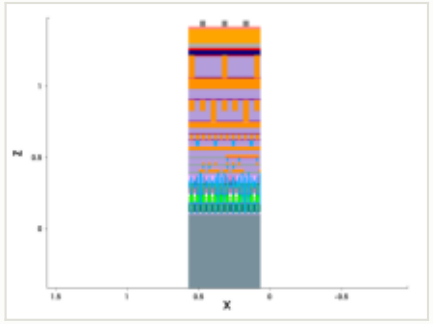
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



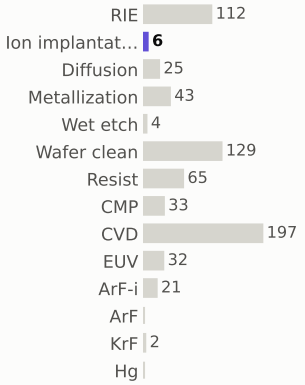
y cut at 0.049 µm



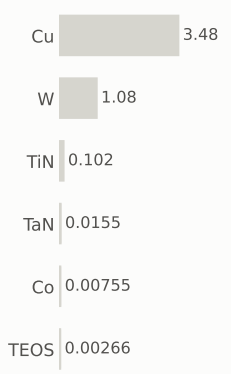
mask for this step

no mask for this step

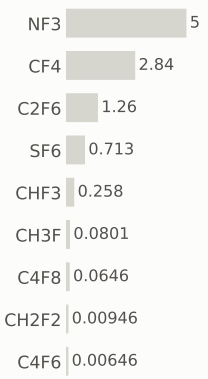
steps by process area



materials deposited (g)



gases emitted (g)



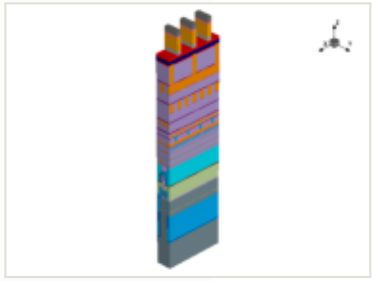
Step #7040: RIE

etch_anisotropic · RIE

1.04	611.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

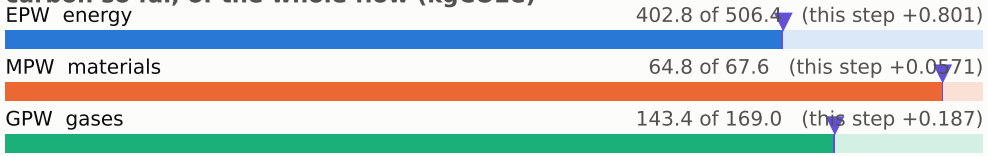


this step

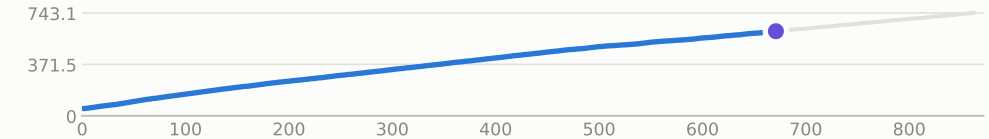
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

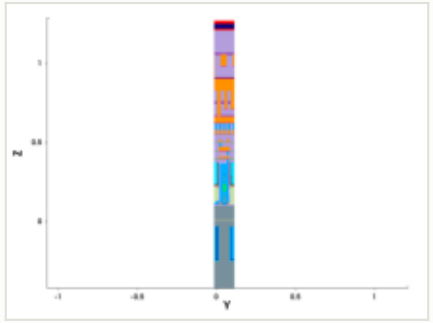
carbon so far, of the whole flow (kgCO2e)



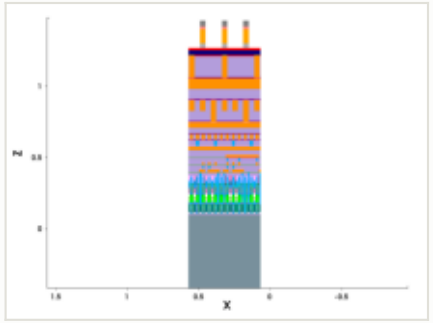
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



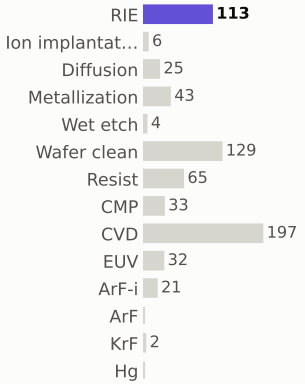
y cut at 0.049 µm



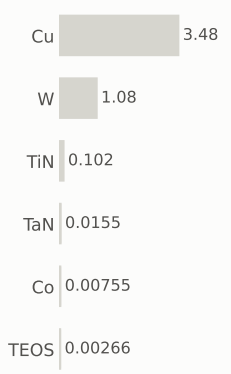
mask for this step

no mask for this step

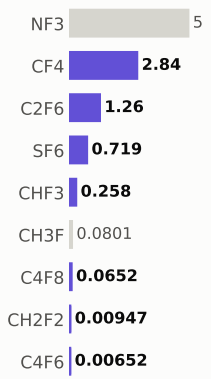
steps by process area



materials deposited (g)



gases emitted (g)



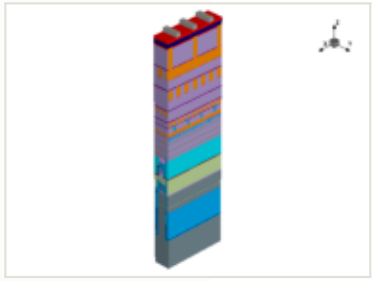
Step #7050: Wafer_clean

etch_strip · Wafer_clean

0.169	611.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

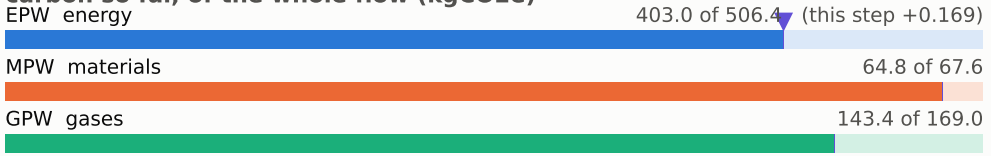


this step

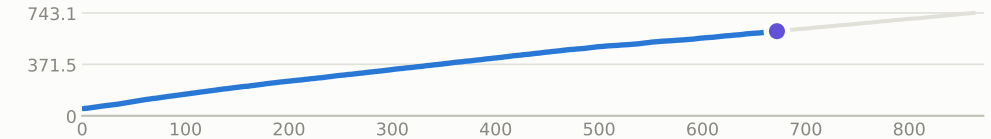
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

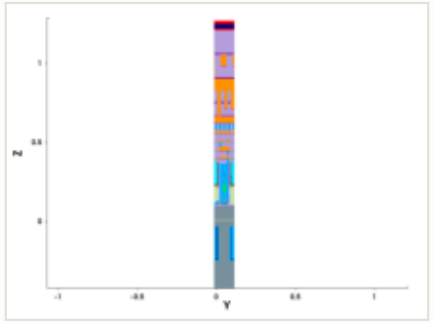
carbon so far, of the whole flow (kgCO2e)



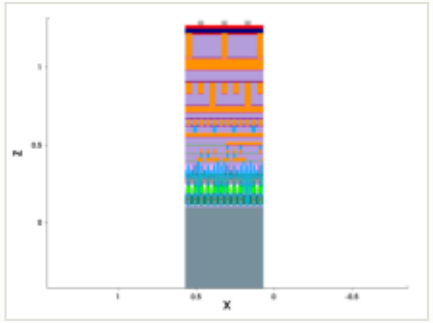
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



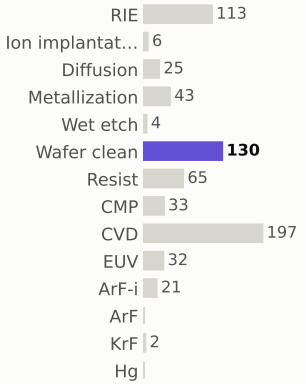
y cut at 0.049 µm



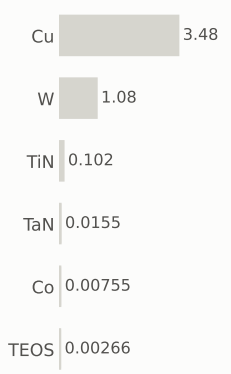
mask for this step

no mask for this step

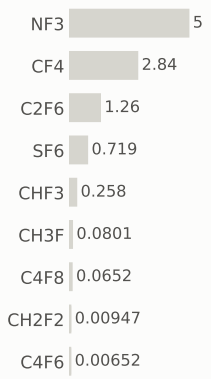
steps by process area



materials deposited (g)



gases emitted (g)



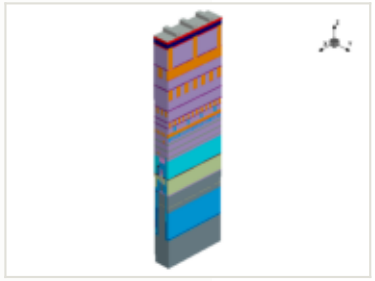
Step #7060: CVD

depo_isotropic · CVD

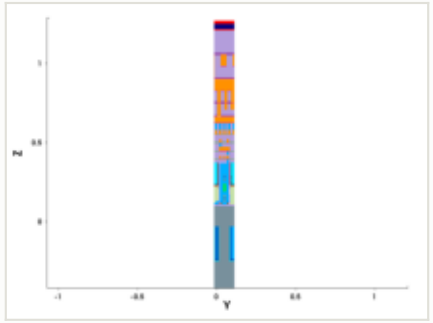
1.31	612.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

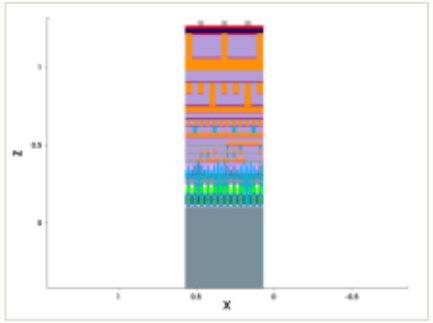
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

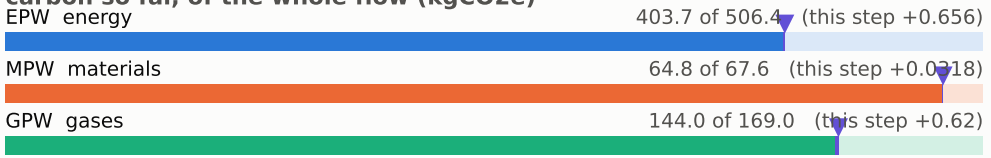
no mask for this step

this step

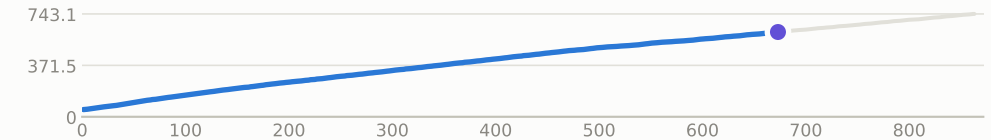
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

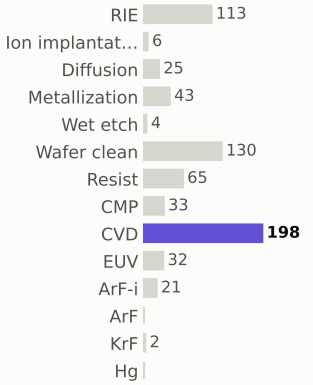
carbon so far, of the whole flow (kgCO2e)



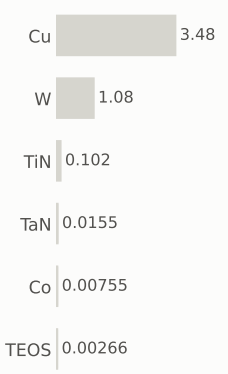
embodied carbon up to each step (kgCO2e per wafer)



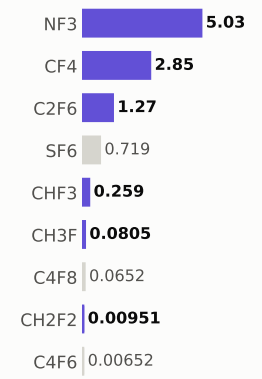
steps by process area



materials deposited (g)



gases emitted (g)



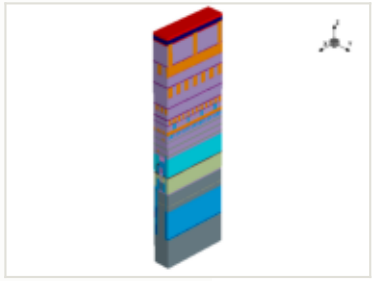
Step #7070: RIE

etch_anisotropic · RIE

1.04	613.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

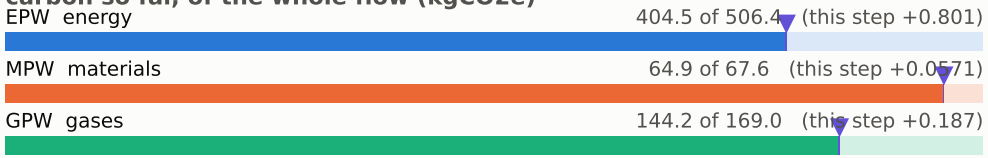


this step

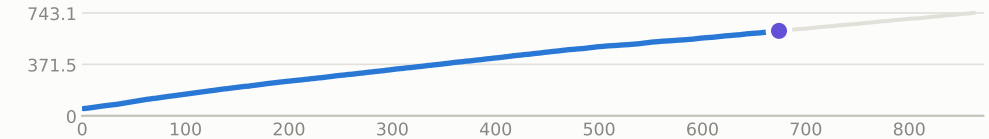
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

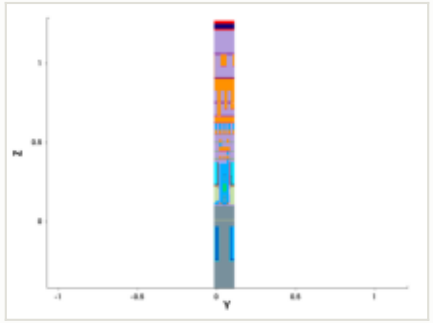
carbon so far, of the whole flow (kgCO2e)



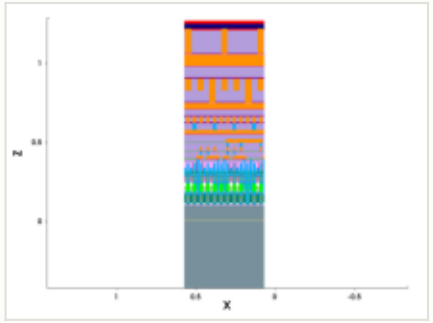
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



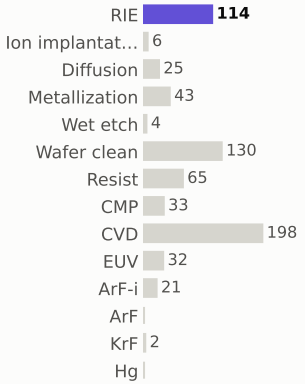
y cut at 0.049 µm



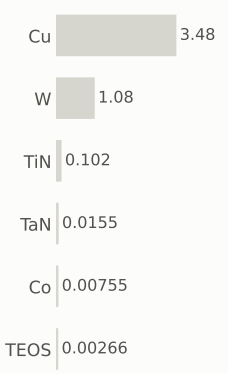
mask for this step

no mask for this step

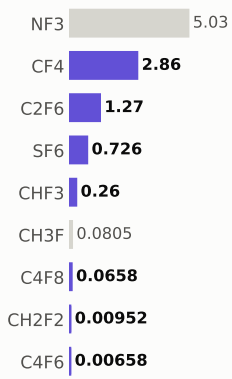
steps by process area



materials deposited (g)



gases emitted (g)



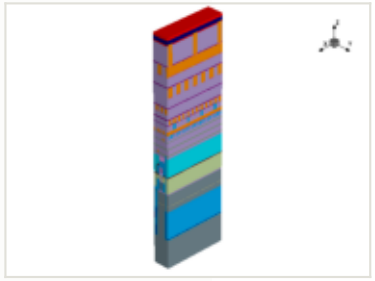
Step #7080: Wafer_clean

etch_strip · Wafer_clean

0.169	613.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

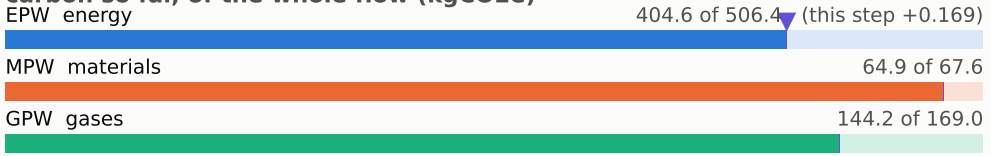


this step

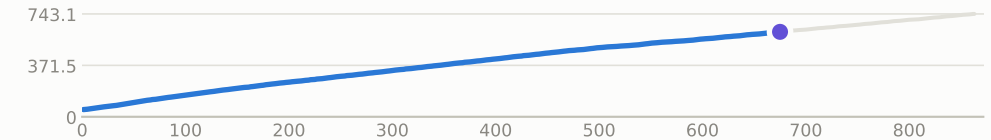
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

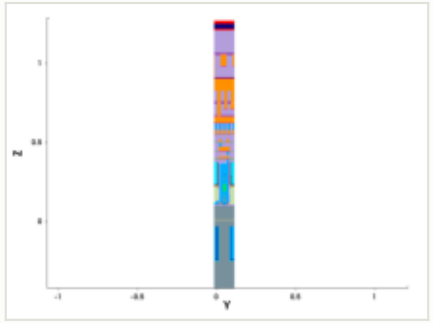
carbon so far, of the whole flow (kgCO2e)



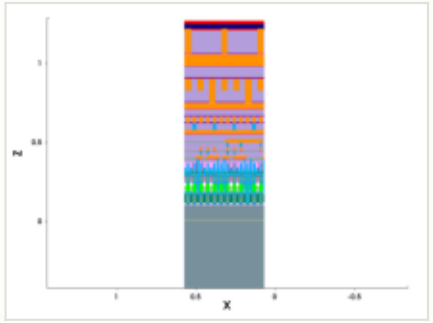
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



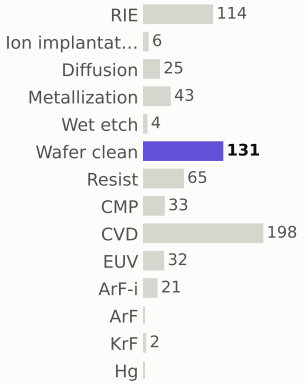
y cut at 0.049 µm



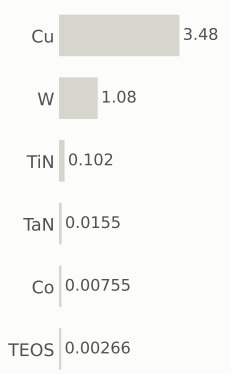
mask for this step

no mask for this step

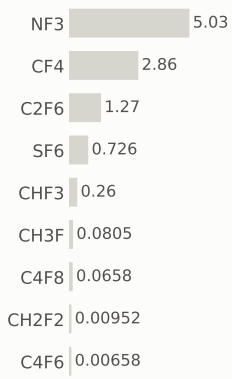
steps by process area



materials deposited (g)



gases emitted (g)



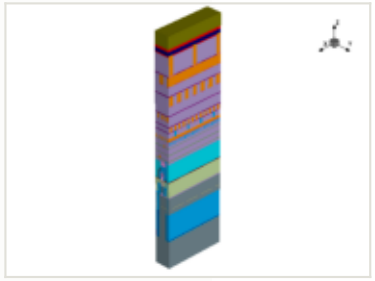
Step #7090: Resist

depo_fill · Resist

0.169	613.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

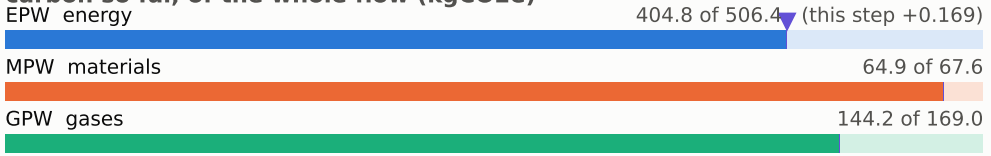
zoomed to fit



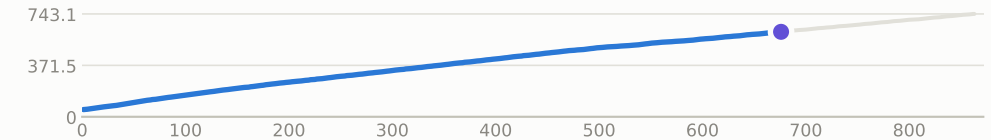
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

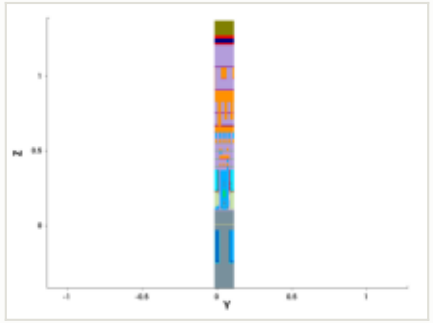
carbon so far, of the whole flow (kgCO2e)



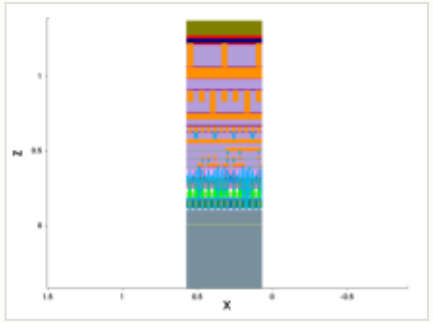
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



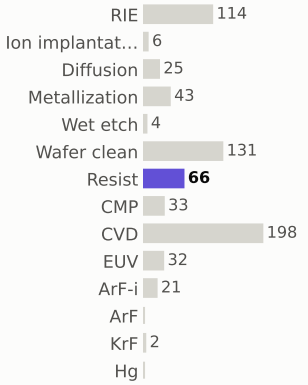
y cut at 0.049 µm



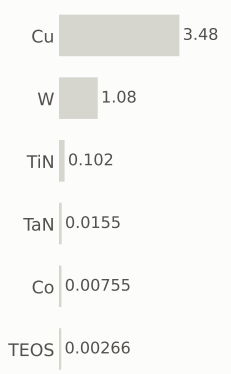
mask for this step

no mask for this step

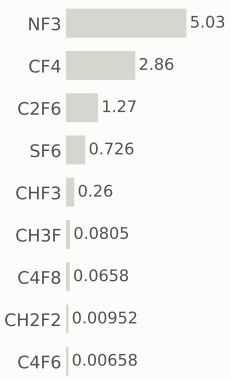
steps by process area



materials deposited (g)



gases emitted (g)



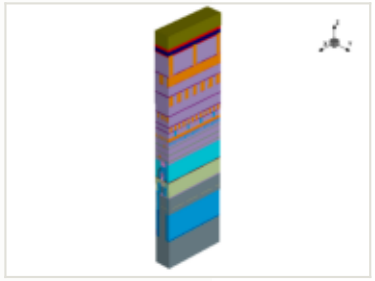
Step #7100: Wafer_clean

clean · Wafer_clean

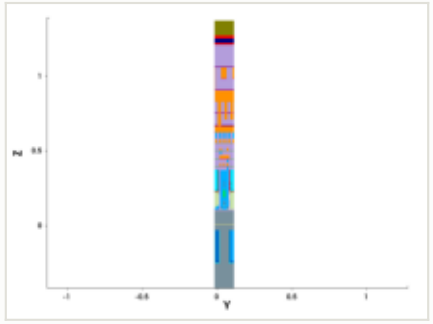
0.169	614.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

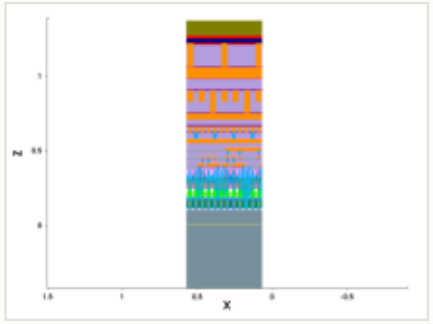
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

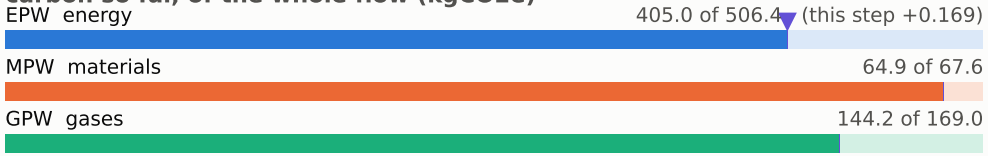
no mask for this step

this step

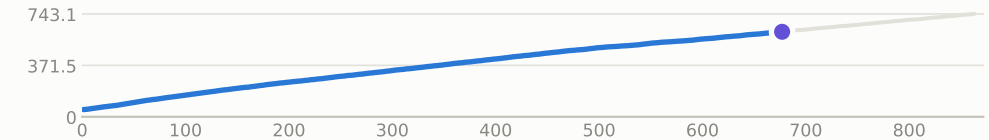
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

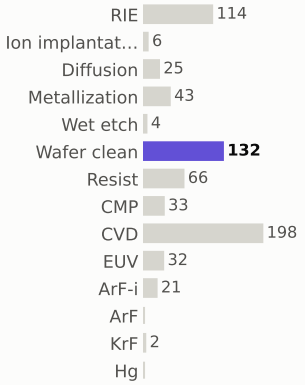
carbon so far, of the whole flow (kgCO2e)



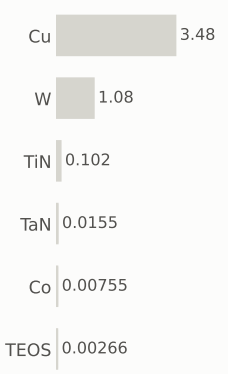
embodied carbon up to each step (kgCO2e per wafer)



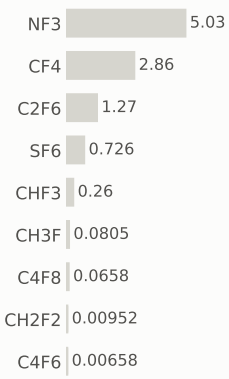
steps by process area



materials deposited (g)



gases emitted (g)



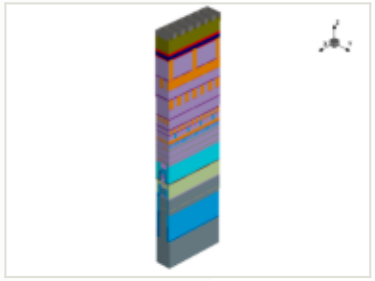
Step #7110: ArF-i

litho_patterning · ArF-i

0.217	614.2	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

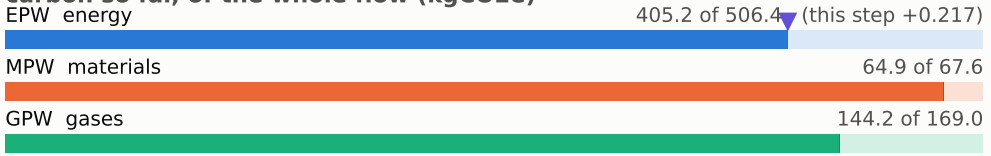
zoomed to fit



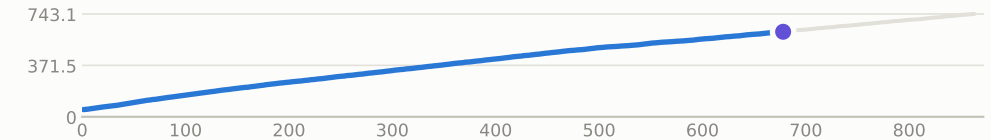
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

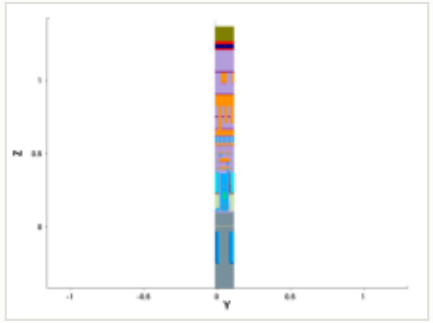
carbon so far, of the whole flow (kgCO2e)



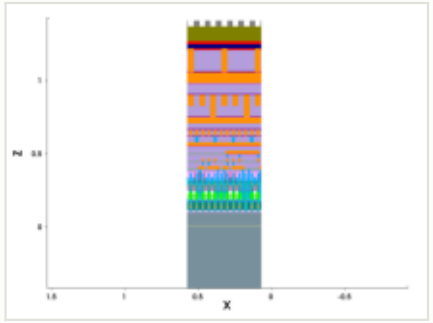
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

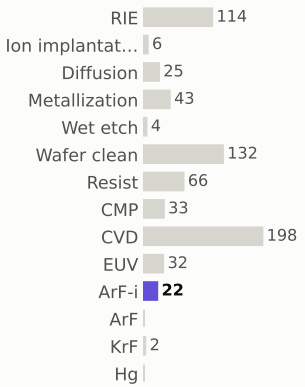


y cut at 0.049 µm

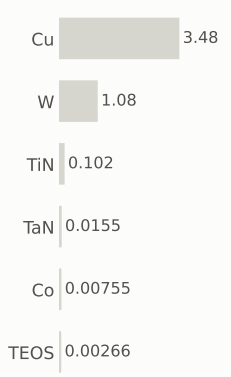


mask: layer 60:0 (GDSII)

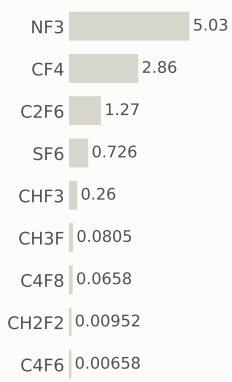
steps by process area



materials deposited (g)



gases emitted (g)



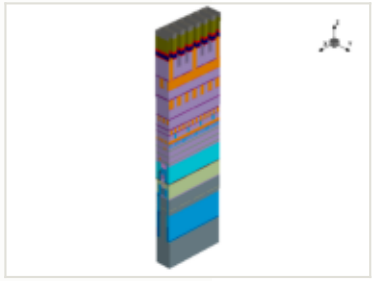
Step #7120: RIE

etch_anisotropic · RIE

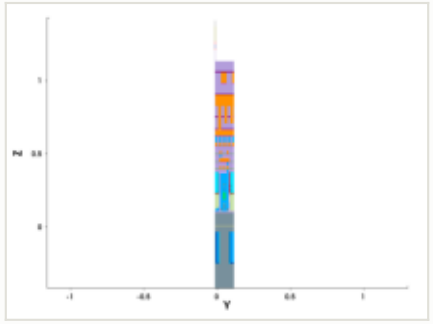
1.04	615.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

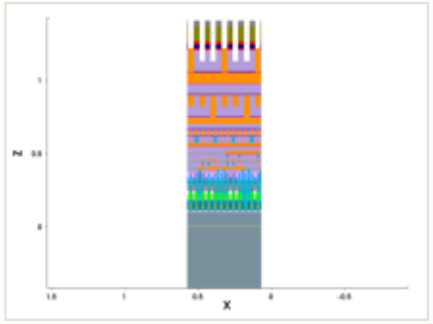
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

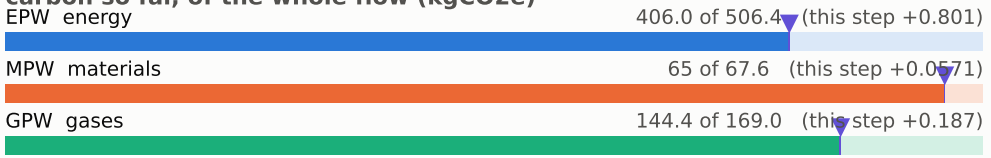
no mask for this step

this step

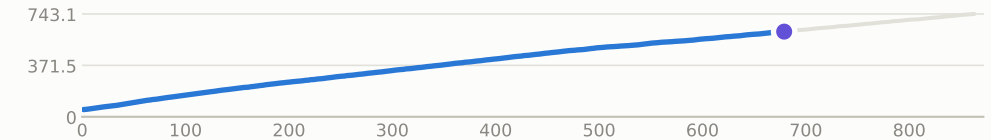
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

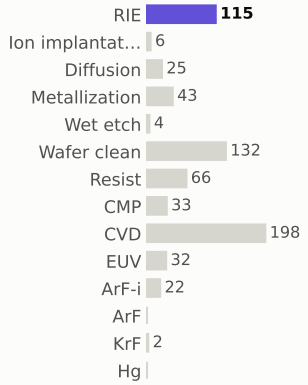
carbon so far, of the whole flow (kgCO2e)



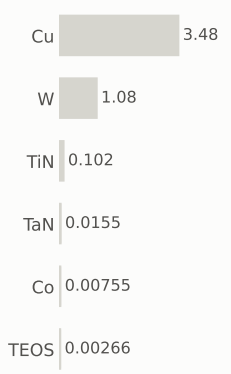
embodied carbon up to each step (kgCO2e per wafer)



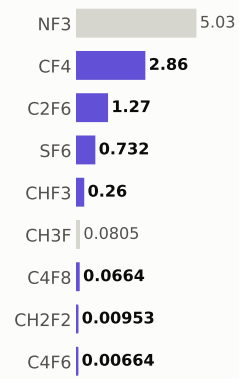
steps by process area



materials deposited (g)



gases emitted (g)



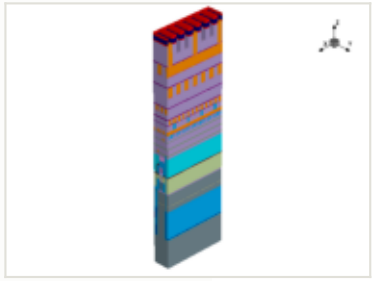
Step #7130: Wafer_clean

etch_strip · Wafer_clean

0.169	615.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

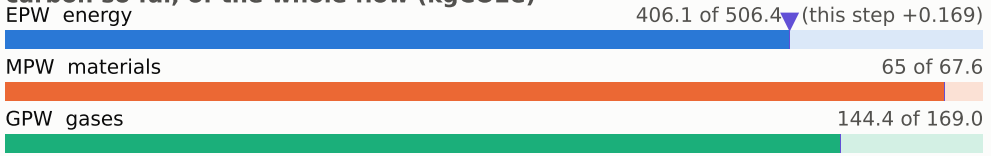


this step

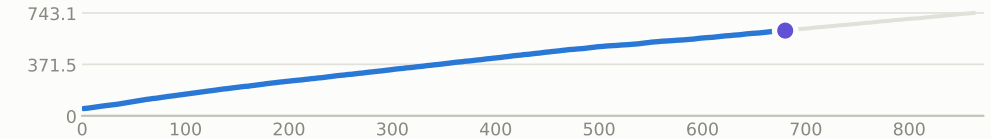
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

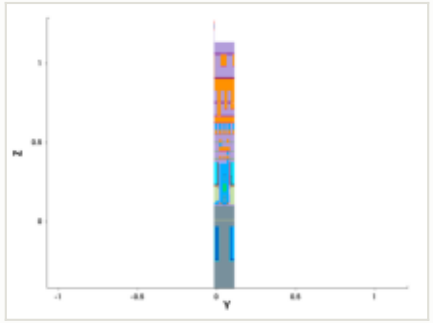
carbon so far, of the whole flow (kgCO2e)



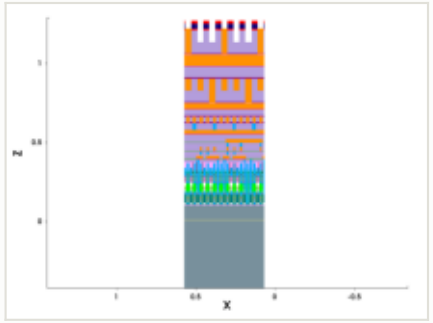
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



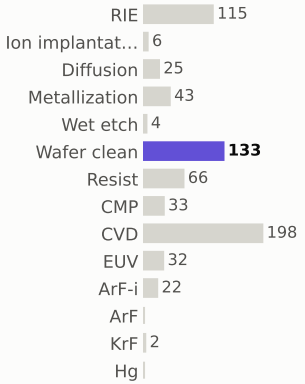
y cut at 0.049 µm



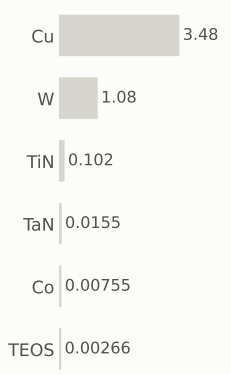
mask for this step

no mask for this step

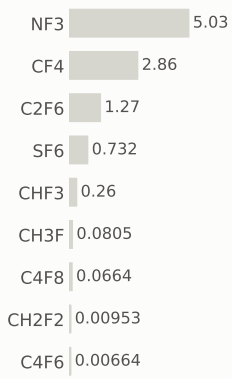
steps by process area



materials deposited (g)



gases emitted (g)



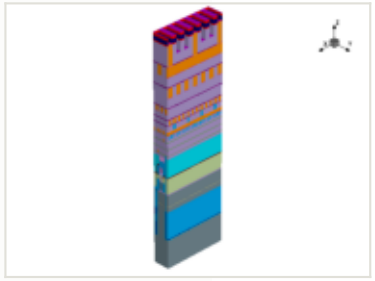
Step #7140: Metallization

depo_ellipsoidal · Metallization

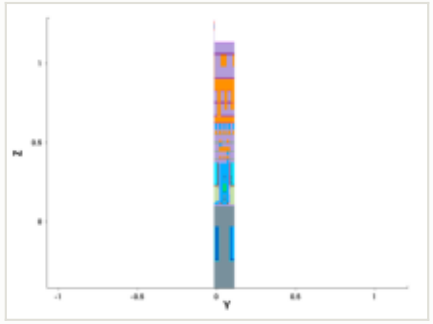
1.2	616.7	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

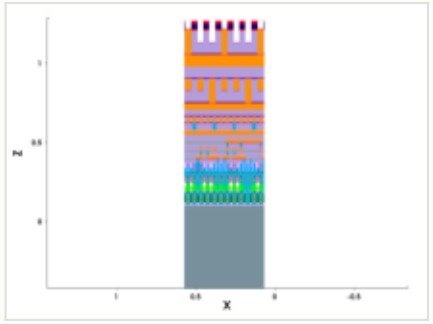
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

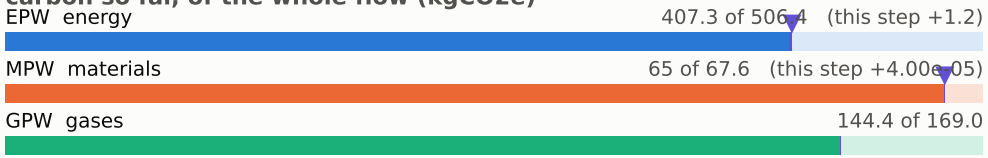
no mask for this step

this step

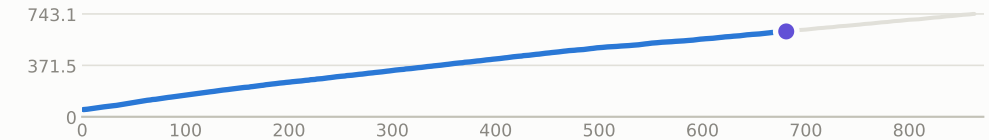
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

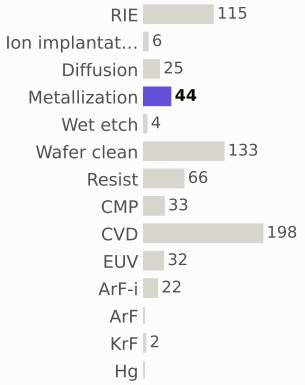
carbon so far, of the whole flow (kgCO2e)



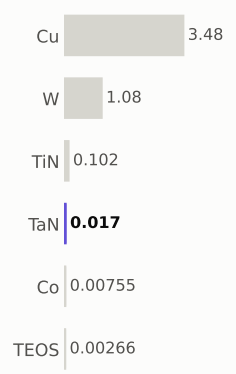
embodied carbon up to each step (kgCO2e per wafer)



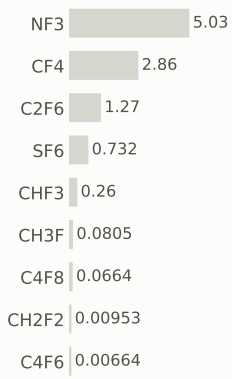
steps by process area



materials deposited (g)



gases emitted (g)



Step #7150: Metallization

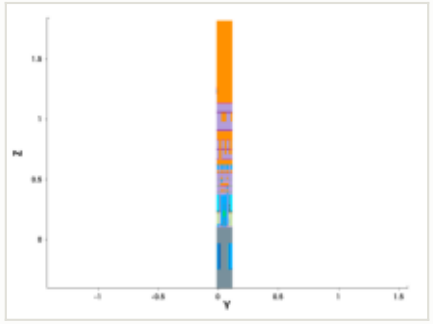
depo_fill · Metallization

1.21	617.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

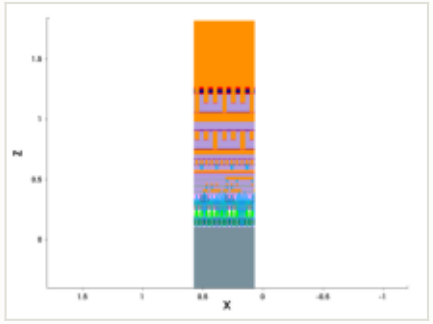
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

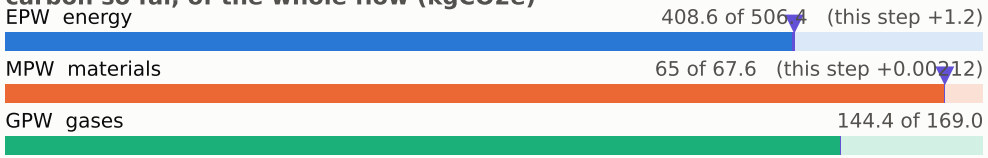
no mask for this step

this step

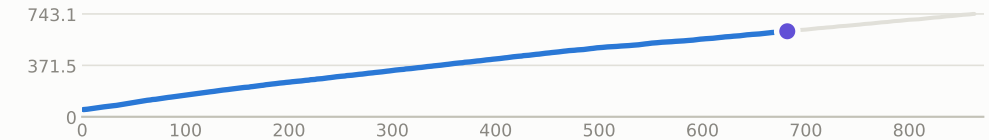
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

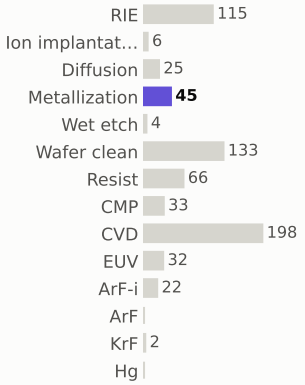
carbon so far, of the whole flow (kgCO2e)



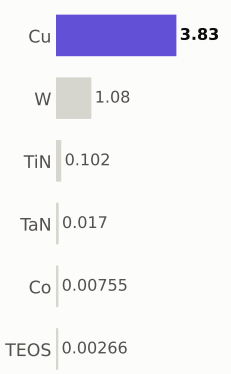
embodied carbon up to each step (kgCO2e per wafer)



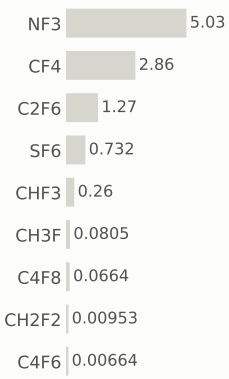
steps by process area



materials deposited (g)



gases emitted (g)



Step #7160: Diffusion

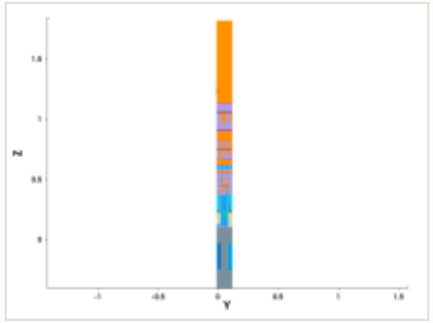
treatment · Diffusion

0.801	618.7	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

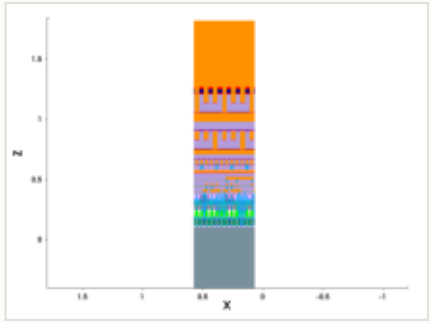
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

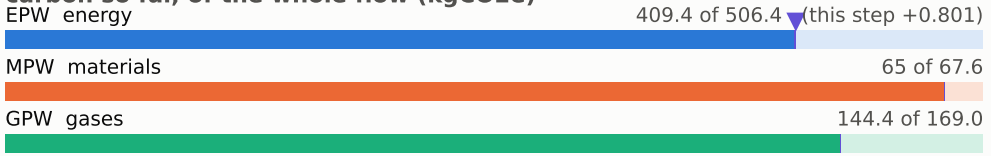
no mask for this step

this step

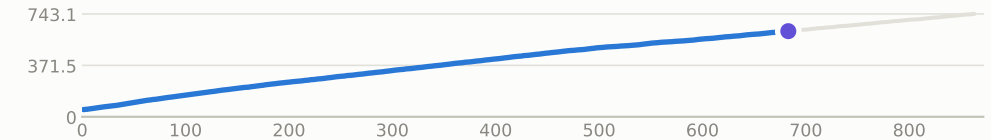
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

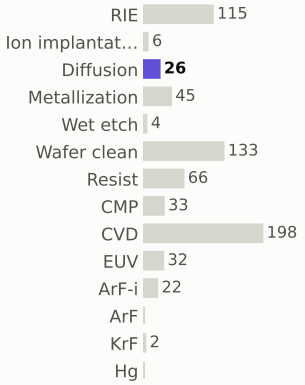
carbon so far, of the whole flow (kgCO2e)



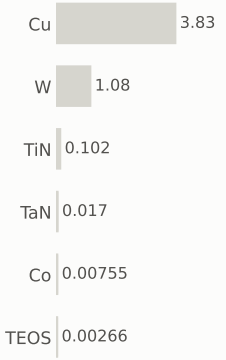
embodied carbon up to each step (kgCO2e per wafer)



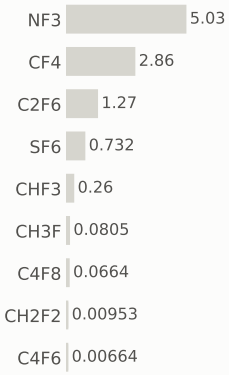
steps by process area



materials deposited (g)



gases emitted (g)



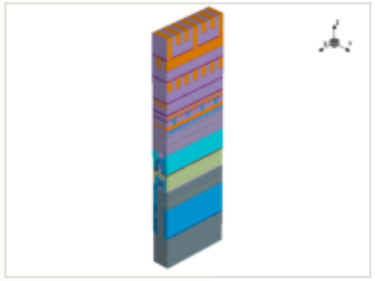
Step #7170: CMP

etch_cmp · CMP

0.169	618.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

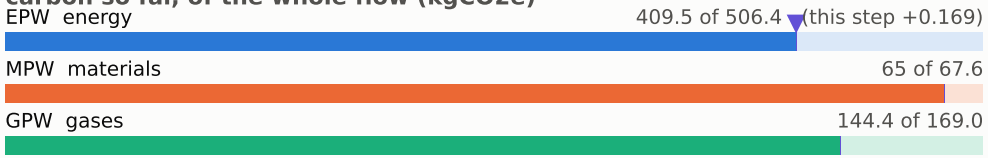


this step

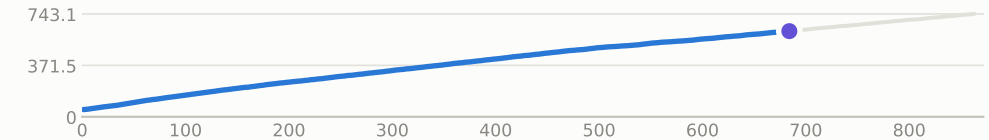
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

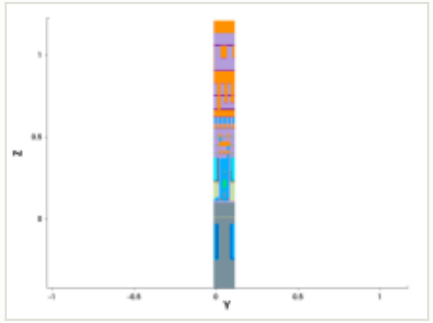
carbon so far, of the whole flow (kgCO2e)



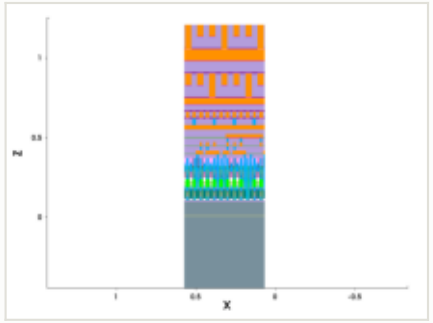
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



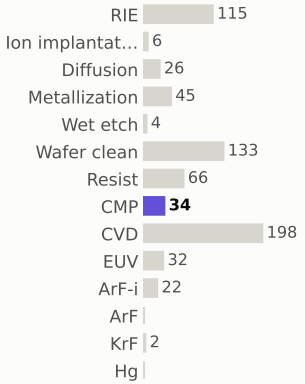
y cut at 0.049 µm



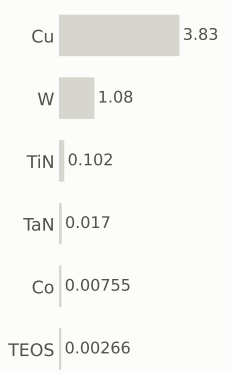
mask for this step

no mask for this step

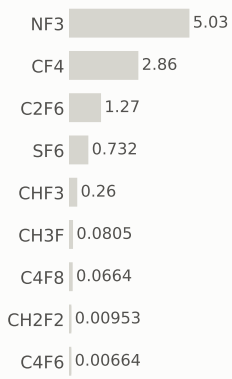
steps by process area



materials deposited (g)



gases emitted (g)



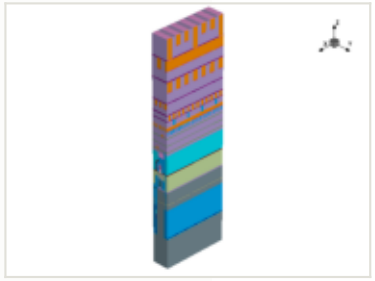
Step #7180: Metallization

depo_isotropic · Metallization

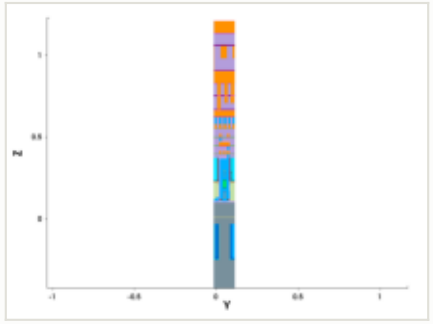
1.2	620.0	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

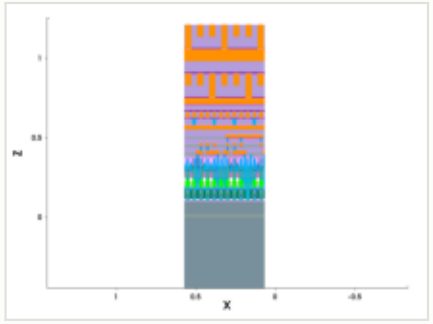
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



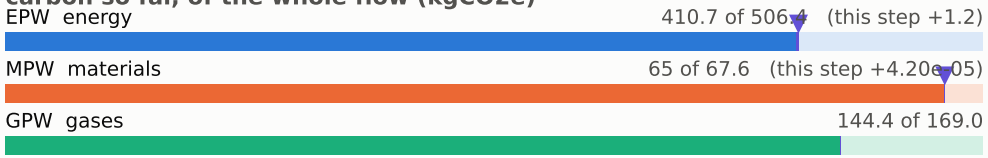
mask for this step

no mask for this step

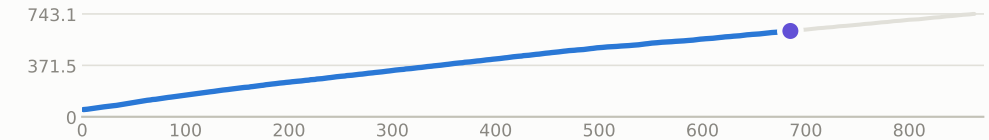
this step

process area	Metallization	EPW	1.2 kgCO2e
material	Co	EPW tool	0.86 kgCO2e
thickness	0.0015 µm	EPW overhead	0.344 kgCO2e
material GWP	44.5 gCO2e/g	MPW	4.20e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

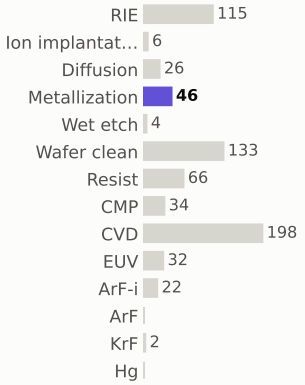
carbon so far, of the whole flow (kgCO2e)



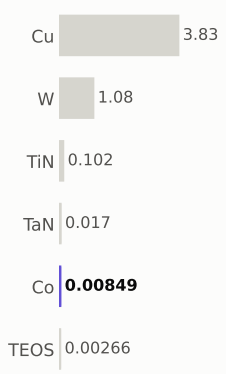
embodied carbon up to each step (kgCO2e per wafer)



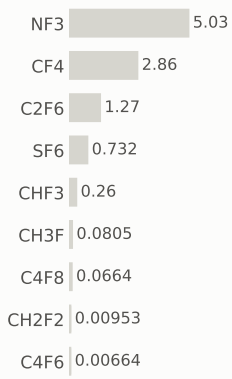
steps by process area



materials deposited (g)



gases emitted (g)



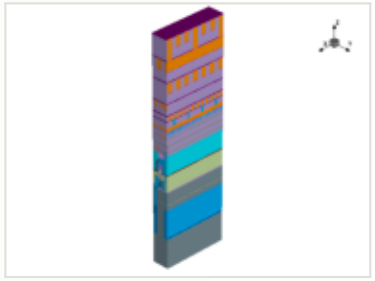
Step #7200: CVD

depo_fill · CVD

1.31	621.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

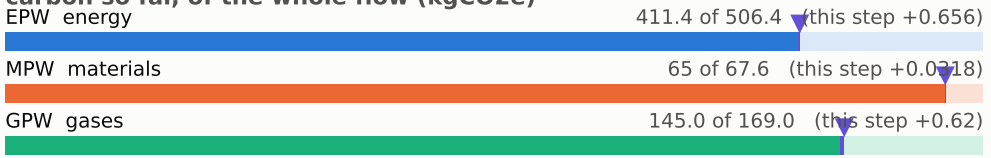


this step

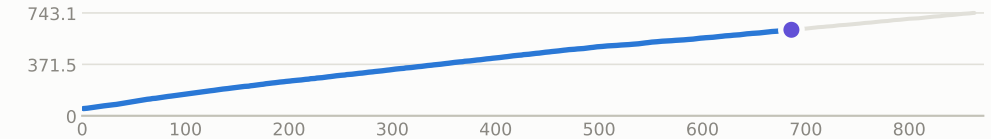
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

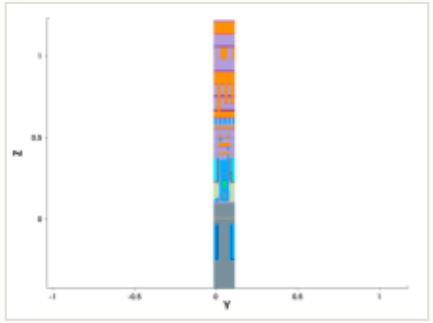
carbon so far, of the whole flow (kgCO2e)



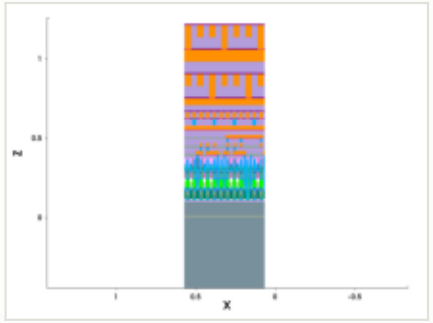
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



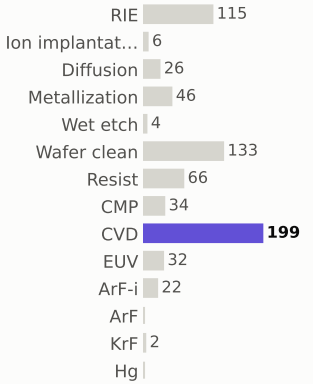
y cut at 0.049 µm



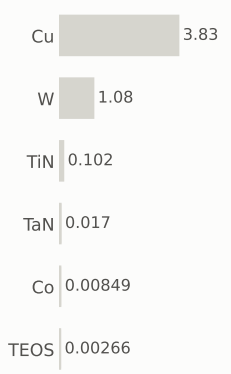
mask for this step

no mask for this step

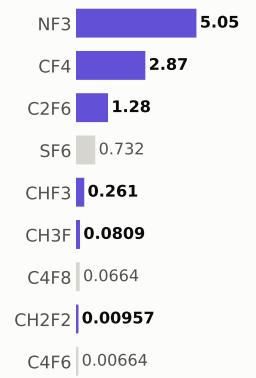
steps by process area



materials deposited (g)



gases emitted (g)



Step #7210: CVD

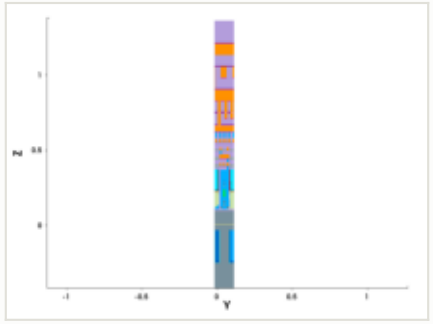
depo_fill · CVD

1.31	622.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

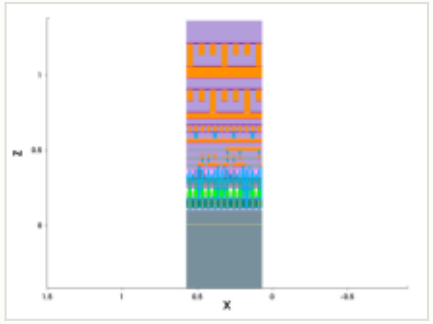
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

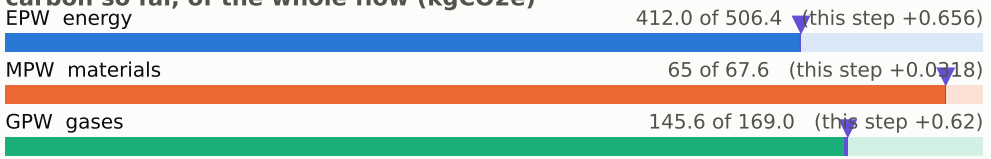
no mask for this step

this step

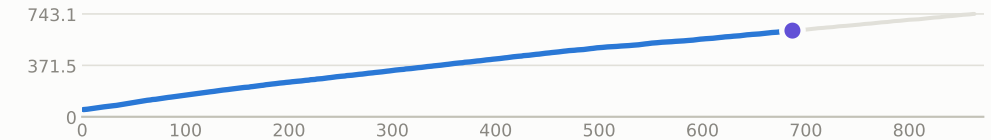
process area	CVD
material	—
thickness	0.146 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

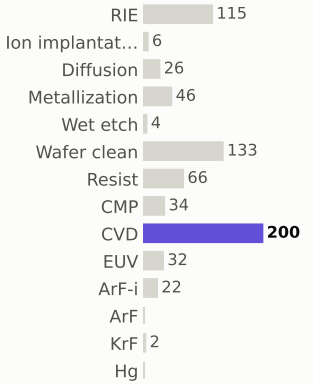
carbon so far, of the whole flow (kgCO2e)



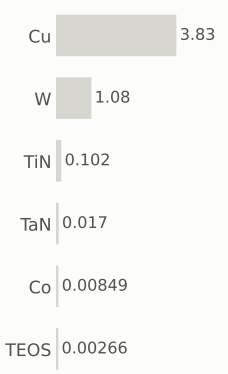
embodied carbon up to each step (kgCO2e per wafer)



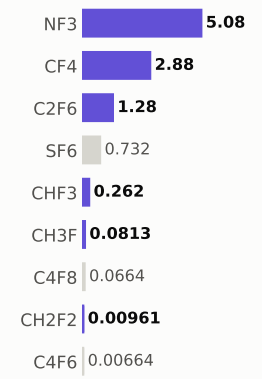
steps by process area



materials deposited (g)



gases emitted (g)



Step #7220: Diffusion

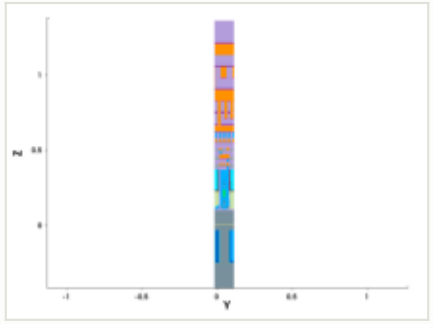
treatment · Diffusion

0.801	623.5	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

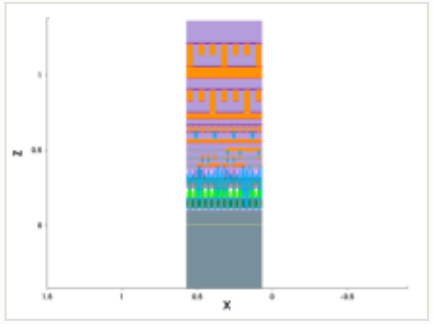
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

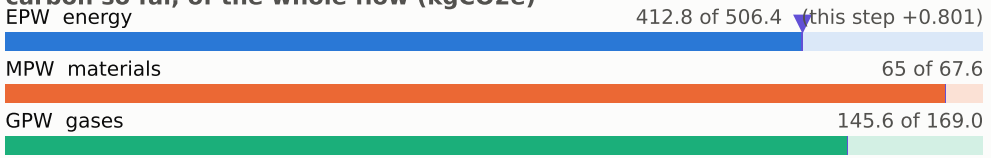
no mask for this step

this step

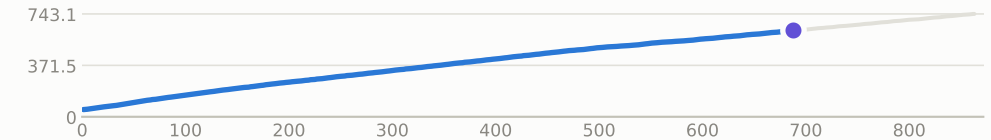
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

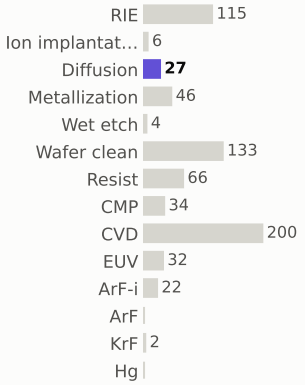
carbon so far, of the whole flow (kgCO2e)



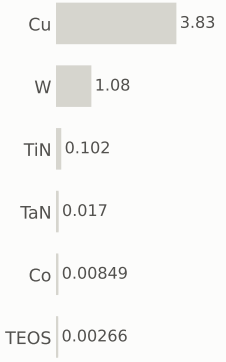
embodied carbon up to each step (kgCO2e per wafer)



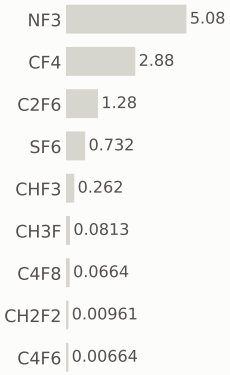
steps by process area



materials deposited (g)



gases emitted (g)



Step #7230: Wafer_clean

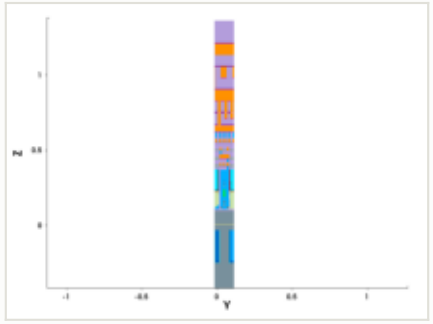
clean · Wafer_clean

0.169	623.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

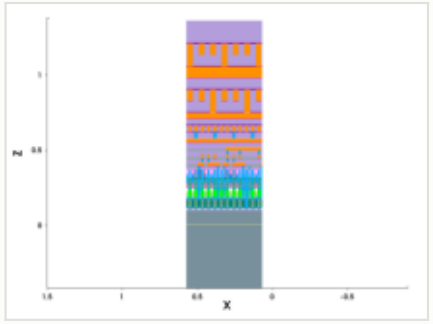
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

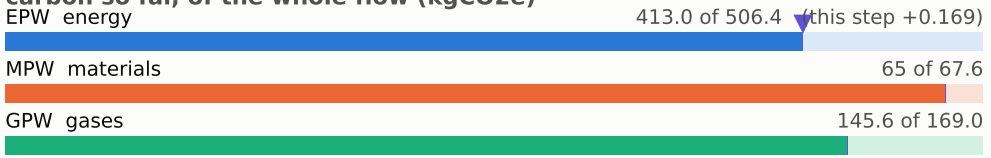
no mask for this step

this step

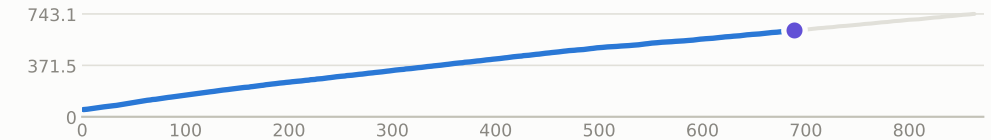
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

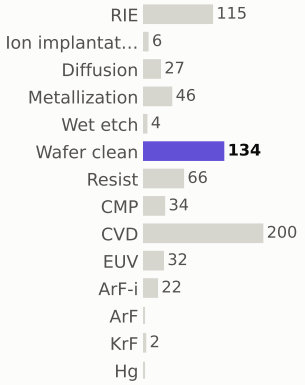
carbon so far, of the whole flow (kgCO2e)



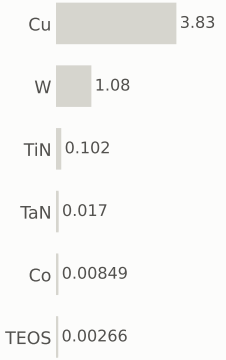
embodied carbon up to each step (kgCO2e per wafer)



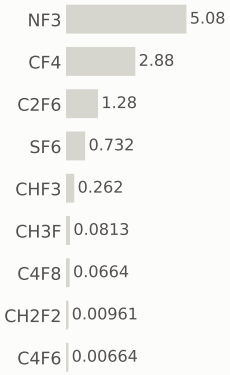
steps by process area



materials deposited (g)



gases emitted (g)



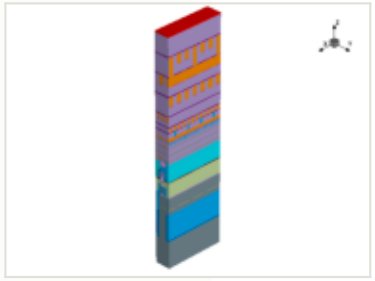
Step #7240: CVD

depo_fill · CVD

1.31	624.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

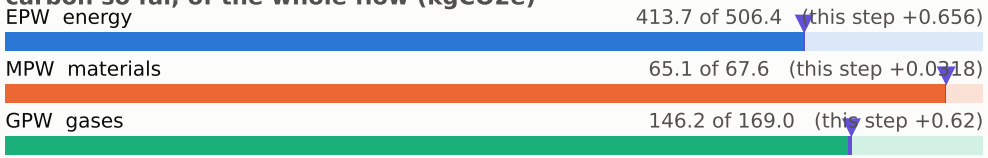


this step

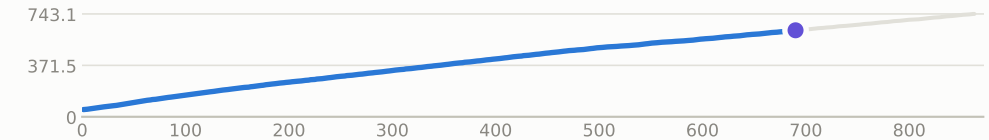
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

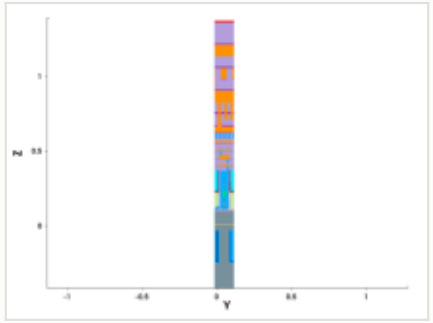
carbon so far, of the whole flow (kgCO2e)



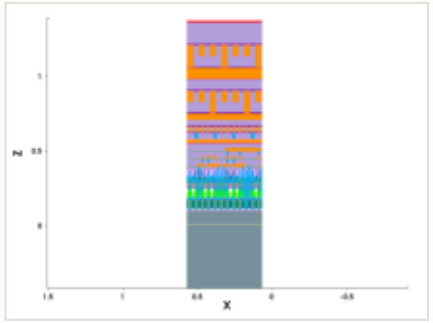
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



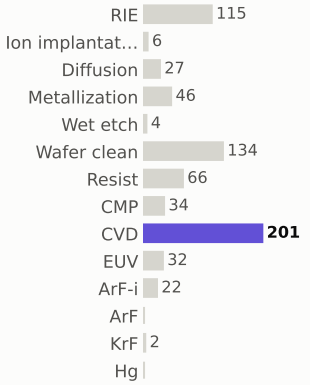
y cut at 0.049 µm



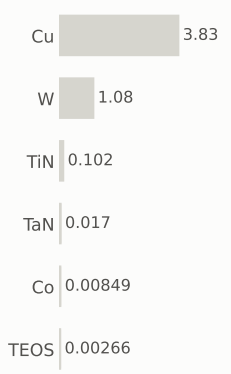
mask for this step

no mask for this step

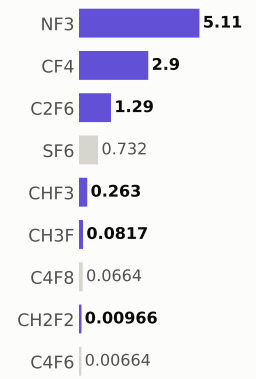
steps by process area



materials deposited (g)



gases emitted (g)



Step #7250: Resist

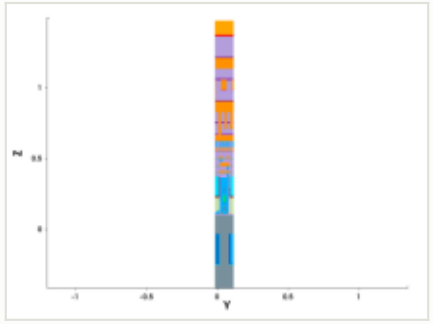
depo_fill · Resist

0.169	625.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

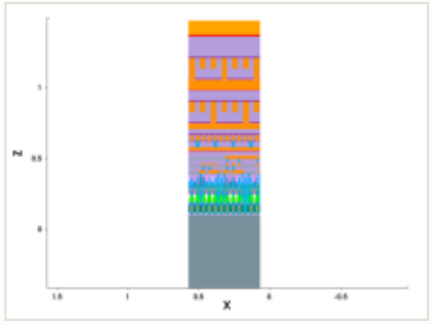
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



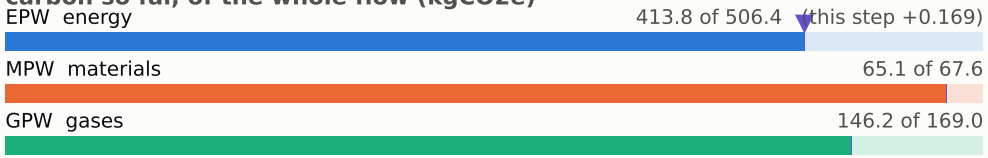
mask for this step

no mask for this step

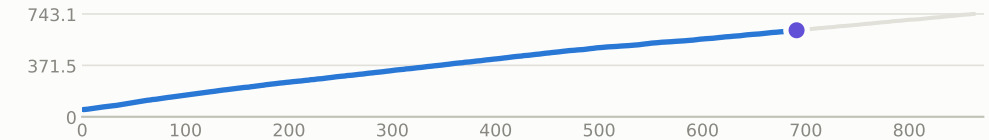
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

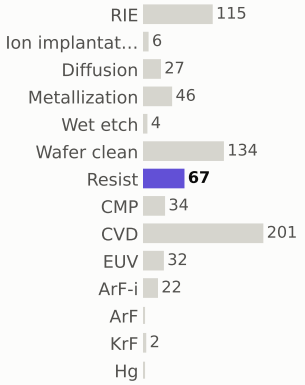
carbon so far, of the whole flow (kgCO2e)



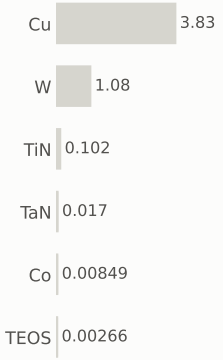
embodied carbon up to each step (kgCO2e per wafer)



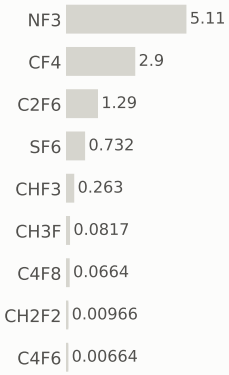
steps by process area



materials deposited (g)



gases emitted (g)



Step #7260: CVD

depo_fill · CVD

1.31	626.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

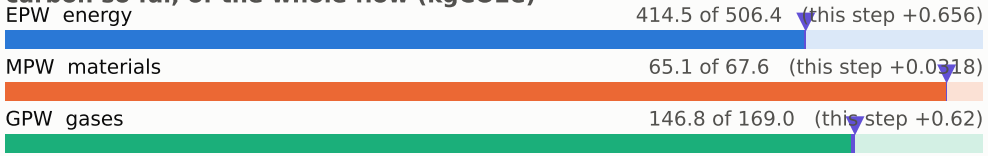
zoomed to fit

this step

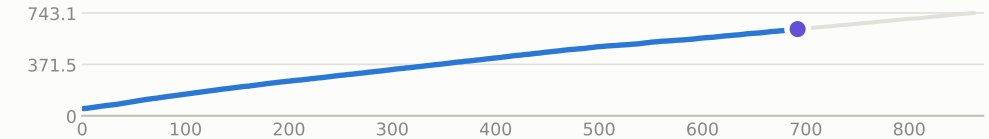
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

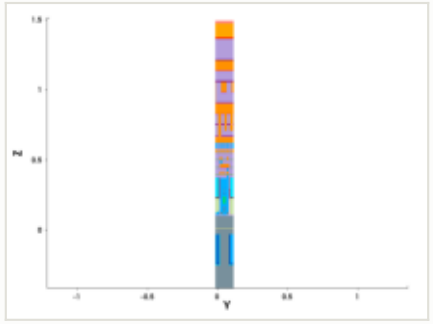
carbon so far, of the whole flow (kgCO2e)



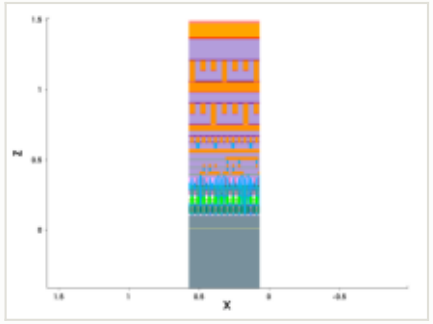
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



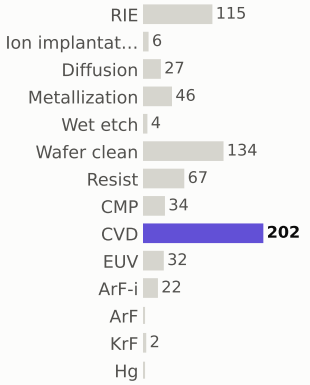
y cut at 0.049 µm



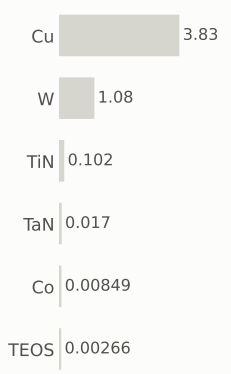
mask for this step

no mask for this step

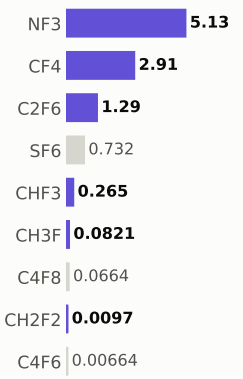
steps by process area



materials deposited (g)



gases emitted (g)



Step #7270: Wafer_clean

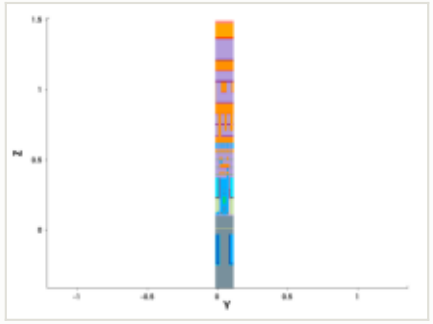
clean · Wafer_clean

0.169	626.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

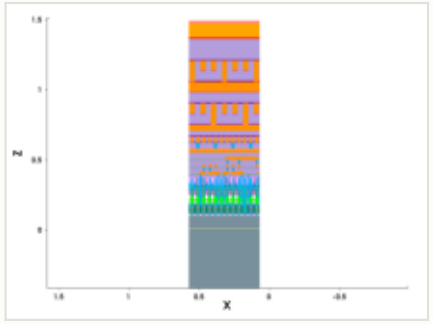
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

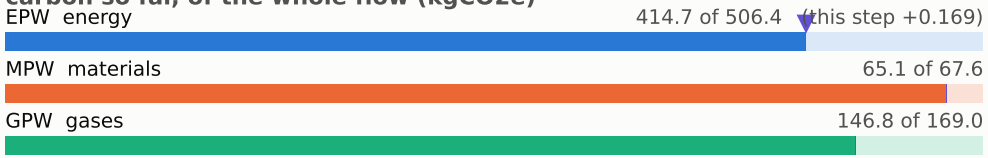
no mask for this step

this step

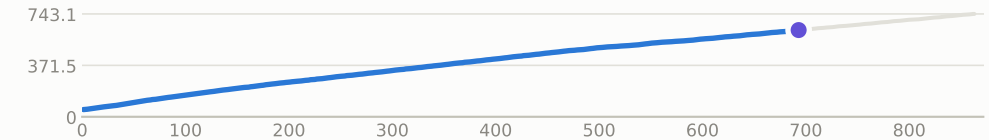
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

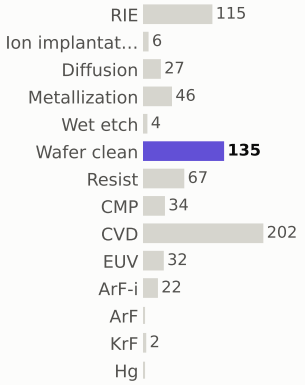
carbon so far, of the whole flow (kgCO2e)



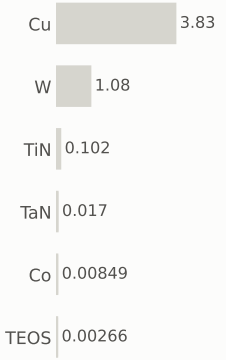
embodied carbon up to each step (kgCO2e per wafer)



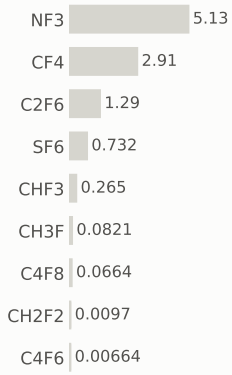
steps by process area



materials deposited (g)



gases emitted (g)



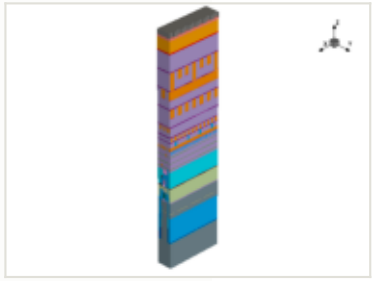
Step #7280: ArF-i

litho_patterning · ArF-i

0.217	626.8	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

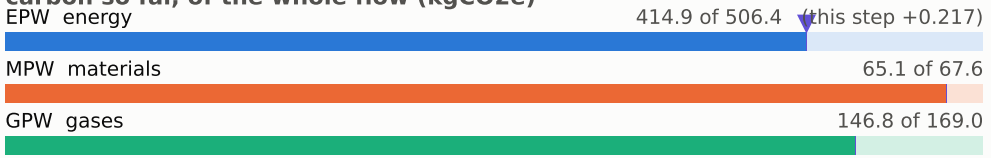


this step

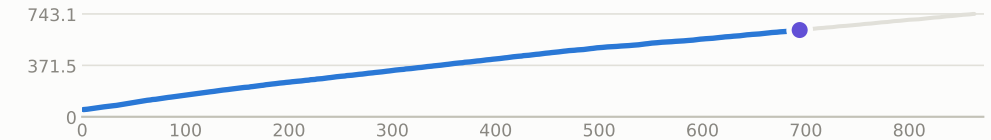
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

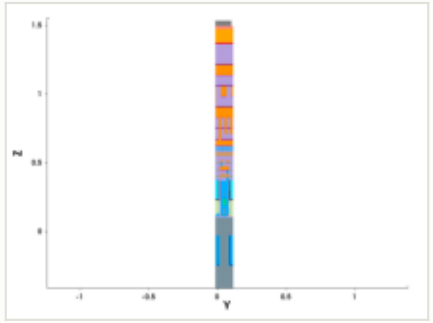
carbon so far, of the whole flow (kgCO2e)



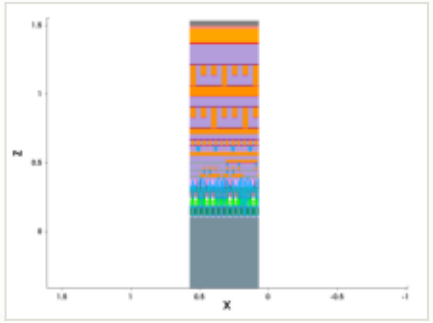
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

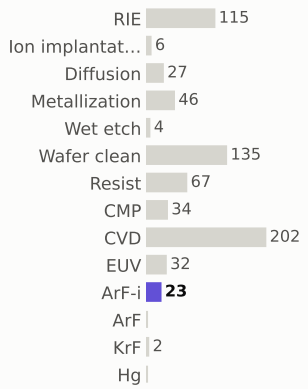


y cut at 0.049 µm

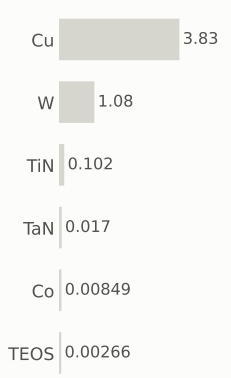


mask: layer 10621:0 (GDSII)

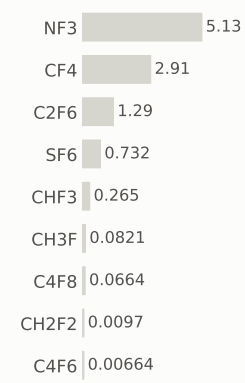
steps by process area



materials deposited (g)



gases emitted (g)



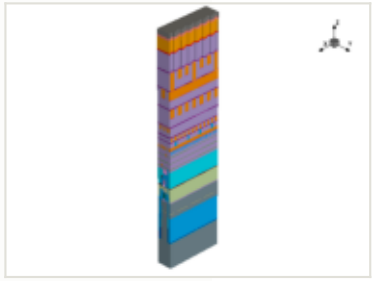
Step #7290: RIE

etch_anisotropic · RIE

1.04	627.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

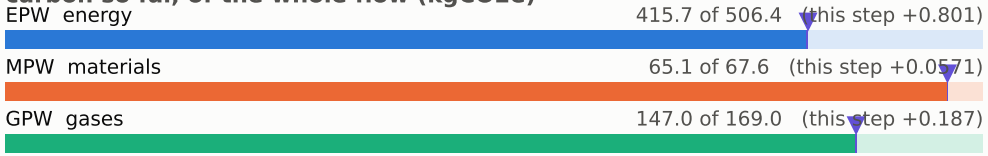


this step

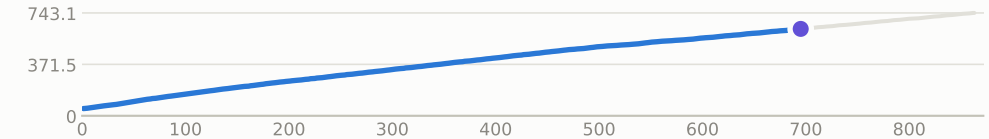
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

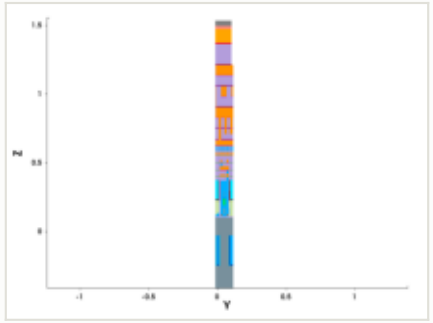
carbon so far, of the whole flow (kgCO2e)



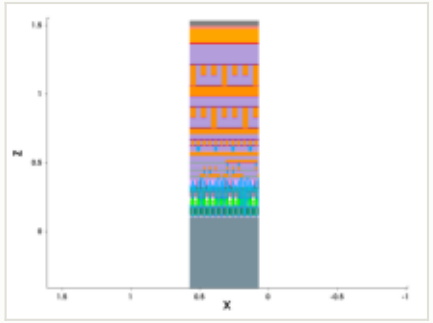
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



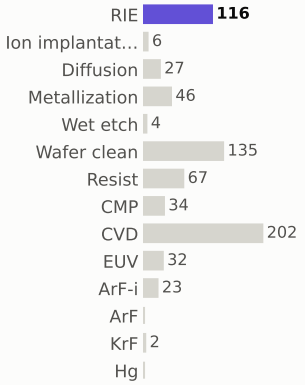
y cut at 0.049 µm



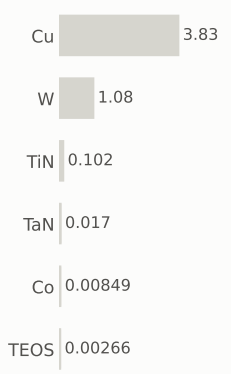
mask for this step

no mask for this step

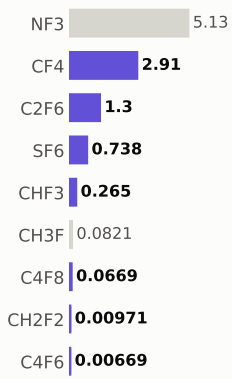
steps by process area



materials deposited (g)



gases emitted (g)



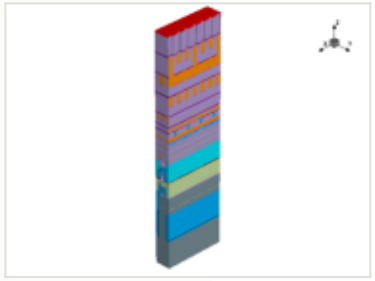
Step #7300: Wafer_clean

etch_strip · Wafer_clean

0.169	628.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

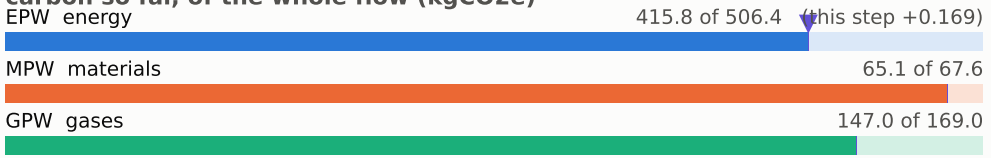


this step

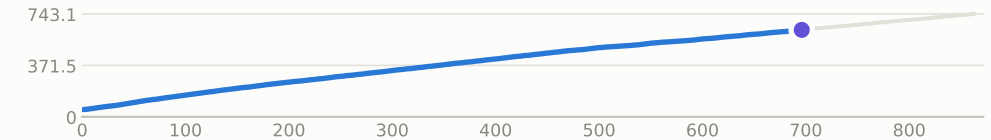
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

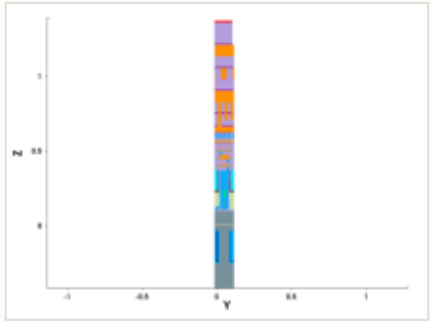
carbon so far, of the whole flow (kgCO2e)



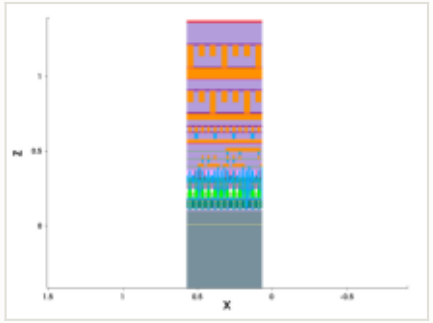
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



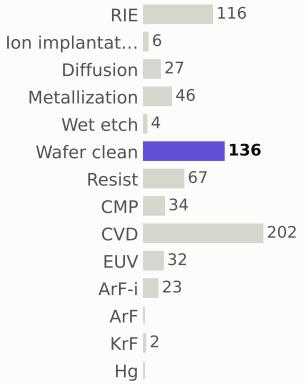
y cut at 0.049 µm



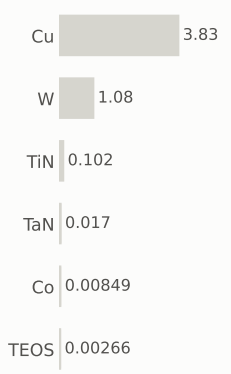
mask for this step

no mask for this step

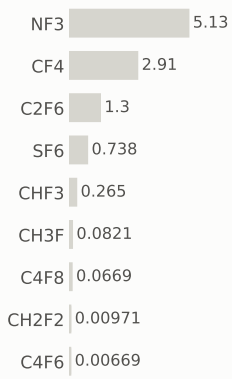
steps by process area



materials deposited (g)



gases emitted (g)



Step #7310: Resist

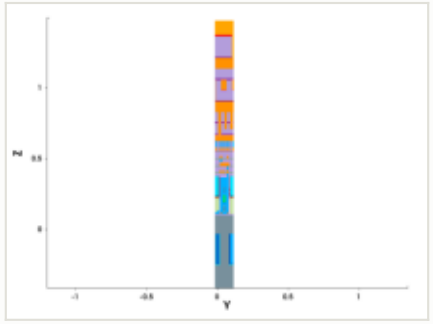
depo_fill · Resist

0.169	628.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

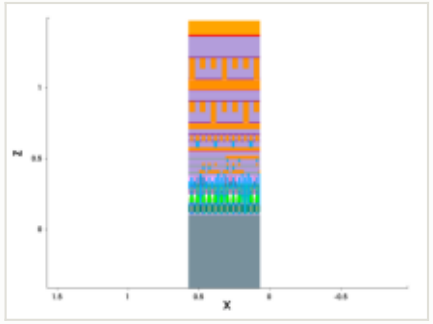
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



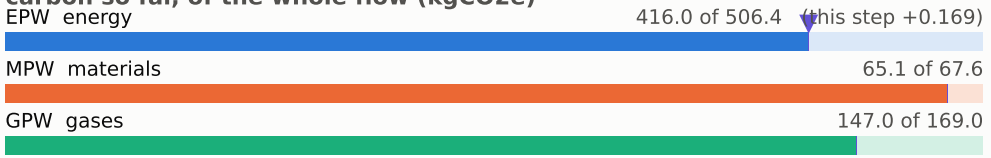
mask for this step

no mask for this step

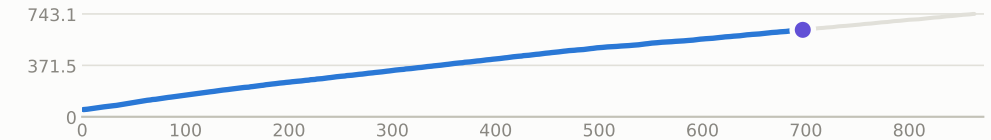
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

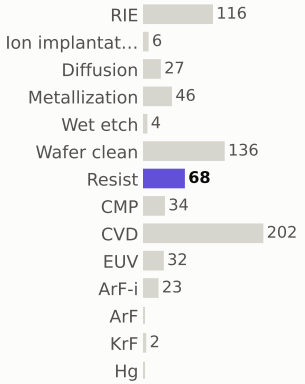
carbon so far, of the whole flow (kgCO2e)



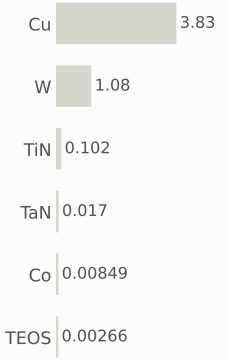
embodied carbon up to each step (kgCO2e per wafer)



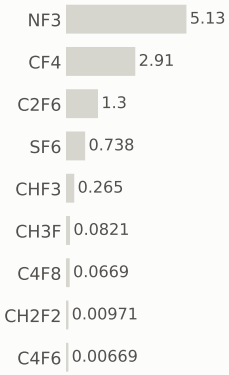
steps by process area



materials deposited (g)



gases emitted (g)



Step #7320: CVD

depo_fill · CVD

1.31	629.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

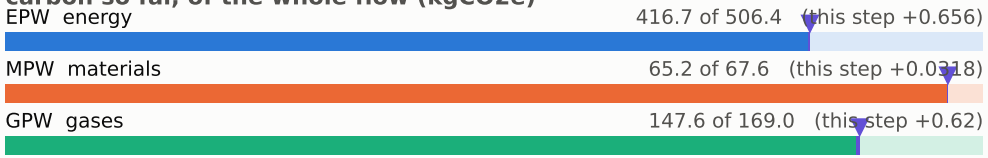
zoomed to fit

this step

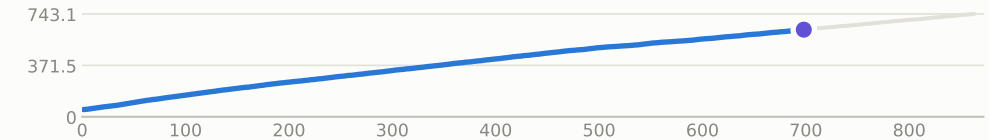
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

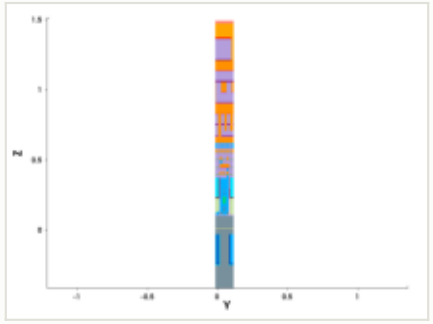
carbon so far, of the whole flow (kgCO2e)



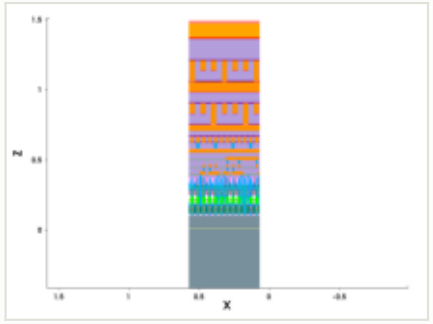
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



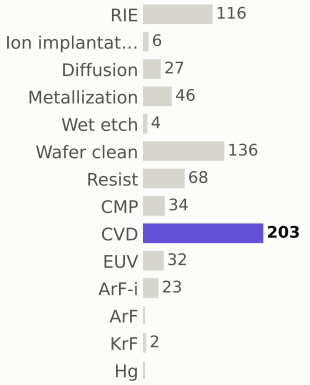
y cut at 0.049 µm



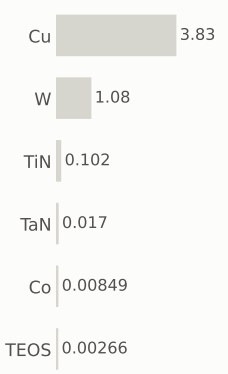
mask for this step

no mask for this step

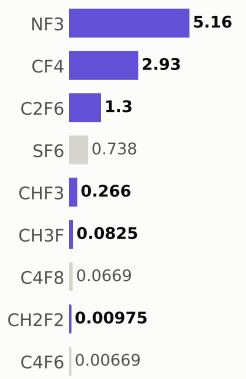
steps by process area



materials deposited (g)



gases emitted (g)



Step #7330: Wafer_clean

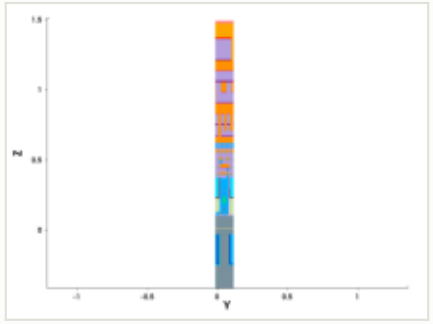
clean · Wafer_clean

0.169	629.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

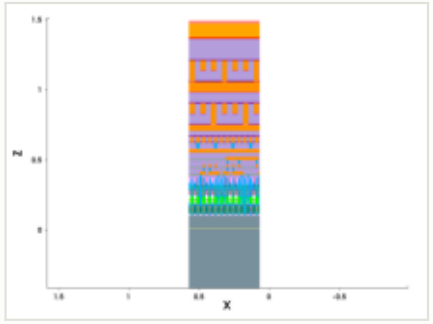
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

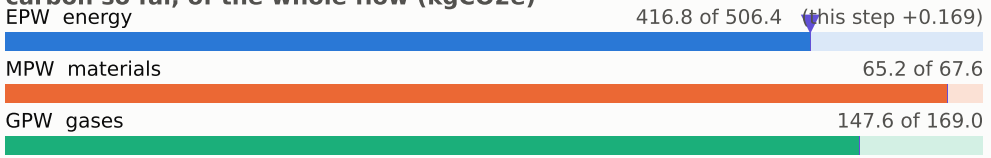
no mask for this step

this step

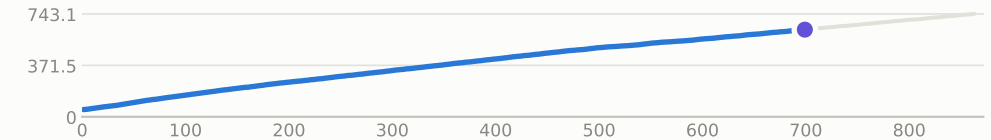
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

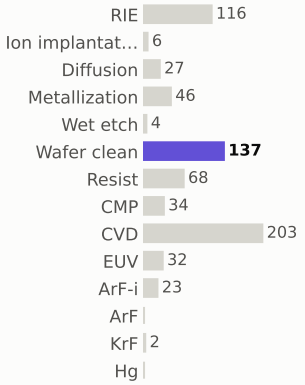
carbon so far, of the whole flow (kgCO2e)



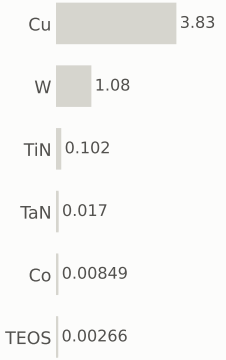
embodied carbon up to each step (kgCO2e per wafer)



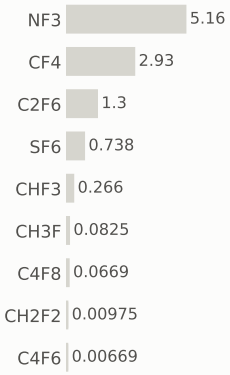
steps by process area



materials deposited (g)



gases emitted (g)



Step #7340: ArF-i

litho_patterning · ArF-i

0.217	629.9	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

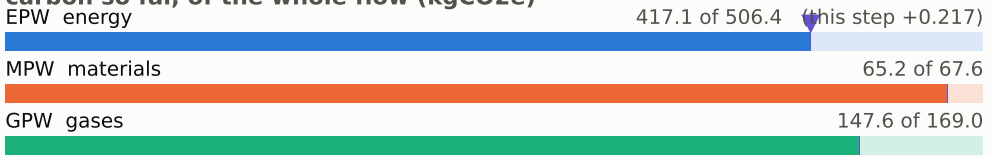
zoomed to fit

this step

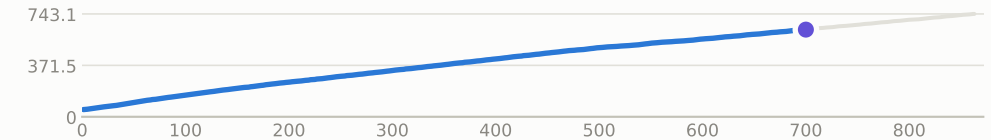
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

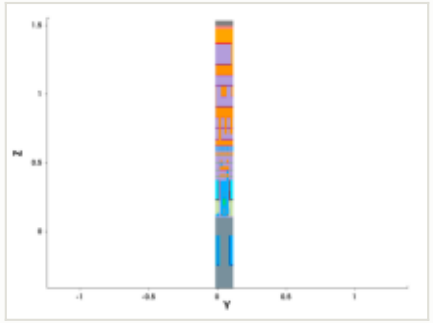
carbon so far, of the whole flow (kgCO2e)



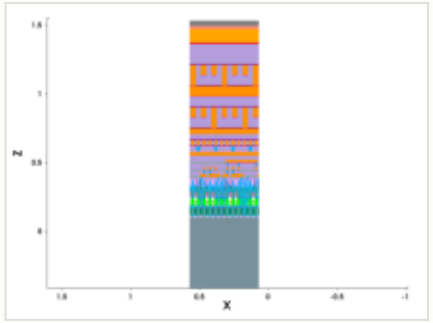
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



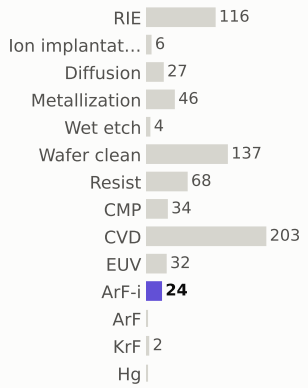
y cut at 0.049 µm



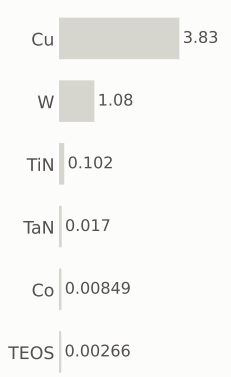
mask: layer 10622:0 (GDSII)

nothing in the clip window

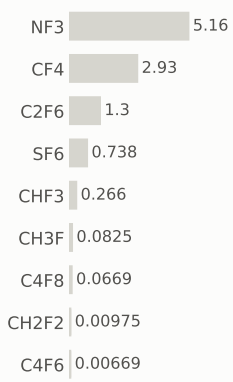
steps by process area



materials deposited (g)



gases emitted (g)



Step #7350: RIE

etch_anisotropic · RIE

1.04	630.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

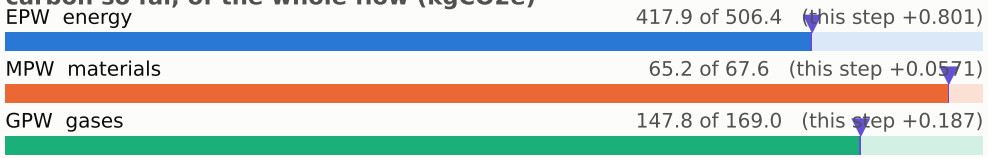
zoomed to fit

this step

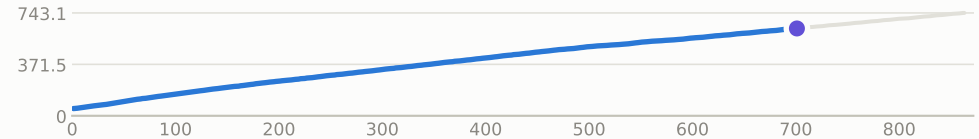
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

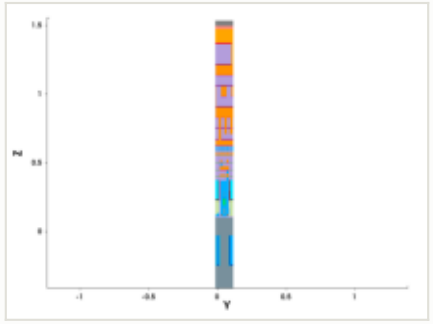
carbon so far, of the whole flow (kgCO2e)



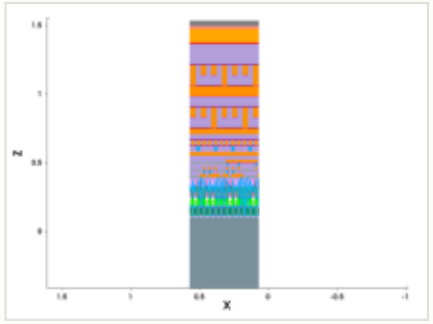
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



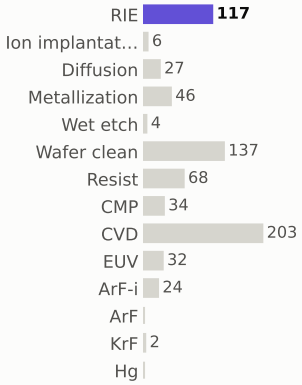
y cut at 0.049 µm



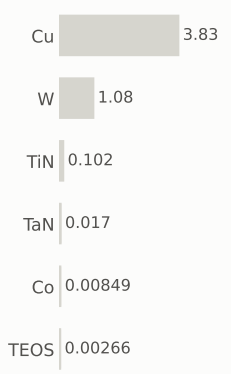
mask for this step

no mask for this step

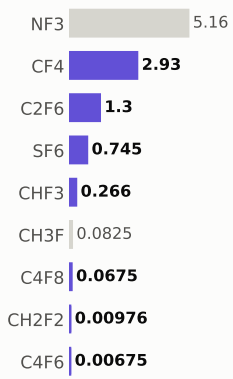
steps by process area



materials deposited (g)



gases emitted (g)



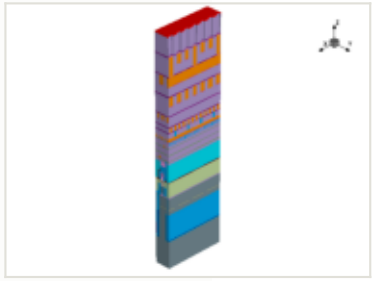
Step #7360: Wafer_clean

etch_strip · Wafer_clean

0.169	631.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

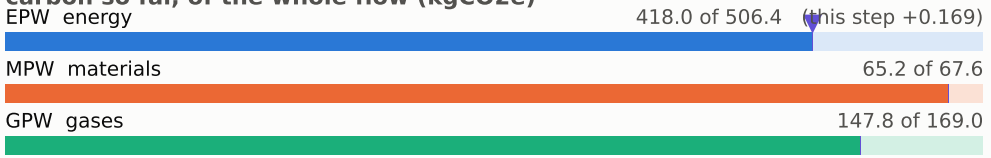


this step

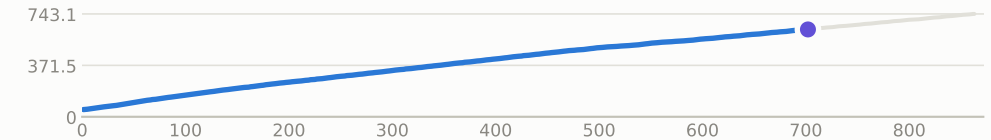
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

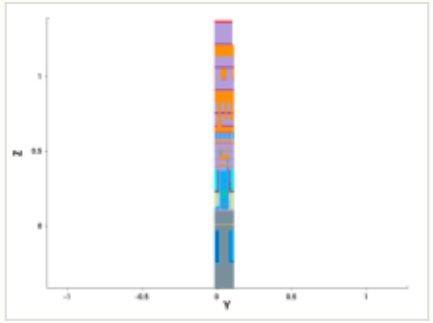
carbon so far, of the whole flow (kgCO2e)



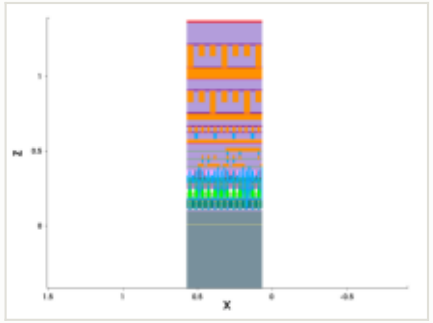
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



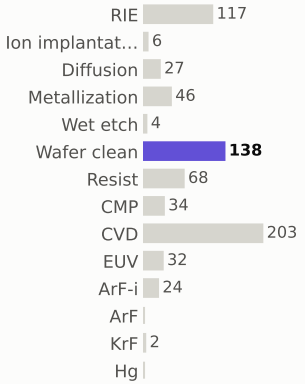
y cut at 0.049 µm



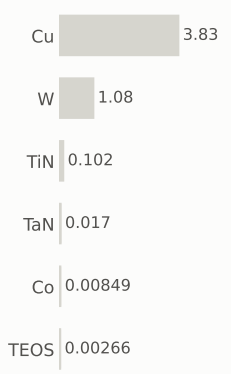
mask for this step

no mask for this step

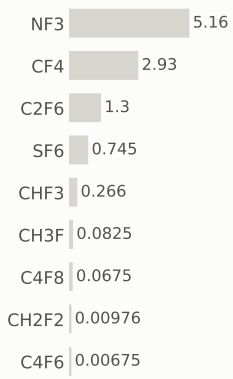
steps by process area



materials deposited (g)



gases emitted (g)



Step #7370: Metallization

depo_ellipsoidal · Metallization

1.2	632.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

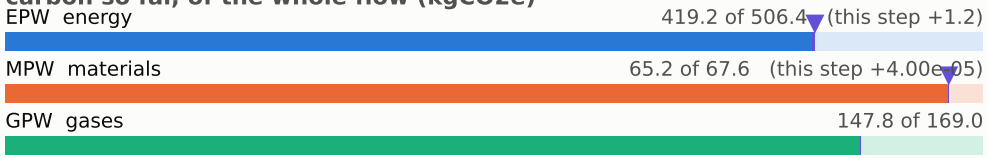
zoomed to fit

this step

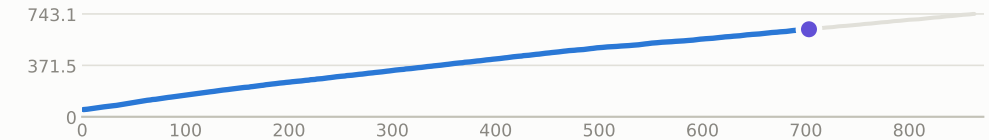
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

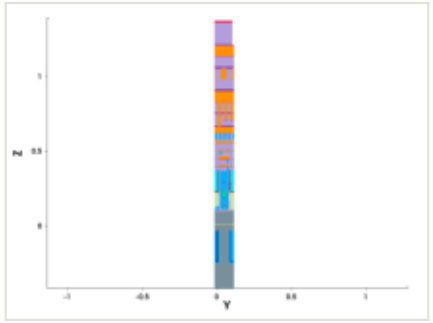
carbon so far, of the whole flow (kgCO2e)



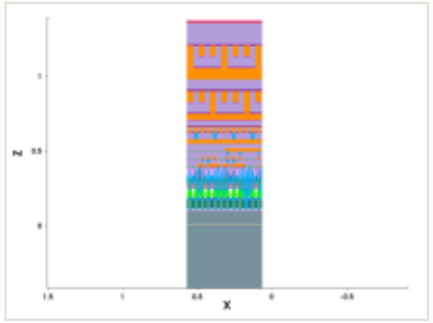
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



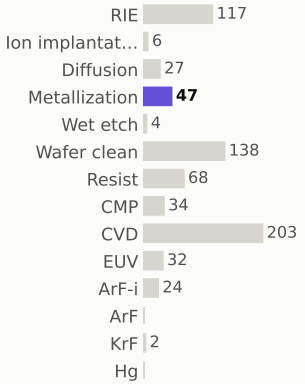
y cut at 0.049 µm



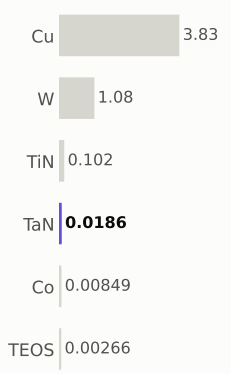
mask for this step

no mask for this step

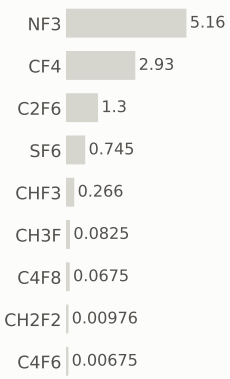
steps by process area



materials deposited (g)



gases emitted (g)



Step #7380: Metallization

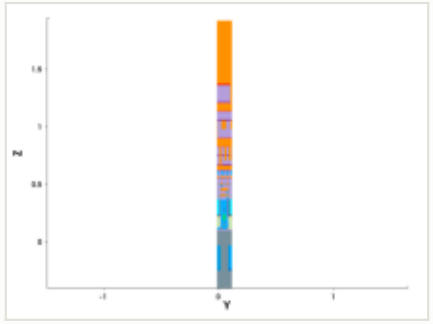
depo_fill · Metallization

1.21	633.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

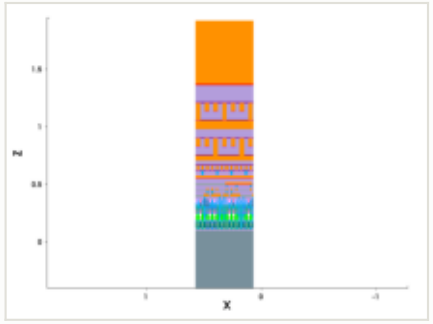
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

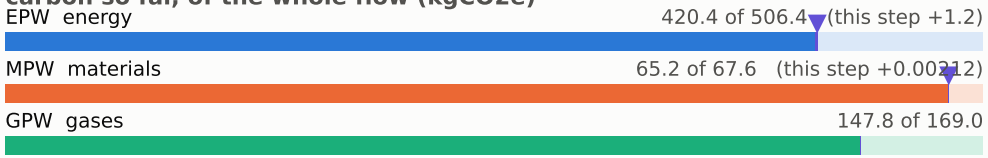
no mask for this step

this step

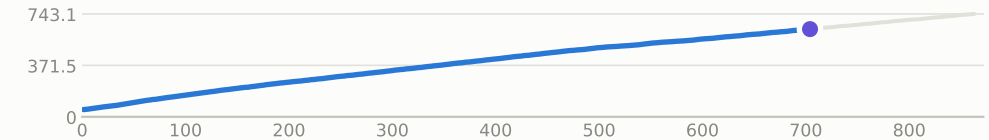
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

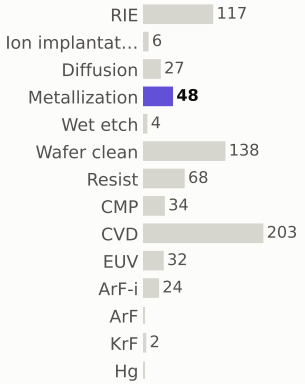
carbon so far, of the whole flow (kgCO2e)



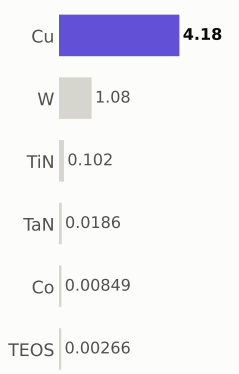
embodied carbon up to each step (kgCO2e per wafer)



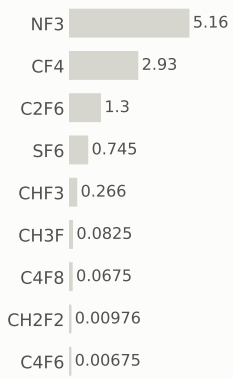
steps by process area



materials deposited (g)



gases emitted (g)



Step #7390: Diffusion

treatment · Diffusion

0.801	634.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

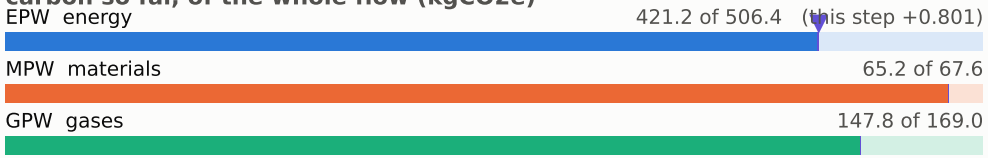
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

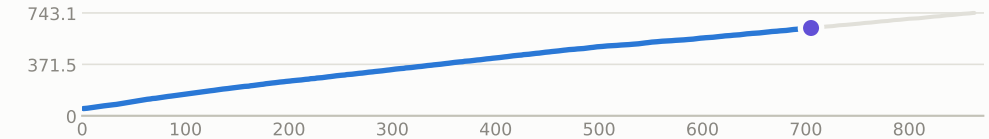
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

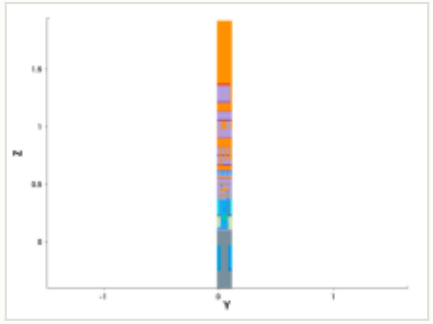
carbon so far, of the whole flow (kgCO2e)



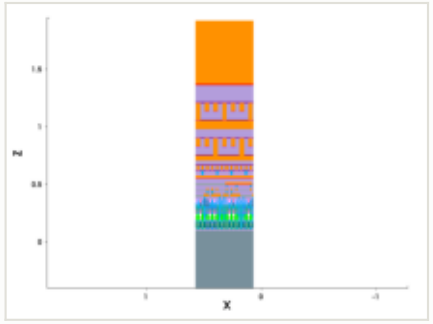
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



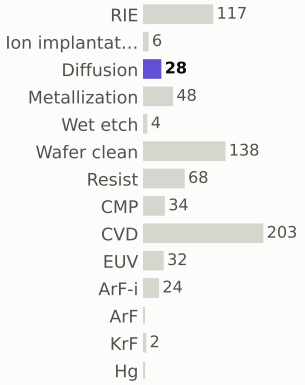
y cut at 0.049 µm



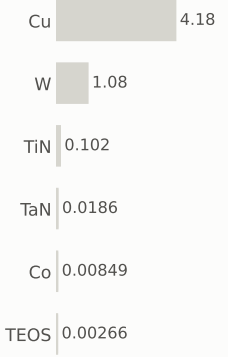
mask for this step

no mask for this step

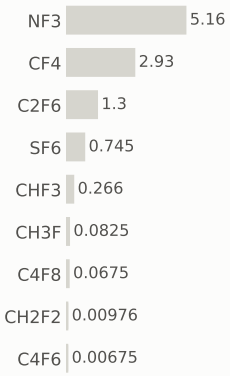
steps by process area



materials deposited (g)



gases emitted (g)



Step #7400: CMP

etch_cmp · CMP

0.169	634.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

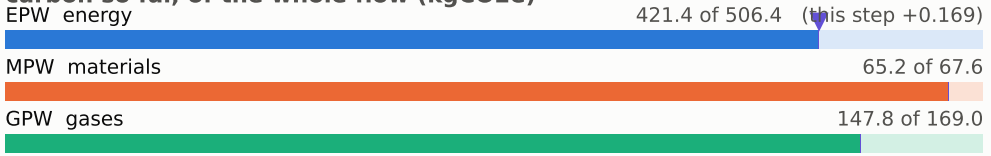
zoomed to fit

this step

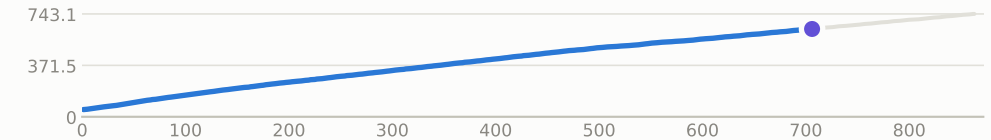
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

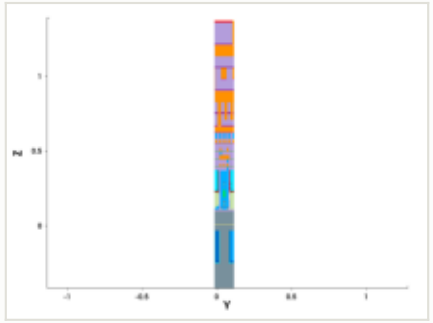
carbon so far, of the whole flow (kgCO2e)



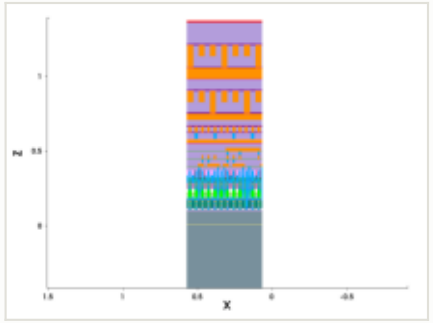
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



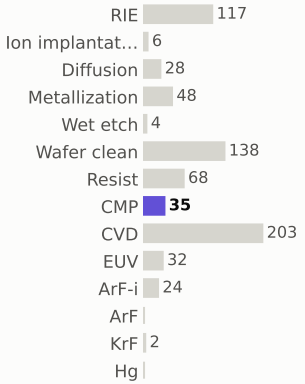
y cut at 0.049 µm



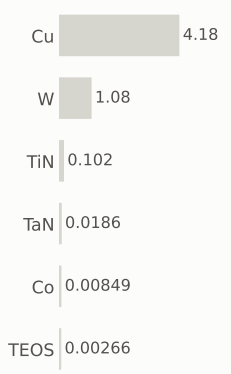
mask for this step

no mask for this step

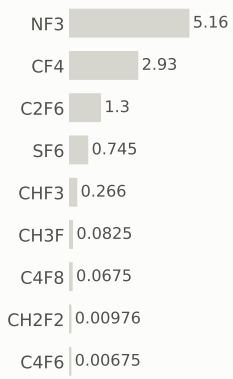
steps by process area



materials deposited (g)



gases emitted (g)



Step #7410: CVD

depo_fill · CVD

1.31	635.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

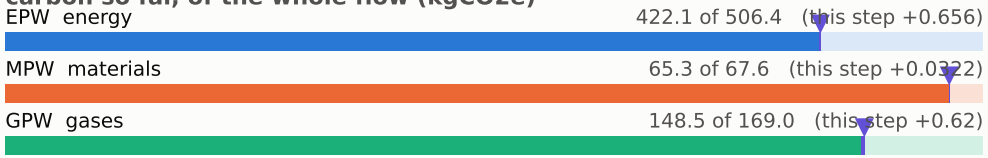
zoomed to fit

this step

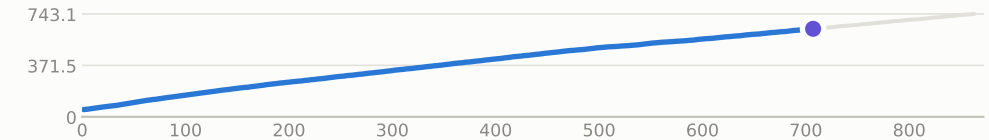
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

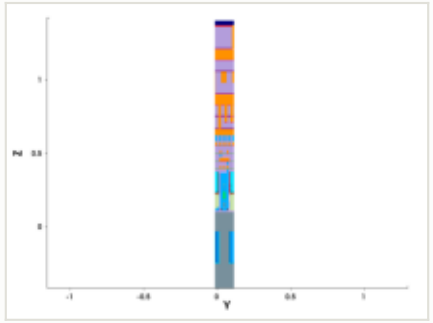
carbon so far, of the whole flow (kgCO2e)



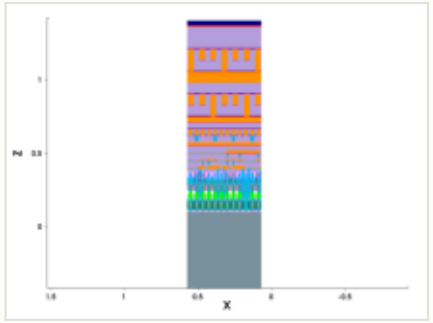
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



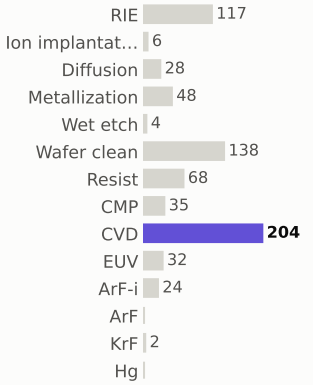
y cut at 0.049 µm



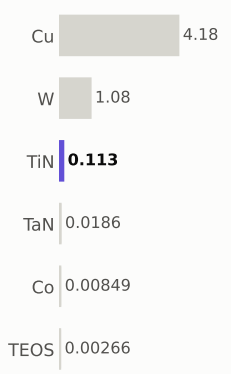
mask for this step

no mask for this step

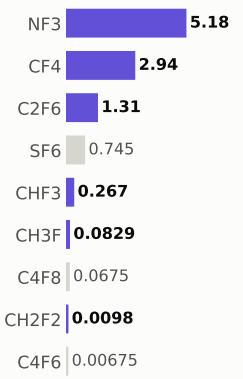
steps by process area



materials deposited (g)



gases emitted (g)



Step #7420: Wafer_clean

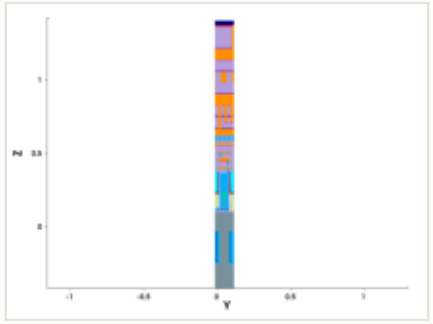
clean · Wafer_clean

0.169	635.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

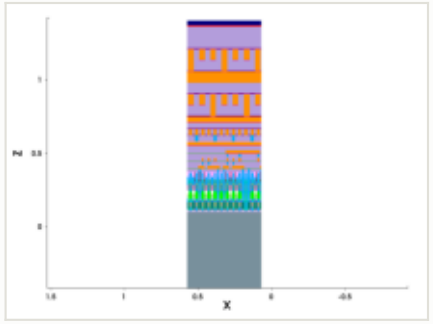
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

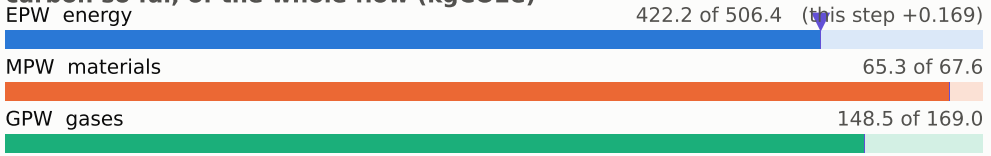
no mask for this step

this step

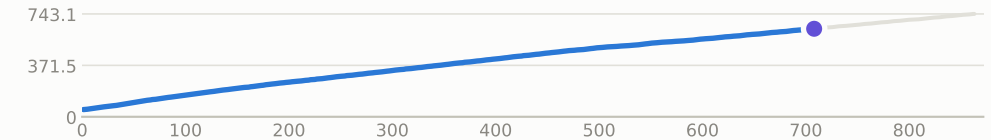
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

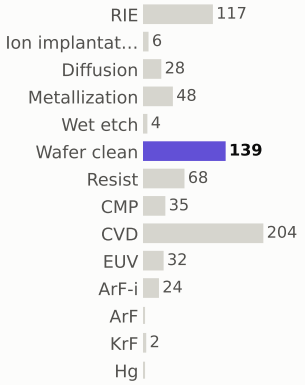
carbon so far, of the whole flow (kgCO2e)



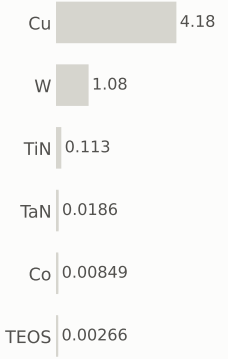
embodied carbon up to each step (kgCO2e per wafer)



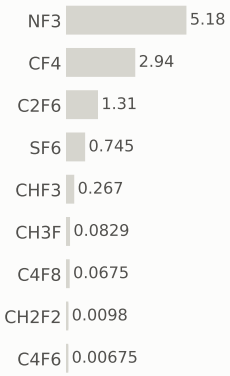
steps by process area



materials deposited (g)



gases emitted (g)



Step #7430: CVD

depo_fill · CVD

1.31	637.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

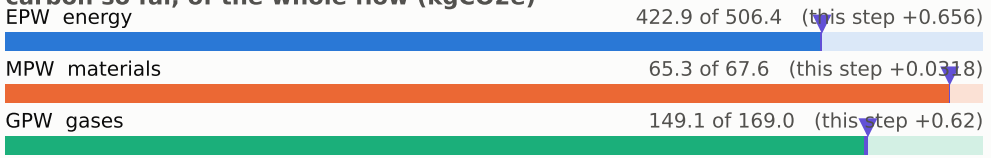
zoomed to fit

this step

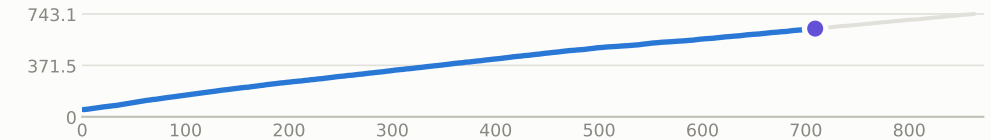
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

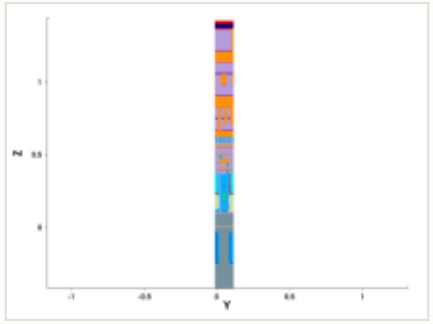
carbon so far, of the whole flow (kgCO2e)



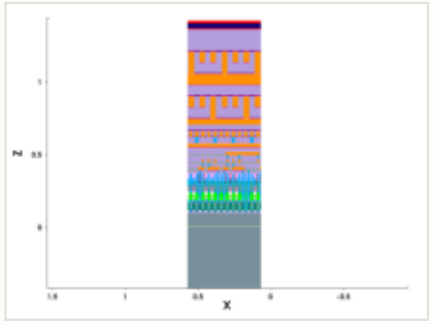
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



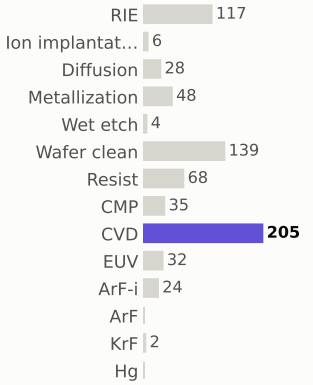
y cut at 0.049 µm



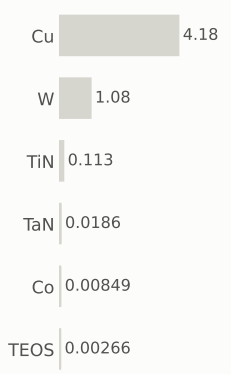
mask for this step

no mask for this step

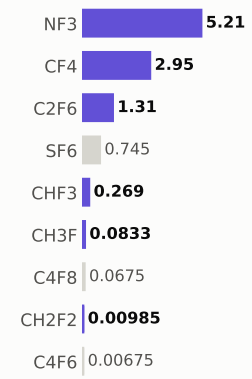
steps by process area



materials deposited (g)



gases emitted (g)



Step #7440: CVD

depo_fill · CVD

1.31	638.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

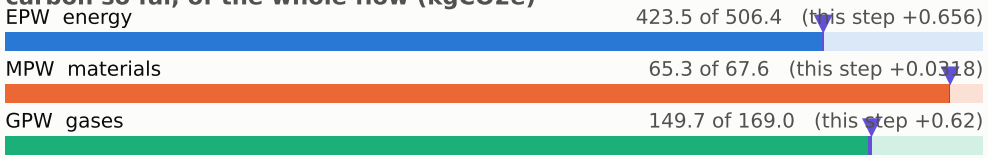
zoomed to fit

this step

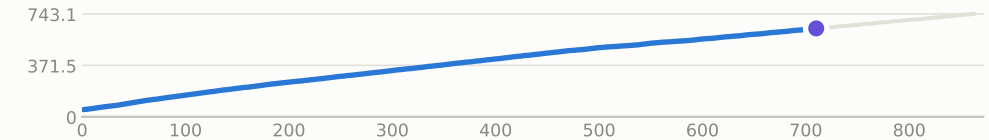
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

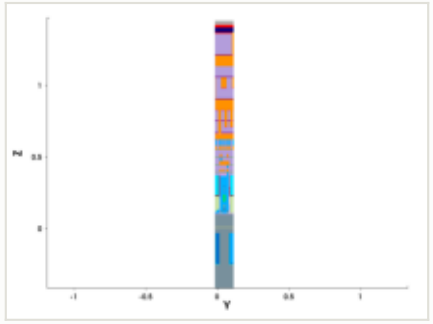
carbon so far, of the whole flow (kgCO2e)



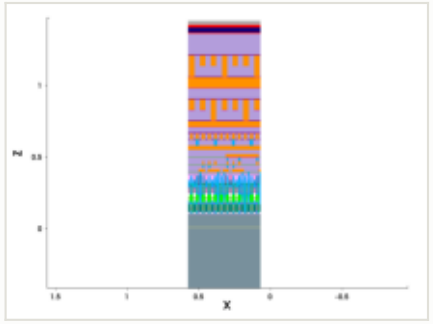
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



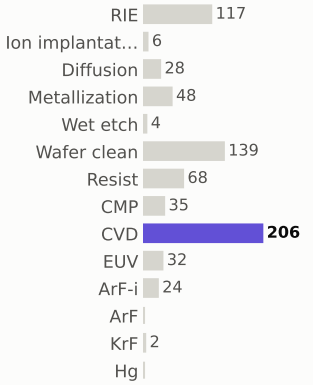
y cut at 0.049 µm



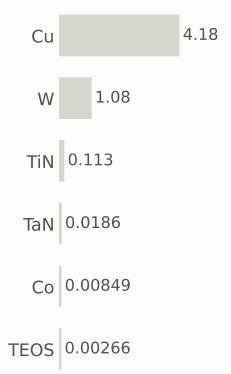
mask for this step

no mask for this step

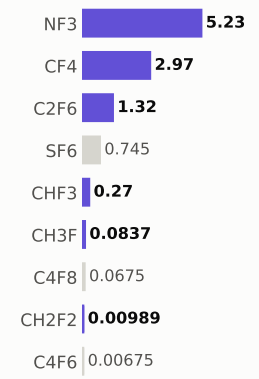
steps by process area



materials deposited (g)



gases emitted (g)



Step #7450: Resist

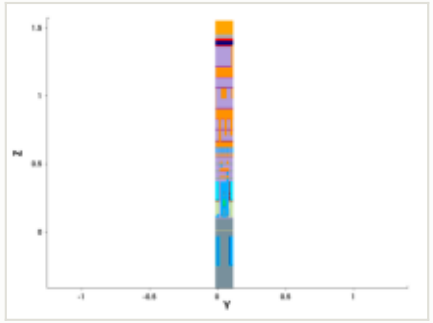
depo_fill · Resist

0.169	638.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

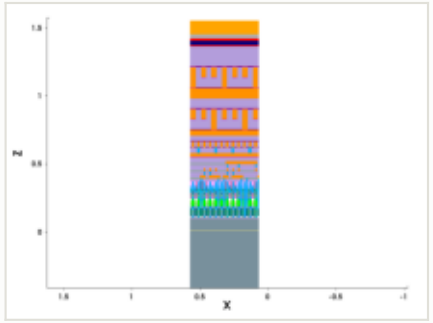
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



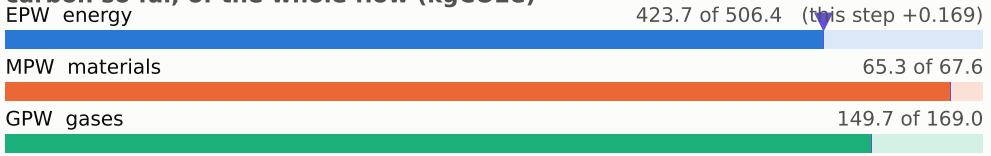
mask for this step

no mask for this step

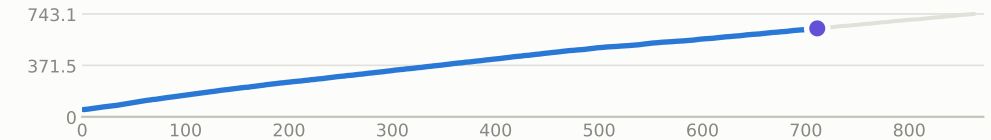
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

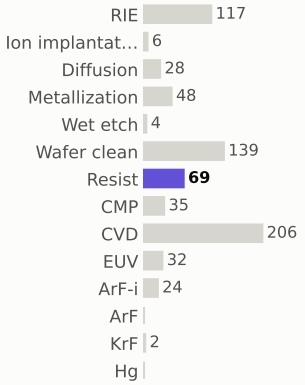
carbon so far, of the whole flow (kgCO2e)



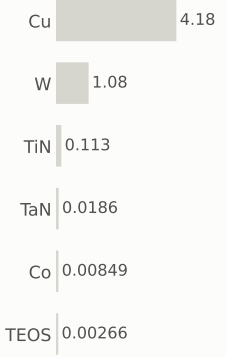
embodied carbon up to each step (kgCO2e per wafer)



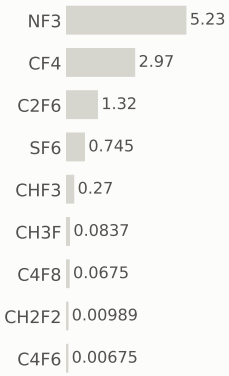
steps by process area



materials deposited (g)



gases emitted (g)



Step #7460: CVD

depo_fill · CVD

1.31	640.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

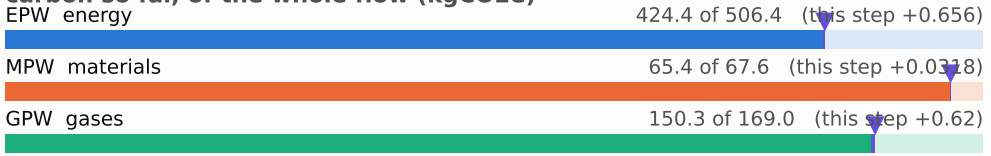
zoomed to fit

this step

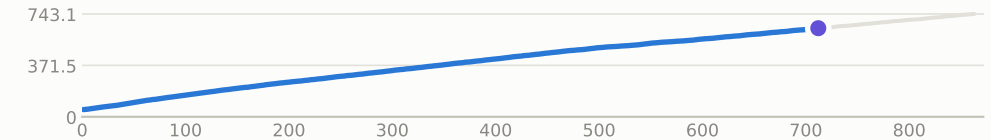
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

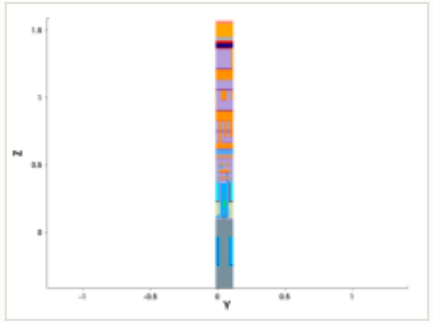
carbon so far, of the whole flow (kgCO2e)



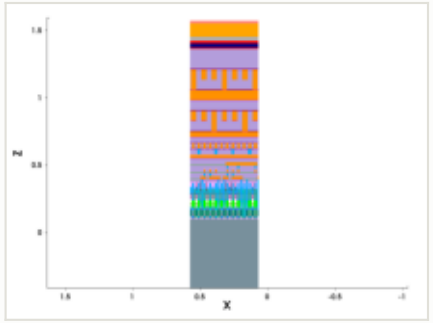
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



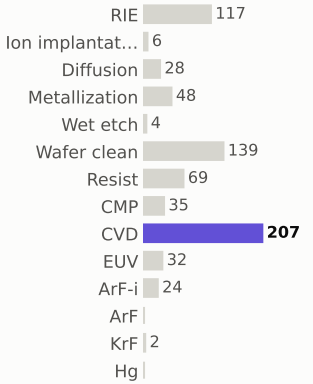
y cut at 0.049 µm



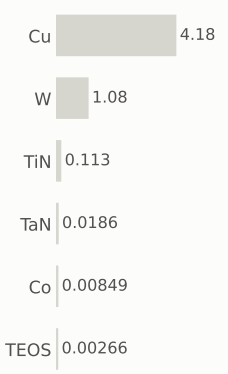
mask for this step

no mask for this step

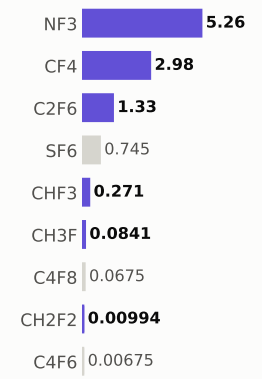
steps by process area



materials deposited (g)



gases emitted (g)



Step #7470: Wafer_clean

clean · Wafer_clean

0.169	640.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

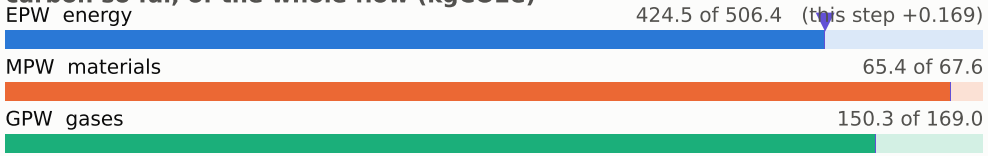
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

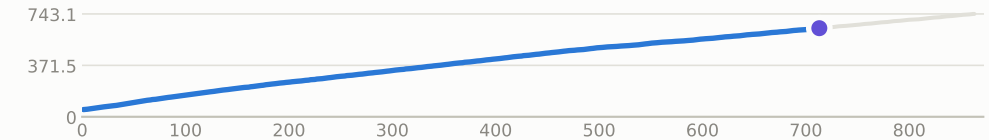
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

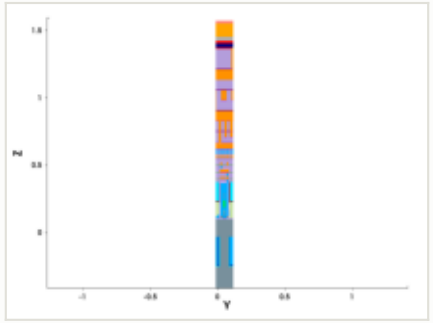
carbon so far, of the whole flow (kgCO2e)



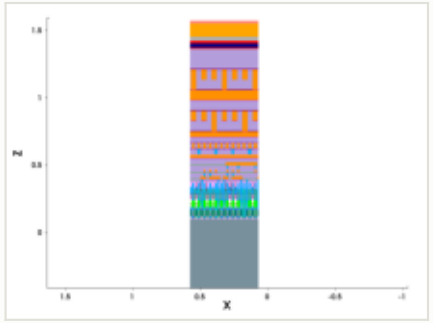
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



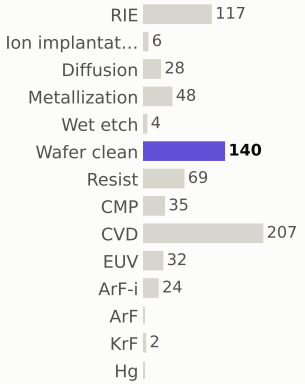
y cut at 0.049 µm



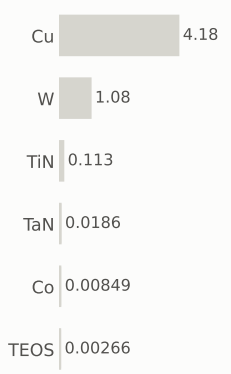
mask for this step

no mask for this step

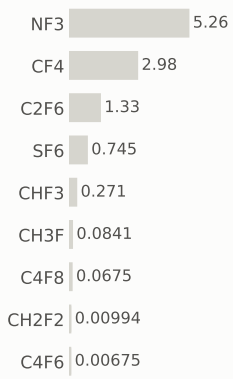
steps by process area



materials deposited (g)



gases emitted (g)



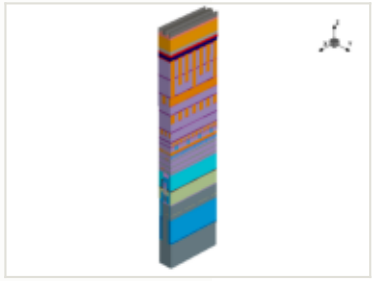
Step #7480: ArF-i

litho_patterning · ArF-i

0.217	640.4	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

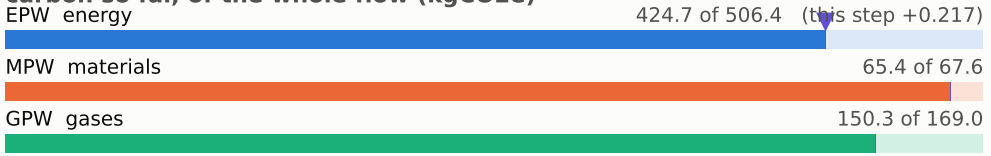


this step

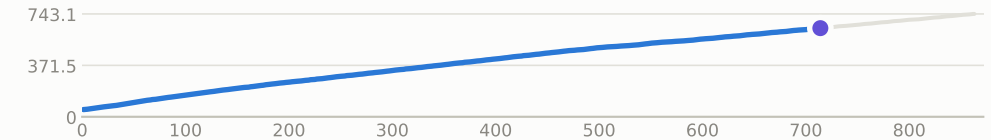
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

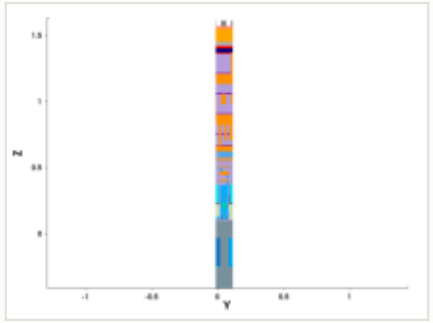
carbon so far, of the whole flow (kgCO2e)



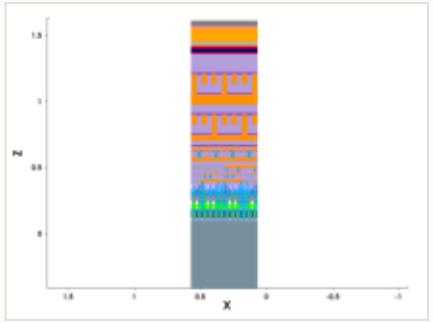
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

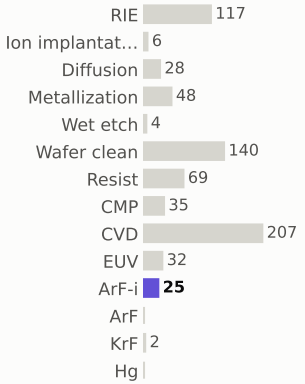


y cut at 0.049 µm

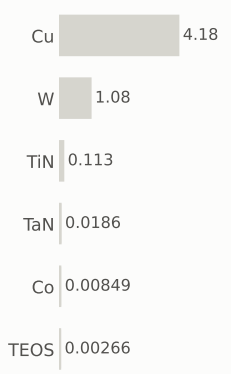


mask: layer 10651:0, positive (GDSII)

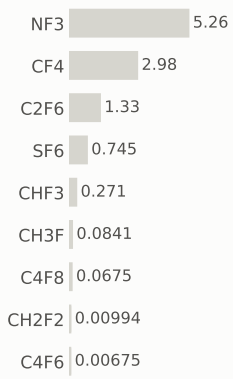
steps by process area



materials deposited (g)



gases emitted (g)



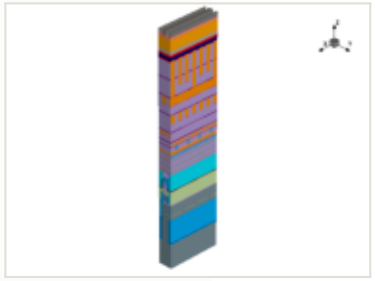
Step #7490: Ion_implantation

treatment · Ion_implantation

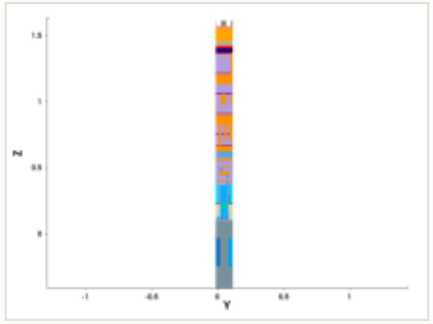
0.801	641.2	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

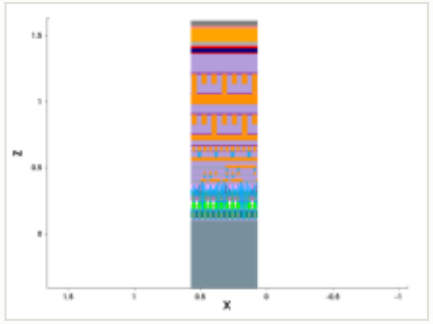
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

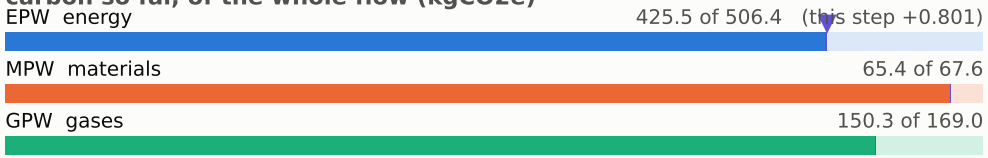
no mask for this step

this step

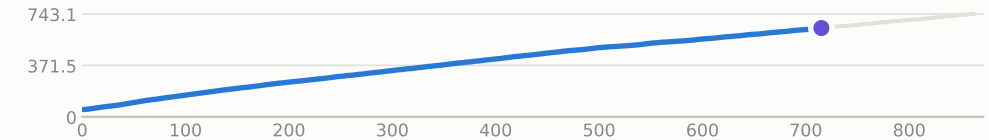
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

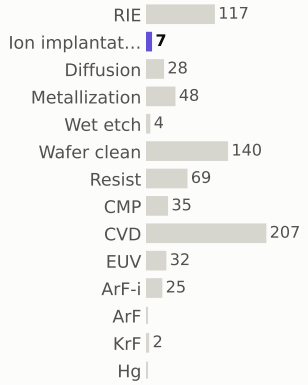
carbon so far, of the whole flow (kgCO2e)



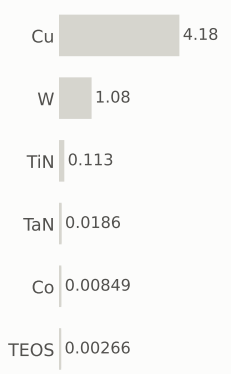
embodied carbon up to each step (kgCO2e per wafer)



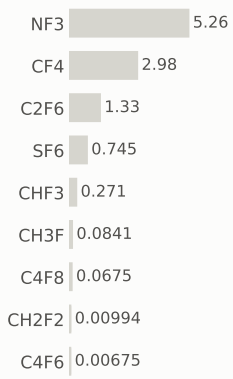
steps by process area



materials deposited (g)



gases emitted (g)



Step #7500: RIE

etch_anisotropic · RIE

1.04	642.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

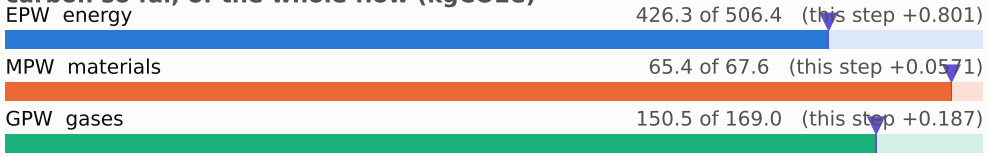
zoomed to fit

this step

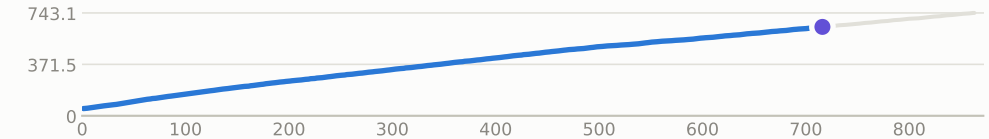
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

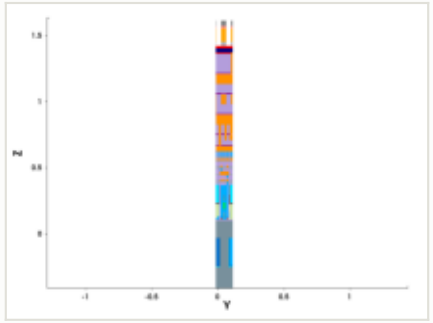
carbon so far, of the whole flow (kgCO2e)



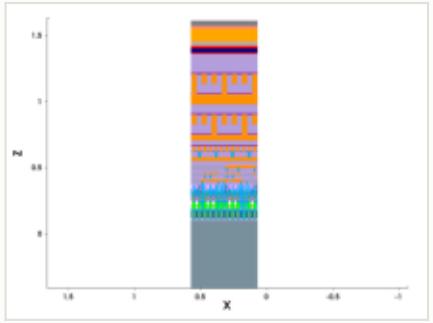
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



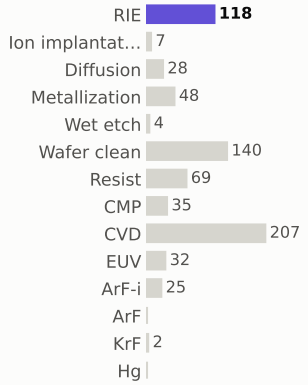
y cut at 0.049 µm



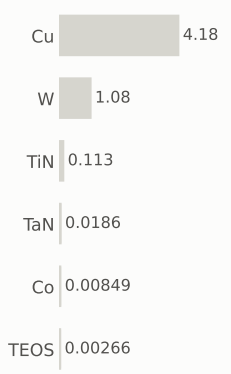
mask for this step

no mask for this step

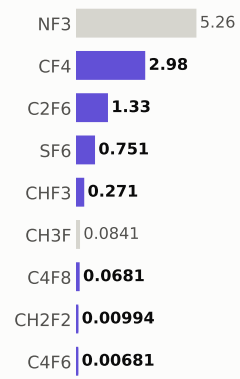
steps by process area



materials deposited (g)



gases emitted (g)



Step #7510: Wafer_clean

etch_strip · Wafer_clean

0.169	642.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

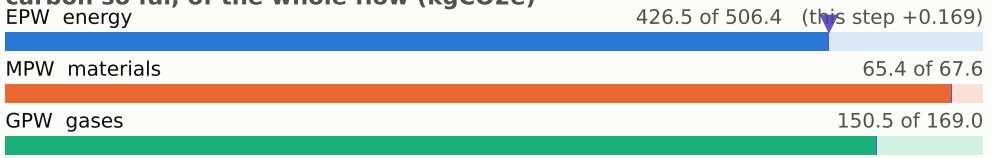
zoomed to fit

this step

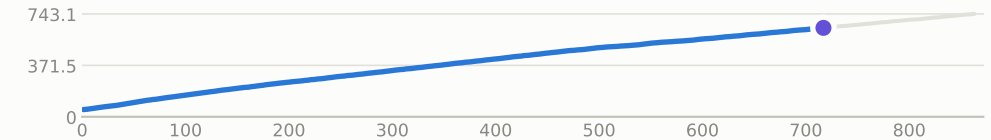
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

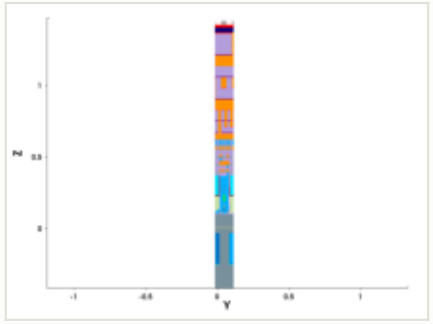
carbon so far, of the whole flow (kgCO2e)



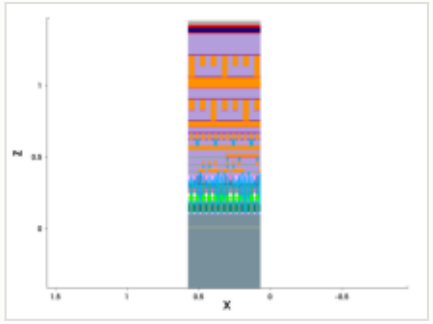
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



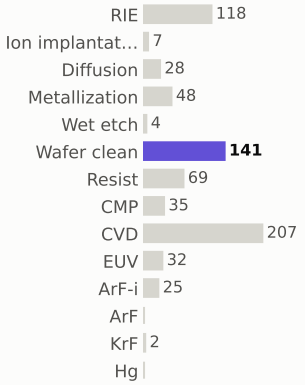
y cut at 0.049 µm



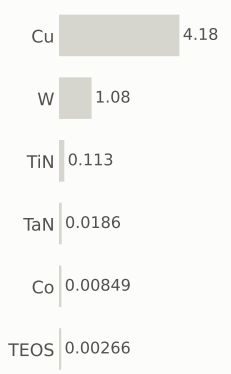
mask for this step

no mask for this step

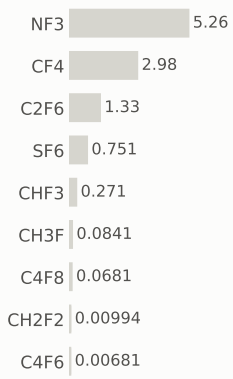
steps by process area



materials deposited (g)



gases emitted (g)



Step #7520: CVD

depo_isotropic · CVD

1.31	643.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

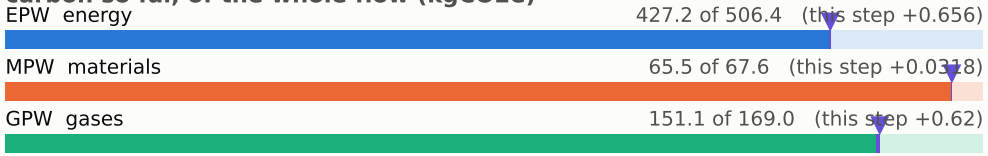
zoomed to fit

this step

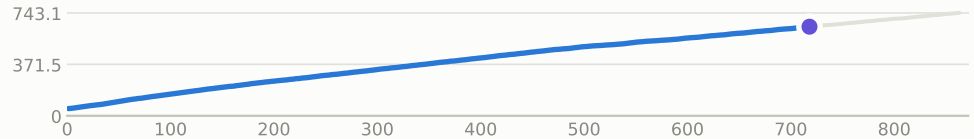
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

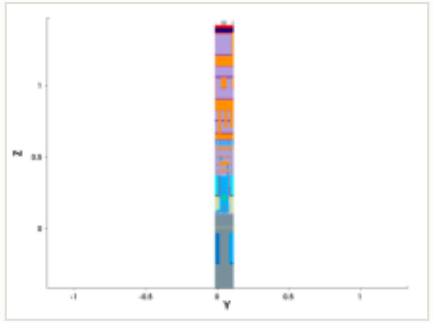
carbon so far, of the whole flow (kgCO2e)



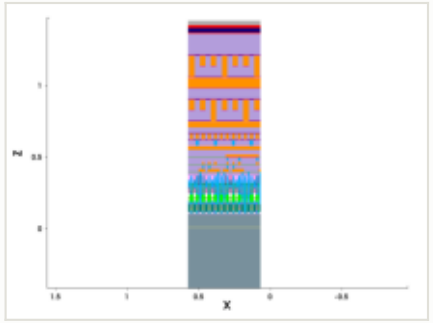
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



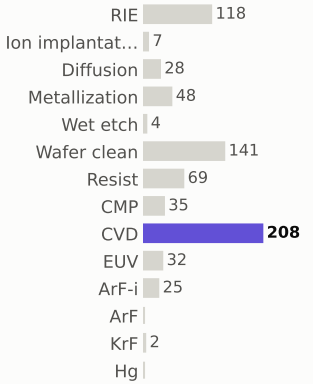
y cut at 0.049 µm



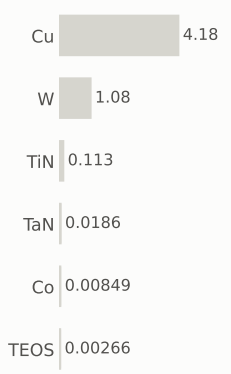
mask for this step

no mask for this step

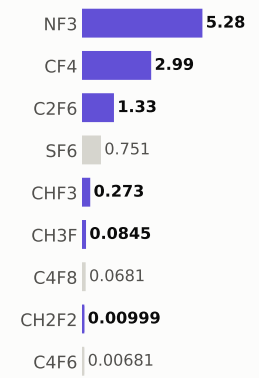
steps by process area



materials deposited (g)



gases emitted (g)



Step #7530: RIE

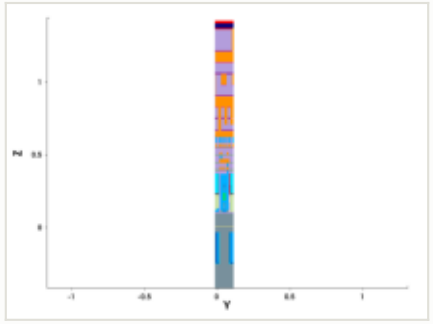
etch_anisotropic · RIE

1.04	644.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

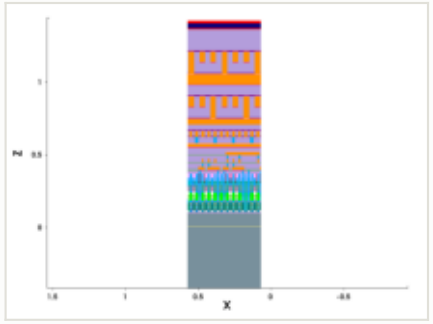
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

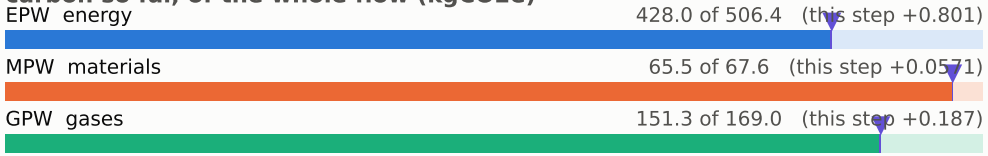
no mask for this step

this step

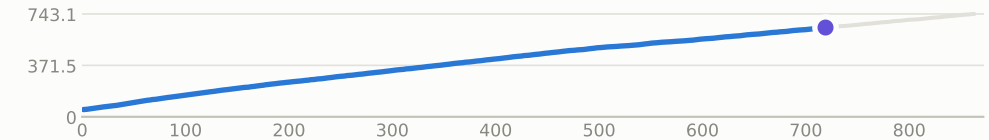
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

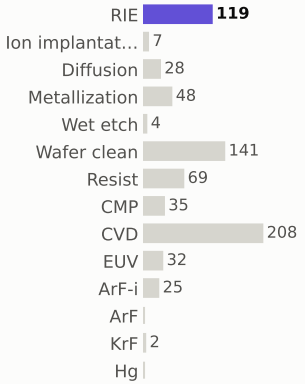
carbon so far, of the whole flow (kgCO2e)



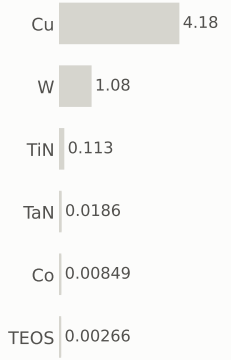
embodied carbon up to each step (kgCO2e per wafer)



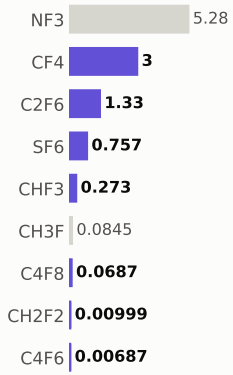
steps by process area



materials deposited (g)



gases emitted (g)



Step #7540: Wafer_clean

etch_strip · Wafer_clean

0.169	645.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

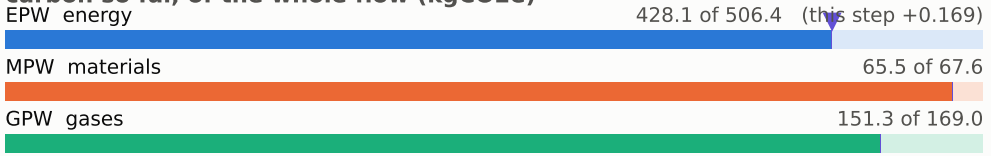
zoomed to fit

this step

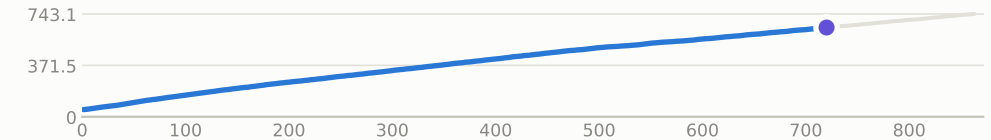
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

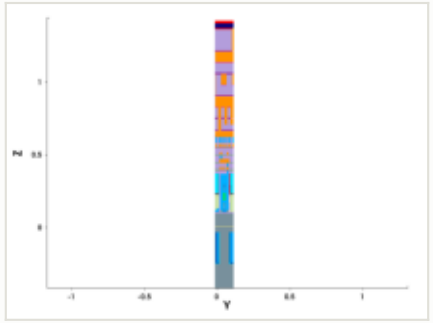
carbon so far, of the whole flow (kgCO2e)



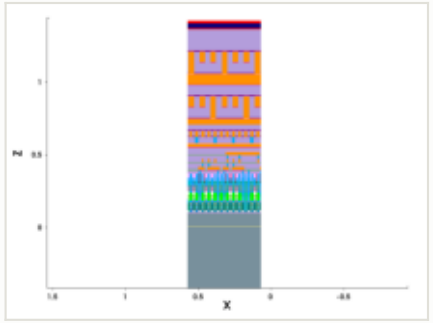
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



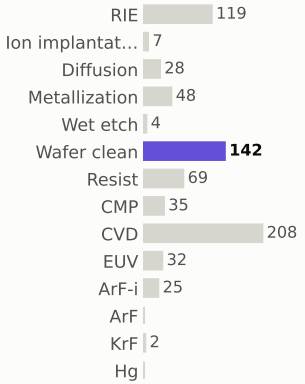
y cut at 0.049 µm



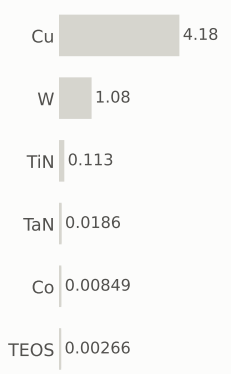
mask for this step

no mask for this step

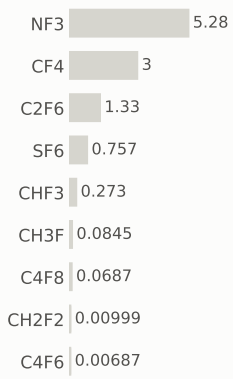
steps by process area



materials deposited (g)



gases emitted (g)



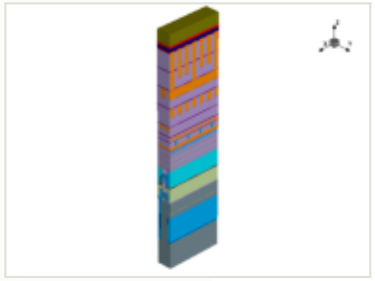
Step #7550: Resist

depo_fill · Resist

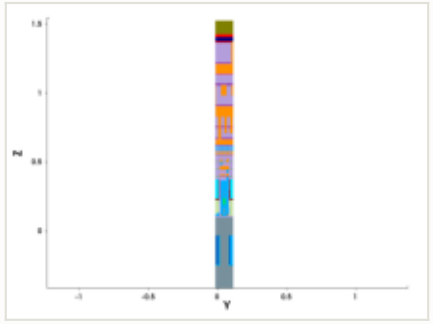
0.169	645.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

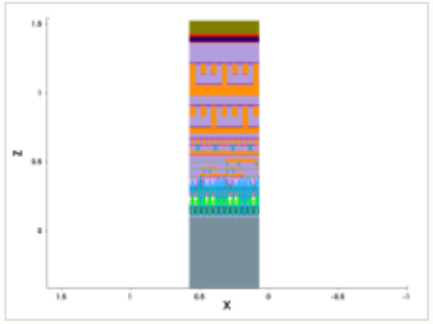
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



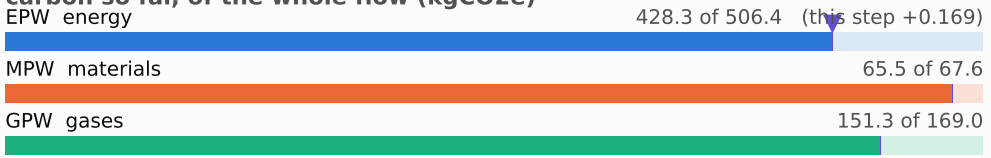
mask for this step

no mask for this step

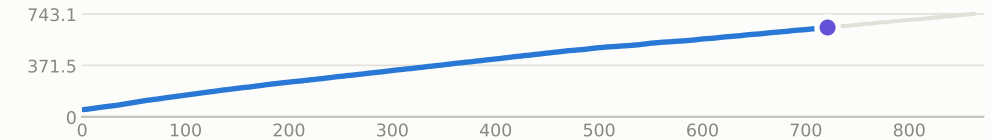
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

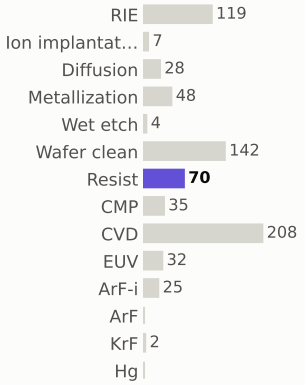
carbon so far, of the whole flow (kgCO2e)



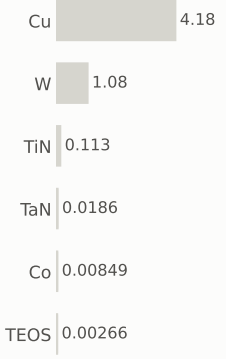
embodied carbon up to each step (kgCO2e per wafer)



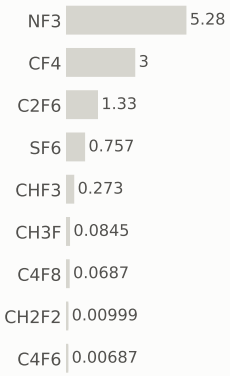
steps by process area



materials deposited (g)



gases emitted (g)



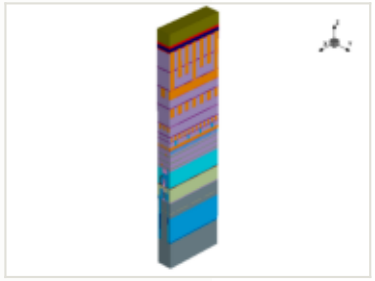
Step #7560: Wafer_clean

clean · Wafer_clean

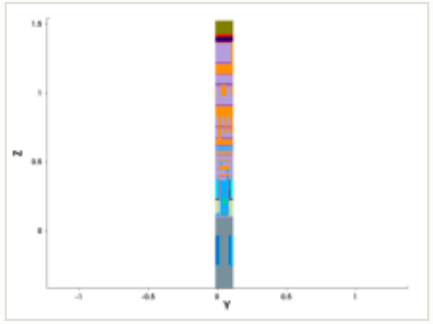
0.169	645.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

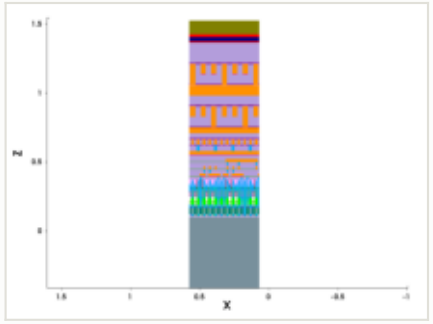
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

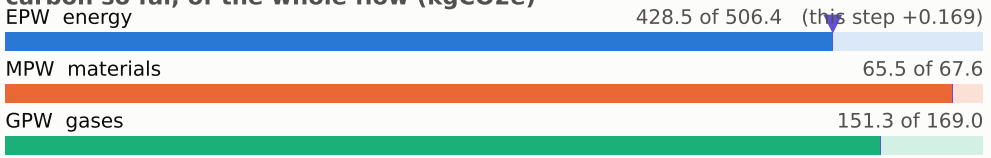
no mask for this step

this step

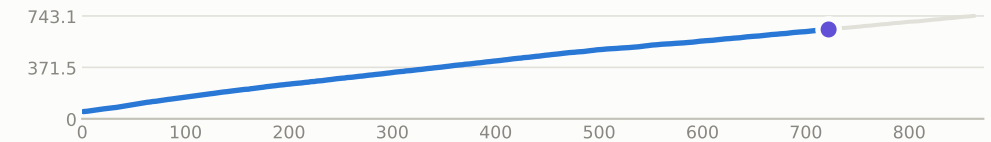
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

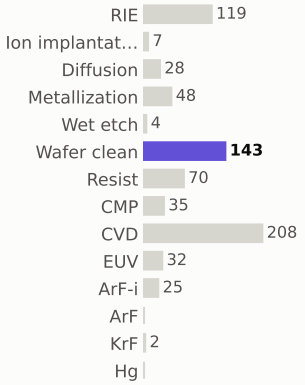
carbon so far, of the whole flow (kgCO2e)



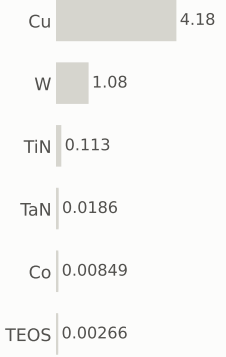
embodied carbon up to each step (kgCO2e per wafer)



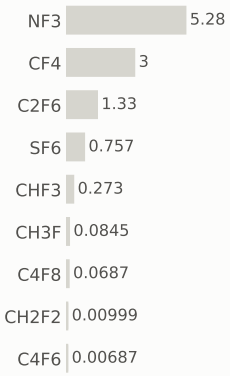
steps by process area



materials deposited (g)



gases emitted (g)



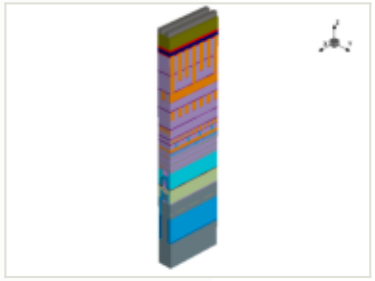
Step #7570: ArF-i

litho_patterning · ArF-i

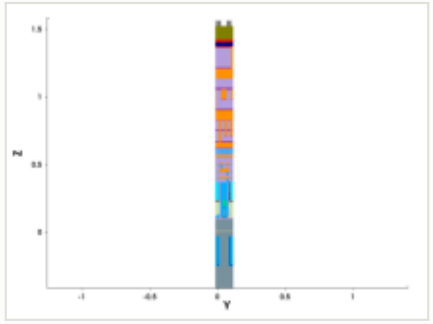
0.217	645.5	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

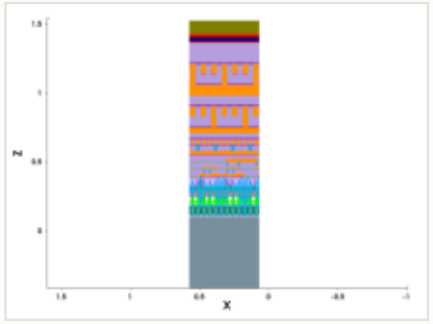
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



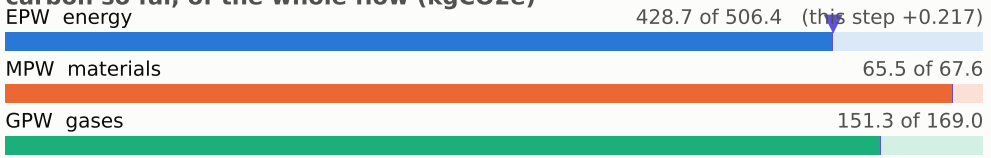
mask: layer 65:0 (GDSII)

this step

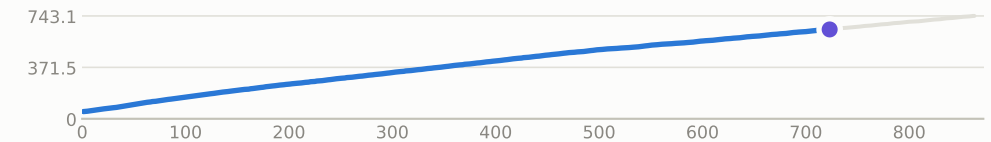
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

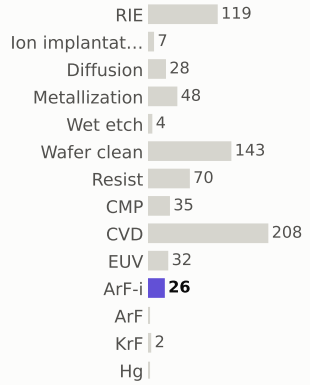
carbon so far, of the whole flow (kgCO2e)



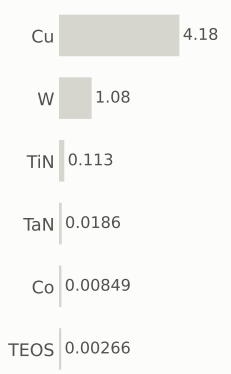
embodied carbon up to each step (kgCO2e per wafer)



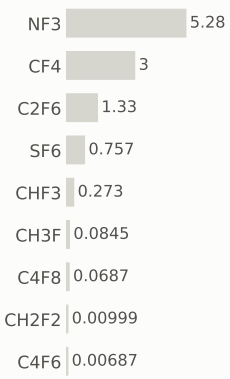
steps by process area



materials deposited (g)



gases emitted (g)



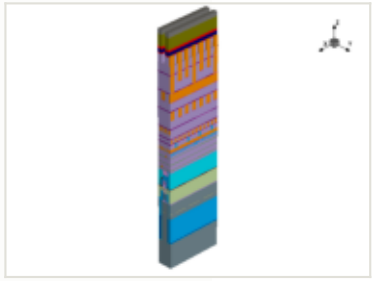
Step #7580: RIE

etch_anisotropic · RIE

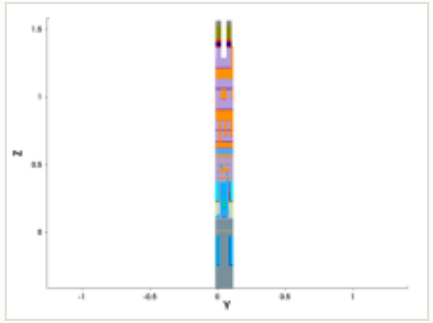
1.04	646.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

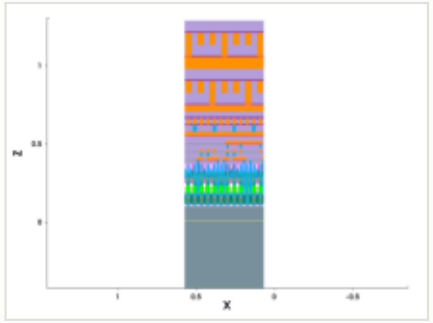
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

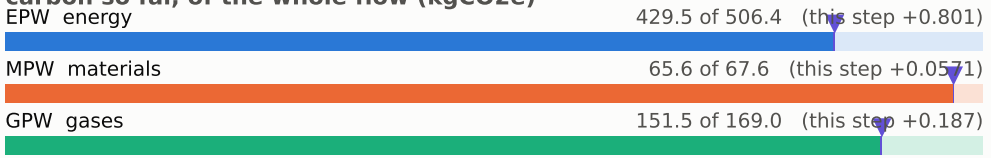
no mask for this step

this step

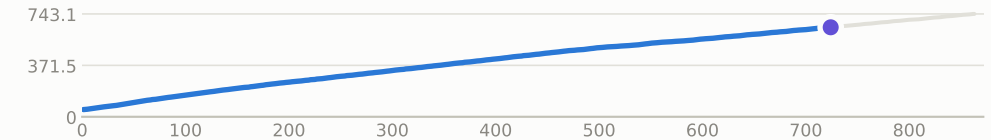
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

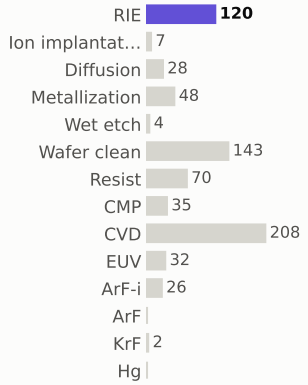
carbon so far, of the whole flow (kgCO2e)



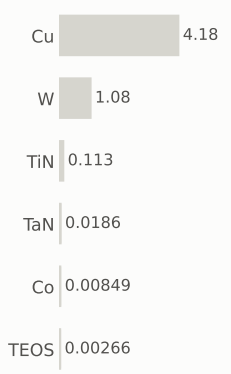
embodied carbon up to each step (kgCO2e per wafer)



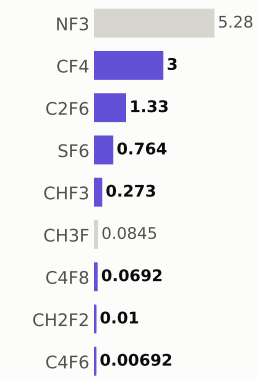
steps by process area



materials deposited (g)



gases emitted (g)



Step #7590: Wafer_clean

etch_strip · Wafer_clean

0.169	646.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

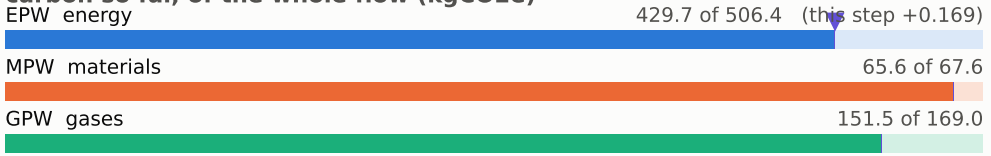
zoomed to fit

this step

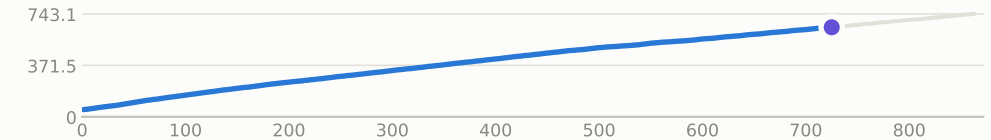
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

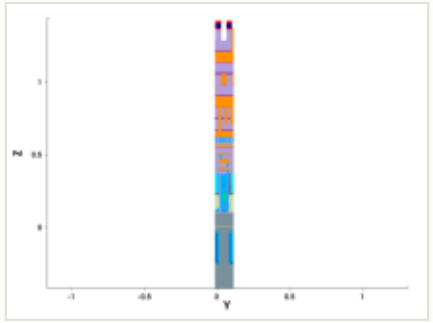
carbon so far, of the whole flow (kgCO2e)



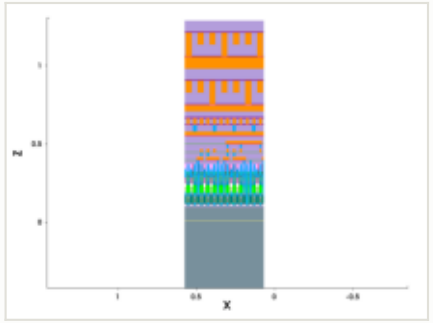
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



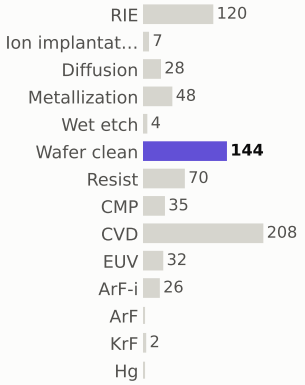
y cut at 0.049 µm



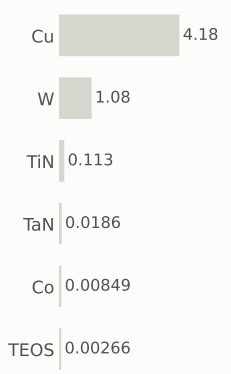
mask for this step

no mask for this step

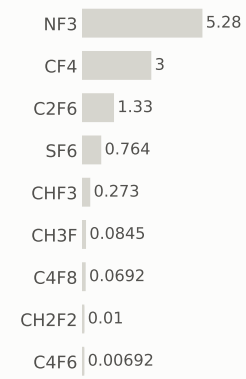
steps by process area



materials deposited (g)



gases emitted (g)



Step #7600: Metallization

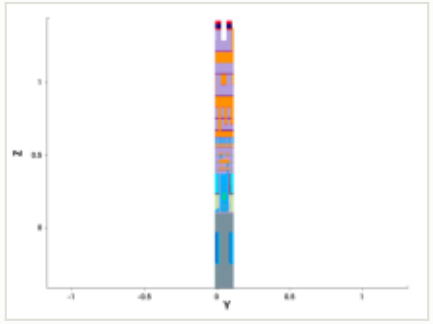
depo_ellipsoidal · Metallization

1.2	647.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

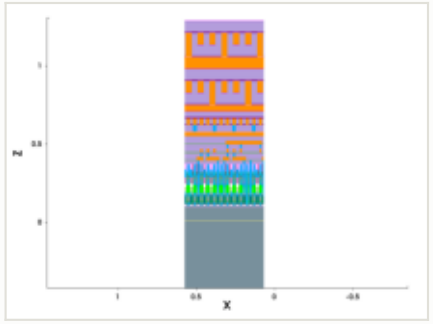
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



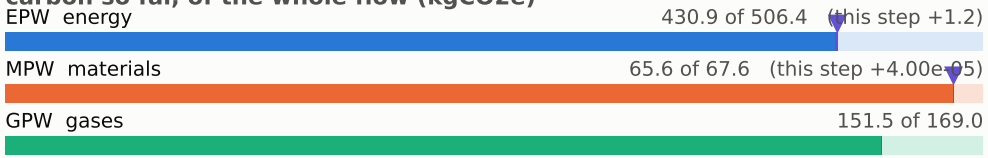
mask for this step

no mask for this step

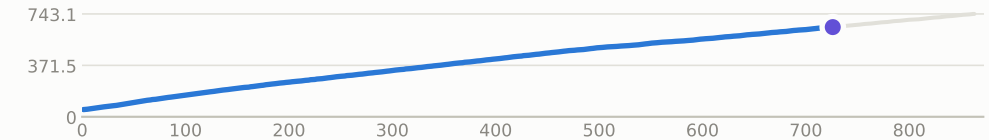
this step

process area	Metallization	EPW	1.2 kgCO2e
material	TaN	EPW tool	0.86 kgCO2e
thickness	0.0016 µm	EPW overhead	0.344 kgCO2e
material GWP	25.8159 gCO2e/g	MPW	4.00e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

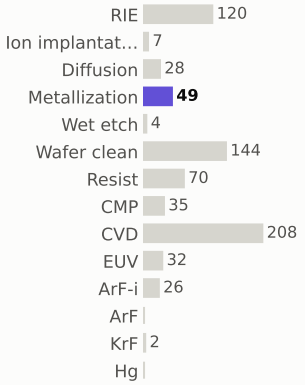
carbon so far, of the whole flow (kgCO2e)



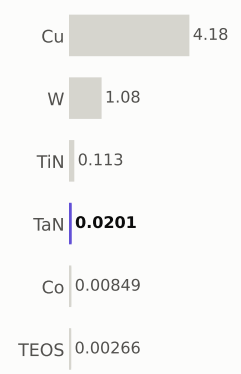
embodied carbon up to each step (kgCO2e per wafer)



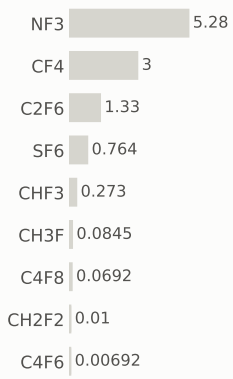
steps by process area



materials deposited (g)



gases emitted (g)



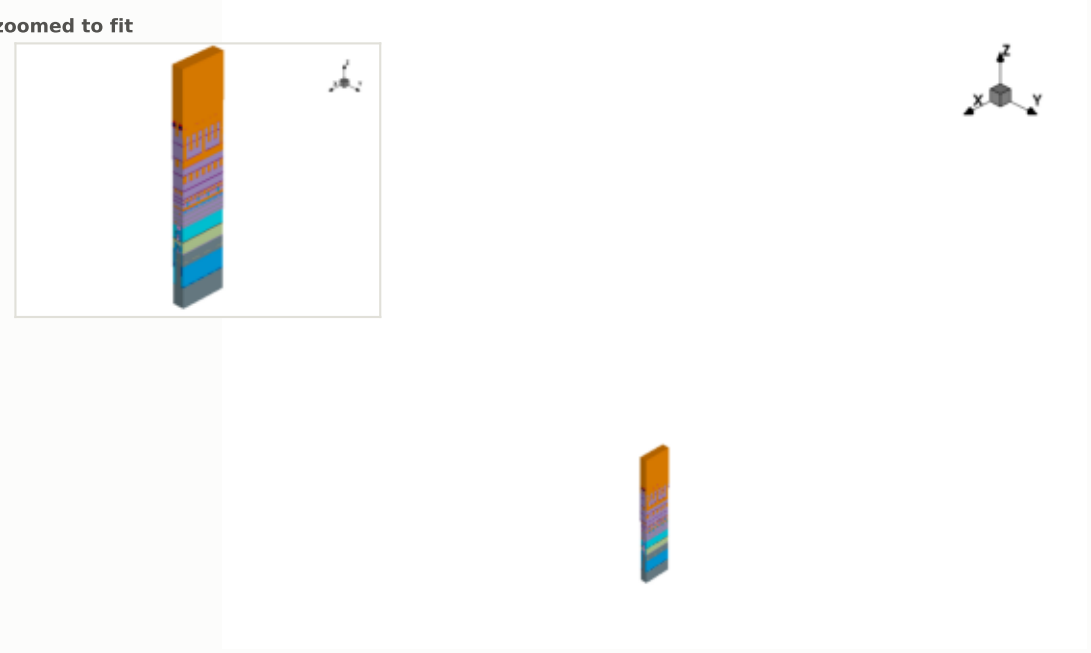
Step #7610: Metallization

depo_fill · Metallization

1.21	649.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

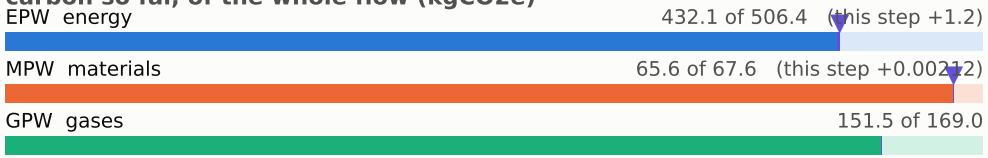
zoomed to fit



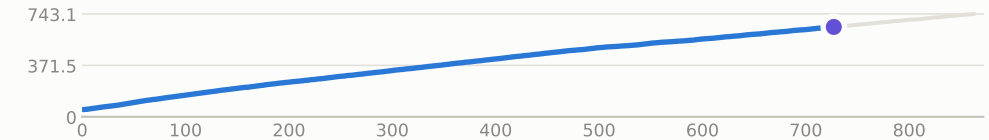
this step

process area	Metallization	EPW	1.2 kgCO2e
material	Cu	EPW tool	0.86 kgCO2e
thickness	0.55 µm	EPW overhead	0.344 kgCO2e
material GWP	6.09011 gCO2e/g	MPW	0.00212 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.21 kgCO2e**

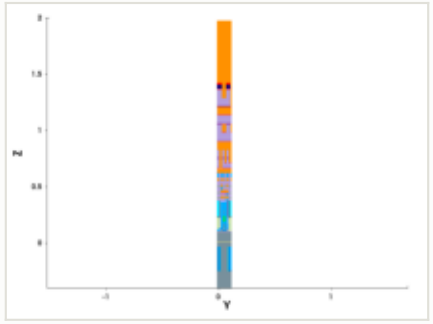
carbon so far, of the whole flow (kgCO2e)



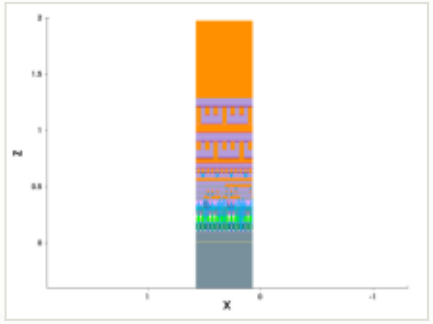
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



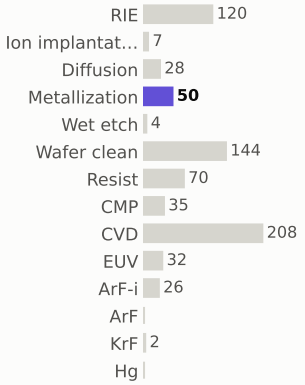
y cut at 0.049 µm



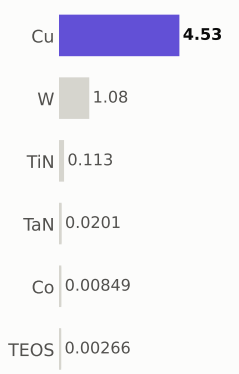
mask for this step

no mask for this step

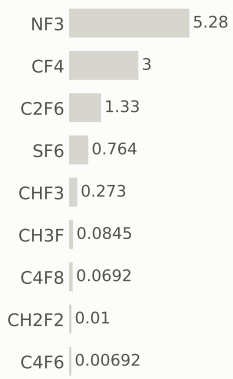
steps by process area



materials deposited (g)



gases emitted (g)



Step #7620: Diffusion

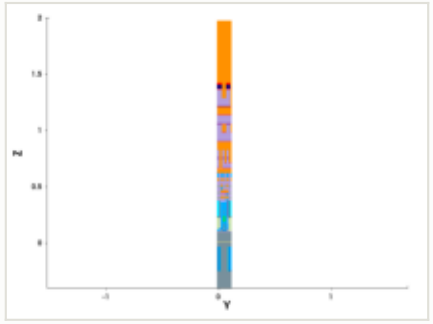
treatment · Diffusion

0.801	649.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

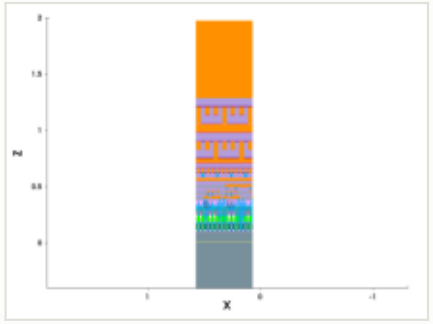
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

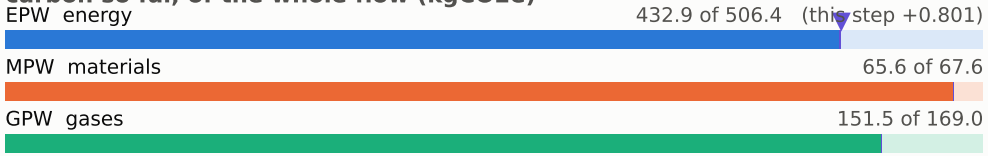
no mask for this step

this step

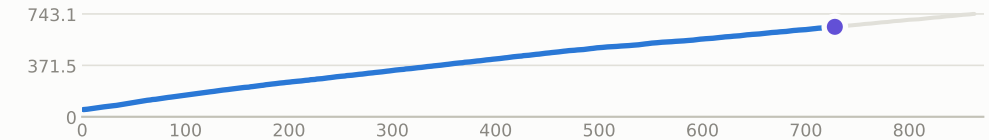
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

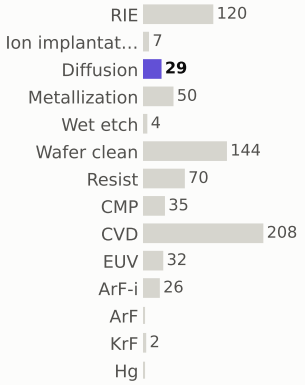
carbon so far, of the whole flow (kgCO2e)



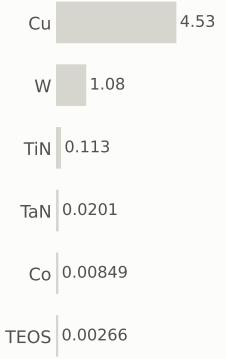
embodied carbon up to each step (kgCO2e per wafer)



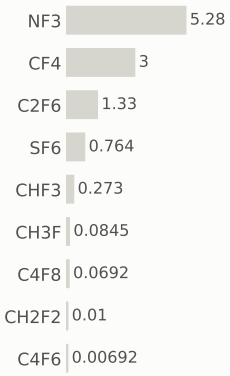
steps by process area



materials deposited (g)



gases emitted (g)



Step #7630: CMP

etch_cmp · CMP

0.169	650.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

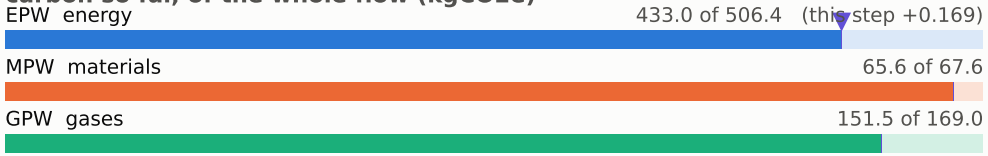
zoomed to fit

this step

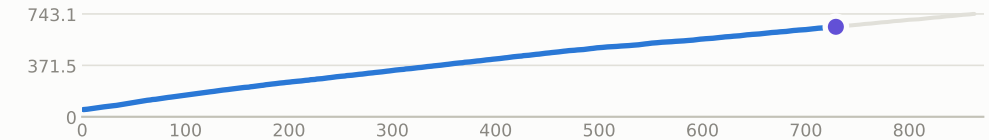
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

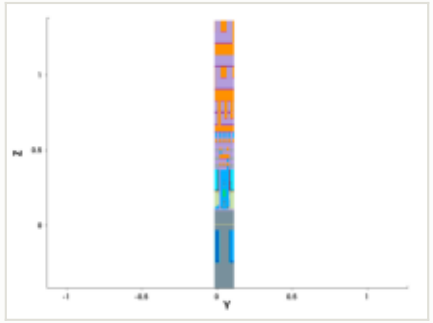
carbon so far, of the whole flow (kgCO2e)



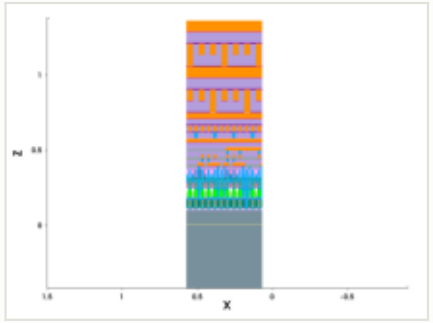
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



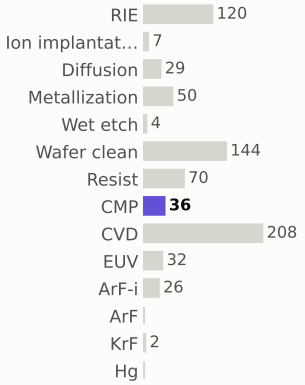
y cut at 0.049 µm



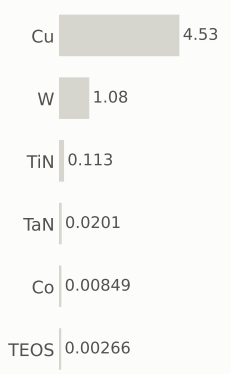
mask for this step

no mask for this step

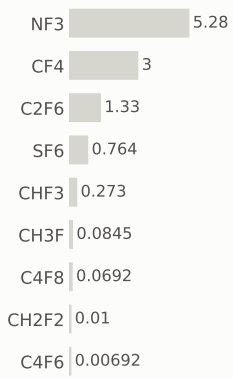
steps by process area



materials deposited (g)



gases emitted (g)



Step #7640: Metallization

depo_isotropic · Metallization

1.2	651.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

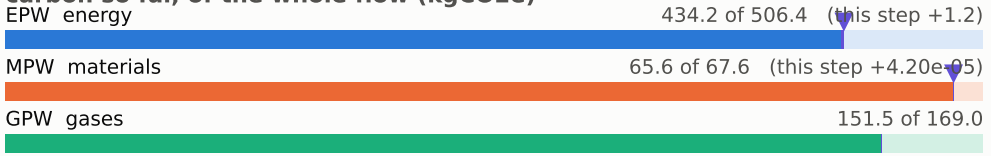
zoomed to fit

this step

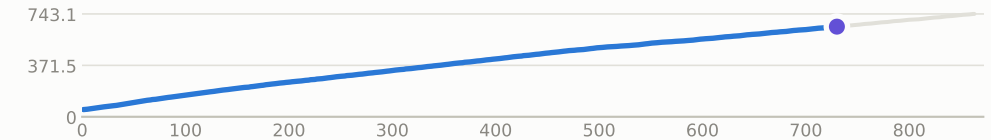
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

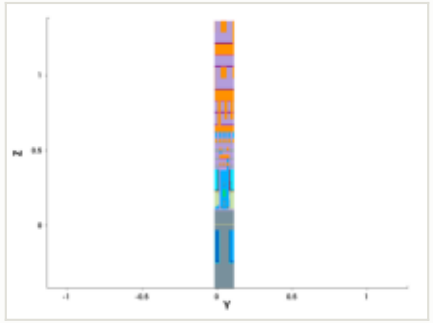
carbon so far, of the whole flow (kgCO2e)



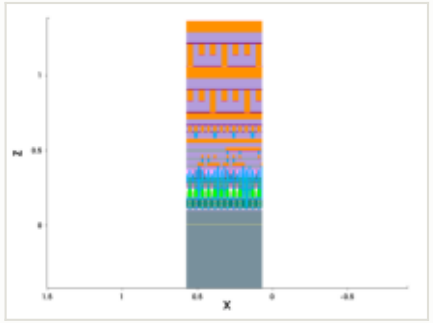
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



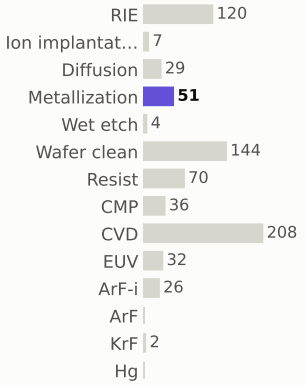
y cut at 0.049 µm



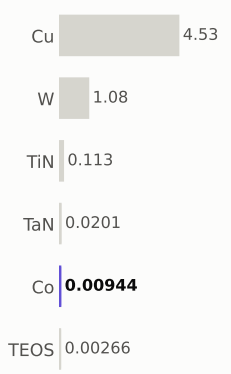
mask for this step

no mask for this step

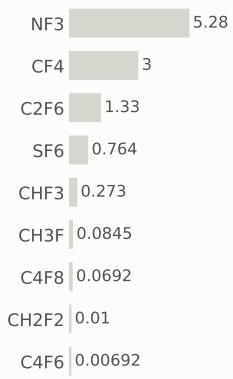
steps by process area



materials deposited (g)



gases emitted (g)



Step #7660: CVD

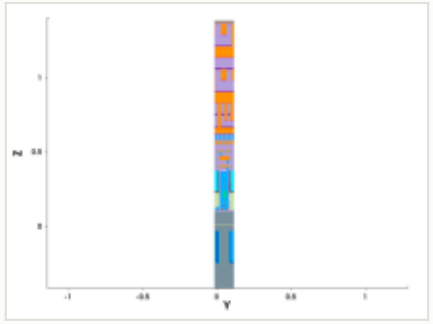
depo_fill · CVD

1.31	652.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

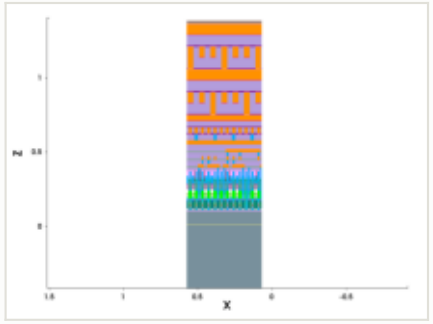
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

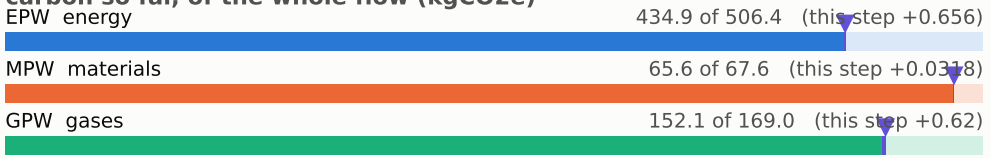
no mask for this step

this step

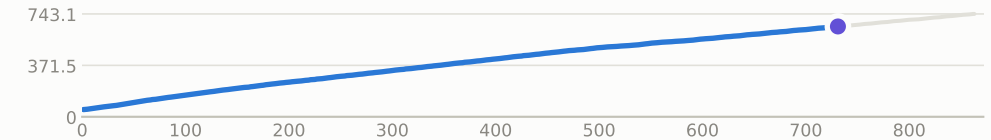
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

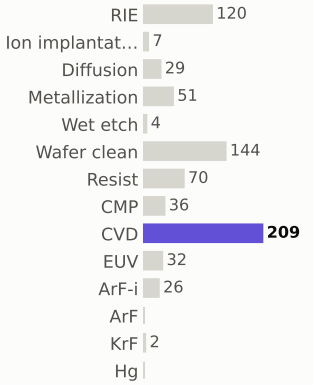
carbon so far, of the whole flow (kgCO2e)



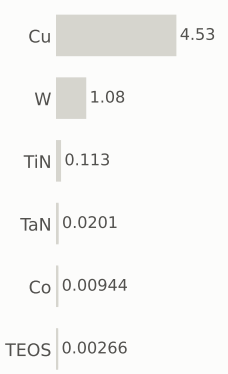
embodied carbon up to each step (kgCO2e per wafer)



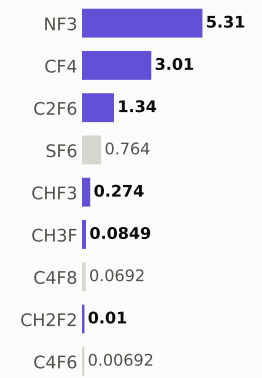
steps by process area



materials deposited (g)



gases emitted (g)



Step #7670: CVD

depo_fill · CVD

1.31	653.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

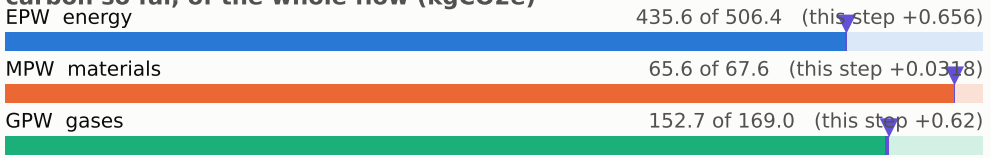
zoomed to fit

this step

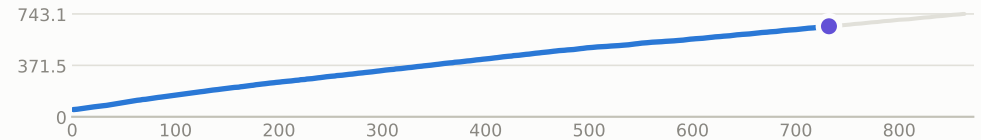
process area	CVD
material	—
thickness	0.204 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

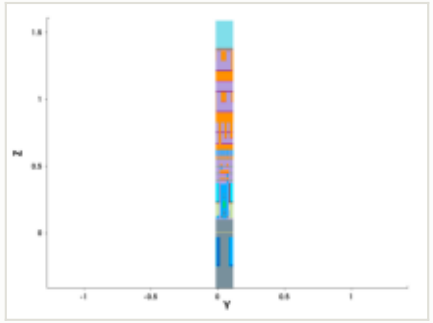
carbon so far, of the whole flow (kgCO2e)



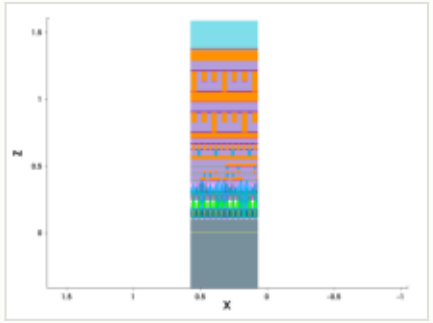
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



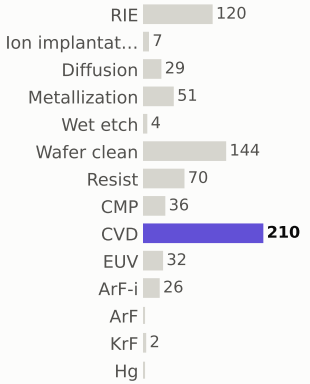
y cut at 0.049 µm



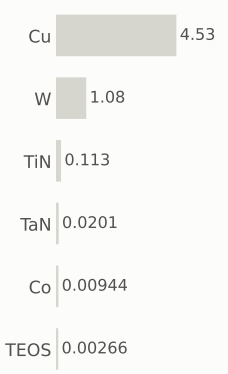
mask for this step

no mask for this step

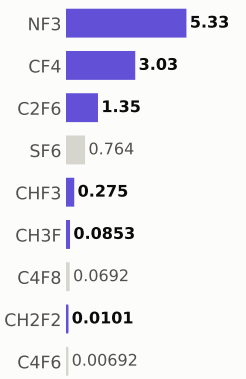
steps by process area



materials deposited (g)



gases emitted (g)



Step #7680: Resist

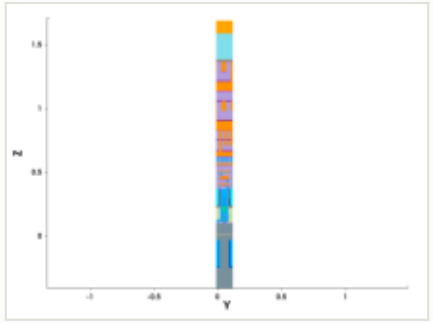
depo_fill · Resist

0.169	654.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

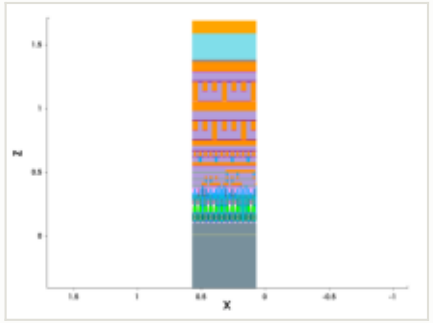
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



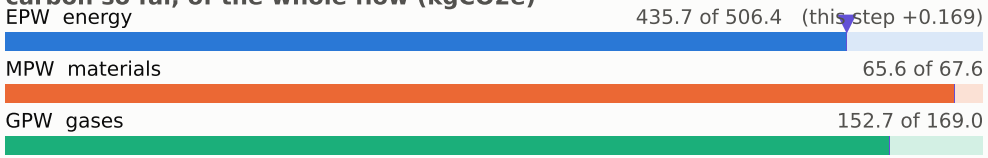
mask for this step

no mask for this step

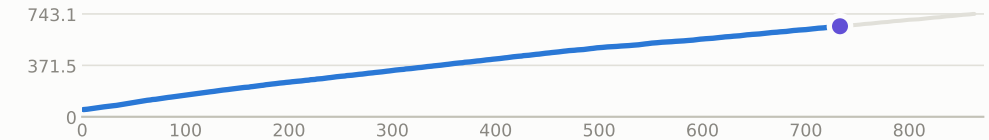
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

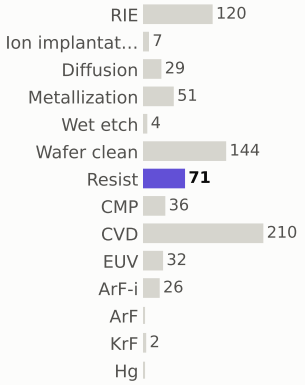
carbon so far, of the whole flow (kgCO2e)



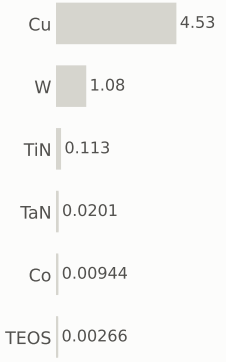
embodied carbon up to each step (kgCO2e per wafer)



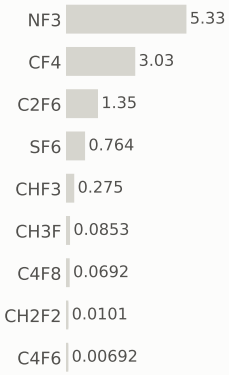
steps by process area



materials deposited (g)



gases emitted (g)



Step #7690: CVD

depo_fill · CVD

1.31	655.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

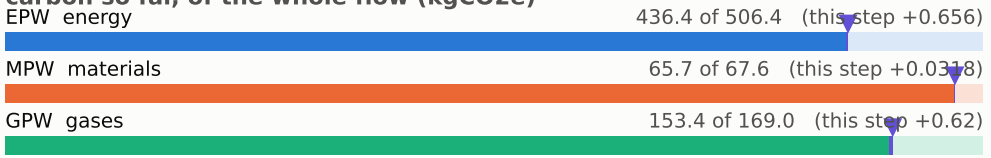
zoomed to fit

this step

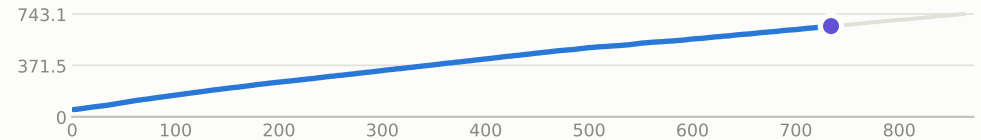
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

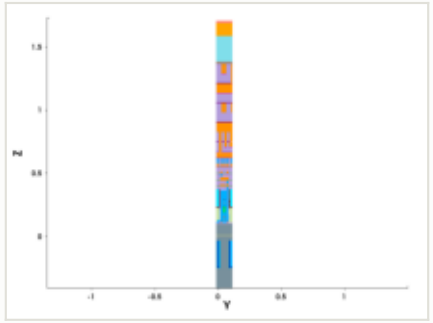
carbon so far, of the whole flow (kgCO2e)



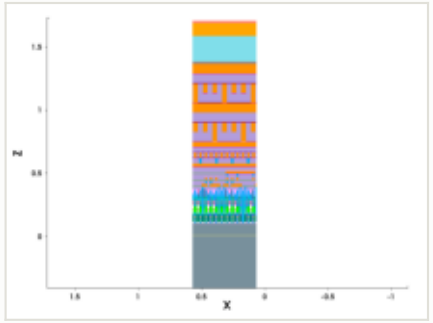
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



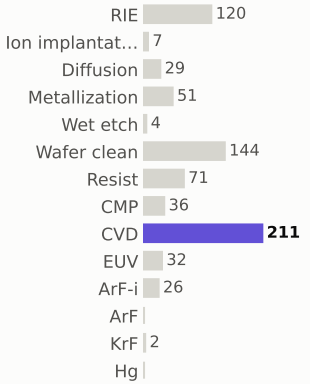
y cut at 0.049 µm



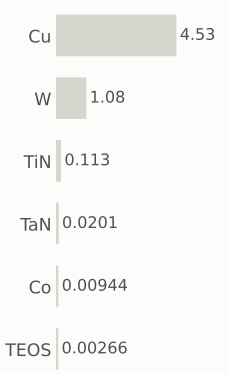
mask for this step

no mask for this step

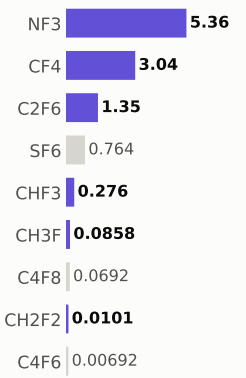
steps by process area



materials deposited (g)



gases emitted (g)



Step #7700: Wafer_clean

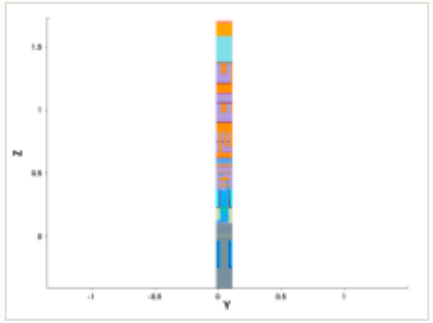
clean · Wafer_clean

0.169	655.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

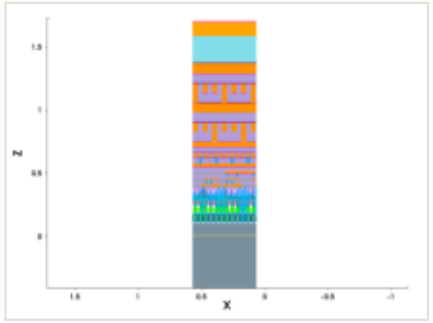
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

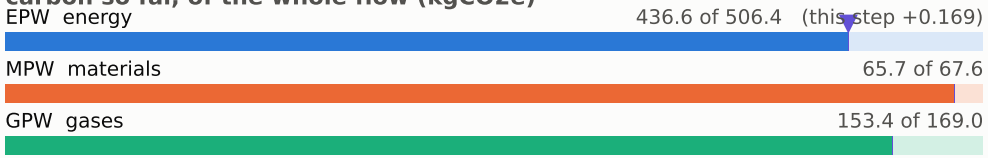
no mask for this step

this step

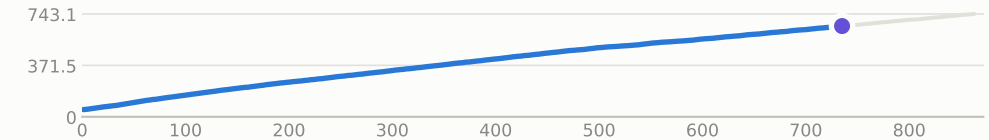
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

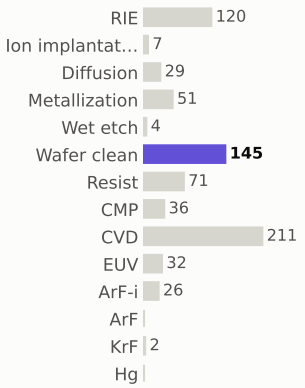
carbon so far, of the whole flow (kgCO2e)



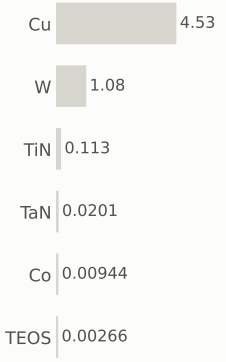
embodied carbon up to each step (kgCO2e per wafer)



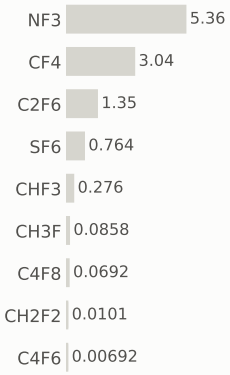
steps by process area



materials deposited (g)



gases emitted (g)



Step #7710: ArF-i

litho_patterning · ArF-i

0.217	655.8	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

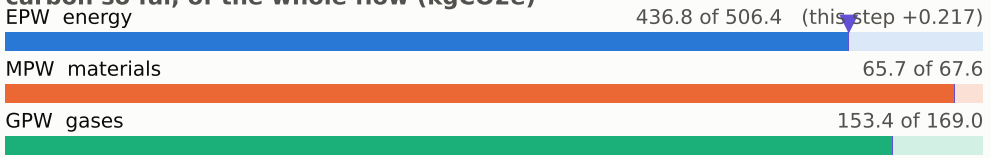
zoomed to fit

this step

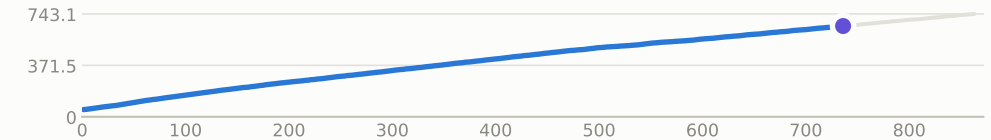
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

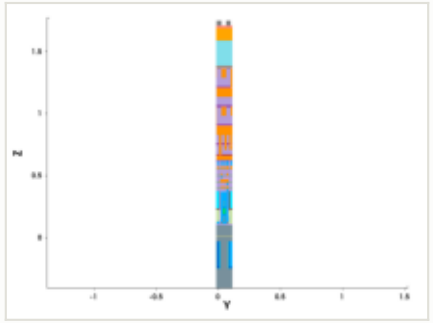
carbon so far, of the whole flow (kgCO2e)



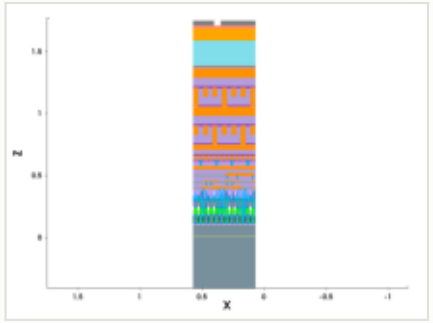
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

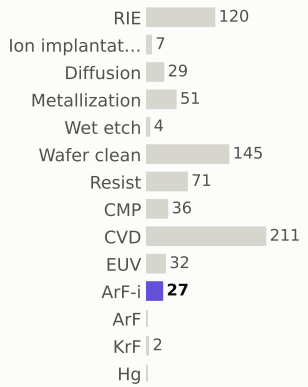


y cut at 0.049 µm

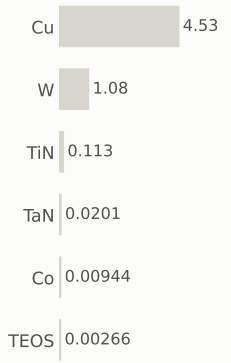


mask: layer 10671:0 (GDSII)

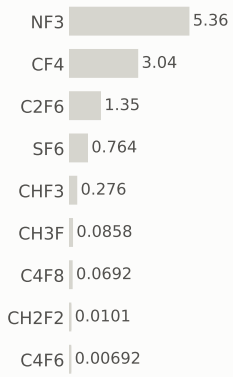
steps by process area



materials deposited (g)



gases emitted (g)



Step #7720: RIE

etch_anisotropic · RIE

1.04	656.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

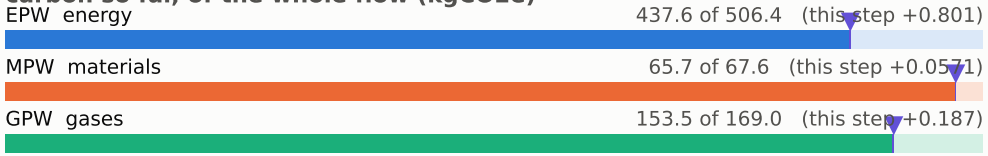
zoomed to fit

this step

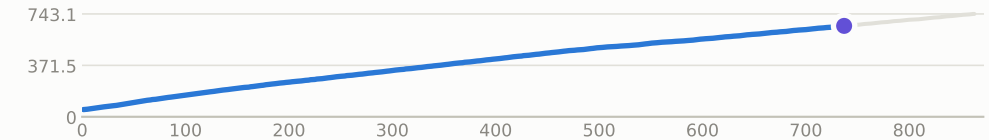
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

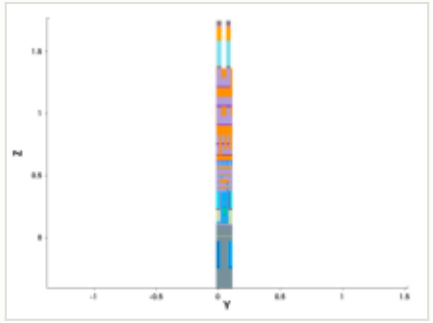
carbon so far, of the whole flow (kgCO2e)



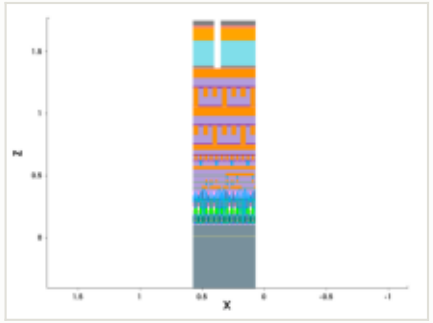
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



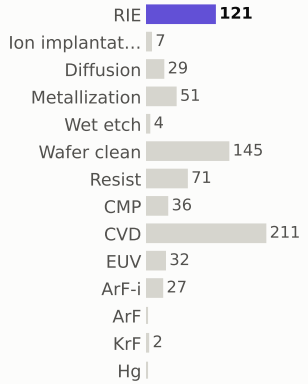
y cut at 0.049 µm



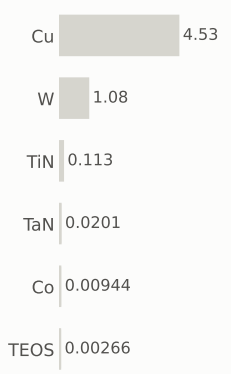
mask for this step

no mask for this step

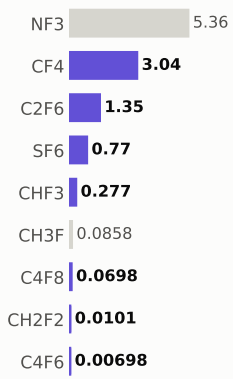
steps by process area



materials deposited (g)



gases emitted (g)



Step #7730: Wafer_clean

etch_strip · Wafer_clean

0.169	657.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

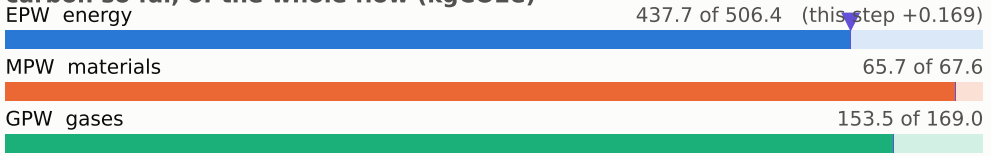
zoomed to fit

this step

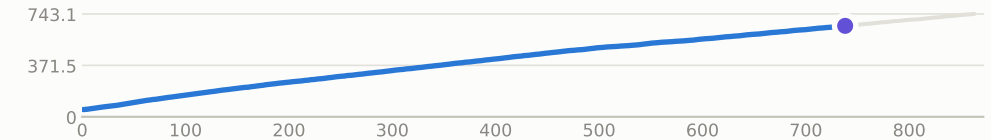
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

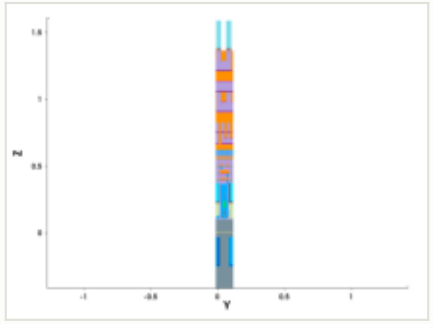
carbon so far, of the whole flow (kgCO2e)



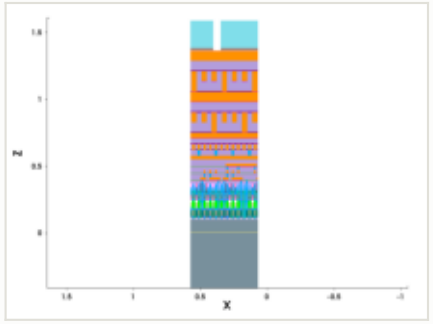
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



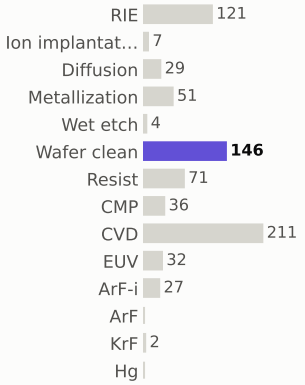
y cut at 0.049 µm



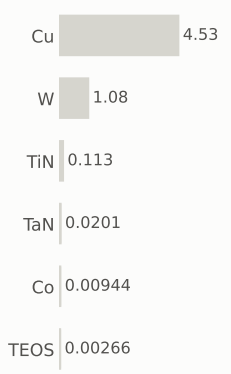
mask for this step

no mask for this step

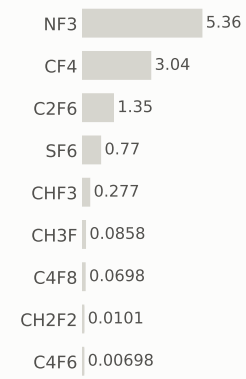
steps by process area



materials deposited (g)



gases emitted (g)



Step #7740: Resist

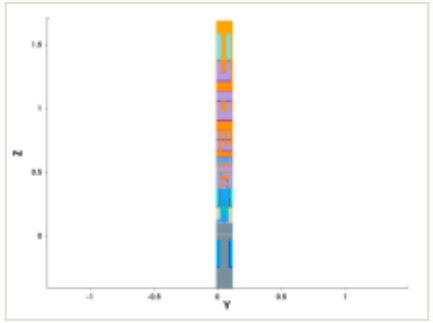
depo_fill · Resist

0.169	657.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

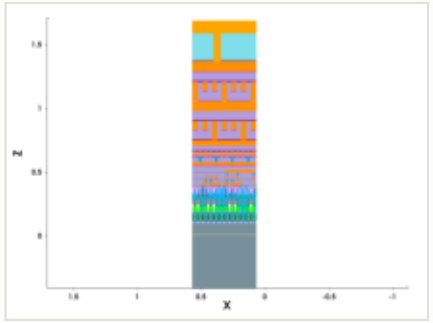
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



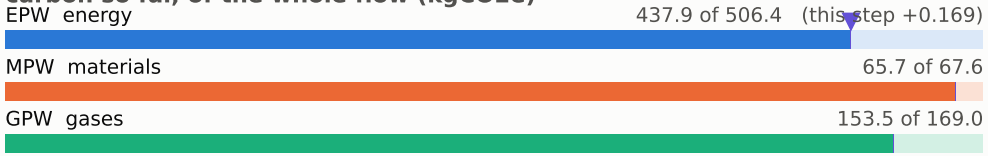
mask for this step

no mask for this step

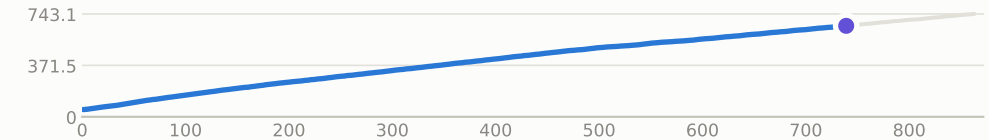
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

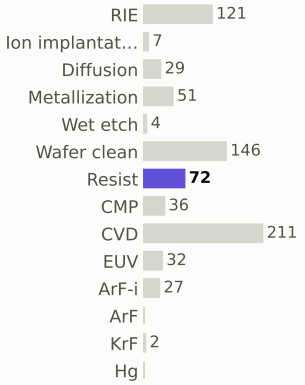
carbon so far, of the whole flow (kgCO2e)



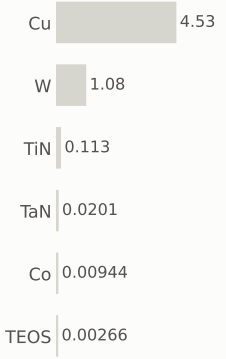
embodied carbon up to each step (kgCO2e per wafer)



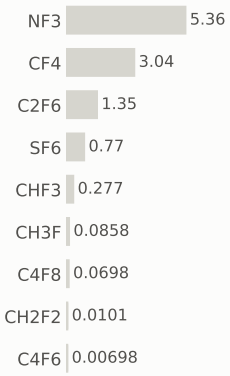
steps by process area



materials deposited (g)



gases emitted (g)



Step #7750: CVD

depo_fill · CVD

1.31	658.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

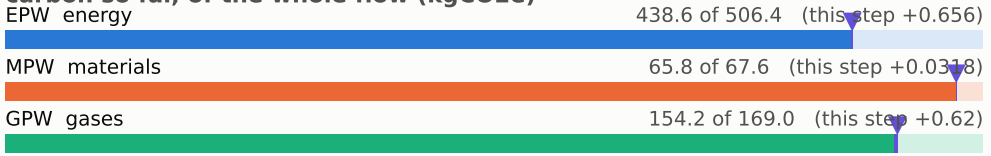
zoomed to fit

this step

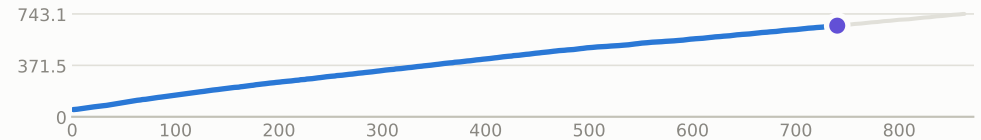
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

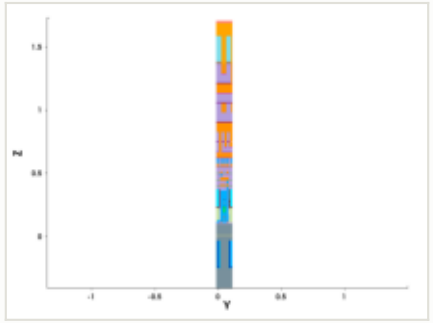
carbon so far, of the whole flow (kgCO2e)



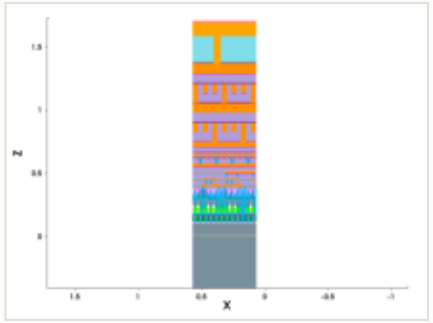
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



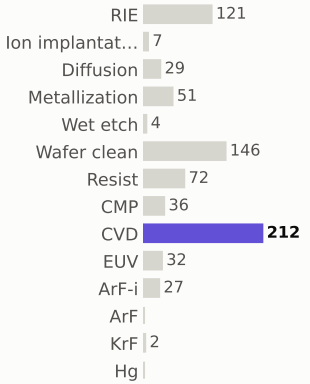
y cut at 0.049 µm



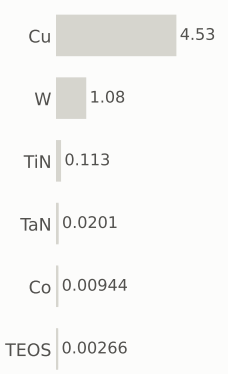
mask for this step

no mask for this step

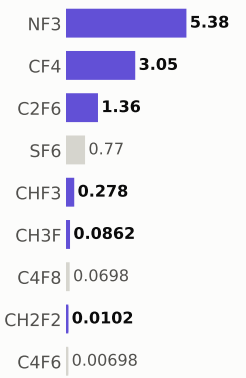
steps by process area



materials deposited (g)



gases emitted (g)



Step #7760: Wafer_clean

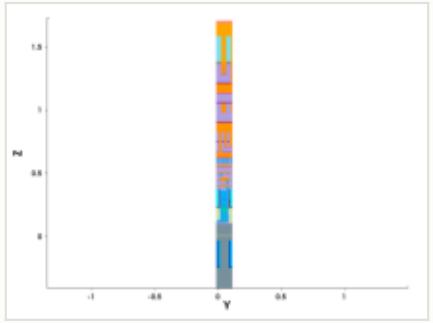
clean · Wafer_clean

0.169	658.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

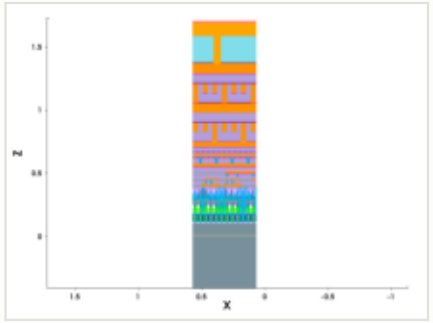
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

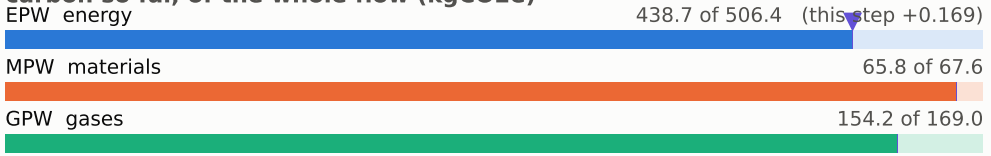
no mask for this step

this step

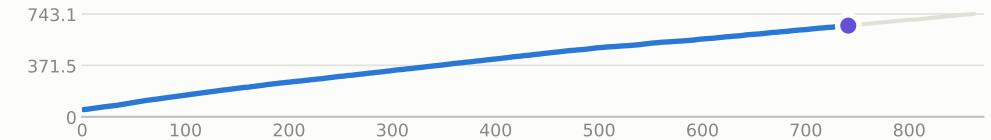
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

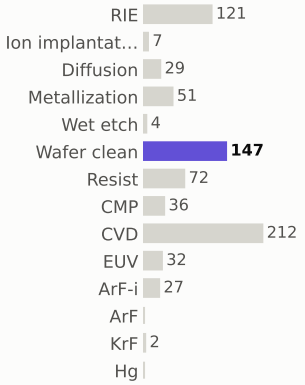
carbon so far, of the whole flow (kgCO2e)



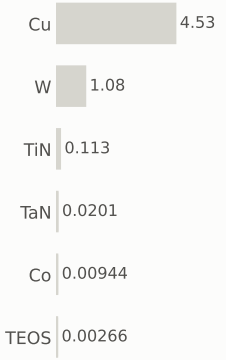
embodied carbon up to each step (kgCO2e per wafer)



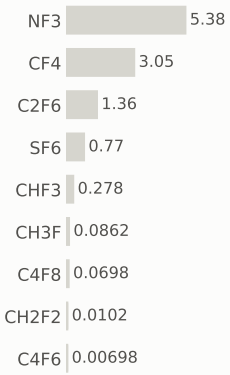
steps by process area



materials deposited (g)



gases emitted (g)



Step #7770: ArF-i

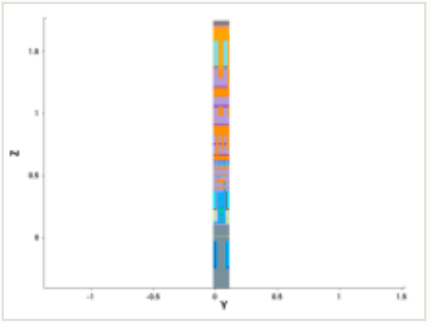
litho_patterning · ArF-i

0.217	658.9	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

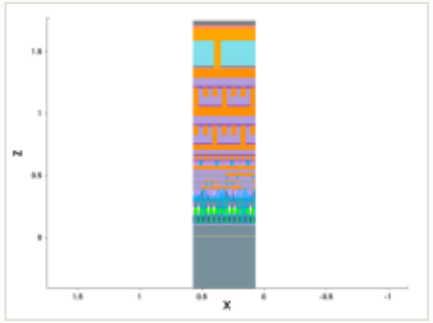
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask: layer 10672:0 (GDSII)

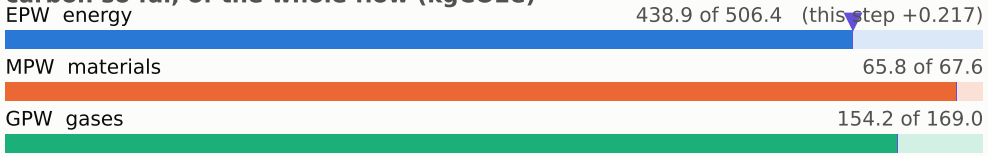
nothing in the clip window

this step

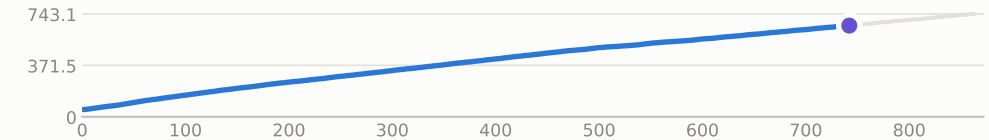
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

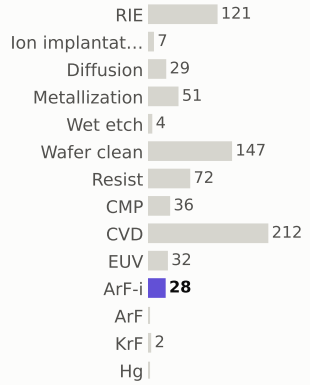
carbon so far, of the whole flow (kgCO2e)



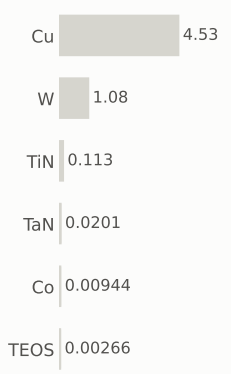
embodied carbon up to each step (kgCO2e per wafer)



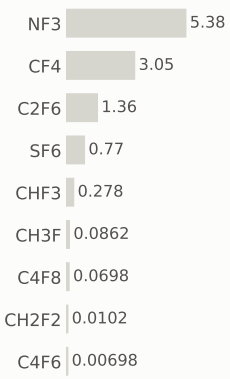
steps by process area



materials deposited (g)



gases emitted (g)



Step #7780: RIE

etch_anisotropic · RIE

1.04	659.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

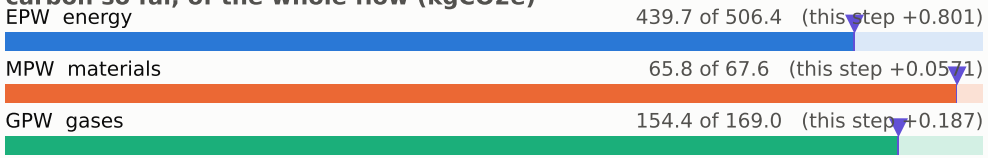
zoomed to fit

this step

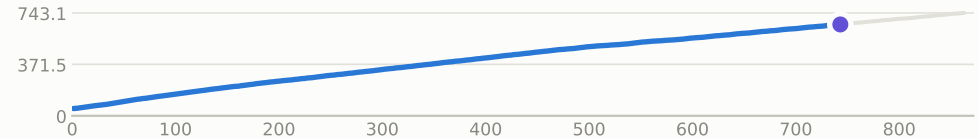
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

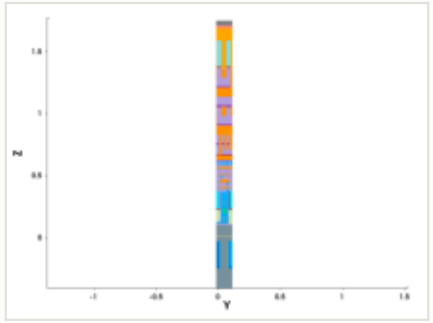
carbon so far, of the whole flow (kgCO2e)



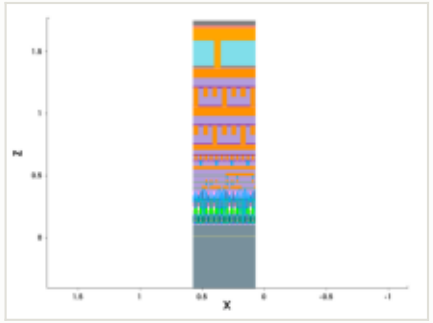
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



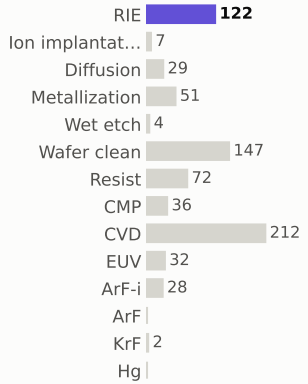
y cut at 0.049 µm



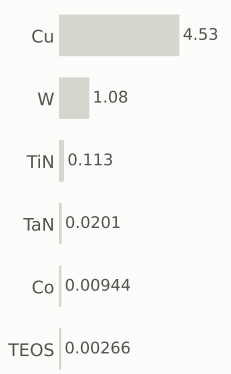
mask for this step

no mask for this step

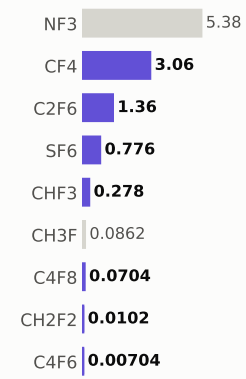
steps by process area



materials deposited (g)



gases emitted (g)



Step #7790: Wafer_clean

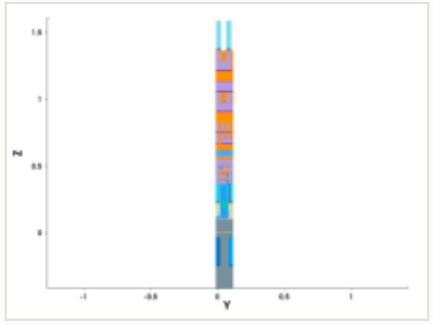
etch_strip · Wafer_clean

0.169	660.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

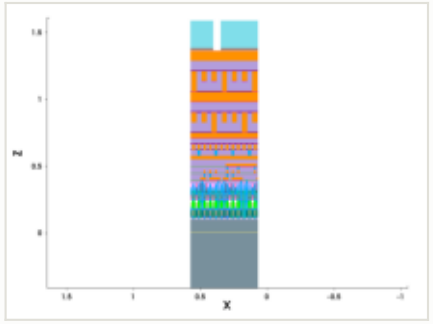
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



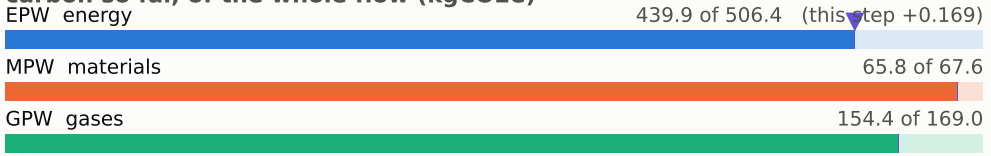
mask for this step

no mask for this step

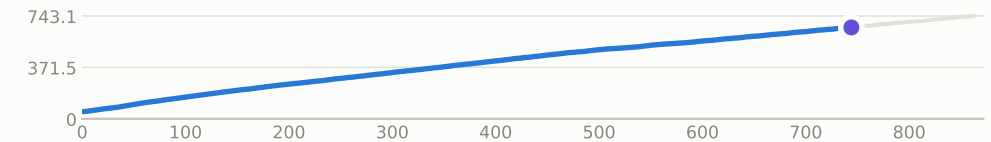
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

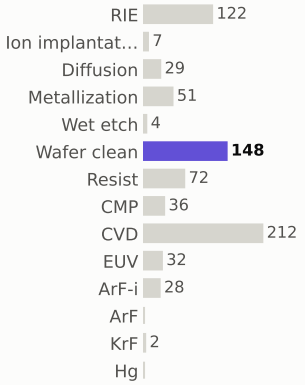
carbon so far, of the whole flow (kgCO2e)



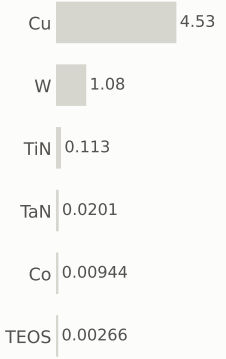
embodied carbon up to each step (kgCO2e per wafer)



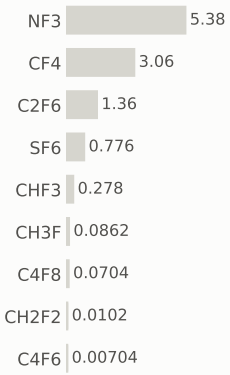
steps by process area



materials deposited (g)



gases emitted (g)



Step #7800: Metallization

depo_ellipsoidal · Metallization

1.2	661.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

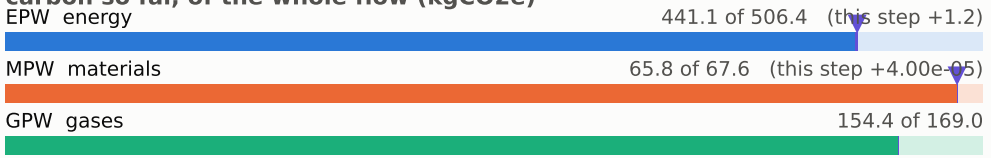
zoomed to fit

this step

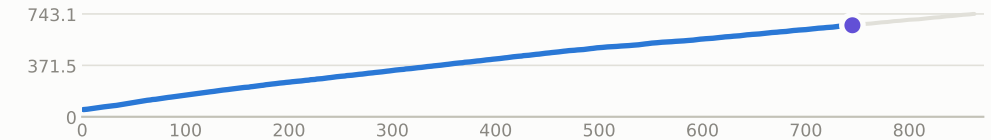
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

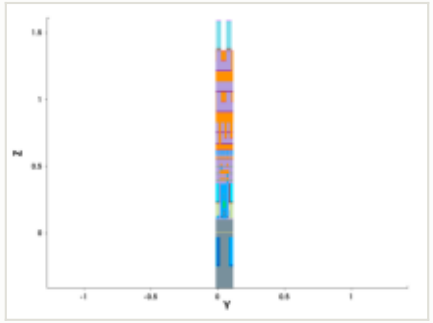
carbon so far, of the whole flow (kgCO2e)



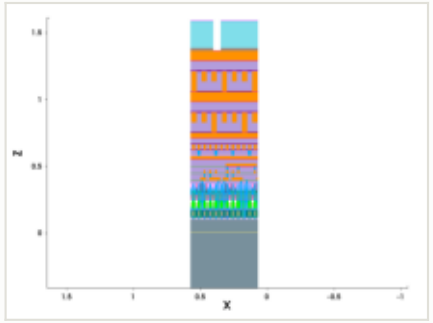
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



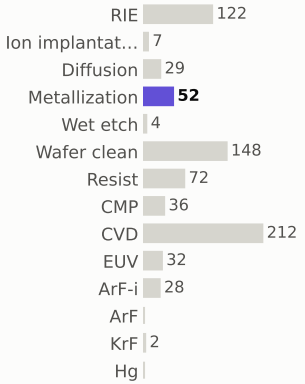
y cut at 0.049 µm



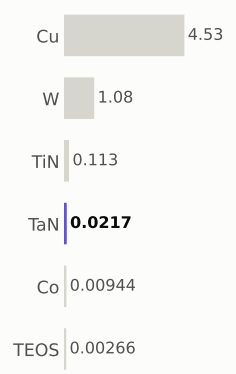
mask for this step

no mask for this step

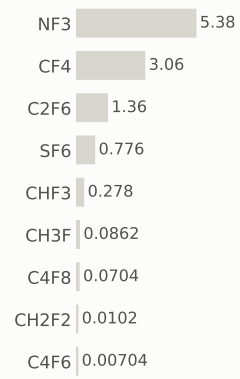
steps by process area



materials deposited (g)



gases emitted (g)



Step #7810: Metallization

depo_fill · Metallization

1.21	662.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

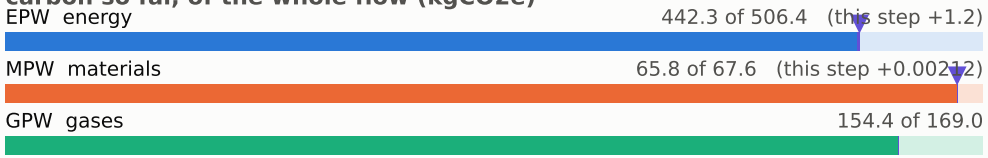
zoomed to fit

this step

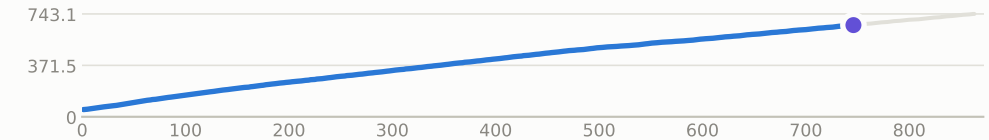
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

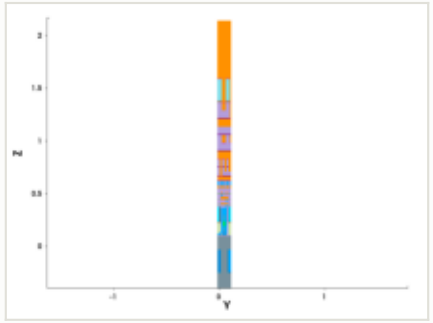
carbon so far, of the whole flow (kgCO2e)



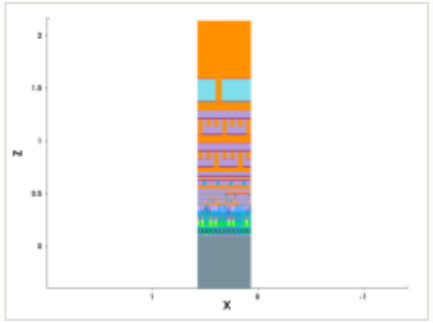
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



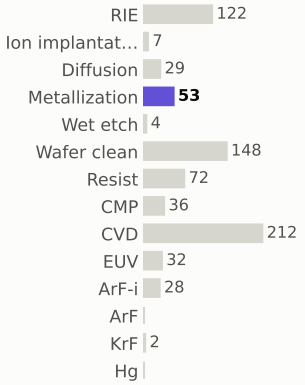
y cut at 0.049 µm



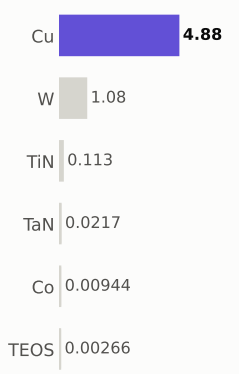
mask for this step

no mask for this step

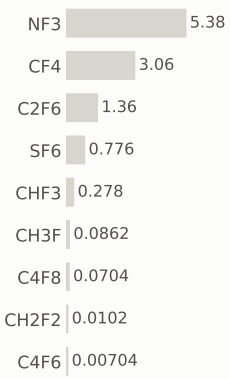
steps by process area



materials deposited (g)



gases emitted (g)



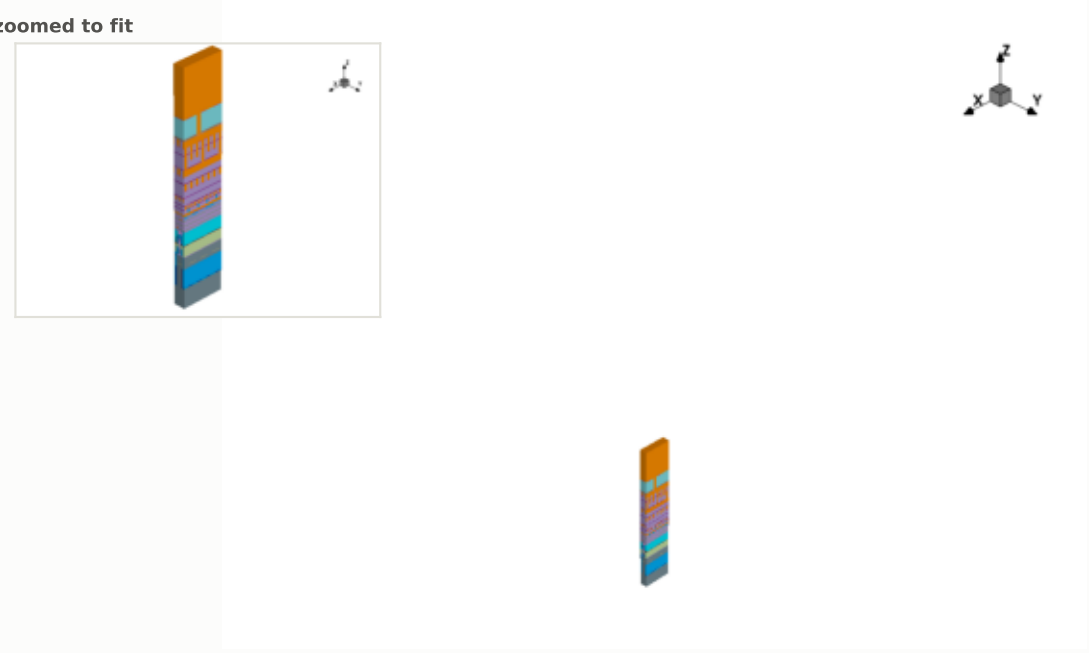
Step #7820: Diffusion

treatment · Diffusion

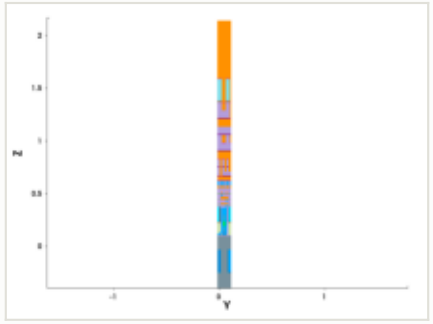
0.801	663.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

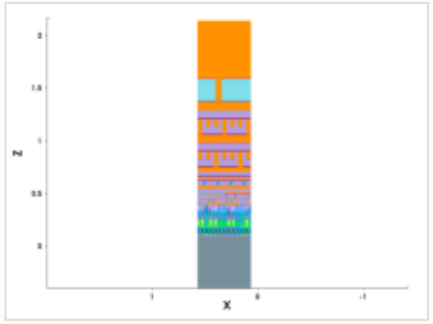
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

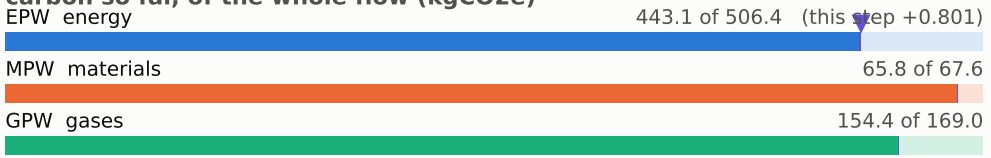
no mask for this step

this step

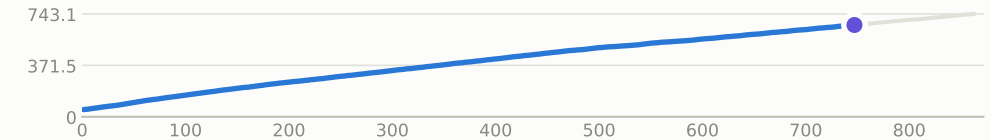
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

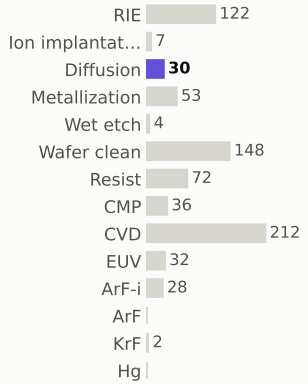
carbon so far, of the whole flow (kgCO2e)



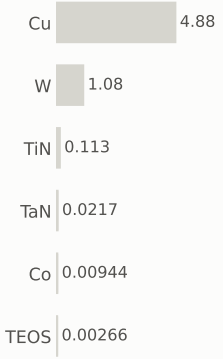
embodied carbon up to each step (kgCO2e per wafer)



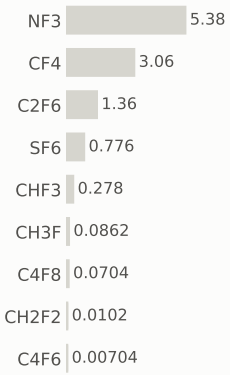
steps by process area



materials deposited (g)



gases emitted (g)



Step #7830: CMP

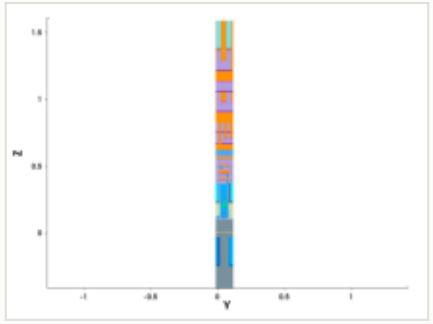
etch_cmp · CMP

0.169	663.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

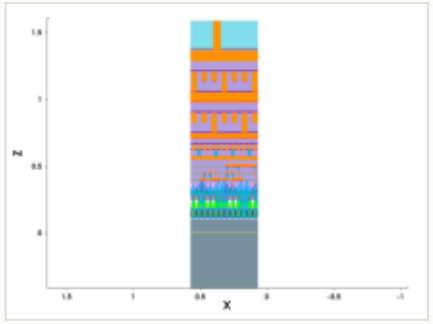
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

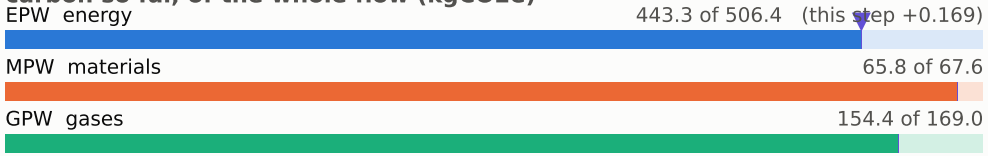
no mask for this step

this step

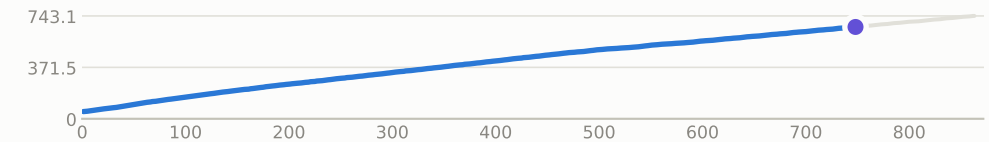
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

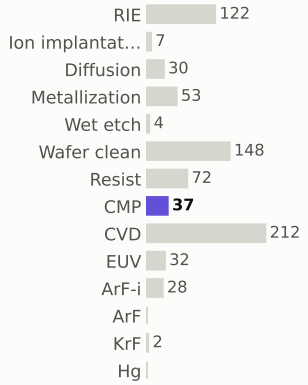
carbon so far, of the whole flow (kgCO2e)



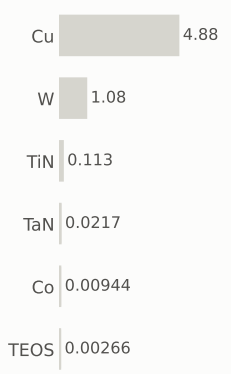
embodied carbon up to each step (kgCO2e per wafer)



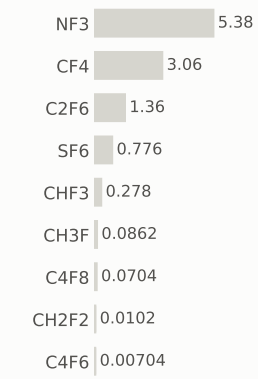
steps by process area



materials deposited (g)



gases emitted (g)



Step #7840: ArF

litho_patterning · ArF

0.199	663.7	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

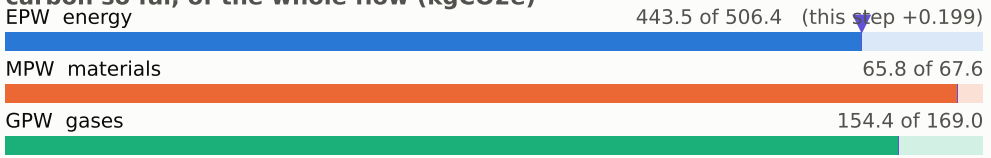
3D structure after this step

zoomed to fit

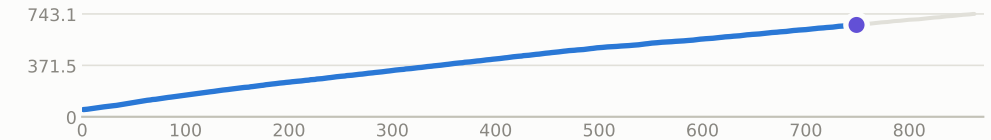
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

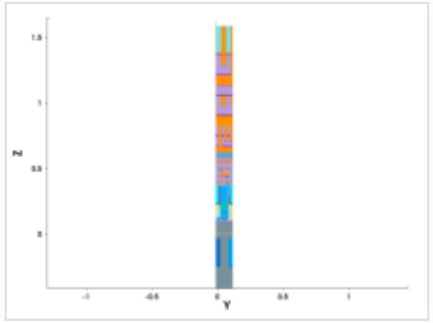
carbon so far, of the whole flow (kgCO2e)



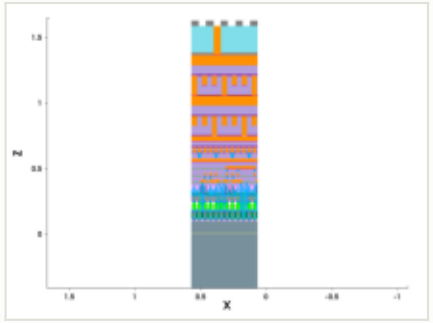
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

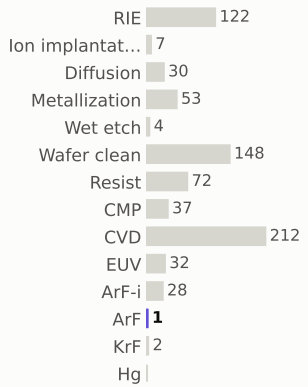


y cut at 0.049 µm

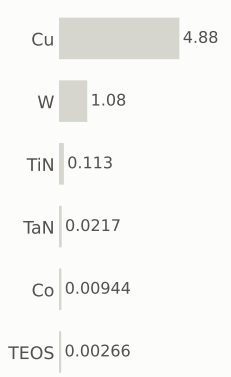


mask: layer 70:0 (GDSII)

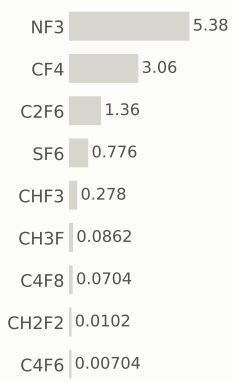
steps by process area



materials deposited (g)



gases emitted (g)



Step #7850: RIE

etch_anisotropic · RIE

1.04	664.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

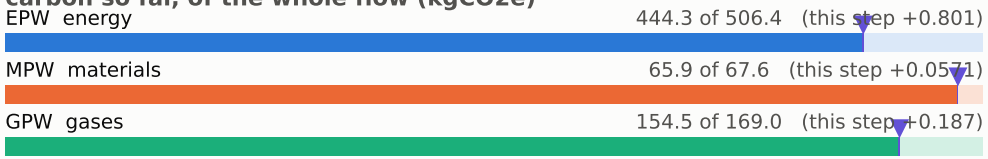
zoomed to fit

this step

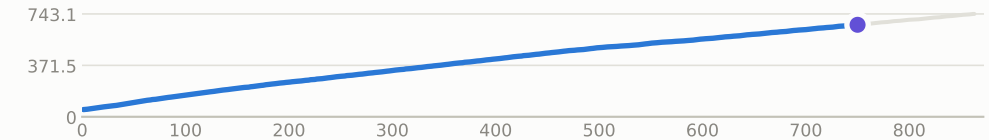
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

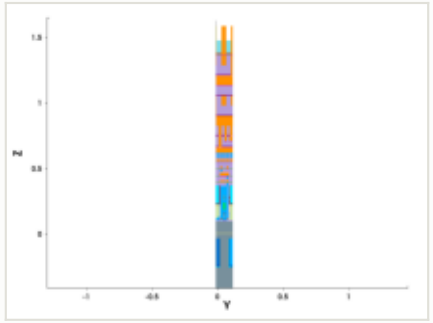
carbon so far, of the whole flow (kgCO2e)



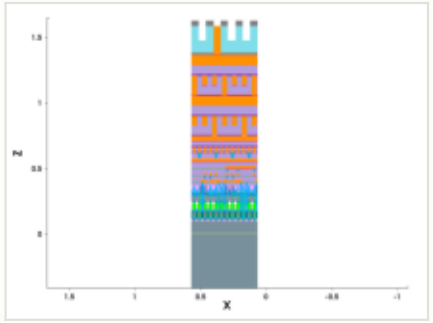
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



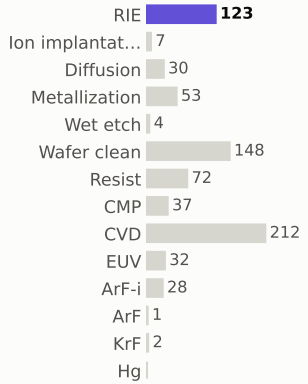
y cut at 0.049 µm



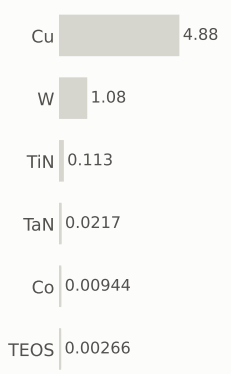
mask for this step

no mask for this step

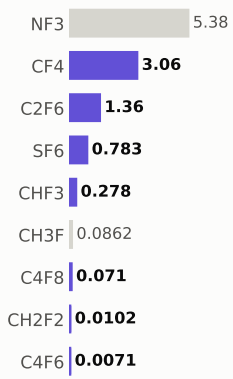
steps by process area



materials deposited (g)



gases emitted (g)



Step #7860: Wafer_clean

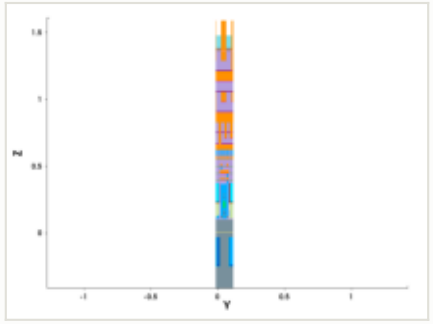
etch_strip · Wafer_clean

0.169	664.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

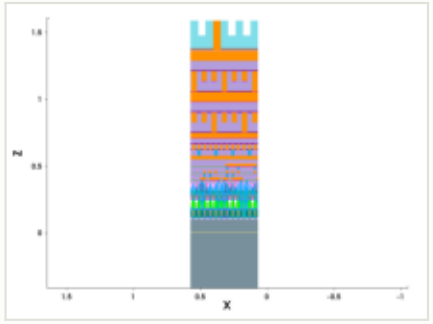
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



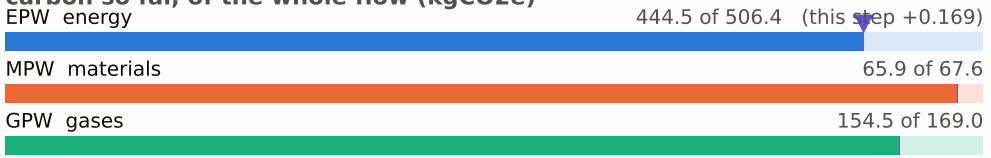
mask for this step

no mask for this step

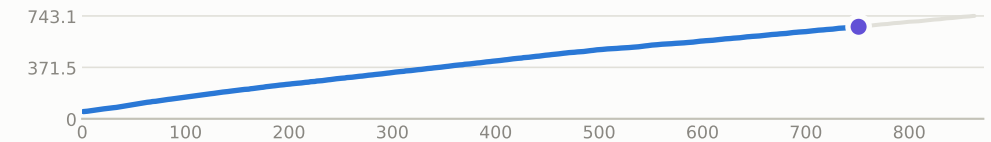
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

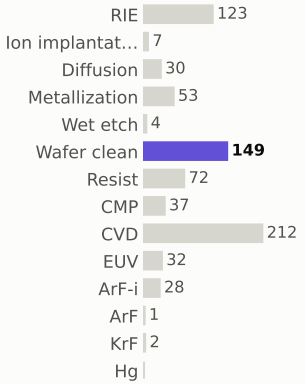
carbon so far, of the whole flow (kgCO2e)



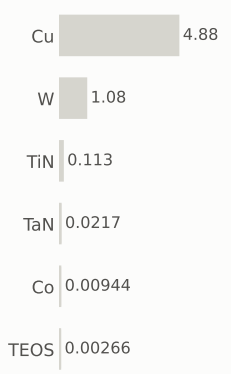
embodied carbon up to each step (kgCO2e per wafer)



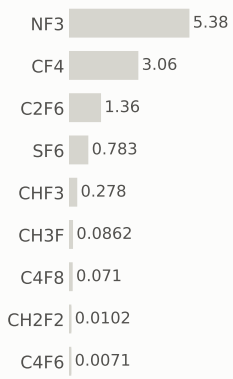
steps by process area



materials deposited (g)



gases emitted (g)



Step #7870: Metallization

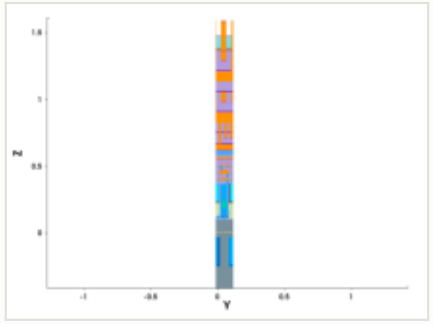
depo_ellipsoidal · Metallization

1.2	666.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

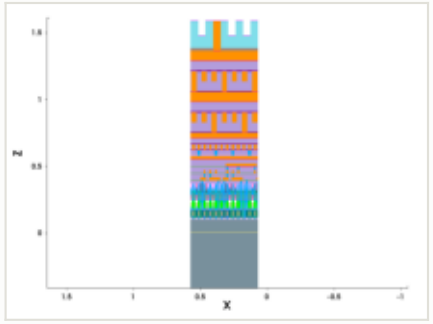
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



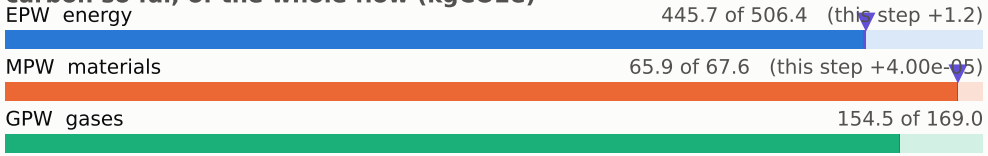
mask for this step

no mask for this step

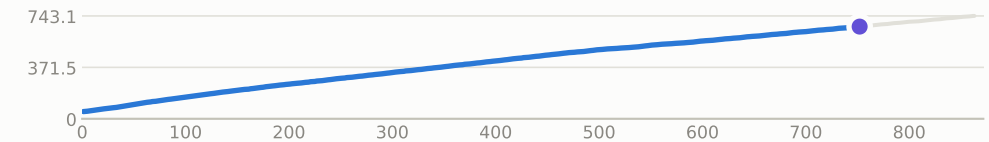
this step

process area	Metallization	EPW	1.2 kgCO2e
material	TaN	EPW tool	0.86 kgCO2e
thickness	0.0016 µm	EPW overhead	0.344 kgCO2e
material GWP	25.8159 gCO2e/g	MPW	4.00e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

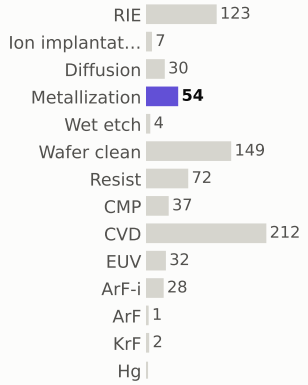
carbon so far, of the whole flow (kgCO2e)



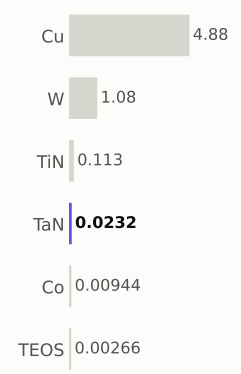
embodied carbon up to each step (kgCO2e per wafer)



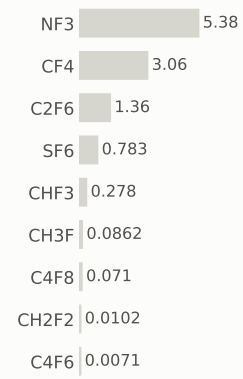
steps by process area



materials deposited (g)



gases emitted (g)



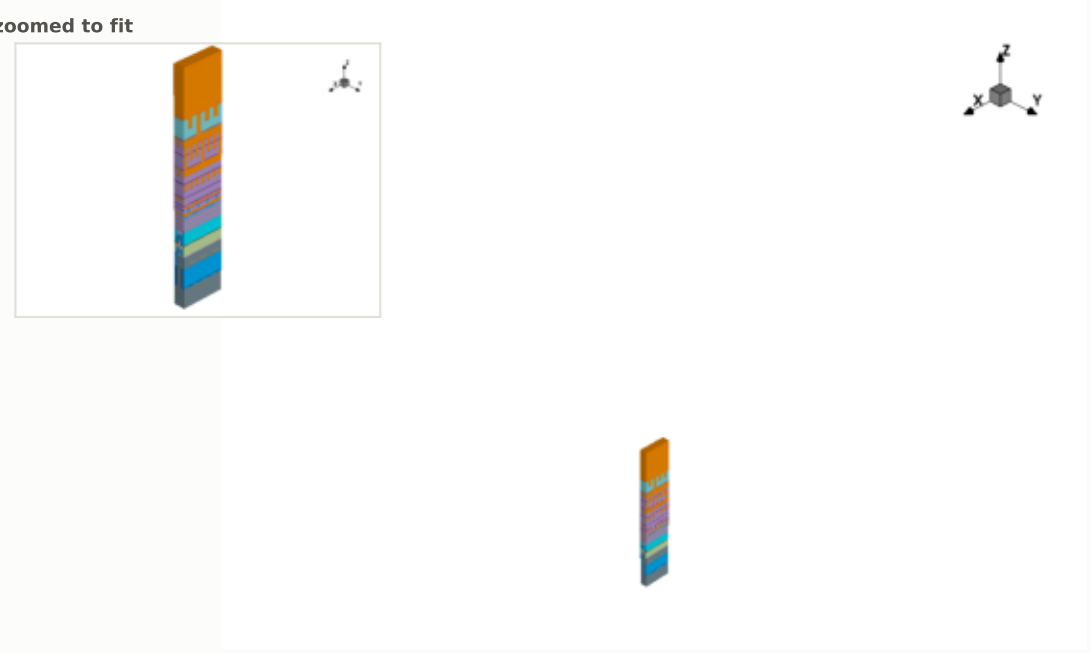
Step #7880: Metallization

depo_fill · Metallization

1.21	667.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

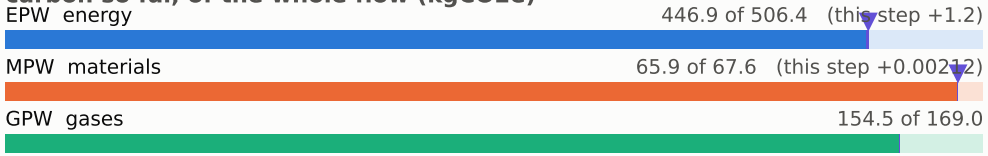


this step

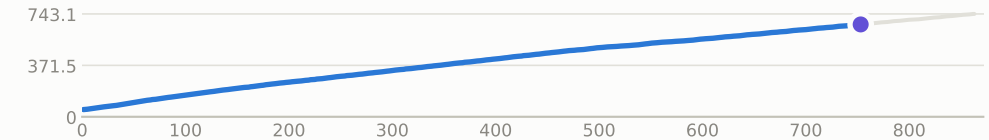
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

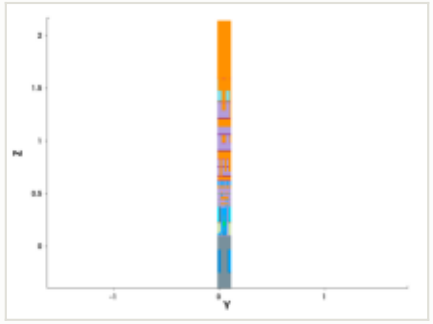
carbon so far, of the whole flow (kgCO2e)



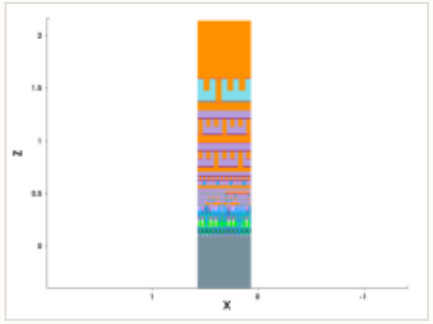
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



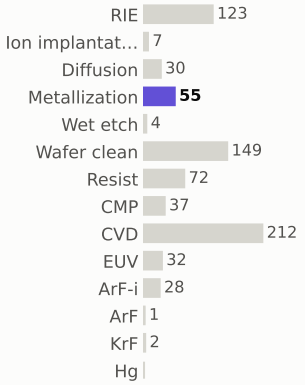
y cut at 0.049 µm



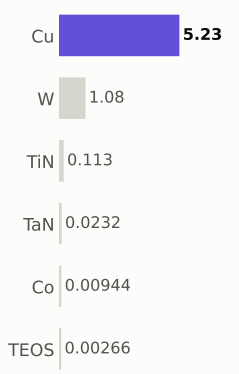
mask for this step

no mask for this step

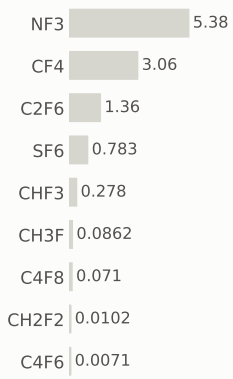
steps by process area



materials deposited (g)



gases emitted (g)



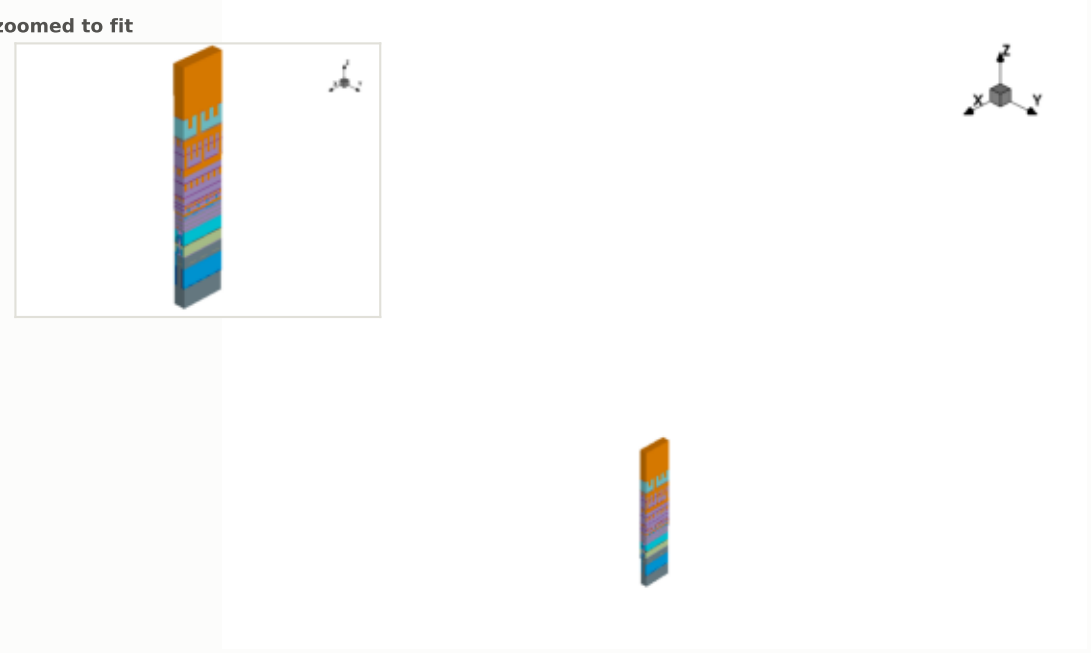
Step #7890: Diffusion

treatment · Diffusion

0.801	668.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

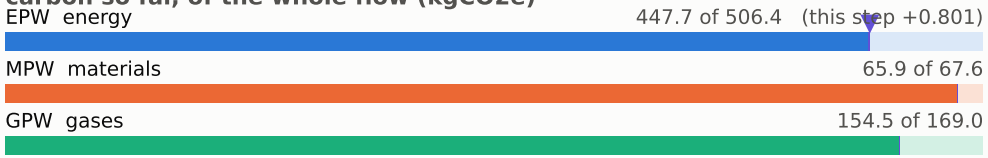


this step

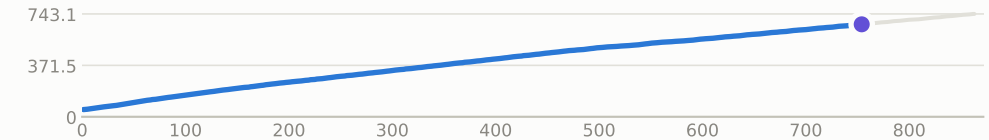
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

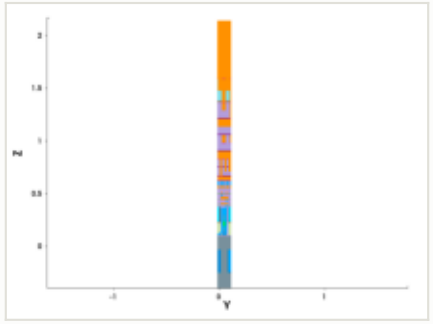
carbon so far, of the whole flow (kgCO2e)



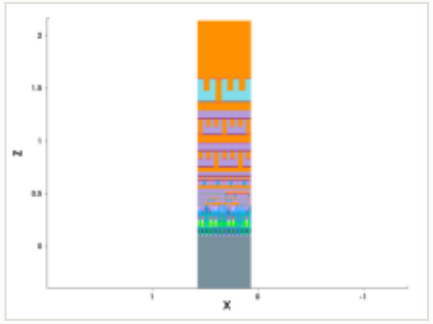
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



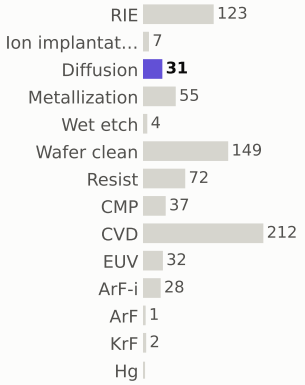
y cut at 0.049 µm



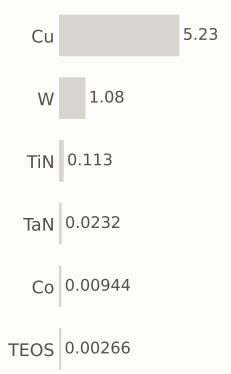
mask for this step

no mask for this step

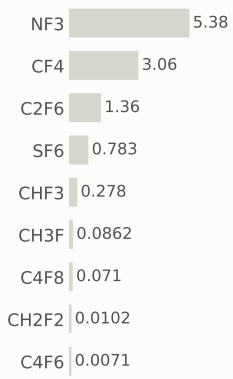
steps by process area



materials deposited (g)



gases emitted (g)



Step #7900: CMP

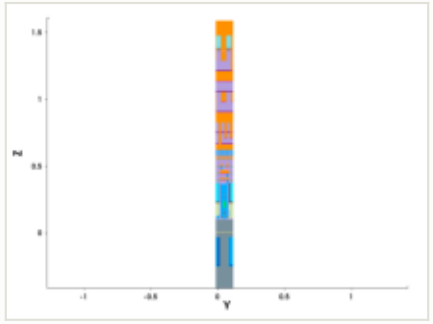
etch_cmp · CMP

0.169	668.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

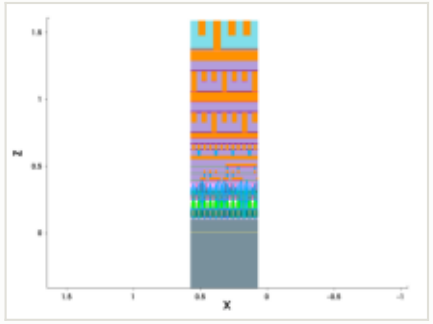
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

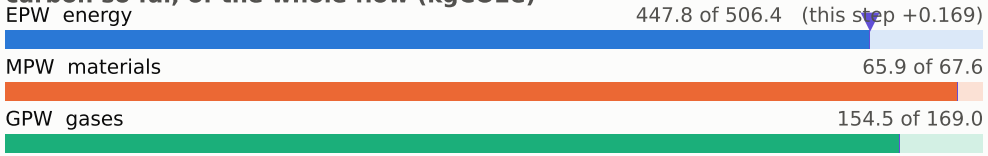
no mask for this step

this step

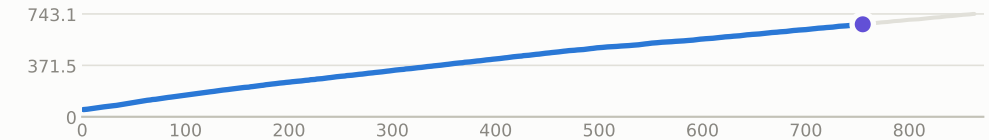
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

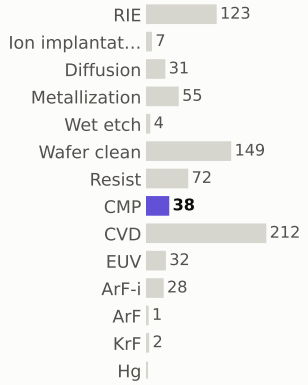
carbon so far, of the whole flow (kgCO2e)



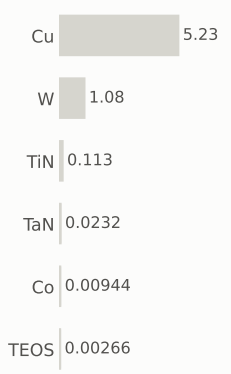
embodied carbon up to each step (kgCO2e per wafer)



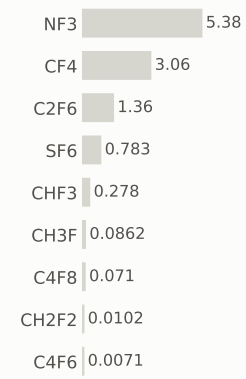
steps by process area



materials deposited (g)



gases emitted (g)



Step #7920: CVD

depo_fill · CVD

1.31	669.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

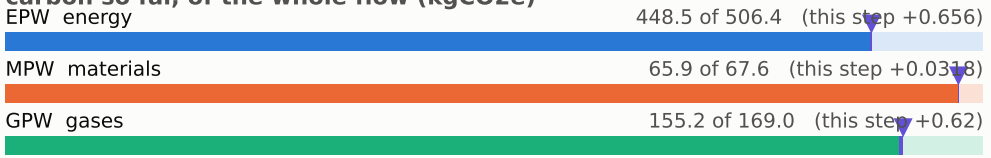
zoomed to fit

this step

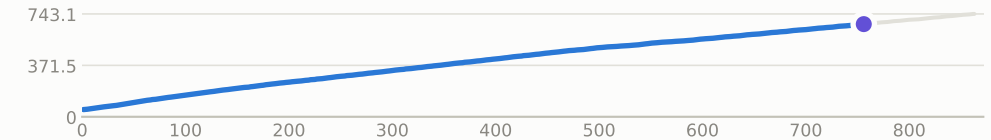
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

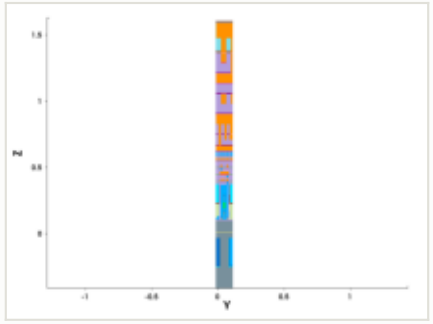
carbon so far, of the whole flow (kgCO2e)



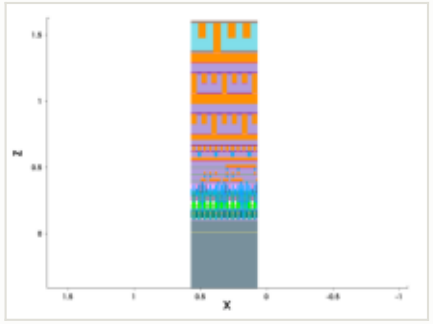
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



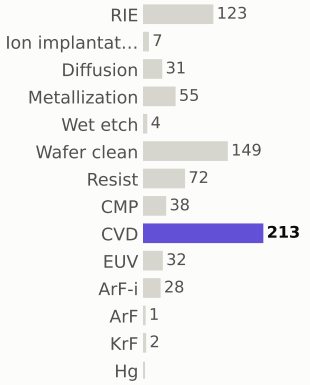
y cut at 0.049 µm



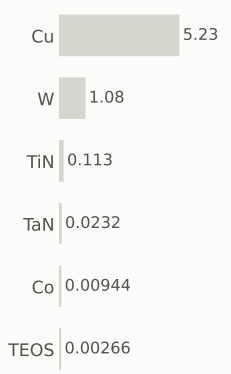
mask for this step

no mask for this step

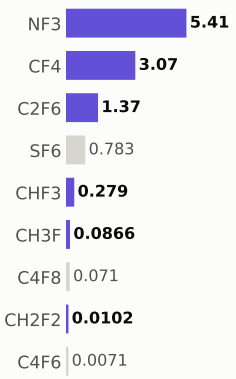
steps by process area



materials deposited (g)



gases emitted (g)



Step #7930: CVD

depo_fill · CVD

1.31	670.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

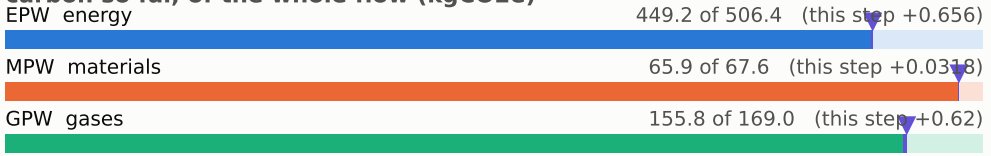
zoomed to fit

this step

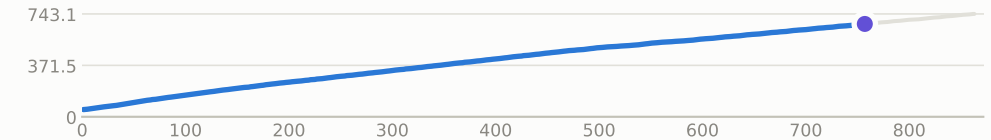
process area	CVD
material	—
thickness	0.204 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

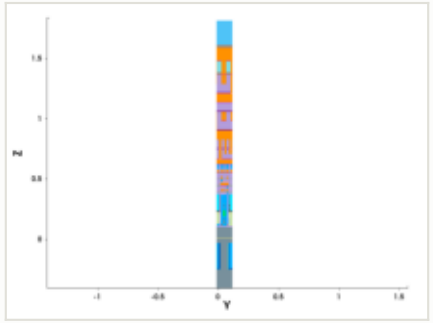
carbon so far, of the whole flow (kgCO2e)



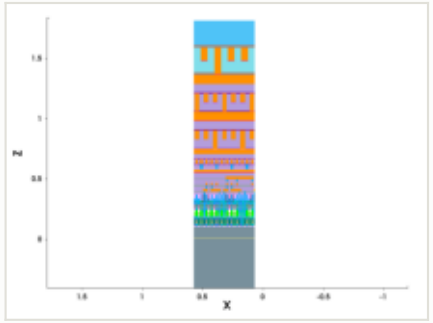
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



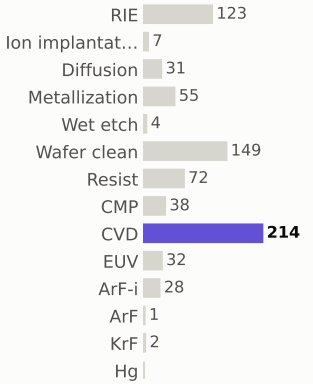
y cut at 0.049 µm



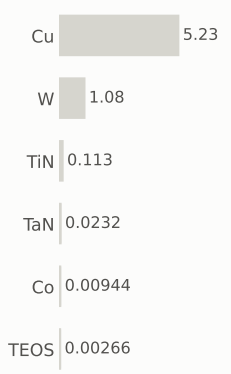
mask for this step

no mask for this step

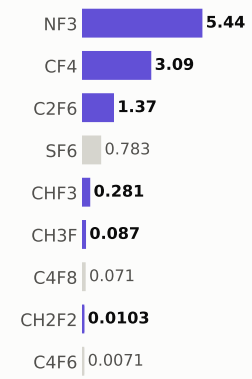
steps by process area



materials deposited (g)



gases emitted (g)



Step #7940: ArF

litho_patterning · ArF

0.199	671.1	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

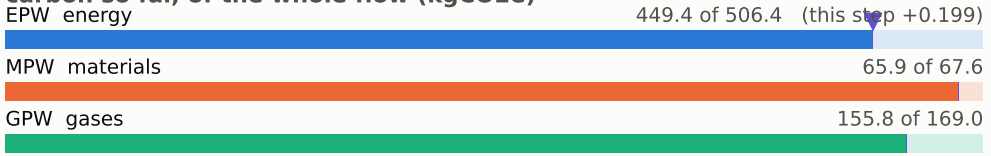
zoomed to fit

this step

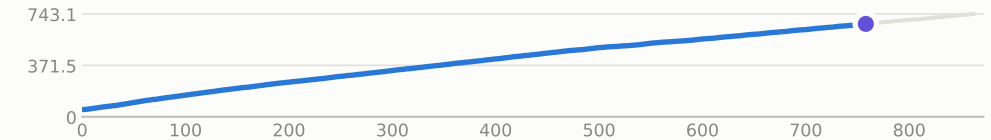
process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

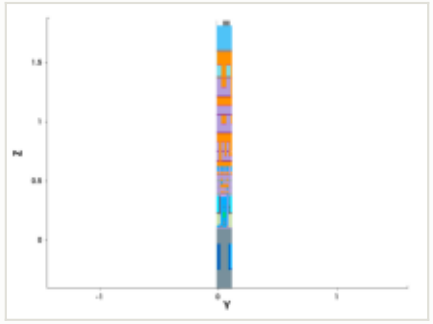
carbon so far, of the whole flow (kgCO2e)



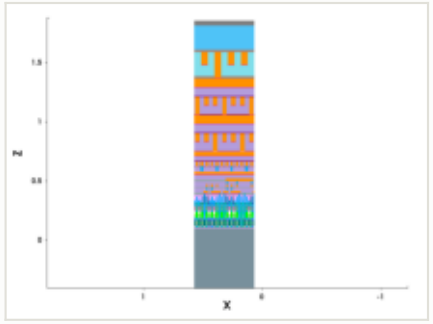
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

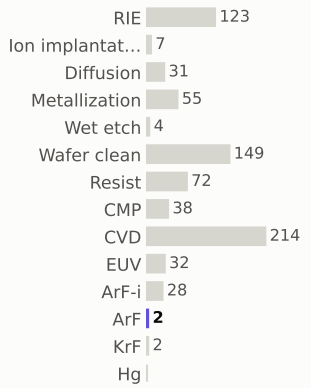


y cut at 0.049 µm

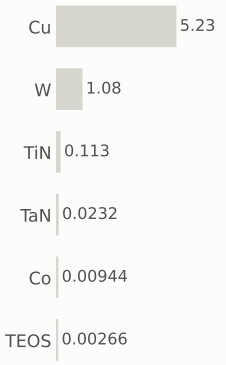


mask: layer 75:0 (GDSII)

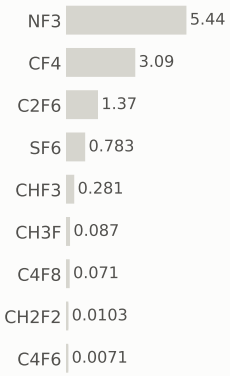
steps by process area



materials deposited (g)



gases emitted (g)



Step #7950: RIE

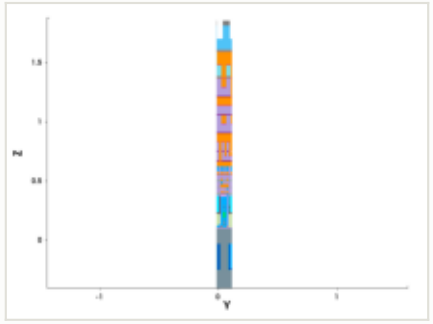
etch_anisotropic · RIE

1.04	672.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

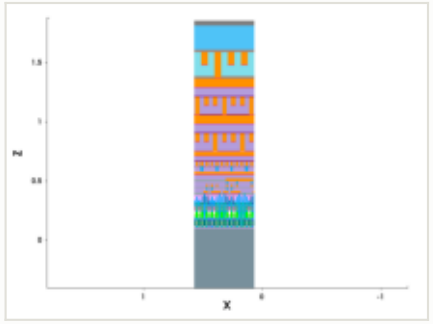
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

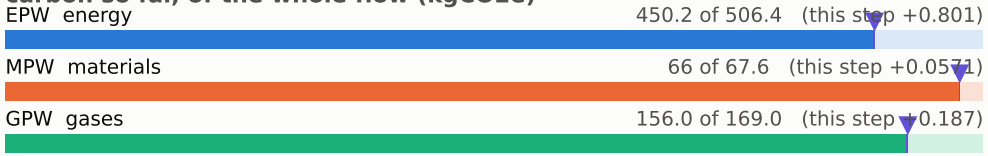
no mask for this step

this step

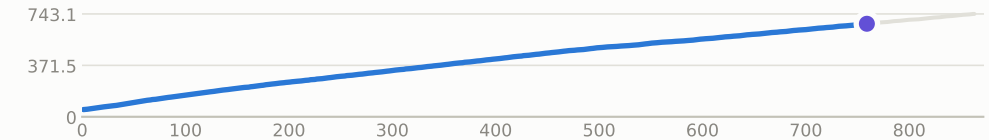
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

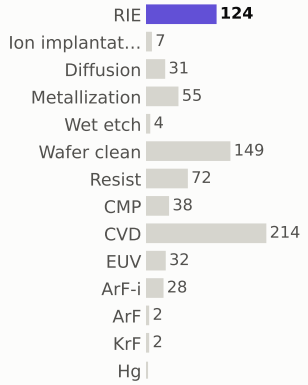
carbon so far, of the whole flow (kgCO2e)



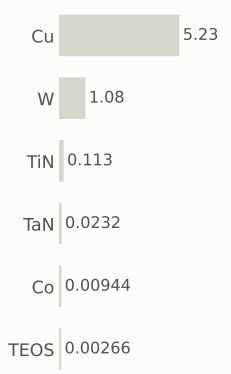
embodied carbon up to each step (kgCO2e per wafer)



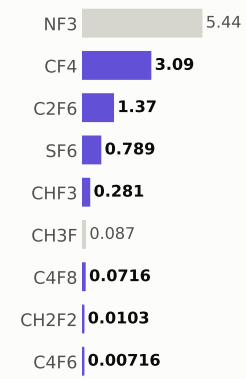
steps by process area



materials deposited (g)



gases emitted (g)



Step #7960: Wafer_clean

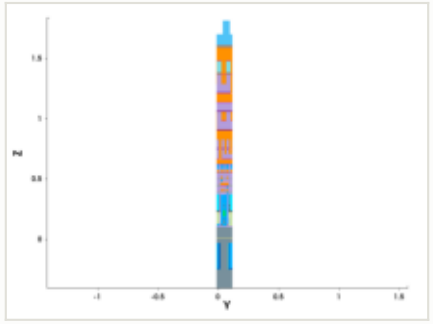
etch_strip · Wafer_clean

0.169	672.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

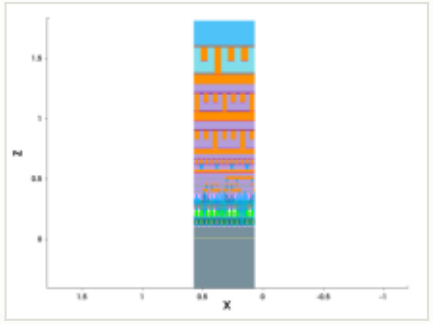
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



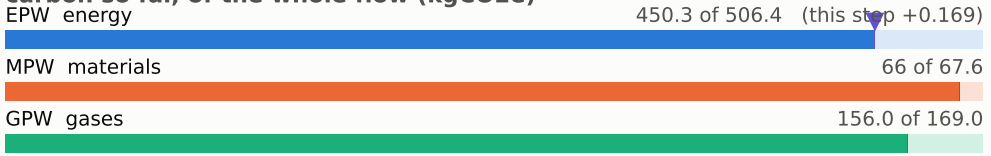
mask for this step

no mask for this step

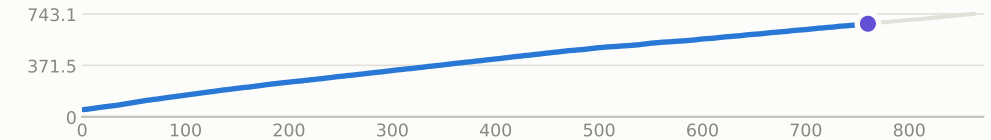
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

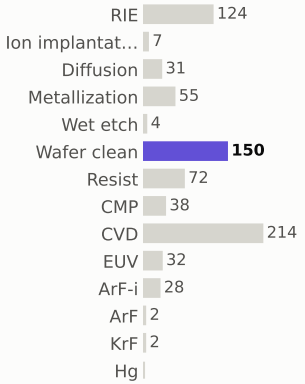
carbon so far, of the whole flow (kgCO2e)



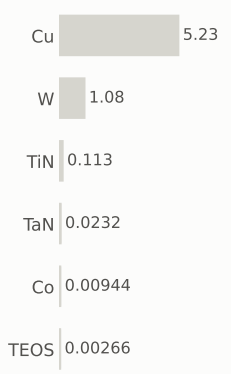
embodied carbon up to each step (kgCO2e per wafer)



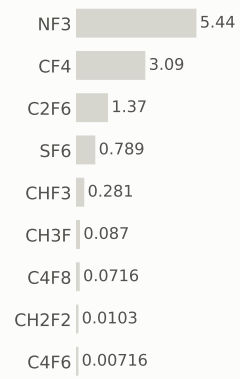
steps by process area



materials deposited (g)



gases emitted (g)



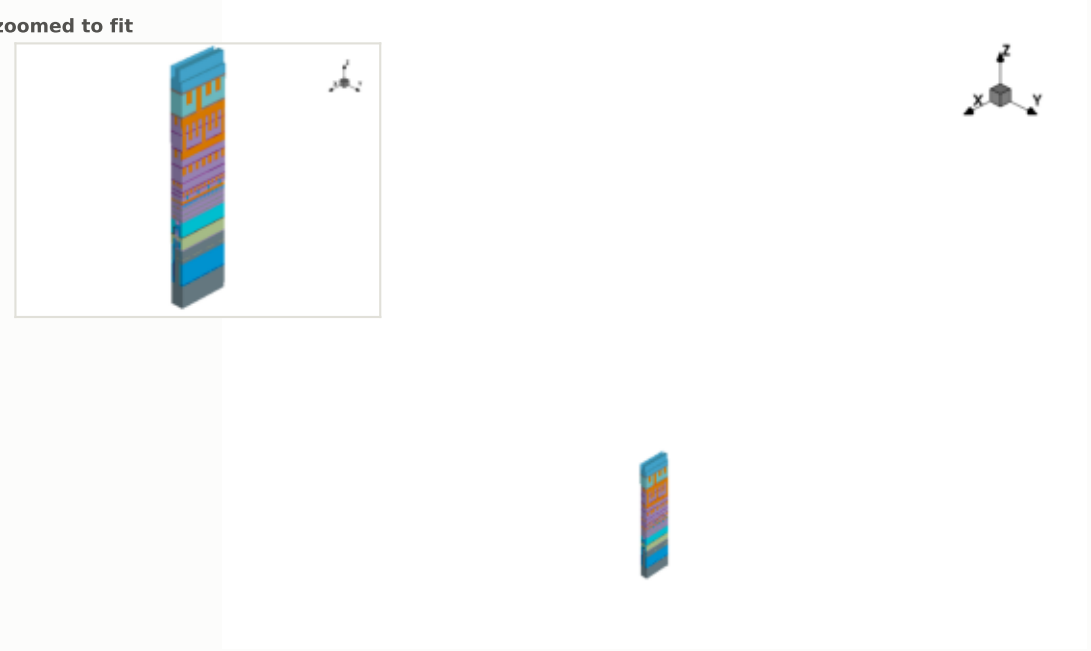
Step #7970: RIE

clean · RIE

1.04	673.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

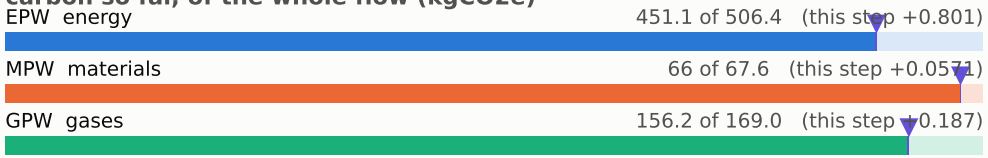


this step

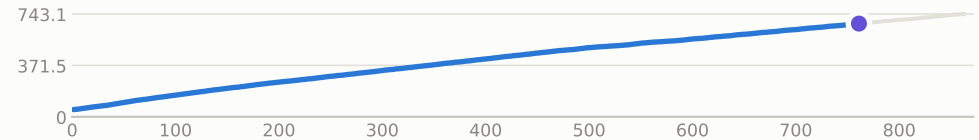
process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

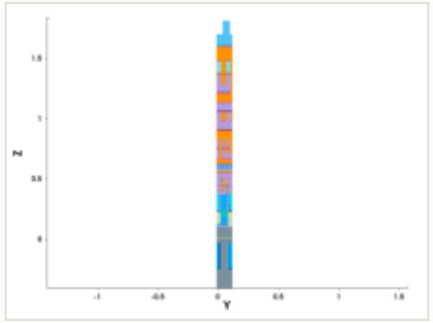
carbon so far, of the whole flow (kgCO2e)



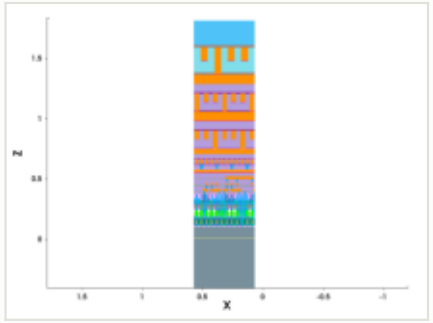
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



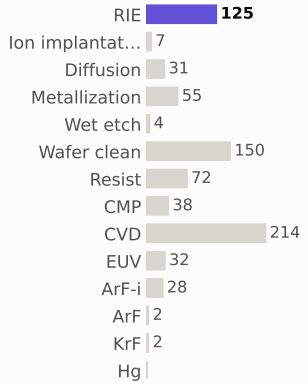
y cut at 0.049 µm



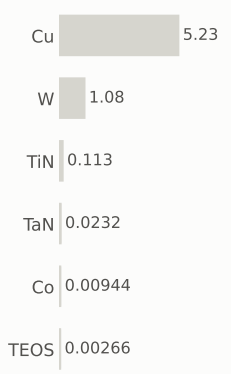
mask for this step

no mask for this step

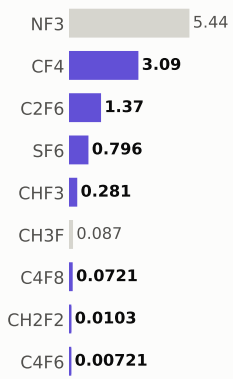
steps by process area



materials deposited (g)



gases emitted (g)



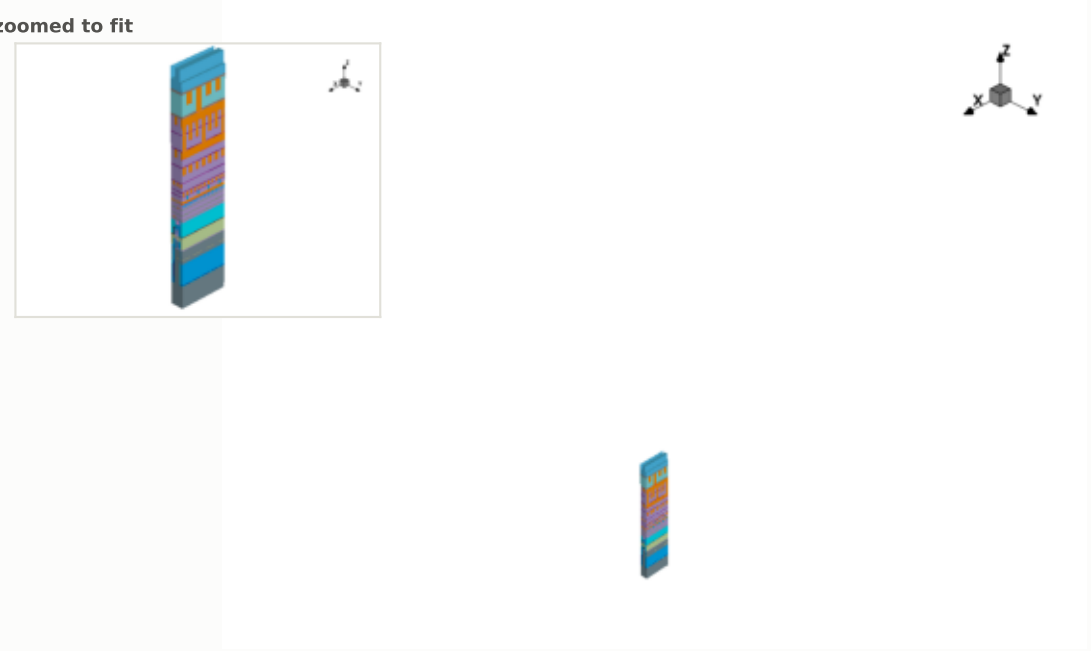
Step #7980: Wafer_clean

clean · Wafer_clean

0.169	673.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

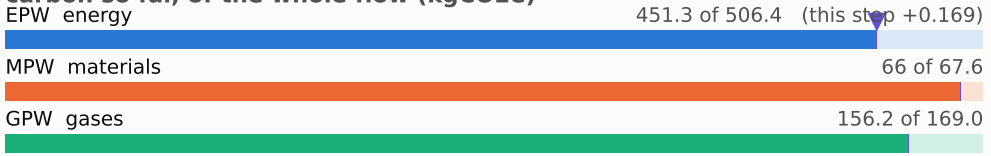


this step

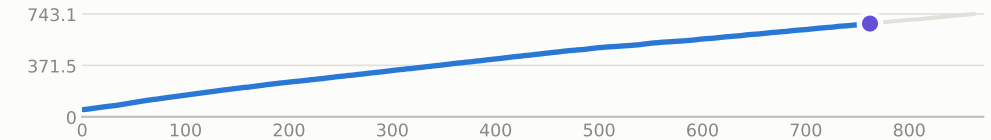
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

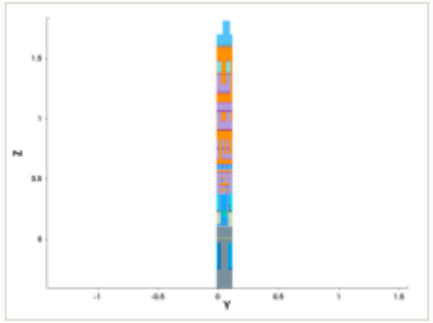
carbon so far, of the whole flow (kgCO2e)



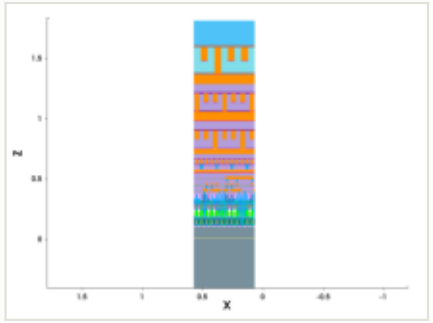
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



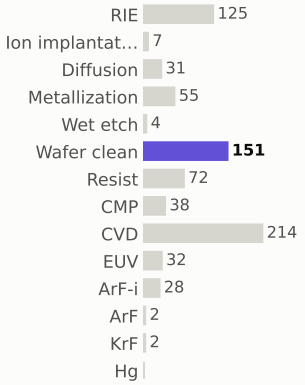
y cut at 0.049 µm



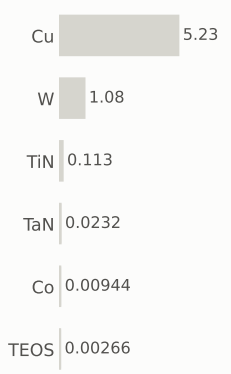
mask for this step

no mask for this step

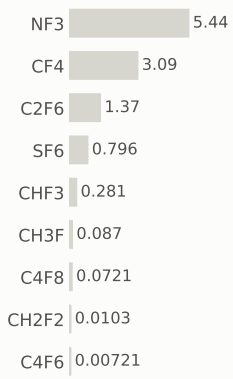
steps by process area



materials deposited (g)



gases emitted (g)



Step #7990: ArF

litho_patterning · ArF

0.199	673.7	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

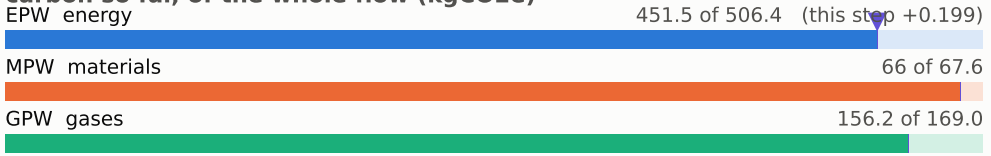
zoomed to fit

this step

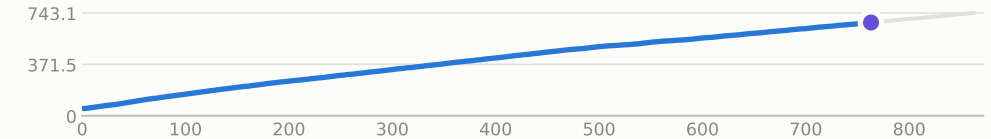
process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

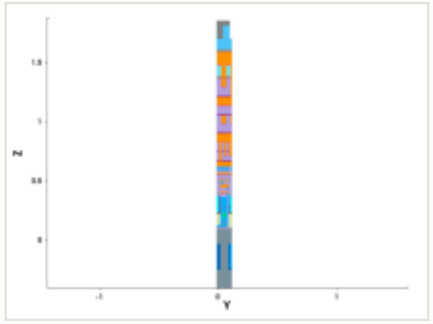
carbon so far, of the whole flow (kgCO2e)



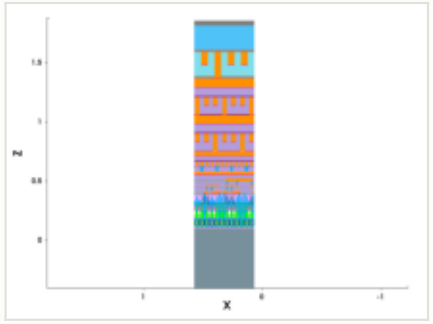
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

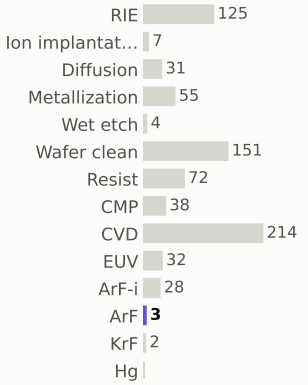


y cut at 0.049 µm

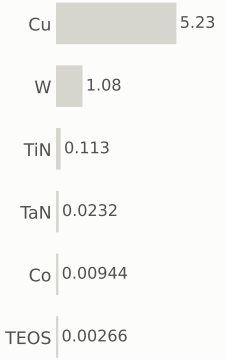


mask: layer 72:0 (GDSII)

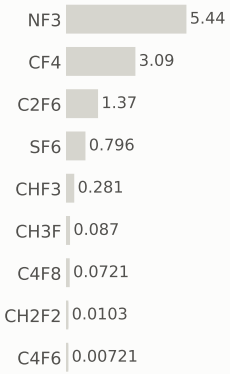
steps by process area



materials deposited (g)



gases emitted (g)



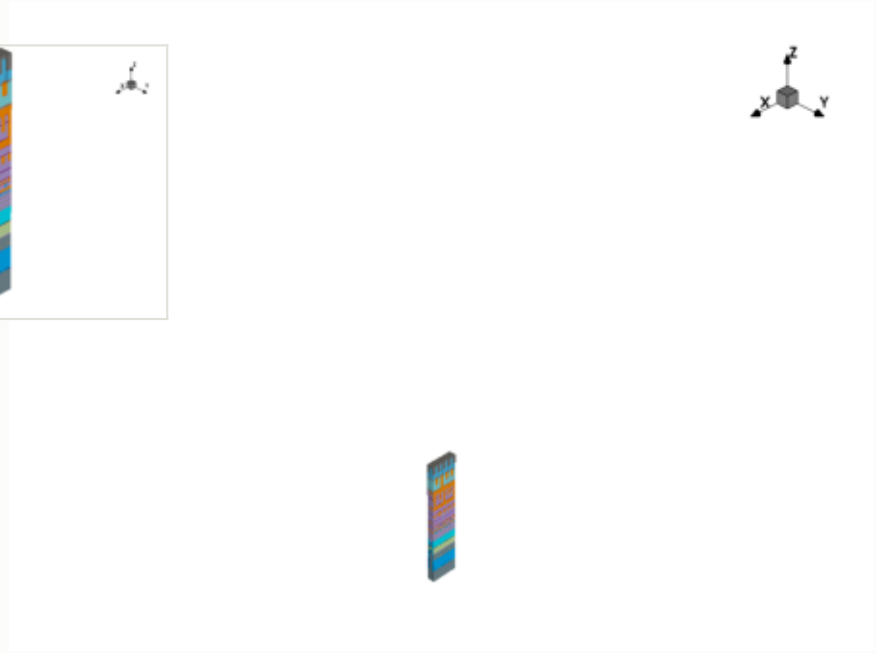
Step #8000: RIE

etch_anisotropic · RIE

1.04	674.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

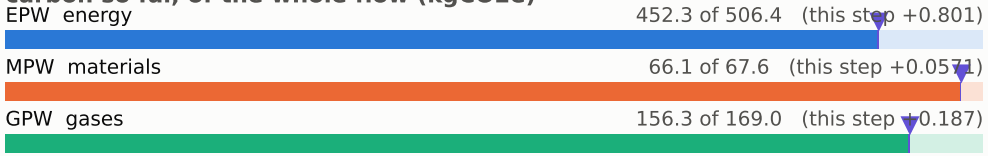


this step

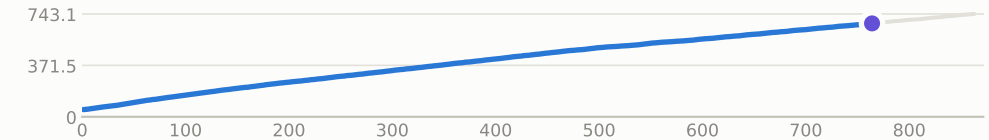
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

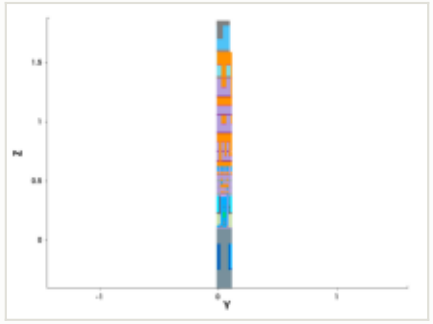
carbon so far, of the whole flow (kgCO2e)



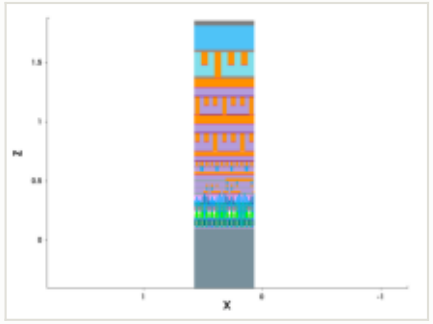
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



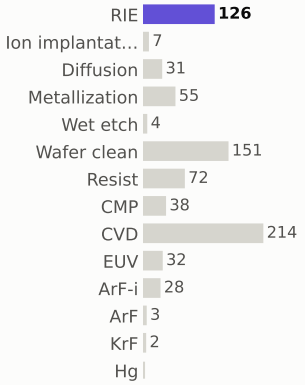
y cut at 0.049 µm



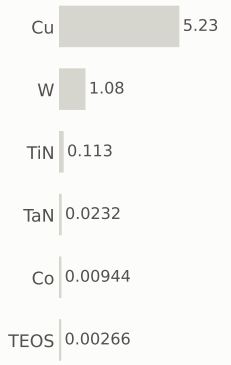
mask for this step

no mask for this step

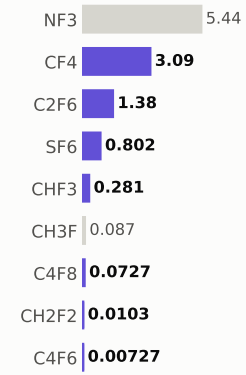
steps by process area



materials deposited (g)



gases emitted (g)



Step #8010: Wafer_clean

etch_strip · Wafer_clean

0.169	674.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

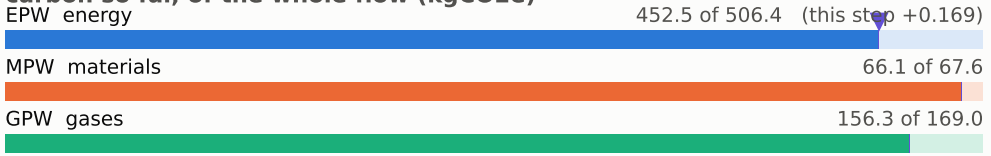
3D structure after this step

zoomed to fit

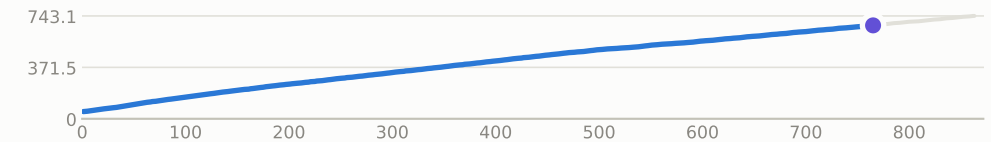
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

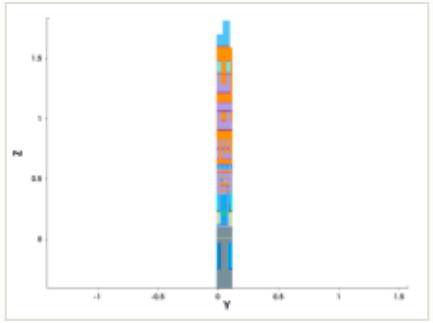
carbon so far, of the whole flow (kgCO2e)



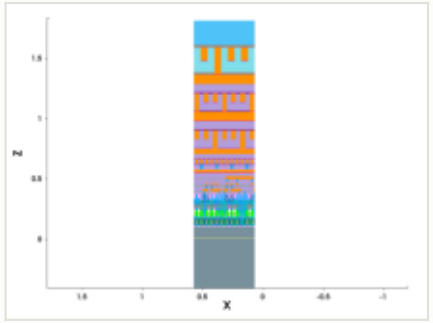
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



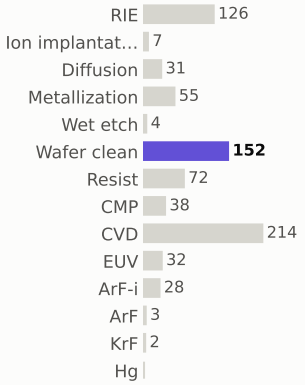
y cut at 0.049 µm



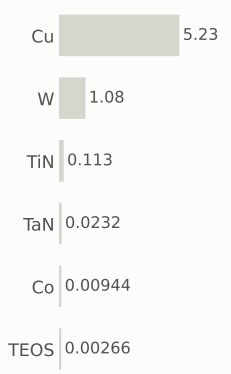
mask for this step

no mask for this step

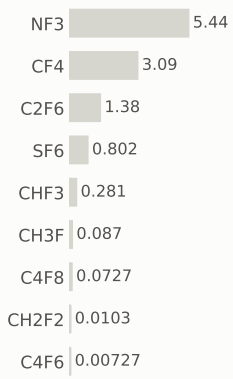
steps by process area



materials deposited (g)



gases emitted (g)



Step #8020: Metallization

depo_ellipsoidal · Metallization

1.2	676.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

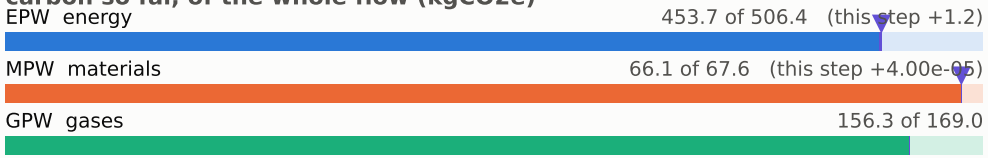
zoomed to fit

this step

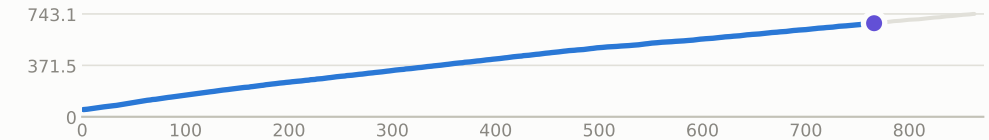
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

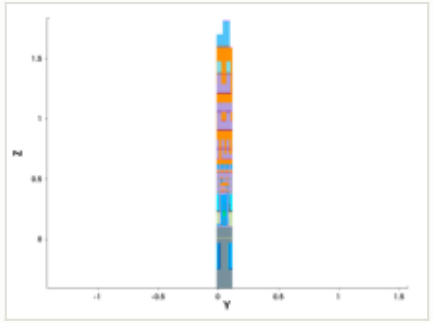
carbon so far, of the whole flow (kgCO2e)



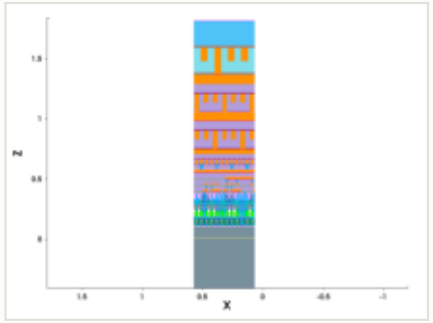
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



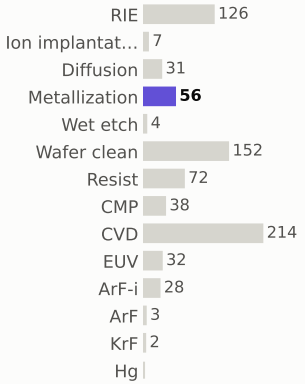
y cut at 0.049 µm



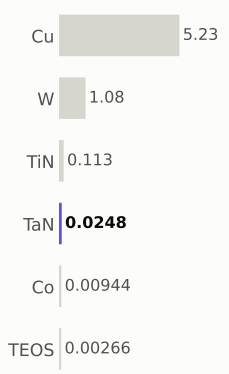
mask for this step

no mask for this step

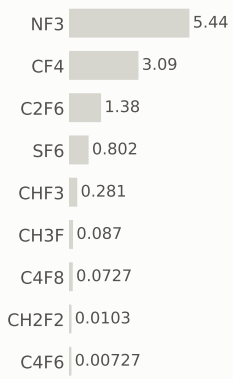
steps by process area



materials deposited (g)



gases emitted (g)



Step #8030: Metallization

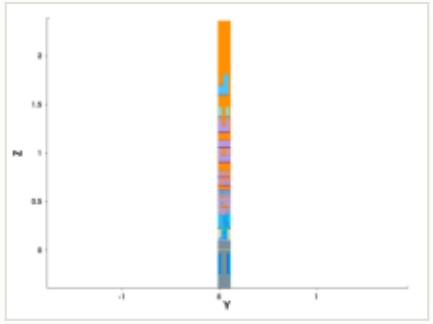
depo_fill · Metallization

1.21	677.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

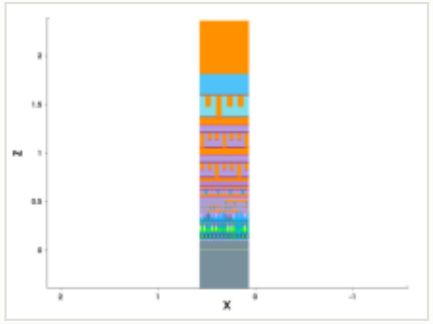
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

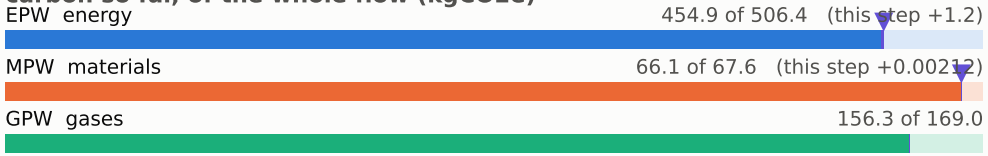
no mask for this step

this step

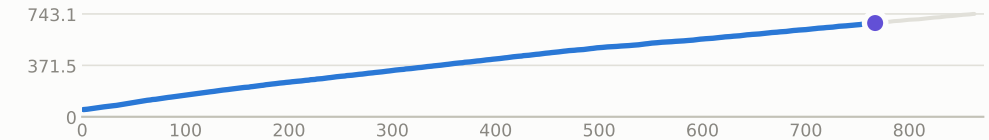
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

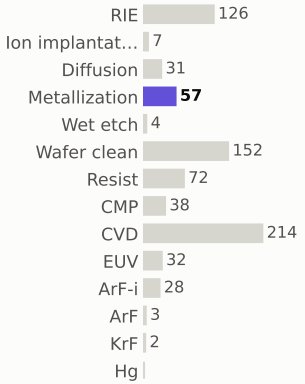
carbon so far, of the whole flow (kgCO2e)



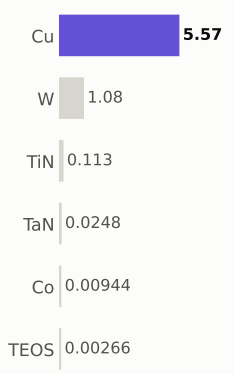
embodied carbon up to each step (kgCO2e per wafer)



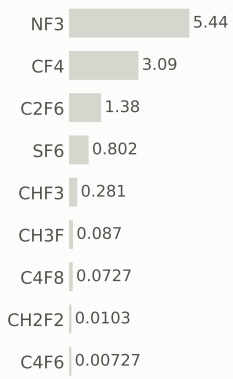
steps by process area



materials deposited (g)



gases emitted (g)



Step #8040: Diffusion

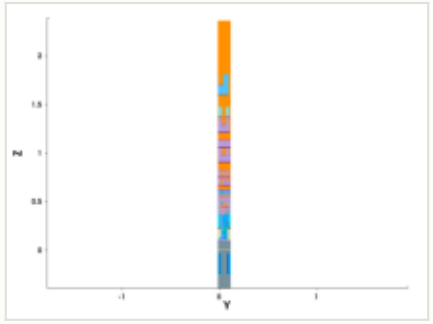
treatment · Diffusion

0.801	678.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

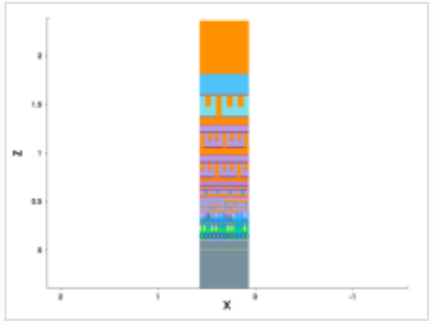
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

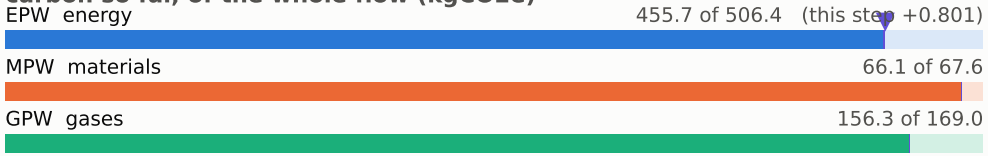
no mask for this step

this step

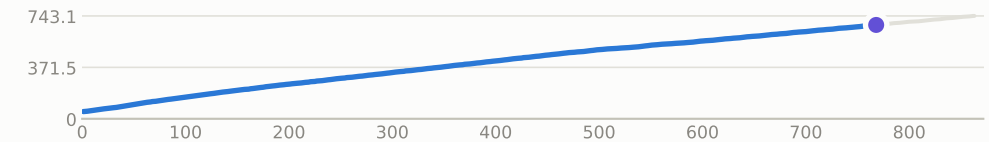
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

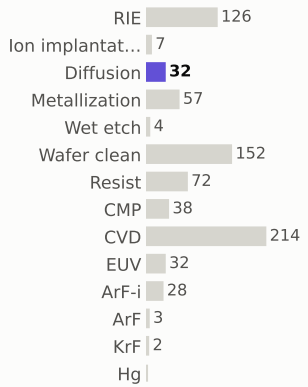
carbon so far, of the whole flow (kgCO2e)



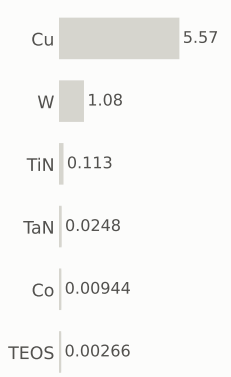
embodied carbon up to each step (kgCO2e per wafer)



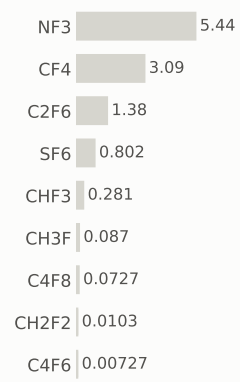
steps by process area



materials deposited (g)



gases emitted (g)



Step #8050: CMP

etch_cmp · CMP

0.169	678.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

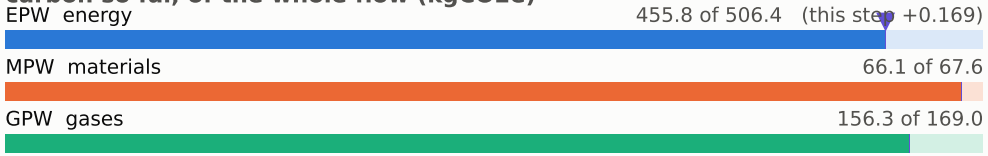
zoomed to fit

this step

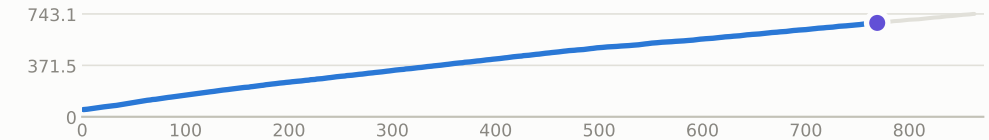
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

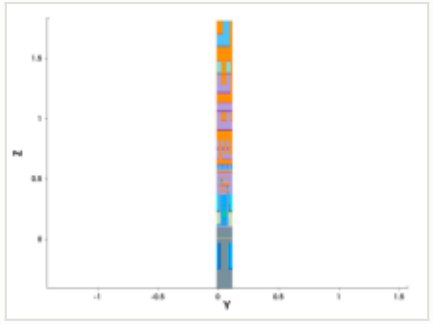
carbon so far, of the whole flow (kgCO2e)



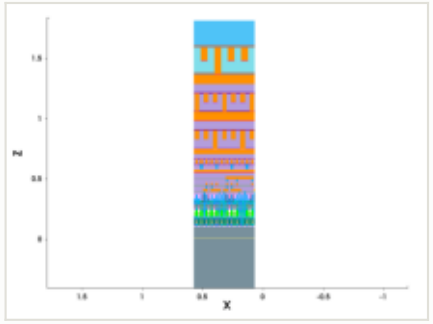
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



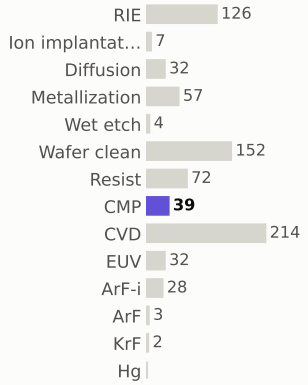
y cut at 0.049 µm



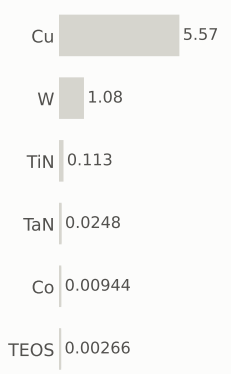
mask for this step

no mask for this step

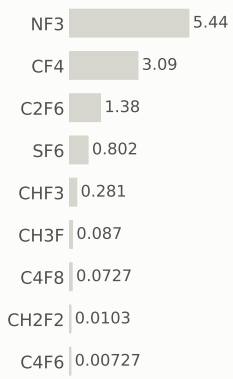
steps by process area



materials deposited (g)



gases emitted (g)



Step #8070: CVD

depo_fill · CVD

1.31	679.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

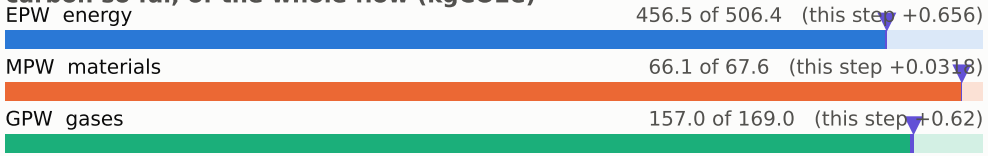
zoomed to fit

this step

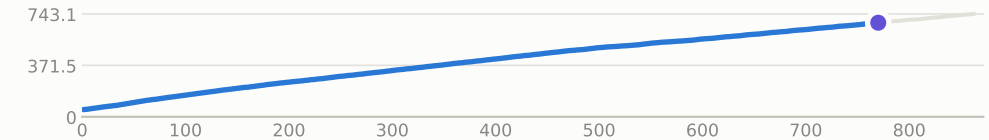
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

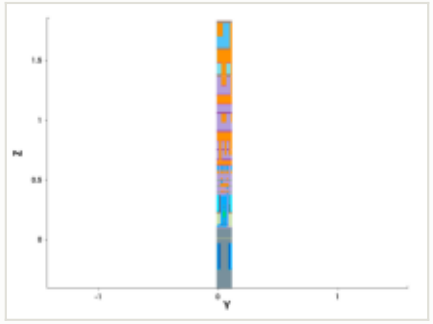
carbon so far, of the whole flow (kgCO2e)



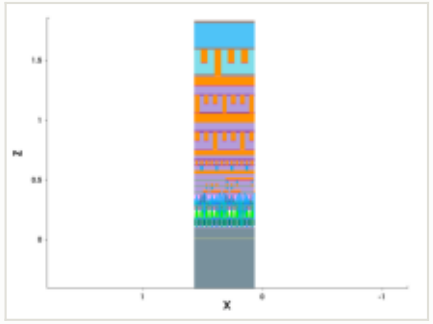
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



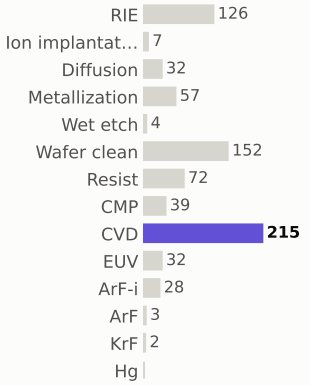
y cut at 0.049 µm



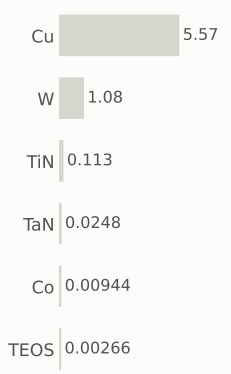
mask for this step

no mask for this step

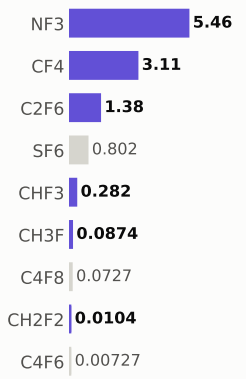
steps by process area



materials deposited (g)



gases emitted (g)



Step #8080: CVD

depo_fill · CVD

1.31	680.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

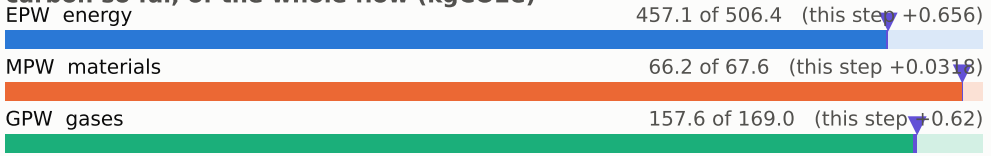
zoomed to fit

this step

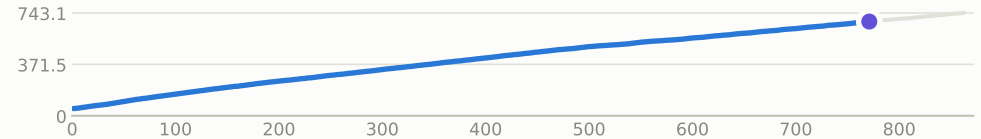
process area	CVD
material	—
thickness	1.42 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

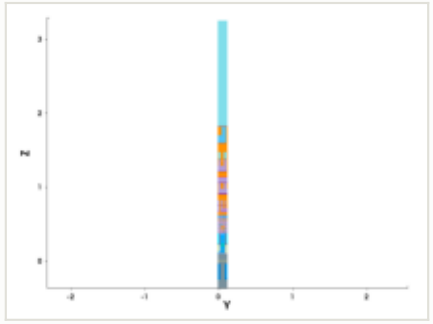
carbon so far, of the whole flow (kgCO2e)



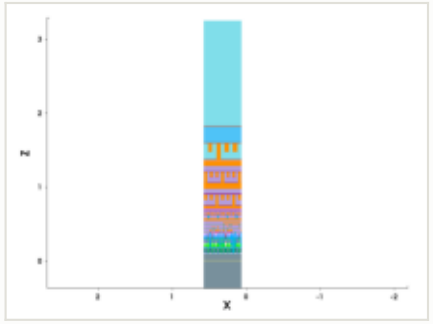
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



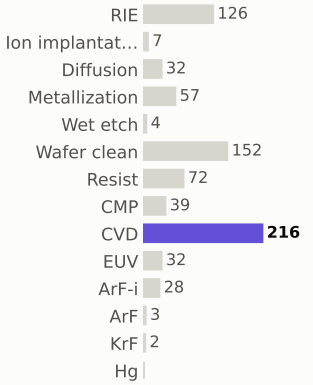
y cut at 0.049 µm



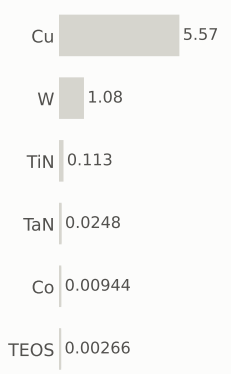
mask for this step

no mask for this step

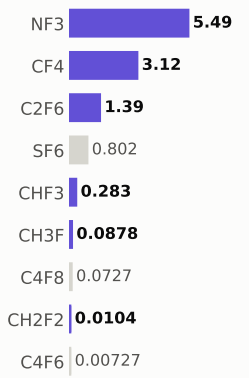
steps by process area



materials deposited (g)



gases emitted (g)



Step #8090: Hg

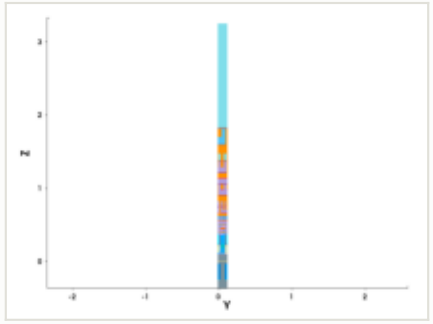
litho_patterning · Hg

0.199	681.1	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

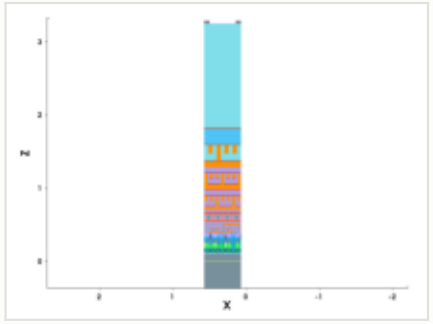
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



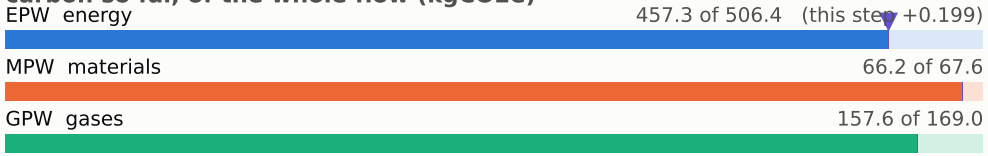
mask: layer 80:0 (GDSII)

this step

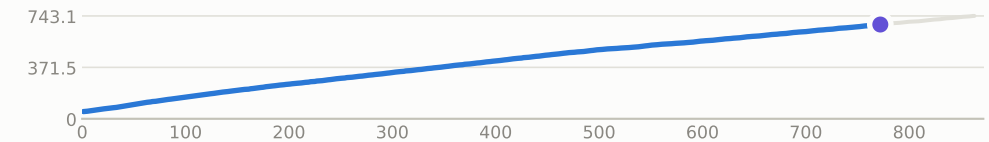
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

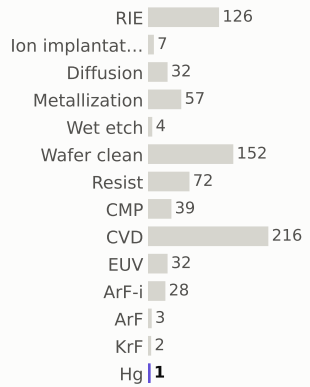
carbon so far, of the whole flow (kgCO2e)



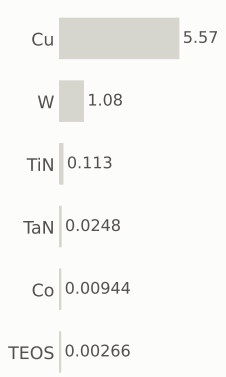
embodied carbon up to each step (kgCO2e per wafer)



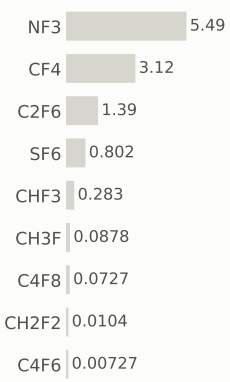
steps by process area



materials deposited (g)



gases emitted (g)



Step #8100: RIE

etch_anisotropic · RIE

1.04	682.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

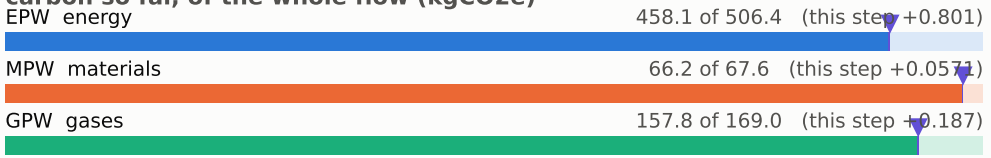
zoomed to fit

this step

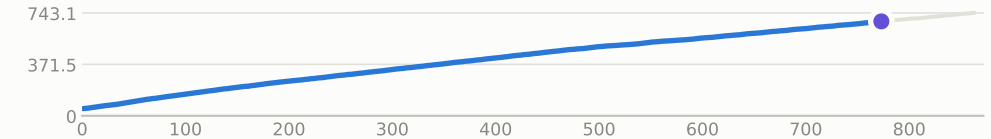
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

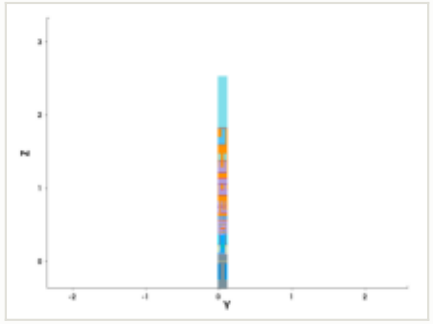
carbon so far, of the whole flow (kgCO2e)



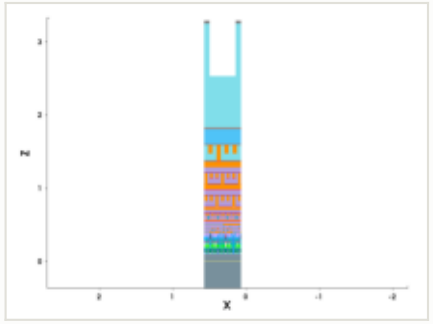
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



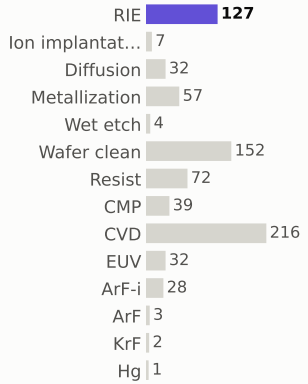
y cut at 0.049 µm



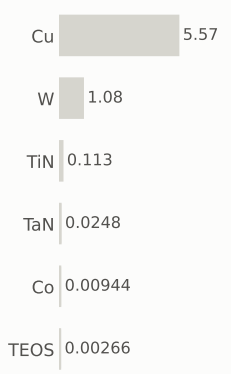
mask for this step

no mask for this step

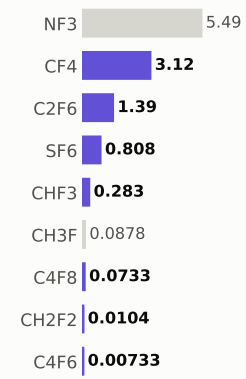
steps by process area



materials deposited (g)



gases emitted (g)



Step #8110: Wafer_clean

etch_strip · Wafer_clean

0.169	682.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

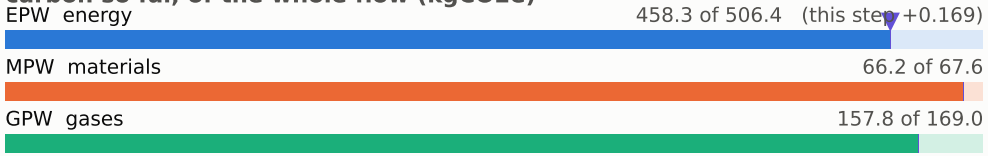
3D structure after this step

zoomed to fit

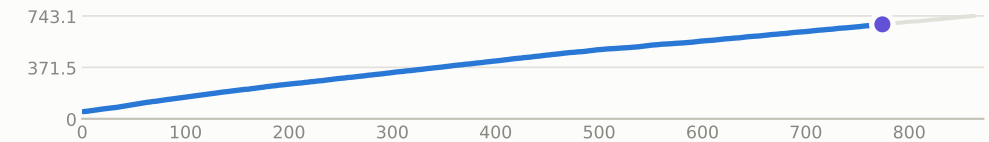
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

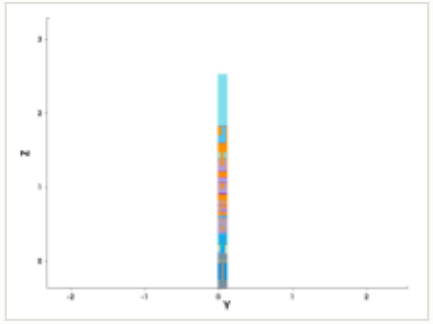
carbon so far, of the whole flow (kgCO2e)



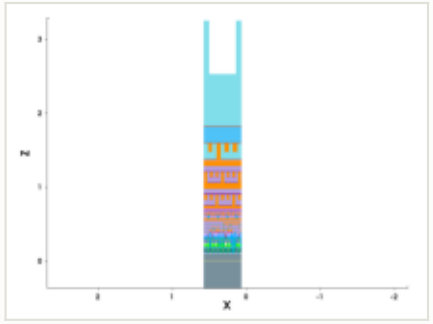
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



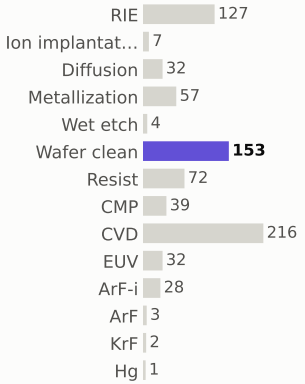
y cut at 0.049 µm



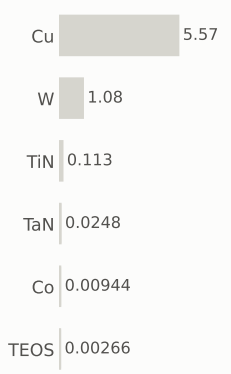
mask for this step

no mask for this step

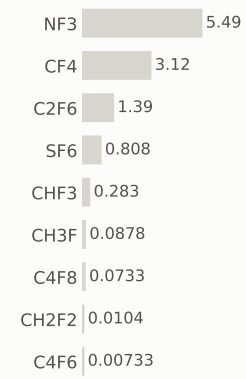
steps by process area



materials deposited (g)



gases emitted (g)



Step #8120: RIE

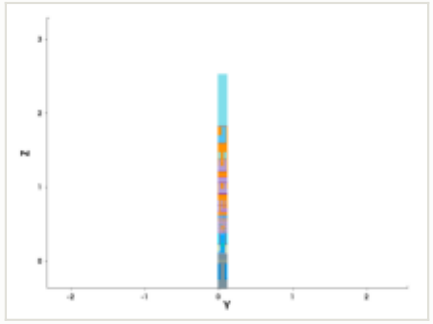
clean · RIE

1.04	683.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

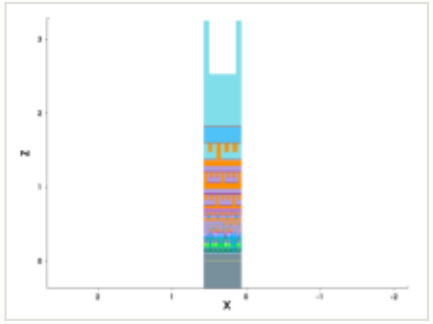
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

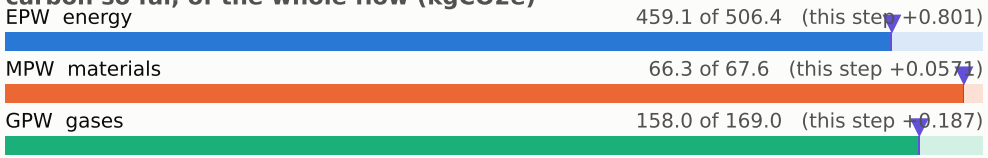
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

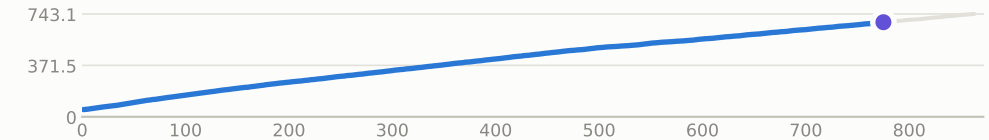
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

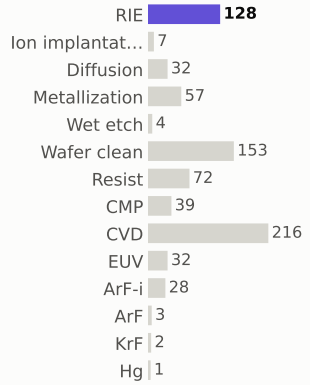
carbon so far, of the whole flow (kgCO2e)



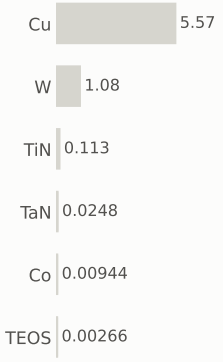
embodied carbon up to each step (kgCO2e per wafer)



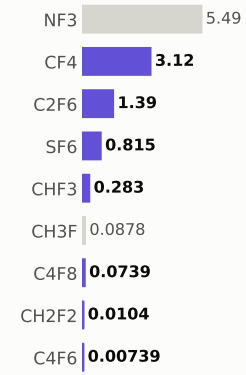
steps by process area



materials deposited (g)



gases emitted (g)



Step #8130: Wafer_clean

clean · Wafer_clean

0.169	683.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

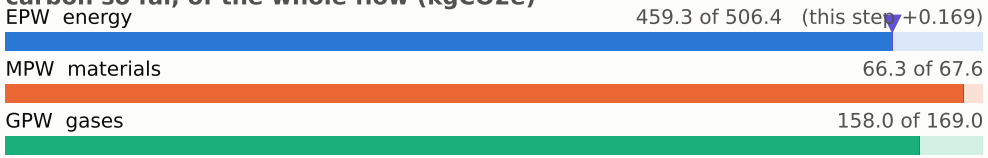
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

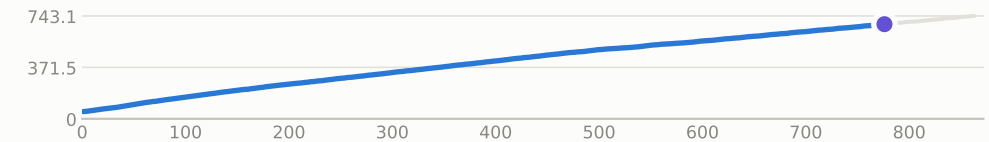
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

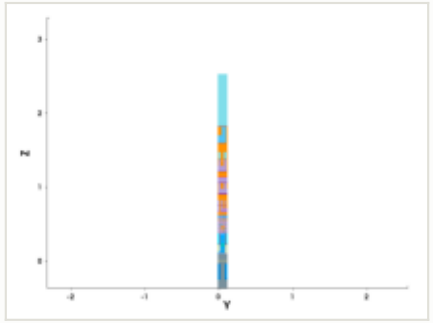
carbon so far, of the whole flow (kgCO2e)



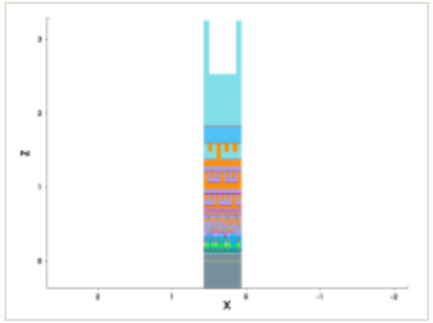
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



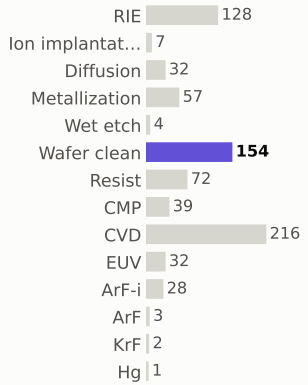
y cut at 0.049 µm



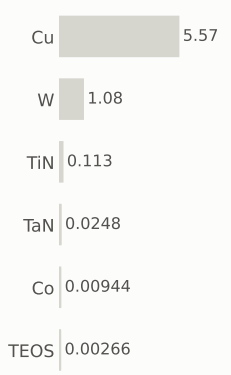
mask for this step

no mask for this step

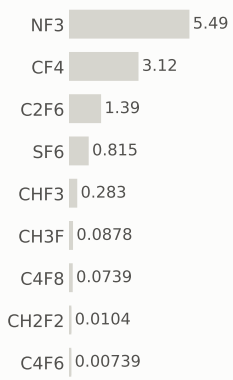
steps by process area



materials deposited (g)



gases emitted (g)



Step #8140: ArF

litho_patterning · ArF

0.199	683.7	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

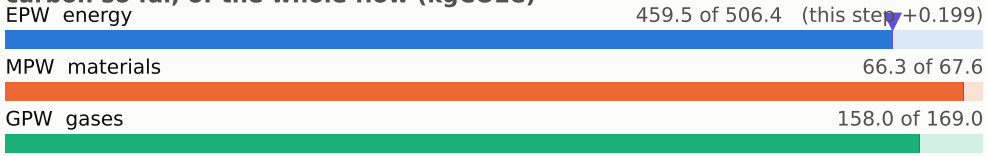
3D structure after this step

zoomed to fit

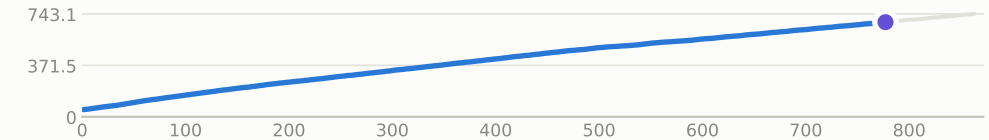
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

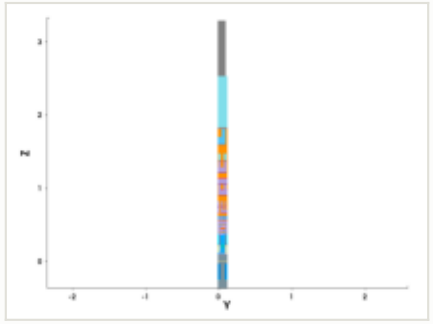
carbon so far, of the whole flow (kgCO2e)



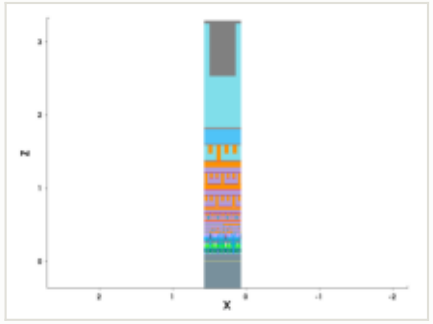
embodied carbon up to each step (kgCO2e per wafer)



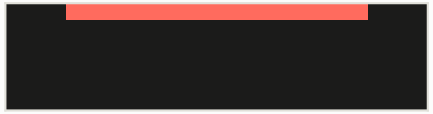
x cut at 0.385 µm



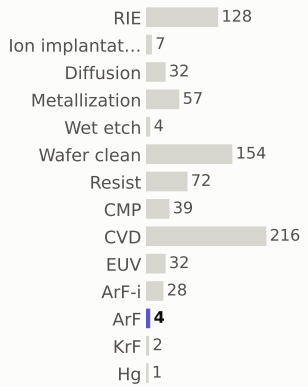
y cut at 0.049 µm



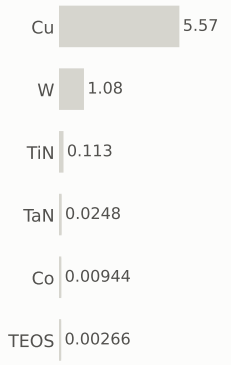
mask: layer 77:0 (GDSII)



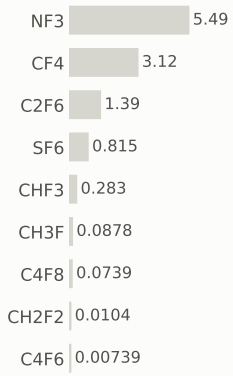
steps by process area



materials deposited (g)



gases emitted (g)



Step #8150: RIE

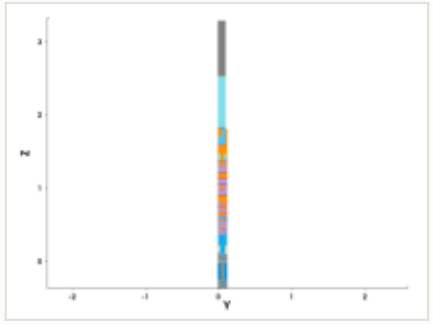
etch_anisotropic · RIE

1.04	684.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

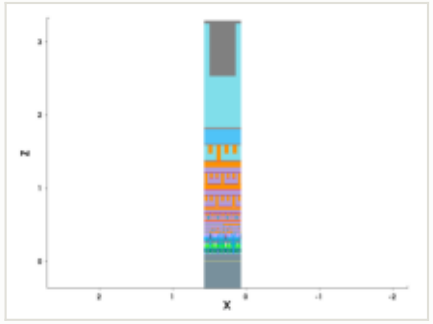
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

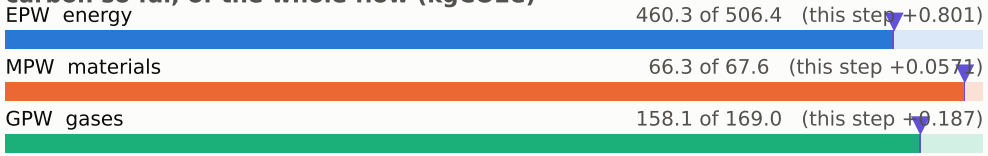
no mask for this step

this step

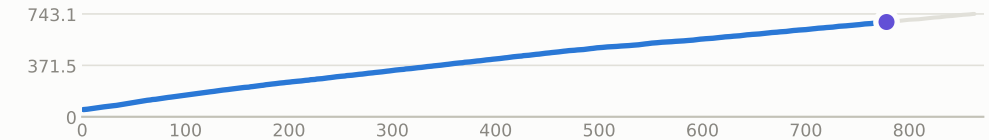
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

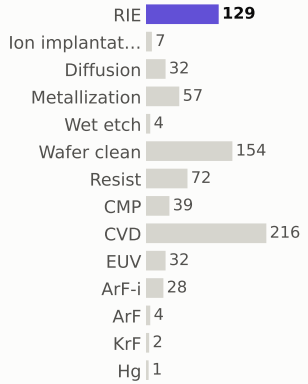
carbon so far, of the whole flow (kgCO2e)



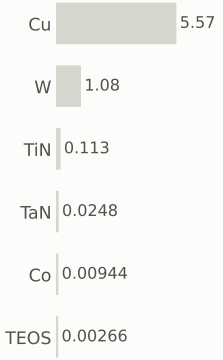
embodied carbon up to each step (kgCO2e per wafer)



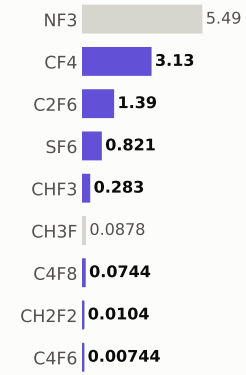
steps by process area



materials deposited (g)



gases emitted (g)



Step #8160: Wafer_clean

etch_strip · Wafer_clean

0.169	684.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

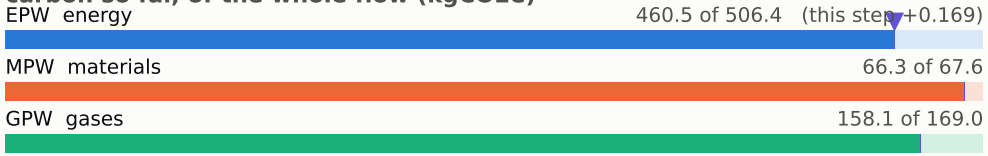
3D structure after this step

zoomed to fit

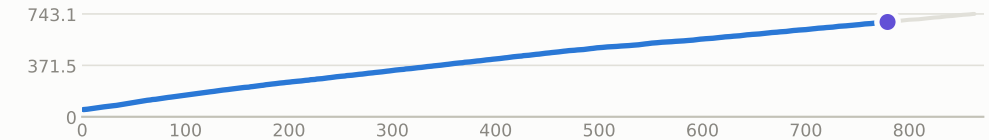
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

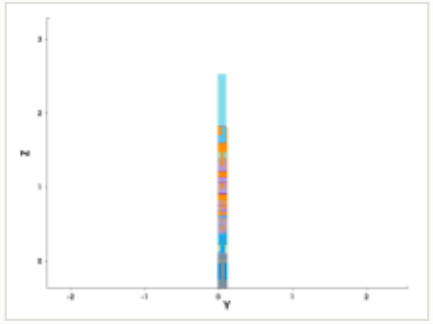
carbon so far, of the whole flow (kgCO2e)



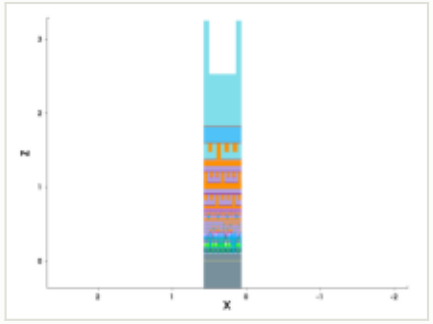
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



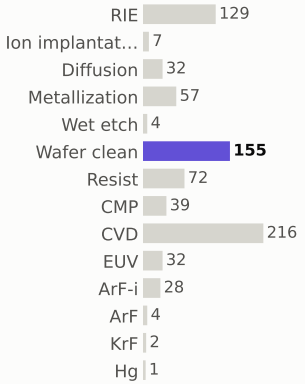
y cut at 0.049 µm



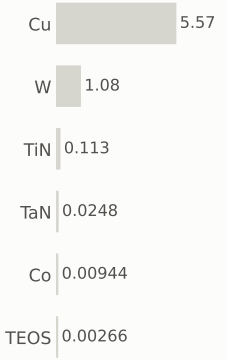
mask for this step

no mask for this step

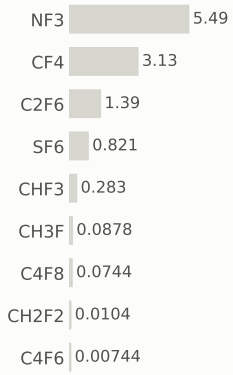
steps by process area



materials deposited (g)



gases emitted (g)



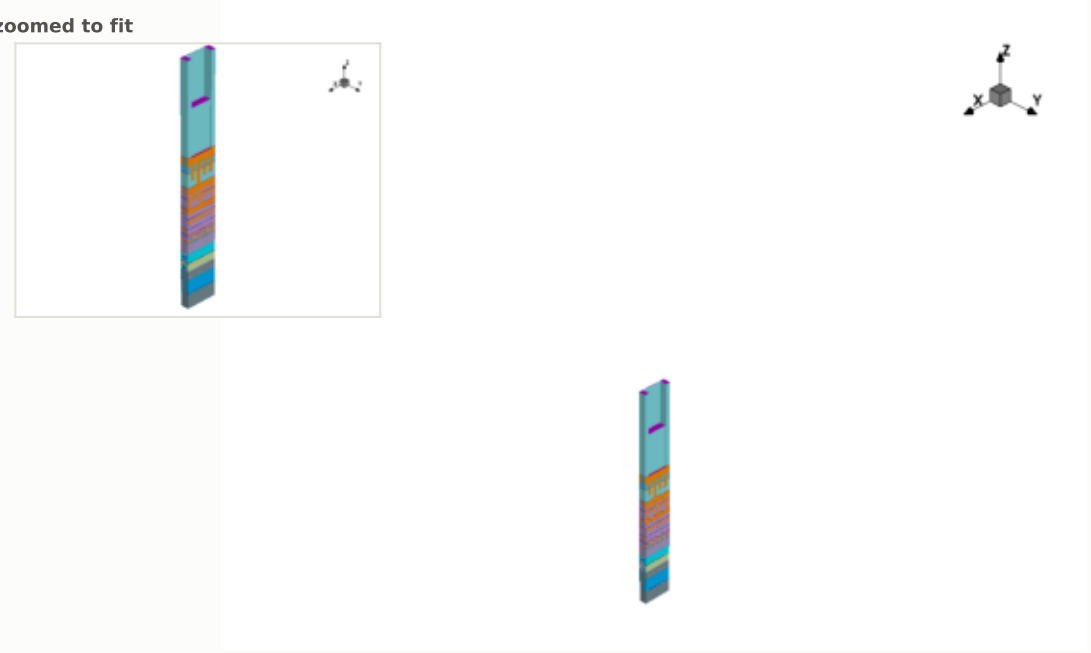
Step #8170: Metallization

depo_ellipsoidal · Metallization

1.2	686.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

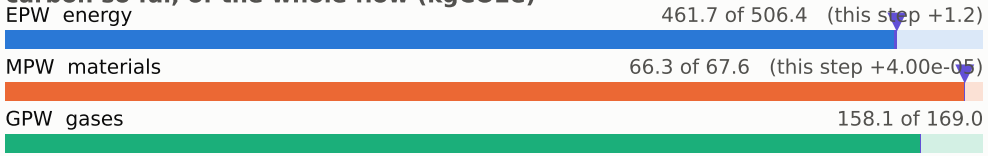
zoomed to fit



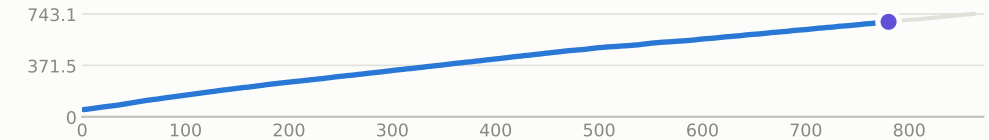
this step

process area	Metallization	EPW	1.2 kgCO2e
material	TaN	EPW tool	0.86 kgCO2e
thickness	0.0016 µm	EPW overhead	0.344 kgCO2e
material GWP	25.8159 gCO2e/g	MPW	4.00e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

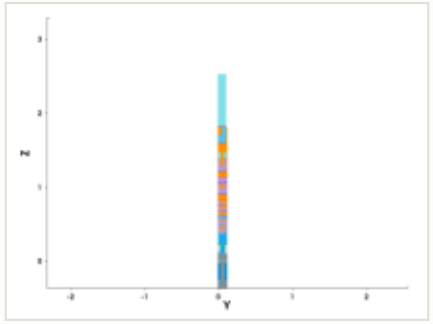
carbon so far, of the whole flow (kgCO2e)



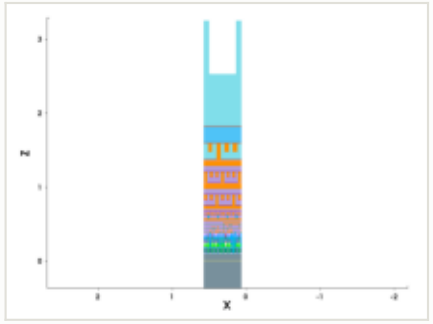
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



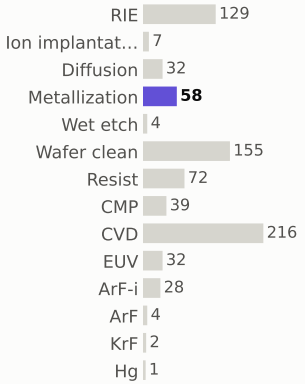
y cut at 0.049 µm



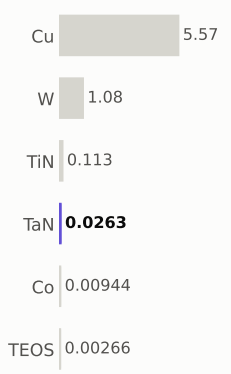
mask for this step

no mask for this step

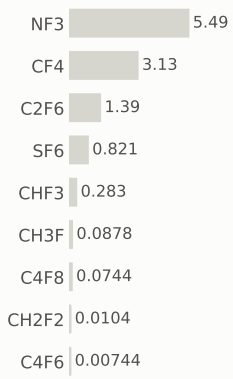
steps by process area



materials deposited (g)



gases emitted (g)



Step #8180: Metallization

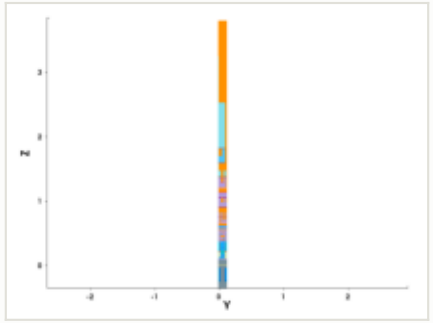
depo_fill · Metallization

1.21	687.4	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

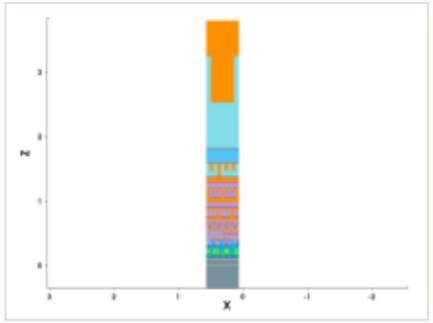
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

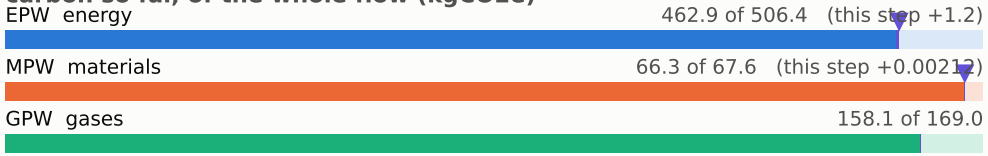
no mask for this step

this step

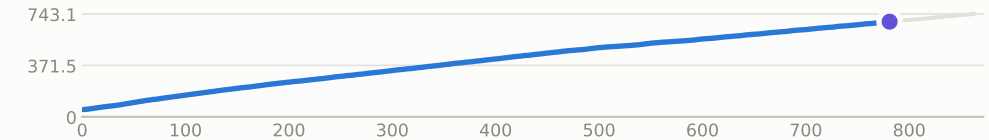
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

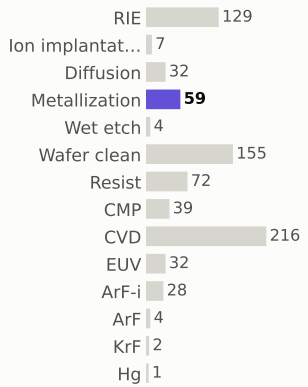
carbon so far, of the whole flow (kgCO2e)



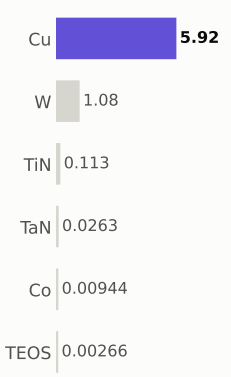
embodied carbon up to each step (kgCO2e per wafer)



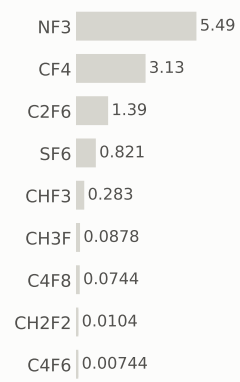
steps by process area



materials deposited (g)



gases emitted (g)



Step #8190: Diffusion

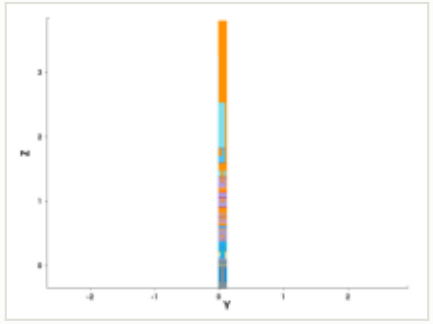
treatment · Diffusion

0.801	688.2	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

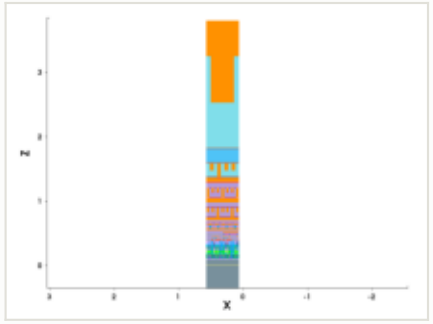
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

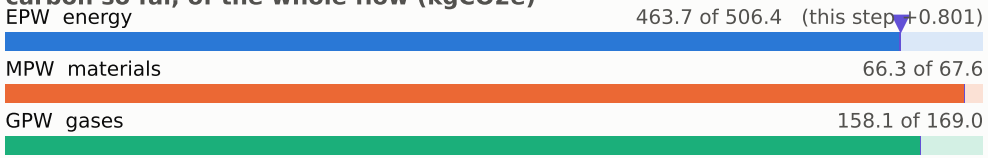
no mask for this step

this step

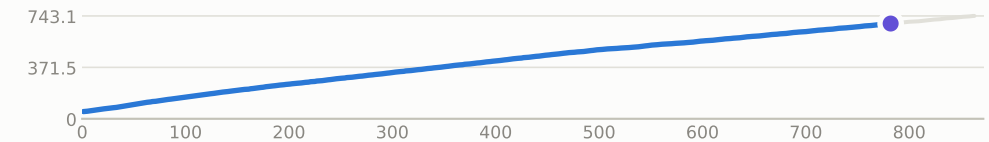
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

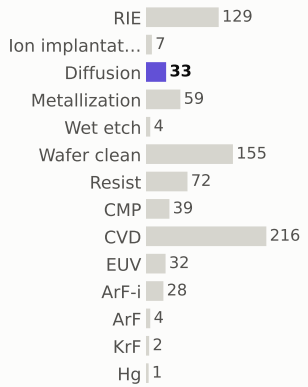
carbon so far, of the whole flow (kgCO2e)



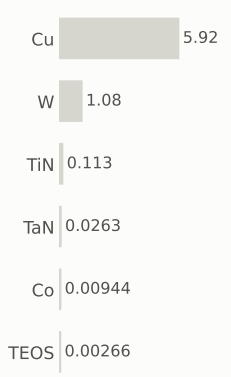
embodied carbon up to each step (kgCO2e per wafer)



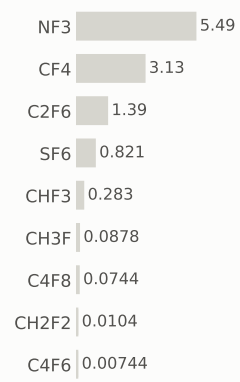
steps by process area



materials deposited (g)



gases emitted (g)



Step #8200: CMP

etch_cmp · CMP

0.169	688.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

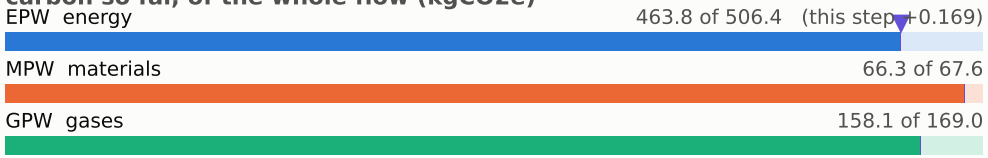
zoomed to fit

this step

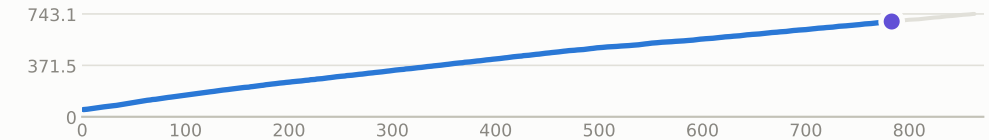
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

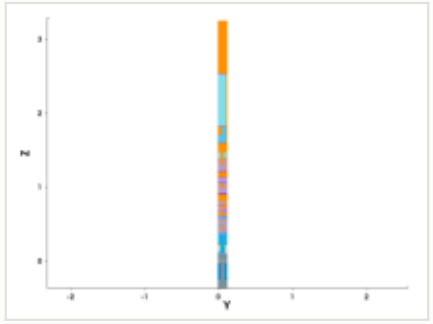
carbon so far, of the whole flow (kgCO2e)



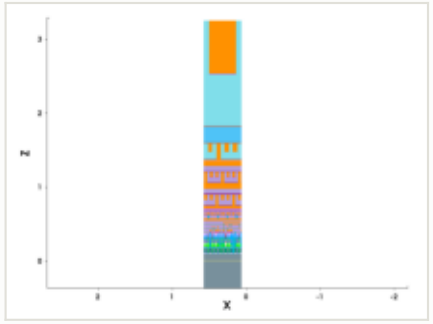
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



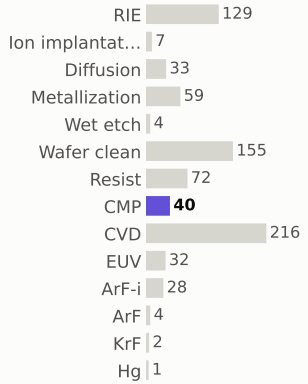
y cut at 0.049 µm



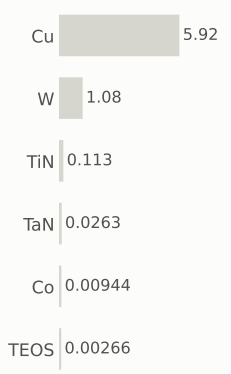
mask for this step

no mask for this step

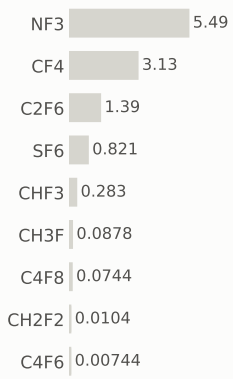
steps by process area



materials deposited (g)



gases emitted (g)



Step #8220: CVD

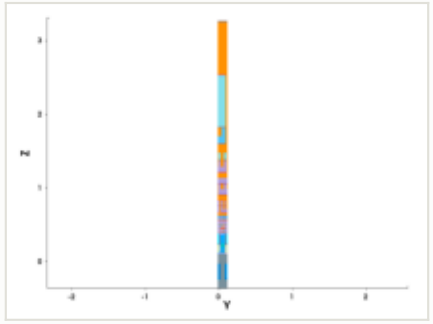
depo_fill · CVD

1.31	689.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

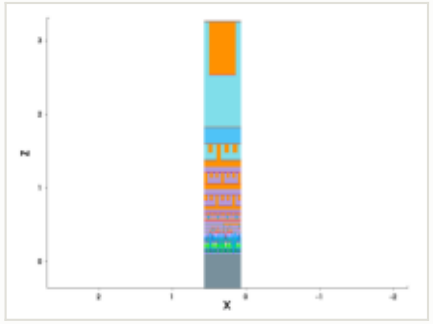
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

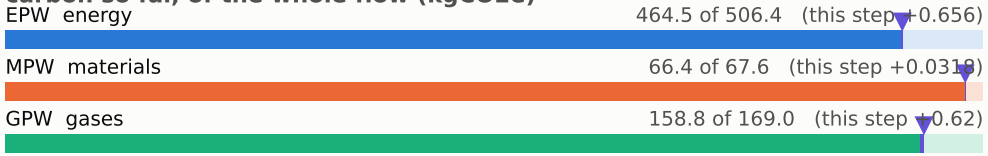
no mask for this step

this step

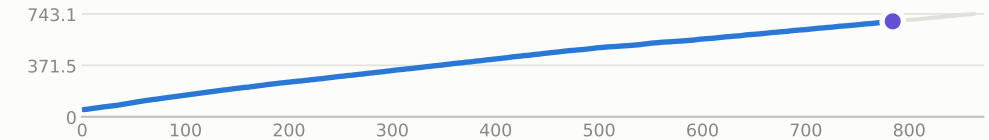
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

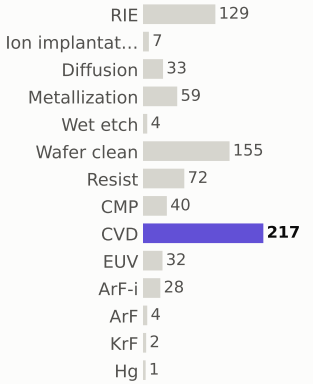
carbon so far, of the whole flow (kgCO2e)



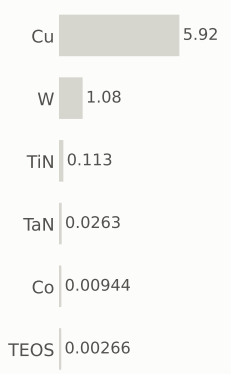
embodied carbon up to each step (kgCO2e per wafer)



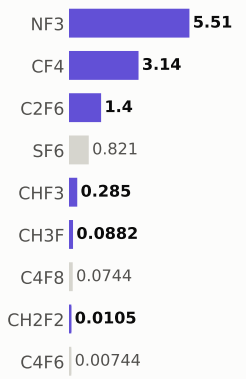
steps by process area



materials deposited (g)



gases emitted (g)



Step #8230: CVD

depo_fill · CVD

1.31	690.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

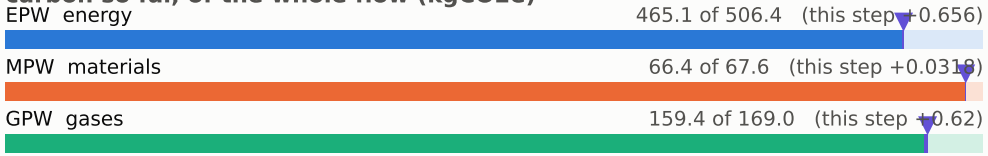
zoomed to fit

this step

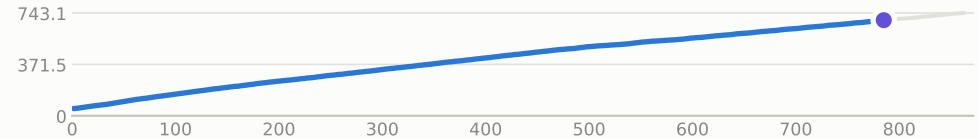
process area	CVD
material	—
thickness	1.42 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

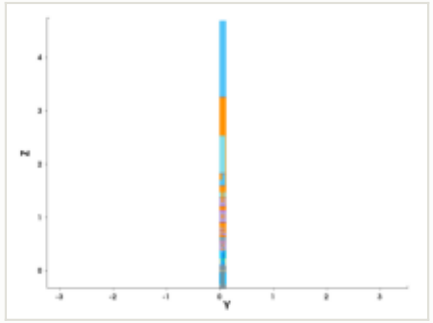
carbon so far, of the whole flow (kgCO2e)



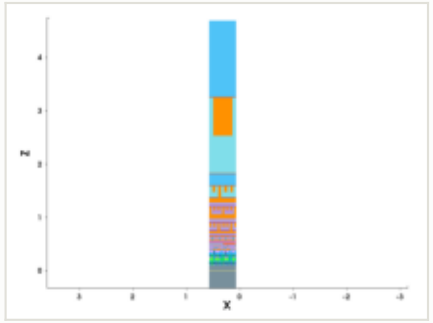
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



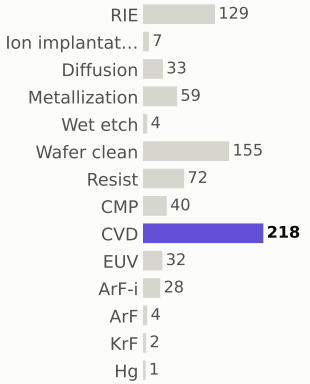
y cut at 0.049 µm



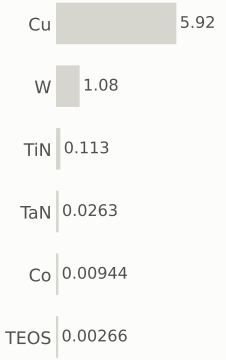
mask for this step

no mask for this step

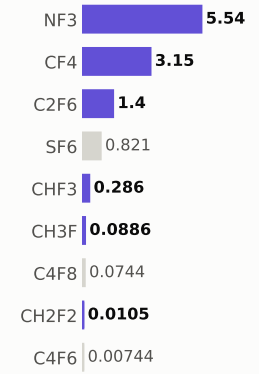
steps by process area



materials deposited (g)



gases emitted (g)



Step #8240: Hg

litho_patterning · Hg

0.199	691.1	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

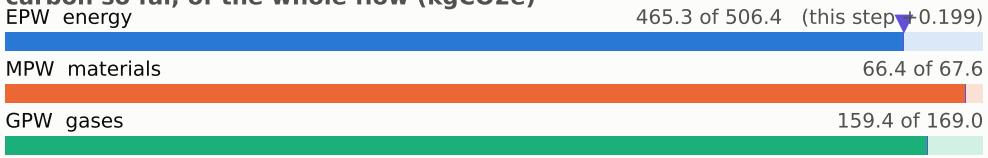
zoomed to fit

this step

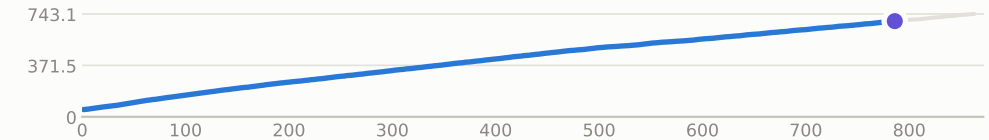
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

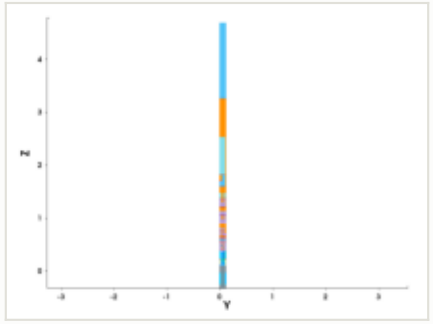
carbon so far, of the whole flow (kgCO2e)



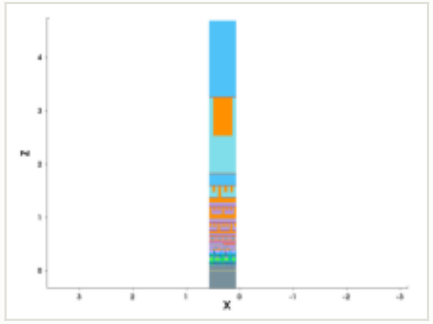
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

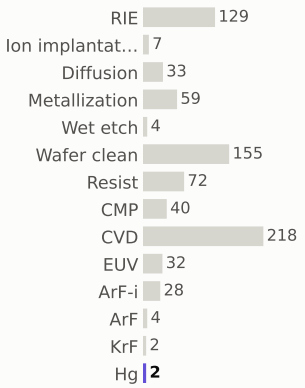


y cut at 0.049 µm

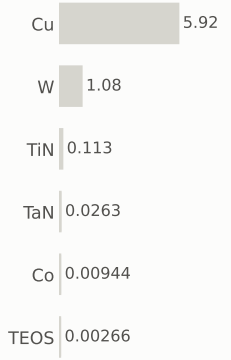


mask: layer 85:0 (GDSII)

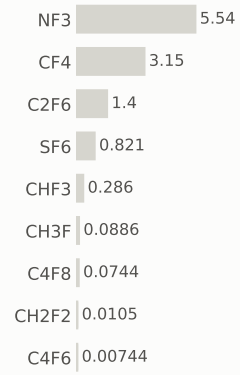
steps by process area



materials deposited (g)



gases emitted (g)



Step #8250: RIE

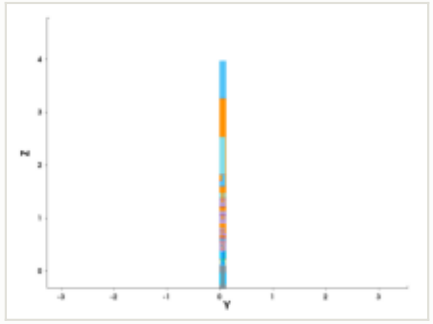
etch_anisotropic · RIE

1.04	692.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

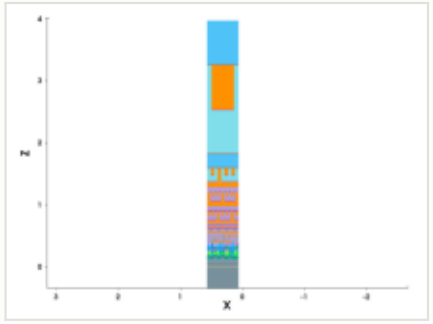
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

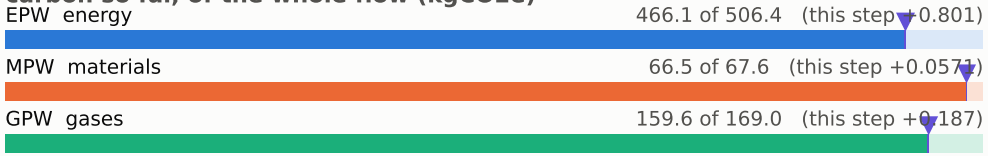
no mask for this step

this step

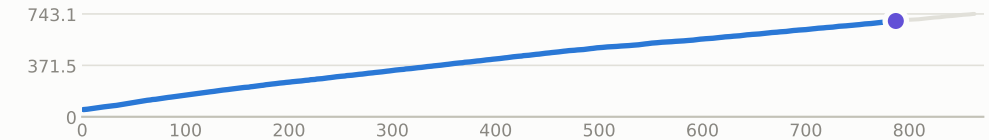
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

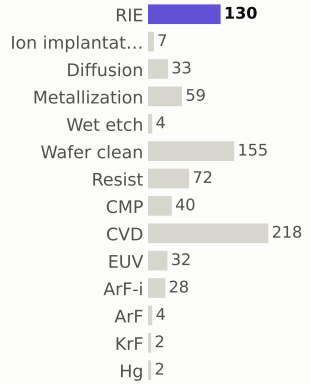
carbon so far, of the whole flow (kgCO2e)



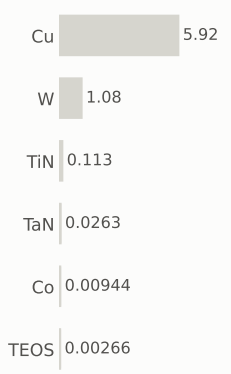
embodied carbon up to each step (kgCO2e per wafer)



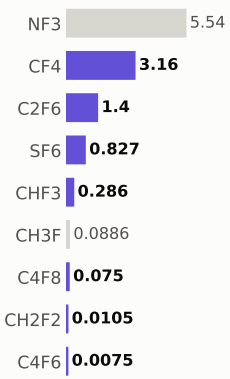
steps by process area



materials deposited (g)



gases emitted (g)



Step #8260: Wafer_clean

etch_strip · Wafer_clean

0.169	692.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

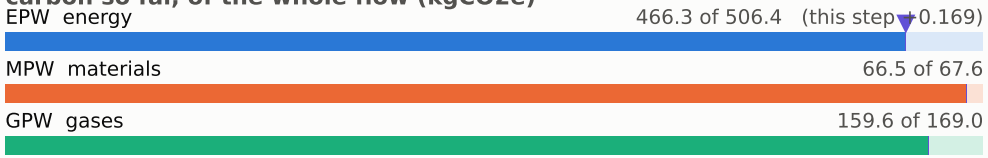
zoomed to fit

this step

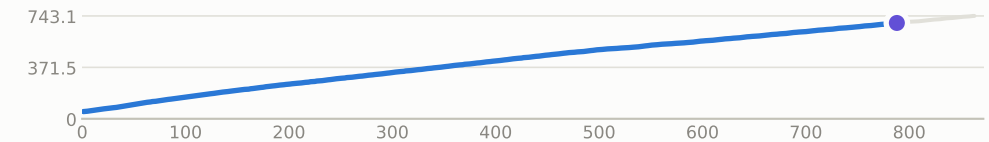
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

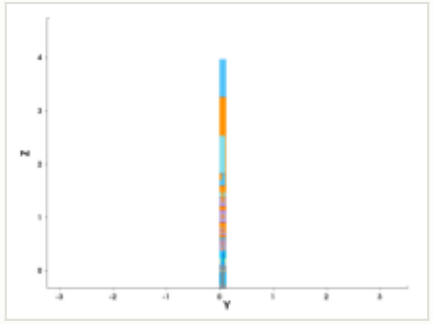
carbon so far, of the whole flow (kgCO2e)



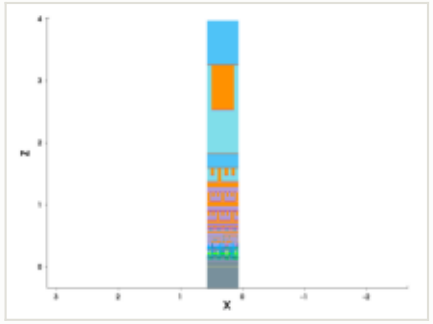
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



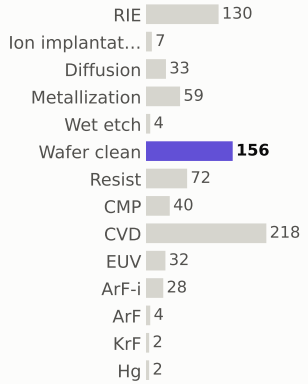
y cut at 0.049 µm



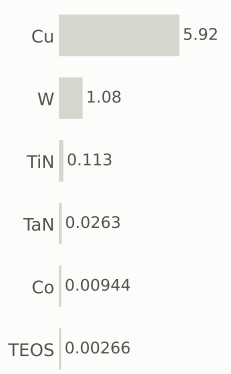
mask for this step

no mask for this step

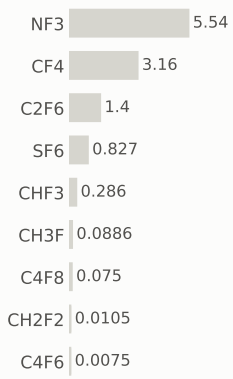
steps by process area



materials deposited (g)



gases emitted (g)



Step #8270: RIE

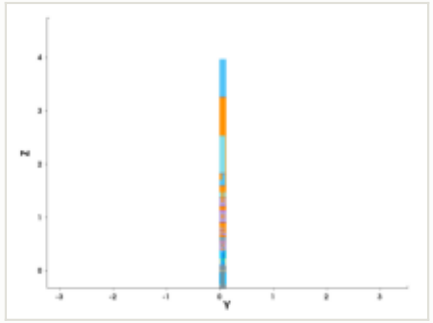
clean · RIE

1.04	693.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

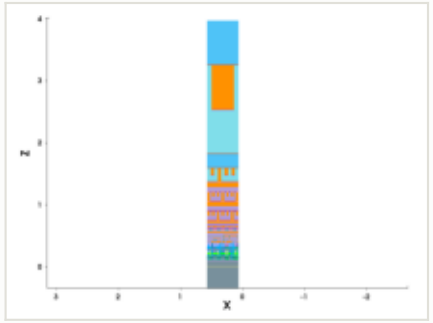
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

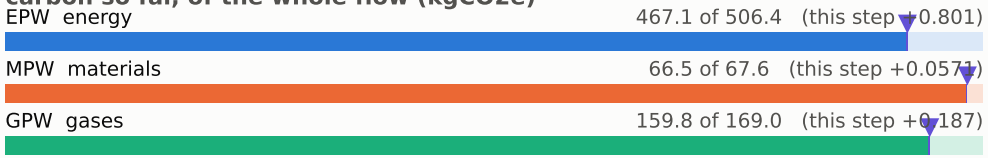
no mask for this step

this step

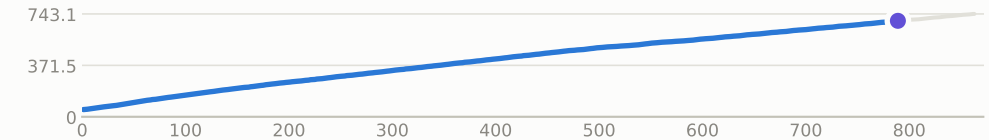
process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

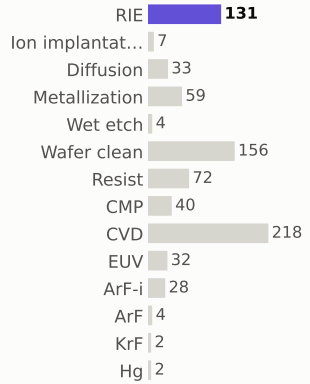
carbon so far, of the whole flow (kgCO2e)



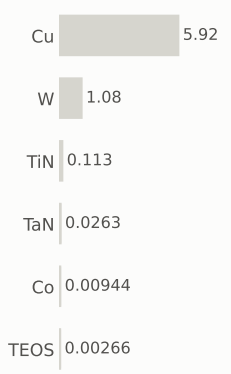
embodied carbon up to each step (kgCO2e per wafer)



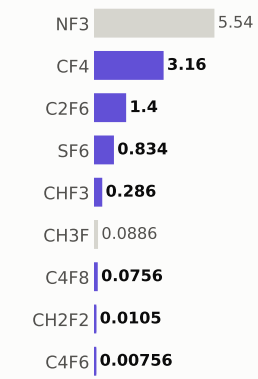
steps by process area



materials deposited (g)



gases emitted (g)



Step #8280: Wafer_clean

clean · Wafer_clean

0.169	693.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

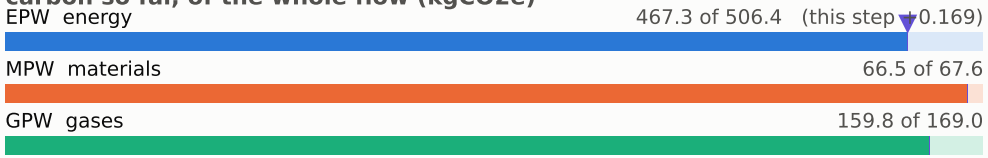
zoomed to fit

this step

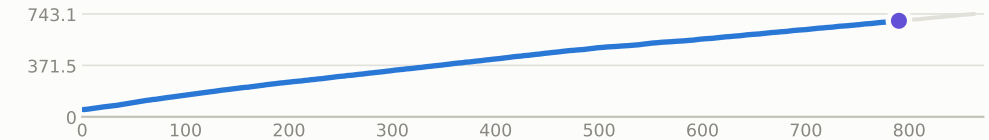
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

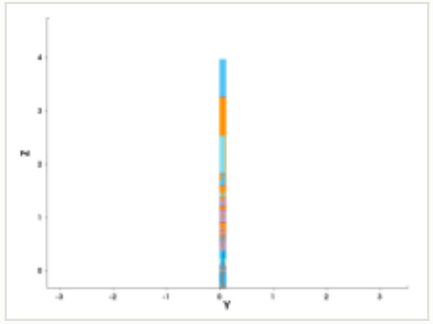
carbon so far, of the whole flow (kgCO2e)



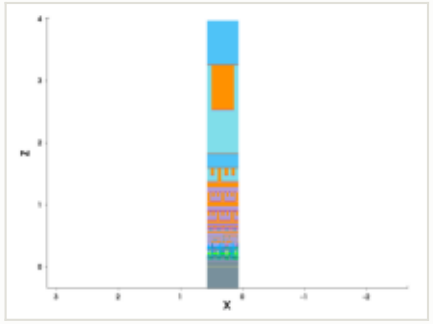
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



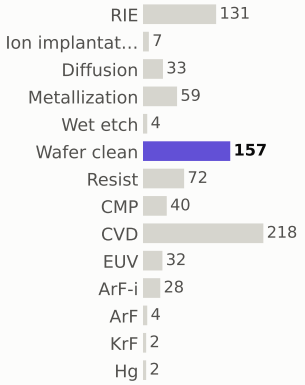
y cut at 0.049 µm



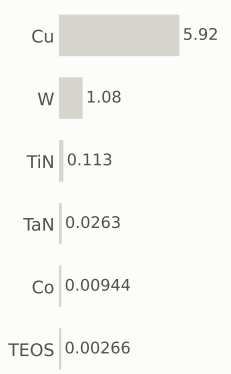
mask for this step

no mask for this step

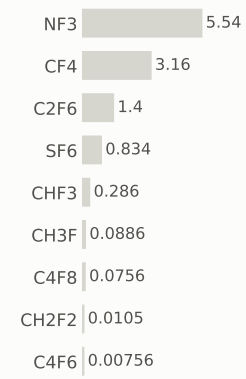
steps by process area



materials deposited (g)



gases emitted (g)



Step #8290: Hg

litho_patterning · Hg

0.199	693.8	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

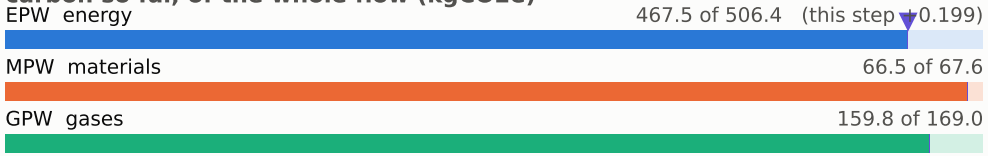
3D structure after this step

zoomed to fit

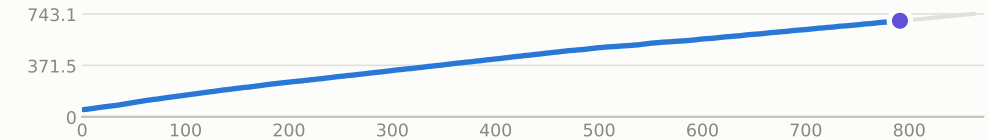
this step

process area	Hg	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

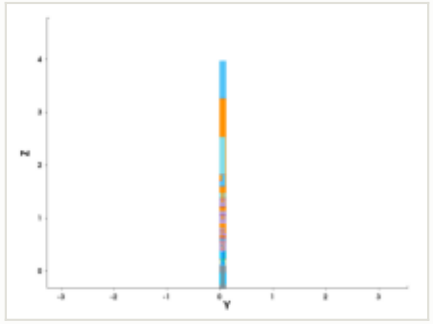
carbon so far, of the whole flow (kgCO2e)



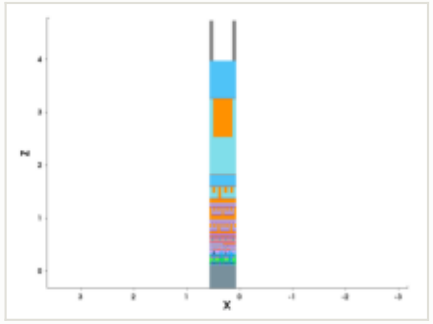
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

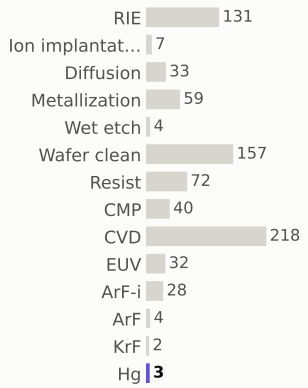


y cut at 0.049 µm

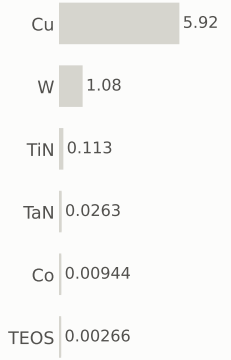


mask: layer 82:0 (GDSII)

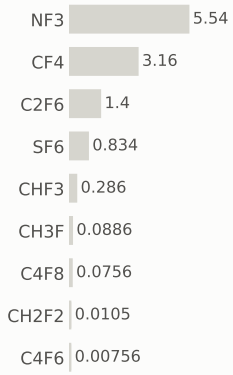
steps by process area



materials deposited (g)



gases emitted (g)



Step #8300: RIE

etch_anisotropic · RIE

1.04	694.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

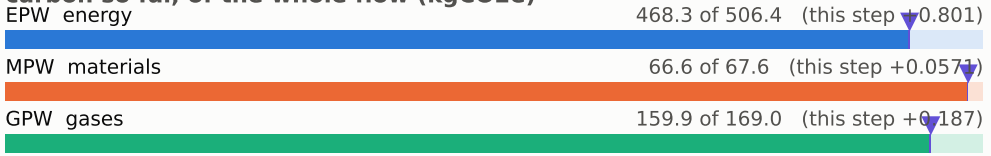
zoomed to fit

this step

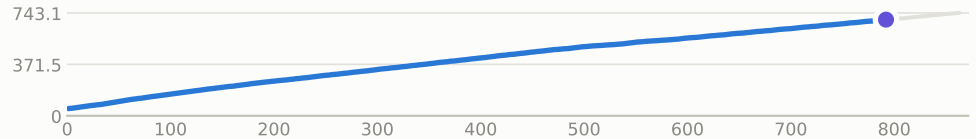
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

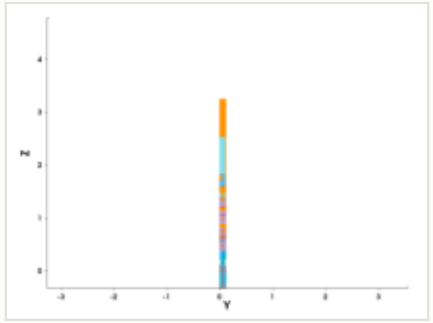
carbon so far, of the whole flow (kgCO2e)



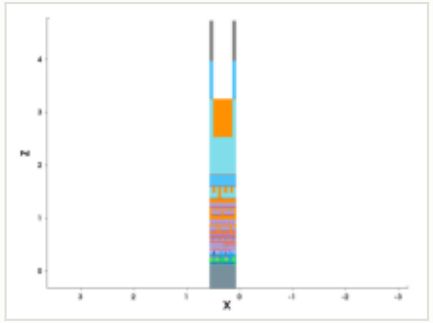
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



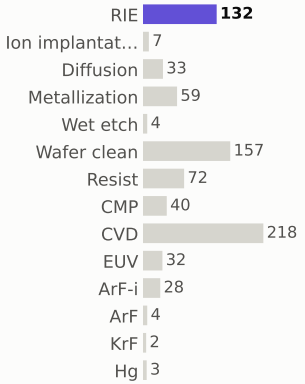
y cut at 0.049 µm



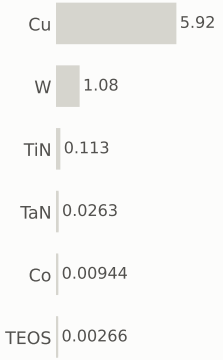
mask for this step

no mask for this step

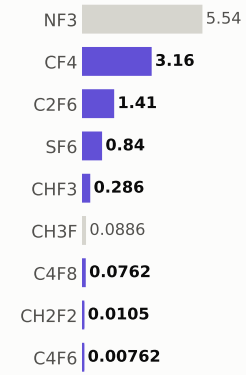
steps by process area



materials deposited (g)



gases emitted (g)



Step #8310: Wafer_clean

etch_strip · Wafer_clean

0.169	695.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

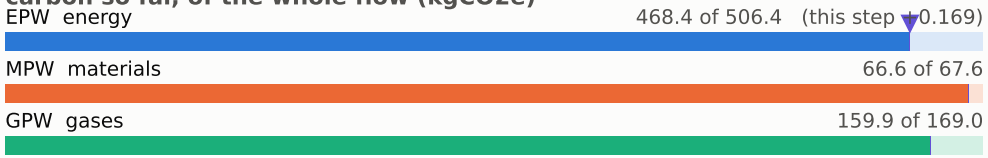
zoomed to fit

this step

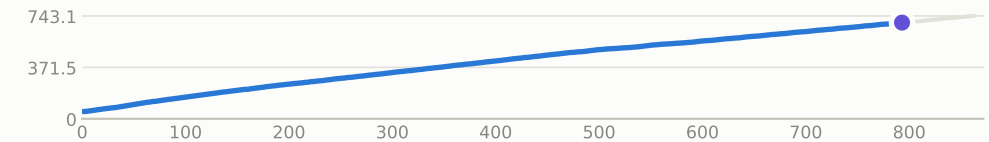
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

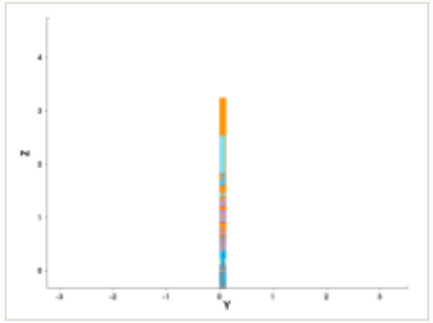
carbon so far, of the whole flow (kgCO2e)



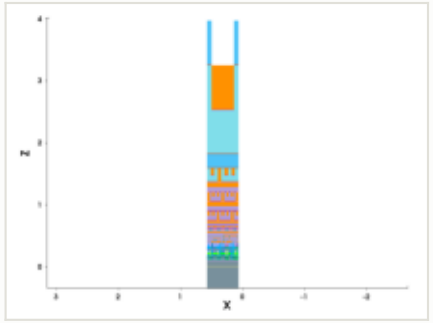
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



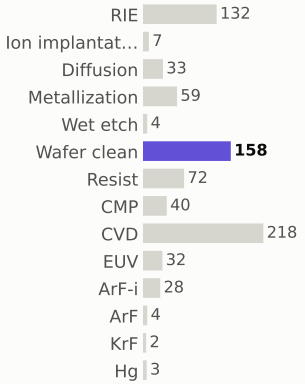
y cut at 0.049 µm



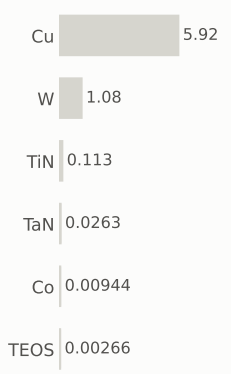
mask for this step

no mask for this step

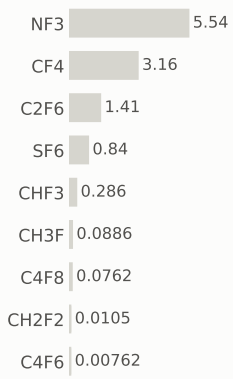
steps by process area



materials deposited (g)



gases emitted (g)



Step #8320: Metallization

depo_ellipsoidal · Metallization

1.2	696.2	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

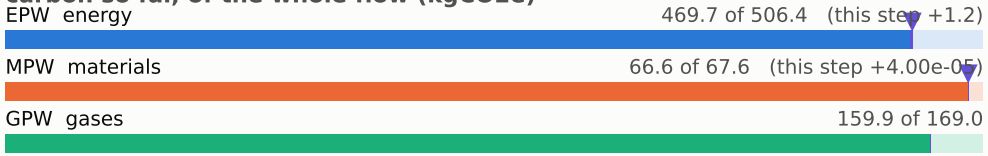
zoomed to fit

this step

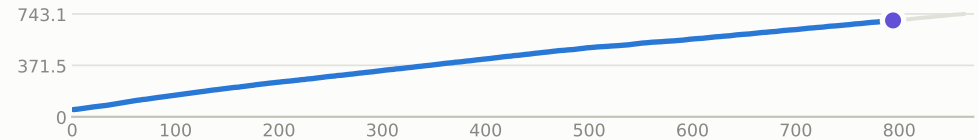
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

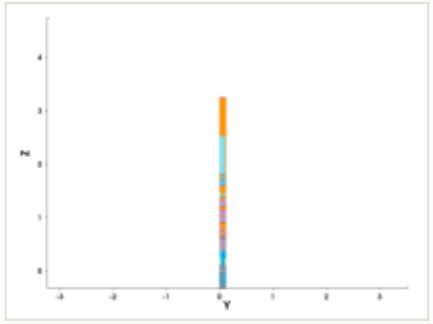
carbon so far, of the whole flow (kgCO2e)



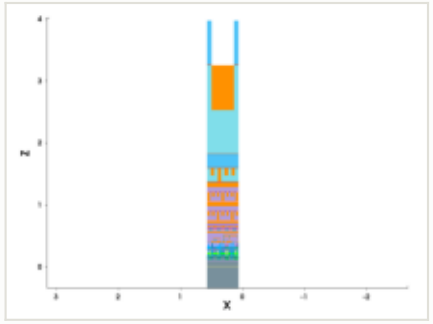
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



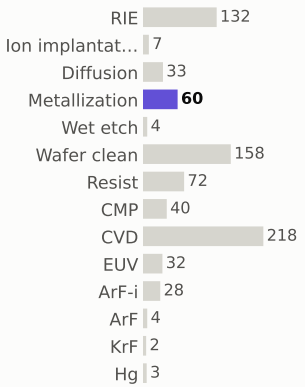
y cut at 0.049 µm



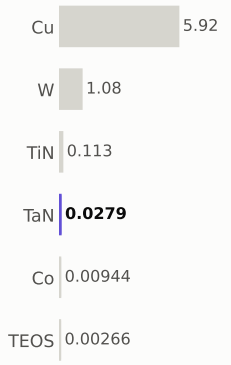
mask for this step

no mask for this step

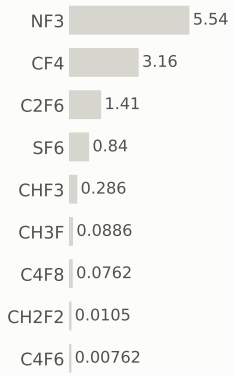
steps by process area



materials deposited (g)



gases emitted (g)



Step #8330: Metallization

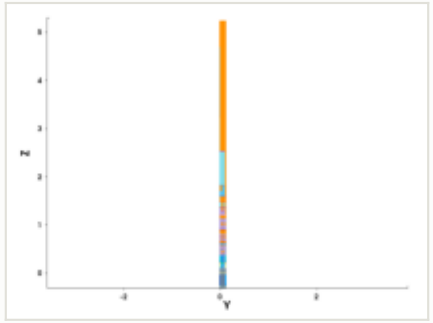
depo_fill · Metallization

1.21	697.4	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

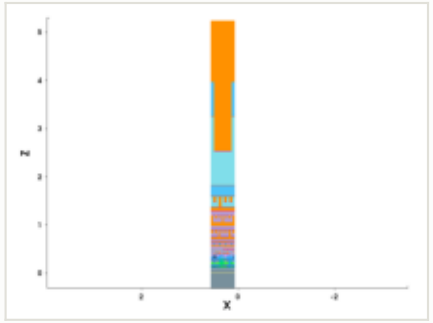
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

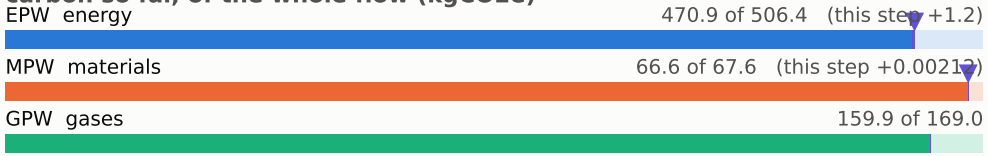
no mask for this step

this step

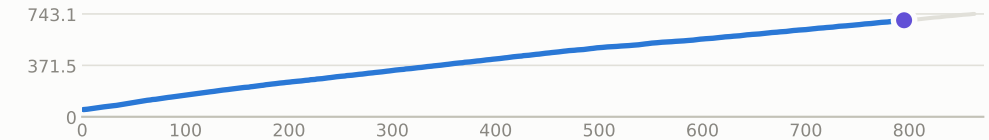
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

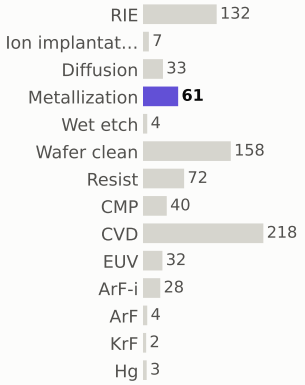
carbon so far, of the whole flow (kgCO2e)



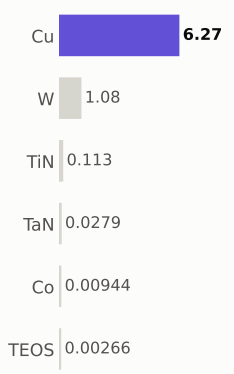
embodied carbon up to each step (kgCO2e per wafer)



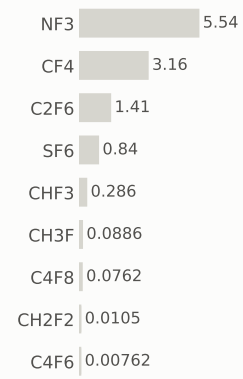
steps by process area



materials deposited (g)



gases emitted (g)



Step #8340: Diffusion

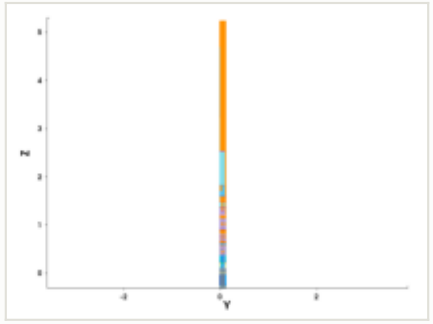
treatment · Diffusion

0.801	698.2	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

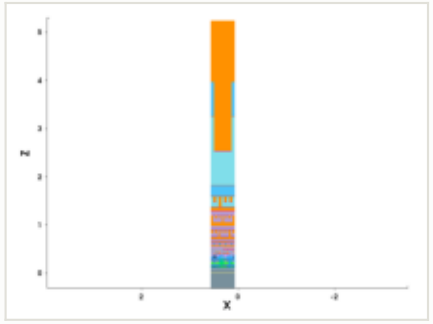
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

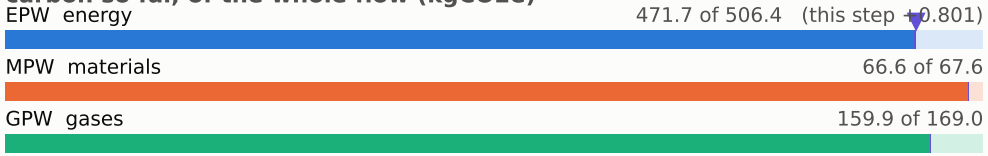
no mask for this step

this step

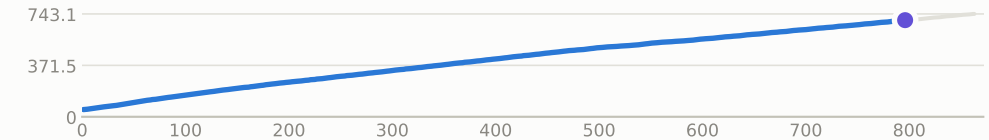
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

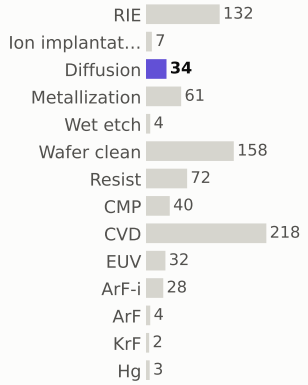
carbon so far, of the whole flow (kgCO2e)



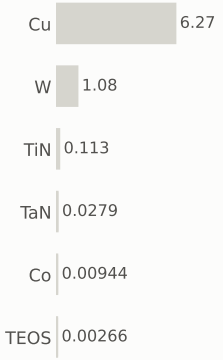
embodied carbon up to each step (kgCO2e per wafer)



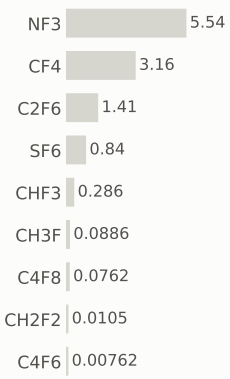
steps by process area



materials deposited (g)



gases emitted (g)



Step #8350: CMP

etch_cmp · CMP

0.169	698.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

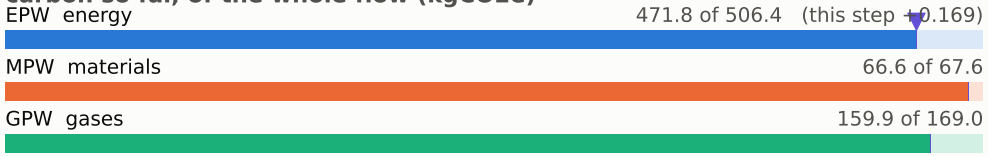
3D structure after this step

zoomed to fit

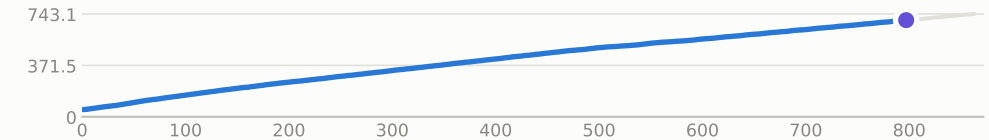
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

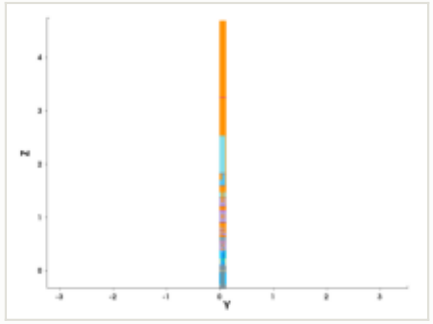
carbon so far, of the whole flow (kgCO2e)



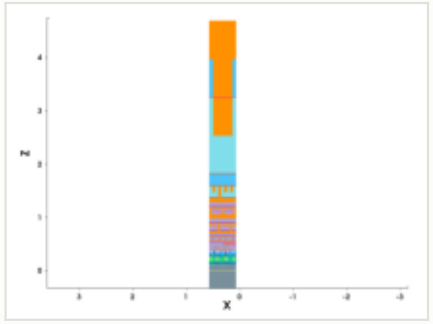
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



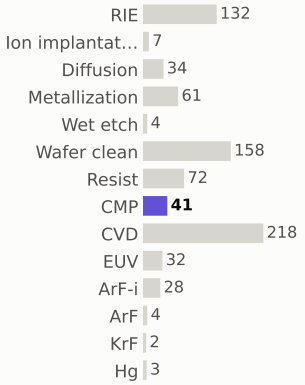
y cut at 0.049 µm



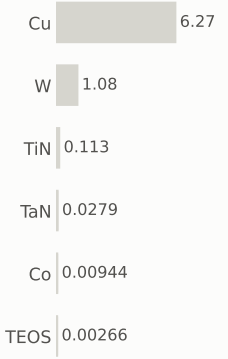
mask for this step

no mask for this step

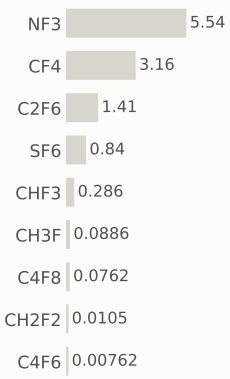
steps by process area



materials deposited (g)



gases emitted (g)



Step #8370: CVD

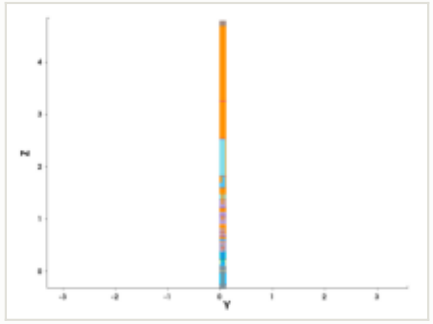
depo_anisotropic · CVD

1.31	699.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

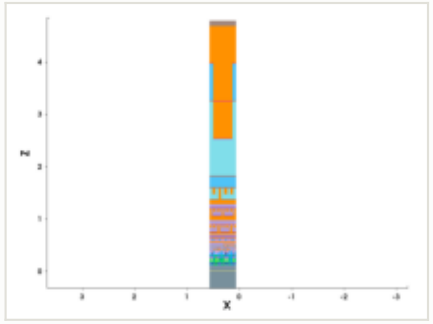
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

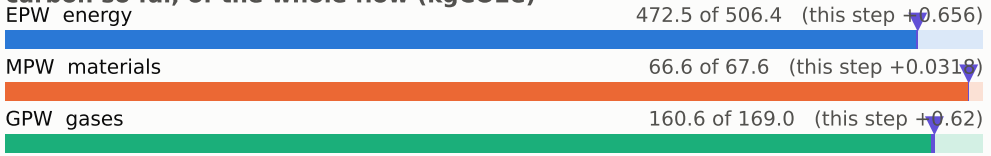
no mask for this step

this step

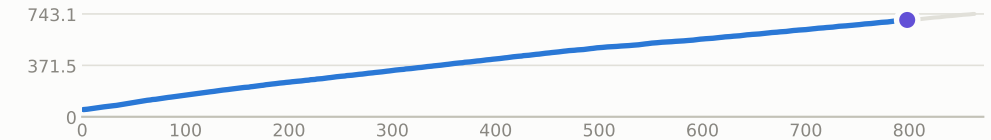
process area	CVD
material	—
thickness	0.1 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

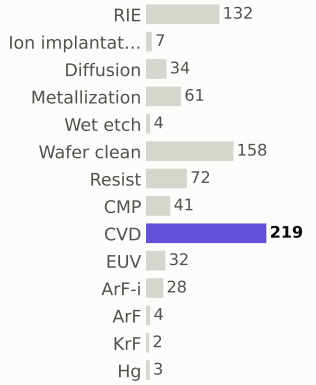
carbon so far, of the whole flow (kgCO2e)



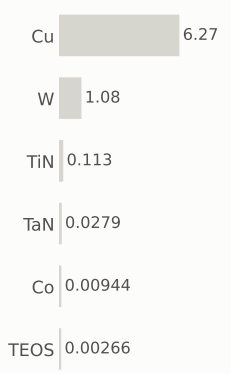
embodied carbon up to each step (kgCO2e per wafer)



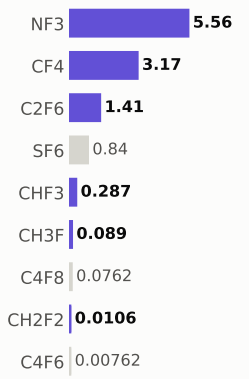
steps by process area



materials deposited (g)



gases emitted (g)



Step #8380: CVD

depo_anisotropic · CVD

1.31	701.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

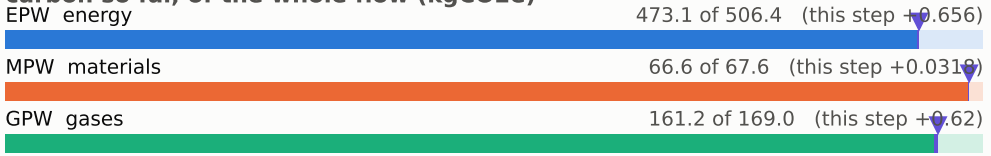
zoomed to fit

this step

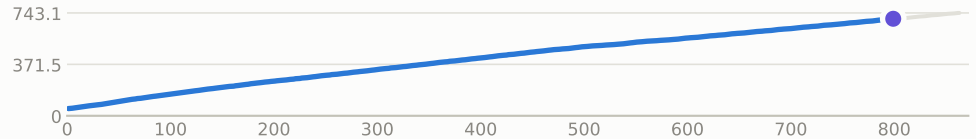
process area	CVD
material	—
thickness	0.1 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

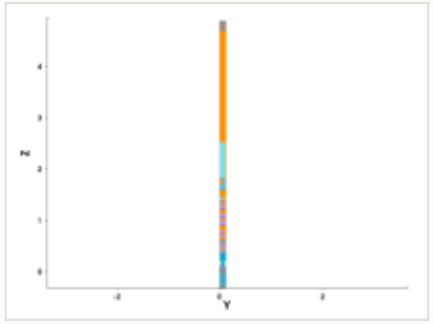
carbon so far, of the whole flow (kgCO2e)



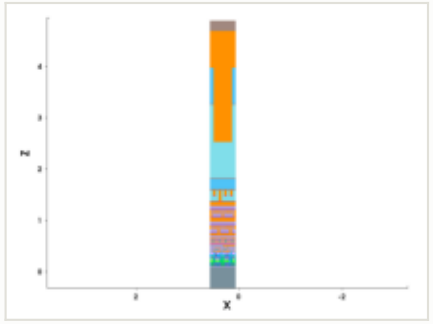
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



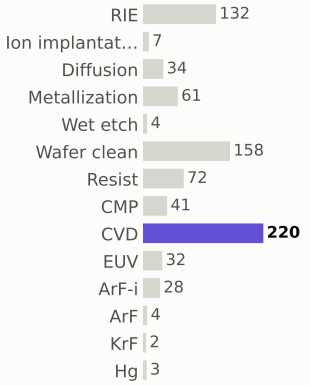
y cut at 0.049 µm



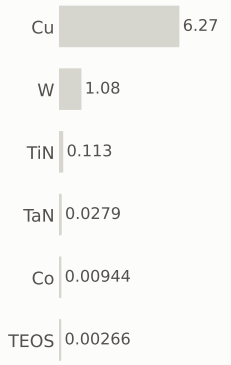
mask for this step

no mask for this step

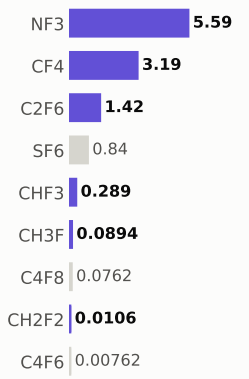
steps by process area



materials deposited (g)



gases emitted (g)



Step #8390: CMP

etch_cmp · CMP

0.169	701.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

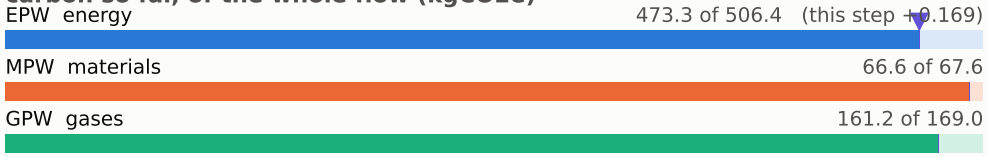
zoomed to fit

this step

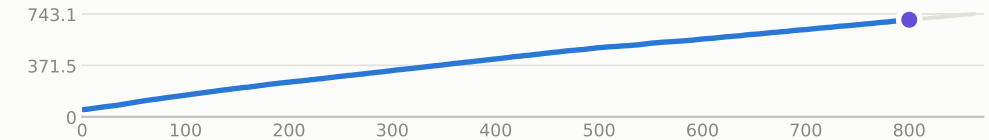
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

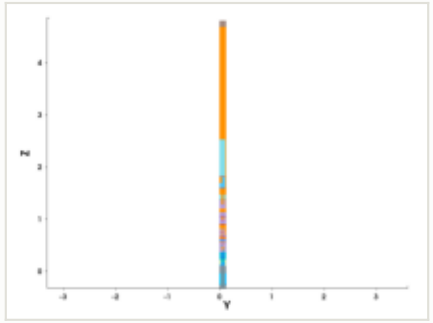
carbon so far, of the whole flow (kgCO2e)



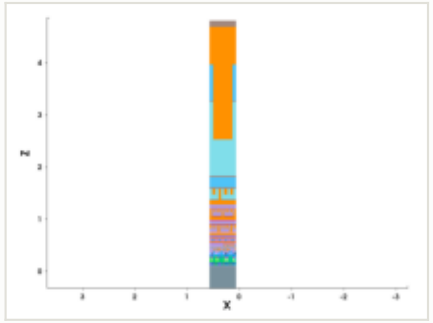
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



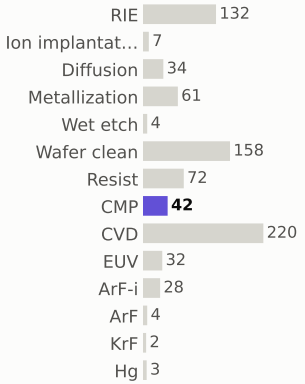
y cut at 0.049 µm



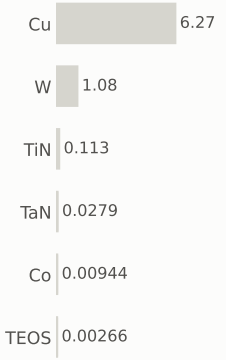
mask for this step

no mask for this step

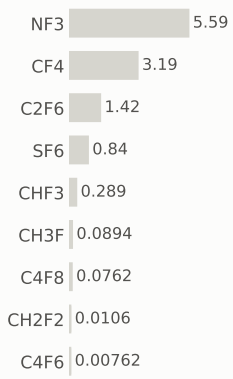
steps by process area



materials deposited (g)



gases emitted (g)



Step #8400: CMP

etch_cmp · CMP

0.169	701.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

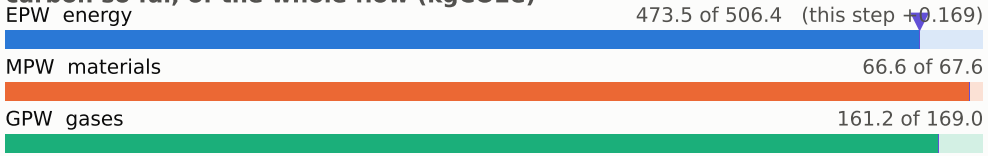
zoomed to fit

this step

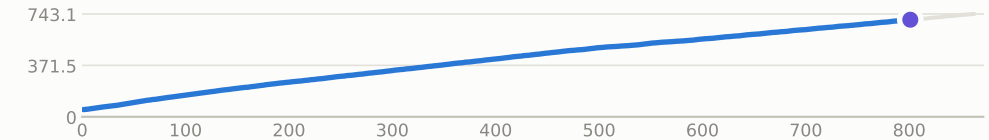
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

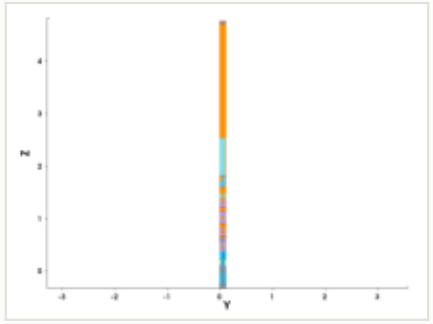
carbon so far, of the whole flow (kgCO2e)



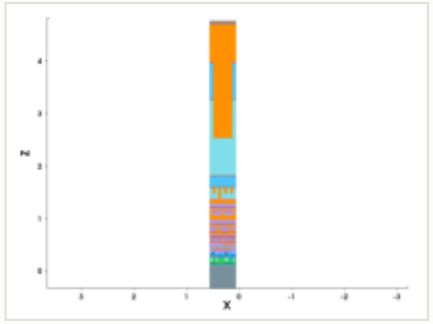
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



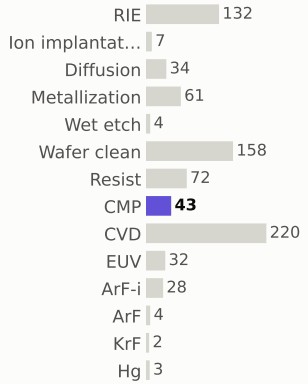
y cut at 0.049 µm



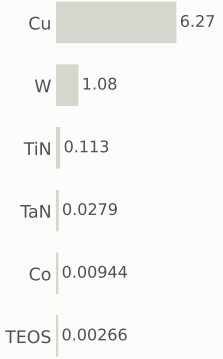
mask for this step

no mask for this step

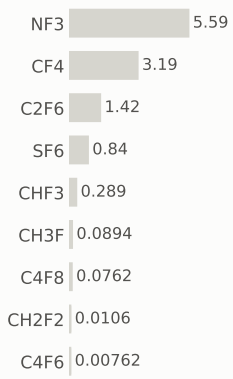
steps by process area



materials deposited (g)



gases emitted (g)



Step #8410: Wafer

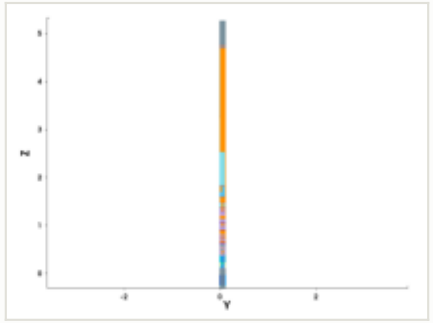
bond_structure · Wafer

0	701.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

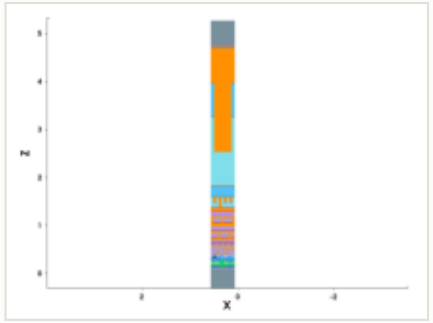
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

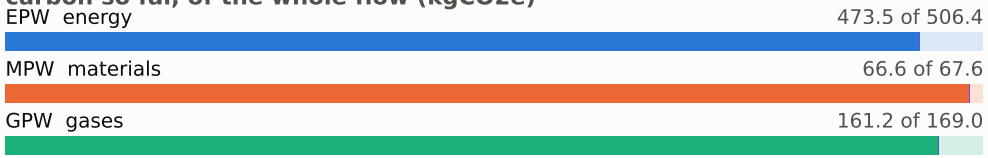
no mask for this step

this step

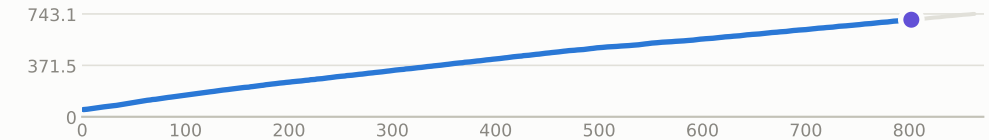
process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

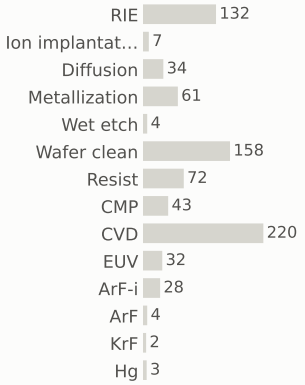
carbon so far, of the whole flow (kgCO2e)



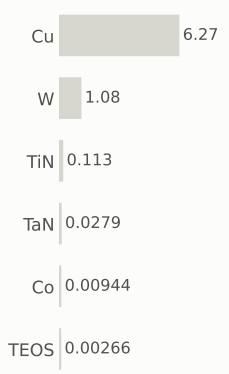
embodied carbon up to each step (kgCO2e per wafer)



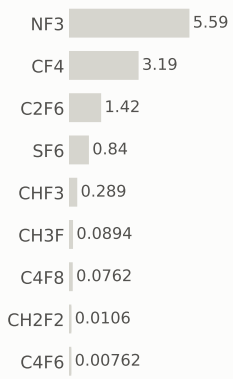
steps by process area



materials deposited (g)



gases emitted (g)



Step #8430: Wafer

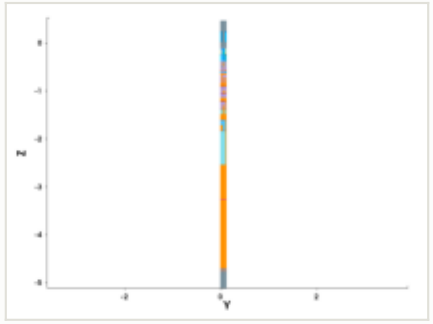
flip_structure · Wafer

0	701.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

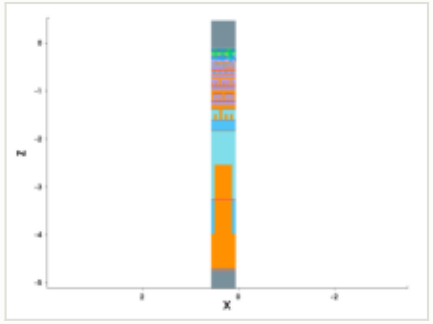
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

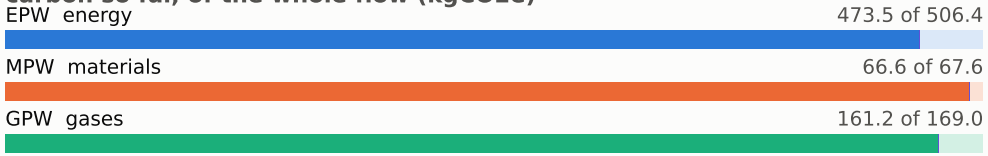
no mask for this step

this step

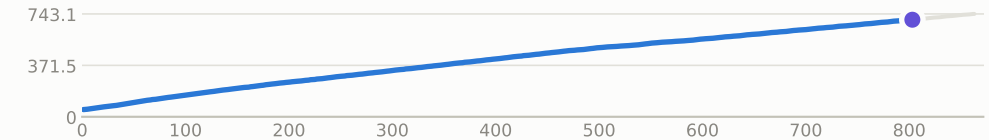
process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

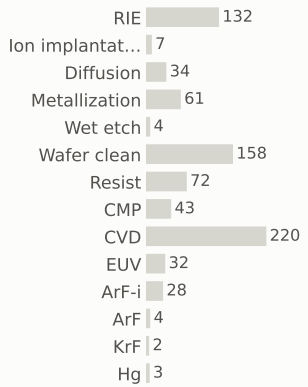
carbon so far, of the whole flow (kgCO2e)



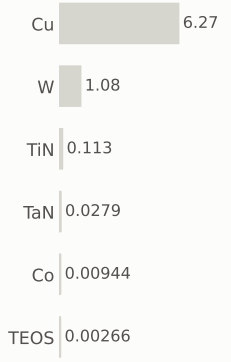
embodied carbon up to each step (kgCO2e per wafer)



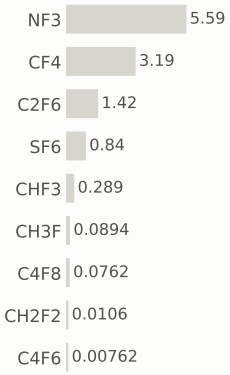
steps by process area



materials deposited (g)



gases emitted (g)



Step #8440: CMP

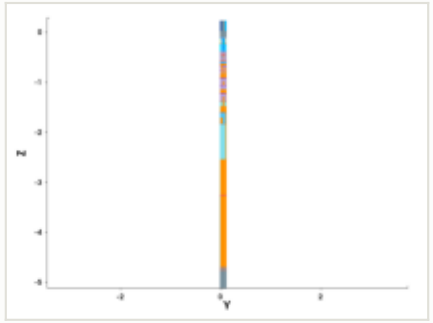
etch_cmp · CMP

0.169	701.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

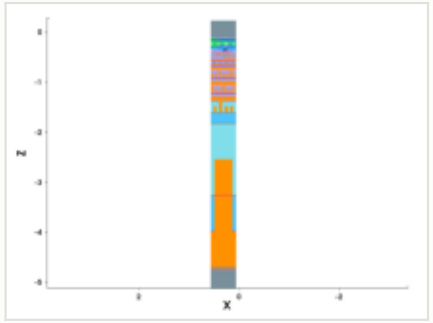
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

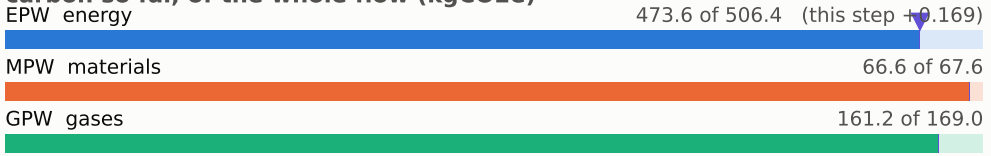
no mask for this step

this step

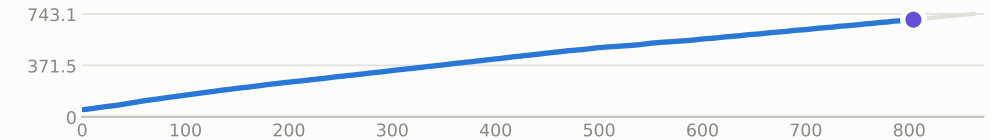
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

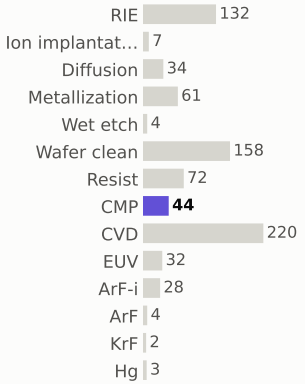
carbon so far, of the whole flow (kgCO2e)



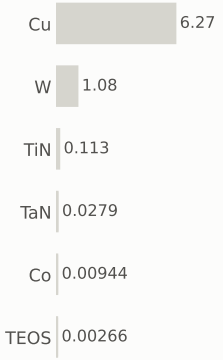
embodied carbon up to each step (kgCO2e per wafer)



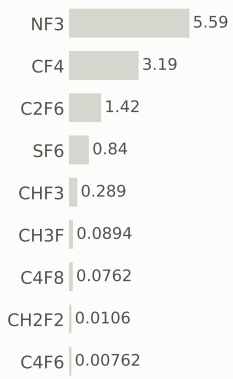
steps by process area



materials deposited (g)



gases emitted (g)



Step #8450: CMP

etch_cmp · CMP

0.169	701.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

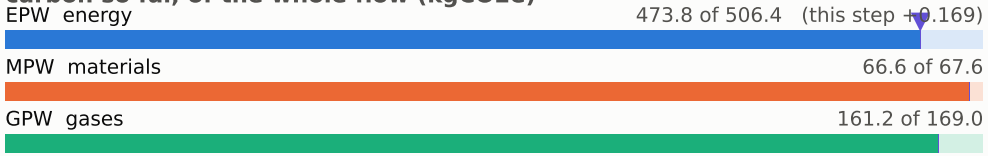
3D structure after this step

zoomed to fit

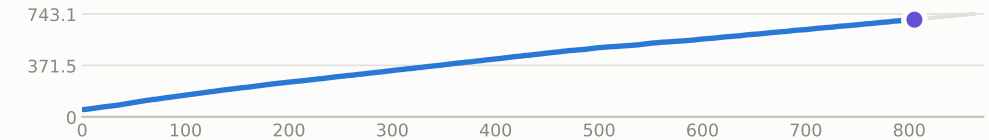
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

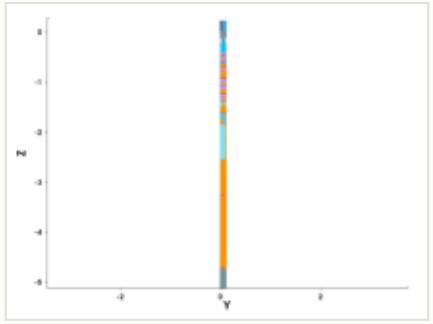
carbon so far, of the whole flow (kgCO2e)



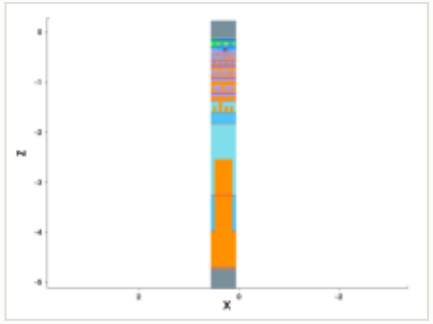
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



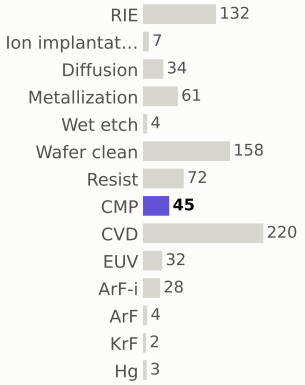
y cut at 0.049 µm



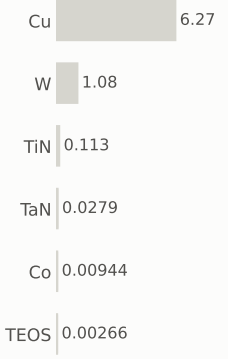
mask for this step

no mask for this step

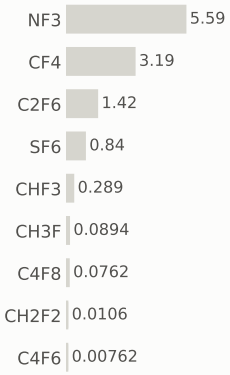
steps by process area



materials deposited (g)



gases emitted (g)



Step #8460: CVD

depo_anisotropic · CVD

1.31	703.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

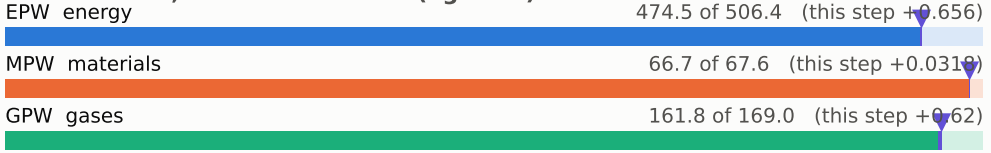
zoomed to fit

this step

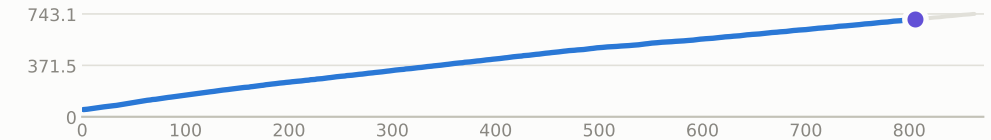
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

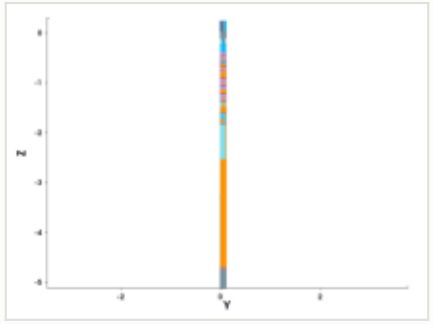
carbon so far, of the whole flow (kgCO2e)



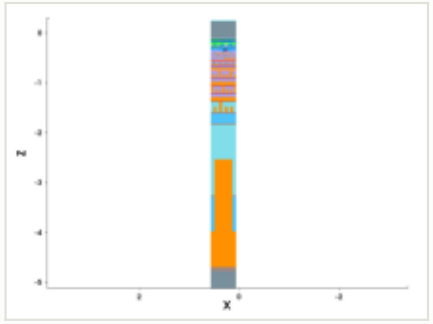
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



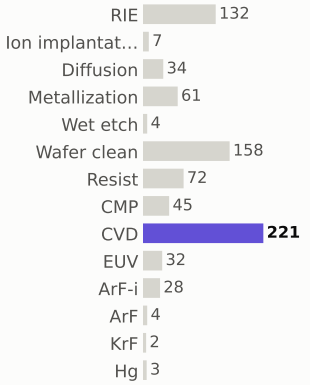
y cut at 0.049 µm



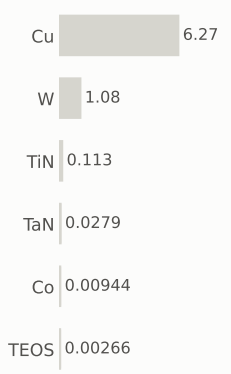
mask for this step

no mask for this step

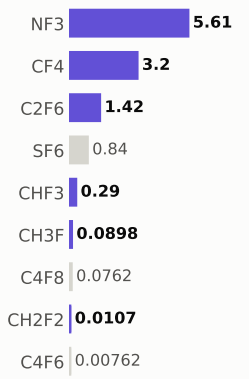
steps by process area



materials deposited (g)



gases emitted (g)



Step #8480: RIE

clean · RIE

1.04	704.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

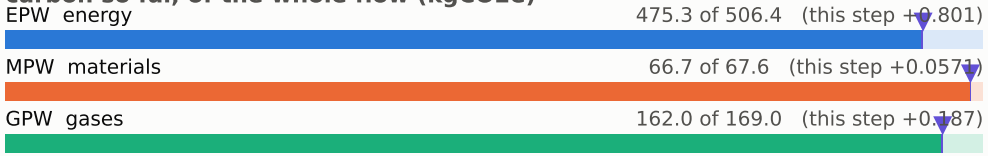
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

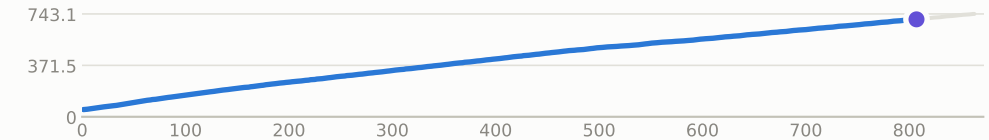
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

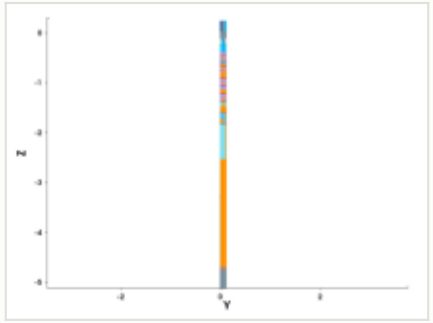
carbon so far, of the whole flow (kgCO2e)



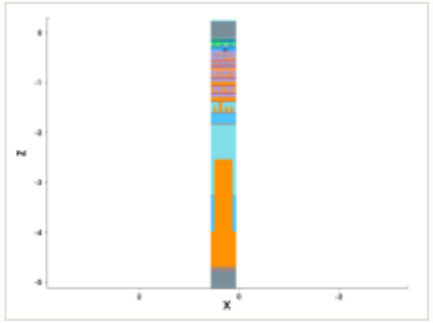
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



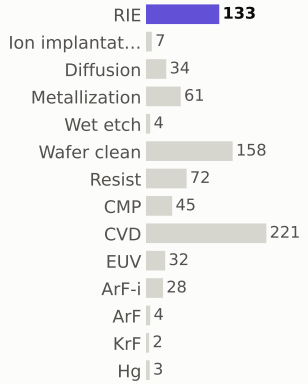
y cut at 0.049 µm



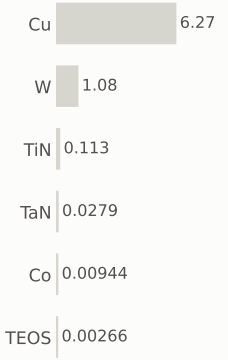
mask for this step

no mask for this step

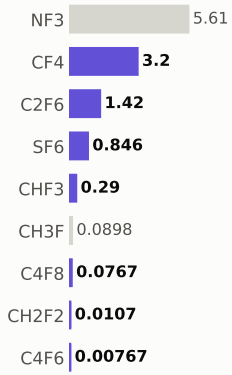
steps by process area



materials deposited (g)



gases emitted (g)



Step #8490: Wafer_clean

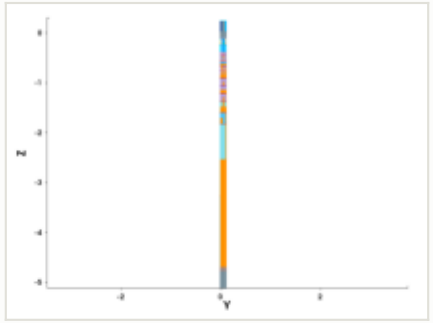
clean · Wafer_clean

0.169	704.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

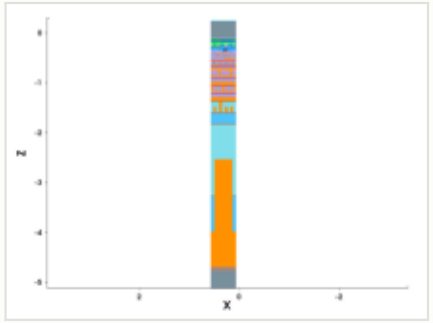
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

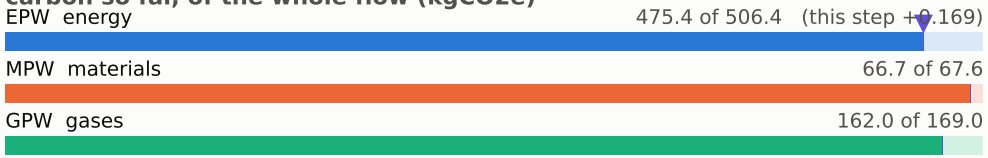
no mask for this step

this step

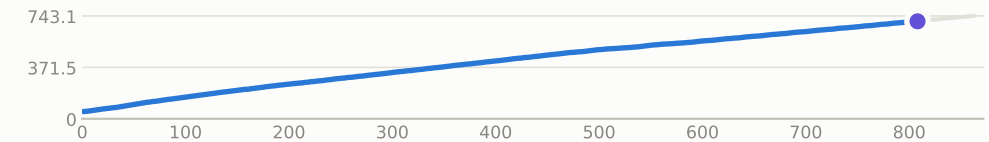
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

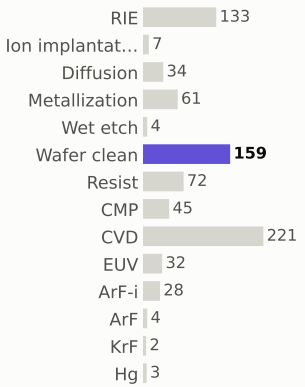
carbon so far, of the whole flow (kgCO2e)



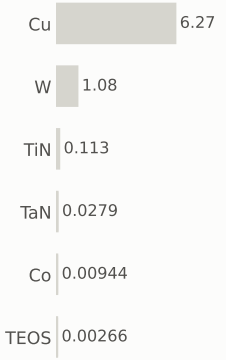
embodied carbon up to each step (kgCO2e per wafer)



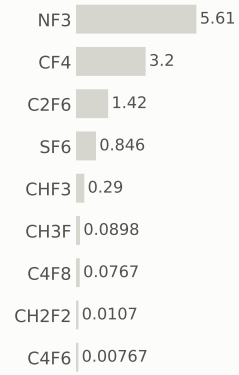
steps by process area



materials deposited (g)



gases emitted (g)



Step #8500: KrF

litho_patterning · KrF

0.199	704.4	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

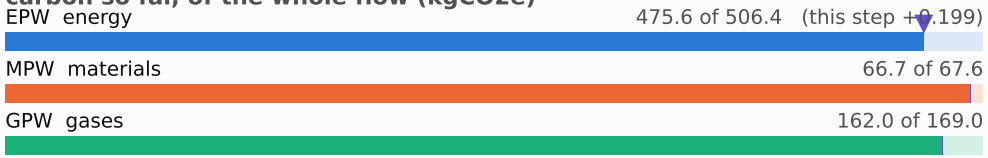
zoomed to fit

this step

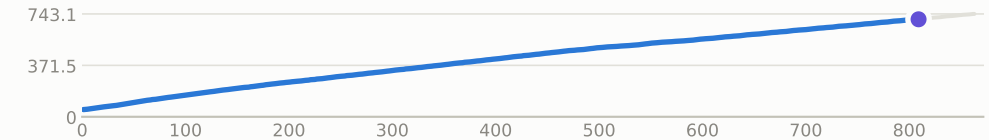
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

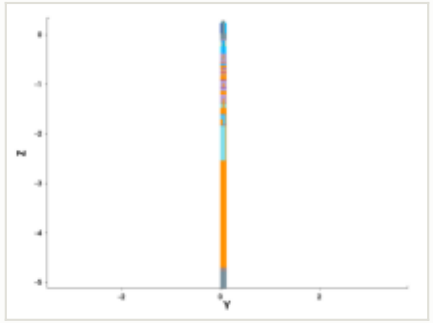
carbon so far, of the whole flow (kgCO2e)



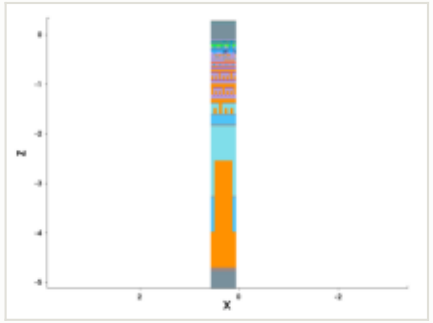
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

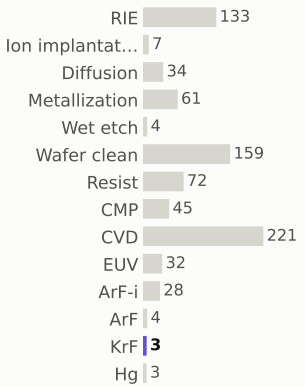


y cut at 0.049 µm

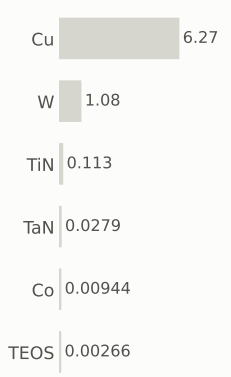


mask: layer 8:0 (GDSII)

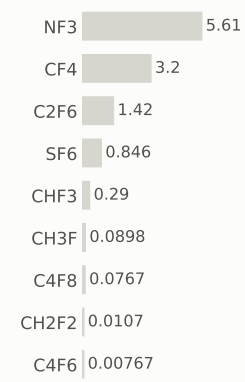
steps by process area



materials deposited (g)



gases emitted (g)



Step #8510: RIE

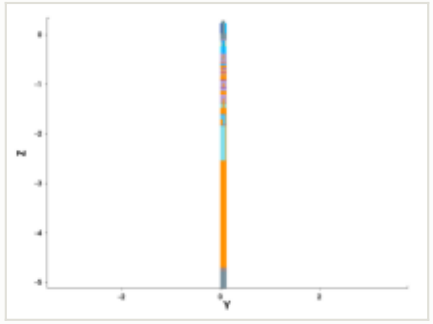
etch_anisotropic · RIE

1.04	705.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

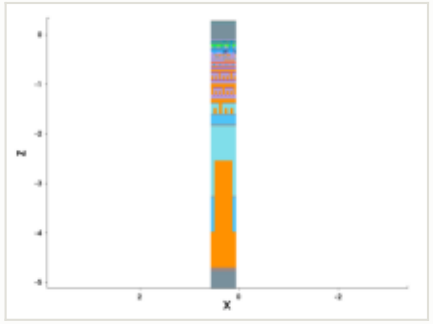
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

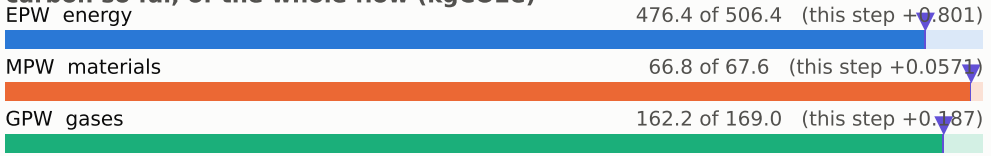
no mask for this step

this step

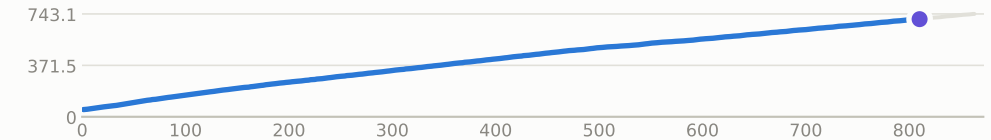
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

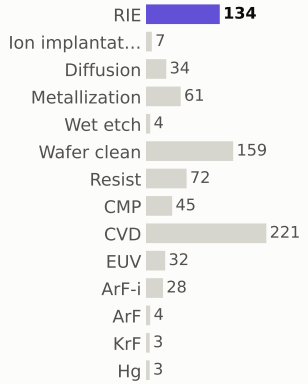
carbon so far, of the whole flow (kgCO2e)



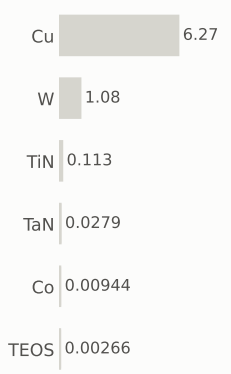
embodied carbon up to each step (kgCO2e per wafer)



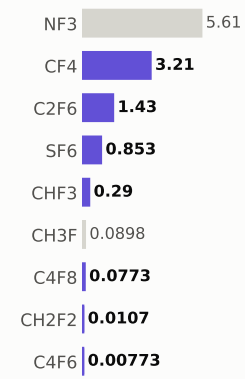
steps by process area



materials deposited (g)



gases emitted (g)



Step #8520: Wafer_clean

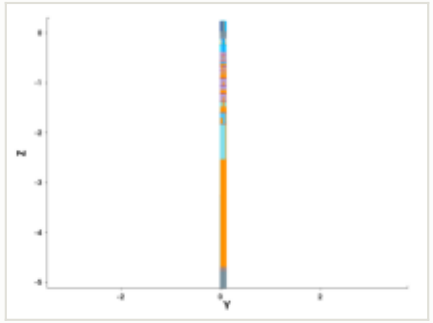
etch_strip · Wafer_clean

0.169	705.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

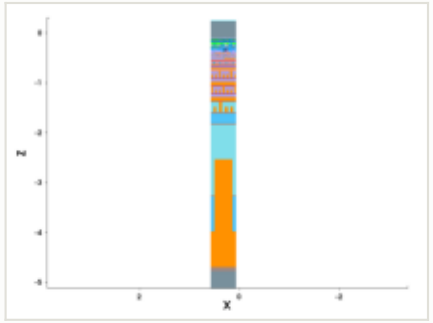
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



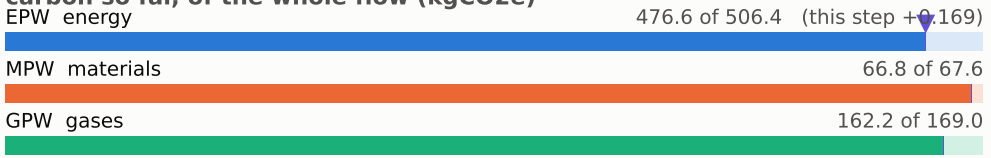
mask for this step

no mask for this step

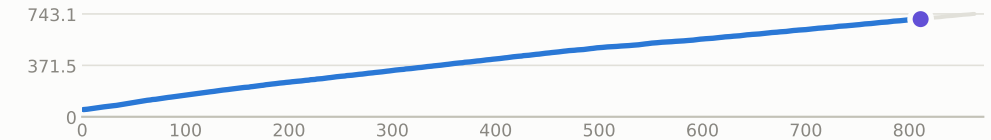
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

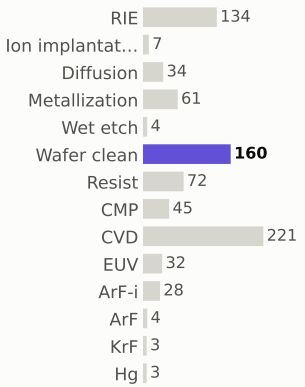
carbon so far, of the whole flow (kgCO2e)



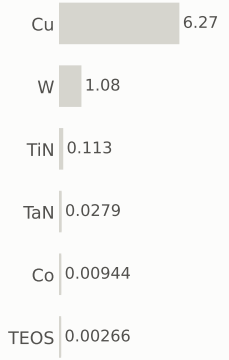
embodied carbon up to each step (kgCO2e per wafer)



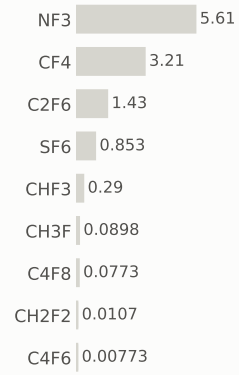
steps by process area



materials deposited (g)



gases emitted (g)



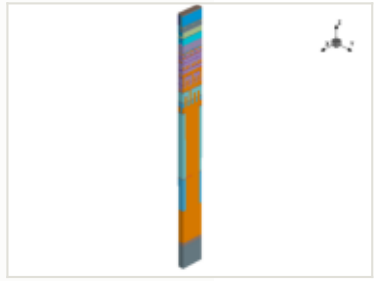
Step #8540: CVD

depo_fill · CVD

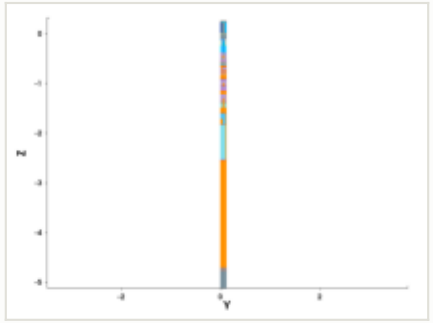
1.31	706.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

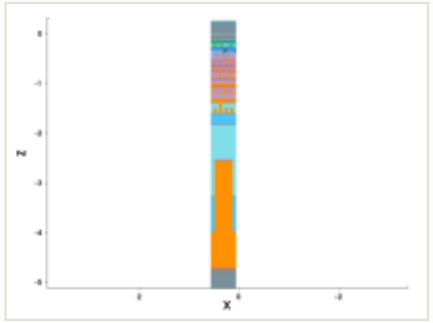
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

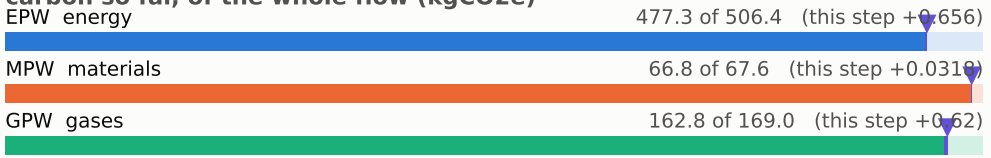
no mask for this step

this step

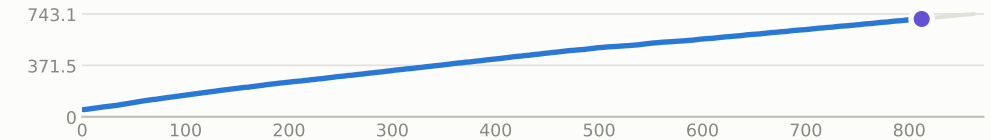
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

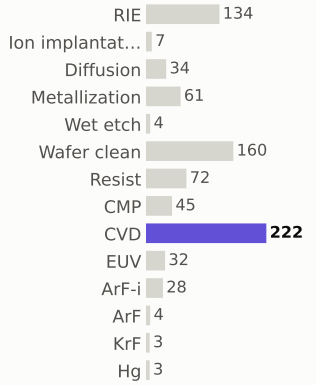
carbon so far, of the whole flow (kgCO2e)



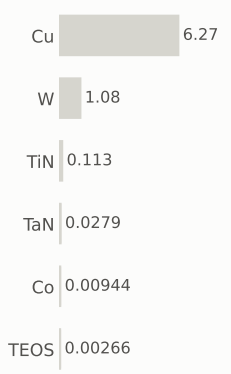
embodied carbon up to each step (kgCO2e per wafer)



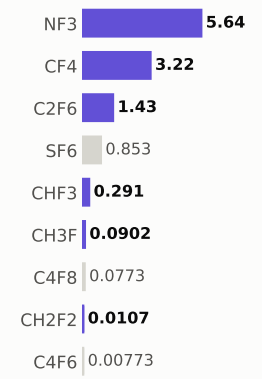
steps by process area



materials deposited (g)



gases emitted (g)



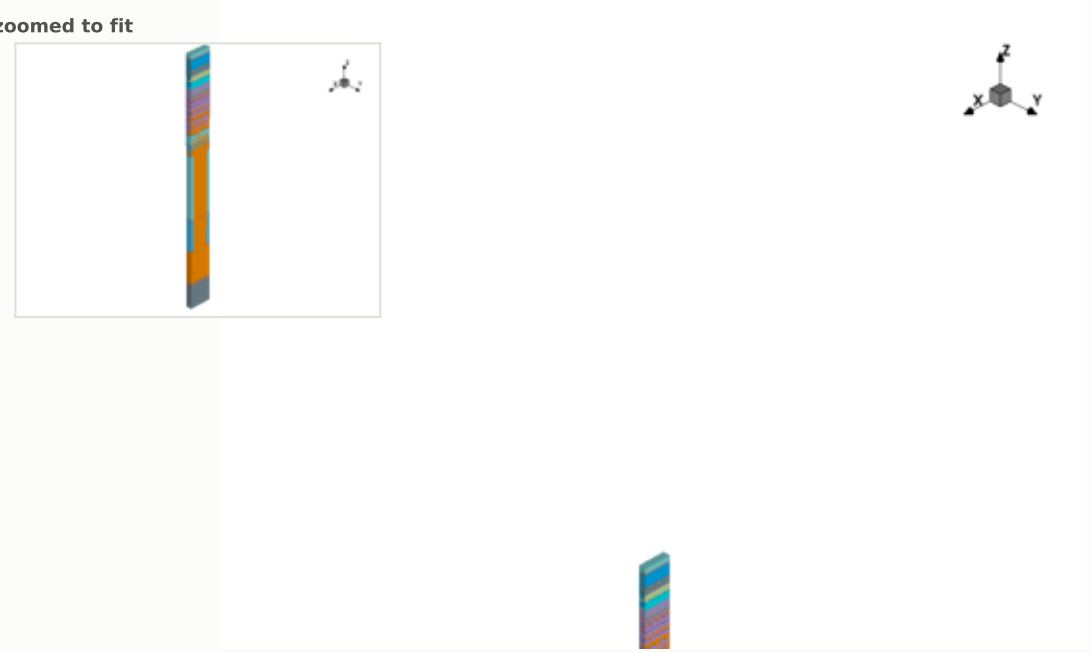
Step #8550: CVD

depo_fill · CVD

1.31	708.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

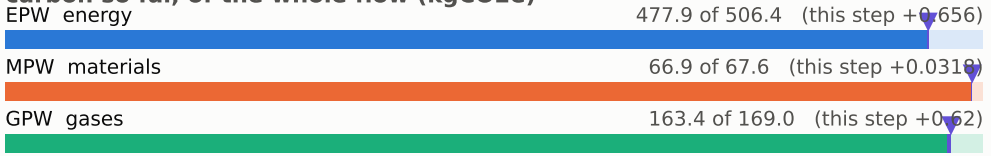
zoomed to fit



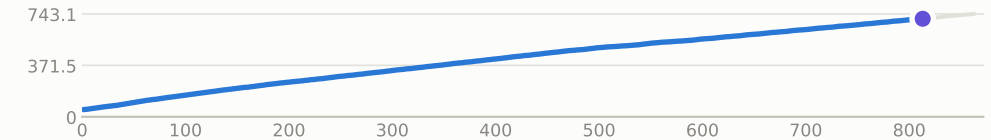
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.092 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

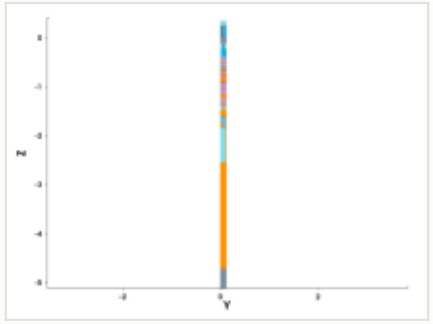
carbon so far, of the whole flow (kgCO2e)



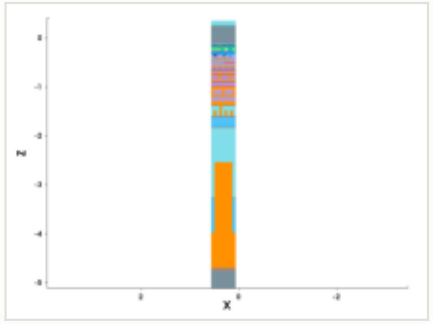
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



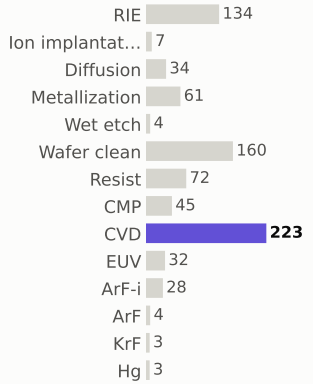
y cut at 0.049 µm



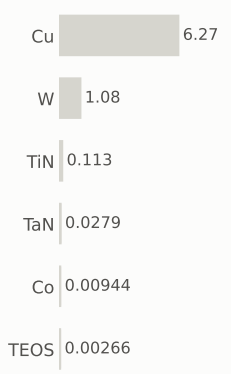
mask for this step

no mask for this step

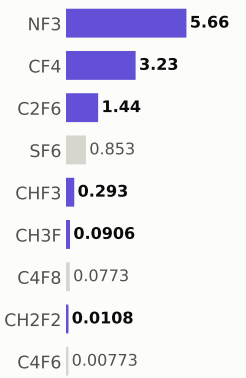
steps by process area



materials deposited (g)



gases emitted (g)



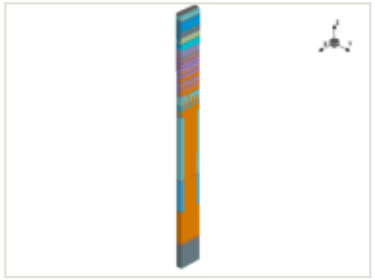
Step #8560: ArF

litho_patterning · ArF

0.199	708.4	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

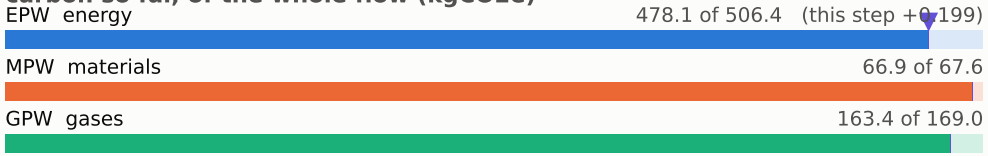
zoomed to fit



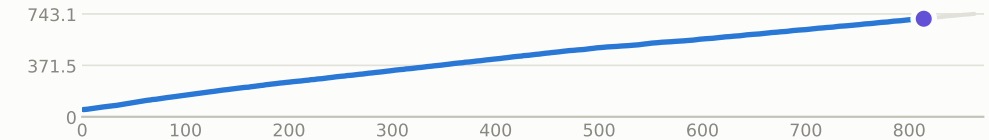
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

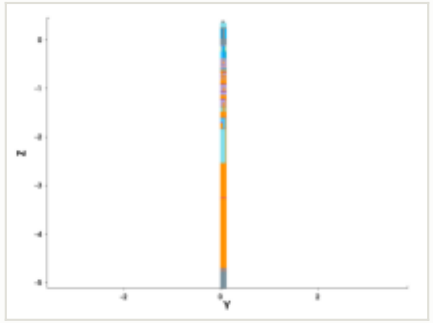
carbon so far, of the whole flow (kgCO2e)



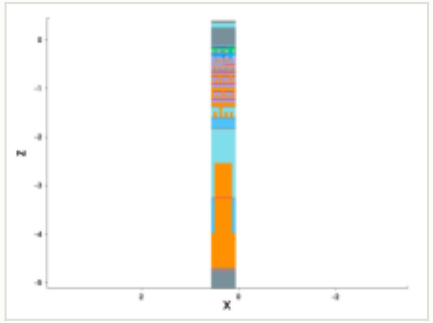
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

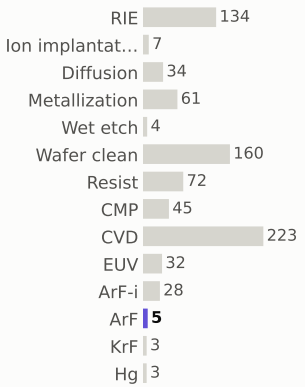


y cut at 0.049 µm

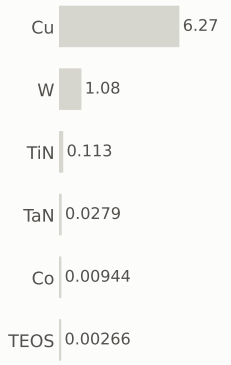


mask: layer 920:0 (GDSII)

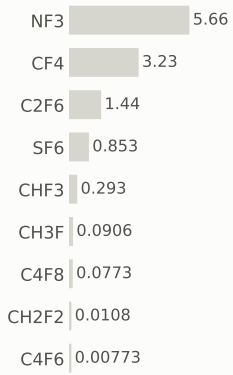
steps by process area



materials deposited (g)



gases emitted (g)



Step #8570: RIE

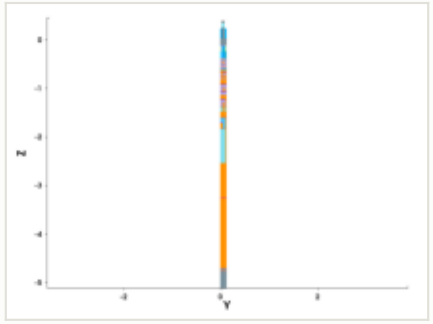
etch_anisotropic · RIE

1.04	709.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

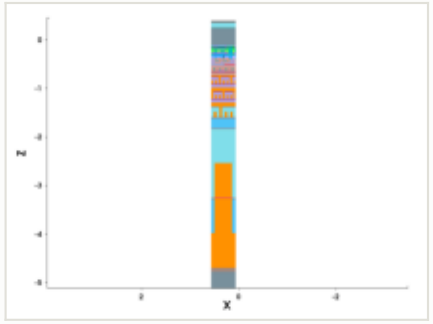
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

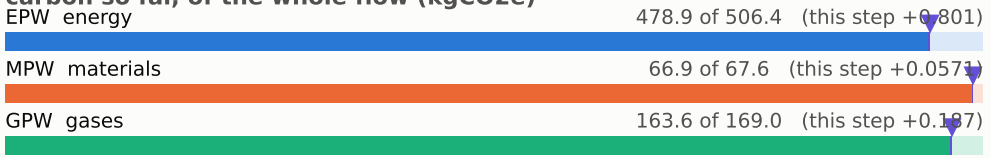
no mask for this step

this step

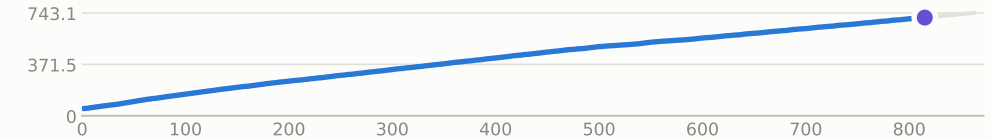
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

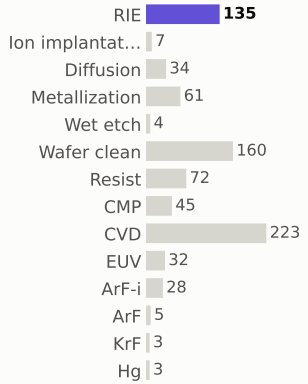
carbon so far, of the whole flow (kgCO2e)



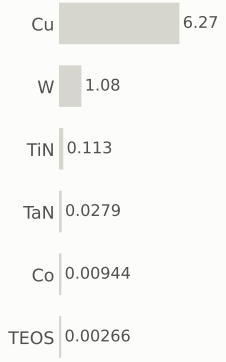
embodied carbon up to each step (kgCO2e per wafer)



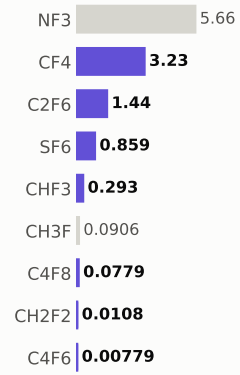
steps by process area



materials deposited (g)



gases emitted (g)



Step #8580: Wafer_clean

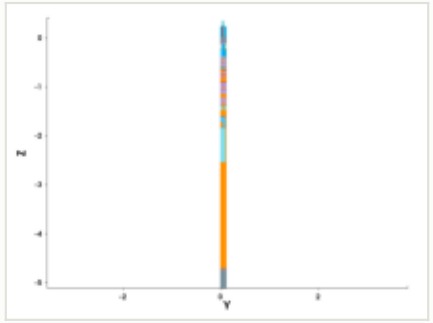
etch_strip · Wafer_clean

0.169	709.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

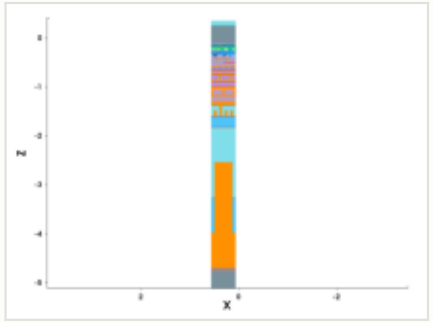
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



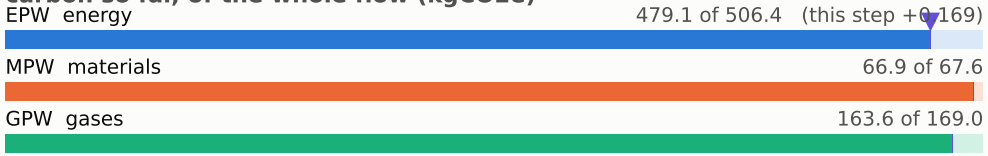
mask for this step

no mask for this step

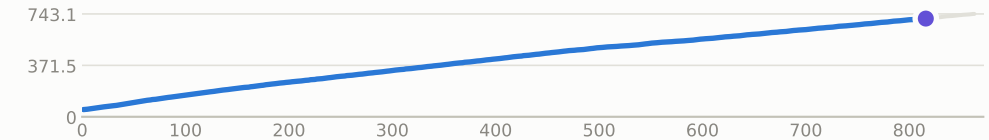
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

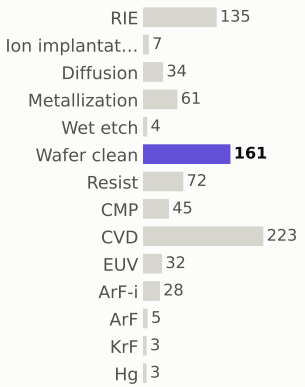
carbon so far, of the whole flow (kgCO2e)



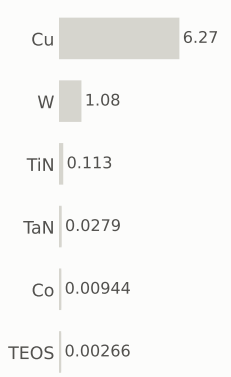
embodied carbon up to each step (kgCO2e per wafer)



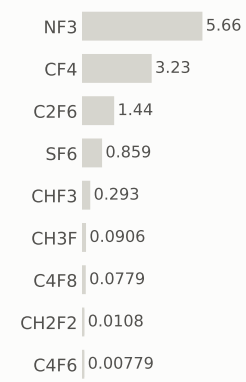
steps by process area



materials deposited (g)



gases emitted (g)



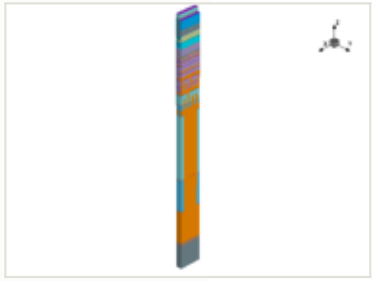
Step #8590: Metallization

depo_ellipsoidal · Metallization

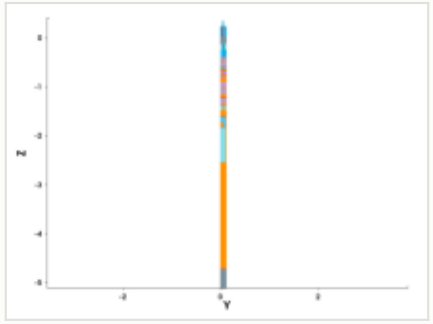
1.2	710.8	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

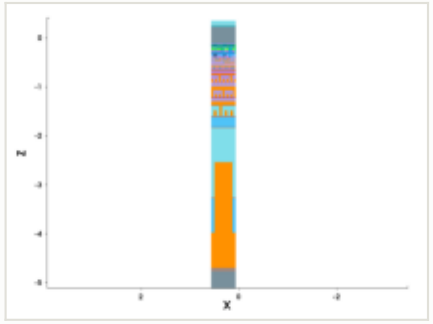
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

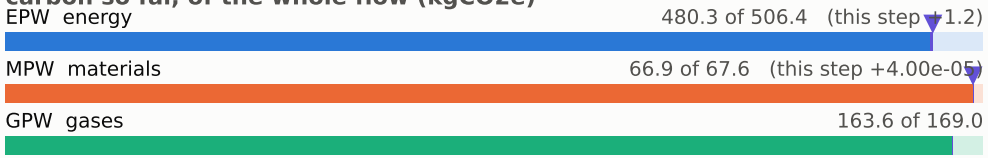
no mask for this step

this step

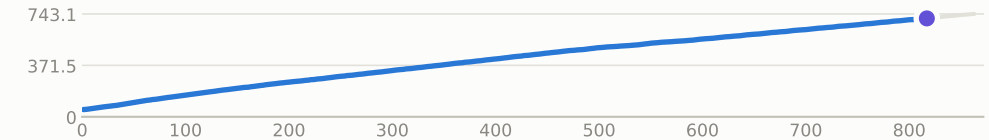
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

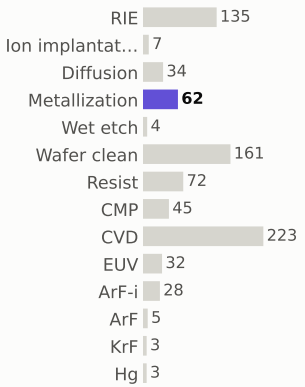
carbon so far, of the whole flow (kgCO2e)



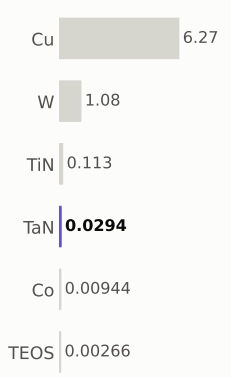
embodied carbon up to each step (kgCO2e per wafer)



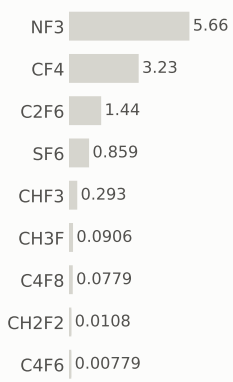
steps by process area



materials deposited (g)



gases emitted (g)



Step #8600: Metallization

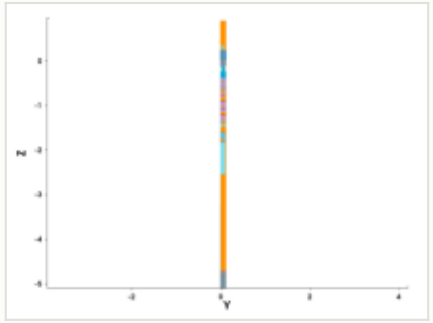
depo_fill · Metallization

1.21	712.0	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

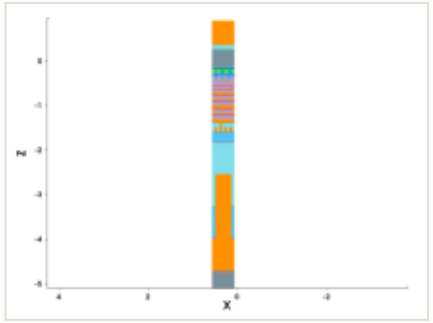
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

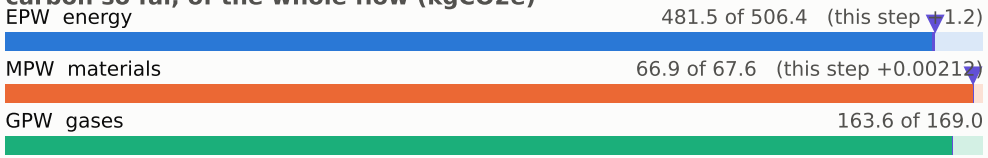
no mask for this step

this step

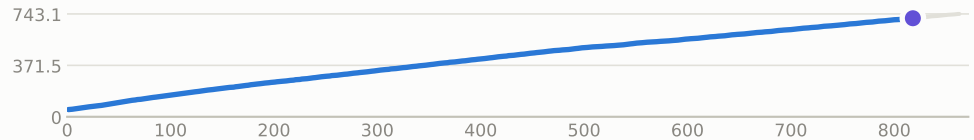
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

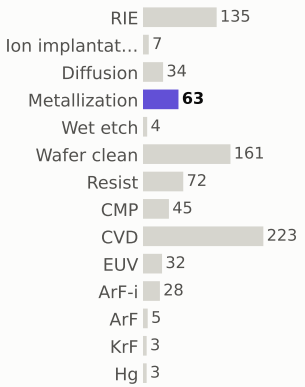
carbon so far, of the whole flow (kgCO2e)



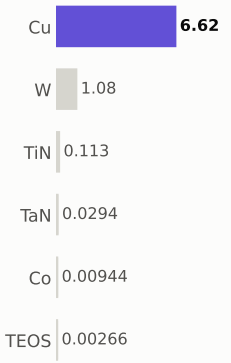
embodied carbon up to each step (kgCO2e per wafer)



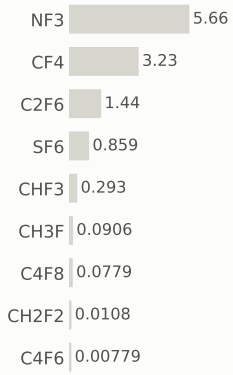
steps by process area



materials deposited (g)



gases emitted (g)



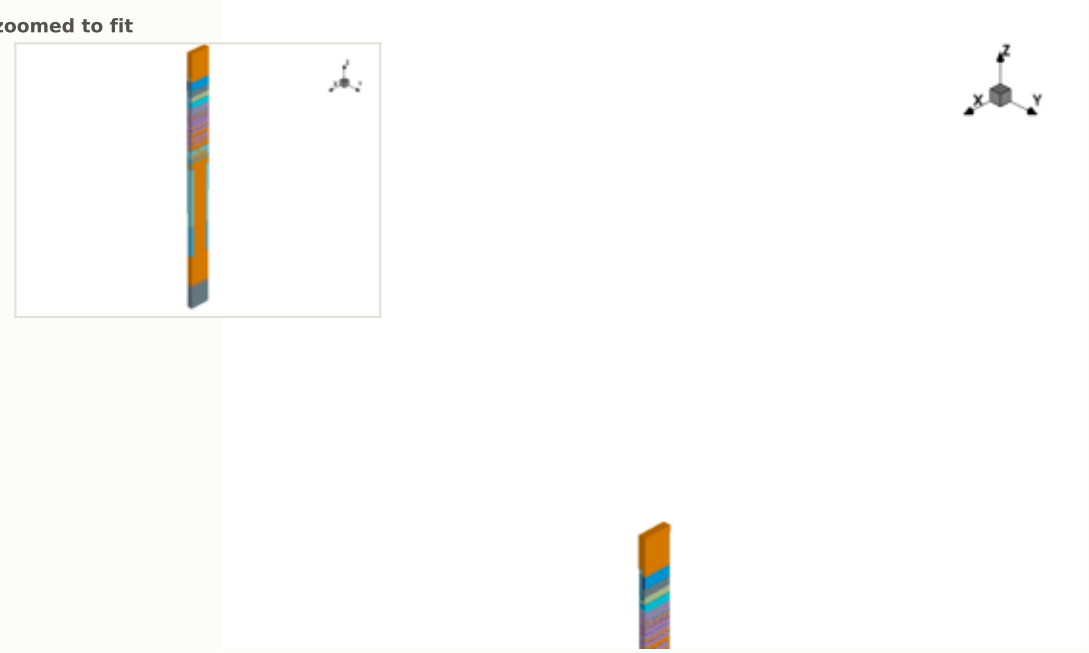
Step #8610: Diffusion

treatment · Diffusion

0.801	712.8	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

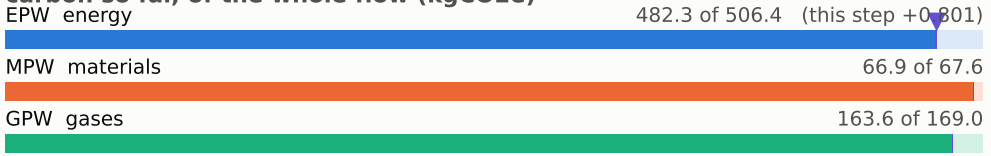


this step

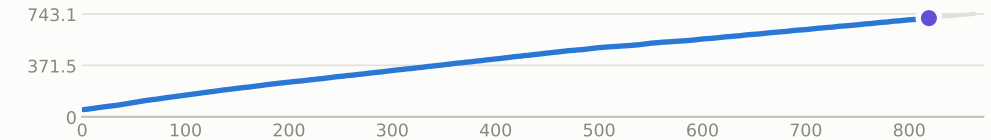
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

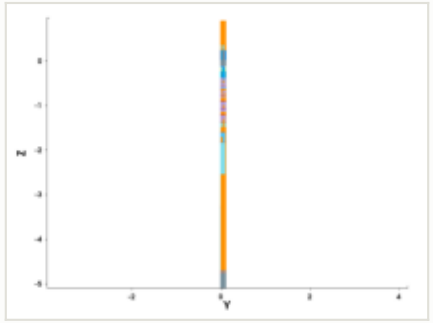
carbon so far, of the whole flow (kgCO2e)



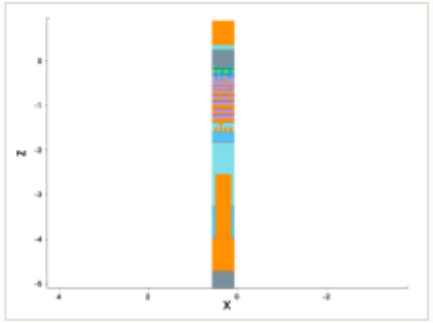
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



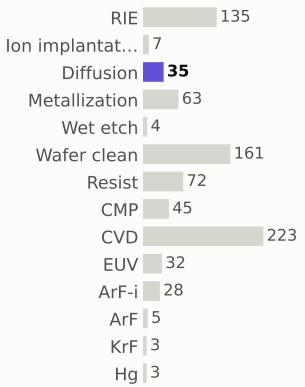
y cut at 0.049 µm



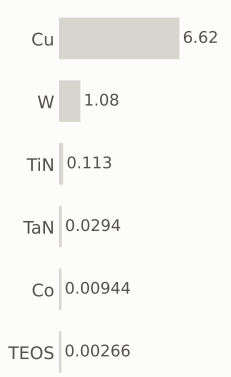
mask for this step

no mask for this step

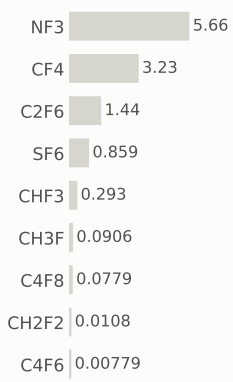
steps by process area



materials deposited (g)



gases emitted (g)



Step #8620: CMP

etch_cmp · CMP

0.169	713.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

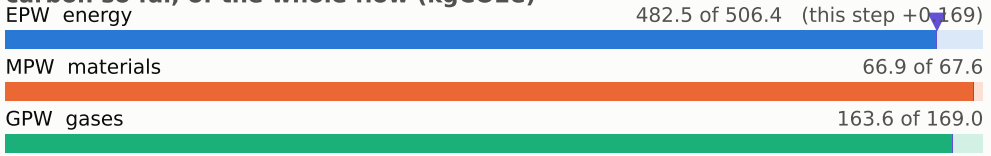
zoomed to fit

this step

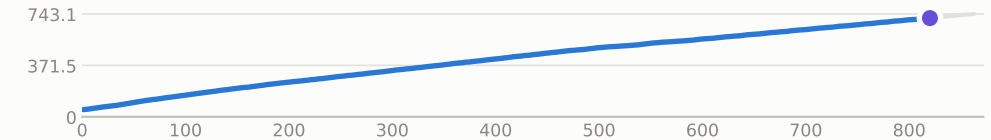
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

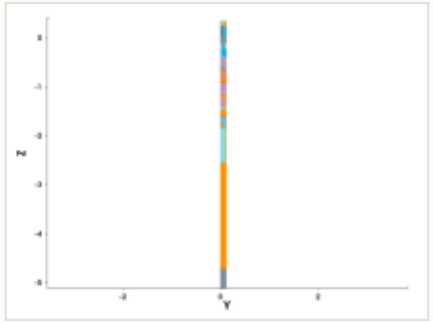
carbon so far, of the whole flow (kgCO2e)



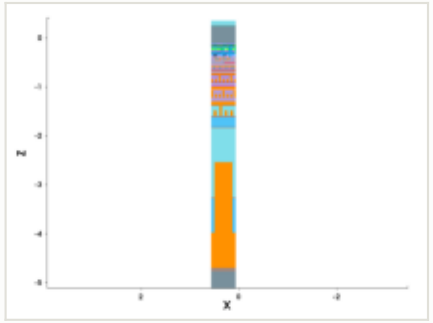
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



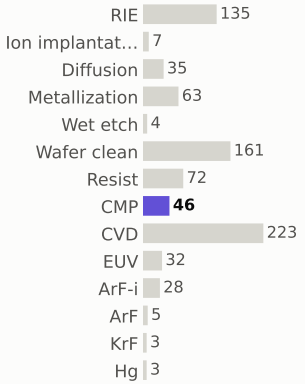
y cut at 0.049 µm



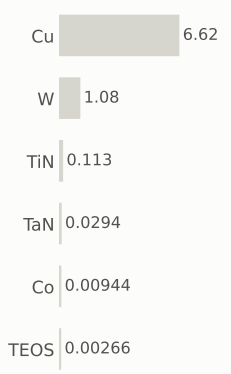
mask for this step

no mask for this step

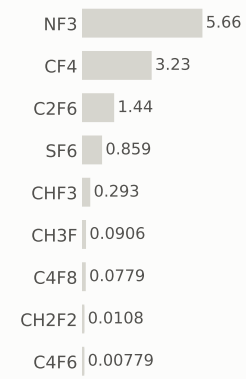
steps by process area



materials deposited (g)



gases emitted (g)



Step #8640: CVD

depo_fill · CVD

1.31	714.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

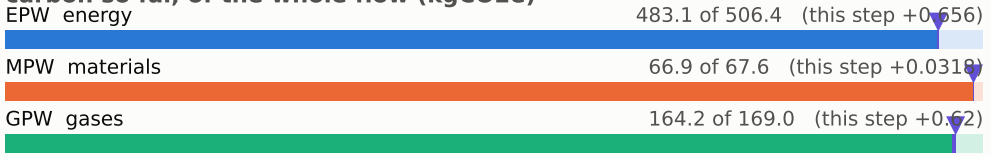
zoomed to fit

this step

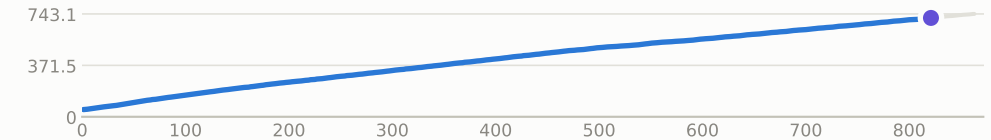
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

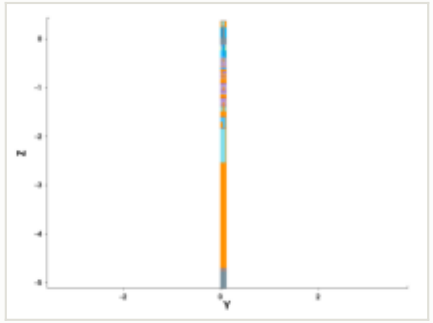
carbon so far, of the whole flow (kgCO2e)



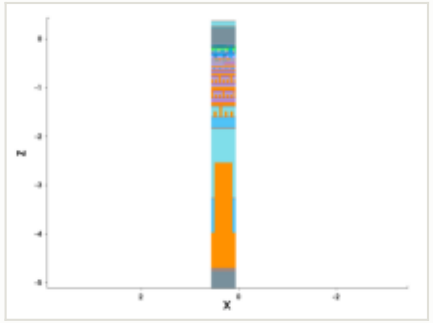
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



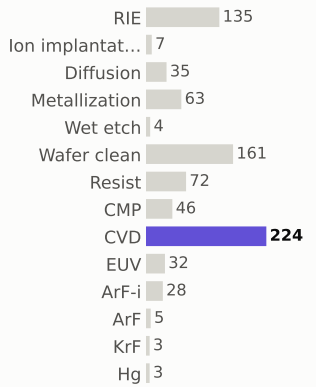
y cut at 0.049 µm



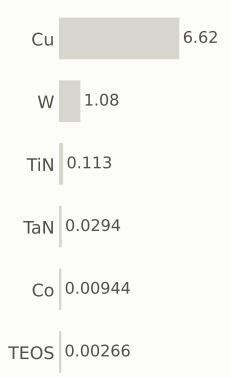
mask for this step

no mask for this step

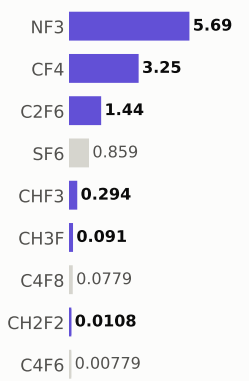
steps by process area



materials deposited (g)



gases emitted (g)



Step #8650: CVD

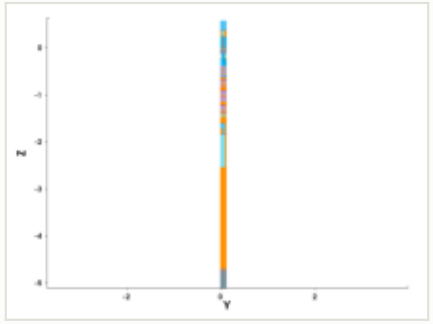
depo_fill · CVD

1.31	715.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

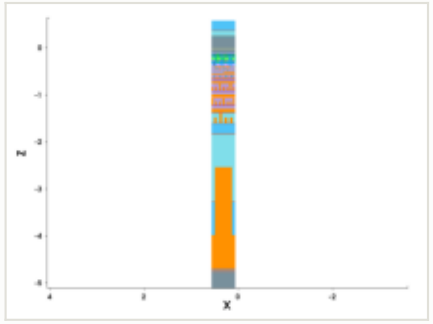
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

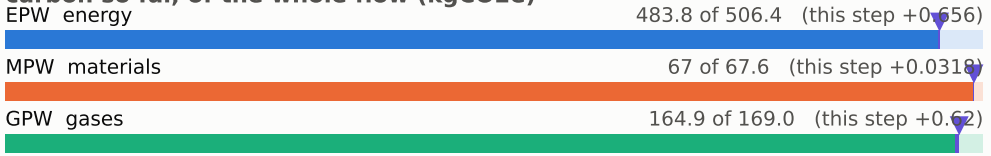
no mask for this step

this step

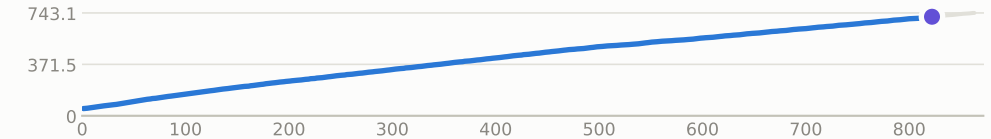
process area	CVD
material	—
thickness	0.204 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

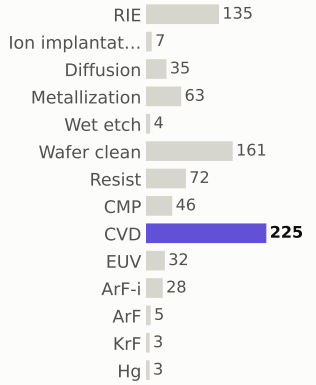
carbon so far, of the whole flow (kgCO2e)



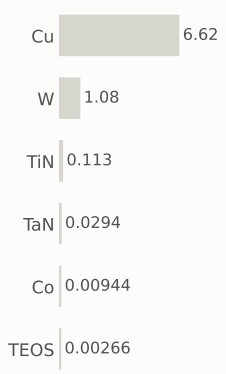
embodied carbon up to each step (kgCO2e per wafer)



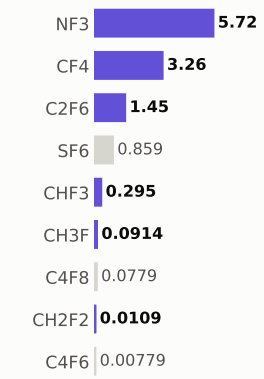
steps by process area



materials deposited (g)



gases emitted (g)



Step #8660: ArF

litho_patterning · ArF

0.199	715.8	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

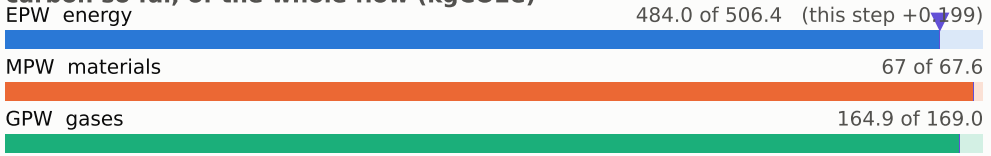
zoomed to fit

this step

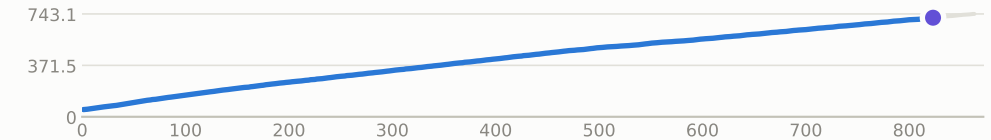
process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

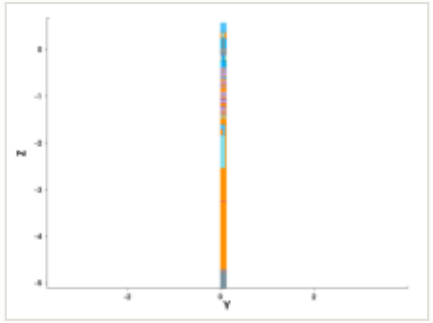
carbon so far, of the whole flow (kgCO2e)



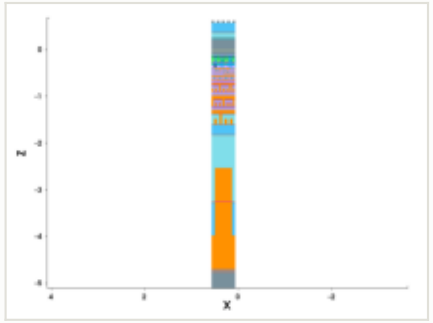
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

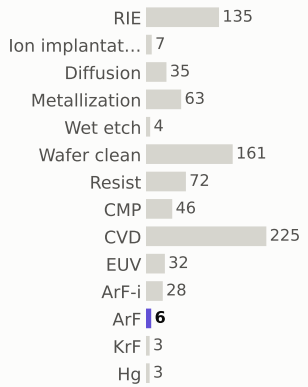


y cut at 0.049 µm

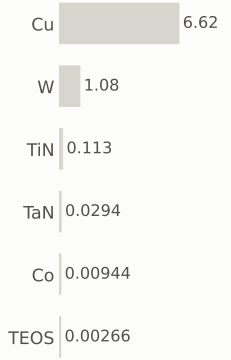


mask: layer 940:0 (GDSII)

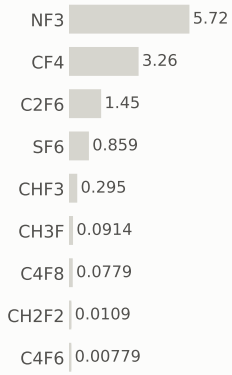
steps by process area



materials deposited (g)



gases emitted (g)



Step #8670: RIE

etch_anisotropic · RIE

1.04	716.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

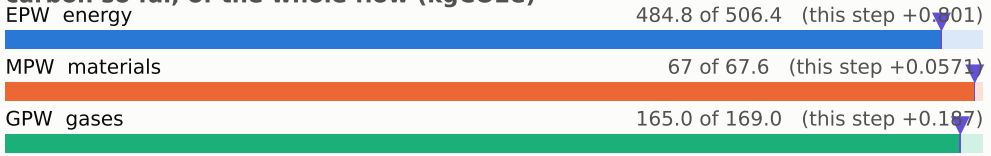
zoomed to fit

this step

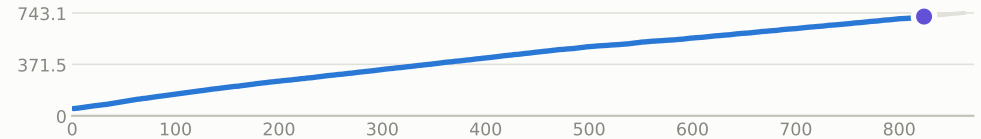
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

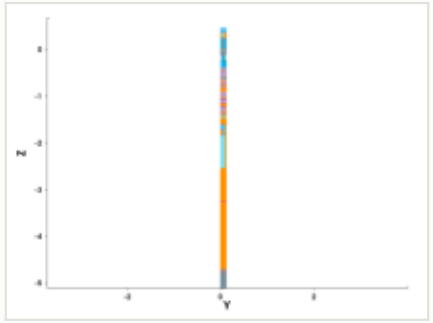
carbon so far, of the whole flow (kgCO2e)



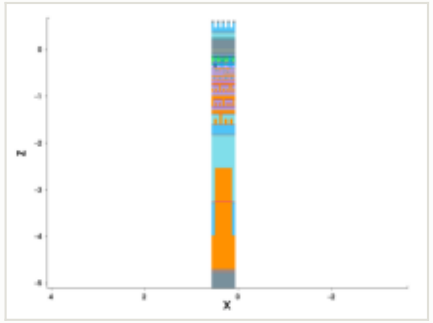
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



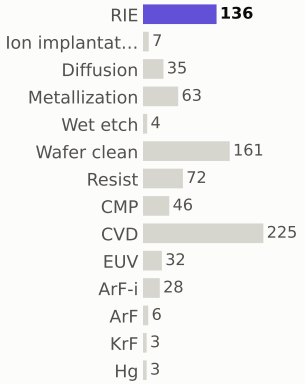
y cut at 0.049 µm



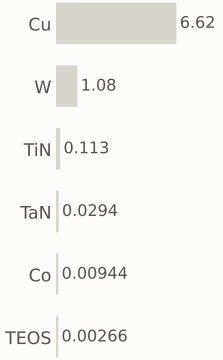
mask for this step

no mask for this step

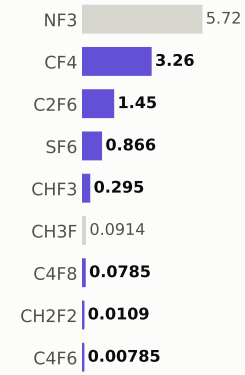
steps by process area



materials deposited (g)



gases emitted (g)



Step #8680: Wafer_clean

etch_strip · Wafer_clean

0.169	717.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

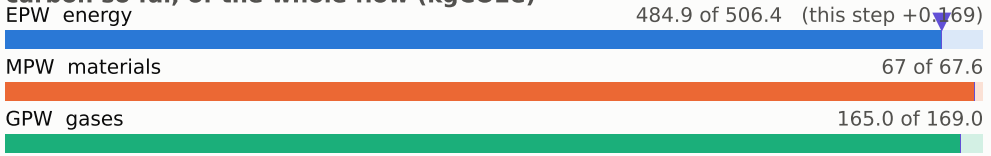
3D structure after this step

zoomed to fit

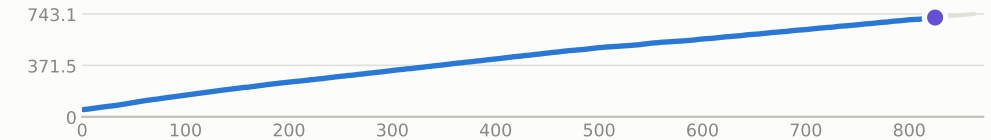
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

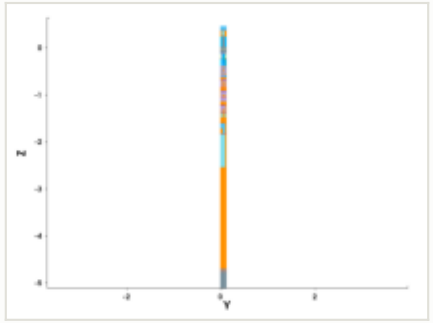
carbon so far, of the whole flow (kgCO2e)



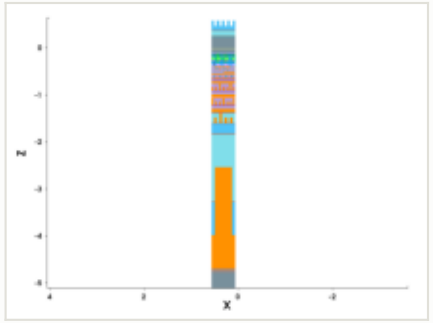
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



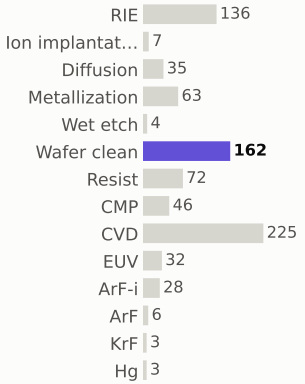
y cut at 0.049 µm



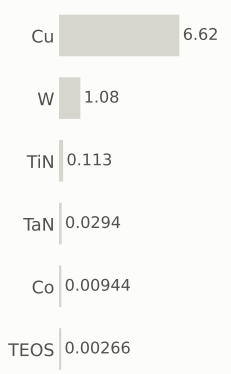
mask for this step

no mask for this step

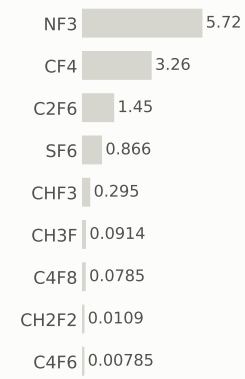
steps by process area



materials deposited (g)



gases emitted (g)



Step #8690: RIE

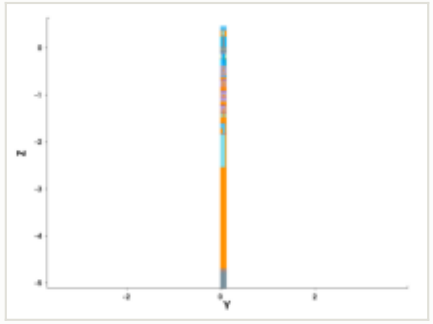
clean · RIE

1.04	718.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

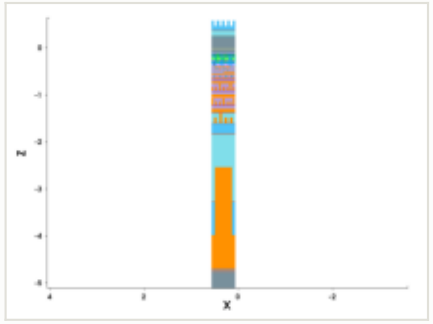
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

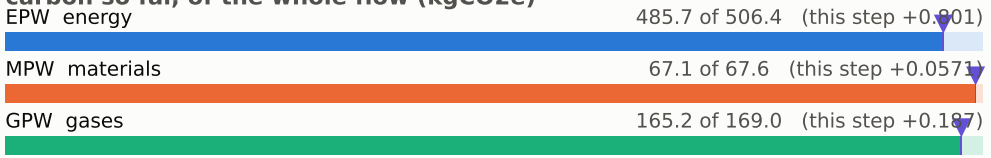
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

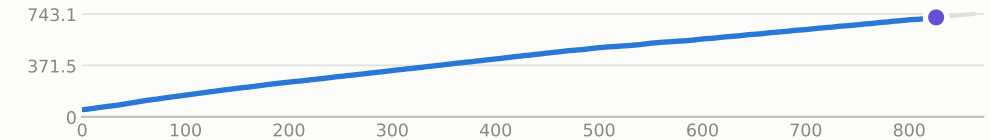
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

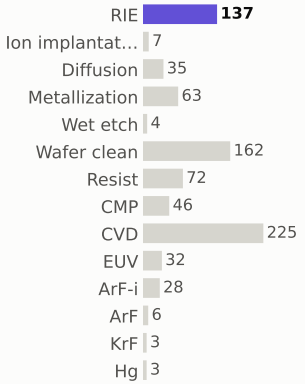
carbon so far, of the whole flow (kgCO2e)



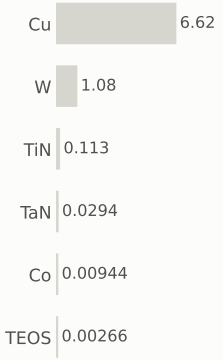
embodied carbon up to each step (kgCO2e per wafer)



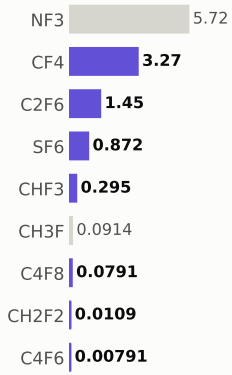
steps by process area



materials deposited (g)



gases emitted (g)



Step #8700: Wafer_clean

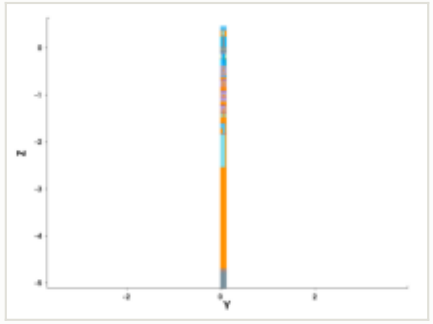
clean · Wafer_clean

0.169	718.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

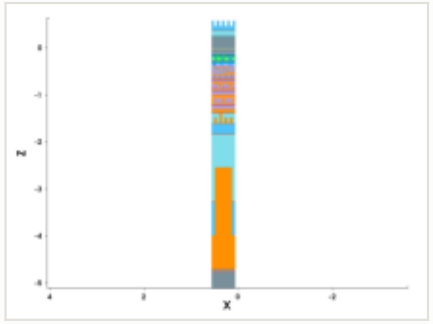
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

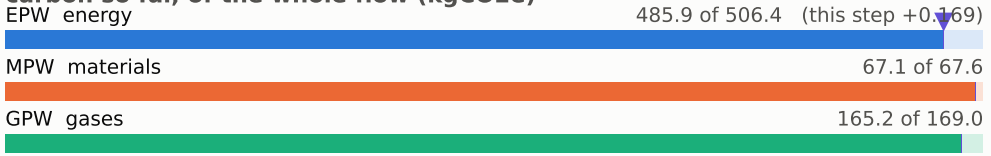
no mask for this step

this step

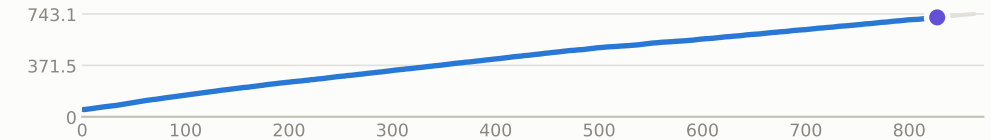
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

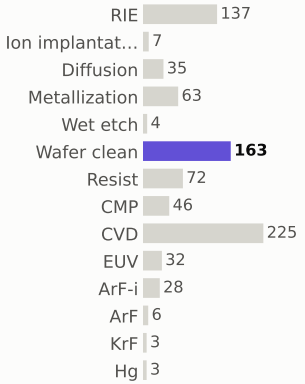
carbon so far, of the whole flow (kgCO2e)



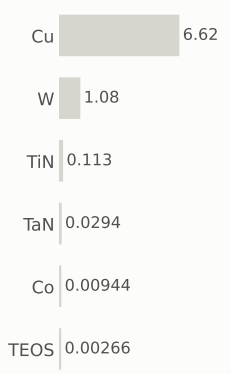
embodied carbon up to each step (kgCO2e per wafer)



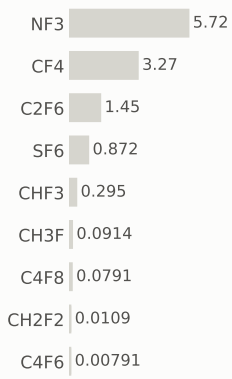
steps by process area



materials deposited (g)



gases emitted (g)



Step #8710: ArF

litho_patterning · ArF

0.199	718.4	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

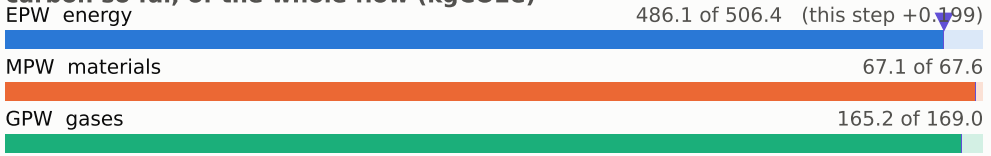
3D structure after this step

zoomed to fit

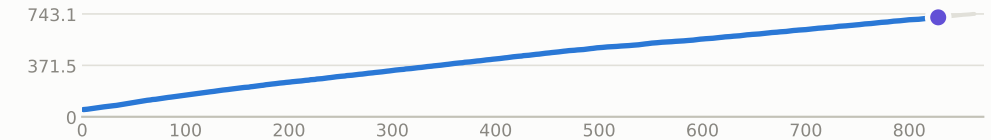
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

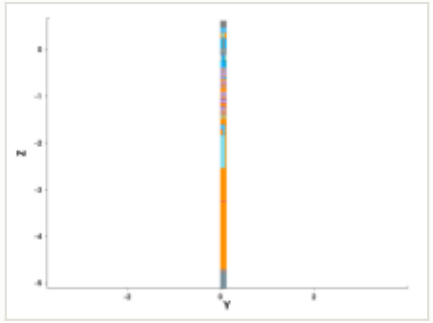
carbon so far, of the whole flow (kgCO2e)



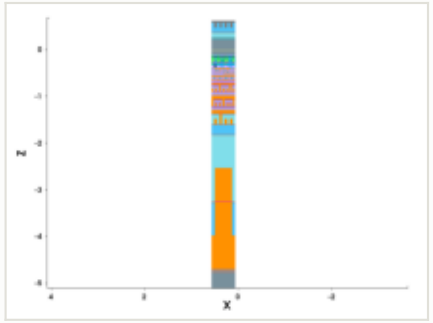
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

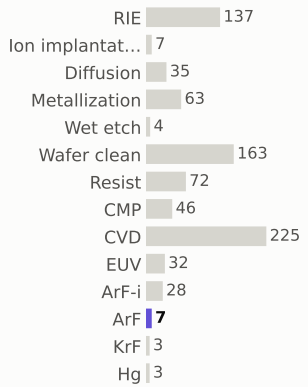


y cut at 0.049 µm

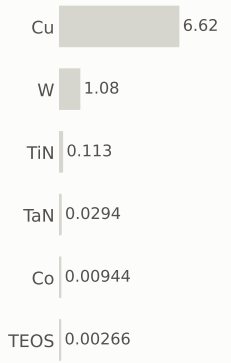


mask: layer 930:0 (GDSII)

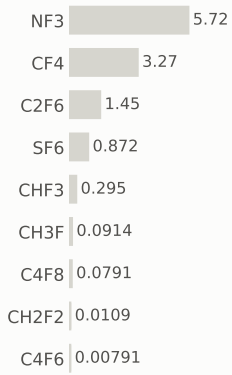
steps by process area



materials deposited (g)



gases emitted (g)



Step #8720: RIE

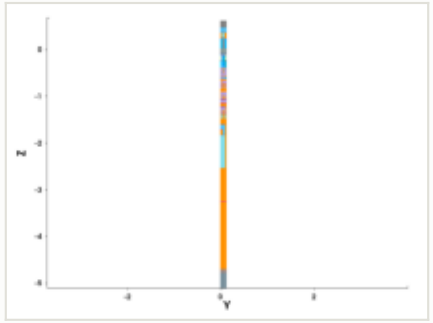
etch_anisotropic · RIE

1.04	719.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

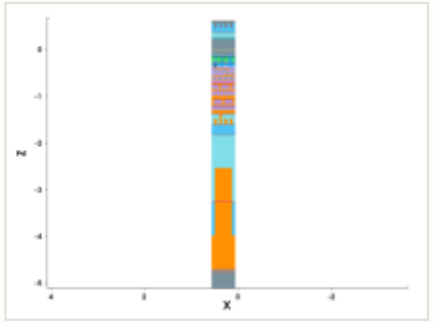
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

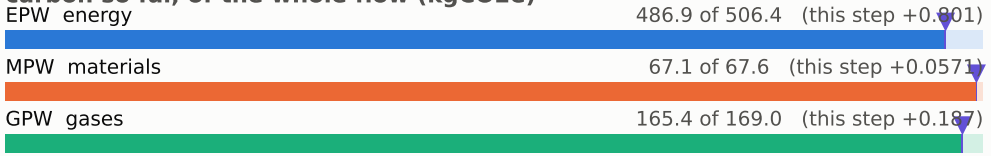
no mask for this step

this step

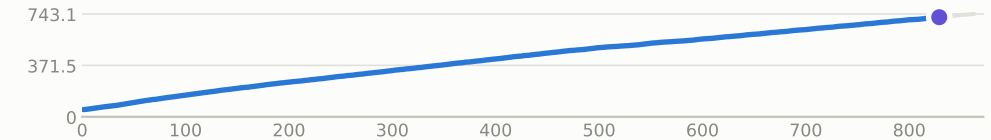
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

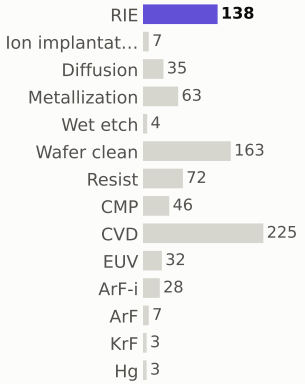
carbon so far, of the whole flow (kgCO2e)



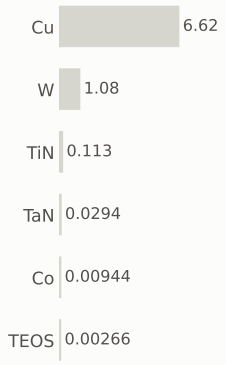
embodied carbon up to each step (kgCO2e per wafer)



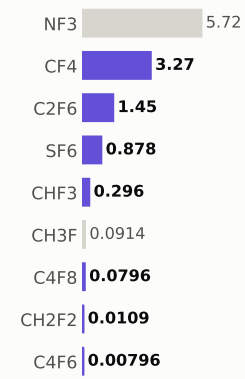
steps by process area



materials deposited (g)



gases emitted (g)



Step #8730: Wafer_clean

etch_strip · Wafer_clean

0.169	719.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

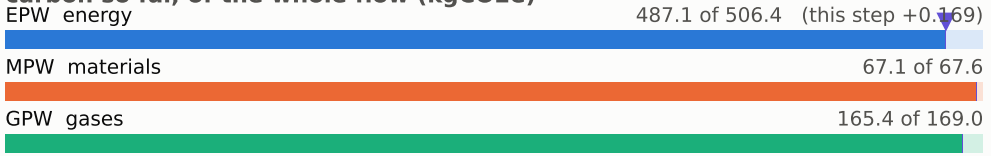
3D structure after this step

zoomed to fit

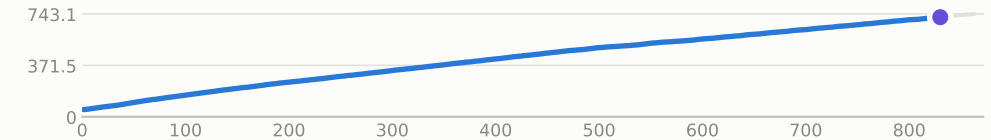
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

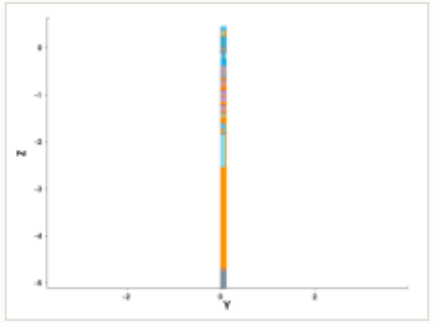
carbon so far, of the whole flow (kgCO2e)



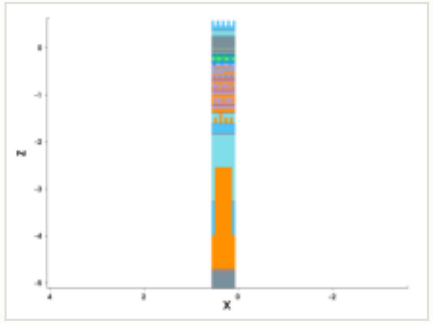
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



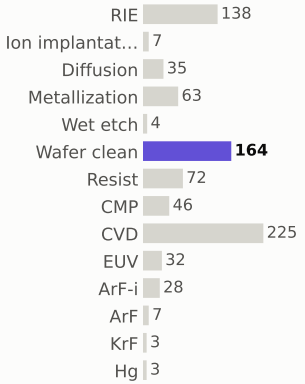
y cut at 0.049 µm



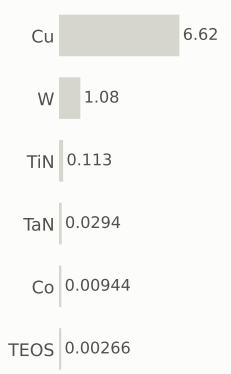
mask for this step

no mask for this step

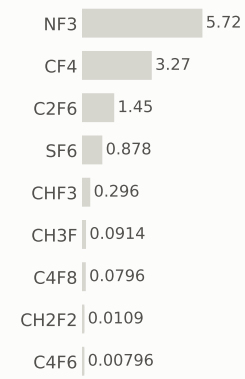
steps by process area



materials deposited (g)



gases emitted (g)



Step #8740: Metallization

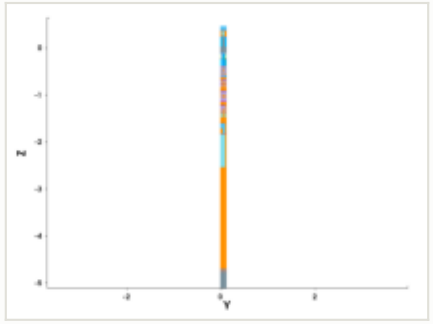
depo_ellipsoidal · Metallization

1.2	720.8	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

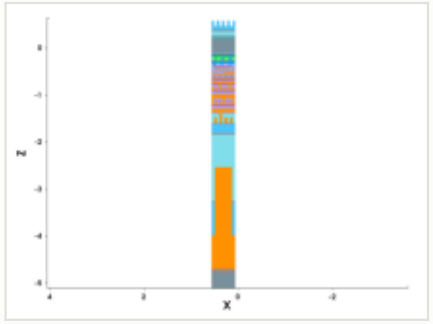
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

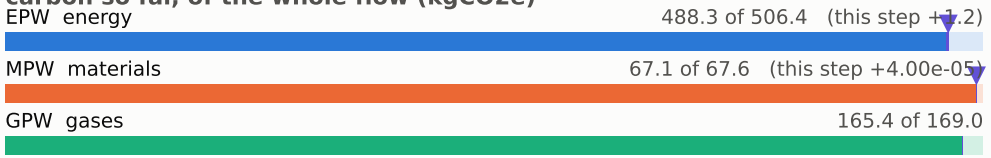
no mask for this step

this step

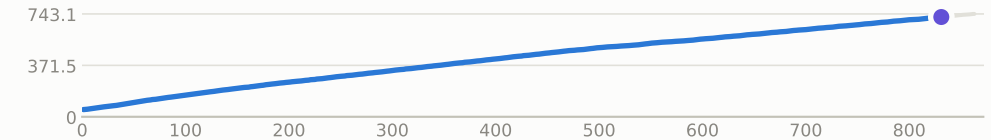
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

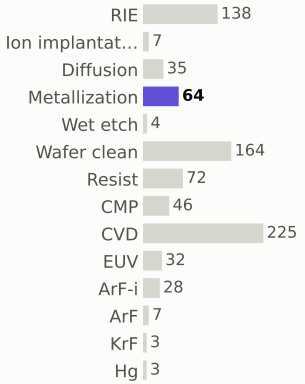
carbon so far, of the whole flow (kgCO2e)



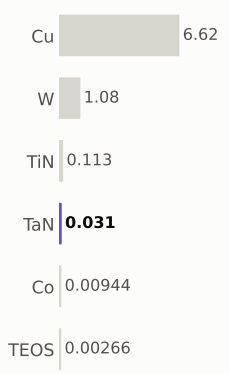
embodied carbon up to each step (kgCO2e per wafer)



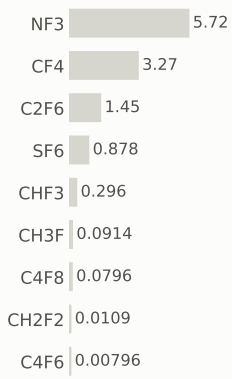
steps by process area



materials deposited (g)



gases emitted (g)



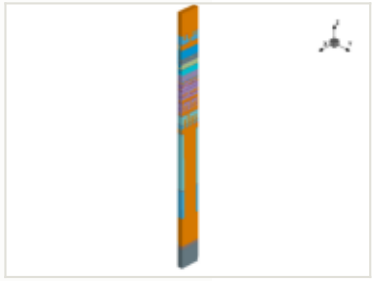
Step #8750: Metallization

depo_fill · Metallization

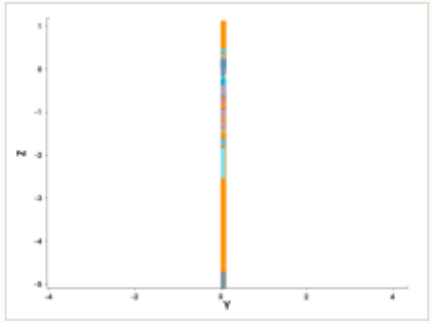
1.21	722.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

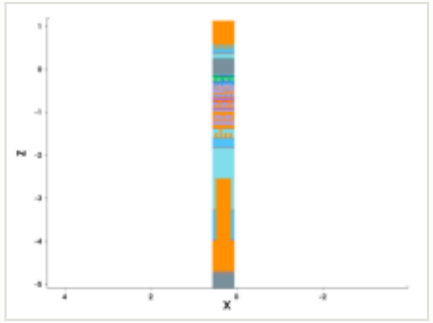
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

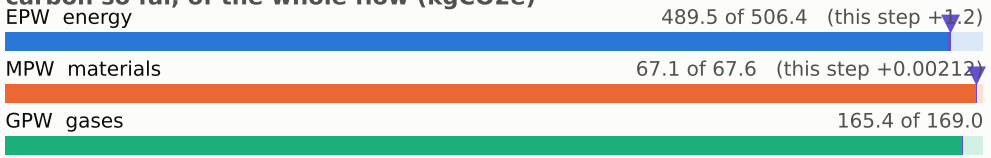
no mask for this step

this step

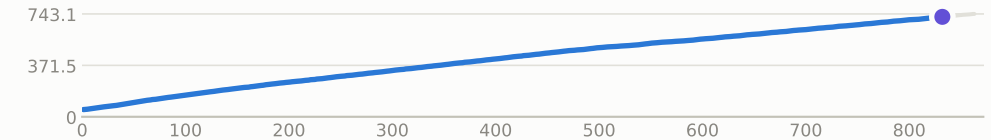
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

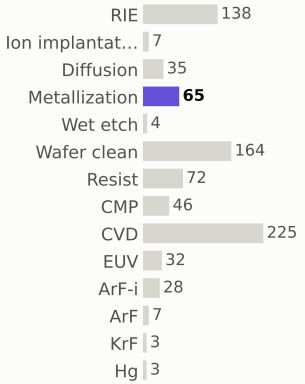
carbon so far, of the whole flow (kgCO2e)



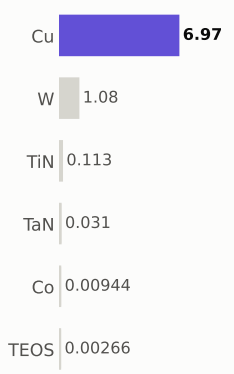
embodied carbon up to each step (kgCO2e per wafer)



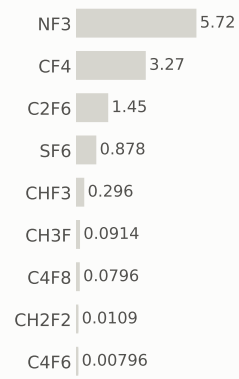
steps by process area



materials deposited (g)



gases emitted (g)



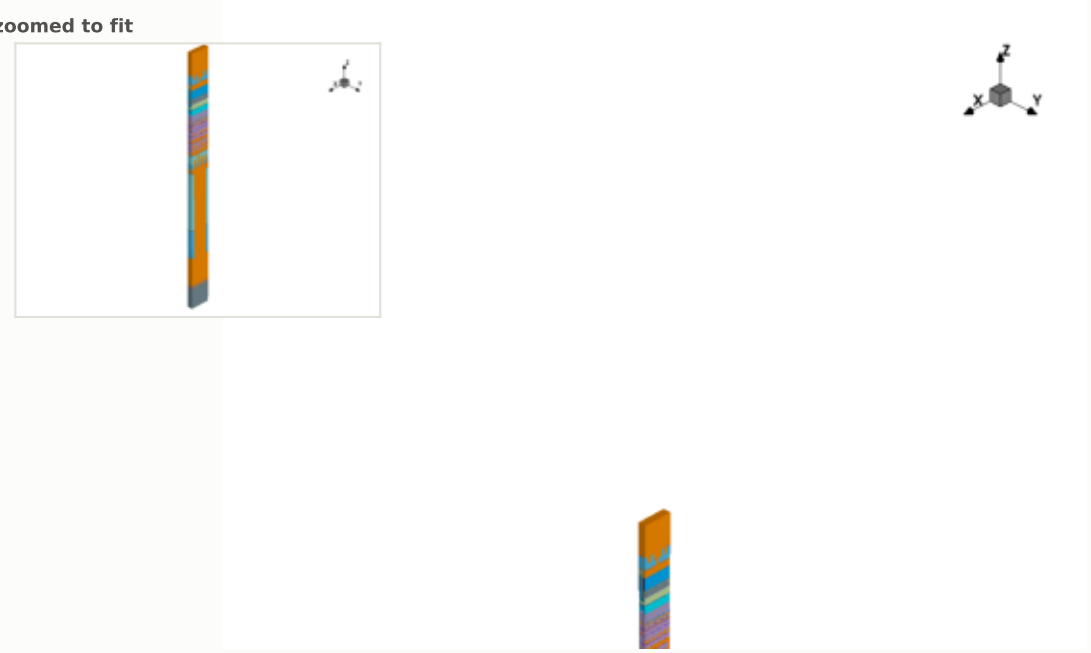
Step #8760: Diffusion

treatment · Diffusion

0.801	722.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

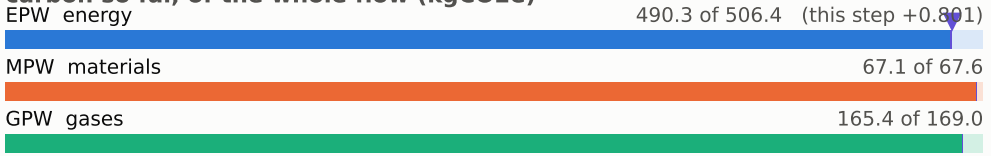


this step

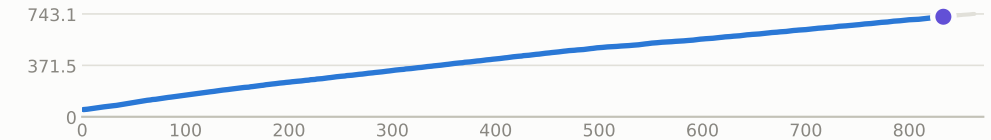
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

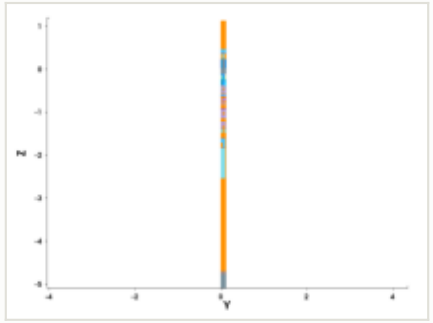
carbon so far, of the whole flow (kgCO2e)



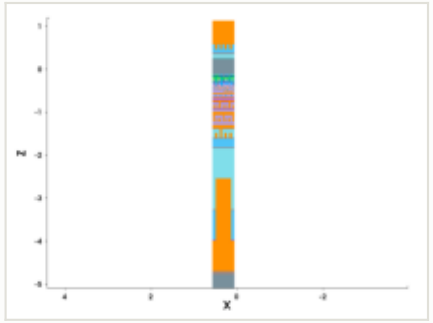
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



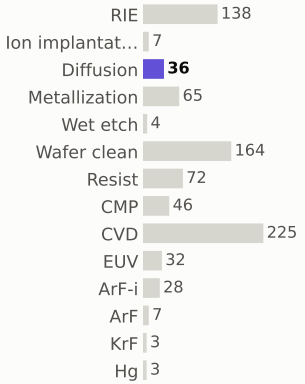
y cut at 0.049 µm



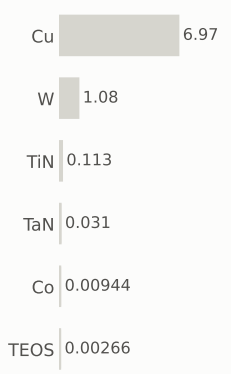
mask for this step

no mask for this step

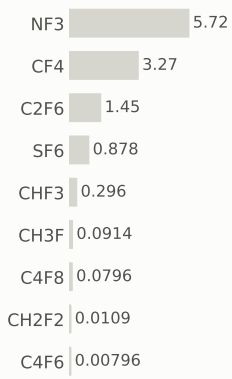
steps by process area



materials deposited (g)



gases emitted (g)



Step #8770: CMP

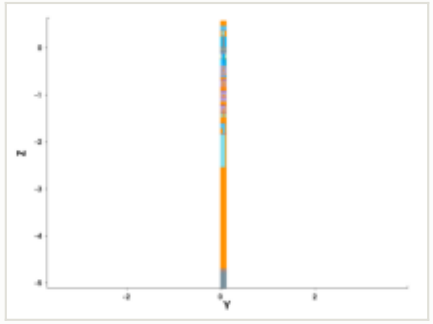
etch_cmp · CMP

0.169	723.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

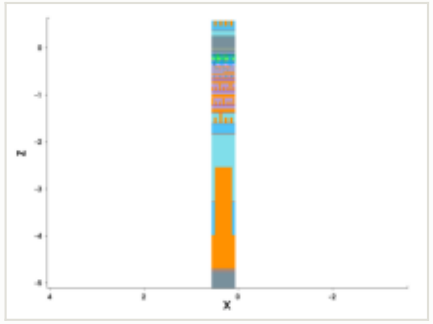
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

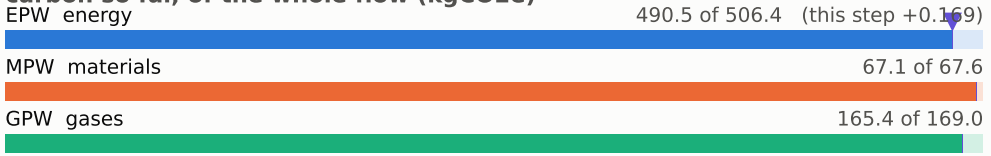
no mask for this step

this step

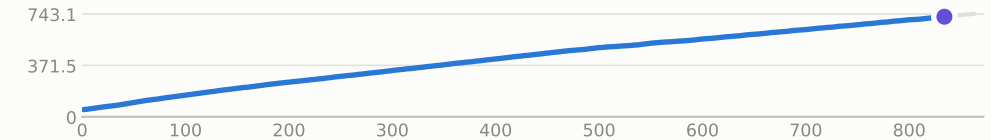
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

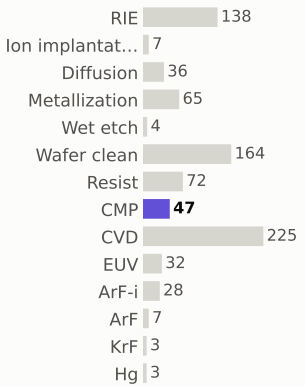
carbon so far, of the whole flow (kgCO2e)



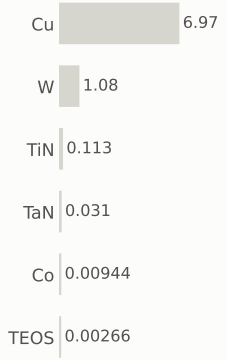
embodied carbon up to each step (kgCO2e per wafer)



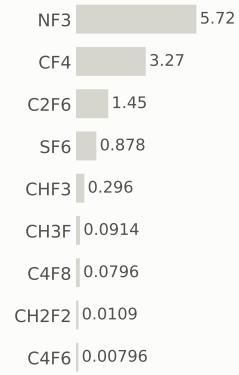
steps by process area



materials deposited (g)



gases emitted (g)



Step #8790: CVD

depo_fill · CVD

1.31	724.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

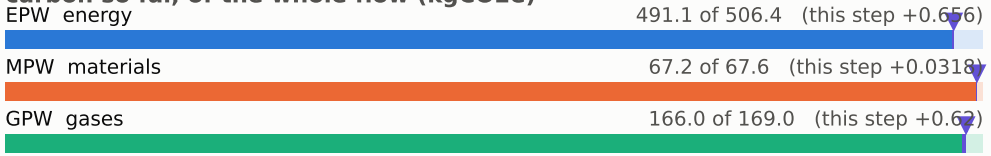
zoomed to fit

this step

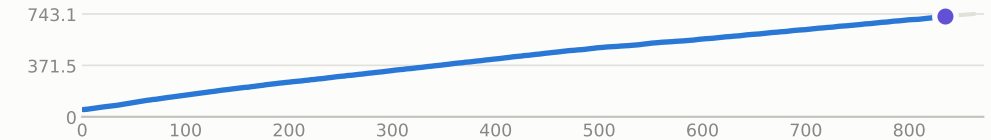
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

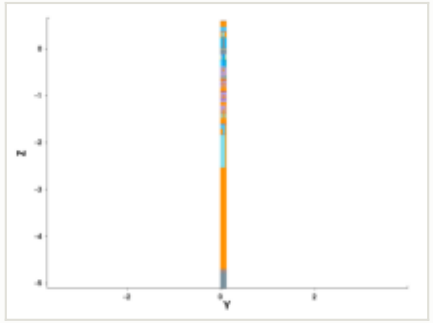
carbon so far, of the whole flow (kgCO2e)



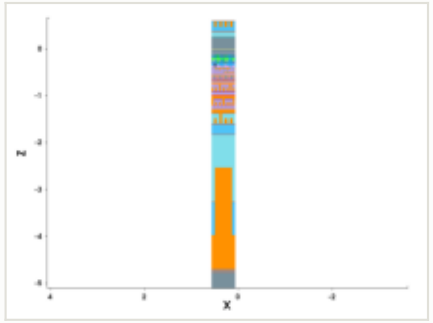
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



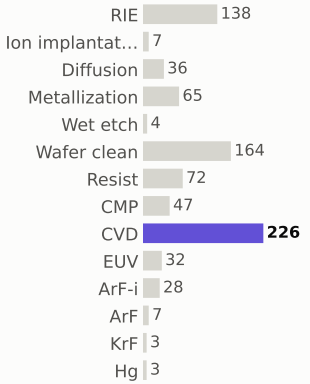
y cut at 0.049 µm



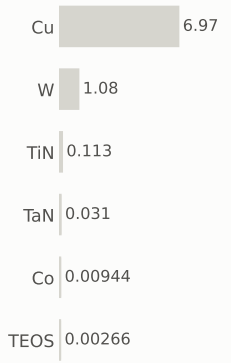
mask for this step

no mask for this step

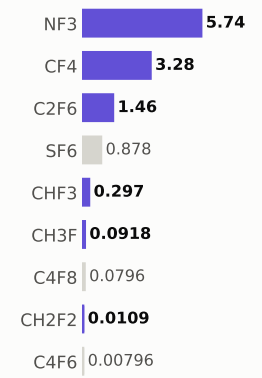
steps by process area



materials deposited (g)



gases emitted (g)



Step #8800: CVD

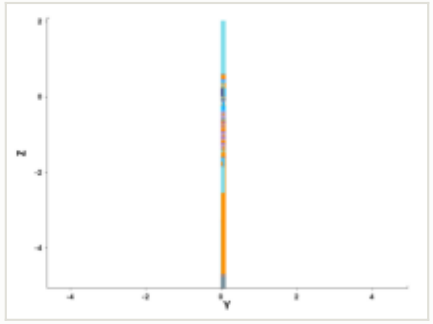
depo_fill · CVD

1.31	725.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

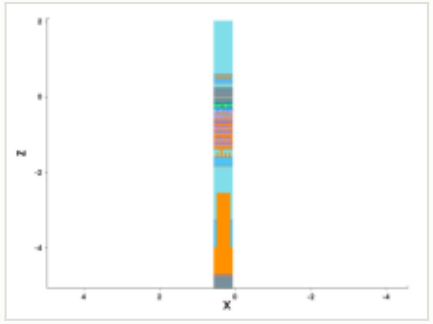
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

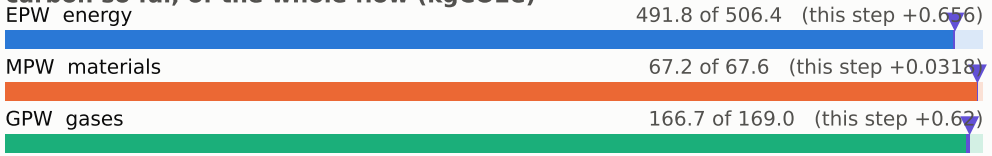
no mask for this step

this step

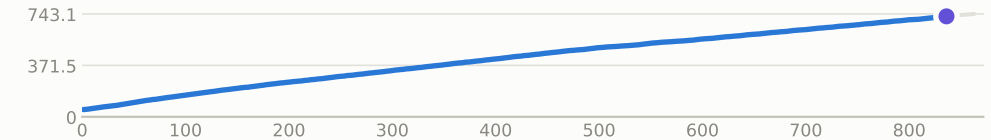
process area	CVD
material	—
thickness	1.42 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

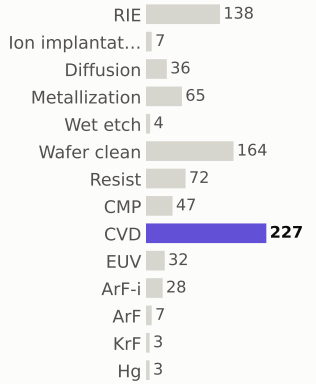
carbon so far, of the whole flow (kgCO2e)



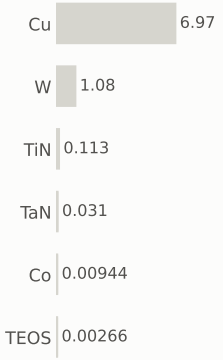
embodied carbon up to each step (kgCO2e per wafer)



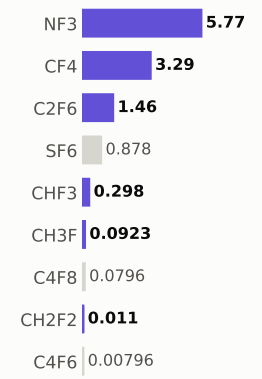
steps by process area



materials deposited (g)



gases emitted (g)



Step #8810: Hg

litho_patterning · Hg

0.199	725.8	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

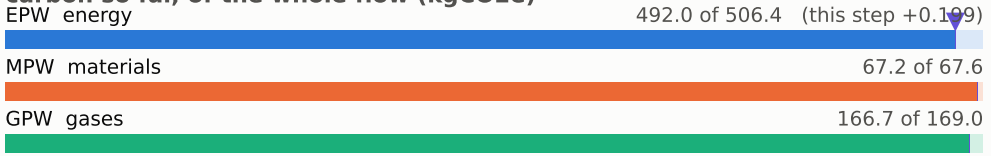
zoomed to fit

this step

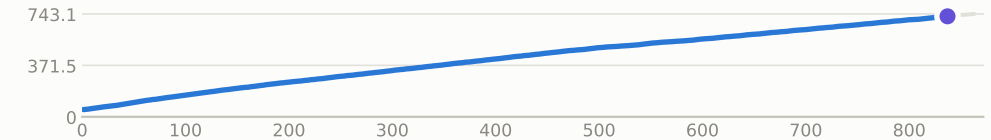
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

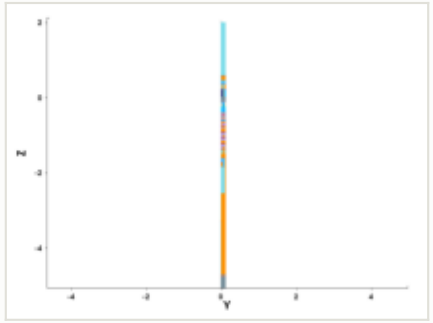
carbon so far, of the whole flow (kgCO2e)



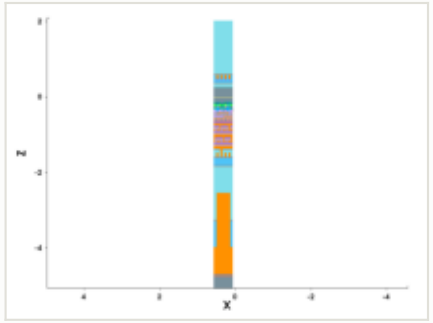
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

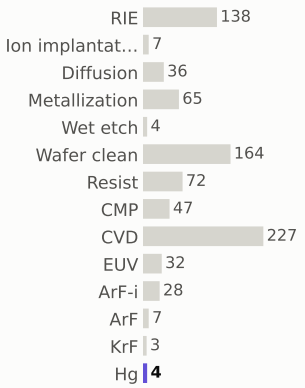


y cut at 0.049 µm

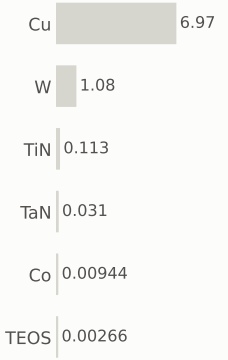


mask: layer 960:0 (GDSII)

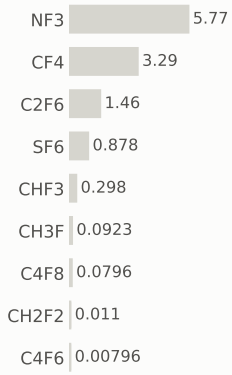
steps by process area



materials deposited (g)



gases emitted (g)



Step #8820: RIE

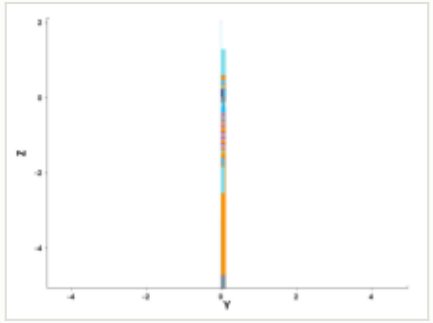
etch_anisotropic · RIE

1.04	726.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

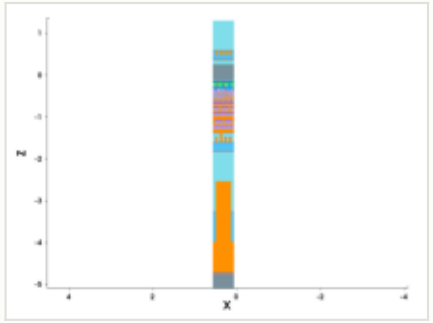
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

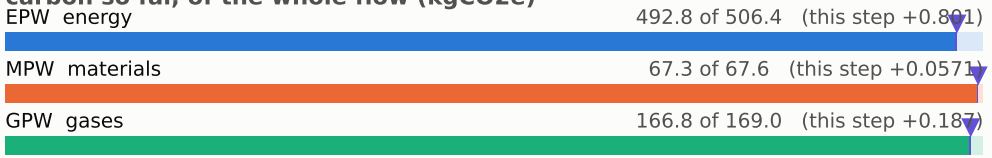
no mask for this step

this step

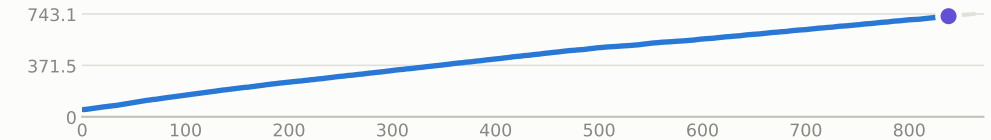
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

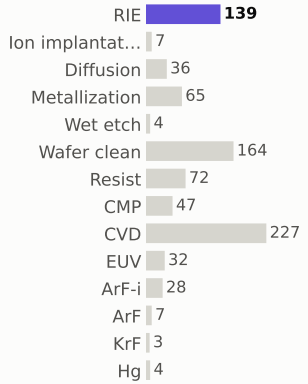
carbon so far, of the whole flow (kgCO2e)



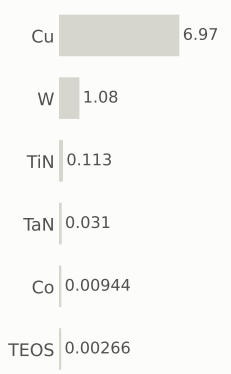
embodied carbon up to each step (kgCO2e per wafer)



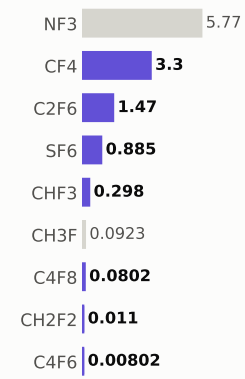
steps by process area



materials deposited (g)



gases emitted (g)



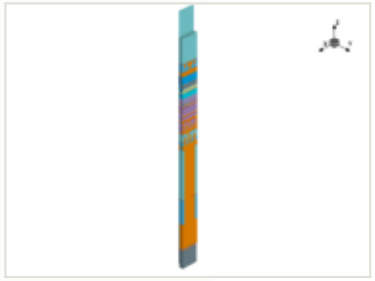
Step #8830: Wafer_clean

etch_strip · Wafer_clean

0.169	727.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

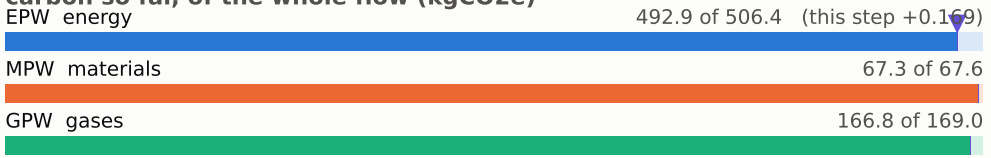


this step

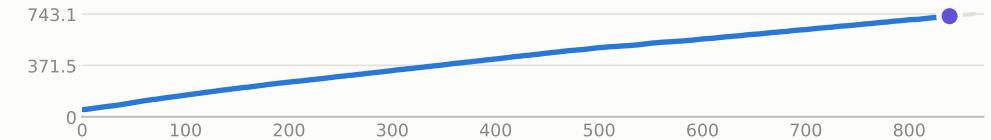
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

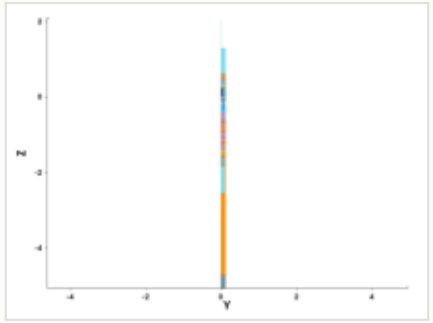
carbon so far, of the whole flow (kgCO2e)



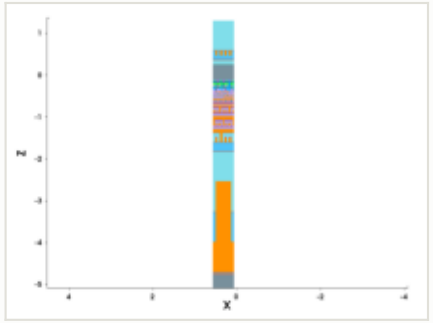
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



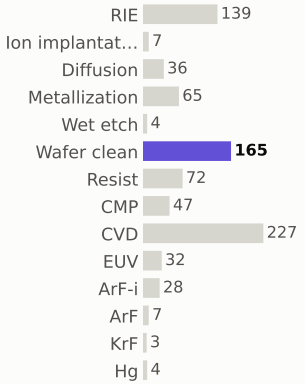
y cut at 0.049 µm



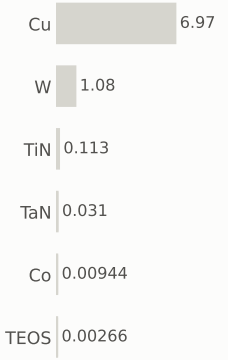
mask for this step

no mask for this step

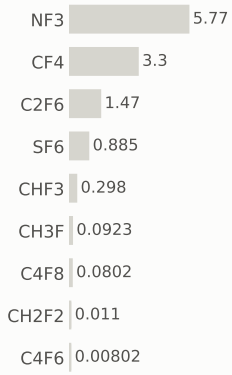
steps by process area



materials deposited (g)



gases emitted (g)



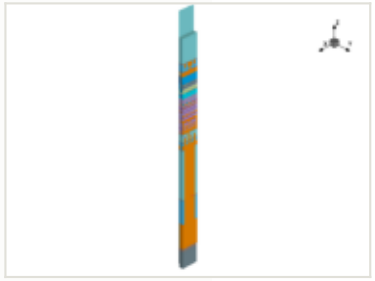
Step #8840: RIE

clean · RIE

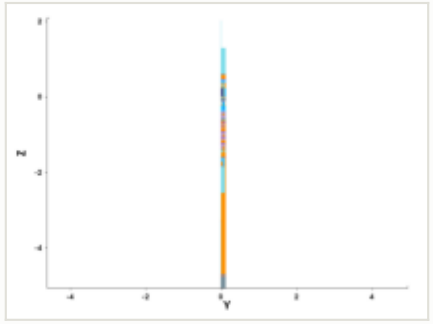
1.04	728.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

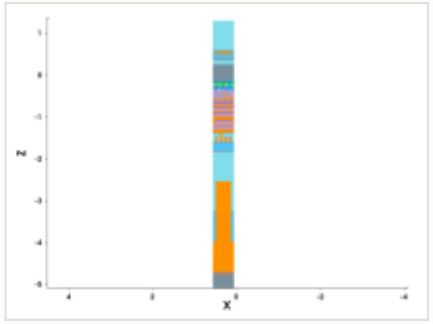
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

no mask for this step

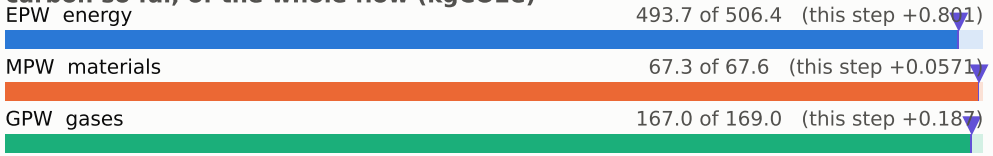
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

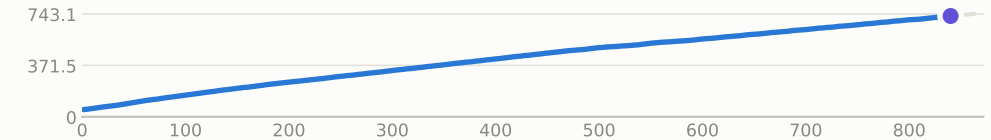
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

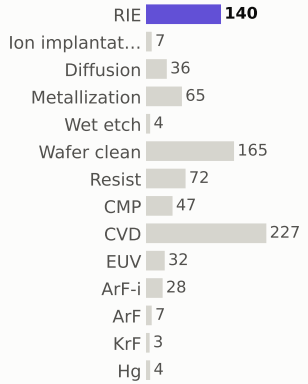
carbon so far, of the whole flow (kgCO2e)



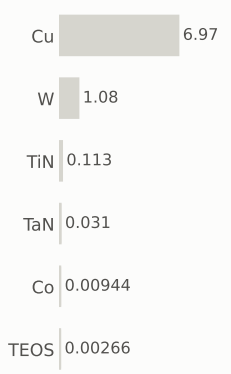
embodied carbon up to each step (kgCO2e per wafer)



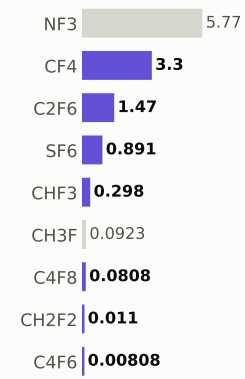
steps by process area



materials deposited (g)



gases emitted (g)



Step #8850: Wafer_clean

clean · Wafer_clean

0.169	728.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

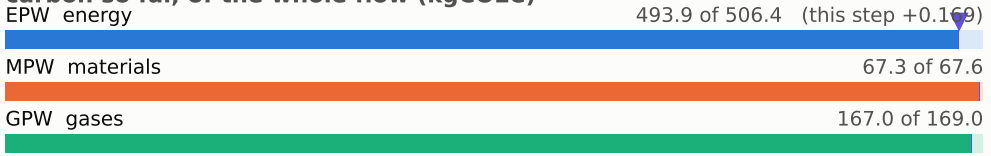
zoomed to fit

this step

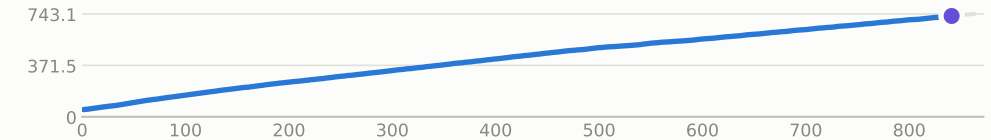
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

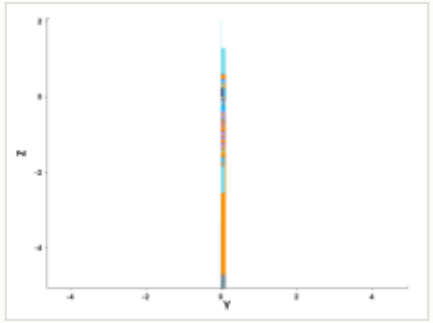
carbon so far, of the whole flow (kgCO2e)



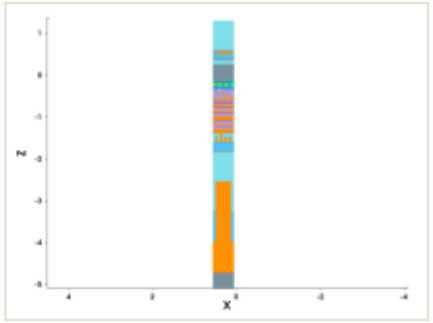
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



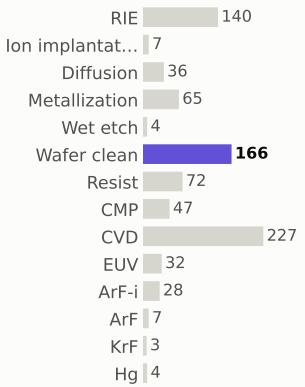
y cut at 0.049 µm



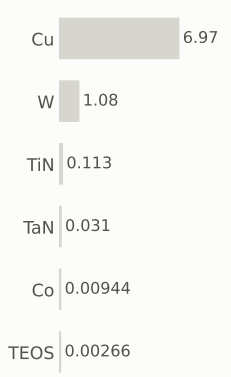
mask for this step

no mask for this step

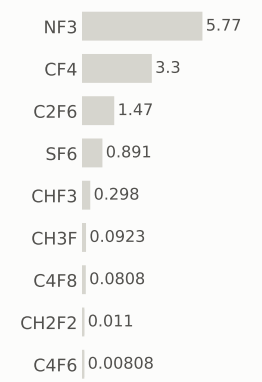
steps by process area



materials deposited (g)



gases emitted (g)



Step #8860: ArF

litho_patterning · ArF

0.199	728.5	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

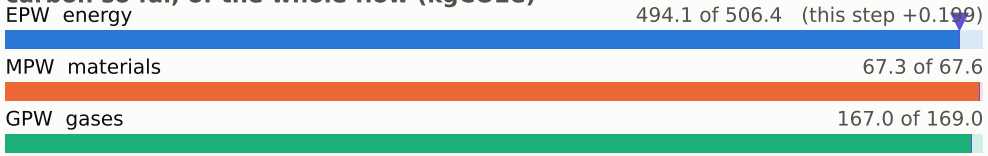
3D structure after this step

zoomed to fit

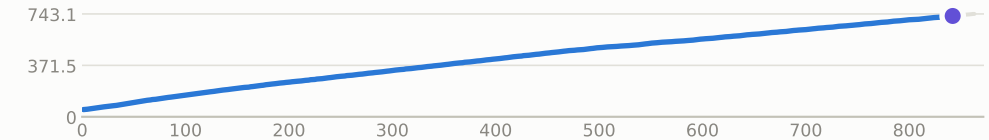
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

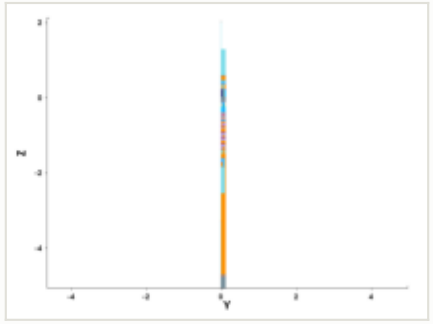
carbon so far, of the whole flow (kgCO2e)



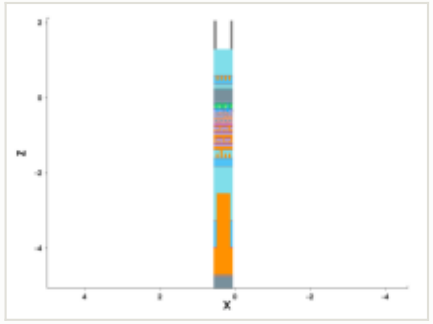
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

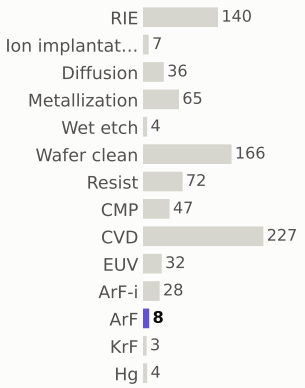


y cut at 0.049 µm

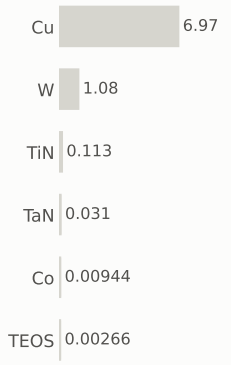


mask: layer 950:0 (GDSII)

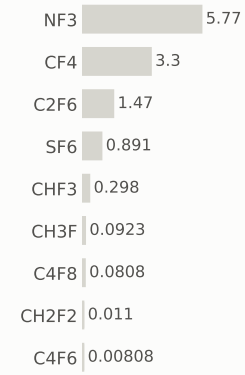
steps by process area



materials deposited (g)



gases emitted (g)



Step #8870: RIE

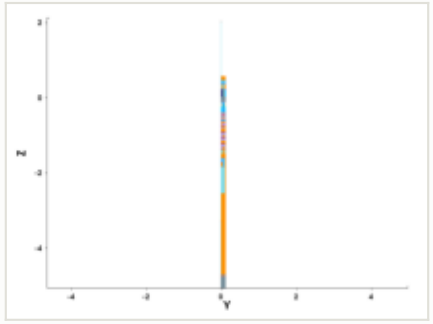
etch_anisotropic · RIE

1.04	729.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

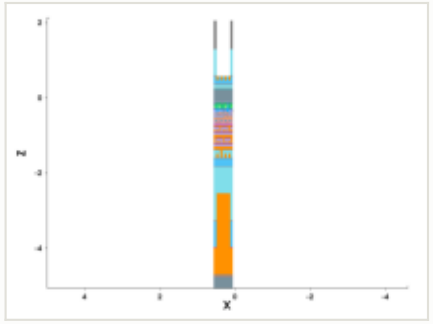
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

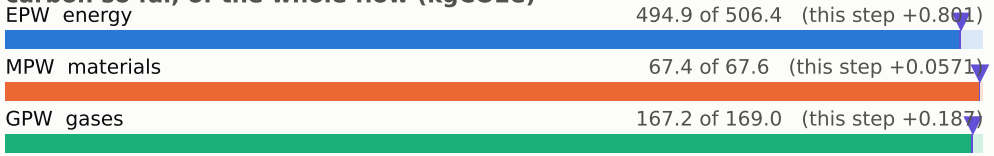
no mask for this step

this step

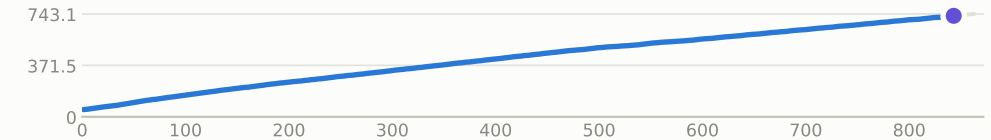
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

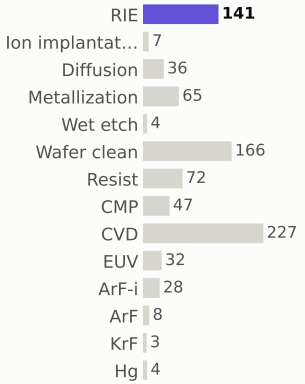
carbon so far, of the whole flow (kgCO2e)



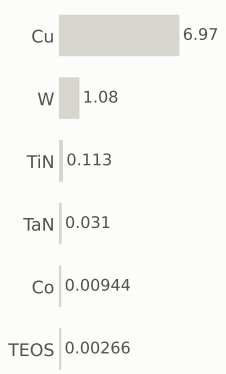
embodied carbon up to each step (kgCO2e per wafer)



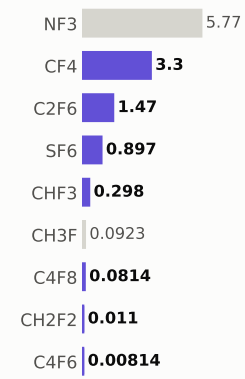
steps by process area



materials deposited (g)



gases emitted (g)



Step #8880: Wafer_clean

etch_strip · Wafer_clean

0.169	729.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

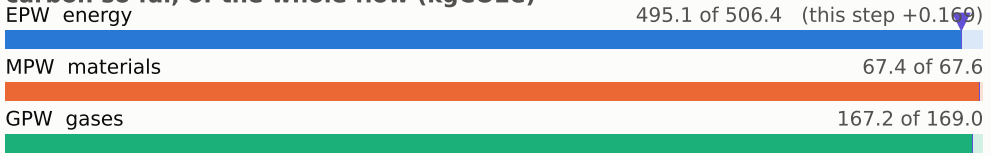
3D structure after this step

zoomed to fit

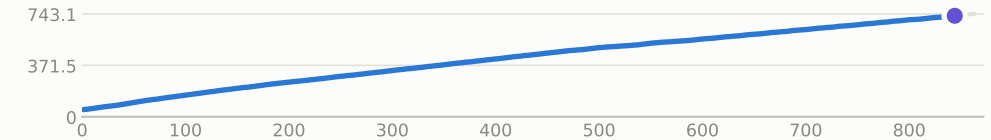
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

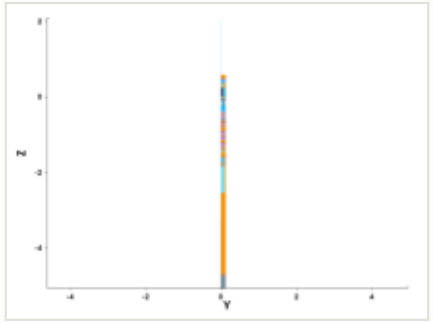
carbon so far, of the whole flow (kgCO2e)



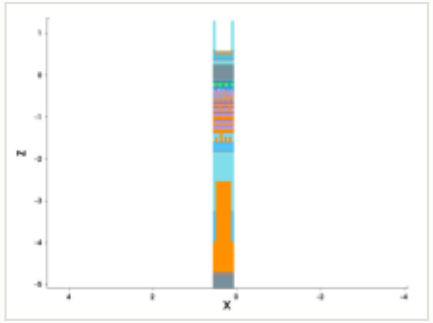
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



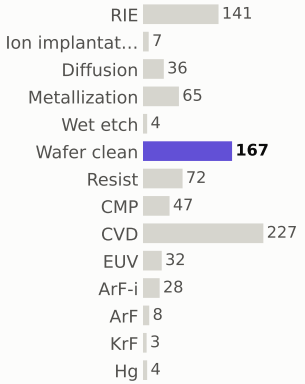
y cut at 0.049 µm



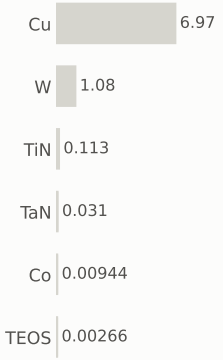
mask for this step

no mask for this step

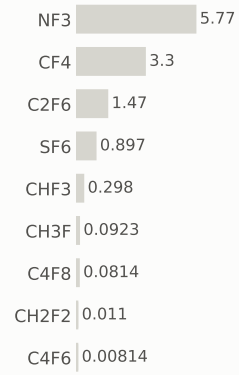
steps by process area



materials deposited (g)



gases emitted (g)



Step #8890: Metallization

depo_ellipsoidal · Metallization

1.2	730.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

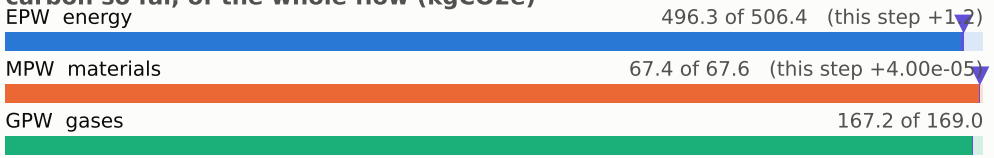
zoomed to fit

this step

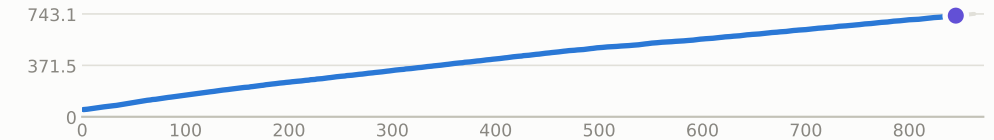
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

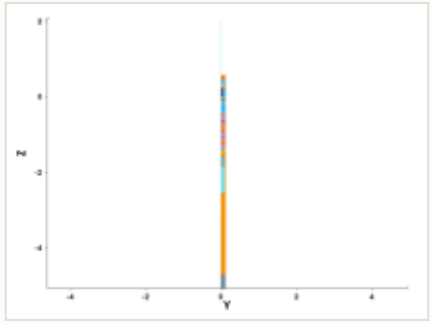
carbon so far, of the whole flow (kgCO2e)



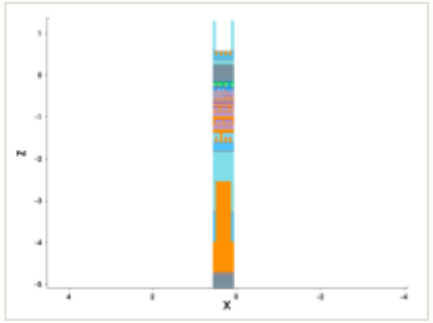
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



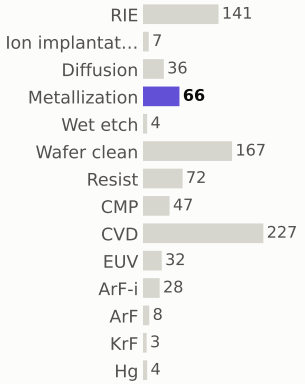
y cut at 0.049 µm



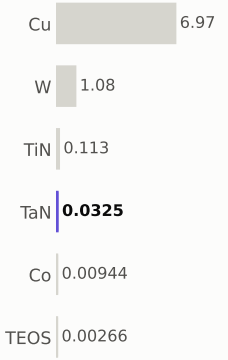
mask for this step

no mask for this step

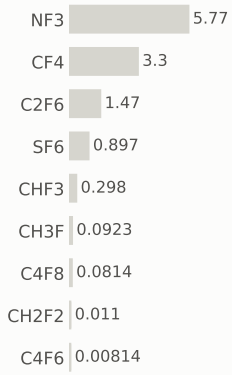
steps by process area



materials deposited (g)



gases emitted (g)



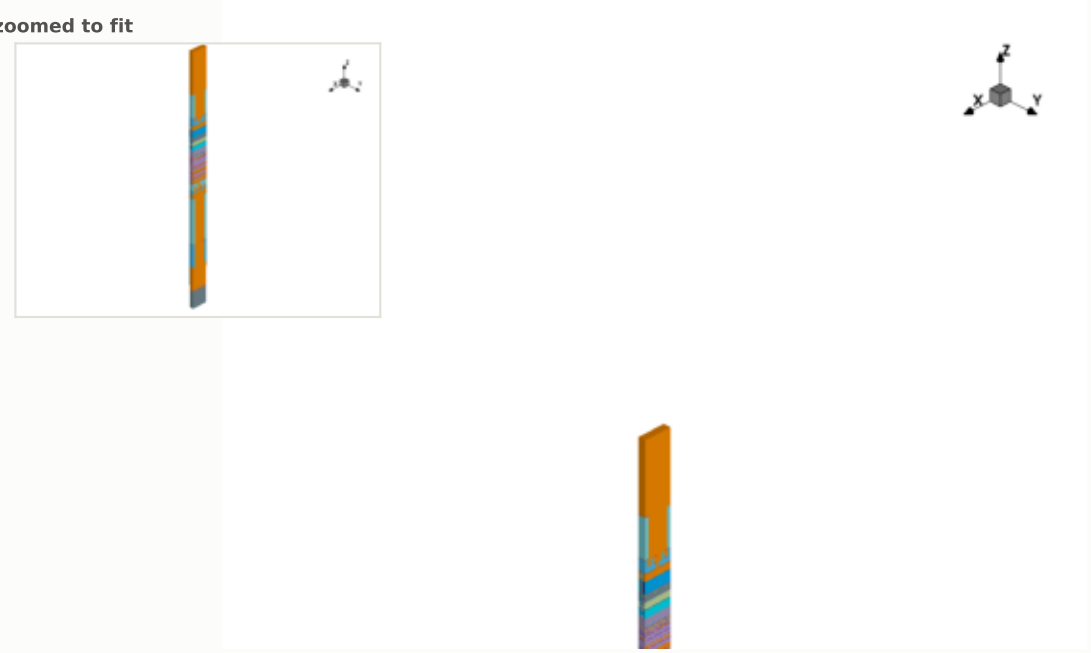
Step #8900: Metallization

depo_fill · Metallization

1.21	732.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

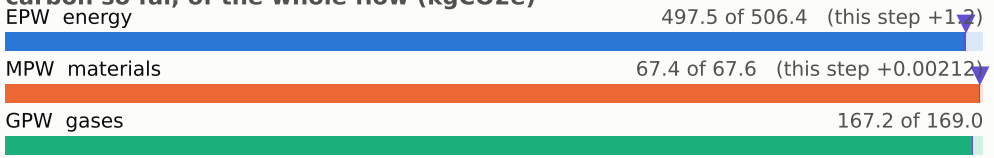
zoomed to fit



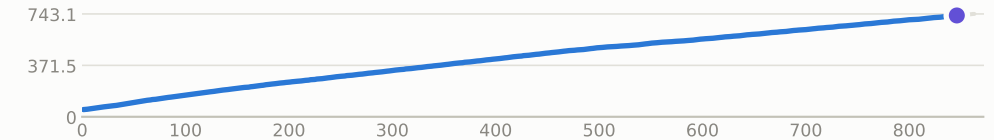
this step

process area	Metallization	EPW	1.2 kgCO2e
material	Cu	EPW tool	0.86 kgCO2e
thickness	0.55 µm	EPW overhead	0.344 kgCO2e
material GWP	6.09011 gCO2e/g	MPW	0.00212 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.21 kgCO2e**

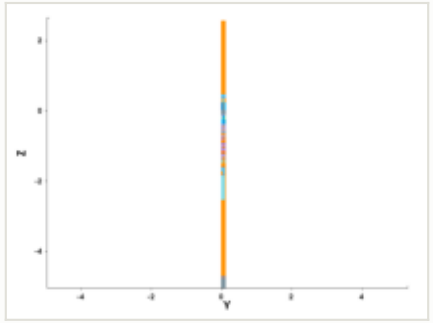
carbon so far, of the whole flow (kgCO2e)



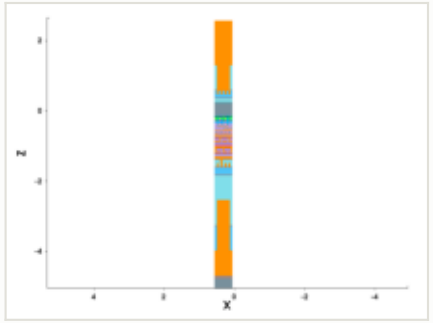
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



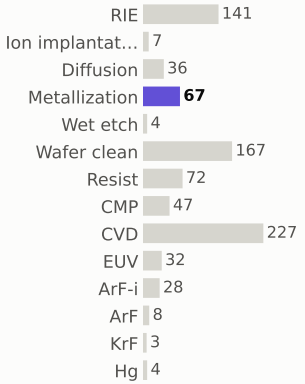
y cut at 0.049 µm



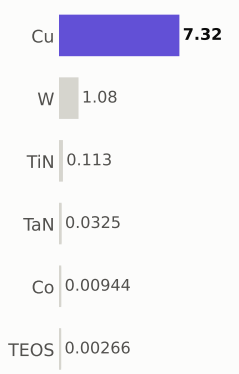
mask for this step

no mask for this step

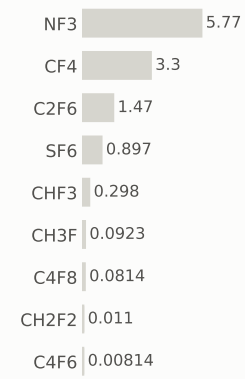
steps by process area



materials deposited (g)



gases emitted (g)



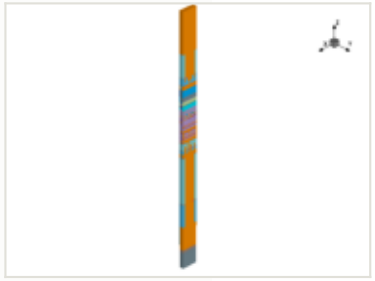
Step #8910: Diffusion

treatment · Diffusion

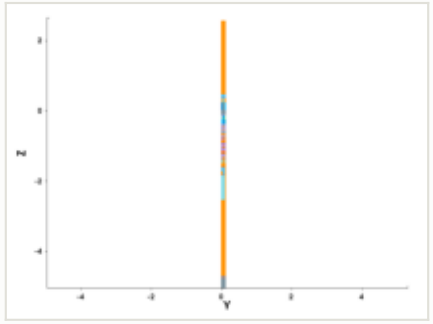
0.801	732.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

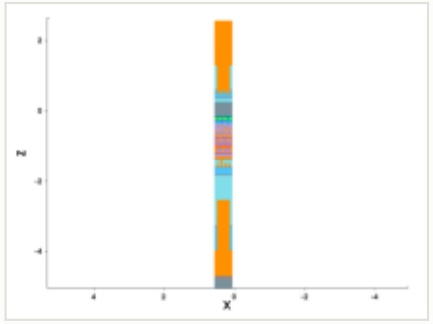
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

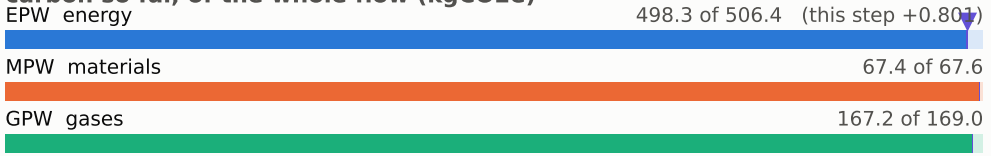
no mask for this step

this step

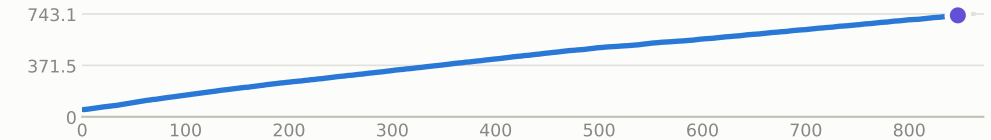
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

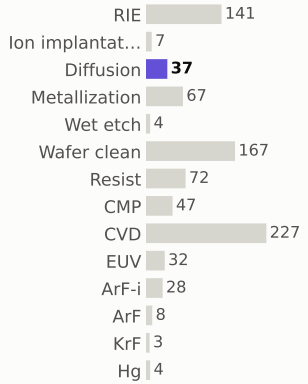
carbon so far, of the whole flow (kgCO2e)



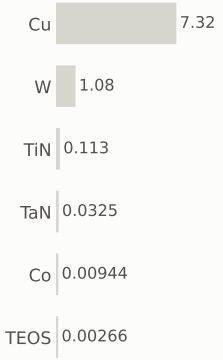
embodied carbon up to each step (kgCO2e per wafer)



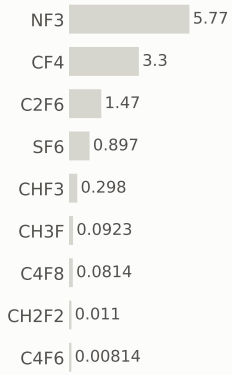
steps by process area



materials deposited (g)



gases emitted (g)



Step #8920: CMP

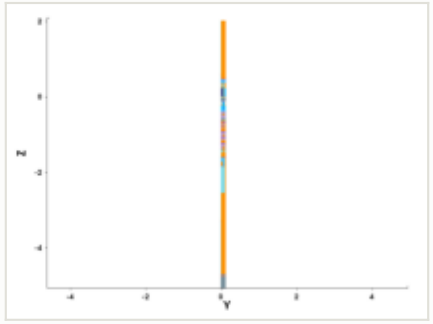
etch_cmp · CMP

0.169	733.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

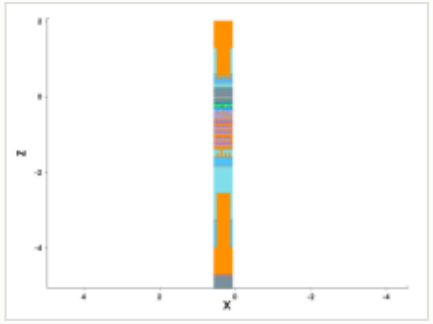
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

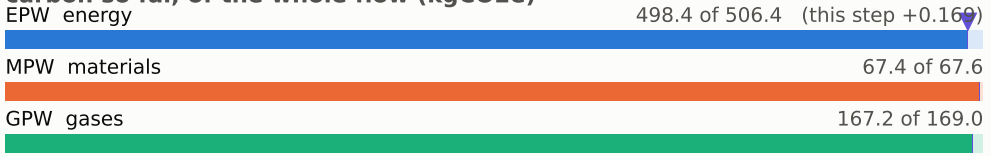
no mask for this step

this step

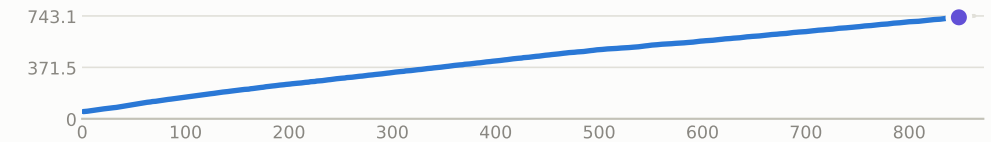
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

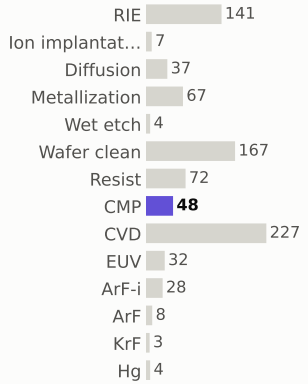
carbon so far, of the whole flow (kgCO2e)



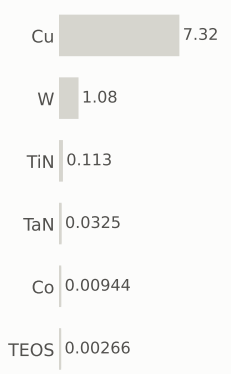
embodied carbon up to each step (kgCO2e per wafer)



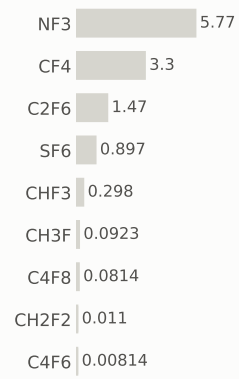
steps by process area



materials deposited (g)



gases emitted (g)



Step #8940: CVD

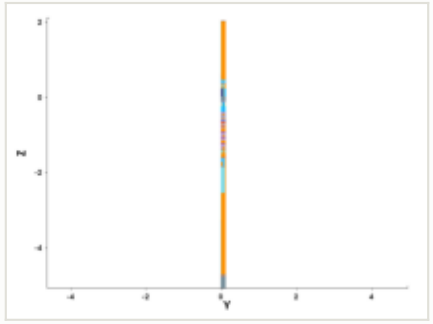
depo_fill · CVD

1.31	734.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

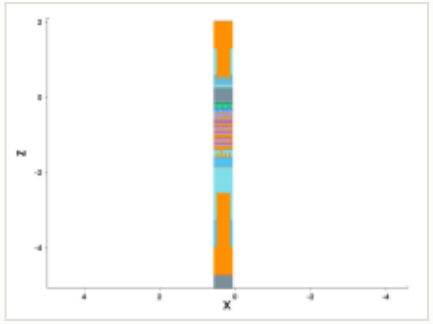
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

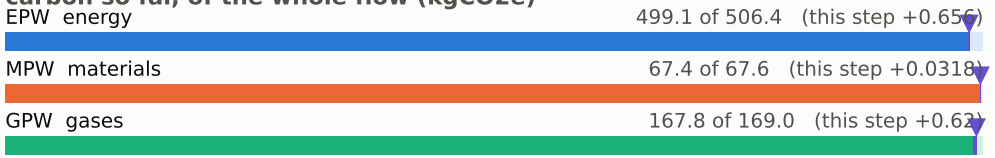
no mask for this step

this step

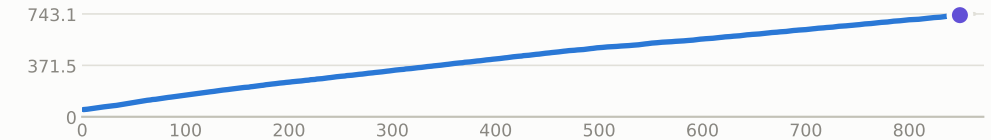
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

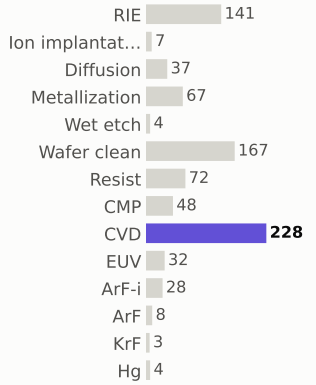
carbon so far, of the whole flow (kgCO2e)



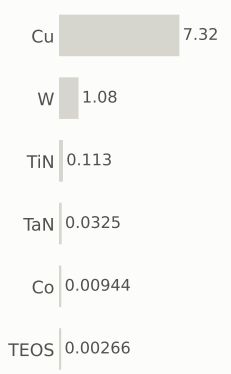
embodied carbon up to each step (kgCO2e per wafer)



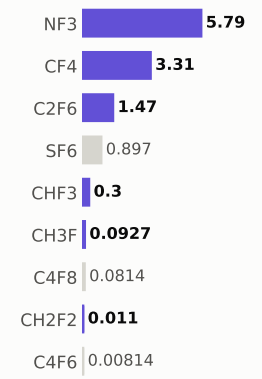
steps by process area



materials deposited (g)



gases emitted (g)



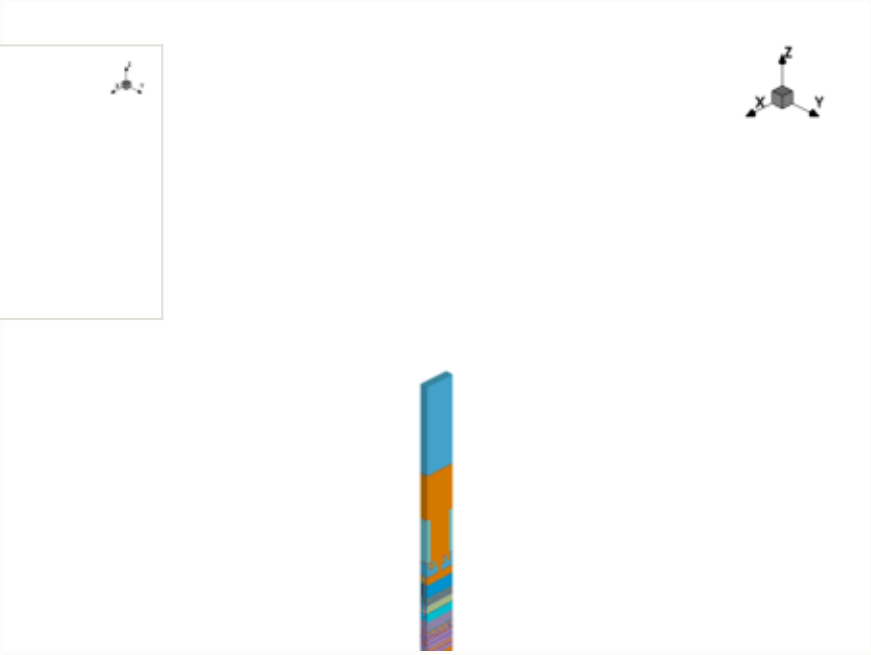
Step #8950: CVD

depo_fill · CVD

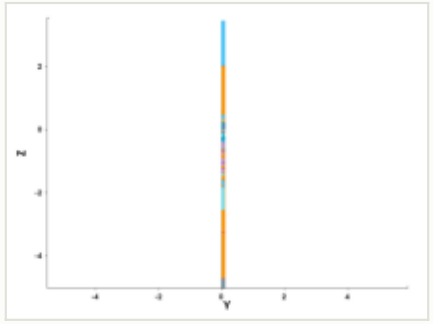
1.31	735.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

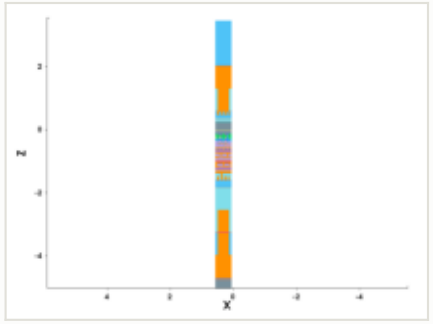
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

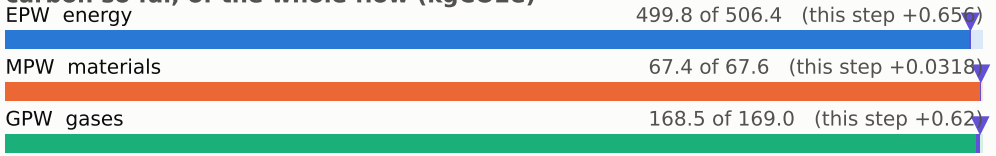
no mask for this step

this step

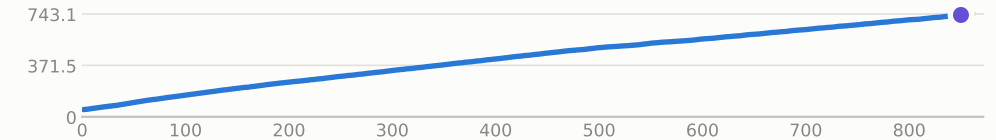
process area	CVD
material	—
thickness	1.42 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

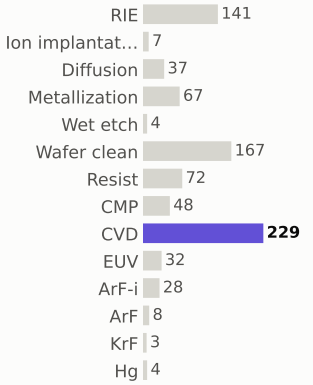
carbon so far, of the whole flow (kgCO2e)



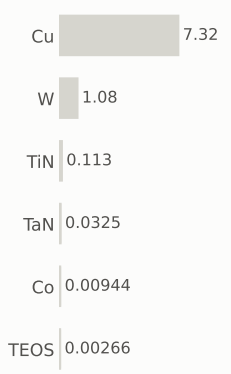
embodied carbon up to each step (kgCO2e per wafer)



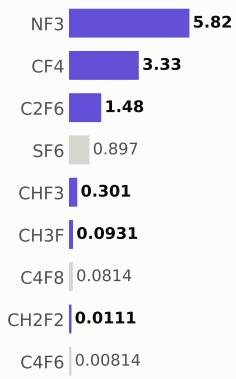
steps by process area



materials deposited (g)



gases emitted (g)



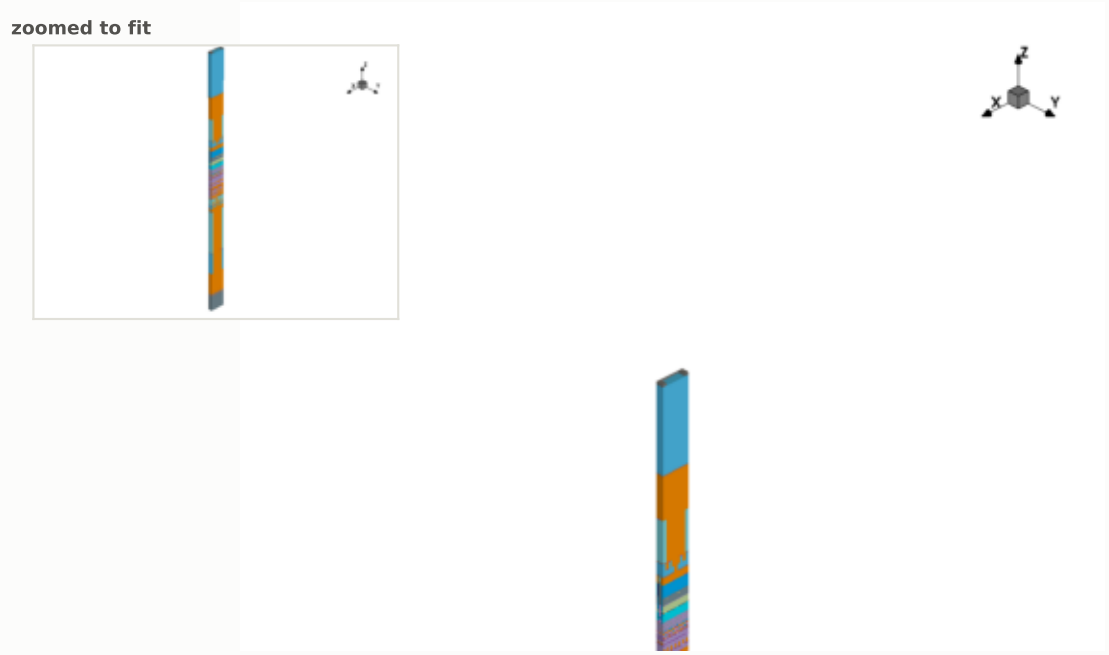
Step #8960: Hg

litho_patterning · Hg

0.199	735.9	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

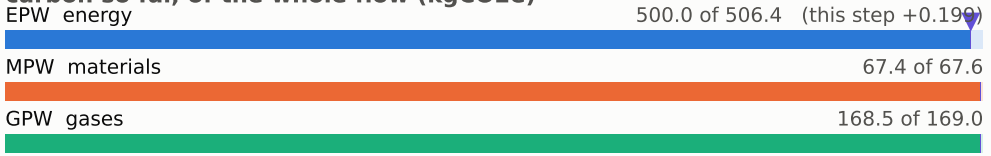


this step

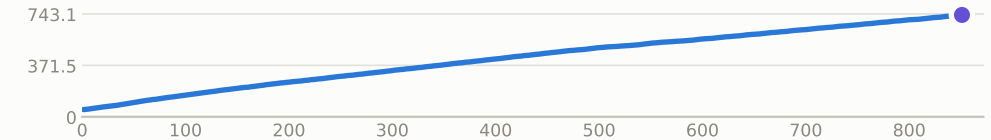
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

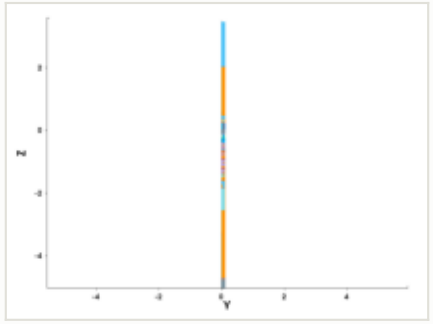
carbon so far, of the whole flow (kgCO2e)



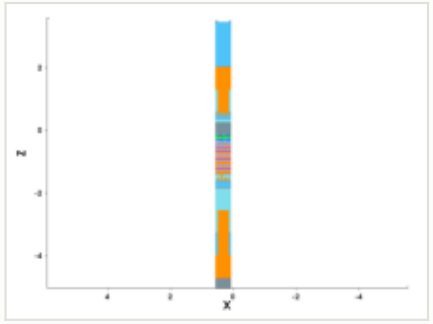
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

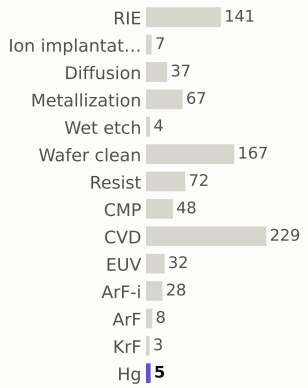


y cut at 0.049 µm

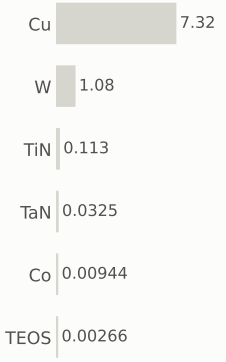


mask: layer 980:0 (GDSII)

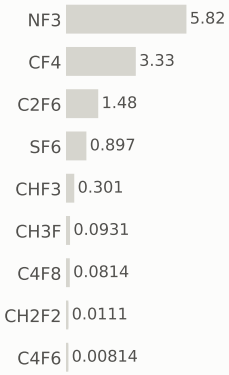
steps by process area



materials deposited (g)



gases emitted (g)



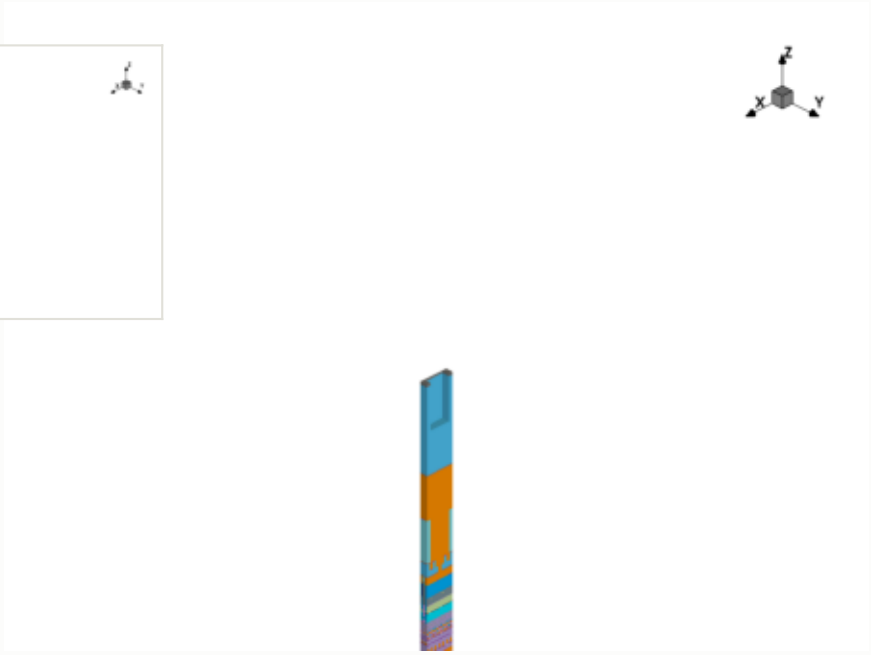
Step #8970: RIE

etch_anisotropic · RIE

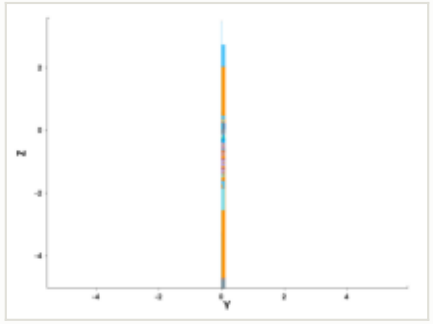
1.04	736.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

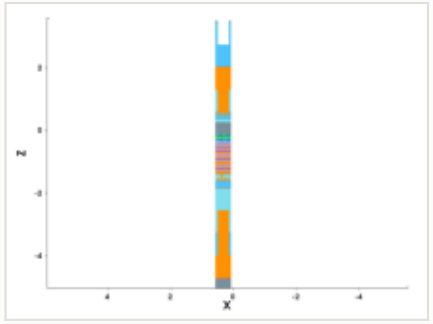
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

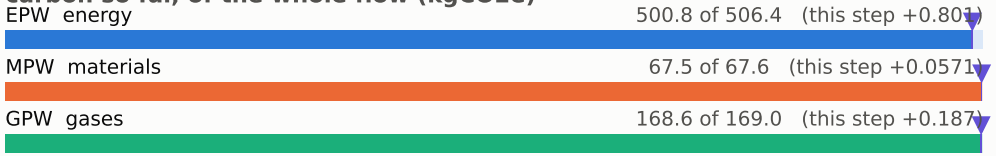
no mask for this step

this step

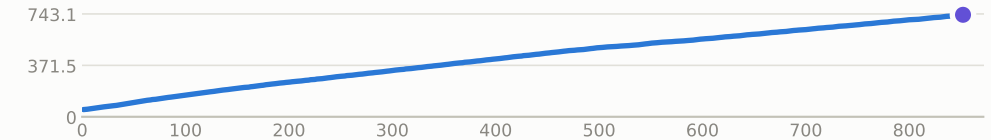
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

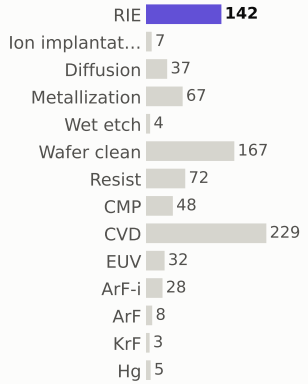
carbon so far, of the whole flow (kgCO2e)



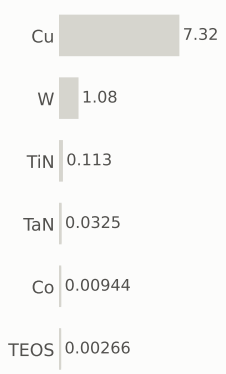
embodied carbon up to each step (kgCO2e per wafer)



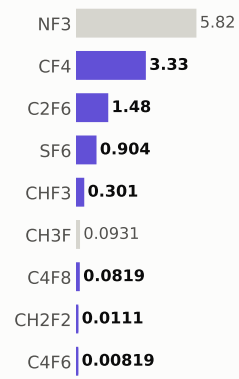
steps by process area



materials deposited (g)



gases emitted (g)



Step #8980: Wafer_clean

etch_strip · Wafer_clean

0.169	737.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

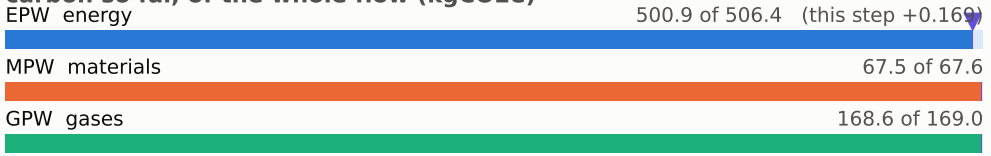
zoomed to fit

this step

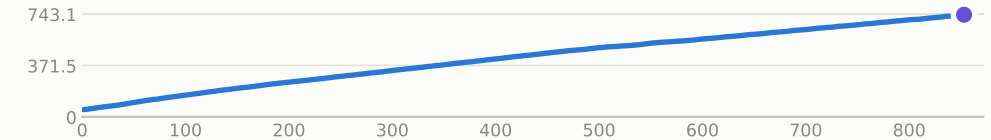
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

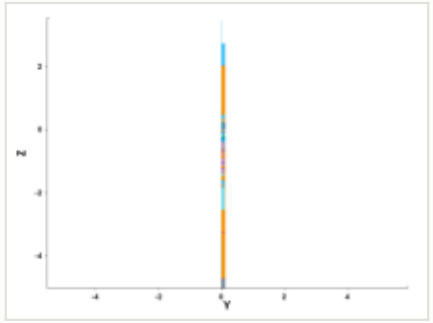
carbon so far, of the whole flow (kgCO2e)



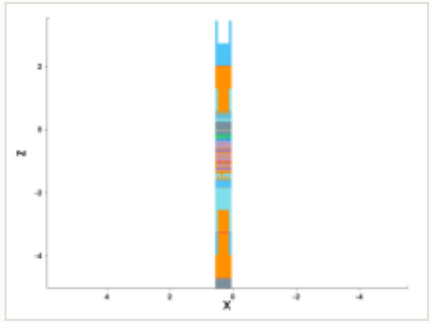
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



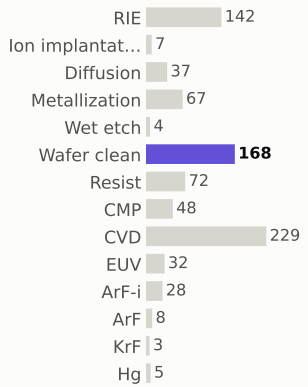
y cut at 0.049 µm



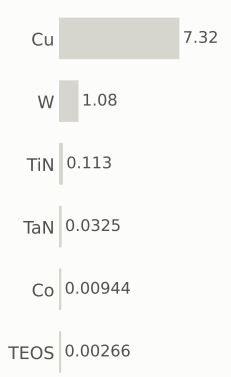
mask for this step

no mask for this step

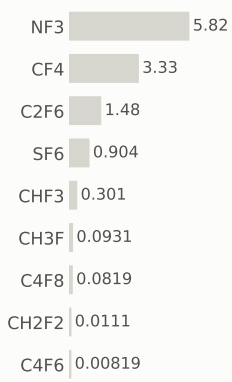
steps by process area



materials deposited (g)



gases emitted (g)



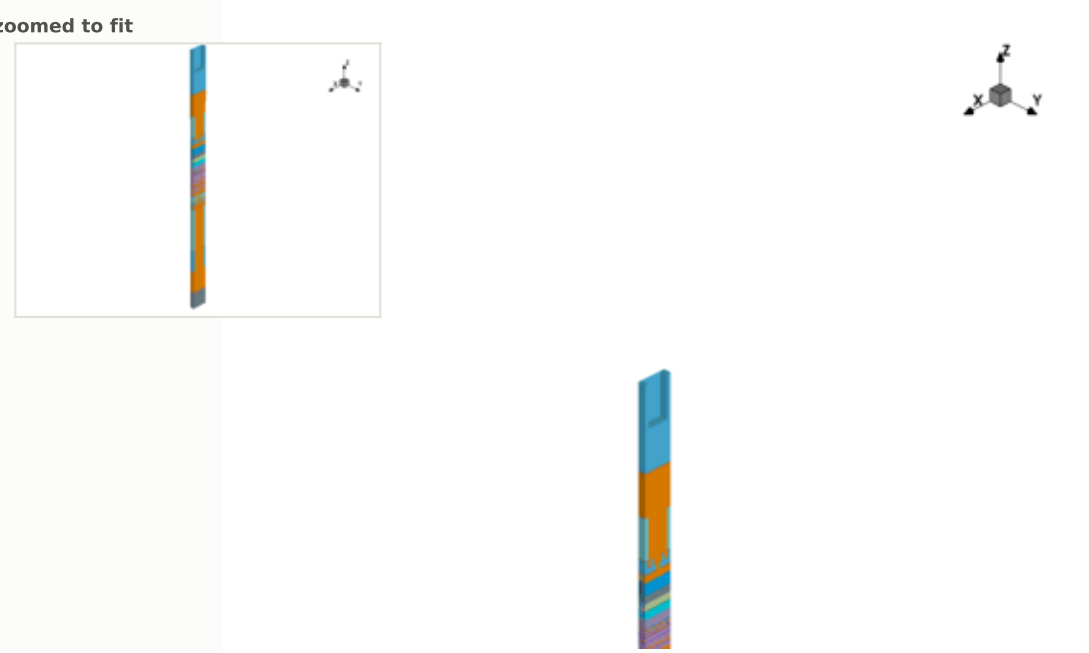
Step #8990: RIE

clean · RIE

1.04	738.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

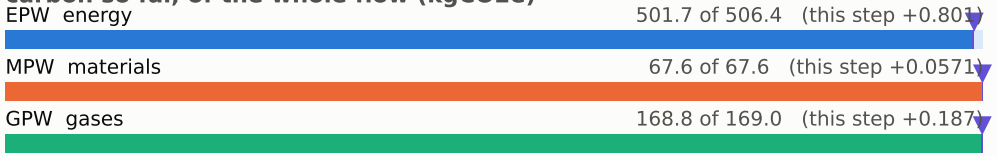


this step

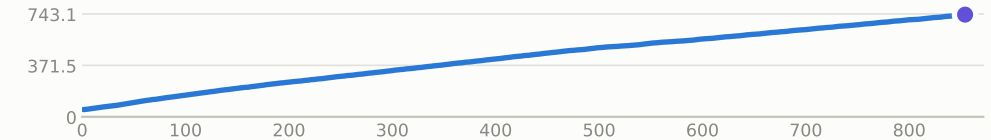
process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

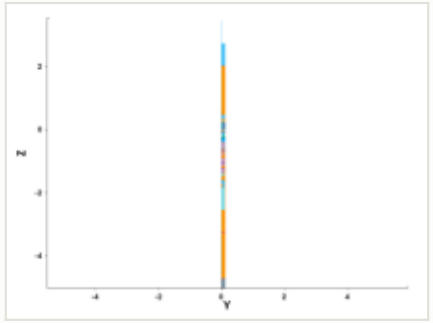
carbon so far, of the whole flow (kgCO2e)



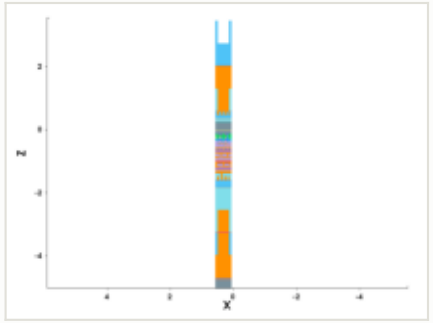
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



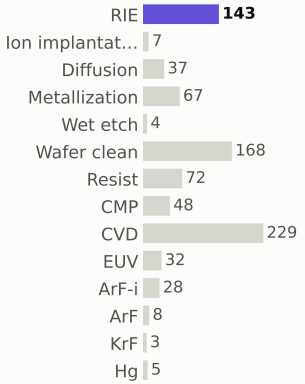
y cut at 0.049 µm



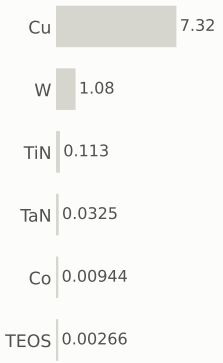
mask for this step

no mask for this step

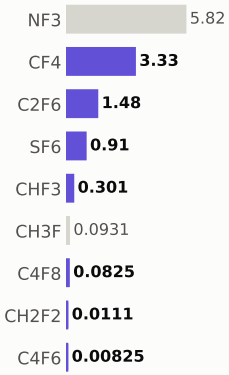
steps by process area



materials deposited (g)



gases emitted (g)



Step #9000: Wafer_clean

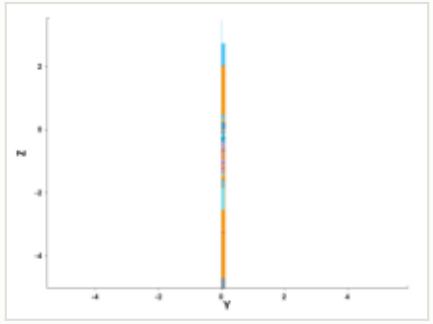
clean · Wafer_clean

0.169	738.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

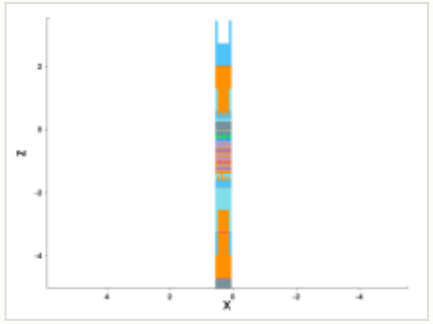
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

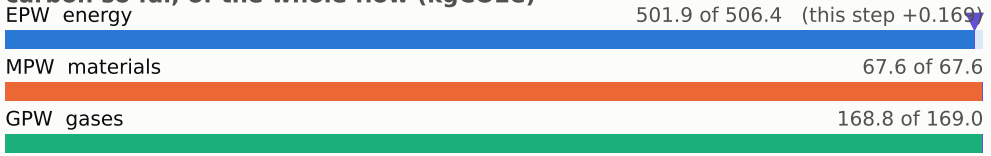
no mask for this step

this step

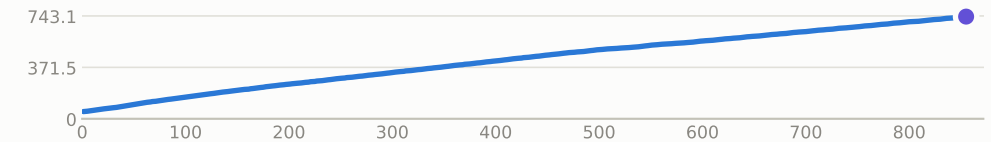
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

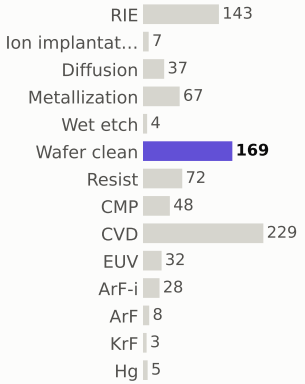
carbon so far, of the whole flow (kgCO2e)



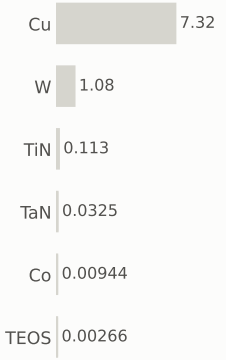
embodied carbon up to each step (kgCO2e per wafer)



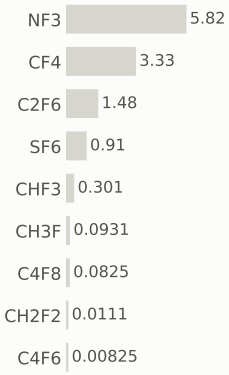
steps by process area



materials deposited (g)



gases emitted (g)



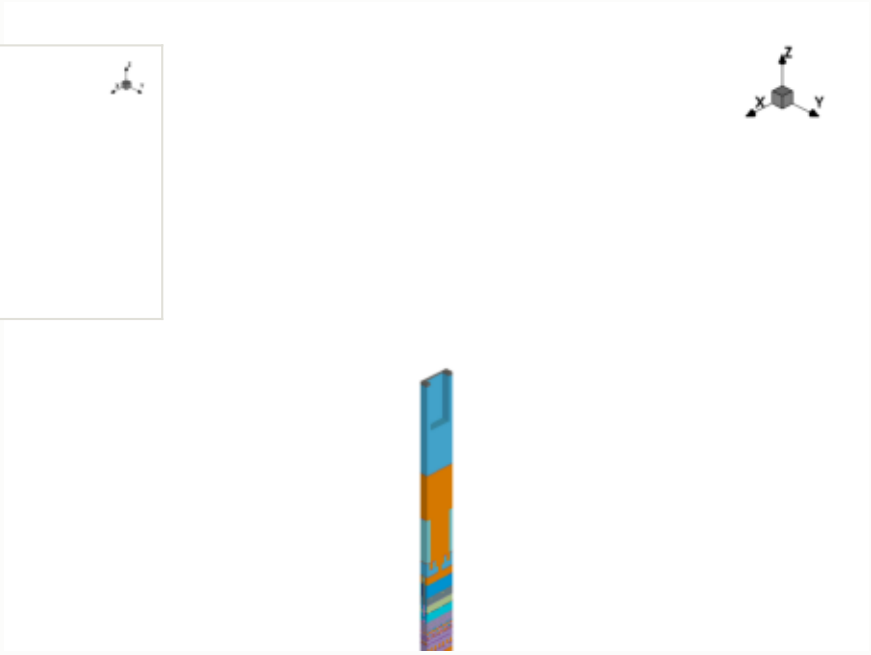
Step #9010: Hg

litho_patterning · Hg

0.199	738.5	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

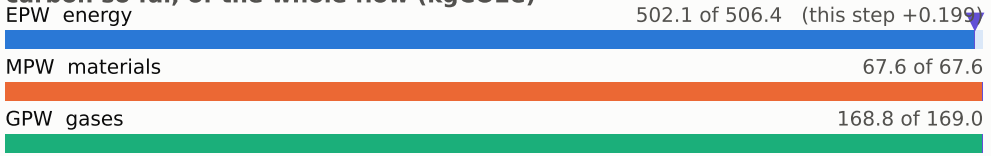
zoomed to fit



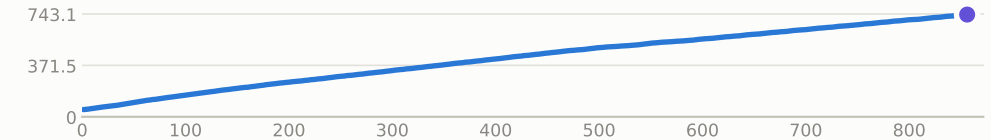
this step

process area	Hg	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

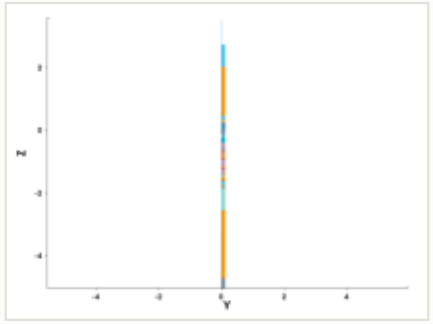
carbon so far, of the whole flow (kgCO2e)



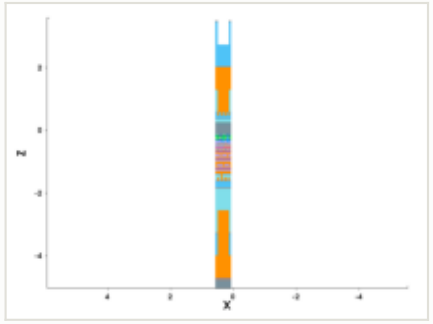
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm

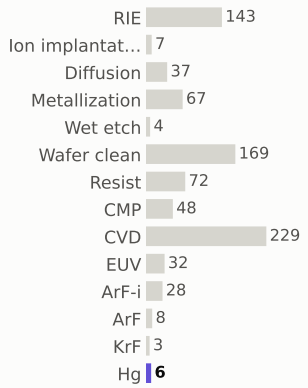


y cut at 0.049 µm

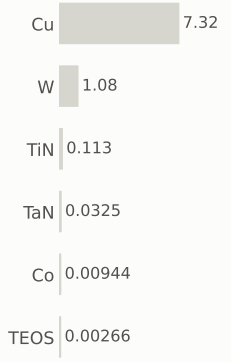


mask: layer 970:0 (GDSII)

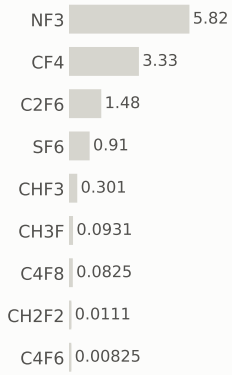
steps by process area



materials deposited (g)



gases emitted (g)



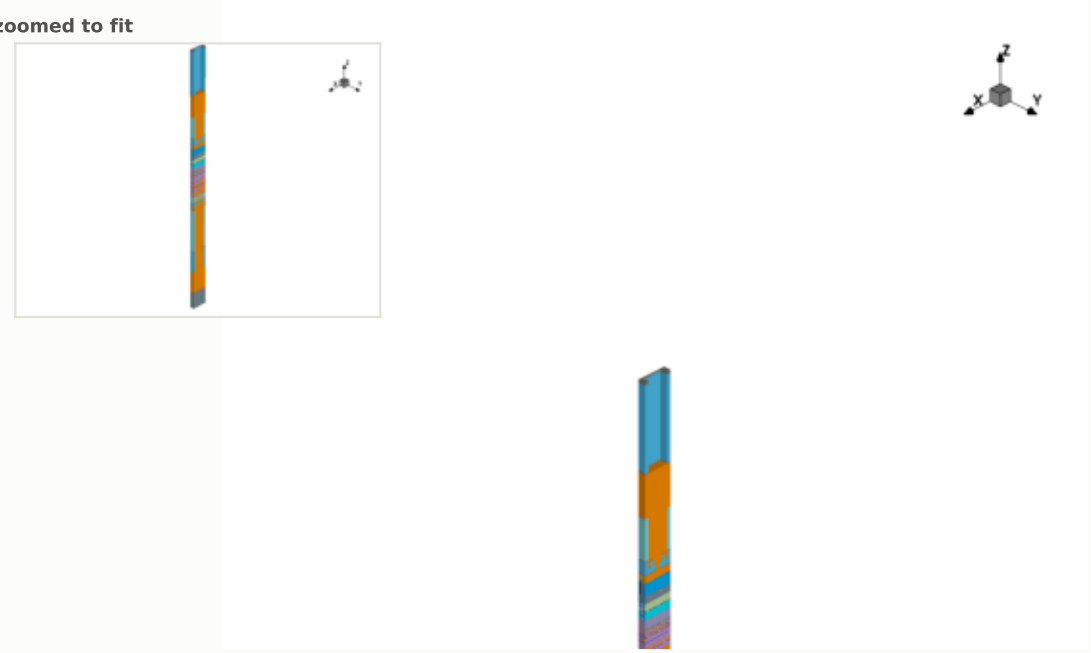
Step #9020: RIE

etch_anisotropic · RIE

1.04	739.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

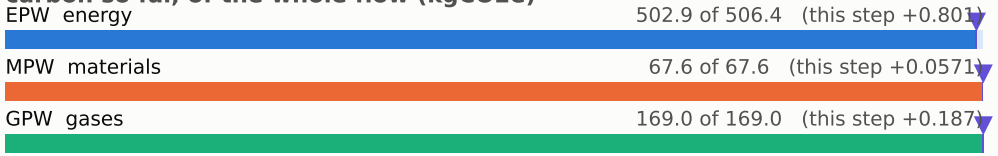
zoomed to fit



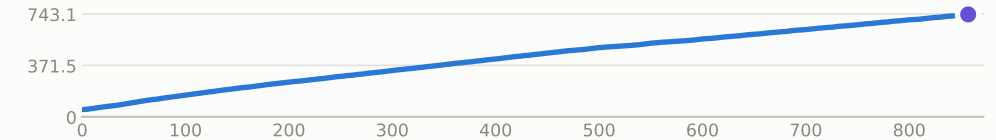
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

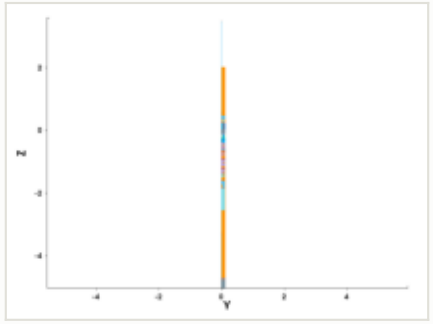
carbon so far, of the whole flow (kgCO2e)



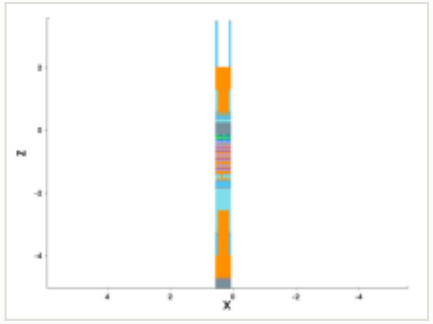
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



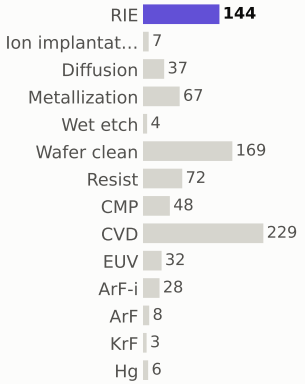
y cut at 0.049 µm



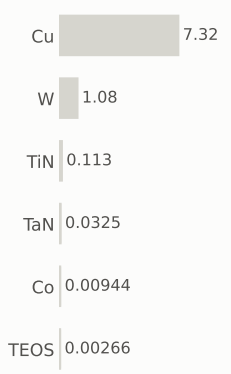
mask for this step

no mask for this step

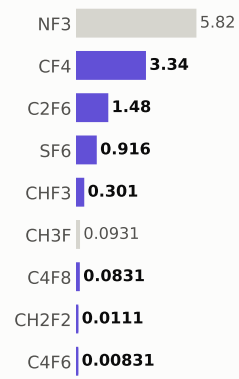
steps by process area



materials deposited (g)



gases emitted (g)



Step #9030: Wafer_clean

etch_strip · Wafer_clean

0.169	739.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

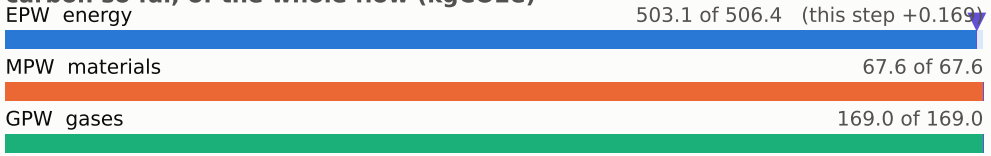
zoomed to fit

this step

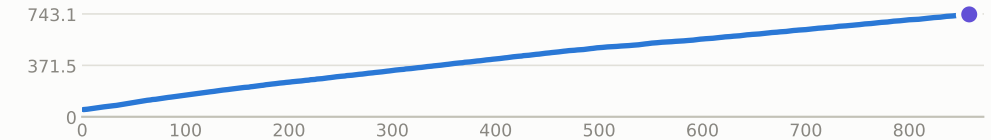
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

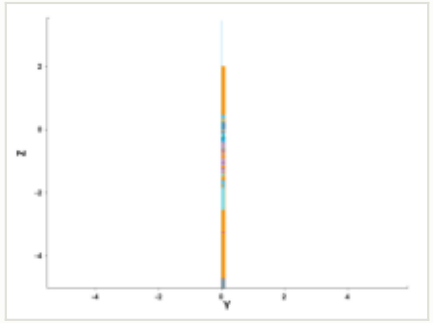
carbon so far, of the whole flow (kgCO2e)



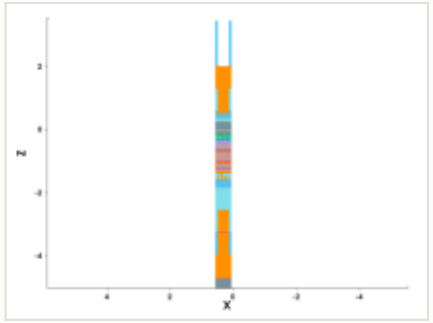
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



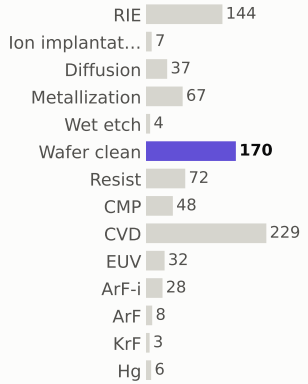
y cut at 0.049 µm



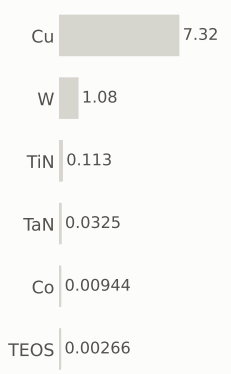
mask for this step

no mask for this step

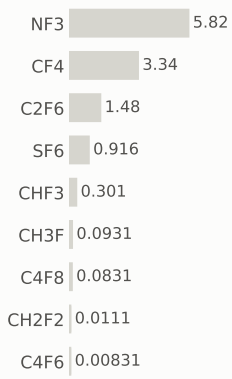
steps by process area



materials deposited (g)



gases emitted (g)



Step #9040: Metallization

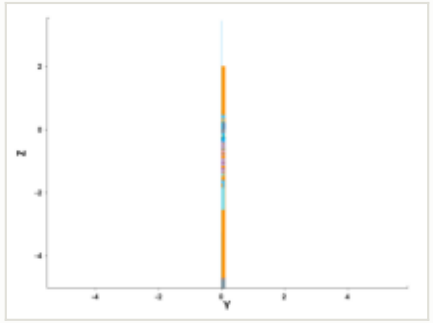
depo_ellipsoidal · Metallization

1.2	740.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

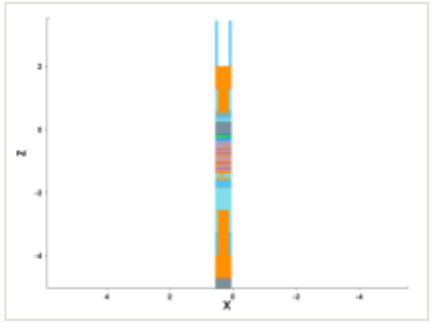
3D structure after this step

zoomed to fit

x cut at 0.385 µm



y cut at 0.049 µm



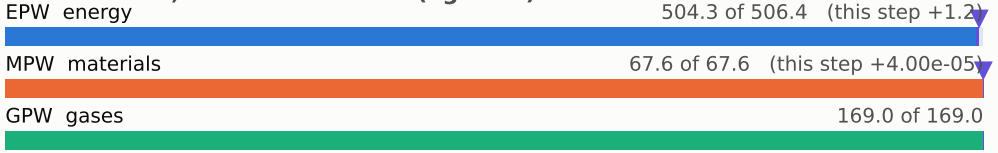
mask for this step

no mask for this step

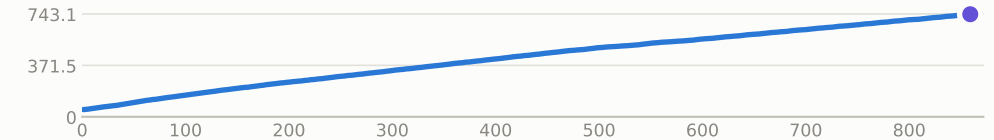
this step

process area	Metallization	EPW	1.2 kgCO2e
material	TaN	EPW tool	0.86 kgCO2e
thickness	0.0016 µm	EPW overhead	0.344 kgCO2e
material GWP	25.8159 gCO2e/g	MPW	4.00e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

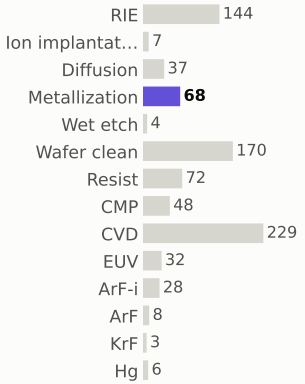
carbon so far, of the whole flow (kgCO2e)



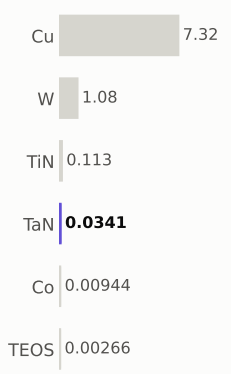
embodied carbon up to each step (kgCO2e per wafer)



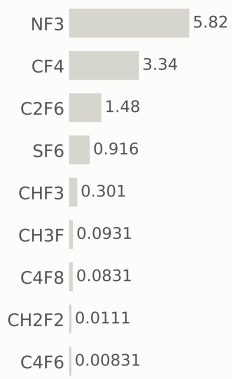
steps by process area



materials deposited (g)



gases emitted (g)



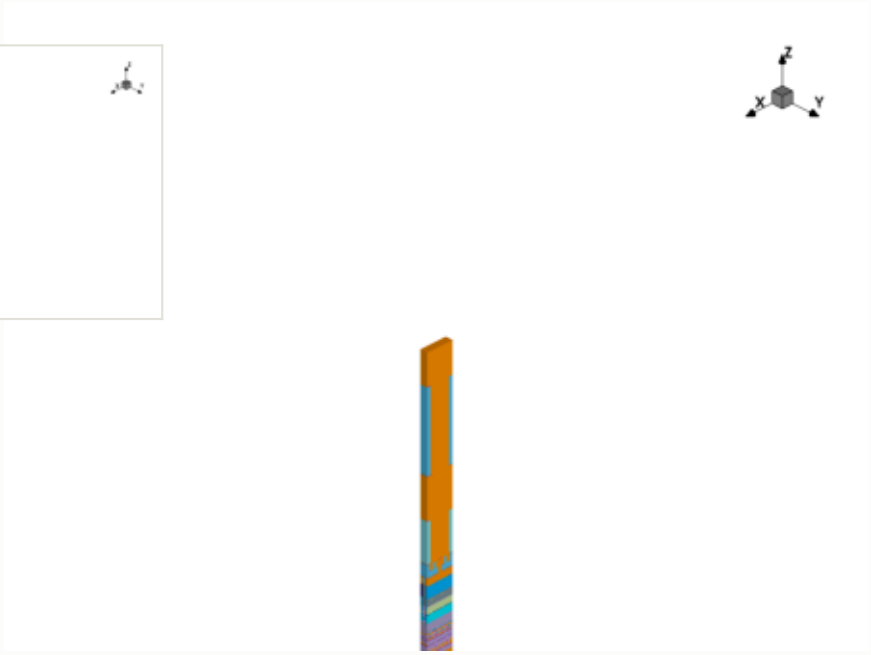
Step #9050: Metallization

depo_fill · Metallization

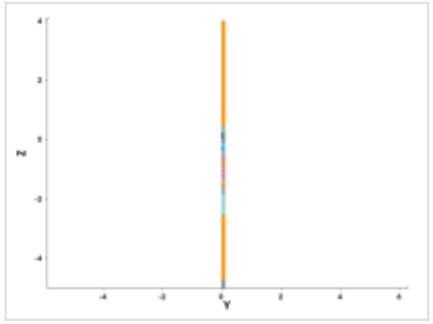
1.21	742.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

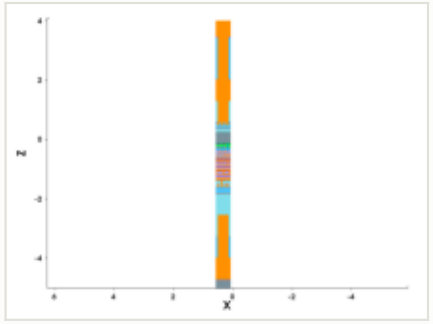
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

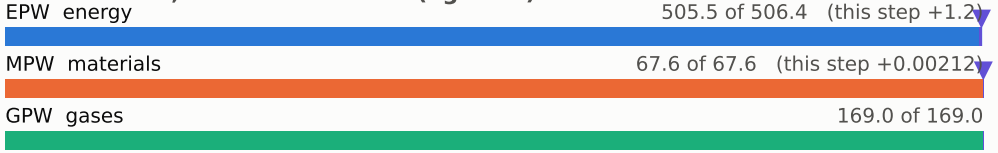
no mask for this step

this step

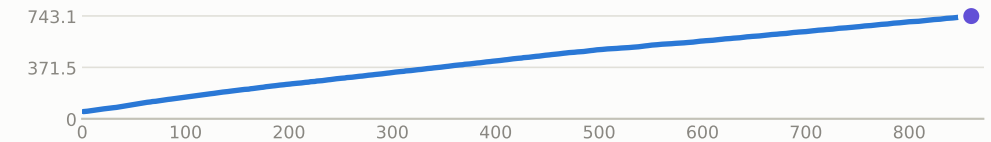
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

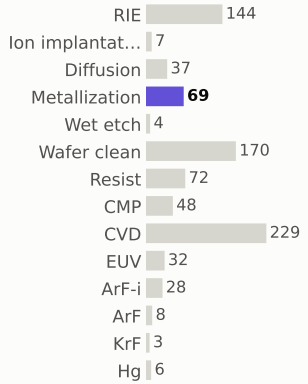
carbon so far, of the whole flow (kgCO2e)



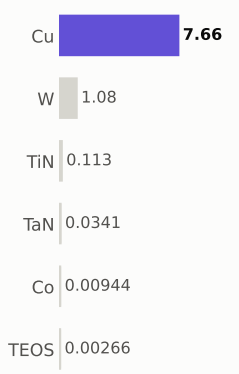
embodied carbon up to each step (kgCO2e per wafer)



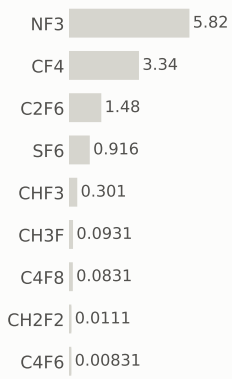
steps by process area



materials deposited (g)



gases emitted (g)



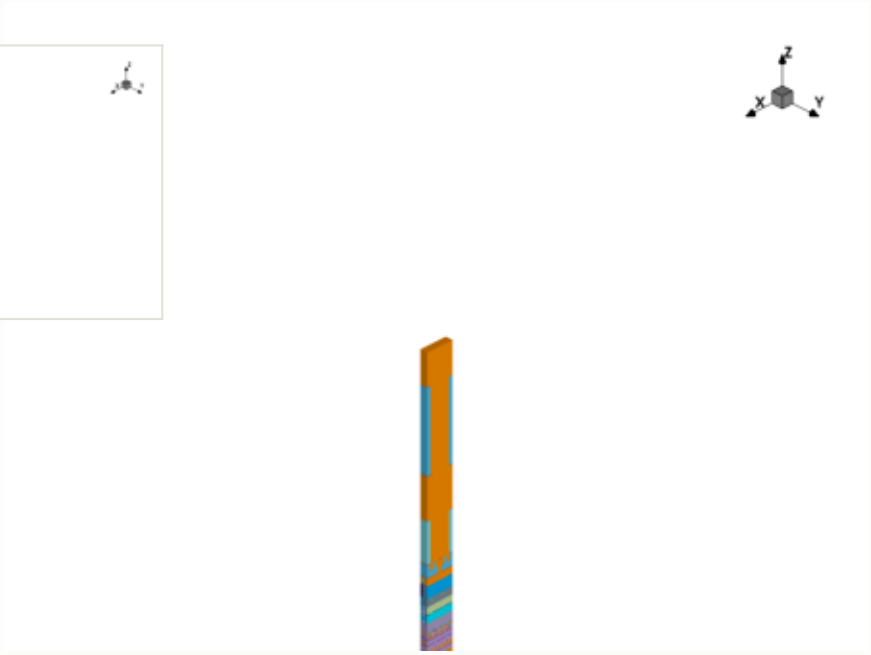
Step #9060: Diffusion

treatment · Diffusion

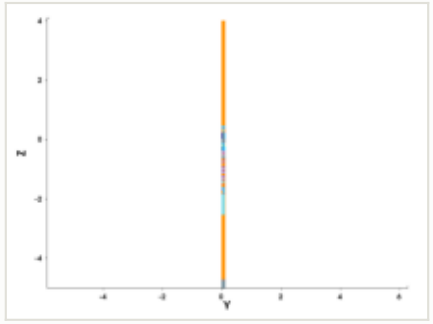
0.801	742.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

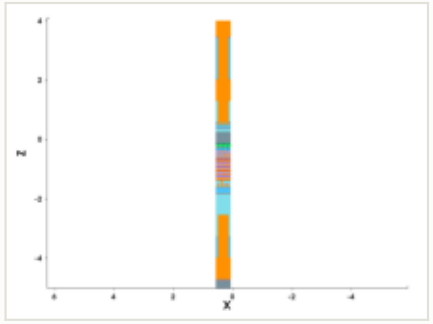
zoomed to fit



x cut at 0.385 µm



y cut at 0.049 µm



mask for this step

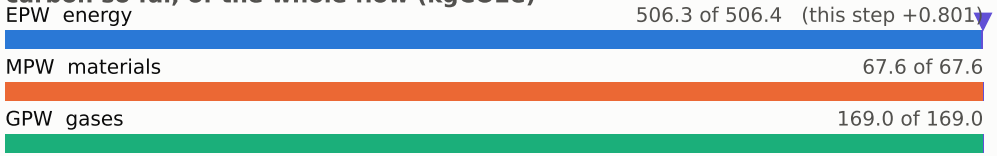
no mask for this step

this step

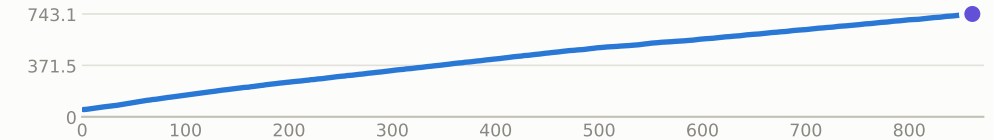
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

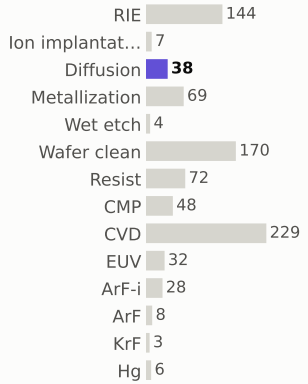
carbon so far, of the whole flow (kgCO2e)



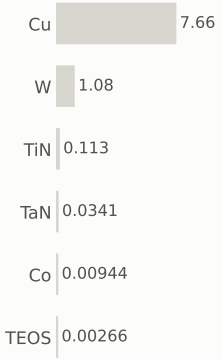
embodied carbon up to each step (kgCO2e per wafer)



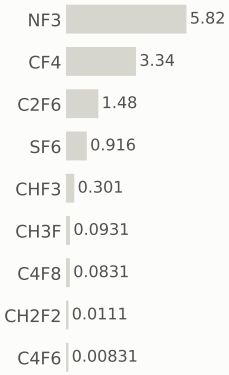
steps by process area



materials deposited (g)



gases emitted (g)



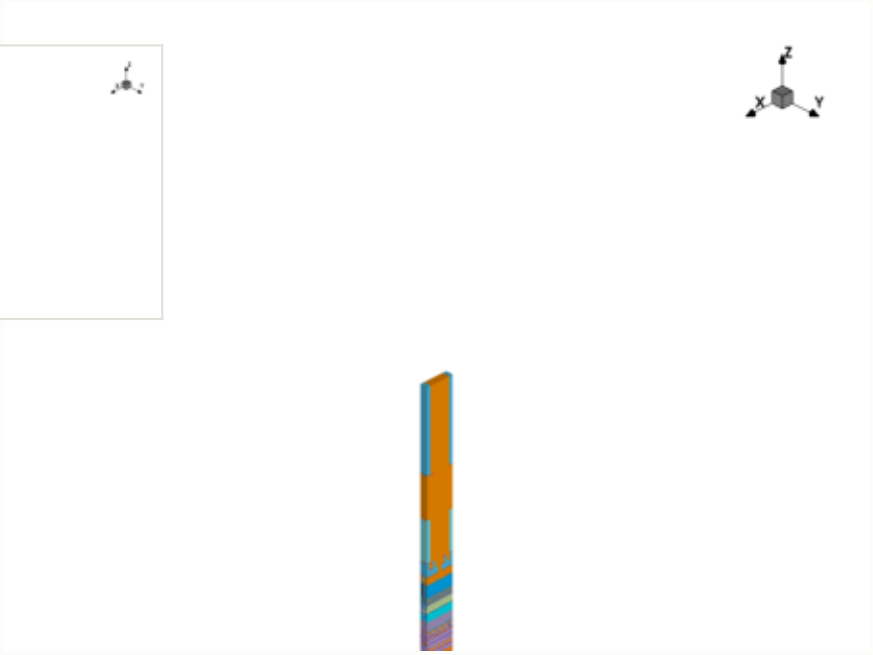
Step #9070: CMP

etch_cmp · CMP

0.169	743.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

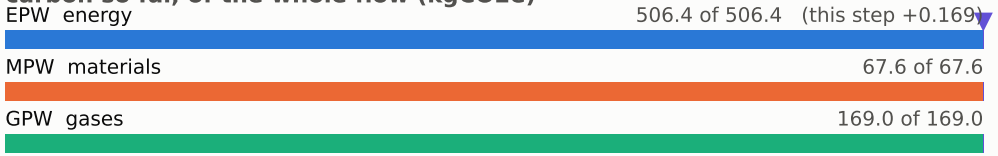
zoomed to fit



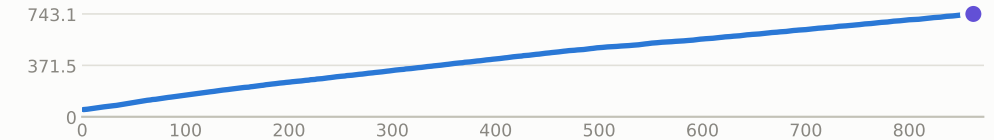
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

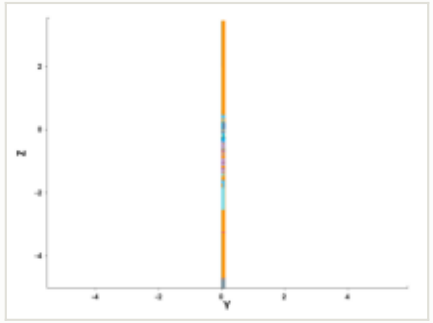
carbon so far, of the whole flow (kgCO2e)



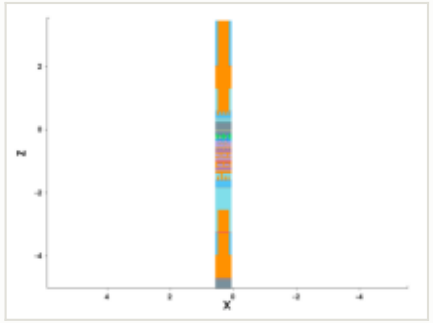
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



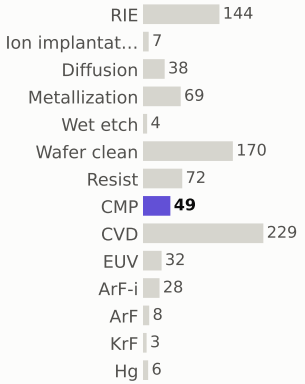
y cut at 0.049 µm



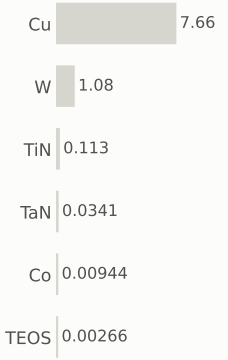
mask for this step

no mask for this step

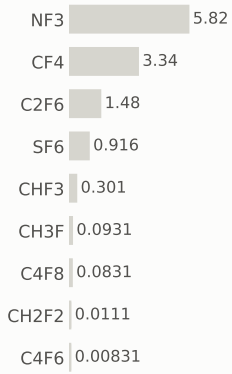
steps by process area



materials deposited (g)



gases emitted (g)



Step #9090: Wafer

flip_structure · Wafer

0	743.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

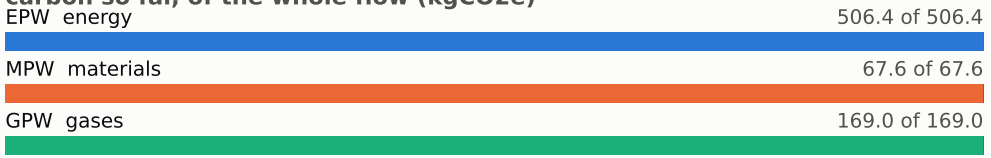
zoomed to fit

this step

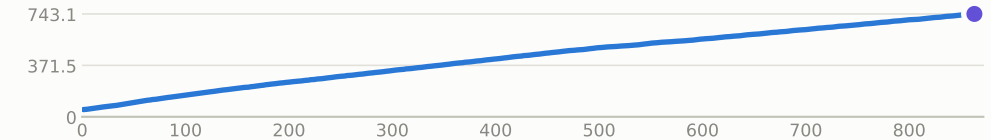
process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

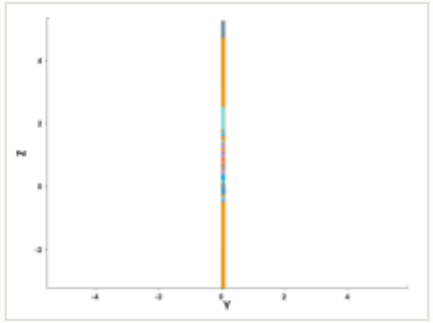
carbon so far, of the whole flow (kgCO2e)



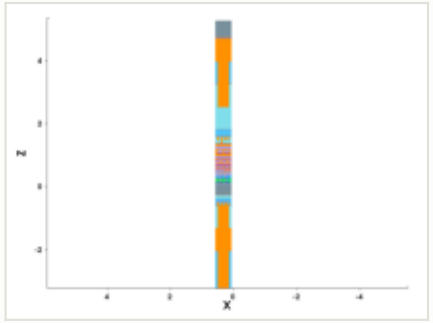
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.385 µm



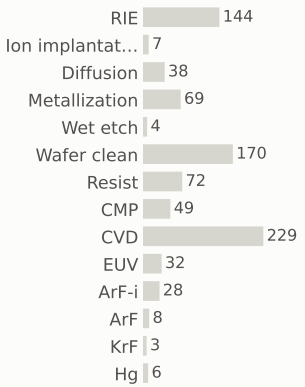
y cut at 0.049 µm



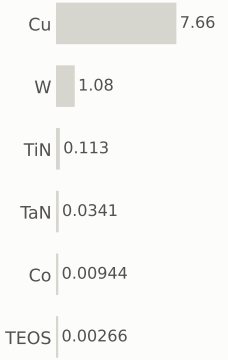
mask for this step

no mask for this step

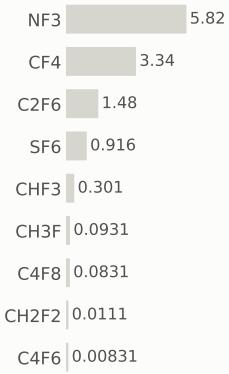
steps by process area



materials deposited (g)

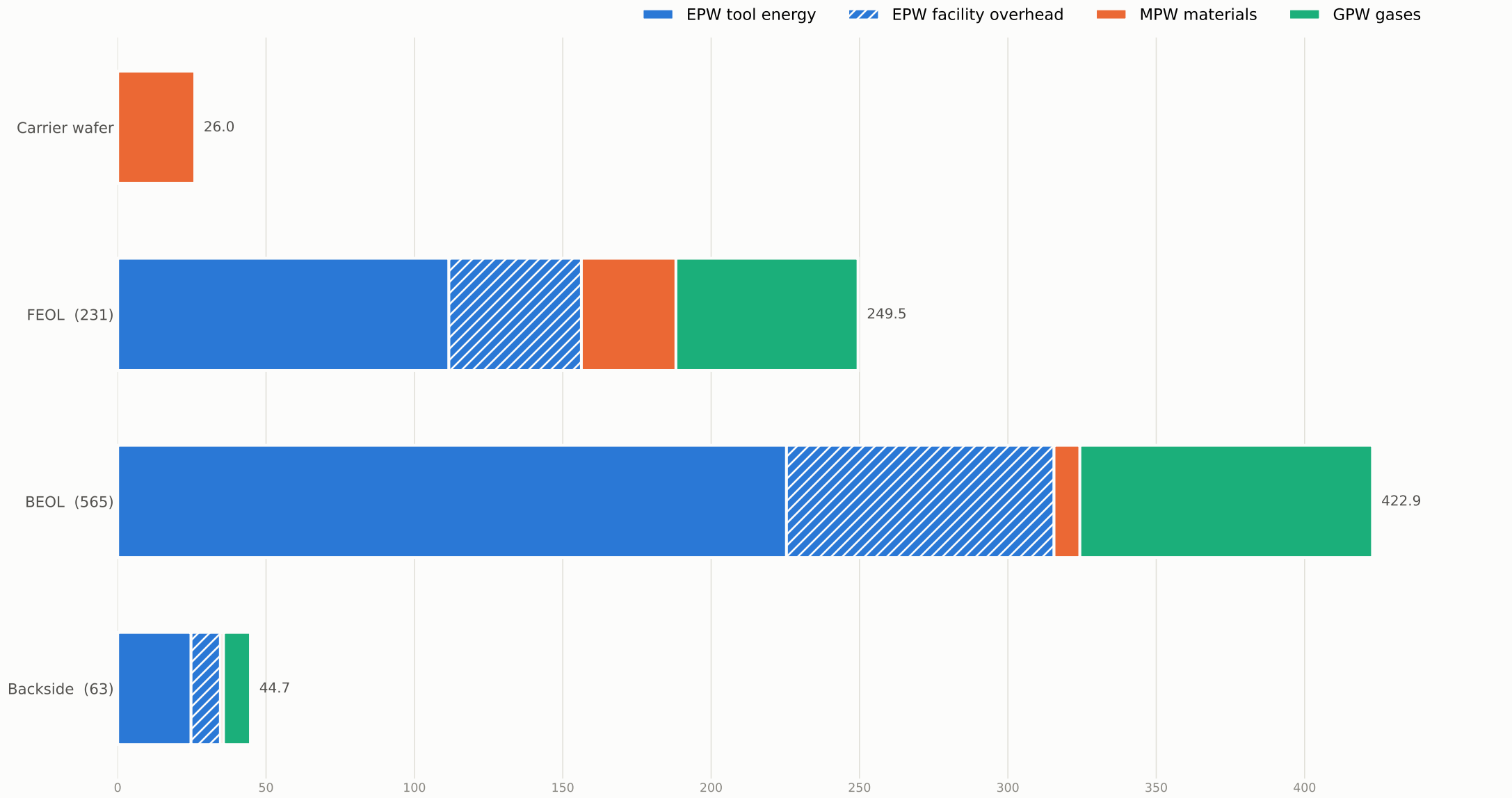


gases emitted (g)



Carbon by module

kgCO2e per wafer, in flow order (steps in brackets)



Totals by process area

kgCO2e per wafer; EPW = tool + facility overhead (overhead energy = tool energy x 0.4)

process area	steps	tool kWh	overhead kWh	EPW tool	EPW overhead	EPW	MPW	GPW	total
CVD	229	249.61	99.84	107.33	42.93	150.27	7.28	142.08	299.63
RIE	144	191.52	76.61	82.35	32.94	115.30	8.22	26.93	150.45
Metallization	69	138.00	55.20	59.34	23.74	83.08	0.13	0.00	83.21
EUV	32	104.00	41.60	44.72	17.89	62.61	0.00	0.00	62.61
Diffusion	38	50.54	20.22	21.73	8.69	30.43	0.00	0.00	30.43
Wafer_clean	170	47.60	19.04	20.47	8.19	28.66	0.00	0.00	28.66
Resist	72	20.16	8.06	8.67	3.47	12.14	0.00	0.00	12.14
CMP	49	13.72	5.49	5.90	2.36	8.26	0.00	0.00	8.26
ArF-i	28	10.08	4.03	4.33	1.73	6.07	0.00	0.00	6.07
Ion_implantation	7	9.31	3.72	4.00	1.60	5.60	0.00	0.00	5.60
ArF	8	2.64	1.06	1.14	0.45	1.59	0.00	0.00	1.59
Hg	6	1.98	0.79	0.85	0.34	1.19	0.00	0.00	1.19
Wet_etch	4	1.12	0.45	0.48	0.19	0.67	0.00	0.00	0.67
KrF	3	0.99	0.40	0.43	0.17	0.60	0.00	0.00	0.60
Wafer		0.00	0.00	0.00	0.00	0.00	52.00	0.00	52.00
all	859	841.27	336.51	361.75	144.70	506.44	67.62	169.02	743.09